

Molecular Dynamics and Learned Propagators

From Newton's equations to equivariant trajectory models,
with electrochemical systems in view

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Symbols and Conventions

The tables below provide a reference for the principal symbols and their units. Symbols are also defined in the text, where they are first needed.

Bold upright letters denote vectors or matrices. Angle brackets denote an ensemble average and an overbar a time average. The step of a numerical integrator is written δt , and the much longer stride of a learned propagator Δt .

Quantities are expressed in the units of `mdlab`: energy in electronvolts, length in ångströms, time in femtoseconds, mass in atomic mass units and temperature in kelvin. Other programs differ in the unit of time: the LAMMPS `metal` set uses picoseconds, and ASE measures time internally in ångströms times $\sqrt{\text{amu}/\text{eV}}$, about 10.18 fs. Times and velocities are converted explicitly wherever data pass between them.

Particles and phase space

Symbol	Meaning	Dimension or unit
N	Number of atoms	dimensionless
$\mathbf{r}_i, \mathbf{v}_i, \mathbf{p}_i$	Position, velocity and momentum of atom i	$\text{\AA}, \text{\AA fs}^{-1}, \text{amu } \text{\AA fs}^{-1}$
$\mathbf{a}_i$	Acceleration of atom i	$\text{\AA fs}^{-2}$
m_i	Mass of atom i	amu
$\mathbf{r}^N, \mathbf{p}^N$	All positions, all momenta	as above
Γ	A point in phase space, $(\mathbf{r}^N, \mathbf{p}^N)$	mixed
$\{A, B\}$	Poisson bracket of two quantities of the state	that of AB per energy $\times$ time
$\hat{L}$	Liouville operator, $\hat{L}A = \{A, H\} = \text{d}A/\text{d}t$	fs^{-1}
$\mathcal{A}$	Area of a region of a phase plane	energy $\times$ time
$\hat{L}_r, \hat{L}_p$	Parts of $\hat{L}$ that move the positions and the momenta	fs^{-1}
$\hat{L}_p^f, \hat{L}_p^s$	Parts of $\hat{L}_p$ that kick with the fast and the slow forces (Ch. 11)	fs^{-1}
$\tilde{H}$	Shadow energy kept by a step map	eV
Ω	Matrix of the symplectic condition $\mathbf{J}^T \Omega \mathbf{J} = \Omega$	dimensionless
$\mathbf{J}$	Jacobian matrix of a map of phase space	mixed
$\mathbf{r}_{ij}, r_{ij}$	Separation vector $\mathbf{r}_j - \mathbf{r}_i$ and its length	$\text{\AA}$

Energies and forces

Symbol	Meaning	Dimension or unit
U	Potential energy	eV
K	Kinetic energy	eV
E	Total energy, $K + U$	eV
H	Hamiltonian, whose value on a trajectory is E	eV
$\mathbf{F}_i = -\nabla_i U$	Force on atom i	$\text{eV } \text{\AA}^{-1}$
∇, ∇_i	Gradient; gradient with respect to the position of atom i	$\text{\AA}^{-1}$
$\mathbf{F}_i^{\text{nc}}$	Non-conservative part of the force on atom i	$\text{eV } \text{\AA}^{-1}$
$\mathbf{F}_i^{\text{c}}$	Constraint force on atom i (Ch. 11)	$\text{eV } \text{\AA}^{-1}$
$\mathbf{F}_i^{\text{f}}, \mathbf{F}_i^{\text{s}}$	Fast and slow parts of the force, split for multiple time steps	$\text{eV } \text{\AA}^{-1}$
W	Work done by a force along a path	eV
$\mathbf{P}, M, \mathbf{R}$	Total momentum, total mass and centre of mass of a group	$\text{amu } \text{\AA} \text{ fs}^{-1}, \text{amu}, \text{\AA}$
$\mathbf{V} = \mathbf{P}/M$	Velocity of the centre of mass (its length V only locally)	$\text{\AA} \text{ fs}^{-1}$
γ	Friction (drag) rate, drag force divided by mass and velocity	fs^{-1}
k	Stiffness of a spring or bond	$\text{eV } \text{\AA}^{-2}$
m_{r}	Reduced mass of a pair	amu
$\Phi, \mathbf{M}$	Force-constant matrix (Hessian of U) and mass matrix	$\text{eV } \text{\AA}^{-2}, \text{amu}$
$\mathbf{L}, \mathbf{G}$	Angular momentum and torque, about a stated point	$\text{amu } \text{\AA}^2 \text{ fs}^{-1}, \text{eV}$
ω	Angular velocity of a rigid turning	$\text{rad } \text{fs}^{-1}$
$I, \mathbf{I}$	Moment of inertia about an axis; inertia tensor	$\text{amu } \text{\AA}^2$
$\mathbf{e}_x, \mathbf{e}_r$	Unit vectors along an axis; away from a centre	dimensionless
δ_{ab}	Kronecker delta, 1 if $a = b$ and 0 otherwise	dimensionless
$\mathcal{L}, \mathcal{S}$	Lagrangian $K - U$; action $\int \mathcal{L} dt$	eV, eV fs
q_k, p_k	Generalised coordinate and its momentum $\partial \mathcal{L} / \partial \dot{q}_k$	that of q_k ; energy $\times$ time per unit of q_k
λ	Lagrange multiplier (an eigenvalue in the toolboxes of Ch. 4; in Ch. 13 the factor by which a thermostat scales the velocities)	force per unit of $\nabla \chi$ ($\text{eV } \text{\AA}^{-2}$ for a bond)
χ	Function whose zero is a constraint	that of the constraint ($\text{\AA}^2$ for a bond)
d	Length at which a bond is held (Ch. 6, 11; in Ch. 16 the number of dimensions, as in $2dDt$)	$\text{\AA}$
n_c	Number of held distances (constraints) (Ch. 11; with its argument, $n_c(r)$ is the running coordination number of Ch. 16)	dimensionless

Energies and forces *continued*

Symbol	Meaning	Dimension or unit
$\tilde{\mathbf{r}}_i$	Provisional position of atom i , before the constraints are applied	Å
η, ζ	Corrections of a held bond: $\delta t^2 \lambda / m_r$ for the positions; for the velocities	dimensionless, fs ⁻¹
$\varphi(r)$	Pair potential, the energy of two atoms at the distance r	eV
ε, σ	Depth and zero-crossing distance of a Lennard-Jones potential	eV, Å
ρ_i	Neighbour density of atom i in a many-body potential	eV ²
$\mathcal{W}$	Virial $\sum_i \mathbf{r}_i \cdot \mathbf{F}_i$	eV
ψ	Dihedral angle of a chain of four atoms	rad
k_e, q_i	Coulomb constant $e^2 / (4\pi\epsilon_0)$; charge of atom i	eV Å, e
$\mathcal{M}$	Madelung constant	dimensionless
α, ϵ_E	Ewald splitting parameter; size of the terms left out (in Ch. 12, α is the cubic coefficient of the chain of Fermi, Pasta, Ulam and Tsingou)	Å ⁻¹ , dimensionless
$\mathcal{C}(\mathbf{G}), \mathcal{S}(\mathbf{G})$	Sums of $q_i \cos(\mathbf{G} \cdot \mathbf{r}_i)$ and $q_i \sin(\mathbf{G} \cdot \mathbf{r}_i)$	e
$\Phi_s(\mathbf{d})$	Smooth part of $1/r$ summed over the copies of a separation (Ewald)	Å ⁻¹
ϵ_m	Machine epsilon, 2.2×10^{-16}	dimensionless

Time

Symbol	Meaning	Dimension or unit
δt	Integrator time step	fs
Δt	Stride of a learned propagator (in Ch. 16 also the interval between saved frames)	fs
ω	Angular speed or angular frequency	rad fs ⁻¹
ω_0	Angular frequency of small undamped swings	rad fs ⁻¹
$\tilde{\omega}$	Angular frequency of an oscillation computed with a step δt	rad fs ⁻¹
t_f	Length of a run	fs
τ	Unit of time of reduced units, $\sigma\sqrt{m/\epsilon}$	fs
$\nu, \tilde{\nu}$	Frequency (in Ch. 13 the collision rate of Andersen's thermostat; in Ch. 16 also the rate of events per molecule); wavenumber	fs ⁻¹ , cm ⁻¹
ϕ	Phase of a sinusoid (in the rotor of Ch. 12, the azimuth of the axis, with θ its angle from z)	rad
$\mathcal{T}$	Period of an oscillation	fs
$\omega_f, \omega_s, \mathcal{T}_f$	Angular frequencies of a fast and a slow spring; period of the fast one (Ch. 11)	rad fs ⁻¹ , fs

Geometry

Symbol	Meaning	Dimension or unit
θ	An angle	rad
$\mathbf{a}, \mathbf{b}, \mathbf{c}$	Lattice vectors of a cell, the columns of $\mathbf{h}$ (an acceleration carries a subscript, $\mathbf{a}_i$)	$\text{\AA}$
$\mathbf{s}$	Fractional coordinates, $\mathbf{h}^{-1}\mathbf{r}$	dimensionless
$\mathbf{d}$	A separation $\mathbf{r}_{ij}$ and its copies $\mathbf{d} + \mathbf{h}\mathbf{n}$ (Ch. 10)	$\text{\AA}$
$w_k, w_{\min}$	Perpendicular widths of a cell; the smallest	$\text{\AA}$
$\mathbf{a}^*, \mathbf{b}^*, \mathbf{c}^*, \mathbf{G}$	Reciprocal vectors; a wave vector of the cell (Ch. 10; $\mathbf{G}$ is the torque in Ch. 5)	$\text{\AA}^{-1}$

Thermodynamics

Symbol	Meaning	Dimension or unit
T, β	Temperature and inverse thermal energy, $\beta = 1/(k_B T)$	K, eV^{-1}
k_B	Boltzmann's constant, $8.617\,333\,262 \times 10^{-5}$	eV K^{-1}
N_f	Number of kinetic degrees of freedom	dimensionless
T_{kin}	Kinetic temperature, $2K/(N_f k_B)$ (Ch. 12)	K
$\mathcal{P}$	Probability, or probability density	dimensionless, or per unit of its variable
σ_x	Standard deviation of x (σ alone is the Lennard-Jones distance; in Chs 13 and 14 the strength of a noise)	that of x
$\Omega(E)$	Number of microstates of energy E , a phase-space volume (not the bold Ω)	dimensionless
S, F	Entropy; free energy $-k_B T \ln Z$ (a force is always bold, $\mathbf{F}_i$)	eV K^{-1} , eV
Z, Z_{NPT}, Ξ	Canonical, isothermal-isobaric and grand canonical partition functions	dimensionless, $\text{\AA}^3$, dimensionless
C_V	Heat capacity at fixed volume	eV K^{-1}
μ	Chemical potential	eV
T_0, τ_T	Target temperature of a thermostat; its coupling time (Ch. 13)	K, fs
ξ, Q	Friction variable of a Nosé-Hoover thermostat and its mass (Ch. 13)	fs^{-1} , eV fs^2
E_{eff}	Effective energy, $K + U$ less the heat a thermostat has added (Ch. 13)	eV
ε_o	Energy of an electronic orbital (Ch. 12)	eV
$\Gamma(a)$	Gamma function, $\int_0^\infty s^{a-1} e^{-s} ds$; distinct from the phase point Γ	dimensionless
ρ	Number density, atoms per unit volume (ρ_i is a neighbour density)	$\text{\AA}^{-3}$
$V, P, \mathbf{P}$	Volume, pressure and pressure tensor	$\text{\AA}^3$, GPa
$\mathcal{W}_{\alpha\beta}$	Virial tensor, $\sum_i r_{i\alpha} F_{i\beta}$, whose trace is $\mathcal{W}$ (Ch. 14)	eV

Thermodynamics *continued*

Symbol	Meaning	Dimension or unit
$\epsilon_{\alpha\beta}, \epsilon$	Strain; in the MTK piston, the strain of every length, $V = V_0 e^{3\epsilon}$ (Ch. 14)	dimensionless
B, κ_T, κ	Bulk modulus; isothermal compressibility $1/B$; a barostat's estimate of it (Ch. 14)	GPa, GPa ⁻¹
C_{11}, C_{12}, C_{33}	Elastic stiffnesses of a crystal (Ch. 14)	GPa
P_0, τ_P	Target pressure of a barostat; its time constant (Ch. 14)	GPa, fs
M_ϵ, p_ϵ	Mass and momentum of the MTK piston (Ch. 14)	eV fs ² , eV fs
A, M_P	Area of a face or wall; mass of a piston (Ch. 14)	Å ² , kg
$U_w, \mathbf{F}_i^w$	Energy of the walls; force of the walls on atom i (Ch. 14)	eV, eV Å ⁻¹
$F_\perp$	Total force of the atoms on a wall, at right angles to it (Ch. 14)	eV Å ⁻¹
D	Diffusion coefficient (Ch. 13)	Å ² fs ⁻¹
n, n_{eff}	Number of samples of a run; the number of independent samples they are worth, n/g (Ch. 15)	dimensionless
s, s'	Spread of samples about their mean: s^2 the sum of squared deviations over $n - 1$, s'^2 over n (Ch. 15)	that of the samples
$\text{cov}, \text{corr}(k)$	Covariance; autocorrelation function at a lag of k samples (k an index here) (Ch. 15)	square of the samples', dimensionless
g, τ_{int}	Statistical inefficiency; integrated correlation time $g\delta t/2$ (Ch. 15)	dimensionless, fs
$g_b, g(t_0)$	Statistical inefficiency of b samples; of the samples from t_0 on (Ch. 15)	dimensionless
τ_c	Correlation time of an exponential memory, $-\delta t / \ln c$ (Ch. 15)	fs
$\bar{A}_j, s_b$	Mean of block j ; spread of the block means, s_b^2 over $n_b - 1$ (Ch. 15)	that of the samples
n'	Number of samples after the start kept (Ch. 15)	dimensionless
S_m, σ_m	Sum of squared misses about a fitted line; spread of the misses (Ch. 15)	square of the samples', that of the samples
$\mathcal{P}(E \beta)$	Density of the energy at the inverse temperature β (Ch. 15)	eV ⁻¹
b, n_b	Length of a block in samples; number of blocks (Ch. 15)	dimensionless
$t_0, \delta A_0, \tau_\tau$	Start of the part of a run kept; offset of a transient at the start; its decay time (Ch. 15)	fs, that of A , fs
$\dot{A}, S_{tt}$	Slope of a drift; $\sum_i (t_i - \bar{t})^2$ (Ch. 15)	that of A per fs, fs ²
σ^*	Error asked of a mean (Ch. 15)	that of the mean
η_s, L	Shear viscosity; side of a cubic box (Ch. 15)	Pa s, Å
$g(r), g_{AB}(r)$	Radial distribution function; partial, of kind B around kind A (Ch. 16; always with its argument, g alone being the statistical inefficiency)	dimensionless

Thermodynamics *continued*

Symbol	Meaning	Dimension or unit
$n_c(r)$	Running coordination number, the partners within r (Ch. 16)	dimensionless
$S(G), G$	Structure factor, always with its argument (S alone is the entropy); length of the wave vector $\mathbf{G}$ (Ch. 16)	dimensionless, $\text{\AA}^{-1}$
$\mathbf{k}, \mathbf{k}'$	Wave vectors of an incoming and a scattered wave, $\mathbf{G} = \mathbf{k}' - \mathbf{k}$ (Ch. 16)	$\text{\AA}^{-1}$
N_r, V_r	Number of atoms in a region; its volume (Ch. 16)	dimensionless, $\text{\AA}^3$
$r_{\text{on}}, r_{\text{off}}$	Distances at which a bond forms and breaks (Ch. 16)	$\text{\AA}$
R_i	Covalent radius of atom i 's element (Ch. 16)	$\text{\AA}$
D_M, a_M	Depth and width of a Morse well, the D and a of Ch. 4 and 8 (Ch. 16)	$\text{eV}, \text{\AA}^{-1}$
X	Exposure, the sum over molecules of the time each could react (Ch. 16)	ps, or $\text{\AA}^3$ ps for pairs
k_b, k_f, k_h	Rate of breaking per bond; rate constant of forming; rate of hops per atom (Ch. 16)	$\text{ps}^{-1}, \text{\AA}^3 \text{ps}^{-1}, \text{ps}^{-1}$
K_{eq}, E_a	Equilibrium constant $n_{\text{AB}}V/n_{\text{A}}n_{\text{B}}$; activation energy of a rate (Ch. 16)	$\text{\AA}^3, \text{eV}$
$C_{vv}(t), C_m(t)$	Velocity autocorrelation function; the same weighted by the mass (Ch. 16)	$\text{\AA}^2 \text{fs}^{-2}, \text{eV}$
$\mathbf{M}_q, \sigma_e, \sigma_{\text{NE}}, H_R$	Charge displacement $\sum_i q_i \mathbf{r}_i$; conductivity; its Nernst-Einstein value; Haven ratio $\sigma_{\text{NE}}/\sigma_e$ (Ch. 16)	$e \text{\AA}, \text{S m}^{-1}, \text{S m}^{-1}$, dimensionless
$D_{\alpha\beta}, f$	Diffusion tensor; correlation factor of hops, $4D/k_{\text{h}}a^2$ on a surface (Ch. 16)	$\text{\AA}^2 \text{fs}^{-1}$, dimensionless
ℓ	Length of a counted hop (Ch. 16); distance between neighbouring sites of lithium in a gallery of graphite, $a/\sqrt{3}$ (Ch. 17)	$\text{\AA}$
a	Side of the hexagonal cell of a sheet of graphite, $2.46 \text{\AA}$, the distance between neighbouring hollows (Ch. 13, 16, 17)	$\text{\AA}$
t_{A}	ASE's unit of time, $\text{\AA}\sqrt{\text{amu}/\text{eV}} = 10.18 \text{fs}$ (Ch. 17)	fs
i_x, i_y, i_z	Image counts of an atom: the cell vectors by which unwrapping moves it (Ch. 17)	dimensionless
$\tilde{y}, \Delta(x)$	Straight line between two values of y ; its miss $y - \tilde{y}$ (Ch. 17)	that of y
j, σ_x	Flow of atoms per unit area and time; spread of a cloud of atoms along x (Ch. 16)	$\text{\AA}^{-2} \text{fs}^{-1}, \text{\AA}$
$\mathcal{E}, F_x$	Electric field; force along x (Ch. 16)	$\text{V } \text{\AA}^{-1}, \text{eV } \text{\AA}^{-1}$
r_{rep}	Range of the repulsion $Ae^{-r/r_{\text{rep}}}$ of the model salt (Ch. 16)	$\text{\AA}$
σ_{xy}	Spread of the shear stress (Ch. 16)	GPa
$i, \bar{z}$	The imaginary unit, $i^2 = -1$; the conjugate of a complex number z (Ch. 16)	dimensionless
$\hat{x}(v), X_k, \nu_{\text{N}}$	Fourier transform of $x(t)$; discrete transform of samples x_n ; Nyquist frequency $1/2\Delta t$ (Ch. 16)	that of x times fs, that of x , fs^{-1}

Thermodynamics *continued*

Symbol	Meaning	Dimension or unit
$N_s, W(N_s)$	Number of samples of a signal; number of operations of a discrete transform of that length (Ch. 16)	dimensionless
X_k^e, X_k^o	Discrete transforms of the even and the odd samples (Ch. 16)	that of x
$\mathcal{D}(\nu), \hat{C}_m(\nu)$	Vibrational density of states, area 3 per atom; transform of C_m (Ch. 16)	fs, eV fs
ω_d, ν_b	Angular frequency of a damped swing (Ch. 4); frequency of a bond's highest peak (Ch. 16)	rad fs ⁻¹ , fs ⁻¹
t_c, t_p	Longest lag of a correlation function used; period of a repeating signal (Ch. 16)	fs
$F(x), r_h$	Free energy along a coordinate x ; distance of an atom from the nearest hollow (Ch. 16)	eV, Å
h	Cell matrix, lattice vectors as columns	Å
$\langle A \rangle, \bar{A}$	Ensemble average and time average of A	that of A

Learning

Symbol	Meaning	Dimension or unit
w, θ	Fitted or trainable parameters (bold, a list)	model dependent
$L, \mathcal{L}$	Loss (Ch. 18; L is a length in earlier chapters)	target units squared for squared-error loss
$X_{ij} = \phi_j(x_i)$	Design matrix: basis function j at reading i	units of the basis function
λ	Ridge penalty (Ch. 18)	dimensionless for the scaled bases used here
A, λ_k	Hessian of the loss and its eigenvalues (Ch. 18)	loss per squared weight, for weights with common units
η	Learning rate (Ch. 18)	squared weight per loss
B	Number of readings in a batch (Ch. 18)	dimensionless
$\mathbf{h}_i^{(t)}$	Features of atom i after layer t	dimensionless
$\mathbf{m}_{ij}$	Message from atom j to atom i	dimensionless
$r_c, r_s, r_{\text{skin}}$	Cutoff radius; start of switching (Ch. 8); skin of a Verlet neighbour list	Å
$Y_\ell^m, D^{(\ell)}(R)$	Real spherical harmonic and Wigner matrix of rotation R	dimensionless

Orientation

0.1 The question this book prepares for

Molecular dynamics follows atoms by integrating Newton's equations in small steps of time. The forces may come from a prescribed function of the atomic positions, an electronic-structure calculation or a function fitted by machine learning. In each case, the step must resolve the fastest vibration retained in the model; Chapter 11 explains how fixing selected bond lengths removes some of these vibrations. In a graphite anode with a carbonate electrolyte, the stretches of carbon-hydrogen and carbon-carbon bonds hold it near a femtosecond, while lithium moving between sites, solvent reorganising around an ion and an interphase growing at the surface are many orders of magnitude slower.

Machine-learned potentials can reduce the cost of each force evaluation, but a cheaper evaluation does not by itself permit a longer stable step. Learned propagators predict the motion over a longer interval. TrajCast maps the positions and velocities at time t to those at $t + \Delta t$, with Δt ten to thirty times the reference step, and repeats the map to build a trajectory¹; ATOM predicts several future configurations in one evaluation and is pretrained across many molecules². TrajCast was tested on a molecule, crystalline quartz and liquid water; ATOM was tested on trajectories of isolated molecules. Whether such models can describe an electrochemical system, with its periodic electrode, mobile ions, long-range electrostatics and rare activated events, is the question this book prepares its reader to investigate.

Answering it draws on four bodies of knowledge, each resting on the one before: classical mechanics and its numerical integration, which define a trajectory and limit its step; statistical mechanics, which ties a trajectory to the ensemble it samples and explains what a thermostat does to it; the practice of molecular dynamics, from converged averages to the observables both papers use as evidence; and machine

learning, as far as the equivariant networks and transformers on which the two models are built. The present edition develops the first three and begins the fourth with regression and optimisation; the course map identifies the planned continuation.

0.2 How the book is organised

This working edition contains Chapters 0–18 and the trajectory gallery in Appendix F. The six-part map in [Figure 1](#) also shows the intended continuation through neural networks and learned propagators, so that the purpose of the earlier material is visible from the outset.

The six parts follow the order introduced above, and two links between them shape the reading. The Hamiltonian formulation in Part I supplies the conservation laws that TrajCast imposes on its predictions and the geometric property that keeps the standard integrator stable, so Part II can locate exactly the limit on the step that learned propagators aim to lift. Part IV ends with a study of lithium in graphite run with a foundation potential, and its trajectories provide data for the planned comparisons in Part VI, where the learned models will be judged on a system the reader has already simulated.

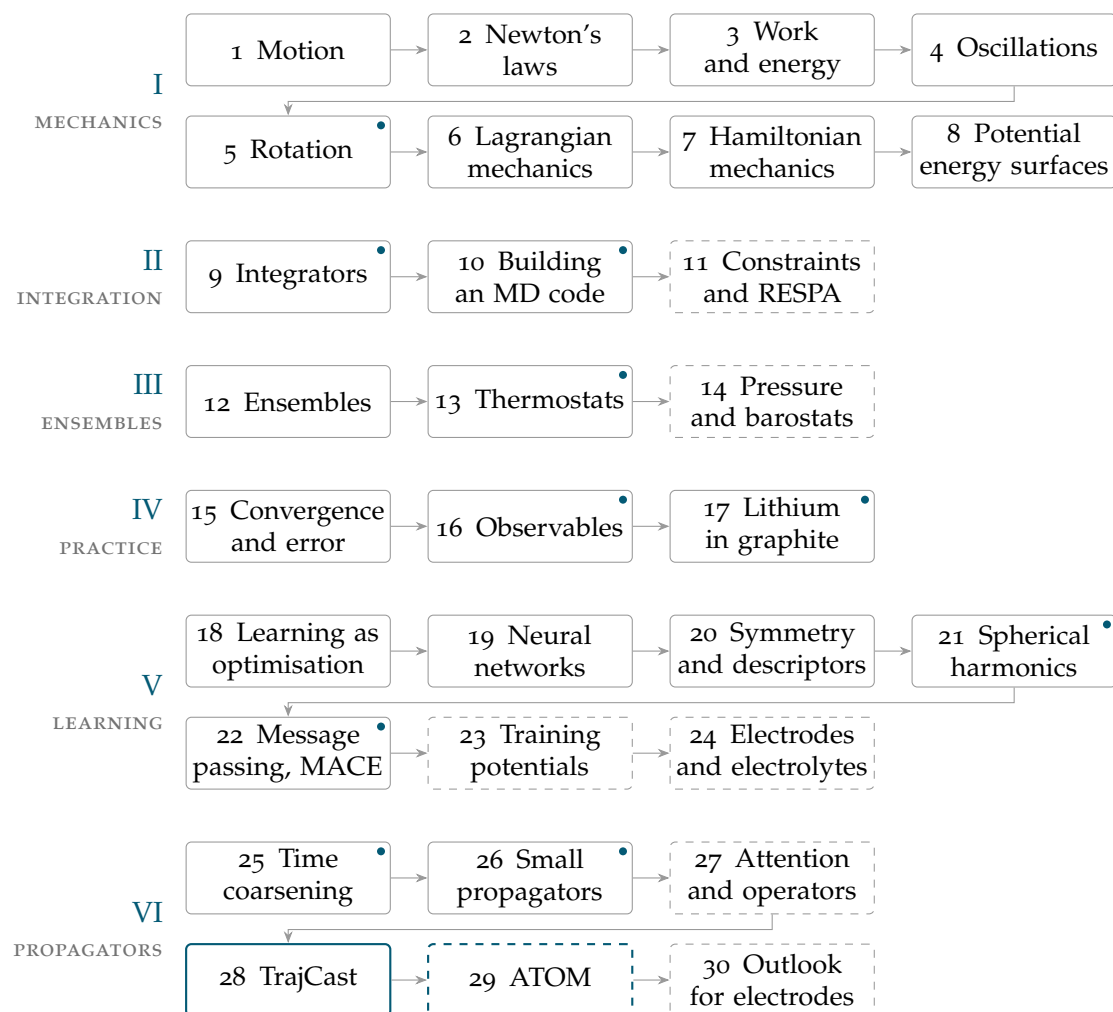


FIGURE 1: The course map, including the planned Chapters 19–30, in reading order. An accent dot marks a chapter on which the TrajCast chapter draws directly; a dashed outline marks a chapter off the fast track, the shortest path to reading the TrajCast code with understanding.

0.3 How to use the book

The book assumes school mathematics and nothing else: algebra, trigonometry, and the differentiation and integration of a function of one variable. It assumes no physics. Every further tool, from vectors and matrices to differential equations, partial derivatives and probability, is explained where it is first needed. Part I starts from what a velocity is and builds, one idea at a time, all the classical mechanics that molecular dynamics uses; each idea is first seen in an everyday system, a thrown ball or a block on a spring, and then applied to atoms by the same argument. Statistical mechanics, molecular simulation and machine learning are built in the same way from their foundations. The notebook demonstrations and chapter exercises also allow the material to be used for teaching; longer sections can be spread over several sessions.

Each chapter has a companion notebook with the same sections. The book carries the argument and the mathematics; the notebook carries the same ideas as code to run and vary, and a pointer in the text says what to explore. No programming is assumed: Notebook 01 opens with the little Python that the early notebooks use.

Make a working copy of each notebook, run its cells in order, then return to an example and change a parameter whose effect you can predict. Compare the result with that prediction and record the reason for any difference. The exercise cells leave space for your answers, with worked solutions folded below them. These calculations are intended to be revisited as new methods give another way to check an earlier result.

NOTEBOOK 01 *the thrown ball: move the time cursor, watch x , v and a*

The code grows chapter by chapter in a small package, `mdlab`, from a single thrown ball to a molecular dynamics engine with thermostats and analysis tools, then to the regression and optimisation routines of Chapter 18. The planned continuation develops equivariant networks. Every listing is taken from it, and every physical routine in it is tested against an analytic result or an independent implementation.

Four boxes set material apart from the argument.

KEY IDEA. The result of a section that later chapters rely on.

FROM ELECTRONIC STRUCTURE. A connection to electronic-structure theory, for readers who know it. The argument never depends on these boxes, and a reader new to the subject can pass over them.

IN PRACTICE. What goes wrong in real simulations, and how to diagnose it.

MATHEMATICS. A mathematical tool, such as the Taylor series, derived in full where it is first needed. A reader who already knows it can pass over the box.

Motion

Molecular dynamics follows the positions of atoms through time, computing step by step how each atom moves under the influence of the others. Before we can ask what causes that motion, we need a precise way of describing it: where an object is at each instant, how quickly its position is changing, and how that rate of change itself varies. This chapter develops the description for everyday motion, using a ball thrown upwards, a projectile and a stone whirled on a string, and then shows that it applies without modification to a system of atoms. The cause of motion, force, is deliberately set aside here and is introduced in Chapter 2.

1.1 Quantities and units

Every quantity in mechanics is a number multiplied by a unit. A height of 7.3 m is 7.3 times the metre, and the same height expressed in centimetres is 730 cm; the number depends on the unit chosen, whereas the height itself does not. This chapter requires only two units, the metre for length and the second for time, and every other unit it uses is built from these two, such as the metre per second (m s^{-1}) for the rate at which distance is covered. In this way of writing units, symbols set side by side are multiplied, and a power -1 means one divided by that unit, so that $\text{s}^{-1} = 1/\text{s}$ and m s^{-1} is the metre divided by the second.

The kind of quantity, as distinct from its size, is its dimension. We write L for length and T for time, so that a rate of covering distance, being a length divided by a time, has dimension LT^{-1} . Since a length cannot meaningfully be added to a time, only quantities of the same dimension can be added, subtracted, equated or compared. The number that a function such as $\sin$ acts on, called its argument, must be a pure number, one without a unit, because the function is defined for numbers: the sine of three metres has no meaning, and [Section 1.5](#) shows that an angle measured in radians is such a

pure number. Every correct equation must therefore balance in dimension, term by term. A car travelling at 20 m s^{-1} for 3 s covers $20 \times 3 = 60 \text{ m}$, and the dimensions of the product balance in the same way as the numbers:

$$\underbrace{\text{LT}^{-1}}_{\text{rate}} \times \underbrace{\text{T}}_{\text{time}} = \text{L}.$$

This check costs nothing to apply. An equation that fails to balance is certainly wrong, although one that balances is not thereby shown to be right, since a missing pure number such as $\frac{1}{2}$ leaves the dimensions unchanged.

Atoms require much smaller units than everyday objects. Neighbouring atoms are a few ångströms apart (Chapter 8), where $1 \text{ Å} = 10^{-10} \text{ m}$, and molecular dynamics advances in steps of about a femtosecond, $1 \text{ fs} = 10^{-15} \text{ s}$, for reasons derived in Chapter 9. A speed of 1 Å fs^{-1} is therefore

$$\frac{10^{-10} \text{ m}}{10^{-15} \text{ s}} = 10^{-10-(-15)} \text{ m s}^{-1} = 10^5 \text{ m s}^{-1}.$$

1.2 Velocity

Consider a ball thrown vertically upwards. We measure its height x above the hand, taking upwards as positive, and start the clock at the instant $t = 0$ at which the ball leaves the hand. Provided that air resistance is negligible, the heights recorded on a film of the throw are described by

$$x(t) = v_0 t - \frac{1}{2} g t^2, \quad (1.1)$$

where v_0 is the speed at which the ball is thrown and g is a constant, the same for every object near the Earth's surface. The value of g varies slightly from place to place, and we use its conventional standard value, 9.80665 m s^{-2} , read 'metres per second squared': a metre divided by a second twice, so that g has dimension LT^{-2} (Section 1.3 explains why such a unit arises). For the present, Equation (1.1) is simply a description of what is observed, and Chapter 2 derives it. Its two terms balance in dimension, as they must: $v_0 t$ has dimension $\text{LT}^{-1} \times \text{T} = \text{L}$, and $g t^2$ has dimension $\text{LT}^{-2} \times \text{T}^2 = \text{L}$. The figures and worked numbers of this chapter use $v_0 = 12 \text{ m s}^{-1}$.

Average velocity

To find how fast the ball is moving at a particular time t_1 , chosen anywhere during the flight, we begin with a finite interval, from t_1 to a slightly later time $t_1 + \tau$. The instant

t_1 is not the moment of the throw, which is $t = 0$; throughout the book a subscript 0 marks a value at that starting moment, as in v_0 , and other subscripts mark instants chosen later. During this interval the ball rises by $x(t_1 + \tau) - x(t_1)$, and dividing the rise by the duration τ gives the average velocity over the interval. On the graph of $x(t)$ the average velocity is the slope of the straight line, or secant, joining the points at t_1 and $t_1 + \tau$ (Figure 1.1a).

For the ball we can compute the average velocity exactly. The height at the later time follows from Equation (1.1) by replacing t with $t_1 + \tau$, and the square is then multiplied out:

$$\begin{aligned} x(t_1 + \tau) &= v_0(t_1 + \tau) - \frac{1}{2}g(t_1 + \tau)^2 && \text{replacing } t \text{ by } t_1 + \tau \\ &= v_0t_1 + v_0\tau - \frac{1}{2}g(t_1^2 + 2t_1\tau + \tau^2) && \text{expanding the square} \\ &= v_0t_1 - \frac{1}{2}gt_1^2 + v_0\tau - gt_1\tau - \frac{1}{2}g\tau^2 && \text{multiplying out and collecting terms.} \end{aligned}$$

The first two terms of the last line are exactly $x(t_1)$, so subtracting $x(t_1)$ leaves only the terms that contain τ ,

$$x(t_1 + \tau) - x(t_1) = v_0\tau - gt_1\tau - \frac{1}{2}g\tau^2,$$

and dividing by τ gives the average velocity,

$$\frac{x(t_1 + \tau) - x(t_1)}{\tau} = v_0 - gt_1 - \frac{1}{2}g\tau. \quad (1.2)$$

The average velocity depends on the length of the interval through its final term, $-\frac{1}{2}g\tau$; the first two terms are fixed once t_1 is chosen. With $v_0 = 12 \text{ m s}^{-1}$ and $t_1 = 0.4 \text{ s}$, for example, when the ball is 4.015 m up and still rising, substituting into Equation (1.2) gives, in metres per second with τ in seconds,

$$v_0 - gt_1 - \frac{1}{2}g\tau = 12 - 9.80665 \times 0.4 - \frac{1}{2} \times 9.80665 \times \tau = 8.077 - 4.903 \tau.$$

Each interval therefore lowers the average below 8.077 m s^{-1} by 4.903τ , and the table lists the result for three intervals, each worked out before rounding:

τ / s	$\frac{1}{2}g\tau / \text{m s}^{-1}$	average velocity / m s^{-1}
1.2	5.884	2.193
0.6	2.942	5.135
0.2	0.981	7.097

The shorter the interval, the closer the average comes to 8.077 m s^{-1} (Figure 1.1b). The reason is that the velocity of the ball, counted positive upwards, decreases throughout each interval: the ball slows as it rises, and in the longest interval, which runs past the top of the flight at 1.22 s (a time found in Section 1.3), it then falls ever faster. Its average velocity is therefore lower than its velocity at the start, t_1 ; a shorter interval leaves less time for the velocity to fall, and the average climbs back towards the velocity at t_1 itself. The value $v_0 - gt_1 = 8.077 \text{ m s}^{-1}$ is therefore the velocity at the instant t_1 , as we now make precise.

Velocity at an instant

The velocity at the single instant t_1 cannot be found by setting $\tau = 0$ in the quotient on the left of Equation (1.2), because both its numerator and its denominator would then vanish and $0/0$ has no value. Instead we ask what value the average approaches as the interval is made shorter and shorter. In Equation (1.2) the answer is plain: the term $-\frac{1}{2}g\tau$ can be made as small as we please by choosing τ small enough, while the first two terms do not depend on τ at all, so the average approaches $v_0 - gt_1$. This value is the limit of the average as τ tends to zero, and we define it to be the velocity at t_1 ,

$$v(t_1) = \lim_{\tau \rightarrow 0} \frac{x(t_1 + \tau) - x(t_1)}{\tau} = \left. \frac{dx}{dt} \right|_{t_1},$$

where the vertical bar with t_1 means that the derivative is evaluated at the instant $t = t_1$. The velocity is therefore the derivative of the position with respect to time. Geometrically, as τ shrinks the secant turns about the point at t_1 until it lies along the

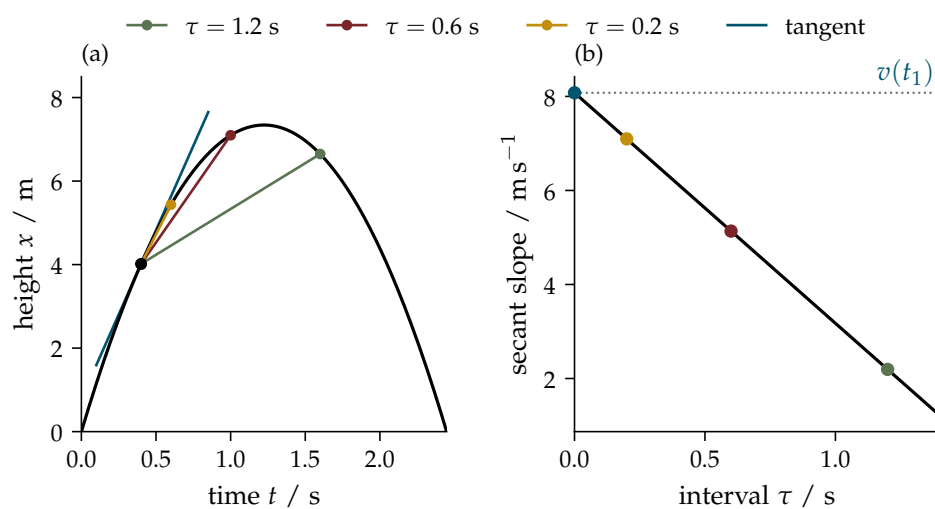


FIGURE 1.1: Velocity as a limit. (a) Height of the ball of Equation (1.1) against time, with secants from $t_1 = 0.4 \text{ s}$ over three intervals τ and the tangent at t_1 . (b) The slope of the secant against τ , which falls short of $v(t_1)$ by $g\tau/2$ and reaches it as $\tau \rightarrow 0$.

tangent there (Figure 1.1a), so the velocity is the slope of the tangent to the graph of $x(t)$. For the ball, since t_1 can be any time,

$$v(t) = v_0 - gt,$$

and at $t_1 = 0.4$ s the ball is moving upwards at 8.077 m s^{-1} .

DERIVATIVES OF POWERS. The calculation above is an instance of a general rule. For a power t^n , with n a whole number $1, 2, 3, \dots$, multiply out $(t + \tau)^n$, the product of n identical factors $(t + \tau)$. Each term of the result takes either t or τ from every factor. Taking t from all n factors gives t^n ; taking τ from exactly one factor, which can be any of the n , gives n terms each equal to $t^{n-1}\tau$; every other choice takes τ at least twice. Hence

$$(t + \tau)^n = t^n + nt^{n-1}\tau + \tau^2 \times (\text{terms in } t \text{ and } \tau).$$

Subtracting t^n , dividing by τ and letting $\tau \rightarrow 0$ removes every term that still contains τ , leaving

$$\frac{d}{dt} t^n = n t^{n-1}.$$

The derivative of a sum is the sum of the derivatives, and a constant factor passes through unchanged, because the difference quotient has both properties for every τ . A constant c has derivative zero, since its difference quotient is $(c - c)/\tau = 0$ for every τ . Applied to Equation (1.1), these rules give $d(v_0 t)/dt = v_0$ and $d(\frac{1}{2}gt^2)/dt = \frac{1}{2}g \times 2t = gt$, and hence $v = v_0 - gt$ once more.

The sign of the velocity carries the direction of motion: it is positive while the ball rises and negative while it falls. Its magnitude $|v|$ is the speed, which says how fast the ball moves without saying in which direction.

NOTEBOOK 01 shrink the interval and watch the secant turn into the tangent

1.3 Acceleration

Because the velocity of the ball changes during its flight, a further quantity is needed to describe how quickly it changes. This is the acceleration, the rate of change of velocity, defined from the velocity in exactly the way that the velocity was defined from the position,

$$a(t) = \lim_{\tau \rightarrow 0} \frac{v(t + \tau) - v(t)}{\tau} = \frac{dv}{dt} = \frac{d^2x}{dt^2}.$$

The last form records that the acceleration is obtained by differentiating the position twice. Derivatives with respect to time occur so often that they are also written with dots over the symbol, one dot for each derivative: $\dot{x} = dx/dt = v$ and $\ddot{x} = d^2x/dt^2 = a$. The dimension of acceleration is that of a velocity divided by a time, $LT^{-1}/T = LT^{-2}$, and it is measured in metres per second per second, $m s^{-2}$. For the ball, differentiating $v(t) = v_0 - gt$ term by term gives zero for the constant v_0 and $-g$ for the term $-gt$, so

$$a(t) = -g$$

at every instant of the flight. Figure 1.2 shows the height, the velocity and the acceleration on a common time axis.

The top of the flight

The ball reaches the top of its flight when it stops rising, that is, when its velocity is zero. Setting $v(t) = 0$ and solving for t ,

$$v_0 - gt = 0 \quad \Rightarrow \quad t = \frac{v_0}{g} = \frac{12}{9.80665} \text{ s} = 1.224 \text{ s},$$

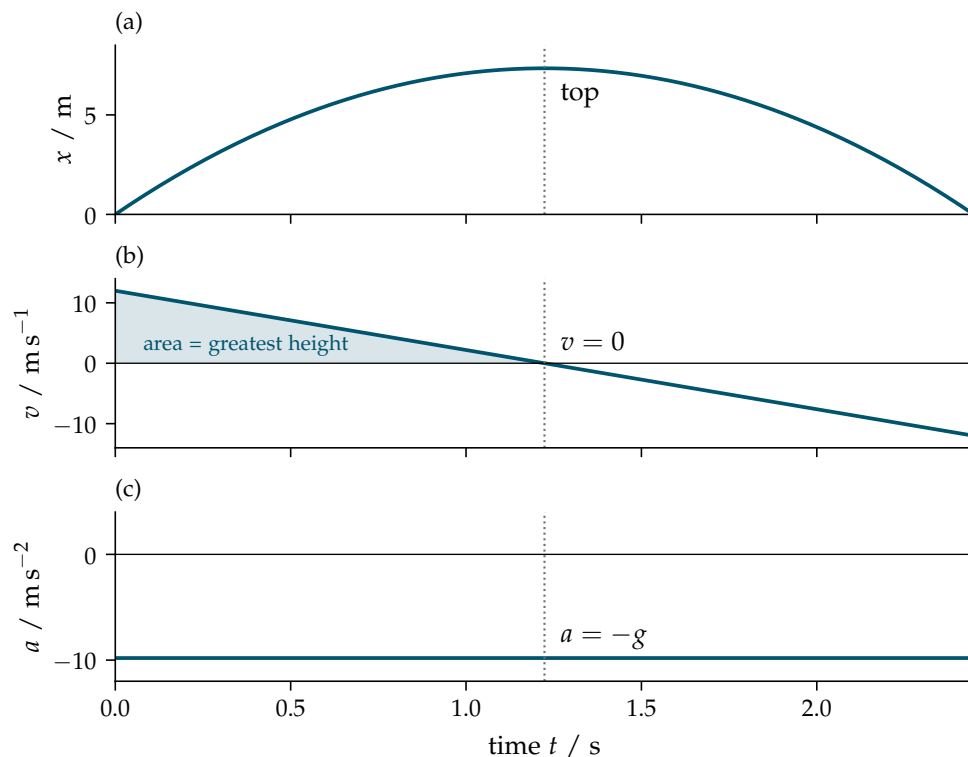


FIGURE 1.2: The thrown ball, $v_0 = 12 \text{ m s}^{-1}$. (a) Height, (b) velocity and (c) acceleration against time. At the top of the flight (dotted line) the velocity is zero and the acceleration is $-g$, as at every other instant. The shaded area under $v(t)$ up to the top equals the greatest height (Section 1.4).

where the arrow $\implies$ reads ‘which implies’: the equation on its right follows from the one on its left. At this instant the ball is momentarily at rest, yet its acceleration is still $-g$. There is no contradiction here, because a function can vanish at an instant without its derivative vanishing at that instant. The simplest example is $f(t) = t$, which is zero at $t = 0$ while its slope there is 1. In the same way the velocity $v(t) = v_0 - gt$ is zero at $t = v_0/g$, while its slope there, the acceleration, is $-g$. It is this non-zero acceleration that carries the velocity from positive values, through zero, to negative values, and it therefore turns the ball round.

KEY IDEA. Velocity is the rate of change of position, and acceleration is the rate of change of velocity. Either can be zero at an instant at which the other is not.

Speeding up and slowing down

The signs of v and a together determine whether a motion is speeding up or slowing down. The speed is the size of the velocity, $|v|$, and its square is $|v|^2 = v^2$. Since the speed is never negative, it grows exactly when its square grows, and the square spares us the absolute value. To see the rule, we therefore ask how v^2 changes in time. The square is a function of v , which is itself a function of t , and the chain rule (below) gives

$$\frac{d}{dt} v^2 = 2v \frac{dv}{dt} = 2va.$$

When v and a have the same sign their product is positive, v^2 grows and the speed increases; when their signs differ the product is negative and the speed decreases. While the ball rises, $v > 0$ and $a < 0$, so it slows down. Once it begins to fall, v and a are both negative, and it speeds up.

THE CHAIN RULE. Let f depend on t through an intermediate quantity $u(t)$. Over a short interval τ , u changes by $\Delta u = u(t + \tau) - u(t)$, and multiplying and dividing by Δu ,

$$\frac{f(u + \Delta u) - f(u)}{\tau} = \frac{f(u + \Delta u) - f(u)}{\Delta u} \times \frac{\Delta u}{\tau}.$$

As $\tau \rightarrow 0$, $\Delta u \rightarrow 0$ too, so the first factor approaches $f'(u)$, the derivative of f with respect to u , and the second approaches du/dt . At $\Delta u = 0$, define the first factor by its limiting value $f'(u)$; its product with $\Delta u/\tau$ is then zero, as is the left side. This also covers intervals on which u does not change. Hence

$$\frac{d}{dt} f(u(t)) = f'(u) \frac{du}{dt}.$$

With $f(u) = u^2$, so that $f'(u) = 2u$, and $u = v$, this gives the rate $2v \, dv/dt$ used above.

NOTEBOOK 01 move the time cursor along x , v and a

1.4 From acceleration back to motion

Differentiation takes us from position to velocity and from velocity to acceleration. To predict a motion we need to travel in the opposite direction, from a known acceleration back to the position, and the tool for this is integration.

Displacement as an area

Divide the time from 0 to t into many short intervals of length Δt , and let t_k be the start of the k th interval. Over an interval short enough that the velocity hardly changes, the object moves through approximately $v(t_k) \Delta t$, a velocity multiplied by a time. The total displacement is the sum of these small contributions,

$$x(t) - x(0) \approx \sum_k v(t_k) \Delta t,$$

where $\sum_k$, a capital Greek sigma for ‘sum’, means that the expression after it is added up for every interval $k = 1, 2, 3, \dots$ in turn. Each term $v(t_k) \Delta t$ is the area of a thin rectangle of height $v(t_k)$ and width Δt standing on the time axis. As $\Delta t \rightarrow 0$ the approximation becomes exact and the sum becomes an integral, the area under the graph of v . The same argument applied to the velocity, whose rate of change is a , gives the companion result, and together

$$v(t) = v(0) + \int_0^t a(t') \, dt', \quad x(t) = x(0) + \int_0^t v(t') \, dt'. \quad (1.3)$$

These are the fundamental theorem of calculus, which states that integrating a derivative from 0 to t returns the change in the original function. The variable of integration is written t' to distinguish it from the upper limit t . Where v is negative the area counts as negative, so the integral measures the signed displacement. For the ball, the area under $v(t)$ from the throw to the top of the flight is therefore the greatest height that it reaches, shaded in [Figure 1.2\(b\)](#).

[Equation \(1.3\)](#) also shows what is needed to determine a motion. Each integration introduces a starting value, the velocity $v(0)$ in the first and the position $x(0)$ in the second, and these do not follow from the acceleration. Two balls subject to the same acceleration but thrown at different speeds therefore follow different paths. When the

acceleration is known at every instant, and the initial position and velocity are given, the two integrals determine the motion uniquely. Predicting a motion thus reduces to finding its acceleration, and Chapter 2 shows that the acceleration is supplied by forces.

Constant acceleration

When the acceleration is a constant a , both integrals in Equation (1.3) can be evaluated directly. Writing $x_0 = x(0)$ and $v_0 = v(0)$, the first gives the velocity,

$$v(t) = v_0 + \int_0^t a \, dt' = v_0 + at,$$

and substituting this velocity into the second gives the position,

$$x(t) = x_0 + \int_0^t (v_0 + at') \, dt' = x_0 + \left[v_0 t' + \frac{1}{2} at'^2 \right]_0^t = x_0 + v_0 t + \frac{1}{2} at^2.$$

Together,

$$v(t) = v_0 + at, \quad x(t) = x_0 + v_0 t + \frac{1}{2} at^2, \quad (1.4)$$

and Equation (1.1) is the particular case $x_0 = 0$ and $a = -g$.

These two equations both contain the time, but many questions concern only where an object goes, not when it gets there: how high does the ball rise, for instance, or how fast is it moving at a given height? A relation free of t answers such questions directly. Provided that $a \neq 0$, the first equation gives $t = (v - v_0)/a$, and substituting this into the second eliminates the time:

$$\begin{aligned} x - x_0 &= v_0 \frac{v - v_0}{a} + \frac{a}{2} \frac{(v - v_0)^2}{a^2} && \text{substituting for } t \\ &= \frac{2v_0(v - v_0) + (v - v_0)^2}{2a} && \text{cancelling } a; \text{ common denominator } 2a \\ &= \frac{(v - v_0)(2v_0 + v - v_0)}{2a} && \text{taking out } (v - v_0) \\ &= \frac{(v - v_0)(v + v_0)}{2a} = \frac{v^2 - v_0^2}{2a} && 2v_0 - v_0 = v_0; \text{ difference of squares.} \end{aligned}$$

Multiplying both sides by $2a$ and rearranging,

$$v^2 = v_0^2 + 2a(x - x_0). \quad (1.5)$$

For the ball, $x_0 = 0$ and $a = -g$, so Equation (1.5) reads $v^2 = v_0^2 - 2gx$. The ball stops rising where $v = 0$, which gives $0 = v_0^2 - 2gx$ and hence the greatest height

$$x_{\max} = \frac{v_0^2}{2g} = \frac{12^2}{2 \times 9.80665} \text{ m} = 7.342 \text{ m}.$$

The ball returns to the hand when $x = 0$ again. Setting $x = 0$ in Equation (1.1) and taking out the common factor t ,

$$0 = v_0 t - \frac{1}{2} g t^2 = t \left(v_0 - \frac{1}{2} g t \right),$$

which is satisfied at the throw, $t = 0$, and again at $t = 2v_0/g = 2.447 \text{ s}$, twice the time to the top. At that moment Equation (1.5) with $x = 0$ gives $v^2 = v_0^2$, so $v = \pm v_0$; since the ball is then falling, $v = -v_0$, and it returns at the speed with which it was thrown.

When the acceleration depends on the position, as it will for an atom pushed by its neighbours, the integrand of Equation (1.3) contains the very motion we are trying to find. Except for special force laws, we cannot evaluate those integrals in closed form. A computer then advances the motion through many short intervals instead, and Section 1.8 develops the mathematics that makes this possible, once the trigonometric functions it relies on are in place.

NOTEBOOK 01 *the area under $v(t)$ up to the cursor*

1.5 Angles and the trigonometric functions

The motions met so far run along straight lines or arcs of a parabola. Many of the motions that matter later turn or repeat: a wheel spinning, a block bouncing on a spring, two bonded atoms vibrating, a molecule rotating. All of them are described with angles and with the sine and cosine functions, and this section builds those functions from the geometry of a circle, together with every property of them that the book uses.

Measuring angles

Consider a mark on the rim of a wheel of radius r . As the wheel turns through some angle, the mark travels a distance s along the rim, the length of the arc. The size of the angle could be given in degrees, but degrees are a convention, 360 to a full turn, chosen for their many divisors. A more natural measure follows from the arc itself. Doubling the angle doubles the arc, and doubling the radius at a fixed angle doubles it

too, so the ratio of arc to radius depends only on the angle. That ratio defines the angle in radians,

$$\theta = \frac{s}{r}.$$

Being a length divided by a length, θ is a pure number, which is what allows it to appear inside $\sin$ and $\cos$ (Section 1.1). One full turn carries the mark once round the circumference, $s = 2\pi r$, so a full turn is 2π radians. The two measures are therefore related by $2\pi \text{ rad} = 360^\circ$, and

$$1 \text{ rad} = \frac{180^\circ}{\pi} \approx 57.3^\circ, \quad 90^\circ = \frac{\pi}{2}, \quad 60^\circ = \frac{\pi}{3}, \quad 45^\circ = \frac{\pi}{4}, \quad 30^\circ = \frac{\pi}{6}.$$

From here on, every angle is in radians unless degrees are stated.

Sine and cosine

Draw a circle of radius 1, the unit circle, centred on the origin, and mark the point P reached by turning anticlockwise through an angle θ from the point $(1, 0)$ on the positive x axis (Figure 1.3a). The coordinates of P define the cosine and the sine of θ ,

$$P = (\cos \theta, \sin \theta).$$

For an angle between 0 and $\pi/2$, the line OP , the perpendicular from P to the x axis and the axis itself form a right-angled triangle with hypotenuse 1, so $\cos \theta$ is the adjacent side divided by the hypotenuse and $\sin \theta$ the opposite side divided by the hypotenuse, the familiar definitions from triangles. The circle definition has the advantage that it works for every angle, including angles beyond a quarter turn, negative angles and angles of more than a full turn. On a wheel of radius r the mark sits at $r(\cos \theta, \sin \theta)$, since scaling the unit circle by r scales both coordinates.

Plotting the two coordinates against the angle, as P goes once round the circle, unrolls the circle into the two waves of Figure 1.3(b). Several properties follow directly from the picture.

- Since P lies on a circle of radius 1, Pythagoras' theorem gives

$$\cos^2 \theta + \sin^2 \theta = 1,$$

where $\cos^2 \theta$ means $(\cos \theta)^2$. Both functions therefore lie between -1 and 1 .

- Turning a further 2π brings P back to where it was, so both functions repeat with period 2π : $\sin(\theta + 2\pi) = \sin \theta$ and $\cos(\theta + 2\pi) = \cos \theta$.

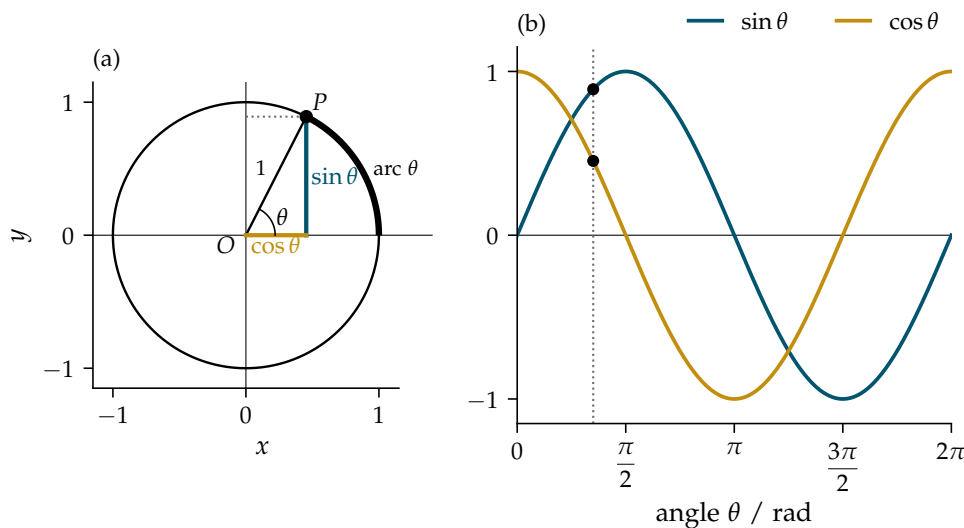


FIGURE 1.3: Sine and cosine from the circle. (a) The point P at angle $\theta = 1.1$ rad (63°) on the unit circle has coordinates $(\cos \theta, \sin \theta)$; the thick arc from $(1, 0)$ to P has length θ . (b) The two coordinates against θ over one full turn, with the values at $\theta = 1.1$ rad marked.

- Turning clockwise instead of anticlockwise reflects P in the x axis, which keeps its x coordinate and reverses its y coordinate: $\cos(-\theta) = \cos \theta$ and $\sin(-\theta) = -\sin \theta$.
- Reflecting P in the line $y = x$ swaps its coordinates. The reflection also turns the angle measured anticlockwise from the x axis into the same angle measured clockwise from the y axis, which itself lies at $\pi/2$, so the reflected point is at angle $\pi/2 - \theta$. Hence $\cos(\pi/2 - \theta) = \sin \theta$ and $\sin(\pi/2 - \theta) = \cos \theta$. Replacing θ by $-\theta$ in the second and using the previous property gives $\sin(\theta + \pi/2) = \cos \theta$: the cosine is the sine shifted by a quarter turn, which is visible in Figure 1.3(b).
- Reflecting P in the y axis keeps its y coordinate, reverses its x coordinate and carries θ to $\pi - \theta$, since the reflected point lies at angle θ measured clockwise from the negative x axis, which is at π . Hence $\sin(\pi - \theta) = \sin \theta$ and $\cos(\pi - \theta) = -\cos \theta$.
- At the quarter turns P sits on an axis, giving $\sin 0 = 0$, $\cos 0 = 1$, $\sin(\pi/2) = 1$, $\cos(\pi/2) = 0$, $\sin \pi = 0$ and $\cos \pi = -1$. At $\theta = \pi/4$ the point lies on the line $y = x$, so its two coordinates are equal, and $\cos^2 + \sin^2 = 1$ becomes $2\cos^2(\pi/4) = 1$; taking the positive root, since P lies in the first quarter, both equal $1/\sqrt{2}$.

The tangent function, $\tan \theta = \sin \theta / \cos \theta$ (defined where $\cos \theta \neq 0$), is the slope of the line OP ; despite its name it is not the tangent line of Section 1.2. Finally, the sine can be undone. For a number s between -1 and 1 , $\arcsin s$ is the angle between $-\pi/2$ and $\pi/2$ whose sine is s . Because $\sin(\pi - \theta) = \sin \theta$, the angle $\pi - \arcsin s$ has the same sine. When $0 < s < 1$ these are the only two angles between 0 and π with sine s , since

the horizontal line $y = s$ meets the upper half of the unit circle at exactly two points; when $s = 1$ the two coincide, since $\pi - \pi/2 = \pi/2$, and the single angle in that range whose sine is 1 is $\pi/2$.

The addition formulae

The sine and cosine of a sum of two angles are needed repeatedly, and they follow from one observation: the distance between two points on a circle does not change when the circle is turned. Let P and Q be the points on the unit circle at angles p and q . By Pythagoras' theorem, the square of the distance between them is

$$\begin{aligned} |PQ|^2 &= (\cos p - \cos q)^2 + (\sin p - \sin q)^2 \\ &= (\cos^2 p + \sin^2 p) + (\cos^2 q + \sin^2 q) - 2(\cos p \cos q + \sin p \sin q) \\ &= 2 - 2(\cos p \cos q + \sin p \sin q), \end{aligned}$$

where the second line multiplies out the two squares and groups the terms, and the third uses $\cos^2 + \sin^2 = 1$ for each angle. Now turn the circle clockwise through q (Figure 1.4). Turning lowers the angle of every point by q , so Q moves to angle 0, the point $(1, 0)$, and P moves to angle $p - q$. A turn moves the whole circle rigidly, so the distance between P and Q is unchanged. Computing it again from the new positions,

$$\begin{aligned} |PQ|^2 &= (\cos(p - q) - 1)^2 + \sin^2(p - q) \\ &= \cos^2(p - q) - 2\cos(p - q) + 1 + \sin^2(p - q) && \text{expanding the square} \\ &= 2 - 2\cos(p - q) && \text{using } \cos^2 + \sin^2 = 1. \end{aligned}$$

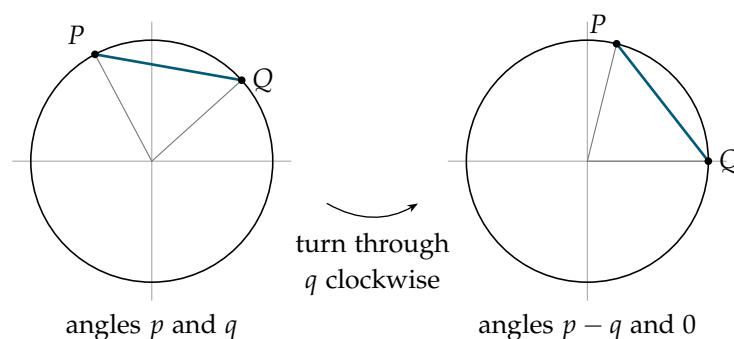


FIGURE 1.4: The chord PQ (teal) before and after the circle is turned clockwise through q . Turning does not change its length, which gives two expressions for $|PQ|^2$.

Equating the two expressions for $|PQ|^2$ gives the first addition formula,

$$\cos(p - q) = \cos p \cos q + \sin p \sin q.$$

The others follow from it. For the cosine of a sum,

$$\begin{aligned}\cos(p + q) &= \cos p \cos(-q) + \sin p \sin(-q) && \text{replacing } q \text{ by } -q \\ &= \cos p \cos q - \sin p \sin q && \text{as } \cos(-q) = \cos q, \sin(-q) = -\sin q.\end{aligned}$$

For the sine,

$$\begin{aligned}\sin(p + q) &= \cos(\pi/2 - (p + q)) && \text{as } \sin \theta = \cos(\pi/2 - \theta) \\ &= \cos((\pi/2 - p) - q) && \text{regrouping the angle} \\ &= \cos(\pi/2 - p) \cos q + \sin(\pi/2 - p) \sin q && \text{first addition formula} \\ &= \sin p \cos q + \cos p \sin q && \text{quarter-turn relations,}\end{aligned}$$

and replacing q by $-q$ as before gives $\sin(p - q) = \sin p \cos q - \cos p \sin q$. Together,

$$\begin{aligned}\cos(p \pm q) &= \cos p \cos q \mp \sin p \sin q, \\ \sin(p \pm q) &= \sin p \cos q \pm \cos p \sin q,\end{aligned}\tag{1.6}$$

with the upper signs taken together and the lower signs together. Setting $q = p$ gives the double-angle formulae $\sin 2p = 2 \sin p \cos p$ and $\cos 2p = \cos^2 p - \sin^2 p$.

The derivatives of sine and cosine

To differentiate $\sin \theta$ we need the limit of the difference quotient $[\sin(\theta + h) - \sin \theta]/h$ as $h \rightarrow 0$. Expanding the first term with [Equation \(1.6\)](#),

$$\frac{\sin(\theta + h) - \sin \theta}{h} = \frac{\sin \theta \cos h + \cos \theta \sin h - \sin \theta}{h} = \cos \theta \frac{\sin h}{h} - \sin \theta \frac{1 - \cos h}{h},$$

where the second step takes out the factor $-\sin \theta$ from the first and third terms of the numerator, $\sin \theta \cos h - \sin \theta = -\sin \theta (1 - \cos h)$, and divides each part by h separately. So the derivative depends on two limits, those of $\sin h/h$ and $(1 - \cos h)/h$ as $h \rightarrow 0$. Neither can be found by setting $h = 0$, which gives $0/0$ in both.

The first limit follows from comparing three areas in [Figure 1.5](#), for an angle h with $0 < h < \pi/2$. The point A is $(1, 0)$, P is the point at angle h on the unit circle, and T

is where the line OP , extended, meets the vertical line through A ; since OP has slope $\tan h$, the length AT is $\tan h$. The triangle OAP has base 1 and height $\sin h$, so its area is $\frac{1}{2} \sin h$. The sector OAP , the slice of the disc bounded by the arc, is the fraction $h/2\pi$ of the whole disc of area π , so its area is $\frac{1}{2}h$; this is where measuring h in radians matters. The larger triangle OAT has base 1 and height $\tan h$, so its area is $\frac{1}{2} \tan h$. Each region contains the one before, so

$$\frac{1}{2} \sin h \leq \frac{1}{2} h \leq \frac{1}{2} \tan h.$$

Multiplying by $2/\sin h$, which is positive, and using $\tan h = \sin h / \cos h$, this becomes

$$1 \leq \frac{h}{\sin h} \leq \frac{1}{\cos h}.$$

As $h \rightarrow 0$, the point P slides along the circle to $A = (1, 0)$, so $\cos h \rightarrow 1$ and $\sin h \rightarrow 0$, and the middle quantity is squeezed between two quantities that both approach 1, and must approach 1 itself; the same then holds for its reciprocal, $\sin h/h \rightarrow 1$. For negative h , $\sin(-h)/(-h) = \sin h/h$, since the sine reverses sign with its angle, so the same limit holds from both sides. For the second limit, multiplying numerator and denominator by $1 + \cos h$ and using $1 - \cos^2 h = \sin^2 h$,

$$\frac{1 - \cos h}{h} = \frac{1 - \cos^2 h}{h(1 + \cos h)} = \frac{\sin h}{h} \cdot \frac{\sin h}{1 + \cos h} \rightarrow 1 \times \frac{0}{2} = 0.$$

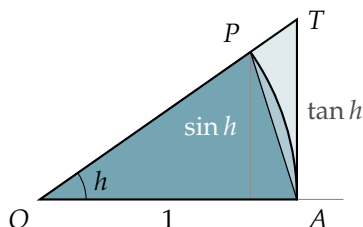


FIGURE 1.5: The three nested regions that bound $\sin h/h$. The triangle OAP (darkest) lies inside the sector OAP , which adds the sliver between the chord PA and the arc (medium); the sector lies inside the triangle OAT (lightest), where T lies on OP extended. Their areas are $\frac{1}{2} \sin h$, $\frac{1}{2} h$ and $\frac{1}{2} \tan h$.

Returning to the difference quotient, the first limit turns $\cos \theta \sin h/h$ into $\cos \theta$ and the second removes the term in $\sin \theta$, so

$$\frac{d}{d\theta} \sin \theta = \cos \theta, \quad \frac{d}{d\theta} \cos \theta = -\sin \theta, \quad (1.7)$$

where the second follows in the same way from the formula for $\cos(\theta + h)$:

$$\frac{\cos(\theta + h) - \cos \theta}{h} = -\sin \theta \frac{\sin h}{h} - \cos \theta \frac{1 - \cos h}{h} \rightarrow -\sin \theta.$$

Had the angle been measured in degrees, the sector area would have carried a factor $\pi/180$ and so would both derivatives, which is why radians are used throughout. The derivatives also agree with [Figure 1.3\(b\)](#): the sine rises most steeply where the cosine is largest, falls most steeply where the cosine is smallest, and is flat where the cosine is zero.

Sinusoids

A quantity that varies in time as

$$x(t) = A \sin(\omega t + \phi) \quad (1.8)$$

is called a sinusoid, and three constants describe it ([Figure 1.6](#)). The amplitude A , taken positive, is the largest value that x reaches, since the sine never exceeds 1 and does reach it. The angular frequency ω , in radians per second, sets how quickly the argument $\omega t + \phi$ advances, and since the sine repeats when its argument grows by 2π , the motion repeats after a time $2\pi/\omega$, the period. The number of repetitions per second is one divided by the period, $\omega/2\pi$, the frequency, measured in hertz, where one hertz is one repetition per second, $1 \text{ Hz} = 1 \text{ s}^{-1}$. The phase ϕ sets where in its cycle the motion starts: writing $\omega t + \phi = \omega(t + \phi/\omega)$ shows that x does at time t what $A \sin \omega t$ does at the later time $t + \phi/\omega$, so every crest arrives earlier by ϕ/ω . With $\phi = \pi/2$ the sinusoid becomes $A \cos \omega t$, by the quarter-turn relation. By the chain rule of [Section 1.3](#) and [Equation \(1.7\)](#), the velocity and acceleration of a sinusoid are

$$\dot{x} = A\omega \cos(\omega t + \phi), \quad \ddot{x} = -A\omega^2 \sin(\omega t + \phi) = -\omega^2 x.$$

The velocity is again a sinusoid. Since $\cos \theta = \sin(\theta + \pi/2)$, the velocity can be written as $A\omega \sin(\omega t + \phi + \pi/2)$, a sinusoid whose phase is larger by $\pi/2$, so its crests come earlier by $(\pi/2)/\omega$, a quarter of the period. The acceleration is always proportional to x and opposite to it. This last property will reappear in [Section 1.7](#) and again in Chapter 4.

NOTEBOOK 01 *turn the unit circle and watch $\sin \theta$ and $\cos \theta$; shape a sinusoid*

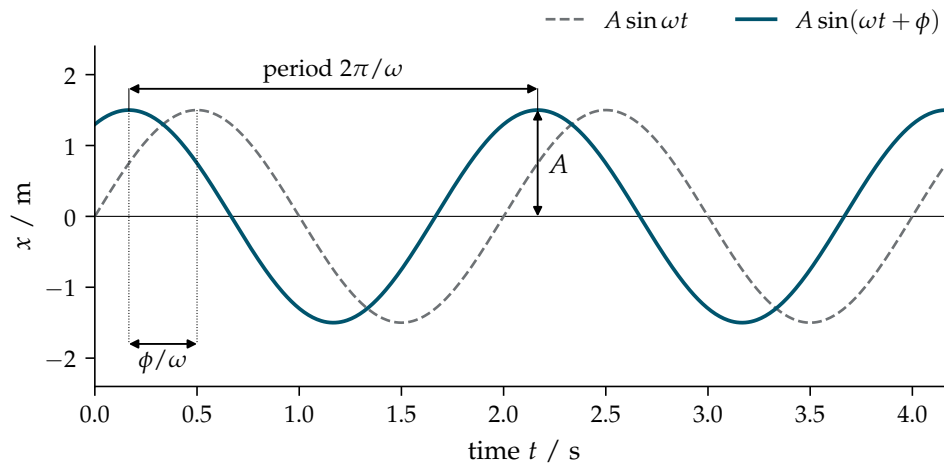


FIGURE 1.6: A sinusoid $x(t) = A \sin(\omega t + \phi)$ with $A = 1.5$ m, $\omega = \pi \text{ rad s}^{-1}$ (period 2 s) and $\phi = \pi/3$, against $A \sin \omega t$ (dashed). Every crest of the shifted wave arrives earlier by $\phi/\omega = 0.333$ s.

1.6 Motion in a plane and in space

Most motion does not take place along a single line, and positions must then be described by vectors. A position in space is a vector $\mathbf{r} = (x, y, z)$ measured from a chosen origin, and a moving object traces out a curve $\mathbf{r}(t)$, its path. The box below collects the properties of vectors that the book uses.

VECTORS. A vector $\mathbf{a} = (a_x, a_y, a_z)$ is added to another, or multiplied by a number, component by component. Drawn as an arrow from the origin to the point (a_x, a_y, a_z) , a vector has a length and a direction. Multiplying it by a positive number scales its length and keeps its direction, and a negative number also reverses the direction. The difference $\mathbf{b} - \mathbf{a}$ is the arrow from the tip of $\mathbf{a}$ to the tip of $\mathbf{b}$, since adding it to $\mathbf{a}$ gives $\mathbf{b}$. Its length, or magnitude, follows from Pythagoras' theorem applied twice, first in the xy plane and then with the z component, $|\mathbf{a}| = (a_x^2 + a_y^2 + a_z^2)^{1/2}$. The scalar product, or dot product, of two vectors is the number

$$\mathbf{a} \cdot \mathbf{b} = a_x b_x + a_y b_y + a_z b_z,$$

so that $\mathbf{a} \cdot \mathbf{a} = |\mathbf{a}|^2$. Expanding $|\mathbf{a} - \mathbf{b}|^2 = (a_x - b_x)^2 + (a_y - b_y)^2 + (a_z - b_z)^2$ gives $|\mathbf{a}|^2 + |\mathbf{b}|^2 - 2\mathbf{a} \cdot \mathbf{b}$, so

$$\mathbf{a} \cdot \mathbf{b} = \frac{1}{2}(|\mathbf{a}|^2 + |\mathbf{b}|^2 - |\mathbf{a} - \mathbf{b}|^2)$$

depends only on lengths, and is therefore unchanged when the axes are turned. We may then turn the axes until both vectors lie in the xy plane, and its geometric meaning follows from the addition formulae of Section 1.5. Write the two vectors

in terms of their lengths and the angles α and β that they make with the x axis, $\mathbf{a} = |\mathbf{a}|(\cos \alpha, \sin \alpha)$ and $\mathbf{b} = |\mathbf{b}|(\cos \beta, \sin \beta)$. Then

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}|(\cos \alpha \cos \beta + \sin \alpha \sin \beta) = |\mathbf{a}||\mathbf{b}| \cos(\alpha - \beta),$$

and $\alpha - \beta$ is the angle between the vectors (or its negative, or that plus a full turn; since the cosine is even and has period 2π , the value is the same). The scalar product is therefore the product of the two lengths and the cosine of the angle between them. In particular, two non-zero vectors are perpendicular exactly when their scalar product is zero, since $\cos(\pi/2) = 0$.

Velocity and acceleration are defined as before, now as vectors,

$$\mathbf{v} = \frac{d\mathbf{r}}{dt} = \left(\frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt} \right), \quad \mathbf{a} = \frac{d\mathbf{v}}{dt} = \frac{d^2\mathbf{r}}{dt^2},$$

so that each component is differentiated separately and obeys the one-dimensional results of the preceding sections. The speed is the length of the velocity vector, $v = |\mathbf{v}|$.

The velocity follows the path

Over a short interval τ the object moves from $\mathbf{r}(t)$ to $\mathbf{r}(t + \tau)$, and the vector $\mathbf{r}(t + \tau) - \mathbf{r}(t)$ joining the two positions is a chord of the path. Dividing it by the positive number τ changes its length but not its direction, and by the definition of the derivative the quotient approaches $\mathbf{v}(t)$ as $\tau \rightarrow 0$. At the same time the chord shrinks towards the point $\mathbf{r}(t)$ and turns into the tangent to the path there; indeed, the direction the chord approaches is what we mean by the tangent to a curve in space. The velocity is therefore directed along the tangent, in the direction of motion, whenever it is non-zero. No such argument applies to the acceleration, which may point in any direction relative to the path.

The projectile

A ball thrown with speed v_0 at an angle α above the horizontal, with $0 < \alpha < \pi/2$ so that it sets off both forwards and upwards, illustrates this distinction. We take x horizontal and y vertical, with the ball launched from the origin. With air resistance negligible, its acceleration is observed to be $\mathbf{a} = (0, -g)$, the same $-g$ as for the ball thrown straight up and nothing horizontally (Chapter 2 explains why), constant and directed downwards (the motion stays in the vertical xy plane, so the z components are zero and are dropped). Its initial velocity has length v_0 and makes the angle α with the x axis, so by the definition of sine and cosine its components are $v_0 \cos \alpha$ horizontally

and $v_0 \sin \alpha$ vertically. Equation (1.4) therefore applies to each component separately, with $a = 0$ horizontally and $a = -g$ vertically,

$$x(t) = v_0 \cos \alpha \, t, \quad y(t) = v_0 \sin \alpha \, t - \frac{1}{2}gt^2.$$

The horizontal motion is uniform, while the vertical motion is that of the ball of Section 1.2 thrown upwards at $v_0 \sin \alpha$, and neither component affects the other.

The shape of the path follows from eliminating t . The first equation gives $t = x/(v_0 \cos \alpha)$, a division that needs $\cos \alpha \neq 0$; a ball thrown straight up, $\alpha = \pi/2$, never leaves the line $x = 0$ and is the ball of Section 1.2. Substituting for t in the second equation,

$$\begin{aligned} y &= v_0 \sin \alpha \frac{x}{v_0 \cos \alpha} - \frac{g}{2} \frac{x^2}{v_0^2 \cos^2 \alpha} && \text{substituting for } t \\ &= x \tan \alpha - \frac{g x^2}{2v_0^2 \cos^2 \alpha} && \text{using } \sin \alpha / \cos \alpha = \tan \alpha, \end{aligned}$$

a parabola opening downwards. The ball returns to its launch height when $y = 0$; taking out the factor t from $y(t)$ gives $t(v_0 \sin \alpha - \frac{1}{2}gt) = 0$, so the flight lasts $t = 2v_0 \sin \alpha / g$. In that time the ball travels the horizontal distance

$$d = v_0 \cos \alpha \times \frac{2v_0 \sin \alpha}{g} = \frac{v_0^2 (2 \sin \alpha \cos \alpha)}{g} = \frac{v_0^2 \sin 2\alpha}{g},$$

its range, where the last step uses the double-angle formula of Section 1.5. Since $\sin 2\alpha$ is at most 1, with that value reached when $2\alpha = \pi/2$, the range is greatest at $\alpha = \pi/4$, or 45° . For $v_0 = 12 \text{ m s}^{-1}$ the range is $144 \sin 120^\circ / 9.80665 = 12.717 \text{ m}$ at a launch angle of 60° , and $144 / 9.80665 = 14.684 \text{ m}$ at 45° .

Figure 1.7(a) shows the velocity turning to follow the path while the acceleration continues to point downwards. The vertical component of the velocity, $v_0 \sin \alpha - gt$, vanishes at $t = v_0 \sin \alpha / g$, halfway through the flight, so at the top of the path the velocity is horizontal, and its scalar product with the downward acceleration is zero.

NOTEBOOK 01 launch a projectile with chosen speed and angle

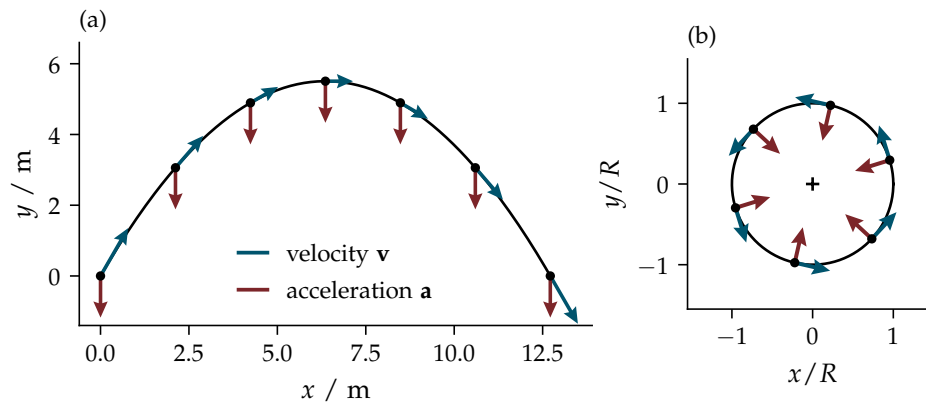


FIGURE 1.7: Velocity (teal) and acceleration (red) along two paths. (a) A projectile launched at 12 m s^{-1} , 60° above the horizontal, at seven equally spaced times. (b) Uniform motion on a circle of radius R (Section 1.7). Arrow lengths are proportional to the quantities they represent, on different scales for velocity and acceleration.

1.7 Circular motion

A stone whirled on a string at a steady rate moves around a circle of radius R , centred on the origin, at constant speed. Its angle from the x axis grows at a steady rate ω , the angular speed in radians per second, and we start the clock as it crosses the positive x axis, so that the angle is ωt at time t . By the definition of sine and cosine in Section 1.5, scaled to a circle of radius R , the position of the stone is

$$\mathbf{r}(t) = R (\cos \omega t, \sin \omega t),$$

and it completes one revolution when ωt has grown by 2π , in the period $2\pi/\omega$.

Velocity and acceleration

Each component is a sinusoid, and the chain rule with the derivatives of Equation (1.7) gives

$$\frac{d}{dt} \cos \omega t = -\omega \sin \omega t, \quad \frac{d}{dt} \sin \omega t = \omega \cos \omega t.$$

Differentiating the position once and then a second time therefore gives

$$\mathbf{v}(t) = R\omega (-\sin \omega t, \cos \omega t), \quad \mathbf{a}(t) = -R\omega^2 (\cos \omega t, \sin \omega t) = -\omega^2 \mathbf{r}(t). \quad (1.9)$$

The scalar product of the velocity with the position is

$$\mathbf{v} \cdot \mathbf{r} = R^2 \omega (-\sin \omega t \cos \omega t + \cos \omega t \sin \omega t) = 0,$$

so the velocity is perpendicular to the radius and hence tangent to the circle, as [Section 1.6](#) requires. Its magnitude is

$$v = R\omega (\sin^2 \omega t + \cos^2 \omega t)^{1/2} = R\omega,$$

which does not change in time. The radian, being a ratio of lengths, drops out of the units, so $R\omega$ is in metres per second. The acceleration $-\omega^2 \mathbf{r}$ points from the stone towards the centre, opposite to the position vector, with magnitude $\omega^2 R$. Since $\omega = v/R$, this magnitude can also be written $\omega^2 R = v^2/R$ ([Figure 1.7b](#)). A stone on a string of 0.5 m going round twice a second, for example, has $\omega = 2 \times 2\pi = 12.57 \text{ rad s}^{-1}$, a speed of 6.28 m s^{-1} and an acceleration of 79 m s^{-2} towards the centre, about eight times g .

Constant speed with non-zero acceleration

Uniform circular motion therefore has a constant speed but a non-zero acceleration at every instant, because acceleration measures any change of velocity, including a change of direction alone. The relation between speed and acceleration found in [Section 1.3](#) extends to vectors. Writing $v^2 = \mathbf{v} \cdot \mathbf{v} = v_x^2 + v_y^2 + v_z^2$ and applying that result to each component,

$$\frac{d}{dt}(v_x^2 + v_y^2 + v_z^2) = 2v_x a_x + 2v_y a_y + 2v_z a_z = 2\mathbf{v} \cdot \mathbf{a}.$$

The speed is therefore constant over an interval exactly when $\mathbf{v} \cdot \mathbf{a} = 0$ at every instant of that interval, which is the case on the circle. On the parabola of [Section 1.6](#), by contrast, $\mathbf{v} \cdot \mathbf{a} = v_0 \cos \alpha \times 0 + (v_0 \sin \alpha - gt)(-g) = -g v_y$, where $v_y = v_0 \sin \alpha - gt$ is the vertical velocity. This is negative while the ball rises ($v_y > 0$) and positive while it falls, so the ball slows as it rises and speeds up as it falls; only at the top is it zero.

The projection onto a line

The x coordinate of the stone, the projection of its motion onto the x axis, moves back and forth along a line as $x(t) = R \cos \omega t$, a sinusoid of amplitude R and angular frequency ω . The x component of [Equation \(1.9\)](#) gives this projection the acceleration $-R\omega^2 \cos \omega t$, which is $-\omega^2$ times $x(t)$ itself:

$$\frac{d^2 x}{dt^2} = -\omega^2 x,$$

the property of every sinusoid found in [Section 1.5](#). The acceleration of the projection is proportional to its distance from the centre and directed back towards it, so it is greatest at the ends of the swing and zero as it passes through the middle. [Section 2.8](#)

shows that a block attached to a spring obeys this equation, and Chapter 4 that the small vibrations of atoms obey it to a good approximation.

NOTEBOOK 01 *circular motion and its projection*

1.8 Short intervals and the Taylor series

For the ball, the acceleration was constant and the integrals of [Section 1.4](#) could be done by hand. For most motions, and for general systems of interacting atoms, the acceleration changes from moment to moment and the integrals cannot be done in closed form. A computer then advances the motion through many short intervals, and on each interval it needs an approximation to the position a short time τ ahead, built from what is known at the start of the interval. The Taylor series provides that approximation, and tells us how large its error is.

Matching a polynomial to a function

Near a time t , we look for a polynomial in τ that matches $x(t + \tau)$ as closely as possible,

$$x(t + \tau) \approx c_0 + c_1\tau + c_2\tau^2 + c_3\tau^3 + \cdots,$$

and we choose its coefficients so that the polynomial and the function have the same value, the same first derivative, the same second derivative, and so on, at $\tau = 0$. Each condition fixes one coefficient.

- Setting $\tau = 0$ leaves only c_0 , so $c_0 = x(t)$.
- Differentiating once with respect to τ gives $c_1 + 2c_2\tau + 3c_3\tau^2 + \cdots$, which at $\tau = 0$ is c_1 . The function's own derivative with respect to τ , with t held fixed, is by the chain rule with inner function $u = t + \tau$, whose derivative with respect to τ is 1, $\dot{x}(t + \tau)$; repeating this, its n th derivative is $x^{(n)}(t + \tau)$, the n th derivative of x evaluated at $t + \tau$. At $\tau = 0$ the first derivative is $\dot{x}(t)$, so matching gives $c_1 = \dot{x}(t)$.
- Differentiating twice gives $2c_2 + 6c_3\tau + \cdots$, which at $\tau = 0$ is $2c_2$, so $c_2 = \ddot{x}(t)/2$.
- Differentiating three times gives $6c_3 + \cdots$, so $c_3 = \dddot{x}(t)/6$, where $\ddot{x}$ is the third derivative.

Each differentiation brings down one more factor from the power of τ , so the n th differentiation of $c_n\tau^n$ produces $n \times (n - 1) \times \cdots \times 1 = n!$ times c_n , a product written

$n!$ and read ‘ n factorial’. After n differentiations the lower powers of τ have become zero and the higher ones still contain τ , so at $\tau = 0$ only $n!c_n$ survives, and matching the n th derivative $x^{(n)}(t)$ gives $c_n = x^{(n)}(t)/n!$. Collecting the coefficients gives the Taylor series,

$$x(t + \tau) = \sum_{n=0}^{\infty} \frac{x^{(n)}(t)}{n!} \tau^n = x(t) + \dot{x}(t) \tau + \frac{\ddot{x}(t)}{2} \tau^2 + \frac{\dddot{x}(t)}{6} \tau^3 + \dots,$$

with $0! = 1$. For the functions met in this book, polynomials, sines, cosines and exponentials, the infinite series converges to $x(t + \tau)$: adding more and more terms brings the sum as close to $x(t + \tau)$ as we please. In general, matching derivatives guarantees only that the series stopped after a finite number of terms has a small error, as described below, and that is all the book relies on. For a motion, $\dot{x} = v$ and $\ddot{x} = a$, so

$$x(t + \tau) = x(t) + v(t) \tau + \frac{1}{2} a(t) \tau^2 + \frac{1}{6} \ddot{x}(t) \tau^3 + \mathcal{O}(\tau^4). \quad (1.10)$$

The size of the error

The symbol $\mathcal{O}(\tau^4)$ stands for everything left out, and it records how quickly that remainder vanishes as the interval shrinks: a quantity is $\mathcal{O}(\tau^4)$, read ‘of order τ^4 ’, if its absolute value is no larger than some fixed constant times $|\tau|^4$ once $|\tau|$ is small enough. More generally, stopping the series after the term in τ^n leaves an error of order τ^{n+1} for a smooth function, one that can be differentiated as often as we like. To see why, call the error $D(\tau)$, the function $x(t + \tau)$ minus the polynomial of degree n , and take $\tau > 0$. By construction D and its first n derivatives with respect to τ vanish at $\tau = 0$. Since $n + 1$ differentiations remove a polynomial of degree n entirely, the $(n + 1)$ th derivative of D is $x^{(n+1)}(t + \tau)$. Let M be the greatest value of $|x^{(n+1)}|$ between t and $t + \tau$. By Equation (1.3) the n th derivative of D , which starts at zero, is the integral of the $(n + 1)$ th from 0 to τ , an area under a curve lying between $-M$ and M , so its size is at most $M\tau$. Integrating again, the size of the $(n - 1)$ th derivative is at most $\int_0^\tau Ms \, ds = M\tau^2/2$. Each further integration raises the power by one and divides by the new power, so after $n + 1$ integrations

$$|D(\tau)| \leq \frac{M |\tau|^{n+1}}{(n + 1)!},$$

and the same argument holds for $\tau < 0$. Halving the interval therefore divides this bound by 2^{n+1} , and the error itself by about that factor.

LOGARITHMIC AXES. Errors such as these span many powers of ten, so they are plotted on logarithmic axes, on which the marks 10^{-3} , 10^{-2} and 10^{-1} are equally spaced. Equal distances along such an axis stand for equal ratios, here a factor of

ten, rather than equal differences: the position of a positive number y along the axis is proportional to its logarithm $\log y$, the power to which 10 must be raised to give y , so that $\log 10^k = k$. Writing $y = 10^a$ and $z = 10^b$, so that $yz = 10^a \times 10^b = 10^{a+b}$, gives $\log(yz) = a + b = \log y + \log z$: the logarithm turns products into sums. Since $(10^a)^p = 10^{pa}$, it also turns a power into a multiple, $\log(\tau^p) = p \log \tau$. An error D that behaves as $D = C\tau^p$ therefore obeys

$$\log D = \log C + p \log \tau,$$

which is a straight line when $\log D$ is plotted against $\log \tau$. Its slope is the power p : a line that rises by one power of ten for every power of ten in τ has $p = 1$, and one that rises by four has $p = 4$. The constant C only moves the line up or down. The slope of an error plot thus gives the power with which the error falls, and two lines of equal slope differ only by a constant factor.

Figure 1.8 shows these properties for $x(t) = \sin t$ expanded about $t_1 = \pi/4$. By Equation (1.7) the derivatives of the sine cycle through $\sin, \cos, -\sin, -\cos$, so at $\pi/4$ every one of them is $\pm 1/\sqrt{2}$, and none is zero, so the leading term $x^{(n+1)}(t_1) \tau^{n+1}/(n+1)!$ of every error is present. The polynomial of degree 1 is the tangent line, the polynomial of degree 2 adds the bending of the graph, and each further degree follows the function over a wider range. Close to t_1 , the error of the polynomial of degree n falls as τ^{n+1} , so on the logarithmic axes of Figure 1.8(b) it is a straight line of slope $n+1$: for the cubic, halving τ from 0.1 to 0.05 divides the error by 16.15, close to $2^4 = 16$.

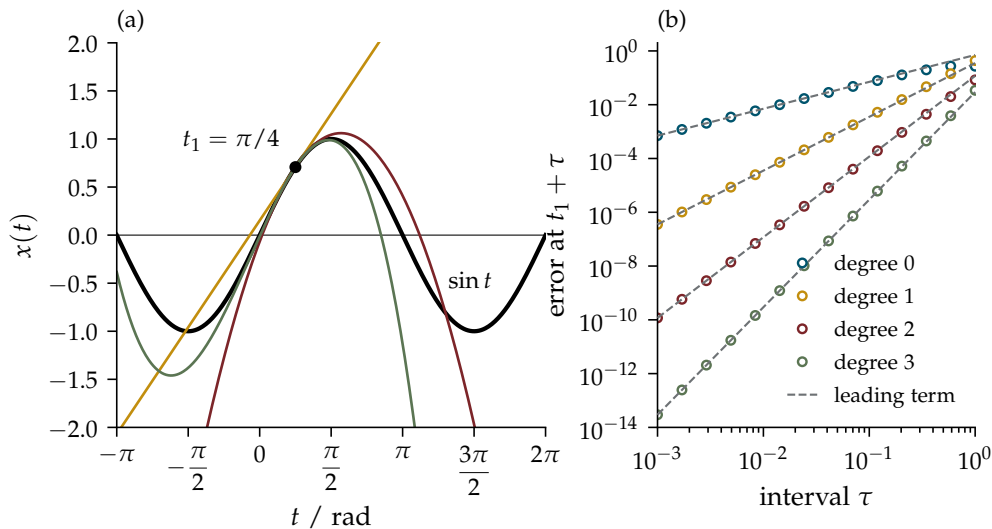


FIGURE 1.8: Taylor polynomials of $x(t) = \sin t$ about $t_1 = \pi/4$. (a) The polynomials of degrees 1, 2 and 3 (colours as in the key of b) follow the function ever more closely near t_1 . (b) The error of the polynomial of degree n at $t_1 + \tau$, including the constant polynomial of degree 0, with its leading term $|x^{(n+1)}(t_1)| \tau^{n+1}/(n+1)!$ (dashed); each error falls as τ^{n+1} .

Two series used later

Setting $t = 0$ in the Taylor series and writing y for τ , so that $x(y) = x(0) + \dot{x}(0)y + \ddot{x}(0)y^2/2! + \dots$, and using $\sin 0 = 0$ and $\cos 0 = 1$, the derivatives of $\sin y$ at $y = 0$ are $0, 1, 0, -1, 0, 1, \dots$ and those of $\cos y$ are $1, 0, -1, 0, 1, \dots$, so

$$\sin y = y - \frac{y^3}{3!} + \frac{y^5}{5!} - \dots, \quad \cos y = 1 - \frac{y^2}{2!} + \frac{y^4}{4!} - \dots \quad (1.11)$$

For small angles these give the approximations $\sin y \approx y$ and $\cos y \approx 1 - y^2/2$, which are used throughout the book; the first agrees with the limit $\sin h/h \rightarrow 1$ found in [Section 1.5](#).

Back to the ball

For the ball, the Taylor series ends after its third term, because $\ddot{x} = da/dt = 0$ when the acceleration is constant, and every higher derivative vanishes too. The expansion of [Section 1.2](#),

$$x(t_1 + \tau) = x(t_1) + (v_0 - gt_1)\tau - \frac{1}{2}g\tau^2,$$

was this series in disguise: its coefficients are $v(t_1)$ and $\frac{1}{2}a$. For a general motion, the first three terms of [Equation \(1.10\)](#) amount to [Equation \(1.4\)](#) with the acceleration held at its value at the start of the interval, and their error is of order τ^3 , small when the interval is short. Advancing a motion through many such intervals is how a computer follows it, and Chapter 9 develops this idea into the methods that molecular dynamics uses.

NOTEBOOK 01 *Taylor polynomials of growing order*

1.9 From balls to atoms

None of the preceding sections depended on the moving object being a ball, and the same description applies directly to atoms. A system of N atoms has a position vector for each atom, $\mathbf{r}_1(t), \dots, \mathbf{r}_N(t)$, and each atom has its own velocity $\mathbf{v}_i = d\mathbf{r}_i/dt$ and acceleration $\mathbf{a}_i = d\mathbf{v}_i/dt$. When all the positions are collected into a single list of $3N$ numbers, $\mathbf{r}^N = (\mathbf{r}_1, \dots, \mathbf{r}_N)$ (the superscript is a label, not a power), the motion of the whole system becomes a single path $\mathbf{r}^N(t)$: just as three numbers locate a point in space, $3N$ numbers locate a point in a space of $3N$ dimensions, which cannot be drawn but obeys the same component-by-component rules, and every result of the preceding sections applies to each of its components. The only change is in the units, which become $\text{\AA}$, fs , $\text{\AA fs}^{-1}$ and $\text{\AA fs}^{-2}$.

Velocities from frames

A molecular dynamics calculation does not record this path continuously. It stores frames instead, the positions of all the atoms at times separated by an interval τ , and the velocities must sometimes be recovered from these frames afterwards. The derivative, a limit as $\tau \rightarrow 0$, must then be replaced by a finite difference with τ fixed, and the Taylor series of Equation (1.10) tells us how large an error this introduces. We treat one coordinate x ; every component of every atom behaves in the same way.

The simplest choice is the forward difference, the secant slope of Section 1.2. Subtracting $x(t)$ from both sides of Equation (1.10) and dividing by τ ,

$$\frac{x(t + \tau) - x(t)}{\tau} = v(t) + \frac{1}{2}a(t)\tau + \mathcal{O}(\tau^2),$$

where the term $\frac{1}{6}\ddot{x}\tau^2$ and the remainder, now $\mathcal{O}(\tau^4)/\tau = \mathcal{O}(\tau^3)$, are both no larger than a constant times τ^2 when τ is small, and so together are $\mathcal{O}(\tau^2)$. So the forward difference returns the velocity with an error whose leading term, $\frac{1}{2}a\tau$, is proportional to τ . For the ball this is the term $-\frac{1}{2}g\tau$ of Equation (1.2).

A better estimate uses the frames on both sides of t . Writing the Taylor series forwards and backwards in time, by replacing τ with $-\tau$ in the second,

$$\begin{aligned} x(t + \tau) &= x + v\tau + \frac{1}{2}a\tau^2 + \frac{1}{6}\ddot{x}\tau^3 + \frac{1}{24}x^{(4)}\tau^4 + \mathcal{O}(\tau^5), \\ x(t - \tau) &= x - v\tau + \frac{1}{2}a\tau^2 - \frac{1}{6}\ddot{x}\tau^3 + \frac{1}{24}x^{(4)}\tau^4 + \mathcal{O}(\tau^5), \end{aligned}$$

where x , v , a , $\ddot{x}$ and $x^{(4)}$ are evaluated at t . The terms with even powers of τ have the same sign in both lines, and those with odd powers have opposite signs. Subtracting the second line from the first therefore cancels the even terms and doubles the odd ones,

$$x(t + \tau) - x(t - \tau) = 2v\tau + \frac{1}{3}\ddot{x}\tau^3 + \mathcal{O}(\tau^5),$$

where the remainder is of order τ^5 because the τ^4 terms also cancel. Dividing by 2τ gives the central difference,

$$\frac{x(t + \tau) - x(t - \tau)}{2\tau} = v(t) + \frac{1}{6}\ddot{x}(t)\tau^2 + \mathcal{O}(\tau^4),$$

whose error is proportional to τ^2 . Halving the interval therefore halves the error of the forward difference but quarters that of the central difference. In `mdlab` the central difference occupies a single line.

```

1 def central_difference(positions: ArrayLike, tau: float) -> NDArray:
2     x = np.asarray(positions, dtype=float)
3     return (x[2:] - x[:-2]) / (2.0 * tau)

```

Listings in this book are taken from `mdlab` without their docstrings, which give the shape and unit of every argument. Python counts from 0, so `x[2:]` holds the frames from the third onwards and `x[:-2]` those up to the third from last; entry k of their difference is $x_{k+2} - x_k$, the central difference at frame $k + 1$, so the first and last frames receive no velocity. `np.asarray` turns the input into a NumPy array of decimal numbers, and the annotations after the colons and the arrow only document the types expected and returned.

A vibrating coordinate

The atoms in molecules and solids vibrate (Chapter 4), and a vibrating coordinate can be modelled by the sinusoid of [Section 1.5](#) with zero phase, $x(t) = A \sin \omega t$, of amplitude A , angular frequency ω and period $2\pi/\omega$. For this motion the central difference can be evaluated exactly. Subtracting the two sine formulae of [Equation \(1.6\)](#),

$$\begin{aligned} \sin(p + q) - \sin(p - q) &= (\sin p \cos q + \cos p \sin q) - (\sin p \cos q - \cos p \sin q) \\ &= 2 \cos p \sin q. \end{aligned}$$

With $p = \omega t$ and $q = \omega \tau$, and multiplying through by $A/2\tau$,

$$\frac{x(t + \tau) - x(t - \tau)}{2\tau} = \frac{A \cos \omega t \sin \omega \tau}{\tau} = \underbrace{A \omega \cos \omega t}_{v(t)} \frac{\sin \omega \tau}{\omega \tau}, \quad (1.12)$$

where the last step multiplies and divides by ω so that the true velocity $v(t) = A\omega \cos \omega t$ appears. Every recovered velocity is therefore too small by the same factor, $\sin(\omega\tau)/(\omega\tau)$. To see how close this factor is to 1, we divide the series for $\sin y$ in [Equation \(1.11\)](#) by y ,

$$\frac{\sin y}{y} = 1 - \frac{y^2}{6} + \mathcal{O}(y^4).$$

The relative error, the error divided by the true velocity, is therefore

$$\frac{v(t) - v(t) \sin(\omega\tau)/(\omega\tau)}{v(t)} = 1 - \frac{\sin \omega\tau}{\omega\tau} \approx \frac{(\omega\tau)^2}{6}.$$

For an amplitude of $0.1 \text{ \AA}$ and a period of 20 fs , the angular frequency is $\omega = 2\pi/20 \text{ fs} = 0.3142 \text{ fs}^{-1}$. Frames saved every 1 fs then give $\omega\tau = 0.3142$ and an error of $0.3142^2/6 = 1.645\%$ by the approximation, against 1.637% exactly. Frames every 0.1 fs give 0.016% either way, and frames every 2 fs give 6.58% by the approximation against 6.45% exactly (Figure 1.9). When τ equals half a period, $\omega\tau = \pi$ and the factor $\sin \pi/\pi$ vanishes altogether: $x(t + \tau)$ and $x(t - \tau)$ are then a whole period apart and therefore equal, and the central difference detects no motion at all.

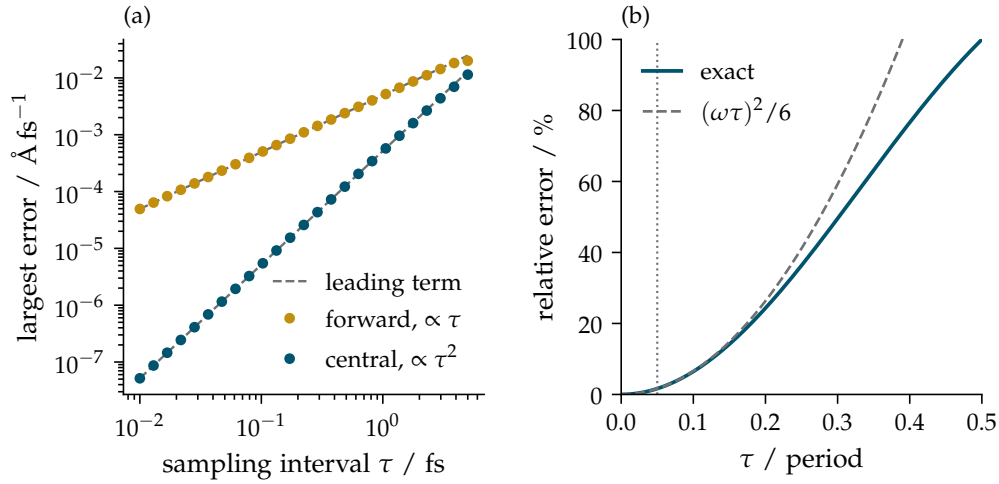


FIGURE 1.9: Velocities recovered from frames. A coordinate $x = A \sin \omega t$ with $A = 0.1 \text{ \AA}$ and period $2\pi/\omega = 20 \text{ fs}$ is sampled every τ . (a) Largest error of the forward and central differences, with their leading terms $A\omega^2\tau/2$ and $A\omega^3\tau^2/6$ (dashed). (b) Relative error of the central difference, $1 - \sin(\omega\tau)/(\omega\tau)$, against its leading term; the dotted line marks $\tau = 1 \text{ fs}$.

IN PRACTICE. Frames must be closely spaced compared with the fastest motion that they are intended to resolve, and Equation (1.12) makes this requirement quantitative. The coordinates of real atoms move as sums of sinusoids of several frequencies (Chapter 4). The central difference of a sum is the sum of the central differences of its terms, so it shrinks the velocity of each sinusoid by that sinusoid's own factor $\sin(\omega\tau)/(\omega\tau)$. If the interval satisfies $(\omega_{\max}\tau)^2/6 \leq \epsilon$ for the highest angular frequency $\omega_{\max}$ present, each sinusoid's velocity is recovered to a relative accuracy ϵ , and the error in the whole velocity is at most ϵ times the sum of the largest speeds $A\omega$ of the sinusoids, since the size of a sum is at most the sum of the sizes of its terms. This bounds the error relative to the typical speed of the motion, not relative to the velocity at each instant: where the sinusoids cancel and the velocity passes through zero, even a small error is a large fraction of it. For $x = \sin t - \frac{1}{2} \sin 2t$, in metres with t in seconds, and $\tau = 0.1$ s, $(\omega_{\max}\tau)^2/6 = 0.67\%$ with $\omega_{\max} = 2$. The true velocity is $\cos t - \cos 2t$, while by Equation (1.12) applied to each term the central difference is $\cos t (\sin 0.1/0.1) - \cos 2t (\sin 0.2/0.2)$. At $t = 0.01$ s, where the true velocity is $1.5 \times 10^{-4} \text{ m s}^{-1}$, the central difference gives $5.1 \times 10^{-3} \text{ m s}^{-1}$, 34 times too large. A path saved less frequently than the criterion allows should store the velocities directly. A related requirement limits the size of the step that molecular dynamics itself can take (Chapter 9).

NOTEBOOK 01 *sample an oscillation and recover its velocity*

What we now have

Position, velocity and acceleration are successive derivatives with respect to time, and a motion follows from its acceleration once the initial position and velocity are known. The sine and cosine are the coordinates of a point on the unit circle, and with angles measured in radians their derivatives are $\cos$ and $-\sin$; and every sinusoid has an acceleration proportional and opposite to its displacement. In two or three dimensions position, velocity and acceleration are vectors; the velocity is always directed along the path, whereas the acceleration may point in any direction, and in uniform circular motion it points towards the centre. The Taylor series approximates a motion over a short interval with an error that falls as a known power of the interval. The same definitions apply to a system of atoms, whose motion is a path in $3N$ dimensions, and velocities recovered from saved frames carry an error determined by the interval between the frames. What determines the acceleration has so far been left open, and Chapter 2 answers that question by introducing force.

Exercises

Full solutions are in ‘[Solutions to the exercises](#)’ at the end of the book, and Notebook 01 checks every numerical answer.

EXERCISE 1.1 (*dimensions*). Let x be a height, t a time, v and v_0 velocities and g an acceleration. Which of the following equations balance in dimension? (i) $x = v_0 t^2$; (ii) $v^2 = v_0^2 - 2gx$; (iii) $t = \sqrt{2x/g}$; (iv) $x = v_0 e^{-gt}$.

EXERCISE 1.2 (*a higher throw*). A ball is thrown vertically upwards at 15 m s^{-1} . Find the time it takes to reach the top, the greatest height it reaches and its speed when it returns to the hand.

EXERCISE 1.3 (*displacement and distance*). An object moves along a line with velocity $v(t) = 3t^2 - 12$, in metres per second for t in seconds. At what time $t \geq 0$ is it at rest, and what is its acceleration at that instant? Find its displacement between $t = 0$ and $t = 3 \text{ s}$, and the total distance it travels over the same interval, $\int |v| dt$.

EXERCISE 1.4 (*radians*). Express 1.1 rad in degrees and 135° in radians, and find the length of the arc cut off by an angle of 135° on a wheel of radius 0.35 m .

EXERCISE 1.5 (*an identity*). Starting from the double-angle formula $\cos 2\theta = \cos^2 \theta - \sin^2 \theta$ and $\cos^2 \theta + \sin^2 \theta = 1$, show that $\sin^2 \theta = \frac{1}{2}(1 - \cos 2\theta)$. Use it to find $\sin(\pi/8)$ exactly, and give its value to four decimal places.

EXERCISE 1.6 (*a sinusoid*). A coordinate moves as $x(t) = 0.2 \sin(5t + \pi/6)$, in metres for t in seconds. Find its amplitude, period and frequency, its largest speed and largest acceleration, and the first time after $t = 0$ at which x is greatest.

EXERCISE 1.7 (*range*). Show that the range $d = v_0^2 \sin 2\alpha / g$ is greatest when $\alpha = 45^\circ$. For $v_0 = 12 \text{ m s}^{-1}$, find the two launch angles that give a range of 10 m .

EXERCISE 1.8 (*an orbit*). A point moves anticlockwise around a circle of radius $1.0 \text{ \AA}$ at a constant speed of $0.02 \text{ \AA fs}^{-1}$. Find the magnitude of its acceleration and its period.

EXERCISE 1.9 (*a Taylor estimate*). Use the series of [Equation \(1.11\)](#) to approximate $\cos 0.1$ by its terms up to y^2 . Estimate the error from the first term omitted, and compare with the exact value.

EXERCISE 1.10 (*sampling*). Show that for $x = A \sin \omega t$ the largest error of the forward difference, relative to the largest speed $A\omega$, is approximately $\omega\tau/2$ when $\omega\tau$ is small. For a coordinate with a period of 20 fs , find the largest frame interval τ for which velocities recovered by central differences are within 1% .

Forces and Newton's laws

Chapter 1 described motion through position, velocity and acceleration. Once the acceleration is known at every instant, the starting position and velocity fix the motion, but we have yet to establish what determines that acceleration. Newton's laws connect it to the pushes and pulls on a body, meaning the object whose motion we follow. We first treat the body as a particle: a point that carries its mass, so that we can study its motion from place to place before considering rotation in Chapter 5.

To use those pushes and pulls in a calculation, we need to measure them as forces and establish the frames of reference in which the laws hold. We then introduce mass and apply the laws to a falling ball, a bead sinking through a thick liquid and a block on a spring. When several bodies interact, adding their equations reveals a conserved quantity, the total momentum, provided that the external forces sum to zero. The same calculation gives the equations of molecular dynamics for a collection of atoms. Chapter 3 develops a further consequence of these equations: the conditions under which the total energy is conserved.

2.1 Force and the first law

In everyday speech a force is a push or a pull: a hand pushing a door, a rope pulling a sledge, a magnet pulling a pin. To build a law of motion on forces we first need to measure them, and this section shows how, before stating the first of Newton's three laws.

Measuring a force

A spring stretches when it is pulled, and the harder the pull, the further it stretches. A spring with a scale beside it, called a spring balance, therefore turns a pull into a

reading. We can define a unit of force as the pull that stretches one particular standard spring by one particular amount. Two identical standard springs pulling side by side, each stretched by that amount, then exert two units, three springs exert three units, and so on (Figure 2.1a). In this way a force acquires a size, a number of units. Section 2.3 replaces this home-made unit by the standard one, the newton.

A force also has a direction, the direction in which it pulls, so it is a vector. When several forces act on one body together, experiment shows that their combined effect on its motion from place to place is that of a single force equal to their vector sum, called the net force,

$$\mathbf{F} = \mathbf{F}_1 + \mathbf{F}_2 + \cdots ,$$

in the sense in which a net profit is what remains when everything is added up. A pull of 4 units towards the east and a pull of 3 units towards the north, for example, are the vectors $(4,0)$ and $(0,3)$ when the x axis points east and the y axis north. Adding component by component (Section 1.6) gives $(4,3)$. This is the same as drawing the second arrow from the tip of the first, since starting at the tip of $(4,0)$ and moving by $(0,3)$ ends at $(4,3)$ (Figure 2.1b). The two pulls therefore act like a single pull of length $(4^2 + 3^2)^{1/2} = 5$ units, by Pythagoras' theorem, directed at an angle θ north of east. Writing $(4,3) = 5(\cos \theta, \sin \theta)$ gives $\tan \theta = \sin \theta / \cos \theta = 3/4$. The tangent can be undone as the sine was in Section 1.5: for any number s , $\arctan s$ is the angle between $-\pi/2$ and $\pi/2$ whose tangent is s , and here $\theta = \arctan(3/4) = 0.644$ rad, or 36.9° . Two forces of equal size pulling in opposite directions add to zero, so a body can be pushed and pulled by several forces while the net force on it vanishes. Such a body may still turn: two equal and opposite pushes at opposite corners of a book lying on a table add to zero, yet set the book spinning, because where a force acts decides how it turns a body. Turning is the subject of Chapter 5, and until then every body is a particle, for which only the net force matters.

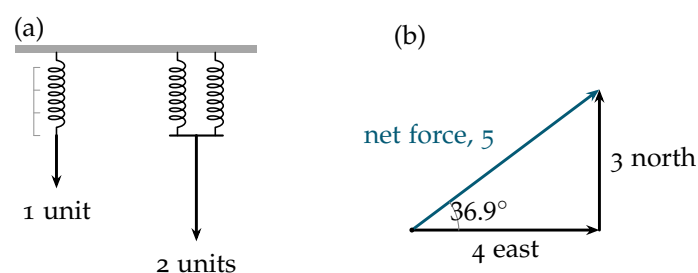


FIGURE 2.1: Measuring and adding forces. (a) A unit of force stretches a standard spring by a set amount; two such springs pulling together exert two units. (b) Pulls of 4 units east and 3 units north, drawn tip to tail, add to a net force of 5 units at 36.9° north of east.

The first law

A book slid across a table slows down and stops. The table and the air exert forces on it: friction, the grip of one surface sliding over another, and drag, the resistance of the air or liquid a body moves through. Where these forces are made small, the book's behaviour changes. On smooth ice, or on an air-hockey table, where a cushion of air lifts the puck clear of the surface, a puck glides on at nearly constant speed in a straight line. We call a surface smooth when the friction it exerts is small enough to neglect.

Since forces acting together behave like their net force, a body whose forces add to zero moves as one with no force on it at all. Newton's first law states the result: a body on which no net force acts keeps moving in a straight line at constant speed, that is, at constant velocity, and a body at rest stays at rest. A force is therefore not needed to keep a body moving, only to change its motion. The next section shows that this law cannot be true for every observer, and uses that fact to pick out the viewpoints from which all three laws hold.

CHECK YOUR UNDERSTANDING. A puck glides across smooth ice in a straight line at a steady 3 m s^{-1} . What is the net force on it?

2.2 Frames of reference

Every position in Chapter 1 was measured from some origin, along some axes, with times read from some clock. That choice is called a frame of reference. [Section 1.9](#) used the word frame for a saved snapshot of positions, so here we always say frame of reference in full. Two observers moving relative to each other use different frames of reference and describe the same motion differently. This section works out how their descriptions are related, which tells us from which frames of reference the first law can hold.

One throw seen from two frames of reference

A train runs along a straight, level track at a constant velocity $\mathbf{u}$. We take the x axis along the track in the direction of travel and the y axis vertically upwards, so that $\mathbf{u} = (5, 0) \text{ m s}^{-1}$. A passenger throws a ball straight up, as the passenger sees it, at 6 m s^{-1} . To the passenger the ball rises and falls along a vertical line and lands back in the hand ([Figure 2.2a](#)). Someone standing on the platform sees the same ball travel forwards with the train while it rises and falls, and so trace out a parabola ([Figure 2.2b](#)). Both descriptions are correct; they differ because they are made from different frames of reference.

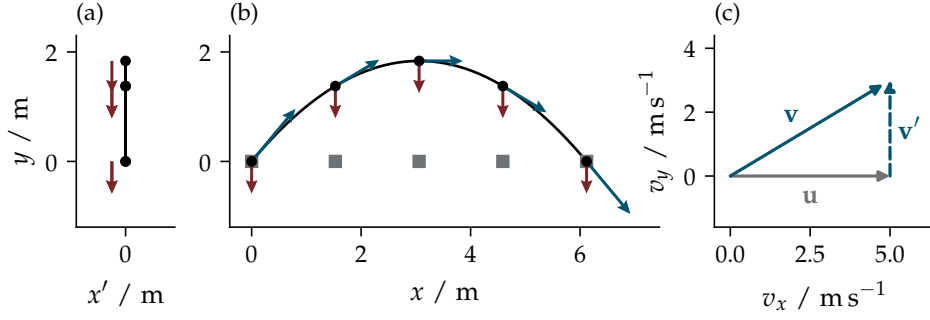


FIGURE 2.2: A ball thrown straight up at 6 m s^{-1} by a passenger in a train moving at $\mathbf{u} = (5, 0) \text{ m s}^{-1}$, at five equally spaced times from the throw to the catch 1.224 s later. (a) In the frame of reference of the train the ball moves on a vertical line; it passes the same heights on the way up and down, so the five positions show as three. Arrows give its acceleration. (b) In the frame of reference of the platform it follows a parabola, with velocity (teal) and acceleration (dark red); squares mark the passenger's hand. (c) The velocities at the second instant, $t = 1.224/4 = 0.306 \text{ s}$, when the passenger sees the ball rising at $6 - 9.80665 \times 0.306 = 3 \text{ m s}^{-1}$: the platform sees $\mathbf{v}$, which is the velocity $\mathbf{v}' = (0, 3) \text{ m s}^{-1}$ seen in the train plus the velocity $\mathbf{u}$ of the train.

To relate the two descriptions, the platform observer measures positions from a fixed mark on the platform, the passenger measures them from the hand, and both sets of axes point the same way. The hand moves at the constant velocity $\mathbf{u}$, so by Equation (1.4) with zero acceleration it is at $\mathbf{r}_0 + \mathbf{u}t$ at time t , where $\mathbf{r}_0$ is its position at $t = 0$. To go from the mark to the ball we can go first from the mark to the hand and then from the hand to the ball (Figure 2.3), and adding these two vectors gives the position $\mathbf{r}$ of the ball seen from the platform in terms of its position $\mathbf{r}'$ seen from the hand,

$$\mathbf{r} = \mathbf{r}_0 + \mathbf{u}t + \mathbf{r}'.$$

Subtracting $\mathbf{r}_0 + \mathbf{u}t$ from both sides gives the position seen in the train,

$$\mathbf{r}' = \mathbf{r} - \mathbf{r}_0 - \mathbf{u}t. \quad (2.1)$$

Here we have assumed that both observers read the same time t from their clocks, which is true to an excellent approximation for every speed far below that of light, about $3 \times 10^8 \text{ m s}^{-1}$, and so for everything in this book.

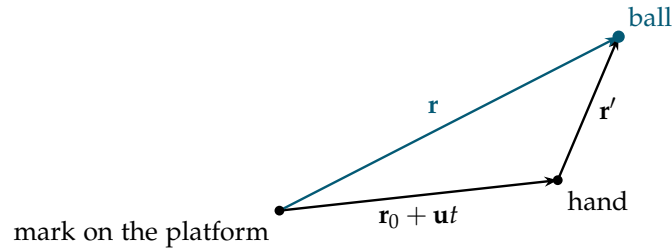


FIGURE 2.3: The position $\mathbf{r}$ of the ball seen from the platform is the position $\mathbf{r}_0 + \mathbf{u}t$ of the hand plus the position $\mathbf{r}'$ of the ball seen from the hand.

Velocities and accelerations

Differentiating Equation (2.1) with respect to time relates the velocities. The starting point $\mathbf{r}_0$ is fixed, so its derivative is zero, and the derivative of $\mathbf{u}t$ is $\mathbf{u}$ because $\mathbf{u}$ is constant:

$$\begin{aligned} \frac{d\mathbf{r}'}{dt} &= \frac{d\mathbf{r}}{dt} - \frac{d\mathbf{r}_0}{dt} - \frac{d(\mathbf{u}t)}{dt} && \text{differentiating each term} \\ &= \mathbf{v} - \mathbf{0} - \mathbf{u} && \mathbf{r}_0 \text{ fixed, } \mathbf{u} \text{ constant,} \end{aligned}$$

so that

$$\mathbf{v}' = \mathbf{v} - \mathbf{u}. \quad (2.2)$$

The velocity seen in the train is the velocity seen from the platform, less the velocity of the train. Differentiating once more, $\mathbf{u}$ is constant and drops out entirely, and the two observers find the same acceleration,

$$\mathbf{a}' = \mathbf{a}.$$

For the throw, the passenger sees the ball leave the hand with velocity $\mathbf{v}' = (0, 6) \text{ m s}^{-1}$, so the platform sees it leave with $\mathbf{v} = \mathbf{v}' + \mathbf{u} = (5, 6) \text{ m s}^{-1}$, a speed of $(5^2 + 6^2)^{1/2} = 7.810 \text{ m s}^{-1}$. Figure 2.2(c) draws this sum at a later instant. Both observers see the same acceleration, $(0, -g)$, and, because $\mathbf{u}$ has no vertical component, the same vertical starting velocity, 6 m s^{-1} , so they see the same vertical motion. By Equations (1.4) and (1.5), the ball rises for $6/g = 0.612 \text{ s}$ to a greatest height of $6^2/2g = 1.835 \text{ m}$ and falls back in the same time, landing after 1.224 s . In that time the platform observer sees it travel $5 \times 1.224 = 6.12 \text{ m}$ along the track, which is exactly how far the hand has moved, so the passenger catches it.

Inertial frames of reference

Positions and velocities depend on the frame of reference, but accelerations do not, for any two frames of reference moving at constant velocity relative to each other. The agreement fails when one of them accelerates. If the train brakes, the hand no longer moves at constant velocity. We write its position as some function $\mathbf{s}(t)$, which for the steady train was $\mathbf{r}_0 + \mathbf{u}t$. The same subtraction then gives $\mathbf{r}' = \mathbf{r} - \mathbf{s}(t)$, and differentiating twice,

$$\mathbf{a}' = \mathbf{a} - \frac{d^2\mathbf{s}}{dt^2}.$$

The two observers now disagree about accelerations, by exactly the acceleration of the train. Picture a cup on a smooth table in the train. If the train slows at 2 m s^{-2} , its acceleration is $d^2\mathbf{s}/dt^2 = (-2, 0) \text{ m s}^{-2}$, pointing backwards. Nothing pushes the cup along the track, so from the platform $\mathbf{a} = \mathbf{0}$, and the passenger sees $\mathbf{a}' = \mathbf{0} - (-2, 0) = (2, 0) \text{ m s}^{-2}$: the cup slides forwards with nothing pushing it, which breaks the first law, while from the platform it simply keeps moving at its old velocity as the train slows beneath it, as the first law says.

The first law therefore cannot hold in every frame of reference, and we turn this round into a definition. A frame of reference in which a body with no net force on it moves at constant velocity is called inertial, from inertia, the tendency of a body to keep the velocity it has. What experiment adds is that such frames of reference exist. A frame of reference fixed to the ground is very nearly inertial. The Earth turns once in 86 164 s measured against the distant stars. This is a little less than the 86 400 s of a 24-hour day, which is measured against the Sun: the Earth's yearly journey round the Sun adds one turn against the stars each year, so $86400 \times 365.24/366.24 = 86\,164 \text{ s}$. Its angular speed is therefore $\omega = 2\pi/86164 = 7.29 \times 10^{-5} \text{ rad s}^{-1}$. In a frame of reference with its origin at the Earth's centre and its axes pointing at fixed distant stars, a point on the equator, carried round a circle of radius $R = 6.38 \times 10^6 \text{ m}$, has by Equation (1.9) an acceleration of only $\omega^2 R = (7.29 \times 10^{-5})^2 \times 6.38 \times 10^6 = 0.034 \text{ m s}^{-2}$, about 0.35 % of g . By $\mathbf{a}' = \mathbf{a} - d^2\mathbf{s}/dt^2$, a body with no net force on it therefore seems, from the ground, to accelerate by at most about this much; the turning of the ground's axes adds effects that are smaller still at everyday speeds. Since accelerations agree between frames of reference moving at constant velocity relative to each other, every frame of reference moving at constant velocity relative to the ground is very nearly inertial as well. All three of Newton's laws are stated for inertial frames of reference, and from here on every motion is described in one.

NOTEBOOK 02 *change the train's speed and watch both views of the throw*

CHECK YOUR UNDERSTANDING. A passenger walks towards the back of the train at 1 m s^{-1} while the train moves forwards at 5 m s^{-1} . What is the passenger's velocity seen from the platform? Is a spinning roundabout an inertial frame of reference?

2.3 Mass and the second law

The first law says what happens when no net force acts. The second says what a net force does, and to state it we need one more quantity, mass, which measures how strongly a body resists being accelerated.

Mass

Pull two different bodies on a smooth level floor with the same force, for instance by stretching the same spring by the same amount, and they accelerate differently: a loaded trolley gathers speed more slowly than an empty one. Experiment shows that the ratio of their accelerations, a_2/a_1 , does not depend on which force is used. Because the ratio does not depend on the force, it belongs to the two bodies alone, and we can use it to give each body a number, its mass, by defining the ratio of their masses as

$$\frac{m_1}{m_2} = \frac{a_2}{a_1}.$$

The body that accelerates less has the larger mass. If, under the same pull, an empty trolley accelerates at 3 m s^{-2} and a loaded one at 1 m s^{-2} , the loaded trolley has three times the mass of the empty one. Choosing one body as a standard, the kilogram (kg), then fixes every other mass. Mass is a new kind of quantity alongside length and time, and we give it the dimension M. It measures the inertia of the body, and experiment also shows that when two bodies are joined their masses add.

The second law

Two experiments now combine into a law. The mass experiment says that one force gives two bodies accelerations in the inverse ratio of their masses, $a_2/a_1 = m_1/m_2$, so the product $m_1a_1 = m_2a_2$ is the same for both bodies: it depends on the force and not on the body it acts on. Experiment also shows that n standard springs pulling together give a body n times the acceleration that one gives it, so the product is n times as large when the force is n units. The product of mass and acceleration is therefore proportional to the force, $F = cma$ for some constant c , and the acceleration points

along the force. We are free to choose the unit of force, and choosing it to make $c = 1$, as the newton below does, gives Newton's second law,

$$\mathbf{F} = m\mathbf{a}, \quad (2.3)$$

where $\mathbf{F}$ is the net force on the body. Because it is a vector equation, it holds for each component separately: $F_x = ma_x$, $F_y = ma_y$ and $F_z = ma_z$.

The law also fixes the unit of force. One newton (N) is the force that gives one kilogram an acceleration of one metre per second per second, $1 \text{ N} = 1 \text{ kg m s}^{-2}$, so force has dimension MLT^{-2} . The spring unit of [Section 2.1](#) was a stopgap: from now on forces are measured in newtons, and spring balances are marked in newtons. A trolley of 20 kg pushed with a net force of 10 N accelerates at $10/20 = 0.5 \text{ m s}^{-2}$, in the direction of the push.

KEY IDEA. In an inertial frame of reference the net force on a body equals its mass times its acceleration, $\mathbf{F} = m\mathbf{a}$, where the net force is the vector sum of every force acting on the body.

CHECK YOUR UNDERSTANDING. A net force of 6 N acts on a 2 kg body. What is its acceleration? What would it be if the mass were doubled?

2.4 The third law

Forces do not appear from nowhere: every force on a body is exerted by some other body. The third law describes what happens at the other end, and it is what later lets us treat several bodies, and many atoms, together.

We write $\mathbf{F}_{AB}$ for the force on body A exerted by body B ; the first subscript names the body that feels the force. The third law states that forces between two bodies come in matched pairs,

$$\mathbf{F}_{AB} = -\mathbf{F}_{BA},$$

equal in size and opposite in direction. Like the second law, this is a fact of experiment: hook two spring balances together and pull them apart, and they always show the same reading, however hard or unevenly you pull. When you push on a wall, the wall pushes back on you with an equal force. When the Earth pulls a falling ball down, the ball pulls the Earth up with a force of the same size. A ball of half a kilogram falls, with air resistance negligible, at the acceleration g of Chapter 1, so by the second law the Earth's pull on it is $0.5 \times 9.80665 = 4.9 \text{ N}$, its weight ([Section 2.5](#)). The Earth's

mass is about 6×10^{24} kg, so by the second law again the Earth's acceleration is $4.9 / (6 \times 10^{24}) = 8 \times 10^{-25} \text{ m s}^{-2}$, far too small to notice.

The two forces of a pair always act on different bodies, so they never cancel in the second law of either body. A book resting on a table shows the commonest confusion (Figure 2.4). The Earth pulls the book down and the table pushes it up, and these two forces are equal and opposite, but they are not a third-law pair: both act on the book, and they are equal only because the book does not accelerate. The partner of the Earth's pull on the book is the book's pull on the Earth, and the partner of the table's push on the book is the book's push on the table.

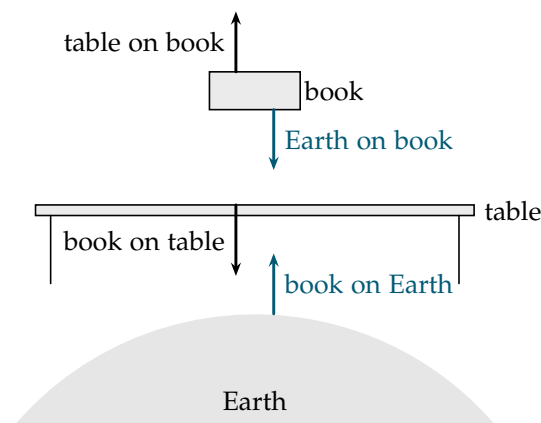


FIGURE 2.4: The forces between a book resting on a table, the table and the Earth, with the three bodies drawn apart for clarity. The black arrows are one third-law pair, the table's push on the book and the book's push on the table; the teal arrows are another, the Earth's pull on the book and the book's pull on the Earth. The two forces on the book itself, one from each pair, balance because the book does not accelerate. Forces between the table and the Earth are not drawn.

CHECK YOUR UNDERSTANDING. The Earth pulls an apple down and the apple pulls the Earth up with an equal force. Why does the apple fall towards the Earth, rather than the Earth towards the apple?

2.5 Weight, free fall and free-body diagrams

With the second law in hand we can explain the first motion of Chapter 1, the thrown ball, which was there described without a cause.

Weight

Near the surface of the Earth every body is pulled downwards by a force, its weight. Hanging a body at rest from a spring balance measures its weight: the body does not accelerate, so by the second law the balance's upward pull equals the Earth's

downward pull (the lamp below works this through). Such measurements show that the weight is proportional to the mass: twice, three times or half the mass gives twice, three times or half the reading. We therefore write the weight as $m\mathbf{g}$, where $\mathbf{g}$ is a vector pointing straight down whose length g is the same for every body. Since weight divided by mass is in newtons per kilogram, which by the definition of the newton is m s^{-2} , g has the units of an acceleration. Its standard value is $g = 9.80665 \text{ m s}^{-2}$, and the free-fall argument below shows that it is the g of Chapter 1. An apple of 100 g weighs $0.1 \times 9.80665 = 0.981 \text{ N}$, about one newton.

In everyday speech weight often means mass, as in ‘a bag of flour weighs a kilogram’. This book keeps them apart. Mass, in kilograms, measures inertia and is the same everywhere; weight, in newtons, is the pull of the Earth, and on the Moon, whose pull is weaker, the same bag would weigh about a sixth as much. A kitchen scale measures the force and divides by g to display a mass.

Free fall

A ball in flight, with air resistance negligible, feels its weight and nothing else, a motion called free fall. The second law gives $m\mathbf{a} = m\mathbf{g}$, and dividing both sides by m ,

$$\mathbf{a} = \mathbf{g}.$$

The mass cancels, so every body falls with the same acceleration, whatever its mass. This is why g in Chapter 1 was the same for every object, and it rests on the experimental fact that weight is proportional to mass. A feather and a coin dropped together in a room do not land together, but only because the drag of the air (Section 2.7) is large compared with the feather’s tiny weight. Without air they fall together, as an astronaut showed on the Moon in 1971 by dropping a hammer and a feather side by side.

We can now derive the motion of the ball that Chapter 1 took from observation. The motion is along a single vertical line, so, as in Equation (1.1), we call the height x and measure it upwards. The component of $\mathbf{a} = \mathbf{g}$ along this line is

$$\frac{d^2x}{dt^2} = -g,$$

the minus sign because $\mathbf{g}$ points down. Integrating once from $t = 0$, as in Section 1.4, gives $dx/dt = v_0 - gt$, where v_0 is the launch velocity. Integrating again, with the ball leaving the hand at $x = 0$, gives

$$x(t) = v_0t - \frac{1}{2}gt^2,$$

which is Equation (1.1). The projectile follows in the same way: the horizontal component of $\mathbf{g}$ is zero, so the horizontal velocity stays constant.

For atoms, gravity hardly matters: Section 2.10 shows that the weight of an atom is some 10^{16} times smaller than the forces its neighbours exert on it.

Free-body diagrams

Applying the second law to a body starts with a free-body diagram: a sketch of the body alone, with an arrow for every force acting on it and for nothing else. Figure 2.5(a) shows the ball in flight, which feels only its weight; its velocity is drawn dashed, because a velocity is not a force. Figure 2.5(b) shows a lamp of 2 kg hanging at rest from a cord. Two forces act on it: its weight, mg downwards, and the pull of the cord, called its tension, upwards, which we write F_{cord} . The lamp does not accelerate, so by the second law the net force on it is zero, and taking upwards as positive,

$$F_{\text{cord}} - mg = 0 \implies F_{\text{cord}} = mg = 2 \times 9.80665 \text{ N} = 19.6 \text{ N}.$$

The arrows of a free-body diagram are labelled with the sizes of the forces, and their directions are shown by the arrows themselves.

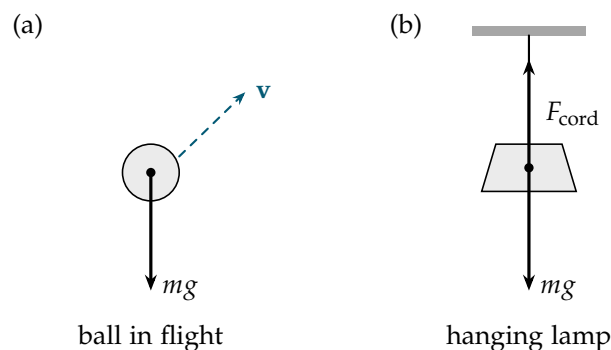


FIGURE 2.5: Free-body diagrams. (a) A ball in flight feels only its weight; its velocity is dashed, because it is not a force. (b) A lamp hanging at rest feels its weight and the tension F_{cord} of the cord, which balance.

CHECK YOUR UNDERSTANDING. A hammer and a feather are dropped together, first in a vacuum, then in air. Which lands first in each case, and why?

2.6 Equations of motion and the state

For the falling ball the force was constant, and two integrations gave the motion. Most forces are not constant: the drag of a liquid grows with the speed of the body moving through it, and the pull of a spring grows with how far it is stretched. This section shows what the second law becomes in general, and what must be known at one instant to predict the whole motion.

The equation of motion

For motion along a line the force may depend on the position x , on the velocity v and on the time t , and we write it $F(x, v, t)$. Since the acceleration is the second derivative of the position, the second law reads

$$m \frac{d^2x}{dt^2} = F(x, v, t). \quad (2.4)$$

An ordinary equation such as $x^2 = 4$ has numbers as its unknowns; Equation (2.4) has a whole function, $x(t)$, as its unknown, and it contains that function through its derivatives. An equation of this kind is called a differential equation. Its order is that of the highest derivative it contains, so Equation (2.4) is of second order. It is the equation of motion of the body, and solving it means finding a function $x(t)$ for which it holds at every instant.

The state

Equation (2.4) can be split into two first-order equations by treating the velocity as a second unknown. The velocity is the rate of change of the position, and the acceleration, which is the rate of change of the velocity, equals F/m by the second law, so

$$\frac{dx}{dt} = v, \quad \frac{dv}{dt} = \frac{F(x, v, t)}{m}. \quad (2.5)$$

These equations say that if the position and the velocity are known at some instant, then the rates at which both are changing are known too. Over a short interval τ , therefore, the position changes by about $v\tau$ and the velocity by about $(F/m)\tau$, which gives the position and velocity at the time $t + \tau$, from which the next step follows in the same way.

Take the thrown ball of Chapter 1, leaving the hand at $x = 0$ with $v_0 = 12 \text{ m s}^{-1}$, where $F/m = -g$, and take one step of $\tau = 0.1 \text{ s}$:

$$x(0.1) \approx x(0) + v(0) \tau = 0 + 12 \times 0.1 = 1.200 \text{ m},$$

$$v(0.1) \approx v(0) - g\tau = 12 - 9.80665 \times 0.1 = 11.019 \text{ m s}^{-1}.$$

The exact height from Equation (1.1) is $12 \times 0.1 - \frac{1}{2} \times 9.80665 \times 0.1^2 = 1.151 \text{ m}$. The step overshoots by $\frac{1}{2}g\tau^2 = 0.049 \text{ m}$, because it used the velocity at the start of the interval throughout, while the ball was in fact slowing down. Halving τ divides this error per step by four but doubles the number of steps needed to cover a given time, so the error at a given time is halved. Shorter steps therefore approach the true motion. Chapter 9 builds on this approximation to derive the integrators used in molecular dynamics.

Once the force law $F(x, v, t)$ is given, the stepping argument suggests, and the theorem below guarantees, that the pair (x, v) at a known instant is everything that needs to be known to predict the future, and it is called the state of the body. The instant matters only when the force depends on time explicitly, that is, through its third argument t and not only through the changing x and v , as a push switched on at a chosen moment does; the forces between atoms in this book do not, and for them the state alone suffices. Neither half is enough on its own. Figure 2.6(a) shows three balls that start from the same height with different velocities and follow three different motions; balls thrown at the same velocity from different heights would differ as well. Figure 2.6(b) draws the same three motions as paths in the plane whose two coordinates are x and v , so that each point is one state. The arrows show, for every state, the direction in which Equation (2.5) moves it, and each path follows the arrows.

That each starting state leads to exactly one motion is a theorem of mathematics, the existence and uniqueness theorem for differential equations, known as the Picard–Lindelöf theorem. It holds provided that F is continuous in time and its slopes with respect to x and v stay bounded in a neighbourhood of the starting state. It then guarantees a single solution for at least some interval of time. Smooth force laws satisfy this condition away from singularities such as two atoms occupying the same position. A discontinuous cutoff requires separate care, which is one reason to examine how forces are truncated in Chapter 8. In three dimensions the same holds with $\mathbf{r}$ and $\mathbf{v}$ in place of x and v , and the state is the pair of vectors $(\mathbf{r}, \mathbf{v})$. This is the state of a particle; a body that can also turn needs, in addition, its orientation and how fast it is turning (Chapter 5).

KEY IDEA. The equation of motion, $m d^2\mathbf{r}/dt^2 = \mathbf{F}$, is a second-order differential equation. Given the force law and the starting time, the position and velocity together, the state, determine the motion that follows.

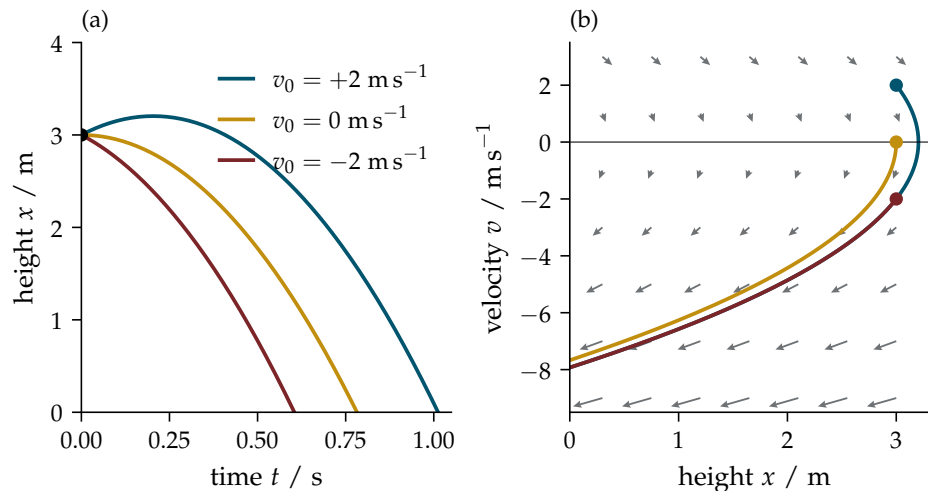


FIGURE 2.6: The state of a falling ball. (a) Three balls released 3 m above the ground with velocities of $+2$, 0 and -2 m s^{-1} : the same position with different velocities gives different motions. (b) The same motions as paths through the states (x, v) . Each grey arrow shows how the state at its tail changes over 0.03 s , and dots mark the starting states.

CHECK YOUR UNDERSTANDING. Two stones are dropped from rest, one from a bridge and one from a window 1 m lower. Do they have the same state at the moment of release? Do they follow the same motion?

2.7 Drag and terminal velocity

Drop a small bead into a jar of honey or a thick oil. It speeds up for a moment and then sinks at a steady speed, however tall the jar. This is the first force we meet that depends on the velocity, and solving its equation of motion needs a new function, the exponential. The same equation returns in Chapter 13, where it describes how a thermostat steadies the temperature of a simulation.

The drag force

The liquid resists the motion of the bead with a drag force that points against the velocity. For small bodies moving slowly through a thick liquid this force is, to a good approximation, proportional to the velocity,

$$\mathbf{F}_{\text{drag}} = -b \mathbf{v},$$

where the drag coefficient b , in kilograms per second, depends on the size of the body and the thickness of the liquid. Fast bodies moving through air feel a drag that grows more nearly as the square of the speed, which we do not treat. The liquid also pushes

the bead upwards with a constant force called buoyancy, equal to the weight of the liquid that the bead displaces. For a bead much denser than the liquid this is a small correction, and we neglect it.

The equation of motion

We measure the depth x downwards, so that the weight mg is positive and the drag $-bv$ opposes a downward velocity. The second law for the bead is

$$m \frac{dv}{dt} = mg - bv,$$

and dividing both sides by m ,

$$\frac{dv}{dt} = g - \gamma v, \quad \gamma = \frac{b}{m}. \quad (2.6)$$

The constant γ , called the drag rate, has units of $(\text{kg s}^{-1})/\text{kg} = \text{s}^{-1}$, an inverse time; we shall see that $1/\gamma$ sets the time scale on which the bead settles into its steady motion.

Before solving Equation (2.6) we can find the steady speed directly. The bead stops speeding up when $dv/dt = 0$, which happens when the drag has grown to balance the weight,

$$0 = g - \gamma v_{\infty} \implies v_{\infty} = \frac{g}{\gamma} = \frac{mg}{b}.$$

This steady speed is called the terminal velocity. We write it v_{∞} , read ‘v infinity’, because the bead approaches it as the time grows without limit.

Measuring the velocity from its terminal value

Equation (2.6) becomes simpler if we follow not the velocity itself but its distance from the terminal velocity, $w = v - v_{\infty}$. Since v_{∞} is a constant, $dw/dt = dv/dt$, and substituting $v = w + v_{\infty}$ into Equation (2.6),

$$\begin{aligned} \frac{dw}{dt} &= g - \gamma(w + v_{\infty}) && \text{substituting for } v \\ &= (g - \gamma v_{\infty}) - \gamma w && \text{multiplying out the bracket} \\ &= -\gamma w && \text{since } \gamma v_{\infty} = g. \end{aligned}$$

The gap between the velocity and its terminal value therefore shrinks at a rate proportional to the gap itself: when the gap is large it closes quickly, and as it narrows it

closes ever more slowly. To solve this equation we need a function whose derivative is a constant times the function, and the boxes below build it.

THE EXPONENTIAL FUNCTION. Define $E(s)$ as the function that equals its own derivative and takes the value 1 at $s = 0$,

$$\frac{dE}{ds} = E(s), \quad E(0) = 1.$$

Differentiating both sides again shows that the second derivative also equals E , and so on for every derivative, so at $s = 0$ every derivative of E equals 1. The Taylor series of [Section 1.8](#) about $s = 0$ is therefore

$$E(s) = 1 + s + \frac{s^2}{2!} + \frac{s^3}{3!} + \cdots$$

Differentiating the series term by term confirms that it is its own derivative: since $n! = n \times (n-1)!$, the term $s^n/n!$ becomes $n s^{n-1}/n! = s^{n-1}/(n-1)!$, the term before it, so the differentiated series is the same series again. Such a function therefore exists, and since $dE/ds = E$ is a first-order differential equation, the uniqueness theorem of [Section 2.6](#) allows only one with $E(0) = 1$. Its value at $s = 1$ is a number called e . Adding the first seven terms, $1 + 1 + 0.5 + 0.1667 + 0.0417 + 0.0083 + 0.0014 = 2.7181$; the next three terms, 0.0002, 0.00002 and 0.000003, raise this to 2.71828, and later terms are too small to change these digits: $e = 2.71828 \dots$

Now let λ be any constant. By the chain rule of [Section 1.3](#), with inner function λt ,

$$\frac{d}{dt} E(\lambda t) = E'(\lambda t) \lambda = \lambda E(\lambda t),$$

the second step using $E' = E$, so $E(\lambda t)$ is a function whose derivative is λ times itself, which is what we need. The series also gives values at negative s , for instance

$$E(-1) = 1 - 1 + 0.5 - 0.1667 + 0.0417 - 0.0083 + 0.0014 - \cdots = 0.368.$$

THE PRODUCT RULE. For two functions $p(s)$ and $q(s)$, the change in their product over a small step h can be split into a change in p and a change in q , by adding and subtracting $p(s)q(s+h)$,

$$p(s+h)q(s+h) - p(s)q(s) = [p(s+h) - p(s)]q(s+h) + p(s)[q(s+h) - q(s)].$$

Dividing by h and letting $h \rightarrow 0$, the brackets divided by h become the derivatives p' and q' , and $q(s+h)$ becomes $q(s)$, so

$$\frac{d}{ds}(pq) = p'q + pq'.$$

The product rule shows why E deserves its usual name. Fix a number c and consider $f(s) = E(c+s)E(-s)$. The derivative of $E(-s)$ is $-E(-s)$, by the chain rule with $\lambda = -1$, and the derivative of $E(c+s)$ is $E(c+s)$, by the chain rule with inner function $c+s$, whose derivative is 1, so

$$\begin{aligned} f'(s) &= E(c+s)E(-s) + E(c+s)[-E(-s)] && \text{by the product rule} \\ &= 0. \end{aligned}$$

A function whose derivative is zero everywhere does not change, by [Equation \(1.3\)](#) (with f in place of v and $f' = 0$ in place of a , it reads $f(s) = f(0) + \int_0^s 0 \, ds' = f(0)$), so f keeps its value at $s = 0$, which is $E(c)E(0) = E(c)$. Setting $c = 0$ gives $E(s)E(-s) = 1$, and dividing $E(c+s)E(-s) = E(c)$ by $E(-s)$, which is the same as multiplying by $E(s)$,

$$E(c+s) = E(c)E(s).$$

This is the rule obeyed by powers, $e^{c+s} = e^c e^s$. Applying it with $c = s = 1$ gives $E(2) = e^2$, then $E(3) = e^3$, and so on, and $E(-s) = 1/E(s)$ matches $e^{-s} = 1/e^s$. Two further facts follow. Since $E(s)E(-s) = 1$, E is never zero, and since $E(s) = E(\frac{s}{2})^2$, by the rule with both c and s set to $\frac{s}{2}$, it is never negative; so $E(s) > 0$ for every s . Its derivative, equal to itself, is then positive too, so E rises steadily and never takes the same value at two different s . Fractions agree with powers as well: $E(\frac{1}{2})^2 = E(\frac{1}{2} + \frac{1}{2}) = E(1) = e$, so $E(\frac{1}{2}) = \sqrt{e} = e^{1/2}$, the positive root. The function E therefore extends the powers of e to every real number, and from now on we write it e^s , the exponential function. Because it never takes a value twice, it can be undone: its inverse is the natural logarithm, $\ln$, defined by $\ln(e^s) = s$. It is not the base-10 logarithm $\log$ of [Section 1.8](#); calculators mark it 'ln'.

The rule for sums also explains how a decaying exponential decays. Lowering s by one multiplies the function by a fixed factor, $e^{s-1} = e^s e^{-1} = 0.368 e^s$. In $e^{\lambda t}$ with negative λ the exponent $s = \lambda t$ falls by $\lambda \times (1/|\lambda|) = -1$ each time t grows by $1/|\lambda|$, so the function falls to about 37% of its value over every such interval, wherever it starts.

Solving the equation

The gap obeys $dw/dt = -\gamma w$, and by the toolbox, with $\lambda = -\gamma$, the function $w = C e^{-\gamma t}$ has derivative $-\gamma C e^{-\gamma t} = -\gamma w$ for any constant C . At $t = 0$ the exponential is 1, so C is the starting gap, $C = v_0 - v_\infty$, where v_0 is the velocity at release. Here the force depends on the velocity alone, so the second of Equation (2.5) involves v alone, and by the uniqueness theorem of Section 2.6 the starting velocity fixes its solution. We have therefore found it, and adding v_∞ back,

$$v(t) = v_\infty + (v_0 - v_\infty) e^{-\gamma t}. \quad (2.7)$$

Whatever the starting velocity, the gap shrinks to $e^{-1} = 37\%$ of its size every interval $1/\gamma$, and the velocity approaches v_∞ from below if the bead starts slower and from above if it starts faster. A bead released from rest has $v_0 = 0$, and then $v(t) = v_\infty (1 - e^{-\gamma t})$.

As an example, take a bead of mass 0.01 kg with a drag coefficient of 0.05 kg s^{-1} . Then $\gamma = 0.05/0.01 = 5 \text{ s}^{-1}$, so $1/\gamma = 0.2 \text{ s}$, and the terminal velocity is $v_\infty = 9.80665/5 = 1.961 \text{ m s}^{-1}$. These values are chosen so that the approach is easy to see on a scale of seconds. A real bead this heavy and this fast would feel a drag that is not proportional to its speed, while for the small, slow beads to which the linear law applies the whole approach is over in a small fraction of a second. Released from rest, the bead's velocity as a fraction of v_∞ is $1 - e^{-\gamma t}$, where $e^{-2} = 0.368^2$, $e^{-3} = 0.368^3$ and $e^{-5} = 0.368^5$ by the rule for sums:

t / s	γt	$e^{-\gamma t}$	$v/v_\infty = 1 - e^{-\gamma t}$
0.2	1	0.368	63.2 %
0.4	2	0.135	86.5 %
0.6	3	0.050	95.0 %
1.0	5	0.007	99.3 %

To find when it reaches 90 % of v_∞ , we need $e^{-\gamma t} = 0.1$. Taking the natural logarithm of both sides gives $-\gamma t = \ln 0.1 = -2.303$, the value of the logarithm coming from a calculator, so $t = 2.303/5 = 0.461 \text{ s}$ (Figure 2.7a).

The distance fallen

The depth follows by integrating the velocity, as in [Section 1.4](#). The function $-e^{-\gamma t'}/\gamma$ has derivative $e^{-\gamma t'}$, by the toolbox with $\lambda = -\gamma$, so

$$\int_0^t e^{-\gamma t'} dt' = \left[-\frac{e^{-\gamma t'}}{\gamma} \right]_0^t = \frac{1 - e^{-\gamma t}}{\gamma},$$

and for a bead released from rest at $x = 0$,

$$x(t) = \int_0^t v_{\infty}(1 - e^{-\gamma t'}) dt' = v_{\infty} t - \frac{v_{\infty}}{\gamma} (1 - e^{-\gamma t}). \quad (2.8)$$

Two limits check this result. Long after release the exponential has died away and $x \approx v_{\infty} t - v_{\infty}/\gamma$: the bead sinks at its terminal velocity, but a fixed distance v_{∞}/γ behind where it would be had it moved at that velocity from the start, because it spent its first moments speeding up. For the example bead this lag is $1.961/5 = 0.392$ m. Just after release, when γt is small, the exponential series with $s = -\gamma t$ gives $e^{-\gamma t} = 1 - \gamma t + \frac{1}{2}\gamma^2 t^2 + \mathcal{O}(t^3)$, so $1 - e^{-\gamma t} = \gamma t - \frac{1}{2}\gamma^2 t^2 + \mathcal{O}(t^3)$, with $\mathcal{O}(t^3)$ as in [Section 1.8](#), and substituting into [Equation \(2.8\)](#),

$$\begin{aligned} x(t) &= v_{\infty} t - \frac{v_{\infty}}{\gamma} \left(\gamma t - \frac{1}{2}\gamma^2 t^2 \right) + \mathcal{O}(t^3) \\ &= v_{\infty} t - v_{\infty} t + \frac{1}{2}v_{\infty}\gamma t^2 + \mathcal{O}(t^3) && \text{multiplying out the bracket} \\ &= \frac{1}{2}gt^2 + \mathcal{O}(t^3) && \text{since } v_{\infty}\gamma = g. \end{aligned}$$

At first the bead is moving too slowly for the drag to matter, and it falls freely. For the example bead, after 0.02 s it has fallen 1.898 mm against 1.961 mm in free fall; after 0.1 s, 41.79 mm against 49.03 mm; and after a second, 1.572 m against 4.903 m ([Figure 2.7b](#)).

NOTEBOOK 02 *change the mass and drag coefficient and watch the approach to terminal velocity*

CHECK YOUR UNDERSTANDING. A bead is fired downwards into the honey faster than its terminal velocity. Does it speed up or slow down? Sketch its velocity against time.

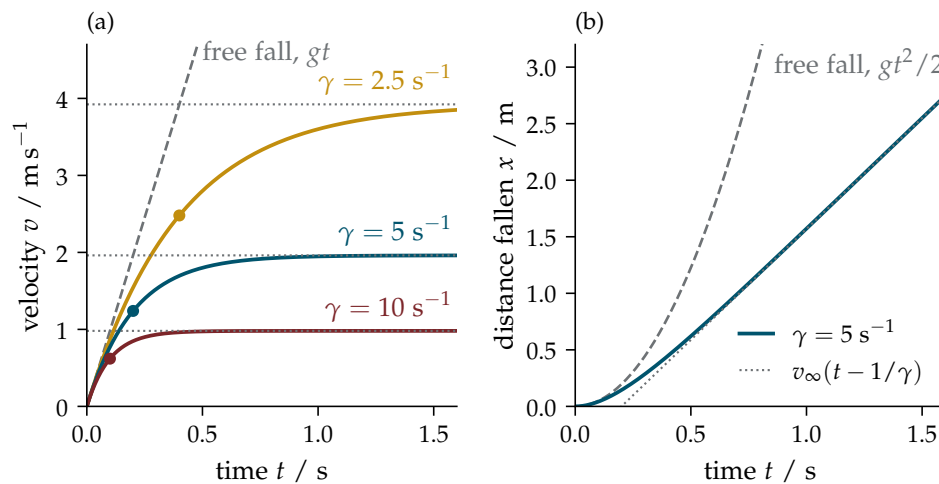


FIGURE 2.7: Falling from rest with a drag proportional to the velocity. (a) Velocity for three drag rates γ , approaching the terminal velocity $v_\infty = g/\gamma$ (dotted), against free fall (dashed); dots mark $t = 1/\gamma$, where 63% of the terminal velocity is reached. (b) Distance fallen for $\gamma = 5 \text{ s}^{-1}$, which follows free fall at first and then the line $v_\infty(t - 1/\gamma)$.

2.8 The spring

A block fixed to a spring and pulled aside swings back and forth about its resting place. This motion recurs throughout the book because an atom displaced slightly from a stable position feels a similar restoring force from its neighbours. Chapter 4 develops that connection; here we derive the motion from the second law.

Hooke's law

Lay a spring on a smooth table, fix one end to a wall and attach a block of mass m to the other. We measure the position x of the block from its resting place, where the spring has its natural length, so that x is the stretch of the spring: positive when it is stretched, negative when it is squashed. A stretched spring pulls the block back towards the wall, and a squashed one pushes it away (Figure 2.8). Measurements with a spring balance show that, for stretches small compared with the length of the spring, the force is proportional to the stretch,

$$F = -kx.$$

This is Hooke's law. The stiffness k , in newtons per metre, measures how hard the spring is to stretch, and the minus sign records that the force always points back towards $x = 0$, against the stretch. A spring with $k = 50 \text{ N m}^{-1}$ needs a force of 1 N to stretch it by $1/50 = 0.02 \text{ m}$, or 2 cm.

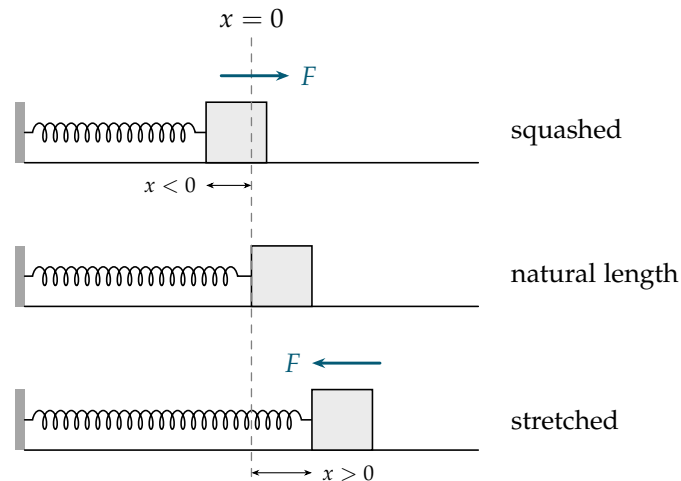


FIGURE 2.8: A block on a spring. Squashed ($x < 0$), the spring pushes the block away from the wall; at its natural length ($x = 0$) it exerts no force; stretched ($x > 0$), it pulls the block back. In each case the force (teal) points towards $x = 0$, against the stretch.

The equation of motion

Vertically, the weight of the block is balanced by the push of the table, called the normal force because it acts at right angles to the surface, which mathematicians call normal to it. Along the table the spring is the only force, so the second law reads $m \, d^2x/dt^2 = -kx$, and dividing by m ,

$$\frac{d^2x}{dt^2} = -\frac{k}{m}x. \quad (2.9)$$

Rather than solve this equation from nothing, we recognise it. In [Section 1.5](#) we found that every sinusoid obeys $d^2x/dt^2 = -\omega^2x$, which is [Equation \(2.9\)](#) if $\omega^2 = k/m$. With $\omega = \sqrt{k/m}$, both $\cos \omega t$ and $\sin \omega t$ are such sinusoids, and the equation is obeyed also by any combination

$$x(t) = c_1 \cos \omega t + c_2 \sin \omega t$$

with constant c_1 and c_2 . Differentiating twice with the chain rule confirms it,

$$\begin{aligned} \frac{dx}{dt} &= -c_1\omega \sin \omega t + c_2\omega \cos \omega t, \\ \frac{d^2x}{dt^2} &= -c_1\omega^2 \cos \omega t - c_2\omega^2 \sin \omega t && \text{differentiating again} \\ &= -\omega^2x = -\frac{k}{m}x && \text{taking out the common factor } -\omega^2, \end{aligned}$$

so [Equation \(2.9\)](#) holds whatever the values of c_1 and c_2 .

Matching the starting state

The two constants are fixed by the state at $t = 0$. Since $\cos 0 = 1$ and $\sin 0 = 0$, setting $t = 0$ in x and dx/dt gives $x_0 = c_1$ and $v_0 = c_2\omega$, where x_0 and v_0 are the starting position and velocity. The motion is therefore

$$x(t) = x_0 \cos \omega t + \frac{v_0}{\omega} \sin \omega t. \quad (2.10)$$

Every starting state (x_0, v_0) is matched in this way, so by the uniqueness of solutions (Section 2.6) Equation (2.10) is the motion for every start. With $x_0 = v_0 = 0$ it gives $x = 0$ for all time: a block at rest at its resting place stays there.

The same motion can be written as a single sinusoid, which shows at a glance how far the block swings. We write it with a cosine,

$$x(t) = A \cos(\omega t + \phi),$$

which by the quarter-turn relation $\cos \theta = \sin(\theta + \pi/2)$ is the sinusoid of Equation (1.8) with its phase larger by $\pi/2$. Expanding it with the addition formula Equation (1.6),

$$A \cos(\omega t + \phi) = A \cos \phi \cos \omega t - A \sin \phi \sin \omega t,$$

which is Equation (2.10) when the coefficients of $\cos \omega t$ and $\sin \omega t$ agree,

$$A \cos \phi = x_0, \quad A \sin \phi = -\frac{v_0}{\omega}.$$

To find A , we square both equations and add them, using $\cos^2 \phi + \sin^2 \phi = 1$:

$$x_0^2 + \left(\frac{v_0}{\omega}\right)^2 = A^2(\cos^2 \phi + \sin^2 \phi) = A^2 \implies A = \sqrt{x_0^2 + (v_0/\omega)^2}.$$

If $A = 0$ the block rests at $x = 0$ and the phase plays no part. Otherwise the two equations give $\cos \phi = x_0/A$ and $\sin \phi = -v_0/(\omega A)$. Together these fix the point $(\cos \phi, \sin \phi)$ on the unit circle, and so fix ϕ up to whole turns of 2π . The arcsine of Section 1.5 returns the angle between $-\pi/2$ and $\pi/2$ with the right sine; the cosine of that angle is never negative, and by $\cos^2 + \sin^2 = 1$ it equals $+\sqrt{1 - \sin^2 \phi}$, which is $\cos \phi$ itself when $\cos \phi \geq 0$. In that case, therefore, $\phi = \arcsin(\sin \phi)$. When $\cos \phi < 0$ the angle is $\pi - \arcsin(\sin \phi)$ instead, which has the same sine, since $\sin(\pi - \theta) = \sin \theta$, and a negative cosine, since $\cos(\pi - \theta) = -\cos \theta$. The sign of the cosine thus selects between the two angles that share a sine.

Take a block of 0.5 kg on the spring of 50 N m^{-1} , so that $\omega = \sqrt{50/0.5} = 10 \text{ rad s}^{-1}$, and start it 3 cm from its resting place, moving outwards, away from the wall, at

0.4 m s^{-1} . In metres, $x_0 = 0.03$ and $v_0/\omega = 0.04$, so $x(t) = 0.03 \cos 10t + 0.04 \sin 10t$ by Equation (2.10), and

$$A = \sqrt{0.03^2 + 0.04^2} = 0.05 \text{ m},$$

$$\cos \phi = 0.03/0.05 = 0.6, \quad \sin \phi = -0.04/0.05 = -0.8.$$

The block swings out to 5 cm before turning back. The cosine is positive, so $\phi = \arcsin(-0.8) = -0.927 \text{ rad}$. Started instead 3 cm on the wall side, $x_0 = -0.03$, with the same velocity, the block has the same amplitude, but $\cos \phi = -0.6$ is negative, so $\phi = \pi - \arcsin(-0.8) = \pi + 0.927 = 4.069 \text{ rad}$.

The period and the motion

The block repeats its motion when ωt grows by 2π , so its angular frequency and period are

$$\omega = \sqrt{\frac{k}{m}}, \quad \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{m}{k}}. \quad (2.11)$$

A heavier block oscillates more slowly and a stiffer spring more quickly. As long as Hooke's law holds, the period does not depend on the amplitude, and there is a simple reason. Pull the block twice as far and the spring pulls twice as hard, so at every stage the acceleration is twice as large; the block covers twice the distance at twice the speed and takes the same time. In the language of the equation, if $x(t)$ solves Equation (2.9) then so does $2x(t)$, because both sides double.

Figure 2.9 shows the simplest start of all, the same block pulled out 4 cm and released from rest, for which $\phi = 0$, since then $x_0 = A$ and $v_0 = 0$. The period is $2\pi/10 = 0.628 \text{ s}$. On the way in, the spring pulls the block in the direction it is moving, so the block speeds up all the way to the centre; past the centre the spring pulls against the motion and slows it. The speed is therefore largest at $x = 0$. Differentiating $x = A \cos(\omega t + \phi)$ with the chain rule gives $dx/dt = -A\omega \sin(\omega t + \phi)$; where $x = 0$ the cosine is zero, so the sine is ± 1 by $\cos^2 + \sin^2 = 1$, and the speed there is $A\omega = 0.04 \times 10 = 0.4 \text{ m s}^{-1}$. At the turning points, $x = \pm A$, the block is momentarily at rest, yet its acceleration there, $-\omega^2 x = \mp A\omega^2 = \mp 4 \text{ m s}^{-2}$ by Equation (2.9), is the largest it ever has, since $|x|$ never exceeds A . This is the top of the throw of Section 1.3 again: zero velocity does not mean zero acceleration.

NOTEBOOK 02 *change the mass, stiffness and starting state of the block*

CHECK YOUR UNDERSTANDING. A block oscillates on a spring with a period of 1 s. What is the period if the mass is made four times larger? What if the block is pulled out twice as far before release?

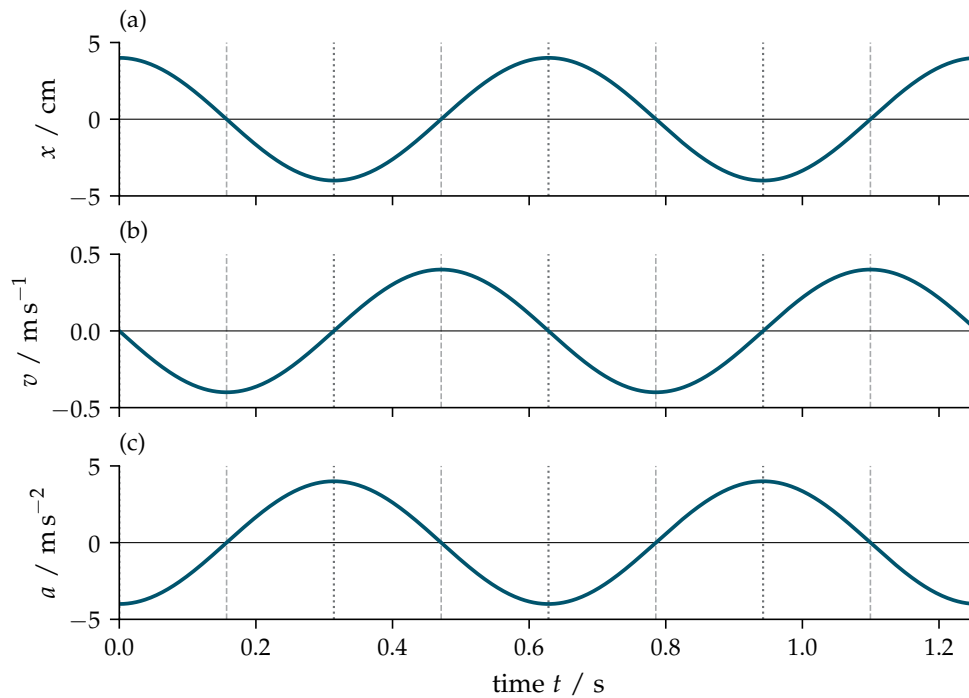


FIGURE 2.9: A block of 0.5 kg on a spring of 50 N m^{-1} , released from rest 4 cm from its resting place, over two periods. (a) Position, (b) velocity and (c) acceleration. At the dotted lines the block turns: its velocity is zero and its acceleration largest. At the dashed lines it passes the centre: its speed is largest and its acceleration zero.

2.9 Several bodies: momentum and the centre of mass

So far each body has moved under forces from its surroundings. When several bodies push and pull on one another, as the atoms of a material do, the third law ties their motions together. This section shows how, using a new quantity, momentum, and a new point, the centre of mass.

Momentum

The momentum of a body is its mass times its velocity,

$$\mathbf{p} = m\mathbf{v},$$

a vector in kg m s^{-1} that points along the velocity. Since the mass is constant, the rate of change of the momentum is $d\mathbf{p}/dt = m d\mathbf{v}/dt = m\mathbf{a}$, and the second law can be written

$$\frac{d\mathbf{p}}{dt} = \mathbf{F} :$$

the net force on a body is the rate at which its momentum changes. A constant force $\mathbf{F}$ acting from $t = 0$ to a time t therefore changes the momentum by $\mathbf{F}t$, since $\mathbf{p}$ then grows at the constant rate $\mathbf{F}$, as v grows by at at the constant rate a in Equation (1.4); so the momentum of a body measures how long a given force must push to bring it from rest to its velocity, or to stop it again. A trolley of 20 kg pushed from rest by 10 N for 2 s gains a momentum of $10 \times 2 = 20 \text{ kg m s}^{-1}$, and so a velocity of $20/20 = 1 \text{ m s}^{-1}$. A lorry and a bicycle moving at the same speed have very different momenta, and the brakes of the lorry must push far harder, or for far longer, to stop it.

Two carts

Two carts, of masses $m_1 = 1 \text{ kg}$ and $m_2 = 3 \text{ kg}$, stand at rest on a level, smooth track with a compressed spring between them, and are released (Figure 2.10). On each cart the weight and the normal force balance, so only the forces along the track matter, and these come from the spring. The spring is light, and we treat it as having no mass. Two steps then fix the forces on the carts. First, the net force on the spring must be zero, since by the second law any net force on a body of no mass would give it an unlimited acceleration; so the two carts push on the two ends of the spring with forces of equal size. Second, by the third law, the spring pushes back on each cart with a force of that same size: $-\mathbf{F}$ on the first cart, backwards, and $+\mathbf{F}$ on the second, forwards.

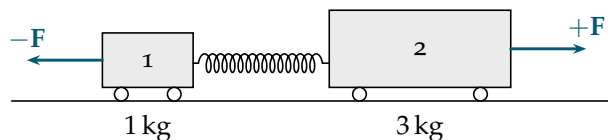


FIGURE 2.10: Two carts at rest with a compressed spring between them. The spring pushes the two carts with forces of equal size in opposite directions (teal); the weights and normal forces, which balance on each cart, are not drawn.

Applying the second law to each cart and adding the two equations,

$$\frac{d\mathbf{p}_1}{dt} = -\mathbf{F}, \quad \frac{d\mathbf{p}_2}{dt} = +\mathbf{F} \quad \Rightarrow \quad \frac{d}{dt}(\mathbf{p}_1 + \mathbf{p}_2) = -\mathbf{F} + \mathbf{F} = \mathbf{0}.$$

The total momentum $\mathbf{P} = \mathbf{p}_1 + \mathbf{p}_2$ therefore does not change, however the force of the spring varies, and since the carts started at rest it stays zero. A quantity that keeps the same value as time passes is said to be conserved, and here the total momentum is conserved. The carts therefore always have equal and opposite momenta, $m_1\mathbf{v}_1 = -m_2\mathbf{v}_2$, and the lighter cart moves faster by the ratio of the masses, three times faster here.

Figure 2.11 shows the motion computed for a spring of stiffness 200 N m^{-1} compressed by 10 cm. The spring is not fixed to either cart, so once it reaches its natural length

it could only stay in contact by pulling, which a loose spring cannot do, and it falls away after about 0.096 s, a time found, like the speeds below, by a computer solution. From then on no force acts along the track, and by the first law the carts coast apart at constant velocities, -1.225 m s^{-1} and $+0.408 \text{ m s}^{-1}$, a ratio of exactly -3 . Their momenta, $\mp 1.225 \text{ kg m s}^{-1}$, add to zero at every instant. Chapter 3 shows how energy gives these speeds directly.

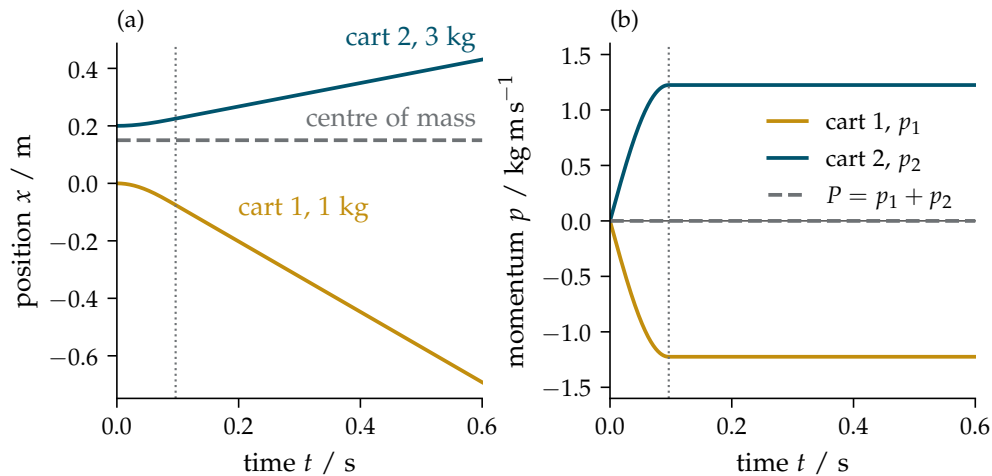


FIGURE 2.11: The carts of Figure 2.10 pushed apart by a spring of 200 N m^{-1} compressed by 10 cm; the spring falls away at the dotted line. (a) Positions of the carts and of their centre of mass (dashed). (b) Momenta of the carts and the component P of their total momentum along the track (dashed), which stays zero.

Many bodies

The same argument works for any number of bodies. The forces on each body are of two kinds. External forces come from outside the group, such as the weight of a cart, and we write the net external force on body i as $\mathbf{F}_i^{\text{ext}}$. Internal forces come from the other members, and we write the force on body i from body j as $\mathbf{F}_{ij}$. For three bodies, with total momentum $\mathbf{P} = \mathbf{p}_1 + \mathbf{p}_2 + \mathbf{p}_3$, adding the three second laws and then regrouping the internal forces by pairs gives

$$\begin{aligned} \frac{d\mathbf{P}}{dt} &= \frac{d\mathbf{p}_1}{dt} + \frac{d\mathbf{p}_2}{dt} + \frac{d\mathbf{p}_3}{dt} \\ &= (\mathbf{F}_1^{\text{ext}} + \mathbf{F}_{12} + \mathbf{F}_{13}) + (\mathbf{F}_2^{\text{ext}} + \mathbf{F}_{21} + \mathbf{F}_{23}) + (\mathbf{F}_3^{\text{ext}} + \mathbf{F}_{31} + \mathbf{F}_{32}) \\ &= \mathbf{F}^{\text{ext}} + (\mathbf{F}_{12} + \mathbf{F}_{21}) + (\mathbf{F}_{13} + \mathbf{F}_{31}) + (\mathbf{F}_{23} + \mathbf{F}_{32}), \end{aligned}$$

where $\mathbf{F}^{\text{ext}} = \mathbf{F}_1^{\text{ext}} + \mathbf{F}_2^{\text{ext}} + \mathbf{F}_3^{\text{ext}}$ is the total external force. By the third law each bracket is zero. With N bodies, numbered $i = 1, \dots, N$, the total momentum is $\mathbf{P} = \mathbf{p}_1 + \dots + \mathbf{p}_N$,

written $\sum_{i=1}^N \mathbf{p}_i$ with the summation sign of [Section 1.4](#), and the regrouping is the same: every internal force $\mathbf{F}_{ij}$ meets its partner $\mathbf{F}_{ji}$, each pair cancels, and

$$\frac{d\mathbf{P}}{dt} = \mathbf{F}^{\text{ext}}. \quad (2.12)$$

The internal forces, however strong and however complicated, cannot change the total momentum. Whenever the total external force is zero, the total momentum is conserved. That happens when the external forces cancel, as the weights and normal forces of the carts do, and also when there are no external forces at all, in which case the group is called isolated.

The centre of mass

The group as a whole has a position of its own, the centre of mass, an average of the positions of its members in which each counts in proportion to its mass,

$$\mathbf{R} = \frac{1}{M} \sum_{i=1}^N m_i \mathbf{r}_i, \quad M = \sum_{i=1}^N m_i,$$

where M is the total mass. For two bodies on a line it is the balance point: a light rod carrying the two carts would balance on a pivot placed there. The centres of the carts start at $x = 0$ and $x = 0.2$ m, so their centre of mass is at $(1 \times 0 + 3 \times 0.2)/4 = 0.15$ m, nearer the heavier cart. Its motion follows in three lines,

$$\begin{aligned} M\mathbf{R} &= \sum_{i=1}^N m_i \mathbf{r}_i && \text{multiplying by } M \\ M \frac{d\mathbf{R}}{dt} &= \sum_{i=1}^N m_i \mathbf{v}_i = \mathbf{P} && \text{differentiating term by term, the masses constant} \\ M \frac{d^2\mathbf{R}}{dt^2} &= \frac{d\mathbf{P}}{dt} = \mathbf{F}^{\text{ext}} && \text{differentiating again, by Equation (2.12).} \end{aligned}$$

The centre of mass moves as a single body of mass M would under the external forces alone. For the carts there is no external force along the track, so the centre of mass stays at 0.15 m while the carts fly apart ([Figure 2.11a](#)). A firework that bursts in flight shows the same thing in two dimensions: the pieces scatter in every direction, but until the first of them lands, their centre of mass carries on along the parabola that the unburst firework would have followed. The only external force is still the weight, $\mathbf{F}^{\text{ext}} = \sum_i m_i \mathbf{g} = M\mathbf{g}$, so $d^2\mathbf{R}/dt^2 = \mathbf{g}$ as before the burst, and the internal forces of the burst change neither $\mathbf{R}$ nor $d\mathbf{R}/dt = \mathbf{P}/M$. Chapter 7 shows that the conservation of momentum follows from a symmetry, that the laws of motion are the same at every place.

KEY IDEA. Internal forces cancel in pairs by the third law, so the total momentum of a group changes only through external forces, and is conserved when they add to zero. The centre of mass moves as a single body under the external forces alone.

NOTEBOOK 02 *change the mass ratio of the carts; the centre of mass never moves*

CHECK YOUR UNDERSTANDING. A skater of 60 kg standing on smooth ice throws a ball of 0.5 kg forwards at 12 m s^{-1} , measured relative to the ice. What is the skater's velocity afterwards?

2.10 From balls to atoms

Every law of this chapter was found with balls, carts and springs. This section applies the same laws to atoms and writes them in the form, and in the units, that molecular dynamics uses.

Atoms are, strictly, quantum objects, and quantum mechanics rather than Newton's laws describes them exactly. The nuclei, however, are thousands of times heavier than the electrons, and this book treats them as classical bodies obeying the second law, while quantum mechanics is used only to compute the forces on them (Chapter 8). Treating the nuclei classically is an approximation. It is least accurate for the lightest nuclei, and for hydrogen the quantum character of the nuclear motion can change measured properties even at room temperature; Chapter 8 discusses when the classical picture is adequate.

The equations of motion of N atoms

Each atom i , numbered from 1 to N , has a mass m_i , a position $\mathbf{r}_i$ and a velocity $\mathbf{v}_i$, and feels a force $\mathbf{F}_i$ from the other atoms. We write $\mathbf{r}^N$ for the whole list of positions, $\mathbf{r}_1, \dots, \mathbf{r}_N$. In the models of this book the force on an atom depends on where every atom is, but not on how fast the atoms move, until the thermostats of Chapter 13 add a drag. Before then it is a function of $\mathbf{r}^N$ alone, and Chapter 8 explains where it comes from. The second law for each atom gives

$$m_i \frac{d^2 \mathbf{r}_i}{dt^2} = \mathbf{F}_i(\mathbf{r}^N), \quad i = 1, \dots, N. \quad (2.13)$$

These are $3N$ second-order differential equations, one for each component of each position, and they are coupled: the equation for one atom contains the positions of all the others through its force, so none can be solved alone. Molecular dynamics is the numerical solution of these equations. By [Section 2.6](#) the state of the system is

the list of all positions together with the list of all velocities, $6N$ numbers in all, and it determines the motion that follows. The learned propagators planned for Part VI, programs trained to predict the state a long step ahead, take this state as their input for that reason.

Newton's law in eV, Å, fs and amu

Equation (2.13) needs one conversion in the units of this book. Forces between atoms are measured in electronvolts per ångström, eV Å^{-1} . The electronvolt is a unit of energy, which Chapter 3 introduces; for now it is enough that $1 \text{ eV} = 1.602\,176\,634 \times 10^{-19} \text{ N m}$, so that eV Å^{-1} is a unit of force,

$$1 \text{ eV Å}^{-1} = \frac{1.602\,176\,634 \times 10^{-19} \text{ N m}}{10^{-10} \text{ m}} = 1.602\,176\,634 \times 10^{-9} \text{ N},$$

since $10^{-19}/10^{-10} = 10^{-19-(-10)} = 10^{-9}$. Masses are measured in atomic mass units, $1 \text{ amu} = 1.660\,539\,068\,92 \times 10^{-27} \text{ kg}$, defined as one twelfth of the mass of an atom of carbon-12, the commonest form of carbon; tables of atomic masses use this unit. A force of 1 eV Å^{-1} acting on a mass of 1 amu therefore produces an acceleration that we first find in m s^{-2} and then convert. For the conversion, multiplying $1 \text{ Å} = 10^{-10} \text{ m}$ by 10^{10} gives $1 \text{ m} = 10^{10} \text{ Å}$, and in the same way $1 \text{ s} = 10^{15} \text{ fs}$, so that $1 \text{ s}^2 = (10^{15} \text{ fs})^2 = 10^{30} \text{ fs}^2$. A newton per kilogram is a metre per second squared, by the definition of the newton, and dividing the powers of ten gives $10^{-9}/10^{-27} = 10^{-9-(-27)} = 10^{18}$, so

$$\frac{1.602\,176\,634 \times 10^{-9} \text{ N}}{1.660\,539\,068\,92 \times 10^{-27} \text{ kg}} = 0.9648533 \times 10^{18} \text{ m s}^{-2} = 9.648\,533 \times 10^{17} \text{ m s}^{-2},$$

the last step moving the decimal point. Converting to ångströms and femtoseconds,

$$9.648533 \times 10^{17} \times \frac{10^{10}}{10^{30}} \text{ Å fs}^{-2} = 9.648\,533 \times 10^{-3} \text{ Å fs}^{-2},$$

since $17 + 10 - 30 = -3$. This number converts a force in eV Å^{-1} divided by a mass in amu into an acceleration in Å fs^{-2} .

A lithium atom, of mass 6.94 amu, its standard atomic weight³ (the masses of atoms in this book are these values), under a force of 1 eV Å^{-1} therefore accelerates at $9.648533 \times 10^{-3}/6.94 = 1.390 \times 10^{-3} \text{ Å fs}^{-2}$. Starting from rest, after 10 fs it is moving at $at = 0.0139 \text{ Å fs}^{-1}$, which, since $1 \text{ Å fs}^{-1} = 10^5 \text{ m s}^{-1}$ (Section 1.1), is 1390 m s^{-1} , about four times the speed of sound in air, 343 m s^{-1} , and it has moved $\frac{1}{2}at^2 = 0.0695 \text{ Å}$, about 2 % of the 3 Å between neighbouring atoms in lithium metal. Atoms are fast, but a few femtoseconds carry them only a small fraction of the way to their neighbours, which is why molecular dynamics can follow them in short steps; Chapter 9 shows how the

step is chosen, and a femtosecond is typical. Under the same force a hydrogen atom, $6.94/1.008 = 6.88$ times lighter, accelerates 6.88 times faster.

The same numbers show why molecular dynamics leaves gravity out. A lithium atom has mass $6.94 \times 1.66054 \times 10^{-27} \text{ kg} = 1.152 \times 10^{-26} \text{ kg}$ and so weighs $1.152 \times 10^{-26} \times 9.80665 = 1.13 \times 10^{-25} \text{ N}$. Dividing by $1.602177 \times 10^{-9} \text{ N}$, the size of $1 \text{ eV } \text{\AA}^{-1}$, gives $7 \times 10^{-17} \text{ eV } \text{\AA}^{-1}$, some 10^{16} times smaller than the force of $1 \text{ eV } \text{\AA}^{-1}$ used above as an example of a force between atoms.

Newton's law in code

In `mdlab` the factor is computed once, from the recommended values of the physical constants (CODATA), and Newton's law for every atom at once is then a few lines of code.

```

1 def acceleration(forces: ArrayLike, masses: ArrayLike) -> NDArray:
2     forces = np.asarray(forces, dtype=float)
3     masses = np.asarray(masses, dtype=float)
4     # Each atom's single mass divides all three components of its force.
5     return forces / masses[..., np.newaxis] * units.FORCE_TO_ACCEL

```

The forces arrive as an array of N rows, one per atom, and three columns, one per component, and the masses as a list of N numbers; `np.asarray` turns whatever is passed in, a list for instance, into arrays of decimal numbers. The dimensions of an array, in this sense, are the directions along which it is indexed: the forces have two, of lengths N and 3, and the masses one, of length N . When NumPy combines two arrays of different shapes it lines up their last dimensions, and a dimension of length 1 is repeated as often as needed to match the other array. Dividing the $N \times 3$ forces by the N masses directly therefore pairs the three columns of the forces with the N masses. For most N the lengths differ and NumPy stops with an error, but for $N = 3$ they agree and the division runs silently with the wrong meaning: the x components of all three forces are divided by the mass of atom 1, the y components by the mass of atom 2 and the z components by the mass of atom 3. Writing `masses[..., np.newaxis]`, where `...` stands for every dimension the array already has and `np.newaxis` adds a new last dimension of length 1, turns the list of masses into a single column, an $N \times 1$ array. Its last dimension has length 1, so NumPy repeats the column across the three components and divides every row of the forces by the mass in the same row, which is Equation (2.13) for every atom at once, whatever the value of N . Multiplying by `FORCE_TO_ACCEL` gives the accelerations in $\text{\AA fs}^{-2}$.

Momentum in a simulation

In an isolated group of atoms every force is internal, and for the models in this book the forces sum to zero, as the forces on the two carts did (Chapter 7 shows why this holds even for forces that do not act between simple pairs). By Equation (2.12) the total momentum $\mathbf{P}$ of the atoms is then conserved, and their centre of mass moves at the constant velocity $\mathbf{P}/M$.

IN PRACTICE. A simulation whose starting velocities have a non-zero total momentum carries the whole system along at $\mathbf{P}/M$ for ever, a drift that has nothing to do with the motion of the atoms relative to one another. Subtracting $\mathbf{P}/M$ from every velocity removes it. By Section 2.2 this is only a change to the frame of reference moving with the centre of mass, so it alters no acceleration and no force, and it is the usual first step when starting velocities are assigned (Section 10.5).

NOTEBOOK 02 *Newton's law for a group of atoms, and removing the drift*

CHECK YOUR UNDERSTANDING. Under the same force, which accelerates faster, a lithium atom or an oxygen atom (15.999 amu), and by what factor?

What we now have

A force is a push or a pull with a size and a direction, and the forces on a body add as vectors to a net force. Accelerations are the same in all frames of reference moving at constant velocity relative to one another, and in the inertial frames among them, to which a frame fixed to the ground is a close approximation, three laws hold: a body with no net force on it keeps its velocity; the net force equals the mass times the acceleration; and the forces between two bodies are equal and opposite. Under its weight alone every body falls with the same acceleration g . The second law turns every motion into a differential equation, whose solution is fixed by the state, the position and velocity at one instant. With a drag proportional to its velocity a body approaches its terminal velocity $v_\infty = g/\gamma$ exponentially, the gap shrinking to 37% every $1/\gamma$; on a spring it oscillates as a sinusoid with $\omega = \sqrt{k/m}$, whatever its amplitude, as long as Hooke's law holds. For several bodies the internal forces cancel in pairs, so the total momentum changes only through external forces and the centre of mass moves as a single body. For atoms the same laws give $3N$ coupled equations, which in eV, Å, fs and amu need the single factor $9.648\,533 \times 10^{-3}$, the acceleration in Å fs⁻² that a force of 1 eV Å⁻¹ gives a mass of 1 amu.

What remains is to understand forces through energy. Chapter 3 shows which forces conserve the total energy of a motion, and why that matters for every simulation.

Answers to the checks

SECTION 2.1. Zero. The puck moves in a straight line at constant speed, so its velocity is constant, and by the first law the net force on it vanishes.

SECTION 2.2. Taking forwards as positive, the passenger's velocity in the train is $v' = -1 \text{ m s}^{-1}$, so adding u to both sides of Equation (2.2), $v' = v - u$, the platform sees $v = v' + u = -1 + 5 = 4 \text{ m s}^{-1}$ forwards. A roundabout is not an inertial frame of reference: each point on it moves in a circle and so accelerates towards the centre (Section 1.7), and a ball rolled across it appears, to someone riding it, to curve with no sideways force.

SECTION 2.3. $a = F/m = 6/2 = 3 \text{ m s}^{-2}$. Doubling the mass to 4 kg halves it, to 1.5 m s^{-2} .

SECTION 2.4. The two forces are equal, but the acceleration each produces is the force divided by the mass of the body it acts on. The apple's mass is about 0.1 kg and the Earth's about $6 \times 10^{24} \text{ kg}$, so the apple accelerates at g and the Earth by an amount far too small to detect.

SECTION 2.5. In a vacuum they land together, because the only force on each is its weight, and $\mathbf{a} = \mathbf{g}$ whatever the mass. In air the hammer lands first: the drag on the feather is comparable to its tiny weight, while the drag on the hammer is negligible beside its weight.

SECTION 2.6. No: both start at rest, so their velocities agree, but their positions differ by 1 m, so their states differ. Their motions differ too, the lower stone staying 1 m below the upper at every instant. Matching velocities alone does not fix the motion.

SECTION 2.7. It slows down. Above the terminal velocity the drag is larger than the weight, so the net force points upwards. By Equation (2.7) the gap $v_0 - v_\infty$ is positive and shrinks exponentially, so the velocity falls towards v_∞ from above, steeply at first and then ever more gently, never crossing the dotted line of Figure 2.7(a).

SECTION 2.8. By Equation (2.11) the period is proportional to $\sqrt{m}$, so four times the mass doubles it, to 2 s. The period does not depend on the amplitude, so pulling twice as far leaves it at 1 s.

SECTION 2.9. The skater and ball start at rest, so their total momentum is zero. On smooth ice no horizontal external force acts, the weight and the push of the ice being vertical and balancing, so by Equation (2.12) the horizontal momentum stays zero:

$60v + 0.5 \times 12 = 0$, giving $v = -6/60 = -0.1 \text{ m s}^{-1}$. The skater slides backwards at 0.1 m s^{-1} .

SECTION 2.10. The lithium atom, which is lighter. Under one force the acceleration is inversely proportional to the mass, so it accelerates $15.999/6.94 = 2.31$ times faster than the oxygen atom.

Exercises

The exercises run from questions of understanding, through calculations, to short derivations marked ‘show that’. Full solutions are in ‘Solutions to the exercises’ at the end of the book, and Notebook 02 checks every numerical answer.

EXERCISE 2.1 (*a lift*). A lift rises at a steady 2 m s^{-1} . What is the net force on a passenger standing in it? Is a frame of reference fixed to the lift inertial? What changes while the lift is speeding up?

EXERCISE 2.2 (*a cyclist*). A cyclist riding at 6 m s^{-1} throws a ball forwards at 4 m s^{-1} relative to the bicycle. How fast does the ball move relative to the ground? How fast if it is thrown backwards at the same speed?

EXERCISE 2.3 (*two forces*). A body of mass 3 kg is pulled by a force of 4 N due east and a force of 3 N due north. Find the size and direction of the net force and of the acceleration.

EXERCISE 2.4 (*measuring mass*). The same stretched spring gives a 2 kg trolley an acceleration of 0.6 m s^{-2} and a second trolley an acceleration of 1.5 m s^{-2} . Find the mass of the second trolley and the force of the spring.

EXERCISE 2.5 (*a horse and cart*). A horse pulls a cart, and by the third law the cart pulls back on the horse with an equal force. Explain why the two can still speed up together. (Draw a free-body diagram for each, and include the push of the ground on the horse’s hooves.)

EXERCISE 2.6 (*a rope*). A 2 kg box is lifted by a vertical rope with an upward acceleration of 1.5 m s^{-2} . Draw its free-body diagram and find the tension in the rope.

EXERCISE 2.7 (*on the Moon*). On the Moon, $g = 1.62 \text{ m s}^{-2}$. Find the mass and the weight of a 70 kg astronaut there, and compare the weight with that on the Earth.

EXERCISE 2.8 (*a state*). At $t = 0$ a ball is 3 m above the ground and moving downwards at 2 m s^{-1} . Ignoring air resistance, when does it reach the ground? (Setting the height

to zero gives a quadratic equation in t , whose positive root is the answer; compare Figure 2.6.)

EXERCISE 2.9 (*shorter steps*). Show that one step of length τ from the start of the throw in Section 2.6 overshoots the true height by exactly $\frac{1}{2}g\tau^2$. Show that two steps of length $\tau/2$, covering the same time, overshoot by $\frac{1}{4}g\tau^2$, half as much.

EXERCISE 2.10 (*faster than terminal*). The bead of Section 2.7, with $\gamma = 5\text{ s}^{-1}$, is fired downwards into the liquid at 4 m s^{-1} . Find its velocity after 0.2 s.

EXERCISE 2.11 (*the approach*). For a bead released from rest with $\gamma = 5\text{ s}^{-1}$, find the time at which its velocity reaches 99 % of the terminal velocity.

EXERCISE 2.12 (*checking a solution*). Show by differentiating Equation (2.8) that dx/dt is the velocity $v_\infty(1 - e^{-\gamma t})$, and that $d^2x/dt^2 = g - \gamma dx/dt$, which is Equation (2.6).

EXERCISE 2.13 (*a stiff spring*). A 0.2 kg block on a spring of stiffness 80 N m^{-1} is pulled 3 cm from its resting place and released. Find the angular frequency, the period, the largest speed and the largest acceleration.

EXERCISE 2.14 (*a kick*). A block with $\omega = 10\text{ rad s}^{-1}$ rests at $x = 0$ and is struck so that it starts with $v_0 = 0.5\text{ m s}^{-1}$. Find A and ϕ , and show that the motion is $x = A \sin \omega t$.

EXERCISE 2.15 (*a hanging spring*). A block of mass m hangs at rest from a vertical spring of stiffness k . Show that the spring is stretched by mg/k . Measuring x downwards from this resting place, show that the block obeys $d^2x/dt^2 = -(k/m)x$, so that gravity shifts the centre of the oscillation but leaves its period unchanged.

EXERCISE 2.16 (*coupling*). On a level, smooth track, a 2 kg cart moving at 3 m s^{-1} runs into a stationary 1 kg cart and the two couple together. Using momentum, find their common velocity afterwards.

EXERCISE 2.17 (*walking on a boat*). A person of 70 kg stands at one end of a 140 kg boat at rest on still water, and walks 3 m along the boat towards the other end. Neglecting the drag of the water, use the centre of mass to find how far the boat moves, and in which direction.

EXERCISE 2.18 (*an oxygen atom*). An oxygen atom (15.999 amu) at rest feels a constant force of $0.5\text{ eV \AA}^{-1}$. Find its acceleration in $\text{\AA fs}^{-2}$ and its velocity after 5 fs, in $\text{\AA fs}^{-1}$ and in m s^{-1} .

EXERCISE 2.19 (*removing the drift*). Show that subtracting $\mathbf{P}/M$ from the velocity of every atom leaves a total momentum of zero, and that it changes no acceleration.

Work and energy

Chapter 2 predicted motion by solving the equation of motion: the forces fix the acceleration, from which we obtain the velocity and position. Some questions can be answered without finding the whole trajectory. A bead sliding into a dip, for example, may have enough speed to climb the next rise; the two carts of [Section 2.9](#) reach definite speeds when their spring falls away. Energy allows us to determine those speeds by comparing the starting and finishing states.

We begin with the work done by a force and show from Newton's second law that it equals the change in kinetic energy. For a force that depends only on position along a line, the work defines a potential energy. The sum of kinetic and potential energy is then conserved, and a graph of the potential shows the accessible positions, turning points and possible equilibria. Friction transfers energy out of the motion we are following. In two or three dimensions, position dependence alone no longer guarantees a potential: the work must also be independent of the path. We develop the line integral, partial derivatives, gradient and curl as they become necessary to test this condition. Applied to atoms, these ideas give a practical check on an isolated simulation with conservative forces: its total energy should remain constant within the numerical error. Chapter 4 uses the shape of the potential to explain oscillations.

3.1 Work done by a constant force

Pulling a sledge across snow is harder work the harder we pull and the further we go. Physics keeps the word work for a quantity built from exactly these two ingredients, a force and the distance moved, and this section defines it for a force that does not change.

A force along the motion

When a constant force of size F pushes a body through a distance d in the direction of the force, the force does the work

$$W = Fd.$$

Its unit is the newton metre, called the joule (J): $1 \text{ J} = 1 \text{ N m}$, and since $1 \text{ N} = 1 \text{ kg m s}^{-2}$ (Section 2.3), $1 \text{ J} = 1 \text{ kg m}^2 \text{ s}^{-2}$. Lifting a box of 2 kg through 1.5 m at a steady speed, for instance, takes an upward push equal to its weight, $mg = 2 \times 9.80665 = 19.6133 \text{ N}$, because a body moving at constant velocity has no acceleration, and so, by the second law, no net force on it (Section 2.3). The push does $19.6133 \times 1.5 = 29.42 \text{ J}$ of work. A force that points against the motion does negative work: while the box rises, its weight points down and does -29.42 J .

A force at an angle

The rope of a sledge usually pulls upwards as well as forwards (Figure 3.1). We split the pull into two perpendicular parts, its components along the motion and across it, as in Section 2.1. For a rope pulling with a force of size F at an angle θ above the level snow, these are $F \cos \theta$ along the snow and $F \sin \theta$ upwards. Only the first is counted as doing work: the work done by a constant force $\mathbf{F}$ while its body moves through a straight displacement $\Delta \mathbf{r}$ is defined as

$$W = \mathbf{F} \cdot \Delta \mathbf{r} = |\mathbf{F}| |\Delta \mathbf{r}| \cos \theta = F_x \Delta x + F_y \Delta y + F_z \Delta z, \quad (3.1)$$

the scalar product of Section 1.6, where θ is now the angle between the force and the displacement and Δx , Δy and Δz are the components of the displacement. The product $|\mathbf{F}| \cos \theta$ is the component of the force along the displacement, so Equation (3.1) is that component times the distance moved. This is the combination of force and distance worth naming, because Sections 3.2 and 3.3 show that it is exactly what the second law turns into a change of speed: the component across the motion bends the path but does not speed the body up or slow it down.

For the sledge, take x along the snow and y upwards, a pull of 40 N at $\theta = 30^\circ$, and a displacement of 10 m along the snow. Then $\mathbf{F} = (40 \cos 30^\circ, 40 \sin 30^\circ) = (34.64, 20.0) \text{ N}$ and $\Delta \mathbf{r} = (10, 0) \text{ m}$, so

$$W = 34.64 \times 10 + 20.0 \times 0 = 346.4 \text{ J}.$$

The upward part of the pull, 20.0 N, does no work, because the sledge does not move upwards.

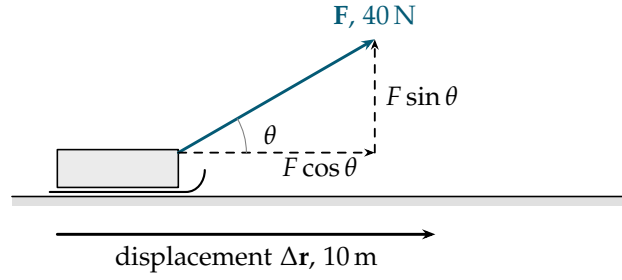


FIGURE 3.1: A sledge pulled along level snow by a rope at an angle θ . The pull $\mathbf{F}$ (teal) has a component $F \cos \theta$ along the snow, which does work, and a component $F \sin \theta$ across the motion, which does none.

Equation (3.1) gives every case at once. A force with a component along the displacement, $0 \leq \theta < \pi/2$, does positive work; a force with a component against it, $\pi/2 < \theta \leq \pi$, does negative work; and a force perpendicular to the displacement, $\theta = \pi/2$, does none, since $\cos(\pi/2) = 0$. The normal force of a level floor on a block sliding across it, and the weight of that block, both point vertically, across the horizontal motion, and so do no work on the block.

Several forces

When several constant forces act on one body, their works add up to the work of the net force. The scalar product is a sum of products of components, and each component of the net force is the sum of the components of the separate forces, so, for two forces,

$$\begin{aligned}
 (\mathbf{F}_1 + \mathbf{F}_2) \cdot \Delta \mathbf{r} &= (F_{1x} + F_{2x}) \Delta x + (F_{1y} + F_{2y}) \Delta y + (F_{1z} + F_{2z}) \Delta z \\
 &= (F_{1x} \Delta x + F_{1y} \Delta y + F_{1z} \Delta z) + (F_{2x} \Delta x + F_{2y} \Delta y + F_{2z} \Delta z) \quad \text{regrouping} \\
 &= \mathbf{F}_1 \cdot \Delta \mathbf{r} + \mathbf{F}_2 \cdot \Delta \mathbf{r},
 \end{aligned}$$

and the same regrouping works for any number of forces. For the box lifted at a steady speed, the push does 29.42 J, the weight -29.42 J, and the net force, which is zero, does none.

CHECK YOUR UNDERSTANDING. A waiter carries a tray at a constant height and a steady speed across a room. How much work does the upward push of the waiter's hand do on the tray?

3.2 Kinetic energy

The work done on a body must go somewhere. This section shows, from the second law, that the work done by the net force changes one particular property of the motion, a combination of mass and speed called the kinetic energy.

Along a line

Let a constant net force F act along a line on a body of mass m . By the second law its acceleration $a = F/m$ is constant, and Equation (1.5) relates its velocity v at position x to its velocity v_0 at position x_0 , $v^2 = v_0^2 + 2a(x - x_0)$. Multiplying both sides by $m/2$,

$$\begin{aligned}\frac{1}{2}mv^2 &= \frac{1}{2}mv_0^2 + ma(x - x_0) \\ &= \frac{1}{2}mv_0^2 + F(x - x_0) \quad \text{by the second law, } ma = F.\end{aligned}$$

The last term is the work done by the net force over the displacement $x - x_0$, so the work has been added to the quantity $\frac{1}{2}mv^2$. This quantity is the kinetic energy of the body,

$$K = \frac{1}{2}mv^2, \quad (3.2)$$

and the result reads

$$W = K - K_0,$$

where $K_0 = \frac{1}{2}mv_0^2$: the work done by the net force equals the change in the kinetic energy. Its unit is the joule, as it must be, since a kilogram times a speed squared is $\text{kg m}^2 \text{s}^{-2}$.

The trolley of Section 2.3, of mass 20 kg, pushed from rest by a net force of 10 N through 1 m, gains $W = 10 \times 1 = 10 \text{ J}$ of kinetic energy. Setting $\frac{1}{2}mv^2 = K$ and solving for the speed,

$$v = \sqrt{\frac{2K}{m}} = \sqrt{\frac{2 \times 10}{20}} = 1 \text{ m s}^{-1}.$$

The second law gives the same result the long way: the acceleration is $10/20 = 0.5 \text{ m s}^{-2}$, the trolley covers 1 m in 2 s, since $\frac{1}{2} \times 0.5 \times 2^2 = 1$, and by then it moves at $0.5 \times 2 = 1 \text{ m s}^{-1}$. The energy argument never needed the time.

In space

The same steps work for a constant force in three dimensions. The acceleration $\mathbf{a} = \mathbf{F}/m$ is constant, so each component obeys Equation (1.5) on its own, $v_x^2 = v_{0x}^2 + 2a_x \Delta x$, and likewise for y and z (a component with no acceleration keeps a constant velocity, so the equation holds for it too, although Equation (1.5) was derived for $a \neq 0$), where $\Delta \mathbf{r} = (\Delta x, \Delta y, \Delta z)$ is the displacement. Adding the three equations, and writing $v^2 = \mathbf{v} \cdot \mathbf{v} = v_x^2 + v_y^2 + v_z^2$ for the speed squared,

$$v^2 = v_0^2 + 2(a_x \Delta x + a_y \Delta y + a_z \Delta z) = v_0^2 + 2\mathbf{a} \cdot \Delta \mathbf{r},$$

and multiplying by $m/2$ and using $m\mathbf{a} = \mathbf{F}$,

$$\frac{1}{2}mv^2 - \frac{1}{2}mv_0^2 = \mathbf{F} \cdot \Delta \mathbf{r} = W.$$

The kinetic energy $\frac{1}{2}mv^2$ depends on the speed alone, not on the direction of motion, and the work is that of Equation (3.1), with $\Delta \mathbf{r}$ the displacement from start to finish even when the path between them curves. The component of the force perpendicular to $\Delta \mathbf{r}$ has indeed changed nothing that K measures.

A projectile shows the result at work. A ball thrown at 12 m s^{-1} at 60° above the horizontal feels only its weight, $\mathbf{F} = (0, -mg)$, with y upwards, so over a rise y the weight does the work $\mathbf{F} \cdot \Delta \mathbf{r} = -mgy$. At the top of the path the velocity is horizontal (Section 1.6) and equal to the unchanging horizontal part, $12 \cos 60^\circ = 6 \text{ m s}^{-1}$. Equating the change of kinetic energy to the work and dividing by m ,

$$\frac{1}{2}(6^2 - 12^2) = -gy_{\text{top}} \implies y_{\text{top}} = \frac{144 - 36}{2 \times 9.80665} = 5.507 \text{ m},$$

the height that the vertical motion alone gives, $(12 \sin 60^\circ)^2 / 2g$, by Equation (1.5), since $(12 \sin 60^\circ)^2 = 12^2(1 - \cos^2 60^\circ) = 144 - 36$. The mass has cancelled, as it did for free fall in Section 2.5.

Since K grows as the square of the speed, doubling the speed makes it four times larger. A car braked by a force of given size F , treated as a single body that slides to rest (its spinning wheels carry a little energy of their own, Chapter 5), stops when that force has done work $-K$ on it. The force points against the motion, so over a distance d it does the work $-Fd$, and $-Fd = -K$ gives $d = K/F$: at twice the speed the car needs four times the distance to stop.

CHECK YOUR UNDERSTANDING. A body's kinetic energy is 50 J. What is it when the speed is halved? What is the kinetic energy of a 2 kg body moving at 3 m s^{-1} ?

3.3 Work done by a varying force

Most forces change as a body moves: a spring pulls harder the further it is stretched, and the forces between atoms change with every step the atoms take. [Section 3.2](#) relied on a constant force, through [Equation \(1.5\)](#). This section defines the work of a force that varies and shows that the change in kinetic energy still equals it.

Work as an area

Let a force $F(x)$ act along a line, with a value that depends on the position x , while its body moves from x_A to x_B . Divide the way into many short steps of length Δx , and let x_k be the middle of the k th step. Over a step short enough that the force hardly changes, the force does about $F(x_k) \Delta x$ of work, the area of a thin rectangle of height $F(x_k)$ and width Δx ([Figure 3.2a](#)). Adding the steps, and letting $\Delta x \rightarrow 0$ as in [Section 1.4](#),

$$W = \lim_{\Delta x \rightarrow 0} \sum_k F(x_k) \Delta x = \int_{x_A}^{x_B} F(x) dx : \quad (3.3)$$

the work is the area under the graph of the force against position, counted as negative where F is negative, and with the sign reversed again when the body moves towards smaller x , $x_B < x_A$, since every step Δx is then negative. The force of [Figure 3.2\(a\)](#) is $F(x) = 3 + 1.5 \sin(1.4x) + 0.4x$ in newtons, with x in metres, from $x_A = 0.5$ m to $x_B = 4.5$ m. The eight rectangles drawn give 15.743 J, and 128 rectangles give 15.748 J, the area itself to this precision: the area is $G(4.5) - G(0.5) = 15.748$ J, with $G(x) = 3x + 0.2x^2 - (1.5/1.4) \cos 1.4x$, whose derivative is F . For a constant force the integral is $F(x_B - x_A)$, which is [Equation \(3.1\)](#) along a line.

Stretching a spring

A hand that stretches a spring slowly, so that the end of the spring does not accelerate, must pull with a force equal and opposite to the pull of the spring, $+kx$ by Hooke's law ([Section 2.8](#)). Stretching from $x = 0$ to x , it does the work

$$\int_0^x kx' dx' = \left[\frac{1}{2} kx'^2 \right]_0^x = \frac{1}{2} kx^2,$$

the area of a triangle of base x and height kx ([Figure 3.2b](#)). The variable of integration is written x' to keep it apart from the upper limit x , as in [Section 1.4](#). For the spring of 50 N m^{-1} in [Section 2.8](#), stretching by 4 cm takes $\frac{1}{2} \times 50 \times 0.04^2 = 0.04$ J. The spring's

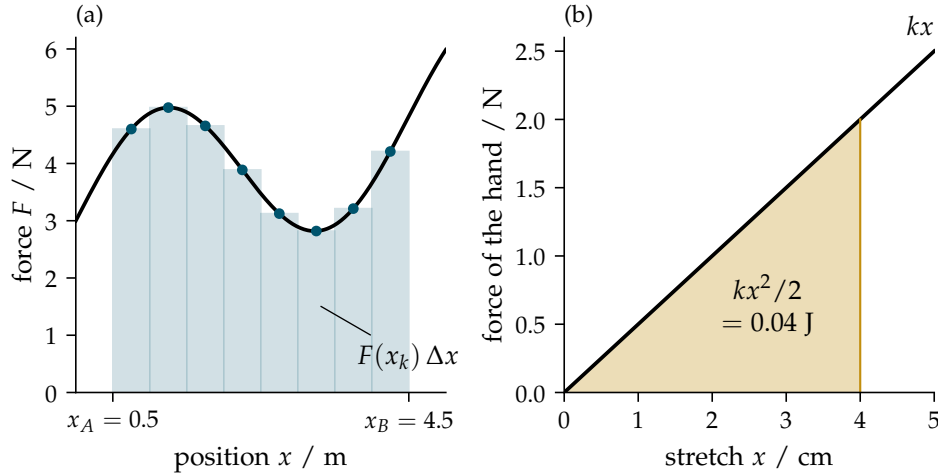


FIGURE 3.2: The work of a varying force. (a) A force that changes along the way; over each short step Δx it does about $F(x_k) \Delta x$ of work, the area of a rectangle, and the rectangles add up to the area under the curve. (b) The force kx that a hand exerts to stretch a spring of stiffness 50 N m^{-1} slowly; the work to stretch it by 4 cm is the shaded triangle, $\frac{1}{2}kx^2 = 0.04 \text{ J}$.

own force, $-kx$, does the opposite work when the stretch grows. When its end moves from x_A to x_B , by whatever means,

$$W_{\text{spring}} = \int_{x_A}^{x_B} (-kx) dx = \left[-\frac{1}{2}kx^2 \right]_{x_A}^{x_B} = \frac{1}{2}kx_A^2 - \frac{1}{2}kx_B^2. \quad (3.4)$$

Power

For a force that varies, we look at the rate at which the kinetic energy changes, in any number of dimensions. Write $K = \frac{1}{2}m(v_x^2 + v_y^2 + v_z^2)$. By the chain rule of [Section 1.3](#), the rate of change of v_x^2 is $2v_x dv_x/dt = 2v_x a_x$, and likewise for the other components, so

$$\begin{aligned} \frac{dK}{dt} &= \frac{1}{2}m(2v_x a_x + 2v_y a_y + 2v_z a_z) && \text{differentiating each square} \\ &= m \mathbf{a} \cdot \mathbf{v} && \text{the scalar product of Section 1.6} \\ &= \mathbf{F} \cdot \mathbf{v} && \text{by the second law, } m\mathbf{a} = \mathbf{F}, \end{aligned}$$

where $\mathbf{F}$ is the net force. Since $\mathbf{F} \cdot \mathbf{v} = |\mathbf{F}| |\mathbf{v}| \cos \theta$, with θ the angle between the force and the velocity, a net force perpendicular to the velocity leaves the speed unchanged at that instant and only bends the path. The quantity $\mathbf{F} \cdot \mathbf{v}$ is the rate at which the force does work. Over a short interval Δt the body moves through $\Delta \mathbf{r} \approx \mathbf{v} \Delta t$, and on so short a stretch the force is nearly constant, so by [Equation \(3.1\)](#) it does about $\mathbf{F} \cdot \mathbf{v} \Delta t$ of work.

The rate at which a force does work is called its power, measured in joules per second, or watts (W). The trolley pushed by 10 N while it moves at 1 m s^{-1} receives 10 W.

The work-energy theorem

Integrating the rate of change of K from a time t_A to a time t_B , by Equation (1.3),

$$K(t_B) - K(t_A) = \int_{t_A}^{t_B} \mathbf{F} \cdot \mathbf{v} dt = W. \quad (3.5)$$

The integral adds up the work $\mathbf{F} \cdot \mathbf{v} dt$ done over every short interval of the motion, so it is the work done by the net force along the way the body actually goes. This is the work-energy theorem: *the change in the kinetic energy of a body equals the work done on it by the net force*. It holds for any force, constant or varying, and for any motion; the second law is its only ingredient.

When the motion is along a line and the force depends on position alone, the integral over time becomes the area of Equation (3.3). Let $G(x)$ be any function whose derivative is the force, $G'(x) = F(x)$. One such function is the area $G(x) = \int_{x_A}^x F(x') dx'$, because moving its upper limit from x to $x + h$ adds a thin strip of area about $F(x)h$, so that $G'(x) = \lim_{h \rightarrow 0} [G(x+h) - G(x)]/h = F(x)$. This is the fundamental theorem of calculus (Section 1.4) read the other way: integrating a function and then differentiating returns the function. Along the motion $x(t)$ the chain rule gives

$$\frac{d}{dt} G(x(t)) = G'(x) \frac{dx}{dt} = F(x) v,$$

so the integrand Fv is the rate of change of $G(x(t))$, and integrating it from t_A to t_B gives the change in G ,

$$\int_{t_A}^{t_B} F(x) v dt = G(x_B) - G(x_A) = \int_{x_A}^{x_B} F(x) dx,$$

where x_A and x_B are the positions at the two times. This holds even if the body goes back and forth on the way, because only the ends enter.

The block of Section 2.8, of mass 0.5 kg, released from rest 4 cm from its resting place, shows what this buys. As it moves to $x = 0$ the spring does, by Equation (3.4), $\frac{1}{2} \times 50 \times 0.04^2 - 0 = 0.04 \text{ J}$ of work on it, and the block's kinetic energy grows from zero to 0.04 J. Its speed at the centre is therefore

$$v = \sqrt{\frac{2 \times 0.04}{0.5}} = 0.4 \text{ m s}^{-1},$$

the largest speed $A\omega$ found in [Section 2.8](#) from the full solution. The energy argument needed neither the sinusoid nor the time taken.

CHECK YOUR UNDERSTANDING. How much work does it take to stretch the spring of 50 N m^{-1} from 4 cm to 8 cm? Why is it more than the work of the first 4 cm?

3.4 Potential energy and the conservation of energy

A ball thrown upwards slows down, stops and falls back, arriving at the hand with the speed it left with. Its kinetic energy has gone and come back, as though it had been stored somewhere on the way up. This section makes the store precise for a position-dependent force along a line, and shows that Newton's second law then keeps the total fixed.

Potential energy

The weight of a body and the pull of a spring both depend only on where the body is, not on how fast it moves or on the time. For such a force $F(x)$ along a line, [Equation \(3.3\)](#) gives the work from x_A to x_B as an integral fixed by the two ends. Choose a fixed reference point x_0 , which need not be where the motion starts, and define the potential energy

$$U(x) = - \int_{x_0}^x F(x') dx', \quad (3.6)$$

minus the work the force does as its body moves from x_0 to x . Two rules for integrals follow from writing each one, as in [Section 3.3](#), as the change $G(b) - G(a)$ in a function with $G' = F$: reversing the limits changes the sign, since $G(a) - G(b) = -[G(b) - G(a)]$, and an integral can be split at any point c , inside the range or not, since $[G(c) - G(a)] + [G(b) - G(c)] = G(b) - G(a)$. Then, splitting the integral at x_0 and reversing the first part,

$$\begin{aligned} W_{A \rightarrow B} &= \int_{x_A}^{x_B} F dx = \int_{x_A}^{x_0} F dx + \int_{x_0}^{x_B} F dx && \text{going by way of } x_0 \\ &= - \int_{x_0}^{x_A} F dx + \int_{x_0}^{x_B} F dx && \text{reversing the first integral} \\ &= U(x_A) - U(x_B) && \text{by Equation (3.6).} \end{aligned}$$

The work done by the force is the drop in potential energy. The force uses up potential energy when it does positive work, and stores it when it does negative work, as the weight does while the ball rises.

Moving the reference point from x_0 to x_1 adds the same constant to U everywhere, since splitting the integral at x_0 gives $-\int_{x_1}^x F dx' = -\int_{x_1}^{x_0} F dx' + U(x)$, whose first term does not depend on x ; this changes no drop in U . Only differences of potential energy have a physical meaning, and we choose x_0 wherever it is convenient. The force follows from U : by the fundamental theorem of calculus, read as in [Section 3.3](#), the derivative of $\int_{x_0}^x F(x') dx'$ with respect to its upper limit is $F(x)$, so

$$F(x) = -\frac{dU}{dx}. \quad (3.7)$$

The force points the way U decreases, and the steeper the graph of U , the stronger the force.

Two potential energies follow at once. For a body near the ground, with x its height measured upwards, the weight is $F = -mg$, and with $x_0 = 0$,

$$U(x) = -\int_0^x (-mg) dx' = mgx.$$

For a spring, $F = -kx$, and with $x_0 = 0$ at the natural length,

$$U(x) = -\int_0^x (-kx') dx' = \frac{1}{2}kx^2,$$

the work of [Section 3.3](#) now stored in the stretched spring.

Conservation of energy

When the net force is of this kind, the work-energy theorem $K_B - K_A = W_{A \rightarrow B}$ and the drop $W_{A \rightarrow B} = U_A - U_B$ combine into $K_B - K_A = U_A - U_B$, or, moving U_B to the left and K_A to the right,

$$K_B + U_B = K_A + U_A.$$

The sum of the two energies is the same at every pair of points along the motion. We call it the total energy, $E = K + U$. The argument can also be run as a rate, which shows exactly where Newton's law enters.

PROPOSITION 3.1. *Let a body of mass m move along a line under a net force $F(x) = -dU/dx$ that depends only on its position. Then its total energy*

$$E = \frac{1}{2}mv^2 + U(x)$$

keeps the same value throughout the motion.

Proof. Differentiate E along the motion, using the chain rule on each term: $\frac{1}{2}mv^2$ changes at the rate $mv \, dv/dt$, and $U(x(t))$ at the rate $(dU/dx)(dx/dt) = -Fv$. Hence

$$\frac{dE}{dt} = mv \frac{dv}{dt} - Fv = v \left(m \frac{dv}{dt} - F \right) = 0,$$

because the bracket vanishes by the second law, $m \, dv/dt = F$. A quantity whose rate of change is zero does not change. $\square$

Newton's second law enters at a single point: it turns the rate of change of the kinetic energy into the power of the force, which the potential energy then exactly cancels. Energy is therefore conserved, in the sense of [Section 2.9](#), and [Proposition 3.1](#) is the conservation of energy. Forces that have a potential energy, and so conserve E , are called conservative.

Two checks

Energy reproduces the results of the earlier chapters with no equation of motion solved. A ball of 0.5 kg thrown up at $12 \, \text{m s}^{-1}$ starts at $x = 0$, where $U = 0$, with $E = K = \frac{1}{2} \times 0.5 \times 12^2 = 36 \, \text{J}$. At the top of its flight $K = 0$, so all of E is potential energy, $mgx_{\max} = E$, and

$$x_{\max} = \frac{E}{mg} = \frac{36}{0.5 \times 9.80665} = 7.342 \, \text{m},$$

the height found in [Section 1.4](#). Halfway up, at 3.671 m, $U = mgx$ is half its value at the top, so half of E is potential energy and half kinetic: $\frac{1}{2}mv^2 = 18 \, \text{J}$, half the kinetic energy of the throw, so v^2 is half of 12^2 and $v = 12/\sqrt{2} = 8.485 \, \text{m s}^{-1}$. [Figure 3.3](#) follows K and U through the flight: they trade with each other while their sum stays at 36 J. Plotted against height, $U = mgx$ is a straight line and so is $K = E - mgx$, which reaches zero exactly at the top.

For the block on a spring, at a turning point $x = \pm A$ the block is at rest and $E = \frac{1}{2}kA^2$; at the centre $U = 0$ is at its smallest, so $K = E - U$ and the speed are at their largest, and $E = \frac{1}{2}mv_{\max}^2$. Equating the two,

$$\frac{1}{2}mv_{\max}^2 = \frac{1}{2}kA^2 \quad \implies \quad v_{\max} = A\sqrt{\frac{k}{m}} = A\omega,$$

with ω from [Equation \(2.11\)](#), as in [Section 2.8](#).

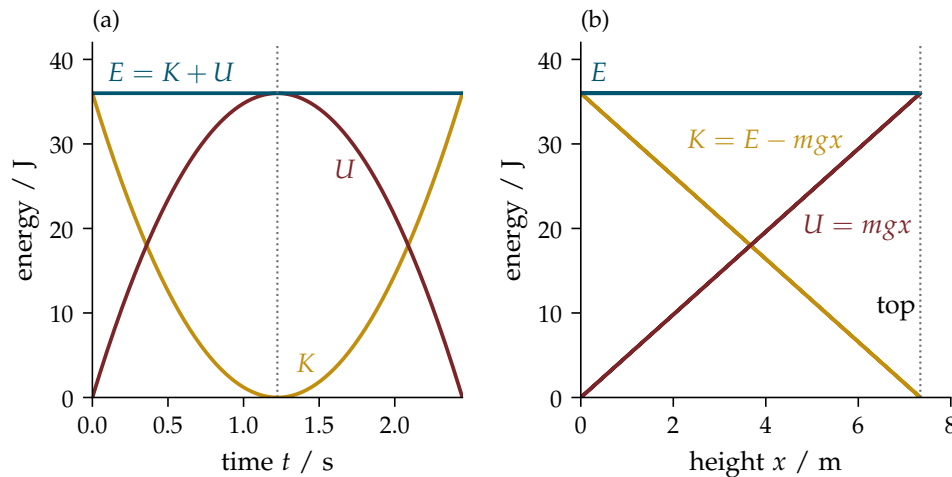


FIGURE 3.3: The energies of a ball of 0.5 kg thrown straight up at 12 m s^{-1} . (a) Kinetic energy K , potential energy $U = mgx$ and their sum E against time; the dotted line marks the top. (b) The same against height: K and U are straight lines, and K reaches zero at the top, 7.342 m.

KEY IDEA. A force that depends only on position along a line has a potential energy $U(x) = -\int_{x_0}^x F(x') dx'$, with $F = -dU/dx$. Newton's second law then keeps the total energy $E = K + U$ constant: kinetic and potential energy are traded, never their sum.

NOTEBOOK 03 follow K , U and E through the throw with a time cursor

CHECK YOUR UNDERSTANDING. A ball is thrown at 12 m s^{-1} from a balcony 5 m above the ground. With what speed does it reach the ground? Does the answer depend on whether it is thrown upwards, sideways or downwards?

3.5 Sliding on a hill-shaped track

A ball thrown upwards moves along a straight line. Many motions follow a curved track instead, as a sledge follows the slopes of a hillside, and energy is most useful there, because the forces along a curved track change from point to point. This section treats an object sliding along a track shaped into hills and valleys, and finds its speed at every point from its height alone.

A bead on a wire

The cleanest version of such a track is a smooth, stiff wire bent into hills and valleys in a vertical plane, with a bead threaded on it. We measure x horizontally and y vertically

upwards, and describe the shape of the wire by its height $h(x)$ above the ground; our example, drawn in [Figure 3.5\(a\)](#) in the next section, is

$$h(x) = 0.15x^4 - 0.6x^2 + 0.15x + 1.0, \quad (3.8)$$

with x and h in metres. It has a deep valley on the left, a hill in the middle and a shallower valley on the right. Three features of this system are chosen so that it obeys the simple laws below, and each is an idealisation that a real track only approaches.

- The bead is threaded on the wire, so it cannot leave the track. A block sliding on top of a track could lose contact over a crest, where the track curves downwards more sharply than the fast-moving block can follow, and then fly through the air instead.
- The bead slides. A ball rolling along a track also spins, and part of its kinetic energy is then in the spinning, so at each height a rolling ball moves more slowly than a sliding body; Chapter 5 treats spinning.
- The wire is smooth. No real wire is perfectly smooth: friction slowly removes energy ([Section 3.7](#)), and a real bead never quite returns to the height from which it was released.

The push of the wire does no work

Two forces act on the bead: its weight, mg downwards, and the push of the wire. A smooth surface exerts no friction, so the push of the wire has no component along the wire: it is a normal force, perpendicular to the wire at the point where the bead is ([Figure 3.4a](#)). Its size and direction change as the bead moves, and we do not need to know them. The velocity of the bead points along its path ([Section 1.6](#)), which is along the wire, so the normal force $\mathbf{F}_n$ is perpendicular to the velocity at every instant, $\mathbf{F}_n \cdot \mathbf{v} = 0$. Its power is therefore zero throughout, so the work it does over any part of the motion, the integral of its power as in [Equation \(3.5\)](#), is zero.

The work of the weight

The weight is constant, but the path is curved. To extend the straight-path argument of [Equation \(3.1\)](#), we replace the wire between two points A and B by a chain of short straight chords ([Figure 3.4b](#)). On the k th chord the displacement is $(\Delta x_k, \Delta y_k)$, and the weight, $(0, -mg)$, does the work

$$(0, -mg) \cdot (\Delta x_k, \Delta y_k) = -mg \Delta y_k.$$

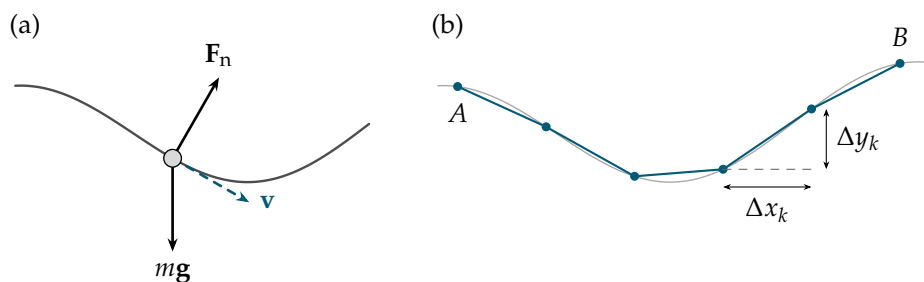


FIGURE 3.4: A bead on a smooth wire. (a) The two forces on the bead: its weight mg and the push F_n of the wire, which is perpendicular to the wire and so to the velocity $\mathbf{v}$. (b) The wire from A to B replaced by short straight chords; on the chord that rises by Δy_k the weight does the work $-mg \Delta y_k$.

Adding the chords, the rises Δy_k add up to the total rise from A to B , since each chord starts where the one before it ends, so

$$W_{\text{weight}} = -mg \sum_k \Delta y_k = -mg (y_B - y_A).$$

The result is the same for any number of chords, so it holds also in the limit in which the chords follow the wire exactly. The weight does work that depends only on the change in height, whatever the shape of the path between.

The speed at every height

By the work-energy theorem [Equation \(3.5\)](#), the change in the bead's kinetic energy is the work of the weight plus the work of the wire, $-mg (y_B - y_A) + 0$. On the wire $y = h(x)$, so

$$\begin{aligned} \frac{1}{2}mv_B^2 - \frac{1}{2}mv_A^2 &= -mg (h(x_B) - h(x_A)) \\ &= -mg h(x_B) + mg h(x_A) \quad \text{multiplying out the bracket,} \end{aligned}$$

$$\frac{1}{2}mv_B^2 + mg h(x_B) = \frac{1}{2}mv_A^2 + mg h(x_A) \quad \text{adding } mg h(x_B) + \frac{1}{2}mv_A^2 \text{ to both sides.}$$

The total energy

$$E = \frac{1}{2}mv^2 + mg h(x) \tag{3.9}$$

is conserved, with the potential energy $U = mgh(x)$ of the weight. It is convenient to write $E = mgh_E$, so that h_E is the height at which the bead would be at rest; a bead

released from rest at height h_E has exactly this energy. Then $\frac{1}{2}mv^2 = mg(h_E - h)$, and dividing by $m/2$ and taking the square root,

$$v = \sqrt{2g(h_E - h(x))}.$$

The mass has cancelled: every bead released from the same height moves at the same speed at the same place. Released from rest at $h_E = 1.3$ m, the bead passes the bottom of the deep valley, where $h = 0.1834$ m, at $\sqrt{2 \times 9.80665 \times (1.3 - 0.1834)} = 4.680 \text{ m s}^{-1}$, crosses the top of the hill, at 1.0094 m, at 2.387 m s^{-1} , and passes the bottom of the shallow valley, at 0.6072 m, at 3.686 m s^{-1} . None of these numbers required the shape of the wire between the points, or the time taken to get there.

Energy gives the speed at each place directly. It does not say which way the bead is moving: a bead passing a point to the left and one passing it to the right have the same speed. Nor does it give the time of arrival at once; that needs the time spent on each short stretch of wire, its length divided by the speed there, added up along the way, or the equation of motion along the wire, which Chapter 6 derives in a form suited to such constrained motion.

NOTEBOOK 03 *release the bead from different heights and read its speed*

CHECK YOUR UNDERSTANDING. A bead is released from rest on the left slope at a height of 0.8 m. Can it reach the shallow valley on the right? What is the least speed it would need at the bottom of the deep valley to cross the hill?

3.6 Energy diagrams

The conservation of energy turns a graph of the potential energy into a map of every possible motion. This section reads such a map.

Allowed regions and turning points

For a body moving along a line with potential energy $U(x)$, or a bead on a wire with $U = mgh(x)$, the kinetic energy at each point is what remains of the total, $K = E - U(x)$. A kinetic energy $\frac{1}{2}mv^2$ cannot be negative, so the body can only be where

$$U(x) \leq E.$$

These points form the allowed region of the motion. Where U rises to meet E , the kinetic energy falls to zero, the body stops for an instant, and the force, which points down the slope of U by Equation (3.7) (for the bead, the part of the weight along

the wire, which points downhill), sends it back the way it came. Such a point, where $U(x) = E$ with a non-zero slope of U , is a turning point. If E instead meets U at an equilibrium, the force also vanishes there: a body placed there at rest stays there, and approaching an unstable equilibrium can take indefinitely long. Away from these special points, the speed follows from $\frac{1}{2}mv^2 = E - U(x)$,

$$v = \sqrt{\frac{2(E - U(x))}{m}}.$$

A graph of U with a horizontal line at the height E is called an energy diagram. For the bead, $U/mg = h$ is the shape of the wire itself, so its energy diagram is a picture of the track, and $E/mg = h_E$ is the height at which the bead would be at rest. Figure 3.5(a) draws three energies. Their turning points, the solutions of $h(x) = h_E$, were found by computer; substituting them into Equation (3.8) checks them, and two are exact, $h(-1) = 0.4$ and $h(2) = 1.3$.

- At $h_E = 0.4$ m the bead is confined to the deep valley, between the turning points $x = -1.831$ m and -1.000 m. It slides back and forth between them for ever.
- At $h_E = 0.8$ m there are two separate allowed regions, one in each valley: from -2.042 to -0.477 m, and from 0.795 to 1.724 m. With the same energy the bead may be in either, but it cannot cross the hill between them, which rises to 1.009 m. Which valley it occupies depends only on where it started.
- At $h_E = 1.3$ m the bead passes over the hill and turns only at the steep ends of the track, at -2.206 and 2.000 m.

Figure 3.5(b) gives the speed along the track at the highest of these energies. It falls to zero at the turning points, is greatest at the bottom of the deep valley, 4.680 m s^{-1} , and dips to 2.387 m s^{-1} as the bead crosses the hill.

Equilibria

A body placed at rest stays at rest only where the net force on it is zero. By Equation (3.7) these are the points where $dU/dx = 0$, where the graph of U is flat: the bottoms of valleys, the tops of hills, and any level shelf. Such a point x_0 is called an equilibrium. For the bead, the push of the wire and the weight balance where the wire is horizontal, $dh/dx = 0$, since only there is the push, which is perpendicular to the wire, vertical, so that it can cancel the weight. Differentiating Equation (3.8) with the power rule,

$$\frac{dh}{dx} = 0.6x^3 - 1.2x + 0.15,$$

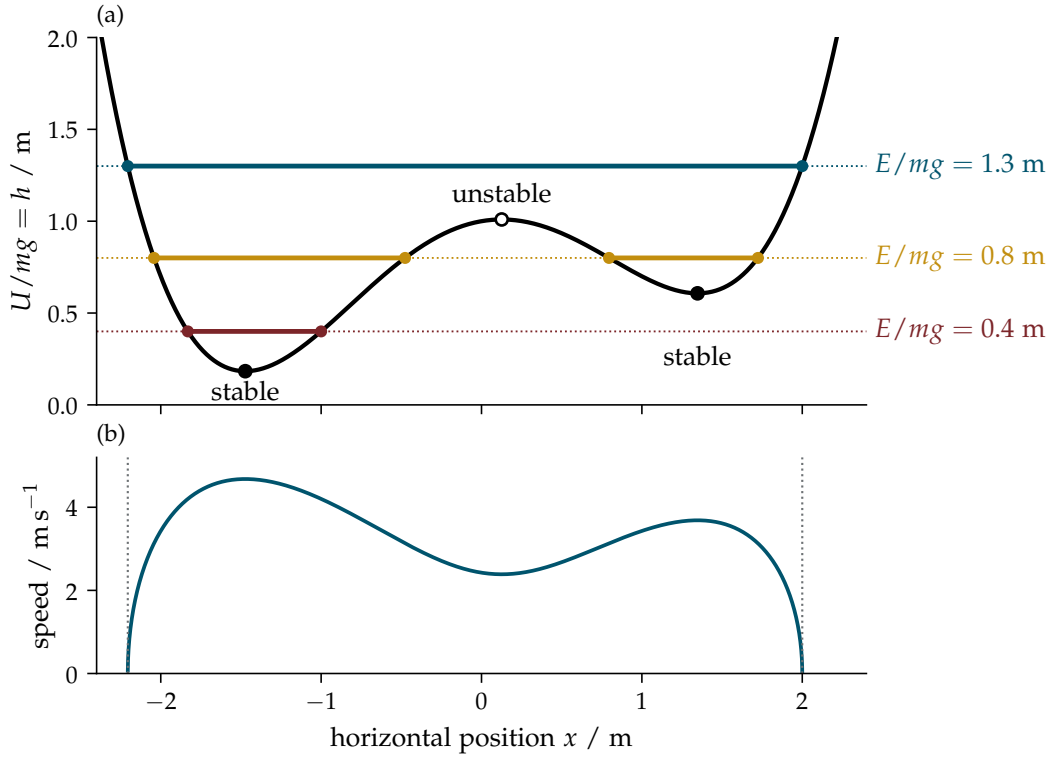


FIGURE 3.5: The track as an energy diagram. (a) The potential energy of the bead divided by mg , which is the height $h(x)$ of the wire, with three total energies E/mg . Each energy line is solid where the bead may be ($U \leq E$) and dotted where it may not; dots mark the turning points. Filled circles mark the stable equilibria at the valley bottoms, and the open circle the unstable one at the top of the hill. (b) The speed along the track at $E/mg = 1.3 m$; the dotted lines are the turning points.

and this cubic is zero at three points, which a computer finds to be $x = -1.473 m$ (the deep valley, $h = 0.183 m$), $x = 0.126 m$ (the hill, $h = 1.009 m$) and $x = 1.347 m$ (the shallow valley, $h = 0.607 m$).

The equilibria differ in what happens after a small nudge. At the bottom of a valley, the slope of U on either side points back down into the valley, so the force on a displaced body points back towards the equilibrium, and the body swings about it: the equilibrium is stable. At the top of a hill, the slope on either side leads away, and a body nudged off it runs away down one side: the equilibrium is unstable.

The second derivative of U tells the two apart without a picture. Near an equilibrium x_0 , the Taylor series of Section 1.8, applied to the force F about x_0 with $x - x_0$ in the role of the interval τ , gives

$$\begin{aligned} F(x) &= F(x_0) + F'(x_0)(x - x_0) + \mathcal{O}((x - x_0)^2) \\ &= -U''(x_0)(x - x_0) + \mathcal{O}((x - x_0)^2) \quad \text{since } F(x_0) = 0 \text{ and } F' = -U'', \end{aligned}$$

where U'' is the second derivative of U with respect to x . If $U''(x_0) > 0$, the graph of U curves upwards, as at a valley bottom, and close to x_0 the force is that of a spring of stiffness $k = U''(x_0)$, $F \approx -k(x - x_0)$: it always points back, the equilibrium is stable. By Section 2.8, with the displacement $x - x_0$ in place of x , the body then oscillates about x_0 , approximately, with angular frequency $\sqrt{U''(x_0)/m}$, and Chapter 4 examines how good the approximation is. If $U''(x_0) < 0$, the force points away on both sides, and the equilibrium is unstable. If $U''(x_0) = 0$, the next terms of the series decide. For the track, $d^2h/dx^2 = 1.8x^2 - 1.2$, which is $+2.706 \text{ m}^{-1}$ and $+2.066 \text{ m}^{-1}$ at the two valley bottoms and -1.171 m^{-1} at the top of the hill, as the picture says.

NOTEBOOK 03 move E on the energy diagram and watch the allowed regions change

CHECK YOUR UNDERSTANDING. At $E/mg = 0.8 \text{ m}$ a bead starts at rest at $x = 0.795 \text{ m}$. Where does it next turn back, where is it fastest, and how fast is it moving there?

3.7 Friction and the loss of energy

A box shoved across a floor slides to a stop, and a real bead on a real wire never quite climbs back to the height it was released from. Both lose energy to friction. This section measures the loss and finds the rule that says how fast the total energy changes when part of the force is not conservative.

Sliding friction

When one surface slides over another, each exerts on the other a friction force along the surfaces, and the force on each points against the direction in which that surface slides over the other. For many pairs of dry surfaces, the size of this force depends only weakly on the speed and on the area in contact, and is roughly proportional to the normal force F_n with which the surfaces are pressed together,

$$F_{\text{fr}} = \mu_k F_n.$$

The number μ_k , the coefficient of sliding friction, depends on the two surfaces and is found by measurement. This rule is an empirical description of limited accuracy, and we use it only to see how energy is lost.

A box of 5 kg slides across a level floor at 2 m s^{-1} , with $\mu_k = 0.3$. Vertically the box does not accelerate, so the normal force balances the weight, $F_n = mg$, and the friction is $0.3 \times 5 \times 9.80665 = 14.71 \text{ N}$, pointing against the motion. The weight and the normal force are perpendicular to the motion and do no work, so over a distance d the work

on the box is that of friction, $-\mu_k mg d$. The box stops when this has removed all of its kinetic energy,

$$\begin{aligned}
 0 - \frac{1}{2}mv_0^2 &= -\mu_k mg d, \\
 d &= \frac{v_0^2}{2\mu_k g} && \text{dividing both sides by } -\mu_k mg \\
 &= \frac{2^2}{2 \times 0.3 \times 9.80665} = 0.680 \text{ m.}
 \end{aligned}$$

The mass cancels again: a heavier box has proportionally more kinetic energy to lose, and it presses harder on the floor and feels a proportionally larger friction.

The 10 J of kinetic energy the box had is not destroyed. The rubbing surfaces warm up, and Chapter 12 shows that warmth is the hidden kinetic and potential energy of the jiggling atoms of the box and the floor. Energy has passed from the motion of the box as a whole, which we follow, into motions of its atoms, which we do not.

Down a rough ramp

A block of 2 kg slides from rest down a straight ramp 3 m long, inclined at 30° , with $\mu_k = 0.2$ (Figure 3.6). Three forces act. The line perpendicular to the ramp is the vertical turned through the same 30° as the ramp is turned from the level, so it makes an angle of 30° with the weight, mg straight down. As for the sledge of Section 3.1, the weight therefore has a component $mg \cos 30^\circ$ into the ramp and a component $mg \sin 30^\circ$ at right angles to that, down the ramp. The normal force balances the first of these, because the block does not accelerate into or away from the ramp, so $F_n = mg \cos 30^\circ = 16.986 \text{ N}$. Friction acts up the ramp with size $\mu_k F_n = 0.2 \times 16.986 = 3.397 \text{ N}$.

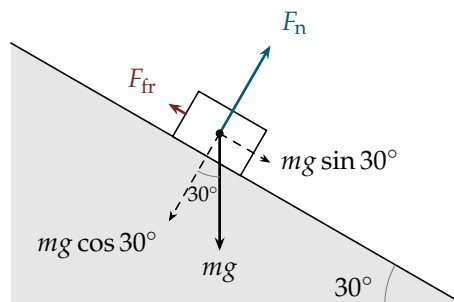


FIGURE 3.6: A block sliding down a rough ramp. Its weight mg is split (dashed) into a part along the ramp and a part into it. The normal force F_n (teal) balances the part into the ramp, and friction F_{fr} (dark red) acts up the ramp, against the motion.

Over the length $L = 3$ m the block drops by $L \sin 30^\circ = 1.5$ m, and the three forces do the work

$$\begin{aligned} W_{\text{weight}} &= -mg \times (-1.5) && \text{by Section 3.5, } -mg \text{ times the rise,} \\ &= 2 \times 9.80665 \times 1.5 = 29.42 \text{ J,} \\ W_{\text{n}} &= 0 && \text{perpendicular to the motion,} \\ W_{\text{fr}} &= -3.397 \times 3 = -10.19 \text{ J} && \text{against the motion.} \end{aligned}$$

The block arrives with $K = 29.42 - 10.19 = 19.23$ J, and so with speed $\sqrt{2 \times 19.23/2} = 4.385 \text{ m s}^{-1}$; on a smooth ramp it would arrive at $\sqrt{2g \times 1.5} = 5.424 \text{ m s}^{-1}$.

The rate of energy loss

Friction and the drag of Section 2.7 always point against the velocity, so on a trip out to a point and back they do negative work on both legs, whereas a force with a potential energy does the work $U(x_A) - U(x_A) = 0$ on any trip that returns to its start x_A . They therefore have no potential energy. To see their effect on E , write the net force on a body moving along a line as a conservative part, $-dU/dx$, plus the rest, F^{nc} , the non-conservative part. The kinetic energy changes at the rate $(-dU/dx + F^{\text{nc}})v$, by Section 3.3, and the potential energy at the rate $(dU/dx)v$, by the chain rule, so their sum changes at the rate

$$\frac{dE}{dt} = F^{\text{nc}} v : \quad (3.10)$$

the total energy changes at exactly the rate at which the non-conservative force does work. Friction always opposes the velocity, so $F^{\text{nc}}v = -\mu_k F_{\text{n}}|v|$, and the energy can only fall. For linear drag, $F^{\text{nc}} = -bv$ and $dE/dt = -bv^2$.

The bead of Section 2.7 sinking at its terminal velocity shows this rate at work. Measuring the depth x downwards, its potential energy is $U = -mgx$, which falls at the rate $mg v_\infty$, while its kinetic energy stays constant. Every second, then, the weight does $mg v_\infty = 0.01 \times 9.80665 \times 1.961 = 0.1923$ J of work, and the drag removes all of it: $bv_\infty^2 = 0.05 \times 1.961^2 = 0.1923$ W, the same, because $bv_\infty = mg$.

CHECK YOUR UNDERSTANDING. A box sliding across a floor stops after 0.68 m. How far would a box of twice the mass, sliding on the same floor at the same starting speed, travel before stopping? Where has the kinetic energy gone?

3.8 The energy of several bodies

Section 2.9 showed that the forces two bodies exert on each other cannot change their total momentum. This section asks the same question of energy, using the two carts of that section, and finds a different answer: forces between bodies can change their total kinetic energy, and they conserve the total energy only when they store what they give in a potential energy.

The pair of forces does work

Two carts, of masses $m_1 = 1 \text{ kg}$ and $m_2 = 3 \text{ kg}$, at positions $x_1 < x_2$ on a level, smooth track, are pushed apart by a light spring of stiffness $k = 200 \text{ N m}^{-1}$ compressed between them (Figure 2.10). Let ζ be the compression of the spring, how much shorter it is than its natural length. The spring touches both carts, so as the gap $x_2 - x_1$ between them grows, ζ shrinks by the same amount:

$$\frac{d\zeta}{dt} = -\frac{d}{dt}(x_2 - x_1) = -(v_2 - v_1).$$

By Section 2.9 the spring pushes the two carts with forces of equal size in opposite directions, and by Hooke's law (Section 2.8) a spring compressed by ζ pushes with a force of size $k\zeta$, so it pushes the first cart with $-k\zeta$ and the second with $+k\zeta$. The weights and normal forces do no work, so by Section 3.3 the kinetic energies change at the rates

$$\frac{dK_1}{dt} = -k\zeta v_1, \quad \frac{dK_2}{dt} = +k\zeta v_2.$$

Adding them,

$$\begin{aligned} \frac{d}{dt}(K_1 + K_2) &= k\zeta(v_2 - v_1) \\ &= -k\zeta \frac{d\zeta}{dt} && \text{using the rate of change of } \zeta \\ &= -\frac{d}{dt}\left(\frac{1}{2}k\zeta^2\right) && \text{since } \frac{d}{dt}\left(\frac{1}{2}k\zeta^2\right) = k\zeta \frac{d\zeta}{dt} \text{ by the chain rule.} \end{aligned}$$

Moving the last term to the left, the sum

$$E = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 + \frac{1}{2}k\zeta^2$$

does not change: the total energy of the two carts and the spring is conserved, and $\frac{1}{2}k\tilde{\zeta}^2$ is the potential energy of the spring, as in [Section 3.4](#). It depends only on how far apart the carts are.

The contrast with momentum lies in the velocities. The two forces are equal and opposite, so they cancel in the total momentum, whose rate of change is their sum. Their powers are $-k\tilde{\zeta} v_1$ and $+k\tilde{\zeta} v_2$, which cancel only if $v_1 = v_2$. When the carts move apart, the first moves backwards, along its push, and the second forwards, along its push, so both forces do positive work and the total kinetic energy grows. A pair of equal and opposite forces does work whenever the distance between the bodies changes. This is how the carts, at rest at first, come to move.

The speeds of the carts

The spring starts compressed by 0.1 m, so it stores $\frac{1}{2} \times 200 \times 0.1^2 = 1$ J, and the carts start at rest. Once the spring has reached its natural length and fallen away, $\tilde{\zeta} = 0$ and all of that energy is kinetic,

$$\frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 = 1 \text{ J.}$$

One equation cannot fix two speeds, but momentum supplies the second. The total momentum stays zero ([Section 2.9](#)), so $m_1v_1 + m_2v_2 = 0$, which gives $v_1 = -(m_2/m_1)v_2 = -3v_2$. Substituting,

$$\frac{1}{2} \times 1 \times (-3v_2)^2 + \frac{1}{2} \times 3 \times v_2^2 = \frac{9}{2}v_2^2 + \frac{3}{2}v_2^2 = 6v_2^2 = 1,$$

in joules, so $v_2^2 = \frac{1}{6}$ and

$$v_2 = +\sqrt{\frac{1}{6}} = +0.408 \text{ m s}^{-1}, \quad v_1 = -3\sqrt{\frac{1}{6}} = -1.225 \text{ m s}^{-1},$$

taking the positive root for v_2 because the spring pushes the second cart forwards. These are the speeds that a computer solution of the equations of motion gave in [Section 2.9](#), found now with two lines of algebra. [Figure 3.7](#) shows the energy passing from the spring to the carts while the total stays at 1 J.

The lighter cart ends with 0.75 J and the heavier with 0.25 J, although their momenta are equal and opposite. Writing the kinetic energy in terms of the momentum $p = mv$ shows why,

$$K = \frac{1}{2}mv^2 = \frac{(mv)^2}{2m} = \frac{p^2}{2m} :$$

for momenta of equal size the kinetic energies are in the inverse ratio of the masses, three to one here.

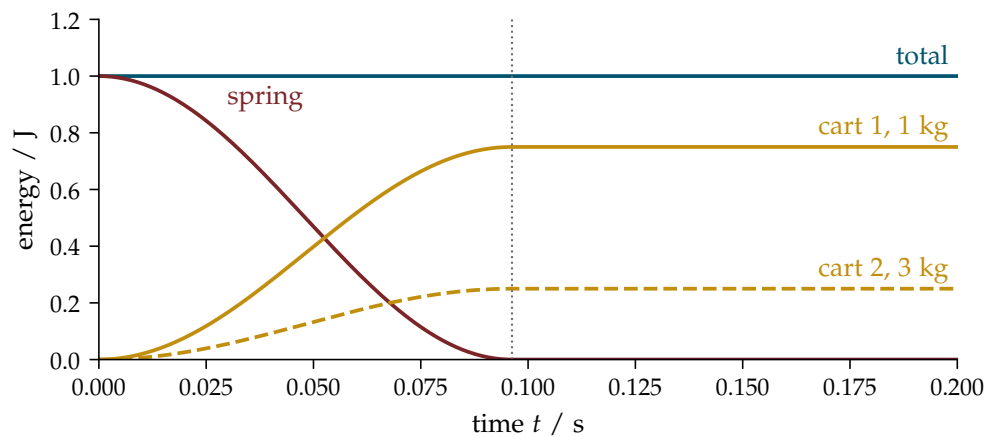


FIGURE 3.7: The energies of the carts of Figure 2.11: the energy stored in the spring, $\frac{1}{2}kx^2$, the kinetic energies of the two carts, and the total, which stays at 1 J. The spring falls away at the dotted line, after 0.096 s.

The same argument extends to any number of bodies that push on one another in third-law pairs, provided that the two forces of each pair act along the line joining its two bodies, with a size that depends only on the distance between them: each pair then has a potential energy, and the kinetic energy plus the sum of the pair energies is conserved. Section 3.12 proves this for atoms, in three dimensions.

NOTEBOOK 03 *change the masses of the carts and the energy they share*

CHECK YOUR UNDERSTANDING. If both carts had a mass of 2 kg, with the same spring and compression, at what speeds would they separate?

3.9 Work along a path in space

On a line, every force that depends only on position has a potential energy, because the integral of Equation (3.6) can always be done. In a plane or in space a body can travel from one point to another by many routes, and the work may differ from route to route. This section defines the work along a curved path, the tool we need, and shows that the route can matter.

The line integral

Section 3.5 found the work of the weight along a curved wire by cutting the wire into short straight chords. The same construction works for any force, constant or not, and the box below turns it into a definition.

LINE INTEGRALS. Let a path C run from a point A to a point B , and let a force $\mathbf{F}(\mathbf{r})$ depend on position. Mark points $\mathbf{r}_0 = A, \mathbf{r}_1, \dots, \mathbf{r}_n = B$ in order along the path, and join each to the next by a straight chord, $\Delta\mathbf{r}_k = \mathbf{r}_k - \mathbf{r}_{k-1}$. On a chord short enough that the force hardly changes along it, the force does about $\mathbf{F}(\bar{\mathbf{r}}_k) \cdot \Delta\mathbf{r}_k$ of work by Equation (3.1), where $\bar{\mathbf{r}}_k = \frac{1}{2}(\mathbf{r}_{k-1} + \mathbf{r}_k)$ is the midpoint of the chord. The work along the path is the limit of the sum as the chords shrink,

$$W[C] = \lim \sum_{k=1}^n \mathbf{F}(\bar{\mathbf{r}}_k) \cdot \Delta\mathbf{r}_k = \int_C \mathbf{F} \cdot d\mathbf{r},$$

called the line integral of $\mathbf{F}$ along C ; the square brackets in $W[C]$ record that the work belongs to the whole path C . Three properties follow from the sum. Reversing the direction of travel reverses every chord, $\Delta\mathbf{r}_k \rightarrow -\Delta\mathbf{r}_k$, and so changes the sign of the work. The sum uses only the points of the path and their order, so it does not matter how fast the path is traversed. And along a path made of one path followed by another, the work is the sum of the works along the two, since the sum over chords splits into two such sums. For a closed path, one that returns to its start, the integral is written $\oint_C$.

To compute a line integral, describe the path by a parameter s that runs from s_A to s_B , so that $\mathbf{r}(s)$ traces the path, and cut the range of s into short steps Δs , with s_k the middle of the k th. Each step moves the point by $\Delta\mathbf{r}_k \approx (d\mathbf{r}/ds) \Delta s$, where $d\mathbf{r}/ds$, the derivative of each component of $\mathbf{r}$, is taken at s_k . Then

$$\begin{aligned} \int_C \mathbf{F} \cdot d\mathbf{r} &= \lim \sum_k \mathbf{F}(\mathbf{r}(s_k)) \cdot \frac{d\mathbf{r}}{ds} \Delta s && \text{putting in the chords} \\ &= \int_{s_A}^{s_B} \mathbf{F}(\mathbf{r}(s)) \cdot \frac{d\mathbf{r}}{ds} ds && \text{a sum of values times } \Delta s, \text{ Section 1.4,} \end{aligned}$$

an ordinary integral in one variable. For example, take the force $\mathbf{F} = c(-y, x)$ in the plane, with c a constant, and the circle of radius R about the origin, traversed once anticlockwise: $\mathbf{r}(s) = R(\cos s, \sin s)$ for s from 0 to 2π . Then

$$\begin{aligned} \frac{d\mathbf{r}}{ds} &= R(-\sin s, \cos s) && \text{differentiating each component} \\ \mathbf{F}(\mathbf{r}(s)) &= c(-R \sin s, R \cos s) && \text{setting } x = R \cos s, y = R \sin s \\ \mathbf{F} \cdot \frac{d\mathbf{r}}{ds} &= cR^2 \sin^2 s + cR^2 \cos^2 s && \text{multiplying components and adding} \\ &= cR^2 && \text{since } \sin^2 s + \cos^2 s = 1, \end{aligned}$$

and the line integral is $\int_0^{2\pi} cR^2 ds = 2\pi cR^2$.

Along the motion itself the time t is a natural parameter, and $d\mathbf{r}/dt$ is the velocity. The work-energy theorem [Equation \(3.5\)](#) therefore reads, in space,

$$K_B - K_A = \int_{t_A}^{t_B} \mathbf{F} \cdot \mathbf{v} dt = \int_{\text{path}} \mathbf{F} \cdot d\mathbf{r},$$

the line integral of the net force along the path the body actually follows.

When the route matters

The weight is the simplest case. With z vertical, $\mathbf{F} = (0, 0, -mg)$, each chord contributes $-mg \Delta z_k$, and the sum is $-mg(z_B - z_A)$ for every path from A to B , as on the wire of [Section 3.5](#).

Friction behaves differently. A box of 5 kg is pushed slowly across a floor with $\mu_k = 0.3$ from A to a point B 5 m away. Friction always points against the motion with size $\mu_k mg = 14.71$ N, so on every chord it does -14.71 newtons times the chord's length, and its work is -14.71 newtons times the length of the path. Straight from A to B that is $-14.71 \times 5 = -73.55$ J; along the other two sides of a right-angled triangle, of 3 m and 4 m, it is $-14.71 \times 7 = -102.97$ J. Friction depends on the direction of motion, so this is no great surprise.

More surprising is that a force depending only on position can also do different work on different routes. The force of the toolbox,

$$\mathbf{F}_{\text{sw}}(\mathbf{r}) = c(-y, x), \tag{3.11}$$

which we call a swirl, has size $c|\mathbf{r}|$ and points around the origin, perpendicular to $\mathbf{r}$, since $(-y, x) \cdot (x, y) = 0$. Once round a circle it does the work $2\pi cR^2$. A closed path starts and ends at the same point, so if the work did not depend on the route, the circle would give the same work as staying put, which is zero. The work of the swirl therefore depends on the route, even though the force depends only on where the body is. No force between atoms has this form; we use the swirl as the plainest example of such a force, and in [Section 3.14](#) to stand for the errors of a faulty force calculation.

The work in code

On a computer a path is a list of points, and the line integral is the sum of the toolbox with the force taken at the midpoint of each chord.

```

1 def work(force: Callable[[NDArray], NDArray], path: ArrayLike) -> float:
2     path = np.asarray(path, dtype=float)
3     chords = np.diff(path, axis=0) # the step from each point to the next
4     midpoints = 0.5 * (path[1:] + path[:-1])
5     # F · Δr on every chord, added over the components and the chords
6     return float(np.sum(force(midpoints) * chords))

```

The first argument, *force*, is itself a function (Callable): given an array of positions, one row per point, it returns the forces there, one row each. The path is an array with one row per point and one column per component. `np.diff` subtracts each row from the next, giving the chords $\Delta \mathbf{r}_k$, one row each, and `path[1:]` and `path[:-1]`, the points without the first and without the last, average to the midpoints. The force is evaluated at all the midpoints at once; multiplying by the chords, component by component, and adding everything gives $\sum_k \mathbf{F}(\bar{\mathbf{r}}_k) \cdot \Delta \mathbf{r}_k$, since adding the products of the components of one row is its scalar product. The tests of `mdlab` confirm that halving the length of every chord divides the error of this sum by four.

NOTEBOOK 03 *bend a path and compare the work of the weight, of friction and of the swirl*

CHECK YOUR UNDERSTANDING. How much work does the swirl of Equation (3.11) do on a body that moves from the origin straight out to the point (3,4)?

3.10 Conservative forces and the gradient

The weight does the same work along every route, and the swirl does not. This section names the forces that behave like the weight, gives each a potential energy, and shows how the force is recovered from that energy by differentiating in several directions at once. The tools are the partial derivative and the gradient, and a hill on a walker's map makes both easy to picture.

Conservative forces

DEFINITION 3.2. A force $\mathbf{F}(\mathbf{r})$ that depends on position is conservative if the work $W[C]$ from *A* to *B* is the same for every path *C* from *A* to *B*, for every pair of points *A* and *B*.

An equivalent statement is that the work around every closed path is zero. Any closed path, starting at *A* and passing through some other point *B*, is made of a path C_1 from *A* to *B* followed by a second path C_2 traversed backwards, from *B* to *A*, and any two paths from *A* to *B* make a closed path in this way. Its work is $W[C_1] - W[C_2]$, since reversing C_2 changes the sign of its work, and this vanishes for every such pair of paths exactly when the work does not depend on the path.

For a conservative force, fix a reference point $\mathbf{r}_0$ and define, as on a line,

$$U(\mathbf{r}) = - \int_{\mathbf{r}_0}^{\mathbf{r}} \mathbf{F} \cdot d\mathbf{r}', \quad (3.12)$$

the line integral along any path from $\mathbf{r}_0$ to $\mathbf{r}$, all of which give the same value. Going from A to B by way of $\mathbf{r}_0$, exactly as in [Section 3.4](#),

$$W_{A \rightarrow B} = U(A) - U(B) :$$

the work is the drop in potential energy. On a line, $F = -dU/dx$ recovered the force. In space, U changes differently in different directions, and we need the slope in each.

Slopes on a map

A hill is described on a map by its height $h(x, y)$ above each point, with x measured east and y north. Our example, drawn in [Figure 3.8](#), is

$$h(x, y) = 300 e^{-(x^2+2y^2)/2},$$

in metres, with x and y in kilometres: a summit 300 m high at the origin, its contours (lines of equal height) ellipses longer in the east-west direction.

PARTIAL DERIVATIVES. A walker heading due east keeps y fixed, and the height changes with x alone. The slope in that direction is the derivative with respect to x with y held constant, the partial derivative

$$\frac{\partial h}{\partial x} = \lim_{\Delta x \rightarrow 0} \frac{h(x + \Delta x, y) - h(x, y)}{\Delta x},$$

written with a rounded ∂ to show that the other variables are held fixed. The slope due north is $\partial h / \partial y$, with x held fixed. Each is found by the ordinary rules of differentiation, treating the other variable as a constant. For the hill, write the exponent as $u = -\frac{1}{2}x^2 - y^2$. Holding y fixed, $-y^2$ is a constant, whose derivative is zero, so $\partial u / \partial x = -x$; holding x fixed, $-\frac{1}{2}x^2$ is the constant, so $\partial u / \partial y = -2y$. Since e^u has derivative e^u ([Section 2.7](#)), the chain rule of [Section 1.3](#) gives

$$\frac{\partial h}{\partial x} = 300 e^u \times (-x) = -x h, \quad \frac{\partial h}{\partial y} = 300 e^u \times (-2y) = -2y h.$$

At the point $P = (0.6, 0.5)$ km, $h = 300 e^{-(0.36+0.5)/2} = 195.2$ m, so $\partial h / \partial x = -0.6 \times 195.2 = -117.1$ m per kilometre and $\partial h / \partial y = -2 \times 0.5 \times 195.2 = -195.2$ m per

kilometre: heading east from P the walker sets off downhill at 117 m per kilometre, and heading north at 195 m per kilometre, rates that hold only near P (Figure 3.8b and c). A function of three variables, or of $3N$, has one partial derivative for each. Partial derivatives can be taken again: $\partial^2 h / \partial x^2$ differentiates $\partial h / \partial x$ with respect to x , and $\partial^2 h / \partial x \partial y$ means $\partial(\partial h / \partial y) / \partial x$.

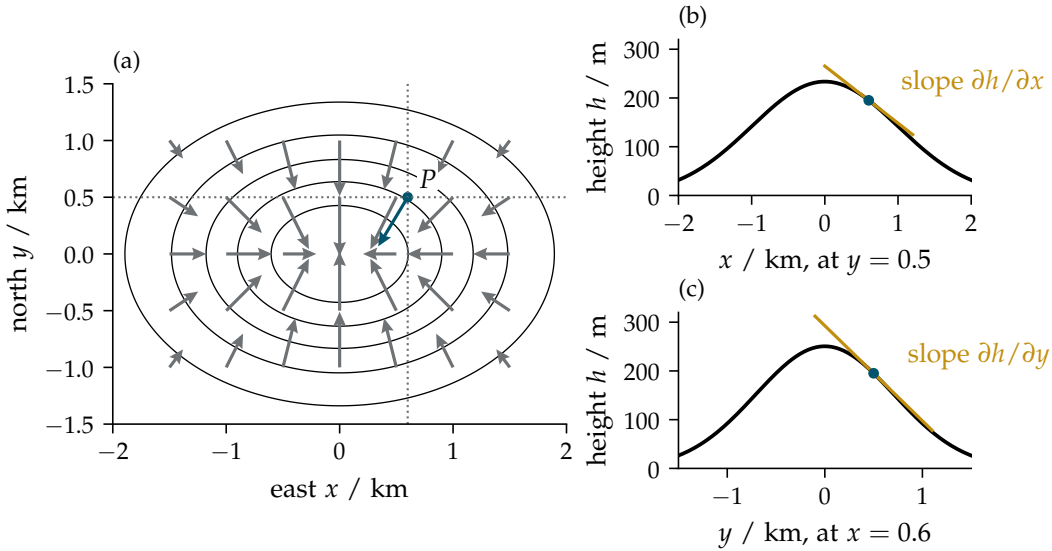


FIGURE 3.8: Partial derivatives and the gradient of a hill, $h = 300e^{-(x^2+2y^2)/2}$ m. (a) Contours every 50 m, with the gradient ∇h at a grid of points (grey) and at $P = (0.6, 0.5)$ km (teal). Each arrow points straight uphill, across the contours. (b) The height along the east-west line through P (dotted in a), whose slope at P is $\partial h / \partial x$. (c) The height along the north-south line through P , whose slope at P is $\partial h / \partial y$.

THE GRADIENT. The two partial derivatives together give the change of height for a short step in any direction. Step from (x, y) to $(x + \Delta x, y + \Delta y)$ in two legs, first east and then north, adding and subtracting the height at the corner $(x + \Delta x, y)$:

$$\begin{aligned} h(x + \Delta x, y + \Delta y) - h(x, y) &= [h(x + \Delta x, y + \Delta y) - h(x + \Delta x, y)] \\ &\quad + [h(x + \Delta x, y) - h(x, y)]. \end{aligned}$$

The second bracket is a step east, and by the definition of $\partial h / \partial x$ it is $(\partial h / \partial x) \Delta x$ plus an error that is small compared with Δx . The first is a step north, taken from the corner, so it is about Δy times the northward slope at the corner, with an error small compared with Δy ; that slope approaches the northward slope at (x, y) as the step shrinks, provided that the slopes change smoothly from place to place, as they do for this hill. Hence

$$\Delta h \approx \frac{\partial h}{\partial x} \Delta x + \frac{\partial h}{\partial y} \Delta y = \nabla h \cdot \Delta \mathbf{r}, \quad \nabla h = \left(\frac{\partial h}{\partial x}, \frac{\partial h}{\partial y} \right),$$

with an error small compared with the length of the step $\Delta \mathbf{r} = (\Delta x, \Delta y)$. The vector ∇h , read 'grad h ', is the gradient of h ; in three dimensions it has a third component, $\partial h / \partial z$, and the step a third leg. Three consequences follow.

- *Steepest ascent.* For a step of length ℓ in the direction of a unit vector $\mathbf{n}$, a vector of length 1, $\Delta \mathbf{r} = \ell \mathbf{n}$ and $\Delta h \approx \ell \nabla h \cdot \mathbf{n}$. Dividing by ℓ , the height changes at the rate $\nabla h \cdot \mathbf{n} = |\nabla h| |\mathbf{n}| \cos \theta = |\nabla h| \cos \theta$ per unit distance, with θ the angle between $\mathbf{n}$ and ∇h . The rate is greatest for $\theta = 0$, where $\cos \theta = 1$: the gradient points up the steepest slope, and its length is that slope.
- *Contours.* Along a contour h does not change, so for $\mathbf{n}$ pointing along the contour $\nabla h \cdot \mathbf{n} = 0$: wherever ∇h is not zero, it is perpendicular to the contour through the point.
- *Along a path.* If a point moves along a path $\mathbf{r}(t)$, then over a short interval Δt it moves by $\Delta \mathbf{r} = \mathbf{r}(t + \Delta t) - \mathbf{r}(t)$ and the height changes by $\Delta h = h(\mathbf{r}(t + \Delta t)) - h(\mathbf{r}(t))$, so

$$\frac{\Delta h}{\Delta t} \approx \nabla h \cdot \frac{\Delta \mathbf{r}}{\Delta t} \quad \text{dividing } \Delta h \approx \nabla h \cdot \Delta \mathbf{r} \text{ by } \Delta t$$

$$\frac{d}{dt} h(\mathbf{r}(t)) = \nabla h \cdot \frac{d\mathbf{r}}{dt} \quad \text{letting } \Delta t \rightarrow 0,$$

the chain rule along a path. The error of the first line is an error small compared with $|\Delta \mathbf{r}|$, divided by Δt ; since $|\Delta \mathbf{r}| / \Delta t$ approaches $|d\mathbf{r}/dt|$, it vanishes in the limit.

At P the gradient is $(-117.1, -195.2)$ metres per kilometre, of length

$$|\nabla h| = \sqrt{117.1^2 + 195.2^2} = 227.6$$

metres per kilometre. That is a rise of $227.6/1000 = 0.2276$ metres per metre, and the angle whose tangent is 0.2276 is 12.8° . Walking straight towards the summit instead, in the direction of the unit vector $\mathbf{n} = (-0.6, -0.5)/0.781$, where $0.781 = \sqrt{0.6^2 + 0.5^2}$ is the length of $(-0.6, -0.5)$, the height rises at the rate

$$\nabla h \cdot \mathbf{n} = \frac{(-117.1)(-0.6) + (-195.2)(-0.5)}{0.781} = \frac{167.86}{0.781} = 214.9$$

metres per kilometre, less than 227.6: on elliptical contours the steepest way up need not point at the summit (Figure 3.8a); it does only along the axes of the ellipses.

The force is minus the gradient

Now let $U(\mathbf{r})$ be the potential energy of a conservative force. For a short straight step $\Delta\mathbf{r}$ from $\mathbf{r}$, the force hardly changes along the step, so, since the work along the step is the drop $U(A) - U(B)$ with $A = \mathbf{r}$ and $B = \mathbf{r} + \Delta\mathbf{r}$, Equation (3.1) makes U change by about $-\mathbf{F} \cdot \Delta\mathbf{r}$. By the gradient toolbox it also changes by about $\nabla U \cdot \Delta\mathbf{r}$. The two agree, to within errors small compared with the step, for steps in every direction, which is possible only if

$$\mathbf{F} = -\nabla U, \quad \text{that is,} \quad F_x = -\frac{\partial U}{\partial x}, \quad F_y = -\frac{\partial U}{\partial y}, \quad F_z = -\frac{\partial U}{\partial z}. \quad (3.13)$$

Taking the step along x , for example, the first gives $-F_x \Delta x$ and the second $(\partial U / \partial x) \Delta x$; dividing both by Δx and letting the step shrink removes the errors, leaving $F_x = -\partial U / \partial x$.

The converse holds as well. Suppose $\mathbf{F} = -\nabla U$ for some U with smoothly varying partial derivatives. Along any path $\mathbf{r}(s)$ from A to B , the chain rule along a path gives $\mathbf{F} \cdot d\mathbf{r}/ds = -\nabla U \cdot d\mathbf{r}/ds = -dU(\mathbf{r}(s))/ds$, so the line integral is

$$\begin{aligned} \int_{s_A}^{s_B} -\frac{d}{ds} U(\mathbf{r}(s)) ds &= -[U(B) - U(A)] \quad \text{by the fundamental theorem of calculus} \\ &= U(A) - U(B), \end{aligned}$$

the same for every path. *A force is conservative exactly when it is minus the gradient of a potential energy.* For the weight, $U = mgz$, and $-\nabla U = (0, 0, -mg)$, as it should be.

Equation (3.13) makes the force a property of the shape of U . By the gradient toolbox, the force points in the direction in which the potential energy falls fastest, its size is the steepest slope of U , and it is perpendicular to the contours of U , which in space are surfaces of equal U . A small block placed on the hillside of Figure 3.8(a), with no friction to hold it, is likewise pushed straight down the slope, opposite to the arrows of ∇h . Where $\nabla U = \mathbf{0}$ the force vanishes, and a body at rest stays at rest. In two or three dimensions most such equilibria are of three kinds: minima, from which every direction leads up; maxima, from which every direction leads down; and saddle points, like a mountain pass, which is the lowest point on the ridge between two valleys and the highest point on the path through the pass. Chapter 4 develops the test that tells them apart.

CHECK YOUR UNDERSTANDING. On a map, the contours on one side of a hill are crowded together and on the other side widely spaced. On which side is $|\nabla h|$ larger? In which direction does ∇h point where it crosses a contour?

3.11 The curl test

Given a force, how can we tell whether it is conservative? Checking the work along every path is impossible. This section derives a test that uses only derivatives of the force at each point, and explains what the test measures: how much the force circulates around a small loop.

A property of every gradient

MIXED PARTIAL DERIVATIVES. For a function $U(x, y)$ whose second partial derivatives change smoothly from place to place, the order of differentiation does not matter,

$$\frac{\partial}{\partial x} \left(\frac{\partial U}{\partial y} \right) = \frac{\partial}{\partial y} \left(\frac{\partial U}{\partial x} \right).$$

For $U = x^2y^3$, for instance, $\partial U/\partial y = 3x^2y^2$, whose derivative with respect to x is $6xy^2$, and $\partial U/\partial x = 2xy^3$, whose derivative with respect to y is also $6xy^2$. To see why it holds in general, take the combination of the values of U at the four corners of a small rectangle,

$$D = U(x + \Delta x, y + \Delta y) - U(x + \Delta x, y) - U(x, y + \Delta y) + U(x, y),$$

and group its four terms in two ways. Grouped as

$$D = [U(x + \Delta x, y + \Delta y) - U(x + \Delta x, y)] - [U(x, y + \Delta y) - U(x, y)],$$

each bracket is a step north, one taken at $x + \Delta x$ and one at x . Holding x fixed, the fundamental theorem of calculus makes each bracket the integral of $\partial U/\partial y$ along its step, so, writing $\frac{\partial U}{\partial y}(a, b)$ for its value at the point (a, b) ,

$$D = \int_y^{y+\Delta y} \left[\frac{\partial U}{\partial y}(x + \Delta x, y') - \frac{\partial U}{\partial y}(x, y') \right] dy'$$

exactly. By the same theorem, now holding y' fixed, the square bracket is the integral of $\partial(\partial U/\partial y)/\partial x$ from x to $x + \Delta x$. Because this second derivative changes smoothly, on a small enough rectangle it stays as close as we like to its value at (x, y) , so D is $\Delta x \Delta y$ times $\partial(\partial U/\partial y)/\partial x$ at (x, y) , with an error small compared with $\Delta x \Delta y$. Grouped instead as

$$D = [U(x + \Delta x, y + \Delta y) - U(x, y + \Delta y)] - [U(x + \Delta x, y) - U(x, y)],$$

each bracket is a step east, and the same argument with x and y exchanged makes D equal to $\Delta x \Delta y$ times $\partial(\partial U/\partial x)/\partial y$ at (x, y) , again with an error small compared with $\Delta x \Delta y$. Both expressions describe the same number D , so dividing by $\Delta x \Delta y$ and letting the rectangle shrink, the two mixed derivatives are equal.

If $\mathbf{F} = -\nabla U$ in the plane, then $F_x = -\partial U/\partial x$ and $F_y = -\partial U/\partial y$, so by the toolbox

$$\frac{\partial F_y}{\partial x} = -\frac{\partial}{\partial x} \left(\frac{\partial U}{\partial y} \right) = -\frac{\partial}{\partial y} \left(\frac{\partial U}{\partial x} \right) = \frac{\partial F_x}{\partial y}.$$

Every conservative force whose components have smoothly changing partial derivatives therefore has

$$\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} = 0 \quad (3.14)$$

at every point. The swirl $c(-y, x)$ of Equation (3.11) fails: $F_y = cx$ gives $\partial F_y/\partial x = c$, $F_x = -cy$ gives $\partial F_x/\partial y = -c$, and the difference is $2c$. The swirl is not conservative, as its work around a circle already showed.

Circulation around a small loop

The combination in Equation (3.14) has a meaning of its own, which the next box finds, and which turns the necessary condition into a test.

THE CURL. Take the small rectangle of Figure 3.9, with corners (x, y) and $(x + \Delta x, y + \Delta y)$, and add up the work of a force $\mathbf{F}$ once around it anticlockwise, taking the force at the middle of each side.

- Along the bottom, eastwards, the displacement is $(\Delta x, 0)$: work $F_x \Delta x$, with F_x taken at height y .
- Up the right side: $F_y \Delta y$, with F_y taken at $x + \Delta x$.
- Along the top, westwards, the displacement is $(-\Delta x, 0)$: work $-F_x \Delta x$, with F_x taken at height $y + \Delta y$.
- Down the left side: $-F_y \Delta y$, with F_y taken at x .

Pairing the two sides that run east and west, and the two that run north and south, the total is

$$\begin{aligned} \oint \mathbf{F} \cdot d\mathbf{r} &\approx [F_y(x + \Delta x) - F_y(x)] \Delta y - [F_x(y + \Delta y) - F_x(y)] \Delta x \\ &\approx \frac{\partial F_y}{\partial x} \Delta x \Delta y - \frac{\partial F_x}{\partial y} \Delta y \Delta x = \left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \right) \Delta A, \end{aligned}$$

where each square bracket is the change of one component over one short step, and $\Delta A = \Delta x \Delta y$ is the area. The combination in brackets is therefore the work around a small loop per unit area enclosed. The work around a closed loop is called its circulation, so the combination is the circulation per unit area; it is called the curl of $\mathbf{F}$ in the plane. In three dimensions a small loop can lie in the xy , the yz or the zx plane. Renaming the letters in turn, x to y , y to z and z to x , carries a loop in the xy plane, traversed anticlockwise as seen from the positive end of the z axis, into one in the yz plane, anticlockwise as seen from the positive end of the x axis, and carries $\partial F_y/\partial x - \partial F_x/\partial y$ into $\partial F_z/\partial y - \partial F_y/\partial z$; renaming once more gives the zx plane and $\partial F_x/\partial z - \partial F_z/\partial x$. The three combinations form the curl vector, written $\nabla \times \mathbf{F}$ and read 'curl $\mathbf{F}$ ' (the $\times$ is the cross product of Chapter 5, not needed here),

$$\nabla \times \mathbf{F} = \left(\frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z}, \frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x}, \frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \right),$$

whose third component is the one found above, for a loop in the xy plane.

A large loop made of the sides of many small rectangles, tiling the region it encloses, has a circulation equal to the sum of the circulations of the small rectangles. Each inner side belongs to two neighbouring rectangles and is traversed once in each direction, so its two contributions cancel, and only the outer boundary is left. Hence the work around the loop is the sum of the curl times ΔA over the rectangles. As the rectangles shrink, such a sum of values times small areas approaches a limit, written $\iint \dots dA$ and called the integral over the area, just as a sum of values times short steps approaches the integral of [Section 1.4](#); when the value is a constant, the sum, and so the integral, is that constant times the whole area. So

$$\oint \mathbf{F} \cdot d\mathbf{r} = \iint \left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \right) dA,$$

a result known as Green's theorem, for a loop traversed anticlockwise. It needs $\mathbf{F}$ and its partial derivatives to change smoothly at every point inside the loop, because every small rectangle of the tiling must have a curl. A smooth curved loop is approached by staircases of ever smaller rectangles. A staircase never becomes as short as the curve, which would matter for friction; but for a force that depends only on position, the work over each short stretch is nearly $\mathbf{F} \cdot \Delta \mathbf{r}$, and the staircase and the stretch of curve beside it share the same displacement $\Delta \mathbf{r}$, so the theorem holds for the curve too.

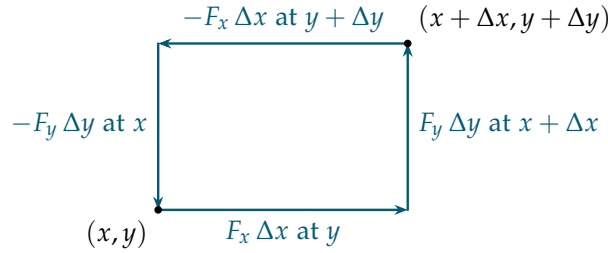


FIGURE 3.9: The work of a force once anticlockwise around a small rectangle, side by side. The two sides that run east and west differ only in their height, and the two that run north and south only in their east-west position.

The test

The curl test now reads in both directions. If a force is conservative, its curl is zero everywhere, by Equation (3.14). Conversely, if the curl is zero everywhere in a region with no holes, every loop in the region encloses only points of the region, so Green's theorem makes the work around it zero; within the region the work then does not depend on the path, and the force is conservative there. In three dimensions the same holds, because any loop can be spanned by a surface tiled with small loops in the same way, a result called Stokes' theorem whose details we do not need. The proviso about holes is real: Exercise 3.18 gives a force whose curl is zero wherever it is defined, but which is not defined at the origin, and whose work around the origin is not zero.

For the swirl, with curl $2c$ everywhere, Green's theorem gives the work anticlockwise around any loop as $2c$ times the area it encloses. On the circle of the line-integral toolbox that is $2c \times \pi R^2 = 2\pi c R^2$, as computed there directly.

When the test is passed, U follows from Equation (3.12) along whichever path is easiest. From the origin, go along the x axis to $(x, 0)$, a leg on which only F_x does work, and then straight north to (x, y) , on which only F_y does:

$$U(x, y) = - \int_0^x F_x(x', 0) dx' - \int_0^y F_y(x, y') dy'.$$

For many atoms the test has the same form, with every pair of the $3N$ coordinates in place of x and y . Numbering the coordinates $q_1, \dots, q_{3N}$, a force is conservative only if

$$\frac{\partial F_a}{\partial q_b} = \frac{\partial F_b}{\partial q_a} \quad \text{for every pair } a \text{ and } b,$$

where F_a is the force component along q_a . For a conservative force these derivatives are minus the second derivatives of U , which Chapter 4 collects into a table called the Hessian.

NOTEBOOK 03 compute the curl of a force symbolically, and the work around loops

CHECK YOUR UNDERSTANDING. Is the force $\mathbf{F} = (y, x)$ conservative? If so, find a potential energy U with $\mathbf{F} = -\nabla U$.

3.12 The energy of atoms

Every result of this chapter was found with boxes, beads and carts. This section carries them to atoms: it introduces the unit in which the energies of atoms are measured, converts the kinetic energy into the units of this book, and proves that Newton's laws conserve the total energy of any number of atoms whose forces come from a potential energy.

The electronvolt

The joule is far too large a unit for a single atom. The lithium atom of [Section 2.10](#), of mass $6.94 \times 1.66054 \times 10^{-27} \text{ kg} = 1.1524 \times 10^{-26} \text{ kg}$, moving at 1390 m s^{-1} after 10 fs under a force of $1 \text{ eV } \text{\AA}^{-1}$, has a kinetic energy of $\frac{1}{2} \times 1.1524 \times 10^{-26} \times 1390^2 = 1.113 \times 10^{-20} \text{ J}$. Energies of atoms are therefore measured in electronvolts, the unit already used for forces in [Section 2.10](#), with

$$1 \text{ eV} = 1.602176634 \times 10^{-19} \text{ J}$$

exactly. The name comes from electricity, which this book does not need; the size of the unit in joules is fixed by the definition of the SI units, and only its size matters here. The lithium atom has $1.113 \times 10^{-20} / 1.602 \times 10^{-19} = 0.0695 \text{ eV}$, and [Section 2.10](#) found that it had moved $0.0695 \text{ \AA}$. The force of $1 \text{ eV } \text{\AA}^{-1}$ has done $1 \times 0.0695 = 0.0695 \text{ eV}$ of work on it, equal to its kinetic energy, as the work-energy theorem requires.

Kinetic energy in eV, $\text{\AA}$, fs and amu

With masses in atomic mass units and velocities in $\text{\AA fs}^{-1}$, the product $\frac{1}{2}mv^2$ comes out in $\text{amu } \text{\AA}^2 \text{ fs}^{-2}$, which is not the electronvolt; Newton's law needed a conversion factor in [Section 2.10](#) for the same reason. We convert one unit of each kind. A mass of 1 amu is $1.66053906892 \times 10^{-27} \text{ kg}$, and a velocity of $1 \text{ \AA fs}^{-1}$ is $10^{-10}/10^{-15} = 10^5 \text{ m s}^{-1}$, whose square is $10^{10} \text{ m}^2 \text{ s}^{-2}$. Their product is, in joules,

$$1 \text{ amu } \text{\AA}^2 \text{ fs}^{-2} = 1.66053906892 \times 10^{-27} \times 10^{10} \text{ J} = 1.66053906892 \times 10^{-17} \text{ J},$$

since $-27 + 10 = -17$. Dividing by the size of an electronvolt in joules,

$$\frac{1.66053906892 \times 10^{-17}}{1.602176634 \times 10^{-19}} = 1.036427 \times 10^2 = 103.6427,$$

where $1.66053906892/1.602176634 = 1.036427$ and $10^{-17}/10^{-19} = 10^{-17-(-19)} = 10^2$. Therefore

$$1 \text{ amu } \text{\AA}^2 \text{ fs}^{-2} = 103.6427 \text{ eV}, \quad K [\text{eV}] = \frac{1}{2} m [\text{amu}] v^2 [\text{\AA}^2 \text{ fs}^{-2}] \times 103.6427. \quad (3.15)$$

This factor and the factor of Newton's law, 9.648533×10^{-3} , are each other's inverse, $1/103.6427 = 9.648533 \times 10^{-3}$, and for a reason. Equation (3.15) says that one electronvolt is $1/103.6427$ of an $\text{amu } \text{\AA}^2 \text{ fs}^{-2}$, so a force of $1 \text{ eV } \text{\AA}^{-1}$, divided by a mass of 1 amu , is $(1/103.6427) \text{ amu } \text{\AA}^2 \text{ fs}^{-2}$ divided by $\text{\AA} \text{ amu}$, which, cancelling the amu and one $\text{\AA}$, is $(1/103.6427) \text{\AA} \text{ fs}^{-2}$: both factors express the electronvolt in atomic mass units, ångströms and femtoseconds. In `mdlab` the kinetic-energy factor is `MV2_T0_EV`, computed from the CODATA values of the constants, and the Newton factor `FORCE_T0_ACCEL` is defined as its inverse, so the two can never disagree. The kinetic energy of a group of atoms is then a few lines.

```

1 def kinetic_energy(masses: ArrayLike, velocities: ArrayLike) -> float:
2     masses = np.asarray(masses, dtype=float)
3     velocities = np.asarray(velocities, dtype=float)
4     speed_squared = np.sum(velocities**2, axis=-1) # v · v, one per atom
5     if masses.ndim > 0 and masses.shape != speed_squared.shape:
6         raise ValueError("give one mass for each row of velocities")
7     in_amu_a2_per_fs2 = 0.5 * np.sum(masses * speed_squared)
8     return float(in_amu_a2_per_fs2) * units.MV2_T0_EV

```

The velocities arrive with one row per atom and one column per component, as the forces did in Section 2.10. Squaring the array squares every component, and adding along the last axis, `axis=-1`, adds the three squared components within each row, which is $\mathbf{v}_i \cdot \mathbf{v}_i$ for each atom i . The check applies when the masses come as a list, for which `masses.ndim`, the number of dimensions of the array, is `1`: it stops the function with a message unless `masses.shape`, the length of each dimension, matches that of the squared speeds, one mass for each atom. A single number, which has no dimensions, skips the check and serves as the mass of every atom. Multiplying the masses by the squared speeds pairs each atom's mass with its own speed, the sum over atoms and the factor one half give $\sum_i \frac{1}{2} m_i v_i^2$ in $\text{amu } \text{\AA}^2 \text{ fs}^{-2}$, and the last line converts to electronvolts. The lithium atom, which moves along a line and so has a velocity of one component, called as `kinetic_energy(6.94, [0.0139])`, returns `0.0695 eV`.

The potential energy of many atoms

For N atoms the potential energy is a function of all their positions, $U(\mathbf{r}^N)$, a function of $3N$ coordinates. Chapter 8 shows where such functions come from; here we need only that the force on each atom is minus the gradient of U with respect to that atom's own coordinates,

$$\mathbf{F}_i = -\nabla_i U = -\left(\frac{\partial U}{\partial x_i}, \frac{\partial U}{\partial y_i}, \frac{\partial U}{\partial z_i}\right), \quad (3.16)$$

where the subscript on ∇_i says which atom's coordinates are differentiated, all the others held fixed. Everything in [Sections 3.9 to 3.11](#) carries over with $3N$ coordinates in place of three: a path moves all the atoms at once, from one arrangement to another; the work done on them along it is the sum over atoms of $\mathbf{F}_i \cdot \Delta \mathbf{r}_i$ added over its short steps, and for these forces it is the drop in U between the two arrangements.

PROPOSITION 3.3. *Let N atoms move according to Newton's second law, $m_i d\mathbf{v}_i/dt = \mathbf{F}_i$, with forces $\mathbf{F}_i = -\nabla_i U(\mathbf{r}^N)$ from a potential energy that does not depend on time except through the positions. Then the total energy*

$$E = \sum_{i=1}^N \frac{1}{2} m_i v_i^2 + U(\mathbf{r}^N)$$

is constant along every motion.

Proof. By [Section 3.3](#), applied to each atom, the kinetic energy changes at the rate $dK/dt = \sum_i \mathbf{F}_i \cdot \mathbf{v}_i$, where $K = \sum_i \frac{1}{2} m_i v_i^2$, the sum of the powers of the forces; this is where the second law enters. The gradient toolbox of [Section 3.10](#), with a step of $3N$ legs, one along each coordinate, in place of two, gives, grouping the legs atom by atom,

$$\Delta U \approx \sum_{i=1}^N \left(\frac{\partial U}{\partial x_i} \Delta x_i + \frac{\partial U}{\partial y_i} \Delta y_i + \frac{\partial U}{\partial z_i} \Delta z_i \right) = \sum_{i=1}^N \nabla_i U \cdot \Delta \mathbf{r}_i,$$

and the chain rule along a path then reads

$$\frac{dU}{dt} = \sum_{i=1}^N \nabla_i U \cdot \frac{d\mathbf{r}_i}{dt} = - \sum_{i=1}^N \mathbf{F}_i \cdot \mathbf{v}_i.$$

The two rates cancel, so $dE/dt = 0$. □

A potential energy that depends only on the distance between two atoms, like the energy $\frac{1}{2} k \zeta^2$ of the spring between the carts of [Section 3.8](#), is of this form, and [Exercise 3.20](#) shows that for it [Equation \(3.16\)](#) produces the equal and opposite forces of the third law by itself.

FROM ELECTRONIC STRUCTURE. In Born–Oppenheimer molecular dynamics, $U(\mathbf{r}^N)$ is the ground-state energy of the electrons with the nuclei held at $\mathbf{r}^N$, plus the repulsion between the nuclei, and the forces are its negative gradient. The Hellmann–Feynman theorem supplies the part that comes from the explicit dependence of the Hamiltonian on the nuclear positions, and the Pulay terms add the part that comes from basis functions moving with the atoms. A force computed without the Pulay terms, or from an incompletely converged density, is not exactly $-\nabla_i U$ of the energy reported with it, and [Section 3.14](#) shows what such a discrepancy does to the total energy.

CHECK YOUR UNDERSTANDING. An oxygen atom (15.999 amu) moves at $0.01 \text{ \AA fs}^{-1}$. What is its kinetic energy in electronvolts?

3.13 A lithium atom on a model surface

The bead on its wire showed what an energy diagram says in one dimension. This section draws the same picture for an atom moving in two: a lithium atom on a layer of carbon atoms arranged in hexagons, as in a sheet of graphite. We build a simple model of its potential energy, check that its force does the same work along every path, and follow the atom as it hops from site to site with its total energy fixed.

The model

A lithium atom resting on a carbon layer feels a potential energy that depends on where it sits relative to the carbon atoms. We follow only its motion parallel to the layer, at a fixed height, with the carbon atoms held still, so U is a function of the two coordinates $\mathbf{r} = (x, y)$. Our model has three kinds of site ([Figure 3.10a](#)). Above the centre of a hexagon, a hollow, we make the energy lowest. Above a carbon atom, a top, we make it highest. Midway between two neighbouring hollows, above the middle of the bond shared by their hexagons, lies a bridge, the lowest point of the ridge between the two hollows, and so a saddle point ([Section 3.10](#)). We take the spacing of neighbouring hollows to be $a = 2.46 \text{ \AA}$, chosen to resemble the hexagons of a layer of graphite, and the rise from a hollow to a bridge, the barrier, to be $U_b = 0.3 \text{ eV}$. Both are chosen to make the model easy to read; neither is fitted to any measurement or calculation.

A simple function with this pattern is a sum of three ripples. A single ripple $\cos(\mathbf{g} \cdot \mathbf{r})$, for a fixed vector $\mathbf{g}$, is constant along every line perpendicular to $\mathbf{g}$, since moving along such a line leaves $\mathbf{g} \cdot \mathbf{r}$ unchanged, and it rises and falls like a sheet of corrugated iron as $\mathbf{r}$ moves along $\mathbf{g}$, with crests $2\pi/|\mathbf{g}|$ apart. Three such ripples whose vectors $\mathbf{g}_1$, $\mathbf{g}_2$ and $\mathbf{g}_3$ point at -30° , 90° and 210° , all of length $|\mathbf{g}| = 4\pi/(\sqrt{3}a) = 2.949 \text{ \AA}^{-1}$, so

that their crests lie $2\pi/|\mathbf{g}| = \sqrt{3}a/2 = 2.130 \text{ \AA}$ apart, the distance $a \sin 60^\circ$ between neighbouring rows of hollows, cross to make a hollow above the centre of every hexagon, and we set

$$U(\mathbf{r}) = \frac{U_b}{4} \left[3 - \cos(\mathbf{g}_1 \cdot \mathbf{r}) - \cos(\mathbf{g}_2 \cdot \mathbf{r}) - \cos(\mathbf{g}_3 \cdot \mathbf{r}) \right]. \quad (3.17)$$

At the origin all three cosines are 1, and $U = 0$: the origin is a hollow. At the bridge $(a/2, 0)$, using $\mathbf{g}_k \cdot \mathbf{r} = |\mathbf{g}| |\mathbf{r}| \cos \theta_k$ with θ_k the angle between them (Section 1.6), and $|\mathbf{g}| a/2 = 2\pi/\sqrt{3}$,

$$\mathbf{g}_1 \cdot \mathbf{r} = \frac{2\pi}{\sqrt{3}} \cos -30^\circ = \pi, \quad \mathbf{g}_2 \cdot \mathbf{r} = \frac{2\pi}{\sqrt{3}} \cos 90^\circ = 0, \quad \mathbf{g}_3 \cdot \mathbf{r} = \frac{2\pi}{\sqrt{3}} \cos 210^\circ = -\pi,$$

since $\cos -30^\circ = \sqrt{3}/2$ and $\cos 210^\circ = -\sqrt{3}/2$. The cosines are $-1, 1$ and -1 , and $U = (U_b/4)(3 - (-1) - 1 - (-1)) = (U_b/4) \times 4 = U_b$. The top $(a/2, a/(2\sqrt{3}))$ lies a distance $\sqrt{a^2/4 + a^2/12} = \sqrt{a^2/3} = a/\sqrt{3} = 1.420 \text{ \AA}$ from the hollow, in the direction at 30° to the x axis, since $\tan \theta = (a/(2\sqrt{3})) / (a/2) = 1/\sqrt{3}$. There $|\mathbf{g}| a/\sqrt{3} = 4\pi/3$, and the angles from $\mathbf{r}$ to $\mathbf{g}_1, \mathbf{g}_2$ and $\mathbf{g}_3$ are $60^\circ, 60^\circ$ and 180° , so the three products are $\frac{4\pi}{3} \cos 60^\circ = 2\pi/3, 2\pi/3$ and $\frac{4\pi}{3} \cos 180^\circ = -4\pi/3$. Each cosine is $-\frac{1}{2}$ (Exercise 3.19 checks this in components), and $U = (U_b/4)(3 + \frac{3}{2}) = 9U_b/8 = 0.3375 \text{ eV}$. Shifting $\mathbf{r}$ by a along x changes $\mathbf{g}_1 \cdot \mathbf{r}$ by 2π , $\mathbf{g}_2 \cdot \mathbf{r}$ by 0 and $\mathbf{g}_3 \cdot \mathbf{r}$ by -2π , which leaves every cosine unchanged, and the same holds for a shift by a at 60° , so the pattern repeats from hollow to hollow.

The force follows from Equation (3.13). Writing $\mathbf{g} \cdot \mathbf{r} = g_x x + g_y y$, the chain rule gives $\partial \cos(\mathbf{g} \cdot \mathbf{r}) / \partial x = -g_x \sin(\mathbf{g} \cdot \mathbf{r})$, and likewise with g_y for y , so $\nabla \cos(\mathbf{g} \cdot \mathbf{r}) = -\sin(\mathbf{g} \cdot \mathbf{r}) \mathbf{g}$, and

$$\nabla U = \frac{U_b}{4} \sum_{k=1}^3 \sin(\mathbf{g}_k \cdot \mathbf{r}) \mathbf{g}_k \quad \text{the 3 is constant, and each } -\cos \text{ gives } +\sin,$$

$$\mathbf{F} = -\nabla U = -\frac{U_b}{4} \sum_{k=1}^3 \sin(\mathbf{g}_k \cdot \mathbf{r}) \mathbf{g}_k.$$

At the hollow and the bridge every sine is zero, since $\sin 0 = \sin(\pm\pi) = 0$. At the top the three sines are equal, because $\sin(-4\pi/3) = \sin(-4\pi/3 + 2\pi) = \sin(2\pi/3)$, so the force is proportional to $\mathbf{g}_1 + \mathbf{g}_2 + \mathbf{g}_3$, which is zero: the x components of the three directions add to $\cos -30^\circ + \cos 90^\circ + \cos 210^\circ = \sqrt{3}/2 + 0 - \sqrt{3}/2 = 0$, and the y components to $-\frac{1}{2} + 1 - \frac{1}{2} = 0$. Every site is an equilibrium. The function `energy.hexagonal_surface` of `mdlab` returns U and $\mathbf{F}$ together, and its tests check the force against central differences of U and the site energies above.

Work on the surface

Figure 3.10(a) shows the contours of U , the force, and two paths from the hollow A at the origin to the top B at $(3.69, 0.710)$ Å, one spacing a along x from the top nearest A : a straight line, and an arc that bulges to its left, lying $1.2 \sin(\pi s)$ Å from the line at the point a fraction s of the way along it, so 1.2 Å at the middle. The function work gives -0.3375 eV along both, equal to $U(A) - U(B)$, as a conservative force requires. In Figure 3.10(b) the swirl of Equation (3.11), centred on A with $c = 0.005 \text{ eV Å}^{-2}$, is added to the force. Along the straight path, which runs directly away from the centre of the swirl, the swirl is perpendicular to the path and does no work, so the work stays at -0.3375 eV. Along the arc it is -0.3662 eV. The arc followed by the straight path backwards is a clockwise loop, and the curl of the total force is $0 + 2c$, the surface force being a gradient, so by Green's theorem the work round this loop, $W_{\text{arc}} - W_{\text{straight}}$, is $-2c$ times the area between the paths. That area is the bulge integrated along the chord, of length $|AB| = \sqrt{3.69^2 + 0.710^2} = 3.758$ Å, on which a step ds in the fraction moves $|AB| ds$, $\int_0^1 1.2 \sin(\pi s) |AB| ds = 2 \times 1.2 \times 3.758 / \pi = 2.871 \text{ Å}^2$, using $\int_0^1 \sin(\pi s) ds = [-\cos(\pi s) / \pi]_0^1 = 2 / \pi$; and $2c \times 2.871 = 0.0287 \text{ eV}$ is exactly the difference between the two works.

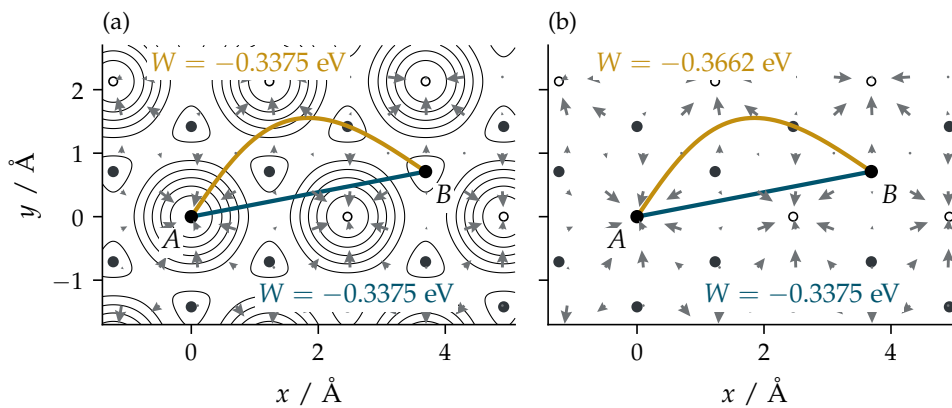


FIGURE 3.10: Work on the model surface. Open circles mark hollows and dark dots the carbon atoms, at the tops. (a) Contours of U at 0.05, 0.10, 0.15, 0.20, 0.25 and 0.32 eV, the force $-\nabla U$ (grey arrows), and two paths from the hollow A to the top B ; the work is -0.3375 eV along both. (b) The same paths after a swirl with $c = 0.005 \text{ eV Å}^{-2}$, centred on A , is added to the force (grey arrows). No potential energy exists to contour, and the works differ by 0.0287 eV.

Hopping with a fixed energy

A lithium atom leaves the hollow at the origin with 0.33 eV of kinetic energy, 10 % more than the barrier U_b , moving at 37.5° to the x axis. By Equation (3.15) its speed is

$$v = \sqrt{\frac{2K}{m \times 103.6427}} = \sqrt{\frac{2 \times 0.33}{6.94 \times 103.6427}} = 0.03029 \text{ Å fs}^{-1},$$

about 3000 m s^{-1} . Its total energy, $E = 0.33 \text{ eV}$, exceeds U everywhere except in small islands around each carbon atom, where U rises to 0.3375 eV : the atom may go anywhere on the surface except into these islands, just as the bead at $h_E = 1.3 \text{ m}$ could go anywhere except up the ends of its track. Figure 3.11 shows the motion over 1.5 ps , or 1500 fs (a picosecond is 1000 fs), computed by a computer solution of the equations of motion, a method from SciPy, a library of scientific routines for Python, that advances the state in short steps, as in Section 2.6, but combines several force evaluations per step and adjusts each step so that the error it estimates for that step stays below about one part in 10^{11} ; over many steps these small errors add up. Chapter 9 develops the methods that molecular dynamics itself uses. The atom crosses from hollow to hollow four times, visiting five hollows, while its kinetic energy rises and falls between 0.0005 and 0.33 eV . Its total energy stays at 0.33 eV to within $2.5 \times 10^{-11} \text{ eV}$, which is the error of the computer solution.

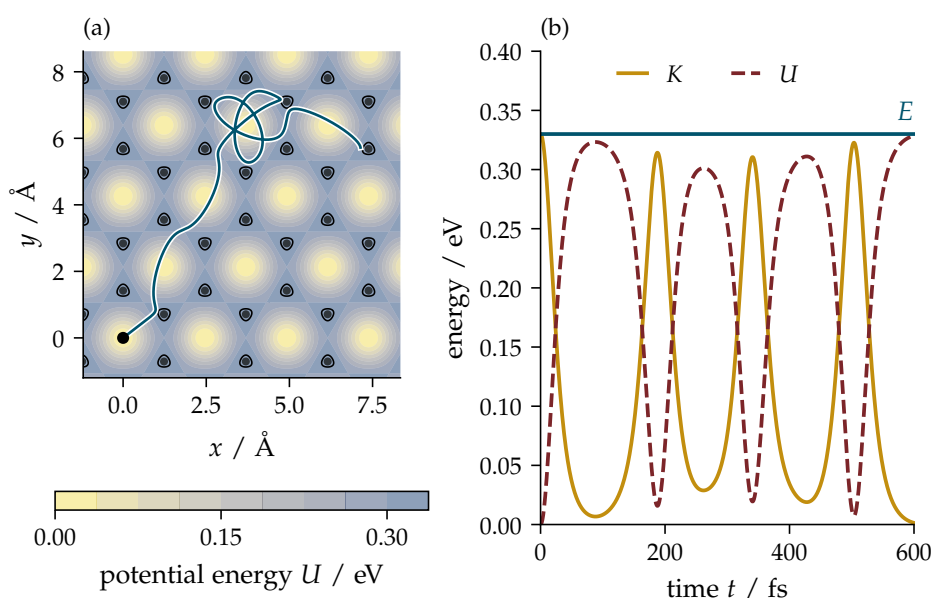


FIGURE 3.11: A lithium atom of 6.94 amu on the model surface, leaving the hollow at the origin with 0.33 eV of kinetic energy. (a) The potential energy, shaded from the hollows (light) to the bridges and tops (dark), the boundary $U = E$ of the islands around the carbon atoms (dots), where the atom may not go, and its path over 1.5 ps . (b) Kinetic, potential and total energy over the first 600 fs .

With less energy the picture changes as it did on the track. Below the barrier, $E < 0.3 \text{ eV}$, the allowed region breaks into separate patches, one around each hollow, and an atom stays in the hollow it starts in, rattling about its bottom. To hop to a neighbouring hollow it needs at least the barrier as kinetic energy at the bottom of the hollow, a speed there of at least $\sqrt{2 \times 0.3 / (6.94 \times 103.6427)} = 0.02888 \text{ Å fs}^{-1}$. Real atoms exchange energy with their neighbours all the time, and Chapter 12 gives the odds that one collects as much as the barrier; Chapters 16 and 17 measure how fast lithium spreads through graphite by such hops.

NOTEBOOK 03 *launch the lithium atom with your own energy and direction*

CHECK YOUR UNDERSTANDING. A lithium atom at the bottom of a hollow of the model surface has 0.25 eV of kinetic energy. Can it reach a neighbouring hollow? How fast is it moving where $U = 0.1$ eV?

3.14 When the energy changes

Proposition 3.3 needs two conditions: every force is minus the gradient of a potential energy, and that potential energy does not change with time on its own. This section works out what happens to the total energy of a group of atoms when either fails, and why that makes the energy the first thing to check in a simulation of atoms isolated from their surroundings.

A force that is not a gradient

Write the force on each atom as a conservative part plus the rest, $\mathbf{F}_i = -\nabla_i U + \mathbf{F}_i^{\text{nc}}$. The rest adds to the power that changes the kinetic energy, $dK/dt = \sum_i \mathbf{F}_i \cdot \mathbf{v}_i$, but not to the rate of change of U , which the chain rule along a path still gives as $\sum_i \nabla_i U \cdot \mathbf{v}_i$. Adding the two rates,

$$\begin{aligned} \frac{dE}{dt} &= \sum_{i=1}^N (-\nabla_i U + \mathbf{F}_i^{\text{nc}}) \cdot \mathbf{v}_i + \sum_{i=1}^N \nabla_i U \cdot \mathbf{v}_i \\ &= \sum_{i=1}^N \mathbf{F}_i^{\text{nc}} \cdot \mathbf{v}_i, \quad \text{the gradient terms cancelling,} \end{aligned} \quad (3.18)$$

the rate at which the non-conservative forces do work, as for the box on the floor in [Equation \(3.10\)](#). A drag on every atom, $\mathbf{F}_i^{\text{nc}} = -\gamma m_i \mathbf{v}_i$ with the drag rate γ of [Section 2.7](#), gives, since $\mathbf{v}_i \cdot \mathbf{v}_i = v_i^2$ and $\sum_i m_i v_i^2 = 2K$,

$$\frac{dE}{dt} = -\gamma \sum_{i=1}^N m_i v_i^2 = -2\gamma K,$$

so the energy can only fall. Chapter 13 adds such a drag on purpose, with random kicks that make up for it, to hold atoms at a chosen temperature.

The swirl shows what a force does that sometimes adds energy and sometimes removes it. [Figure 3.12](#) repeats the motion of [Figure 3.11](#) with the swirl of [Figure 3.10\(b\)](#) added to the force. The total energy now wanders, from 0.0185 eV below its starting value to 0.0117 eV above it within 1.5 ps. The power of the swirl is $\mathbf{F}_{\text{sw}} \cdot \mathbf{v} = c(xv_y - yv_x)$, so the atom gains energy while it moves anticlockwise about the origin and loses it while it moves clockwise: on the positive x axis, for instance, $y = 0$ and the power is

$c x v_y$, positive when the atom moves up, which is anticlockwise there. Its change agrees with the work done by the swirl, the integral of Equation (3.18) over time, to within 1×10^{-8} eV.

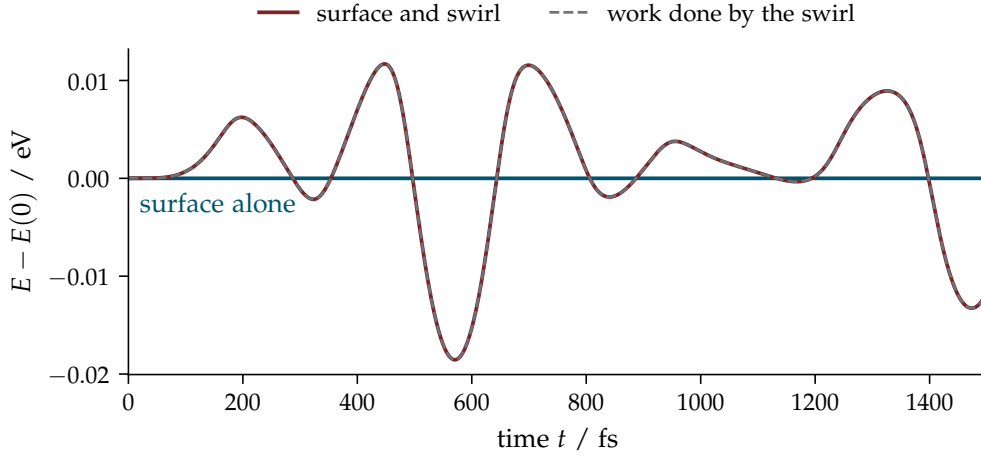


FIGURE 3.12: The lithium atom of Figure 3.11, launched in the same way on the surface alone (teal) and with the swirl of Figure 3.10(b) added to the force (dark red). The change of the total energy $E = K + U$ from its starting value is shown against time, with the work done by the swirl, $\int_0^t \mathbf{F}_{\text{sw}} \cdot \mathbf{v} dt'$ (dashed), which it follows to within 1×10^{-8} eV.

A potential energy that changes with time

If the potential energy also depends on time, $U(\mathbf{r}^N, t)$, because something outside the atoms is being changed while they move, then U changes even when the atoms stand still. A step in time, Δt , is then one more leg of the gradient toolbox's step, and the chain rule along a path gains one term, the partial derivative with respect to time,

$$\frac{dU}{dt} = \sum_{i=1}^N \nabla_i U \cdot \mathbf{v}_i + \frac{\partial U}{\partial t},$$

so that, with conservative forces, $dE/dt = \partial U/\partial t$. A hand that moves the far end of a spring does work on the body at the near end in this way: with the hand at $X(t)$, measured so that the spring is slack when the body is at $x = X$, the body has the potential energy $U = \frac{1}{2}k(x - X(t))^2$, which changes even while the body stands still. So does a wall pushed in to squeeze a group of atoms, the picture behind the pistons with which Chapter 14 controls their pressure.

IN PRACTICE. The exact motion of an isolated group of atoms, one that exchanges no energy with its surroundings, keeps E constant. A simulation of such a group whose total energy drifts is therefore not following the exact motion, and there are two common causes. The first is the finite time step with which the equations of motion are advanced, the subject of Chapter 9. The second is a force that is not exactly $-\nabla_i U$ of the potential energy reported with it, for which Equation (3.18) gives the drift directly: an error in the code that computes the forces, a potential energy cut off abruptly at some distance (Chapter 8), or a machine-learned model, a program fitted to examples of computed forces (Part V), that predicts the forces directly instead of differentiating an energy, so that nothing constrains their curl. Bigi, Langer and Ceriotti found that models of the last kind can make several types of molecular dynamics unstable and the convergence of geometry optimisation, the search for a minimum of U , ill-defined, and that their departure from energy conservation is difficult to monitor⁴. Even so, recording E in a short run of the isolated system is the first check of any new model or code.

KEY IDEA. For atoms whose forces are $\mathbf{F}_i = -\nabla_i U$, with U fixed in time, Newton's laws keep $E = K + U$ constant. Any other part of the force changes E at the rate at which it does work, $dE/dt = \sum_i \mathbf{F}_i^{\text{nc}} \cdot \mathbf{v}_i$, and a change of U with time changes E at the rate $\partial U/\partial t$.

NOTEBOOK 03 vary the strength of the swirl and watch the energy wander

CHECK YOUR UNDERSTANDING. In a simulation of an isolated group of atoms, the total energy rises steadily. Name two possible causes, and say which equation of this section describes the second.

What we now have

A force does work $\mathbf{F} \cdot \Delta \mathbf{r}$ on a straight step, and along a curved path the line integral $\int \mathbf{F} \cdot d\mathbf{r}$; the second law makes the work of the net force equal to the change in the kinetic energy $K = \frac{1}{2}mv^2$, along a motion obeying that law.

A force is conservative when its work depends only on where a path starts and ends. It then has a potential energy U , the force is $-\nabla U$, pointing straight down the slope of U and across its contours, and Newton's laws keep the total energy $E = K + U$ constant when U has no explicit time dependence, for one body or for N atoms. Energy alone gives the speed of a body at every place, and the energy diagram shows where it can go, where it turns back and where it can rest: at minima of U stably, at maxima unstably.

Friction, drag and every other force that is not a gradient change E at the rate at which they do work, $\sum_i \mathbf{F}_i^{\text{nc}} \cdot \mathbf{v}_i$. Forces between bodies can change the total kinetic energy, though not the total momentum, and energy and momentum together gave the speeds of the carts of Chapter 2.

A force has zero curl if it is conservative, and in a region without holes zero curl is enough. For atoms, energies are measured in electronvolts, and $\frac{1}{2}mv^2$ in amu and $\text{\AA} \text{fs}^{-1}$ becomes electronvolts on multiplying by 103.6427, the inverse of the factor of Newton's law. On a model surface, a lithium atom with more energy than the barrier between sites can hop from hollow to hollow, its total energy fixed, and a force that is not a gradient makes that energy wander.

Chapter 4 uses energy to study the motion that connects springs to atomic vibrations: the oscillation of a body about a stable minimum of its potential energy.

Answers to the checks

SECTION 3.1. None. The hand pushes vertically and the tray moves horizontally, so the push is perpendicular to the displacement and $\mathbf{F} \cdot \Delta \mathbf{r} = 0$.

SECTION 3.2. K is proportional to v^2 , so halving the speed divides it by four, to 12.5 J. A body of 2 kg at 3 m s^{-1} has $K = \frac{1}{2} \times 2 \times 3^2 = 9 \text{ J}$.

SECTION 3.3. By Equation (3.4) with the signs reversed, the hand does $\frac{1}{2}k(x_B^2 - x_A^2) = \frac{1}{2} \times 50 \times (0.08^2 - 0.04^2) = 0.12 \text{ J}$, three times the 0.04 J of the first 4 cm. The force grows with the stretch, so over the second 4 cm the hand pulls harder all the way: the area under kx between 4 and 8 cm is a trapezium three times the first triangle.

SECTION 3.4. Taking the ground as $x = 0$, the ball starts with $E = \frac{1}{2}m \times 12^2 + mg \times 5$ and reaches the ground with $U = 0$, so $\frac{1}{2}mv^2 = \frac{1}{2}m \times 144 + 5mg$, and $v = \sqrt{144 + 2 \times 9.80665 \times 5} = 15.56 \text{ m s}^{-1}$. The direction of the throw does not matter: only the weight does work, and by Section 3.2 its work is $-mg$ times the rise whatever the path.

SECTION 3.5. No. Starting at rest at 0.8 m, the bead can never rise above 0.8 m, and the hill is 1.009 m high. To cross it the bead must at least reach the top, where $K \geq 0$ requires $h_E \geq 1.009 \text{ m}$, so at the bottom of the deep valley it needs $\frac{1}{2}v^2 \geq g(1.009 - 0.183)$, or $v \geq \sqrt{2 \times 9.80665 \times 0.826} = 4.025 \text{ m s}^{-1}$. This is the limiting speed: with exactly the barrier energy the bead approaches the crest ever more slowly. Crossing in finite time needs energy strictly above the barrier.

SECTION 3.6. It slides right, down into the shallow valley and up the far side, and turns back at the other turning point, $x = 1.724$ m. It is fastest at the bottom of the valley, $x = 1.347$ m, where $h = 0.6072$ m, with $v = \sqrt{2 \times 9.80665 \times (0.8 - 0.6072)} = 1.945 \text{ m s}^{-1}$.

SECTION 3.7. The same distance. The friction is $\mu_k mg$ and the kinetic energy $\frac{1}{2}mv_0^2$, both proportional to the mass, so the stopping distance $v_0^2/2\mu_k g$ does not depend on it. The kinetic energy has become the hidden kinetic and potential energy of the jiggling atoms of the box and the floor, which warm up.

SECTION 3.8. With equal masses, momentum gives $v_1 = -v_2$, and energy $\frac{1}{2} \times 2 \times v_1^2 + \frac{1}{2} \times 2 \times v_2^2 = 2v_2^2 = 1 \text{ J}$, so $v_2^2 = 0.5$: the carts separate at 0.7071 m s^{-1} each, in opposite directions, and share the energy equally.

SECTION 3.9. None. Along the line from the origin to $(3, 4)$ the displacement is always along $\mathbf{r}$, and the swirl is perpendicular to $\mathbf{r}$, so $\mathbf{F}_{\text{sw}} \cdot d\mathbf{r} = 0$ on every chord.

SECTION 3.10. Where the contours are crowded, the height changes by the same amount over a shorter distance, so the slope, and $|\nabla h|$, is larger there. Where ∇h crosses a contour it is perpendicular to it, pointing uphill.

SECTION 3.11. Yes. $\partial F_y / \partial x = \partial x / \partial x = 1$ and $\partial F_x / \partial y = \partial y / \partial y = 1$, so the curl is zero everywhere, and the whole plane has no holes. Taking the path of Section 3.11 from the origin along the x axis, where $F_x = y = 0$, and then straight up to (x, y) , where $F_y = x$, gives $U = -(0 + xy) = -xy$, and it works: $-\partial U / \partial x = y = F_x$ and $-\partial U / \partial y = x = F_y$.

SECTION 3.12. By Equation (3.15), $K = \frac{1}{2} \times 15.999 \times 0.01^2 \times 103.6427 = 0.0829 \text{ eV}$.

SECTION 3.13. No: the bridge between hollows is at 0.3 eV , more than its total energy of 0.25 eV , so it stays in its own hollow. Where $U = 0.1 \text{ eV}$ its kinetic energy is 0.15 eV , so $v = \sqrt{2 \times 0.15 / (6.94 \times 103.6427)} = 0.0204 \text{ Å fs}^{-1}$.

SECTION 3.14. A time step too long for the step-by-step solution to follow the motion accurately (Chapter 9), and forces that are not exactly minus the gradient of the potential energy being reported, such as forces predicted directly by a machine-learned model, or a force switched off abruptly at some distance. Equation (3.18) describes the second: the energy changes at the rate at which the non-conservative part of the force does work.

Exercises

The exercises run from questions of understanding, through calculations, to short derivations marked ‘show that’. Full solutions are in ‘[Solutions to the exercises](#)’ at the end of the book, and Notebook 03 checks every numerical answer.

EXERCISE 3.1 (*a puck on a string*). A puck on a smooth, level table is tied by a string to a peg and slides round the peg in a circle at a constant speed. Does the pull of the string do work on the puck? What does your answer say about the puck’s kinetic energy?

EXERCISE 3.2 (*a suitcase*). A suitcase is pulled 25 m along a level floor by its handle, with a force of 60 N directed 50° above the horizontal. How much work does the pull do?

EXERCISE 3.3 (*stairs*). A person of 70 kg climbs a staircase that rises 3 m, at a steady speed, in 5 s. How much work do they do against their weight, and at what average power?

EXERCISE 3.4 (*braking*). A car of 1200 kg travels at 50 km h^{-1} . Find its kinetic energy, and the distance in which a braking force of 6000 N stops it. How far does the same force take to stop it from 100 km h^{-1} ?

EXERCISE 3.5 (*the second stretch*). How much work does it take to stretch a spring of 200 N m^{-1} from its natural length to 10 cm, and how much more to stretch it from 10 cm to 20 cm?

EXERCISE 3.6 (*a launcher*). A spring of 500 N m^{-1} , compressed by 8 cm, launches a puck of 0.1 kg along a smooth level table. Find the puck’s speed once it leaves the spring, and the height it reaches if it then slides up a smooth ramp.

EXERCISE 3.7 (*a pendulum*). A small bob hangs from a string 1.2 m long and is released from rest with the string at 40° to the vertical. Explain why the string does no work on the bob, and find the bob’s speed at the bottom of its swing.

EXERCISE 3.8 (*back on the track*). The bead on the track of [Equation \(3.8\)](#) passes the bottom of the deep valley at 3 m s^{-1} . What is the greatest height it reaches? Can it cross the hill? Use [Figure 3.5](#) to estimate its turning points, and Notebook 03 to find them.

EXERCISE 3.9 (*a double well*). A body moves along a line with potential energy $U(x) = U_0 [(x/a)^2 - 1]^2$, where U_0 and a are positive constants. Show that it has equilibria at $x = 0$ and $x = \pm a$, and decide from U'' which are stable. For $E = U_0/2$, show that the turning points are at $x = \pm a \sqrt{1 \pm 1/\sqrt{2}}$, and describe the allowed regions.

EXERCISE 3.10 (*on ice*). A puck slides onto rough ice at 3 m s^{-1} , with $\mu_k = 0.05$. How far does it slide before it stops?

EXERCISE 3.11 (*a rough ramp*). A block slides from rest down a straight ramp inclined at 25° , starting 2 m above the bottom, with $\mu_k = 0.15$. Find its speed at the bottom.

EXERCISE 3.12 (*other carts*). Carts of 1 kg and 2 kg, at rest on a smooth track with the lighter on the left, are pushed apart by a spring of 200 N m^{-1} compressed by 5 cm. Find their velocities, with x to the right, once the spring has fallen away.

EXERCISE 3.13 (*coupling*). In Exercise 2.16 a cart of 2 kg moving at 3 m s^{-1} couples to a cart of 1 kg at rest, and the pair moves on at 2 m s^{-1} . Compare the kinetic energies before and after. Where has the difference gone, and why does momentum not show the same loss?

EXERCISE 3.14 (*two routes*). A force in the plane is $\mathbf{F} = a(y, x)$, with $a = 0.5 \text{ eV } \text{\AA}^{-2}$. Compute its work from the origin to $(2, 1) \text{ \AA}$ along the straight line, and along the route that goes first along the x axis to $(2, 0)$ and then straight up. Find a potential energy U with $\mathbf{F} = -\nabla U$.

EXERCISE 3.15 (*an elliptical well*). An atom in the plane has potential energy $U = \frac{1}{2}k(x^2 + 4y^2)$, with $k = 2 \text{ eV } \text{\AA}^{-2}$. Find the force on it at $(1, 1) \text{ \AA}$. Show that the force is perpendicular to the contour of U through that point, and that it does not point at the origin, where U is least.

EXERCISE 3.16 (*the curl test*). Show that $\mathbf{F} = a(2xy, x^2 + z^2, 2yz)$, with a in $\text{eV } \text{\AA}^{-3}$, has zero curl, and find the potential energy U with $U(\mathbf{0}) = 0$. Evaluate U at $(1, 2, 3) \text{ \AA}$ for $a = 1 \text{ eV } \text{\AA}^{-3}$.

EXERCISE 3.17 (*round a square*). Compute the work of the swirl $\mathbf{F} = c(-y, x)$ once anticlockwise around the square with corners $(0, 0)$, $(L, 0)$, (L, L) and $(0, L)$, side by side, and show that it is $2c$ times the area of the square.

EXERCISE 3.18 (*a field with a hole*). In the plane with the origin removed, let $\mathbf{F} = b(-y, x)/(x^2 + y^2)$, with b in eV. Show that $\partial F_y/\partial x = \partial F_x/\partial y$ wherever $\mathbf{F}$ is defined (differentiate $(x^2 + y^2)q = 1$ by the product rule to find the partial derivatives of $q = 1/(x^2 + y^2)$), and compute the work once anticlockwise around the circle of radius R about the origin. Explain why a non-zero answer does not contradict the curl test, and find the work around a circle that does not enclose the origin.

EXERCISE 3.19 (*the top site*). For the model surface of Equation (3.17), write $\mathbf{g}_1$, $\mathbf{g}_2$ and $\mathbf{g}_3$ in components, and show by the scalar product in components that at the top

$\mathbf{r} = (a/2, a/(2\sqrt{3}))$ the three products $\mathbf{g}_k \cdot \mathbf{r}$ are $2\pi/3$, $2\pi/3$ and $-4\pi/3$, and hence that $U = 9U_b/8$.

EXERCISE 3.20 (*pair forces*). Two atoms have potential energy $U = \phi(r_{12})$, a function of their distance $r_{12} = |\mathbf{r}_2 - \mathbf{r}_1|$ alone. Show that $\partial r_{12}/\partial x_2 = (x_2 - x_1)/r_{12}$ (differentiate r_{12}^2 , a sum of three squares, with respect to x_2), and hence that $\mathbf{F}_2 = -\phi'(r_{12}) (\mathbf{r}_2 - \mathbf{r}_1)/r_{12}$ and $\mathbf{F}_1 = -\mathbf{F}_2$: the forces are equal and opposite and act along the line joining the atoms.

EXERCISE 3.21 (*stopping an oxygen atom*). An oxygen atom (15.999 amu) moving at $0.01 \text{ \AA fs}^{-1}$ enters a region where a constant force of $0.5 \text{ eV \AA}^{-1}$ opposes its motion. Use the work-energy theorem to find how far it travels before it stops.

EXERCISE 3.22 (*a drag on an atom*). An atom moves with no potential energy under the drag $-\gamma m \mathbf{v}$. Use [Equation \(3.18\)](#) to show that $K(t) = K(0) e^{-2\gamma t}$. For a lithium atom (6.94 amu) starting at $0.02 \text{ \AA fs}^{-1}$ with $\gamma = 0.005 \text{ fs}^{-1}$, find K after 100 fs, in electronvolts.

Oscillations

Chapter 2 solved the motion of a block on a spring, and Chapter 3 showed that near a smooth potential-energy minimum with positive curvature the restoring force is approximately that of a spring. This connects an ordinary spring to the vibrations of atoms near a stable arrangement. We now examine how the motion depends on the shape of the well and on the forces supplied by the surroundings.

For the harmonic spring, circular motion, the exchange of kinetic and potential energy, and the path through the plane of states give three ways to follow the same oscillation. Larger displacements explore terms beyond the quadratic approximation, so the period can depend on the energy; in the pendulum and pair potentials studied here, it increases. Drag removes energy and damps the motion, while a periodic force can supply energy most effectively near resonance. These effects return in thermostats and in the response of molecules to light.

When several atoms move, their displacements are coupled because moving one changes the forces on its neighbours. We first reduce the two-body problem to a single relative coordinate, then find the independent patterns of a coupled system, its normal modes, using the eigenvalues of a matrix. Molecular examples connect these frequencies to periods of less than ten femtoseconds, establishing the scale that a simulation's time step must resolve. Chapter 5 then turns to rotation.

4.1 The shadow of a circle

A block of mass m on a spring of stiffness k moves, by [Section 2.8](#), as

$$x(t) = A \cos(\omega t + \phi), \quad \omega = \sqrt{\frac{k}{m}}, \quad (4.1)$$

where the amplitude A and the phase ϕ are fixed by the position and velocity at the start. This motion is called simple harmonic motion, and the system that performs it a harmonic oscillator. This section gives the formula a picture that makes its phase, its velocity and its period easy to read.

The reference circle

Section 1.7 followed a point going round a circle at a steady angular speed, and found that its projection on the x axis moves back and forth as a sinusoid. Now run the argument the other way. Let a point P go anticlockwise round a circle of radius A centred on the origin, at the angular speed ω , starting at the angle ϕ from the x axis, so that at time t its angle is

$$\theta(t) = \omega t + \phi.$$

By the definition of the cosine, scaled to a circle of radius A (Section 1.5), the x coordinate of P is $A \cos(\omega t + \phi)$, which is Equation (4.1). The block therefore moves exactly as the shadow of P on the x axis, the point directly below or above it (Figure 4.1). The circle is called the reference circle of the oscillation, and the phase ϕ is the angle at which P starts.

The velocity follows the same picture. Equation (1.9) differentiated $R(\cos \omega t, \sin \omega t)$; here the radius is A and the angle is $\omega t + \phi$, which also grows at the rate ω , so the chain rule gives the same factor ω and the velocity of P is $A\omega(-\sin \theta, \cos \theta)$. Its x component, $-A\omega \sin(\omega t + \phi)$, is the derivative of $x(t)$: the velocity of the block is the shadow of the velocity of P . The point P always moves at the speed $A\omega$, tangent to the circle. Where P crosses the top or the bottom of the circle it moves parallel to the x axis, so the shadow moves at the full speed $A\omega$; this is the centre of the swing. Where P crosses the x axis it moves straight up or down, so the shadow stands still for an instant; these are the turning points, $x = \pm A$. Likewise the acceleration of P points towards the centre with size $\omega^2 A$ (Section 1.7), and its x component is $-\omega^2 x$, the acceleration of the block.

An example

Take the block of Section 2.8, with $m = 0.5 \text{ kg}$ and $k = 50 \text{ N m}^{-1}$, so that $\omega = 10 \text{ rad s}^{-1}$, started 3 cm from its resting place and moving outwards at 0.4 m s^{-1} . That section found $A = 5 \text{ cm}$, $\cos \phi = x(0)/A = 0.6$ and $\phi = \arcsin(-0.8) = -0.927 \text{ rad}$. On the reference circle, P starts 0.927 rad , or $0.927 \times 180^\circ/\pi = 53.1^\circ$, below the positive x axis, where its shadow is at $A \cos \phi = 5 \times 0.6 = 3 \text{ cm}$. Moving anticlockwise, P climbs towards the axis, so the shadow moves outwards, as the block does. P reaches the

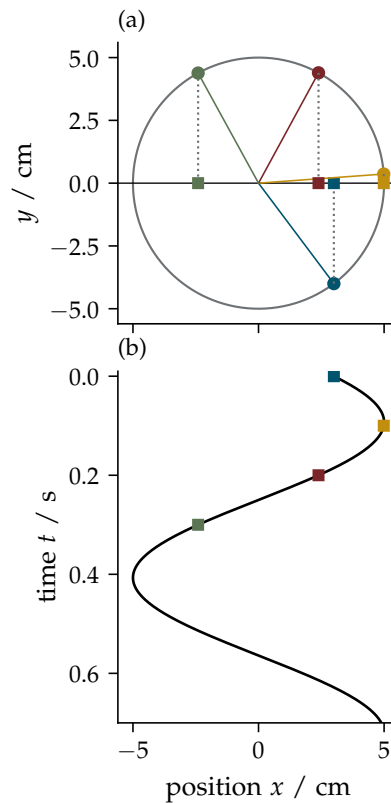


FIGURE 4.1: The oscillation as the shadow of a circle. A block with $\omega = 10 \text{ rad s}^{-1}$ starts 3 cm out, moving outwards at 0.4 m s^{-1} . (a) The point P , going anticlockwise round a circle of radius $A = 5 \text{ cm}$ from $\phi = -0.927 \text{ rad}$, at $t = 0, 0.1, 0.2$ and 0.3 s (dots), and its shadow on the x axis (squares), one colour per moment. (b) The shadow's position against time, with time running downwards so that each square lies directly below the square of the same colour in panel (a).

axis when $\omega t + \phi = 0$, at $t = 0.927/10 = 0.0927 \text{ s}$, and there the block turns back, at $x = A = 5 \text{ cm}$.

P goes once round the circle while ωt grows by 2π , so the period is $2\pi/\omega = 2\pi/10 = 0.628 \text{ s}$, and the frequency, the number of oscillations per second (Section 1.5), is

$$\nu = \frac{\omega}{2\pi} = \frac{10}{2\pi} = 1.592 \text{ Hz},$$

written with the Greek letter nu. The angular frequency ω is the rate at which P turns, in radians per second, and is 2π times the frequency because each oscillation is one full turn of 2π radians.

CHECK YOUR UNDERSTANDING. A block oscillates with $\omega = 10 \text{ rad s}^{-1}$ and $A = 5 \text{ cm}$. Where is the block, and which way and how fast is it moving, when the angle $\omega t + \phi$ of its point P equals $\pi/2$?

4.2 Energy in the oscillator

An oscillating block has its largest kinetic energy at the centre of the swing and none at the ends, and its potential energy does the opposite. This section follows the trade and shows that, averaged over a period, the energy is shared equally between the two forms, a result that returns when Chapter 12 relates kinetic energy to temperature.

The trade

With $x = A \cos(\omega t + \phi)$ and $v = -A\omega \sin(\omega t + \phi)$ (Section 4.1), and writing $\theta = \omega t + \phi$ for short, the two energies of Section 3.4 are

$$K = \frac{1}{2}mv^2 = \frac{1}{2}mA^2\omega^2 \sin^2 \theta = \frac{1}{2}kA^2 \sin^2 \theta \quad \text{since } \omega^2 = k/m \text{ gives } m\omega^2 = k,$$

$$U = \frac{1}{2}kx^2 = \frac{1}{2}kA^2 \cos^2 \theta.$$

Taking out the common factor $\frac{1}{2}kA^2$ and using $\sin^2 \theta + \cos^2 \theta = 1$, their sum is $E = \frac{1}{2}kA^2(\sin^2 \theta + \cos^2 \theta) = \frac{1}{2}kA^2$, constant, as Proposition 3.1 requires. The energy therefore fixes the amplitude, $A = \sqrt{2E/k}$: an oscillation with four times the energy swings twice as far. For the same block released instead from rest 4 cm out, as in Figure 2.9, $E = \frac{1}{2} \times 50 \times 0.04^2 = 0.04$ J. At $x = 2$ cm the potential energy is $\frac{1}{2} \times 50 \times 0.02^2 = 0.01$ J, and the remaining 0.03 J is kinetic (Figure 4.2b). As the block swings, K and U rise and fall twice in every period. By Equation (1.6) with $\cos \pi = -1$ and $\sin \pi = 0$, $\sin(\theta + \pi) = -\sin \theta$ and $\cos(\theta + \pi) = -\cos \theta$, and squaring removes the minus signs, so $\sin^2 \theta$ and $\cos^2 \theta$ repeat when θ grows by π , half a turn (Figure 4.2a).

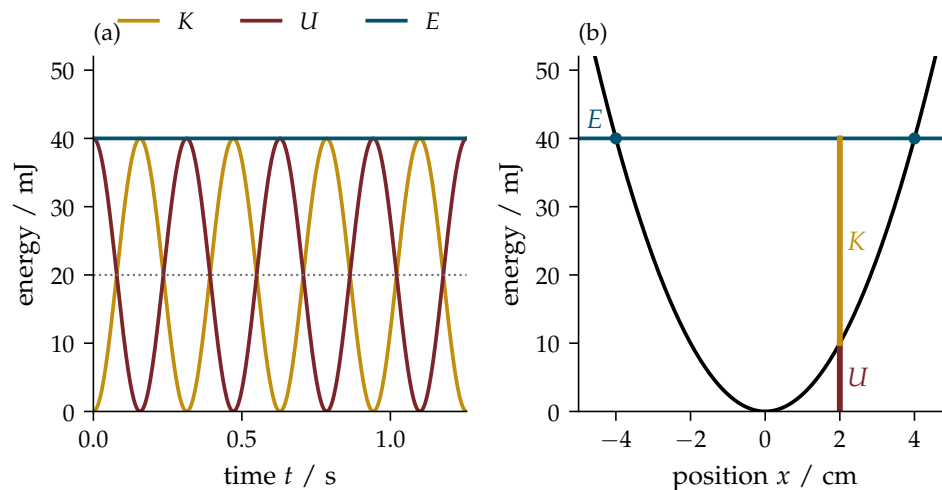


FIGURE 4.2: Energy in the oscillator: the block of Section 4.1, released instead from rest 4 cm out, so that $E = 0.04$ J. (a) Kinetic energy K , potential energy U and their sum against time over two periods; the dotted line is $E/2$, the average of each. (b) The energy diagram, $U = \frac{1}{2}kx^2$ with the line E ; at $x = 2$ cm the energy splits into $U = 0.01$ J and $K = 0.03$ J, and the dots mark the turning points.

Equal shares on average

THE AVERAGE OF A FUNCTION. The average of n numbers is their sum divided by n . To average a function $f(t)$ over an interval from 0 to t_1 , sample it at n equally spaced times t_k , a step $\tau = t_1/n$ apart, and average the samples:

$$\frac{1}{n} \sum_k f(t_k) = \frac{1}{t_1} \sum_k f(t_k) \tau \longrightarrow \frac{1}{t_1} \int_0^{t_1} f(t) dt \quad \text{as } n \rightarrow \infty,$$

the first step multiplying and dividing by τ and using $n\tau = t_1$, the second the definition of the integral as a limit of such sums (Section 1.4). The average of f over the interval is therefore $\frac{1}{t_1} \int_0^{t_1} f dt$, the area under its graph divided by the length of the interval. We write it with an overbar, $\overline{f}$.

To average $\cos^2 \theta$ over a period, rewrite it. The addition formula of Section 1.5 with both angles equal to θ gives $\cos 2\theta = \cos^2 \theta - \sin^2 \theta$, and

$$\cos 2\theta = \cos^2 \theta - (1 - \cos^2 \theta) = 2\cos^2 \theta - 1 \quad \text{replacing } \sin^2 \theta,$$

$$\cos^2 \theta = \frac{1}{2} + \frac{1}{2} \cos 2\theta \quad \text{adding 1 to both sides and halving.}$$

The second term averages to zero over a period, $t_1 = 2\pi/\omega$. With $\theta = \omega t + \phi$, so that $2\theta = 2\omega t + 2\phi$, the function $\sin(2\omega t + 2\phi)/(2\omega)$ has derivative $\cos(2\omega t + 2\phi)$ by the

chain rule, so the fundamental theorem of calculus (Section 1.4) turns the integral into the change in that function,

$$\int_0^{t_1} \cos(2\omega t + 2\phi) dt = \frac{\sin(2\omega t_1 + 2\phi) - \sin 2\phi}{2\omega} = 0,$$

because $2\omega t_1 = 4\pi$, two whole turns, after which the sine takes its old value. Averaging the two terms of $\cos^2 \theta$ separately,

$$\begin{aligned} \overline{\cos^2 \theta} &= \frac{1}{t_1} \int_0^{t_1} \frac{1}{2} dt + \frac{1}{t_1} \int_0^{t_1} \frac{1}{2} \cos(2\omega t + 2\phi) dt \quad \text{the integral of a sum,} \\ &= \frac{1}{t_1} \times \frac{t_1}{2} + 0 = \frac{1}{2} \quad \text{a rectangle, and the integral above,} \end{aligned}$$

and, since $\sin^2 \theta = 1 - \cos^2 \theta$, the same steps give $\overline{\sin^2 \theta} = 1 - \frac{1}{2} = \frac{1}{2}$. The trade above gave $K = E \sin^2 \theta$ and $U = E \cos^2 \theta$, with E a constant that comes outside each integral, so

$$\overline{K} = \overline{U} = \frac{1}{2}E : \quad (4.2)$$

over a period, a harmonic oscillator holds half its energy as kinetic energy and half as potential energy. For the block, each averages 0.02 J, the dotted line of Figure 4.2(a).

NOTEBOOK 04 *the spring with an animation, energy bars and its phase portrait*

CHECK YOUR UNDERSTANDING. At which positions does the block of Figure 4.2 have equal kinetic and potential energy?

4.3 The phase portrait

Section 2.6 showed that the state of a body moving along a line is the pair (x, v) , and drew motions as paths through the plane of states (Figure 2.6b). This plane is called the phase plane, and a drawing of the paths through it the phase portrait of the system. For the oscillator, energy alone gives the shape of every path.

Ellipses of constant energy

Along any motion of the block, $\frac{1}{2}mv^2 + \frac{1}{2}kx^2 = E$. Dividing both sides by E , which is positive unless the block rests at $x = 0$, gives $kx^2/2E + mv^2/2E = 1$, and dividing the top and bottom of the first fraction by k and of the second by m ,

$$\frac{x^2}{2E/k} + \frac{v^2}{2E/m} = 1, \quad \text{or} \quad \left(\frac{x}{a}\right)^2 + \left(\frac{v}{b}\right)^2 = 1 \quad \text{with } a = \sqrt{\frac{2E}{k}}, \quad b = \sqrt{\frac{2E}{m}}. \quad (4.3)$$

The second form says that the point $(x/a, v/b)$ lies on the circle of radius 1, so the state (x, v) lies on that circle stretched by the factor a along the x axis and b along the v axis. A stretched circle is an ellipse, and a and b are its semi-axes, half its width and half its height. Here $a = \sqrt{2E/k}$ is the amplitude A (Section 4.2), and $b = \sqrt{2E/m} = \sqrt{(2E/k)(k/m)} = A\omega$, multiplying and dividing by k under the root, is the largest speed. For the block with $E = 0.04 \text{ J}$ they are 4 cm and 0.4 m s^{-1} .

Figure 4.3(a) draws the ellipses for three energies. Every state lies on exactly one such ellipse, the one of its own energy, and the block goes round it for ever, because energy is conserved. Two paths can never cross: if they met at one state, that state would start two different motions, which Section 2.6 rules out. The direction of travel is clockwise: in the upper half of the plane $v > 0$, so x grows and the state moves to the right; in the lower half it moves to the left. The panel is drawn with 1 cm of position as long as 0.1 m s^{-1} of velocity, that is, with the velocity divided by $\omega = 10 \text{ rad s}^{-1}$, since $0.1 \text{ m s}^{-1} / 10 \text{ rad s}^{-1} = 0.01 \text{ m}$. On that scale the ellipses are circles, since $(x, v/\omega) = A(\cos \theta, -\sin \theta)$ by Section 4.1: the reference circle again, reflected in the x axis, which reverses the sign of the second coordinate and so turns the anticlockwise motion of P into a clockwise one.

Phase portraits return throughout the book. Chapter 7 uses momentum in place of velocity and studies the flow of whole regions of the phase plane, and Chapter 9 judges methods of computing motion by whether their paths close, as these do, or spiral outwards. Panel (b), the pendulum, belongs to Section 4.5; there θ is the angle of the pendulum, not $\omega t + \phi$. Its teal paths seem to cross at the upright position, but that state is a path of its own, a pendulum balanced at rest, which a pendulum on the teal paths approaches ever more slowly and never reaches, so no state starts two motions.

CHECK YOUR UNDERSTANDING. A block with $\omega = 10 \text{ rad s}^{-1}$ starts at $x = 0$ with $v = 0.3 \text{ m s}^{-1}$. What are the semi-axes of its ellipse in the phase plane, and where on the ellipse does it start?

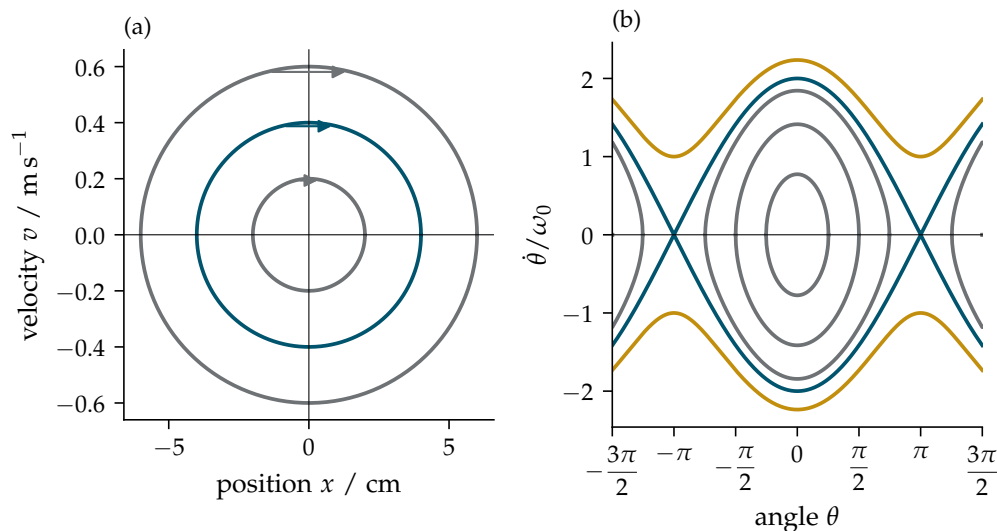


FIGURE 4.3: Phase portraits. (a) The block of Section 4.1 at the energies 0.01, 0.04 (teal) and 0.09 J, traversed clockwise; with 1 cm drawn as long as 0.1 m s^{-1} the ellipses are circles. (b) The pendulum of Section 4.5, with the rate of change of its angle, $\dot{\theta}$, in units of its small-swing angular frequency ω_0 : closed paths while it swings (grey), the path through the upright position (teal), and open paths when it goes over the top (ochre).

4.4 Small oscillations about any minimum

Wells are everywhere, springs far fewer: a pendulum hangs at the bottom of a well of gravitational energy, and an atom sits at the bottom of a well made by its neighbours. This section shows that small swings about a minimum with positive curvature are approximately those of a spring, and finds the stiffness of that spring for a pendulum, for the bead of Chapter 3 and for a lithium atom on the model surface.

The spring approximation near a minimum

Let $U(x)$ have a minimum at x_0 , with $U'(x_0) = 0$ and positive curvature, $U''(x_0) > 0$. This last condition matters: a flatter minimum, such as $U(x) = x^4$ at the origin, has no quadratic restoring term. The Taylor series of Section 1.8, with $x - x_0$ in the role of the interval, gives

$$\begin{aligned} U(x) &= U(x_0) + U'(x_0)(x - x_0) + \frac{1}{2}U''(x_0)(x - x_0)^2 + \mathcal{O}((x - x_0)^3) \\ &= U(x_0) + \frac{1}{2}k(x - x_0)^2 + \mathcal{O}((x - x_0)^3), \quad k = U''(x_0), \end{aligned}$$

since $U'(x_0) = 0$. Near its bottom such a well is approximately a parabola, the potential energy of a spring of stiffness $k = U''(x_0)$ stretched by $x - x_0$; the constant $U(x_0)$

changes no force. As long as the swing is small enough that the cubic and higher terms can be neglected, the motion is simple harmonic, with

$$\omega_0 = \sqrt{\frac{U''(x_0)}{m}}, \quad (4.4)$$

whatever the shape of the well further out. This is the result of [Section 3.6](#), now read from U itself rather than from the force, and the subscript 0 marks the angular frequency of small swings. How small is small enough, and what happens beyond, is the subject of [Section 4.5](#).

Motion along a curved path

A pendulum bob and a bead on a wire move along curves. Measuring the position by the distance s along the curve from its lowest point, the speed is $|ds/dt|$ and the energy is $E = \frac{1}{2}m\dot{s}^2 + U(s)$, where the dot denotes the rate of change with time, as in [Section 1.3](#). Energy is conserved, because the push of the rod or the wire is at right angles to the motion ([Section 3.5](#)), so differentiating E with the chain rule,

$$0 = \frac{dE}{dt} = m\dot{s}\ddot{s} + \frac{dU}{ds}\dot{s} = \dot{s}\left(m\ddot{s} + \frac{dU}{ds}\right).$$

Whenever the body is moving, $\dot{s} \neq 0$, and the bracket must vanish: $m\ddot{s} = -dU/ds$, and since both sides change smoothly this holds also at the instants when the body stops. Along its curve the body therefore moves as a body on a line with potential energy $U(s)$, and [Equation \(4.4\)](#) applies with s in place of x .

The pendulum

A pendulum's bob, a small heavy weight of mass m , is fixed to a light rigid rod of length L , light meaning that its own mass is negligible, which turns freely about a pivot ([Figure 4.4](#)). A rod, unlike a string, can push as well as pull, so the bob stays on its circle even above the pivot, as the large swings of [Section 4.5](#) need. When the rod makes an angle θ with the vertical, the bob has moved $s = L\theta$ along its arc, by the definition of the radian ([Section 1.5](#)), and it is $L \cos \theta$ below the pivot, so $L - L \cos \theta$ above its lowest point. Its potential energy is therefore

$$U = mgL(1 - \cos \theta) = mgL\left(1 - \cos \frac{s}{L}\right).$$

Differentiating twice with respect to s , by the chain rule,

$$\frac{dU}{ds} = mg \sin \frac{s}{L}, \quad \frac{d^2U}{ds^2} = \frac{mg}{L} \cos \frac{s}{L},$$

which is mg/L at $s = 0$, since $\cos 0 = 1$, so by Equation (4.4)

$$\omega_0 = \sqrt{\frac{g}{L}}.$$

The mass has cancelled, as in free fall. A pendulum 1.2 m long swings with $\omega_0 = \sqrt{9.80665/1.2} = 2.859 \text{ rad s}^{-1}$ and a period of $2\pi/\omega_0 = 2.198 \text{ s}$, for small swings.

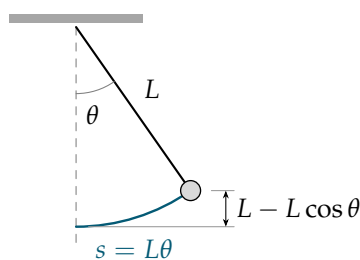


FIGURE 4.4: A pendulum, a bob on a light rod of length L , at the angle θ . The bob has moved $s = L\theta$ along its arc (teal) and has risen $L - L \cos \theta$ above its lowest point.

The bead in its valley

The bead of Section 3.5 has $U = mgh(x)$, but it moves along the wire, so we need d^2U/ds^2 at the bottom of a valley, not d^2U/dx^2 . By the chain rule, with x a function of s ,

$$\frac{dU}{ds} = \frac{dU}{dx} \frac{dx}{ds}, \quad \frac{d^2U}{ds^2} = \frac{d^2U}{dx^2} \left(\frac{dx}{ds} \right)^2 + \frac{dU}{dx} \frac{d^2x}{ds^2},$$

the second by the product rule. At the bottom of a valley the wire is level, so $dU/dx = 0$ and a short step along the wire is a step along x , $dx/ds = 1$. Hence $d^2U/ds^2 = mgh''(x_0)$ there, and $\omega_0 = \sqrt{gh''(x_0)}$. The deep valley lies at $x_0 = -1.473 \text{ m}$ (Section 3.6), where $h'' = 1.8 \times 1.473^2 - 1.2 = 2.7055 \text{ m}^{-1}$, so $\omega_0 = \sqrt{9.80665 \times 2.7055} = 5.151 \text{ rad s}^{-1}$, a period of $2\pi/5.151 = 1.220 \text{ s}$.

A lithium atom in a hollow

On the model surface of Section 3.13, take the line through a hollow and a bridge, the x axis, along which a hop runs. There $\mathbf{r} = (x, 0)$, and the three products in Equation (3.17)

are $\mathbf{g}_1 \cdot \mathbf{r} = |\mathbf{g}|x \cos -30^\circ = 2\pi x/a$, using $|\mathbf{g}| = 4\pi/(\sqrt{3}a)$ and $\cos -30^\circ = \sqrt{3}/2$; $\mathbf{g}_2 \cdot \mathbf{r} = 0$; and $\mathbf{g}_3 \cdot \mathbf{r} = -2\pi x/a$. Since the cosine is even and $\cos 0 = 1$,

$$U(x,0) = \frac{U_b}{4} \left[3 - 2 \cos \frac{2\pi x}{a} - 1 \right] = \frac{U_b}{2} \left(1 - \cos \frac{2\pi x}{a} \right), \quad (4.5)$$

a well of the same shape as the pendulum's. Differentiating twice by the chain rule, as for the pendulum with s/L replaced by $2\pi x/a$, gives $U'' = \frac{1}{2}U_b(2\pi/a)^2 \cos(2\pi x/a)$, and $\cos 0 = 1$, so

$$k = U''(0) = \frac{U_b}{2} \left(\frac{2\pi}{a} \right)^2 = \frac{2\pi^2 U_b}{a^2} = \frac{19.7392 \times 0.3}{6.0516} = 0.97854 \text{ eV } \text{\AA}^{-2}$$

with $U_b = 0.3 \text{ eV}$ and $a = 2.46 \text{ \AA}$. The force in $\text{eV } \text{\AA}^{-1}$ divided by the mass in amu becomes an acceleration in $\text{\AA fs}^{-2}$ on multiplying by 9.648533×10^{-3} (Section 2.10), so for lithium, $\omega_0 = \sqrt{0.97854 \times 9.648533 \times 10^{-3}/6.94} = 0.036884 \text{ rad fs}^{-1}$, and the period is $2\pi/\omega_0 = 170.3 \text{ fs}$. The model is illustrative, but the scale is the point: an atom rattles in its site a few times in a picosecond. Exercise 4.8 shows that the well has the same curvature along y as along x .

KEY IDEA. Near a minimum with $U''(x_0) > 0$, $U \approx U(x_0) + \frac{1}{2}U''(x_0)(x - x_0)^2$: small swings are approximately those of a spring of stiffness $U''(x_0)$, with angular frequency $\sqrt{U''(x_0)/m}$.

CHECK YOUR UNDERSTANDING. A pendulum is made four times as long. What happens to the period of its small swings? Does the mass of the bob matter?

4.5 Larger swings: anharmonic wells

Real wells are not parabolas. The pendulum's well rises ever more slowly than its parabola as the bob swings towards the horizontal, and the well that holds two atoms together rises steeply when they are pushed together but levels off when they are pulled apart. A well that is not a parabola is called anharmonic. This section computes the period of a swing of any size from the energy alone, and finds that in every well of interest here it grows with the energy.

The period from the energy

Section 3.5 noted that energy fixes the time of a journey through the speed: a short step dx taken at speed v lasts dx/v . In a well, with total energy E and turning points $x_1 < x_2$, the speed at x is $v(x) = \sqrt{2(E - U(x))/m}$, so the journey from x_1 to x_2 takes

$\int_{x_1}^{x_2} dx/v(x)$. The way back takes as long, since the speed at each point is the same in either direction, and the period is

$$\mathcal{T}(E) = 2 \int_{x_1}^{x_2} \frac{dx}{\sqrt{2(E - U(x))/m}}. \quad (4.6)$$

At the turning points the speed is zero and the integrand is infinite. The integral then means the limit as its ends approach the turning points, and it is finite when, as in the harmonic case below, the function whose derivative is the integrand tends to finite values there. To evaluate it we need the derivative of the arcsine.

THE DERIVATIVE OF AN INVERSE FUNCTION. For u between -1 and 1 , $\theta = \arcsin u$ is the angle between $-\pi/2$ and $\pi/2$ whose sine is u (Section 1.5), so $\sin \theta(u) = u$ for every such u . Differentiating both sides with respect to u , the left by the chain rule,

$$\cos \theta \frac{d\theta}{du} = 1, \quad \text{so} \quad \frac{d}{du} \arcsin u = \frac{1}{\cos \theta} = \frac{1}{\sqrt{1 - u^2}},$$

since $\cos \theta = +\sqrt{1 - \sin^2 \theta}$ for an angle between $-\pi/2$ and $\pi/2$, where the cosine is not negative. With $u = x/A$ and the chain rule, $du/dx = 1/A$, so

$$\frac{d}{dx} \arcsin \frac{x}{A} = \frac{1}{A} \frac{1}{\sqrt{1 - x^2/A^2}} = \frac{1}{\sqrt{A^2(1 - x^2/A^2)}} = \frac{1}{\sqrt{A^2 - x^2}},$$

the positive A having gone inside the root as A^2 . The same method, differentiating the equation that defines the inverse, gives the derivative of any inverse function.

For the spring, $U = \frac{1}{2}kx^2$ and $E = \frac{1}{2}kA^2$, so $E - U = \frac{1}{2}k(A^2 - x^2)$ and $v = \omega\sqrt{A^2 - x^2}$. Then

$$\begin{aligned} \mathcal{T} &= \frac{2}{\omega} \int_{-A}^A \frac{dx}{\sqrt{A^2 - x^2}} = \frac{2}{\omega} \left[\arcsin \frac{x}{A} \right]_{-A}^A \quad \text{by the toolbox and Equation (1.3)} \\ &= \frac{2}{\omega} \left(\frac{\pi}{2} - \left(-\frac{\pi}{2} \right) \right) = \frac{2\pi}{\omega}, \end{aligned}$$

the period of Section 4.1, whatever the amplitude. The integrand is infinite at $\pm A$, yet $\arcsin(x/A)$ rises only to $\pm\pi/2$ there, so the area under it is finite: the spike is tall but very narrow. For other wells the integral is evaluated by computer. The function `oscillators.period` of `mdlab` finds the turning points, then writes $x = c - \eta \cos \vartheta$, with c the centre and η the half-width of the allowed region, so that ϑ runs from 0 at x_1 to π at x_2 . A short step $d\vartheta$ moves x by $dx = \eta \sin \vartheta d\vartheta$, the derivative of $c - \eta \cos \vartheta$ times the step, so each time dx/v becomes $\eta \sin \vartheta d\vartheta/v$, and these are added up over ϑ from 0 to π . Near x_1 , where $U(x_1) = E$, the Taylor series gives $E - U(x) \approx -U'(x_1)(x - x_1)$, and $x - x_1 = \eta(1 - \cos \vartheta) \approx \frac{1}{2}\eta\vartheta^2$, so v grows in proportion to ϑ ; so does $\sin \vartheta \approx \vartheta$,

and their ratio, the new integrand, stays finite, as it does near x_2 . The integral over ϑ is then evaluated by Gauss–Legendre quadrature, a weighted sum of the integrand at chosen points, which converges quickly for a smooth integrand. Its tests check it against the spring, against closed forms for the pendulum and for the Morse well below, and against motions timed by `solve_newton`.

The pendulum and the hop

The pendulum's well, $U = mgL(1 - \cos \theta)$, and the hop coordinate of the lithium atom, Equation (4.5), have the same shape,

$$U = \frac{1}{2}D \left(1 - \cos \frac{2\pi x}{\Lambda} \right),$$

a cosine well of depth D that repeats every Λ : for the pendulum x is the distance $s = L\theta$ along the arc, $\Lambda = 2\pi L$ and $D = 2mgL$, the potential energy of the bob held upright; for the atom $\Lambda = a$ and $D = U_b$, the barrier. Away from its bottom the cosine well rises more slowly than its parabola (Figure 4.5a), so the restoring force there is weaker than a spring's, and in a swing of the same amplitude the body moves more slowly and takes longer. Equation (4.6) gives, for the pendulum,

amplitude	10°	45°	90°	150°	179°
period / small-swing period	1.0019	1.0400	1.1803	1.7622	3.9011

For swings of 10° the period is within 0.2 % of $2\pi\sqrt{L/g}$, and it changes only slowly with the amplitude: a clock whose swing grows from 10° to 11° loses about 35 s a day, which is why clockmakers keep the swing small and steady. Beyond 90° the bob ends each swing above the pivot, where only a rod, which can push, keeps it on its circle; a string would go slack. As the amplitude approaches 180° the bob lingers ever longer near the top, where the force on it is nearly zero, and the period grows without limit (Figure 4.6).

The lithium atom behaves the same way. With 0.25 eV above the bottom of the hollow, below the 0.3 eV barrier, it rattles in its hollow with a period of 253.8 fs, half as long again as the 170.3 fs of small swings, and with 0.297 eV the period is 400.8 fs. With exactly the barrier as energy, the atom would creep ever more slowly towards the bridge without ever reaching it. The phase portrait of the cosine well (Figure 4.3b) shows all of this at once: closed paths for swings or rattles, the path through the top that separates them from the open paths of a pendulum going over the top or an atom hopping on from hollow to hollow.

Two wells for a pair of atoms

The energy of two atoms depends on the distance r between them, and two standard shapes model it; Chapter 8 discusses where they apply. The Morse well, used for a chemical bond, is

$$U_M(r) = D (1 - e^{-a(r-r_0)})^2,$$

with three constants. At $r = r_0$ the exponential is 1 and $U_M = 0$, its least value, since it is a square; when the atoms are pulled far apart the exponential vanishes and U_M levels off at D , the energy needed to separate them; when they are pushed together, $r < r_0$, the exponential grows rapidly and so does U_M . The constant a , an inverse length, sets the width of the well; here it is not the spacing of the surface. Near the bottom, with $y = r - r_0$ and the exponential series of [Section 2.7](#), $e^{-ay} = 1 - ay + \mathcal{O}(y^2)$, so $1 - e^{-ay} = ay + \mathcal{O}(y^2)$ and, squaring,

$$U_M = Da^2y^2 + \mathcal{O}(y^3), \quad \text{a spring of stiffness } k = 2Da^2.$$

The Lennard-Jones well, often used for atoms that form no bond, such as those of the noble gases (helium, neon, argon and their family, which hardly react), is

$$U_{LJ}(r) = 4\epsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right],$$

a steep repulsion, r^{-12} , that dominates at short range and a weaker attraction, $-r^{-6}$, that dominates at long range. The power rule of [Section 1.2](#) was found for whole $n \geq 1$, but it holds for $r^{-n} = 1/r^n$ too: differentiating $r^{-n} r^n = 1$ by the product rule gives $r^n d(r^{-n})/dr + r^{-n} nr^{n-1} = 0$, so $d(r^{-n})/dr = -n r^{-n-1}$. Differentiating with this rule,

$$\frac{dU_{LJ}}{dr} = 4\epsilon \left[-12 \frac{\sigma^{12}}{r^{13}} + 6 \frac{\sigma^6}{r^7} \right] = \frac{24\epsilon}{r} \left[\left(\frac{\sigma}{r} \right)^6 - 2 \left(\frac{\sigma}{r} \right)^{12} \right],$$

which vanishes when the bracket, $(\sigma/r)^6 [1 - 2(\sigma/r)^6]$, does, so when $(\sigma/r)^6 = \frac{1}{2}$, that is $(r/\sigma)^6 = 2$, at $r_{\min} = 2^{1/6}\sigma = 1.1225\sigma$, where $U_{LJ} = 4\epsilon(\frac{1}{4} - \frac{1}{2}) = -\epsilon$. Differentiating once more,

$$\frac{d^2U_{LJ}}{dr^2} = \frac{4\epsilon}{r^2} \left[156 \left(\frac{\sigma}{r} \right)^{12} - 42 \left(\frac{\sigma}{r} \right)^6 \right] = \frac{4\epsilon}{r_{\min}^2} \left[\frac{156}{4} - \frac{42}{2} \right] = \frac{4 \times 18\epsilon}{2^{1/3}\sigma^2} = 57.15 \frac{\epsilon}{\sigma^2}$$

at the minimum, the middle steps putting in $(\sigma/r)^{12} = \frac{1}{4}$, $(\sigma/r)^6 = \frac{1}{2}$ and $r_{\min}^2 = 2^{2/6}\sigma^2 = 2^{1/3}\sigma^2$, with $39 - 21 = 18$.

Both wells are lopsided ([Figure 4.5b, c](#)): steeper than their parabola on the inside, softer on the outside. A swing therefore reaches less far in and much further out than a

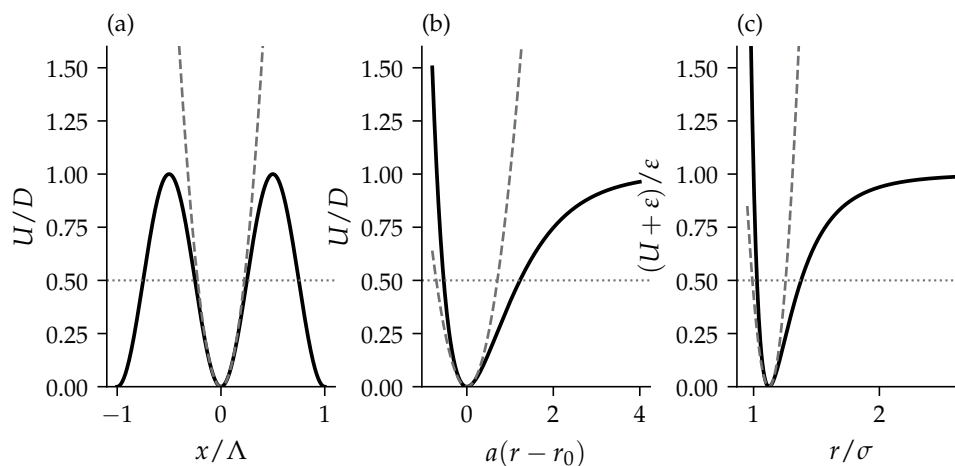


FIGURE 4.5: Three anharmonic wells (black) against the parabolas of the same curvature at their bottoms (dashed), each in its own natural units; the dotted lines mark half the depth. (a) The cosine well of the pendulum and of the lithium hop. (b) The Morse well of a bond. (c) The Lennard-Jones well, drawn from its bottom.

spring's would; in the Lennard-Jones well with half the depth as energy, the turning points are at 1.027σ and 1.377σ , against 0.990σ and 1.255σ for the parabola, whose turning points lie $\sqrt{2 \times 0.5/57.15} = 0.132\sigma$ either side of the minimum at 1.1225σ . The soft outer side wins, and the period grows with the energy (Figure 4.6). Measured from the bottom, at a tenth, half and nine tenths of the depth the period is 1.054, 1.414 and 3.162 times that of small swings for the Morse well, which are $1/\sqrt{1 - E/D}$ exactly, the closed form the tests use, and 1.069, 1.554 and 4.399 times for the Lennard-Jones well. These ratios hold for any constants and any mass, because in the units of the figures the wells have no constants left, and by Equation (4.6) every period is proportional to $\sqrt{m}$. A bond with more energy therefore vibrates more slowly, and the period of the small swings, the one that sets the steps of a simulation, is the shortest.

NOTEBOOK 04 explore the period in each well as the energy grows

CHECK YOUR UNDERSTANDING. Why does a pendulum swinging out to 90° on each side take longer for each swing than one swinging out to 10° , even though it moves faster at the bottom?

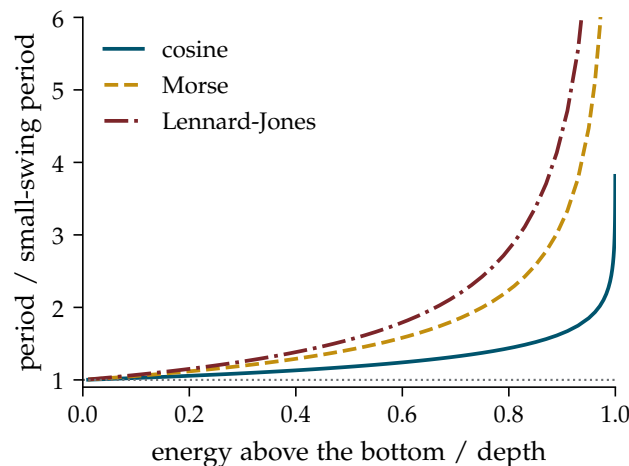


FIGURE 4.6: The period of the motion in each well of Figure 4.5, divided by the period of small swings, against the energy above the bottom as a fraction of the depth, computed from Equation (4.6). The cosine well's period grows without limit as the energy approaches the barrier, though so slowly that it is only 3.8 at 0.9999 of the barrier, where its curve stops; the others' grows without limit as the energy approaches that needed to separate the atoms. A spring keeps the ratio 1 (dotted).

4.6 Damping

A block on a spring in air swings for a long time; the same block in honey creeps back to its resting place without swinging at all. This section adds the drag of Section 2.7 to the spring and solves the motion exactly, with one substitution that removes the drag from the equation.

The equation and a substitution

With the drag force $-bv$ of Section 2.7 added to the spring's $-kx$, the second law reads $m\ddot{x} = -kx - b\dot{x}$, and dividing by m ,

$$\ddot{x} = -\omega_0^2 x - \gamma \dot{x}, \quad \omega_0 = \sqrt{\frac{k}{m}}, \quad \gamma = \frac{b}{m}, \quad (4.7)$$

where ω_0 is the angular frequency the block would have without drag and γ is the drag rate. The drag removes energy, so we expect the swing to shrink, and we write

$$x(t) = e^{-\gamma t/2} y(t),$$

which separates a shrinking factor from a new unknown y . The factor $\frac{1}{2}\gamma$ is chosen because it makes the drag term disappear, as we now show. By the product rule and the chain rule, with $de^{-\gamma t/2}/dt = -\frac{1}{2}\gamma e^{-\gamma t/2}$ by the exponential toolbox of Section 2.7,

and for $\ddot{x}$ the product rule applied to $\dot{x}$, the last step multiplying out the second bracket and collecting the two terms in $\dot{y}$,

$$\begin{aligned}\dot{x} &= e^{-\gamma t/2}(\dot{y} - \tfrac{1}{2}\gamma y), \\ \ddot{x} &= e^{-\gamma t/2}(\ddot{y} - \tfrac{1}{2}\gamma \dot{y}) - \tfrac{1}{2}\gamma e^{-\gamma t/2}(\dot{y} - \tfrac{1}{2}\gamma y) = e^{-\gamma t/2}(\ddot{y} - \gamma \dot{y} + \tfrac{1}{4}\gamma^2 y).\end{aligned}$$

Putting these into Equation (4.7), dividing by $e^{-\gamma t/2}$, which is never zero, and multiplying out the bracket on the right,

$$\begin{aligned}\ddot{y} - \gamma \dot{y} + \tfrac{1}{4}\gamma^2 y &= -\omega_0^2 y - \gamma \dot{y} + \tfrac{1}{2}\gamma^2 y \\ \ddot{y} &= -\omega_0^2 y + \tfrac{1}{2}\gamma^2 y - \tfrac{1}{4}\gamma^2 y \quad \text{adding } \gamma \dot{y} - \tfrac{1}{4}\gamma^2 y \text{ to both sides} \\ &= -(\omega_0^2 - \tfrac{1}{4}\gamma^2) y.\end{aligned}$$

The terms in $\dot{y}$ have cancelled: y obeys an equation without drag, whose solution depends on the sign of $\omega_0^2 - \frac{1}{4}\gamma^2$.

Three kinds of motion

Light drag, $\frac{1}{2}\gamma < \omega_0$. Then $\omega_0^2 - \frac{1}{4}\gamma^2$ is positive, and y is a spring with angular frequency

$$\omega_d = \sqrt{\omega_0^2 - \tfrac{1}{4}\gamma^2},$$

so $y = c_1 \cos \omega_d t + c_2 \sin \omega_d t$, as in Section 2.8, and

$$x(t) = e^{-\gamma t/2}(c_1 \cos \omega_d t + c_2 \sin \omega_d t) :$$

an oscillation whose swing shrinks by the factor $e^{-\gamma t/2}$. Setting $t = 0$ gives $c_1 = x_0$, and the formula for $\dot{x}$ above gives $v_0 = \dot{y}(0) - \frac{1}{2}\gamma x_0 = c_2 \omega_d - \frac{1}{2}\gamma x_0$, so $c_2 = (v_0 + \frac{1}{2}\gamma x_0)/\omega_d$. The block of Section 4.1, with $\omega_0 = 10 \text{ rad s}^{-1}$, in a liquid with $\gamma = 4 \text{ s}^{-1}$, swings with $\omega_d = \sqrt{100 - 4} = 9.798 \text{ rad s}^{-1}$, a period of $2\pi/9.798 = 0.641 \text{ s}$, a little longer than the $2\pi/10 = 0.628 \text{ s}$ without drag. Its swing halves when $e^{-\gamma t/2} = 0.5$; taking the natural logarithm of both sides, $-\frac{1}{2}\gamma t = \ln 0.5 = -0.693$, so the swing halves every $0.693/(\frac{1}{2}\gamma) = 0.347 \text{ s}$. This case is called under-damped.

Heavy drag, $\frac{1}{2}\gamma > \omega_0$. Then $\ddot{y} = \kappa^2 y$ with $\kappa = \sqrt{\frac{1}{4}\gamma^2 - \omega_0^2}$. By the exponential toolbox, $e^{\kappa t}$ has second derivative $\kappa^2 e^{\kappa t}$, and so has $e^{-\kappa t}$, so $y = c_1 e^{\kappa t} + c_2 e^{-\kappa t}$ and

$$x(t) = c_1 e^{-(\gamma/2 - \kappa)t} + c_2 e^{-(\gamma/2 + \kappa)t}.$$

Since $\kappa^2 = \frac{1}{4}\gamma^2 - \omega_0^2$ is less than $\frac{1}{4}\gamma^2$, $\kappa < \frac{1}{2}\gamma$, so both exponents are negative and both terms die away. Multiplying $x = 0$ by $e^{(\gamma/2+\kappa)t}$ gives $c_1 e^{2\kappa t} = -c_2$, which holds at one instant at most, since $e^{2\kappa t}$ never takes a value twice: the block passes its resting place at most once and never swings to and fro, but creeps back. This case is called over-damped. With $\gamma = 40 \text{ s}^{-1}$, $\kappa = \sqrt{400 - 100} = 17.32 \text{ s}^{-1}$, and the slower term decays at only $20 - 17.32 = 2.68 \text{ s}^{-1}$, so the creep is slow.

Critical drag, $\frac{1}{2}\gamma = \omega_0$. Then $\ddot{y} = 0$, so $y = c_1 + c_2 t$ by two integrations, and $x = (c_1 + c_2 t) e^{-\gamma t/2}$. Since $c_1 + c_2 t$ is zero at one instant at most, this critically damped motion passes the resting place at most once, and not at all when released from rest, where $c_1 = x_0$ and, by the formula for $\dot{x}$, $c_2 = \frac{1}{2}\gamma x_0$, of the same sign.

For the block released from rest at $x_0 = 0.04 \text{ m}$, critical drag, $\gamma = 20 \text{ s}^{-1}$, gives $c_2 = 10 \times 0.04 = 0.4 \text{ m s}^{-1}$, and half a second after release the block is $(0.04 + 0.4 \times 0.5) e^{-5} = 0.16 \text{ cm}$ out. With $\gamma = 40 \text{ s}^{-1}$, $y(0) = c_1 + c_2 = 0.04$ and, since $\dot{x}(0) = \dot{y}(0) - \frac{1}{2}\gamma x_0 = 0$, $\dot{y}(0) = \kappa(c_1 - c_2) = \frac{1}{2}\gamma x_0 = 0.8$; these give $c_1 = 0.0431 \text{ m}$ and $c_2 = -0.0031 \text{ m}$, and at half a second $x = 0.0431 e^{-2.68 \times 0.5} = 1.13 \text{ cm}$, the second term being negligible. The critically damped block returns sooner (Figure 4.7).

In each case two constants match any starting position and velocity, so by the uniqueness of solutions (Section 2.6) these formulas are the motion. The energy falls throughout, at the rate $-bv^2$ of Section 3.7.

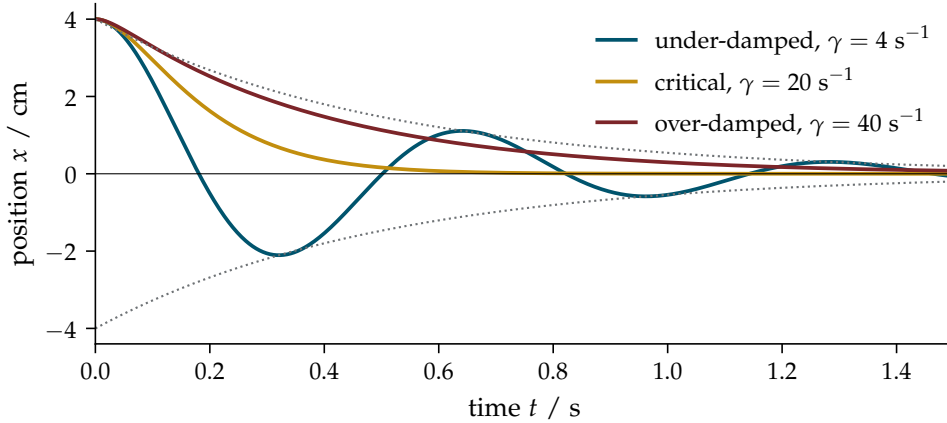


FIGURE 4.7: The block of Section 4.1, with $\omega_0 = 10 \text{ rad s}^{-1}$, released from rest 4 cm out in liquids with three drag rates. The under-damped block turns back on the dotted curves $\pm 4 \text{ cm} \times e^{-\gamma t/2}$; the critically damped one returns without swinging past zero; the over-damped one creeps back more slowly.

Chapter 13 adds this drag to the motion of atoms, together with random kicks that make up for the energy it removes, to hold a simulation at a chosen temperature; the drag rate γ there plays the same role as here.

CHECK YOUR UNDERSTANDING. For the block with $\omega_0 = 10 \text{ rad s}^{-1}$, what drag rate makes the motion critically damped? With $\gamma = 8 \text{ s}^{-1}$, what is the period of the swing?

4.7 Driving and resonance

A child on a swing, pushed gently at just the right moment of each swing, soon swings high; the same pushes at the wrong rhythm achieve little. This section drives the damped oscillator with a force that rises and falls as a sinusoid and finds how the size of the response depends on the rhythm of the push.

The steady response

Add to the block of [Section 4.6](#) a force $F_0 \cos \omega t$, where ω is now the angular frequency of the push, chosen freely, and ω_0 remains that of the spring alone. The second law becomes $m\ddot{x} = -kx - b\dot{x} + F_0 \cos \omega t$; dividing by m , writing $f = F_0/m$ and moving the spring and drag terms to the left,

$$\ddot{x} + \gamma\dot{x} + \omega_0^2 x = f \cos \omega t. \quad (4.8)$$

After the start has been forgotten, we expect the block to swing in step with the push, at the same angular frequency but lagging behind it, so we try

$$x = A \cos(\omega t - \delta),$$

with an amplitude A and a lag δ to be found. Writing $\psi = \omega t - \delta$ for short, the chain rule with $d\psi/dt = \omega$ gives $\dot{x} = -A\omega \sin \psi$ and $\ddot{x} = -A\omega^2 \cos \psi$, so the left side of [Equation \(4.8\)](#) is

$$A(\omega_0^2 - \omega^2) \cos \psi - A\gamma\omega \sin \psi.$$

On the right, $\omega t = \psi + \delta$, and the addition formula of [Section 1.5](#) gives $f \cos \omega t = f \cos \delta \cos \psi - f \sin \delta \sin \psi$. Both sides are now combinations of $\cos \psi$ and $\sin \psi$, and they must agree at every instant. At the instants when ψ is a whole number of turns, $\cos \psi = 1$ and $\sin \psi = 0$; at those a quarter turn later, $\cos \psi = 0$ and $\sin \psi = 1$. Comparing the two sides at these instants,

$$A(\omega_0^2 - \omega^2) = f \cos \delta, \quad A\gamma\omega = f \sin \delta.$$

Squaring both, adding them and using $\cos^2 \delta + \sin^2 \delta = 1$ gives $A^2[(\omega_0^2 - \omega^2)^2 + \gamma^2 \omega^2] = f^2$, and dividing the second equation by the first gives $\tan \delta$:

$$A = \frac{f}{\sqrt{(\omega_0^2 - \omega^2)^2 + \gamma^2 \omega^2}}, \quad \tan \delta = \frac{\gamma \omega}{\omega_0^2 - \omega^2}. \quad (4.9)$$

Since $\sin \delta = A\gamma\omega/f$ is positive, δ lies between 0 and π : the block always lags behind the push. Below resonance, where $\omega < \omega_0$, $\cos \delta$ is positive too, and $\delta = \arctan(\gamma\omega/(\omega_0^2 - \omega^2))$ lies between 0 and $\pi/2$, inside the range of the arctangent. Above it $\cos \delta$ is negative and δ lies between $\pi/2$ and π ; since $\pi - \delta$ has the same sine as δ and the opposite cosine (Section 1.5), $\tan(\pi - \delta) = -\tan \delta$, and $\delta = \pi - \arctan(\gamma\omega/(\omega_0^2 - \omega^2))$.

This is one motion that obeys Equation (4.8), but it does not start from an arbitrary position and velocity. Adding to it any motion x_h of the undriven damped block, for which $\ddot{x}_h + \gamma\dot{x}_h + \omega_0^2 x_h = 0$, gives another, since the derivative of a sum is the sum of the derivatives: putting $x + x_h$ into the left side of Equation (4.8) gives the left side for x , which is $f \cos \omega t$, plus that for x_h , which is zero. The two constants of x_h (Section 4.6) can match any starting state, so by the uniqueness of solutions (Section 2.6) every driven motion is the steady response plus a damped motion, and the damped part dies away, as $e^{-\gamma t/2}$ for light drag. Combining solutions by adding them works only because Equation (4.8) contains x and its derivatives to the first power alone; such an equation is called linear.

Resonance

The amplitude is largest when the denominator of Equation (4.9) is smallest, which, for light drag, is close to $\omega = \omega_0$, where the first term vanishes. Driving a system at its own angular frequency is called driving it at resonance. There $\omega_0^2 - \omega^2 = 0$, so the first of the two equations above gives $\cos \delta = 0$; with $\sin \delta$ positive, $\delta = \pi/2$ and $\sin \delta = 1$, and the second gives $A\gamma\omega_0 = f$:

$$A = \frac{f}{\gamma\omega_0}, \quad \delta = \frac{\pi}{2}.$$

Push the block of Section 4.1, with $m = 0.5 \text{ kg}$, $k = 50 \text{ N m}^{-1}$ and so $\omega_0 = 10 \text{ rad s}^{-1}$, with a force of amplitude 0.5 N , so that $f = 0.5/0.5 = 1 \text{ m s}^{-2}$. Held still, that force would stretch the spring by $F_0/k = 0.5/50 = 1 \text{ cm}$, which is also the amplitude for a very slow push, $\omega \rightarrow 0$, where $A \rightarrow f/\omega_0^2 = (F_0/m)/(k/m) = F_0/k$. At resonance, with $\gamma = 2 \text{ s}^{-1}$, the amplitude is $1/(2 \times 10) = 5 \text{ cm}$, five times larger; with $\gamma = 1 \text{ s}^{-1}$ it is 10 cm , and with $\gamma = 4 \text{ s}^{-1}$, 2.5 cm (Figure 4.8a). The ratio of the amplitude at resonance to that for a very slow push, $(f/(\gamma\omega_0))/(f/\omega_0^2) = \omega_0/\gamma$, is called the

quality factor of the oscillator: the lighter the drag, the higher and narrower the peak. The peak's width, measured where A exceeds its largest value divided by $\sqrt{2}$, is close to γ . Near the peak, $\omega_0^2 - \omega^2 = (\omega_0 - \omega)(\omega_0 + \omega) \approx 2\omega_0(\omega_0 - \omega)$ and $\gamma\omega \approx \gamma\omega_0$, so the square root in Equation (4.9) is about $\omega_0\sqrt{4(\omega_0 - \omega)^2 + \gamma^2}$, which is $\sqrt{2}$ times its value at ω_0 when $4(\omega_0 - \omega)^2 = \gamma^2$, at $\omega = \omega_0 \pm \frac{1}{2}\gamma$, a distance γ apart. Solved exactly by computer, the widths are 1.00, 2.02 and 4.19 rad s^{-1} for $\gamma = 1, 2$ and 4 s^{-1} .

The lag explains why the rhythm matters (Figure 4.8b). At resonance $\delta = \pi/2$: the block is at the centre of its swing, moving fastest, when the push is strongest, so the push always acts along the velocity and its power Fv is never negative. Pushed slowly, the block follows the push, $\delta \rightarrow 0$; pushed fast, it moves against it, $\delta \rightarrow \pi$, and in both cases the push spends half of each cycle working against the motion.

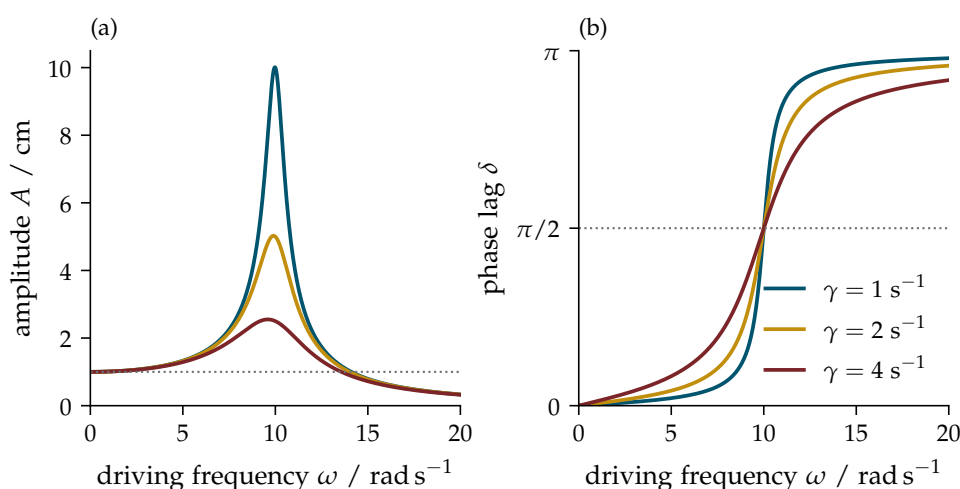


FIGURE 4.8: The block of Section 4.1 driven by a force of amplitude 0.5 N at angular frequency ω , for three drag rates. (a) The steady amplitude; the dotted line is the 1 cm by which the same force would stretch the spring if held still. (b) The lag of the motion behind the push, which passes through $\pi/2$ at resonance.

Resonance returns several times. Molecules absorb infrared light at the frequencies of many of their vibrations (Section 4.10); the width of a resonance reflects the damping, which Chapter 16 meets in the spectra of atomic motion; and Chapter 11 shows that a simulation that updates some forces only every few steps pushes the atoms in a rhythm that can hit a resonance.

NOTEBOOK 04 *drive the block at any rhythm and watch the steady state emerge*

CHECK YOUR UNDERSTANDING. For the block with $\gamma = 0.5 \text{ s}^{-1}$ and $f = 1 \text{ m s}^{-2}$, what is the amplitude when it is pushed at $\omega = 10 \text{ rad s}^{-1}$, and what is it for a very slow push?

4.8 Two bodies and the reduced mass

So far one end of every spring has been fixed to a wall. Two atoms joined by a bond, or the two carts of Chapter 2, have no wall: both ends move. This section shows that their motion splits into a steady drift of the pair and an oscillation of the distance between them, with a mass smaller than either.

The relative motion

Two bodies of masses m_1 and m_2 , at positions $x_1 < x_2$ on a line, are joined by a spring of stiffness k and natural length ℓ . Its stretch is $x_2 - x_1 - \ell$, so it pulls the first body forwards and the second backwards with forces of size $k(x_2 - x_1 - \ell)$:

$$m_1 \ddot{x}_1 = +k(x_2 - x_1 - \ell), \quad m_2 \ddot{x}_2 = -k(x_2 - x_1 - \ell).$$

No external force acts along the line, so by [Section 2.9](#) the centre of mass moves at a constant velocity. For the separation $r = x_2 - x_1$, divide each equation by its mass and subtract the first from the second:

$$\ddot{r} = \ddot{x}_2 - \ddot{x}_1 = -\frac{k}{m_2}(r - \ell) - \frac{k}{m_1}(r - \ell) = -k\left(\frac{1}{m_1} + \frac{1}{m_2}\right)(r - \ell).$$

Defining the reduced mass m_r by $1/m_r = 1/m_1 + 1/m_2$, or, putting the fractions over a common denominator and inverting,

$$m_r = \frac{m_1 m_2}{m_1 + m_2}, \tag{4.10}$$

the separation obeys $m_r \ddot{r} = -k(r - \ell)$. Writing $s = r - \ell$ for the stretch, $\ddot{s} = \ddot{r}$ since ℓ is constant, so $m_r \ddot{s} = -ks$: the stretch oscillates about zero like a single body of mass m_r on a spring fixed to a wall ([Section 2.8](#)), with $\omega = \sqrt{k/m_r}$.

The carts of [Section 2.9](#), of 1 kg and 3 kg, have $m_r = 3/4 = 0.75$ kg, and with the spring of 200 N m^{-1} the separation would oscillate with $\omega = \sqrt{200/0.75} = 16.33 \text{ rad s}^{-1}$. Released at rest with the spring compressed, the separation starts at a turning point and reaches the natural length a quarter of a period later, at $\frac{1}{4} \times 2\pi/\omega = 0.0962 \text{ s}$. This is the moment, found in [Section 2.9](#) by computer, at which the spring fell away.

Light partners do the moving

The reduced mass is smaller than either mass, since $m_r = m_1 \times m_2 / (m_1 + m_2)$ with $m_2 / (m_1 + m_2) < 1$, and likewise $m_r = m_2 \times m_1 / (m_1 + m_2)$ with $m_1 / (m_1 + m_2) < 1$.

When one body is much heavier, it is close to the lighter mass: for an oxygen atom and a hydrogen atom,

$$m_r = \frac{1.008 \times 15.999}{1.008 + 15.999} = 0.9483 \text{ amu},$$

94 % of the hydrogen mass. For a molecule that is not drifting, the centre of mass stays still, so $m_H x_H + m_O x_O$ keeps its value and any changes obey $m_H \Delta x_H = -m_O \Delta x_O$: the hydrogen moves $15.999/1.008 = 15.9$ times as far as the oxygen, the other way: the vibration of an O–H bond is mostly the hydrogen atom moving against an almost still partner. For the same spring, a smaller mass means a higher frequency, $\omega = \sqrt{k/m_r}$, which is why bonds to hydrogen vibrate fastest (Section 4.10). For two atoms in a Morse or Lennard-Jones well (Section 4.5), the same argument applies with r the distance between them and m_r in place of m , for motion along their line of centres without rotation.

CHECK YOUR UNDERSTANDING. Two equal masses m are joined by a spring of stiffness k . What is the reduced mass, and with what angular frequency does the separation oscillate?

4.9 Coupled oscillators and normal modes

For small displacements from equilibrium, the forces in a molecule or crystal can be approximated by coupled springs. When one atom moves, its neighbours feel it, so the motion of one atom cannot be solved on its own. Within this harmonic approximation the system has simple collective motions, its normal modes, in which the atoms oscillate at a common angular frequency. Their sum describes a general motion of the coupled system. Finding the modes needs two new tools, matrices and their eigenvalues.

Two coupled carts

Two carts of mass m stand on a smooth track between two walls, each joined to its wall by a spring of stiffness k , and to each other by a third such spring; all three springs are at their natural lengths when the carts rest. Let x_1 and x_2 be the carts' displacements from rest. The left spring is stretched by x_1 and pulls cart 1 with $-kx_1$; the middle spring is stretched by $x_2 - x_1$ and pulls cart 1 forwards with $k(x_2 - x_1)$ and cart 2 backwards with $-k(x_2 - x_1)$; the right spring is squashed by x_2 and pushes cart 2 with $-kx_2$. Adding the forces on each cart,

$$m\ddot{x}_1 = -2kx_1 + kx_2, \quad m\ddot{x}_2 = kx_1 - 2kx_2. \quad (4.11)$$

Each equation contains both displacements: the carts are coupled. Started by pulling one cart aside, the swing passes back and forth between them (Figure 4.9c), and neither moves as a single spring. A table of numbers organises such equations.

MATRICES. A matrix is a rectangular table of numbers. A square matrix $\mathbf{A}$ with entries A_{ab} , in row a and column b , acts on a vector $\mathbf{u} = (u_1, \dots, u_n)$ to give the vector $\mathbf{A}\mathbf{u}$ whose a th component is the scalar product of row a with $\mathbf{u}$,

$$(\mathbf{A}\mathbf{u})_a = \sum_{b=1}^n A_{ab} u_b; \quad \text{for example} \quad \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} 2u_1 - u_2 \\ -u_1 + 2u_2 \end{pmatrix}.$$

This product is linear: $\mathbf{A}(\mathbf{u} + \mathbf{w}) = \mathbf{A}\mathbf{u} + \mathbf{A}\mathbf{w}$ and $\mathbf{A}(c\mathbf{u}) = c\mathbf{A}\mathbf{u}$ for a number c , since each component is a sum of products. A number times a matrix multiplies every entry, so $(c\mathbf{A})\mathbf{u} = c(\mathbf{A}\mathbf{u})$. A diagonal matrix $\mathbf{B}$ has zeros everywhere off its diagonal, the entries B_{aa} from top left to bottom right, so it multiplies each component by its own entry, $(\mathbf{B}\mathbf{u})_a = B_{aa}u_a$. A matrix with $A_{ab} = A_{ba}$ for every pair is symmetric, and the identity matrix $\mathbf{I}$, with ones on the diagonal and zeros elsewhere, leaves every vector unchanged, $\mathbf{I}\mathbf{u} = \mathbf{u}$.

With the displacements as a vector $\mathbf{x} = (x_1, x_2)$, Equation (4.11) reads

$$m\ddot{\mathbf{x}} = -\Phi\mathbf{x}, \quad \Phi = k \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}.$$

The matrix Φ , called the force-constant matrix, is the table of second derivatives of the energy stored in the springs, $U = \frac{1}{2}kx_1^2 + \frac{1}{2}k(x_2 - x_1)^2 + \frac{1}{2}kx_2^2$. By the chain rule, $\partial U/\partial x_1 = kx_1 - k(x_2 - x_1) = 2kx_1 - kx_2$, minus the force on cart 1, so $\partial^2 U/\partial x_1^2 = 2k$ and $\partial^2 U/\partial x_2 \partial x_1 = -k$, the first row of Φ ; the second row follows in the same way. Such a table of all the second derivatives of an energy is called its Hessian, the table promised in Section 3.11, and the equality of mixed partial derivatives proved there makes it symmetric.

Normal modes

Look for a motion in which both carts swing as sinusoids with the same angular frequency and phase, $\mathbf{x} = \mathbf{u} \cos(\omega t + \phi)$, with $\mathbf{u}$ a fixed vector, the pattern of the motion. Then $\ddot{\mathbf{x}} = -\omega^2 \mathbf{x}$, and the equation of motion becomes $-m\omega^2 \mathbf{u} \cos(\omega t + \phi) = -\Phi \mathbf{u} \cos(\omega t + \phi)$, which holds at all times only if

$$\Phi \mathbf{u} = m\omega^2 \mathbf{u}. \quad (4.12)$$

The matrix must turn the pattern into a multiple of itself. A motion of this kind is called a normal mode, and Equation (4.12) is an eigenvalue problem.

EIGENVALUES AND EIGENVECTORS. A non-zero vector $\mathbf{u}$ with $\mathbf{A}\mathbf{u} = \lambda\mathbf{u}$ for some number λ is an eigenvector of $\mathbf{A}$, and λ its eigenvalue. For a 2×2 matrix with entries a, b in the first row and c, d in the second, the condition is the pair of equations

$$(a - \lambda)u_1 + bu_2 = 0, \quad cu_1 + (d - \lambda)u_2 = 0.$$

Multiply the first by $d - \lambda$ and the second by b , and subtract: the u_2 terms cancel, leaving $[(a - \lambda)(d - \lambda) - bc]u_1 = 0$. Multiplying instead by c and by $a - \lambda$ and subtracting leaves the same bracket times u_2 . Since u_1 and u_2 are not both zero, the bracket must vanish:

$$(a - \lambda)(d - \lambda) - bc = 0.$$

This quadratic equation, the characteristic equation, gives the eigenvalues, and putting each back into either of the pair gives its eigenvector, up to a factor: any multiple $c\mathbf{u}$ of an eigenvector is another, since $\mathbf{A}(c\mathbf{u}) = c\mathbf{A}\mathbf{u} = \lambda(c\mathbf{u})$. For a symmetric matrix, $c = b$, multiplying out gives $\lambda^2 - (a + d)\lambda + (ad - b^2) = 0$, whose discriminant, the quantity under the square root in the formula for the roots of a quadratic, is

$$(a + d)^2 - 4(ad - b^2) = a^2 - 2ad + d^2 + 4b^2 = (a - d)^2 + 4b^2,$$

expanding $(a + d)^2$ and collecting $2ad - 4ad = -2ad$. A sum of squares is never negative, so the eigenvalues are real. An $n \times n$ symmetric matrix likewise has n real eigenvalues, with eigenvectors that can be chosen at right angles to one another; we use this general result without proof.

For the carts, dividing Equation (4.12) by k gives the eigenvalue problem, with $\lambda = m\omega^2/k$, of the matrix with rows $(2, -1)$ and $(-1, 2)$, whose characteristic equation is $(2 - \lambda)^2 - 1 = 0$, so $2 - \lambda = \pm 1$ and $\lambda = 1$ or 3 .

- $\lambda = 1$: the first equation, $(2 - 1)u_1 - u_2 = 0$, gives $\mathbf{u} = (1, 1)$. Both carts move together, the middle spring is never stretched, and $\omega = \sqrt{k/m}$.
- $\lambda = 3$: $(2 - 3)u_1 - u_2 = 0$ gives $\mathbf{u} = (1, -1)$. The carts move in opposite directions, and $\omega = \sqrt{3k/m}$.

With $m = 1 \text{ kg}$ and $k = 100 \text{ N m}^{-1}$ the two modes have $\omega = 10$ and 17.32 rad s^{-1} (Figure 4.9a, b). The two patterns are at right angles, $(1, 1) \cdot (1, -1) = 0$.

Because the equations are linear, a sum of normal modes is also a motion, as in Section 4.7. Any starting displacements split between the two patterns, $(x_1, x_2) = \frac{1}{2}(x_1 + x_2)(1, 1) + \frac{1}{2}(x_1 - x_2)(1, -1)$, and the starting velocities likewise, so an amplitude and a phase for each mode can match any starting state, and by the uniqueness of solutions (Section 2.6) every motion of the carts is a sum of normal modes. (A mode with $\omega = 0$, met below, contributes a steady drift $(c_1 + c_2 t)\mathbf{u}$ in place of a sinusoid.) Pulling cart 1 out by 4 cm and releasing both carts at rest starts the two modes equally, since $(4, 0) = 2(1, 1) + 2(1, -1)$, each at a turning point, with phase zero:

$$\mathbf{x}(t) = 2(1, 1) \cos 10t + 2(1, -1) \cos 17.32t \quad \text{in centimetres.}$$

The fast mode gains on the slow one by $17.32 - 10 = 7.32$ radians every second. When it has gained π , its cosine is minus the slow one's, so cart 1 stands nearly still and cart 2 swings 4 cm out; when it has gained 2π the two are back in step. The swing therefore passes from one cart to the other and back every $2\pi/7.32 = 0.858$ s (Figure 4.9c).

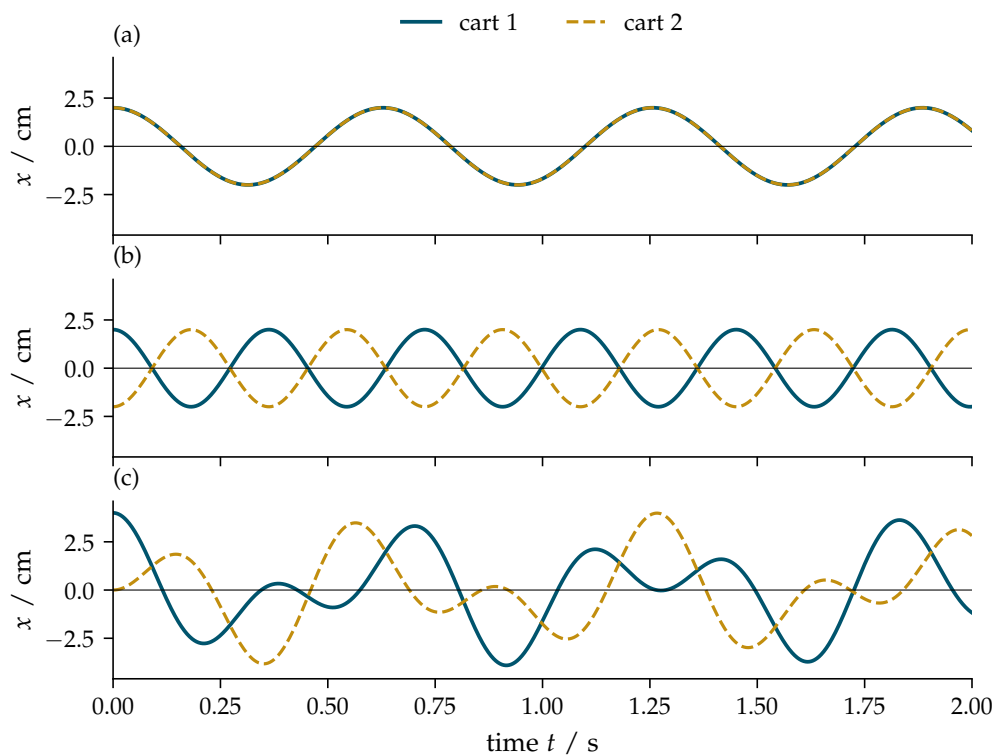


FIGURE 4.9: Two carts of 1 kg between walls, joined by three springs of 100 N m^{-1} . (a) The slow mode: both start 2 cm out and swing together at 10 rad s^{-1} . (b) The fast mode: cart 1 starts 2 cm out one way and cart 2 2 cm the other, and they swing against each other at 17.32 rad s^{-1} . (c) Only cart 1 displaced: the sum of the two modes, in which the swing passes back and forth between the carts. Cart 1 is solid and cart 2 dashed, so both remain identifiable where they move together.

Many atoms

Near a minimum of the potential energy $U(\mathbf{r}^N)$, let $q_1, \dots, q_n$, with $n = 3N$, be the displacements of the coordinates from the minimum. Along the straight line from the minimum in the direction of a displacement $\mathbf{q}$, the energy is $f(s) = U(\text{minimum} + s\mathbf{q})$. The chain rule along a path of [Section 3.10](#), with $d(\text{minimum} + s\mathbf{q})/ds = \mathbf{q}$, gives $f'(s) = \sum_a (\partial U / \partial q_a) q_a$. Each $\partial U / \partial q_a$ is itself a function of position, so the same rule gives its rate of change along the line as $\sum_b (\partial^2 U / \partial q_b \partial q_a) q_b$, and since the q_a are fixed numbers,

$$f''(s) = \sum_a \sum_b \frac{\partial^2 U}{\partial q_b \partial q_a} q_a q_b = \sum_{a,b} \Phi_{ab} q_a q_b,$$

where $\sum_{a,b}$ runs over every pair a, b , n^2 terms in all, and $\Phi_{ab} = \partial^2 U / \partial q_a \partial q_b$, the same in either order ([Section 3.11](#)), is the Hessian. At the minimum $f'(0) = 0$, so the Taylor series of f about $s = 0$, stopped after the term in s^2 and taken at $s = 1$, gives

$$U = U_{\min} + \frac{1}{2} \sum_{a,b} \Phi_{ab} q_a q_b + \mathcal{O}(q^3),$$

with Φ_{ab} taken at the minimum and q the length of $\mathbf{q}$. The step $s = 1$ is not short, but by [Section 1.8](#) the error is at most the largest size of f''' divided by $3! = 6$, and f''' , from the chain rule a third time, is a sum of third derivatives of U times products of three components of $\mathbf{q}$, so the error shrinks as q^3 . This is the many-dimensional version of [Section 4.4](#).

The same expansion gives the test promised in [Section 3.10](#). Let $\mathbf{u}_1, \dots, \mathbf{u}_n$ be eigenvectors of Φ of length 1, at right angles to one another, with eigenvalues $\lambda_1, \dots, \lambda_n$, and write a displacement as $\mathbf{q} = \sum_k c_k \mathbf{u}_k$. Since $\sum_{a,b} \Phi_{ab} q_a q_b = \mathbf{q} \cdot (\Phi \mathbf{q})$, and $\mathbf{u}_k \cdot \mathbf{u}_l$ is 1 for $k = l$ and 0 otherwise, the quadratic term is $\frac{1}{2} \sum_k \lambda_k c_k^2$. Where the gradient vanishes, U therefore has a minimum if every eigenvalue of Φ is positive, a maximum if every one is negative, and a saddle point if there are some of each. Zero eigenvalues need separate attention: higher terms may decide the behaviour, or a symmetry may allow free motion, as in the translation below.

Differentiating the expansion with respect to q_c , with the other displacements held fixed, the product rule gives each term $\Phi_{ab} q_a q_b$ the derivative $\Phi_{cb} q_b$ if $a = c$, plus $\Phi_{ac} q_a$ if $b = c$, and nothing if neither index is c , so

$$\frac{\partial U}{\partial q_c} = \frac{1}{2} \sum_b \Phi_{cb} q_b + \frac{1}{2} \sum_a \Phi_{ac} q_a = \sum_b \Phi_{cb} q_b,$$

the second step renaming a as b and using $\Phi_{bc} = \Phi_{cb}$. Neglecting terms of order q^2 , the force is therefore $F_c = -\sum_b \Phi_{cb} q_b$, and Newton's law for each coordinate,

$m_c \ddot{q}_c = F_c$, reads $\mathbf{M}\ddot{\mathbf{q}} = -\Phi\mathbf{q}$, with $\mathbf{M}$ the diagonal matrix of the masses, each atom's mass appearing for its x , y and z . A normal mode $\mathbf{q} = \mathbf{u} \cos(\omega t + \phi)$ now needs $\Phi\mathbf{u} = \omega^2 \mathbf{M}\mathbf{u}$. When the masses differ, this is not yet an eigenvalue problem of a symmetric matrix, but the substitution $w_a = \sqrt{m_a} u_a$ makes it one. The a th row reads $\sum_b \Phi_{ab} u_b = \omega^2 m_a u_a$; putting $u_b = w_b / \sqrt{m_b}$ on the left and $m_a u_a = \sqrt{m_a} w_a$ on the right, and dividing by $\sqrt{m_a}$,

$$\sum_b \frac{\Phi_{ab}}{\sqrt{m_a m_b}} w_b = \omega^2 w_a,$$

the eigenvalue problem of the mass-weighted Hessian, which is symmetric. Its n eigenvalues are the squared angular frequencies of the n normal modes. The function `normal_modes` of `mdlab` solves it with NumPy's routine for symmetric matrices, `eigh`:

```

1 def normal_modes(
2     hessian: ArrayLike, masses: ArrayLike, *, force_to_accel: float
3 ) -> tuple[NDArray, NDArray]:
4     hessian = np.asarray(hessian, dtype=float)
5     root_m = np.sqrt(np.asarray(masses, dtype=float))
6     weighted = hessian / np.outer(root_m, root_m) #  $\Phi_{ab} / \sqrt{m_a m_b}$ 
7     omega_squared, vectors = np.linalg.eigh(weighted)
8     modes = vectors / root_m[:, np.newaxis] #  $w_a / \sqrt{m_a}$  back to  $u_a$ 
9     return omega_squared * force_to_accel, modes

```

The lone `*` in the first line makes the argument after it a keyword argument, passed by name as `force_to_accel=1.0`. `np.outer(root_m, root_m)` is the table of products $\sqrt{m_a m_b}$, so the division forms the mass-weighted Hessian entry by entry; `eigh` returns its eigenvalues in increasing order and its eigenvectors as columns, each of length 1 and of either sign. `root_m[:, np.newaxis]`, in which `:` takes the whole of the one dimension that `root_m` has, is the column of the $\sqrt{m_a}$, as in [Section 2.10](#), so the next division divides row a by $\sqrt{m_a}$ and turns each $\mathbf{w}$ back into a pattern of displacements $\mathbf{u}$. The factor `force_to_accel` is 1 in SI units and the Newton factor $9.648\,533 \times 10^{-3}$ of [Section 2.10](#) for a Hessian in $\text{eV \AA}^{-2}$ and masses in amu, giving ω^2 in fs^{-2} , as for the lithium atom of [Section 4.4](#).

A linear molecule of three atoms

A molecule like carbon dioxide, an oxygen atom on either side of a carbon atom in a line, can be modelled along its axis by two springs of stiffness k between the three atoms, with masses m_O , m_C , m_O . Its energy is $U = \frac{1}{2}k(x_2 - x_1)^2 + \frac{1}{2}k(x_3 - x_2)^2$, with no walls, and differentiating twice as for the carts gives a Hessian with rows $k(1, -1, 0)$, $k(-1, 2, -1)$ and $k(0, -1, 1)$. It has three modes ([Figure 4.10](#)). The first, $\mathbf{u} = (1, 1, 1)$, moves the whole molecule without stretching a spring, so $\Phi\mathbf{u} = \mathbf{0}$ and $\omega = 0$: no force

resists this pattern, and the molecule drifts along it at a steady velocity, a motion of the whole called a translation. In the second, $\mathbf{u} = (1, 0, -1)$, the carbon stays still while the oxygens move in and out together; $\Phi\mathbf{u} = k(1, 0, -1)$ and $\mathbf{M}\mathbf{u} = (m_{\text{O}}, 0, -m_{\text{O}})$, so $\omega^2 = k/m_{\text{O}}$. In the third, the oxygens move one way and the carbon the other, so that the centre of mass stays still, $m_{\text{O}}u_1 + m_{\text{C}}u_2 + m_{\text{O}}u_3 = 0$ with $u_1 = u_3 = 1$, which gives $\mathbf{u} = (1, -2m_{\text{O}}/m_{\text{C}}, 1)$, and [Exercise 4.17](#) shows that $\omega^2 = (k/m_{\text{O}})(1 + 2m_{\text{O}}/m_{\text{C}})$. With $m_{\text{O}} = 15.999 \text{ amu}$ and $m_{\text{C}} = 12.011 \text{ amu}$, the third mode is faster than the second by $\sqrt{1 + 2 \times 15.999/12.011} = 1.914$, whatever k is.

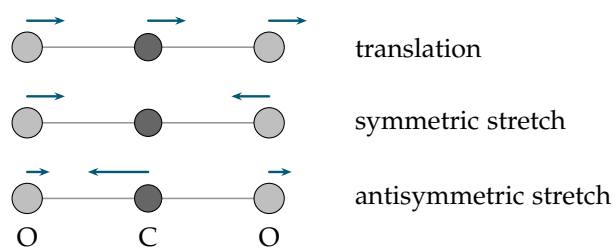


FIGURE 4.10: The three normal modes of a linear O–C–O chain along its axis. Arrows show the pattern $\mathbf{u}$ of displacements, to scale within each mode. In the antisymmetric stretch the carbon moves $2m_{\text{O}}/m_{\text{C}} = 2.66$ times as far as each oxygen, in the opposite direction, so that the centre of mass stays still.

NOTEBOOK 04 *animate the normal modes of the carts and the triatomic chain*

CHECK YOUR UNDERSTANDING. In the slow mode of the two carts, why does the middle spring not affect the angular frequency?

4.10 Molecular vibrations and the time step

The oscillations of this chapter are the motions of atoms in molecules. This section turns measured vibrational frequencies into periods in femtoseconds and draws the consequence for simulation: a computer that advances the motion in steps must take steps short compared with the fastest of these periods.

Frequencies from light

Light is an oscillation that travels, at the speed of light c ; its wavelength is the distance from one crest to the next, and infrared light has wavelengths too long for the eye to see. A molecule absorbs infrared light at frequencies close to those of many of its normal modes, much as a driven oscillator responds most strongly at resonance ([Section 4.7](#)), and other kinds of spectrum reveal most of the rest, so measured spectra give vibrational frequencies. Spectroscopists quote them as wavenumbers $\tilde{\nu}$, the number of wavelengths of the light per centimetre, in cm^{-1} . In one second the light travels a distance c , and

so passes $c\tilde{\nu}$ crests, each one cycle of the oscillation, so its frequency is $\nu = c\tilde{\nu}$. In the units of this book, $c = 2.997\,924\,58 \times 10^8 \text{ m s}^{-1} = 2.997\,924\,58 \times 10^{10} \text{ cm s}^{-1} = 2.997\,924\,58 \times 10^{-5} \text{ cm fs}^{-1}$, the last step dividing by 10^{15} femtoseconds per second, so

$$\nu = 2.99792458 \times 10^{-5} \tilde{\nu}, \quad \text{period} = \frac{1}{\nu}, \quad \omega = 2\pi\nu,$$

with $\tilde{\nu}$ in cm^{-1} , ν in cycles per femtosecond and the period in femtoseconds. In `mdlab` this speed is `units.C_CM_PER_FS`, computed from the exact SI value.

The fastest motions

The vibrational frequencies of small molecules have been measured carefully and collected in tables. [Table 4.1](#) lists a few from Shimanouchi's compilation⁵, as given by the NIST Chemistry WebBook⁶, with their periods. A water molecule's O–H stretches repeat every 9 fs, and the C–H stretches of methane every 11 fs; the bend of water takes 21 fs, more than twice as long as its stretches, and the stretch of the C–C bond in ethane takes 34 fs. Stretches of bonds to hydrogen are the fastest motions in molecules that contain hydrogen, for the reason found in [Section 4.8](#): the hydrogen does almost all the moving, so the reduced mass is small and $\omega = \sqrt{k/m_r}$ is large.

TABLE 4.1: Measured vibrational wavenumbers of three free (gas-phase) molecules^{5,6}, with the frequency and period they imply.

molecule	motion	$\tilde{\nu} / \text{cm}^{-1}$	ν / fs^{-1}	period / fs
water	antisymmetric O–H stretch	3756	0.11260	8.88
water	symmetric O–H stretch	3657	0.10963	9.12
methane	C–H stretch	3019	0.09051	11.05
methane	C–H stretch, all in step	2917	0.08745	11.44
water	bend	1595	0.04782	20.91
ethane	C–C stretch	995	0.02983	33.52

No comparably simple measured value is collected here for a lithium atom in graphite. The model surface of [Section 4.4](#) gives a period of 170.3 fs, a wavenumber of 196 cm^{-1} , but it was built for illustration; Chapters 16 and 17 measure the motion of lithium in graphite with a fitted model of the forces.

The step of a simulation

A computer that follows atoms advances their positions and velocities by repeated short steps, as in [Section 2.6](#). [Section 1.9](#) showed that a sinusoid sampled every τ is described accurately only when $\omega\tau$ is small, and the same holds for a motion advanced

in steps. A simulation of water must therefore take steps τ with $\omega\tau$ small for its fastest vibration, of period 8.88 fs and so $\omega = 2\pi/8.88 = 0.708 \text{ rad fs}^{-1}$: steps short compared with $1/\omega = 1.41 \text{ fs}$, even if every other motion in it is a hundred times slower. Chapter 9 works out how much shorter, and Chapter 11 shows how the fastest vibrations can be frozen out so that longer steps become safe. The learned propagators planned for Part VI take a different route: trained on such simulations, they predict the state of the atoms a single long step ahead.

FROM ELECTRONIC STRUCTURE. The vibrational frequencies computed by electronic-structure codes come from the eigenvalue problem of [Section 4.9](#): the Hessian of the Born–Oppenheimer energy, usually by finite differences of the forces, is mass-weighted and diagonalised. The harmonic frequencies so obtained neglect the anharmonicity of [Section 4.5](#), which is one reason they differ from the measured fundamentals of [Table 4.1](#).

CHECK YOUR UNDERSTANDING. A vibration has a wavenumber of 1000 cm^{-1} . What is its period in femtoseconds?

What we now have

A harmonic oscillator, $x = A \cos(\omega t + \phi)$, moves as the shadow of a point going round a circle of radius A at the angular speed ω , with its phase the starting angle on that circle. Its energy, $\frac{1}{2}kA^2$, is traded between kinetic and potential forms, which over a period hold half each, and in the phase plane its states trace ellipses, circles when the velocity is divided by ω .

Near a smooth minimum with positive curvature, small swings are approximately those of a spring of stiffness U'' , which gave the periods of a pendulum, of the bead of Chapter 3 and of a lithium atom in a model hollow. The period of a swing of any size follows from its energy as $2 \int dx/v$, and in the cosine, Morse and Lennard-Jones wells it grows with the energy, so small swings are the fastest.

A drag proportional to the velocity makes the swing shrink by $e^{-\gamma t/2}$, or, if heavy enough, prevents any swing; a sinusoidal push gives a steady response that peaks at resonance, with a height set by the quality factor ω_0/γ .

Two bodies on a spring oscillate in their separation with the reduced mass $m_1 m_2 / (m_1 + m_2)$, and many coupled bodies in the harmonic approximation move as sums of normal modes, whose patterns come from the eigenvectors of the mass-weighted Hessian and whose squared angular frequencies are its eigenvalues. The measured vibrations of [Table 4.1](#) repeat every 9 to 34 fs, fastest for bonds to hydrogen, and the fastest vibration present sets how short the steps of a simulation must be.

Chapter 5 adds rotation, and with it the angular momentum that the forces between atoms conserve.

Answers to the checks

SECTION 4.1. When $\omega t + \phi = \pi/2$ the point P is at the top of the reference circle, so the block is at the centre, $x = A \cos(\pi/2) = 0$. Its velocity is $-A\omega \sin(\pi/2) = -0.05 \times 10 = -0.5 \text{ m s}^{-1}$: it moves at its greatest speed towards negative x , as P , at the top and going anticlockwise, moves to the left.

SECTION 4.2. Where $U = K$, each is $E/2$, so $\frac{1}{2}kx^2 = \frac{1}{2} \times \frac{1}{2}kA^2$ and $x = \pm A/\sqrt{2} = \pm 4/\sqrt{2} = \pm 2.83 \text{ cm}$.

SECTION 4.3. The amplitude is $A = \sqrt{x_0^2 + (v_0/\omega)^2} = 0.3/10 = 0.03 \text{ m} = 3 \text{ cm}$ by Section 2.8, and the largest speed is $A\omega = 0.3 \text{ m s}^{-1}$, so the semi-axes are 3 cm and 0.3 m s^{-1} . It starts at the top of the ellipse, $(0, 0.3 \text{ m s}^{-1})$, and moves clockwise.

SECTION 4.4. The period $2\pi\sqrt{L/g}$ is proportional to $\sqrt{L}$, so four times the length doubles it. The mass cancelled from $\omega_0 = \sqrt{g/L}$, so it does not matter.

SECTION 4.5. Beyond small angles the restoring force, proportional to $\sin \theta$, grows more slowly than θ , so over most of a large swing the bob is pulled back less strongly than a spring would pull it. At the ends of a 90° swing the pull along the arc is $mg \sin 90^\circ = mg$, against $mg \pi/2$ for the spring of stiffness mg/L stretched by $L\pi/2$, so the bob turns round more slowly and spends longer there. The extra time spent near the ends outweighs the higher speed at the bottom; Equation (4.6) gives 1.180 times the small-swing period, against 1.002 for a 10° swing.

SECTION 4.6. Critical damping needs $\frac{1}{2}\gamma = \omega_0$, so $\gamma = 20 \text{ s}^{-1}$. With $\gamma = 8 \text{ s}^{-1}$, $\omega_d = \sqrt{\omega_0^2 - \frac{1}{4}\gamma^2} = \sqrt{100 - 16} = 9.165 \text{ rad s}^{-1}$, and the period is $2\pi/9.165 = 0.686 \text{ s}$.

SECTION 4.7. At $\omega = \omega_0$, $A = f/(\gamma\omega_0) = 1/(0.5 \times 10) = 0.2 \text{ m} = 20 \text{ cm}$. For a very slow push, $A \rightarrow f/\omega_0^2 = 1/100 \text{ m} = 1 \text{ cm}$, the stretch of the spring under the force held still.

SECTION 4.8. $m_r = m^2/(2m) = m/2$, and the separation oscillates with $\omega = \sqrt{k/m_r} = \sqrt{2k/m}$.

SECTION 4.9. In the slow mode both carts move by the same amount at every instant, so the distance between them never changes and the middle spring is never stretched: it exerts no force, and each cart swings on its outer spring alone, with $\omega = \sqrt{k/m}$.

SECTION 4.10. $\nu = 2.99792458 \times 10^{-5} \times 1000 = 0.02998 \text{ fs}^{-1}$, so the period is $1/\nu = 33.36 \text{ fs}$.

Exercises

The exercises run from questions of understanding, through calculations, to short derivations marked ‘show that’. Full solutions are in ‘Solutions to the exercises’ at the end of the book, and Notebook 04 checks every numerical answer.

EXERCISE 4.1 (*a phase of π*). An oscillating block has phase $\omega t + \phi = \pi$. Where is the point P on its reference circle, where is the block, and what is it about to do?

EXERCISE 4.2 (*a stiff spring’s energy*). A block of 0.2 kg on a spring of 80 N m^{-1} oscillates with an amplitude of 3 cm . Find its energy, its largest speed, and its speed at $x = 1.5 \text{ cm}$.

EXERCISE 4.3 (*the average of a sine squared*). Show that the average of $\sin^2 \omega t$ over one period is $\frac{1}{2}$, directly from $\cos 2\theta = 1 - 2\sin^2 \theta$.

EXERCISE 4.4 (*a state on the ellipse*). The block of Section 4.1 ($m = 0.5 \text{ kg}$, $k = 50 \text{ N m}^{-1}$) is at $x = 3 \text{ cm}$ moving at $v = -0.4 \text{ m s}^{-1}$. Find its energy and the semi-axes of its ellipse in the phase plane, and say whether x is about to grow or shrink.

EXERCISE 4.5 (*a pendulum for a clock*). How long must a pendulum be for its small swings to have a period of 2 s ?

EXERCISE 4.6 (*a time from the arcsine*). Show, from Equation (4.6) and the toolbox ‘The derivative of an inverse function’ of Section 4.5, that a harmonic oscillator takes $\pi/(6\omega)$ to move from its centre to half its amplitude, and evaluate this for $\omega = 10 \text{ rad s}^{-1}$.

EXERCISE 4.7 (*the bottom of a double well*). For the double well $U = U_0[(x/a)^2 - 1]^2$ of Exercise 3.9, show that small oscillations about $x = \pm a$ have $\omega_0 = \sqrt{8U_0/(ma^2)}$.

EXERCISE 4.8 (*a round hollow*). On the model surface, show that along the line $x = 0$ the potential energy is $U(0, y) = \frac{1}{4}U_b[3 - 2\cos(\frac{1}{2}|\mathbf{g}|y) - \cos(|\mathbf{g}|y)]$, and that its second derivative at the hollow, $\frac{3}{8}U_b|\mathbf{g}|^2$, equals the $2\pi^2U_b/a^2$ found along x in Section 4.4.

EXERCISE 4.9 (*the steepest attraction*). At what distance is the attractive force of the Lennard-Jones well strongest? (It is where the slope of U_{LJ} is greatest, so where its second derivative is zero.)

EXERCISE 4.10 (*the curvature of the Morse well*). Show that $d^2U_{\text{M}}/dr^2 = 2Da^2$ at $r = r_0$ by differentiating twice, without the series.

EXERCISE 4.11 (*a swing in a liquid*). The block with $\omega_0 = 10 \text{ rad s}^{-1}$ swings in a liquid with $\gamma = 12 \text{ s}^{-1}$. Find the period of its swing and how long its swing takes to shrink to a tenth.

EXERCISE 4.12 (*critical damping*). Show by substituting into Equation (4.7) that $x = (c_1 + c_2 t) e^{-\omega_0 t}$ is a motion when $\gamma = 2\omega_0$.

EXERCISE 4.13 (*off resonance*). The block with $\omega_0 = 10 \text{ rad s}^{-1}$, $\gamma = 2 \text{ s}^{-1}$ and $f = 1 \text{ m s}^{-2}$ is pushed at $\omega = 5$ and at 15 rad s^{-1} . Find the amplitude and the lag in each case.

EXERCISE 4.14 (*adding motions*). Show that if $x_1(t)$ and $x_2(t)$ are motions of the damped oscillator, Equation (4.7), then so is $c_1 x_1 + c_2 x_2$ for any numbers c_1 and c_2 .

EXERCISE 4.15 (*heavy hydrogen*). Deuterium is hydrogen with a mass of 2.014 amu. If an O–H bond and an O–D bond have the same stiffness, what is the ratio of the angular frequency of the O–D vibration to that of the O–H vibration, and by what factor is the O–D period longer?

EXERCISE 4.16 (*a different middle spring*). The two carts of Section 4.9 are joined by a middle spring of stiffness k' , the outer springs keeping k . Show that the normal modes are still $(1, 1)$ and $(1, -1)$, with $\omega^2 = k/m$ and $(k + 2k')/m$.

EXERCISE 4.17 (*the antisymmetric stretch*). For the O–C–O chain of Section 4.9, show that $\mathbf{u} = (1, -2m_{\text{O}}/m_{\text{C}}, 1)$ satisfies $\Phi \mathbf{u} = \omega^2 \mathbf{M} \mathbf{u}$ with $\omega^2 = (k/m_{\text{O}})(1 + 2m_{\text{O}}/m_{\text{C}})$.

EXERCISE 4.18 (*a free pair*). Two masses m_1 and m_2 on a line are joined by one spring of stiffness k , so that the Hessian has rows $k(1, -1)$ and $k(-1, 1)$. Show that the mass-weighted Hessian has eigenvalues 0 and k/m_r , with m_r the reduced mass, and interpret both.

EXERCISE 4.19 (*the stiffness of an O–H bond*). Treating the symmetric stretch of water, 3657 cm^{-1} , as an O–H bond vibrating with the reduced mass of Section 4.8, find its angular frequency in rad fs^{-1} and the stiffness of the bond in $\text{eV \AA}^{-2}$.

EXERCISE 4.20 (*a step of a tenth*). If a simulation had to take steps no longer than a tenth of the shortest period present, what would be the longest step for water, and how many steps would a picosecond of simulated time take?

Rotation and angular momentum

Chapters 2 to 4 treated a body as a particle when its motion could be specified by a single position. A molecule also changes its orientation as it moves, so we now need to describe rotation and distinguish it from motion of the centre of mass. Angular momentum supplies the corresponding conservation law: for an isolated group with central pair forces, the internal torques cancel and the total angular momentum stays fixed.

We begin with a puck moving in a plane, where the geometry can be followed through its distance and angle from a fixed point. Extending that argument to space introduces the cross product; applying it to many particles then leads to the inertia tensor, which relates angular velocity to angular momentum. These tools let us separate a molecule's drift and rotation from its internal motion, and explain the count of $3N - 6$ vibrational degrees of freedom, or $3N - 5$ for a linear molecule. Chapter 6 uses the same distinction between allowed motions to choose coordinates for constrained systems.

5.1 Turning in a plane

A puck slides without friction on a smooth level table, an air table that floats it on a thin cushion of air. A string tied to the puck runs to a small hole in the middle of the table and through it to a hand underneath. Set going sideways with the string taut, the puck circles the hole; pull the string down, and the puck circles closer and faster. This section finds the quantity that stays fixed while the speed changes, and the reason it does.

Distance and angle

Put the origin at the hole and describe the puck's position by its distance r from the hole and the angle θ that the line from the hole makes with the x axis, so that, by the definition of sine and cosine scaled to a circle of radius r (Section 1.5),

$$x = r \cos \theta, \quad y = r \sin \theta.$$

These are called polar coordinates. As the puck moves, both r and θ change with time, at the rates $\dot{r}$ and $\dot{\theta}$, the dot marking the rate of change with time as in Section 1.3. Differentiating x and y with the product rule, and $\cos \theta$ and $\sin \theta$ with the chain rule, so that $d \cos \theta / dt = -\sin \theta \dot{\theta}$ and $d \sin \theta / dt = \cos \theta \dot{\theta}$,

$$\dot{x} = \dot{r} \cos \theta - r \dot{\theta} \sin \theta, \quad \dot{y} = \dot{r} \sin \theta + r \dot{\theta} \cos \theta.$$

Two vectors of length 1, called unit vectors, make this easier to read: $\mathbf{e}_r = (\cos \theta, \sin \theta)$, pointing straight away from the hole, and $\mathbf{e}_\theta = (-\sin \theta, \cos \theta)$, at right angles to it in the direction of increasing θ , anticlockwise; their scalar product is $-\cos \theta \sin \theta + \sin \theta \cos \theta = 0$. Collecting the terms in $\dot{r}$ and in $r \dot{\theta}$ in each component, $(\dot{x}, \dot{y}) = \dot{r} (\cos \theta, \sin \theta) + r \dot{\theta} (-\sin \theta, \cos \theta)$, so the velocity is

$$\mathbf{v} = \dot{r} \mathbf{e}_r + r \dot{\theta} \mathbf{e}_\theta, \tag{5.1}$$

a part $v_r = \dot{r}$ straight towards or away from the hole and a part $v_\theta = r \dot{\theta}$ across the line to it. On a circle r is constant, $\dot{\theta}$ is the angular speed ω , and $v_\theta = r \dot{\theta}$ is the speed $R\omega$ of Section 1.7.

Angular momentum and torque

For a body of mass m moving in the plane, define

$$L = m (xv_y - yv_x), \tag{5.2}$$

its angular momentum about the origin. In polar coordinates,

$$\begin{aligned} xv_y - yv_x &= r \cos \theta (\dot{r} \sin \theta + r \dot{\theta} \cos \theta) - r \sin \theta (\dot{r} \cos \theta - r \dot{\theta} \sin \theta) \\ &= r^2 \dot{\theta} (\cos^2 \theta + \sin^2 \theta) = r^2 \dot{\theta} \quad \text{the terms in } \dot{r} \text{ cancel; } \cos^2 \theta + \sin^2 \theta = 1, \end{aligned}$$

so $L = mr^2\dot{\theta} = mr v_\theta$: the distance from the origin times the part of the momentum across the line to it. Moving straight towards or away from the origin gives no angular momentum; going round it anticlockwise gives a positive L , clockwise a negative one. Its unit is $\text{kg m}^2 \text{s}^{-1}$.

How does L change? Differentiating Equation (5.2) by the product rule,

$$\begin{aligned}\frac{dL}{dt} &= m(\dot{x}v_y + x\dot{v}_y - \dot{y}v_x - y\dot{v}_x) \\ &= m(xa_y - ya_x) && \text{since } \dot{x}v_y = v_xv_y = \dot{y}v_x \text{ and } \dot{\mathbf{v}} = \mathbf{a}, \\ &= xF_y - yF_x && \text{by the second law, } m\mathbf{a} = \mathbf{F}.\end{aligned}$$

The combination on the right is the torque of the force about the origin,

$$G = xF_y - yF_x, \quad \text{and} \quad \frac{dL}{dt} = G. \quad (5.3)$$

Torque is to angular momentum what force is to momentum. Writing the force as $\mathbf{F} = F_r\mathbf{e}_r + F_\theta\mathbf{e}_\theta$, the same algebra as for L , with F_r in place of $\dot{r}$ and F_θ in place of $r\dot{\theta}$, gives $G = rF_\theta$: only the part of the force across the line from the origin turns the body, and it turns it more the further out it acts. Its unit is the newton metre.

This is the lever. A door turns about its hinge (Figure 5.1a). A push of 10 N at right angles to the door at the handle, 0.8 m from the hinge, gives a torque of $0.8 \times 10 = 8 \text{ N m}$; the same torque 0.2 m from the hinge needs $8/0.2 = 40 \text{ N}$, and a push straight towards the hinge, with no part across the door, does not turn it at all.

Chapter 2 noted that two equal and opposite pushes at opposite corners of a book set it spinning although they add to zero. Put the origin at the centre of the book, with a push $(0, F)$ at the corner (a, b) and a push $(0, -F)$ at the opposite corner $(-a, -b)$ (Figure 5.1b). The net force is zero, but the torques are $aF - b \times 0 = aF$ and $(-a)(-F) - (-b) \times 0 = aF$, a total of $2aF$. Such a pair of forces, adding to no force but to a torque, is called a couple.

Pulling the string

The string pulls the puck straight towards the hole, so its force has no part across the line to the hole, $F_\theta = 0$, and its torque about the hole is zero. The weight of the puck and the push of the table are vertical and turn nothing in the plane. By Equation (5.3), therefore, L does not change: the angular momentum of the puck is conserved, whatever the hand does with the string. A force that always points towards or away from one fixed point is called a central force, and every central force conserves the angular momentum about its centre.

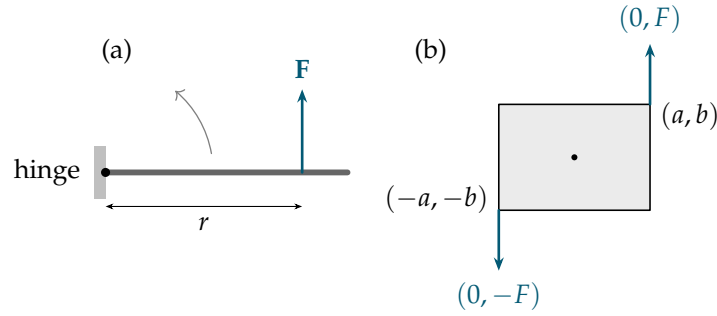


FIGURE 5.1: Torque. (a) A door seen from above: a force $\mathbf{F}$ at right angles to the door, a distance r from the hinge, has the torque rF about the hinge. (b) A book pushed at two opposite corners: the forces add to zero, but their torques about the centre add to $2aF$, and the book spins.

Let the puck, of mass 0.2 kg , circle at $r = 0.5 \text{ m}$ with speed 1.2 m s^{-1} , so that $L = mrv_\theta = 0.2 \times 0.5 \times 1.2 = 0.12 \text{ kg m}^2 \text{ s}^{-1}$, and pull the string down at a steady 0.05 m s^{-1} until $r = 0.25 \text{ m}$. Since $L = mrv_\theta$ stays fixed, $v_\theta = L/(mr)$ grows as r shrinks: at 0.25 m the puck goes round at $0.12/(0.2 \times 0.25) = 2.4 \text{ m s}^{-1}$, twice as fast at half the distance (Figure 5.2). Since $\mathbf{e}_r$ and $\mathbf{e}_\theta$ are at right angles and of length 1, Pythagoras' theorem applied to Equation (5.1) gives $|\mathbf{v}|^2 = \dot{r}^2 + v_\theta^2$, so the kinetic energy splits into a part across the line, $\frac{1}{2}mv_\theta^2$, and a part along it, $\frac{1}{2}m\dot{r}^2$. The part across the line rises from $\frac{1}{2} \times 0.2 \times 1.2^2 = 0.144 \text{ J}$ to $\frac{1}{2} \times 0.2 \times 2.4^2 = 0.576 \text{ J}$, while the part along the line, with $\dot{r} = -0.05 \text{ m s}^{-1}$, stays a constant $\frac{1}{2} \times 0.2 \times 0.05^2 = 0.25 \text{ mJ}$. The extra 0.432 J is the work of the hand: by Section 3.2 the change in kinetic energy is the work done on the puck, the vertical forces do none, and the string pulls inwards while the puck moves inwards, so its work is positive.

Equal areas in equal times

The line from the hole to the puck sweeps out area as the puck moves. A whole circle, of area πr^2 , has the angle 2π , so a sector of angle $\Delta\theta$ has the area $\pi r^2 \Delta\theta / (2\pi) = \frac{1}{2}r^2 \Delta\theta$. In a short time Δt the angle grows by about $\dot{\theta} \Delta t$, and the line sweeps a thin sector of area $\Delta A \approx \frac{1}{2}r^2 \dot{\theta} \Delta t$; the change of r in that time alters the area only by a part proportional to $(\Delta t)^2$. Dividing by Δt and letting $\Delta t \rightarrow 0$, as in Section 1.2,

$$\frac{dA}{dt} = \frac{1}{2}r^2\dot{\theta} = \frac{L}{2m} \quad \text{since } L = mr^2\dot{\theta}.$$

While L is constant the line sweeps equal areas in equal times, here $0.12/(2 \times 0.2) = 0.3 \text{ m}^2$ every second: long thin sectors far out, short wide ones close in (Figure 5.2a). A planet is pulled by the Sun's gravity towards the Sun, a central force, and the same argument gives the law of equal areas that Kepler found in the motions of the planets.

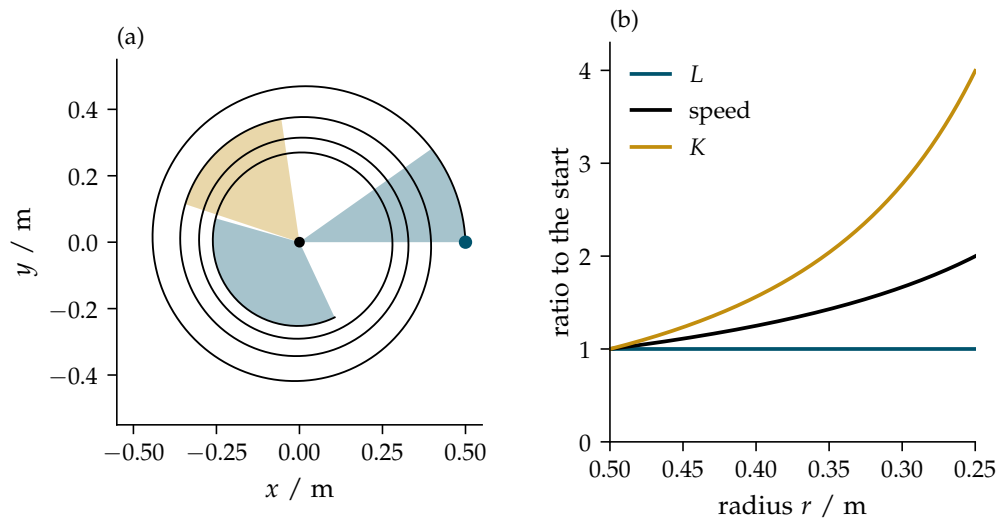


FIGURE 5.2: A puck of 0.2 kg on a string through a hole, circling at 0.5 m and 1.2 m s^{-1} and then pulled in at a steady 0.05 m s^{-1} to 0.25 m , computed by stepping Newton's second law forward in short steps of time (`solve_newton` in `mdl`lab). (a) The path seen from above, from the dot; the shaded sectors are swept in 0.25 s each, at the start, the middle and the end of the pull, and all three have the area 0.075 m^2 . (b) The angular momentum L , the speed and the kinetic energy K , each divided by its value at the start, against the distance from the hole.

KEY IDEA. In a plane, the angular momentum $L = m(xv_y - yv_x) = mrv_\theta$ changes at the rate of the torque, $G = xF_y - yF_x = rF_\theta$. A central force has no torque about its centre, so it conserves L : pulled inwards, a body goes round faster.

NOTEBOOK 05 *pull the puck in and out and watch L , the speed and the energy*

CHECK YOUR UNDERSTANDING. The puck circling at 0.5 m and 1.2 m s^{-1} is let out slowly to 1 m . How fast does it then go round, and is the work done on it by the string positive or negative?

5.2 Angular momentum in space

The puck turned in the xy plane, about the z axis, and the combination $xv_y - yv_x$ measured that turning. A body moving in space can turn about any axis. Seen from far out along the x axis, with the axes drawn as in the toolbox below, the y and z axes sit as the x and y axes do seen from above the table, so the same combination with the letters renamed, $x \rightarrow y \rightarrow z \rightarrow x$, measures the turning about the x axis, $yv_z - zv_y$, and renaming once more gives $zv_x - xv_z$ about the y axis. The three together form a vector, made from the two vectors $\mathbf{r}$ and $\mathbf{v}$ by a new product.

THE CROSS PRODUCT. The cross product of $\mathbf{a} = (a_x, a_y, a_z)$ and $\mathbf{b} = (b_x, b_y, b_z)$ is the vector

$$\mathbf{a} \times \mathbf{b} = (a_y b_z - a_z b_y, a_z b_x - a_x b_z, a_x b_y - a_y b_x),$$

each component built from the other two letters in the order $x \rightarrow y \rightarrow z \rightarrow x$. Its properties follow from the components:

- Swapping the vectors swaps the two products in every component, so $\mathbf{b} \times \mathbf{a} = -\mathbf{a} \times \mathbf{b}$, and $\mathbf{a} \times \mathbf{a} = \mathbf{0}$, since each of its components is a product minus the same product, such as $a_y a_z - a_z a_y = 0$. A number factor comes outside, $(c\mathbf{a}) \times \mathbf{b} = c(\mathbf{a} \times \mathbf{b})$, and a sum splits, $\mathbf{a} \times (\mathbf{b} + \mathbf{c}) = \mathbf{a} \times \mathbf{b} + \mathbf{a} \times \mathbf{c}$, since each component is a sum of products.
- It is at right angles to both vectors: multiplying out,

$$\begin{aligned} \mathbf{a} \cdot (\mathbf{a} \times \mathbf{b}) &= a_x(a_y b_z - a_z b_y) + a_y(a_z b_x - a_x b_z) + a_z(a_x b_y - a_y b_x) \\ &= a_x a_y b_z - a_x a_z b_y + a_y a_z b_x - a_x a_y b_z + a_x a_z b_y - a_y a_z b_x, \end{aligned}$$

where the first and fourth products cancel, the second and fifth, and the third and sixth, so the scalar product is zero; the same holds for $\mathbf{b}$, and a zero scalar product means a right angle ([Section 1.6](#)).

- With the unit vectors along the axes, $\mathbf{e}_x = (1, 0, 0)$, $\mathbf{e}_y = (0, 1, 0)$ and $\mathbf{e}_z = (0, 0, 1)$, the definition gives $\mathbf{e}_x \times \mathbf{e}_y = \mathbf{e}_z$, and likewise $\mathbf{e}_y \times \mathbf{e}_z = \mathbf{e}_x$ and $\mathbf{e}_z \times \mathbf{e}_x = \mathbf{e}_y$. We always draw the axes so that, curling the fingers of the right hand from $\mathbf{e}_x$ towards $\mathbf{e}_y$, the thumb points along $\mathbf{e}_z$. For $\mathbf{a}$ and $\mathbf{b}$ in the xy plane, at the angles θ_a and θ_b from the x axis, $\mathbf{a} \times \mathbf{b} = (0, 0, a_x b_y - a_y b_x)$, and by [Equation \(1.6\)](#) $a_x b_y - a_y b_x = |\mathbf{a}||\mathbf{b}|(\cos \theta_a \sin \theta_b - \sin \theta_a \cos \theta_b) = |\mathbf{a}||\mathbf{b}| \sin(\theta_b - \theta_a)$, positive when $\mathbf{b}$ lies less than half a turn anticlockwise from $\mathbf{a}$. So the same right-hand rule, fingers curling from $\mathbf{a}$ towards $\mathbf{b}$, gives the direction of $\mathbf{a} \times \mathbf{b}$ ([Figure 5.3](#)); for vectors in any other plane we take it on trust, since turning the axes to bring them into the xy plane needs more than we have shown.
- Differentiating each component by the product rule, the x component for instance as $d(a_y b_z - a_z b_y)/dt = (\dot{a}_y b_z - \dot{a}_z b_y) + (a_y \dot{b}_z - a_z \dot{b}_y)$, gives $d(\mathbf{a} \times \mathbf{b})/dt = \dot{\mathbf{a}} \times \mathbf{b} + \mathbf{a} \times \dot{\mathbf{b}}$, with the order of each product kept.

For example, $(1, 2, 0) \times (3, -1, 2) = (2 \times 2 - 0 \times (-1), 0 \times 3 - 1 \times 2, 1 \times (-1) - 2 \times 3) = (4, -2, -7)$, and its scalar products with $(1, 2, 0)$ and $(3, -1, 2)$ are $4 - 4 + 0 = 0$ and $12 + 2 - 14 = 0$.

THE LENGTH OF A CROSS PRODUCT. Squaring the three components, each by $(p - q)^2 = p^2 - 2pq + q^2$, and adding,

$$\begin{aligned} |\mathbf{a} \times \mathbf{b}|^2 &= a_y^2 b_z^2 + a_z^2 b_y^2 + a_z^2 b_x^2 + a_x^2 b_z^2 + a_x^2 b_y^2 + a_y^2 b_x^2 \\ &\quad - 2(a_y a_z b_y b_z + a_z a_x b_z b_x + a_x a_y b_x b_y). \end{aligned}$$

The product $|\mathbf{a}|^2 |\mathbf{b}|^2 = (a_x^2 + a_y^2 + a_z^2)(b_x^2 + b_y^2 + b_z^2)$ has the same six squares plus $a_x^2 b_x^2 + a_y^2 b_y^2 + a_z^2 b_z^2$, and $(\mathbf{a} \cdot \mathbf{b})^2 = (a_x b_x + a_y b_y + a_z b_z)^2$ is those three squares plus twice the bracket above. Subtracting,

$$\begin{aligned} |\mathbf{a} \times \mathbf{b}|^2 &= |\mathbf{a}|^2 |\mathbf{b}|^2 - (\mathbf{a} \cdot \mathbf{b})^2 \\ &= |\mathbf{a}|^2 |\mathbf{b}|^2 (1 - \cos^2 \theta) && \text{as } \mathbf{a} \cdot \mathbf{b} = |\mathbf{a}| |\mathbf{b}| \cos \theta \\ &= |\mathbf{a}|^2 |\mathbf{b}|^2 \sin^2 \theta && \text{as } \cos^2 \theta + \sin^2 \theta = 1, \end{aligned}$$

with θ the angle between the vectors, between 0 and π , by the scalar product of [Section 1.6](#). Since $\sin \theta \geq 0$ there, $|\mathbf{a} \times \mathbf{b}| = |\mathbf{a}| |\mathbf{b}| \sin \theta$, the area of the parallelogram with sides $\mathbf{a}$ and $\mathbf{b}$, whose base is $|\mathbf{a}|$ and height $|\mathbf{b}| \sin \theta$. For the example above, $|\mathbf{a} \times \mathbf{b}| = \sqrt{4^2 + 2^2 + 7^2} = \sqrt{69} = 8.3066$, and with $|\mathbf{a}|^2 = 1 + 4 = 5$, $|\mathbf{b}|^2 = 9 + 1 + 4 = 14$ and $\mathbf{a} \cdot \mathbf{b} = 3 - 2 + 0 = 1$, $\sqrt{5 \times 14 - 1^2}$ is the same.

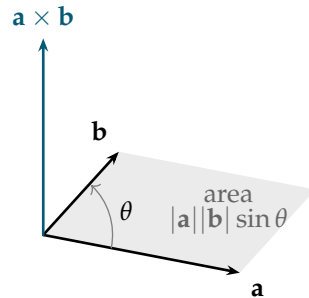


FIGURE 5.3: The cross product, drawn in perspective: $\mathbf{a}$ and $\mathbf{b}$ lie in a flat horizontal sheet seen from above at a slant, and $\mathbf{a} \times \mathbf{b}$ stands straight up from it, at right angles to both, along the thumb of a right hand whose fingers curl from $\mathbf{a}$ towards $\mathbf{b}$. Its length is the area of the parallelogram they span (shaded).

Angular momentum and torque in space

The angular momentum about the origin of a body at $\mathbf{r}$ with momentum $\mathbf{p} = m\mathbf{v}$, and the torque about the origin of a force $\mathbf{F}$ acting at $\mathbf{r}$, are

$$\mathbf{L} = \mathbf{r} \times \mathbf{p}, \quad \mathbf{G} = \mathbf{r} \times \mathbf{F}. \quad (5.4)$$

By the product rule of the toolbox,

$$\frac{d\mathbf{L}}{dt} = \mathbf{v} \times m\mathbf{v} + \mathbf{r} \times m\mathbf{a} = \mathbf{r} \times \mathbf{F} = \mathbf{G}, \quad (5.5)$$

since $\mathbf{v} \times \mathbf{v} = \mathbf{0}$ and $m\mathbf{a} = \mathbf{F}$: the rate of change of the angular momentum is the torque, now as vectors. For the puck, $\mathbf{r} = (x, y, 0)$ and $\mathbf{v} = (v_x, v_y, 0)$, so the first two components of $\mathbf{L}$ vanish and $\mathbf{L} = (0, 0, m(xv_y - yv_x))$: the L of Section 5.1 is the z component L_z of the vector, and $\mathbf{L}$ points straight up out of the table when the puck goes round anticlockwise. In general $|\mathbf{L}| = m|\mathbf{r}||\mathbf{v}|\sin\theta$, with θ here the angle between $\mathbf{r}$ and $\mathbf{v}$, as in the toolbox, not the polar angle of Section 5.1; this is the distance times the part of the momentum at right angles to $\mathbf{r}$, as in the plane.

A central force, $\mathbf{F} = c\mathbf{r}$ for some number c that may change as the body moves (for the puck, minus the pull of the string divided by r), has $\mathbf{G} = c\mathbf{r} \times \mathbf{r} = \mathbf{0}$ and conserves the whole vector $\mathbf{L}$, direction as well as size. Then $\mathbf{r} \cdot \mathbf{L} = m\mathbf{r} \cdot (\mathbf{r} \times \mathbf{v}) = 0$ at every instant, by the second property in the toolbox, so $\mathbf{r}$ is always at right angles to the fixed $\mathbf{L}$, as the spokes of a wheel are to its axle, and a body with $\mathbf{L} \neq \mathbf{0}$ stays in the plane through the centre at right angles to $\mathbf{L}$ (Exercise 5.5). This is why a planet's orbit stays in one plane. The puck is held in its plane by the table, but its $\mathbf{L}$ is fixed too: the weight and the push of the table cancel, leaving only the central pull of the string.

KEY IDEA. The cross product $\mathbf{a} \times \mathbf{b}$ is at right angles to both vectors, by the right-hand rule, with length $|\mathbf{a}||\mathbf{b}|\sin\theta$. Angular momentum $\mathbf{L} = \mathbf{r} \times \mathbf{p}$ changes at the rate of the torque $\mathbf{G} = \mathbf{r} \times \mathbf{F}$; a central force conserves it.

CHECK YOUR UNDERSTANDING. Find $\mathbf{e}_z \times \mathbf{e}_x$ from the components. A record turntable turns clockwise when seen from above. Which way does its angular momentum point?

5.3 The angular momentum of many bodies

Section 2.9 added the momenta of a group of bodies and found that the forces between them cancel in pairs, so that only outside forces change the total. This section does the same for angular momentum, and finds one extra condition: the forces between two bodies must act along the line joining them.

Internal torques cancel

For N bodies, at $\mathbf{r}_i$ with momenta $\mathbf{p}_i$, the total angular momentum about the origin is

$$\mathbf{L} = \sum_{i=1}^N \mathbf{r}_i \times \mathbf{p}_i,$$

and by Equation (5.5) for each body, $d\mathbf{L}/dt = \sum_i \mathbf{r}_i \times \mathbf{F}_i$, with $\mathbf{F}_i$ the total force on body i . As in Section 2.9, split each force into the external force $\mathbf{F}_i^{\text{ext}}$, from outside the group, and the forces $\mathbf{F}_{ij}$ on body i from the other bodies j . The torques of the forces between bodies i and j appear as a pair,

$$\begin{aligned} \mathbf{r}_i \times \mathbf{F}_{ij} + \mathbf{r}_j \times \mathbf{F}_{ji} &= \mathbf{r}_i \times \mathbf{F}_{ij} - \mathbf{r}_j \times \mathbf{F}_{ij} \quad \text{by the third law, } \mathbf{F}_{ji} = -\mathbf{F}_{ij}, \\ &= (\mathbf{r}_i - \mathbf{r}_j) \times \mathbf{F}_{ij} \quad \text{a sum splits, in the first vector as in the second.} \end{aligned}$$

The third law alone does not make this zero: two equal and opposite forces side by side, like the pushes on the book, form a couple. But if the force between two bodies acts along the line joining them, $\mathbf{F}_{ij} = c(\mathbf{r}_i - \mathbf{r}_j)$ for some number c , the pair gives $c(\mathbf{r}_i - \mathbf{r}_j) \times (\mathbf{r}_i - \mathbf{r}_j) = \mathbf{0}$. Then every pair drops out, and

$$\frac{d\mathbf{L}}{dt} = \sum_{i=1}^N \mathbf{r}_i \times \mathbf{F}_i^{\text{ext}} = \mathbf{G}^{\text{ext}} : \quad (5.6)$$

only the external torque changes the total angular momentum, and when it is zero, the angular momentum of the group is conserved.

Forces between atoms

Atoms whose potential energy is a sum of terms $\varphi(r_{ij})$, each depending only on the distance r_{ij} between two atoms, push and pull along the lines joining them. Differentiating $r_{ij}^2 = (x_j - x_i)^2 + (y_j - y_i)^2 + (z_j - z_i)^2$ with respect to x_i by the chain rule gives $2r_{ij} \partial r_{ij} / \partial x_i = -2(x_j - x_i)$, so $\partial r_{ij} / \partial x_i = -(x_j - x_i) / r_{ij}$, and likewise for y_i and z_i . The force on atom i from this term is then

$$-\nabla_i \varphi(r_{ij}) = -\varphi'(r_{ij}) \nabla_i r_{ij} = \varphi'(r_{ij}) \frac{\mathbf{r}_j - \mathbf{r}_i}{r_{ij}},$$

along the line from atom i to atom j , as Exercise 3.20 found. An isolated group of such atoms therefore conserves its angular momentum as well as its momentum. Many of the forces of Chapter 8 depend on three or more atoms at once and cannot be split into pairs, but Chapter 7 shows that, whenever the potential energy is unchanged when the whole group is turned, the internal torques still add to zero.

Turning about the centre of mass

Unlike momentum, angular momentum depends on the point it is measured about: about the point $\mathbf{a}$, each position is measured from $\mathbf{a}$, so that $\sum_i (\mathbf{r}_i - \mathbf{a}) \times \mathbf{p}_i$ takes the place of $\mathbf{L}$. Measure each position and velocity from the centre of mass $\mathbf{R}$ and its velocity $\mathbf{V} = \mathbf{P}/M$ (Section 2.9), $\mathbf{r}_i = \mathbf{R} + \mathbf{r}'_i$ and $\mathbf{v}_i = \mathbf{V} + \mathbf{v}'_i$. The primed quantities satisfy $\sum_i m_i \mathbf{r}'_i = \sum_i m_i \mathbf{r}_i - M\mathbf{R} = \mathbf{0}$ and, differentiating, $\sum_i m_i \mathbf{v}'_i = \mathbf{0}$. Then

$$\begin{aligned}
 \mathbf{L} &= \sum_i m_i (\mathbf{R} + \mathbf{r}'_i) \times (\mathbf{V} + \mathbf{v}'_i) \\
 &= \left(\sum_i m_i \right) \mathbf{R} \times \mathbf{V} + \mathbf{R} \times \sum_i m_i \mathbf{v}'_i \quad \text{multiplying out} \\
 &\quad + \left(\sum_i m_i \mathbf{r}'_i \right) \times \mathbf{V} + \sum_i m_i \mathbf{r}'_i \times \mathbf{v}'_i \\
 &= \mathbf{R} \times \mathbf{P} + \mathbf{L}', \quad \mathbf{L}' = \sum_i m_i \mathbf{r}'_i \times \mathbf{v}'_i \quad M\mathbf{V} = \mathbf{P}; \text{ the middle two vanish.}
 \end{aligned}$$

The angular momentum of a group is that of its total momentum carried at the centre of mass, plus $\mathbf{L}'$, its angular momentum about its own centre of mass, which does not depend on where the origin is, since moving the origin shifts each $\mathbf{r}_i$ and $\mathbf{R}$ alike and leaves $\mathbf{r}'_i = \mathbf{r}_i - \mathbf{R}$ unchanged. For a thrown spinning ball, the first part belongs to the flight and the second to the spin; for a molecule, $\mathbf{L}'$ is the turning of the molecule itself. For an isolated group both parts are conserved separately: $\mathbf{P}$ is constant, so $d(\mathbf{R} \times \mathbf{P})/dt = \mathbf{V} \times \mathbf{P} = \mathbf{0}$, since $\mathbf{P} = M\mathbf{V}$ is parallel to $\mathbf{V}$; and $\mathbf{L}$ is constant by Equation (5.6), so $\mathbf{L}' = \mathbf{L} - \mathbf{R} \times \mathbf{P}$ is constant too.

KEY IDEA. Forces between pairs of bodies along the lines joining them cancel in the total torque, so only external torques change the angular momentum of a group, and an isolated group of atoms keeps it. The total splits into the part carried by the centre of mass and the part $\mathbf{L}'$ about it.

CHECK YOUR UNDERSTANDING. Two bodies of 1 kg at $(\pm 0.5, 0, 0)$ m move with velocities $(0, \mp 1, 0)$ m s⁻¹. Find their total momentum and their angular momentum about the origin and about the point $(0, 1, 0)$ m.

5.4 Rigid rotation

A wheel, a spinning top and, to a good approximation, a molecule are rigid: the distances between their parts do not change as they turn. This section finds the velocity of every part of a turning rigid body, its angular momentum and its kinetic energy, and then keeps the promise of [Section 3.5](#) by racing a rolling disc and a rolling hoop against a sliding block; a ball joins the race in [Exercise 5.10](#).

Turning about a fixed axis

Let a rigid body turn about the z axis at the angular speed ω . Each part moves round a circle about the axis, of radius ρ , its distance from the axis, at the speed $\omega\rho$ ([Section 1.7](#)). Define the angular velocity as the vector $\boldsymbol{\omega} = (0, 0, \omega)$, along the axis, of length the angular speed, and pointing by the right-hand rule: fingers curled along the motion, thumb along $\boldsymbol{\omega}$. Then the velocity of the part at $\mathbf{r} = (x, y, z)$ is

$$\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r} = (-\omega y, \omega x, 0), \quad (5.7)$$

by the components of the cross product; for a part at $x = \rho \cos \omega t$, $y = \rho \sin \omega t$ this is $\rho\omega(-\sin \omega t, \cos \omega t, 0)$, the velocity of [Equation \(1.9\)](#). For an axis through the origin in any direction, with θ the angle between $\boldsymbol{\omega}$ and $\mathbf{r}$, the length $\omega|\mathbf{r}| \sin \theta$ of $\boldsymbol{\omega} \times \mathbf{r}$ is $\omega\rho$, since $|\mathbf{r}| \sin \theta$ is the distance from the axis, and its direction, at right angles to the axis and to $\mathbf{r}$ by the right-hand rule, is along the circle the way the body turns. Neither refers to the x , y and z axes, so $\boldsymbol{\omega} \times \mathbf{r}$ gives the velocity for any such axis.

The angular momentum about the axis is the sum over the parts of $m_i(x_i v_{y,i} - y_i v_{x,i})$, and with $v_{x,i} = -\omega y_i$ and $v_{y,i} = \omega x_i$,

$$L_z = \sum_i m_i \omega (x_i^2 + y_i^2) = I\omega, \quad I = \sum_i m_i \rho_i^2, \quad (5.8)$$

where $\rho_i^2 = x_i^2 + y_i^2$. The number I , the masses weighted by their squared distances from the axis, is the moment of inertia about the axis. The kinetic energy is likewise $\sum_i \frac{1}{2} m_i (\omega \rho_i)^2 = \frac{1}{2} I \omega^2$. Set beside $p = mv$ and $K = \frac{1}{2} m v^2$, these say that I does for turning what the mass does for moving: the larger I , the harder the body is to set turning or to stop, since a torque G_z about the fixed axis changes $L_z = I\omega$ at the rate G_z ([Equation \(5.6\)](#)), and so changes ω at the rate G_z/I .

Some moments of inertia

Two bodies on a rod. Masses m_1 and m_2 a distance d apart turn about an axis through their centre of mass at right angles to the line joining them. The centre of mass is at distances d_1 from m_1 and d_2 from m_2 , with $d_1 + d_2 = d$. Measured along the line from the centre of mass, m_1 is at $-d_1$ and m_2 at $+d_2$, and by [Section 2.9](#) $-m_1d_1 + m_2d_2 = 0$, so $m_1d_1 = m_2d_2$. Putting $d_2 = d - d_1$ gives $(m_1 + m_2)d_1 = m_2d$, so $d_1 = m_2d/(m_1 + m_2)$, and likewise $d_2 = m_1d/(m_1 + m_2)$. Then

$$I = m_1d_1^2 + m_2d_2^2 = \frac{m_1m_2^2 + m_2m_1^2}{(m_1 + m_2)^2} d^2 = \frac{m_1m_2}{m_1 + m_2} d^2 = m_r d^2, \quad (5.9)$$

since the top is $m_1m_2(m_2 + m_1)$ and one factor $m_1 + m_2$ cancels with the bottom: the reduced mass of [Section 4.8](#) times the square of the length. A diatomic molecule turns with this moment of inertia ([Section 5.8](#)).

A hoop of mass M and radius R , about its axis, has all its mass at the distance R , so $I = MR^2$.

A disc of mass M and radius R , of even thickness, about its axis: cut it into thin rings. The ring between ρ and $\rho + d\rho$ has the area $2\pi\rho d\rho$, its length times its width, out of the disc's πR^2 , so its mass is $M \times 2\pi\rho d\rho / (\pi R^2) = (2M/R^2)\rho d\rho$, all at the distance ρ . Adding the rings,

$$I = \int_0^R \rho^2 \frac{2M}{R^2} \rho d\rho = \frac{2M}{R^2} \int_0^R \rho^3 d\rho = \frac{2M}{R^2} \left[\frac{\rho^4}{4} \right]_0^R = \frac{2M}{R^2} \frac{R^4}{4} = \frac{1}{2}MR^2,$$

taking the constant outside and integrating ρ^3 by [Equation \(1.3\)](#); half the hoop's, since much of the disc's mass lies near the axis.

Pulling in the arms

A person sits on a stool that turns freely about a vertical axis, holding a weight of 2 kg in each hand at arm's length, 0.8 m from the axis. Say that the person and the stool have $I_0 = 1.5 \text{ kg m}^2$ about the axis, the same whether the arms are out or in. Then, treating the weights as points, $I = 1.5 + 2 \times 2 \times 0.8^2 = 4.06 \text{ kg m}^2$; set turning at 1 rad s^{-1} , the stool has $L_z = 4.06 \text{ kg m}^2 \text{ s}^{-1}$. Nothing outside exerts a torque about the axis: each weight is vertical, with $F_x = F_y = 0$, so its $G_z = xF_y - yF_x$ is zero, and the bearing turns freely; by the z component of [Equation \(5.6\)](#), L_z is conserved. Pulling the weights in to 0.2 m reduces I to $1.5 + 2 \times 2 \times 0.2^2 = 1.66 \text{ kg m}^2$, and since $L_z = I\omega$ is conserved, ω rises to $4.06/1.66 = 2.446 \text{ rad s}^{-1}$. The kinetic energy $\frac{1}{2}I\omega^2$ rises from $\frac{1}{2} \times 4.06 \times 1^2 = 2.030 \text{ J}$ to $\frac{1}{2} \times 1.66 \times (4.06/1.66)^2 = \frac{1}{2} \times 4.06^2/1.66 = 4.965 \text{ J}$; the arms

supply the difference, 2.935 J, by pulling the weights inwards, as the hand pulled the puck. A spinning skater who draws in the arms speeds up for the same reason.

Rolling

[Section 3.5](#) let a bead slide and noted that a rolling ball also spins, and moves more slowly. A wheel, a disc or a ball of radius R rolls without slipping when the point touching the ground is momentarily at rest. Seen from its centre, which moves at the velocity $\mathbf{V}$, the body only turns, so every part has the velocity $\mathbf{V}$ plus the turning velocity of [Equation \(5.7\)](#) measured from the centre, velocities adding as in [Section 2.2](#). For a body rolling in the $+x$ direction, with $\mathbf{V} = (V, 0, 0)$ and turning clockwise as seen from the $+z$ side, x to the right and y up, $\boldsymbol{\omega} = (0, 0, -\omega)$, and the bottom point, at $(0, -R, 0)$ from the centre, has the turning velocity $(0, 0, -\omega) \times (0, -R, 0) = (-\omega R, 0, 0)$. Its total velocity is $(V - \omega R, 0, 0)$, and rolling without slipping means

$$V = \omega R.$$

The kinetic energy splits like the angular momentum of [Section 5.3](#). With $\mathbf{v}_i = \mathbf{V} + \mathbf{v}'_i$ and $\sum_i m_i \mathbf{v}'_i = \mathbf{0}$,

$$K = \sum_i \frac{1}{2} m_i |\mathbf{V} + \mathbf{v}'_i|^2 = \frac{1}{2} M V^2 + \mathbf{V} \cdot \sum_i m_i \mathbf{v}'_i + \sum_i \frac{1}{2} m_i |\mathbf{v}'_i|^2 = \frac{1}{2} M V^2 + \frac{1}{2} I \omega^2,$$

expanding the square as a scalar product; the middle term vanishes, and the last is the energy of turning about the centre, $\frac{1}{2} I \omega^2$, since each $\mathbf{v}'_i$ is the turning velocity. With $\omega = V/R$ and $I = \kappa M R^2$, where κ is 1 for a hoop and $\frac{1}{2}$ for a disc,

$$K = \frac{1}{2} M V^2 + \frac{1}{2} \kappa M R^2 \frac{V^2}{R^2} = \frac{1}{2} (1 + \kappa) M V^2.$$

To roll, the body needs a grip on the ground, a friction force along the ground at the point of contact; but that point is at rest, so the power $\mathbf{F} \cdot \mathbf{v}$ of the friction, and of the ground's push at the same point, is zero and neither does work. The forces holding the body together do none either: a pair $\mathbf{F}_{ij} = c(\mathbf{r}_i - \mathbf{r}_j)$ has the power $\mathbf{F}_{ij} \cdot (\mathbf{v}_i - \mathbf{v}_j) = \frac{1}{2} c d|\mathbf{r}_i - \mathbf{r}_j|^2/dt$, zero since the distance is fixed ([Section 3.8](#)). Only the weight works, and a body rolling from rest down a height h has $\frac{1}{2}(1 + \kappa) M V^2 = Mgh$, so

$$V = \sqrt{\frac{2gh}{1 + \kappa}},$$

whatever its mass and radius. After a drop of 1 m, a block sliding without friction ($\kappa = 0$) moves at 4.429 m s^{-1} , a rolling disc at 3.616 m s^{-1} and a rolling hoop at 3.132 m s^{-1} :

a third of the disc's kinetic energy and half the hoop's are in the turning, $\kappa/(1+\kappa)$. On a slope at the angle α , the height fallen after a distance s is $s \sin \alpha$, so $V^2 = 2gs \sin \alpha / (1 + \kappa)$; differentiating with respect to time, by the chain rule on the left, $2V\dot{V} = 2g \sin \alpha \dot{s} / (1 + \kappa)$, and with $\dot{s} = V$, dividing both sides by $2V$, the acceleration is the constant

$$\dot{V} = \frac{g \sin \alpha}{1 + \kappa}.$$

On a slope of 30° , with $\sin 30^\circ = \frac{1}{2}$, it is $g/2 = 4.903$, $g/3 = 3.269$ and $g/4 = 2.452 \text{ m s}^{-2}$. From rest at constant acceleration each covers $s = \frac{1}{2}\dot{V}t^2$ (Equation (1.4)), so 2 m takes $t = \sqrt{2 \times 2/\dot{V}}$: 0.903, 1.106 and 1.277 s (Figure 5.4). Real wheels and balls lose a little energy as they and the ground flex, which we neglect. The spinning wheels of the braking car of Section 3.2 carry kinetic energy of this kind, a small part of the whole because the wheels are light compared with the car.

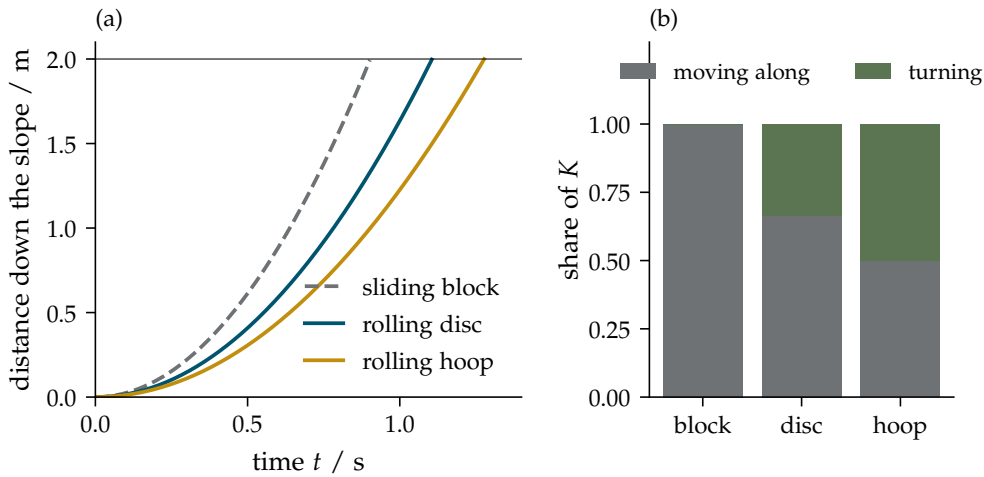


FIGURE 5.4: A block sliding without friction, a disc and a hoop rolling without slipping, all released from rest on a slope of 30° . (a) Distance down the slope against time; the line marks 2 m. (b) The share of each one's kinetic energy in moving along the slope, $\frac{1}{2}MV^2$, and in turning, $\frac{1}{2}I\omega^2$. Neither panel depends on the mass or the radius.

KEY IDEA. A rigid body turning at ω about a fixed axis moves with $\mathbf{v} = \omega \times \mathbf{r}$, and has $L_z = I\omega$ and $K = \frac{1}{2}I\omega^2$, with $I = \sum m\rho^2$. Two bodies a distance d apart have $I = m_r d^2$. A rolling body keeps part of its kinetic energy in its turning.

NOTEBOOK 05 race a block, a disc and a hoop down a slope

CHECK YOUR UNDERSTANDING. A hoop and a disc of the same mass and radius spin about their axes at the same angular speed. Which has more kinetic energy, and by what factor?

5.5 The inertia tensor

Section 5.4 found the angular momentum along the axis of turning, $L_z = I\omega$. A body can also have angular momentum across that axis: two weights fixed to the ends of a rod, the rod clamped at a slant to an axle, keep pulling sideways on the axle as they go round. This section finds the whole vector $\mathbf{L}$ for a rigid body turning in any direction, and with it the matrix that takes the place of the mass.

Angular momentum of a turning body

Let a rigid body turn with angular velocity $\boldsymbol{\omega}$ about an axis through its centre of mass, and measure each position $\mathbf{r}_i$ from the centre of mass. Each part moves with $\mathbf{v}_i = \boldsymbol{\omega} \times \mathbf{r}_i$, so

$$\mathbf{L} = \sum_i m_i \mathbf{r}_i \times (\boldsymbol{\omega} \times \mathbf{r}_i).$$

Two rules for products of three vectors turn this into something computable.

TRIPLE PRODUCTS. *The vector triple product.* For any three vectors,

$$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = \mathbf{b}(\mathbf{a} \cdot \mathbf{c}) - \mathbf{c}(\mathbf{a} \cdot \mathbf{b}).$$

To check the x component, use $(\mathbf{b} \times \mathbf{c})_z = b_x c_y - b_y c_x$ and $(\mathbf{b} \times \mathbf{c})_y = b_z c_x - b_x c_z$:

$$\begin{aligned} [\mathbf{a} \times (\mathbf{b} \times \mathbf{c})]_x &= a_y(b_x c_y - b_y c_x) - a_z(b_z c_x - b_x c_z) \\ &= b_x(a_y c_y + a_z c_z) - c_x(a_y b_y + a_z b_z) \quad \text{multiplying out} \\ &= b_x(\mathbf{a} \cdot \mathbf{c}) - c_x(\mathbf{a} \cdot \mathbf{b}) \quad \text{adding and subtracting } a_x b_x c_x. \end{aligned}$$

The y and z components follow by renaming $x \rightarrow y \rightarrow z \rightarrow x$.

The scalar triple product. Writing out

$$\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = a_x(b_y c_z - b_z c_y) + a_y(b_z c_x - b_x c_z) + a_z(b_x c_y - b_y c_x)$$

gives six products, each of one component of every vector. Writing out

$$\mathbf{b} \cdot (\mathbf{c} \times \mathbf{a}) = b_x(c_y a_z - c_z a_y) + b_y(c_z a_x - c_x a_z) + b_z(c_x a_y - c_y a_x)$$

in the same way gives the same six with the same signs, in a different order. So the three vectors may be moved round in a cycle, $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = \mathbf{b} \cdot (\mathbf{c} \times \mathbf{a}) = \mathbf{c} \cdot (\mathbf{a} \times \mathbf{b})$.

By the vector triple product with $\mathbf{a} = \mathbf{c} = \mathbf{r}_i$ and $\mathbf{b} = \boldsymbol{\omega}$, $\mathbf{r}_i \times (\boldsymbol{\omega} \times \mathbf{r}_i) = \boldsymbol{\omega} |\mathbf{r}_i|^2 - \mathbf{r}_i (\mathbf{r}_i \cdot \boldsymbol{\omega})$. Its component a , for $a = x, y$ or z , is $\omega_a |\mathbf{r}_i|^2 - (\mathbf{r}_i)_a \sum_b (\mathbf{r}_i)_b \omega_b$, with $(\mathbf{r}_i)_a$ the component a of $\mathbf{r}_i$. Writing $\omega_a = \sum_b \delta_{ab} \omega_b$, where the Kronecker delta δ_{ab} is 1 when $a = b$ and 0 otherwise, so that only the term with $b = a$ survives, both terms become sums over b ; multiplying by m_i , adding over the atoms and taking the sum over b outside,

$$L_a = \sum_b I_{ab} \omega_b, \quad I_{ab} = \sum_i m_i (|\mathbf{r}_i|^2 \delta_{ab} - (\mathbf{r}_i)_a (\mathbf{r}_i)_b), \quad (5.10)$$

or $\mathbf{L} = \mathbf{I}\boldsymbol{\omega}$ with the matrix product of [Section 4.9](#). The 3×3 matrix $\mathbf{I}$, set in sans-serif as this book sets tensors, apart from the identity $\mathbf{I}$, is the inertia tensor of the body about its centre of mass. It is symmetric, $I_{ab} = I_{ba}$, since swapping a and b changes neither δ_{ab} nor $(\mathbf{r}_i)_a (\mathbf{r}_i)_b$. Its diagonal entries are the moments of inertia about the three axes, for instance $I_{zz} = \sum_i m_i (x_i^2 + y_i^2 + z_i^2 - z_i^2) = \sum_i m_i (x_i^2 + y_i^2)$, as in [Equation \(5.8\)](#); its other entries, such as $I_{xy} = -\sum_i m_i x_i y_i$, are the source of the sideways angular momentum. For $\boldsymbol{\omega} = (0, 0, \omega)$, $\mathbf{L} = \omega (I_{xz}, I_{yz}, I_{zz})$, along the axis only when I_{xz} and I_{yz} vanish.

The kinetic energy follows with the scalar triple product. Since $\mathbf{v}_i = \boldsymbol{\omega} \times \mathbf{r}_i$,

$$|\mathbf{v}_i|^2 = \mathbf{v}_i \cdot (\boldsymbol{\omega} \times \mathbf{r}_i) = \boldsymbol{\omega} \cdot (\mathbf{r}_i \times \mathbf{v}_i),$$

moving the vectors round in a cycle, so that multiplying by $\frac{1}{2}m_i$ and adding, with $\sum_i m_i \mathbf{r}_i \times \mathbf{v}_i = \mathbf{L}$,

$$K = \frac{1}{2} \boldsymbol{\omega} \cdot \mathbf{L} = \frac{1}{2} \boldsymbol{\omega} \cdot \mathbf{I}\boldsymbol{\omega}. \quad (5.11)$$

Principal axes

Because $\mathbf{I}$ is symmetric, it has three real eigenvalues $I_1 \leq I_2 \leq I_3$ with eigenvectors at right angles to one another ([Section 4.9](#)). The eigenvectors are the principal axes of the body and the eigenvalues its principal moments of inertia. Turning about a principal axis, $\mathbf{I}\boldsymbol{\omega} = I_k \boldsymbol{\omega}$, and the angular momentum lies along the axis; turning about any other axis, it does not. Write $\boldsymbol{\omega} = \omega_1 \mathbf{e}_1 + \omega_2 \mathbf{e}_2 + \omega_3 \mathbf{e}_3$, with $\mathbf{e}_k$ the principal axes as vectors of length 1 and ω_k the parts of $\boldsymbol{\omega}$ along them. The product is linear, so $\mathbf{I}\boldsymbol{\omega} = I_1 \omega_1 \mathbf{e}_1 + I_2 \omega_2 \mathbf{e}_2 + I_3 \omega_3 \mathbf{e}_3$, and since $\mathbf{e}_j \cdot \mathbf{e}_k$ is 1 for $j = k$ and 0 otherwise, [Equation \(5.11\)](#) becomes $K = \frac{1}{2} (I_1 \omega_1^2 + I_2 \omega_2^2 + I_3 \omega_3^2)$, as for the quadratic term in [Section 4.9](#). With the principal axes as the coordinate axes, $\mathbf{e}_1 = (1, 0, 0)$ and so on, and column k of the tensor, $\mathbf{I}\mathbf{e}_k = I_k \mathbf{e}_k$, holds I_k on the diagonal and zeros elsewhere: the tensor is diagonal.

A wheel turning steadily about an axle that is not a principal axis has an angular momentum that goes round with the wheel: worked out in axes fixed in the wheel,

with the axle as their z axis, $\mathbf{L} = \omega (I_{xz}, I_{yz}, I_{zz})$ does not change, so its sideways part turns with the wheel. Its direction changes, so by Equation (5.6) the bearings must supply an external torque all the time, and the axle shakes. Car wheels are balanced by clipping small weights to their rims until the axle passes through the centre of mass and is a principal axis.

A water molecule

The measured equilibrium geometry of water⁷ has O–H bonds of 0.958 Å at an angle of 104.4776°. Place the oxygen atom at the origin, one hydrogen atom on the x axis, at (0.958, 0, 0), and the other in the xy plane, at

$$(0.958 \cos 104.4776^\circ, 0.958 \sin 104.4776^\circ, 0) = (-0.2395, 0.9276, 0) \text{ Å}$$

(Figure 5.5), with the masses 15.999 and 1.008 amu. The centre of mass is at $1.008 \times (0.7185, 0.9276, 0) / 18.015 = (0.0402, 0.0519, 0) \text{ Å}$, and about it Equation (5.10) gives, in amu Å^2 ,

$$\mathbf{I} = \begin{pmatrix} 0.8188 & 0.2615 & 0 \\ 0.2615 & 0.9538 & 0 \\ 0 & 0 & 1.7726 \end{pmatrix}.$$

The atoms lie in the plane $z = 0$, so $I_{xz} = I_{yz} = 0$ and the z axis, at right angles to the molecule, is a principal axis, with $I_3 = 1.7726$. The other two come from the 2×2 block in the top left, by the characteristic equation of Section 4.9, multiplied out with $0.8188 + 0.9538 = 1.7726$ and $0.8188 \times 0.9538 - 0.2615^2 = 0.7126$:

$$(0.8188 - \lambda)(0.9538 - \lambda) - 0.2615^2 = \lambda^2 - 1.7726\lambda + 0.7126 = 0,$$

so $\lambda = \frac{1}{2}(1.7726 \pm \sqrt{1.7726^2 - 4 \times 0.7126}) = \frac{1}{2}(1.7726 \pm 0.5401)$, giving $I_1 = 0.616$ and $I_2 = 1.156 \text{ amu Å}^2$. The axis of I_2 is the line that halves the angle between the bonds, an axis of symmetry of the molecule: putting $\lambda = 1.1564$ into the first equation of the pair in Section 4.9, $(0.8188 - 1.1564)u_1 + 0.2615u_2 = 0$, gives $u_2/u_1 = 0.3376/0.2615 = 1.291$, a line at $\arctan 1.291 = 52.24^\circ$ to the x axis, half of 104.4776° . The axis of I_1 is at right angles to it in the plane. The two roots, $\frac{1}{2}(1.7726 \pm 0.5401)$, add to $1.7726 = 0.8188 + 0.9538 = I_{xx} + I_{yy}$, and for a flat body $I_{xx} + I_{yy} = I_{zz}$ (Exercise 5.13), so the largest moment, neither of the other two being negative, each a sum of masses times squared distances, is the sum of the other two; computed without rounding, $I_1 = 0.6162$ and $I_2 = 1.1564$.

A linear molecule, with every atom on one line, has a zero principal moment about that line, since every ρ_i is zero, and two equal ones about the axes at right angles to it.

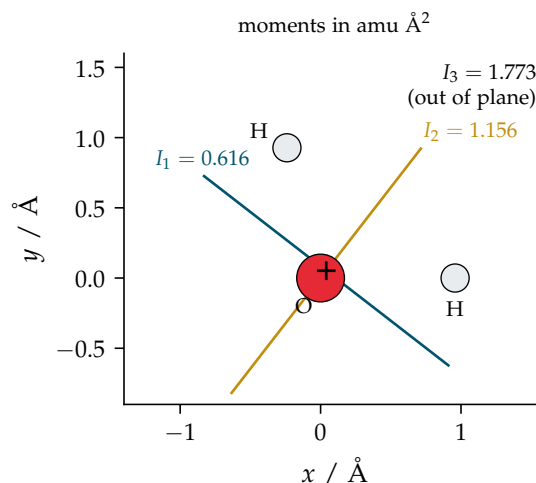


FIGURE 5.5: A water molecule at its measured geometry, one bond along x , with its centre of mass (cross) and the two principal axes in its plane, labelled with their principal moments in $\text{amu } \text{\AA}^2$. The third principal axis points out of the page, with $I_3 = 1.773 \text{ amu } \text{\AA}^2$.

KEY IDEA. For a rigid body turning about its centre of mass, $\mathbf{L} = I\boldsymbol{\omega}$ and $K = \frac{1}{2}\boldsymbol{\omega} \cdot I\boldsymbol{\omega}$, with the symmetric inertia tensor $I_{ab} = \sum_i m_i (|\mathbf{r}_i|^2 \delta_{ab} - (\mathbf{r}_i)_a (\mathbf{r}_i)_b)$. Its eigenvectors, the principal axes, are the directions in which $\mathbf{L}$ is parallel to $\boldsymbol{\omega}$.

NOTEBOOK 05 the inertia tensor of water and its principal axes

CHECK YOUR UNDERSTANDING. Two atoms of mass m sit on the z axis at $z = \pm d/2$. Write down their inertia tensor about their midpoint, and its principal moments.

5.6 Removing translation and rotation

A simulation of a single molecule starts from velocities chosen at random. Their total momentum is not zero, so the molecule drifts, and their angular momentum is not zero, so it tumbles. Section 2.10 removed the drift by a change of frame. This section removes the tumbling too, and shows that what remains is the motion of the atoms relative to one another, with a kinetic energy of its own. The angular velocity of the tumbling must be found from the angular momentum, which needs one more tool.

LINEAR SYSTEMS AND THE INVERSE MATRIX. Three equations for three unknowns, such as

$$2x + y = 5, \quad x + 3y + z = 8, \quad y + 2z = 7,$$

are solved by elimination: use one equation to remove an unknown from the others, and repeat. The first gives $x = (5 - y)/2$; putting this into the second and

multiplying by 2 gives $5 - y + 6y + 2z = 16$, or $5y + 2z = 11$; subtracting the third, $y + 2z = 7$, removes z and leaves $4y = 4$, so $y = 1$, then $z = 3$ from the third and $x = 2$ from the first. In matrix form the equations are $\mathbf{A}\mathbf{u} = \mathbf{b}$, with $\mathbf{A}$ the matrix of the numbers multiplying the unknowns, rows $(2, 1, 0)$, $(1, 3, 1)$ and $(0, 1, 2)$, $\mathbf{u} = (x, y, z)$ and $\mathbf{b} = (5, 8, 7)$.

When elimination gives one answer for every $\mathbf{b}$, the matrix has an inverse $\mathbf{A}^{-1}$, the matrix that undoes it, $\mathbf{A}^{-1}(\mathbf{A}\mathbf{u}) = \mathbf{u}$, and multiplying both sides of $\mathbf{A}\mathbf{u} = \mathbf{b}$ by $\mathbf{A}^{-1}$ gives the solution $\mathbf{u} = \mathbf{A}^{-1}\mathbf{b}$. When elimination instead removes a whole equation, leaving $0 = 0$ or $0 = \text{a number that is not zero}$, the matrix has no inverse and there are many solutions or none. With $\mathbf{b} = \mathbf{0}$ there is always the solution $\mathbf{u} = \mathbf{0}$, so a matrix with no inverse turns some vector $\mathbf{u} \neq \mathbf{0}$ into $\mathbf{0}$. NumPy solves $\mathbf{A}\mathbf{u} = \mathbf{b}$ with `np.linalg.solve`; `np.linalg.lstsq` also handles a matrix with no inverse, returning, when there are many solutions, the shortest.

The recipe

Take the masses m_i , positions $\mathbf{r}_i$ and velocities $\mathbf{v}_i$ of the atoms, and measure positions from the centre of mass, $\mathbf{r}'_i = \mathbf{r}_i - \mathbf{R}$.

1. Subtract the velocity of the centre of mass, $\mathbf{V} = \mathbf{P}/M$, from every velocity: $\mathbf{v}'_i = \mathbf{v}_i - \mathbf{V}$.
2. Find the angular momentum about the centre of mass, $\mathbf{L}' = \sum_i m_i \mathbf{r}'_i \times \mathbf{v}'_i$, and the inertia tensor $\mathbf{I}$ about it, [Equation \(5.10\)](#), and solve $\mathbf{I}\boldsymbol{\omega} = \mathbf{L}'$ for $\boldsymbol{\omega}$, the angular velocity of the rigid turning that carries this angular momentum.
3. Subtract that turning from every velocity: $\mathbf{u}_i = \mathbf{v}'_i - \boldsymbol{\omega} \times \mathbf{r}'_i$.

The velocities $\mathbf{u}_i$ that remain have no total momentum and no angular momentum. For the momentum, $\sum_i m_i \mathbf{u}_i = \sum_i m_i \mathbf{v}'_i - \boldsymbol{\omega} \times \sum_i m_i \mathbf{r}'_i = \mathbf{0} - \mathbf{0}$, by the two sums of [Section 5.3](#). For the angular momentum,

$$\sum_i m_i \mathbf{r}'_i \times \mathbf{u}_i = \mathbf{L}' - \sum_i m_i \mathbf{r}'_i \times (\boldsymbol{\omega} \times \mathbf{r}'_i) = \mathbf{L}' - \mathbf{I}\boldsymbol{\omega} = \mathbf{0},$$

by the calculation that led to [Equation \(5.10\)](#). The inertia tensor has an inverse unless all the atoms lie on one line: if $\mathbf{I}\mathbf{n} = \mathbf{0}$ with $\mathbf{n} \neq \mathbf{0}$, turning with $\boldsymbol{\omega} = \mathbf{n}$ has, by [Equation \(5.11\)](#), $\sum_i \frac{1}{2} m_i |\mathbf{n} \times \mathbf{r}'_i|^2 = \frac{1}{2} \mathbf{n} \cdot \mathbf{I}\mathbf{n} = 0$, so every $\mathbf{n} \times \mathbf{r}'_i = \mathbf{0}$ and every atom lies on the line through the centre of mass along $\mathbf{n}$. For a linear molecule along the z axis, $\mathbf{I}$ is diagonal with entries J , J and 0 , where $J = \sum_i m_i z_i'^2$, and $\mathbf{L}'$ has no z part (every $\mathbf{r}'_i \times \mathbf{v}'_i$ is at right angles to $\mathbf{r}'_i$, which lies along z). Then $\omega_x = L'_x/J$ and $\omega_y = L'_y/J$,

while ω_z may be anything, since turning about the line moves no atom. We take $\omega_z = 0$, the shortest of these solutions, which is what `lstsq` returns.

Three kinds of kinetic energy

Every velocity is now a sum of three parts, $\mathbf{v}_i = \mathbf{V} + \boldsymbol{\omega} \times \mathbf{r}'_i + \mathbf{u}_i$: the drift, the turning and the rest. Expanding $|\mathbf{v}_i|^2 = \mathbf{v}_i \cdot \mathbf{v}_i$ term by term,

$$|\mathbf{v}_i|^2 = V^2 + |\boldsymbol{\omega} \times \mathbf{r}'_i|^2 + |\mathbf{u}_i|^2 + 2\mathbf{V} \cdot (\boldsymbol{\omega} \times \mathbf{r}'_i) + 2\mathbf{V} \cdot \mathbf{u}_i + 2(\boldsymbol{\omega} \times \mathbf{r}'_i) \cdot \mathbf{u}_i,$$

and adding $\frac{1}{2}m_i$ times this over the atoms gives three squares and three cross terms, each with $\frac{1}{2} \times 2 = 1$, and the cross terms all vanish:

- $\sum_i m_i \mathbf{V} \cdot (\boldsymbol{\omega} \times \mathbf{r}'_i) = \mathbf{V} \cdot (\boldsymbol{\omega} \times \sum_i m_i \mathbf{r}'_i) = 0$;
- $\sum_i m_i \mathbf{V} \cdot \mathbf{u}_i = \mathbf{V} \cdot \sum_i m_i \mathbf{u}_i = 0$;
- $\sum_i m_i (\boldsymbol{\omega} \times \mathbf{r}'_i) \cdot \mathbf{u}_i = \boldsymbol{\omega} \cdot \sum_i m_i \mathbf{r}'_i \times \mathbf{u}_i = 0$, moving the vectors round in a cycle (Section 5.5).

With Equation (5.11) for the turning, therefore,

$$K = \frac{1}{2}MV^2 + \frac{1}{2}\boldsymbol{\omega} \cdot \mathbf{I}\boldsymbol{\omega} + \sum_i \frac{1}{2}m_i|\mathbf{u}_i|^2 : \quad (5.12)$$

the kinetic energy of the drift, of the turning and of the motion of the atoms relative to one another, each separate from the others.

Figure 5.6 applies the recipe to a model ring of six carbon atoms, 1.4 Å from its centre, given velocities that drift, turn and jostle. At the start $\mathbf{P} = (0.4524, 0.1738, 0)$ amu Å fs⁻¹ and $L'_z = 0.7616$ amu Å² fs⁻¹. Removing the drift leaves L'_z unchanged, since $\sum_i m_i \mathbf{r}'_i \times \mathbf{V} = (\sum_i m_i \mathbf{r}'_i) \times \mathbf{V} = \mathbf{0}$. The ring lies flat with its velocities in its plane, so, as for water, $I_{xz} = I_{yz} = 0$ and $\mathbf{L}'$ points along z , and $\boldsymbol{\omega} = (0, 0, L'_z/I_{zz})$. With $M = 6 \times 12.011 = 72.066$ amu and $I_{zz} = 6 \times 12.011 \times 1.4^2 = 141.25$ amu Å², $\omega_z = 0.7616/141.25 = 0.00539$ rad fs⁻¹, and removing that turning leaves both at zero. The kinetic energy of 0.4317 eV splits into $|\mathbf{P}|^2/2M = 0.2349/(2 \times 72.066) \times 103.6427 = 0.1689$ eV of drift, $L_z'^2/2I_{zz} = 0.7616^2/(2 \times 141.25) \times 103.6427 = 0.2128$ eV of turning, and the remaining 0.0500 eV of motion within the ring.

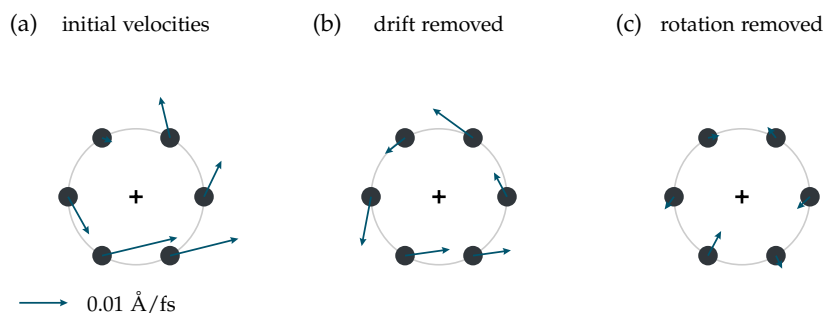


FIGURE 5.6: A model ring of six carbon atoms with velocities in its plane (arrows) made of a drift, a turn and a random part. (a) As made. (b) After subtracting the velocity of the centre of mass (cross): no drift, but the ring still turns. (c) After also subtracting the rigid turning $\omega \times \mathbf{r}'$: what remains moves the atoms relative to one another.

In code

The function `remove_rigid_motion` of `mdlab` follows the recipe:

```

1 def remove_rigid_motion(
2     masses: ArrayLike,
3     positions: ArrayLike,
4     velocities: ArrayLike,
5     rotation: bool = True,
6 ) -> NDArray:
7     masses = np.asarray(masses, dtype=float)
8     positions = np.asarray(positions, dtype=float)
9     velocities = np.asarray(velocities, dtype=float)
10    total = masses.sum()
11    drift = masses @ velocities / total #  $V = P/M$ 
12    velocities = velocities - drift
13    if rotation:
14        centre = masses @ positions / total #  $R$ 
15        omega = angular_velocity(masses, positions, velocities)
16        velocities = velocities - np.cross(omega, positions - centre)
17    return velocities

```

`masses @ velocities` is NumPy's matrix product of the list of N masses with the table of velocities, N rows of three: it multiplies each row by its mass and adds the rows, giving $\sum_i m_i \mathbf{v}_i = \mathbf{P}$. `velocities - drift` takes the three numbers of drift from every row: NumPy lines up the last dimensions ([Section 2.10](#)) and repeats the shorter array down the N rows, and `positions - centre` works the same way. `angular_velocity` computes $\mathbf{L}'$ and $\mathbf{I}$ about the centre of mass and solves for ω with `np.linalg.lstsq`, and `np.cross` takes the cross product of ω with every row of `positions - centre`. With `rotation=False` only the drift is removed. The tests check the result against ASE, an independent library for atomistic simulation, and check [Equation \(5.12\)](#).

IN PRACTICE. Most simulations follow a box of atoms repeated periodically in every direction (Chapter 10). There the forces still sum to zero, so the total momentum is conserved and its drift is removed, with `rotation=False`; but turning the atoms while the box and its copies stay fixed changes their energy, so (Chapter 7) the angular momentum is not conserved, and removing it would be wrong. For an isolated molecule both are removed. The learned propagator TrajCast (discussed in the planned Chapter 28) applies the same correction to the velocities it predicts: it adds the one velocity that restores the total momentum to its value before the step, and, for isolated molecules, the rigid turning $\boldsymbol{\omega} \times (\mathbf{r}_i - \mathbf{R})$ with $\boldsymbol{\omega}$ found from the inertia tensor, at the predicted positions, and the change in angular momentum.

NOTEBOOK 05 *switch the drift and the turning off and on, and watch the three energies*

CHECK YOUR UNDERSTANDING. Two atoms of 1 amu at $(\mp 1, 0, 0)$ Å move with velocities $(0, 0.01, 0)$ and $(0, 0.03, 0)$ Å fs⁻¹. Find $\mathbf{V}$ and $\boldsymbol{\omega}$. What is left?

5.7 Degrees of freedom

The positions of N atoms are $3N$ numbers, so a molecule can move in $3N$ independent ways, called its degrees of freedom. Some of these move the molecule as a whole, without changing any distance inside it. This section counts them, shows that they appear as zero frequencies among the normal modes of [Section 4.9](#), and finds how many ways a molecule has left to vibrate.

Counting

A rigid motion of the whole molecule is a drift in some direction, a combination of drifts along x , y and z , three ways, plus a turning about some axis through the centre of mass, a combination of turnings about x , y and z , three more. For a linear molecule, turning about its own line moves no atom ([Section 5.6](#)), so it has only two. What is left changes the shape of the molecule:

$$3N - 6 \text{ internal degrees of freedom, or } 3N - 5 \text{ for a linear molecule.}$$

Water, with $N = 3$, has $9 - 6 = 3$, the two stretches and the bend of [Table 4.1](#); carbon dioxide, linear, has $9 - 5 = 4$; methane, CH₄, has $15 - 6 = 9$, and benzene, C₆H₆, $36 - 6 = 30$.

Rigid motions as zero modes

Near its minimum the motion of a molecule is a sum of normal modes, with squared angular frequencies the eigenvalues of the mass-weighted Hessian (Section 4.9). A rigid motion stretches nothing, so we expect nothing to pull it back: a mode of zero frequency. We show this for a molecule held together by springs between pairs of atoms.

The Hessian of one spring. A spring of stiffness k joins atoms i and j and stores $\frac{1}{2}k(r_{ij} - d)^2$, with d its natural length; we take every spring to be at its natural length at the minimum, $r_{ij} = d$, with $\mathbf{e} = (\mathbf{r}_j - \mathbf{r}_i)/d$ the unit vector along it. Displace the atoms by small amounts $\mathbf{q}_i$ and $\mathbf{q}_j$. The new separation is $(\mathbf{r}_j + \mathbf{q}_j) - (\mathbf{r}_i + \mathbf{q}_i) = d\mathbf{e} + \mathbf{q}_j - \mathbf{q}_i$, since $\mathbf{r}_j - \mathbf{r}_i = d\mathbf{e}$, and the new length is its length. For a vector $\mathbf{a}$ and a small vector $\mathbf{s}$ of length s , expand $|\mathbf{a} + \mathbf{s}|^2 = (\mathbf{a} + \mathbf{s}) \cdot (\mathbf{a} + \mathbf{s})$ and set beside it the square of a guess for the length, chosen so that the first two terms agree:

$$|\mathbf{a} + \mathbf{s}|^2 = |\mathbf{a}|^2 + 2\mathbf{a} \cdot \mathbf{s} + s^2,$$

$$\left(|\mathbf{a}| + \frac{\mathbf{a} \cdot \mathbf{s}}{|\mathbf{a}|}\right)^2 = |\mathbf{a}|^2 + 2\mathbf{a} \cdot \mathbf{s} + \frac{(\mathbf{a} \cdot \mathbf{s})^2}{|\mathbf{a}|^2}.$$

The two squares differ by $s^2 - (\mathbf{a} \cdot \mathbf{s})^2/|\mathbf{a}|^2$, of order s^2 since $|\mathbf{a} \cdot \mathbf{s}| = |\mathbf{a}|s|\cos\theta| \leq |\mathbf{a}|s$, with θ the angle between them. Two positive numbers X and Y differ by $(X^2 - Y^2)/(X + Y)$, and here $X + Y$ is close to $2|\mathbf{a}|$, so the numbers themselves differ by order s^2 too:

$$|\mathbf{a} + \mathbf{s}| = |\mathbf{a}| + \frac{\mathbf{a} \cdot \mathbf{s}}{|\mathbf{a}|} + \mathcal{O}(s^2).$$

With $\mathbf{a} = d\mathbf{e}$ and $\mathbf{s} = \mathbf{q}_j - \mathbf{q}_i$, the stretch is $r_{ij} - d = \mathbf{e} \cdot (\mathbf{q}_j - \mathbf{q}_i) + \mathcal{O}(q^2)$, with q the size of the displacements. Squaring, the correction of order q^2 appears only multiplied by $\mathbf{e} \cdot (\mathbf{q}_j - \mathbf{q}_i)$, of order q , or by itself, so the energy, to second order in the displacements, is

$$U = \frac{1}{2}k [\mathbf{e} \cdot (\mathbf{q}_j - \mathbf{q}_i)]^2 :$$

only the part of the relative displacement along the spring stretches it. By the chain rule, since $\mathbf{e} \cdot \mathbf{q}_j = \sum_b e_b (\mathbf{q}_j)_b$ with $(\mathbf{q}_j)_b$ the component b of $\mathbf{q}_j$, $\partial U / \partial (\mathbf{q}_j)_a = k e_a \mathbf{e} \cdot (\mathbf{q}_j - \mathbf{q}_i)$ and $\partial U / \partial (\mathbf{q}_i)_a = -k e_a \mathbf{e} \cdot (\mathbf{q}_j - \mathbf{q}_i)$. Differentiating once more, $\partial^2 U / \partial (\mathbf{q}_j)_a \partial (\mathbf{q}_j)_b = \partial^2 U / \partial (\mathbf{q}_i)_a \partial (\mathbf{q}_i)_b = k e_a e_b$ and $\partial^2 U / \partial (\mathbf{q}_i)_a \partial (\mathbf{q}_j)_b = -k e_a e_b$. The Hessian of the spring is made of these 3×3 blocks, and that of a molecule is the sum over its springs. The function `spring_hessian` of `mdlab` builds it, and its tests check it against second differences of the energy.

Rigid displacements cost no energy. Move every atom by the same small vector $\mathbf{c}$: then $\mathbf{q}_j - \mathbf{q}_i = \mathbf{0}$ for every spring, and U does not change. Turn the molecule through a

small angle about an axis along the unit vector $\mathbf{n}$ through the origin: by Equation (5.7), turning at the angular speed ω for a short time t moves each atom by its velocity times t , $\mathbf{q}_i = \omega t \mathbf{n} \times \mathbf{r}_i = \theta \mathbf{n} \times \mathbf{r}_i$, with $\theta = \omega t$ the small angle turned. Then

$$\mathbf{e} \cdot (\mathbf{q}_j - \mathbf{q}_i) = \theta \mathbf{e} \cdot (\mathbf{n} \times (\mathbf{r}_j - \mathbf{r}_i)) = \theta d \mathbf{e} \cdot (\mathbf{n} \times \mathbf{e}) = 0,$$

since $\mathbf{n} \times \mathbf{e}$ is at right angles to $\mathbf{e}$. Again no spring is stretched. The force of a spring on atom j is minus the derivative of its energy, by the chain rule $-k \mathbf{e} [\mathbf{e} \cdot (\mathbf{q}_j - \mathbf{q}_i)]$, and that on atom i is its opposite, so for both kinds of rigid displacement every force vanishes. The force is $-\Phi \mathbf{q}$ (Section 4.9), so $\Phi \mathbf{q} = \mathbf{0}$. With $w_b = \sqrt{m_b} q_b$, where a and b now run over all $3N$ coordinates as in Section 4.9, row a of the mass-weighted Hessian gives

$$\sum_b \frac{\Phi_{ab}}{\sqrt{m_a m_b}} \sqrt{m_b} q_b = \frac{1}{\sqrt{m_a}} \sum_b \Phi_{ab} q_b = 0,$$

cancelling $\sqrt{m_b}$: $\mathbf{w}$ is an eigenvector with eigenvalue zero, a normal mode of zero frequency (Exercise 5.18 checks a case by hand). The three drifts and three turnings are independent, none a combination of the others, and a symmetric matrix has as many zero eigenvalues as it has independent vectors that it turns into zero (a general result we use without proof, like that of Section 4.9), so they give six zero modes, five for a linear molecule. The other $3N - 6$ ($3N - 5$) eigenvalues are positive only if every change of shape costs energy, as at the minimum of a real molecule.

The same holds for any potential energy that does not change when the whole molecule is moved or turned, as for every isolated molecule, provided the atoms sit at a minimum. A moved or turned copy of the minimum is then a minimum too, where every force vanishes. A small rigid displacement $\eta \mathbf{u}$ lands within $\mathcal{O}(\eta^2)$ of such a copy, so the force there, $-\eta \Phi \mathbf{u}$ to first order, is of order η^2 ; dividing by η and letting η shrink leaves $\Phi \mathbf{u} = \mathbf{0}$.

A spring model of water

Model water, at the measured geometry and with the masses of Section 5.5, by three springs: two O–H bonds of stiffness k_{OH} , and an H–H spring of stiffness k_{HH} that stands in for the molecule's resistance to bending; without it the bend would cost no energy and give a seventh zero eigenvalue. Its Hessian is 9×9 , and with the masses of the atoms normal_modes gives nine eigenvalues. Six are zero to within the rounding error of the computer's arithmetic, which keeps about 16 significant figures, all below $1 \times 10^{-15} \text{ fs}^{-2}$ against 0.09 fs^{-2} for the slowest vibration: the three drifts and the three turnings. In the antisymmetric stretch one hydrogen atom moves out as the other moves in, and the H–H distance does not change to first order, so k_{OH} alone sets its frequency.

Choosing $k_{\text{OH}} = 48.48 \text{ eV } \text{\AA}^{-2}$ to match the measured antisymmetric stretch of Table 4.1, and then $k_{\text{HH}} = 17.66 \text{ eV } \text{\AA}^{-2}$ to match the bend, makes the other three eigenvalues 0.09028, 0.5006 and 0.7338 fs^{-2} ; with ω the square root of each and $\tilde{\nu} = \omega / (2\pi c)$ from Section 4.10, they are 1595, 3756 and 4548 cm^{-1} , the last, the symmetric stretch, 24% above the measured 3657 cm^{-1} . The count of modes is exact; their frequencies are only as good as the model, and three springs are a crude model of a molecule. A single spring between two atoms likewise gives five zero eigenvalues and one vibration.

FROM ELECTRONIC STRUCTURE. A vibrational analysis from electronic-structure calculations builds the Hessian, usually by finite differences of the forces, and diagonalises its mass-weighted form (Section 4.10). The six rigid modes then come out close to zero but not at zero: the geometry is never exactly at the minimum, and the forces left over give the turnings a small stiffness of either sign; the finite differences add errors of their own; and a real-space integration grid is not quite unchanged when the atoms are moved or turned against it. The rigid motions can then be projected out of the Hessian before it is diagonalised, the matrix counterpart of the recipe of Section 5.6.

The count returns in Chapter 12, which measures temperature by the kinetic energy shared among the degrees of freedom in which the atoms are free to move. A simulation that removes the drift has three fewer, and one of an isolated molecule that also removes the turning has six fewer, five if the molecule is linear.

KEY IDEA. A molecule of N atoms has $3N$ degrees of freedom: three drifts, three turnings (two if it is linear) and $3N - 6$ ($3N - 5$) internal motions. The rigid motions stretch nothing, and appear as zero eigenvalues of the mass-weighted Hessian.

NOTEBOOK 05 *the nine modes of the spring model of water*

CHECK YOUR UNDERSTANDING. How many internal degrees of freedom has ethane, C_2H_6 , and how many has acetylene, C_2H_2 , whose four atoms lie on a line?

5.8 Spinning molecules

Molecules in a gas tumble as they fly, and the turning of a molecule is slow compared with its vibration. This section puts numbers to that for a molecule of lithium fluoride, LiF, one lithium atom bonded to one fluorine atom.

Lithium fluoride

The two atoms of a free LiF molecule are $1.5639 \text{ \AA}$ apart at the bottom of their well⁶. With the masses 6.94 and 18.998 amu, the reduced mass is $6.94 \times 18.998 / 25.938 =$

5.0831 amu, and by Equation (5.9) the moment of inertia about an axis through the centre of mass at right angles to the bond is

$$I = m_{\text{r}} d^2 = 5.0831 \times 1.5639^2 = 12.432 \text{ amu } \text{\AA}^2.$$

We use the average mass of lithium atoms; the measured constants belong to molecules made with the commonest, slightly heavier kind, for which I is about 1 % larger. The lighter lithium atom lies $18.998/25.938 \times 1.5639 = 1.1455 \text{ \AA}$ from the centre of mass and the fluorine atom $0.4184 \text{ \AA}$, so the lithium atom does most of the moving.

How fast does it turn? Chapter 12 shows that at a temperature T a freely turning molecule of two atoms carries, on average, the energy $k_{\text{B}}T$ in its turning, where $k_{\text{B}} = 8.617333 \times 10^{-5} \text{ eV K}^{-1}$ is Boltzmann's constant; at room temperature, 300 K (the kelvin is the degree of the Celsius scale, counted from -273.15°C , so 300 K is about 27°C), that is $8.617333 \times 10^{-5} \times 300 = 0.025852 \text{ eV}$. In the units of atoms an energy in eV becomes one in $\text{amu } \text{\AA}^2 \text{ fs}^{-2}$ on multiplying by 9.648533×10^{-3} (Section 3.12), and for a molecule with this average energy $K = \frac{1}{2}I\omega^2$ gives

$$\omega = \sqrt{\frac{2K}{I}} = \sqrt{\frac{2 \times 0.025852 \times 9.648533 \times 10^{-3}}{12.432}} = 0.006335 \text{ rad fs}^{-1},$$

one turn in $2\pi/\omega = 992 \text{ fs}$, about a picosecond, with the angular momentum $I\omega = 0.0788 \text{ amu } \text{\AA}^2 \text{ fs}^{-1}$ along the axis of turning (Figure 5.7). The lithium atom then circles at $0.006335 \times 1.1455 = 0.00726 \text{ \AA fs}^{-1}$, or 726 m s^{-1} .

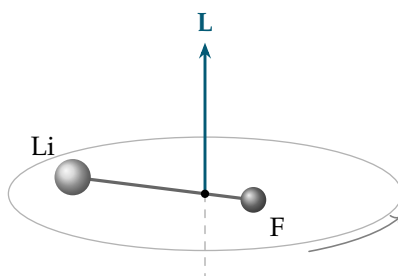


FIGURE 5.7: A LiF molecule turning about an axis through its centre of mass (dot), at right angles to the bond. Its angular momentum $\mathbf{L} = I\boldsymbol{\omega}$ points along the axis by the right-hand rule. The lithium atom, the lighter, lies $1.145 \text{ \AA}$ from the centre and the fluorine atom $0.418 \text{ \AA}$.

Fast vibration, slow turning

The bond of LiF has the harmonic wavenumber⁶ 910.34 cm^{-1} , that of small swings about the bottom of its well, so by Section 4.10 its period is

$$\frac{1}{2.99792458 \times 10^{-5} \times 910.34} = 36.64 \text{ fs}.$$

The molecule therefore vibrates 27 times in each turn, and turns only $360/27 = 13^\circ$ in each vibration. Over one vibration it is nearly still; over one turn its length swings back and forth 27 times about the same average, so the turning feels only that average. This separation of time scales is why a turning molecule is often treated as rigid, and why the step of a simulation is set by the fastest vibrations (Section 4.10). Turning does stretch the bond a little, since the atoms need a pull towards the centre to go round, but for LiF at room temperature only by about $0.002 \text{ \AA}$ (Exercise 5.19).

A lithium atom on the model surface of Section 3.13 is a different case. The surface pushes it with forces that point towards no single centre, so in general they exert a torque about any point, and the atom's angular momentum is not conserved. Among groups of atoms, only an isolated one, a cluster in empty space or a molecule in a gas between collisions, is sure to keep its angular momentum.

NOTEBOOK 05 *a turning LiF molecule with its angular momentum*

CHECK YOUR UNDERSTANDING. At the same turning energy, how would the period of turning change for a molecule of the same length with twice the reduced mass?

What we now have

A body turning about a point has angular momentum $\mathbf{L} = \mathbf{r} \times \mathbf{p}$, which changes at the rate of the torque $\mathbf{G} = \mathbf{r} \times \mathbf{F}$; in a plane these are $L = mrv_\theta$ and $G = rF_\theta$, the lever. A central force exerts no torque, so it conserves $\mathbf{L}$: a puck pulled in on a string goes round faster, sweeping equal areas in equal times. For a group of bodies whose forces act along the lines joining them, as forces between atoms built from pairs do, the internal torques cancel and only external torques change the total, which splits into a part carried by the centre of mass and a part about it.

A rigid body turning with angular velocity $\boldsymbol{\omega}$ moves with $\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}$; about a fixed axis $L_z = I\omega$ and $K = \frac{1}{2}I\omega^2$, with $I = m_r d^2$ for two bodies, and a rolling body keeps part of its energy in turning.

In general $\mathbf{L} = \mathbf{I}\boldsymbol{\omega}$ and $K = \frac{1}{2}\boldsymbol{\omega} \cdot \mathbf{I}\boldsymbol{\omega}$ with the symmetric inertia tensor, whose eigenvectors are the principal axes. Subtracting the drift $\mathbf{V} = \mathbf{P}/M$ and the turning $\boldsymbol{\omega} \times \mathbf{r}'$ with $\mathbf{I}\boldsymbol{\omega} = \mathbf{L}'$ removes the momentum and the angular momentum from a set of velocities and splits the kinetic energy into three separate parts.

The rigid motions are zero modes of the Hessian, which leaves a molecule $3N - 6$ ways to vibrate ($3N - 5$ if it is linear), and at room temperature a molecule such as LiF turns once in the time it vibrates 27 times.

Chapter 6 rebuilds mechanics from the energy alone, in a form that handles constraints, such as the rod of a pendulum, without ever finding the forces they exert.

Answers to the checks

SECTION 5.1. The string still pulls only towards the hole, so $L = 0.12 \text{ kg m}^2 \text{ s}^{-1}$ is unchanged, and at 1 m the puck goes round at $v_\theta = L/(mr) = 0.12/(0.2 \times 1) = 0.6 \text{ m s}^{-1}$. Its kinetic energy falls from 0.144 J to $\frac{1}{2} \times 0.2 \times 0.6^2 = 0.036 \text{ J}$, so the string does the work -0.108 J : negative, since it pulls inwards while the puck moves outwards.

SECTION 5.2. By the components, $\mathbf{e}_z \times \mathbf{e}_x = (0 \times 0 - 1 \times 0, 1 \times 1 - 0 \times 0, 0 \times 0 - 0 \times 1) = (0, 1, 0) = \mathbf{e}_y$. Curling the fingers of the right hand clockwise as seen from above, the thumb points down: the turntable's angular momentum points straight down.

SECTION 5.3. The total momentum is $1 \times (0, -1, 0) + 1 \times (0, 1, 0) = \mathbf{0}$. About the origin, $(0.5, 0, 0) \times (0, -1, 0) = (0, 0, -0.5)$ and $(-0.5, 0, 0) \times (0, 1, 0) = (0, 0, -0.5)$, so $\mathbf{L} = (0, 0, -1) \text{ kg m}^2 \text{ s}^{-1}$, a clockwise turn seen from above. About $(0, 1, 0)$ the arms are $(0.5, -1, 0)$ and $(-0.5, -1, 0)$, and $(0.5, -1, 0) \times (0, -1, 0) = (0, 0, -0.5)$ and $(-0.5, -1, 0) \times (0, 1, 0) = (0, 0, -0.5)$: the same $\mathbf{L}$, since moving the point changes $\mathbf{L}$ by a term proportional to the total momentum (Exercise 5.6), here zero.

SECTION 5.4. The hoop has $K = \frac{1}{2}MR^2\omega^2$ and the disc $\frac{1}{2} \times \frac{1}{2}MR^2\omega^2$, so the hoop has twice as much: more of its mass is far from the axis.

SECTION 5.5. With the atoms at $(0, 0, \pm d/2)$, $I_{xx} = \sum m(y^2 + z^2) = 2m(d/2)^2 = \frac{1}{2}md^2$, likewise I_{yy} , and $I_{zz} = \sum m(x^2 + y^2) = 0$; every product such as $-\sum mxz$ is zero since $x = y = 0$. The tensor is diagonal, so the principal moments are $\frac{1}{2}md^2$, $\frac{1}{2}md^2$ and 0, the first two equal to $m_{\text{r}}d^2$ with $m_{\text{r}} = m/2$.

SECTION 5.6. $\mathbf{V} = (0, 0.02, 0) \text{ Å fs}^{-1}$, so $\mathbf{v}'_1 = (0, -0.01, 0)$ and $\mathbf{v}'_2 = (0, 0.01, 0)$. Then $\mathbf{L}' = (-1, 0, 0) \times (0, -0.01, 0) + (1, 0, 0) \times (0, 0.01, 0) = (0, 0, 0.02) \text{ amu Å}^2 \text{ fs}^{-1}$, and about the centre of mass the inertia tensor is diagonal with $I_{xx} = 0$, both atoms lying on the x axis, and $I_{yy} = I_{zz} = 1 \times 1^2 + 1 \times 1^2 = 2 \text{ amu Å}^2$. It has no inverse, so, as in Section 5.6, take $\boldsymbol{\omega}$ with no part along the line: $\boldsymbol{\omega} = (0, 0, 0.02/2) = (0, 0, 0.01) \text{ rad fs}^{-1}$. The turning velocity of atom 1 is $(0, 0, 0.01) \times (-1, 0, 0) = (0, -0.01, 0) = \mathbf{v}'_1$, and likewise for atom 2, so nothing is left: the pair was only drifting and turning.

SECTION 5.7. Ethane has 8 atoms and is not linear: $24 - 6 = 18$. Acetylene has 4 atoms on a line: $12 - 5 = 7$.

SECTION 5.8. With $I = m_r d^2$, twice the reduced mass doubles I , and at the same energy $\omega = \sqrt{2K/I}$ falls by $\sqrt{2}$, so the period of turning is $\sqrt{2} = 1.414$ times longer.

Exercises

The exercises run from questions of understanding, through calculations, to short derivations marked ‘show that’. Full solutions are in ‘Solutions to the exercises’ at the end of the book, and Notebook 05 checks every numerical answer.

EXERCISE 5.1 (*a door pushed at a slant*). Why are door handles placed far from the hinges? A push of 10 N at the handle, 0.8 m from the hinge, is made at 30° to the face of the door instead of at right angles. What is its torque about the hinge?

EXERCISE 5.2 (*pulling the puck further*). The puck of Section 5.1 ($m = 0.2$ kg, circling at 0.5 m and 1.2 m s^{-1}) is pulled in to 0.1 m. Find its speed of going round, its kinetic energy and the work done by the hand, neglecting the slow inward motion.

EXERCISE 5.3 (*a straight line*). Show that a body moving at constant velocity, with no force on it, has a constant angular momentum $m(\mathbf{r} - \mathbf{a}) \times \mathbf{v}$ about any point $\mathbf{a}$, its position measured from $\mathbf{a}$, and explain, with the triangles swept in equal times, why it sweeps equal areas in equal times.

EXERCISE 5.4 (*a cross product*). Find $(2, 0, 1) \times (1, 3, -1)$, check that it is at right angles to both vectors, and find the area of the parallelogram they span in two ways.

EXERCISE 5.5 (*motion in a plane*). A body moves with a constant angular momentum $\mathbf{L} \neq \mathbf{0}$ about the origin. Show that it stays in the plane through the origin at right angles to $\mathbf{L}$.

EXERCISE 5.6 (*changing the point*). Show that the angular momentum of a group of bodies about the point $\mathbf{a}$, $\sum_i (\mathbf{r}_i - \mathbf{a}) \times \mathbf{p}_i$, equals $\mathbf{L} - \mathbf{a} \times \mathbf{P}$, with $\mathbf{L}$ its angular momentum about the origin and $\mathbf{P}$ its total momentum. When is it the same about every point?

EXERCISE 5.7 (*a rod*). Show that a thin uniform rod of mass M and length ℓ has the moment of inertia $M\ell^2/12$ about an axis through its middle at right angles to it, and $M\ell^2/3$ about a parallel axis through one end.

EXERCISE 5.8 (*the parallel axis*). A body has the moment of inertia I_{cm} about an axis through its centre of mass. Show that about a parallel axis a distance s away its moment of inertia is $I_{\text{cm}} + Ms^2$, and check the result against Exercise 5.7.

EXERCISE 5.9 (*the stool again*). The stool of Section 5.4, now with $I_0 = 2 \text{ kg m}^2$ and weights of 3 kg held at 0.7 m, turns at 1.5 rad s^{-1} . The weights are pulled in to 0.3 m. Find the new angular speed and the work done by the arms.

EXERCISE 5.10 (*a rolling ball*). A solid ball has $I = \frac{2}{5}MR^2$ about any axis through its centre. Rolling from rest, how fast is it moving after a drop of 1 m, and how long does it take to roll 2 m down a slope of 30° ? Where does it finish in the race of Figure 5.4?

EXERCISE 5.11 (*the grip a disc needs*). Using Newton's second law for the centre of mass along the slope, find the friction force that the slope must supply to a disc of 1 kg rolling down at 30° , and show that in general it is $\kappa Mg \sin \alpha / (1 + \kappa)$.

EXERCISE 5.12 (*a square of atoms*). Four atoms of 1 amu sit at $(\pm 1, \pm 1, 0) \text{ \AA}$. Find their inertia tensor about the origin and its principal moments. Why is every axis through the origin in the plane of the square a principal axis?

EXERCISE 5.13 (*a flat molecule*). Show that for a body whose parts all lie in the plane $z = 0$, $I_{zz} = I_{xx} + I_{yy}$, and that $I_{xz} = I_{yz} = 0$, so that the z axis is a principal axis.

EXERCISE 5.14 (*a diatomic molecule*). Two atoms of masses m_1 and m_2 lie on the z axis a distance d apart. Show that their inertia tensor about their centre of mass is diagonal, with entries $m_1 d^2$, $m_2 d^2$ and 0.

EXERCISE 5.15 (*elimination*). Solve, by elimination, $4\omega_x + \omega_y = 6$, $\omega_x + 4\omega_y + \omega_z = 12$, $\omega_y + 4\omega_z = 14$.

EXERCISE 5.16 (*what is left*). Two atoms of 1 amu at $(\mp 1, 0, 0) \text{ \AA}$ move with velocities $(0.01, 0.01, 0)$ and $(-0.01, 0.03, 0) \text{ \AA fs}^{-1}$. Remove the drift and the turning, describe the motion that remains, check Equation (5.12) in $\text{amu \AA}^2 \text{ fs}^{-2}$, and give the total kinetic energy in eV.

EXERCISE 5.17 (*counting*). How many internal degrees of freedom have ammonia, NH_3 , and hydrogen peroxide, H_2O_2 , neither of them linear?

EXERCISE 5.18 (*a zero mode by hand*). A single spring of stiffness k joins an atom at the origin to one at $(0, 0, d)$. Write down its 6×6 Hessian, and show directly that it turns the displacements of a small turning about the x axis into zero. Why does a turning about the z axis not count?

EXERCISE 5.19 (*a spinning bond stretches*). The separation of a turning LiF molecule goes round a circle, which needs a pull $m_r \omega^2 r$ towards the centre, supplied by the bond as a spring, $k(r - d)$. Show that the bond stretches by about $d (\omega / \omega_0)^2$, with

$\omega_0 = \sqrt{k/m_r}$ the angular frequency of the vibration, and evaluate it for the turning of [Section 5.8](#).

EXERCISE 5.20 (*a hotter molecule*). At 1000 K a LiF molecule carries on average $k_B T = 0.08617$ eV in its turning. How long does one turn take?

Lagrangian mechanics

Newton's laws require the total force on each body, but a constraint force is often known only through the motion it must maintain. The push of a wire keeps a bead on its curve, and the force of a pendulum's rod keeps the bob at a fixed distance from the pivot. Their magnitudes depend on the motion we are trying to find. When these forces do no work, we can instead describe the allowed motion with independent coordinates and write the kinetic and potential energies in those coordinates.

This leads to the Lagrangian formulation of mechanics. Hamilton's principle compares whole paths through time, and the Euler–Lagrange equations turn that comparison into equations of motion. We apply them to the bead on the wire of Chapter 3, then use the dependence of the Lagrangian on its coordinates to identify conserved momenta. When the constraint force itself is needed, a Lagrange multiplier supplies it; Chapter 11 uses this method to hold molecular bonds at fixed lengths. The conjugate momenta introduced here also prepare the Hamiltonian formulation in Chapter 7.

6.1 Coordinates that fit the problem

The bead of [Section 3.5](#) slides without friction along a wire bent into the curve $y = h(x)$. Its position has two coordinates, x and y , but the wire allows only one of them to be chosen: given x , the bead is at height $h(x)$. This section describes such motions by the coordinates they actually have, and writes their energies in those coordinates.

The bead and the wire

Newton's second law for the bead needs every force on it, and one of them, the push of the wire, is unknown. It is whatever force keeps the bead on the wire, at right angles to the wire, and its size depends on the part of the weight across the wire

as well as on how fast the bead goes and how sharply the wire bends. [Section 3.5](#) sidestepped it with energy: the push does no work, so the speed at each height follows from $\frac{1}{2}mv^2 + mgh = E$. That gave the speed at every point, but not directly when the bead gets there.

The energy is simplest in the one coordinate that the wire leaves free, x . The bead is at $(x, h(x))$, so by the chain rule its velocity is

$$\mathbf{v} = \left(\dot{x}, \frac{dh}{dx} \dot{x} \right) = \dot{x} (1, h'(x)),$$

the dot marking the rate of change with time ([Section 1.3](#)) and h' the slope of the wire. The square of its length, the speed squared, is $\dot{x}^2 + h'(x)^2 \dot{x}^2 = \dot{x}^2 (1 + h'^2)$ by Pythagoras' theorem, so

$$K = \frac{1}{2}m (1 + h'(x)^2) \dot{x}^2, \quad U = mgh(x).$$

Both energies are now written with x and $\dot{x}$ alone, and the push of the wire appears in neither.

The pendulum

A pendulum's bob on a light rod of length L ([Section 4.4](#)) is fixed by the angle θ of the rod from the vertical. With the pivot at the origin and y up, the bob is at $(L \sin \theta, -L \cos \theta)$. Differentiating each component with the chain rule, $d \sin \theta / dt = \cos \theta \dot{\theta}$ and $d \cos \theta / dt = -\sin \theta \dot{\theta}$ ([Section 5.1](#)),

$$\mathbf{v} = (L \cos \theta \dot{\theta}, -L \sin \theta \dot{\theta}) = L \dot{\theta} (\cos \theta, -\sin \theta),$$

of length $L|\dot{\theta}|$ by Pythagoras' theorem, since $\cos^2 \theta + \sin^2 \theta = 1$. The bob is $L \cos \theta$ below the pivot, so $L - L \cos \theta$ above the bottom of the swing ([Section 4.4](#)), and

$$K = \frac{1}{2}mL^2\dot{\theta}^2, \quad U = mgL(1 - \cos \theta).$$

Generalised coordinates

The fewest numbers that fix the position of every part of a system, once its constraints are obeyed, are called its generalised coordinates; each is free to change whatever the others do, and we write them $q_1, q_2, \dots, q_n$. Their rates of change, $\dot{q}_1, \dots, \dot{q}_n$, are the generalised velocities. The bead has one, x ; the pendulum one, θ ; the puck of [Section 5.1](#) on its table two, r and θ ; and N free atoms have $3N$, their Cartesian coordinates. The number n is the number of degrees of freedom of [Section 5.7](#). A constraint that fixes

one coordinate in terms of the others, as the wire fixes $y = h(x)$ or the rod fixes the distance of the bob from the pivot, is called holonomic, and it removes one degree of freedom. The wire and the rod do no work, since they push at right angles to every motion they allow; all the constraints in this book are holonomic and do no work.

KEY IDEA. Generalised coordinates are the fewest numbers that fix the position of a system once its constraints are obeyed. In them the kinetic and potential energies can be written without the forces of the constraints, which do no work.

CHECK YOUR UNDERSTANDING. How many generalised coordinates does a double pendulum have, one bob hanging by a rod from a second bob that hangs by a rod from a fixed pivot, moving in a plane, and how many a rigid diatomic molecule free in space?

6.2 The action and Hamilton's principle

Newton's laws work forwards from a starting state, one instant to the next. There is a second way to single out the true motion: compare whole paths that start at one place at one time and end at another place at a later time, and pick the one that makes a certain number stationary. This section defines that number and tests the statement on a thrown ball; [Section 6.3](#) shows that it is equivalent to Newton's laws.

The Lagrangian and the action

The Lagrangian of a system is its kinetic energy minus its potential energy,

$$\mathcal{L}(q, \dot{q}) = K - U, \quad (6.1)$$

written, as in [Section 6.1](#), in terms of the generalised coordinates and velocities, q standing for all of $q_1, \dots, q_n$ and $\dot{q}$ for all of $\dot{q}_1, \dots, \dot{q}_n$. For a path $q(t)$ from the time t_1 to the time t_2 , its action is the integral

$$\mathcal{S} = \int_{t_1}^{t_2} \mathcal{L}(q(t), \dot{q}(t)) dt, \quad (6.2)$$

one number for the whole path, in joule seconds. Each path, true or not, has an action: put its $q(t)$ and $\dot{q}(t)$ into $\mathcal{L}$ and integrate.

Hamilton's principle states that, among all paths that start at q_A at the time t_1 and end at q_B at the time t_2 , the path the system actually follows makes the action stationary: changing the path from $q(t)$ to $q(t) + \zeta\eta(t)$, with η any bump that is zero at t_1 and

t_2 and ζ a small number, changes $\mathcal{S}$ by no amount proportional to ζ itself, only by amounts of order ζ^2 (Section 1.8), as at the bottom of a valley or the top of a hill of $\mathcal{S}$ against ζ .

A thrown ball

A ball of 0.145 kg is thrown straight up at 12 m s^{-1} , as in Chapters 1 to 3. Its height is $x(t) = 12t - \frac{1}{2}gt^2$ (Section 1.4), and after 2 s it is 4.387 m up and falling. Now invent other paths between the same two events (an event is a place at a given time), starting at $x = 0$ at $t = 0$ and ending at 4.387 m at 2 s: add to the true height $\zeta\eta(t)$, with $\eta(t) = \sin(\pi t/\tau)$ and $\tau = 2 \text{ s}$, a bump that is zero at both ends. Each trial path has its own action, with $\mathcal{L} = \frac{1}{2}m\dot{x}^2 - mgx$, and the function action of mdlab computes it by splitting the path into short straight steps (Figure 6.1). For the true path, $\zeta = 0$, the integral can also be done by hand. With $\dot{x} = 12 - gt$,

$$\begin{aligned}\mathcal{L} &= \frac{1}{2}m(12 - gt)^2 - mg(12t - \frac{1}{2}gt^2) \\ &= 72m - 12mg t + \frac{1}{2}mg^2t^2 - 12mg t + \frac{1}{2}mg^2t^2 \\ &= 72m - 24mg t + mg^2t^2,\end{aligned}$$

expanding the square and the bracket, then collecting like terms, and, integrating each term by the power rule and the fundamental theorem (Section 1.4),

$$\mathcal{S} = \left[72m t - 12mg t^2 + \frac{1}{3}mg^2t^3 \right]_0^2 = 20.8800 - 68.2543 + 37.1859 = -10.1884 \text{ J s},$$

as action also finds. The paths with $\zeta = \pm 1 \text{ m}$ have -10.0095 J s , and those with $\pm 2 \text{ m}$ have -9.4729 J s . The excess grows as ζ^2 : 0.1789 and 0.7155, four times as much for twice the change, even for bumps of metres, which are not small; the next paragraph shows why. Among these paths the true one has the least action.

The excess can be found exactly. The change of path changes the velocity by $\zeta\dot{\eta}$ and the height by $\zeta\eta$, so

$$\begin{aligned}\mathcal{S}(\zeta) &= \int_0^\tau \left[\frac{1}{2}m(\dot{x} + \zeta\dot{\eta})^2 - mg(x + \zeta\eta) \right] dt \\ &= \int_0^\tau \left[\left(\frac{1}{2}m\dot{x}^2 - mgx \right) + \zeta(m\dot{x}\dot{\eta} - mg\eta) + \zeta^2 \frac{1}{2}m\dot{\eta}^2 \right] dt \\ &= \mathcal{S}(0) + \zeta \int_0^\tau (m\dot{x}\dot{\eta} - mg\eta) dt + \zeta^2 \frac{1}{2}m \int_0^\tau \dot{\eta}^2 dt,\end{aligned}$$

expanding the square, grouping by powers of ζ , and splitting the integral. There are no higher powers of ζ , which is why even large bumps follow ζ^2 exactly. Section 6.3

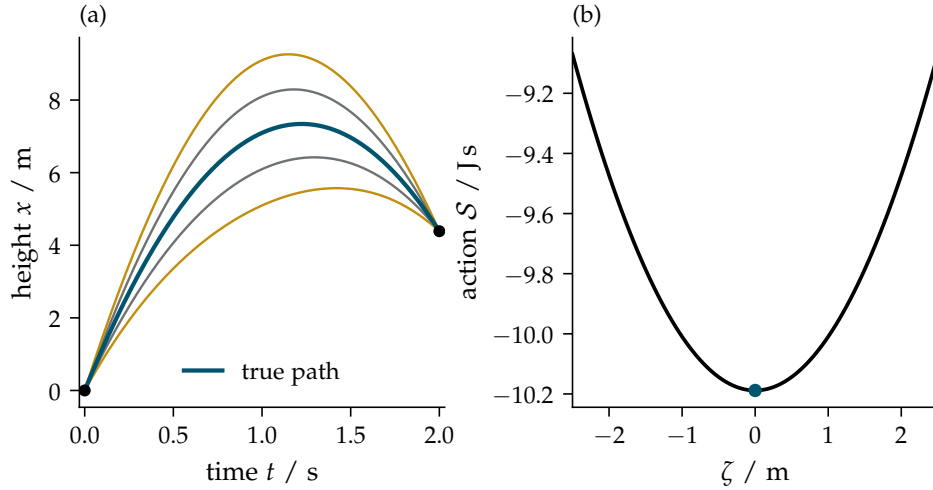


FIGURE 6.1: Trial paths of a ball of 0.145 kg thrown up at 12 m s^{-1} , all starting at $x = 0$ at $t = 0$ and ending 4.387 m up at 2 s . (a) The true path (teal) and the trial paths with $\zeta = \pm 1$ (grey) and $\pm 2 \text{ m}$ (ochre). (b) The action of each trial path against ζ , least at $\zeta = 0$.

shows that the middle integral vanishes for the true path and any bump η . The last needs $\dot{\eta}$ and the identity $\cos^2 \theta = \frac{1}{2} + \frac{1}{2} \cos 2\theta$ of [Section 4.2](#):

$$\begin{aligned}
 \int_0^\tau \dot{\eta}^2 dt &= \left(\frac{\pi}{\tau}\right)^2 \int_0^\tau \cos^2 \frac{\pi t}{\tau} dt & \dot{\eta} &= \frac{\pi}{\tau} \cos \frac{\pi t}{\tau} \text{ by the chain rule,} \\
 &= \left(\frac{\pi}{\tau}\right)^2 \int_0^\tau \left(\frac{1}{2} + \frac{1}{2} \cos \frac{2\pi t}{\tau}\right) dt & & \text{the identity, with } \theta = \pi t/\tau, \\
 &= \left(\frac{\pi}{\tau}\right)^2 \left[\frac{t}{2} + \frac{\tau}{4\pi} \sin \frac{2\pi t}{\tau}\right]_0^\tau & & \text{the fundamental theorem,} \\
 &= \left(\frac{\pi}{\tau}\right)^2 \frac{\tau}{2} = \frac{\pi^2}{2\tau} & & \text{as } \sin 2\pi = \sin 0 = 0.
 \end{aligned}$$

The term in ζ^2 is therefore $\zeta^2 \times \frac{1}{2}m \times \pi^2/(2\tau) = \zeta^2 m\pi^2/(4\tau)$, and $m\pi^2/(4\tau) = 0.145 \times 9.8696/8 = 0.17889 \text{ J s m}^{-2}$, the excess found above for $\zeta = 1 \text{ m}$.

When the action is not least

The true path is not always the one of least action. Take the block of [Section 4.1](#), with $m = 0.5 \text{ kg}$ and $k = 50 \text{ N m}^{-1}$, so that $\omega = \sqrt{k/m} = 10 \text{ rad s}^{-1}$, oscillating for $\tau = 0.5 \text{ s}$, longer than half its period, $\pi/\omega = 0.3142 \text{ s}$. Now $\mathcal{L} = \frac{1}{2}m\dot{x}^2 - \frac{1}{2}kx^2$, and the potential energy of the changed path, $\frac{1}{2}k(x + \zeta\eta)^2 = \frac{1}{2}kx^2 + \zeta kx\eta + \zeta^2 \frac{1}{2}k\eta^2$, brings a second

term in ζ^2 . The term in ζ again vanishes for the true path (Section 6.3), whatever its amplitude, so for $\eta = \sin(\pi t/\tau)$ the excess is

$$\begin{aligned}\mathcal{S}(\zeta) - \mathcal{S}(0) &= \zeta^2 \left(\frac{1}{2}m \int_0^\tau \dot{\eta}^2 dt - \frac{1}{2}k \int_0^\tau \eta^2 dt \right) \quad \text{expanding as for the ball,} \\ &= \zeta^2 \left(\frac{1}{2}m \left(\frac{\pi}{\tau} \right)^2 \frac{\tau}{2} - \frac{1}{2}k \frac{\tau}{2} \right) \quad \text{as above, with } \sin^2 \theta = \frac{1}{2} - \frac{1}{2} \cos 2\theta, \\ &= \zeta^2 \frac{\tau}{4} \left(m \left(\frac{\pi}{\tau} \right)^2 - k \right) \quad \text{taking out } \tau/4.\end{aligned}$$

The bracket is negative whenever $\pi/\tau < \sqrt{k/m} = \omega$, that is whenever τ is longer than π/ω : here it is $0.125 \times (0.5 \times 39.48 - 50) \zeta^2 = -3.783 \zeta^2$, and the action falls. For the faster bump $\sin(2\pi t/\tau)$ the same steps with 2π in place of π give $0.125 \times (0.5 \times 157.91 - 50) \zeta^2 = +3.620 \zeta^2$, and the action rises. Against the sizes of the two bumps the true path sits at a saddle of the action, like the saddle points of Section 3.10: stationary, which is all Hamilton's principle claims.

KEY IDEA. The Lagrangian is $\mathcal{L} = K - U$ and the action of a path is $\mathcal{S} = \int \mathcal{L} dt$. Among all paths with the same ends in space and time, the true motion makes the action stationary: a small change of path, of size ζ , changes it only by an amount of order ζ^2 .

NOTEBOOK 06 bend the path of the ball and watch its action

CHECK YOUR UNDERSTANDING. For a body with no forces on it, $U = 0$, the true path between two events is a straight line at constant speed. Using the expansion above, why is its action less than that of every other path with the same ends?

6.3 The Euler–Lagrange equation

Hamilton's principle picks out the true path among all paths with the same ends. This section turns it into an equation that the true path obeys at every instant, using two tools, and shows that for a body moving in a potential the equation is Newton's second law.

INTEGRATION BY PARTS. The product rule, $(fg)' = f'g + fg'$, integrated from a to b by the fundamental theorem of calculus (Section 1.4), gives $[fg]_a^b = \int_a^b f'g dt + \int_a^b fg' dt$, where $[fg]_a^b = f(b)g(b) - f(a)g(a)$, so

$$\int_a^b fg' dt = [fg]_a^b - \int_a^b f'g dt :$$

an integral of f times the derivative of g becomes one of the derivative of f times g , at the cost of the bracket. For example, with $f = t$ and $g = \sin t$, so that $f' = 1$ and $g' = \cos t$,

$$\int_0^{\pi/2} t \cos t \, dt = \left[t \sin t \right]_0^{\pi/2} - \int_0^{\pi/2} \sin t \, dt = \frac{\pi}{2} - \left[-\cos t \right]_0^{\pi/2} = \frac{\pi}{2} - 1,$$

since $\sin \frac{\pi}{2} = 1$ and $\sin 0 = 0$, and $-\cos t$, whose derivative is $\sin t$, rises from -1 to 0 .

The first-order change of the action

Change the path $q(t)$ to $q(t) + \zeta \eta(t)$, with η any smooth function that is zero at both ends, $\eta(t_1) = \eta(t_2) = 0$, so that the changed path keeps the same ends. At each instant the Lagrangian changes from $\mathcal{L}(q, \dot{q})$ to $\mathcal{L}(q + \zeta \eta, \dot{q} + \zeta \dot{\eta})$.

THE FIRST-ORDER CHANGE OF A FUNCTION OF A PATH. Read $\mathcal{L}(q + \zeta \eta, \dot{q} + \zeta \dot{\eta})$, at a fixed instant, as a function of ζ alone: as ζ grows from 0 , the point $(q + \zeta \eta, \dot{q} + \zeta \dot{\eta})$ moves in a straight line in the plane of the two variables of $\mathcal{L}$, changing at the rate $(\eta, \dot{\eta})$ per unit of ζ : at a fixed instant η and $\dot{\eta}$ are two fixed numbers, and ζ plays the part that time played in [Section 3.10](#). The chain rule along a path of [Section 3.10](#), $d\mathcal{L}/d\zeta = \nabla \mathcal{L} \cdot (\eta, \dot{\eta})$, gives at $\zeta = 0$

$$\frac{d\mathcal{L}}{d\zeta} = \frac{\partial \mathcal{L}}{\partial q} \eta + \frac{\partial \mathcal{L}}{\partial \dot{q}} \dot{\eta},$$

where $\partial \mathcal{L} / \partial \dot{q}$ is the partial derivative with respect to the second variable, the first held fixed ([Section 3.10](#)). At a single instant q and $\dot{q}$ are two numbers, and $\mathcal{L}$ is an ordinary function of the two, like the height $h(x, y)$ of a map; holding q fixed here does not make $\dot{q}$ zero, because the link between them, that $\dot{q}$ is the rate of change of q , appears only once a whole path is put in. By the Taylor series ([Section 1.8](#)), $\mathcal{L}$ at ζ is $\mathcal{L}$ at 0 plus ζ times this, plus a remainder no larger than a constant times ζ^2 .

Integrating over the path, these remainders add up to at most $(t_2 - t_1)$ times the largest such constant times ζ^2 , still of order ζ^2 , so the action changes by

$$S(\zeta) - S(0) = \zeta \int_{t_1}^{t_2} \left(\frac{\partial \mathcal{L}}{\partial q} \eta + \frac{\partial \mathcal{L}}{\partial \dot{q}} \dot{\eta} \right) dt + \mathcal{O}(\zeta^2).$$

The integral multiplying ζ contains both η and $\dot{\eta}$. Along the path, $\partial \mathcal{L} / \partial \dot{q}$ takes one value at each instant, found by putting that instant's $q(t)$ and $\dot{q}(t)$ into it, so it is a function of t alone; its rate of change with time is written $d(\partial \mathcal{L} / \partial \dot{q}) / dt$, with

an ordinary d because nothing is now held fixed (for $\mathcal{L} = \frac{1}{2}m\dot{x}^2 - U$ below it is $d(m\dot{x})/dt = m\ddot{x}$). Integrating the second term by parts, with $f = \partial\mathcal{L}/\partial\dot{q}$ and $g = \eta$,

$$\int_{t_1}^{t_2} \frac{\partial\mathcal{L}}{\partial\dot{q}} \dot{\eta} dt = \left[\frac{\partial\mathcal{L}}{\partial\dot{q}} \eta \right]_{t_1}^{t_2} - \int_{t_1}^{t_2} \frac{d}{dt} \left(\frac{\partial\mathcal{L}}{\partial\dot{q}} \right) \eta dt,$$

and the bracket is zero, because η is zero at both ends. Putting what is left in place of the second term, and gathering the two integrals into one, since both contain the factor η ,

$$\mathcal{S}(\zeta) - \mathcal{S}(0) = \zeta \int_{t_1}^{t_2} \left[\frac{\partial\mathcal{L}}{\partial q} - \frac{d}{dt} \left(\frac{\partial\mathcal{L}}{\partial\dot{q}} \right) \right] \eta dt + \mathcal{O}(\zeta^2).$$

The equation

For the action to be stationary, its change must be of order ζ^2 (Section 6.2), so the term in ζ must vanish for every bump η . Call the square bracket $B(t)$. If B were positive at some instant t^* , it would be positive on a short interval around t^* , since it changes smoothly; choose η positive on that interval and zero everywhere else, such as $\eta = \sin^2 u$ with $u = \pi(t - a)/(b - a)$ between the ends a and b of the interval, whose slope $(2\pi/(b - a)) \sin u \cos u$ is zero at both ends, so that it joins the zero outside smoothly; the integral of $B\eta$ is positive, not zero. The same argument rules out a negative B . So $B(t) = 0$ at every instant:

$$\frac{d}{dt} \left(\frac{\partial\mathcal{L}}{\partial\dot{q}} \right) = \frac{\partial\mathcal{L}}{\partial q}. \quad (6.3)$$

This is the Euler–Lagrange equation. Conversely, on a path that obeys it at every instant $B = 0$, so the term in ζ vanishes for every bump: the equation and Hamilton’s principle pick out the same paths. A system with several generalised coordinates has one such equation for each, $d(\partial\mathcal{L}/\partial\dot{q}_k)/dt = \partial\mathcal{L}/\partial q_k$, since each coordinate can be given its own bump while the others are left alone.

Newton’s law again

For a body on a line in a potential energy $U(x)$, $\mathcal{L} = \frac{1}{2}m\dot{x}^2 - U(x)$. The partial derivatives are $\partial\mathcal{L}/\partial\dot{x} = m\dot{x}$, with x held fixed, and $\partial\mathcal{L}/\partial x = -U'(x)$, with $\dot{x}$ held fixed, so Equation (6.3) reads

$$\frac{d}{dt}(m\dot{x}) = -U'(x), \quad \text{that is} \quad m\ddot{x} = F(x),$$

Newton’s second law with the force $F = -U'$ of Section 3.4. For the thrown ball, $U = mgx$, so $B = -mg - m\ddot{x}$, which is zero on the true path, where $\ddot{x} = -g$. The middle integral of Section 6.2 is the term in ζ above, which by parts is $\int B\eta \, dt$, and this is why it vanished. For N atoms with $\mathcal{L} = \sum_i \frac{1}{2}m_i|\dot{\mathbf{r}}_i|^2 - U(\mathbf{r}^N)$, the $3N$ equations are $m_i\ddot{\mathbf{r}}_i = -\nabla_i U$, the equations of motion of Section 3.12. Hamilton’s principle therefore contains Newton’s laws for every system whose forces come from a potential energy, and from here on either can be used.

FROM ELECTRONIC STRUCTURE. The Euler–Lagrange equation is the condition that a functional, a number computed from a whole function, be stationary: the bracket $\partial\mathcal{L}/\partial q - d(\partial\mathcal{L}/\partial\dot{q})/dt$ is the functional derivative $\delta\mathcal{S}/\delta q(t)$. It is the same operation as the derivative $\delta E[n]/\delta n(\mathbf{r})$ of the density functional with respect to the electron density; setting it equal to a Lagrange multiplier that holds the number of electrons fixed gives the Euler equation of density-functional theory, $\delta E/\delta n(\mathbf{r}) = \mu$, with the chemical potential μ as the multiplier⁸ (Section 6.6).

KEY IDEA. Hamilton’s principle holds for a path exactly when, at every instant, $d(\partial\mathcal{L}/\partial\dot{q}_k)/dt = \partial\mathcal{L}/\partial q_k$ for each generalised coordinate. For forces from a potential this is Newton’s second law.

CHECK YOUR UNDERSTANDING. Write down the Euler–Lagrange equation for a block on a spring, $\mathcal{L} = \frac{1}{2}m\dot{x}^2 - \frac{1}{2}kx^2$.

6.4 Constrained motion

The Euler–Lagrange equation holds in any generalised coordinates. Applied to the bead’s single coordinate x , it gives the equation of motion along the wire without the wire’s push ever appearing, and with it the times that Section 3.5 left to this chapter.

The bead on the wire

From Section 6.1, $\mathcal{L} = \frac{1}{2}m(1 + h'^2)\dot{x}^2 - mgh(x)$, with h , h' and h'' all functions of x . The two partial derivatives are

$$\frac{\partial\mathcal{L}}{\partial\dot{x}} = m(1 + h'^2)\dot{x}, \quad \frac{\partial\mathcal{L}}{\partial x} = m h' h'' \dot{x}^2 - m g h',$$

the second by the chain rule, with $\dot{x}$ held fixed, since $d(h'^2)/dx = 2h'h''$. The rate of change of the first along the motion needs the product rule, and the chain rule for h'^2 , which changes at the rate $2h'h''\dot{x}$:

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{x}} = m(1 + h'^2) \ddot{x} + 2m h' h'' \dot{x}^2.$$

Setting the two equal, by Equation (6.3),

$$\begin{aligned} m(1 + h'^2) \ddot{x} + 2m h' h'' \dot{x}^2 &= m h' h'' \dot{x}^2 - m g h' \\ (1 + h'^2) \ddot{x} &= -h' h'' \dot{x}^2 - g h' && \text{dividing by } m, \text{ moving } 2h' h'' \dot{x}^2 \text{ across} \\ &= -h' (g + h'' \dot{x}^2) && \text{taking out } -h', \end{aligned}$$

and dividing by $1 + h'^2$,

$$\ddot{x} = -\frac{h' (g + h'' \dot{x}^2)}{1 + h'^2}. \quad (6.4)$$

The mass has cancelled. On a straight slope at the angle α , $h = x \tan \alpha$, $h' = \tan \alpha$ and $h'' = 0$. Dividing $\cos^2 \alpha + \sin^2 \alpha = 1$ by $\cos^2 \alpha$ gives $1 + \tan^2 \alpha = 1/\cos^2 \alpha$, so Equation (6.4) gives $\ddot{x} = -g \tan \alpha \cos^2 \alpha = -g \sin \alpha \cos \alpha$, with $\tan \alpha = \sin \alpha / \cos \alpha$. The distance along the slope is the hypotenuse of a right-angled triangle with horizontal side x , $x/\cos \alpha$, so its acceleration is $\ddot{x}/\cos \alpha = -g \sin \alpha$: the part $mg \sin \alpha$ of the weight down the ramp of Section 3.7, divided by m , with no friction. On a curved wire the term in $h'' \dot{x}^2$ adds the effect of the bending.

Why did the push of the wire never appear? It acts at right angles to the wire, so it does no work in any motion along the wire; it has no potential energy, and $\mathcal{L} = K - U$, made from the bead's speed and height alone, does not contain it. Describing the bead by x alone compares only paths that keep it on the wire, and Newton's laws show that Hamilton's principle holds among those. In x and y the bead obeys $m\ddot{\mathbf{r}} = -\nabla U + \mathbf{N}$, with $\mathbf{N}$ the push. Change the path by $\zeta \boldsymbol{\eta}(t)$ along the wire, zero at both ends. The steps of Section 6.3, once for x and once for y , change the action by $\zeta \int (-\nabla U - m\ddot{\mathbf{r}}) \cdot \boldsymbol{\eta} dt = -\zeta \int \mathbf{N} \cdot \boldsymbol{\eta} dt$ plus terms in ζ^2 , and this is zero because $\mathbf{N}$ is at right angles to the wire and $\boldsymbol{\eta}$ along it. The true path therefore makes the action stationary among the paths that stay on the wire, which are the paths of x alone, and the push never enters. Section 6.6 runs the same argument backwards for the pendulum. This is the reason for writing mechanics this way: constraints that do no work are obeyed by choosing the coordinates, and their forces drop out.

The bead against the clock

The function `solve_wire` of `mdlab` solves Equation (6.4). Released from rest on the wire of Section 3.5 at its left turning point for $h_E = 1.3$ m, $x = -2.206$ m (Section 3.6), the bead reaches the bottom of the deep valley after 0.554 s, the top of the hill after 1.097 s and the bottom of the shallow valley after 1.533 s, and turns at $x = 2$ m after 2.016 s, coming back to its start after 4.032 s (Figure 6.2). Its speed along the wire, $\sqrt{1 + h'^2} |\dot{x}|$, equals $\sqrt{2g(1.3 - h)}$ at every point, the speed that Section 3.5 found from energy alone: 4.680, 2.387 and 3.686 m s⁻¹ at the three places. The same times follow from the energy, by adding up the time $\sqrt{1 + h'^2} dx / \sqrt{2g(1.3 - h)}$, length over speed, of each short stretch of wire, as Section 4.5 did for the period; the equation of motion is the route that still works with more than one coordinate.

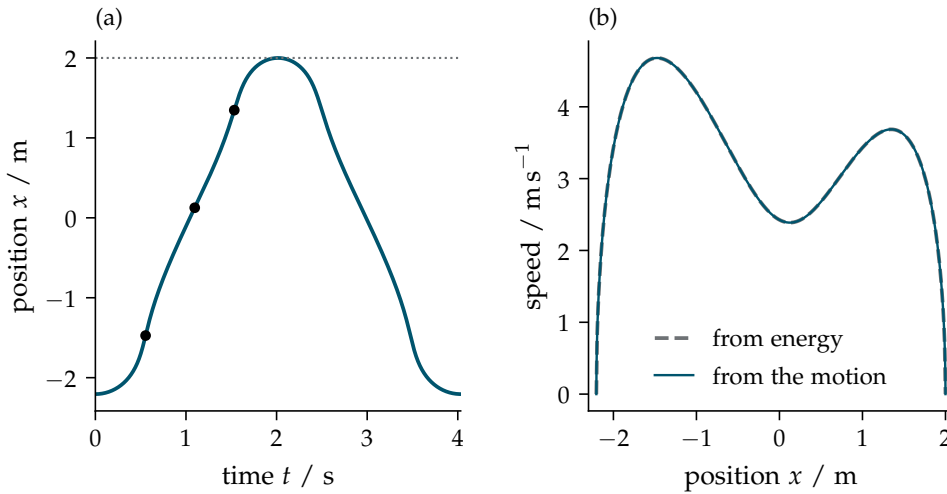


FIGURE 6.2: The bead on the wire of Section 3.5, released from rest at $x = -2.206$ m, 1.3 m high, its motion found from Equation (6.4). (a) Its position along x against time, through one swing to $x = 2$ m (dotted) and back; the dots mark the bottom of the deep valley, the top of the hill and the bottom of the shallow valley on the way out. (b) Its speed along the wire against x on the way out, lying on the speed from energy (dashed), which it hides.

The pendulum

With $\mathcal{L} = \frac{1}{2}mL^2\dot{\theta}^2 - mgL(1 - \cos \theta)$, $\partial\mathcal{L}/\partial\dot{\theta} = mL^2\dot{\theta}$ and $\partial\mathcal{L}/\partial\theta = -mgL \sin \theta$, so Equation (6.3) gives $mL^2\ddot{\theta} = -mgL \sin \theta$, or

$$\ddot{\theta} = -\frac{g}{L} \sin \theta, \quad (6.5)$$

in one line, with the rod's pull nowhere. For small swings $\sin \theta \approx \theta$ (Section 1.8), and $\ddot{\theta} = -(g/L)\theta$ is the spring of Section 2.8 with $\omega_0 = \sqrt{g/L}$, as Section 4.4 found from the energy.

KEY IDEA. Constraints that do no work are obeyed by choosing generalised coordinates, and the Euler–Lagrange equations in those coordinates give the motion without the constraining forces: for the bead, $\ddot{x} = -h'(g + h''\dot{x}^2)/(1 + h'^2)$.

NOTEBOOK 06 put the bead on any wire and time its journey

CHECK YOUR UNDERSTANDING. How long does the bead released at 1.3 m take to go from the bottom of the deep valley to the bottom of the shallow one?

6.5 Polar coordinates and conserved momenta

Section 5.1 found that a central force conserves angular momentum, by computing the torque. In polar coordinates the Lagrangian shows it at a glance, and the same reading gives a general rule: a coordinate that the Lagrangian does not contain carries a conserved quantity.

A body pulled towards a centre

A puck on a smooth table is pulled towards a fixed point, the origin, by a force that depends only on its distance r , with potential energy $U(r)$. By **Equation (5.1)** its velocity has the parts $\dot{r}$ along the line to the origin and $r\dot{\theta}$ across it, at right angles, so its speed squared is $\dot{r}^2 + r^2\dot{\theta}^2$ and

$$\mathcal{L} = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2) - U(r).$$

For the coordinate θ , $\partial\mathcal{L}/\partial\dot{\theta} = mr^2\dot{\theta}$, and $\partial\mathcal{L}/\partial\theta = 0$, since θ appears nowhere in $\mathcal{L}$. **Equation (6.3)** therefore reads

$$\frac{d}{dt}(mr^2\dot{\theta}) = 0 :$$

the angular momentum $L = mr^2\dot{\theta}$ of **Section 5.1** (not the rod length L of the pendulum) is conserved.

Conserved momenta

For any generalised coordinate q_k , the quantity

$$p_k = \frac{\partial\mathcal{L}}{\partial\dot{q}_k} \tag{6.6}$$

is called the momentum belonging to q_k , or its conjugate momentum. For x in $\mathcal{L} = \frac{1}{2}m\dot{x}^2 - U$ it is $m\dot{x}$, the momentum of [Section 2.9](#); for θ above it is the angular momentum. By [Equation \(6.3\)](#), $\dot{p}_k = \partial\mathcal{L}/\partial q_k$, so whenever $\mathcal{L}$ does not contain q_k itself, only its rate $\dot{q}_k$, the momentum p_k is conserved. The Lagrangian of the puck does not contain θ because turning the whole picture about the origin changes nothing; that of a free body does not contain x because moving it changes nothing. Chapter 7 turns this observation into Noether's theorem, which ties every such symmetry to a conserved quantity.

The effective potential

For r , $\partial\mathcal{L}/\partial\dot{r} = m\dot{r}$ and $\partial\mathcal{L}/\partial r = mr\dot{\theta}^2 - U'(r)$, so

$$\begin{aligned} m\ddot{r} &= mr\dot{\theta}^2 - U'(r) \\ &= mr \frac{L^2}{m^2 r^4} - U'(r) && \text{putting in } \dot{\theta} = L/(mr^2), \text{ from } L = mr^2\dot{\theta}, \\ &= \frac{L^2}{mr^3} - U'(r) && \text{cancelling } m \text{ and } r. \end{aligned}$$

Since L is constant, the right side is a function of r alone, and it is minus the derivative of

$$U_{\text{eff}}(r) = U(r) + \frac{L^2}{2mr^2}, \quad (6.7)$$

because $d(L^2/(2mr^2))/dr = -L^2/(mr^3)$ by the power rule for negative powers ([Section 4.5](#)). The distance from the origin therefore moves as a body on a line in the effective potential U_{eff} , and the energy diagrams of [Section 3.6](#) apply to it: $\frac{1}{2}m\dot{r}^2 + U_{\text{eff}}(r)$ keeps a fixed value E , and r turns back where $U_{\text{eff}}(r) = E$. This E is the whole energy $K + U$, since $L^2/(2mr^2) = \frac{1}{2}mr^2\dot{\theta}^2$ is the kinetic energy of the motion across the line. The extra term grows without limit as $r \rightarrow 0$, so for a potential that stays finite at the centre, as the spring's does, a body that is going round, $L \neq 0$, never reaches the centre.

Join the puck of 0.2 kg to the hole by a spring of 2 N m^{-1} and natural length zero, so that $U = \frac{1}{2}kr^2$, and start it 0.5 m out, moving across the line to the hole at 0.5 m s^{-1} . Then $L = mrv_{\theta} = 0.2 \times 0.5 \times 0.5 = 0.05 \text{ kg m}^2 \text{ s}^{-1}$ and $E = \frac{1}{2} \times 0.2 \times 0.5^2 + \frac{1}{2} \times 2 \times 0.5^2 = 0.275 \text{ J}$. The turning points solve $\frac{1}{2}kr^2 + L^2/(2mr^2) = E$. Multiplying by r^2 and moving

Er^2 across gives $\frac{1}{2}kr^4 - Er^2 + L^2/(2m) = 0$, a quadratic in the unknown r^2 , with $\frac{1}{2}k = 1$ and $L^2/(2m) = 0.05^2/(2 \times 0.2) = 0.00625$:

$$r^4 - 0.275r^2 + 0.00625 = 0,$$

$$\begin{aligned} r^2 &= \frac{1}{2}(0.275 \pm \sqrt{0.275^2 - 4 \times 0.00625}) \\ &= \frac{1}{2}(0.275 \pm 0.225) = 0.25 \text{ or } 0.025, \end{aligned}$$

by the formula for the roots of a quadratic, so r swings between $\sqrt{0.025} = 0.1581$ m and $\sqrt{0.25} = 0.5$ m (Figure 6.3). The path, computed with the function `solve_newton`, is an ellipse centred on the hole between these two distances.

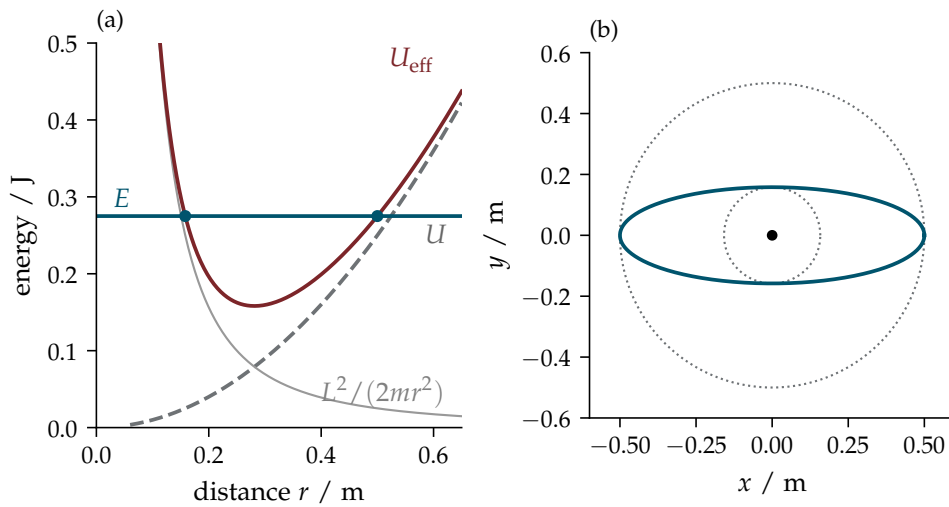


FIGURE 6.3: A puck of 0.2 kg joined to a hole by a spring of 2 N m^{-1} , started 0.5 m out at 0.5 m s^{-1} across the line to the hole. (a) The potential energy U , the term $L^2/(2mr^2)$ and their sum, the effective potential, with the energy E ; the dots are the turning points. (b) The path seen from above, between circles at the two turning distances.

The r equation also gives the pull of the string in Figure 5.2. A pull of size F towards the hole is the force $-F$ along $\mathbf{e}_r$ and takes the place of $-U'(r)$: a pull $F(t)$ acts as the potential $F(t)r$ at each instant, and the derivation of Section 6.3 compares paths instant by instant, so it still applies, so $m\ddot{r} = mr\dot{\theta}^2 - F$. There the puck was drawn in at a steady speed, $\ddot{r} = 0$, so $F = mr\dot{\theta}^2 = L^2/(mr^3)$, the pull with which Figure 5.2 was computed.

KEY IDEA. The momentum belonging to a coordinate is $p_k = \partial\mathcal{L}/\partial\dot{q}_k$, and it is conserved when $\mathcal{L}$ does not contain q_k . For a central force, θ is such a coordinate and its momentum is the angular momentum; the distance r then moves in the effective potential $U + L^2/(2mr^2)$.

NOTEBOOK 06 explore the effective potential and the paths it gives

CHECK YOUR UNDERSTANDING. For $\mathcal{L} = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2) - mgy$, a ball thrown in a vertical plane, which momentum is conserved, and why?

6.6 Constraints and Lagrange multipliers

Choosing coordinates that obey a constraint makes its force disappear, which is what we want when only the motion matters. Sometimes the force itself is wanted, such as the tension of a pendulum's rod, and sometimes no convenient coordinates exist, as for the bonds of a large molecule held at fixed lengths. Then we keep the Cartesian coordinates and let the constraint enter through a new unknown, a Lagrange multiplier. This section introduces the tool, uses it on the pendulum, and compares the result with Newton's laws.

LAGRANGE MULTIPLIERS. Suppose we want the largest or smallest value of $f(x, y)$ among the points on the curve $\chi(x, y) = 0$, written with the Greek chi since g is the acceleration of free fall. Move along the curve, in the direction of a short arrow $\mathbf{u}$ along it. By the chain rule along a path (Section 3.10), f changes at the rate $\nabla f \cdot \mathbf{u}$, and χ at the rate $\nabla \chi \cdot \mathbf{u} = 0$, since χ stays zero on the curve. At a largest or smallest value of f on the curve its rate of change along the curve is zero too, $\nabla f \cdot \mathbf{u} = 0$. In the plane there is only one direction at right angles to $\mathbf{u}$, so ∇f and $\nabla \chi$, the latter not zero, point along the same line there:

$$\nabla f = \lambda \nabla \chi, \quad \chi = 0,$$

for some number λ , called the Lagrange multiplier: three equations for the three unknowns x , y and λ . For example, the rectangle of largest area $f = xy$ with a perimeter of 4 m, $\chi = 2x + 2y - 4 = 0$, has $(y, x) = \lambda(2, 2)$, so $x = y$, and the perimeter gives $x = y = 1$ m: a square. With more variables the same argument, applied to every arrow $\mathbf{u}$ along the surface $\chi = 0$, again puts ∇f along $\nabla \chi$; with several constraints $\chi_c = 0$, one multiplier λ_c each, it becomes $\nabla f = \sum_c \lambda_c \nabla \chi_c$, which we take on trust.

Here λ is a multiplier, not the eigenvalue of Section 4.9.

The pendulum in x and y

Describe the bob by its Cartesian position $\mathbf{r} = (x, y)$, with the pivot at the origin and y up, and impose the rod as the constraint $\chi = \frac{1}{2}(x^2 + y^2 - L^2) = 0$. Its gradient is $\nabla \chi = (x, y) = \mathbf{r}$, along the rod and at right angles to the circle $\chi = 0$, as a gradient is to its contour (Section 3.10). The rod does no work (Section 6.1), so its force too is at

right angles to the circle, along $\nabla\chi$: it is $\lambda \nabla\chi = \lambda \mathbf{r}$ for some multiplier λ that may change with time, and Newton's second law reads

$$m\ddot{\mathbf{r}} = m\mathbf{g} + \lambda\mathbf{r},$$

with $\mathbf{g} = (0, -g)$ the acceleration of free fall.

Hamilton's principle gives the same equation, and shows where the multiplier comes from. Here $\mathcal{L} = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2) - mgy$. Changing the path by $\zeta\boldsymbol{\eta}(t)$, with $\boldsymbol{\eta} = (\eta_x, \eta_y)$ zero at both ends, the steps of [Section 6.3](#), once for x and once for y , change the action by $\zeta \int \mathbf{B} \cdot \boldsymbol{\eta} dt$ plus terms of order ζ^2 , where

$$\mathbf{B} = \left(\frac{\partial \mathcal{L}}{\partial x} - \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{x}}, \frac{\partial \mathcal{L}}{\partial y} - \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{y}} \right) = (-m\ddot{x}, -mg - m\ddot{y}) = m\mathbf{g} - m\ddot{\mathbf{r}}.$$

Among all paths this would give $\mathbf{B} = \mathbf{0}$, the bob falling freely. Only paths that keep the bob on its circle are allowed, so at each instant $\boldsymbol{\eta}$ must point along the circle, $\boldsymbol{\eta} = \eta(t) \mathbf{e}_\theta$ with $\mathbf{e}_\theta$ the arrow of length one along the circle at the bob ([Equation \(5.1\)](#)); such a change alters the length only by terms in ζ^2 , which do not touch the term in ζ . The change of action is then $\zeta \int (\mathbf{B} \cdot \mathbf{e}_\theta) \eta dt$, and the bump argument of [Section 6.3](#) gives $\mathbf{B} \cdot \mathbf{e}_\theta = 0$ at every instant: $\mathbf{B}$ has no part along the circle. In the plane only the direction of $\mathbf{r}$ is left, as in the toolbox, so $\mathbf{B} = -\lambda\mathbf{r}$ for some number λ at each instant, the minus sign only a choice of name, and $m\mathbf{g} - m\ddot{\mathbf{r}} = -\lambda\mathbf{r}$ is the second law above. The constraint leaves the principle silent about one force, along $\nabla\chi$, and that force is the rod's. Equivalently, these are the Euler–Lagrange equations of $\mathcal{L} + \lambda\chi$ in x and y treated as free, the usual way of writing a multiplier, with one λ for each instant.

The multiplier is fixed by the constraint itself. Differentiating $\mathbf{r} \cdot \mathbf{r} = x^2 + y^2 = L^2$ twice with respect to time, by the product rule on each component as in [Section 1.7](#),

$$2\mathbf{r} \cdot \dot{\mathbf{r}} = 0,$$

$$\dot{\mathbf{r}} \cdot \dot{\mathbf{r}} + \mathbf{r} \cdot \ddot{\mathbf{r}} = 0 \quad \text{differentiating again and halving,}$$

$$|\mathbf{v}|^2 + \mathbf{r} \cdot \mathbf{g} + \frac{\lambda}{m} L^2 = 0 \quad \text{putting in } \ddot{\mathbf{r}} = \mathbf{g} + (\lambda/m)\mathbf{r} \text{ and } \mathbf{r} \cdot \mathbf{r} = L^2,$$

so

$$\lambda = -\frac{m(|\mathbf{v}|^2 + \mathbf{r} \cdot \mathbf{g})}{L^2}. \quad (6.8)$$

The function `rod_multiplier` of `mdlab` evaluates it. The rod's force $\lambda\mathbf{r}$ is $-\lambda L$ times $-\mathbf{r}/L$, the arrow of length one from the bob to the pivot, so the rod pulls the bob towards the pivot with the force $F_{\text{rod}} = -\lambda L$, which is negative when the rod pushes.

With the bob at the angle θ from the vertical, $\mathbf{r} = L(\sin \theta, -\cos \theta)$, so $\mathbf{r} \cdot \mathbf{g} = gL \cos \theta$, and $|\mathbf{v}| = L|\dot{\theta}|$ (Section 6.1). Then

$$F_{\text{rod}} = -\lambda L = \frac{m(L^2\dot{\theta}^2 + gL \cos \theta)}{L} = m(g \cos \theta + L\dot{\theta}^2) :$$

the part of the weight along the rod plus the inward pull $mL\dot{\theta}^2$ that going round a circle needs (Section 1.7).

Released from rest at the angle θ_0 , the bob keeps its energy $K + U$ of Section 6.1 at its starting value $mgL(1 - \cos \theta_0)$, so $\frac{1}{2}mL^2\dot{\theta}^2 = mgL(1 - \cos \theta_0) - mgL(1 - \cos \theta) = mgL(\cos \theta - \cos \theta_0)$; multiplying by $2/(mL)$, $L\dot{\theta}^2 = 2g(\cos \theta - \cos \theta_0)$ and

$$F_{\text{rod}} = mg(3 \cos \theta - 2 \cos \theta_0).$$

Figure 6.4 follows a bob of 0.5 kg on a rod of 1.2 m released from 120° , computed in x and y with the multiplier: the length stays 1.2 m to within 1×10^{-11} m, and the force agrees with the formula at every angle: at the bottom it is $mg(3 - 2 \cos 120^\circ) = mg(3 + 1) = 4mg = 4 \times 0.5 \times 9.80665 = 19.613$ N. It is negative where $3 \cos \theta < 2 \cos \theta_0$, beyond $\cos \theta = \frac{2}{3} \cos \theta_0 = -\frac{1}{3}$, here beyond 109.47° : the rod pushes the bob outwards. A string, which can only pull, could not follow this swing at all: at the release point the force needed is $mg \cos 120^\circ = -\frac{1}{2}mg$, a push, so a string is slack from the start. Exercise 6.15 shows that a string stays taut only for swings of at most 90° , the limit of Section 4.5. The angle 109.47° belongs to this swing of 120° : between 90° and 109.47° the bob, though above the pivot, moves fast enough that the rod still pulls.

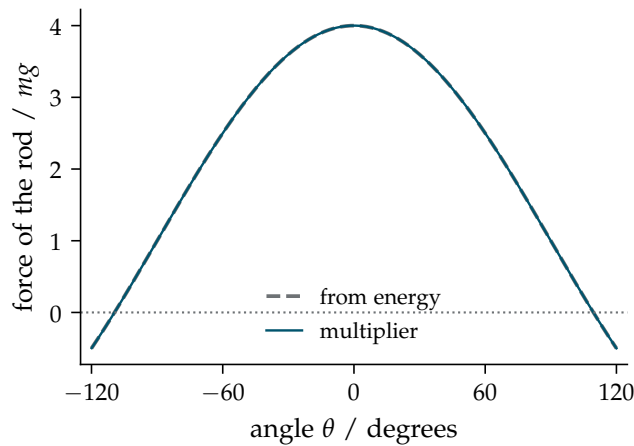


FIGURE 6.4: The force of the rod on a pendulum's bob of 0.5 kg on a rod of 1.2 m, released from rest at 120° , divided by the bob's weight, against the angle over one swing: from the Lagrange multiplier of the constraint (solid) and from energy, $3 \cos \theta - 2 \cos \theta_0$ (dashed), which the solid line hides. Below the dotted line the rod pushes.

Bonds of fixed length

A bond held at the length d between atoms i and j is the constraint $\chi = \frac{1}{2}(|\mathbf{r}_j - \mathbf{r}_i|^2 - d^2) = 0$. Its gradient with respect to $\mathbf{r}_j$ is $\mathbf{r}_j - \mathbf{r}_i$, and with respect to $\mathbf{r}_i$ it is $-(\mathbf{r}_j - \mathbf{r}_i)$ (Exercise 6.16), so the multiplier gives the two atoms equal and opposite forces along the bond, as the third law requires. Freezing the fastest bonds of a molecule this way, those to hydrogen, whose periods of 9 fs for O–H and 11 fs for C–H limit the step of a simulation (Section 4.10), lets the steps grow; Chapter 11 finds the multipliers at every step with the method called SHAKE.

KEY IDEA. To find the largest or smallest value of f subject to $\chi = 0$, solve $\nabla f = \lambda \nabla \chi$ with $\chi = 0$. In mechanics a constraint $\chi = 0$ that does no work enters as the force $\lambda \nabla \chi$, and λ , fixed by differentiating the constraint twice, measures the force needed to keep it: for a pendulum it gives the force of the rod, $-\lambda L$.

NOTEBOOK 06 *the pendulum in x and y with its multiplier*

CHECK YOUR UNDERSTANDING. A pendulum is released from rest at 90° . What force does the rod exert at the bottom of the swing, in units of the bob's weight?

What we now have

Mechanics can be written in any coordinates that fix the positions of a system once its constraints are obeyed, its generalised coordinates, with the Lagrangian $\mathcal{L} = K - U$ expressed in them. Among all paths with the same ends in space and time, the true one makes the action $S = \int \mathcal{L} dt$ stationary, which is not always least. Changing a path by a small bump and integrating by parts turns this into the Euler–Lagrange equations, $d(\partial \mathcal{L} / \partial \dot{q}_k) / dt = \partial \mathcal{L} / \partial q_k$, one for each coordinate, which for forces from a potential are Newton's second law.

Constraints that do no work drop out of $\mathcal{L}$: solving the bead's equation along its wire gives the times of the journey that Chapter 3 described by its speeds, and the pendulum's equation takes one line.

The momentum $p_k = \partial \mathcal{L} / \partial \dot{q}_k$ of a coordinate that $\mathcal{L}$ does not contain is conserved, which is why a central force conserves angular momentum; the distance from the centre then moves in the effective potential $U + L^2 / (2mr^2)$.

When a constraining force is wanted, or no convenient coordinates exist, a Lagrange multiplier keeps the constraint: for a pendulum it gives the force of the rod, $-\lambda L$, and for bonds held at fixed length the multipliers are what Chapter 11 computes at every step.

Chapter 7 trades the generalised velocities for the momenta and arrives at Hamilton's equations, the form of mechanics on which the methods of simulation are built.

Answers to the checks

SECTION 6.1. The double pendulum in a plane needs the two angles of its rods, 2 coordinates. The rigid diatomic molecule needs the three coordinates of its centre of mass and two angles for the direction of its bond, 5, the $3N - 1$ left when one distance of $N = 2$ atoms is fixed.

SECTION 6.2. With $U = 0$ the expansion of the action has only the kinetic terms: $\mathcal{S}(\zeta) = \mathcal{S}(0) + \zeta \int m\dot{x}\dot{\eta} dt + \zeta^2 \frac{1}{2}m \int \dot{\eta}^2 dt$. On the straight path $\dot{x}$ is constant, so the middle term is $m\dot{x} [\eta]_{t_1}^{t_2} = 0$, the bump being zero at both ends. The last term is positive unless $\dot{\eta} = 0$ throughout, which with η zero at the ends means no change at all. Any other path with the same ends is the straight one plus such a change, with $\zeta = 1$ and η the difference, and with $U = 0$ the expansion is exact, so every other path has more action.

SECTION 6.3. $\partial\mathcal{L}/\partial\dot{x} = m\dot{x}$ and $\partial\mathcal{L}/\partial x = -kx$, so $m\ddot{x} = -kx$, the equation of Section 2.8.

SECTION 6.4. From 0.554 s to 1.533 s, that is $1.533 - 0.554 = 0.979$ s.

SECTION 6.5. $\mathcal{L}$ does not contain x , only $\dot{x}$, so $p_x = \partial\mathcal{L}/\partial\dot{x} = m\dot{x}$ is conserved: nothing pushes the ball sideways. It does contain y , through mgy , so $p_y = m\dot{y}$ changes, at the rate $\partial\mathcal{L}/\partial y = -mg$, the weight.

SECTION 6.6. With $\theta_0 = 90^\circ$, $\cos\theta_0 = 0$, and at the bottom $\cos\theta = 1$, so $F_{\text{rod}} = mg(3 - 0) = 3mg$, three times the weight.

Exercises

The exercises run from questions of understanding, through calculations, to short derivations marked 'show that'. Full solutions are in 'Solutions to the exercises' at the end of the book, and Notebook 06 checks every numerical answer.

EXERCISE 6.1 (*counting coordinates*). How many generalised coordinates do (a) a bead threaded on a fixed circular hoop, (b) two atoms joined by a rigid bond and moving in a plane, and (c) a rigid water molecule free in space have?

EXERCISE 6.2 (by parts). Find $\int_0^1 x e^x dx$ by integration by parts.

EXERCISE 6.3 (a saddle of the action). For the block of Section 6.2 ($m = 0.5 \text{ kg}$, $k = 50 \text{ N m}^{-1}$) over $\tau = 0.5 \text{ s}$, show that adding the bump $\zeta \sin(n\pi t/\tau)$ to the true path changes the action by $\zeta^2(\tau/4)(m(n\pi/\tau)^2 - k)$, and evaluate the coefficient of ζ^2 for $n = 1$ and 2 .

EXERCISE 6.4 (free fall). Write the Euler–Lagrange equation for $\mathcal{L} = \frac{1}{2}m\dot{x}^2 - mgx$.

EXERCISE 6.5 (two coordinates). Show that for $\mathcal{L} = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2) - U(x, y)$ the two Euler–Lagrange equations are Newton’s second law for the two components of the force $-\nabla U$.

EXERCISE 6.6 (the bottom of a valley). Show that near the bottom x_0 of a valley of the wire, where $h'(x_0) = 0$, Equation (6.4) becomes $\ddot{x} \approx -g h''(x_0)(x - x_0)$, and hence that small swings have $\omega_0 = \sqrt{g h''(x_0)}$, as Section 4.4 found.

EXERCISE 6.7 (a bead in a bowl). A bead slides in a circular bowl, $h(x) = R - \sqrt{R^2 - x^2}$. Show that with $x = R \sin \theta$ its Lagrangian becomes that of a pendulum of length R .

EXERCISE 6.8 (energy from the equation). Multiplying Equation (6.5) by $\dot{\theta}$, show that $\frac{1}{2}mL^2\dot{\theta}^2 + mgL(1 - \cos \theta)$ does not change.

EXERCISE 6.9 (the energy of a Lagrangian). For a Lagrangian $\mathcal{L}(q, \dot{q})$ of one coordinate that does not contain the time, show from Equation (6.3) that $\dot{q} \partial \mathcal{L} / \partial \dot{q} - \mathcal{L}$ does not change along the motion, and that for the bead it equals $K + U$.

EXERCISE 6.10 (two carts). Two carts of masses m_1 and m_2 on a smooth track are joined by a spring, $\mathcal{L} = \frac{1}{2}m_1\dot{x}_1^2 + \frac{1}{2}m_2\dot{x}_2^2 - \frac{1}{2}k(x_2 - x_1 - \ell)^2$. Rewrite it with the centre of mass $X = (m_1x_1 + m_2x_2)/(m_1 + m_2)$ and the separation $r = x_2 - x_1$, and show that the momentum belonging to X is conserved and that r moves with the reduced mass of Section 4.8.

EXERCISE 6.11 (turning points). The puck of Figure 6.3 is started 0.4 m out at 0.3 m s^{-1} across the line to the hole. Find L , E and the two distances between which it swings.

EXERCISE 6.12 (going round in a circle). For $U = \frac{1}{2}kr^2$, show that the puck goes round in a circle when $r^4 = L^2/(mk)$, and that it then turns at the angular speed $\sqrt{k/m}$, whatever L . Evaluate the radius for the puck of Figure 6.3, of 0.2 kg on a spring of 2 N m^{-1} , with $L = 0.05 \text{ kg m}^2 \text{ s}^{-1}$.

EXERCISE 6.13 (on the unit circle). Find the largest value of xy on the circle $x^2 + y^2 = 1$, and where it is reached.

EXERCISE 6.14 (*a deep swing*). A bob of 0.5 kg on a rod is released from rest at 60° . What is the force of the rod when the bob passes the bottom, and when it is released?

EXERCISE 6.15 (*when a string goes slack*). Show that a pendulum on a string released from rest at θ_0 keeps its string taut throughout only if $\theta_0 \leq 90^\circ$.

EXERCISE 6.16 (*the forces of a bond*). For $\chi = \frac{1}{2}(|\mathbf{r}_j - \mathbf{r}_i|^2 - d^2)$, show that $\nabla_j \chi = \mathbf{r}_j - \mathbf{r}_i$ and $\nabla_i \chi = -(\mathbf{r}_j - \mathbf{r}_i)$, so that the forces $\lambda \nabla_i \chi$ and $\lambda \nabla_j \chi$ on the two atoms are equal, opposite and along the bond.

EXERCISE 6.17 (*a smaller swing*). The bead of Section 6.4 is released from rest on the left slope at a height of 0.8 m. Using `solve_wire` in Notebook o6, find where it turns and how long it takes to come back.

Hamiltonian mechanics and phase space

Chapter 6 described motion through generalised coordinates and their velocities, and introduced the conjugate momentum belonging to each coordinate. We now use those momenta in place of the velocities. This change leads to Hamilton's equations and makes the state of a system a point in phase space, the space of all allowed coordinates and momenta. Following many starting states at once reveals properties of the motion that are difficult to see from a single trajectory.

For the time-independent systems considered here, the flow preserves energy as well as the area, or in more dimensions the volume, of a region of states. We derive this volume conservation as Liouville's theorem, which Chapter 12 needs for statistical mechanics. We then connect spatial symmetries to conserved momenta and establish the conditions under which reversing momenta retraces a trajectory. These results give us concrete tests for a numerical method: the final section compares two simple step maps, preparing the integration methods of Chapter 9.

7.1 From velocities to momenta

The state of the bead on its wire ([Section 6.1](#)) is its position x and its rate $\dot{x}$, but it can equally be given by x and the momentum $p = \partial\mathcal{L}/\partial\dot{x}$ of [Equation \(6.6\)](#), which for the bead is $m(1 + h'^2)\dot{x}$ ([Section 6.4](#)): since $1 + h'^2$ is positive, each $\dot{x}$ gives one p , and $\dot{x} = p/(m(1 + h'^2))$ gives the rate back. Hamilton's form of mechanics describes the state by the coordinates and their momenta. [Section 7.4](#) shows that the motion keeps areas in the plane of q and p , which it does not in general do in the plane of q and $\dot{q}$. The new description needs a function of q and p that does the work that $\mathcal{L}$ does

for q and $\dot{q}$. The tool that exchanges a variable for the slope of a function of it is the Legendre transform.

THE LEGENDRE TRANSFORM. Let $f(v)$ be a function whose slope $f'(v)$ grows steadily with v , that is $f''(v) > 0$ (its graph curves upwards). Then each slope p is reached at no more than one value of v , and we take p among the slopes that f has. The straight line pv through the origin has that slope (Figure 7.1a). The gap between line and curve, $pv - f(v)$, changes at the rate $p - f'(v)$ as v grows: the rate is positive while $f'(v) < p$ and negative once $f'(v) > p$, so the gap rises to its largest value at the v where $f'(v) = p$ and falls after it. That largest gap is the Legendre transform of f ,

$$f^*(p) = pv - f(v), \quad \text{with } v \text{ such that } f'(v) = p.$$

The tangent to f at that v has slope p and passes through $(v, f(v))$, so at the vertical axis, a distance v to the left, it is at $f(v) - pv = -f^*(p)$: the transform is where the tangent of slope p meets the axis, with its sign reversed.

As p changes, v changes with it. By the product rule and the chain rule,

$$\begin{aligned} \frac{df^*}{dp} &= v + p \frac{dv}{dp} - f'(v) \frac{dv}{dp} && \text{differentiating } pv \text{ and } f(v(p)) \\ &= v && \text{since } f'(v) = p \text{ cancels the last two terms.} \end{aligned}$$

The slope of f^* is v , as the slope of f is p (Figure 7.1b). This slope grows with p : differentiating $f'(v(p)) = p$ with respect to p by the chain rule gives $f''(v) dv/dp = 1$, so $dv/dp = 1/f''(v) > 0$, as for the inverse functions of Section 4.5. So f^* can be transformed in its turn. Its transform at the slope v is $vp - f^*(p)$, with p the slope of f at which the slope of f^* equals v , which is the $p = f'(v)$ we started from, and

$$vp - f^*(p) = vp - (pv - f(v)) = f(v) :$$

the transform of the transform is the function we started from, so f^* carries the same information as f , written in terms of the slope. For example, $f(v) = \frac{1}{2}mv^2$ has $f'(v) = mv$, so $p = mv$ and $v = p/m$, and

$$\begin{aligned} f^*(p) &= p \frac{p}{m} - \frac{1}{2}m \left(\frac{p}{m} \right)^2 && \text{putting } v = p/m \text{ into } pv - f(v) \\ &= \frac{p^2}{m} - \frac{p^2}{2m} = \frac{p^2}{2m}, \end{aligned}$$

whose slope p/m is indeed v .

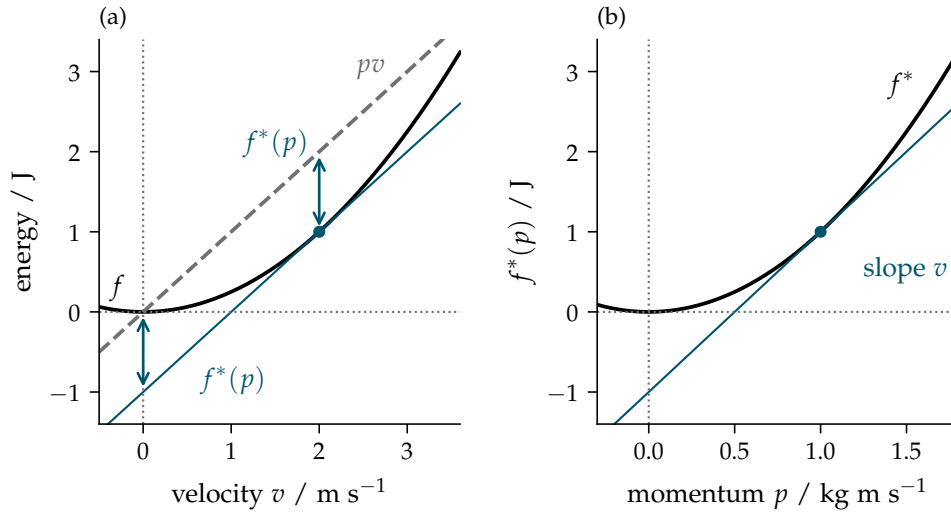


FIGURE 7.1: The Legendre transform of $f(v) = \frac{1}{2}mv^2$ for a body of 0.5 kg. (a) The line pv with $p = 1 \text{ kg m s}^{-1}$ (dashed) differs from f by $pv - f(v)$, which is largest, $f^*(p) = 1 \text{ J}$, at $v = p/m = 2 \text{ m s}^{-1}$, where the slope of f is p ; the tangent there (teal) meets the vertical axis at $-f^*(p)$. (b) The transform $f^*(p) = p^2/(2m)$, whose slope at p is v .

The Hamiltonian

For one coordinate, the Lagrangian $\mathcal{L}(q, \dot{q})$ is a function of $\dot{q}$ at each fixed q , and its slope in $\dot{q}$ is the momentum. Its Legendre transform in $\dot{q}$, with q held fixed throughout, is the Hamiltonian,

$$H(q, p) = p\dot{q} - \mathcal{L}(q, \dot{q}), \quad \text{with } \dot{q} \text{ such that } \frac{\partial \mathcal{L}}{\partial \dot{q}} = p, \quad (7.1)$$

written in terms of q and p alone by putting in $\dot{q}$ found from p ; for several coordinates it is $H = \sum_k p_k \dot{q}_k - \mathcal{L}$. The transform needs $\partial \mathcal{L} / \partial \dot{q}$ to grow steadily with $\dot{q}$, which holds for every system met so far, because the kinetic energy is $K = \frac{1}{2}a(q)\dot{q}^2$ with a positive a : m for a body on a line, $m(1 + h'^2)$ for the bead, mL^2 for the pendulum.

For such a system $p = \partial \mathcal{L} / \partial \dot{q} = a\dot{q}$, since U does not contain $\dot{q}$, so $p\dot{q} = a\dot{q}^2 = 2K$, and

$$H = 2K - (K - U) = K + U :$$

the Hamiltonian is the energy, now written in terms of the momentum. It is the quantity $\dot{q} \partial \mathcal{L} / \partial \dot{q} - \mathcal{L}$ that [Exercise 6.9](#) found to be conserved. With several coordinates, when K is a sum of terms $K_k = \frac{1}{2}a_k \dot{q}_k^2$, one for each, every term gives $p_k \dot{q}_k = a_k \dot{q}_k^2 = 2K_k$, and

adding them again $H = K + U$; [Exercise 7.2](#) shows that a kinetic energy with a product of two rates gives the same result. The examples of Chapter 6 become

$$\begin{aligned} \text{a block on a spring:} \quad p &= m\dot{x}, & H &= \frac{p^2}{2m} + \frac{1}{2}kx^2, \\ \text{the pendulum:} \quad p &= mL^2\dot{\theta}, & H &= \frac{p^2}{2mL^2} + mgL(1 - \cos\theta), \\ \text{the bead:} \quad p &= m(1 + h'^2)\dot{x}, & H &= \frac{p^2}{2m(1 + h'(x)^2)} + mgh(x), \end{aligned}$$

each kinetic energy following from $K = \frac{1}{2}a\dot{q}^2$ with $\dot{q} = p/a$, which gives $K = \frac{1}{2}a(p/a)^2 = p^2/(2a)$. The pendulum's momentum is its angular momentum about the pivot ([Section 5.1](#)). The bead's kinetic energy now depends on x as well as on p , because the same momentum means a faster bead where the wire is flat than where it is steep.

For N atoms, with $\mathbf{p}_i = m_i\dot{\mathbf{r}}_i$,

$$H(\mathbf{r}^N, \mathbf{p}^N) = \sum_{i=1}^N \frac{|\mathbf{p}_i|^2}{2m_i} + U(\mathbf{r}^N),$$

and the state is the point $\Gamma = (\mathbf{r}^N, \mathbf{p}^N)$ of $6N$ numbers. For the puck of [Section 6.5](#), [Exercise 7.5](#) finds $H = p_r^2/(2m) + p_\theta^2/(2mr^2) + U(r)$, in which the effective potential of Chapter 6 appears with p_θ as the angular momentum L of [Section 6.5](#), not the rod length.

FROM ELECTRONIC STRUCTURE. The same transform links the thermodynamic potentials. The free energy of the electrons at a fixed number N_e has the slope μ , the chemical potential, with respect to N_e ; its Legendre transform in N_e , with the sign reversed, is the grand potential, the free energy minus μN_e , which is a function of μ . Grand-canonical DFT holds the electrode's chemical potential fixed and lets N_e follow, so it minimises the grand potential, the transform, at fixed μ , as Hamilton's form of mechanics holds p as the variable in place of $\dot{q}$.

KEY IDEA. The Hamiltonian $H(q, p) = \sum_k p_k \dot{q}_k - \mathcal{L}$ is the Legendre transform of the Lagrangian in the velocities, written in terms of the coordinates and the momenta. When the kinetic energy is quadratic in the velocities it is the energy, $H = K + U$.

CHECK YOUR UNDERSTANDING. Write the Hamiltonian of a block of 2 kg on a spring of 8 N m^{-1} . If its energy is 1 J, what is its momentum as it passes $x = 0$?

7.2 Hamilton's equations

The Hamiltonian is built from the Lagrangian, so the Euler–Lagrange equation must be expressible through it. Finding how needs the partial derivatives of H , and these differ from those of $\mathcal{L}$ in what is held fixed: $\partial\mathcal{L}/\partial q$ is taken at fixed $\dot{q}$, while $\partial H/\partial q$ is taken at fixed p , and at fixed p the rate $\dot{q}$ changes with q . The chain rule of [Section 3.10](#) keeps track of this once it is stated for functions of several variables that themselves depend on several others.

THE CHAIN RULE IN GENERAL. Let $f(x, y)$ depend on two variables, and let these depend in turn on two others, $x(s, t)$ and $y(s, t)$. Holding s fixed and letting t change moves the point (x, y) along a path, so the chain rule along a path ([Section 3.10](#)) gives

$$\frac{\partial f}{\partial t} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial t},$$

where every derivative with respect to t is taken at fixed s , and $\partial f/\partial x$ and $\partial f/\partial y$ are evaluated at the point $(x(s, t), y(s, t))$. Holding t fixed instead gives the same with s in place of t , and with more variables there is one term for each variable of f . For example, $f = x^2y$ with $x = s + t$ and $y = s - t$ has

$$\frac{\partial f}{\partial t} = 2xy \times 1 + x^2 \times (-1) = 2(s+t)(s-t) - (s+t)^2,$$

which is what differentiating $f = (s+t)^2(s-t)$ directly with respect to t gives, by the product rule. A partial derivative is only defined once we say which variables are held fixed: $\partial f/\partial x$ at fixed y is $2xy$, but if y were written as $s - t$ and x as $s + t$ and we asked how f changes with x at fixed s , so that $y = 2s - x$ changes too, then $f = x^2(2s - x)$ and the answer would be $2x(2s - x) - x^2 = 2xy - x^2$, which differs from $2xy$.

The equations

In [Equation \(7.1\)](#), $\dot{q}$ is a function of q and p , found from $p = \partial\mathcal{L}/\partial\dot{q}$, so

$$H(q, p) = p\dot{q}(q, p) - \mathcal{L}(q, \dot{q}(q, p)).$$

The variable p enters through the factor p and through $\dot{q}$, and $\mathcal{L}$ depends on p only through its second variable $\dot{q}$. At fixed q , by the product rule for the first term and the chain rule for the second,

$$\begin{aligned}\frac{\partial H}{\partial p} &= \dot{q} + p \frac{\partial \dot{q}}{\partial p} - \frac{\partial \mathcal{L}}{\partial \dot{q}} \frac{\partial \dot{q}}{\partial p} && \text{differentiating at fixed } q \\ &= \dot{q} && \text{since } \partial \mathcal{L} / \partial \dot{q} = p.\end{aligned}$$

The variable q enters $p \dot{q}$ through $\dot{q}(q, p)$, and $\mathcal{L}$ twice, as its first variable, whose rate of change with q is 1, and through $\dot{q}(q, p)$. At fixed p , the general chain rule, with $\mathcal{L}$ for f , q and $\dot{q}$ for x and y , and q for t , gives one term for each,

$$\begin{aligned}\frac{\partial H}{\partial q} &= p \frac{\partial \dot{q}}{\partial q} - \frac{\partial \mathcal{L}}{\partial q} - \frac{\partial \mathcal{L}}{\partial \dot{q}} \frac{\partial \dot{q}}{\partial q} && \text{differentiating at fixed } p \\ &= -\frac{\partial \mathcal{L}}{\partial q} && \text{since } \partial \mathcal{L} / \partial \dot{q} = p \text{ again,}\end{aligned}$$

where $\partial \mathcal{L} / \partial q$ is the derivative of [Section 6.3](#), at fixed $\dot{q}$. The Euler–Lagrange equation [Equation \(6.3\)](#) says $\dot{p} = d(\partial \mathcal{L} / \partial \dot{q}) / dt = \partial \mathcal{L} / \partial q$, which is $-\partial H / \partial q$. Together,

$$\dot{q} = \frac{\partial H}{\partial p}, \quad \dot{p} = -\frac{\partial H}{\partial q}. \quad (7.2)$$

These are Hamilton's equations, one pair for each coordinate, found in the same way for each k when there are several. In place of one equation in the second derivative $\ddot{q}$ there are two in first derivatives, so the state (q, p) at one instant fixes the motion, as [Section 2.6](#) found for (x, v) .

For the block, $H = p^2 / (2m) + \frac{1}{2} kx^2$ and [Equation \(7.2\)](#) give $\dot{x} = p/m$ and $\dot{p} = -kx$, so $m\ddot{x} = \dot{p} = -kx$, the spring of [Section 2.8](#). For the pendulum, $\dot{\theta} = p / (mL^2)$ and $\dot{p} = -mgL \sin \theta$, which together give $mL^2 \ddot{\theta} = \dot{p} = -mgL \sin \theta$, [Equation \(6.5\)](#). For N atoms, $\dot{\mathbf{r}}_i = \mathbf{p}_i / m_i$ and $\dot{\mathbf{p}}_i = -\nabla_i U = \mathbf{F}_i$: Newton's second law in the first-order form of [Equation \(2.5\)](#), with the momentum in place of the velocity. [Exercise 7.6](#) recovers the bead's equation [Equation \(6.4\)](#) from its Hamiltonian.

Phase space and the flow

For one coordinate the states (q, p) fill a plane, the phase plane of [Section 4.3](#) with the momentum in place of the velocity; for n coordinates they fill a space of $2n$ dimensions, and for N atoms one of $6N$. This is phase space. At each of its points Hamilton's equations give the velocity of the state, $(\dot{q}, \dot{p}) = (\partial H / \partial p, -\partial H / \partial q)$, a velocity attached to every point, as the current of a river has a velocity at every point of its surface. A

state moves as a leaf carried by such a current, and the motions of the system are the paths of the leaves, which, by the uniqueness of [Section 2.6](#), never cross when H does not contain the time; the separatrix of the pendulum below only seems to cross itself at the upright states, which are motions of their own ([Section 4.3](#)). This motion of all states at once is called the Hamiltonian flow.

The flow keeps the energy. Along any motion, by the chain rule along a path,

$$\frac{dH}{dt} = \frac{\partial H}{\partial q} \dot{q} + \frac{\partial H}{\partial p} \dot{p} = \frac{\partial H}{\partial q} \frac{\partial H}{\partial p} - \frac{\partial H}{\partial p} \frac{\partial H}{\partial q} = 0.$$

Equivalently, the velocity of the flow is at right angles to the gradient $(\partial H/\partial q, \partial H/\partial p)$, since their dot product is the same zero, and so runs along the contours of H ([Section 3.10](#)). The phase portrait is therefore the contour map of H . When H contains the time itself, the chain rule has a third term, and $dH/dt = \partial H/\partial t$. The block pushed by the force $F_0 \cos \omega t$ of [Section 4.7](#), without drag, has $U = \frac{1}{2}kx^2 - F_0x \cos \omega t$, whose slope gives the spring's force and the push, and $\partial H/\partial t = \omega F_0x \sin \omega t$, which is not zero: H , which now includes the push's term $-F_0x \cos \omega t$, rises and falls as the push feeds energy in and takes it out.

The pendulum's portrait

For the pendulum of [Section 6.6](#), a bob of 0.5 kg on a rod of 1.2 m, $mL^2 = 0.72 \text{ kg m}^2$ and $mgL = 5.884 \text{ J}$. The contours of H ([Figure 7.2a](#)) are of three kinds. Below the energy $2mgL$ of the bob held upright, they are closed loops around $\theta = 0$: the bob swings, and p changes sign at each end of the swing. Above $2mgL$ they are open, wavy lines on which p never vanishes: the bob goes over the top. The contour at $2mgL$ itself passes through the upright states $(\pm\pi, 0)$ and separates the two kinds; it is called the separatrix. On it, at the bottom, $p^2/(2mL^2) = 2mgL$, so

$$p = \sqrt{4m^2gL^3} = 2mL^2\sqrt{\frac{g}{L}} = 2mL^2\omega_0 = 2 \times 0.72 \times 2.8587 = 4.117 \text{ kg m}^2 \text{ s}^{-1},$$

with $\omega_0 = \sqrt{g/L}$ of [Section 4.4](#).

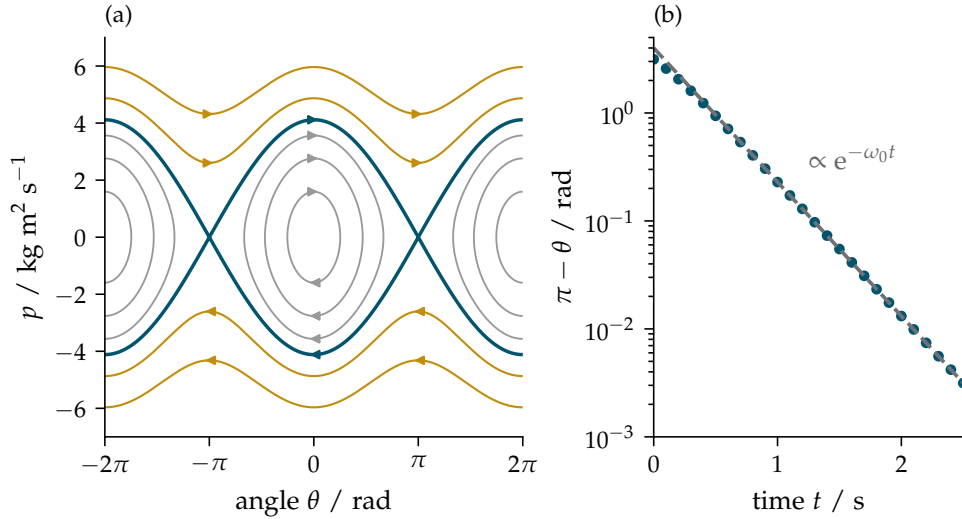


FIGURE 7.2: The pendulum of 0.5 kg on a rod of 1.2 m. (a) Contours of H in the phase plane, with arrowheads showing the direction of the flow: swinging, below the upright energy $2mgL$ (grey); the separatrix, at $2mgL$ (teal); going over the top (ochre). (b) Started at the bottom with the momentum of the separatrix, the angle still to go, $\pi - \theta$, from Hamilton's equations (dots), on a logarithmic axis, against a line that falls as $e^{-\omega_0 t}$ (dashed).

A bob given exactly this momentum at the bottom creeps towards the top and never arrives, as Section 4.3 said. On the separatrix the energy is $2mgL$ at every angle, $\frac{1}{2}mL^2\dot{\theta}^2 + mgL(1 - \cos \theta) = 2mgL$, so

$$\begin{aligned} \dot{\theta}^2 &= \frac{2g}{L} (1 + \cos \theta) && \text{subtracting } mgL(1 - \cos \theta), \text{ dividing by } \frac{1}{2}mL^2 \\ &= \frac{4g}{L} \cos^2 \frac{\theta}{2} && \text{by } \cos \theta = 2 \cos^2 \frac{\theta}{2} - 1 \\ \dot{\theta} &= 2\omega_0 \cos \frac{\theta}{2} && \text{the root for motion towards } +\pi, \end{aligned}$$

taking the positive root because $\cos \frac{\theta}{2} > 0$ for $-\pi < \theta < \pi$, and where the second line uses the double-angle formula of Section 1.5, $\cos \theta = \cos^2 \frac{\theta}{2} - \sin^2 \frac{\theta}{2}$, with $\sin^2 = 1 - \cos^2$. Near the top, we write $\varphi = \pi - \theta$ for the angle still to go. Then $\cos \frac{\theta}{2} = \cos(\frac{\pi}{2} - \frac{\varphi}{2}) = \sin \frac{\varphi}{2}$ (Section 1.5), and $\sin \frac{\varphi}{2} \approx \frac{\varphi}{2}$ for small φ (Section 1.8), and $\dot{\varphi} = -\dot{\theta}$, so

$$\dot{\varphi} \approx -\omega_0 \varphi.$$

This is the equation of the drag of Section 2.7 with ω_0 in place of γ and nothing to approach but zero, so φ falls as $e^{-\omega_0 t}$: it halves every $\ln 2 / \omega_0 = 0.2425$ s, however small it is, and never reaches zero. Computed from Hamilton's equations (Figure 7.2b), the bob started at the bottom is 0.229 rad short of the top after 1 s and 0.0132 rad, 0.75° , short after 2 s, and from then on the gap halves every 0.2425 s.

KEY IDEA. Hamilton's equations, $\dot{q} = \partial H / \partial p$ and $\dot{p} = -\partial H / \partial q$, move every state of phase space along a flow that runs along the contours of H , so H is conserved unless it contains the time.

NOTEBOOK 07 launch a pendulum anywhere on its phase portrait

CHECK YOUR UNDERSTANDING. A ball thrown straight up has $H = p^2 / (2m) + mgq$, with q its height. Write Hamilton's equations for it, and describe the contours of H in its phase plane.

7.3 Poisson brackets and the Liouville operator

Besides the coordinates, a simulation is asked for quantities computed from the state: the energy, a momentum, the distance between two atoms, the angle of a bond. This section finds how any such quantity changes along the motion, and turns the answer into a series that gives the motion from the starting state, at every time in simple cases and over a short time in general, the form in which Chapter 9 will build methods of computing motion.

The rate of change of any quantity

Let $A(q, p)$ be any quantity computed from the state. Along a motion, by the chain rule along a path and Hamilton's equations,

$$\frac{dA}{dt} = \frac{\partial A}{\partial q} \dot{q} + \frac{\partial A}{\partial p} \dot{p} = \frac{\partial A}{\partial q} \frac{\partial H}{\partial p} - \frac{\partial A}{\partial p} \frac{\partial H}{\partial q}.$$

The combination on the right is called the Poisson bracket of A and H . For any two quantities A and B of the state it is defined as

$$\{A, B\} = \frac{\partial A}{\partial q} \frac{\partial B}{\partial p} - \frac{\partial A}{\partial p} \frac{\partial B}{\partial q}, \quad (7.3)$$

with a sum of such terms, one for each pair q_k, p_k , when there are several coordinates. So

$$\frac{dA}{dt} = \{A, H\}.$$

Exchanging A and B in Equation (7.3) exchanges the two products, so $\{B, A\} = -\{A, B\}$; with $B = A$ this reads $\{A, A\} = -\{A, A\}$, so the bracket of any quantity with itself is zero. Four cases follow at once. With $A = q$ and $B = p$, $\partial q / \partial q = 1$, $\partial p / \partial p = 1$ and the cross derivatives are zero, so $\{q, p\} = 1$. With $A = q$, $\{q, H\} =$

$\partial H/\partial p$, and with $A = p$, $\{p, H\} = -\partial H/\partial q$: Hamilton's equations themselves. With $A = H$, $\{H, H\} = 0$: the energy is conserved. More generally, a quantity that does not contain the time is conserved when its bracket with H is zero, and only then. For the puck, $H = p_r^2/(2m) + p_\theta^2/(2mr^2) + U(r)$ does not contain θ . In the sum over the pairs (r, p_r) and (θ, p_θ) the only derivative of p_θ that is not zero is $\partial p_\theta/\partial p_\theta = 1$, so $\{p_\theta, H\} = -\partial H/\partial \theta = 0$, and the angular momentum p_θ is conserved; [Exercise 7.10](#) finds the same in Cartesian coordinates.

FROM ELECTRONIC STRUCTURE. Dirac found that the equations of quantum mechanics follow from the classical ones when the Poisson bracket $\{A, B\}$ is replaced by the commutator $(AB - BA)/(\hbar i)$ of the corresponding operators⁹. Then $\{q, p\} = 1$ becomes $qp - pq = \hbar i$, and $dA/dt = \{A, H\}$ becomes Heisenberg's equation of motion. A quantity that commutes with the Hamiltonian is conserved in both.

The Liouville operator

The bracket with H is an instruction that turns any quantity into its rate of change. Writing it as $\hat{L}$,

$$\hat{L}A = \{A, H\} = \dot{q} \frac{\partial A}{\partial q} + \dot{p} \frac{\partial A}{\partial p},$$

where $\dot{q} = \partial H/\partial p$ and $\dot{p} = -\partial H/\partial q$ are written in terms of q and p . An instruction of this kind, which acts on a function to give another, is called an operator, as d/dx is the operator that gives a derivative; $\hat{L}$ is the Liouville operator. The result $\hat{L}A$ is again a quantity of the state, so its own rate of change is $\hat{L}(\hat{L}A)$, written $\hat{L}^2 A$, and this is the second derivative of A along the motion. Repeating, the n th derivative of A along the motion is $\hat{L}^n A$. The Taylor series of [Section 1.8](#), taken about the start of the motion, then gives

$$A(q(t), p(t)) = A + t \hat{L}A + \frac{t^2}{2!} \hat{L}^2 A + \frac{t^3}{3!} \hat{L}^3 A + \cdots = \sum_{n=0}^{\infty} \frac{t^n}{n!} \hat{L}^n A, \quad (7.4)$$

with every term on the right computed at the starting state (q_0, p_0) and $\hat{L}^0 A = A$. For the functions of [Section 1.8](#) the infinite series converges at every t ; in general only the series stopped after a few terms can be relied on, and only over a short time.

For a ball thrown straight up, $H = p^2/(2m) + mgq$, so $\dot{q} = p/m$, $\dot{p} = -mg$ and $\hat{L} = (p/m)\partial/\partial q - mg\partial/\partial p$. Applying it to q repeatedly,

$$\hat{L}q = \frac{p}{m}, \quad \hat{L}^2 q = \hat{L} \frac{p}{m} = -mg \times \frac{1}{m} = -g, \quad \hat{L}^3 q = \hat{L}(-g) = 0,$$

since $-g$ contains neither q nor p . Every later term is zero too, and Equation (7.4) gives $q(t) = q_0 + p_0 t/m - \frac{1}{2} g t^2$, the ball of Section 1.2.

For the block on a spring, $\dot{q} = p/m$, $\dot{p} = -kq$ and $\hat{L} = (p/m) \partial/\partial q - kq \partial/\partial p$. With $\omega^2 = k/m$,

$$\begin{aligned}\hat{L}q &= \frac{p}{m}, & \hat{L}^2 q &= \hat{L} \frac{p}{m} = -\frac{kq}{m} = -\omega^2 q, \\ \hat{L}^3 q &= -\omega^2 \hat{L}q = -\omega^2 \frac{p}{m}, & \hat{L}^4 q &= -\omega^2 \hat{L}^2 q = \omega^4 q,\end{aligned}$$

since $\hat{L}$ of a constant times a quantity is the constant times $\hat{L}$ of the quantity. The pattern repeats every two steps, each pair bringing a factor $-\omega^2$. Term by term, Equation (7.4) gives

$$q(t) = q_0 + t \frac{p_0}{m} - \frac{t^2}{2!} \omega^2 q_0 - \frac{t^3}{3!} \omega^2 \frac{p_0}{m} + \frac{t^4}{4!} \omega^4 q_0 + \dots$$

Gathering the terms in q_0 and in p_0 separately, and writing each power of t with the matching power of ω , which for the terms in p_0 means writing $p_0/m = (p_0/(m\omega)) \omega$,

$$\begin{aligned}q(t) &= q_0 \left(1 - \frac{(\omega t)^2}{2!} + \frac{(\omega t)^4}{4!} - \dots \right) + \frac{p_0}{m\omega} \left(\omega t - \frac{(\omega t)^3}{3!} + \dots \right) \\ &= q_0 \cos \omega t + \frac{p_0}{m\omega} \sin \omega t,\end{aligned}$$

by the series of Equation (1.11). With $p_0 = mv_0$ this is Equation (2.10). For the pendulum the terms $\hat{L}^n \theta$ grow more complicated, and Notebook 07 shows that a few of them follow the motion well over a short time but not over a long one. Chapter 9 writes the series Equation (7.4) as an exponential of $t\hat{L}$ and splits $\hat{L}$ into a part that moves only the positions and a part that moves only the momenta, each of which can be followed exactly; that splitting is how the standard method of molecular dynamics is derived.

KEY IDEA. Any quantity of the state changes at the rate $dA/dt = \{A, H\} = \hat{L}A$, and is conserved when its Poisson bracket with H is zero. Repeating the Liouville operator $\hat{L}$ gives every derivative of A , and the Taylor series $\sum_n (t^n/n!) \hat{L}^n A$ gives its value at later times from the starting state: at every time for the ball and the spring, over a short time in general.

NOTEBOOK 07 the series of the Liouville operator, term by term

CHECK YOUR UNDERSTANDING. For the block on a spring, find $\hat{L}(qp)$. At which states is the product qp momentarily not changing?

7.4 Liouville's theorem

A drop of ink in a shallow, slow stream, whose water moves alike at every depth, is carried downstream, stretched and folded by the current; the water cannot be squeezed, and none rises from below or sinks, so the patch of ink keeps its area while its shape changes. The Hamiltonian flow does the same to a patch of states. We start many copies of the pendulum from slightly different states, filling a small region of the phase plane, a blob, and let each move by Hamilton's equations. This section shows that the blob keeps its area, whatever the system, and that drag makes it shrink.

Figure 7.3(a) follows such a blob of pendulums: an ellipse of starting states centred on 1.6 rad at rest, reaching 0.4 rad either side in angle and $0.8 \text{ kg m}^2 \text{ s}^{-1}$ either way in momentum, a circle of radius 1 stretched by 0.4 along one axis and 0.8 along the other. Swings of different size take different times (Section 4.5), so the blob is sheared into a curved streak; yet its area, $\pi \times 0.4 \times 0.8 = 1.00531 \text{ J s}$, is the same at every time drawn, to within 2.6 parts in 10^7 , the error of drawing its edge with 6000 points. An area in this plane is an angle times an angular momentum, and has the unit of the action of Section 6.2.

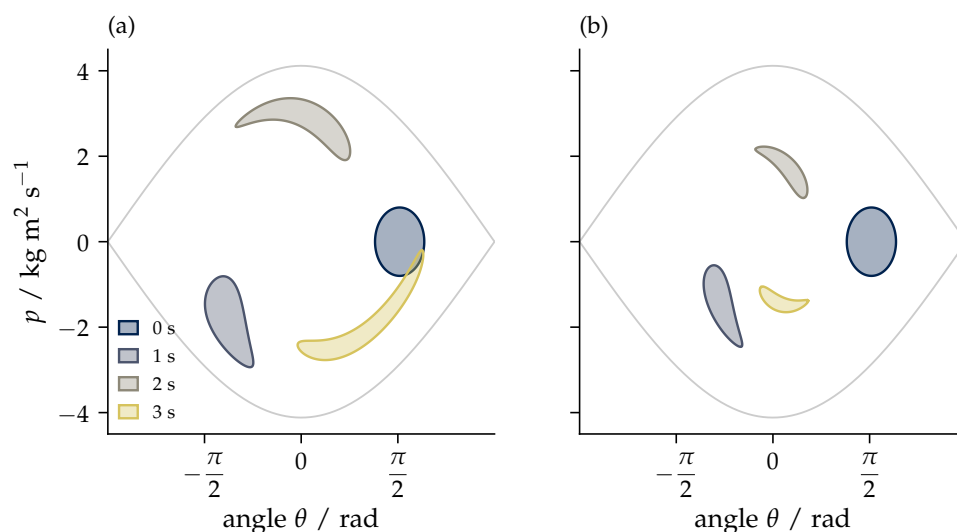


FIGURE 7.3: A blob of states of the pendulum of Figure 7.2, centred on $\theta = 1.6$ rad at rest, carried by the flow and drawn at the times given, inside the separatrix (grey). (a) Without drag: sheared, with its area unchanged. (b) With the drag rate $\gamma = 0.5 \text{ s}^{-1}$: shrinking towards the state of rest at the bottom.

Why the area is kept

We write the velocity of the flow at the point (q, p) as (u, w) , with $u = \dot{q}$ and $w = \dot{p}$. Each short piece of the edge of the blob, of length Δs , moves with the flow, and in a short time τ it sweeps out a thin strip. Only the part of the velocity at right angles to

the piece sweeps area, so if $\mathbf{n}$ is the arrow of length one at right angles to the piece, pointing out of the blob, the strip has the area $(u, w) \cdot \mathbf{n} \Delta s \tau$, counted positive when the edge moves outwards. Adding the strips around the whole edge, the area $\mathcal{A}$ of the blob changes at the rate

$$\frac{d\mathcal{A}}{dt} = \oint (u, w) \cdot \mathbf{n} ds,$$

the flow out through the edge. Going round the edge anticlockwise, a piece $(\Delta q, \Delta p)$ has the blob on its left, and turning it a right angle clockwise gives $(\Delta p, -\Delta q)$, which points out of the blob and has the length Δs ; so $\mathbf{n} \Delta s = (\Delta p, -\Delta q)$, and

$$\frac{d\mathcal{A}}{dt} = \oint (u dp - w dq).$$

Green's theorem (Section 3.11) turns a loop integral into an integral over the area inside: for any smooth (F_q, F_p) , $\oint (F_q dq + F_p dp) = \iint (\partial F_p / \partial q - \partial F_q / \partial p) dA$, around the loop anticlockwise. With $F_q = -w$ and $F_p = u$,

$$\frac{d\mathcal{A}}{dt} = \iint \left(\frac{\partial u}{\partial q} + \frac{\partial w}{\partial p} \right) dA. \quad (7.5)$$

The sum in the large brackets is called the divergence of the flow: by Equation (7.5), applied to a blob so small that the sum hardly changes across it, it is the rate at which the blob's area grows, divided by that area. For the Hamiltonian flow, $u = \partial H / \partial p$ and $w = -\partial H / \partial q$, so

$$\frac{\partial u}{\partial q} + \frac{\partial w}{\partial p} = \frac{\partial^2 H}{\partial q \partial p} - \frac{\partial^2 H}{\partial p \partial q} = 0,$$

because the two mixed partial derivatives of a smooth function are equal (Section 3.11). The integrand is zero everywhere, so $d\mathcal{A}/dt = 0$: the Hamiltonian flow keeps the area of every region it carries. This is Liouville's theorem.

With the drag of Section 4.6, $\dot{p} = -\partial H / \partial q - \gamma p$, so w gains the term $-\gamma p$, and $\partial w / \partial p$ gains $-\gamma$. The divergence is then $-\gamma$ everywhere, and Equation (7.5) gives $d\mathcal{A}/dt = -\gamma \mathcal{A}$, the equation of Section 2.7, so

$$\mathcal{A}(t) = \mathcal{A}_0 e^{-\gamma t}.$$

With $\gamma = 0.5 \text{ s}^{-1}$, the blob of Figure 7.3(b) keeps 0.60653, 0.36788 and 0.22313 of its area after 1, 2 and 3 s, the values of $e^{-\gamma t}$ to the five figures printed. Every state ends at rest at the bottom: the drag erases the differences between starting states, which a Hamiltonian flow never does.

The stretching of a small square

Liouville's theorem can also be read off a single state and its close neighbours, which is the form Chapter 9 uses for methods that move in steps.

THE JACOBIAN DETERMINANT. The flow over a time t carries each starting state (q_0, p_0) to the state (q, p) reached at time t , so q and p are functions of q_0 and p_0 . We take a small square of starting states with one corner at (q_0, p_0) and sides of length ε along the two axes (Figure 7.4). By the Taylor series to first order, the corner at $(q_0 + \varepsilon, p_0)$ goes to $(q, p) + \varepsilon \mathbf{a}$, with $\mathbf{a} = (\partial q / \partial q_0, \partial p / \partial q_0)$, and the corner at $(q_0, p_0 + \varepsilon)$ goes to $(q, p) + \varepsilon \mathbf{b}$, with $\mathbf{b} = (\partial q / \partial p_0, \partial p / \partial p_0)$. The far corner $(q_0 + \varepsilon, p_0 + \varepsilon)$ is a step ε in both variables, and the first-order rule $\Delta h \approx (\partial h / \partial x) \Delta x + (\partial h / \partial y) \Delta y$ of Section 3.10, applied to q and to p , takes it to $(q, p) + \varepsilon \mathbf{a} + \varepsilon \mathbf{b}$. To first order, then, the square becomes the parallelogram on the arrows $\varepsilon \mathbf{a}$ and $\varepsilon \mathbf{b}$. The area of the parallelogram on $\mathbf{a} = (a_1, a_2)$ and $\mathbf{b} = (b_1, b_2)$ is the length of their cross product (Section 5.2); with third components zero, the cross product has only the third component $a_1 b_2 - a_2 b_1$, so the area is $|a_1 b_2 - a_2 b_1|$. The four derivatives form the Jacobian matrix of the flow,

$$\mathbf{J} = \begin{pmatrix} \partial q / \partial q_0 & \partial q / \partial p_0 \\ \partial p / \partial q_0 & \partial p / \partial p_0 \end{pmatrix},$$

whose columns are $\mathbf{a}$ and $\mathbf{b}$, and the number $a_1 b_2 - a_2 b_1$ made from a 2×2 matrix in this way, the product of the diagonal entries minus the product of the other two, is its determinant, $\det \mathbf{J}$. The square of area ε^2 therefore becomes a parallelogram of area $|\det \mathbf{J}| \varepsilon^2$: the determinant is the factor by which the flow stretches small areas. For a body in free flight, $q = q_0 + p_0 t / m$ and $p = p_0$, so

$$\mathbf{J} = \begin{pmatrix} 1 & t/m \\ 0 & 1 \end{pmatrix}, \quad \det \mathbf{J} = 1 \times 1 - \frac{t}{m} \times 0 = 1,$$

a shear that tilts the square without changing its area (Exercise 7.12). For a map of n variables, the $n \times n$ Jacobian matrix has a determinant too, the factor by which the map stretches small volumes in n dimensions; its definition for larger matrices, and this property, are standard results of linear algebra that we take on trust.

Since the Hamiltonian flow keeps the area of every region, a small square keeps its area ε^2 , while the toolbox gives $|\det \mathbf{J}| \varepsilon^2$ with an error that shrinks faster than ε^2 ; letting ε shrink, $|\det \mathbf{J}| = 1$. At $t = 0$ the map does nothing and $\det \mathbf{J} = 1 \times 1 - 0 \times 0 = 1$, and $\det \mathbf{J}$ changes smoothly with t , so it cannot jump to -1 : $\det \mathbf{J} = 1$ at every time. With drag, the same reasoning gives $\det \mathbf{J} = e^{-\gamma t}$. Computed with `flow_jacobian`, which follows four copies started a small step either side of the state along each axis

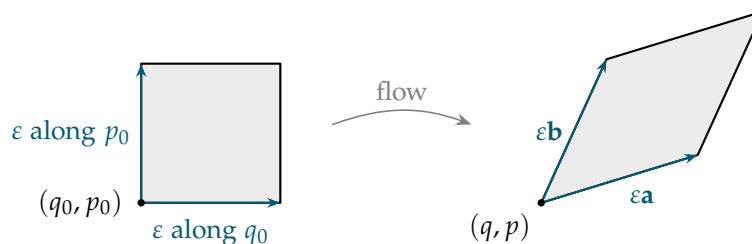


FIGURE 7.4: The flow over a time t carries a small square of starting states into a parallelogram on the arrows $\epsilon \mathbf{a}$ and $\epsilon \mathbf{b}$, the columns of the Jacobian matrix times the side ϵ ; its area is $|\det \mathbf{J}|$ times that of the square.

and takes central differences (Section 1.9) of where they arrive, the Jacobian of the pendulum's flow from $(\theta, p) = (1 \text{ rad}, 0.5 \text{ kg m}^2 \text{ s}^{-1})$ over 2.5 s has a determinant that differs from 1 by about 10^{-10} , although the map is far from doing nothing: its entries are 1.057 and 0.213 in the first row and 1.157 and 1.179 in the second; rounded, they give $1.057 \times 1.179 - 0.213 \times 1.157 = 0.9998$, which is 1 to the accuracy of the rounding.

For N atoms the same holds in the $6N$ dimensions of phase space. The divergence of the flow is the sum, over all $6N$ coordinates, of the derivative of each coordinate's rate with respect to that coordinate, $\sum_k (\partial \dot{q}_k / \partial q_k + \partial \dot{p}_k / \partial p_k)$; for Hamilton's equations it is $\sum_k (\partial^2 H / \partial q_k \partial p_k - \partial^2 H / \partial p_k \partial q_k)$, zero term by term, and the form of Green's theorem in many dimensions, the divergence theorem¹⁰, then shows that every region of phase space keeps its volume and $\det \mathbf{J} = 1$. A cloud of states can be stretched and folded by the flow but never compressed. In particular, the number of copies in a small region that moves with the flow does not change, and neither does its volume, so the density of copies seen by an observer moving with any one state stays the same. Since each state also keeps its energy, a cloud whose density, the number of copies per unit volume, is a function $f(H)$ of the energy alone does not change at all: the copies now at a state Γ came from a state of the same energy, where the density was $f(H(\Gamma))$, and have kept that density. Chapter 12 builds statistical mechanics on such clouds.

KEY IDEA. The Hamiltonian flow has zero divergence, because the mixed partial derivatives of H are equal, so it keeps the area, in many dimensions the volume, of every region of phase space it carries: the Jacobian determinant of the flow is 1. Drag at the rate γ shrinks areas as $e^{-\gamma t}$.

NOTEBOOK 07 watch a blob of states flow, with and without drag

CHECK YOUR UNDERSTANDING. A blob of states of the pendulum with drag at $\gamma = 0.5 \text{ s}^{-1}$ has an area of 1 J s. What is its area 4 s later?

7.5 Symmetries and conserved quantities

Two carts joined by a spring on a straight track (Section 2.9) have an energy that depends only on the distance between them, so sliding both along the track by the same amount changes nothing. Chapter 2 found that their total momentum is conserved, because the spring pushes them equally and oppositely. This section shows that the conservation follows from the symmetry itself, that the energy is unchanged by the shift, whatever the forces are like, and that the same holds for turning and for the passing of time.

Moving everything at once

Let the potential energy of N atoms be unchanged when every atom is moved by the same displacement $\mathbf{c}$:

$$U(\mathbf{r}_1 + \mathbf{c}, \dots, \mathbf{r}_N + \mathbf{c}) = U(\mathbf{r}_1, \dots, \mathbf{r}_N) \quad \text{for every } \mathbf{c}.$$

With $\mathbf{c} = (c, 0, 0)$ along x , we differentiate both sides with respect to c . On the left, each x_i is replaced by $x_i + c$, which changes at the rate 1 as c grows, so the general chain rule (Section 7.2) gives one term for each atom; the right does not contain c . At $c = 0$,

$$\sum_{i=1}^N \frac{\partial U}{\partial x_i} = 0, \quad \text{so} \quad \sum_{i=1}^N F_{i,x} = 0,$$

and the same for y and z : the forces on the atoms add to zero. By Hamilton's equations $\dot{\mathbf{p}}_i = \mathbf{F}_i$, so the total momentum $\mathbf{P} = \sum_i \mathbf{p}_i$ changes at the rate $\sum_i \mathbf{F}_i = 0$ and is conserved. Nothing here assumed that the forces act between pairs. The atoms of Figure 7.5(a) have an energy that depends only on the angle at the middle one, $U = \frac{1}{2}k_\theta(\theta - \theta_0)^2$, a simple model of the bending of a molecule. The forces, 0.890, 1.430 and 0.685 eV Å⁻¹, are not along the lines that join the atoms (the force on atom 1 is at right angles to its bond), yet they add to zero to within 5×10^{-10} eV Å⁻¹, the error of the central differences that computed them. This is the reason promised in Section 2.10.

Turning everything at once

Now let U be unchanged when the whole group is turned by any angle about any axis through the origin. Turning at the angular velocity $\mathbf{n}$, an arrow of length one along the axis, that is at one radian per unit of time, for a time α turns the group by the angle α , and to first order each atom moves by its velocity times α , $\alpha \mathbf{n} \times \mathbf{r}_i$ (Section 5.4).

Differentiating U with respect to α at $\alpha = 0$ by the general chain rule, one term for each coordinate of each atom,

$$\begin{aligned}
 0 &= \sum_{i=1}^N \nabla_i U \cdot (\mathbf{n} \times \mathbf{r}_i) && \text{since } U \text{ does not change} \\
 &= \sum_{i=1}^N \mathbf{n} \cdot (\mathbf{r}_i \times \nabla_i U) && \text{moving the three vectors round (Section 5.5)} \\
 &= -\mathbf{n} \cdot \sum_{i=1}^N \mathbf{r}_i \times \mathbf{F}_i = -\mathbf{n} \cdot \mathbf{G} && \text{with } \mathbf{F}_i = -\nabla_i U.
 \end{aligned}$$

This holds for every axis $\mathbf{n}$, so every component of the total torque $\mathbf{G}$ is zero. The angular momentum $\mathbf{L} = \sum_i \mathbf{r}_i \times \mathbf{p}_i$ changes, by the product rule, at the rate

$$\frac{d\mathbf{L}}{dt} = \sum_{i=1}^N \left(\frac{\mathbf{p}_i}{m_i} \times \mathbf{p}_i + \mathbf{r}_i \times \mathbf{F}_i \right) = \mathbf{G} = 0,$$

since the cross product of an arrow with itself is zero; so $\mathbf{L}$ is conserved. For the three atoms of Figure 7.5(a) the torque is below 2×10^{-9} eV, about the origin and about the point (3,2) Å, again the error of the differences. This is the result promised in Section 5.3 for forces that cannot be split into pairs.

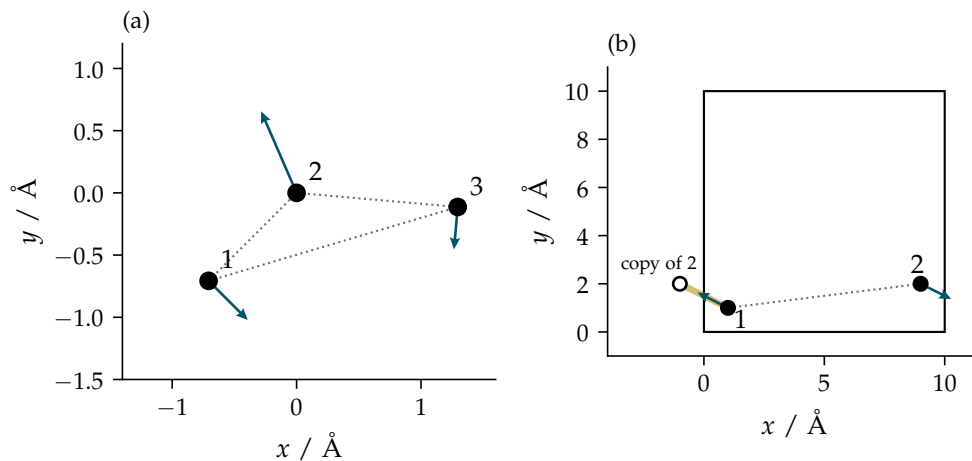


FIGURE 7.5: (a) Three atoms whose energy depends only on the angle at atom 2, $\frac{1}{2}k_\theta(\theta - \theta_0)^2$ with $k_\theta = 2 \text{ eV rad}^{-2}$ and $\theta_0 = 104.5^\circ$, held at 130° with bonds of 1.0 and 1.3 Å. The forces (arrows, $0.5 \text{ Å per eV Å}^{-1}$) are not along the lines joining the atoms (dotted), yet add to zero with no torque. (b) Two atoms in a box 10 Å wide repeated in every direction, joined by a spring to the nearest copy of each other (ochre); the forces ($6 \text{ Å per eV Å}^{-1}$) add to zero but are not along the line joining the atoms in the box (dotted), so they exert a torque.

Noether's theorem

Both results, and the conservation of a momentum whose coordinate is missing from $\mathcal{L}$ (Section 6.5), are cases of one theorem, proved most directly from the Lagrangian. Suppose that $\mathcal{L}$ does not change, to first order in a small number α , when every coordinate is changed by $\alpha s_k(q)$, with s_k any functions of the coordinates, and every rate by the matching $\alpha ds_k/dt$, the rate of $q_k + \alpha s_k$ being $\dot{q}_k + \alpha ds_k/dt$. Then, by the chain rule,

$$\begin{aligned}
 0 &= \sum_k \left(\frac{\partial \mathcal{L}}{\partial q_k} s_k + \frac{\partial \mathcal{L}}{\partial \dot{q}_k} \frac{ds_k}{dt} \right) && \text{the change of } \mathcal{L} \text{ per unit of } \alpha \\
 &= \sum_k \left(\dot{p}_k s_k + p_k \frac{ds_k}{dt} \right) && \text{by Equation (6.3) and } p_k = \partial \mathcal{L} / \partial \dot{q}_k \\
 &= \frac{d}{dt} \sum_k p_k s_k && \text{by the product rule,}
 \end{aligned}$$

so $\sum_k p_k s_k$ is conserved. Moving every atom along x has $s = 1$ for every x_i and 0 for the other coordinates, and leaves the kinetic energy unchanged, so $\sum_i p_{i,x}$ is conserved. Turning about $\mathbf{n}$ has the change $\mathbf{n} \times \mathbf{r}_i$ for atom i , and leaves $\frac{1}{2} m_i |\dot{\mathbf{r}}_i|^2$ unchanged, since the rate changes by $\alpha \mathbf{n} \times \dot{\mathbf{r}}_i$, at right angles to $\dot{\mathbf{r}}_i$: $|\dot{\mathbf{r}}_i + \alpha \mathbf{n} \times \dot{\mathbf{r}}_i|^2 = |\dot{\mathbf{r}}_i|^2 + 2\alpha \dot{\mathbf{r}}_i \cdot (\mathbf{n} \times \dot{\mathbf{r}}_i) + \mathcal{O}(\alpha^2)$, and the middle term is zero (Section 5.2). So $\sum_i \mathbf{p}_i \cdot (\mathbf{n} \times \mathbf{r}_i) = \mathbf{n} \cdot \mathbf{L}$ is conserved, by the triple product again. A coordinate missing from $\mathcal{L}$ has $s = 1$ for itself and 0 for the others, and its own momentum is conserved. The passing of time is the remaining symmetry, though not of this form: Section 7.2 showed directly that a Hamiltonian that does not contain t is conserved. Each symmetry of $\mathcal{L}$ under a change that can be made as small as we like, and the passing of time for H , comes with a conserved quantity: this is Noether's theorem, named after Emmy Noether.

A box repeated in every direction

Most simulations follow the atoms in a box repeated periodically in every direction (Chapter 10), and each atom feels the nearest copies of the others. Moving every atom by the same displacement moves every copy too, and changes nothing, so the forces still add to zero and the total momentum is conserved. Turning the atoms while the box and the lattice of its copies stay fixed is not a symmetry. In Figure 7.5(b) two atoms at (1,1) and (9,2) Å in a box 10 Å wide are joined by a spring of $1 \text{ eV } \text{\AA}^{-2}$ and natural length 2 Å to the nearest copy of each other. The nearest copy of atom 2, one box width to the left, is at $(9 - 10, 2) = (-1, 2)$ Å, at the separation $(-2, 1)$ and the distance $\sqrt{5} = 2.2361$ Å, so the spring is stretched by 0.2361 Å and pulls atom 1 towards that copy with $\mathbf{F}_1 = k \times 0.2361 \times (-2, 1) / \sqrt{5} = (-0.2111, 0.1056) \text{ eV } \text{\AA}^{-1}$; atom 2 feels the opposite force, from the copy of atom 1. The forces add to zero, and moving both

atoms by any displacement leaves the energy at 0.027 864 eV. They act along the line from atom 1 to the copy, at an angle to the line joining the two atoms in the box, and their torque about the origin is

$$G_z = (x_1 - x_2) F_{1,y} - (y_1 - y_2) F_{1,x} = (-8)(0.1056) - (-1)(-0.2111) = -1.056 \text{ eV},$$

using $\mathbf{F}_2 = -\mathbf{F}_1$ in $G_z = \sum_i (x_i F_{i,y} - y_i F_{i,x})$. Turning both atoms about the origin changes the energy at the rate 1.056 eV per radian, so the angular momentum of the atoms in the box is not conserved, and Section 5.6 removed only the drift of a periodic system.

A learned model of the forces keeps these conservation laws when its forces are the slopes of an energy with the same symmetries. A learned model of the motion predicts the next state without Hamilton's equations, so the theorem does not cover it even when it respects the symmetries, which is why TrajCast corrects its predicted momenta, as the planned Chapter 28 will discuss.

KEY IDEA. Every symmetry of the Lagrangian under a change that can be made as small as we like gives a conserved quantity (Noether's theorem), and so does the passing of time: unchanged by moving everything, the total momentum; by turning everything, the angular momentum; by the passing of time, the energy. No assumption about pairs of forces is needed.

NOTEBOOK 07 *forces of a bending molecule and of atoms in a periodic box*

CHECK YOUR UNDERSTANDING. A fixed flat wall at $x = 0$ that repels atoms is added to a periodic box. Which components of the total momentum are still conserved?

7.6 Running the motion backwards

A film of a swinging pendulum looks as natural when it is run backwards; a film of a book sliding to rest across a table does not, because books are never seen to gather speed from a table that has cooled a little. This section shows that the difference lies in the drag, and that motion under a time-independent Hamiltonian can be reversed when reversing every momentum leaves that Hamiltonian unchanged.

The reversed motion

Let $q(t)$ and $p(t)$ be a motion. The reversed motion visits the same positions in the opposite order, with every momentum reversed:

$$Q(t) = q(-t), \quad P(t) = -p(-t).$$

Suppose that H does not contain t and that $H(q, -p) = H(q, p)$, as it is whenever K depends on the momenta through their squares. Differentiating this identity with respect to p , by the chain rule on the left, gives $-(\partial H/\partial p)(q, -p) = (\partial H/\partial p)(q, p)$, so reversing the momentum reverses the sign of $\partial H/\partial p$; differentiating it with respect to q gives $(\partial H/\partial q)(q, -p) = (\partial H/\partial q)(q, p)$, so reversing the momentum leaves $\partial H/\partial q$ unchanged. Then, by the chain rule with the inner function $-t$, whose rate is -1 , and Hamilton's equations for the original motion at the time $-t$,

$$\begin{aligned}\frac{dQ}{dt} &= -\dot{q}(-t) = -\frac{\partial H}{\partial p}(q(-t), p(-t)) = \frac{\partial H}{\partial p}(Q, P), \\ \frac{dP}{dt} &= \dot{p}(-t) = -\frac{\partial H}{\partial q}(q(-t), p(-t)) = -\frac{\partial H}{\partial q}(Q, P),\end{aligned}$$

where the last step of each line uses $Q = q(-t)$, $P = -p(-t)$ and the two results just found. The reversed motion obeys Hamilton's equations too. Since H does not contain t , shifting the time changes nothing in these equations, so $q(\tau - t)$, $-p(\tau - t)$ obeys them as well. It starts at $(q(\tau), -p(\tau))$, and a state fixes the motion that follows it (Section 2.6), so if a motion is stopped at a time τ , every momentum is reversed, and the motion is followed for another time τ , the system retraces its path and arrives where it started, with its starting momenta reversed.

Figure 7.6(a) shows this for a puck of 0.2 kg on smooth ice, held near a point by springs that are stiffer north–south (5 N m^{-1}) than east–west (2 N m^{-1}), so that it traces a looping path. Followed for 4 s, reversed and followed for another 4 s, it returns to within $3 \times 10^{-14} \text{ m}$ of its start, with the momentum $(-0.02, -0.10) \text{ kg m s}^{-1}$, the reverse of the one it started with.

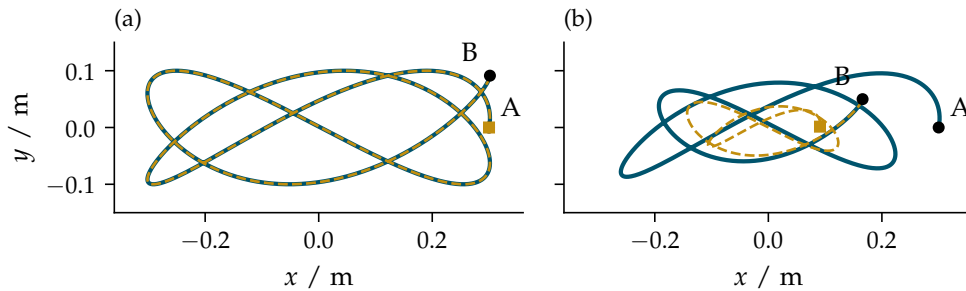


FIGURE 7.6: A puck of 0.2 kg on springs of 2 (east–west) and 5 N m^{-1} (north–south), started at A with the velocity $(0.1, 0.5) \text{ m s}^{-1}$, followed for 4 s to B (teal), then with every momentum reversed for another 4 s (ochre, dashed), ending at the square. (a) Without drag it retraces its path to A. (b) With drag at $\gamma = 0.3 \text{ s}^{-1}$ it does not.

Drag breaks the symmetry

With drag, $\dot{p} = -\partial H/\partial q - \gamma p$. Repeating the second line above, the reversed motion has $dP/dt = \dot{p}(-t) = -\partial H/\partial q - \gamma p(-t) = -\partial H/\partial q + \gamma P$: it would need a force

that pushes the puck along its motion, in place of one that resists it, and the dragged equations do not supply one. Figure 7.6(b) shows the result: after the reversal the puck keeps losing energy and ends 0.21 m from its start. Sliding friction, like drag, always opposes the motion, so the reversed slide would need it to push instead; it is what makes the book's slide irreversible, and the motion of atoms under forces from a potential has no such term.

Chapter 9 asks the same of a method of computing motion: one step forwards, a reversal of the momenta and one more step should bring the system back to where it began, as the exact motion does.

KEY IDEA. When H is time-independent and unchanged by reversing the momenta, the reversed motion $q(-t)$, $-p(-t)$ obeys Hamilton's equations, so reversing every momentum makes a system retrace its path. Drag breaks this.

CHECK YOUR UNDERSTANDING. A ball thrown straight up at 12 m s^{-1} reaches its highest point after 1.224 s. Using time reversal, and without solving any equation, say when and how fast it comes back to the hand.

7.7 Steps that keep area

A computer follows a motion in steps. A method of computing motion takes the state (q, p) at one instant and returns the state (q', p') a time δt later, so it is a map of the phase plane into itself, a step map, and like the flow it has a Jacobian determinant. The exact flow over δt keeps both the energy and areas. This section shows, for the spring and the two simplest methods, that a step map which inflates areas makes the energy grow without limit, while the area-preserving method holds the energy within a band for the stable step used here.

The forward Euler method

The simplest step takes the rates at the start of the step as if they held throughout it, so that the change over the step is the rate times δt , the secant of Section 1.2 read backwards:

$$q' = q + \delta t \frac{\partial H}{\partial p}(q, p), \quad p' = p - \delta t \frac{\partial H}{\partial q}(q, p).$$

This is the forward Euler method. For the block of [Section 4.1](#), 0.5 kg on a spring of 50 N m^{-1} , it reads $q' = q + \delta t p/m$ and $p' = p - \delta t kq$. Its energy after the step is

$$\begin{aligned}
 H' &= \frac{p'^2}{2m} + \frac{1}{2}kq'^2 = \frac{(p - \delta t kq)^2}{2m} + \frac{1}{2}k\left(q + \frac{\delta t p}{m}\right)^2 \\
 &= \frac{p^2}{2m} - \frac{\delta t kqp}{m} + \frac{\delta t^2 k^2 q^2}{2m} + \frac{1}{2}kq^2 + \frac{\delta t kqp}{m} + \frac{\delta t^2 kp^2}{2m^2} \quad \text{expanding both squares} \\
 &= \frac{p^2}{2m} + \frac{1}{2}kq^2 + \frac{k}{m} \delta t^2 \left(\frac{1}{2}kq^2 + \frac{p^2}{2m}\right) \quad \text{collecting terms} \\
 &= (1 + \omega^2 \delta t^2) H \quad \text{with } \omega^2 = k/m.
 \end{aligned}$$

In the third line the two terms in qp have cancelled and $k\delta t^2/m$ has been taken out of the two terms in δt^2 . Every step multiplies the energy by $1 + \omega^2 \delta t^2$, whatever the state, so the computed motion spirals outwards in the phase plane ([Figure 7.7](#)). With $\omega = 10 \text{ rad s}^{-1}$ and $\delta t = 0.02 \text{ s}$, so that $\omega \delta t = 0.2$, the factor is 1.04, and after 50 steps, one second, the energy is 7.107 times its starting value; after 10 s it is 3.3×10^8 times. A shorter step only slows the growth: the energy doubles after the n steps for which $(1 + \omega^2 \delta t^2)^n = 2$. Writing $1 + \omega^2 \delta t^2 = e^{\ln(1 + \omega^2 \delta t^2)}$ and $2 = e^{\ln 2}$ ([Section 2.7](#)), this is $n \ln(1 + \omega^2 \delta t^2) = \ln 2$, so $n = \ln 2 / \ln(1 + \omega^2 \delta t^2)$, which is 17.7 at $\omega \delta t = 0.2$, 69.7 at 0.1 and 6932 at 0.01, about 11 periods of $2\pi/(\omega \delta t) = 628$ steps each.

The step map also stretches areas. Its Jacobian matrix, from the derivatives of q' and p' with respect to q and p , and its determinant are

$$\mathbf{J} = \begin{pmatrix} 1 & \delta t/m \\ -k\delta t & 1 \end{pmatrix}, \quad \det \mathbf{J} = 1 \times 1 - \frac{\delta t}{m} \times (-k\delta t) = 1 + \omega^2 \delta t^2,$$

the same factor: every step inflates every area by it.

The symplectic Euler method

A small change to the order of the updates gives a map that keeps area. The momentum is moved first, with the force at the old position, and then the position, with the new momentum:

$$p' = p - \delta t \frac{\partial H}{\partial q}(q), \quad q' = q + \delta t \frac{\partial H}{\partial p}(p'),$$

for a Hamiltonian $H = K(p) + U(q)$ whose kinetic energy depends only on the momentum and potential energy only on the position, as for atoms in Cartesian

coordinates. For the spring, $p' = p - \delta t k q$, and putting this into $q' = q + \delta t p' / m$ gives $q' = (1 - \omega^2 \delta t^2) q + \delta t p / m$. So

$$\mathbf{J} = \begin{pmatrix} 1 - \omega^2 \delta t^2 & \delta t / m \\ -k \delta t & 1 \end{pmatrix}, \quad \det \mathbf{J} = 1 - \omega^2 \delta t^2 - \frac{\delta t}{m} \times (-k \delta t) = 1.$$

The same holds for any $H = K(p) + U(q)$. The step is two maps in turn: a kick, $(q, p) \rightarrow (q, p - \delta t U'(q))$, which changes only the momentum, by an amount that depends only on the position, and then a drift, $(q, p') \rightarrow (q + \delta t K'(p'), p')$, which changes only the position, by an amount that depends only on the momentum. Their Jacobian matrices are

$$\begin{pmatrix} 1 & 0 \\ -\delta t U''(q) & 1 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 1 & \delta t K''(p') \\ 0 & 1 \end{pmatrix},$$

each with determinant $1 \times 1 - 0 = 1$: each is a shear. A small square taken through the kick keeps its area, and the parallelogram it becomes keeps its area through the drift, so the whole step keeps area. This is the symplectic Euler method; a step map that keeps area in the phase plane is called symplectic. With more coordinates, keeping volume is necessary but the full condition is stronger, and Chapter 9 states it¹¹. The function `symplectic_euler_step` of `mdlab` takes one step:

```

1 def symplectic_euler_step(
2     dh_dq: Callable[[NDArray, NDArray], NDArray],
3     dh_dp: Callable[[NDArray, NDArray], NDArray],
4     q: ArrayLike,
5     p: ArrayLike,
6     dt: float,
7 ) -> tuple[NDArray, NDArray]:
8     q = np.asarray(q, dtype=float)
9     p = np.asarray(p, dtype=float)
10    p_new = p - dt * dh_dq(q, p)
11    return q + dt * dh_dp(q, p_new), p_new

```

The two derivatives of H are passed in as functions, called with the state as it stands: the kick with the old q , the drift with the new momentum `p_new`. For the pendulum of [Section 7.2](#), whose step is not linear, the determinant computed by central differences with $\delta t = 0.05$ s differs from 1 by less than 10^{-9} at each of the states tried.

With the same step of 0.02 s, the symplectic Euler method keeps the block's energy between 0.9091 and 1.1111 times its starting value, over 10 s and over 2000 s alike ([Figure 7.7b](#)): its error oscillates and does not grow. Halving the step to 0.01 s narrows the band to between 0.9524 and 1.0526. Chapter 9 explains why: a step map that keeps area keeps a slightly different energy, exactly for the spring and very nearly in

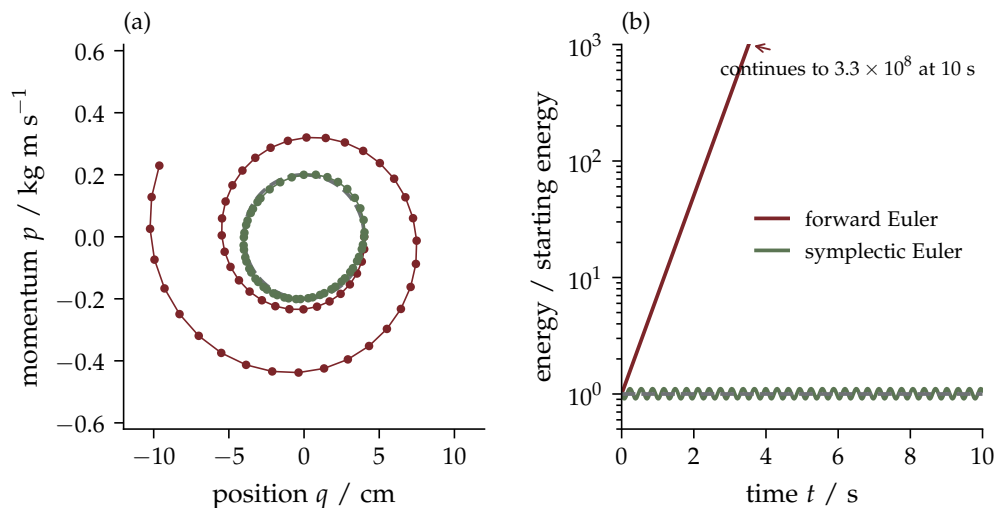


FIGURE 7.7: The block of 0.5 kg on a spring of 50 N m^{-1} ($\omega = 10 \text{ rad s}^{-1}$), released from rest at 4 cm, stepped with $\delta t = 0.02 \text{ s}$ by the forward Euler method (red) and the symplectic Euler method (green). (a) The states after each step for 1 s, with the true ellipse (dashed), which the green points nearly hide. (b) The energy over 10 s as a multiple of its starting value, on a logarithmic axis; the forward Euler energy leaves the axis, at 1000 times its starting value, after 3.5 s.

general¹¹, and so stays close to the true one. It also finds methods whose band is much narrower for the same step.

IN PRACTICE. The total energy of an isolated system is the first diagnostic of a method of computing motion and of its step. A steady growth or fall shows a method that does not keep area, as here, a step too long for the method, or forces that are not the slopes of a single potential energy (Section 3.14); for smooth forces and sufficiently short stable steps, the symplectic method shown here has bounded fluctuations that shrink as the step is shortened. Chapter 9 sizes the step from these fluctuations.

KEY IDEA. A step map keeps area when its Jacobian determinant is 1. The forward Euler method multiplies the spring's energy and every area by $1 + \omega^2 \delta t^2$ at each step; the symplectic Euler method, a kick then a drift, keeps area and holds the energy within a band.

NOTEBOOK 07 *step the spring with both methods and watch a square of states*

CHECK YOUR UNDERSTANDING. With $\omega \delta t = 0.05$, by what factor does the forward Euler method multiply the energy of a spring in 100 steps?

What we now have

The Legendre transform of the Lagrangian in the velocities is the Hamiltonian $H(q, p) = \sum_k p_k \dot{q}_k - \mathcal{L}$, which for a kinetic energy quadratic in the velocities is the energy $K + U$ written in terms of the coordinates and their momenta. Hamilton's equations, $\dot{q} = \partial H / \partial p$ and $\dot{p} = -\partial H / \partial q$, move every state of phase space along a flow that runs on the contours of H ; for the pendulum the contours show the swinging and the circulating motions and the separatrix between them, along which the bob creeps towards the top as $e^{-\omega_0 t}$. Any quantity changes at the rate $\{A, H\} = \hat{L}A$, and the Taylor series in the Liouville operator gives the motion from the starting state, at every time for the ball and the spring and over a short time in general.

The flow has no divergence, so it keeps the area of every region of phase space, and its Jacobian determinant is 1 (Liouville's theorem); drag shrinks areas as $e^{-\gamma t}$.

Every symmetry of the Lagrangian under a change that can be made as small as we like, and the passing of time, gives a conserved quantity (Noether's theorem): the energy from the passing of time, the momentum from moving everything, the angular momentum from turning everything, with no assumption about pairs of forces; in a periodic box the momentum survives and the angular momentum does not. For a time-independent Hamiltonian that is even in the momenta, reversing every momentum makes the system retrace its path. Drag breaks this reversal symmetry.

A method of computing motion is a step map: the forward Euler method inflates the areas and the energy of a spring by $1 + \omega^2 \delta t^2$ at every step, while the symplectic Euler method, a kick followed by a drift, keeps area and holds the energy within a band.

Chapter 8 asks where the potential energy U of atoms comes from, and Chapter 9 returns to step maps and builds the methods with which molecular dynamics is done.

Answers to the checks

SECTION 7.1. $H = p^2 / (2 \times 2) + \frac{1}{2} \times 8 x^2 = p^2 / 4 + 4x^2$, in joules for p in kg m s^{-1} and x in metres. At $x = 0$ all the energy is kinetic, $p^2 / 4 = 1$, so $p = \pm 2 \text{ kg m s}^{-1}$.

SECTION 7.2. $\dot{q} = \partial H / \partial p = p / m$ and $\dot{p} = -\partial H / \partial q = -mg$. On a contour $H = E$, $q = E / (mg) - p^2 / (2m^2g)$: parabolas with their tips at the greatest height $E / (mg)$, where $p = 0$, opening towards smaller heights. Since $\dot{p} = -mg$ everywhere, every state moves steadily towards smaller p along its parabola, rising while $p > 0$ and falling after.

SECTION 7.3. $\hat{L}(qp) = \dot{q}p + q\dot{p} = p^2/m - kq^2$, by the product rule. It is zero where $p^2/m = kq^2$, that is $p = \pm\sqrt{mk}q$, the two lines through the origin on which the kinetic and potential energies are equal.

SECTION 7.4. $\mathcal{A} = 1 \text{ J s} \times e^{-0.5 \times 4} = e^{-2} \text{ J s} = 0.135 \text{ J s}$.

SECTION 7.5. The y and z components. Moving every atom along y or z carries the atoms and their copies along the wall, changing nothing, so the y and z components of the forces still add to zero. Moving them along x changes their distances from the fixed wall, and so their energy: the wall's push is a force from outside the atoms, and P_x is not conserved.

SECTION 7.6. At the top $p = 0$, and reversing a zero momentum changes nothing, so the motion after the top is the reversed motion before it: the ball comes back to the hand 1.224 s after passing the top, about 2.45 s after it was thrown, moving downwards at 12 m s^{-1} .

SECTION 7.7. $(1 + 0.05^2)^{100} = 1.0025^{100} = 1.284$.

Exercises

The exercises run from questions of understanding, through calculations, to short derivations marked 'show that'. Full solutions are in 'Solutions to the exercises' at the end of the book, and Notebook 07 checks every numerical answer.

EXERCISE 7.1 (*a transform with a logarithm*). Show that the Legendre transform of $f(v) = e^v$ is $f^*(p) = p \ln p - p$, for $p > 0$, and that its slope is v .

EXERCISE 7.2 (*two rates in one term*). A kinetic energy $K = \frac{1}{2}(a\dot{q}_1^2 + 2b\dot{q}_1\dot{q}_2 + c\dot{q}_2^2)$, with constants a , b and c , contains a product of two rates. Show that $p_1\dot{q}_1 + p_2\dot{q}_2 = 2K$, so that again $H = K + U$.

EXERCISE 7.3 (*a hanging spring*). A block of mass m hangs from a spring of stiffness k , with q its distance below the point where the spring has its natural length, so that $U = \frac{1}{2}kq^2 - mgq$. Write H and Hamilton's equations, and find the state that does not change.

EXERCISE 7.4 (*a ball in a vertical plane*). For $\mathcal{L} = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2) - mgy$, find p_x , p_y and H , write Hamilton's equations, and say which momentum is conserved and why.

EXERCISE 7.5 (*the puck*). Show that the puck of Section 6.5, with $\mathcal{L} = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2) - U(r)$, has $H = p_r^2/(2m) + p_\theta^2/(2mr^2) + U(r)$, and that Hamilton's equations give the conservation of p_θ and the motion of r in the effective potential Equation (6.7).

EXERCISE 7.6 (*the bead from its Hamiltonian*). Show that Hamilton's equations for $H = p^2/(2m(1+h'^2)) + mgh(x)$ give the bead's equation Equation (6.4).

EXERCISE 7.7 (*over the top*). For the pendulum of Figure 7.2, find the momentum and the speed of the bob at the bottom of the separatrix.

EXERCISE 7.8 (*the last ten degrees*). Using $\dot{\phi} \approx -\omega_0\phi$, how long does the bob of Figure 7.2, on its separatrix, take to go from 10° short of the top to 1° short?

EXERCISE 7.9 (*a product rule for brackets*). Show that $\{A, BC\} = \{A, B\}C + B\{A, C\}$.

EXERCISE 7.10 (*angular momentum in Cartesian coordinates*). A body moves in a plane with $H = (p_x^2 + p_y^2)/(2m) + U(r)$, where $r = \sqrt{x^2 + y^2}$. Show that $L_z = xp_y - yp_x$ has $\{L_z, H\} = 0$.

EXERCISE 7.11 (*the momentum of the spring*). Use the series Equation (7.4) to find $p(t)$ for the block on a spring.

EXERCISE 7.12 (*a shear*). A body of 1 kg flies freely. Where does the flow over 2 s carry the four corners of the square of starting states $0 \leq q_0 \leq 0.1$ m, $0 \leq p_0 \leq 0.1$ kg m s⁻¹, and what is the area of the figure they make?

EXERCISE 7.13 (*half the area*). With drag at $\gamma = 0.5$ s⁻¹, how long does a blob of states take to lose half its area?

EXERCISE 7.14 (*two carts*). For the two carts of Exercise 6.10, $\mathcal{L} = \frac{1}{2}m_1\dot{x}_1^2 + \frac{1}{2}m_2\dot{x}_2^2 - \frac{1}{2}k(x_2 - x_1 - \ell)^2$, use Noether's theorem to show that $m_1\dot{x}_1 + m_2\dot{x}_2$ is conserved.

EXERCISE 7.15 (*under gravity*). A group of atoms with an energy U_{int} that is unchanged by moving or turning the whole group falls in a uniform field of gravity along $-z$, so $U = U_{\text{int}} + \sum_i m_i g z_i$. Which components of the total momentum, and of the angular momentum about the origin, are conserved?

EXERCISE 7.16 (*the torque about any point*). Show that when the forces on a group add to zero, their torque is the same about every point, and find the torque of the two atoms of Figure 7.5(b) about the centre of the box, (5, 5) Å.

EXERCISE 7.17 (*which can be reversed?*). For which of $H_1 = p^2/(2m) + U(q)$, $H_2 = (p - c)^2/(2m)$ with a constant $c \neq 0$, and $H_3 = p^2/(2m) + \frac{1}{2}kq^2 + bq p$ with a constant

$b \neq 0$ does the argument of [Section 7.6](#) show that reversing the momentum retraces the path?

EXERCISE 7.18 (*doubling the energy*). The forward Euler method follows the block of [Figure 7.7](#) with $\delta t = 0.005$ s. After how many steps, and how long, has the energy doubled?

EXERCISE 7.19 (*the other order*). The other symplectic Euler method moves the position first, $q' = q + \delta t p/m$, then the momentum with the force at the new position, $p' = p - \delta t kq'$. Show that for the spring its Jacobian determinant is 1.

EXERCISE 7.20 (*the energy the method keeps*). Show that a step of the symplectic Euler method of [Section 7.7](#) leaves $\tilde{H} = p^2/(2m) + \frac{1}{2}kq^2 - (k\delta t/(2m))qp$ unchanged for the spring.

Potential energy surfaces and interatomic models

The preceding chapters treated the potential energy $U(\mathbf{r}^N)$ as a known function and used its derivatives to determine the motion. For an atomistic simulation we must also choose how to obtain that function. Within the Born–Oppenheimer approximation, the electrons remain in their ground state as the nuclei move, and their energy together with the nuclear repulsion defines the potential energy surface. Evaluating an electronic-structure calculation at every step is costly, which motivates simpler interatomic models.

We begin by examining when classical motion of the nuclei is an appropriate approximation. Pair potentials then give a direct route from an energy model to forces, their numerical checks and the treatment of a cutoff. Their limitations motivate environment-dependent interactions for metals and carbon, and bonded force fields for molecules. Finally, the slow decay of the Coulomb interaction shows why charges need a separate summation method. Chapter 9 uses these forces to advance the motion, while Chapter 18 introduces the learning methods used to represent a potential energy surface. Later chapters on these models are planned for the continuation of Part V.

8.1 Where the potential energy comes from

An atom is a small, heavy nucleus surrounded by much lighter electrons. The forces between atoms are the electrical attractions and repulsions of these charges: nuclei repel nuclei, electrons repel electrons, and each attracts the other. This section explains how the motion of the electrons can be set aside, leaving a potential energy for the nuclei alone, and what it costs to compute it.

Heavy nuclei, light electrons

A proton is 1836 times as heavy as an electron, and a lithium atom, of mass 6.94 amu (Section 2.10), 12 651 times, almost all of it in the nucleus. Pushed by forces of similar size, a body's acceleration is inversely proportional to its mass, and Section 4.4 showed that a body held by a spring of stiffness k swings with $\omega_0 = \sqrt{k/m}$: with comparable stiffnesses, the electrons respond $\sqrt{1836} = 43$ times faster than a proton and 112 times faster than lithium. On the time scale on which the nuclei move, the electrons have time to settle into the arrangement of lowest energy for wherever the nuclei happen to be, and to follow them as they move.

This is the picture of Born and Oppenheimer¹². With the nuclei held still at the positions $\mathbf{r}^N$, the electrons settle into their state of lowest energy for those positions, and that energy, together with the repulsion of the nuclei for one another, depends only on $\mathbf{r}^N$; we call it $U(\mathbf{r}^N)$. When the nuclei move, the electrons follow, and the nuclei move as if pushed by the forces $-\nabla_i U$ (Section 3.12). The function U is called the potential energy surface: for a single coordinate it is a curve, for two a landscape like that of the lithium atom on its model layer (Section 3.13), and for N atoms a surface in $3N$ dimensions. The picture fails when the motion of the nuclei carries the electrons from one state into another close in energy, as after light is absorbed or in some chemical reactions; every system in this book stays in its lowest state.

The cost of computing it

Finding the lowest state of the electrons is a problem of quantum mechanics. Density-functional theory, one such method, turns it into the eigenvalues and eigenvectors (Section 4.9) of a matrix whose size n grows in proportion to the number of atoms. The work of finding all of them grows as n^3 , and Figure 8.1(a) measures it: with NumPy, doubling the size of a dense symmetric matrix multiplied the time to find its eigenvalues and eigenvectors by about 7.5, close to $2^3 = 8$, and the times for the larger matrices lie near a line of slope 3 on logarithmic axes (Section 1.8): a time proportional to n^3 has a logarithm that grows by 3 times the logarithm of n . The energy of a model that adds up the energies of pairs of atoms needs one term for each pair: each of the N atoms pairs with $N - 1$ others, which counts every pair twice, so there are $N(N - 1)/2 \approx N^2/2$ pairs for large N , a number that doubling N multiplies by $2^2 = 4$; doubling N multiplied its time by about 4 (Figure 8.1b); with the cutoffs of Section 8.5 and the neighbour lists of Chapter 10 it grows only as N . Computing U from the electrons at every step of a simulation therefore limits it to far smaller systems and shorter times than a model allows, and most simulations use a model of U fitted to such calculations, which is the subject of the rest of this chapter.

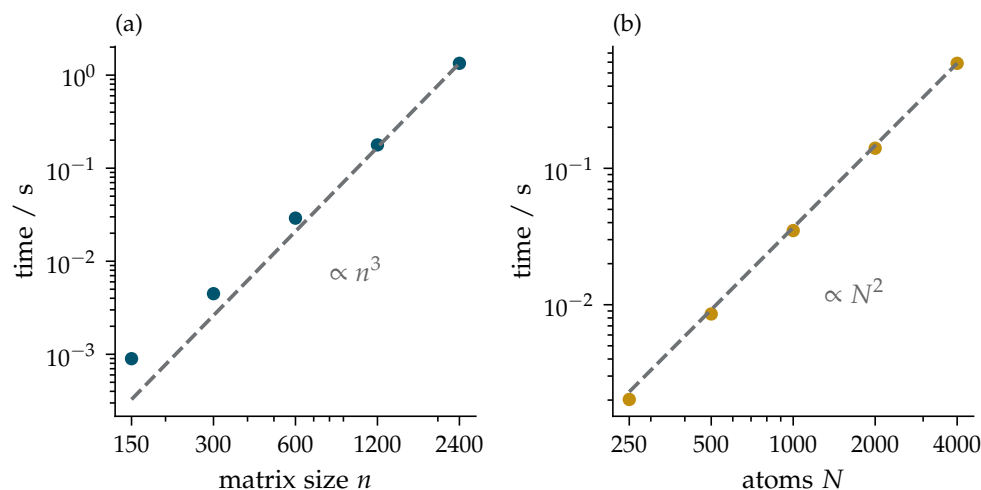


FIGURE 8.1: What computing the energy costs, with NumPy on one processor core; the times depend on the computer, the slope at large sizes does not. (a) Finding all the eigenvalues and eigenvectors of a dense symmetric matrix of size n (one with no zero entries to exploit), against n , with a line of slope 3 (dashed). (b) The energy and forces of N atoms interacting in pairs, every pair computed, against N , with a line of slope 2 (dashed).

FROM ELECTRONIC STRUCTURE. The forces on the nuclei follow from the electronic energy by the Hellmann–Feynman theorem¹³: at the converged density they are the electrostatic forces of the electrons and the other nuclei, plus Pulay corrections when the basis functions move with the atoms. The cubic cost is that of a dense eigensolver; linear-scaling methods use the nearsightedness of the density matrix to reach a cost proportional to N ¹⁴, and machine-learned potentials (planned for the continuation of Part V) build the same locality into a model fitted to the expensive calculations.

KEY IDEA. Because the electrons are light and fast, they settle at every arrangement of the nuclei, and the energy of that settled state, with the repulsion of the nuclei, is the potential energy surface $U(\mathbf{r}^N)$ on which the nuclei move. Computing it from the electrons costs, in the usual methods, work growing as the cube of the system’s size, which is why most simulations fit a model of it.

CHECK YOUR UNDERSTANDING. Finding all the eigenvalues and eigenvectors of a matrix of size 1000 takes 0.1 s. Roughly how long does it take for one of size 4000?

8.2 When the nuclei may be treated as classical

The electrons need quantum mechanics; this book moves the nuclei by Newton's laws, which [Section 2.10](#) called an approximation. This section finds when it is a good one, from a comparison of two energies.

Quantum mechanics changes the motion of a vibration in one way that matters here: a harmonic oscillator ([Section 4.4](#)) of angular frequency ω cannot have any energy it likes, only the values $(n + \frac{1}{2})\hbar\omega$, with $n = 0, 1, 2, \dots$ and $\hbar = 0.6582 \text{ eV fs}$ the reduced Planck constant¹⁰. Its energy therefore comes in steps of $\hbar\omega$, and even in its lowest state, $n = 0$, it keeps the energy $\frac{1}{2}\hbar\omega$, its zero-point energy, which a classical oscillator lacks. At a temperature T , the motion of each vibration carries a typical energy of order $k_B T$, where k_B is Boltzmann's constant and $k_B T = 0.02585 \text{ eV}$ at 300 K (Chapter 12 makes this precise). When $\hbar\omega \ll k_B T$, a typical vibration spans many steps, too small to notice, and the classical description is good. When $\hbar\omega \gg k_B T$, the vibration sits in its lowest state, and a classical model, which lets it lose its zero-point energy, describes it wrongly.

With $\omega = 2\pi\nu = 2\pi c\tilde{\nu}$ for a wavenumber $\tilde{\nu}$ ([Section 4.10](#)), $\hbar\omega = 2\pi\hbar c\tilde{\nu}$, and $2\pi\hbar c = 1.2398 \times 10^{-4} \text{ eV cm}$. At 300 K:

- the symmetric O–H stretch of water, 3657 cm^{-1} ([Table 4.1](#)), has $\hbar\omega = 0.4534 \text{ eV}$, 17.5 times $k_B T$: thoroughly quantum;
- the stretch of LiF, 910.34 cm^{-1} ([Section 5.8](#)), has $\hbar\omega = 0.1129 \text{ eV}$, 4.4 times $k_B T$;
- the model lithium atom rattling in its hollow with a period $\mathcal{T} = 170.3 \text{ fs}$ ([Section 4.4](#)) has $\hbar\omega = \hbar 2\pi/\mathcal{T} = 0.0243 \text{ eV}$, 0.94 times $k_B T$: borderline at room temperature, and 2.8 times $k_B T$ at 100 K.

The lighter the atom and the stiffer its bonds, the larger ω and the stronger the quantum character, so the motion of hydrogen is where classical nuclei fail first: quantum effects of the nuclei can change measured properties of water and other hydrogen-bonded systems at room temperature¹⁵. A bond between heavier atoms can still be quantum, as LiF shows, and the rattle of lithium in its hollow is borderline. The motions this book measures, the turning and drifting of whole molecules and the hops of lithium and other ions heavier than hydrogen, we follow with classical nuclei, accepting that the energy of the fast stretches is then wrong. Methods that keep the quantum character of the nuclei exist; they are beyond the scope of this book.

KEY IDEA. A vibration may be treated classically when its quantum step $\hbar\omega = 2\pi\hbar c \tilde{\nu}$ is small compared with $k_{\text{B}}T$. At room temperature this holds for slow motions, is borderline for the rattle of lithium in its hollow, and fails for the stretches of chemical bonds, most of all bonds to hydrogen.

CHECK YOUR UNDERSTANDING. Is the C–H stretch, near 3000 cm^{-1} , classical at 300 K?

8.3 Pair potentials

The simplest model of U adds up an energy for each pair of atoms that depends only on the distance between them,

$$U(\mathbf{r}^N) = \sum_{i < j} \varphi(r_{ij}), \quad (8.1)$$

where the sum runs over every pair once: i takes each value from 1 to N and, for each i , j each value from $i + 1$ to N , so $\varphi(r_{12})$ enters but not $\varphi(r_{21})$; φ is called a pair potential. It ignores any effect that a third atom has on the energy of a pair, which makes it a reasonable first model for atoms that keep their electrons to themselves, such as those of the noble gases, and a poor one for metals and for atoms joined by chemical bonds (Sections 8.6 and 8.7). Section 4.5 met the two most common shapes; this section adds a third and compares them.

Three shapes

The Lennard-Jones potential,

$$\varphi_{\text{LJ}}(r) = 4\varepsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right],$$

has its minimum, of depth ε , at $r_{\text{m}} = 2^{1/6}\sigma = 1.1225\sigma$ (written r_{min} in Section 4.5). Its attraction falls as r^{-6} , the form of the weak attraction between neutral atoms far apart, and its repulsion, r^{-12} , is steep and cheap to compute as the square of r^{-6} ; inverse powers of this kind go back to Jones¹⁶. For argon, $\varepsilon/k_{\text{B}} = 120\text{ K}$, so $\varepsilon = 0.01034\text{ eV}$, and $\sigma = 3.4\text{ Å}$: the values with which Rahman made the first simulation of a real liquid by molecular dynamics¹⁷.

The Morse potential of Section 4.5 describes a chemical bond between two atoms¹⁸. Written there with its least value 0, it is lowered here by D , so that, like every pair potential in a sum Equation (8.1), it is zero for atoms far apart:

$$\varphi_M(r) = D(1 - e^{-a(r-r_0)})^2 - D = D(e^{-2a(r-r_0)} - 2e^{-a(r-r_0)}),$$

expanding the square, $1 - 2e^{-a(r-r_0)} + e^{-2a(r-r_0)}$, multiplying by D and cancelling the resulting D against $-D$. Its minimum is $-D$ at r_0 , and its stiffness there is $2Da^2$, as before.

The Buckingham potential replaces the r^{-12} wall by an exponential,

$$\varphi_B(r) = A e^{-r/\rho} - \frac{C}{r^6},$$

with a strength A and a range ρ for the repulsion (a local constant, unrelated to the ρ_i of Section 8.6) and a strength C for the attraction¹⁹. Written with the position r_m and depth ε of its minimum and a steepness α , as $A = 6\varepsilon e^\alpha/(\alpha - 6)$, $\rho = r_m/\alpha$ and $C = \alpha\varepsilon r_m^6/(\alpha - 6)$ (Exercise 8.5), it is called the exp-6 potential.

Figure 8.2 compares the three with the same minimum, $-\varepsilon$ at r_m . The Lennard-Jones stiffness there is $57.15\varepsilon/\sigma^2$ (Section 4.5), which with $\sigma^2 = r_m^2/2^{1/3}$ is $57.15 \times 2^{1/3} \varepsilon/r_m^2 = 72\varepsilon/r_m^2$; the Morse width $ar_m = 6$ matches it, since $2Da^2 = 2\varepsilon \times 36/r_m^2$, and the exp-6 steepness $\alpha = 14$ gives $73.5\varepsilon/r_m^2$, within 2% of it. Near the minimum they agree; at $r = 2r_m$ the Lennard-Jones and exp-6 energies are -0.031ε and -0.027ε , falling as r^{-6} , while the Morse energy, -0.005ε , falls exponentially; and close in they differ completely: the Lennard-Jones wall rises without limit; the Morse energy stays finite, $1.6 \times 10^5 \varepsilon$ at $r = 0$; and the exp-6 energy rises to a maximum of $2.8 \times 10^4 \varepsilon$ at $0.203 r_m$ and then falls to $-\infty$, because as $r \rightarrow 0$ the term $-C/r^6$ grows without limit while $A e^{-r/\rho}$ approaches A . With $\alpha = 14$ that maximum is far out of thermal reach, but it falls steeply as α is lowered, and for some fitted parameter sets a very hot simulation can push two atoms past it, after which they fall into each other.

Reduced units

For atoms that interact only by a Lennard-Jones potential, the constants ε , σ and the mass m can be removed from the equations of motion altogether. We measure lengths in units of σ , energies in units of ε and times in units of a time τ still to be chosen, so we write $\mathbf{r}_i = \sigma \mathbf{r}_i^*$, $U = \varepsilon U^*$ and $t = \tau t^*$, where the starred quantities are pure numbers. By the chain rule, with $t^* = t/\tau$, $d\mathbf{r}_i/dt = \sigma (d\mathbf{r}_i^*/dt^*)(dt^*/dt) = (\sigma/\tau) d\mathbf{r}_i^*/dt^*$,

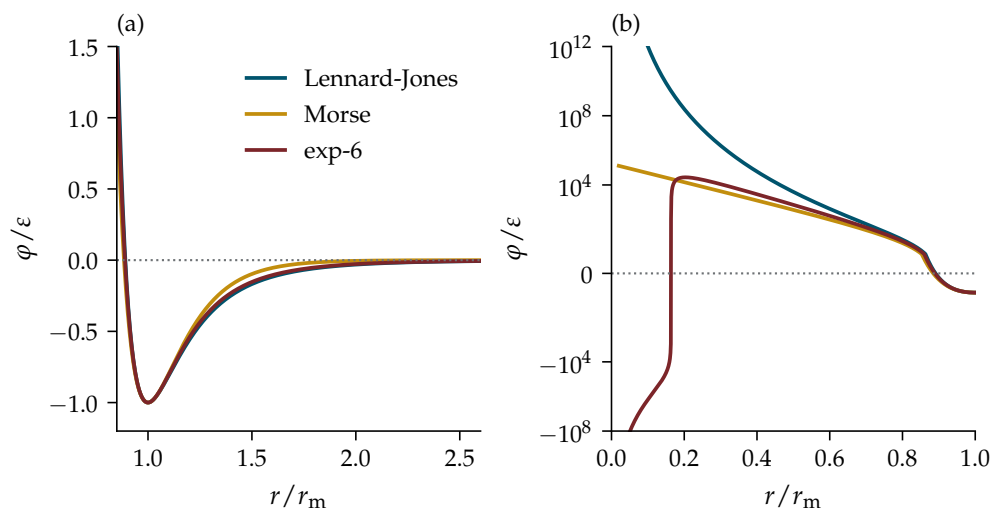


FIGURE 8.2: The Lennard-Jones, Morse ($ar_m = 6$) and exp-6 ($\alpha = 14$) pair potentials with the same minimum $-\epsilon$ at r_m . (a) Near the well. (b) The wall, on a scale that is logarithmic beyond ± 1 : the Lennard-Jones energy rises without limit, the Morse energy rises only exponentially, a straight line here, to a finite value at $r = 0$, and the exp-6 energy turns over and falls.

and once more $\ddot{\mathbf{r}}_i = (\sigma/\tau^2) d^2\mathbf{r}_i^*/dt^{*2}$; likewise $\partial U/\partial x_i = \epsilon(\partial U^*/\partial x_i^*)(dx_i^*/dx_i) = (\epsilon/\sigma)\partial U^*/\partial x_i^*$, so $\nabla_i U = (\epsilon/\sigma)\nabla_i^* U^*$, and the second law $m\ddot{\mathbf{r}}_i = -\nabla_i U$ becomes

$$\begin{aligned} \frac{m\sigma}{\tau^2} \frac{d^2\mathbf{r}_i^*}{dt^{*2}} &= -\frac{\epsilon}{\sigma} \nabla_i^* U^* \\ \frac{d^2\mathbf{r}_i^*}{dt^{*2}} &= -\frac{\epsilon\tau^2}{m\sigma^2} \nabla_i^* U^* \end{aligned} \quad \text{multiplying by } \tau^2/(m\sigma),$$

and, since $\sigma/r_{ij} = 1/r_{ij}^*$, $U^* = \sum 4[(r_{ij}^*)^{-12} - (r_{ij}^*)^{-6}]$ contains no constants. Choosing

$$\tau = \sigma \sqrt{\frac{m}{\epsilon}} \quad (8.2)$$

makes the factor 1, and the equation is the same for every substance of this kind: one simulation in reduced units describes argon, krypton or any other, each with its own ϵ , σ and m , converting back by σ , ϵ and τ , as far as the real atoms interact by a Lennard-Jones potential and move classically. For argon, of mass 39.948 amu, $\epsilon = 0.01034 \times 9.648533 \times 10^{-3} = 9.977 \times 10^{-5}$ in $\text{amu} \text{ \AA}^2/\text{fs}^2$ (Section 2.10), so $\tau = 3.4 \times \sqrt{39.948/9.977 \times 10^{-5}} = 2151$ fs, about 2.15 ps.

KEY IDEA. A pair potential adds up an energy $\phi(r_{ij})$ for every pair. The Lennard-Jones, Morse and exp-6 shapes agree near their minimum and differ in the tail and the wall; the exp-6 wall turns over. In units of σ , ϵ and $\tau = \sigma\sqrt{m/\epsilon}$, all Lennard-Jones substances obey the same equations.

NOTEBOOK 08 compare the three pair potentials and their forces

CHECK YOUR UNDERSTANDING. Krypton has $\varepsilon/k_B \approx 160$ K, $\sigma \approx 3.6$ Å and a mass of 83.8 amu. Is its τ longer or shorter than argon's, and why?

8.4 Forces, the virial, and checking forces

A simulation needs the force on every atom at every step, and a few other sums computed alongside it. This section derives them for a pair potential, shows the code that computes them, and shows how to check any force routine against its energy.

Forces

Atom i appears in the sum [Equation \(8.1\)](#) in the $N - 1$ terms $\varphi(r_{ij})$ with $j \neq i$. [Section 5.3](#) found the gradient of one such term, $-\nabla_i \varphi(r_{ij}) = \varphi'(r_{ij}) \mathbf{r}_{ij}/r_{ij}$ with $\mathbf{r}_{ij} = \mathbf{r}_j - \mathbf{r}_i$, so

$$\mathbf{F}_i = \sum_{j \neq i} \varphi'(r_{ij}) \frac{\mathbf{r}_{ij}}{r_{ij}}. \quad (8.3)$$

Each term is the force on i from j , along the line joining them, towards j when $\varphi' > 0$, where the energy rises with distance and the pair attracts. The force on j from i is the same with i and j exchanged, and since $\mathbf{r}_{ji} = -\mathbf{r}_{ij}$ it is equal and opposite: Newton's third law ([Section 2.4](#)). A program therefore computes each pair once and adds its force to one atom and subtracts it from the other, which is what `pair_energy_forces` of `mdlab` does:

```

1 def pair_energy_forces(
2     positions: ArrayLike, pair: PairFunction
3 ) -> tuple[float, NDArray, float]:
4     r = np.asarray(positions, dtype=float)
5     n = len(r)
6     i, j = np.triu_indices(n, k=1) # every pair once
7     rij = r[i] - r[j]
8     dist = np.linalg.norm(rij, axis=1)
9     phi, dphi = pair(dist)
10    f_on_i = -(dphi / dist)[:, None] * rij # force on i from j
11    forces = np.zeros_like(r)
12    np.add.at(forces, i, f_on_i)
13    np.add.at(forces, j, -f_on_i)
14    return float(phi.sum()), forces, float(-(dist * dphi).sum())

```

`np.triu_indices(n, k=1)` lists the pairs (i, j) with $i < j$ as two arrays of the same length; `rij` holds $\mathbf{r}_i - \mathbf{r}_j = -\mathbf{r}_{ij}$ for every pair at once, so the minus sign makes `f_on_i` the force of [Equation \(8.3\)](#); and `np.add.at` adds each pair's force into the row of its atom, even when an atom appears in many pairs; the plainer `forces[i] += f_on_i`

would keep only one force for an atom listed several times in $\mathbf{i}$. The argument pair is any function that returns φ and φ' for an array of distances, such as `lennard_jones`.

The virial

The last number returned is the virial, $\mathcal{W} = \sum_i \mathbf{r}_i \cdot \mathbf{F}_i$, from which Chapter 14 computes the pressure. For a pair potential it takes a simple form. The pair (i, j) contributes $\mathbf{F}_{ij}$ to atom i and $-\mathbf{F}_{ij}$ to atom j , and so $\mathbf{r}_i \cdot \mathbf{F}_{ij} - \mathbf{r}_j \cdot \mathbf{F}_{ij} = -\mathbf{r}_{ij} \cdot \mathbf{F}_{ij}$ to $\mathcal{W}$. With $\mathbf{F}_{ij} = \varphi'(r_{ij}) \mathbf{r}_{ij}/r_{ij}$ and $\mathbf{r}_{ij} \cdot \mathbf{r}_{ij} = r_{ij}^2$,

$$\mathcal{W} = - \sum_{i < j} r_{ij} \varphi'(r_{ij}). \quad (8.4)$$

It depends only on distances, so it does not change when the origin is moved. That holds for any potential whose forces add to zero: moving the origin by $\mathbf{a}$ replaces each $\mathbf{r}_i$ by $\mathbf{r}_i - \mathbf{a}$ and changes $\mathcal{W}$ by $-\mathbf{a} \cdot \sum_i \mathbf{F}_i = 0$ (Section 7.5). A pair pushed closer than its minimum repels, $\varphi' < 0$, and contributes positively; a stretched pair attracts and contributes negatively.

Checking forces

A force routine is easy to get wrong and hard to check by eye, and a force that is not exactly $-\nabla U$ makes the energy drift (Section 3.14). The standard check compares it with central differences of the energy, $F \approx -[U(x+h) - U(x-h)]/(2h)$ for each coordinate in turn (Section 1.9). Two errors limit the check. The Taylor series of Section 1.8 gives $U(x \pm h) = U \pm hU' + \frac{1}{2}h^2U'' \pm \frac{1}{6}h^3U''' + \frac{1}{24}h^4U'''' + \mathcal{O}(h^5)$; subtracting cancels the even powers, which have the same sign in both, as in Section 1.9, so

$$\frac{U(x+h) - U(x-h)}{2h} = U' + \frac{h^2}{6} U''' + \mathcal{O}(h^4),$$

an error that shrinks as h^2 . But a computer stores a number with about 16 significant decimal digits: rounding to the nearest stored number changes it by at most half the machine epsilon, $\epsilon_m = 2.2 \times 10^{-16}$ (the gap between 1 and the next stored number), times its size, so a computed energy, a sum of many rounded terms, is uncertain by about $\epsilon_m|U|$ or more. The difference of two such energies divided by $2h$ carries an error of order $\epsilon_m|U|/h$, which grows as h shrinks. The total,

$$\text{error} \approx \frac{h^2}{6} |U'''| + \frac{\epsilon_m|U|}{h}, \quad (8.5)$$

is least where its slope with respect to h , $h|U'''|/3 - \epsilon_m|U|/h^2$, is zero, that is at $h^3 = 3\epsilon_m|U|/|U'''|$: a step of about the cube root of ϵ_m , 6×10^{-6} , times the length on which

U changes (if U changes by about its own size over a length L , then $|U'''| \approx |U|/L^3$ and $h \approx (3\epsilon_m)^{1/3}L$); near the steep repulsion L is much shorter than σ (Exercise 8.9). Figure 8.3 measures it for twelve Lennard-Jones atoms. The error falls as h^2 (fitted slope 2.03) for large steps, grows as $1/h$ (slope -1.01) for small ones, and is least, $1.0 \times 10^{-8} \epsilon/\sigma$, at $h = 10^{-6} \sigma$, against forces of up to $129 \epsilon/\sigma$.

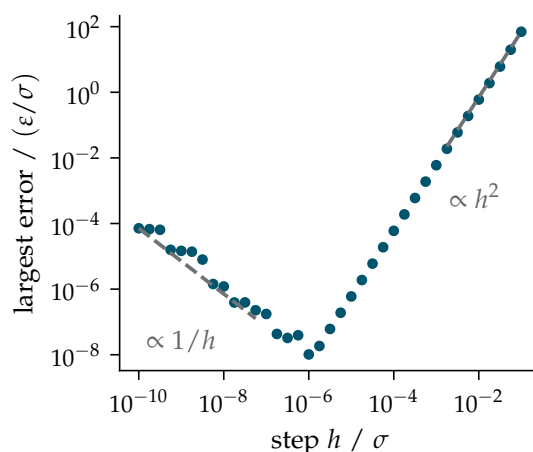


FIGURE 8.3: The largest error of forces from central differences of the energy, for twelve Lennard-Jones atoms, against the step h : falling as h^2 (dashed) from the Taylor series and rising as $1/h$ (dashed) from the rounding of the energies.

IN PRACTICE. Every new force routine, and every change to one, should be checked against central differences of its own energy, on an arrangement of atoms that is not symmetric, so that no error cancels. A step near 10^{-6} times the distance between atoms, the best step of Figure 8.3, is a good start; the check should be repeated at a step ten times larger and smaller, and the difference should behave as Equation (8.5) predicts. The tests of `mdlab` compare every force routine of this chapter with central differences, at one step.

KEY IDEA. For a pair potential, $\mathbf{F}_i = \sum_j \phi'(r_{ij}) \mathbf{r}_{ij}/r_{ij}$, computed once per pair and applied with opposite signs, and the virial is $\mathcal{W} = -\sum r_{ij} \phi'(r_{ij})$. Forces are checked against central differences of the energy, whose error is least at a step near $\epsilon_m^{1/3}$ times the length scale.

NOTEBOOK 08 break a force routine on purpose and catch it

CHECK YOUR UNDERSTANDING. Two Lennard-Jones atoms sit at $r = \sigma$. Is their virial positive or negative?

8.5 Cutoffs

A pair potential such as the Lennard-Jones energy never reaches zero as r grows, but beyond a few σ it is very small: at $r_c = 2.5\sigma$ it is -0.0163ϵ , less than 2% of the depth of the well. Ignoring every pair farther apart than such a cutoff distance r_c leaves each atom with a fixed number of partners, on average, at a given density, however large the system, and with the neighbour lists of Chapter 10 makes the cost grow as N in place of N^2 . This section compares three ways of cutting it off.

Three ways to cut

The simplest choice, truncation, keeps φ unchanged inside r_c and sets it to zero outside. The energy then jumps by $\varphi(r_c)$ whenever a pair crosses the cutoff, while the force computed from Equation (8.3), which uses only φ' , knows nothing of the jump: the force is not the slope of the energy reported with it at r_c , and the total energy changes by $\mp\varphi(r_c)$ at every crossing, as Section 3.14 warned.

Truncating and shifting subtracts the constant $\varphi(r_c)$ inside the cutoff,

$$\varphi_s(r) = \begin{cases} \varphi(r) - \varphi(r_c), & r < r_c, \\ 0, & r \geq r_c, \end{cases}$$

so that the energy is continuous. A constant has no slope, so the forces are unchanged, and the force still jumps at r_c , from a pull of $\varphi'(r_c) = 0.039\epsilon/\sigma$ for the Lennard-Jones potential to nothing.

Switching multiplies φ by a function S that falls smoothly from 1 to 0 between a switching distance r_s and r_c . With $x = (r - r_s)/(r_c - r_s)$ running from 0 to 1 across that range,

$$S(x) = 1 - 10x^3 + 15x^4 - 6x^5 \quad (8.6)$$

is 1 at $x = 0$ and 0 at $x = 1$ with its first and second derivatives zero at both ends (Exercise 8.11); these six conditions fix the six coefficients of a polynomial of degree five, and none of lower degree meets them all, so the energy and the force both pass smoothly to zero. The price is a changed force between r_s and r_c . Since S depends on r through x , the chain rule gives $dS/dr = (dS/dx)/(r_c - r_s)$, and the product rule then gives the force

$$-\frac{d(\varphi S)}{dr} = -\varphi' S - \frac{\varphi}{r_c - r_s} \frac{dS}{dx}.$$

There $\varphi < 0$ and $dS/dx = -30x^2(1-x)^2 < 0$, so the second term is negative: a pull that the original potential does not have (Figure 8.4b).

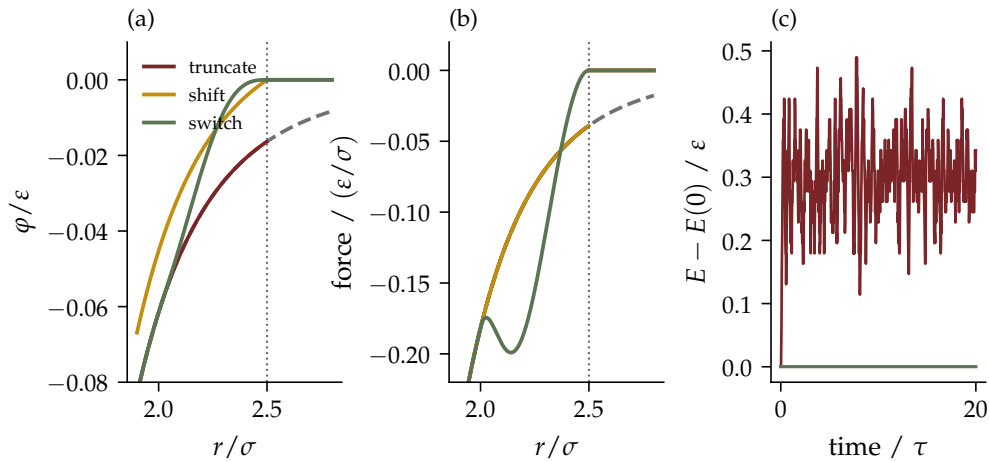


FIGURE 8.4: The Lennard-Jones potential cut at $r_c = 2.5\sigma$ by truncation, truncation and shifting, and switching between 2σ and 2.5σ , with the full potential dashed. (a) The energy; the truncated energy jumps at r_c . (b) The force, the same for truncation and shifting, which jump at r_c . (c) The total energy of 32 atoms followed for 20τ by `solve_newton` of `mdlab`, an adaptive solver, as its change from the start: truncated (oxblood), and without a cutoff, shifted and switched, all on the line at zero.

What each does to the motion

Figure 8.4(c) follows a cluster of 32 Lennard-Jones atoms for 20τ with `solve_newton` of `mdlab`, which steps Newton's equations forward with a high-order method and chooses each step to hold its error below a set tolerance, once with each potential, and records the total energy every 0.025τ . Without a cutoff it changes by at most $1.3 \times 10^{-10}\epsilon$, the error of the solver. With truncation it wanders by up to 0.49ϵ , in jumps of up to 0.11ϵ between successive records, each the net effect of several crossings (seven for the largest) of 0.0163ϵ each. The motion is that of the shifted potential, since the forces are the same, but the energy reported with it is not conserved. Shifted and switched, the energy is conserved to $6 \times 10^{-8}\epsilon$, some 500 times less well than without a cutoff: S of Equation (8.6) has a third derivative that jumps at both ends, so a higher derivative of the force still jumps whenever a pair crosses r_s or r_c .

The shifted potential pays for its jumping force in another way. The solver chooses its own steps to keep its error below a set tolerance, and to do so it called for the forces 26 465 times without a cutoff and 28 925 times with switching, but 1 221 089 times with truncation or shifting, 46 times as often. Its error, like that of any method that advances the motion step by step, rests on the Taylor series of the motion over a step (Section 1.8), which assumes that the force changes smoothly; a jump in the force breaks that assumption, and the solver can only keep its accuracy by taking tiny steps

across every jump. The fixed-step methods of Chapter 9 cannot do that, and Chapter 9 measures what a jump in the force costs them.

Machine-learned potentials face the same choice and make the smooth one. Behler and Parrinello multiplied each neighbour's contribution by $\frac{1}{2}[\cos(\pi r/r_c) + 1]$, which falls to zero with zero slope at the cutoff distance²⁰, and TrajCast (the planned Chapter 28) multiplies by the polynomial envelope of DimeNet²¹, $1 - \frac{1}{2}(p+1)(p+2)x^p + p(p+2)x^{p+1} - \frac{1}{2}p(p+1)x^{p+2}$, with $x = r/r_c$ and $p = 6$, which is zero at the cutoff with its first and second derivatives (Exercise 8.12).

KEY IDEA. A cutoff should take both the energy and the force smoothly to zero. Truncation makes the energy jump, so the total energy is not conserved. Shifting makes the energy continuous but leaves the force jumping, which an adaptive solver can follow only with far shorter steps, 46 times as many force calls here. Switching, or a smooth envelope as in machine-learned potentials, removes both jumps, though a jump left in a higher derivative still costs an accurate solver some accuracy.

NOTEBOOK 08 *cut a potential three ways and watch the energy*

CHECK YOUR UNDERSTANDING. For argon ($\varepsilon = 0.010\,34\text{ eV}$), how large is each jump of the total energy when a pair crosses a truncated cutoff at 2.5σ ?

8.6 Many-body potentials for metals and carbon

In a pair potential every bond has the same strength whatever else its atoms are bonded to. That fails for two materials used for the negative electrodes of lithium batteries, lithium metal and graphite. In both, the strength of a bond depends on how many other bonds its atoms have. This section builds a model with that property, the simplest many-body potential, and describes the forms used for metals and carbon.

Bonds that weaken as they multiply

In a metal the outer electrons are shared by all the atoms, and an atom with more neighbours spreads its share of them over more bonds, each weaker. A model that captures this writes the energy of atom i with a square root,

$$U = \sum_i \left[\sum_{j \neq i} A e^{-\eta(r_{ij}/r_0-1)} - \sqrt{\rho_i} \right], \quad \rho_i = \sum_{j \neq i} \zeta^2 e^{-2\zeta(r_{ij}/r_0-1)}, \quad (8.7)$$

with a pair repulsion of strength A and range set by η , and a bond energy $-\sqrt{\rho_i}$ built from a sum ρ_i over the neighbours, each contributing ζ^2 at the distance r_0 and less farther away. Finnis and Sinclair proposed this square root for metals such as iron and

tungsten²², and the exponential forms used here are those of Cleri and Rosato²³; it is called the second-moment model.

For an atom with z neighbours, all at r_0 , $\rho_i = z\tilde{\zeta}^2$ and its bond energy is $-\tilde{\zeta}\sqrt{z}$: it grows only as the square root of the number of bonds, so each bond is worth $-\tilde{\zeta}/\sqrt{z}$, weaker the more there are. In a pair potential the bond energy is z times that of one bond. In a close-packed crystal, its atoms stacked like oranges on a market stall, each atom touches 12 others; in the body-centred cubic structure of lithium each sits at the centre of a cube of 8; and in a simple cubic crystal each has 6, along the three axes. [Figure 8.5](#) compares the two models, scaled to agree for close-packed metals, $z = 12$. A single bond of the second-moment model is worth $1/\sqrt{12} = 0.289$ of a lone bond ($z = 1$) when the atom has 12 neighbours, and $1/\sqrt{8} = 0.354$ in body-centred cubic lithium, with 8. One consequence is that an atom at a surface, with fewer neighbours, holds those it has more tightly, and [Exercise 8.16](#) shows that, counting only the bond energy, making a vacancy, by moving an atom from inside a close-packed crystal to its surface, costs about half the energy that holds an atom in the crystal, where a pair potential makes the two equal.

The forces follow from the chain rule. The square root couples every neighbour of i : moving atom k changes ρ_i through the term r_{ik} , and $\sqrt{\rho_i}$ changes by $\frac{1}{2}\rho_i^{-1/2}$ times that. The force on an atom therefore depends on the neighbours of its neighbours, and is no longer a sum of forces along pairs with fixed strengths. The forces still add to zero and exert no total torque, because U depends only on distances and so does not change when all the atoms are moved or turned together ([Section 7.5](#)); the function `second_moment` of `mdlab` computes it, and its test checks it against central differences.

The embedded-atom method generalises [Equation \(8.7\)](#): the energy of atom i is $\mathcal{F}(\bar{\rho}_i) + \frac{1}{2}\sum_j \varphi(r_{ij})$, with $\bar{\rho}_i$ the density of the electrons, their charge per unit volume, that its

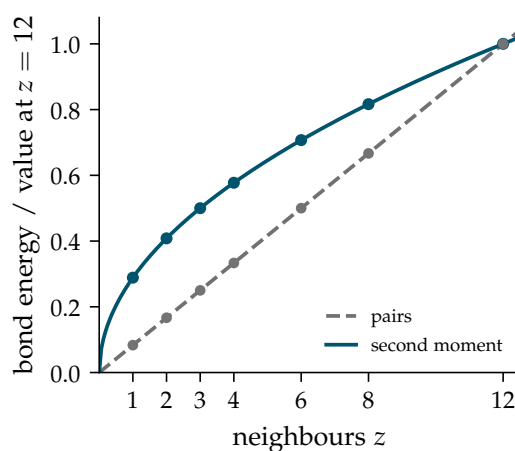


FIGURE 8.5: The bond energy of an atom with z neighbours, as a fraction of its value at $z = 12$: proportional to z for a pair potential (dashed), and to $\sqrt{z}$ for the second-moment model (teal), at a pair of atoms (1), a chain (2), a graphite layer (3), diamond (4), simple cubic (6), body-centred cubic lithium (8) and close-packed metals (12).

neighbours produce where it sits and $\mathcal{F}$, the embedding energy, a function fitted to measurements or to calculations²⁴. The second-moment model is the case $\mathcal{F}(\bar{\rho}) = -\sqrt{\bar{\rho}}$, with $\bar{\rho}_i = \rho_i$ and $\varphi(r) = 2A e^{-\eta(r/r_0-1)}$, the 2 making up for the $\frac{1}{2}$.

Carbon and the order of a bond

Carbon forms directional bonds, three to a layer in graphite and four in diamond, and its bonds too weaken as they multiply. Tersoff's potential writes the energy as a sum over bonds, $\frac{1}{2} \sum_i \sum_{j \neq i} f_c(r_{ij}) [f_R(r_{ij}) - b_{ij} f_A(r_{ij})]$, with a repulsion f_R , an attraction f_A , a smooth cutoff f_c , and a bond order b_{ij} that falls as atom i gains other neighbours, depending also on the angles between its bonds^{25,26}. The reactive bond-order potential of Brenner and co-workers extends this to hydrocarbons and describes the strong bonds within a graphite layer²⁷. It has no interaction between the layers, which are held together by the weak attraction between neutral atoms; a version adds a Lennard-Jones term for it²⁸. Lithium between the layers of graphite needs, in addition, lithium-carbon and lithium-lithium interactions, and models fitted for this are among the machine-learned potentials planned for the continuation of Part V.

KEY IDEA. In metals and in carbon the strength of a bond falls as an atom's bonds multiply. The second-moment model, with bond energy $-\sqrt{\rho_i}$, makes the bond energy grow as $\sqrt{z}$; the embedded-atom method and bond-order potentials refine the same idea. Their forces couple the neighbours of neighbours but still add to zero and exert no total torque.

NOTEBOOK 08 *the second-moment model: bonds against neighbours*

CHECK YOUR UNDERSTANDING. In the second-moment model, by what factor is one bond of an atom in a chain ($z = 2$) stronger than one bond in close-packed metal?

8.7 Bonded force fields

The solvents of a lithium-ion battery's electrolyte are small organic molecules, carbonates such as ethylene carbonate, a ring of five atoms, and chains such as dimethyl carbonate. While no reaction takes place, each atom of a molecule keeps the same bonded partners, and its energy can be written as a sum of terms for the bonds, the angles between bonds and the twists about them, with the interactions of atoms farther apart, in the same molecule or in different ones, added as pairs. This section sets out that form, called a force field, and the one new piece of geometry it needs, the dihedral angle.

The terms

In the all-atom form, in which every hydrogen is a site of its own, of the force field OPLS (Optimized Potentials for Liquid Simulations) of Jorgensen and co-workers²⁹,

$$\begin{aligned}
 U = & \sum_{\text{bonds}} \kappa_r (r - r_{\text{eq}})^2 + \sum_{\text{angles}} \kappa_\theta (\theta - \theta_{\text{eq}})^2 + \sum_{\text{dihedrals}} \sum_{n=1}^3 \frac{\kappa_n}{2} [1 + (-1)^{n+1} \cos n\psi] \\
 & + \sum_{i < j} f_{ij} \left[\frac{k_e q_i q_j}{r_{ij}} + 4\varepsilon_{ij} \left(\frac{\sigma_{ij}^{12}}{r_{ij}^{12}} - \frac{\sigma_{ij}^6}{r_{ij}^6} \right) \right].
 \end{aligned} \tag{8.8}$$

The first sum holds each bond near its length r_{eq} by a spring; κ_r is half the stiffness k of the spring energy $\frac{1}{2}kx^2$ (Section 3.3), since the factor $\frac{1}{2}$ is absorbed into it. The second holds each angle θ between two bonds at an atom near θ_{eq} , as the three atoms of Figure 7.5(a) were held, with κ_θ half the angular stiffness for the same reason. The third gives the energy of a twist ψ about each bond as a short series of cosines, written out $\frac{1}{2}\kappa_1(1 + \cos \psi) + \frac{1}{2}\kappa_2(1 - \cos 2\psi) + \frac{1}{2}\kappa_3(1 + \cos 3\psi)$ with constants κ_n , and the last adds the Coulomb energy of charges q_i placed on the atoms (Section 8.8) and a Lennard-Jones energy, for every pair. The factor f_{ij} is 0 for atoms joined by one bond or two, whose interaction the bond and angle terms already describe, $\frac{1}{2}$ for atoms three bonds apart, and 1 for all others, including every pair in different molecules. Each constant depends on the kinds of atom involved, and a force field is a table of them, fitted to calculations of the electrons and to measured properties of liquids.

The dihedral angle

Four atoms bonded in a chain, $a-b-c-d$, can twist about the middle bond $b-c$ (Figure 8.6). The twist is measured by the angle between the plane of a, b, c and the plane of b, c, d , seen along the middle bond. Each plane is fixed by the arrow at right angles to it, which the cross product of two arrows in the plane gives (Section 5.2). With $\mathbf{b}_1 = \mathbf{r}_b - \mathbf{r}_a$, $\mathbf{b}_2 = \mathbf{r}_c - \mathbf{r}_b$ and $\mathbf{b}_3 = \mathbf{r}_d - \mathbf{r}_c$ along the three bonds,

$$\mathbf{n}_1 = \mathbf{b}_1 \times \mathbf{b}_2, \quad \mathbf{n}_2 = \mathbf{b}_2 \times \mathbf{b}_3$$

are at right angles to the first and second planes. Seen along $\mathbf{b}_2$, the two planes appear as lines from the middle bond towards a and towards d , and ψ is the angle between them. The part of $\mathbf{b}_1$ along $\mathbf{b}_2$ drops out of $\mathbf{n}_1$, since $\mathbf{b}_2 \times \mathbf{b}_2 = \mathbf{0}$, and the part left points from a towards the bond, so $\mathbf{n}_1$ is $\mathbf{b}_2$ crossed with an arrow from the bond towards a ; likewise $\mathbf{n}_2$ is $\mathbf{b}_2$ crossed with an arrow from the bond towards d . Each is

its line turned through a right angle about $\mathbf{b}_2$, the same way for both, so the angle between them is ψ , and by the dot product

$$\cos \psi = \frac{\mathbf{n}_1 \cdot \mathbf{n}_2}{|\mathbf{n}_1| |\mathbf{n}_2|}. \quad (8.9)$$

The cosine does not tell a twist one way from the same twist the other way. The cross product $\mathbf{n}_1 \times \mathbf{n}_2$ is along $\mathbf{b}_2$, the one direction at right angles to both, pointing along $\mathbf{b}_2$ or against it according to the sense of the twist, so the sign of $(\mathbf{n}_1 \times \mathbf{n}_2) \cdot \mathbf{b}_2$ gives the sign of ψ , which then runs from $-\pi$ to π ; ψ is positive when, seen from b looking towards c , the bond to a turns clockwise to reach the bond to d . For a flat zigzag, a and d on opposite sides of the middle bond, $\mathbf{n}_1$ and $\mathbf{n}_2$ point in opposite directions and $\psi = 180^\circ$, the arrangement called *trans*; with a and d on the same side, $\psi = 0$, called *cis*. The function `dihedral_angle` of `mdlab` computes ψ in this way.

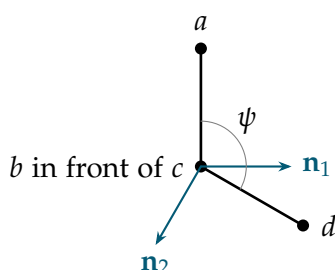


FIGURE 8.6: Four atoms bonded in a chain, seen along the middle bond with b in front of c . The planes of a, b, c and of b, c, d are seen edge-on as the lines towards a and d , at the dihedral angle ψ ; $\mathbf{n}_1$ and $\mathbf{n}_2$ are those lines turned through a right angle the same way, so they meet at the same angle.

KEY IDEA. A force field writes the energy of molecules as springs for bonds and angles, a cosine series for each twist, and Coulomb and Lennard-Jones energies between atoms more than two bonds apart. The twist is the dihedral angle between the two planes of a chain of four atoms, found from two cross products.

NOTEBOOK 08 *twist a chain of four atoms and watch its torsion energy*

CHECK YOUR UNDERSTANDING. In Equation (8.8), which values of ψ make the κ_1 and the κ_3 terms smallest, when both constants are positive?

8.8 Charges and the slow decay of $1/r$

Ions in an electrolyte, the charged atoms of a crystal such as rock salt and the charges placed on the atoms of a force field interact by Coulomb's law. Two charges q_i and q_j ,

in units of the elementary charge e , the charge of a proton (an electron carries $-e$, and an ion is an atom that has lost or gained electrons), at a distance r have the energy

$$\varphi_C(r) = \frac{k_e q_i q_j}{r}, \quad k_e = \frac{e^2}{4\pi\epsilon_0} = 14.3996 \text{ eV } \text{\AA}, \quad (8.10)$$

where ϵ_0 is the electric constant; k_e is computed from its value in `mdlab.units`. This section shows that this energy cannot be cut off as the earlier ones were.

A slow decay

Two opposite unit charges at $1.5639 \text{ \AA}$, the bond length of LiF (Section 5.8), have $-14.3996/1.5639 = -9.21 \text{ eV}$; at $10 \text{ \AA}$ they still have -1.44 eV , where two argon atoms have $-6.4 \times 10^{-5} \text{ eV}$. Take an ion inside a large region filled with ions at the number density n , that is n ions in each unit of volume. The ions at distances between r and $r + \Delta r$ fill a shell of area $4\pi r^2$ and thickness Δr , so there are about $4\pi r^2 n \Delta r$ of them, a number that grows as r^2 . Each contributes an energy of size k_e/r , so the shell as a whole contributes an amount of size $4\pi k_e n r \Delta r$, which grows with r . For a Lennard-Jones attraction the same shell contributes $4\pi r^2 n \Delta r \times r^{-6}$, falling as r^{-4} , and the far shells hardly matter; for charges, every shell counts more than the last, and only the cancellation between positive and negative ions keeps the total finite.

A sum that depends on its order

Rock salt, NaCl, shows what this means. Its ions sit on a cubic grid of spacing d , the sign of the charge alternating from each site to its neighbours, so the ion at $(a, b, c)d$, with a, b, c integers, positive, negative or zero, carries the charge $(-1)^{a+b+c}$ relative to the ion at the origin. The energy of the ion at the origin with all the others is $-\mathcal{M}k_e/d$, with the Madelung constant

$$\mathcal{M} = - \sum_{(a,b,c) \neq (0,0,0)} \frac{(-1)^{a+b+c}}{\sqrt{a^2 + b^2 + c^2}}.$$

Figure 8.7 adds up this sum in two orders. Over the ions inside a growing sphere, the partial sums do not settle, out to $12d$: the six nearest ions alone give the partial sum 6, adding the twelve next give -2.49 , and between radii $6d$ and $12d$ the sums swing between -10.3 and $+14.7$, because each new shell of ions has a net charge of its own. Over the ions inside a growing cube, with the ions on its faces, edges and corners counted with the fractions $\frac{1}{2}$, $\frac{1}{4}$ and $\frac{1}{8}$ of a small cube of side d around each that lie inside, every cube is neutral: the weighted charge is a product of three equal sums, one along each axis, each zero, as $-\frac{1}{2} + 1 - \frac{1}{2} = 0$ for the smallest cube. The sums settle

quickly, to 1.747567 at a half-side of $12d$, within 2×10^{-6} of the limit 1.747565³⁰. The answer depends on the order in which the terms are added. That cannot happen when the sizes of the terms, all taken positive, add up to a finite total; here they do not, since the sizes in one shell add up to an amount that grows as r . Physically, the arrangement of charge at the surface of the region summed over affects the energy of every ion inside it. For a crystal of this kind with a spacing $d = 2.82 \text{ \AA}$, the electrostatic energy of one ion with all the others is $-1.7476 \times 14.3996/2.82 = -8.92 \text{ eV}$.

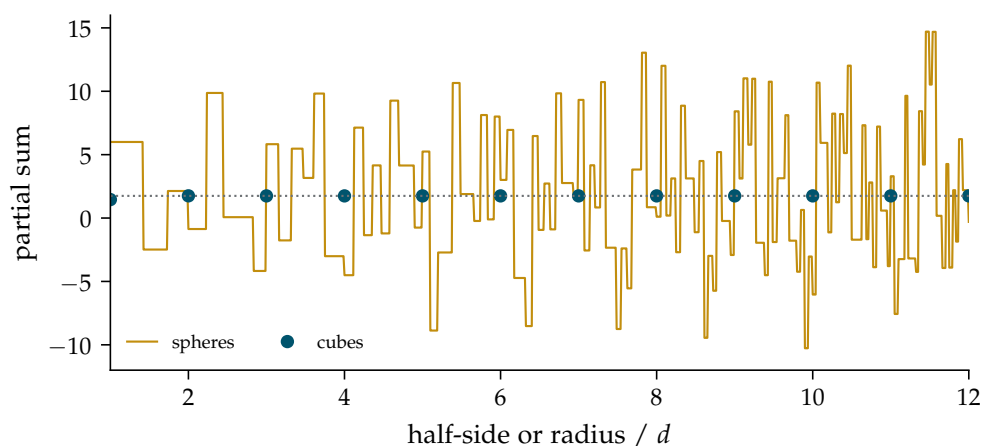


FIGURE 8.7: The Madelung sum of rock salt over the ions inside a growing sphere (ochre), which never settles, and inside a growing neutral cube (teal dots), which settles to 1.7476 (dotted), against the radius or half-side in units of the nearest-neighbour distance d .

A cutoff therefore cannot be applied to charges as it was to the Lennard-Jones energy: whatever its radius, the charge inside it is not zero in general and the energy depends on where it falls. Electrolytes, full of ions, and electrodes, whose surfaces carry charge, need the electrostatic energy of an infinite, periodic arrangement of charges summed correctly. Chapter 10, once periodic boxes are in place, derives the Ewald method, one way of doing that.

KEY IDEA. The Coulomb energy $k_e q_i q_j / r$ falls so slowly that each shell of charges matters more than the last, and the sum for a crystal of ions depends on the order in which its terms are added, which sets the charge at the surface of the region summed. Charges cannot be cut off like a Lennard-Jones energy, and periodic systems need a sum built for them, such as the Ewald sum of Chapter 10.

NOTEBOOK 08 *sum the charges of rock salt over spheres and cubes*

CHECK YOUR UNDERSTANDING. How many ions lie at the distance $\sqrt{2}d$ from an ion in rock salt, and what is their charge relative to it?

What we now have

The potential energy on which atoms move is the energy of their electrons, settled at each arrangement of the nuclei, plus the repulsion of the nuclei: the potential energy surface of Born and Oppenheimer. The dense eigensolver considered here has a cost that grows as the cube of its matrix size, motivating cheaper models of the energy. Treating the nuclei themselves as classical is good when the quantum step $\hbar\omega$ of their vibrations is small beside $k_B T$, which holds for heavy atoms and slow motions and fails for the stretches of bonds to hydrogen.

The simplest models add up pair potentials, the Lennard-Jones, Morse and exp-6 shapes, whose forces are computed once per pair and applied with opposite signs, with the virial $\mathcal{W} = -\sum r_{ij}\varphi'(r_{ij})$ alongside; in reduced units every Lennard-Jones substance obeys the same equations. Every force routine is checked against central differences of its energy, at a step near the cube root of the machine epsilon times the length scale.

A cutoff must take both energy and force smoothly to zero: truncation breaks the conservation of energy, shifting leaves a jump in the force that an adaptive solver follows only with far shorter steps, and switching or a smooth envelope removes both jumps.

In metals and carbon the strength of a bond falls as an atom's bonds multiply, which the second-moment model, the embedded-atom method and bond-order potentials describe; molecules are described by force fields of bonds, angles, dihedral twists and non-bonded pairs; and the Coulomb energy of charges falls so slowly that its sum depends on its order.

Chapter 9 uses these forces to follow the motion step by step, and Chapter 10 builds a complete simulation of atoms in a periodic box, with neighbour lists for the cutoffs and the Ewald sum for the charges.

Answers to the checks

SECTION 8.1. The size grows by a factor of 4, so the time by $4^3 = 64$: about 6.4 s.

SECTION 8.2. No. Its quantum step is $\hbar\omega = 2\pi\hbar c \tilde{\nu} = 1.2398 \times 10^{-4} \times 3000 = 0.372$ eV, 14.4 times $k_B T = 0.02585$ eV.

SECTION 8.3. $\tau = \sigma\sqrt{m/\epsilon}$. Krypton's σ is larger by a factor $3.6/3.4 = 1.06$ and its m/ϵ by $(83.8/39.948)/(160/120) = 1.57$, whose square root is 1.25, so its τ is about 1.33 times argon's, longer: it is heavier, and the extra depth of its well does not make up for that.

SECTION 8.4. At $r = \sigma$, closer than the minimum at 1.12σ , the pair repels, $\phi'(\sigma) = 4\epsilon(-12 + 6)/\sigma = -24\epsilon/\sigma < 0$, so $\mathcal{W} = -\sigma\phi'(\sigma) = +24\epsilon$: positive.

SECTION 8.5. $|\phi(2.5\sigma)| = 0.0163 \times 0.01034 \text{ eV} = 1.69 \times 10^{-4} \text{ eV}$ per crossing.

SECTION 8.6. Each bond is worth $\xi/\sqrt{z}$, so the ratio is $\sqrt{12}/\sqrt{2} = \sqrt{6} = 2.45$.

SECTION 8.7. $1 + \cos \psi$ is least, 0, at $\psi = 180^\circ$; $1 + \cos 3\psi$ is 0 where $\cos 3\psi = -1$, that is $3\psi = 180^\circ, 540^\circ$ or -180° , so $\psi = 60^\circ, 180^\circ$ or -60° . Both are least at 180° , the trans arrangement.

SECTION 8.8. The sites (a, b, c) with $a^2 + b^2 + c^2 = 2$ have two of a, b, c equal to ± 1 and one 0: three choices of the zero and four of the signs, 12 ions. Their $a + b + c$ is even, so they carry the same charge as the ion at the origin.

Exercises

The exercises run from questions of understanding, through calculations, to short derivations marked ‘show that’. Full solutions are in ‘Solutions to the exercises’ at the end of the book, and Notebook 08 checks every numerical answer.

EXERCISE 8.1 (*fast electrons*). With the same stiffness, how many times faster than a lithium atom of 6.94 amu does an electron vibrate? (An electron’s mass is 5.486×10^{-4} amu.)

EXERCISE 8.2 (*when quantum steps matter*). At what temperature is $k_B T$ equal to the quantum step $\hbar\omega$ of the model lithium hollow of Section 8.2, and of the O–H stretch?

EXERCISE 8.3 (*the strongest pull*). Show that two Lennard-Jones atoms attract most strongly at $r = (26/7)^{1/6}\sigma = 1.2445\sigma$, and find that greatest pull.

EXERCISE 8.4 (*the lowered Morse well*). Show that $\phi_M(r) = D(e^{-2a(r-r_0)} - 2e^{-a(r-r_0)})$ has its minimum $-D$ at r_0 and the stiffness $2Da^2$ there.

EXERCISE 8.5 (*the exp-6 form*). Show that, for $\alpha > 7$, the Buckingham potential with $A = 6\epsilon e^\alpha/(\alpha - 6)$, $\rho = r_m/\alpha$ and $C = \alpha\epsilon r_m^6/(\alpha - 6)$ has its minimum $-\epsilon$ at r_m .

EXERCISE 8.6 (*an argon pair*). Two argon atoms vibrate about the Lennard-Jones minimum with the reduced mass $m/2$ (Section 4.8). Find the period of small vibrations in units of τ and in femtoseconds, and how many steps of 0.005τ it spans.

EXERCISE 8.7 (*no total force*). Show from Equation (8.3) that the forces of any pair potential add to zero.

EXERCISE 8.8 (*the virial of a pair*). Find the virial of two Lennard-Jones atoms at $r = 1.5\sigma$, and say for which distances it is positive.

EXERCISE 8.9 (*the best step*). For the twelve atoms of Figure 8.3, $|U| = 10.75\epsilon$; take $|U'''| \approx 10^4 \epsilon/\sigma^3$. Use Equation (8.5) to estimate the best step.

EXERCISE 8.10 (*a one-sided difference*). Show that the forward difference $[U(x+h) - U(x)]/h$ has an error of about $\frac{1}{2}h|U''| + \epsilon_m|U|/h$, and that its best step is of order $\sqrt{\epsilon_m}$ times the length on which U changes and its least error of order $\sqrt{\epsilon_m}$ times the forces, much worse than for central differences.

EXERCISE 8.11 (*the switching function*). Show that $S(x) = 1 - 10x^3 + 15x^4 - 6x^5$ has $S(0) = 1$, $S(1) = 0$, and its first and second derivatives zero at $x = 0$ and $x = 1$.

EXERCISE 8.12 (*TrajCast's envelope*). Show that the envelope $u(x) = 1 - 28x^6 + 48x^7 - 21x^8$, the case $p = 6$ of Section 8.5, has $u(1) = 0$ and $u'(1) = u''(1) = 0$.

EXERCISE 8.13 (*the energy beyond the cutoff*). Atoms beyond r_c are spread evenly at the number density n , the number of atoms per unit volume. Show that the Lennard-Jones energy per atom that a cutoff leaves out is $\frac{8}{3}\pi n\epsilon\sigma^3[\frac{1}{3}(\sigma/r_c)^9 - (\sigma/r_c)^3]$, and evaluate it for $n = 0.8\sigma^{-3}$ and $r_c = 2.5\sigma$.

EXERCISE 8.14 (*counting crossings*). In Figure 8.4(c) the truncated energy rises by up to 0.49ϵ above its starting value. How many fewer pairs lie inside the cutoff than at the start?

EXERCISE 8.15 (*the force of a square root*). Show that the bond energy $-\sqrt{\rho_i}$ of Equation (8.7) acts on atom $k \neq i$ with the force $\frac{1}{2}\rho_i^{-1/2} \nabla_k \rho_i$, and write $\nabla_k \rho_i$ out.

EXERCISE 8.16 (*a vacancy*). In a crystal where every atom has z neighbours at the same distance, an atom is removed from inside and put back elsewhere on the surface of the crystal; putting it there gains, on average, the energy that holds one atom in the crystal. Counting only the bond energy, show that a pair potential makes the energy of the vacancy equal to that energy, and that the second-moment model makes it $z(1 - \sqrt{1 - 1/z})$ times it; evaluate this for $z = 12$.

EXERCISE 8.17 (*a right-angled twist*). Find the dihedral angle of the atoms at $(1, 0, 0)$, $(0, 0, 0)$, $(0, 0, 1)$ and $(0, 1, 1)$ (in Å), and the torsion energy of Equation (8.8) there with $\kappa_1 = 0.1$, $\kappa_2 = 0.02$ and $\kappa_3 = 0.05$ eV.

EXERCISE 8.18 (*as weak as heat*). At what distance is the Coulomb energy of two unit charges in vacuum equal to $k_{\text{B}}T$ at 300 K?

EXERCISE 8.19 (*the first shells*). Count the ions of rock salt at distances $\sqrt{3}d$ and $2d$ from an ion, give their charges, and find the sphere sums for $\mathcal{M}$ up to radii d , $\sqrt{2}d$, $\sqrt{3}d$ and $2d$.

EXERCISE 8.20 (*the smallest neutral cube*). Find by hand the cube sum for $\mathcal{M}$ with half-side d , counting the ions on the faces, edges and corners with weights $\frac{1}{2}$, $\frac{1}{4}$ and $\frac{1}{8}$.

Integrating the equations of motion

Once the forces are known, a simulation must use them to advance the positions and velocities. The computer cannot follow every instant: it computes a sequence of states separated by a finite time step. Chapter 2 used the forward Euler method, Chapter 3 used a more accurate solver, and Chapter 7 showed that two methods with similar errors over one step can behave quite differently over a long run.

This chapter develops velocity Verlet, a method widely used in molecular dynamics. Its construction from Taylor series gives the error of a step; its construction from kicks and drifts explains its reversibility and preservation of phase-space volume. For a spring it keeps a slightly different energy exactly, which lets us derive both its energy error and its stability limit. These results give a practical way to choose the step for a molecule, including the water example left open in Chapters 1, 2 and 4. Long runs then test the method against RK4 and show why the smooth forces of Chapter 8 matter. Chapter 10 assembles these parts into a complete simulation in a periodic box.

9.1 Steps and their errors

A method of computing motion is a step map ([Section 7.7](#)): it takes the positions and velocities at a time t and returns them at $t + \delta t$. Applied over and over it produces the motion at the times $\delta t, 2\delta t, 3\delta t, \dots$, and the question is how far that computed motion strays from the true one. This section measures it for the simplest method and defines the order of a method, the number that says how fast its error shrinks as the step is shortened.

The forward Euler method again

Section 2.6 advanced a motion by holding the velocity and the acceleration at their values at the start of each step:

$$x_{n+1} = x_n + \delta t v_n, \quad v_{n+1} = v_n + \delta t a(x_n),$$

where x_n and v_n are the computed position and velocity after n steps, at the time $t_n = n\delta t$. The Taylor series of the true position over one step (Section 1.8), and the same series for the velocity, whose derivatives are $\dot{v} = a$ and $\ddot{v} = \dot{a}$, the rate of change of the acceleration,

$$x(t + \delta t) = x(t) + v(t) \delta t + \frac{1}{2} a(t) \delta t^2 + \mathcal{O}(\delta t^3),$$

$$v(t + \delta t) = v(t) + a(t) \delta t + \frac{1}{2} \dot{a}(t) \delta t^2 + \mathcal{O}(\delta t^3),$$

show what one step misses: started from the true state, the computed position is wrong by $\frac{1}{2}a\delta t^2$ and the computed velocity by $\frac{1}{2}\dot{a}\delta t^2$, each plus smaller terms. The error made in one step from a correct start is called the local error; for the forward Euler method it is of order δt^2 .

From one step to a whole run

To follow the motion for a fixed time t_f needs $t_f/\delta t$ steps. Each makes a local error of order δt^2 , and each error is carried forward by the later steps. For a smooth force, an error carried through one step grows by a factor of at most $1 + c \delta t$, with c a bound on how fast the rates of change depend on the state, so through all the later steps by at most

$$(1 + c \delta t)^{t_f/\delta t} \leq (e^{c \delta t})^{t_f/\delta t} = e^{ct_f},$$

since $1 + y \leq 1 + y + \frac{1}{2}y^2 + \dots = e^y$ for $y \geq 0$. This factor depends on t_f and on the force but not on δt , and the error at t_f is at most it times the sum of the local errors¹¹: $(t_f/\delta t) \times \mathcal{O}(\delta t^2) = \mathcal{O}(\delta t)$. A method whose error over a fixed time, its global error, is of order δt is said to be of order 1, one whose global error is of order δt^2 of order 2, and so on. For a method that computes each state from the one before it alone, as the forward Euler method does, the local error is one power of δt higher than the global error. The forward Euler method is of order 1: halving the step halves the error at a given time, as Section 2.6 found.

Figure 9.1 measures this for the pendulum of Chapter 7, $H = p^2/(2mL^2) + mgL(1 - \cos \theta)$, with energies in units of mgL and times in units of $\sqrt{L/g}$, which amounts to setting $m = L = g = 1$, as the reduced units of Section 8.3 do for atoms. Writing q for

the angle, $H = p^2/2 + 1 - \cos q$ and $\ddot{q} = -\sin q$. Started at the angle $q = 1$ with the momentum $p = 0.5$, the error of the angle after a time of 4 falls with the step along a line of slope 1.03 on logarithmic axes (Section 1.8), and its error after a single step along a line of slope 2.01. The figure also shows the two methods of the next sections, whose errors fall much faster.

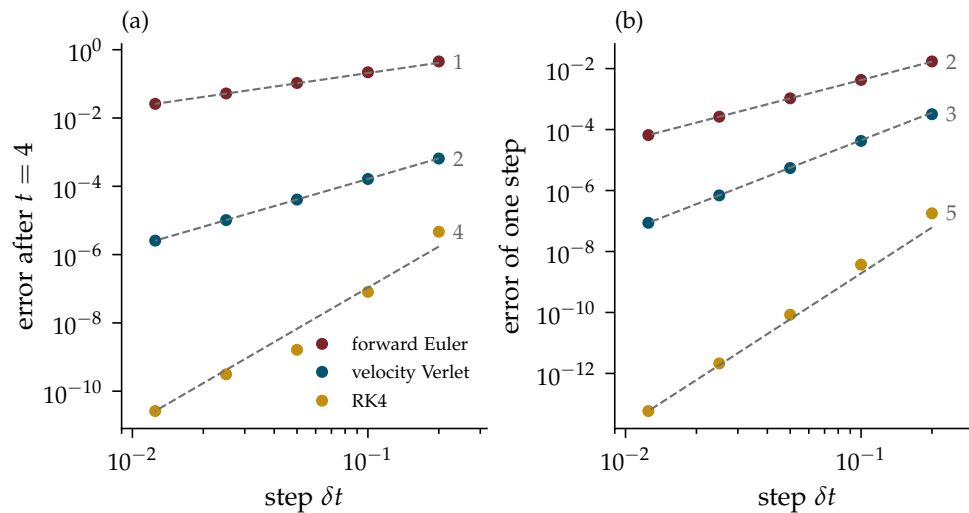


FIGURE 9.1: The errors of three methods for the pendulum $H = p^2/2 + 1 - \cos q$ in reduced units, started at $q = 1$, $p = 0.5$, against the step δt , on logarithmic axes. (a) The error of the angle after a time of 4, with lines of slope 1, 2 and 4 (dashed). (b) The error of the angle after one step from the same start, with lines of slope 2, 3 and 5.

KEY IDEA. A method's local error is the error of one step from a correct start. For a method that steps from the present state alone, the errors of the $t_f/\delta t$ steps add up over a fixed time to a global error with one power of δt fewer, and that power is the order of the method: the forward Euler method, with local and global errors of order δt^2 and δt , is of order 1.

CHECK YOUR UNDERSTANDING. Halving the step of a method divides its error at a given time by 4. What is its order, and how does its local error depend on δt ?

9.2 The Verlet method

Verlet's method, which Verlet introduced in 1967 for his simulations of a liquid of Lennard-Jones atoms³¹, gains a whole order over the forward Euler method at the same cost of one force evaluation per step. This section derives it from the Taylor series, writes it in the form that simulations use, and checks its order.

Positions from two Taylor series

The Taylor series of the motion a step ahead and a step behind are

$$\begin{aligned}x(t + \delta t) &= x + v \delta t + \frac{1}{2}a \delta t^2 + \frac{1}{6}\dot{a} \delta t^3 + \frac{1}{24}\ddot{a} \delta t^4 + \mathcal{O}(\delta t^5), \\x(t - \delta t) &= x - v \delta t + \frac{1}{2}a \delta t^2 - \frac{1}{6}\dot{a} \delta t^3 + \frac{1}{24}\ddot{a} \delta t^4 + \mathcal{O}(\delta t^5),\end{aligned}$$

with x, v, a and the derivatives of a taken at t ; the second is the first with $-\delta t$ in place of δt , which changes the sign of every odd power. Adding them, the odd powers cancel, the terms in δt^5 within $\mathcal{O}(\delta t^5)$ among them, so the remainder is of order δt^6 :

$$\begin{aligned}x(t + \delta t) + x(t - \delta t) &= 2x + a \delta t^2 + \frac{1}{12}\ddot{a} \delta t^4 + \mathcal{O}(\delta t^6) & \frac{1}{24} + \frac{1}{24} = \frac{1}{12} \\x(t + \delta t) &= 2x(t) - x(t - \delta t) + \delta t^2 a(t) + \mathcal{O}(\delta t^4) & \text{subtracting } x(t - \delta t),\end{aligned}$$

the term in δt^4 joining the remainder. Dropping the last term gives the Störmer–Verlet method,

$$x_{n+1} = 2x_n - x_{n-1} + \delta t^2 a(x_n), \quad (9.1)$$

which advances the positions with an error of order δt^4 in one step from two correct positions, using only the force at the present position. Its global error is nevertheless of order δt^2 , as for velocity Verlet below. An error made in one position, with the one before it correct, changes the difference between successive positions, which every later step carries forward, so over a fixed time each local error grows by a factor of order $t_f/\delta t$, and the $t_f/\delta t$ of them give $(t_f/\delta t)^2 \times \mathcal{O}(\delta t^4) = \mathcal{O}(\delta t^2)$. The rule of [Section 9.1](#) that relates the two orders holds only for methods that step from the present state alone. The velocities do not appear, but can be recovered by the central difference of [Section 1.9](#), $v_n = (x_{n+1} - x_{n-1})/(2\delta t)$, with an error of order δt^2 .

Velocity Verlet

Simulations need the velocities at the same times as the positions, for the kinetic energy and to start a run from a state. Expanding the position and the velocity over one step,

$$\begin{aligned}x(t + \delta t) &= x + v \delta t + \frac{1}{2}a \delta t^2 + \mathcal{O}(\delta t^3), \\v(t + \delta t) &= v + a \delta t + \frac{1}{2}\dot{a} \delta t^2 + \mathcal{O}(\delta t^3).\end{aligned}$$

The derivative $\dot{a}$ is not known, but $a(t + \delta t) = a + \dot{a} \delta t + \mathcal{O}(\delta t^2)$, so $\dot{a} \delta t = a(t + \delta t) - a(t) + \mathcal{O}(\delta t^2)$, and putting this into the velocity series,

$$\begin{aligned} v(t + \delta t) &= v + a \delta t + \frac{1}{2} [a(t + \delta t) - a(t)] \delta t + \mathcal{O}(\delta t^3) \\ &= v + \frac{1}{2} [a(t) + a(t + \delta t)] \delta t + \mathcal{O}(\delta t^3) : \end{aligned}$$

the velocity changes by the average of the accelerations at the two ends times the step. Dropping the error terms gives the velocity Verlet method³²,

$$x_{n+1} = x_n + \delta t v_n + \frac{1}{2} \delta t^2 a_n, \quad v_{n+1} = v_n + \frac{1}{2} \delta t (a_n + a_{n+1}), \quad (9.2)$$

with $a_n = a(x_n)$. The method uses $a(x_{n+1})$ in place of the true $a(t + \delta t)$; the two differ by order δt^3 , as x_{n+1} and $x(t + \delta t)$ do, which changes v_{n+1} only at order δt^4 . Its local error is of order δt^3 in both the position and the velocity. Written with the velocity at the middle of the step, $v_{n+\frac{1}{2}} = v_n + \frac{1}{2} \delta t a_n$, it is three moves in turn:

$$v_{n+\frac{1}{2}} = v_n + \frac{1}{2} \delta t a(x_n), \quad x_{n+1} = x_n + \delta t v_{n+\frac{1}{2}}, \quad v_{n+1} = v_{n+\frac{1}{2}} + \frac{1}{2} \delta t a(x_{n+1}), \quad (9.3)$$

a half kick of the velocity, a drift of the position, and another half kick, in the language of [Section 7.7](#). The force at x_{n+1} serves both the end of one step and the start of the next, so each step evaluates the forces once ([Figure 9.2](#)).

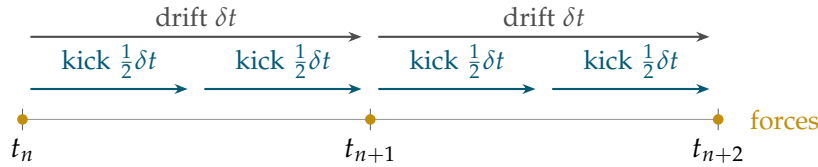


FIGURE 9.2: Velocity Verlet as a kick of the velocities over half a step, a drift of the positions over a whole step with the half-step velocities, and another half kick. The forces (dots) are evaluated once per step, at the new positions, and serve the last half kick of one step and the first of the next.

Started from the same first two positions, velocity Verlet and Störmer–Verlet produce the same positions after them. From [Equation \(9.2\)](#), two successive steps give

$$\begin{aligned} x_{n+1} - x_n &= \delta t v_n + \frac{1}{2} \delta t^2 a_n, \\ x_n - x_{n-1} &= \delta t v_{n-1} + \frac{1}{2} \delta t^2 a_{n-1}, \end{aligned}$$

and subtracting the second from the first,

$$\begin{aligned}
 x_{n+1} - 2x_n + x_{n-1} &= \delta t (v_n - v_{n-1}) + \frac{1}{2}\delta t^2(a_n - a_{n-1}) \\
 &= \frac{1}{2}\delta t^2(a_{n-1} + a_n) + \frac{1}{2}\delta t^2(a_n - a_{n-1}) & v_n - v_{n-1} &= \frac{1}{2}\delta t (a_{n-1} + a_n) \\
 &= \delta t^2 a_n & \text{the terms in } a_{n-1} &\text{cancel,}
 \end{aligned}$$

which is [Equation \(9.1\)](#). The leapfrog method, which keeps only the half-step velocities, $v_{n+\frac{1}{2}} = v_{n-\frac{1}{2}} + \delta t a_n$ and $x_{n+1} = x_n + \delta t v_{n+\frac{1}{2}}$, gives the same positions too ([Exercise 9.6](#)). Started consistently, the three are one method written three ways, and the tests of `mdlab` confirm that they agree to round-off.

In code

The function `velocity_verlet_step` of `mdlab` takes one step for N atoms:

```

1 def velocity_verlet_step(
2     positions: ArrayLike,
3     velocities: ArrayLike,
4     masses: ArrayLike,
5     force: ForceFunction,
6     dt: float,
7     forces: ArrayLike | None = None,
8     force_to_accel: float = FORCE_TO_ACCEL,
9 ) -> tuple[NDArray, NDArray, NDArray]:
10     r = np.asarray(positions, dtype=float)
11     v = np.asarray(velocities, dtype=float)
12     m = np.asarray(masses, dtype=float)
13     f = force(r) if forces is None else np.asarray(forces, dtype=float)
14     v_half = v + 0.5 * dt * _accel(f, m, force_to_accel)
15     r_new = r + dt * v_half
16     f_new = force(r_new)
17     v_new = v_half + 0.5 * dt * _accel(f_new, m, force_to_accel)
18     return r_new, v_new, f_new

```

The helper `_accel` divides each row of forces by its atom's mass and multiplies by the factor `FORCE_TO_ACCEL` of [Section 2.10](#). `ForceFunction` names the kind of argument force is, a function that takes the positions and returns the forces. The argument `forces` may be left out, and then takes the value `None`, Python's word for nothing; the line that sets `f` then computes the forces at the starting positions, and otherwise uses those passed in. The forces returned are passed back in at the next step, so that after the first step force is called once per step.

Its order, and a claim checked

In [Figure 9.1](#), the global error of velocity Verlet falls along a line of slope 2.00, so it is a method of order 2, and the error of one step along a line of slope 2.96: halving the step divides the error at a given time by 4 and the error of one step by about 8. The introduction of the TrajCast paper attributes the typical step of 0.5 to 1.0 fs of molecular dynamics to ‘the cubic scaling of the discretization error with the integration time step’¹. That is the local error of velocity Verlet, of order δt^3 , as derived here; over a fixed time it becomes the global error of order δt^2 .

KEY IDEA. Adding the Taylor series a step ahead and a step behind cancels the odd powers and gives the Störmer–Verlet method. Velocity Verlet, the same method with the velocities kept in step, is a half kick, a drift and a half kick, costs one force evaluation per step, and is of order 2, with a local error of order δt^3 .

NOTEBOOK 09 *measure the order of each method*

CHECK YOUR UNDERSTANDING. Show that velocity Verlet follows a body falling under a constant acceleration $-g$ exactly, whatever the step.

9.3 Reversal and volume

The exact motion can be run backwards ([Section 7.6](#)) and keeps the volume of every region of phase space ([Section 7.4](#)). This section shows that velocity Verlet keeps both properties exactly, for any step, which the forward Euler method does not.

Running a step backwards

We take a step from (x, v) to (x', v') by [Equation \(9.3\)](#), with $v_{\frac{1}{2}}$ the half-step velocity and a, a' the accelerations at x and x' , then reverse the velocity and take one more step from $(x', -v')$:

$$-v' + \frac{1}{2}\delta t a' = -(v' - \frac{1}{2}\delta t a') = -v_{\frac{1}{2}} \quad \text{the half kick undoes the last half kick,}$$

$$x' + \delta t (-v_{\frac{1}{2}}) = x' - (x' - x) = x \quad \text{the drift runs back to } x,$$

$$-v_{\frac{1}{2}} + \frac{1}{2}\delta t a = -(v_{\frac{1}{2}} - \frac{1}{2}\delta t a) = -v \quad \text{the last half kick undoes the first,}$$

using $v' = v_{\frac{1}{2}} + \frac{1}{2}\delta t a'$, $x' = x + \delta t v_{\frac{1}{2}}$ and $v_{\frac{1}{2}} = v + \frac{1}{2}\delta t a$. The step lands on $(x, -v)$: the starting state with its velocity reversed, as for the exact motion. Since every step can be undone this way, a whole run of velocity Verlet, reversed, retraces itself; the tests of `mdlab` run 200 steps of three atoms in nonlinear wells forwards and back and recover

the start to within 10^{-15} , the rounding of the arithmetic. A forward Euler step does not undo itself (Exercise 9.7).

Keeping volume

For one coordinate, the half kick $(x, v) \rightarrow (x, v + \frac{1}{2}\delta t a(x))$ changes the velocity by an amount that depends only on the position, and the drift $(x, v) \rightarrow (x + \delta t v, v)$ changes the position by an amount that depends only on the velocity: each is a shear, with Jacobian determinant 1 (Section 7.7). Velocity Verlet is three such shears in turn, so it keeps area in the phase plane, as Figure 9.3 will show: its states for the spring lie on a closed curve, where those of forward Euler spiral outwards (Figure 7.7), the test of Section 4.3.

For N atoms the same holds in the $6N$ dimensions of phase space. In a kick, the new momenta are the old plus amounts that depend only on the positions, and the positions do not change; its Jacobian matrix, with the positions listed before the momenta, has 1 on its diagonal and zeros everywhere above the diagonal. A square matrix with only zeros on one side of its diagonal has as determinant the product of its diagonal entries, a property of determinants we take on trust, here 1. The drift is the same with the roles exchanged, so every step of velocity Verlet keeps volume in phase space, as the exact flow does (Section 7.4).

The full condition

With more than one coordinate, keeping volume is not all that the exact flow keeps, and saying what else it keeps needs products of matrices.

PRODUCTS OF MATRICES. The product $\mathbf{AB}$ of two $n \times n$ matrices is the matrix whose column b is $\mathbf{A}$ times column b of $\mathbf{B}$, so that its entry in row a and column b is $(\mathbf{AB})_{ab} = \sum_c A_{ac}B_{cb}$, the entries of row a of $\mathbf{A}$ times those of column b of $\mathbf{B}$, added. For example,

$$\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix} = \begin{pmatrix} 7 & 2 \\ 3 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 3 & 7 \end{pmatrix},$$

so the order of a product matters. Since each column of $\mathbf{AB}$ is $\mathbf{A}$ applied to a column of $\mathbf{B}$, $(\mathbf{AB})\mathbf{u} = \mathbf{A}(\mathbf{B}\mathbf{u})$ for every vector $\mathbf{u}$: the product is the map $\mathbf{B}$ followed by the

map $\mathbf{A}$. The transpose $\mathbf{A}^\top$ is $\mathbf{A}$ with rows and columns exchanged, $(\mathbf{A}^\top)_{ab} = A_{ba}$, and exchanging the rows and columns of a product reverses its order, since

$$((\mathbf{AB})^\top)_{ab} = (\mathbf{AB})_{ba} = \sum_c A_{bc} B_{ca} = \sum_c (\mathbf{B}^\top)_{ac} (\mathbf{A}^\top)_{cb} = (\mathbf{B}^\top \mathbf{A}^\top)_{ab}.$$

A large matrix may be written in blocks, smaller matrices set side by side and one above another. The sum $\sum_c A_{ac} B_{cb}$ then splits into sums over the columns of each block, and two such matrices multiply block by block as 2×2 matrices of numbers do, keeping the order of every product of blocks. The trace of a square matrix is the sum of its diagonal entries; for the 2×2 matrix of [Section 4.9](#), with a, b in the first row and c, d in the second, the characteristic equation $(a - \lambda)(d - \lambda) - bc = 0$ multiplies out to $\lambda^2 - (a + d)\lambda + (ad - bc) = 0$, whose coefficients are the trace and the determinant.

We write a point of phase space as $\Gamma = (\mathbf{q}, \mathbf{p})$ and the Jacobian matrix of a step map as $\mathbf{J}$, so that, by the first-order rule of [Section 3.10](#), a small change $\delta\Gamma$ of the starting point becomes the change $\mathbf{J} \delta\Gamma$ after the step. The flow of Hamilton's equations over any time, and so any map that is the flow of some Hamiltonian, satisfies (a result we take on trust)

$$\mathbf{J}^\top \Omega \mathbf{J} = \Omega, \quad \Omega = \begin{pmatrix} \mathbf{0} & \mathbf{I} \\ -\mathbf{I} & \mathbf{0} \end{pmatrix}, \quad (9.4)$$

where $\mathbf{I}$ and $\mathbf{0}$ are the identity and zero matrices of the size of $\mathbf{q}$, placed as four blocks. A map that satisfies [Equation \(9.4\)](#) is called symplectic¹¹. For one coordinate the condition reduces to $\det \mathbf{J} = 1$ ([Exercise 9.8](#)), which is why [Section 7.7](#) could use keeping area as its definition; with more coordinates it implies keeping volume but is stronger. Half kicks and drifts are flows, over a time $\frac{1}{2}\delta t$ or δt , of the Hamiltonians $U(\mathbf{q})$ and $K(\mathbf{p})$ alone, so they satisfy it. So does any sequence of them: two maps in turn take $\delta\Gamma$ to $\mathbf{J}_1 \delta\Gamma$ and then to $\mathbf{J}_2 \mathbf{J}_1 \delta\Gamma$, so the Jacobian matrix of the pair is the product $\mathbf{J}_2 \mathbf{J}_1$, and

$$(\mathbf{J}_2 \mathbf{J}_1)^\top \Omega \mathbf{J}_2 \mathbf{J}_1 = \mathbf{J}_1^\top (\mathbf{J}_2^\top \Omega \mathbf{J}_2) \mathbf{J}_1 = \mathbf{J}_1^\top \Omega \mathbf{J}_1 = \Omega.$$

Velocity Verlet, a half kick, a drift and a half kick, is therefore symplectic.

KEY IDEA. Reversing the velocities after a step of velocity Verlet and stepping again returns the start, with its velocities reversed, without error, and every step, being kicks and a drift, keeps volume in phase space and satisfies the full symplectic condition $\mathbf{J}^\top \Omega \mathbf{J} = \Omega$.

CHECK YOUR UNDERSTANDING. For a single coordinate, write the Jacobian matrix of the half kick $(x, v) \rightarrow (x, v + \frac{1}{2}\delta t a(x))$ and find its determinant.

9.4 Splitting the Liouville operator

Velocity Verlet was derived above from Taylor series. A second derivation, from the Liouville operator of [Section 7.3](#), explains why it is built of kicks and drifts, why each of these moves is exact, and how other methods are made, including the thermostats of Chapter 13.

The motion as an exponential

[Section 7.3](#) found that, over a short time, any quantity A of the state changes along the motion as

$$A(t) = \sum_{n=0}^{\infty} \frac{t^n}{n!} \hat{L}^n A,$$

with $\hat{L}A = \{A, H\}$, every term evaluated at the start. This is the series of the exponential, $e^s = \sum s^n/n!$ ([Section 2.7](#)), with $t\hat{L}$ in place of s , and it is written $e^{t\hat{L}}A$: the exponential of an operator is defined by this series, and applying it to A carries A forward by the time t . For N atoms with $H = \sum_i |\mathbf{p}_i|^2/(2m_i) + U(\mathbf{r}^N)$, Hamilton's equations give $\dot{\mathbf{r}}_i = \mathbf{p}_i/m_i$ and $\dot{\mathbf{p}}_i = \mathbf{F}_i$, so

$$\hat{L}A = \sum_i \left(\frac{\mathbf{p}_i}{m_i} \cdot \nabla_{\mathbf{r}_i} A + \mathbf{F}_i \cdot \nabla_{\mathbf{p}_i} A \right) = \hat{L}_r A + \hat{L}_p A,$$

where $\nabla_{\mathbf{p}_i}$ is the gradient with respect to the momentum of atom i , $\hat{L}_r$ is the first sum, the part that moves the positions, and $\hat{L}_p$ the second, the part that moves the momenta. Many books write the Liouville operator iL , with a factor i taken from quantum mechanics; nothing here depends on it.

Each part alone is exact

Applied to a position, $\hat{L}_r \mathbf{r}_i = \mathbf{p}_i/m_i$, and applied again, $\hat{L}_r(\mathbf{p}_i/m_i) = 0$, since $\mathbf{p}_i/m_i$ does not depend on any position. The series stops after two terms:

$$e^{\delta t \hat{L}_r} \mathbf{r}_i = \mathbf{r}_i + \delta t \frac{\mathbf{p}_i}{m_i}, \quad e^{\delta t \hat{L}_r} \mathbf{p}_i = \mathbf{p}_i,$$

the second because $\hat{L}_r \mathbf{p}_i = 0$. This is an exact drift. In the same way $\hat{L}_p \mathbf{p}_i = \mathbf{F}_i(\mathbf{r}^N)$, and $\hat{L}_p \mathbf{F}_i = 0$, since the forces depend only on the positions, so

$$e^{\delta t \hat{L}_p} \mathbf{p}_i = \mathbf{p}_i + \delta t \mathbf{F}_i, \quad e^{\delta t \hat{L}_p} \mathbf{r}_i = \mathbf{r}_i :$$

an exact kick. Each part of the motion alone can be followed without error, and the error of a method built from them comes only from combining them. Since $\hat{L}_p A = \{A, U\}$ and $\hat{L}_r A = \{A, K\}$, they are the Liouville operators of U alone and of K alone, so by Equation (7.4) $e^{\delta t \hat{L}_p}$ turns any quantity into its value after a kick, and $e^{\delta t \hat{L}_r}$ into its value after a drift.

Putting the parts together

If $\hat{L}_r$ and $\hat{L}_p$ behaved like numbers, then $e^{\delta t(\hat{L}_r + \hat{L}_p)}$ would equal $e^{\delta t \hat{L}_r} e^{\delta t \hat{L}_p}$. They do not: applying $\hat{L}_r$ and then $\hat{L}_p$ is not the same as the other order, because moving the positions changes the forces. For one coordinate, $\hat{L}_p$ applied to p gives F , and $\hat{L}_r$ applied to that gives $(p/m) F'(q)$, while $\hat{L}_r$ applied to p gives 0, and $\hat{L}_p$ applied to that gives 0. We write $X = \hat{L}_p$ and $Y = \hat{L}_r$ for short; XY means Y applied first and then X , so the order of every product must be kept, and 1 is the operator that leaves a quantity unchanged. Expanding each exponential to second order in δt ,

$$\begin{aligned} e^{\delta t(X+Y)} &= 1 + \delta t(X+Y) + \frac{1}{2}\delta t^2(X+Y)^2 + \mathcal{O}(\delta t^3) \\ &= 1 + \delta t(X+Y) + \frac{1}{2}\delta t^2(X^2 + XY + YX + Y^2) + \mathcal{O}(\delta t^3), \end{aligned}$$

where $(X+Y)^2 = (X+Y)(X+Y)$ is multiplied out keeping XY and YX apart. The symmetric product of half a kick, a drift and half a kick is

$$\begin{aligned} e^{\frac{1}{2}\delta t X} e^{\delta t Y} e^{\frac{1}{2}\delta t X} &= \left(1 + \frac{1}{2}\delta t X + \frac{1}{8}\delta t^2 X^2\right) \left(1 + \delta t Y + \frac{1}{2}\delta t^2 Y^2\right) \\ &\quad \times \left(1 + \frac{1}{2}\delta t X + \frac{1}{8}\delta t^2 X^2\right) + \mathcal{O}(\delta t^3). \end{aligned}$$

Multiplying out, the terms of first order are $\frac{1}{2}\delta t X + \delta t Y + \frac{1}{2}\delta t X = \delta t(X+Y)$, and the terms of second order are $\frac{1}{8}\delta t^2 X^2$ and $\frac{1}{8}\delta t^2 X^2$ from the outer factors alone, $\frac{1}{2}\delta t^2 Y^2$ from the middle one, and the products of first-order terms from two factors, $\frac{1}{2}\delta t X \delta t Y$, $\delta t Y \frac{1}{2}\delta t X$ and $\frac{1}{2}\delta t X \frac{1}{2}\delta t X$, in the order in which the factors stand:

$$\frac{1}{8}X^2 + \frac{1}{8}X^2 + \frac{1}{2}Y^2 + \frac{1}{2}XY + \frac{1}{2}YX + \frac{1}{4}X^2 = \frac{1}{2}(X^2 + XY + YX + Y^2),$$

times δt^2 . The two agree through the second order:

$$e^{\delta t \hat{L}} = e^{\frac{1}{2}\delta t \hat{L}_p} e^{\delta t \hat{L}_r} e^{\frac{1}{2}\delta t \hat{L}_p} + \mathcal{O}(\delta t^3). \quad (9.5)$$

The right side, applied to the state, is a half kick, a drift and a half kick: velocity Verlet, with the local error of order δt^3 found in Section 9.2. Splitting the exponential of a sum into a symmetric product of exponentials in this way is called a Trotter factorisation,

and Tuckerman develops this derivation in full¹⁰. In a product of such exponentials the one written last acts first on the quantity, and the one written first acts first on the state: in $e^{\delta t \hat{L}_p} e^{\delta t \hat{L}_r} A$, the drift turns $A(\mathbf{r}, \mathbf{p})$ into $A(\mathbf{r} + \delta t \mathbf{p}/m, \mathbf{p})$, and the kick then replaces $\mathbf{p}$ in this by $\mathbf{p} + \delta t \mathbf{F}(\mathbf{r})$ everywhere it appears, so the state is kicked first and then drifts with the new momentum (Exercise 9.10). That unsymmetric product agrees with $e^{\delta t \hat{L}}$ only to first order (Exercise 9.9), and is the symplectic Euler method of Section 7.7; the symmetric product reads the same in either order.

The same idea builds other methods. Splitting the forces themselves into fast and slow parts gives methods that kick with the slow forces less often¹⁰ (Chapter 11), and Chapter 13 adds to $\hat{L}_r$ and $\hat{L}_p$ a third part, the action of a thermostat, and splits all three.

KEY IDEA. The motion over a step is $e^{\delta t \hat{L}}$, and the Liouville operator splits into a part that moves the positions and a part that moves the momenta, each of whose exponentials is exact: a drift and a kick. The symmetric product of half a kick, a drift and half a kick agrees with $e^{\delta t \hat{L}}$ to second order in δt , and is velocity Verlet.

NOTEBOOK 09 the exponentials of the Liouville operator in SymPy

CHECK YOUR UNDERSTANDING. For one coordinate with $H = p^2/(2m)$, apply $e^{\delta t \hat{L}}$ to q^2 and check that it gives $(q + \delta t p/m)^2$.

9.5 The energy the method keeps

Section 7.7 found that the symplectic Euler method keeps the spring's energy within a band, and promised the reason. This section finds it for velocity Verlet: for a spring the method keeps exactly an energy slightly different from the true one. With a stable step, this bounds the variation of the true energy. We then state the conditions under which a similar argument applies beyond the spring.

The spring

For a body of mass m on a spring of stiffness k , with $\omega^2 = k/m$, a step of Equation (9.3) in terms of the momentum $p = mv$ is

$$p_{\frac{1}{2}} = p - \frac{1}{2} \delta t k q, \quad q' = q + \delta t p_{\frac{1}{2}}/m, \quad p' = p_{\frac{1}{2}} - \frac{1}{2} \delta t k q'.$$

We show that the step keeps

$$\tilde{H} = \frac{p^2}{2m} + \frac{1}{2} k q^2 \left(1 - \frac{\omega^2 \delta t^2}{4} \right), \quad (9.6)$$

which differs from the energy H by a term of order δt^2 . Written in terms of q and the half-step momentum, using $p = p_{\frac{1}{2}} + \frac{1}{2}\delta t k q$, expanding the square, and writing $\frac{1}{2}kq^2 \omega^2 \delta t^2 / 4 = \delta t^2 k^2 q^2 / (8m)$,

$$\begin{aligned}\tilde{H} &= \frac{p_{\frac{1}{2}}^2}{2m} + \frac{\delta t k}{2m} q p_{\frac{1}{2}} + \frac{\delta t^2 k^2 q^2}{8m} + \frac{1}{2}kq^2 - \frac{\delta t^2 k^2 q^2}{8m} \\ &= \frac{p_{\frac{1}{2}}^2}{2m} + \frac{1}{2}kq^2 + \frac{\delta t k}{2m} q p_{\frac{1}{2}}.\end{aligned}$$

Written instead in terms of q' and the same half-step momentum, using $q = q' - \delta t p_{\frac{1}{2}} / m$,

$$\begin{aligned}\frac{1}{2}kq^2 + \frac{\delta t k}{2m} q p_{\frac{1}{2}} &= \frac{1}{2}kq'^2 - \frac{\delta t k}{m} q' p_{\frac{1}{2}} + \frac{\delta t^2 k p_{\frac{1}{2}}^2}{2m^2} && \text{expanding } \frac{1}{2}kq^2, \\ &+ \frac{\delta t k}{2m} q' p_{\frac{1}{2}} - \frac{\delta t^2 k p_{\frac{1}{2}}^2}{2m^2} && \text{and the product,} \\ &= \frac{1}{2}kq'^2 - \frac{\delta t k}{2m} q' p_{\frac{1}{2}} && \text{the terms in } p_{\frac{1}{2}}^2 \text{ cancel,}\end{aligned}$$

so $\tilde{H} = p_{\frac{1}{2}}^2 / (2m) + \frac{1}{2}kq'^2 - (\delta t k / (2m)) q' p_{\frac{1}{2}}$, the mirror image of the first form. Putting $p_{\frac{1}{2}} = p' + \frac{1}{2}\delta t k q'$ into it and expanding as in the first step gives back [Equation \(9.6\)](#) with q' and p' ([Exercise 9.12](#)). So $\tilde{H}(q', p') = \tilde{H}(q, p)$: every step of velocity Verlet keeps $\tilde{H}$ exactly, whatever the step, and $\tilde{H}$ is called the shadow energy of the method.

[Figure 9.3\(a\)](#) shows the consequence for velocity Verlet. With $\omega \delta t = 0.8$, started at rest, the states after each step lie on the curve $\tilde{H} = \text{constant}$, an ellipse whose height in p is $\sqrt{1 - \omega^2 \delta t^2 / 4} = 0.917$ of that of the true one through the same start, and the computed motion goes round it for ever. The true energy is $H = \tilde{H} + \frac{1}{2}kq^2 \omega^2 \delta t^2 / 4$. Where $q = 0$ it equals $\tilde{H}$; at a turning point $p = 0$, so $\tilde{H} = (1 - \omega^2 \delta t^2 / 4) H$, and H is largest there. The true energy therefore oscillates between $\tilde{H}$ and $\tilde{H} / (1 - \omega^2 \delta t^2 / 4)$, a band of width $\omega^2 \delta t^2 / 4$ relative to its top, and does not drift: with $\omega \delta t = 0.5$ it stays between 0.9375 and 1 times its starting value ([Figure 9.3b](#)), $1 - 0.5^2 / 4 = 0.9375$.

The computed motion also runs slightly fast. Velocity Verlet gives the positions of [Equation \(9.1\)](#), which for the spring, with $a = -\omega^2 q$, read $q_{n+1} + q_{n-1} = (2 - \omega^2 \delta t^2) q_n$, so q_0 and q_1 fix every later q_n . Started at rest at $q_0 = 1$, [Equation \(9.2\)](#) gives $q_1 = 1 - \frac{1}{2}\omega^2 \delta t^2$, so we try $q_n = \cos n\theta$ with $\cos \theta = 1 - \frac{1}{2}\omega^2 \delta t^2$. Adding $\cos(n\theta + \theta) = \cos n\theta \cos \theta - \sin n\theta \sin \theta$ and $\cos(n\theta - \theta) = \cos n\theta \cos \theta + \sin n\theta \sin \theta$ ([Equation \(1.6\)](#)) gives $q_{n+1} + q_{n-1} = 2 \cos \theta q_n$, as required, and $\sin n\theta$ satisfies the same rule, so every start gives the same angular frequency. The computed positions are therefore $\cos \tilde{\omega} t_n$, with $\tilde{\omega} \delta t = \theta$, and the computed angular frequency $\tilde{\omega}$ satisfies $\cos \tilde{\omega} \delta t = 1 - \omega^2 \delta t^2 / 2$: 1.07% above ω at $\omega \delta t = 0.5$, and 0.042% at $\omega \delta t = 0.1$.

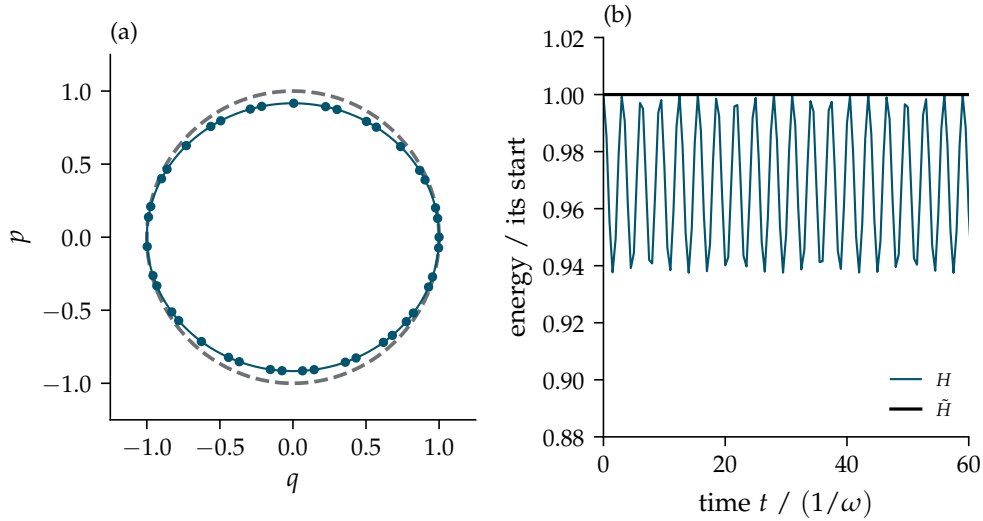


FIGURE 9.3: Velocity Verlet for a spring with $m = k = 1$, started at $q = 1$ at rest. (a) Forty steps with $\omega\delta t = 0.8$ (dots) lie on the ellipse of the shadow energy $\tilde{H}$ (solid), slightly lower than the true curve (dashed). (b) With $\omega\delta t = 0.5$, the energy H of the computed states oscillates in a band, and $\tilde{H}$ stays constant.

The same algebra settles the band that [Section 7.7](#) found for the symplectic Euler method. Its step, $p' = p - \delta t k q$ and then $q' = q + \delta t p' / m$, keeps exactly

$$\frac{p^2}{2m} + \frac{1}{2}kq^2 - \frac{\delta t k}{2m} qp$$

([Exercise 9.13](#)), whose extra term is of first order in δt , where that of $\tilde{H}$ is of second. Along a curve on which it is constant, the true energy ranges between $1/(1 + \omega\delta t/2)$ and $1/(1 - \omega\delta t/2)$ times it, so for a start at rest, where the two are equal, between 0.9091 and 1.1111 times its starting value at $\omega\delta t = 0.2$: the band of [Figure 7.7](#). At the same step velocity Verlet keeps a band of $0.2^2/4 = 1\%$, the narrower band that Chapter 7 promised.

Beyond the spring

For the spring the shadow energy is found exactly. For a Hamiltonian whose Taylor series converge, a motion that stays within a bounded region, and a sufficiently small step, a symplectic method keeps, very nearly and for times that grow exponentially with $1/\delta t$, a shadow energy that differs from H by terms of the order of its global error, δt^2 for velocity Verlet, and so the true energy stays within a band of that width without drifting¹¹. This is the reason, promised in Chapter 7, why a symplectic step map keeps the energy close. Over long times no method follows the true motion closely; a symplectic method follows, exactly for the spring and very nearly in general, the motion of a slightly different Hamiltonian.

KEY IDEA. Velocity Verlet keeps the shadow energy $\tilde{H} = p^2/(2m) + \frac{1}{2}kq^2(1 - \omega^2\delta t^2/4)$ of a spring exactly, so the true energy oscillates in a band of relative width $\omega^2\delta t^2/4$ without drift, and the motion runs slightly fast. For smooth Hamiltonians and a sufficiently small stable step, a nearby shadow energy is kept very nearly over long times, which explains the small energy fluctuations.

NOTEBOOK 09 *the true energy and the shadow energy*

CHECK YOUR UNDERSTANDING. At $\omega\delta t = 0.2$, what is the relative width of the energy band, and how fast does the computed motion run?

9.6 Stability and the size of the step

Chapters 1, 2 and 4 promised to say how long a step molecular dynamics can take. Two limits answer it: a step beyond which the computed motion grows without bound, and, well inside it, the step at which the energy fluctuates by an acceptable amount. This section derives the first, measures the second, and turns both into a procedure.

The stability limit

The shadow energy of Equation (9.6) contains the factor $1 - \omega^2\delta t^2/4$. For $\omega\delta t < 2$ it is positive, $\tilde{H}$ is a sum of two positive squares, and its curves are ellipses: the computed motion stays on one and is bounded. For $\omega\delta t > 2$ the factor is negative, and $\tilde{H} = p^2/(2m) - \frac{1}{2}kq^2(\omega^2\delta t^2/4 - 1)$ is a difference of two squares, which stays fixed while p and q both grow without limit. Its curves are hyperbolas, open curves that run off to infinity, and the computed motion follows one outwards, its size multiplied, after the first few steps, by a fixed factor at every step (Exercise 9.15). Velocity Verlet is therefore stable for the spring only when

$$\omega\delta t < 2, \tag{9.7}$$

that is, with more than $2\pi/2 = \pi$ steps per period. Figure 9.4 measures it: in 200 steps $|q|$ stays at 1 up to $\omega\delta t = 1.99$, reaches 1.2×10^{17} at 2.01 and 3×10^{38} at 2.05.

Atoms near a minimum of U vibrate in normal modes (Section 4.9). We write the displacements from the minimum as $\mathbf{q} = \sum_k c_k \mathbf{u}_k$, with $\mathbf{u}_k$ the patterns of Section 4.9, which satisfy $\Phi \mathbf{u}_k = \omega_k^2 \mathbf{M} \mathbf{u}_k$. When the forces are linear, pattern $\mathbf{u}_k$ feels the force $-\Phi \mathbf{u}_k = -\omega_k^2 \mathbf{M} \mathbf{u}_k$, so its acceleration is $-\omega_k^2 \mathbf{u}_k$: a half kick changes the rate of change of each c_k by $-\frac{1}{2}\delta t \omega_k^2 c_k$, a drift changes each c_k by δt times its own rate, and velocity Verlet steps every c_k as it steps a spring of angular frequency ω_k . Each mode must therefore satisfy Equation (9.7): the fastest vibration in the system sets the limit, however

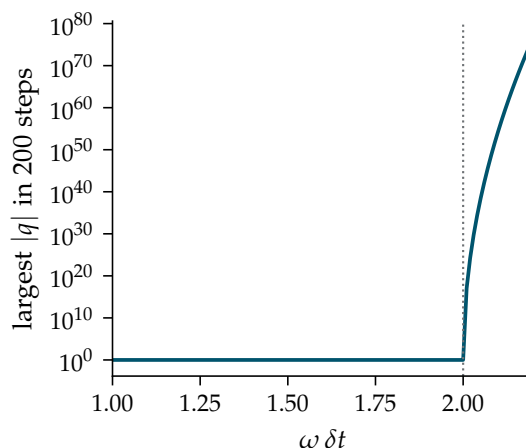


FIGURE 9.4: The largest $|q|$ reached in 200 steps of velocity Verlet for a spring started at $q = 1$ at rest, against $\omega\delta t$, on a logarithmic axis; the limit $\omega\delta t = 2$ is dotted.

slow everything else is. For the faster O–H stretch of water, the antisymmetric one, with a period of 8.88 fs (Section 4.10), $\omega = 2\pi/8.88 = 0.708 \text{ rad fs}^{-1}$, and the limit is $\delta t < 2/\omega = 2.83 \text{ fs}$.

Accuracy

Inside the limit, the energy band of Section 9.5 sets how long a step is acceptable: for a vibration of angular frequency ω it fluctuates by a fraction $\omega^2\delta t^2/4$ of the vibration's energy. For the O–H stretch, a step of 0.5 fs gives $\omega\delta t = 0.354$ and a band of 3.1%; 1 fs gives 12.5%. These are fractions of the energy of the fastest vibration only, but they show why a step must be short compared with $1/\omega = 1.41 \text{ fs}$, as Section 4.10 anticipated.

Figure 9.5 measures this on the spring model of water of Section 5.7, whose fastest vibration, the symmetric stretch, which the model puts 24% above the measured value, has $\omega^2 = 0.7338 \text{ fs}^{-2}$, so $\omega = 0.857 \text{ rad fs}^{-1}$ and $2/\omega = 2.33 \text{ fs}$. Given 0.1 eV of kinetic energy and followed for 2000 fs, its total energy spreads by $1.3 \times 10^{-3} \text{ eV}$, 1.3% of the energy put in, with $\delta t = 0.5 \text{ fs}$, and by 9.6% with 1 fs. The spread grows as δt^2 for the shorter steps (fitted slope 2.02 from 0.05 to 0.5 fs) and faster beyond, by a factor of 7.5 from 0.5 to 1 fs, where $\omega\delta t = 0.86$ is no longer small. The run blows up at $\delta t = 2.3 \text{ fs}$, just inside the limit for a perfect spring. There $1 - \omega^2\delta t^2/4 = 0.030$, so by Equation (9.6) the computed swing of the fastest vibration is $1/\sqrt{0.030} = 5.8$ times the true one, and over so wide a swing the springs between atoms are no longer linear in the positions; with 0.01 eV the same step stays bounded.

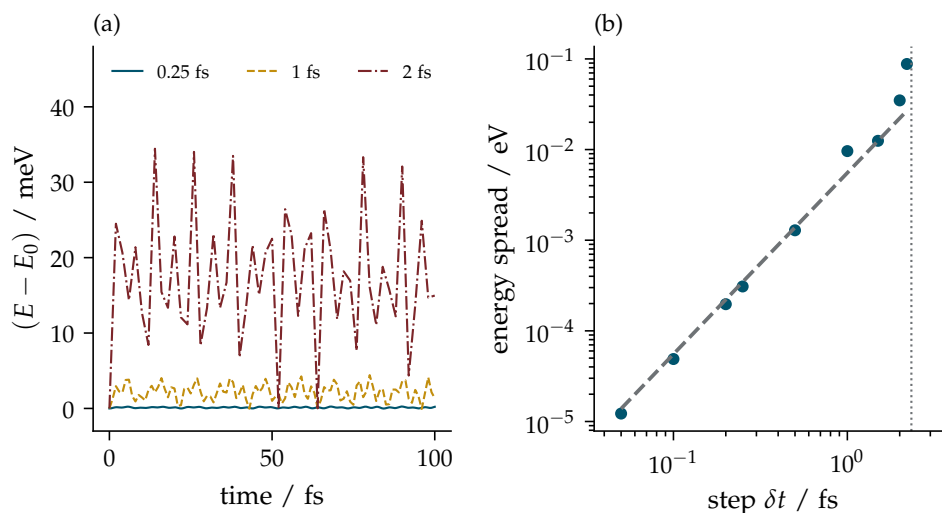


FIGURE 9.5: The spring model of water (Section 5.7) followed by velocity Verlet from its equilibrium geometry with 0.1 eV of kinetic energy. (a) The total energy relative to its initial value $E_0 = 0.1$ eV, over the first 100 fs for three steps. (b) The spread of the total energy over 2000 fs against δt , with a line of slope 2 (dashed) and the stability limit $2/\omega$ of its fastest vibration (dotted); the run at 2.3 fs blew up and is not shown.

IN PRACTICE. The step is chosen by measurement. Without a thermostat, the device of Chapter 13 that holds the temperature, the total energy should stay constant, and short runs at two or three steps, such as 0.25, 0.5 and 1 fs, each covering many periods of the vibrations, record how well it does. Its fluctuation should fall close to δt^2 between the shorter steps, as expected for smooth forces consistent with the reported energy; it should show no steady drift; and it should be small compared with the fluctuation of the kinetic energy, the amount that passes back and forth between kinetic and potential energy. For the water of Figure 9.5 the ratio of their standard deviations is 0.9% at 0.5 fs and 6.7% at 1 fs; how small is acceptable depends on what is to be measured, and is a judgement. The TrajCast paper gives 0.5 to 1.0 fs as the typical step of molecular dynamics with forces from electronic structure¹, and the band of the O–H stretch above, 3.1% at 0.5 fs and 12.5% at 1 fs, shows why bonds to hydrogen hold it there. Without hydrogen the fastest vibrations are slower, and longer steps pass the same test (for the lithium model alone, Exercise 9.17; in graphite the vibrations of the carbon atoms must pass it too). At half the limit, $\omega\delta t = 1$, the band of the fastest vibration is already 25%. Chapter 11 removes the fastest vibrations by holding bonds at fixed length.

The stability limit has a cousin in Section 1.9: saving the positions of an oscillation needs enough frames per period to recover its velocity, and computing the motion needs more than π steps per period to remain bounded at all.

KEY IDEA. Velocity Verlet is stable only for $\omega\delta t < 2$ for every vibration, so the fastest one sets the limit, 2.83 fs for the O–H stretch. Accuracy demands a much shorter step: the energy of a vibration fluctuates by $\omega^2\delta t^2/4$, and the step is chosen by measuring the energy fluctuation of short runs.

NOTEBOOK 09 *push the step to the stability limit*

CHECK YOUR UNDERSTANDING. The model lithium atom rattling in its hollow (Section 4.4) has a period of 170.3 fs. What is the stability limit for the step?

9.7 Long runs and rough forces

A method of higher order makes a smaller error over a fixed time. A simulation, however, runs for very many periods of its fastest motion, and over such times what matters is whether the energy, and with it the statistics of Chapter 12, stays right. This section compares velocity Verlet with a method of fourth order over a long run, and measures what a force that jumps costs, as Section 8.5 promised.

A fourth-order method

The classical Runge–Kutta method of fourth order, RK4, advances the pair $\mathbf{w} = (x, v)$, whose rate of change is $\mathbf{g}(\mathbf{w}) = (v, a(x))$, by four samples of that rate, each taken at a point reached with the one before:

$$\begin{aligned} \mathbf{s}_1 &= \mathbf{g}(\mathbf{w}_n), & \mathbf{s}_2 &= \mathbf{g}(\mathbf{w}_n + \tfrac{1}{2}\delta t \mathbf{s}_1), \\ \mathbf{s}_3 &= \mathbf{g}(\mathbf{w}_n + \tfrac{1}{2}\delta t \mathbf{s}_2), & \mathbf{s}_4 &= \mathbf{g}(\mathbf{w}_n + \delta t \mathbf{s}_3), \end{aligned}$$

$$\mathbf{w}_{n+1} = \mathbf{w}_n + \tfrac{1}{6}\delta t (\mathbf{s}_1 + 2\mathbf{s}_2 + 2\mathbf{s}_3 + \mathbf{s}_4).$$

Its derivation, matching this to the Taylor series up to δt^4 , is long, and we take it on trust. In Figure 9.1 its error after a time of 4 is 1.6×10^{-9} at $\delta t = 0.05$, against 4.1×10^{-5} for velocity Verlet, and falls roughly as δt^4 : the error changes sign between $\delta t = 0.1$ and 0.05, so the points scatter about the line of slope 4, with a fitted slope of 4.3. It calls for the forces four times per step, and it is neither symplectic nor reversible: for the spring a step multiplies areas by $c^2 + s^2 < 1$ (Exercise 9.19), and reversing the velocity and stepping again returns the start only to within that factor.

Figure 9.6(a) follows the 32-atom Lennard-Jones cluster of Section 8.5 for 200τ , where $\tau = \sigma\sqrt{m/\varepsilon}$ is the unit of time of Equation (8.2), with velocity Verlet at $\delta t = 0.005\tau$ and RK4 at $\delta t = 0.02\tau$, so that each calls for the forces 40 000 times. The energy of velocity

Verlet stays within a band of $3.4 \times 10^{-3} \varepsilon$; that of RK4 falls steadily, by 0.64ε in all, as if a drag acted on the atoms. For a spring, [Exercise 9.19](#) shows that every step of RK4 multiplies the energy by a number slightly less than 1. Over a long run the symplectic method keeps the energy better: it keeps a shadow energy ([Section 9.5](#)), while RK4 shrinks areas and keeps no such energy.

Forces that jump

[Figure 9.6\(b\)](#) repeats the velocity Verlet run with the potential cut off at 2.5σ , once shifted, so that the force jumps when a pair crosses the cutoff, and once switched smoothly between 2σ and 2.5σ . With switching the energy stays in a band of $3.4 \times 10^{-3} \varepsilon$, as without a cutoff. With shifting the centre of the band wanders: the mean energy over the second half of the run is $1.4 \times 10^{-3} \varepsilon$ below that over the first, against $4 \times 10^{-6} \varepsilon$ with switching, and the range over the run grows to $6.3 \times 10^{-3} \varepsilon$. Each jump of the force inside a step breaks the Taylor series on which the method's accuracy rests, and leaves an error that the next steps do not undo. A force that is not smooth lets the energy of a fixed-step method wander, which is why the smooth cutoffs of [Section 8.5](#), and the smooth envelopes of machine-learned potentials, matter.

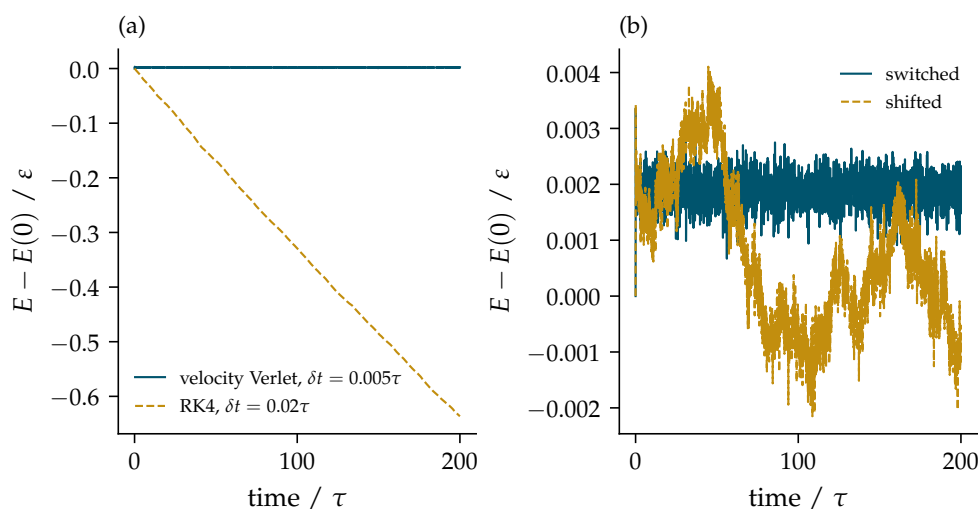


FIGURE 9.6: The 32-atom Lennard-Jones cluster of [Section 8.5](#) over 200τ , as the change of its total energy. (a) Velocity Verlet with $\delta t = 0.005\tau$ and RK4 with $\delta t = 0.02\tau$, which call for the forces equally often. (b) Velocity Verlet with $\delta t = 0.005\tau$ and the potential cut at 2.5σ by switching and by shifting.

KEY IDEA. Over a long run a symplectic method of second order keeps the energy in a band, while RK4, more accurate over a short time and four times as costly per step, lets it drift. Forces that jump inside a step let the energy wander out of its band, so the forces of a simulation must be smooth.

CHECK YOUR UNDERSTANDING. To cover 1 ps, how many force evaluations does velocity Verlet need with $\delta t = 0.5$ fs, and how many does RK4 need with $\delta t = 2$ fs?

What we now have

A method of computing motion is a step map, and its order says how fast its error over a fixed time shrinks with the step: the forward Euler method, whose error over a fixed time shrinks as δt and over one step as δt^2 , is of order 1. Adding the Taylor series a step ahead and a step behind gives the Verlet method, which in its velocity form is a half kick, a drift and a half kick, costs one force evaluation per step and is of order 2; its local error, of order δt^3 , is the ‘cubic scaling’ quoted by TrajCast.

Reversing the velocities after a step and stepping again returns the start, with its velocities reversed, without error, and each step keeps volume in phase space and satisfies the full symplectic condition. The same method follows from splitting the Liouville operator into the parts that move positions and momenta, whose exponentials are a drift and a kick with no error; the symmetric product agrees with the true motion to second order.

For a spring, velocity Verlet keeps the shadow energy $p^2/(2m) + \frac{1}{2}kq^2(1 - \omega^2\delta t^2/4)$ exactly, so the true energy oscillates in a band of relative width $\omega^2\delta t^2/4$; the symplectic Euler method keeps one that differs from the energy at first order in δt , hence its wider band, and for other Hamiltonians a nearby shadow energy is kept very nearly for sufficiently small steps and smooth forces.

The method is stable only for $\omega\delta t < 2$ for the fastest vibration, 2.83 fs for the O–H stretch, and accurate enough well inside that; the step is chosen by measuring the energy fluctuation. Over long runs it keeps the energy better than RK4 at the same cost, provided the forces are smooth.

Chapter 10 uses velocity Verlet to build a complete simulation: atoms in a periodic box, neighbour lists for the cutoffs, and the Ewald sum for charges.

Answers to the checks

SECTION 9.1. Dividing by $4 = 2^2$ means the global error is of order δt^2 , so the method is of order 2, and its local error is of order δt^3 .

SECTION 9.2. With $a_n = a_{n+1} = -g$, Equation (9.2) gives $x_{n+1} = x_n + \delta t v_n - \frac{1}{2}g\delta t^2$ and $v_{n+1} = v_n - g\delta t$, the exact motion of Section 1.4 over a time δt , so no error is made at any step.

SECTION 9.3. $\partial x'/\partial x = 1$, $\partial x'/\partial v = 0$, $\partial v'/\partial x = \frac{1}{2}\delta t \, da/dx$ and $\partial v'/\partial v = 1$, so the determinant is $1 \times 1 - 0 \times \frac{1}{2}\delta t \, da/dx = 1$.

SECTION 9.4. With $\hat{L} = (p/m)\partial/\partial q$, $\hat{L}q^2 = 2qp/m$, $\hat{L}^2q^2 = 2p^2/m^2$ and $\hat{L}^3q^2 = 0$, so $e^{\delta t \hat{L}}q^2 = q^2 + 2qp\delta t/m + p^2\delta t^2/m^2 = (q + \delta t p/m)^2$.

SECTION 9.5. $\omega^2\delta t^2/4 = 0.04/4 = 0.01$, a band of 1%; $\cos \tilde{\omega}\delta t = 1 - 0.02 = 0.98$ gives $\tilde{\omega}\delta t = 0.2003$, so the motion runs 0.17% fast.

SECTION 9.6. $\omega = 2\pi/170.3 = 0.0369 \text{ rad fs}^{-1}$, so $\delta t < 2/\omega = 54.2 \text{ fs}$.

SECTION 9.7. Velocity Verlet: $1000/0.5 = 2000$ steps of one force evaluation each, 2000. RK4: $1000/2 = 500$ steps of four, also 2000.

Exercises

The exercises run from questions of understanding, through calculations, to short derivations marked ‘show that’. Full solutions are in ‘[Solutions to the exercises](#)’ at the end of the book, and Notebook 09 checks every numerical answer.

EXERCISE 9.1 (*Euler and a falling ball*). Show that forward Euler steps of δt for a body falling from x_0 with velocity v_0 under the acceleration $-g$ give, after n steps, $x_n = x_0 + v_0 n\delta t - \frac{1}{2}g\delta t^2 n(n-1)$, and that at the time $t_f = n\delta t$ the error is $\frac{1}{2}gt_f\delta t$, of order δt .

EXERCISE 9.2 (*reading an order*). Halving the step divides a method’s error at a given time by 16. What is its order, and its local error?

EXERCISE 9.3 (*the error of Störmer–Verlet*). Show that the term of order δt^4 in $x(t + \delta t) + x(t - \delta t)$ is $\frac{1}{12}\ddot{a}\delta t^4$, so that the local error of [Equation \(9.1\)](#) is $\frac{1}{12}\ddot{a}\delta t^4$.

EXERCISE 9.4 (*velocities from positions*). With $x_{n\pm 1}$ the true positions $x(t \pm \delta t)$, show that $(x_{n+1} - x_{n-1})/(2\delta t)$ differs from the true velocity $v(t)$ by $\frac{1}{6}\dot{a}\delta t^2$ plus terms of higher order.

EXERCISE 9.5 (*the local error of velocity Verlet*). Show that the position after one step of [Equation \(9.2\)](#) from a correct start differs from the true one by $\frac{1}{6}\dot{a}\delta t^3$ plus terms of higher order.

EXERCISE 9.6 (*leapfrog*). Show that the positions of the leapfrog method, $v_{n+\frac{1}{2}} = v_{n-\frac{1}{2}} + \delta t a_n$ and $x_{n+1} = x_n + \delta t v_{n+\frac{1}{2}}$, obey [Equation \(9.1\)](#).

EXERCISE 9.7 (*Euler does not run backwards*). For a spring with $m = k = 1$, take one forward Euler step from $(q, v) = (1, 0)$, reverse the velocity and take another. Where does it land?

EXERCISE 9.8 (*one coordinate*). Show that for a 2×2 matrix $\mathbf{J}$, $\mathbf{J}^T \boldsymbol{\Omega} \mathbf{J} = (\det \mathbf{J}) \boldsymbol{\Omega}$, with $\boldsymbol{\Omega} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, so that for one coordinate the symplectic condition is $\det \mathbf{J} = 1$.

EXERCISE 9.9 (*the unsymmetric product*). Show that $e^{\delta t X} e^{\delta t Y}$ and $e^{\delta t (X+Y)}$ differ at second order by $\frac{1}{2} \delta t^2 (XY - YX)$.

EXERCISE 9.10 (*which acts first*). For one coordinate, show that $e^{\delta t \hat{L}_p} e^{\delta t \hat{L}_r}$ applied to q gives $q + \delta t (p + \delta t F(q)) / m$: a kick, then a drift.

EXERCISE 9.11 (*a drift of a function*). With $\hat{L}_r = (p/m) \partial / \partial q$, show from the series that $e^{\delta t \hat{L}_r} q^3 = (q + \delta t p/m)^3$.

EXERCISE 9.12 (*the shadow energy*). Complete the derivation of Section 9.5: put $p_{\frac{1}{2}} = p' + \frac{1}{2} \delta t k q'$ into $p_{\frac{1}{2}}^2 / (2m) + \frac{1}{2} k q'^2 - (\delta t k / (2m)) q' p_{\frac{1}{2}}$ and show that the result is $\tilde{H}(q', p')$.

EXERCISE 9.13 (*the band of symplectic Euler*). Show that the symplectic Euler step for the spring, $p' = p - \delta t k q$ and then $q' = q + \delta t p' / m$, keeps $p^2 / (2m) + \frac{1}{2} k q^2 - (\delta t k / (2m)) q p$. Writing $q = A \cos \phi$ and $p / (m\omega) = A \sin \phi$, as for the circle of Section 4.1, show that along a curve on which this is constant the energy H ranges between $1 / (1 + \omega \delta t / 2)$ and $1 / (1 - \omega \delta t / 2)$ times it.

EXERCISE 9.14 (*the step's matrix*). Show that one step of velocity Verlet for the spring is $q' = (1 - \frac{1}{2} \omega^2 \delta t^2) q + (\delta t / m) p$ and $p' = -k \delta t (1 - \frac{1}{4} \omega^2 \delta t^2) q + (1 - \frac{1}{2} \omega^2 \delta t^2) p$, and that its matrix, the Jacobian matrix of the step, has determinant 1 and trace, the sum of its diagonal entries, $2 - \omega^2 \delta t^2$.

EXERCISE 9.15 (*growth beyond the limit*). Beyond the limit the computed motion grows by a factor $|\lambda|$ per step, with λ the eigenvalue of larger size of the step's matrix, a root of $\lambda^2 - (2 - \omega^2 \delta t^2) \lambda + 1 = 0$, the characteristic equation with the trace and determinant of Exercise 9.14 (the toolbox of Section 9.3). Find the factor at $\omega \delta t = 2.05$.

EXERCISE 9.16 (*a step for water*). What is the longest step for which the energy of the antisymmetric O–H stretch, of period 8.88 fs, fluctuates by no more than 3%?

EXERCISE 9.17 (*a step for lithium*). For the model lithium atom in its hollow, of period 170.3 fs, find the step that keeps its energy band to 1%, and compare it with the stability limit.

EXERCISE 9.18 (*steps per period*). What is $2\pi/(\omega\delta t)$, the number of steps per period, for which velocity Verlet keeps an energy band of 1%?

EXERCISE 9.19 (*RK4 and a spring*). For the spring, one RK4 step in the variables q and $u = v/\omega$ is $q' = cq + su$ and $u' = -sq + cu$, with $c = 1 - \frac{1}{2}x^2 + \frac{1}{24}x^4$, $s = x - \frac{1}{6}x^3$ and $x = \omega\delta t$. Show that it multiplies $q^2 + u^2$, and so the energy, by $c^2 + s^2 = 1 - \frac{1}{72}x^6 + \frac{1}{576}x^8$, and find the fraction of the energy lost in 10 000 steps at $x = 0.2$.

EXERCISE 9.20 (*a learned long step*). A learned propagator predicts the state of water 5 fs ahead, ten steps of 0.5 fs. What is $\omega\delta t$ for the O–H stretch at a step of 5 fs, and could velocity Verlet take it?

EXERCISE 9.21 (*a shifted line*). A simulation of water with $\delta t = 0.7$ fs runs the O–H stretch slightly fast. At what wavenumber does the antisymmetric stretch, 3756 cm^{-1} (period 8.88 fs), appear in it?

Building a molecular dynamics code

The earlier chapters followed a handful of atoms in empty space. To represent a liquid or a crystal, we need a sample packed at the material's density whose atoms have neighbours on every side. Repeating a simulation cell supplies those neighbours, but also changes how we find separations and sum the forces. We develop these operations first, including the Ewald sum for charges, then use neighbour lists to avoid testing every pair at every step.

With the forces defined, we can prepare the positions and velocities and advance them with the velocity Verlet loop of Chapter 9. Comparisons with an independent library check the implementation, and timings identify where compiled code can reduce its cost. Chapter 11 starts from this loop and asks how far the step can be lengthened by constraining stiff bonds and evaluating slowly changing forces less often.

10.1 A box repeated in every direction

A sample small enough to simulate is mostly surface. This section counts how much, removes the surface by repeating a box of atoms in every direction, and sets up the coordinates that describe such a box.

Most of a small sample is surface

We place $N = n^3$ atoms on a cubic grid, n along each edge. The atoms that are not on a face form a smaller cube of $(n - 2)^3$, so the fraction on the surface is

$$1 - \frac{(n - 2)^3}{n^3} = 1 - \left(1 - \frac{2}{n}\right)^3$$

$$= \frac{6}{n} - \frac{12}{n^2} + \frac{8}{n^3} \quad \text{expanding } (1 - x)^3 = 1 - 3x + 3x^2 - x^3 \text{ with } x = 2/n,$$

close to $6/n$ when n is large. A thousand atoms ($n = 10$) have 49% of them on the surface, about ten thousand ($n = 22$) 25%, and a million ($n = 100$) still 5.9%. A surface atom has fewer neighbours than one inside, and a sample this small would measure the properties of its surface as much as those of its bulk.

The periodic box

The surface is removed by repeating the box. The atoms sit in a cell, and the cell and its contents are copied without end in every direction, so that space is filled with identical cells. An atom that leaves through one face of the cell comes in through the opposite face, as its copy from the neighbouring cell enters, and every atom has neighbours on all sides; a ship in an old arcade game that leaves the screen at the right edge and reappears at the left lives in such a box, in two dimensions. Each atom feels the others and their copies. The price is an imposed periodicity: the sample repeats with the period of the cell, no wave longer than the cell, such as a long sound wave, can occur, and for a short-ranged potential the cell must be large enough that an atom does not feel its own copies (Section 10.2); the Coulomb energy of Section 10.3 includes them by design.

The cell is a parallelepiped, a box whose six faces are parallelograms, spanned by three lattice vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$, which we place as the columns of the cell matrix

$$\mathbf{h} = \begin{pmatrix} \mathbf{a} & \mathbf{b} & \mathbf{c} \end{pmatrix}. \quad (10.1)$$

For a cubic cell of side L , $\mathbf{a} = (L, 0, 0)$, $\mathbf{b} = (0, L, 0)$, $\mathbf{c} = (0, 0, L)$ and $\mathbf{h}$ is L times the identity. Any position can be written as $\mathbf{r} = s_1\mathbf{a} + s_2\mathbf{b} + s_3\mathbf{c}$, which is $\mathbf{h}$ times the vector $\mathbf{s} = (s_1, s_2, s_3)$: row x of $\mathbf{h}$ is (a_x, b_x, c_x) , so the x component of $\mathbf{h}\mathbf{s}$ is $a_x s_1 + b_x s_2 + c_x s_3$, the x component of $s_1\mathbf{a} + s_2\mathbf{b} + s_3\mathbf{c}$, and likewise for y and z . So

$$\mathbf{r} = \mathbf{h}\mathbf{s}, \quad \mathbf{s} = \mathbf{h}^{-1}\mathbf{r}, \quad (10.2)$$

with the inverse matrix of [Section 5.6](#). The inverse exists: a matrix without one turns some $\mathbf{s} \neq \mathbf{0}$ into $\mathbf{0}$, and $s_1\mathbf{a} + s_2\mathbf{b} + s_3\mathbf{c} = \mathbf{0}$ would put the three lattice vectors in one plane, a flat cell with no volume. The numbers s_1 , s_2 and s_3 are the fractional coordinates of the position: the point lies inside the cell when each is between 0 and 1, and its copies are at $\mathbf{h}(\mathbf{s} + \mathbf{n})$, with $\mathbf{n}$ any vector of three whole numbers, positive, negative or zero.

The volume of the cell is its base times its height. The face spanned by $\mathbf{b}$ and $\mathbf{c}$ is a parallelogram of area $|\mathbf{b} \times \mathbf{c}|$ ([Section 5.2](#)), and $\mathbf{b} \times \mathbf{c}$ is at right angles to that face. The height of the cell above it is the size of the part of $\mathbf{a}$ along that direction, the scalar product of $\mathbf{a}$ with the vector of length one $(\mathbf{b} \times \mathbf{c})/|\mathbf{b} \times \mathbf{c}|$ ([Section 1.6](#)), taken without its sign, since $\mathbf{a}$ may lie on either side of the face. Then

$$\begin{aligned} V &= |\mathbf{b} \times \mathbf{c}| \frac{|\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})|}{|\mathbf{b} \times \mathbf{c}|} && \text{base times height} \\ &= |\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})|, && \text{cancelling } |\mathbf{b} \times \mathbf{c}| \end{aligned} \quad (10.3)$$

the scalar triple product of [Section 5.5](#). For the cubic cell, $V = L^3$.

Wrapping and unwrapping

To put a position back into the cell, we take its fractional coordinates and subtract from each its floor,

$$\mathbf{s} \rightarrow \mathbf{s} - \lfloor \mathbf{s} \rfloor,$$

where the floor $\lfloor x \rfloor$ of a number x is the largest whole number not above it, and $\lfloor \mathbf{s} \rfloor$ takes the floor of each coordinate: $\lfloor 2.3 \rfloor = 2$ and $\lfloor -0.4 \rfloor = -1$, so 2.3 becomes 0.3 and -0.4 becomes 0.6. This is wrapping. The function `wrap` of `mdlab.cell` does it for N atoms:

```

1 def wrap(positions: ArrayLike, cell: ArrayLike) -> NDArray:
2     s = to_fractional(positions, cell)
3     return to_cartesian(s - np.floor(s), cell)

```

where `to_fractional` solves $\mathbf{h}\mathbf{s} = \mathbf{r}$ for each atom and `to_cartesian` multiplies back.

The integration itself does not need wrapped positions: the forces depend only on separations, and [Section 10.2](#) reduces each separation to its nearest copy whichever copy of each atom the code holds. `mdlab` therefore follows each atom's path without wrapping, so that its displacement over a run can be read off directly, and wraps the positions only when it writes them to a file, as files of periodic systems usually hold them. To recover a continuous path from wrapped frames, each frame's position is

replaced by its copy nearest to the previous frame's, which is right as long as no atom moves more than half the narrowest width of the cell between frames (Section 10.2): a longer move is mistaken for a shorter one the other way, as too few frames per period hid an oscillation in Section 1.9.

Momentum in a periodic box

Moving every atom by the same displacement moves each of its copies with it and changes no separation, so the forces still add to zero and the total momentum is conserved, as Section 7.5 found, whether a pair interacts through its nearest copy (Section 10.2) or through all of them (Section 10.3). Turning the atoms is not a symmetry of the box, and the angular momentum is not conserved. In the run of 500 atoms of Section 10.6 the largest component of the total momentum after 2000 steps is 3.8×10^{-15} amu Å/fs, the rounding of the arithmetic.

KEY IDEA. A cell repeated in every direction removes the surface that would dominate any sample small enough to simulate. The cell is described by the matrix $\mathbf{h}$ of its lattice vectors; positions have fractional coordinates $\mathbf{s} = \mathbf{h}^{-1}\mathbf{r}$, the cell has volume $|\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})|$, and wrapping subtracts from each fractional coordinate its floor.

NOTEBOOK 10 *wrap atoms into a cubic and a hexagonal cell*

CHECK YOUR UNDERSTANDING. In the cell with $\mathbf{a} = (4, 0, 0)$, $\mathbf{b} = (0, 5, 0)$ and $\mathbf{c} = (0, 0, 6)$ Å, what are the fractional coordinates of $\mathbf{r} = (9, -1, 3)$ Å, and where does wrapping put it?

10.2 The nearest image

Two atoms in a periodic box are separated by $\mathbf{r}_{ij} = \mathbf{r}_j - \mathbf{r}_i$, written $\mathbf{d}$ in this section to keep subscripts short, and by every copy of it, $\mathbf{d} + \mathbf{h}\mathbf{n}$. A copy is also called an image, and the nearest copy the minimum image. A pair potential cut off at r_c needs the copies within r_c , and when the cell is large enough there is at most one, the nearest. This section finds it, first in a rectangular cell and then in a skewed one, and derives how large the cell must be.

Rounding the fractional coordinates

The fractional coordinates of the separation, $\mathbf{s} = \mathbf{h}^{-1}\mathbf{d}$, and those of a copy, $\mathbf{h}^{-1}(\mathbf{d} + \mathbf{hn}) = \mathbf{s} + \mathbf{n}$, differ by whole numbers. Rounding each to the nearest whole number and subtracting,

$$\mathbf{s} \rightarrow \mathbf{s} - \text{round}(\mathbf{s}), \quad (10.4)$$

gives the copy whose fractional coordinates all lie between $-\frac{1}{2}$ and $\frac{1}{2}$. The function `minimum_image` of `mdlab.cell` does this for many separations at once:

```

1 def minimum_image(separations: ArrayLike, cell: ArrayLike) -> NDArray:
2     s = to_fractional(separations, cell)
3     return to_cartesian(s - np.round(s), cell)

```

In a rectangular cell, with $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ along the axes and of lengths a , b and c , a copy is $\mathbf{d} + (n_1\mathbf{a}, n_2\mathbf{b}, n_3\mathbf{c})$ and its squared length is

$$(d_x + n_1a)^2 + (d_y + n_2b)^2 + (d_z + n_3c)^2.$$

Each term depends on one whole number only, so each is made as small as it can be on its own, by the n_1 that brings $d_x + n_1a$ between $-\frac{1}{2}a$ and $\frac{1}{2}a$, and likewise for the others. That is [Equation \(10.4\)](#): in a rectangular cell, rounding gives the nearest copy of every separation.

A skewed cell

Many crystals have cells that are not rectangular. Graphite's is hexagonal: in each layer $\mathbf{a}$ and $\mathbf{b}$ have the same length, $a = 2.464 \text{ \AA}$, at 120° to each other, and $\mathbf{c}$, of length $6.711 \text{ \AA}$, is at right angles to the layers³³. Rounding in such a cell returns the copy inside the parallelogram of [Figure 10.1\(a\)](#), centred on the first atom, while the copies that are nearest fill the hexagon around it: a separation is its own nearest copy when it ends nearer to the first atom than to any copy of that atom, and in the plane of a layer the six closest copies, at $\pm\mathbf{a}$, $\pm\mathbf{b}$ and $\pm(\mathbf{a} + \mathbf{b})$, cut off a hexagon whose sides halve the lines to them. The two regions overlap but differ: in about 17% of the parallelogram, shaded, a shorter copy exists, and rounding misses it.

Rounding cannot fail for a short enough separation, whatever the cell. Since $\mathbf{b}$ and $\mathbf{c}$ are at right angles to $\mathbf{b} \times \mathbf{c}$, taking the scalar product of $\mathbf{d} = s_1\mathbf{a} + s_2\mathbf{b} + s_3\mathbf{c}$ with $\mathbf{b} \times \mathbf{c}$ leaves only the first term,

$$\mathbf{d} \cdot (\mathbf{b} \times \mathbf{c}) = s_1 \mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}).$$

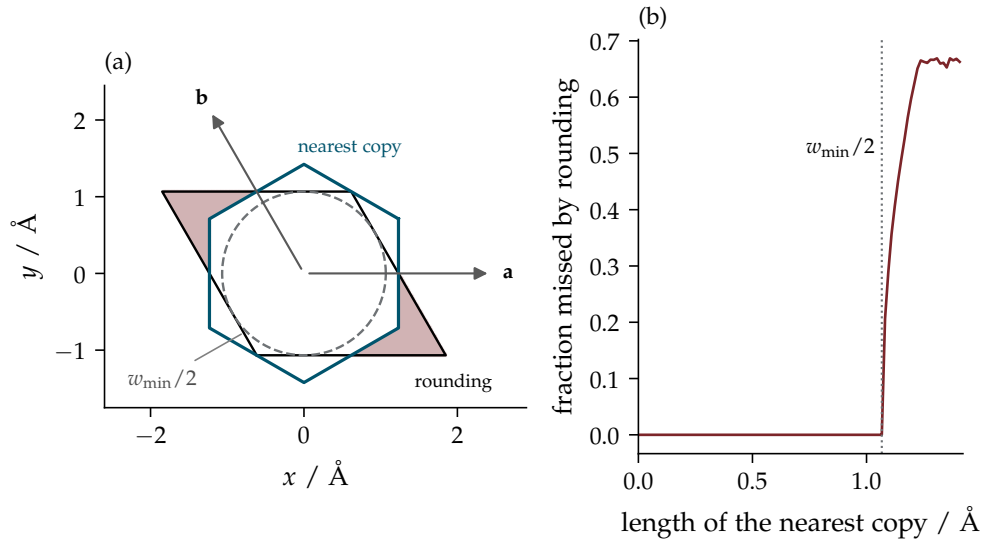


FIGURE 10.1: Rounding and the nearest copy in the hexagonal cell of graphite, in the plane of a layer. (a) Rounding returns separations in the parallelogram centred on the first atom, spanned by $\mathbf{a}$ and $\mathbf{b}$; the separations that are their own nearest copy fill the hexagon; in the shaded corners a shorter copy exists. The circle of radius $\frac{1}{2}w_{\min}$, half the smallest perpendicular width, lies inside both. (b) Of the separations that are their own nearest copy, the fraction that rounding returns wrongly, against their length: none below $\frac{1}{2}w_{\min}$ (dotted).

Dividing by $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$, whose size is V by Equation (10.3),

$$\begin{aligned}
 |s_1| &= \frac{|\mathbf{d} \cdot (\mathbf{b} \times \mathbf{c})|}{V} && \text{dividing and taking sizes} \\
 &\leq \frac{|\mathbf{d}| |\mathbf{b} \times \mathbf{c}|}{V} && \text{a scalar product is at most the product of the lengths} \\
 &= \frac{|\mathbf{d}|}{w_1}, \quad w_1 = \frac{V}{|\mathbf{b} \times \mathbf{c}|} && \text{naming the width.}
 \end{aligned} \tag{10.5}$$

The length w_1 is the height of the cell above the face spanned by $\mathbf{b}$ and $\mathbf{c}$ (Section 10.1), the perpendicular width of the cell in that direction; w_2 and w_3 follow in the same way. Two consequences follow, with $w_{\min}$ the smallest of the three widths.

- A copy shorter than $\frac{1}{2}w_{\min}$ has every fractional coordinate between $-\frac{1}{2}$ and $\frac{1}{2}$, so rounding returns it: rounding finds the nearest copy of every separation shorter than half the smallest width, in any cell. Figure 10.1(b) measures this for graphite: rounding first misses just beyond $\frac{1}{2}w_{\min} = 1.067 \text{ \AA}$.
- A copy $\mathbf{hn}$ of the origin, with $\mathbf{n}$ not zero, has a fractional coordinate of size at least 1, so by Equation (10.5) it is at least $w_{\min}$ long. Two copies of one separation differ by such a vector, and a difference of two vectors is no longer than the sum of their lengths (one side of a triangle is no longer than the other two together), so they cannot both be shorter than $\frac{1}{2}w_{\min}$.

A cutoff no larger than half the smallest width therefore makes rounding exact for every pair within it, in any cell, and each pair has at most one copy within it:

$$r_c \leq \frac{1}{2}w_{\min}. \quad (10.6)$$

For a rectangular cell the widths are the side lengths. For graphite they are 2.134 Å in the layers and 6.711 Å across them, so a single cell allows a cutoff of only 1.067 Å. Repeating the cell n_1 times along **a**, n_2 times along **b** and n_3 times along **c** makes a larger cell, a supercell, of volume $n_1n_2n_3V$, whose face $n_2\mathbf{b} \times n_3\mathbf{c}$ has area $n_2n_3|\mathbf{b} \times \mathbf{c}|$, so its first width is n_1w_1 , and likewise n_2w_2 and n_3w_3 . A cutoff of 10 Å then needs $n_1, n_2 \geq 20/2.134 = 9.4$ and $n_3 \geq 20/6.711 = 2.98$: a supercell of $10 \times 10 \times 3$ of these cells, 1200 atoms at the four atoms per cell of [Section 10.5](#).

IN PRACTICE. Equation (10.6) decides the smallest cell that rounding can serve. A skewed cell is narrower than its lattice vectors are long: graphite's in-plane vectors are 2.464 Å long, and its cell is 2.134 Å wide. When a cell must be narrower than twice the cutoff, as for a single cell of a crystal, the copies within the cutoff are found by searching over several neighbouring cells instead, as the Ewald sum of [Section 10.3](#) does. `nearest_image` of `mdlab.cell` searches in the same way but keeps only the shortest copy, the reference against which rounding is tested.

KEY IDEA. Rounding the fractional coordinates of a separation gives the copy inside the cell centred on the first atom. In a rectangular cell that is always the nearest copy; in any cell it is the nearest for every separation shorter than half the smallest perpendicular width $w_{\min}$, the smallest of $V/|\mathbf{b} \times \mathbf{c}|$ and its two partners, so a cutoff $r_c \leq \frac{1}{2}w_{\min}$ makes the minimum image exact.

NOTEBOOK 10 compare rounding with the nearest copy in a hexagonal cell

CHECK YOUR UNDERSTANDING. A cubic cell of side 20 Å holds atoms that interact with a cutoff of 12 Å. Is rounding enough to find their pairs, and how large must the cell be?

10.3 Charges in a periodic box: the Ewald sum

The electrostatic energy of charges q_i in a periodic box is

$$U = \frac{1}{2} \sum_{i,j} \sum_{\mathbf{n}}' \frac{k_e q_i q_j}{|\mathbf{r}_j - \mathbf{r}_i + \mathbf{h}\mathbf{n}|}, \quad (10.7)$$

summed over every pair and every copy, the prime leaving out an atom with itself, $i = j$ at $\mathbf{n} = \mathbf{0}$. [Section 8.8](#) found that sums of this kind converge so slowly that their

value depends on the order of the terms. Ewald's method³⁴ splits each $1/r$ into a part that dies away quickly, summed over the near copies, and a smooth part, summed as a series of waves. This section derives it with two new tools.

Two tools

GAUSSIAN INTEGRALS. A Gaussian is the bell-shaped curve e^{-ax^2} , with $a > 0$. An integral to infinity is the limit of the integral to X as X grows. For $I = \int_{-\infty}^{\infty} e^{-x^2} dx$, the square I^2 is the limit of $\int_{-X}^X e^{-x^2} dx \int_{-X}^X e^{-y^2} dy$. Written as sums of values times short steps, the product of the two is a sum over small rectangles of $e^{-x^2} e^{-y^2} = e^{-(x^2+y^2)}$ times their areas $dx dy$, so I^2 is the integral of $e^{-(x^2+y^2)}$ over the whole plane (Section 3.11); the square of side $2X$ lies between the discs of radius X and $\sqrt{2} X$, and the integrand is positive, so discs give the same limit. Likewise an integral over a volume, written $\int \dots d^3r$, is the limit of a sum of values times the volumes d^3r of small boxes, and over all space the integral of a product $f_1(x)f_2(y)f_3(z)$ is the product of three ordinary integrals. The integrand $e^{-(x^2+y^2)}$ depends only on the distance $r = \sqrt{x^2 + y^2}$ from the origin, so we cut the plane into rings of radius r and width dr , of area $2\pi r dr$:

$$I^2 = \int_0^\infty e^{-r^2} 2\pi r dr = \left[-\pi e^{-r^2} \right]_0^\infty = \pi, \quad I = \sqrt{\pi}.$$

Changing the variable. If $u = g(x)$ and $F' = f$, the chain rule (Section 1.3) gives $\frac{d}{dx} F(g(x)) = f(g(x)) g'(x)$, so by the fundamental theorem of calculus (Section 1.4) both $\int_{x_1}^{x_2} f(g(x)) g'(x) dx$ and $\int_{g(x_1)}^{g(x_2)} f(u) du$ equal $F(g(x_2)) - F(g(x_1))$: an integral in u becomes one in x on writing $du = g'(x) dx$ and moving the limits to match. With $u = \sqrt{a} x$, for $a > 0$,

$$\int_{-\infty}^{\infty} e^{-ax^2} dx = \frac{1}{\sqrt{a}} \int_{-\infty}^{\infty} e^{-u^2} du = \sqrt{\frac{\pi}{a}}.$$

The error function is $\operatorname{erf}(x) = (2/\sqrt{\pi}) \int_0^x e^{-u^2} du$, which rises from 0 at $x = 0$ to 1 as x grows, since e^{-u^2} takes the same value at u and $-u$, so that its integral from 0 to infinity is half of I , $\frac{1}{2}\sqrt{\pi}$; its complement is $\operatorname{erfc}(x) = 1 - \operatorname{erf}(x) = (2/\sqrt{\pi}) \int_x^\infty e^{-u^2} du$. For small x the integrand is close to 1, so $\operatorname{erf}(x) \approx 2x/\sqrt{\pi}$. For large x , since $u^2 \geq x^2 + 2x(u-x)$, $\int_x^\infty e^{-u^2} du \leq e^{-x^2} \int_x^\infty e^{-2x(u-x)} du = e^{-x^2}/(2x)$, so $\operatorname{erfc}(x) \leq e^{-x^2}/(\sqrt{\pi}x)$: it falls roughly as e^{-x^2} .

FOURIER SERIES ON A PERIODIC CELL. A smooth function $f(x)$ that repeats with period L can be written as a sum of waves that repeat with it, $\cos(2\pi mx/L)$ and $\sin(2\pi mx/L)$ for $m = 0, 1, 2, \dots$, each with its own coefficient; we take this on

trust. The coefficients are found by averaging. Adding the two cosine formulae of Equation (1.6) gives $\cos A \cos B = \frac{1}{2}[\cos(A - B) + \cos(A + B)]$, and adding the two sine formulae gives $\sin A \cos B = \frac{1}{2}[\sin(A + B) + \sin(A - B)]$. A cosine or a sine averages to zero over whole periods, so the average over one period of $\cos(2\pi m x/L) \cos(2\pi m' x/L)$ is $\frac{1}{2}$ if $m = m' \neq 0$ and zero if $m \neq m'$, and that of $\sin(2\pi m x/L) \cos(2\pi m' x/L)$ is zero: multiplying the series by one cosine and averaging leaves half that wave's coefficient (the whole of it for $m = 0$).

In three dimensions the waves are $\cos(\mathbf{G} \cdot \mathbf{r})$ and $\sin(\mathbf{G} \cdot \mathbf{r})$, and they repeat with the cell when $\mathbf{G} \cdot \mathbf{h}\mathbf{n}$ is a multiple of 2π for every $\mathbf{n}$. This holds for $\mathbf{G} = m_1 \mathbf{a}^* + m_2 \mathbf{b}^* + m_3 \mathbf{c}^*$, with whole numbers m_1, m_2 and m_3 and reciprocal vectors $\mathbf{a}^*, \mathbf{b}^*$ and $\mathbf{c}^*$ chosen so that $\mathbf{a} \cdot \mathbf{a}^* = \mathbf{b} \cdot \mathbf{b}^* = \mathbf{c} \cdot \mathbf{c}^* = 2\pi$ and every other such product, such as $\mathbf{a} \cdot \mathbf{b}^*$, is zero, since then $\mathbf{G} \cdot \mathbf{h}\mathbf{n} = \mathbf{G} \cdot (n_1 \mathbf{a} + n_2 \mathbf{b} + n_3 \mathbf{c}) = 2\pi(m_1 n_1 + m_2 n_2 + m_3 n_3)$. Since $\mathbf{h}^{-1}\mathbf{h}$ is the identity, row j of $\mathbf{h}^{-1}$ times column i of $\mathbf{h}$ is 1 for $i = j$ and 0 otherwise, so the reciprocal vectors are 2π times the rows of $\mathbf{h}^{-1}$; and as $\mathbf{s}_j$ is row j of $\mathbf{h}^{-1}$ times $\mathbf{r}$, $\mathbf{G} \cdot \mathbf{r} = 2\pi(m_1 s_1 + m_2 s_2 + m_3 s_3) = 2\pi \mathbf{m} \cdot \mathbf{s}$, with $\mathbf{m} = (m_1, m_2, m_3)$. For a cubic cell of side L the reciprocal vectors have length $2\pi/L$ along the axes. The vectors $\mathbf{G}$ are said to lie in reciprocal space.

The average of a wave over the cell is zero unless $\mathbf{G} = \mathbf{0}$. In fractional coordinates the cell is the unit cube, whose image under $\mathbf{r} = \mathbf{h}\mathbf{s}$ has volume V , so each small box of volume d^3s becomes one of volume $V d^3s$ and the average is $\int \cos(2\pi \mathbf{m} \cdot \mathbf{s}) d^3s$ over the unit cube. Taking $m_1 \neq 0$ (otherwise any other m_j that is not zero serves), the integral over s_1 alone, with s_2 and s_3 held fixed, is $[\sin(2\pi \mathbf{m} \cdot \mathbf{s}) / (2\pi m_1)]$ between $s_1 = 0$ and 1, which is zero because the sine repeats. For a function with $f(-\mathbf{r}) = f(\mathbf{r})$ only cosines are needed, since $f(\mathbf{r}) = \frac{1}{2}[f(\mathbf{r}) + f(-\mathbf{r})]$ and every sine cancels between the two series; summing over all $\mathbf{G}$ with $c_{-\mathbf{G}} = c_{\mathbf{G}}$,

$$f(\mathbf{r}) = \sum_{\mathbf{G}} c_{\mathbf{G}} \cos(\mathbf{G} \cdot \mathbf{r}), \quad c_{\mathbf{G}} = \frac{1}{V} \int_{\text{cell}} f(\mathbf{r}) \cos(\mathbf{G} \cdot \mathbf{r}) d^3r,$$

because the cell average of $\cos(\mathbf{G} \cdot \mathbf{r}) \cos(\mathbf{G}' \cdot \mathbf{r})$, by the product formula, is $\frac{1}{2}$ for $\mathbf{G}' = \mathbf{G}$, $\frac{1}{2}$ for $\mathbf{G}' = -\mathbf{G}$, and zero otherwise (both halves at once when $\mathbf{G} = \mathbf{0}$), so that the cell average of $f(\mathbf{r}) \cos(\mathbf{G} \cdot \mathbf{r})$ is $\frac{1}{2}c_{\mathbf{G}} + \frac{1}{2}c_{-\mathbf{G}} = c_{\mathbf{G}}$.

The cosine integral of a Gaussian follows from these. Let $F(G) = \int_{-\infty}^{\infty} e^{-ax^2} \cos(Gx) dx$ for a number G , so that $F(0) = \sqrt{\pi/a}$. The integral is a limit of sums, and the derivative of a sum is the sum of the derivatives, so $F'(G)$ is the integral of the derivative of $e^{-ax^2} \cos(Gx)$ with respect to G , $-x e^{-ax^2} \sin(Gx)$ by the chain rule. Integrating by parts

(Section 6.3) with $\sin(Gx)$ and $xe^{-ax^2} = \frac{d}{dx}(-e^{-ax^2}/(2a))$, whose product vanishes at both ends,

$$F'(G) = - \int x e^{-ax^2} \sin(Gx) dx = -\frac{G}{2a} \int e^{-ax^2} \cos(Gx) dx = -\frac{G}{2a} F(G).$$

By the product rule (Section 2.7) and the chain rule,

$$\frac{d}{dG} \left[e^{G^2/(4a)} F(G) \right] = e^{G^2/(4a)} \left[\frac{G}{2a} F(G) + F'(G) \right] = 0,$$

so $e^{G^2/(4a)} F(G)$ keeps its value $F(0)$ at $G = 0$:

$$\int_{-\infty}^{\infty} e^{-ax^2} \cos(Gx) dx = \sqrt{\frac{\pi}{a}} e^{-G^2/(4a)}. \quad (10.8)$$

A narrow Gaussian, large a , has a wide spread of waves, and a wide one a narrow spread.

Splitting the Coulomb energy

With $u = rt$, the first Gaussian integral gives

$$\frac{2}{\sqrt{\pi}} \int_0^{\infty} e^{-r^2 t^2} dt = \frac{2}{\sqrt{\pi}} \frac{1}{r} \int_0^{\infty} e^{-u^2} du = \frac{1}{r},$$

and splitting the range of t at a chosen α splits $1/r$ in two,

$$\frac{1}{r} = \frac{\operatorname{erfc}(\alpha r)}{r} + \frac{\operatorname{erf}(\alpha r)}{r}, \quad (10.9)$$

the first from $t > \alpha$ and the second from $t < \alpha$, by the same change of variable. By the bound in the toolbox, the first part falls roughly as $e^{-\alpha^2 r^2}$, so its sum over the copies converges quickly and in any order:

$$U_{\text{real}} = \frac{1}{2} \sum_{i,j} \sum'_{\mathbf{n}} k_e q_i q_j \frac{\operatorname{erfc}(\alpha r_{ij\mathbf{n}})}{r_{ij\mathbf{n}}}, \quad r_{ij\mathbf{n}} = |\mathbf{r}_j - \mathbf{r}_i + \mathbf{h}\mathbf{n}|,$$

taken over the copies closer than a cutoff at which erfc has fallen to nothing that matters. The second part of Equation (10.9) tends to the finite value $2\alpha/\sqrt{\pi}$ as $r \rightarrow 0$, by the small- x form of erf , and falls only as $1/r$: it carries the long range, and it is smooth.

The smooth part as waves

Summed over the copies, the second part is, by its integral over t ,

$$\Phi_s(\mathbf{d}) = \sum_{\mathbf{n}} \frac{\text{erf}(\alpha|\mathbf{d} + \mathbf{h}\mathbf{n}|)}{|\mathbf{d} + \mathbf{h}\mathbf{n}|} = \frac{2}{\sqrt{\pi}} \int_0^\alpha g_t(\mathbf{d}) dt, \quad g_t(\mathbf{d}) = \sum_{\mathbf{n}} e^{-|\mathbf{d} + \mathbf{h}\mathbf{n}|^2 t^2}.$$

Neither side is finite as it stands, since the terms fall only as $1/|\mathbf{d} + \mathbf{h}\mathbf{n}|$ (Section 8.8); the trouble sits in a single wave, found below. The function g_t , a Gaussian placed at every copy of the origin, repeats with the cell and is unchanged by $\mathbf{d} \rightarrow -\mathbf{d}$. Its Fourier coefficients are

$$c_{\mathbf{G}} = \frac{1}{V} \int_{\text{cell}} \sum_{\mathbf{n}} e^{-|\mathbf{r} + \mathbf{h}\mathbf{n}|^2 t^2} \cos(\mathbf{G} \cdot \mathbf{r}) d^3r = \frac{1}{V} \int_{\text{all space}} e^{-|\mathbf{r}|^2 t^2} \cos(\mathbf{G} \cdot \mathbf{r}) d^3r,$$

because, writing $\mathbf{r}' = \mathbf{r} + \mathbf{h}\mathbf{n}$, the term of copy $\mathbf{n}$ is the integral of $e^{-|\mathbf{r}'|^2 t^2} \cos(\mathbf{G} \cdot \mathbf{r}')$ over the cell shifted by $\mathbf{h}\mathbf{n}$, as $\cos(\mathbf{G} \cdot \mathbf{r}' - \mathbf{G} \cdot \mathbf{h}\mathbf{n}) = \cos(\mathbf{G} \cdot \mathbf{r}')$, and the shifted cells, one for every $\mathbf{n}$, fill space once. The Gaussian depends only on $|\mathbf{r}|$, so turning the axes changes nothing, and with the x axis along $\mathbf{G}$ the integral is a product of three, one by Equation (10.8) with $a = t^2$ and $G = |\mathbf{G}|$, and two plain Gaussian integrals:

$$c_{\mathbf{G}} = \frac{1}{V} \frac{\sqrt{\pi}}{t} e^{-G^2/(4t^2)} \times \frac{\sqrt{\pi}}{t} \times \frac{\sqrt{\pi}}{t} = \frac{\pi^{3/2}}{Vt^3} e^{-G^2/(4t^2)}.$$

Integrating over t from 0 to α , with $u = G^2/(4t^2)$, so that $du = -G^2 dt/(2t^3)$ and the limits $t = 0$ and α become $u = \infty$ and $G^2/(4\alpha^2)$, whose exchange cancels the minus sign, gives for $\mathbf{G} \neq \mathbf{0}$

$$\frac{2}{\sqrt{\pi}} \frac{\pi^{3/2}}{V} \int_0^\alpha \frac{e^{-G^2/(4t^2)}}{t^3} dt = \frac{2\pi}{V} \frac{2}{G^2} \int_{G^2/(4\alpha^2)}^\infty e^{-u} du = \frac{4\pi}{VG^2} e^{-G^2/(4\alpha^2)},$$

so that

$$\Phi_s(\mathbf{d}) = \frac{4\pi}{V} \sum_{\mathbf{G} \neq \mathbf{0}} \frac{e^{-G^2/(4\alpha^2)}}{G^2} \cos(\mathbf{G} \cdot \mathbf{d}) + (\text{the } \mathbf{G} = \mathbf{0} \text{ term}).$$

The terms die away as $e^{-G^2/(4\alpha^2)}$: a smooth function needs only a few waves. For $\mathbf{G} = \mathbf{0}$ the coefficient is $\pi^{3/2}/(Vt^3)$, whose integral over t grows without limit as $t \rightarrow 0$. Starting the integral at a small t_0 in place of 0 gives $(2/\sqrt{\pi}) \int_{t_0}^\alpha \pi^{3/2}/(Vt^3) dt = \pi/(Vt_0^2) - \pi/(V\alpha^2)$, a constant that does not depend on $\mathbf{d}$; in the energy it is multiplied by $\sum_{i,j} q_i q_j = (\sum_i q_i)^2$, which for a neutral cell is zero whatever t_0 , and the term is dropped. What is dropped is the part of the sum that depended on its order: dropping it gives the energy of the periodic sample surrounded by a conductor, a material in

which charge moves freely, and other surroundings add a term in the cell's dipole moment $\sum_i q_i \mathbf{r}_i$, which measures how far its positive and negative charges sit apart³⁵.

The energy

The smooth parts of all pairs give $\frac{1}{2}k_e \sum_{i,j} q_i q_j \Phi_s(\mathbf{r}_j - \mathbf{r}_i)$, with every pair taken twice and each atom with itself. By the addition formula, $\cos(\mathbf{G} \cdot \mathbf{r}_j - \mathbf{G} \cdot \mathbf{r}_i) = \cos(\mathbf{G} \cdot \mathbf{r}_j) \cos(\mathbf{G} \cdot \mathbf{r}_i) + \sin(\mathbf{G} \cdot \mathbf{r}_j) \sin(\mathbf{G} \cdot \mathbf{r}_i)$, and the double sum separates into squares of single sums,

$$\sum_{i,j} q_i q_j \cos(\mathbf{G} \cdot \mathbf{r}_j - \mathbf{G} \cdot \mathbf{r}_i) = \mathcal{C}(\mathbf{G})^2 + \mathcal{S}(\mathbf{G})^2,$$

$$\mathcal{C}(\mathbf{G}) = \sum_i q_i \cos(\mathbf{G} \cdot \mathbf{r}_i), \quad \mathcal{S}(\mathbf{G}) = \sum_i q_i \sin(\mathbf{G} \cdot \mathbf{r}_i),$$

the structure factor, which costs N operations for each $\mathbf{G}$. The smooth part of each atom with itself at $\mathbf{n} = \mathbf{0}$ is not in Equation (10.7) and must be taken out: it is $\frac{1}{2}k_e q_i^2$ times the limit $2\alpha/\sqrt{\pi}$. Altogether,

$$U = U_{\text{real}} + \frac{2\pi k_e}{V} \sum_{\mathbf{G} \neq \mathbf{0}} \frac{e^{-G^2/(4\alpha^2)}}{G^2} (\mathcal{C}(\mathbf{G})^2 + \mathcal{S}(\mathbf{G})^2) - k_e \frac{\alpha}{\sqrt{\pi}} \sum_i q_i^2, \quad (10.10)$$

the real-space sum, over copies; the reciprocal-space sum, over the waves; and the self term. The forces are the slopes of each part of Equation (10.10) (Exercise 10.10). The function `ewald_energy_forces` of `mdlab.ewald` computes them; its reciprocal sum is

```

1 weight = np.exp(-g2 / (4 * alpha**2)) / g2
2 phase = r @ g.T
3 cos, sin = np.cos(phase), np.sin(phase)
4 c_sum, s_sum = q @ cos, q @ sin
5 e_rec = 2 * np.pi / volume * np.sum(weight * (c_sum**2 + s_sum**2))

```

with `g` the vectors $\mathbf{G}$ kept, one per row, `g2` their squared lengths, and k_e left out: the function multiplies every part by it at the end. It also wraps the positions into the cell first, since its search over copies in real space starts from there.

A cell that is not neutral, with total charge $Q = \sum_i q_i$, has no finite energy in a periodic box: the $\mathbf{G} = \mathbf{0}$ term, $\frac{1}{2}k_e Q^2 [\pi/(Vt_0^2) - \pi/(V\alpha^2)]$, grows without limit as $t_0 \rightarrow 0$. It is made neutral by a uniform background of charge $-Q$ spread over the cell. The background has only the $\mathbf{G} = \mathbf{0}$ wave and meets the charges and itself through the whole of $1/r$, the integral over t from t_0 to infinity, $\pi/(Vt_0^2)$; its energy with the charges, $-k_e Q^2 \pi/(Vt_0^2)$, and with itself, $\frac{1}{2}k_e Q^2 \pi/(Vt_0^2)$, cancel the growing part and leave $-\pi k_e Q^2/(2V\alpha^2)$. With it, the energy of five charges adding to 1.7 in a cubic cell

of side $6 \text{ \AA}$ is $-11.1048563570 \text{ eV}$ at $\alpha = 0.5, 0.8$ and $1.2 \text{ \AA}^{-1}$, to $3 \times 10^{-10} \text{ eV}$; without it, it moves by 1.0 eV over the same range.

Choosing α and the cutoffs

The total does not depend on α , which only moves work between the two sums. Figure 10.2(a) shows this for rock salt in its cubic cell of side $2d$: as α goes from $0.6/d$ to $5/d$ the real-space part changes by about 1, the reciprocal-space part by about 4 and the self part by about 5, in units of k_e/d , and their sum stays at $-1.7475646 k_e/d$ per ion pair, $-\mathcal{M}k_e/d$ with Chapter 8's Madelung constant³⁰, to $2 \times 10^{-12} k_e/d$. Each sum is cut where its terms have fallen to about a chosen size ϵ_E , the real-space terms falling roughly as $e^{-\alpha^2 r^2}$: $e^{-\alpha^2 r_c^2} = \epsilon_E$ in real space and $e^{-G_c^2/(4\alpha^2)} = \epsilon_E$ in reciprocal space, so

$$r_c = \frac{\sqrt{-\ln \epsilon_E}}{\alpha}, \quad G_c = 2\alpha \sqrt{-\ln \epsilon_E},$$

and the error of the energy follows ϵ_E (Figure 10.2b), in steps because the copies and the waves come in shells. Careful estimates of these errors are due to Kolafa and Perram³⁶.

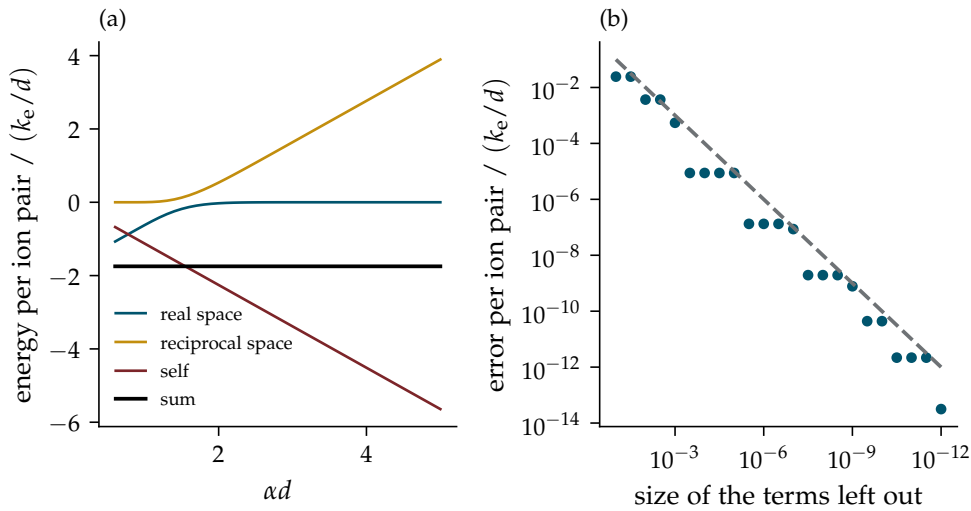


FIGURE 10.2: The Ewald sum of rock salt in its cubic cell of side $2d$, with $k_e = 1$. (a) The real-space, reciprocal-space and self parts of the energy per ion pair against α , and their sum, -1.7475646 at every α . (b) The error of the sum against the size of the terms left out, at $\alpha = 2/d$, with a line of slope 1 (dashed).

Three further checks pass. The same Madelung constant comes from a supercell of $2 \times 2 \times 2$ cubic cells and from the skewed cell of two ions spanned by $(0, 1, 1)d$, $(1, 0, 1)d$ and $(1, 1, 0)d$, 1.747564595 in each to ten figures; in a skewed cell of six charges the forces agree with central differences of the energy (Section 8.4) to $7.4 \times 10^{-10} \text{ eV/\AA}$, against forces of up to $15 \text{ eV \AA}^{-1}$, and add to zero to $1.2 \times 10^{-14} \text{ eV/\AA}$.

The cost of the method grows faster than N : at a fixed density, with α chosen to balance the two sums, it grows as $N^{3/2}$ (Exercise 10.12), provided the real-space sum finds its pairs through a cell list (Section 10.4). `mdlab` chooses α this way by default but tests every pair against every nearby copy, so its own cost grows as N^2 , which is fine for the small cells of this chapter. Particle mesh Ewald methods spread the charges onto a grid and sum the waves with the fast Fourier transform of Chapter 16, at a cost that grows as $N \ln N$ ^{37,38}, and are what large simulations of charged systems use.

FROM ELECTRONIC STRUCTURE. Many electronic-structure calculations in periodic cells, those that expand in plane waves among them, find the electrostatic (Hartree) potential of the electron density with the same Fourier series, one coefficient $4\pi\rho_{\mathbf{G}}/G^2$ per wave of the density ρ in atomic units, find the energy of the nuclei with one another by an Ewald sum, and leave out the $\mathbf{G} = \mathbf{0}$ terms, which for a charged cell amounts to the uniform neutralising background. The background brings in spurious energies of the charge with its copies and with the background, which fall only as $1/L$ with the size of the cell and need corrections³⁹. A slab with a different charge or adsorbate on each face carries a dipole, and because the potential must take the same value at both ends of a periodic cell, the step in potential across the dipole is undone by an artificial field across the vacuum, which a dipole correction removes^{40,41}: the situation of a charged electrode in a periodic cell.

KEY IDEA. Ewald's method splits $1/r$ into $\text{erfc}(\alpha r)/r$, summed over near copies, and $\text{erf}(\alpha r)/r$, smooth, whose sum over copies is a Fourier series with coefficients $4\pi e^{-G^2/(4\alpha^2)}/(VG^2)$. With the self term removed and the $\mathbf{G} = \mathbf{0}$ term dropped for a neutral cell, the energy does not depend on α , and it reproduces the Madelung constant of rock salt.

NOTEBOOK 10 move the work between the two Ewald sums with α

CHECK YOUR UNDERSTANDING. For $\epsilon_E = 10^{-8}$ and $\alpha = 0.3 \text{ \AA}^{-1}$, what are the real-space cutoff and G_c ?

10.4 Neighbour lists

A pair potential cut off at r_c needs only the pairs closer than r_c , and in a dense system each atom has a few dozen of them whatever the size of the system. Testing every pair to find them costs work that grows as N^2 , which Section 8.5 promised to reduce to N . This section does so with two lists.

How many pairs matter

In a liquid of number density ρ (atoms per unit volume), the neighbours of an atom within r_c fill a sphere of volume $\frac{4}{3}\pi r_c^3$, so there are about $\frac{4}{3}\pi r_c^3 \rho$ of them, and half as many pairs per atom, since each pair has two atoms. For a dense Lennard-Jones liquid, $\rho = 0.8\sigma^{-3}$, and $r_c = 2.5\sigma$, that is 52.4 neighbours and 26 pairs per atom, against $N(N-1)/2$ pairs to test. Measured on atoms placed at random at this density, every N from 500 to 32 000 gives 26.0 to 26.4 pairs per atom.

The cell list

The cell is divided into small cells at least r_c wide: along the direction of each perpendicular width w_k of [Section 10.2](#), there are $n_k = \lfloor w_k/r_c \rfloor$ of them, each a slice $1/n_k$ thick in the fractional coordinate s_k . Each atom is sorted, in one pass over the atoms, into the small cell numbered $\lfloor n_k s_k \rfloor$ along each direction, with $\mathbf{s}$ wrapped into the cell. A partner within r_c changes s_k by less than $r_c/w_k \leq 1/n_k$ ([Equation \(10.5\)](#)), less than one slice, so it lies in the atom's own small cell or in one of the $3^3 - 1 = 26$ that touch it, and only those are searched. A small cell holds on average the same number of atoms whatever N , so the work per atom does not grow with N and the total grows as N . With fewer than three small cells along some direction, the 26 neighbours would include one cell twice, and `mdlab` tests all pairs instead. The method goes back to Quentrec and Brot⁴².

[Figure 10.3](#) measures both: testing all pairs grows with a slope of 2.02 on logarithmic axes, the cell list, from $N = 2000$ on, with a slope of 0.98. The two cost the same near $N = 1000$, and at $N = 4000$ the cell list is four times faster. At $N = 32\,000$ the cell list takes 0.78 s; testing all pairs, which takes 0.42 s at $N = 4000$, would by its slope of 2 take $8^2 = 64$ times as long there, about 27 s.

The Verlet list

Atoms move little in a step, so the pairs within r_c change little from one step to the next. Verlet's list³¹ keeps the pairs within $r_c + r_{\text{skin}}$, where the skin r_{skin} is a margin added to the cutoff, and reuses them for many steps, computing at each step only the distances of the listed pairs. It must be rebuilt before a pair missing from it can come within r_c . Suppose that since the list was built every atom has moved less than $\frac{1}{2}r_{\text{skin}}$, atom i by $\Delta \mathbf{r}_i$. A pair not on the list was then at least $r_c + r_{\text{skin}}$ apart. The straight line from i 's old position to j 's old position is no longer than the detour through i 's new

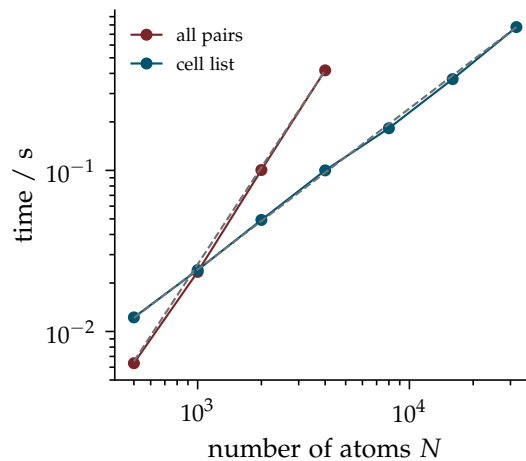


FIGURE 10.3: The time to find every pair within $r_c = 2.5\sigma$ of N atoms placed at random at $\rho = 0.8\sigma^{-3}$, by testing all pairs and through a cell list, against N , with lines of slope 2 and 1 (dashed).

position and j 's new position, since two sides of a triangle are together at least as long as the third, used once for each triangle of the detour:

$$r_{ij}^{\text{then}} \leq |\Delta \mathbf{r}_i| + r_{ij}^{\text{now}} + |\Delta \mathbf{r}_j|,$$

$$r_{ij}^{\text{now}} \geq r_{ij}^{\text{then}} - |\Delta \mathbf{r}_i| - |\Delta \mathbf{r}_j| > (r_c + r_{\text{skin}}) - \frac{1}{2}r_{\text{skin}} - \frac{1}{2}r_{\text{skin}} = r_c \quad \text{rearranging.}$$

The list is therefore complete until some atom has moved half the skin, and is rebuilt then. The class `VerletList` of `mdlab.neighbours` keeps the positions at the last build and tests this at every step:

```

1  def needs_rebuild(self, positions: ArrayLike) -> bool:
2      if self.reference is None:
3          return True
4      if not np.array_equal(self.cell, self.built_cell):
5          return self._strained_needs_rebuild(positions)
6      moved = minimum_image(
7          np.asarray(positions, dtype=float) - self.reference, self.cell
8      )
9      return bool(
10         np.max(np.einsum("ix,ix->i", moved, moved))
11         > (0.5 * self.skin) ** 2
12     )

```

The second test applies only when the cell itself has changed since the list was built, as it does under the barostats of Chapter 14 (Section 14.5); with the cell fixed the method goes on to the rule just derived. The displacements are reduced to their nearest copy so that the test stays right if the positions have been wrapped back into the cell, when an atom that crossed a face would otherwise seem to have moved a whole cell; `einsum` adds the squares of the three components of each displacement, and comparing

squared lengths with $(\frac{1}{2}r_{\text{skin}})^2$ avoids a square root per atom. The list holds pairs out to $r_c + r_{\text{skin}}$, so Equation (10.6) applies to that distance, $r_c + r_{\text{skin}} \leq \frac{1}{2}w_{\text{min}}$, and `mdlab` refuses a smaller cell. A thicker skin means fewer rebuilds and more pairs to check at each step. In the argon run of Section 10.6, with a skin of 1 Å, the list is built 187 times in 2000 steps, about every eleventh step, each time by testing all pairs: its box of 26.3 Å holds only $\lfloor 26.3/9.5 \rfloor = 2$ small cells of $r_c + r_{\text{skin}} = 9.5$ Å along each side.

KEY IDEA. Each atom has about $\frac{4}{3}\pi r_c^3 \rho$ neighbours within the cutoff. A cell list finds them by searching only neighbouring small cells, at a cost that grows as N ; a Verlet list with a skin r_{skin} reuses the pairs within $r_c + r_{\text{skin}}$ until some atom has moved $\frac{1}{2}r_{\text{skin}}$.

NOTEBOOK 10 *time all pairs against the cell list*

CHECK YOUR UNDERSTANDING. With $r_c = 2.5\sigma$ and a skin of 0.3σ , how far may the atoms move before the Verlet list must be rebuilt, and about how many pairs per atom does the list hold at $\rho = 0.8\sigma^{-3}$?

10.5 Starting a simulation

A run needs a starting state: positions inside the cell and a velocity for every atom. This section builds crystals from their cells, places them at the spacing their model prefers, and gives the atoms velocities with no drift, as Section 2.10 promised.

Positions from a crystal

A crystal is a cell and the fractional coordinates of the atoms in it. The face-centred cubic lattice of argon has a cubic cell of side a with an atom at each corner and at the centre of each face. A corner is shared by the eight cells that meet there and a face by two, so each cell holds $8 \times \frac{1}{8} + 6 \times \frac{1}{2} = 4$ atoms, at

$$(0,0,0), \quad (0, \frac{1}{2}, \frac{1}{2}), \quad (\frac{1}{2}, 0, \frac{1}{2}), \quad (\frac{1}{2}, \frac{1}{2}, 0),$$

and graphite's hexagonal cell of Section 10.2 has four, at $(0,0,\frac{1}{4})$, $(0,0,\frac{3}{4})$, $(\frac{1}{3}, \frac{2}{3}, \frac{1}{4})$ and $(\frac{2}{3}, \frac{1}{3}, \frac{3}{4})$, two in each layer, with the layers shifted against each other³³. A supercell of $n_1 \times n_2 \times n_3$ cells has the lattice vectors multiplied by n_1 , n_2 and n_3 , and an atom at $\mathbf{s}$ in the cell numbered (k_1, k_2, k_3) has the fractional coordinates $((s_1 + k_1)/n_1, (s_2 + k_2)/n_2, (s_3 + k_3)/n_3)$ in the supercell. Libraries such as ASE⁴³ build such structures too, and Notebook 10 checks the positions built here against ASE's.

The spacing matters. A crystal started away from the spacing at which its model's energy is least is squeezed or stretched, and pushes on its cell; Chapter 14 measures that push, the pressure. For the Lennard-Jones model with the energy switched off between 2σ and 2.5σ (Section 8.5), the energy per atom of the face-centred cubic lattice is least at $a_0 = 1.54916\sigma$, with neighbours $a_0/\sqrt{2} = 1.0954\sigma$ apart (from $(0,0,0)$ to $(0, \frac{1}{2}, \frac{1}{2})a_0$, a distance $a_0\sqrt{\frac{1}{4} + \frac{1}{4}}$), and an energy of -7.819ϵ per atom, found by minimising the sum over pairs with respect to a . With $\sigma = 3.4 \text{ \AA}$ that is $a_0 = 5.267 \text{ \AA}$, the spacing used for argon below.

Velocities

Temperature, which will set how much kinetic energy to give, is defined in Chapter 12. Until then we choose the kinetic energy directly. The function `starting_velocities` of `mdlab.md` draws each component at random, removes the drift of the centre of mass, and scales the velocities to the chosen kinetic energy:

```

1     m = np.asarray(masses, dtype=float)
2     v = rng.normal(size=(len(m), 3)) / np.sqrt(m)[: , None]
3     v -= (m @ v) / m.sum()
4     kinetic = 0.5 * mv2_to_energy * float(m @ (v * v).sum(1))
5     return v * np.sqrt(kinetic_energy / kinetic)

```

Line 2 draws numbers from the normal distribution of width 1, the bell-shaped spread of `rng.normal` that Chapter 12 derives, and divides each atom's by $\sqrt{m_i}$: a component $x/\sqrt{m_i}$ carries the kinetic energy $\frac{1}{2}m_i(x/\sqrt{m_i})^2 = \frac{1}{2}x^2$, whatever the mass, so light and heavy atoms carry the same kinetic energy on average. Line 3 subtracts the velocity of the centre of mass, $\mathbf{P}/M$ (Section 2.10). Line 4 is the kinetic energy, with `mv2_to_energy` the factor `MV2_T0_EV` of Equation (3.15), or 1 in reduced units, and line 5 scales every velocity by the same factor, which keeps the total momentum zero. In a periodic cell the angular momentum is not conserved (Section 10.1) and is left alone.

Where the energy goes

Started on its lattice with only kinetic energy, a crystal does not keep it. Figure 10.4(a) of the next section follows 500 argon atoms started at a_0 with 0.02 eV of kinetic energy per atom: within 0.1 ps the kinetic energy falls to half, it overshoots to 0.0057 eV at 0.14 ps, and between 0.5 and 1 ps it averages 0.0104 eV, as it does over the last 10 ps of a 20 ps run. Near its minimum the crystal vibrates in normal modes (Section 4.9), and a vibration shares its energy equally, on average, between kinetic and potential (Section 4.2). Three of the modes move every atom together and have no restoring force, but the velocities carry no drift, so they hold no energy. All the energy started

as kinetic, so on average half moves into the potential energy. The share left as kinetic energy is 0.518 of the start at 0.02 eV per atom, 0.506 at 0.005 eV and 0.502 at 0.002 eV: the excess over a half shrinks with the energy, as the wells look more nearly parabolic to smaller vibrations. A run started this way therefore needs a first stretch, discarded before measuring, in which the energy settles between its two forms.

KEY IDEA. Positions come from a crystal's cell and fractional coordinates, replicated into a supercell at the spacing the model prefers. Velocities are drawn at random, stripped of the drift of the centre of mass, and scaled to the chosen kinetic energy; started on its lattice, a crystal then passes about half of that energy to its potential energy.

NOTEBOOK 10 *build a supercell and give it velocities*

CHECK YOUR UNDERSTANDING. How many atoms has a supercell of $5 \times 5 \times 5$ face-centred cubic cells, and how long is its side for $a_0 = 5.267 \text{ \AA}$?

10.6 The loop

The pieces now fit together: a model that returns the energy and forces of atoms in a cell, the velocity Verlet step of [Section 9.2](#), and a record of what happens. This section writes the loop, checks it against an independent library, and runs it on argon and on a model salt.

A model and a loop

A model is anything that takes the positions and returns the potential energy and the forces. For a pair potential in a cell, `PairModel` of `mdlab.md` gets the pairs within r_c from its Verlet list and adds up their forces:

```

1  def __call__(self, positions: ArrayLike) -> tuple[float, NDArray]:
2      r = np.asarray(positions, dtype=float)
3      i, j, d, dist = self.neighbours.pairs(r)
4      phi, dphi = self.pair(dist)
5      push = (dphi / dist)[: , None] * d # the force on i from j
6      forces = np.zeros_like(r)
7      np.add.at(forces, i, push)
8      np.add.at(forces, j, -push)
9      return float(np.sum(phi)), forces

```

With $\mathbf{d} = \mathbf{r}_j - \mathbf{r}_i$, the force on atom i from the pair is $\phi'(r) \mathbf{d}/r$ ([Section 8.4](#)), and the force on j its opposite; `np.add.at` adds each pair's push into the rows of its two atoms, correctly when an atom appears in many pairs. `EwaldModel` returns, in the same form,

the energy and forces of charges by the Ewald sum of [Section 10.3](#), and `sum_of` adds the energies and forces of several models.

The loop of run is velocity Verlet over any such model:

```

1     u, f = model(r)
2     for step in range(n_steps + 1):
3         if step > 0:
4             v += 0.5 * dt * force_to_accel * f / m
5             r += dt * v
6             u, f = model(r)
7             v += 0.5 * dt * force_to_accel * f / m
8         if step % every == 0:
9             kinetic = 0.5 * to_energy * float(np.sum(m * v * v))

```

Each step is a half kick, a drift and a half kick, with one call of the model, whose forces serve the next step's first half kick, as in [Figure 9.2](#). Here `m` holds the masses as a column, one row per atom, and `to_energy` is `MV2_T0_EV`, or 1 in reduced units. Every every steps the time, positions, velocities and both energies are recorded. If a file is named, the frames are written to it in the extended XYZ format, which ASE reads: for each frame, the number of atoms; a line with the lattice vectors, the names of the columns and the potential and kinetic energies; and a line per atom with its element, position and velocity. ASE takes a value named `energy` as the potential energy, as the tests of `mdlab` check, so the potential energy is written under that name.

Checked against ASE

ASE, the Atomic Simulation Environment⁴³, has its own Lennard-Jones forces and its own velocity Verlet. For 256 argon atoms in a face-centred cubic cell, with the energy shifted at the cutoff in both, the two give the same starting energy to every printed figure. After 200 steps of 5 fs the positions at first differed by 1.6×10^{-8} Å. The cause lay in the units: ASE converts femtoseconds into its own unit of time with the 2014 values of the physical constants, and `mdlab` with the 2022 values that SciPy ships, and the two femtoseconds differ by 4.6 parts in 10^9 . An argon atom carrying 0.02 eV moves at about 0.0031 Å fs^{-1} , and over 1000 fs a clock off by that fraction misplaces it by about $0.0031 \times 1000 \times 4.6 \times 10^{-9} = 1.4 \times 10^{-8}$ Å, the size found ([Exercise 10.19](#)). Given the same femtosecond, the positions after 200 steps agree to 1.8×10^{-14} Å and the velocities to 1.1×10^{-16} Å/fs.

IN PRACTICE. A comparison of two codes needs the same units, constants and starting state, and only a few hundred steps: the motion of many atoms is chaotic, two paths that start a little apart separating faster and faster, and any difference, even of rounding, grows over a long run until the paths share nothing but their statistics⁴⁴. A difference that grows steadily from the first step, as this one did, from 6×10^{-13} Å after 2 steps to 4.8×10^{-9} Å after 50 and 1.6×10^{-8} Å after 200, points to something systematic before the chaos takes over.

Two runs

Figure 10.4(a) follows 500 argon atoms (Section 10.5) for 20 ps with $\delta t = 10$ fs, the Lennard-Jones energy switched off between 2σ and 2.5σ and a skin of 1 Å. The total energy stays within 3.7×10^{-5} eV per atom, and over the last 10 ps its standard deviation is 0.3% of that of the kinetic energy, the test of Section 9.6. About 10 meV per atom passes from kinetic to potential energy in the first half picosecond, after which the kinetic energy swings by 1.6 meV; the total momentum at the end of the run is that of Section 10.1, the rounding of the arithmetic.

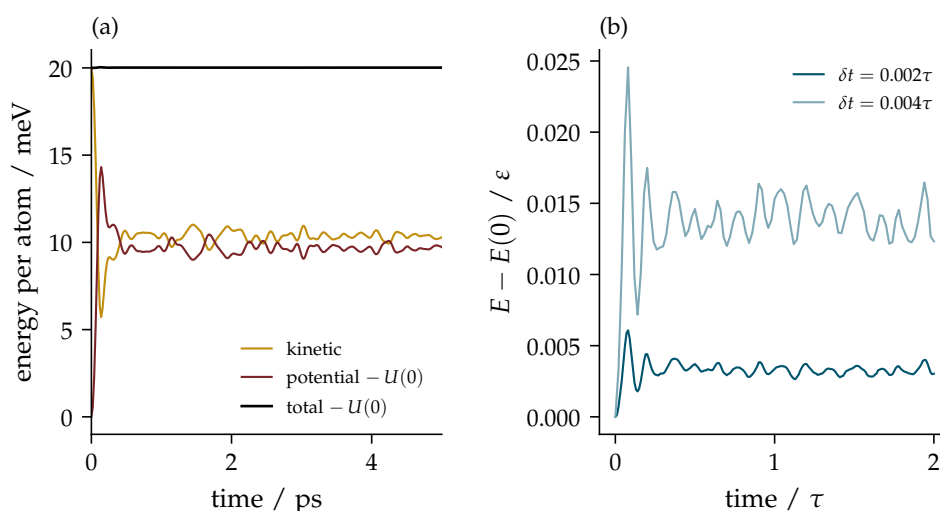


FIGURE 10.4: Two periodic runs. (a) 500 argon atoms started on their face-centred cubic lattice with 0.02 eV of kinetic energy per atom, $\delta t = 10$ fs: the kinetic energy, the potential energy and the total, each per atom and the last two less the starting potential energy, over the first 5 ps of 20 ps. (b) The model salt of 64 ions with Ewald electrostatics: the change of the total energy over 2τ with $\delta t = 0.002\tau$ and 0.004τ .

Figure 10.4(b) runs a model salt, chosen to exercise the Ewald sum in a loop. Its 64 ions, of charge ± 1 and mass 1 in reduced units, sit on a rock-salt lattice, repel each other at contact through the Lennard-Jones energy cut off at its minimum, $2^{1/6}\sigma$, and shifted there to zero, and attract and repel through the Coulomb energy with $k_e = 10\epsilon\sigma$, a value chosen for this model, summed by `EwaldModel`. Its energy is least with neighbouring ions 1.0908σ apart, at -15.810ϵ per ion pair, found as for argon.

Given 0.5ε of kinetic energy per ion and followed for 2τ , the total energy of the 64 ions spreads by $6.08 \times 10^{-3} \varepsilon$ with $\delta t = 0.002\tau$ and by $2.46 \times 10^{-2} \varepsilon$ with 0.004τ , a ratio of 4.04 for a doubled step: the δt^2 of [Section 9.6](#), the test that the Ewald forces are the slopes of the Ewald energy and are smooth.

KEY IDEA. A model returns the energy and the forces; the loop is velocity Verlet over it with one call per step. The code agrees with ASE to the rounding of the arithmetic once both use the same constants, keeps the energy of a periodic crystal with a fluctuation 0.3% of that of its kinetic energy and its momentum to rounding, and its Ewald forces pass the δt^2 test.

NOTEBOOK 10 *plot the argon and model-salt runs, check the loop against ASE, and read a trajectory file back*

CHECK YOUR UNDERSTANDING. With $\delta t = 10$ fs, how many steps are 20 ps, and how many force calls did the argon run make, counting the first?

10.7 Making it fast

A run of a million steps calls the model a million times, and the time a call takes decides what can be simulated. This section measures where the time goes and rewrites one force loop in three faster ways, each checked against the plain one.

The profile

A profiler records how long a program spends in each of its functions. Python's cProfile, run on 200 steps of 4000 Lennard-Jones atoms in reduced units with the NumPy pair model of [Section 10.6](#), finds three quarters of the time inside the neighbour list: 36% in its 14 rebuilds by the cell list, whose loops over small cells are written in Python, and 38% in picking out at every step the listed pairs now within r_c , which reduces each of them to its nearest copy by solving $\mathbf{h}\mathbf{s} = \mathbf{d}$ although the cell is a cube. The forces take most of the remaining quarter, `np.add.at` alone 16%. The first gains would come from the neighbour list.

One loop, four ways

The force loop runs over every pair at every step, and it shows the choices well. A processor runs only machine code, its own short list of simple instructions. A compiler can translate a numerical loop into machine code before it runs. In the ordinary Python loop, the interpreter instead carries out Python operations at each iteration, including the work needed to handle objects and call NumPy functions. The kernel takes the

positions, the side lengths of a rectangular box and a list of pairs, and returns the Lennard-Jones energy, shifted to zero at r_c , and the forces. `mdlab.kernels` has it four times:

- `lj_loop`, a plain Python loop over the pairs;
- `lj_numpy`, the same arithmetic on whole arrays, every pair in each operation;
- `lj_numba`, the same loop with the three components of each pair written as a loop of their own, compiled to machine code by Numba when it is first called;
- `lj_fortran`, the loop written in Fortran and compiled through NumPy's `f2py`, which builds the code that lets Python call it.

The Fortran loop over the pairs is

```

1 do p = 1, n_pairs
2   i = pair_i(p) + 1
3   j = pair_j(p) + 1
4   sep = positions(:, j) - positions(:, i)
5   sep = sep - box * anint(sep / box)
6   r2 = sum(sep * sep)
7   if (r2 >= r_cut2) cycle
8   s6 = (sigma2 / r2)**3
9   push = 24.0_dp * well_depth * (2.0_dp * s6 * s6 - s6) / r2
10  energy = energy + 4.0_dp * well_depth * (s6 * s6 - s6)      &
11         - phi_cut
12  forces(:, i) = forces(:, i) - push * sep
13  forces(:, j) = forces(:, j) + push * sep
14 end do

```

Fortran counts from 1 where Python counts from 0, hence the +1 of lines 2 and 3; `anint` rounds to the nearest whole number, the minimum image of [Section 10.2](#) in a rectangular box; and the array `positions(3, n_atoms)` holds one atom per column. Fortran stores an array column after column and NumPy row after row, so both keep the numbers in memory in the order $x_1, y_1, z_1, x_2, \dots$, and NumPy's array of one atom per row, passed as its transpose, reaches Fortran without a copy. `push` is $-\phi'(r)/r$, so the force on j is `push` times the separation and the force on i its opposite; `push` times the separation is thus the opposite of the push of `PairModel` in [Section 10.6](#), which was the force on i . The Numba version follows the Python loop with each operation on three numbers written as a loop over the components, as in the Fortran.

Figure 10.5 times one call of each on atoms on a simple cubic grid shaken by 0.05σ , at $\rho = 0.8\sigma^{-3}$, with the pairs within $r_c + 0.3\sigma$ listed. At $N = 2000$, with 74 684 listed pairs, the plain loop takes 172 ms, NumPy 5.2 ms, Numba 0.34 ms and Fortran 0.28 ms: NumPy is 33 times faster than the loop, and the two compiled versions another 15 and 18 times faster than NumPy. Every version grows roughly as N , with the number of

listed pairs, the compiled ones less evenly. All three agree with the plain loop to about a part in 10^{12} in the energy and to 3×10^{-12} in forces as large as 1024, in reduced units, the differences of rounding in a different order.

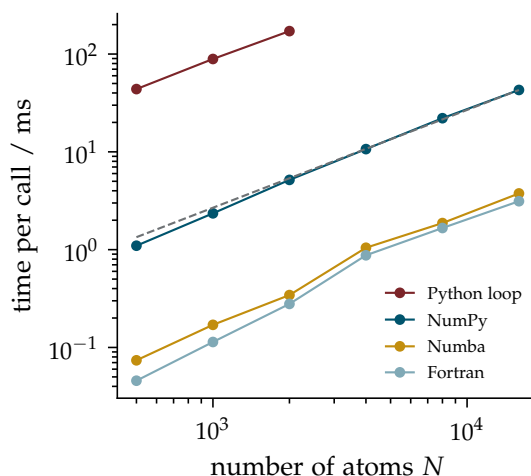


FIGURE 10.5: The time of one call of the Lennard-Jones force kernel against N , for atoms on a simple cubic grid shaken by 0.05σ at $\rho = 0.8\sigma^{-3}$, with $r_c = 2.5\sigma$, written as a plain Python loop, with NumPy, with Numba and in Fortran, with a line of slope 1 (dashed). Each time is the best of five, on one thread, one sequence of instructions on one core of the processor.

The reasons are in how each runs. The plain loop takes $172 \text{ ms} / 74\,684 = 2.3 \mu\text{s}$ for each pair, and the Fortran loop 3.7 ns for the same arithmetic, so almost all of the plain loop's time goes on something other than arithmetic: each pair makes about a dozen calls of NumPy on arrays of three numbers, and every call first finds out what its operands are and makes a new array for its result. NumPy on whole arrays makes those decisions once per operation for all the pairs and does the arithmetic in compiled code, but it passes over the pairs once for every operation, writing a new array each time. A compiled loop keeps each pair's few numbers in the processor's registers, its fastest storage, and touches memory only for the indices, positions and forces.

IN PRACTICE. The profile comes before any rewriting: here the slow part was the neighbour list. Only the loop that dominates is rewritten, the plain version is kept as the reference, and every fast version is checked against it, as the tests of `mdlab` do. Numba compiles a Python loop with few changes once every operation on a small array is written as a loop over its numbers, as in `lj_numba`; the Fortran loop took 18% less time here.

KEY IDEA. A profiler shows where the time goes, and in this code it was the neighbour list. The force loop written in Python, with NumPy on each pair, is hundreds of times slower than the loop compiled by Numba or written in Fortran; NumPy on whole arrays lies between. Every fast version is checked against the plain one to the rounding of the arithmetic.

NOTEBOOK 10 *race the four kernels*

CHECK YOUR UNDERSTANDING. At $0.2\ \mu\text{s}$ per atom per call, how long does the force kernel alone take for 1 ns of 10^4 atoms with $\delta t = 1\ \text{fs}$?

What we now have

A sample small enough to simulate is mostly surface, and repeating a cell of atoms in every direction removes the surface. The cell is the matrix $\mathbf{h}$ of its lattice vectors; fractional coordinates $\mathbf{s} = \mathbf{h}^{-1}\mathbf{r}$ wrap positions into it and, rounded, give the nearest copy of a separation, in any cell, for every separation shorter than half the smallest perpendicular width $w_{\min}$, the least of widths such as $V/|\mathbf{b} \times \mathbf{c}|$, which bounds the cutoff.

Charges in a periodic cell need the Ewald sum: $1/r$ split by Gaussian integrals into a short-ranged part summed over near copies and a smooth part summed as a Fourier series over the reciprocal vectors, less the self term, with the $\mathbf{G} = \mathbf{0}$ term dropped for a neutral cell; it reproduces the Madelung constant of rock salt for any splitting and any equivalent cell.

Cell lists find the pairs within the cutoff at a cost that grows as N , and Verlet lists with a skin reuse them until an atom has moved half the skin. A run starts from a crystal at its model's own spacing, with random velocities stripped of drift, and the loop is velocity Verlet over a model that returns energy and forces. The code agrees with ASE to the rounding of the arithmetic once both use the same physical constants, conserves energy and momentum in a periodic box, and passes the δt^2 test with Ewald forces. A profiler found the time in the neighbour list; the force loop, compiled by Numba or written in Fortran, runs hundreds of times faster than the Python loop that does each pair's arithmetic with NumPy, and agrees with it to rounding.

Chapter 11 holds the fastest vibrations fixed with constraints, so that the step can be longer, and splits the forces by how fast they change.

Answers to the checks

SECTION 10.1. $\mathbf{s} = (9/4, -1/5, 3/6) = (2.25, -0.2, 0.5)$. Wrapping removes the floors 2, -1 and 0, leaving $(0.25, 0.8, 0.5)$, so the atom is put at $(1, 4, 3)\ \text{\AA}$.

SECTION 10.2. No. In a cubic cell rounding does return the nearest copy, but half the side is $10\ \text{\AA}$, less than the cutoff, so a pair can have a second copy within $12\ \text{\AA}$, as

$(\pm 10, 0, 0)$ Å do, and rounding misses it. By Equation (10.6) the side must be at least 24 Å.

SECTION 10.3. $\sqrt{-\ln 10^{-8}} = 4.292$, so $r_c = 4.292/0.3 = 14.31$ Å and $G_c = 2 \times 0.3 \times 4.292 = 2.575$ Å⁻¹.

SECTION 10.4. Half the skin, 0.15σ . The list holds the pairs within 2.8σ : $\frac{1}{2} \times \frac{4}{3}\pi(2.8)^3 \times 0.8 = 36.8$ per atom.

SECTION 10.5. $4 \times 125 = 500$ atoms, in a cube of side $5 \times 5.267 = 26.34$ Å.

SECTION 10.6. $20\,000/10 = 2000$ steps, and 2001 force calls: one before the first step and one in each step.

SECTION 10.7. 10^6 steps of $10^4 \times 0.2 \mu\text{s} = 2$ ms each: 2000 s, about half an hour.

Exercises

The exercises run from questions of understanding, through calculations, to short derivations marked ‘show that’. Full solutions are in ‘Solutions to the exercises’ at the end of the book, and Notebook 10 checks every numerical answer.

EXERCISE 10.1 (*a large enough sample*). How many atoms must a cube of n^3 atoms hold before fewer than 1% of them are on its surface?

EXERCISE 10.2 (*fractional coordinates in graphite*). Graphite’s cell has $\mathbf{a} = (a, 0, 0)$, $\mathbf{b} = (-\frac{1}{2}a, \frac{\sqrt{3}}{2}a, 0)$ and $\mathbf{c} = (0, 0, c)$. Find the fractional coordinates of $\mathbf{r} = (\frac{1}{2}a, \frac{\sqrt{3}}{2}a, \frac{1}{4}c)$, and the position to which wrapping moves it.

EXERCISE 10.3 (*graphite’s volume and widths*). Show that graphite’s cell has volume $\frac{\sqrt{3}}{2}a^2c$ and perpendicular widths $\frac{\sqrt{3}}{2}a$, $\frac{\sqrt{3}}{2}a$ and c , and evaluate them for $a = 2.464$ Å and $c = 6.711$ Å.

EXERCISE 10.4 (*the primitive cell of a face-centred lattice*). The face-centred cubic lattice also has a cell of one atom, with $\mathbf{a} = \frac{1}{2}a_0(0, 1, 1)$, $\mathbf{b} = \frac{1}{2}a_0(1, 0, 1)$ and $\mathbf{c} = \frac{1}{2}a_0(1, 1, 0)$. Find its volume and its perpendicular widths, and the largest cutoff a single such cell allows for $a_0 = 5.267$ Å.

EXERCISE 10.5 (*rounding in a rectangular cell*). In the cell with sides 2, 3 and 4 Å along the axes, find the nearest copy of $\mathbf{d} = (1.8, -2.9, 0.1)$ Å by rounding, and check that no other copy is shorter.

EXERCISE 10.6 (*a normalised Gaussian*). Show that $\int_{-\infty}^{\infty} e^{-(x-x_0)^2/(2s^2)} dx = s\sqrt{2\pi}$ for any x_0 and any $s > 0$.

EXERCISE 10.7 (*the error function near zero*). From the Taylor series of e^{-u^2} , show that $\text{erf}(x) = (2/\sqrt{\pi})(x - x^3/3 + \dots)$, and hence that $\text{erf}(\alpha r)/r$ tends to $2\alpha/\sqrt{\pi}$ as $r \rightarrow 0$.

EXERCISE 10.8 (*waves that do not mix*). Show that the average over one period L of $\cos(2\pi m x/L) \cos(2\pi m' x/L)$, for m and m' among $0, 1, 2, \dots$ and not both zero, is $\frac{1}{2}$ if $m = m'$ and zero if $m \neq m'$.

EXERCISE 10.9 (*graphite's reciprocal vectors*). Show that the reciprocal vectors of graphite's cell are $\mathbf{a}^* = \frac{2\pi}{a}(1, \frac{1}{\sqrt{3}}, 0)$, $\mathbf{b}^* = \frac{2\pi}{a}(0, \frac{2}{\sqrt{3}}, 0)$ and $\mathbf{c}^* = \frac{2\pi}{c}(0, 0, 1)$, and find their lengths for $a = 2.464 \text{ \AA}$ and $c = 6.711 \text{ \AA}$.

EXERCISE 10.10 (*the Ewald forces*). Show that the reciprocal-space force on atom i is

$$\mathbf{F}_i = \frac{4\pi k_e}{V} q_i \sum_{\mathbf{G} \neq 0} \frac{e^{-G^2/(4a^2)}}{G^2} \mathbf{G} [\mathcal{C}(\mathbf{G}) \sin(\mathbf{G} \cdot \mathbf{r}_i) - \mathcal{S}(\mathbf{G}) \cos(\mathbf{G} \cdot \mathbf{r}_i)],$$

and that the real-space pair term $k_e q_i q_j \text{erfc}(\alpha r)/r$ has the slope $-k_e q_i q_j [\text{erfc}(\alpha r)/r + (2\alpha/\sqrt{\pi}) e^{-\alpha^2 r^2}]/r$.

EXERCISE 10.11 (*the waves of rock salt*). Rock salt in its cubic cell of side 2 , with $d = 1$, has cations at $(0, 0, 0)$, $(0, 1, 1)$, $(1, 0, 1)$ and $(1, 1, 0)$ and anions shifted from them by $(1, 0, 0)$. Show that its waves are $\mathbf{G} = \pi(m_1, m_2, m_3)$, that $\mathcal{S}(\mathbf{G}) = 0$ for all of them, and that $\mathcal{C}(\mathbf{G}) = 8$ when m_1, m_2 and m_3 are all odd and zero otherwise.

EXERCISE 10.12 (*the cost of Ewald*). At fixed density $\rho = N/V$, the real-space sum costs about N times the number of copies within $r_c \propto 1/\alpha$, and the reciprocal sum N times the number of waves within $G_c \propto \alpha$. Show that the cheapest α is proportional to $(N/V^2)^{1/6}$ and that the total cost then grows as $N^{3/2}$.

EXERCISE 10.13 (*choosing the Ewald parameters*). For 1000 ions in a cubic cell of side $30 \text{ \AA}$, with $\epsilon_E = 10^{-6}$ and the real-space cutoff at half the side, find α , G_c and roughly how many waves the reciprocal sum needs.

EXERCISE 10.14 (*neighbours in water*). Liquid water holds about 0.0334 molecules per $\text{\AA}^3$. How many molecules lie within $10 \text{ \AA}$ of a given one?

EXERCISE 10.15 (*counting small cells*). With $r_c + r_{\text{skin}} = 9.5 \text{ \AA}$, how many small cells does a cell list use in cubes of side 40 , 30 and $25 \text{ \AA}$, and in which does `mdlab` test all pairs instead?

EXERCISE 10.16 (*how often to rebuild*). Argon atoms with 0.0104 eV of kinetic energy each move at a root-mean-square speed $\sqrt{2K/m}$, with the factor of Equation (3.15) to put it in $\text{\AA fs}^{-1}$. How many steps of 10 fs does such an atom take to move half a skin of 1 $\text{\AA}$, and why does the run of Section 10.6 rebuild its list about twice as often?

EXERCISE 10.17 (*no drift*). Show that subtracting $\mathbf{P}/M$ from every velocity leaves the total momentum zero, and that multiplying every velocity by one factor then keeps it zero.

EXERCISE 10.18 (*half the energy*). A single harmonic vibration is started at its minimum with kinetic energy K_0 . Show that its kinetic energy averages $\frac{1}{2}K_0$ over a period.

EXERCISE 10.19 (*two femtoseconds*). ASE's femtosecond differs from mdlab's by 4.6 parts in 10^9 . Roughly how far apart would two runs of 1000 fs put an atom moving at $0.0031 \text{\AA fs}^{-1}$, and how does this compare with the $1.6 \times 10^{-8} \text{\AA}$ of Section 10.6?

Constraints and multiple time steps

The code of Chapter 10 advances every atom with the same step. Chapter 9 showed why that step must be short compared with the fastest vibration: in water, an oxygen–hydrogen bond stretches with a period of about 9 fs, restricting the step to around 0.5 fs. Molecular rotation and rearrangement take picoseconds, so resolving the bond vibration requires thousands of steps to follow these slower motions.

We examine three ways to reduce that cost. Holding bonds at fixed lengths removes their vibrations; SHAKE and RATTLE calculate the constraint forces needed at each step. Splitting fast and slow forces allows the costly slow forces to be evaluated less often, although resonance limits the outer step. Moving some mass from the heavy atoms to their hydrogens slows the fastest vibrations without changing the potential energy. We compare these choices on a box of water and use its trajectories to identify their limits and common implementation faults. The constraints also change the number of degrees of freedom, which Chapter 12 needs to relate kinetic energy to temperature.

11.1 Holding a bond fixed

The fastest motions in most molecules are the stretches of bonds to hydrogen ([Table 4.1](#)), and [Section 8.2](#) found them thoroughly quantum: for the O–H stretch of water $\hbar\omega$ is 17.5 times $k_B T$ at room temperature, so a simulation with classical nuclei gets the energy of this vibration wrong whatever its step. Yet the same vibration, with its period of 9 fs, is what holds the step down. Replacing each such bond by a rod of fixed length

(Section 6.6) removes the vibration altogether. The bond between atoms i and j , held at the length d , is the constraint

$$\chi = \frac{1}{2}(|\mathbf{r}_{ij}|^2 - d^2) = 0, \quad \mathbf{r}_{ij} = \mathbf{r}_j - \mathbf{r}_i,$$

whose force, found in Section 6.6, is $\lambda \mathbf{r}_{ij}$ on atom j and $-\lambda \mathbf{r}_{ij}$ on atom i : along the bond, equal and opposite, with a size set by the multiplier λ . This section finds λ at every step.

The constrained step

The Störmer–Verlet step of Equation (9.1), written for atom i with its force $\mathbf{F}_i$ and the constraint force $\mathbf{F}_i^c$ added, is

$$\mathbf{r}_i(t + \delta t) = 2\mathbf{r}_i(t) - \mathbf{r}_i(t - \delta t) + \frac{\delta t^2}{m_i}(\mathbf{F}_i + \mathbf{F}_i^c).$$

Without the constraint the atom would go to the provisional position $\tilde{\mathbf{r}}_i = 2\mathbf{r}_i(t) - \mathbf{r}_i(t - \delta t) + (\delta t^2/m_i)\mathbf{F}_i$, so the constrained step is the provisional one corrected, $\mathbf{r}_i(t + \delta t) = \tilde{\mathbf{r}}_i + (\delta t^2/m_i)\mathbf{F}_i^c$. With one bond, and its force taken at time t , the two atoms move to

$$\begin{aligned} \mathbf{r}_j(t + \delta t) &= \tilde{\mathbf{r}}_j + \frac{\delta t^2 \lambda}{m_j} \mathbf{r}_{ij}, \\ \mathbf{r}_i(t + \delta t) &= \tilde{\mathbf{r}}_i - \frac{\delta t^2 \lambda}{m_i} \mathbf{r}_{ij}, \end{aligned}$$

with $\mathbf{r}_{ij}$ the bond at time t . Subtracting the second line from the first, the bond at $t + \delta t$ is

$$\begin{aligned} \mathbf{r}_{ij}(t + \delta t) &= \tilde{\mathbf{r}}_{ij} + \delta t^2 \lambda \left(\frac{1}{m_i} + \frac{1}{m_j} \right) \mathbf{r}_{ij} \\ &= \tilde{\mathbf{r}}_{ij} + \eta \mathbf{r}_{ij}, \quad \eta = \frac{\delta t^2 \lambda}{m_r}, \quad \frac{1}{m_i} + \frac{1}{m_j} = \frac{m_i + m_j}{m_i m_j} = \frac{1}{m_r}, \end{aligned}$$

where $\tilde{\mathbf{r}}_{ij} = \tilde{\mathbf{r}}_j - \tilde{\mathbf{r}}_i$ is the provisional bond and m_r the reduced mass of Section 4.8. The new bond is the provisional one plus a multiple of the old, as Figure 11.1 draws. The constraint must hold at $t + \delta t$, $|\tilde{\mathbf{r}}_{ij} + \eta \mathbf{r}_{ij}|^2 = d^2$. Expanding the square as $(\mathbf{a} + \mathbf{s}) \cdot (\mathbf{a} + \mathbf{s})$ was expanded in Section 5.7 gives $|\tilde{\mathbf{r}}_{ij}|^2 + 2\eta \tilde{\mathbf{r}}_{ij} \cdot \mathbf{r}_{ij} + \eta^2 |\mathbf{r}_{ij}|^2 = d^2$, and with $|\mathbf{r}_{ij}|^2 = d^2$, since the bond held at t ,

$$d^2 \eta^2 + 2(\tilde{\mathbf{r}}_{ij} \cdot \mathbf{r}_{ij}) \eta + |\tilde{\mathbf{r}}_{ij}|^2 - d^2 = 0, \quad (11.1)$$

a quadratic equation for the one unknown η .

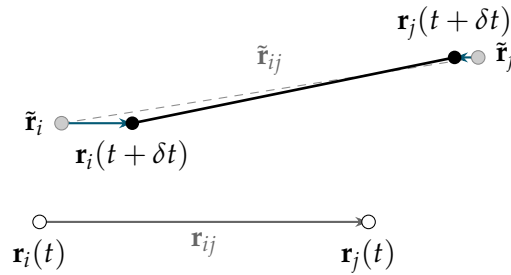


FIGURE 11.1: One step of a bond held at the length d , here 3. Atoms i (mass 1) and j (mass 3) start at the open circles. Without the constraint they would reach the grey provisional positions, a bond $\tilde{r}_{ij}$ too long (dashed); the correction (arrows) moves each along the old bond $\mathbf{r}_{ij}$, the lighter atom three times as far, to the black positions, where the bond has length 3 again.

One bond

Two atoms, i of mass 1 and j of mass 3, joined by a bond of length 1 along x , drift apart to 1.2 in a step, so $\mathbf{r}_{ij} = (1, 0, 0)$, $\tilde{\mathbf{r}}_{ij} = (1.2, 0, 0)$, $\tilde{\mathbf{r}}_{ij} \cdot \mathbf{r}_{ij} = 1.2$ and $|\tilde{\mathbf{r}}_{ij}|^2 - d^2 = 0.44$:

$$\eta^2 + 2.4\eta + 0.44 = 0,$$

$$\eta = \frac{-2.4 \pm \sqrt{2.4^2 - 4 \times 0.44}}{2} = -1.2 \pm 1, \quad \text{the quadratic formula,}$$

so $\eta = -0.2$ or -2.2 . The first shortens the bond to $1.2 - 0.2 = 1$. The second gives $1.2 - 2.2 = -1$, the bond turned round, a solution of the equation that no short step can reach: as δt shrinks, $\tilde{\mathbf{r}}_{ij}$ approaches $\mathbf{r}_{ij}$ and the correction must vanish, which only the root nearer zero does: with $\tilde{\mathbf{r}}_{ij} = \mathbf{r}_{ij}$, Equation (11.1) becomes $d^2\eta^2 + 2d^2\eta = 0$, with roots 0 and -2 . With $\eta = -0.2$ and $m_r = 1 \times 3 / (1 + 3) = 0.75$, atom j moves by $\delta t^2 \lambda / m_j = \eta m_r / m_j = -0.05$ along the bond and atom i by $+0.15$: the lighter atom moves three times as far, and the centre of mass, $1 \times 0.15 + 3 \times (-0.05) = 0$, does not move at all.

The quadratic has real roots only if the number under the square root of the quadratic formula, $4(\tilde{\mathbf{r}}_{ij} \cdot \mathbf{r}_{ij})^2 - 4d^2(|\tilde{\mathbf{r}}_{ij}|^2 - d^2)$, is not negative, that is, if $(\tilde{\mathbf{r}}_{ij} \cdot \mathbf{r}_{ij})^2 \geq d^2(|\tilde{\mathbf{r}}_{ij}|^2 - d^2)$. Writing ϕ for the angle between the provisional and the old bond, so that $\tilde{\mathbf{r}}_{ij} \cdot \mathbf{r}_{ij} = |\tilde{\mathbf{r}}_{ij}|d \cos \phi$, and dividing by d^2 , the condition reads $|\tilde{\mathbf{r}}_{ij}|^2 \cos^2 \phi \geq |\tilde{\mathbf{r}}_{ij}|^2 - d^2$, that is, $d^2 \geq |\tilde{\mathbf{r}}_{ij}|^2(1 - \cos^2 \phi) = |\tilde{\mathbf{r}}_{ij}|^2 \sin^2 \phi$, or, taking square roots, $|\tilde{\mathbf{r}}_{ij}| \sin \phi \leq d$. The part of the provisional bond at right angles to the old one must be no longer than the bond: a correction along the old bond cannot shorten that part. A bond stretched by 50% and turned in one step by more than $\arcsin(1/1.5) = 41.8^\circ$, and less than $180^\circ - 41.8^\circ = 138.2^\circ$, where $\sin \phi$ is again $1/1.5$, cannot be corrected at all, a sign of a step far too long.

Many bonds

The bonds of a molecule share atoms. The three distances held in a rigid water molecule join its three atoms in a triangle, and moving two atoms to correct one distance changes the other two, so the quadratics of the three bonds are coupled. Ryckaert, Ciccotti and Berendsen avoided solving them together⁴⁵: their method, SHAKE, corrects the bonds one at a time, each by an approximate solution of its own equation with the others' atoms where they are, and sweeps through all the bonds again and again until every one holds. The approximation drops the η^2 of Equation (11.1), which is small for a short step:

$$\eta \approx \frac{d^2 - |\tilde{\mathbf{r}}_{ij}|^2}{2\tilde{\mathbf{r}}_{ij} \cdot \mathbf{r}_{ij}}. \quad (11.2)$$

For the example this gives $-0.44/2.4 = -0.1833$, and the bond becomes 1.0167 long. Applied again, with the corrected bond in place of $\tilde{\mathbf{r}}_{ij}$ and $\mathbf{r}_{ij}$ still the old bond, it gives -0.0165 , and the bond is 1.000137. The repeated step is a method for solving any equation.

NEWTON'S METHOD. To solve $f(y) = 0$, we start from a guess y_n and replace the curve $f(y)$ near it by its tangent, the first two terms of its Taylor series (Section 1.8), $f(y_n) + f'(y_n)(y - y_n)$. The tangent is zero at

$$y_{n+1} = y_n - \frac{f(y_n)}{f'(y_n)},$$

the next guess. To see how fast the guesses close in, we write the root as y^* and the error as $e_n = y_n - y^*$, and expand $f(y^*)$ about y_n , with $y^* - y_n = -e_n$:

$$0 = f(y^*) = f(y_n) - e_n f'(y_n) + \frac{1}{2} e_n^2 f''(y_n) + \mathcal{O}(e_n^3),$$

$$\frac{f(y_n)}{f'(y_n)} = e_n - \frac{f''(y_n)}{2f'(y_n)} e_n^2 + \mathcal{O}(e_n^3),$$

the second line by dividing the first by $f'(y_n)$ and rearranging. Subtracting y^* from both sides of the rule gives $e_{n+1} = e_n - f(y_n)/f'(y_n)$, and so

$$e_{n+1} = \frac{f''(y_n)}{2f'(y_n)} e_n^2 + \mathcal{O}(e_n^3).$$

Each error is about a fixed multiple of the square of the one before, so once the error is small the number of correct digits roughly doubles at every step. For $f(y) = y^2 - 2$ from $y_0 = 1$, the guesses 1.5, 1.4166667, 1.4142157 and 1.4142135624 have errors of 0.086, 0.0025, 2.1×10^{-6} and 1.6×10^{-12} against $\sqrt{2}$. The method

needs $f'(y_n) \neq 0$ and a start close enough to the root; far from it, the tangent can point anywhere.

For one bond, $f(\eta) = |\tilde{\mathbf{r}}_{ij} + \eta \mathbf{r}_{ij}|^2 - d^2$, the left side of Equation (11.1), so $f'(\eta) = 2d^2\eta + 2\tilde{\mathbf{r}}_{ij} \cdot \mathbf{r}_{ij} = 2\mathbf{r}_{ij} \cdot (\tilde{\mathbf{r}}_{ij} + \eta \mathbf{r}_{ij})$, which at $\eta = 0$ is $2\tilde{\mathbf{r}}_{ij} \cdot \mathbf{r}_{ij}$, so Equation (11.2) is the first step of Newton's method from $\eta = 0$. Once the atoms have moved, the corrected bond takes the place of $\tilde{\mathbf{r}}_{ij}$, and the same formula is the next step. Here $f''(\eta) = 2d^2$ and, near the root, $f'(\eta)$ is twice the dot product of the old bond with the corrected one, $2d^2 \cos \theta$ if the two bonds are θ apart, so each error in η is $2d^2 / (2 \times 2d^2 \cos \theta)$ times the square of the one before. An error e in η changes the length by about $e d \cos \theta$, so the relative error of the length is $1/(2 \cos^2 \theta)$ times the square of the one before: about half, when the bond has not turned far. Figure 11.2(a) measures it: a bond stretched by 10% has a relative error of 0.1, then 5.0×10^{-3} , 1.4×10^{-5} and 1.1×10^{-10} , each 0.50 to 0.57 times the square of the one before; the corrected bond lies 19.9° off the old one, and $1/(2 \cos^2 19.9^\circ) = 0.565$.

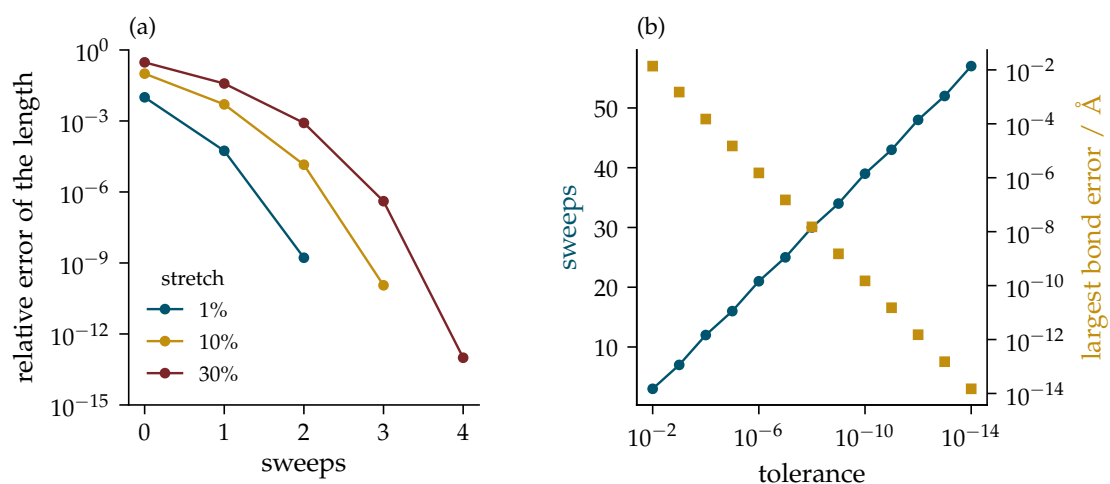


FIGURE 11.2: How SHAKE converges. (a) One bond of length 1 between equal masses, stretched by 1%, 10% and 30% in a direction 18° off the old bond: the relative error of its length after each sweep, until it reaches the rounding of the arithmetic. (b) The 64 rigid water molecules of Section 11.3 after one step of 2 fs without constraints: the sweeps needed against the tolerance (circles) and the largest error of a held distance that is left (squares, right axis).

The function `shake` of `mdlab.constraints` carries out the sweeps. The bonds are sorted beforehand into groups that share no atom, so that every bond of a group can be corrected at once with NumPy. For water, whose bonds are listed molecule by molecule, this applies the same corrections in the same order as taking the bonds one at a time; for other lists two bonds that share an atom can change places within a sweep, which changes the path of the sweeps but not the conditions they end on. The loop is

```

1   for sweep in range(1, max_sweeps + 1):
2       moved = False
3       for g in groups:
4           i, j = pairs[g, 0], pairs[g, 1]
5           now = r[j] - r[i] + copy[g]
6           eta = 0.5 * (d2[g] - np.sum(now * now, 1)) / np.sum(
7               old[g] * now, 1
8           )
9           eta[np.abs(eta) <= tolerance] = 0.0
10          moved = moved or bool(np.any(eta))
11          push = (m_r[g] * eta)[:, None] * old[g]
12          r[j] += push / m[j, None]
13          r[i] -= push / m[i, None]
14      if not moved:
15          return r, sweep

```

Here old holds the bonds $\mathbf{r}_{ij}$ at time t , now the current ones, d2 the squared lengths and m_r the reduced masses; push is $m_r \eta \mathbf{r}_{ij}$, which divided by m_j is atom j 's move. The array copy holds, for each bond, the multiple of the lattice vectors that turns $\mathbf{r}_j - \mathbf{r}_i$ into its nearest copy in the provisional positions (Section 10.2), so a molecule that lies across a face of the cell is corrected as one piece. A sweep that leaves every $|\eta|$ below the tolerance moves nothing, and ends the loop. Since η is about the fraction by which a bond is too long or too short (a bond $1 + s$ times its length along the old one gives, by Equation (11.2), $\eta = -s(1 + \frac{1}{2}s)/(1 + s) \approx -s$), the tolerance is a relative error of the lengths: 10^{-10} in `mdlab` unless another is asked for.

Several bonds converge more slowly than one. Figure 11.2(b) takes the box of 64 rigid water molecules of Section 11.3 through one step of 2 fs with no constraint, which leaves the largest held distance in error by 0.030 Å, 3.1% of an O–H bond, and corrects it by SHAKE. A tolerance of 10^{-10} needs 39 sweeps. Each tenfold reduction of the tolerance costs 4.5 more sweeps, so each sweep reduces the error by a steady factor q with $q^{4.5} = 10$, $q = 10^{1/4.5} = 1.67$, where a single bond squared it: correcting one side of a triangle disturbs the other two, and the errors are passed round it. The largest error left is 1.5 times the tolerance in Å, as it should be for a relative error of distances up to the H–H length of 1.51 Å.

KEY IDEA. A held bond adds a force along the old bond to the step, so the new bond is the provisional one plus η times the old. The constraint at the end of the step is a quadratic in η , and its first-order solution is a step of Newton's method. SHAKE takes the bonds one at a time with that step, and sweeps until every bond holds to a tolerance.

NOTEBOOK 11 *drift a bond and follow SHAKE sweep by sweep*

CHECK YOUR UNDERSTANDING. In the example with masses 1 and 3, the masses are changed to 2 and 2. Does η change, and how far does each atom move?

11.2 Velocities too: RATTLE

SHAKE corrects positions, and the Störmer–Verlet step it was built on carries no velocities. The loop of [Section 10.6](#) uses velocity Verlet, which needs the velocities at the same times as the positions, for the kinetic energy and to start the next step. Andersen’s RATTLE gives velocity Verlet the same constraints⁴⁶; this section derives it, checks it against an independent code, and counts the motions that are left.

Two corrections

Velocity Verlet is a half kick, a drift and a half kick ([Section 9.4](#)). With the constraint force added to the first half kick, the step of atom i begins

$$\begin{aligned}\mathbf{v}_i(t + \tfrac{1}{2}\delta t) &= \mathbf{v}_i(t) + \frac{\delta t}{2m_i}(\mathbf{F}_i + \mathbf{F}_i^c), \\ \mathbf{r}_i(t + \delta t) &= \mathbf{r}_i(t) + \delta t \mathbf{v}_i(t + \tfrac{1}{2}\delta t) = \tilde{\mathbf{r}}_i + \frac{\delta t^2}{2m_i} \mathbf{F}_i^c,\end{aligned}$$

with the provisional position now $\tilde{\mathbf{r}}_i = \mathbf{r}_i(t) + \delta t \mathbf{v}_i(t) + (\delta t^2/2m_i)\mathbf{F}_i$. This has the form of the constrained Störmer step with $\delta t^2/2$ in place of δt^2 , a factor that $\eta = \delta t^2 \lambda / (2m_r)$ absorbs, so shake corrects the positions unchanged. The half-step velocity drove the drift, so it must take the same correction, divided by δt ; equivalently, $\mathbf{v}_i(t + \tfrac{1}{2}\delta t) = [\mathbf{r}_i(t + \delta t) - \mathbf{r}_i(t)]/\delta t$ once the positions are corrected.

The second half kick, with the forces at the new positions, leaves velocities that would in general stretch or shrink the bonds. The length of a bond stays d only if it is not changing, and, differentiating component by component as for $|\mathbf{v}|^2$ in [Section 1.7](#),

$$\frac{d}{dt} \frac{1}{2} |\mathbf{r}_{ij}|^2 = \mathbf{r}_{ij} \cdot \mathbf{v}_{ij}, \quad \mathbf{v}_{ij} = \mathbf{v}_j - \mathbf{v}_i,$$

which must be zero at $t + \delta t$: the two atoms may move side by side or turn about each other, but not approach or separate along the bond. A second constraint force, along the new bond $\mathbf{r}_{ij}$, changes $\mathbf{v}_j$ by $(m_r \zeta / m_j) \mathbf{r}_{ij}$ and $\mathbf{v}_i$ by $-(m_r \zeta / m_i) \mathbf{r}_{ij}$, for some number ζ , so that, as for the positions, the relative velocity changes by $m_r \zeta (1/m_j + 1/m_i) \mathbf{r}_{ij} = \zeta \mathbf{r}_{ij}$. Then

$$\begin{aligned}(\mathbf{v}_{ij} + \zeta \mathbf{r}_{ij}) \cdot \mathbf{r}_{ij} &= 0, \\ \zeta &= -\frac{\mathbf{v}_{ij} \cdot \mathbf{r}_{ij}}{d^2} \quad \text{as } |\mathbf{r}_{ij}|^2 = d^2.\end{aligned}$$

This condition is linear in ζ , so one correction makes a single bond exact, and bonds that share atoms are swept until all hold, as in SHAKE. For atom i of mass 1 and atom j of mass 3, joined along x with $d = 1$, so that $\mathbf{r}_{ij} = (1, 0, 0)$, and with $\mathbf{v}_i = \mathbf{0}$ and $\mathbf{v}_j = (0.3, 0.4, 0)$, $\zeta = -0.3$, and the corrected velocities are $(0.225, 0, 0)$ and $(0.225, 0.4, 0)$: the relative motion along the bond is gone, the sideways motion of atom j is untouched.

Every correction, of positions or of velocities, changes $m_j \mathbf{v}_j$ by $m_j \zeta \mathbf{r}_{ij}$ and $m_i \mathbf{v}_i$ by the opposite (or the positions likewise), so the total momentum and the centre of mass are unchanged by the constraints, as [Section 2.9](#) requires of forces between the atoms of a system. In the example, the momentum along x is $3 \times 0.3 = 0.9$ before and $0.225 + 3 \times 0.225 = 0.9$ after, and along y it stays $3 \times 0.4 = 1.2$.

The function `rattle_step` of `mdlab.constraints` is the whole step:

```

1  v_half = v + 0.5 * dt * force_to_accel * np.asarray(forces) / m
2  r, _ = shake(r0 + dt * v_half, r0, m[:, 0], bonds, lengths, **keep)
3  v_half = (r - r0) / dt
4  u, f = model(r)
5  v = v_half + 0.5 * dt * force_to_accel * f / m
6  v, _ = rattle_velocities(r, v, m[:, 0], bonds, lengths, **keep)
7  return r, v, u, f

```

with `force_to_accel` the factor of [Section 2.10](#) that turns a force in $\text{eV } \text{\AA}^{-1}$ over a mass in amu into an acceleration in $\text{\AA fs}^{-2}$, and keep passing on the cell and the tolerance. The model is called once per step, as in velocity Verlet; the constraints add only the sweeps, and for the water box of [Section 11.3](#) one SHAKE to a tolerance of 10^{-10} after a step of 2 fs takes about a tenth of the time of one call of the model.

Running the step forwards and then backwards from the reversed velocities retraces it. Fifty steps of 2 fs of the water box of [Section 11.3](#), out and back with a tolerance of 10^{-14} , recover the starting positions to $5.2 \times 10^{-12} \text{ \AA}$ and the velocities to $1.9 \times 10^{-13} \text{ \AA/fs}$. Leimkuhler and Skeel showed that RATTLE, like velocity Verlet, is symplectic for motion confined to the constraints⁴⁷, so its energy should stay in a band, as [Section 11.3](#) measures.

Checked against ASE

ASE carries out the same two corrections: its `VelocityVerlet`, given the constraint `FixBondLengths`, corrects the positions and half-step velocities after the drift and the velocities after the second half kick, with the update of [Equation \(11.2\)](#) for the positions and the ζ above for the velocities. Given the forces of the `mdlab` water model, the same start and the same femtosecond ([Section 10.6](#)), 50 steps of 1 fs of the box of 64 rigid molecules agree to $2.1 \times 10^{-12} \text{ \AA}$ in the positions and $1.5 \times 10^{-13} \text{ \AA/fs}$ in the velocities, with both codes' bonds held to $1.5 \times 10^{-13} \text{ \AA}$ at a tolerance of 10^{-13} .

The motions that are left

Each independent distance constraint removes one way of moving. A system of N atoms with n_c independent held distances has $3N - n_c$ degrees of freedom (Sections 5.7 and 6.1), and $3N - n_c - 3$ once the drift of the centre of mass is removed and kept at zero. A water molecule with all three of its distances held has $9 - 3 = 6$: the three drifts and three turnings of Section 5.7, and no way left to vibrate. For 64 molecules, 192 atoms, flexible water has $576 - 3 = 573$ freedoms, water with its 128 O–H bonds held has $573 - 128 = 445$, and rigid water, with $3 \times 64 = 192$ distances held, $573 - 192 = 381$. Chapter 12 needs these counts, since the kinetic energy is shared among the freedoms.

KEY IDEA. RATTLE is velocity Verlet with two corrections along the bonds: after the drift, the positions are corrected by SHAKE and the half-step velocities with them; after the second half kick, each relative velocity loses its part along its bond. Both corrections keep the total momentum; the step is reversible and symplectic, and agrees with ASE to the rounding of the arithmetic. Each held distance removes one freedom.

NOTEBOOK 11 *correct a pair of velocities, and run RATTLE beside ASE*

CHECK YOUR UNDERSTANDING. Atoms of masses 2 and 2, joined along x with $d = 1$, have $\mathbf{v}_i = (0.1, 0, 0)$ and $\mathbf{v}_j = (-0.1, 0.2, 0)$. What are the velocities after the correction?

11.3 Rigid water

Holding bonds fixed is meant to lengthen the step, and how much it lengthens it is a question of measurement. This section builds a small box of liquid water, runs it flexible, with only its O–H bonds held, and rigid, and compares the energy fluctuation of the three at a range of steps by the test of Section 9.6.

A model of water

The TIP3P model of Jorgensen and co-workers⁴⁸, in the form ASE implements it⁴³, places a charge of $-0.834e$ on each oxygen atom and $+0.417e$ on each hydrogen, gives the oxygen atoms a Lennard-Jones energy between one another with $\sigma = 3.15061 \text{ \AA}$ and $\varepsilon = 6.596 \text{ meV}$ (which ASE stores as 0.1521 kilocalories per mole, a chemists' unit that `mdlab.units` converts), and fixes the shape of the molecule, with O–H $0.9572 \text{ \AA}$ and the H–O–H angle 104.52° , so that H–H is $1.5139 \text{ \AA}$; ASE's own tests run it with all three distances held by `FixBondLengths`. The charges describe how one molecule pulls on another; the Coulomb energy of the pairs within a molecule is left out, since the

bonds stand in for it, and would otherwise draw each hydrogen onto its own oxygen with an energy of $k_e q_O q_H / r = -5.2 \text{ eV}$ at $0.9572 \text{ \AA}$.

The class `WaterModel` of `mdlab.water` computes the Lennard-Jones energy between oxygen atoms, switched off smoothly between 5 and $6 \text{ \AA}$ (Section 8.5), and the Coulomb energy of all the charges and their copies by the Ewald sum of Section 10.3, less the Coulomb energy of each molecule's own three pairs. For the flexible model it adds the three springs of the spring model of Section 5.7, $k_{\text{OH}} = 48.48 \text{ eV \AA}^{-2}$ along each O–H and $k_{\text{HH}} = 17.66 \text{ eV \AA}^{-2}$ between the hydrogens, at rest at the TIP3P shape, whose fastest vibration, the symmetric stretch of the model, has $\omega = 0.857 \text{ rad fs}^{-1}$ and a period of 7.33 fs . Its tests check the forces against differences of the energy, for the rigid and the flexible model, and its parameters against ASE's.

Sixty-four molecules, turned at random, are placed on a $4 \times 4 \times 4$ grid in a cube of side $12.43 \text{ \AA}$, which gives the density of liquid water at 298.15 K and 0.1 MPa , 0.997 g cm^{-3} ; half the side, $6.21 \text{ \AA}$, exceeds the cutoff, as Section 10.2 requires. A grid is a poor arrangement for a liquid, and the molecules release energy as they leave it, which heats the box. The script that prepares it therefore follows the rigid molecules for ten rounds of 1 ps with $\delta t = 2 \text{ fs}$, each round starting from new random velocities of the kind of Section 10.5, which discard the heat the last round released, with no drift, no part along a held bond, and a kinetic energy of 12.93 meV for each freedom; Chapter 12 shows that this is the share of each freedom at 300 K . The potential energy falls from -2.37 eV on the grid to -25.70 eV after the tenth round, by 0.12 eV in that last round, a slow change whose end Chapter 15 learns to detect; the runs below start from that state. The flexible molecules start from the same positions, with velocities drawn for all nine freedoms of each molecule, and are prepared for two rounds of 0.5 ps with $\delta t = 0.25 \text{ fs}$.

The step each model allows

Figure 11.3(a) runs the three models at a range of steps, flexible water by velocity Verlet and the other two by RATTLE, and divides the standard deviation of the total energy, the square root of the mean of its squared differences from its mean, by that of the kinetic energy, the yardstick of Section 9.6. All three ratios grow as δt^2 at short steps, with fitted slopes of 2.05 (flexible), 2.10 (O–H held) and 2.01 (rigid) over their three shortest steps. At equal steps they differ greatly: at 1 fs the ratio is 6.7% for flexible water, 1.1% with the O–H bonds held and 0.23% for rigid water.

How small the ratio must be is a judgement, depending on what is to be measured (Section 9.6). Taking 1% as the level, and reading where each curve crosses it between measured points, the flexible model allows 0.41 fs , the model with its O–H bonds held 0.97 fs , and the rigid model 1.98 fs : holding the O–H bonds buys a factor of 2.4 , and

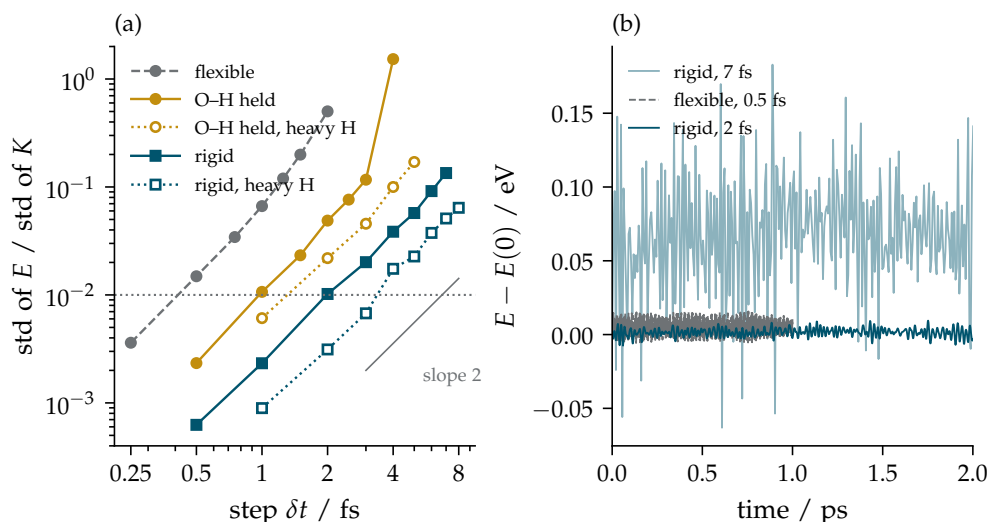


FIGURE 11.3: The step each model of water allows: 64 molecules, flexible (by velocity Verlet), with the O–H bonds held, and rigid (both by RATTLE); open markers for the last two with hydrogen made three times heavier (Section 11.6). (a) The standard deviation of the total energy over that of the kinetic energy, against the step, over 1 ps (flexible, O–H held) or 2 ps (rigid); dotted, the 1% level. (b) The total energy less its starting value for rigid water at 2 and 7 fs and flexible water at 0.5 fs.

holding the angle too a factor of 4.8. With the O–H bonds held, the fastest motion left is the bend, with a period of 17.0 fs in this model, shorter than the 20.9 fs of the free spring model, whose bonds stretch as it bends; with all three distances held, the molecule cannot vibrate at all, and the fastest motions left are the molecules' turning back and forth in the cage of their neighbours. These steps are read from single runs of 1 or 2 ps: the rigid run at 2 fs gives 1.02% over 2 ps and 0.93% over 10 ps (Section 11.8), and since the ratio grows as δt^2 a difference of 10% in it moves the step at 1% by about 5%, a spread that Chapter 15 turns into an error bar. Figure 11.3(b) shows that even at 7 fs, where the ratio is 13%, the energy of rigid water stays in a band over 2 ps, its mean moving by 0.007 eV ps^{-1} , as a symplectic method's should; at 2 fs the band is 0.017 eV wide for the whole box, 0.26 meV per molecule.

The rigid model is a different model, however long its step. Water held rigid cannot stretch or bend, and any property that depends on those motions, such as its spectrum at high frequencies, is lost; the properties of liquid water that depend on how molecules move as wholes are what a rigid model is built to give. A rigid model is therefore a choice of model as much as a means of lengthening the step, and the planned comparisons with learned propagators in Part VI use the flexible model as the reference when those vibrations are retained.

IN PRACTICE. The tolerance of the constraints matters. The step test above holds the bonds to 10^{-10} ; a tolerance of 10^{-3} leaves errors of 1.5×10^{-3} Å and an energy that drifts (Section 11.8). A tolerance may be stated as a relative or an absolute error of the length or of its square, so the numbers to check in a run are the largest error of a held distance, which `mdlab` records as `bond_error`, and whether a tighter tolerance changes the energy.

KEY IDEA. Measured on 64 molecules of TIP3P water, at an energy fluctuation of 1% of the kinetic energy's, holding the O–H bonds lengthens the step from 0.41 to 0.97 fs, and holding all three distances to 1.98 fs: factors of 2.4 and 4.8, bought by removing the fastest vibrations, and with them the motions that depended on them.

NOTEBOOK 11 *compare the three models' energy at any step*

CHECK YOUR UNDERSTANDING. How many steps of 0.41 fs and of 1.98 fs cover a nanosecond?

11.4 Multiple time steps

Constraints remove the fast motions. A second route keeps every motion and recognises that the forces behind them differ, in how fast they change and in what they cost. In the flexible water of Section 11.3, the springs inside each molecule change over the 7.33 fs period of its fastest vibration and cost almost nothing to compute: one call for all 64 molecules takes 0.04 ms. The Lennard-Jones and Ewald forces between molecules change mainly as whole molecules move and turn, and one call takes between 23 and 26 ms in our measurements, some 600 times as long. Velocity Verlet computes both at every step of 0.41 fs. This section builds a method that computes the costly forces less often.

A splitting within a splitting

Section 9.4 split the Liouville operator into a part $\hat{L}_r$ that moves the positions and a part $\hat{L}_p$ that moves the momenta by the forces. The forces are a sum, fast plus slow, $\mathbf{F}_i = \mathbf{F}_i^f + \mathbf{F}_i^s$, and since $\hat{L}_p A = \sum_i \mathbf{F}_i \cdot \nabla_{\mathbf{p}_i} A$ contains each force to the first power, it splits in the same way, $\hat{L}_p = \hat{L}_p^f + \hat{L}_p^s$, each part a kick by its own forces alone. The derivation of Equation (9.5) used nothing about X and Y beyond keeping the order of their products, so with $X = \hat{L}_p^s$ and $Y = \hat{L}_r + \hat{L}_p^f$ it gives

$$e^{\delta t \hat{L}} = e^{\frac{1}{2} \delta t \hat{L}_p^s} e^{\delta t (\hat{L}_r + \hat{L}_p^f)} e^{\frac{1}{2} \delta t \hat{L}_p^s} + \mathcal{O}(\delta t^3) :$$

half a kick by the slow forces, the motion for δt under the fast forces alone, and half a kick by the slow forces. The middle factor is the motion of a system with only the cheap forces, which n velocity Verlet steps of $\delta t/n$ follow,

$$e^{\delta t(\hat{L}_r + \hat{L}_p)} = \left(e^{\frac{\delta t}{2n}\hat{L}_p} e^{\frac{\delta t}{n}\hat{L}_r} e^{\frac{\delta t}{2n}\hat{L}_p} \right)^n + \mathcal{O}(\delta t^3/n^2),$$

the error of n steps each in error by order $(\delta t/n)^3$, $n(\delta t/n)^3 = \delta t^3/n^2$ in all. The outer step is δt , taken with one call of the slow forces; the inner step is $\delta t/n$, taken n times with one call of the fast forces each. Every factor is a kick or a drift, so the product, like velocity Verlet, keeps the full symplectic condition (Section 9.3), and since it reads the same from either end, reversing the velocities and stepping again undoes it factor by factor, as for velocity Verlet in Section 9.3: it is reversible. Tuckerman, Berne and Martyna derived the method in this way, as the reversible reference system propagator algorithm, RESPA⁴⁹. With $n = 1$ it is velocity Verlet with the two forces added (Exercise 11.9).

The function `respa_step` of `mdlab.respa` is the outer step:

```

1      v += 0.5 * dt * force_to_accel * np.asarray(slow_forces) / m
2      u_fast = 0.0
3      for _ in range(n_inner):
4          v += 0.5 * h * force_to_accel * f_fast / m
5          r += h * v
6          u_fast, f_fast = fast(r)
7          v += 0.5 * h * force_to_accel * f_fast / m
8      u_slow, f_slow = slow(r)
9      v += 0.5 * dt * force_to_accel * f_slow / m

```

where h is the inner step $\delta t/n$ and `f_fast` holds the fast forces at the start of the step, passed in from the step before. Its test checks that with one inner step it gives the positions and velocities of velocity Verlet to 10^{-12} . For water, `WaterModel` supplies the two parts: intramolecular, the springs, as the fast forces, and intermolecular, Lennard-Jones and Ewald, as the slow ones.

What it buys

Figure 11.4 runs the flexible box with inner steps of 0.25 fs, at which velocity Verlet alone keeps the energy fluctuation at 0.36% of the kinetic energy's, and outer steps from 0.25 fs to 4.5 fs. At an outer step of 1 fs the fluctuation is 1.8%, where velocity Verlet at the same step gives 6.7%. At the 1% level RESPA reaches an outer step of 0.72 fs against velocity Verlet's 0.41 fs, 1380 calls of the slow forces per picosecond against 2430 (from the steps before rounding), a factor of 1.76; at 2% the factor is 1.82 (Figure 11.4b). Since the slow forces take nearly all the time, the run is faster by close to that factor.

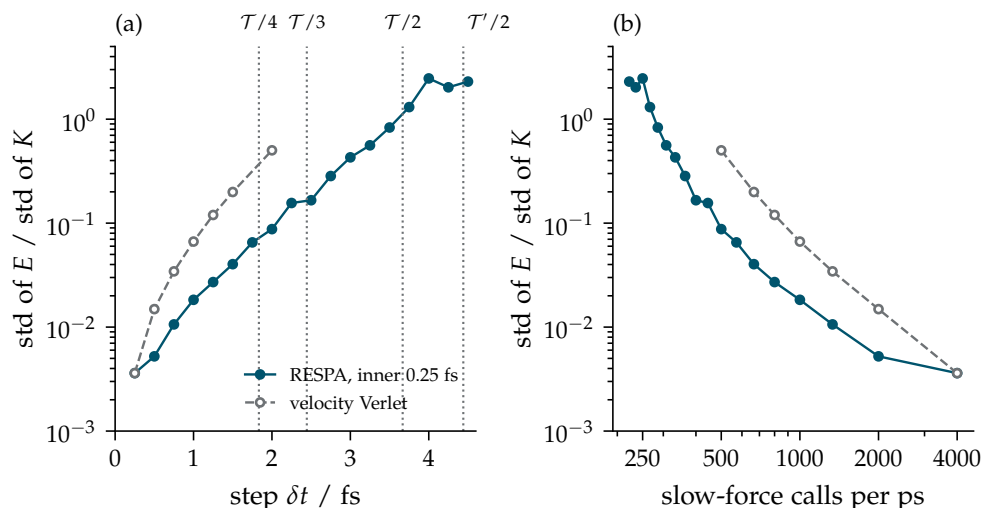


FIGURE 11.4: Multiple time steps for 64 molecules of flexible water: the springs are the fast forces, followed with inner steps of 0.25 fs; Lennard-Jones and Ewald forces are the slow ones, applied once per outer step δt . (a) The standard deviation of the total energy over that of the kinetic energy, against δt , for RESPA and for velocity Verlet taking all forces at every step δt (dashed); dotted, a quarter, a third and a half of the period $\tau = 7.33$ fs of the fastest vibration, and half the next, $\tau' = 8.88$ fs. (b) The same fluctuation against the number of calls of the slow forces per picosecond, which sets the cost of a run.

The gain stops early. Beyond an outer step of about 1.8 fs, a quarter of the fastest period, the fluctuation climbs much faster than δt^2 , to 8.8% at 2 fs and 43% at 3 fs, and the run blows up at 4.5 fs. Following the fast motion more closely does not help: inner steps of 0.125 fs with an outer step of 2 fs give 10.4%, no better than 8.8%. What fails is the outer step itself, and [Section 11.5](#) finds why.

KEY IDEA. RESPA splits the forces into fast and slow parts and the Liouville operator with them: half a kick by the slow forces, n velocity Verlet steps of $\delta t/n$ under the fast forces alone, and half a kick by the slow forces. It is reversible and symplectic. On flexible water it cuts the calls of the costly forces by a factor of about 1.8 at equal accuracy, and fails beyond outer steps of about a quarter of the fastest period.

NOTEBOOK 11 *split the forces of water and choose the outer step*

CHECK YOUR UNDERSTANDING. With 24 ms and 0.04 ms per call, about how long does one picosecond of flexible water take by velocity Verlet at 0.41 fs, and by RESPA at an outer step of 0.72 fs with three inner steps?

11.5 Resonance

RESPA applies the slow forces as sudden kicks, once per outer step, to atoms that are vibrating under the fast forces in between. A swing pushed in step with its motion goes higher, and without drag higher and higher (Section 4.7); kicks that come in step with a vibration, once in every half swing or every swing, can pump it in the same way. This section finds the outer steps at which that happens for the simplest system that shows it, and compares the result with water.

Two springs

One coordinate q , of unit mass, is pulled back by a fast spring and a slow one, with angular frequencies ω_f and ω_s , so its force is $-(\omega_f^2 + \omega_s^2)q$. RESPA follows the fast spring with many inner steps, closely enough to take its motion as exact: by the solution of Section 2.8, over a time δt it carries the state (q, v) to

$$q' = q \cos \theta + \frac{v}{\omega_f} \sin \theta, \quad v' = -\omega_f q \sin \theta + v \cos \theta, \quad \theta = \omega_f \delta t,$$

and half a kick by the slow spring changes v by $-\frac{1}{2}\delta t \omega_s^2 q$ and leaves q . Both are linear, so as matrices acting on the column (q, v) ,

$$\mathbf{R} = \begin{pmatrix} \cos \theta & \sin \theta / \omega_f \\ -\omega_f \sin \theta & \cos \theta \end{pmatrix}, \quad \mathbf{K} = \begin{pmatrix} 1 & 0 \\ -\kappa & 1 \end{pmatrix}, \quad \kappa = \frac{1}{2}\delta t \omega_s^2,$$

and the outer step is $\mathbf{M} = \mathbf{K}\mathbf{R}\mathbf{K}$, the half kick $\mathbf{K}$ acting first. Multiplying, with $C = \cos \theta$ and $S = \sin \theta$,

$$\mathbf{R}\mathbf{K} = \begin{pmatrix} C - \kappa S / \omega_f & S / \omega_f \\ -\omega_f S - \kappa C & C \end{pmatrix},$$

$$\mathbf{K}\mathbf{R}\mathbf{K} = \begin{pmatrix} C - \kappa S / \omega_f & S / \omega_f \\ -\omega_f S - 2\kappa C + \kappa^2 S / \omega_f & C - \kappa S / \omega_f \end{pmatrix},$$

so the trace, the sum of the diagonal entries (Section 9.3), is $2C - 2\kappa S / \omega_f$, which with $2\kappa = \delta t \omega_s^2$ is

$$\text{tr } \mathbf{M} = 2 \cos \theta - \frac{\omega_s^2 \delta t}{\omega_f} \sin \theta. \quad (11.3)$$

The fast motion $\mathbf{R}$, which turns the point $(q, v/\omega_f)$ through the angle θ , and the kick $\mathbf{K}$ each keep area in the phase plane (determinants $\cos^2 \theta + \sin^2 \theta = 1$ and 1), so their

product does too, and $\det \mathbf{M} = 1$ (Section 7.7). Whether repeated outer steps stay bounded depends on the trace alone.

POWERS OF A 2×2 MATRIX. Any 2×2 matrix $\mathbf{M}$, with M_{11}, M_{12} in its first row and M_{21}, M_{22} in its second, satisfies

$$\mathbf{M}^2 = (\text{tr } \mathbf{M}) \mathbf{M} - (\det \mathbf{M}) \mathbf{I},$$

as multiplying out shows: $\mathbf{M}^2$ has $M_{11}^2 + M_{12}M_{21}$ and $M_{11}M_{12} + M_{12}M_{22}$ in its first row and $M_{21}M_{11} + M_{22}M_{21}$ and $M_{21}M_{12} + M_{22}^2$ in its second, while $(M_{11} + M_{22})\mathbf{M}$ has $M_{11}^2 + M_{11}M_{22}$, $M_{11}M_{12} + M_{12}M_{22}$ and $M_{21}M_{11} + M_{21}M_{22}$, $M_{11}M_{22} + M_{22}^2$, and subtracting $(M_{11}M_{22} - M_{12}M_{21})\mathbf{I}$ from the latter gives the former. With $\det \mathbf{M} = 1$, multiplying by $\mathbf{M}^{n-1}$ gives $\mathbf{M}^{n+1} = (\text{tr } \mathbf{M})\mathbf{M}^n - \mathbf{M}^{n-1}$, so each entry x_n of the state after n steps, $\mathbf{M}^n$ times the start, obeys

$$x_{n+1} = (\text{tr } \mathbf{M}) x_n - x_{n-1}.$$

This is the recurrence of Section 9.5. If $|\text{tr } \mathbf{M}| < 2$, then $\text{tr } \mathbf{M} = 2 \cos \phi$ for some angle ϕ , and $\cos n\phi$ and $\sin n\phi$ solve it, as shown there: the motion stays bounded. If $|\text{tr } \mathbf{M}| > 2$, the quadratic $\Lambda^2 - (\text{tr } \mathbf{M})\Lambda + 1 = 0$ has, by the quadratic formula, a real root Λ with $|\Lambda| > 1$, and dividing it by Λ gives $\Lambda + 1/\Lambda = \text{tr } \mathbf{M}$. Then $\Lambda^{n+1} = (\Lambda + 1/\Lambda)\Lambda^n - \Lambda^{n-1}$, so $x_n = \Lambda^n$ solves the recurrence, as does Λ^{-n} , and a general start, a mixture of the two, grows by the factor $|\Lambda|$ at every step. At $|\text{tr } \mathbf{M}| = 2$ exactly the growth, if any, is slower, by a fixed amount at each step.

Without the slow spring, $\text{tr } \mathbf{M} = 2 \cos \theta$, which never leaves $[-2, 2]$: exact inner motion is stable at any outer step. The slow spring adds $-b\theta \sin \theta$ to the trace, with $b = (\omega_s/\omega_f)^2$, since $\omega_s^2 \delta t / \omega_f = b \omega_f \delta t$. Just below $\theta = \pi$ we write $\theta = \pi - \epsilon$ with ϵ small and positive; then $\cos \theta = -\cos \epsilon \approx -1 + \frac{1}{2}\epsilon^2$ and $\sin \theta = \sin \epsilon \approx \epsilon$ (Section 1.8), and

$$\begin{aligned} \frac{1}{2} \text{tr } \mathbf{M} &\approx -1 + \frac{1}{2}\epsilon^2 - \frac{1}{2}b\pi\epsilon && \text{with } \theta \approx \pi \text{ in the last term,} \\ &< -1 && \text{when } \frac{1}{2}\epsilon^2 < \frac{1}{2}b\pi\epsilon, \text{ that is, } 0 < \epsilon < b\pi. \end{aligned}$$

The motion therefore grows for outer steps in a band just below $\theta = \pi$, half the fast period $\mathcal{T}_f = 2\pi/\omega_f$, of width $b\pi$ in θ , or $\frac{1}{2}b\mathcal{T}_f$ in δt ; just below $\theta = 2\pi$, where $\sin \theta$ is negative, the same argument gives a band twice as wide below $\mathcal{T}_f$ (Exercise 11.12). Near $\delta t = \frac{1}{2}\mathcal{T}_f$ the kicks come every half vibration, when the fast motion has carried q to minus its value, and each one, proportional to q , meets the vibration in the same phase as the last: a resonance. Together the two springs shorten the period to $\mathcal{T}_f/\sqrt{1+b}$, so the band lies just below $\frac{1}{2}\mathcal{T}_f$, around half that period. At $\delta t = \frac{1}{2}\mathcal{T}_f$ exactly, $\theta = \pi$, the

trace is -2 , and $\mathbf{M}$ has rows $(-1, 0)$ and $(2\kappa, -1)$: q comes back to minus itself at every step while v gains $2\kappa q$ each time, so the motion grows steadily rather than by a factor.

Figure 11.5 measures it for $\omega_s = 0.3\omega_f$, so $b = 0.09$ and the predicted width below $\frac{1}{2}\mathcal{T}_f$ is $0.045\mathcal{T}_f$. The trace of Equation (11.3) leaves $[-2, 2]$ for δt from 0.459 to $0.500\mathcal{T}_f$, a width of 0.041 , since the term $\frac{1}{2}b\epsilon^2$ dropped with $\theta \approx \pi$ narrows the band by the factor $1/(1+b)$, and from 0.919 to $1.000\mathcal{T}_f$. RESPA run for 300 steps from $q = 1$ at rest reaches a size $\sqrt{q^2 + (v/\omega_f)^2}$ of 2.9×10^{17} at $0.48\mathcal{T}_f$, grows steadily to 23 at $0.50\mathcal{T}_f$, and stays near its start, below 2, at 0.45 and $0.52\mathcal{T}_f$.

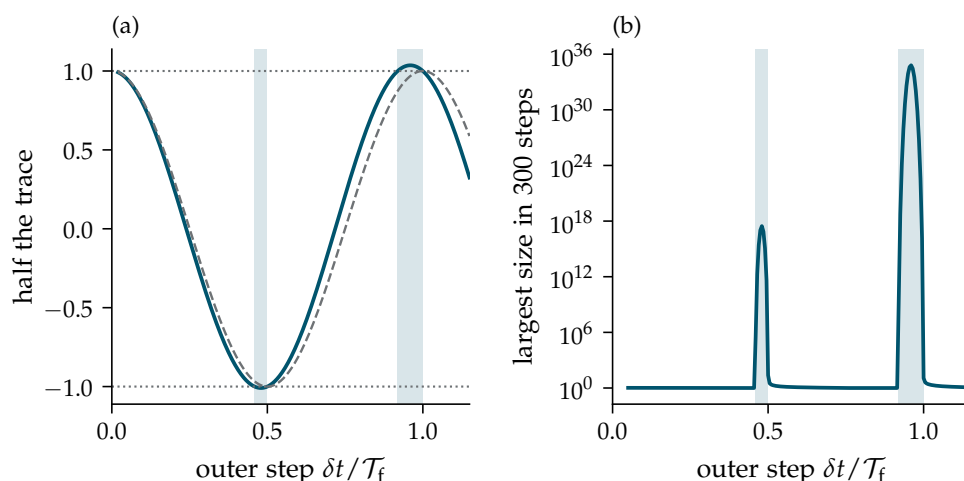


FIGURE 11.5: Resonance of a fast spring kicked by a slow one, $\omega_s = 0.3\omega_f$, stepped by RESPA with 50 inner steps. (a) Half the trace of the outer step, Equation (11.3), against the outer step in units of the fast period; the motion grows where it passes ± 1 (dotted), in the shaded bands, and the dashed curve, $\cos \theta$, is half the trace without the slow spring. (b) The largest size of the state, $\sqrt{q^2 + (v/\omega_f)^2}$, in 300 outer steps from $q = 1$ at rest.

In water

Water has many fast vibrations, and its slow forces are not springs, but the same picture explains part of Figure 11.4. The energy drifts most for outer steps around half the periods of the model's two stretches, 3.67 and 4.44 fs: by $+0.19$, $+1.14$, $+2.25$ and $+0.33 \text{ eV ps}^{-1}$ at 3.5, 3.75, 4 and 4.25 fs, and the run blows up at 4.5 fs. The fluctuation grows too fast well before those steps. From an outer step of 1 to 2 fs it grows by a factor of 4.8, a little more than the factor of 4 of δt^2 , and from 2 to 3 fs by 4.9, against 2.25. Ma, Izaguirre and Skeel found the same in flexible water and proteins, and traced it to resonances of a kind that the linear model above lacks, with instabilities when the outer step is a third or possibly a quarter of the fastest period⁵⁰: here 2.44 and 1.83 fs. The outer step of RESPA is therefore kept below a third of the fastest period, and for this water a quarter, whatever the cost of the slow forces.

KEY IDEA. Kicks by the slow forces once per outer step can pump a fast vibration. For two springs the outer step's matrix has the trace $2 \cos \theta - (\omega_s^2 \delta t / \omega_f) \sin \theta$, and the motion grows where it leaves $[-2, 2]$, in bands just below multiples of half the fast period. In water the fluctuation climbs faster than δt^2 beyond about a quarter of the fastest period, where Ma, Izaguirre and Skeel found resonances beyond this linear picture.

NOTEBOOK 11 *move the outer step across the resonance bands*

CHECK YOUR UNDERSTANDING. For $\omega_s = 0.1 \omega_f$ and a fast period of 9 fs, where is the band of growth below half the period, and how wide is it?

11.6 Heavier hydrogens

The potential energy depends only on where the atoms are; the masses enter only through Newton's law, and set how fast the atoms respond to the forces. A third way to lengthen the step changes the masses so that the fastest motions slow down, while keeping the energy surface, and with it the forces, as it was. This section works out what the change does to the vibrations of water and measures the step it buys.

Moving mass

A bond between two atoms vibrates with $\omega = \sqrt{k/m_r}$ (Section 4.8), and for a bond to hydrogen the reduced mass is close to the hydrogen's own: for O–H, $m_r = 15.999 \times 1.008 / (15.999 + 1.008) = 0.9483$ amu. Hydrogen mass repartitioning makes each hydrogen heavier and takes the same mass from the atom it is bonded to, so that every molecule keeps its total mass. Tripling the hydrogen's mass to 3.024 amu takes 2×2.016 from the oxygen of water, leaving 11.967 amu; the molecule still weighs 18.015 amu, so its centre of mass, now a little nearer the hydrogens (Exercise 11.16), has the same acceleration, the total force over the total mass, as before. The O–H reduced mass becomes $11.967 \times 3.024 / (11.967 + 3.024) = 2.4140$ amu, 2.55 times larger, and the stretch slows by $\sqrt{2.55} = 1.60$. The function `repartition_masses` of `mdlab.constraints` makes the change for any list of bonds to hydrogen.

The normal modes of the spring model of water (Section 5.7) with the new masses show a similar slowing, by 1.68 for the fastest vibration, which falls from $\omega = 0.857$ to $0.509 \text{ rad fs}^{-1}$, its period rises from 7.33 to 12.35 fs, and the stability limit $2/\omega$ of Section 9.6 from 2.33 to 3.93 fs; the bend slows from a period of 20.9 to 32.2 fs, and with the O–H bonds held from 17.0 to 27.2 fs. A rigid molecule cannot vibrate, but it turns, and moving mass outwards from the oxygen to the hydrogens raises its moments of inertia (Section 5.5): the principal moments, at the TIP3P shape, go from 0.615, 1.155

and $1.770 \text{ amu } \text{\AA}^2$ to 1.379, 3.465 and $4.844 \text{ amu } \text{\AA}^2$, so its turning in the cage of its neighbours slows too.

What it buys

The open markers of Figure 11.3(a) measure it on the box of 64 molecules. At the 1% level, water with its O–H bonds held goes from 0.97 to 1.31 fs with heavier hydrogens, a factor of 1.35, although its bend, the fastest motion left, slows by 1.60, so the bend is not all that sets its step; rigid water goes from 1.98 to 3.38 fs, a factor of 1.71, and is then 8.2 times the step of the flexible model with ordinary masses. Hopkins and co-workers, citing earlier studies in which repartitioning let the step be increased by up to a factor of two, ran a short chain of three amino acids and a protein with repartitioned masses and steps of 4 fs⁵¹.

What it keeps, and what it changes

Repartitioning leaves the potential energy and the forces unchanged, and Chapter 12 shows that the probability of finding the atoms in a given arrangement, at a given temperature, depends on the potential energy alone and not on the masses. Averages over the arrangements of the atoms, such as their mean potential energy or how the molecules are packed, are therefore unchanged, exactly so for molecules that are free or held wholly rigid; with only some distances held, as with the O–H bonds alone, the held distances make the masses enter the probability of the angles left free, a small effect that this book does not pursue. Everything that depends on how fast the atoms move is changed: heavier hydrogens move more slowly, so the rates at which molecules turn and diffuse, and the frequencies of their vibrations, belong to a system with different masses. Hopkins and co-workers report no significant difference in the rates and the equilibrium properties they computed with 4 fs steps and repartitioned masses⁵¹; that is a finding about their systems and properties, and a study of motion at the atomic scale should check its own.

KEY IDEA. Moving mass from heavy atoms to their hydrogens keeps the total mass and the forces, and raises the reduced masses of the fastest bonds: tripling hydrogen's mass in water slows the O–H stretch by a factor of 1.60. On 64 molecules it lengthens the step at 1% by 1.35 with only the O–H bonds held and by 1.71 for rigid water. Averages over arrangements are kept, exactly for free or rigid molecules; rates of motion are not.

NOTEBOOK 11 *choose the hydrogen mass and watch the vibrations slow*

CHECK YOUR UNDERSTANDING. In a C–H bond, carbon 12.011 amu and hydrogen 1.008 amu, the hydrogen is made three times heavier at the carbon’s expense. By what factor does the stretch slow?

11.7 How far these methods go

Table 11.1 collects the steps measured in this chapter on the same 64 molecules of water, each at the energy fluctuation of 1% of the kinetic energy’s that Section 11.3 took as the level. The cost of a run is set by the calls of the costly forces between molecules, one per step, or per outer step for RESPA.

TABLE 11.1: The step at which the energy fluctuation of 64 water molecules reaches 1% of the kinetic energy’s, with each method of this chapter; the gain over flexible water by velocity Verlet; and the calls of the forces between molecules per picosecond, from the unrounded steps. Heavy H: hydrogen three times as heavy, at the oxygen’s expense.

model and method	fastest motion left	step / fs	gain	calls per ps
flexible, velocity Verlet	O–H stretch	0.41	1	2430
flexible, RESPA	O–H stretch	0.72	1.8	1380
O–H held	bend	0.97	2.4	1030
O–H held, heavy H	bend, slower	1.31	3.2	765
rigid	turning	1.98	4.8	505
rigid, heavy H	turning, slower	3.38	8.2	296

Each method removes the fastest motion or slows it, and the next fastest then sets the step. Holding the O–H bonds leaves the bend; holding the angle too leaves the turning of whole molecules; heavier hydrogens slow whatever is left. RESPA keeps every motion; at the 1% level its outer step is 0.72 fs, and resonance holds it below about a quarter of the fastest period, 1.8 fs, whatever level is accepted (Section 11.5). The factors are of a few each, and they combine: rigid water with heavier hydrogens takes steps 8.2 times as long as the flexible model, but it is no longer the same model. For a simulation that keeps every motion, the gain here is the 1.8 of RESPA.

A factor of thirty, which TrajCast reaches on quartz, would need steps of 12 fs for the flexible model, longer than the period of its fastest vibration, 7.33 fs. Such steps lie outside the range of the methods developed here. Velocity Verlet becomes unstable beyond $2/\omega$ (Section 9.6), 2.33 fs here, and every method of this chapter must still follow, with steps short compared with its period, the fastest motion it keeps. The authors of TrajCast make the same point about methods with multiple time steps, which are ‘fundamentally limited by the need for small time steps in parts of the system’¹. TrajCast itself, run on 64 molecules of flexible water in a box of nearly the same size as this chapter’s, though with a different model of water, takes strides of

$\Delta t = 5$ fs, ten times the 0.5 fs step of the simulations it learned from, and on quartz strides of 30 fs, thirty times the 1 fs step of its simulations¹. A stride of 5 fs already exceeds the stability limit of velocity Verlet for the faster O–H stretch of real water, of period 8.88 fs, which is 2.83 fs (Section 9.6), as well as the 2.33 fs of this chapter's model. A learned map that predicts the state a whole stride ahead is not bound by that limit, because it does not step through the vibrations in between; what it must get right instead, and how it can be trusted, is the subject of the planned Part VI.

KEY IDEA. Constraints, heavier hydrogens and multiple time steps each lengthen the step by a factor of a few, by removing or slowing the fastest motion, until the next fastest sets the limit. For velocity Verlet, stability requires a step below $2/\omega$ for the fastest vibration retained; this is a limit of that integrator, not of every possible numerical method. A learned propagator is not bound by the same limit, but its accuracy and sampled distribution still need independent tests.

CHECK YOUR UNDERSTANDING. From Table 11.1, how many calls of the forces between molecules does a nanosecond of rigid water with heavier hydrogens take, and how many does flexible water by velocity Verlet take?

11.8 What the trajectory looks like

A run produces a record: energies at every step, positions and velocities every so often. Before any of it is used, the record should be read for signs that the run went wrong, and many faults show in it clearly when the reader knows where to look. This section shows a healthy run of rigid water and the same run broken in eight ways, each caught by a check of `mdlab.diagnostics` or of the record itself. Later chapters add the checks of temperature, pressure and thermostats, and Chapter 17 collects them all.

A healthy run

Figure 11.6 follows the 64 rigid molecules for 10 ps, 5000 steps of 2 fs by RATTLE. The kinetic and potential energies swing by up to about 13 meV per molecule over the first 2 ps, in opposite directions, so that their sum, the total energy, stays in a band 0.31 meV wide per molecule over the whole run, 0.020 eV for the box: the mirror image of K and U is the first sign of a healthy run without a thermostat. The standard deviation of the total energy is 0.93% of the kinetic energy's, just inside the 1% level chosen in Section 11.3. The mean total energy over the second half of the run, less that over the first, is 1.1×10^{-5} eV, and divided by the 5 ps between the centres of the halves it gives a drift of 2.2×10^{-6} eV ps⁻¹, far below the fluctuation; this is what `energy_drift` computes. Every held distance stays within 1.5×10^{-10} Å of its length, the tolerance of

10^{-10} times the H–H length, and the centre of mass moves 1.8×10^{-10} Å in 10 ps, the rounding of the arithmetic. The root-mean-square displacement of the oxygen atoms, the square root of the mean over them of the squared distance each has moved from its start, grows steadily, 1.41, 4.14 and 5.76 Å after 1, 5 and 10 ps, the wandering of molecules in a liquid that Chapter 16 turns into a diffusion coefficient. Over the run the kinetic energy averages 4.99 eV, 13.10 meV per freedom, with a standard deviation of 0.29 eV; the potential energy averages -0.403 eV per molecule. These are averages over one run of 10 ps, given with their spread; how far such averages can be trusted, their error bars, is the subject of Chapter 15. Figure 11.7 shows three frames, with three molecules marked to follow them.

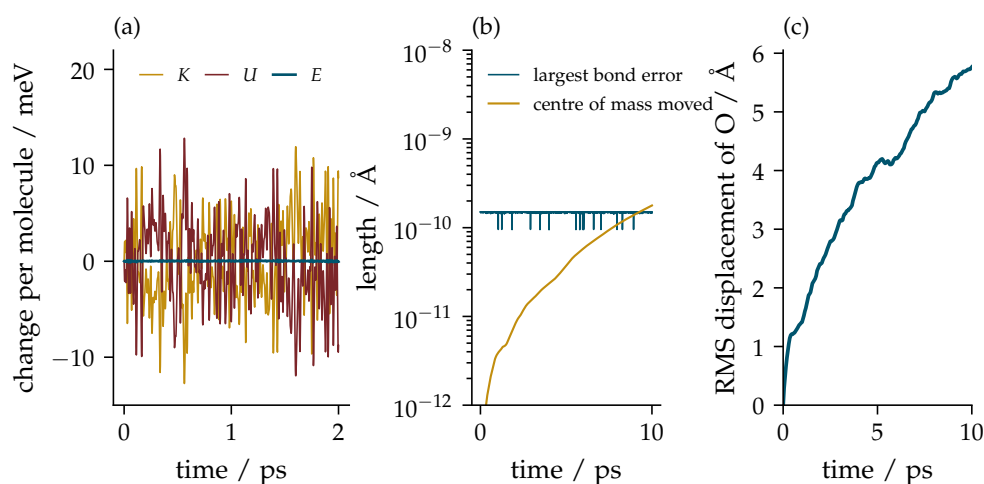


FIGURE 11.6: A healthy run: 64 rigid water molecules followed by RATTLE for 10 ps with $\delta t = 2$ fs. (a) The changes of the kinetic, potential and total energies from their starting values, per molecule, over the first 2 ps. (b) The largest error of a held distance and the distance the centre of mass has moved, on a logarithmic axis. (c) The root-mean-square distance the oxygen atoms have moved from their starts.

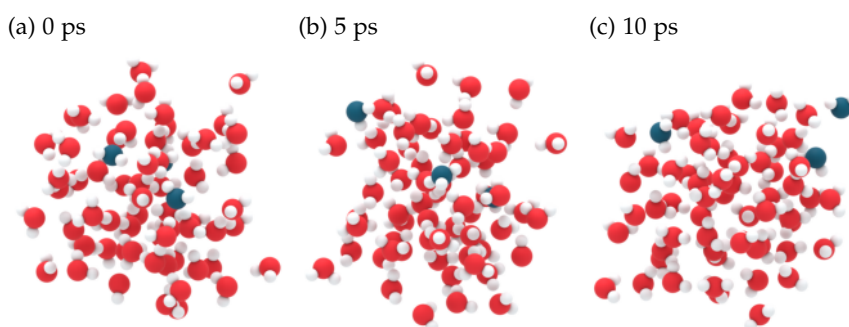


FIGURE 11.7: Frames of the healthy run at 0, 5 and 10 ps, each molecule moved whole into the cell by its oxygen and drawn with its two O–H bonds (oxygen red, hydrogen white). Three molecules have their oxygen drawn in teal so that their motion can be followed. All frames are seen from the same direction at the same scale.

The same run, broken

Figure 11.8 breaks the run in eight ways, each a mistake that is easy to make, and Table 11.2 lists the check that catches each, with its value in the healthy and the broken run.

TABLE 11.2: The faults of Figure 11.8, the check of `mdlab.diagnostics` or of the record that catches each, and its value in the healthy run and in the broken one; runs of 2 ps, except the flexible run at 0.5 fs and both RESPA runs, of 1 ps, and runs that failed sooner.

fault	check	healthy	broken
(a) step too long	energy_fluctuation	1.5%	47%
(b) jumping cutoff	energy_fluctuation	1.0%	2100%
(c) wrong units	$K(0)$ per freedom	12.9 meV	0.125 meV
(d) wrapped wrongly	bond_error	1.5×10^{-10} Å	15.8 Å
(e) overlap	largest_force	2.9 eV Å^{-1}	$3.0 \times 10^5 \text{ eV Å}^{-1}$
(f) loose constraints	energy_drift	$4 \times 10^{-4} \text{ eV ps}^{-1}$	$-0.011 \text{ eV ps}^{-1}$
(g) drift kept	centre_of_mass_path	8×10^{-12} Å	0.99 Å
(h) resonance	energy_drift	$2 \times 10^{-4} \text{ eV ps}^{-1}$	2.2 eV ps^{-1}

(a) *Too long a step.* Flexible water run at 2 fs, the step that suits rigid water, has an energy band 47% of the kinetic energy's spread, against 1.5% at 0.5 fs, and a mean that moves by 0.016 eV ps^{-1} . The check is the energy fluctuation, and the step test of Section 9.6 confirms the cause: halving the step should cut the fluctuation by about four, but from 2 to 1 fs it cuts it by 7.5, so 2 fs lies beyond the range where it grows as δt^2 .

(b) *A cutoff that jumps.* Replacing the Ewald sum by the Coulomb energy of pairs closer than 6 Å, cut off atom by atom, makes the energy jump whenever a pair crosses the cutoff: an oxygen and a hydrogen of different molecules have $k_e q_O q_H / r = -0.835 \text{ eV}$ there. The total energy leaps by 48 eV in the first 18 fs and then wanders over some 130 eV, its standard deviation 21 times the kinetic energy's, while its mean moves by -8.2 eV ps^{-1} : the fault of Section 8.5 at its worst. A band or a drift that does not shrink as the step is shortened points to forces that are not the slopes of a smooth energy.

(c) *Wrong units.* ASE measures time in its own unit, ångströms times $\sqrt{\text{amu}/\text{eV}}$, which is $1/\sqrt{\text{FORCE_T0_ACCEL}} = 10.18 \text{ fs}$ with the factor of Section 2.10. Velocities written in Å fs^{-1} and read by a program that takes them in that unit are 10.18 times too small, and the kinetic energy $10.18^2 = 104$ times too small: 0.125 meV per freedom at the start against the 12.93 meV asked for. Nothing else looks wrong: the energy is conserved as well as in the healthy run, with a fluctuation of 0.6%, and the kinetic energy rises within a tenth of a picosecond as the warm arrangement gives up energy, settling near 8.2 meV per freedom. Conservation shows only that a run is consistent with itself; whether it is the run asked for is shown by comparing the kinetic energy at the start with the value requested.

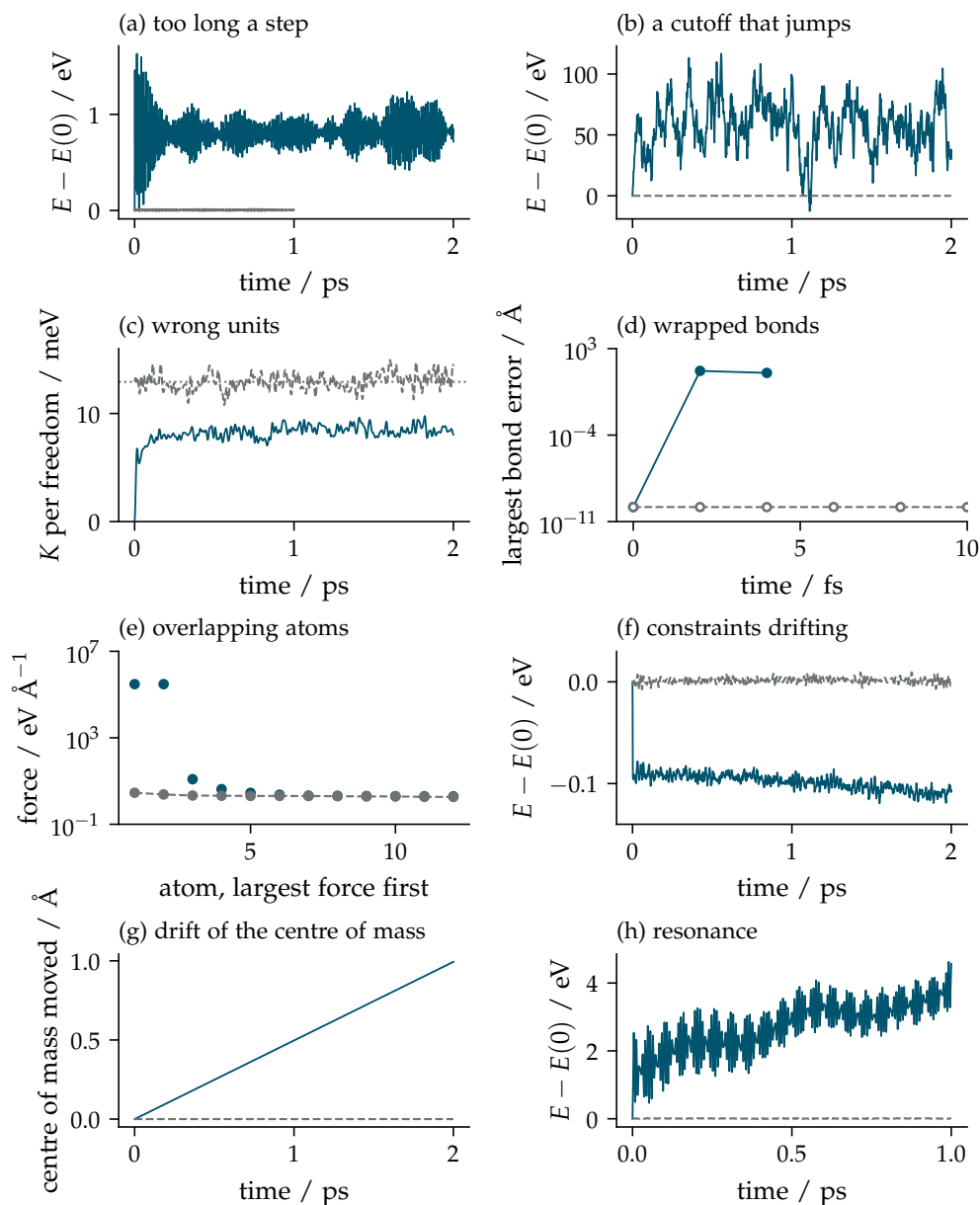


FIGURE 11.8: The run of Figure 11.6 broken in eight ways (teal), beside the healthy run (grey dashed), each through the quantity that catches the fault. (a) Flexible water at 2 fs against 0.5 fs. (b) The Coulomb energy cut off atom by atom at 6 Å against the Ewald sum. (c) Velocities in $\text{\AA fs}^{-1}$ read as ångströms per ASE time unit: the kinetic energy per freedom against the value asked for (dotted). (d) Positions wrapped into the cell at every step while the bonds are measured without their nearest copies: the largest bond error. (e) One molecule placed with its oxygen 1 Å from another's: the forces at the start, largest first. (f) Constraints held to 10^{-3} against 10^{-10} . (g) Starting velocities with their total momentum left in. (h) RESPA with outer steps of 4 fs against 1 fs.

(d) *Wrapped where it must not be.* Wrapping each atom into the cell at every step is harmless for the forces, which take nearest copies, but not for a code that measures a bond as $\mathbf{r}_j - \mathbf{r}_i$ directly: a molecule that lies across a face of the cell then seems to have a bond as long as the cell, or longer if it lies across an edge. Here the prepared positions spread over 2.2 cell widths, so the first wrap splits molecules, the largest bond error jumps from 1.5×10^{-10} to $15.8 \text{ \AA}$ at the first step, and the run fails at its third. The opposite fault, positions left outside the cell where a code needs them inside it, is why the Ewald sum of [Section 10.3](#) wraps them first. Displacements, by contrast, must be measured from positions that have not been wrapped: `largest_jump` reports the largest move between saved frames as a fraction of the cell's width, 0.16 in the unwrapped record of the healthy run, and near 1 when an atom has been wrapped between them.

(e) *Overlapping atoms at the start.* One molecule placed with its oxygen $1 \text{ \AA}$ from another's brings two atoms $0.886 \text{ \AA}$ apart, against $1.52 \text{ \AA}$ for the closest atoms of different molecules in the healthy start, a potential energy of $+25\,192 \text{ eV}$ and a largest force of $3.0 \times 10^5 \text{ eV \AA}^{-1}$ against $2.9 \text{ eV \AA}^{-1}$. The first step fails. The checks belong before the run: `closest_approach`, leaving out bonded pairs, and `largest_force` on the starting frame.

(f) *Constraints drifting.* Stopping SHAKE and RATTLE at a tolerance of 10^{-3} leaves bond errors of $1.5 \times 10^{-3} \text{ \AA}$; the energy falls by 0.093 eV in the first step, as every molecule is distorted a little, and then drifts by $-0.011 \text{ eV ps}^{-1}$, against $4 \times 10^{-4} \text{ eV ps}^{-1}$ in the healthy run over the same 2 ps. The energy gives the alarm, and the largest bond error, $1.5 \times 10^{-3} \text{ \AA}$ against $1.5 \times 10^{-10} \text{ \AA}$, the cause: a bond error at the tolerance asked for is healthy only if a tighter tolerance leaves the energy unchanged.

(g) *Drift of the centre of mass.* Starting velocities drawn without removing their total momentum ([Section 10.5](#)) leave the whole box moving: here $\mathbf{P} = (0.28, 0.29, 0.40) \text{ amu \AA/fs}$, so the centre of mass moves at $|\mathbf{P}|/M = 0.50 \text{ \AA ps}^{-1}$ and has gone $0.99 \text{ \AA}$ in 2 ps, against $8 \times 10^{-12} \text{ \AA}$ in the healthy run. The energy gives no sign, since momentum is conserved and its kinetic energy, 0.015 eV , never changes; that kinetic energy belongs to no motion of the molecules relative to one another. `centre_of_mass_path` catches it.

(h) *Resonance.* RESPA with outer steps of 4 fs, near half the period of the model's stretches, gains 2.2 eV ps^{-1} , against $2 \times 10^{-4} \text{ eV ps}^{-1}$ with outer steps of 1 fs, and its band is 2.5 times the kinetic energy's spread ([Section 11.5](#)).

IN PRACTICE. A run is read in three passes. Before it starts: the closest approach and the largest force on the starting frame, the kinetic energy against the value asked for, and the total momentum. While it runs: the total energy, whose fluctuation should be small against the kinetic energy's and fall by about four when the step is halved, and whose mean should not move; the bond errors, at a tolerance tight enough that tightening it leaves the energy unchanged; and the centre of mass, still. After it: the positions used for displacements must be continuous, with every move between frames a small fraction of the cell. These checks find faults; whether the run samples what it should is the subject of Chapters 12 to 15.

KEY IDEA. In a healthy run without a thermostat, K and U mirror each other and their sum stays in a narrow band with no drift, held distances stay at the tolerance and the centre of mass stays still. Each common fault leaves its own mark: a wide band (too long a step, a jumping cutoff, resonance), a drift (loose constraints, resonance), a burst (wrapping, overlapping atoms), a still band with the wrong kinetic energy (units), or a moving centre of mass (drift kept).

NOTEBOOK 11 *watch the healthy run, then break it in each of the eight ways*

CHECK YOUR UNDERSTANDING. A run's kinetic energy at the start is 0.97% of the value asked for, and its energy is conserved well. What is the likely fault, and what factor was missed?

What we now have

A bond held at a fixed length adds a force along the old bond to each step, and the bond at the end of the step is the provisional one plus a multiple η of the old: a quadratic in η whose first-order solution is a step of Newton's method. SHAKE corrects the bonds one at a time with that step and sweeps until all hold, the error of a single bond squaring at each sweep and that of a water molecule's triangle falling by a steady factor.

RATTLE gives velocity Verlet the same constraints, correcting the positions and half-step velocities after the drift and removing each relative velocity's part along its bond after the second half kick; both corrections keep the total momentum, the step is reversible and symplectic, and it agrees with ASE to the rounding of the arithmetic. Each independent held distance removes one freedom. On 64 molecules of TIP3P water, at an energy fluctuation of 1% of the kinetic energy's, holding the O-H bonds lengthens the step from 0.41 to 0.97 fs and holding all three distances to 1.98 fs.

RESPA splits the forces and the Liouville operator into fast and slow parts and applies the costly slow forces once per outer step; it saves a factor of 1.8 in their calls for flexible

water, and its outer step is limited by resonance: for two springs the outer step's matrix has the trace $2 \cos \theta - (\omega_s^2 \delta t / \omega_f) \sin \theta$, unstable in bands just below multiples of half the fast period, and in water beyond about a quarter of the fastest period.

Moving mass from heavy atoms to their hydrogens slows the fastest motions by a factor of about 1.6 to 1.7 while keeping the forces and, for free or rigid molecules, every average over arrangements, and lengthens the step of rigid water by a further 1.7. All these gains are factors of a few: the Verlet-based methods used here still need a step short enough to resolve the fastest motion they keep.

A healthy run shows K and U mirroring each other, a narrow band of total energy without drift, bonds at the tolerance and a still centre of mass; eight common faults each leave their own mark, caught by the checks of `mdlab.diagnostics` and of the record.

Chapter 12 turns from single trajectories to the statistics of many states: what temperature is, why the kinetic energy is shared equally among the freedoms counted here, and what ensemble a simulation samples.

Answers to the checks

SECTION 11.1. No: Equation (11.1) contains no mass, so $\eta = -0.2$ again. Now $m_r = 1$, and each atom moves by $m_r |\eta| / 2 = 0.1$ along the bond, towards the other.

SECTION 11.2. $\mathbf{v}_{ij} = (-0.2, 0.2, 0)$ and $\mathbf{r}_{ij} = (1, 0, 0)$, so $\zeta = 0.2$; with $m_r = 1$, $\mathbf{v}_j$ gains 0.1 along x and $\mathbf{v}_i$ loses it: $\mathbf{v}_i = \mathbf{0}$ and $\mathbf{v}_j = (0, 0.2, 0)$. The total momentum is $(0, 0.4, 0)$ before and after: zero along x , 2×0.2 along y .

SECTION 11.3. $10^6 / 0.41 = 2.44 \times 10^6$ and $10^6 / 1.98 = 5.05 \times 10^5$ steps.

SECTION 11.4. Velocity Verlet makes 2430 steps, each calling both forces, $24 + 0.04 = 24.04$ ms: about 58 s. RESPA makes 1380 calls of the slow forces, 33 s, and 3×1380 calls of the springs, 0.17 s more.

SECTION 11.5. $b = 0.01$, so the band runs from $(1 - b) \times 4.5 \text{ fs} = 4.455 \text{ fs}$ to 4.5 fs , 0.045 fs wide.

SECTION 11.6. Carbon keeps $12.011 - 2.016 = 9.995 \text{ amu}$; the reduced mass goes from 0.9300 to 2.3216 amu, 2.50 times larger, and the stretch slows by $\sqrt{2.50} = 1.58$.

SECTION 11.7. $296 \times 1000 = 2.96 \times 10^5$ for rigid water with heavier hydrogens, and $2430 \times 1000 = 2.43 \times 10^6$ for flexible water by velocity Verlet; the table's calls come from the unrounded steps, so $10^6 / 0.41 = 2.44 \times 10^6$ differs in the last figure.

SECTION 11.8. Velocities about ten times too small, since $1/0.0097 = 103$ is close to 10.18^2 : velocities in $\text{\AA fs}^{-1}$ taken as ångströms per ASE time unit, the factor of 10.18 fs per ASE time unit missed.

Exercises

The exercises run from questions of understanding, through calculations, to short derivations marked ‘show that’. Full solutions are in ‘Solutions to the exercises’ at the end of the book, and Notebook 11 checks every numerical answer.

EXERCISE 11.1 (*freedoms of methanol*). Methanol, CH_3OH , has six atoms and four bonds to hydrogen. How many freedoms has one molecule with those four bonds held, and how many have 100 such molecules once the drift of their centre of mass is removed?

EXERCISE 11.2 (*one bond solved exactly*). Atoms i and j , of masses 2 and 5 amu, are joined by a bond held at $1.5 \text{\AA}$, with $\mathbf{r}_{ij} = (1.5, 0, 0) \text{\AA}$ at the start of a step. The provisional bond is $\tilde{\mathbf{r}}_{ij} = (1.6, 0.3, 0) \text{\AA}$. Solve Equation (11.1) for η , choose the root, and find how far each atom moves.

EXERCISE 11.3 (*Newton’s method for a square root*). Apply Newton’s method to $f(y) = y^2 - 3$ from $y_0 = 2$, and give y_1, y_2, y_3 and their errors against $\sqrt{3}$.

EXERCISE 11.4 (*the error squares*). Show that for $f(y) = y^2 - a$, with $a > 0$, Newton’s method gives $y_{n+1} = \frac{1}{2}(y_n + a/y_n)$, and that $y_{n+1} - \sqrt{a} = (y_n - \sqrt{a})^2 / (2y_n)$ exactly.

EXERCISE 11.5 (*SHAKE keeps the centre of mass*). Show that a SHAKE correction of one bond, for any η , leaves the centre of mass of its two atoms where it was.

EXERCISE 11.6 (*what the velocity correction removes*). For two atoms, show that their kinetic energy is $\frac{1}{2}M|\mathbf{V}|^2 + \frac{1}{2}m_r|\mathbf{v}_{ij}|^2$, with M and $\mathbf{V}$ their total mass and the velocity of their centre of mass. Hence show that the velocity correction of RATTLE leaves the total momentum unchanged and lowers the kinetic energy by $\frac{1}{2}m_r(\mathbf{v}_{ij} \cdot \mathbf{e})^2$, with $\mathbf{e}$ the unit vector along the bond.

EXERCISE 11.7 (*correcting an O–H pair*). An oxygen atom of mass 16 amu and a hydrogen of mass 1 amu are joined by a bond held at $0.96 \text{\AA}$, with $\mathbf{r}_\text{H} - \mathbf{r}_\text{O} = (0.96, 0, 0) \text{\AA}$, and velocities $\mathbf{v}_\text{O} = (0.001, 0, 0)$ and $\mathbf{v}_\text{H} = (0.02, 0.01, 0) \text{\AA/fs}$. Find ζ , the corrected velocities and the kinetic energy removed, in $\text{amu \AA}^2/\text{fs}^2$.

EXERCISE 11.8 (*a larger box*). How many freedoms have 216 rigid water molecules once the drift is removed, and what kinetic energy, in eV, gives each of them 12.93 meV?

EXERCISE 11.9 (*one inner step*). Show that RESPA with one inner step is velocity Verlet with the force $\mathbf{F}^f + \mathbf{F}^s$.

EXERCISE 11.10 (*the cost of a nanosecond*). With 24 ms per call of the forces between molecules and 0.04 ms per call of the springs, how long do velocity Verlet at 0.41 fs and RESPA with outer steps of 0.72 fs and three inner steps take for a nanosecond of flexible water?

EXERCISE 11.11 (*the square of a matrix*). Check that $\mathbf{M}^2 = (\text{tr } \mathbf{M})\mathbf{M} - (\det \mathbf{M})\mathbf{I}$ for the matrix with rows (1, 2) and (3, 4).

EXERCISE 11.12 (*the second band*). Show that, for small $b = (\omega_s/\omega_f)^2$, the trace of Equation (11.3) exceeds 2 for θ from $2\pi(1 - b)$ to 2π , a band of width $b \mathcal{T}_f$ in the outer step.

EXERCISE 11.13 (*predicted and measured bands*). For $\omega_s = 0.3 \omega_f$, give the predicted edges of the two bands of Figure 11.5, in units of $\mathcal{T}_f$, and compare them with 0.459 to 0.500 and 0.919 to 1.000 from the full trace.

EXERCISE 11.14 (*a finer inner step*). With outer steps of 2 fs, RESPA on flexible water gives an energy fluctuation of 8.8% with inner steps of 0.25 fs and 10.4% with 0.125 fs. Explain why the finer inner step does not help.

EXERCISE 11.15 (*hydrogen doubled*). The hydrogen atoms of water are made twice as heavy, 2.016 amu, at the oxygen's expense. Find the new mass of the oxygen, the O–H reduced mass, and the factor by which the stretch slows.

EXERCISE 11.16 (*where the centre of mass goes*). How far from the oxygen, along the line bisecting the H–O–H angle, is the centre of mass of a TIP3P molecule with ordinary masses, and with the hydrogens three times as heavy?

EXERCISE 11.17 (*what repartitioning changes*). Which of these change when hydrogen masses are repartitioned: the mean potential energy, the distribution of distances between oxygen atoms, the rate at which molecules diffuse, the frequencies of the vibrations?

EXERCISE 11.18 (*a nanosecond of drift*). The faulty start of Section 11.8 gives 64 water molecules of 18.015 amu a total momentum $\mathbf{P}$ with components 0.2817, 0.2936 and 0.4033 amu Å/fs. How far does the centre of mass move in a nanosecond, and how much kinetic energy does the drift carry?

EXERCISE 11.19 (*a step without its unit*). ASE measures time in units of 10.18 fs. A step written as 2 without converting it to that unit is how long, and what is $\omega \delta t$ for the O–H stretch of period 8.88 fs? Is velocity Verlet stable?

EXERCISE 11.20 (*naming the fault*). A run of rigid water at 2 fs has a mean total energy that falls steadily by 0.5 eV ps^{-1} , by the same amount at 1 fs; its bond errors are $10^{-10} \text{ \AA}$ and its centre of mass is still. Which fault of Table 11.2 fits, and what test of the model would confirm it?

Ensembles

A trajectory records where the atoms move, but the quantities we seek are usually averages: the temperature of a liquid, its mean density or the rate at which an atom hops between sites. To interpret these averages we need statistical mechanics, which describes the probabilities of the states available to a system. We begin with probability and ask when an average along one trajectory can represent an average over those states. Examples for which it cannot make the limitation explicit.

Counting the states of an isolated system then gives definitions of entropy and temperature. Allowing the system to exchange energy with its surroundings leads to the canonical distribution, from which we obtain the velocity distribution and the sharing of kinetic energy among the degrees of freedom counted in Chapters 5 and 11. Liquid-argon runs let us compare these predictions with measured temperatures and fluctuations. Chapter 13 uses the resulting distributions to test thermostats: holding the mean temperature is only one part of sampling the required ensemble.

12.1 Probability

A liquid of a few hundred atoms passes through more arrangements in a picosecond than could ever be listed, and what can be said about it is what fraction of the time, or of many copies of it, has some property. That is the language of probability. This section sets out the few rules of it that the chapter uses, on a pair of dice first and then on a quantity that varies continuously.

PROBABILITY. A trial, such as a throw of two dice, has outcomes. Repeated many times, the fraction of trials with a given outcome settles towards a number, its probability $\mathcal{P}$, which lies between 0 and 1; the probabilities of all the distinct outcomes add up to 1, since every trial has one of them. When the outcomes are

equally likely, a probability is a count: two dice have $6 \times 6 = 36$ equally likely outcomes, of which $6 - |s - 7|$ give the total s , for s from 2 to 12: the first die may show any face that leaves the second a face from 1 to 6, which is $s - 1$ faces for $s \leq 7$ and $13 - s$ for $s \geq 7$. So $\mathcal{P}(s) = (6 - |s - 7|)/36$, largest at $s = 7$, with $6/36$. Two outcomes of separate trials are independent if knowing one does not change the probability of the other; then the probability of both is the product of the two, as the 36 outcomes of two dice are the 6×6 pairs, each with probability $\frac{1}{6} \times \frac{1}{6}$.

Figure 12.1(a) throws two dice by computer. After 100 throws the fractions miss the probabilities by up to 0.051; after 10 000 by up to 0.0033. A fraction is only an estimate of a probability, and a better one the more trials it rests on.

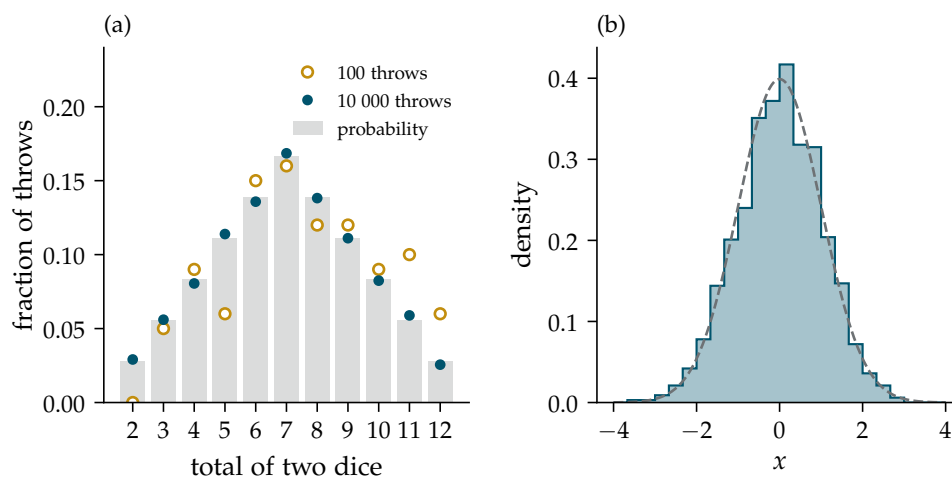


FIGURE 12.1: Frequencies settle on probabilities. (a) The fraction of throws of two dice giving each total, after 100 throws (open) and 10 000 throws (filled), against the probabilities (bars). (b) The histogram of 1000 numbers drawn from the normal distribution, as a density, against the Gaussian of mean 0 and standard deviation 1 (dashed).

Averages and spreads

EXPECTATION AND VARIANCE. A quantity x that takes the value x_k with probability $\mathcal{P}_k$ has the expectation, or mean, $\langle x \rangle = \sum_k x_k \mathcal{P}_k$, the value its average over many trials settles towards, and any function of it the expectation $\langle f(x) \rangle = \sum_k f(x_k) \mathcal{P}_k$. The variance measures the spread about the mean,

$$\begin{aligned} \text{var}(x) &= \langle (x - \langle x \rangle)^2 \rangle = \langle x^2 - 2\langle x \rangle x + \langle x \rangle^2 \rangle \\ &= \langle x^2 \rangle - 2\langle x \rangle \langle x \rangle + \langle x \rangle^2 = \langle x^2 \rangle - \langle x \rangle^2, \end{aligned}$$

expanding the square and then averaging term by term, which uses three properties of $\sum_k f(x_k) \mathcal{P}_k$: the expectation of a sum is the sum of the expectations, a constant

factor such as $\langle x \rangle$ comes outside, and a constant c averages to $c \sum_k \mathcal{P}_k = c$. Its square root is the standard deviation σ_x , in the units of x (written with the quantity as a subscript, to keep it apart from the Lennard-Jones distance σ). For independent x and y the probability of each pair is the product, so

$$\langle xy \rangle = \sum_{k,l} x_k y_l \mathcal{P}(x_k) \mathcal{P}(y_l) = \left(\sum_k x_k \mathcal{P}(x_k) \right) \left(\sum_l y_l \mathcal{P}(y_l) \right) = \langle x \rangle \langle y \rangle,$$

and by the same expansion with $x + y$ in place of x ,

$$\begin{aligned} \text{var}(x + y) &= \langle x^2 \rangle + 2\langle xy \rangle + \langle y^2 \rangle - \langle x \rangle^2 - 2\langle x \rangle \langle y \rangle - \langle y \rangle^2 \\ &= \text{var}(x) + \text{var}(y) + 2(\langle xy \rangle - \langle x \rangle \langle y \rangle) = \text{var}(x) + \text{var}(y) : \end{aligned}$$

the variances of independent quantities add.

One die has the mean $(1 + 2 + \cdots + 6)/6 = 3.5$ and the variance $35/12$ ([Exercise 12.1](#)); two independent dice have twice each, 7 and $35/6 = 5.8333$, and the 10 000 throws above give 6.997 and 5.846.

Densities

A velocity or a position takes a continuum of values, and the probability of any exact value is zero. It has instead a probability density $\mathcal{P}(x)$: the probability that x lies between a and b is $\int_a^b \mathcal{P}(x) dx$, so a narrow range of width Δx at x holds the probability $\mathcal{P}(x) \Delta x$. Cutting all the values into such ranges and letting them narrow, the sums above become integrals: $\int \mathcal{P} dx = 1$ over all values, and $\langle f(x) \rangle = \int f(x) \mathcal{P}(x) dx$. A histogram estimates a density: the fraction of samples in a bin of width Δx , divided by Δx , approaches $\mathcal{P}(x)$ as the samples grow many and the bins narrow. The most important density of the book is the Gaussian, or normal, density,

$$\mathcal{P}(x) = \frac{1}{\sigma_x \sqrt{2\pi}} e^{-(x - \langle x \rangle)^2 / (2\sigma_x^2)}. \quad (12.1)$$

Its integral is 1, by the Gaussian integral $\int e^{-au^2} du = \sqrt{\pi/a}$ of [Section 10.3](#) with $u = x - \langle x \rangle$ and $a = 1/(2\sigma_x^2)$ ([Exercise 10.6](#)). It is symmetric about $\langle x \rangle$: the values $\langle x \rangle + u$ and $\langle x \rangle - u$ have the same density, so their deviations from $\langle x \rangle$ cancel in the average, and $\langle x \rangle$ is its mean. Its variance follows by integration by parts ([Section 6.3](#)), with

$u = x - \langle x \rangle$, $s = \sigma_x$, $f = u$ and $g' = u e^{-u^2/(2s^2)}$, whose integral is $g = -s^2 e^{-u^2/(2s^2)}$, as the chain rule confirms:

$$\begin{aligned} \int u^2 e^{-u^2/(2s^2)} du &= \left[-s^2 u e^{-u^2/(2s^2)} \right]_{-\infty}^{\infty} + s^2 \int e^{-u^2/(2s^2)} du \\ &= 0 + s^2 \times s\sqrt{2\pi}, \end{aligned}$$

and the bracket vanishes at both ends, where the exponential falls faster than u grows; dividing by $s\sqrt{2\pi}$, the variance is σ_x^2 . The fraction of the probability within k standard deviations of the mean follows with $w = u/(\sigma_x\sqrt{2})$, so that $du = \sigma_x\sqrt{2}dw$ and the limits $\pm k\sigma_x$ become $\pm k/\sqrt{2}$:

$$\frac{1}{\sigma_x\sqrt{2\pi}} \int_{-k\sigma_x}^{k\sigma_x} e^{-u^2/(2\sigma_x^2)} du = \frac{1}{\sqrt{\pi}} \int_{-k/\sqrt{2}}^{k/\sqrt{2}} e^{-w^2} dw = \text{erf}(k/\sqrt{2}),$$

the last step since e^{-w^2} takes the same value at w and $-w$, with the error function of [Section 10.3](#): 68.27%, 95.45% and 99.73% for $k = 1, 2$ and 3 . The function `rng.normal` that drew the starting velocities of [Section 10.5](#) returns numbers with this density, mean 0 and standard deviation 1; [Figure 12.1\(b\)](#) draws 1000 of them, whose mean is 0.0029 and standard deviation 1.0115, and whose histogram departs from [Equation \(12.1\)](#) by up to 0.042 per unit. Why velocities should be Gaussian at all is the subject of [Section 12.5](#).

KEY IDEA. A probability is the long-run fraction of trials with an outcome; the probabilities of independent outcomes multiply. A quantity has a mean $\langle x \rangle$ and a variance $\langle x^2 \rangle - \langle x \rangle^2$, and the variances of independent quantities add. A continuous quantity has a density, which a histogram estimates; the Gaussian density of [Equation \(12.1\)](#) has mean $\langle x \rangle$ and variance σ_x^2 .

NOTEBOOK 12 *throw dice and draw normal numbers, and watch the histograms settle*

CHECK YOUR UNDERSTANDING. What is the probability that two dice total at least 10?

12.2 Many states, one system

A simulation follows one system along one trajectory, and its averages are averages over time. Statistical mechanics computes averages over the states the system could be in, weighted by their probabilities. This section sets out the two kinds of average and the assumption that they agree, and then shows systems for which they do not.

Microstates, macrostates and ensembles

The state of N atoms is a point $\Gamma = (\mathbf{r}^N, \mathbf{p}^N)$ of phase space (Section 7.2), called a microstate. What is measured of a macroscopic sample, its energy, volume and number of atoms, or its temperature and pressure, fixes a macrostate, which an enormous number of microstates share. An ensemble is the cloud of copies of Section 7.4: many imagined copies of the system, all in the same macrostate and spread over its microstates, with a probability density $\mathcal{P}(\Gamma)$ in phase space: a small box of phase space, of volume $d\Gamma$, the product of its $6N$ sides, holds the fraction $\mathcal{P}(\Gamma) d\Gamma$ of the copies. The ensemble average of a quantity $A(\Gamma)$ is

$$\langle A \rangle = \int A(\Gamma) \mathcal{P}(\Gamma) d\Gamma,$$

an integral over all $6N$ coordinates and momenta, the limit of a sum over such boxes, as for the volume integrals of Section 10.3, and the time average along one trajectory is

$$\bar{A} = \lim_{t_f \rightarrow \infty} \frac{1}{t_f} \int_0^{t_f} A(\Gamma(t)) dt.$$

Section 7.4 showed that a cloud whose density depends on the state only through its energy, $\mathcal{P} = f(H)$, does not change as the copies move. The ensembles of statistical mechanics are such clouds, and the hypothesis that makes them useful to a simulation is the ergodic hypothesis: that a single trajectory, followed long enough, passes near every microstate of its energy, spending equal times near equal volumes of phase space at that energy, so that $\bar{A}$ equals $\langle A \rangle$ for the ensemble spread evenly over those states (Section 12.3). It can be tested, and it fails for some systems.

A chain that never shares

N atoms of unit mass sit in a line between two fixed walls, each joined to its neighbours by springs of unit stiffness. With q_j the displacement of atom j and $q_0 = q_{N+1} = 0$ at the walls, the spring on the left of atom j is stretched by $q_j - q_{j-1}$ and pulls it back with $-(q_j - q_{j-1})$, and the spring on its right is stretched by $q_{j+1} - q_j$ and pulls it on with $q_{j+1} - q_j$, as for the carts of Section 4.9; together they pull it with the force $q_{j-1} - 2q_j + q_{j+1}$. A normal mode has every atom moving as $u_j \cos(\omega t + \phi)$, with the acceleration $-\omega^2 u_j \cos(\omega t + \phi)$, which for unit mass must equal the force; cancelling the common factor $\cos(\omega t + \phi)$, this needs

$$-\omega^2 u_j = u_{j-1} - 2u_j + u_{j+1}.$$

With $u_j = \sin j\theta$, the addition formulae (Equation (1.6)) give $u_{j-1} + u_{j+1} = \sin(j\theta - \theta) + \sin(j\theta + \theta) = 2 \sin j\theta \cos \theta$, so the condition holds with

$$\omega^2 = 2 - 2 \cos \theta = 4 \sin^2 \frac{1}{2} \theta,$$

using $1 - \cos \theta = 2 \sin^2 \frac{1}{2} \theta$ (Exercise 1.5). The wall at $j = 0$ is met, since $\sin 0 = 0$, and the one at $j = N + 1$ needs $\sin(N + 1)\theta = 0$, so $\theta = k\pi/(N + 1)$ for a whole number k . The values $k = 1$ to N give N different patterns, as many as the chain has modes (Section 4.9); $k = N + 1$ gives $\sin j\pi = 0$ for every atom, no motion at all, and larger k give the same patterns again, up to sign. The modes have the angular frequencies $\omega_k = 2 \sin(k\pi/(2N + 2))$, the positive root, since $k\pi/(2N + 2)$ lies between 0 and $\pi/2$. For $N = 32$, the slowest repeats every 66.0 units of time and the fastest every 3.15. As Section 4.9 showed, every motion of the harmonic chain is a sum of normal modes, each keeping its own amplitude, so a chain started with its energy in one mode keeps it there for ever: two starts with the same energy E , one in the first mode and one in the second, give the energy in the first mode the time averages E and 0. The harmonic chain is not ergodic.

Real bonds are not perfect springs. Adding an energy $\frac{1}{3}\alpha x^3$ to each spring stretched by x couples the modes, and the question is whether the coupling spreads the energy among them. Fermi, Pasta and Ulam asked it in what Dauxois calls the first numerical experiment, programmed and run by Mary Tsingou⁵². Figure 12.2(a) repeats it with `mdlabs.chain`, for 32 atoms with $\alpha = 0.25$ and the energy 0.0724 put into the first mode, followed by velocity Verlet with steps of 0.05, over which the total energy varies by no more than 4.9×10^{-5} of itself. The first mode's share falls below 0.1 by $t = 3950$, the energy passing to modes 2, 3 and 4 and back, and by $t = 10\,225$ it has returned to 0.981. Over the whole run, to $t = 20\,000$, about 300 periods of the first mode, modes 5 to 32 never hold more than 0.198 of the energy; spread equally, they would hold $28/32 = 0.875$. A trajectory started in this special state keeps its energy in the few slowest modes, passing it among them and back to the first, for the whole run; a run a hundred times longer, to $t = 2 \times 10^6$, still leaves only 0.086 of it on average in modes 5 to 32, and never more than 0.220.

Time averages in a liquid

The liquid provides a contrasting test over a finite observation time. Figure 12.2(b) follows 256 atoms of liquid argon, the model of Section 10.6 at the density $0.8\sigma^{-3}$ and about 135 K (a temperature in the sense of Section 12.6), prepared as in Section 11.3, for 50 ps at fixed energy. The running time average of one atom's kinetic energy wanders at first and then settles towards the average over all atoms and saved times. If the atoms sample the same stationary distribution, this pooled average estimates the ensemble

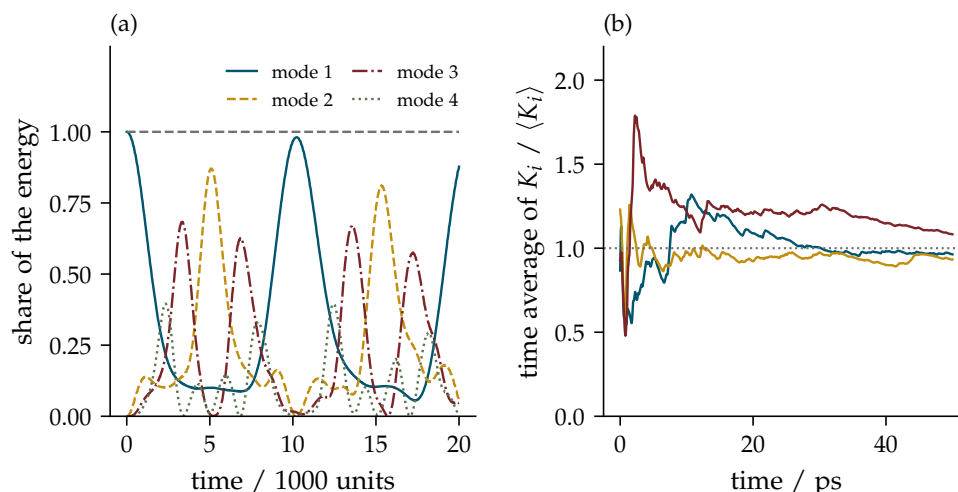


FIGURE 12.2: Time averages. (a) A chain of 32 atoms started with its energy in the first normal mode: the share of the energy in modes 1 to 4 against time, with the coupling $\alpha = 0.25$ (the four labelled curves), and for the harmonic chain, whose first mode keeps all of it (grey dashed). (b) Liquid argon, 256 atoms: the running time average of the kinetic energy of three single atoms, divided by the average over all atoms and times (dotted at 1).

average: after 50 ps the time averages of the 256 atoms lie between 0.807 and 1.249 of it, with a standard deviation of 0.076. They are still settling, and how long a run must be for an average to be trusted is the question of Chapter 15. A crystal at a low temperature, whose vibrations are nearly harmonic, is closer to the chain than to the liquid, and sharing in it can be correspondingly slow.

IN PRACTICE. A run that starts in a special state, such as a crystal on its lattice or energy given to one motion only, may keep the memory of that start. The start should be one the system could reach from many others, and its averages should not depend on which of several such starts is used.

KEY IDEA. An ensemble is a cloud of copies of a system with a density $\mathcal{P}(\Gamma)$ in phase space, and $\langle A \rangle$ its average; a simulation measures the time average $\bar{A}$. The ergodic hypothesis, that the two agree, is consistent with liquid argon over 50 ps; it fails for the harmonic chain, whose modes never share energy. The chain of Fermi, Pasta, Ulam and Tsingou also fails to share its energy equally over the observation times tested here, returning much of it to the first mode; that finite-time result does not establish its behaviour at infinite time.

NOTEBOOK 12 run the chain with any coupling and watch the modes share or not

CHECK YOUR UNDERSTANDING. Why is the period of the fastest mode of a long chain close to π ?

12.3 Counting states at fixed energy

A system that exchanges nothing with its surroundings keeps its energy, as the runs of Chapters 9 to 11 did. This section gives the ensemble of such a system, defines its entropy by counting its states, and finds that temperature is what two systems sharing energy come to have in common.

The microcanonical ensemble

Of all the clouds $\mathcal{P} = f(H)$ that the motion leaves unchanged, an isolated system with energy E calls for the one spread evenly over the states of that energy, those with H between E and $E + \delta E$ for a small δE , and zero elsewhere. If the system is ergodic, its trajectory spends equal times near equal volumes of that shell, so the even spread is the one its time averages sample. This is the microcanonical ensemble. The volume of the shell, $\Omega(E)$, measured in some fixed unit of phase-space volume, whose choice adds only a constant to S , counts the microstates of energy E (it is not the bold matrix Ω of [Section 9.3](#)), and Boltzmann's entropy is

$$S = k_B \ln \Omega, \quad (12.2)$$

with Boltzmann's constant $k_B = 8.617333262 \times 10^{-5} \text{ eV K}^{-1}$, the value SciPy gives. Taking the logarithm turns the counts of two independent systems, which multiply, into entropies, which add.

Sharing energy

Two systems placed in contact exchange energy, the total $E = E_1 + E_2$ staying fixed. Every microstate of the first combines with every microstate of the second, so the pair has $\Omega_1(E_1) \Omega_2(E - E_1)$ microstates in which the first holds E_1 , and in the microcanonical ensemble of the pair the probability of that division is proportional to this number. The most probable division makes it largest, or its logarithm, $\ln \Omega_1(E_1) + \ln \Omega_2(E - E_1)$, largest, which is the same division since the logarithm rises whenever its argument does; this needs its derivative with respect to E_1 to vanish:

$$\frac{d \ln \Omega_1}{dE_1} - \frac{d \ln \Omega_2}{dE_2} = 0,$$

by the chain rule, with $E_2 = E - E_1$ and $dE_2/dE_1 = -1$. The two systems end where each has the same value of $d \ln \Omega / dE$, which is therefore what they have in common in equilibrium. We call it $1/k_B T$:

$$\frac{1}{k_B T} = \frac{d \ln \Omega}{dE}, \quad \text{or} \quad \frac{1}{T} = \frac{dS}{dE}, \quad (12.3)$$

which defines the temperature T , in kelvin. When the first system's $d \ln \Omega / dE$ is the larger, moving energy into it raises the product, so energy flows from the system with the smaller $1/T$, the hotter one, to the colder, as it is found to.

The ideal gas

N atoms of mass m in a box of volume V , with no forces between them, have the energy $E = \sum_i |\mathbf{p}_i|^2 / 2m$ whatever their positions. Each atom can be anywhere in the box, so the positions contribute the factor V^N . The momenta with energy below E are those with $p_1^2 + p_2^2 + \cdots + p_{3N}^2 \leq 2mE$, listing all $3N$ components; by Pythagoras' theorem extended to $3N$ directions, these are the points within the distance $R = \sqrt{2mE}$ of the origin, the inside of a sphere of radius R in $3N$ dimensions. That sphere is the sphere of radius 1 with every coordinate stretched by the factor R , and stretching all $3N$ coordinates by R multiplies the volume of every small box, the product of its $3N$ sides, by R^{3N} , so the volume inside it is cR^{3N} , with c the volume inside the sphere of radius 1, a number that depends only on the dimension. The number of states with energy below E is therefore $cV^N(2mE)^{3N/2}$, and those between E and $E + \delta E$ number its derivative times δE :

$$\Omega(E) = c V^N (2m)^{3N/2} \frac{3N}{2} E^{3N/2-1} \delta E,$$

$$\ln \Omega = \left(\frac{3N}{2} - 1\right) \ln E + \text{a constant},$$

$$\frac{1}{k_B T} = \frac{3N/2 - 1}{E} \approx \frac{3N}{2E} \quad \text{by Equation (12.3), for large } N.$$

The third line uses the slope $1/E$ of $\ln E$: differentiating $e^{\ln E} = E$ by the chain rule gives $E d(\ln E)/dE = 1$. The energy of the gas is $E = \frac{3}{2} N k_B T$: $\frac{3}{2} k_B T$ for each atom, $\frac{1}{2} k_B T$ for each direction of its motion, a sharing that [Section 12.6](#) finds to hold far more widely.

How sharp the sharing is

Two such gases of N atoms each, sharing E , have $\Omega_1 \Omega_2 \propto E_1^{3N/2} (E - E_1)^{3N/2}$ for large N , so the probability that the first holds the fraction $x = E_1/E$ of the energy

is proportional to $x^{3N/2}(1-x)^{3N/2}$. It is largest at $x = \frac{1}{2}$, equal energies and equal temperatures, and Figure 12.3 shows how it narrows as N grows. Near the peak, the Taylor series of its logarithm (Section 1.8) gives its width:

$$\begin{aligned}\ln \mathcal{P} &= \frac{3N}{2} [\ln x + \ln(1-x)] + \text{a constant}, \\ \frac{d \ln \mathcal{P}}{dx} &= \frac{3N}{2} \left[\frac{1}{x} - \frac{1}{1-x} \right] = 0 && \text{at } x = \frac{1}{2}, \\ \frac{d^2 \ln \mathcal{P}}{dx^2} &= -\frac{3N}{2} \left[\frac{1}{x^2} + \frac{1}{(1-x)^2} \right] = -12N && \text{at } x = \frac{1}{2}, \\ \ln \mathcal{P} &\approx \text{a constant} - \frac{1}{2} \times 12N \left(x - \frac{1}{2} \right)^2,\end{aligned}$$

the chain rule giving the slope $-1/(1-x)$ of $\ln(1-x)$, and the last line being the Taylor series to second order, whose first-order term is zero at the peak. That last line is the logarithm of a Gaussian, Equation (12.1), with $1/(2\sigma_x^2) = 6N$: the share has the standard deviation $\sigma_x = 1/\sqrt{12N}$. For 100 atoms in each gas that is 0.0289, against 0.0288 for the curve of Figure 12.3, which counts the states in full, computed numerically; for a thousand, 0.0091; for a macroscopic sample of 10^{23} atoms, 9×10^{-13} . A sample large enough to see is never found far from its most probable division, which is why its temperature is a definite number.

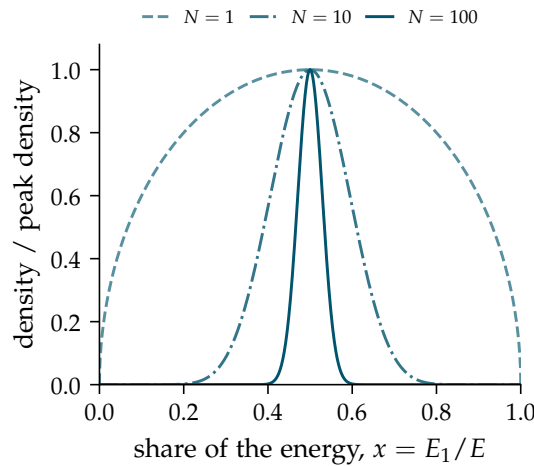


FIGURE 12.3: Two gases of N atoms each sharing a fixed energy: the probability density for the first to hold the fraction x of it, proportional to $x^{3N/2-1}(1-x)^{3N/2-1}$ by the full count $\Omega \propto E^{3N/2-1}$, each curve divided by its largest value, for $N = 1, 10$ and 100 .

KEY IDEA. An isolated system, if it is ergodic, samples its energy shell evenly; the volume $\Omega(E)$ of the shell counts its states, and $S = k_B \ln \Omega$ is its entropy. Two systems sharing energy settle at the division with the most states, where $d \ln \Omega / dE$, which defines $1/k_B T$, is the same for both. An ideal gas holds $\frac{3}{2}k_B T$ per atom, and the division of energy between large systems is sharp, with a relative spread of order $1/\sqrt{N}$.

NOTEBOOK 12 *share energy between two gases of any size*

CHECK YOUR UNDERSTANDING. For two gases of 10 atoms each, what does $1/\sqrt{12N}$ give for the standard deviation of the share?

12.4 A system and its bath

Most systems of interest are not isolated: a sample exchanges energy with its container, and a few hundred atoms in a simulation stand for a small part of a much larger piece of matter. This section finds the probabilities of the states of a system kept in contact with a much larger one, its bath, and with them the partition function from which averages and fluctuations follow.

The Boltzmann factor

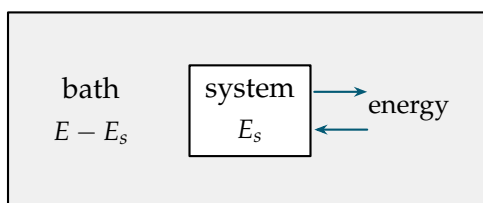


FIGURE 12.4: A small system inside a large bath. The two together are isolated, with energy E ; the system holds E_s and the bath the rest, and energy passes between them.

The system and its bath (Figure 12.4) together are isolated, with the energy E , so the pair is microcanonical: all its microstates of energy E are equally likely. We fix one microstate of the system, with energy E_s . The bath then holds $E - E_s$, in any of its $\Omega_B(E - E_s)$ microstates, so the probability of that one microstate of the system is proportional to $\Omega_B(E - E_s)$. The system's energy is small against the bath's, so we expand about E in a Taylor series. Ω_B itself changes enormously with its energy, roughly as $E^{3N/2}$ for an ideal gas (Section 12.3), but its logarithm changes slowly, so it is the logarithm we expand:

$$\begin{aligned}\ln \Omega_B(E - E_s) &= \ln \Omega_B(E) - E_s \frac{d \ln \Omega_B}{dE} + \frac{1}{2} E_s^2 \frac{d^2 \ln \Omega_B}{dE^2} - \dots \\ &= \ln \Omega_B(E) - \frac{E_s}{k_B T} + \dots,\end{aligned}$$

the second line by Equation (12.3), with T the temperature of the bath. The second derivative is the rate at which $d \ln \Omega_B / dE = 1/k_B T$ changes, $-(1/k_B T^2) dT/dE$ by the chain rule, so the next term is $-\frac{1}{2} E_s^2 (dT/dE) / k_B T^2$; divided by the term kept, $-E_s/k_B T$, it gives the ratio $\frac{1}{2} E_s (dT/dE) / T$, in which dT/dE is how far the bath's temperature

risks per unit of energy it gains. For a bath that is an ideal gas, $T = 2E/3Nk_B$ gives $dT/dE = 2/3Nk_B = T/E$, and the ratio is $E_s/2E$, half the system's share of the energy: negligible for a bath much larger than the system (Exercise 12.5), and the dropped term itself, $(E_s/k_B T)(E_s/2E)$, is small when $E \gg E_s^2/k_B T$. The factor $\Omega_B(E)$ is the same for every microstate of the system and joins the constant of proportionality, so, taking the exponential of both sides,

$$\mathcal{P}(\text{microstate}) \propto e^{-E_s/k_B T} = e^{-\beta E_s}, \quad \beta = \frac{1}{k_B T} : \quad (12.4)$$

the probability of a state of the system falls exponentially with its energy, by the Boltzmann factor. The bath enters only through its temperature. The ensemble of a system at fixed N , V and T with these probabilities is the canonical ensemble.

The partition function

The probabilities must add up to one, so the probability of microstate s is $\mathcal{P}_s = e^{-\beta E_s}/Z$, with Z the sum over all the system's microstates, or for the continuous phase space of classical atoms the integral over it, the partition function:

$$Z = \sum_s e^{-\beta E_s}, \quad \text{or} \quad Z = \int e^{-\beta H(\Gamma)} d\Gamma. \quad (12.5)$$

with $d\Gamma$ measured in the unit of phase-space volume of Section 12.3, so that Z is a pure number; changing that unit adds a constant to S and $k_B T$ times a constant to F , which drop out of every difference and derivative below. Averages follow from it by differentiation with respect to β :

$$\begin{aligned} -\frac{d \ln Z}{d\beta} &= -\frac{1}{Z} \frac{dZ}{d\beta} = \frac{1}{Z} \sum_s E_s e^{-\beta E_s} = \langle E \rangle, \\ \frac{d^2 \ln Z}{d\beta^2} &= -\frac{d\langle E \rangle}{d\beta} = \frac{1}{Z} \sum_s E_s^2 e^{-\beta E_s} - \left(\frac{1}{Z} \sum_s E_s e^{-\beta E_s} \right)^2 = \langle E^2 \rangle - \langle E \rangle^2, \end{aligned}$$

the first by the chain rule, with $dZ/d\beta = -\sum_s E_s e^{-\beta E_s}$, and the second by the product rule (Section 2.7) applied to $\langle E \rangle = (1/Z) \sum_s E_s e^{-\beta E_s}$: the sum has the derivative $-\sum_s E_s^2 e^{-\beta E_s}$, and $1/Z$, by the chain rule, the derivative $-(1/Z^2) dZ/d\beta = (1/Z^2) \sum_s E_s e^{-\beta E_s}$. The energy of a system in contact with a bath fluctuates, and its variance is the rate at which $-\langle E \rangle$ changes with β . The heat capacity $C_V = d\langle E \rangle/dT$ is the energy taken up per degree, at fixed volume. By the chain rule $C_V = (d\langle E \rangle/d\beta)(d\beta/dT)$, and $d\beta/dT = -1/(k_B T^2)$, so

$$\langle E^2 \rangle - \langle E \rangle^2 = -\frac{d\langle E \rangle}{d\beta} = k_B T^2 C_V. \quad (12.6)$$

The size of the energy's fluctuations measures how much energy the system takes up when warmed. The free energy is defined by

$$F = -k_B T \ln Z,$$

and equals $\langle E \rangle - TS$ with the entropy $S = -k_B \sum_s \mathcal{P}_s \ln \mathcal{P}_s$, which for the equal probabilities $1/\Omega$ of the microcanonical ensemble is Equation (12.2) (Exercise 12.7).

A spring, and the odds of a hop

One coordinate on a spring, $H = p^2/2m + \frac{1}{2}kx^2$, has, by the Gaussian integral of Section 10.3 twice,

$$Z = \int e^{-\beta p^2/2m} dp \int e^{-\beta kx^2/2} dx = \sqrt{\frac{2\pi m}{\beta}} \sqrt{\frac{2\pi}{\beta k}} = \frac{2\pi}{\beta \omega},$$

with $\omega = \sqrt{k/m}$. Then $\ln Z = -\ln \beta + \text{a constant}$, and $\langle E \rangle = -d \ln Z / d\beta = 1/\beta = k_B T$: a classical vibration holds $k_B T$ on average, whatever its stiffness, which makes precise the 'typical energy of order $k_B T$ ' of Section 8.2. At 300 K, $k_B T = 0.02585$ eV.

The Boltzmann factor also answers the question of Section 3.13, how often an atom reaches the top of its barrier. The kinetic energy depends on the momenta alone and U on the positions alone, so $e^{-\beta H}$ is $e^{-\beta U}$ times a factor of the momenta, and integrating over the momenta turns that factor into a number that is the same at every position. The positions are left with the density $e^{-\beta U}$, in which the masses no longer appear, as Section 11.6 anticipated: of two places whose potential energies differ by ΔU , an atom is found near the higher, per unit volume, $e^{-\Delta U/k_B T}$ times as often as near the lower. For the lithium atom of that model, which moves over the surface, the density is per unit area, and at the bridge, 0.3 eV above its hollow, the odds are $e^{-0.3/0.02585} = 9.1 \times 10^{-6}$ at 300 K, and 3.0×10^{-3} at 600 K, 331 times larger: a barrier that is rarely reached at room temperature is reached hundreds of times as often when the temperature doubles. For an atom moving in two directions the same number is also the probability that its kinetic energy is at least the barrier, since its kinetic energy has the density $e^{-K/k_B T}/k_B T$ (Section 12.7). How often an atom actually hops depends also on how often it tries, which Chapters 16 and 17 measure.

KEY IDEA. A system in contact with a bath at temperature T is found in a microstate of energy E_s with a probability proportional to $e^{-E_s/k_B T}$. The partition function Z normalises the probabilities; $\langle E \rangle = -d \ln Z / d\beta$, the energy's variance is $k_B T^2 C_V$, and $F = -k_B T \ln Z$. A classical vibration holds $k_B T$ on average.

NOTEBOOK 12 set a barrier and a temperature and read off the odds

CHECK YOUR UNDERSTANDING. What are the odds of the 0.3 eV barrier at 100 K?

12.5 The distribution of velocities

Section 10.5 drew starting velocities from the normal distribution and promised a reason. This section derives the distribution of the velocities of atoms at a temperature T from the Boltzmann factor, then the distributions of their speeds and kinetic energies, and measures all three in liquid argon.

Each component is Gaussian

We first take an unconstrained canonical system, with no condition on its total momentum. The energy of N atoms is $H = \sum_{i,x} p_{ix}^2/2m_i + U(\mathbf{r}^N)$, where the sum runs over the atoms i and, with x standing for each in turn, the three directions x , y and z . The exponential of a sum is a product, so the Boltzmann factor of Equation (12.4) splits into one factor for the positions and one for each component of momentum:

$$e^{-\beta H} = e^{-\beta U(\mathbf{r}^N)} \prod_{i,x} e^{-\beta p_{ix}^2/2m_i}.$$

A probability density that is a product of factors, each depending on one quantity alone, describes quantities that are independent, so every component of every momentum is independent of the others and of the positions. With $p_{ix} = m_i v_{ix}$, the density of one component of velocity is proportional to $e^{-m_i v_{ix}^2/2k_B T}$, which is the Gaussian of Equation (12.1) with mean zero and variance

$$\sigma_v^2 = \frac{k_B T}{m_i}. \quad (12.7)$$

This is the Maxwell-Boltzmann distribution of velocities. Two things in it matter for simulation. It holds whatever the forces: the velocities of atoms in a gas, a liquid or a crystal at the same temperature have the same distribution, in classical mechanics. And the spread depends on the mass: a lithium atom at 300 K has $\sigma_v = \sqrt{k_B T/m} = \sqrt{0.025852/(6.94 \times 103.6427)} = 0.005995 \text{ Å fs}^{-1}$, the factor 103.6427 of Equation (3.15) turning $\text{amu Å}^2 \text{ fs}^{-2}$ into eV, and an argon atom $0.002499 \text{ Å fs}^{-1}$, smaller by $\sqrt{39.948/6.94} = 2.399$, the ratio of the square roots of their masses. Dividing the normal numbers by $\sqrt{m_i}$, as starting velocities of Section 10.5 did, gives each atom this spread up to a common factor, which the scaling to the chosen energy then fixed.

`mdlab.statmech.thermal_velocities` draws velocities at a temperature directly:

```

1  m = np.asarray(masses, dtype=float)
2  xi = rng.standard_normal((len(m), 3))
3  v = xi * np.sqrt(kb * temperature / (m * mv2_to_energy))[:, None]
4  if remove_drift:
5      v = v - (m @ v) / m.sum()
6  return v

```

Line 2 draws $3N$ numbers of mean 0 and variance 1, and line 3 multiplies each atom's by its σ_v , with `mv2_to_energy` the factor of Equation (3.15) that turns mv^2 into eV. Line 5 removes the drift of the centre of mass, as in Section 10.5. ASE's `thermalize_momenta` draws the numbers in the same order, so the same starting number of the random generator, its seed, gives the same velocities: for 50 atoms of random masses at 300 K they differ by at most 1.7×10^{-7} of their size. That is half the relative difference of 3.4×10^{-7} between the two libraries' values of k_B , ASE's being the value recommended for the constants in 2014, since σ_v goes as $\sqrt{k_B}$ and a small relative change in a quantity changes its square root by half as much.

Speeds and kinetic energies

The speed $v = |\mathbf{v}|$ of an atom combines three independent components. The probability that the velocity lies in a small box $dv_x dv_y dv_z$ is the product of the three Gaussians, each with the factor $1/(\sigma_v \sqrt{2\pi}) = (m/2\pi k_B T)^{1/2}$, times the box, and with $v_x^2 + v_y^2 + v_z^2 = v^2$ it is

$$\left(\frac{m}{2\pi k_B T}\right)^{3/2} e^{-mv^2/2k_B T} dv_x dv_y dv_z,$$

which depends on the velocity only through its size. The velocities with speeds between v and $v + dv$ fill a spherical shell of area $4\pi v^2$ and thickness dv , so the density of the speed is

$$\mathcal{P}(v) = 4\pi v^2 \left(\frac{m}{2\pi k_B T}\right)^{3/2} e^{-mv^2/2k_B T}. \quad (12.8)$$

The factor v^2 makes it zero at $v = 0$, since few velocities have all three components near zero. It peaks at $\sqrt{2} \sigma_v$ and has the mean $\sqrt{8/\pi} \sigma_v$ (Exercise 12.10); the mean speed of lithium at 300 K is $0.00957 \text{ Å fs}^{-1}$, or 957 m s^{-1} , with 1 Å fs^{-1} equal to 10^5 m s^{-1} , and that of argon 399 m s^{-1} .

An atom's kinetic energy $K_i = \frac{1}{2}mv^2$ follows by a change of variable. Between K_i and $K_i + dK_i$ lie the speeds between v and $v + dv$ with $dK_i = mv dv$, and the two hold

the same probability, $\mathcal{P}(K_i) dK_i = \mathcal{P}(v) dv$, so $\mathcal{P}(K_i) = \mathcal{P}(v)/(mv)$. With $mv^2/2k_B T = K_i/k_B T$ and $v\sqrt{m} = \sqrt{2K_i}$,

$$\frac{\mathcal{P}(v)}{mv} = \frac{4\pi v\sqrt{m}}{(2\pi k_B T)^{3/2}} e^{-K_i/k_B T} = \frac{4\pi\sqrt{2}}{(2\pi)^{3/2}} (k_B T)^{-3/2} \sqrt{K_i} e^{-K_i/k_B T},$$

and $4\pi\sqrt{2}/(2\pi)^{3/2} = 4\pi\sqrt{2}/(2\sqrt{2}\pi^{3/2}) = 2/\sqrt{\pi}$, so

$$\mathcal{P}(K_i) = \frac{2}{\sqrt{\pi}} (k_B T)^{-3/2} \sqrt{K_i} e^{-K_i/k_B T}, \quad (12.9)$$

the same for every mass. Its mean is $\frac{3}{2}k_B T$ and its variance $\frac{3}{2}(k_B T)^2$, as [Section 12.7](#) shows for any number of freedoms.

Measured in a liquid

[Figure 12.5](#) tests the three densities on the liquid argon of [Figure 12.2\(b\)](#), 256 atoms over 50 ps, with the velocities saved every 50 fs: 768 768 components from 1001 frames. The run is at fixed energy, not in contact with a bath, but each atom has the other 255 around it, which act as its bath. The kinetic temperature of the run, defined in [Section 12.6](#), is 134.63 K, at which [Equation \(12.7\)](#) gives $\sigma_v = 0.001\,674\,\text{\AA}\,\text{fs}^{-1}$; the components have the standard deviation $0.001\,671\,\text{\AA}\,\text{fs}^{-1}$, and [Section 12.6](#) accounts for the difference. The mean speed is $0.002\,668\,\text{\AA}\,\text{fs}^{-1}$ against $0.002\,671$ from [Equation \(12.8\)](#), and the kinetic energies of single atoms have the mean $1.494\,k_B T$ and the variance $1.479\,(k_B T)^2$, against $\frac{3}{2}$ for both.

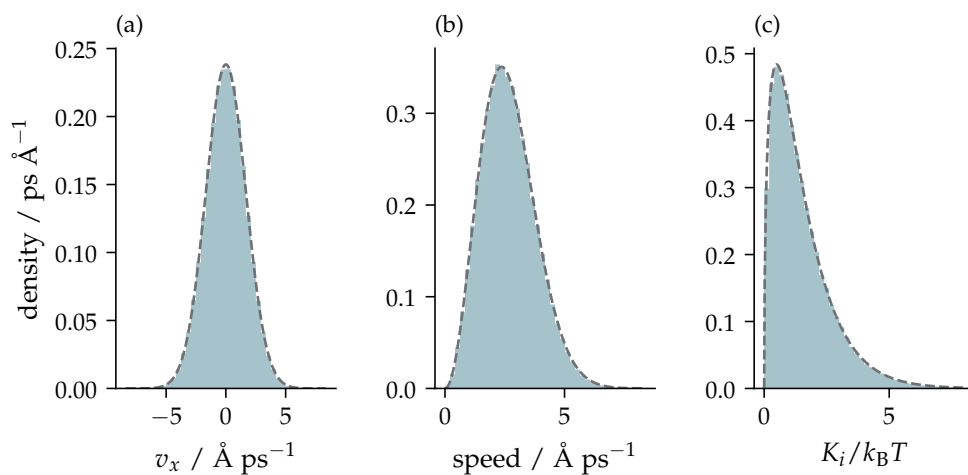


FIGURE 12.5: Velocities in liquid argon at 134.6 K: histograms of 256 atoms over 50 ps, with velocities in $\text{\AA}\,\text{ps}^{-1}$, a thousand times their values in $\text{\AA}\,\text{fs}^{-1}$. (a) Every component of velocity, against the Gaussian of variance $k_B T/m$ (dashed). (b) The speeds, against [Equation \(12.8\)](#). (c) The kinetic energies of single atoms, in units of $k_B T$, against [Equation \(12.9\)](#), a density per unit of $k_B T$.

A Gaussian has a property that the histogram shows only roughly and a single number shows well. The fourth moment of the Gaussian of mean zero is $\langle v^4 \rangle = 3\sigma_v^4$ (Exercise 12.11), so the excess kurtosis

$$\frac{\langle u^4 \rangle}{\langle u^2 \rangle^2} - 3,$$

computed over the mass-weighted components $u = \sqrt{m}v$, which share one variance $k_B T$ whatever the masses, is zero for velocities with the Maxwell-Boltzmann distribution; pooled components of different variances would give more than zero even if each were Gaussian. Components spread evenly over an interval give -1.2 (Exercise 12.11). Over the 768 768 components of the liquid it is -0.0072 , small against the -1.2 of an even spread. `mdlab.diagnostics.excess_kurtosis` computes it, and Section 12.9 uses it to catch velocities that are not yet Maxwellian.

KEY IDEA. In the unconstrained canonical ensemble each component of each velocity is independent and Gaussian, with mean zero and variance $k_B T / m$, whatever the forces. The speed has the density $4\pi v^2 (m/2\pi k_B T)^{3/2} e^{-mv^2/2k_B T}$, and an atom's kinetic energy a density with mean $\frac{3}{2}k_B T$ for every mass. The excess kurtosis of the mass-weighted components is zero for Maxwellian velocities. Removing the total momentum or holding bond lengths constrains these components, as the next section explains.

NOTEBOOK 12 draw velocities at any temperature and mass, and check them against ASE

CHECK YOUR UNDERSTANDING. What is σ_v for a lithium atom at 600 K?

12.6 Equal shares and the kinetic temperature

Section 3.7 said that the warmth of rubbing surfaces is the hidden energy of their jiggling atoms. This section finds how that energy is shared among the ways the atoms can move, and from the share of the kinetic energy defines the temperature that a simulation measures.

The equipartition theorem

We take x to be any one coordinate or momentum of a system in the canonical ensemble, whose range runs from x_1 to x_2 , and require $x e^{-\beta H}$ to vanish at both ends, as it does for a momentum, whose range is unbounded and where the exponential falls faster

than x grows, or for a coordinate held in by a potential that rises without limit, walls included. The average of x times the slope of H along it is

$$\left\langle x \frac{\partial H}{\partial x} \right\rangle = \frac{1}{Z} \int x \frac{\partial H}{\partial x} e^{-\beta H} d\Gamma,$$

and the integral over x , with every other coordinate and momentum held fixed, can be done by parts, since by the chain rule $\partial(e^{-\beta H})/\partial x = -\beta (\partial H/\partial x) e^{-\beta H}$:

$$\begin{aligned} \int_{x_1}^{x_2} x \frac{\partial H}{\partial x} e^{-\beta H} dx &= -\frac{1}{\beta} \int_{x_1}^{x_2} x \frac{\partial(e^{-\beta H})}{\partial x} dx \\ &= -\frac{1}{\beta} \left[x e^{-\beta H} \right]_{x_1}^{x_2} + \frac{1}{\beta} \int_{x_1}^{x_2} e^{-\beta H} dx \quad \text{parts, with } f = x, \\ &= k_B T \int_{x_1}^{x_2} e^{-\beta H} dx \quad \text{the bracket vanishes.} \end{aligned}$$

Integrating over the other coordinates and momenta too, and dividing by $Z = \int e^{-\beta H} d\Gamma$,

$$\left\langle x \frac{\partial H}{\partial x} \right\rangle = k_B T, \quad (12.10)$$

whatever x is and whatever the rest of H . When H contains x only through a term $\frac{1}{2}ax^2$, then $x \partial H/\partial x = ax^2$, twice that term, and the term has the average $\frac{1}{2}k_B T$. Every term of the energy that is the square of a coordinate or momentum appearing in no other term holds, on average, $\frac{1}{2}k_B T$, whatever its coefficient: this is the equipartition of energy. A square whose variable appears elsewhere too, as x does in $\frac{1}{2}x^2 + \frac{1}{2}y^2 + cxy$, need not. Chapter 14 uses Equation (12.10) for the positions to find the pressure.

Three cases recur in this book.

- Each component of momentum enters H as $p_{ix}^2/2m_i$, so, when the total momentum is free, $\langle \frac{1}{2}m_i v_{ix}^2 \rangle = \frac{1}{2}k_B T$ for every atom and direction, whatever its mass and whatever the forces on it. At 300 K that is 12.93 meV, the share given to each freedom of the water of Section 11.3.
- A vibration of small amplitude has $H = p^2/2m + \frac{1}{2}kx^2$, two squares, and holds $\frac{1}{2}k_B T$ as kinetic and $\frac{1}{2}k_B T$ as potential energy, $k_B T$ in all, as Section 12.4 found from its partition function. Section 4.2 found the same equal division as an average over one period of one oscillator; here it is an average over the ensemble.
- A molecule of two atoms turning freely has, in the angle θ of its axis from the z axis and the angle ϕ about it, $H = p_\theta^2/2I + p_\phi^2/(2I \sin^2 \theta)$ (Exercise 12.13). Each term is the square of a momentum, so each holds $\frac{1}{2}k_B T$, and the turning holds $k_B T$ in all, as Section 5.8 assumed.

A term that is not a square does not hold $\frac{1}{2}k_B T$: the potential energy of a liquid, made of Lennard-Jones terms, has no simple share. The kinetic energy is always a sum of squares, which is why it serves as the thermometer.

The kinetic temperature

With N_f freedoms, each holding $\frac{1}{2}k_B T$ of kinetic energy on average, $\langle K \rangle = \frac{1}{2}N_f k_B T$, and the instantaneous kinetic temperature

$$T_{\text{kin}} = \frac{2K}{N_f k_B} \quad (12.11)$$

has the temperature T as its average. It is what a simulation reports as ‘the temperature’ at each step; it fluctuates, by an amount [Section 12.7](#) finds. In an isolated system the kinetic temperature’s average differs from the T of [Equation \(12.3\)](#) by terms of order $1/N$: for the ideal gas of [Section 12.3](#), $k_B T = E / (\frac{3}{2}N - 1)$, while $2K / (3N k_B)$ with $K = E$ gives $k_B T_{\text{kin}} = E / \frac{3}{2}N$, smaller by the factor $1 - 2/(3N)$.

Counting N_f is where errors enter. N atoms have $3N$ components of velocity, and each independent restriction on them removes one.

- A periodic run whose drift has been removed keeps its total momentum at zero ([Section 10.5](#)), so its velocities satisfy three conditions, $\sum_i m_i \mathbf{v}_i = \mathbf{0}$, and the kinetic energy is shared among $3N - 3$ freedoms.
- Each distance held by RATTLE removes the component of relative velocity along it ([Section 11.2](#)): n_c independent held distances remove n_c freedoms.
- An isolated molecule whose turning has also been removed has three more conditions, or two if it is linear ([Section 5.7](#)), leaving $3N - 6$, or $3N - 5$.

In the mass-weighted components $u_k = \sqrt{m_k} v_k$, the kinetic energy is $K = \frac{1}{2} \sum_k u_k^2$, half the squared length of $\mathbf{u}$, and each restriction is a condition that $\mathbf{u}$ be at right angles to a direction fixed by the positions at that instant: for the drift along x , $\sum_i \sqrt{m_i} u_{ix} = 0$. Writing $\mathbf{u} = \sum_k c_k \mathbf{e}_k$ along $3N$ directions of length 1 at right angles to one another, the first n chosen to span the directions that n restrictions fix, gives $K = \frac{1}{2} \sum_k c_k^2$ as in [Section 4.9](#), with c_1 to c_n held at zero: a sum of $3N - n$ squares, each holding $\frac{1}{2}k_B T$. So a periodic run with its drift removed has $N_f = 3N - n_c - 3$, which `mdlab.statmech.degrees_of_freedom` returns: 765 for 256 argon atoms, and 381 for the 64 rigid water molecules of Chapter 11. `kinetic_temperature` then applies [Equation \(12.11\)](#).

The count shows in the measurements of [Section 12.5](#). The kinetic energy of 765 freedoms is spread over 768 components of velocity, so the components’ mean square is

$765/768$ of $k_B T/m$, and their standard deviation $\sqrt{765/768} = 0.99804$ of σ_v : $0.99804 \times 0.001674 = 0.001671 \text{ Å fs}^{-1}$, the value measured. ASE's `get_temperature` divides by $3N$ less the freedoms its constraints remove, and does not take off three for the drift. On the last frame of the liquid, `mdlab` gives 137.7961 K and ASE 137.2579 K, in the ratio 0.996094, which is $765/768$; the two agree once they count alike.

Shares between species, and with the potential

Equipartition gives every atom the same mean kinetic energy whatever its mass, so the heavier atoms of a mixture move more slowly and are no colder. A liquid of 256 atoms with argon's forces, half of mass 39.948 amu and half of 83.798 amu, the mass of krypton, started with velocities drawn at 130 K, has its light atoms at 126 K and its heavy atoms at 124 K over the last 10 ps of 20 ps, each temperature counting the three components of each atom of the species. Whether a gap of 2 K is a real difference or the noise of a short run is the question of Chapter 15.

Equipartition also explains what [Section 10.5](#) saw. A crystal started on its lattice has no potential energy above its minimum, and its vibrations, nearly harmonic, take half of the kinetic energy they are given: each ends with $\frac{1}{2}k_B T$ of each kind. The 500 argon atoms of that section lost half of their kinetic energy within 0.1 ps, and 256 atoms on their lattice at the spacing of ASE's reference value⁴³, $a = 5.26 \text{ Å}$, a little below the model's 5.267 Å , which squeezes the cell slightly but leaves the lattice, by its symmetry, where every force vanishes, given velocities at 40 K (a kinetic temperature of 41.5 K, by the scatter of [Section 12.7](#)), settle at 21.0 K, 0.506 of the start. A liquid has no such simple ratio, since its potential energy is not a sum of squares.

IN PRACTICE. Velocities drawn at a temperature set only the kinetic energy. A crystal started on its lattice ends near half the temperature asked for, and a system started anywhere else ends somewhere that has to be found by running. A run is at the temperature asked for only after that temperature has been measured over the run; a thermostat (Chapter 13) holds it there.

KEY IDEA. Every term of the energy that is the square of a coordinate or momentum appearing in no other term holds $\frac{1}{2}k_B T$ on average. The kinetic energy is such a sum, which makes $T_{\text{kin}} = 2K/(N_f k_B)$ a thermometer, with $N_f = 3N - n_c - 3$ in a periodic run whose drift is removed. Atoms of every mass share alike; a harmonic crystal shares equally between kinetic and potential energy.

NOTEBOOK 12 *count the freedoms of a run and compare the temperature with ASE's*

CHECK YOUR UNDERSTANDING. A box of 100 rigid water molecules, drift removed, has a kinetic energy of 7.4 eV. What is its kinetic temperature?

12.7 How much the temperature fluctuates

The kinetic temperature of a run is never still. This section derives the distribution of the kinetic energy in the canonical ensemble, finds that a run at fixed energy fluctuates less, and turns the difference into a measurement of the heat capacity. Chapter 13 tests every thermostat against the distribution found here.

The kinetic energy in the canonical ensemble

In the N_f free components c_k of Section 12.6, the kinetic energy is $K = \frac{1}{2} \sum_k c_k^2$. The velocities with kinetic energy below K fill the inside of a sphere of radius $\sqrt{2K}$ in N_f dimensions, whose volume, as for the ideal gas of Section 12.3, is a constant times $K^{N_f/2}$, and those between K and $K + dK$ fill a shell whose volume is the derivative of that volume times dK , a constant times $K^{N_f/2-1} dK$. Each velocity in the shell has the Boltzmann factor $e^{-\beta K}$, so

$$\mathcal{P}(K) \propto K^{N_f/2-1} e^{-K/k_B T}. \quad (12.12)$$

This is the gamma distribution. For one atom, $N_f = 3$, it is the $\sqrt{K} e^{-K/k_B T}$ of Equation (12.9); for an atom moving over a surface, $N_f = 2$, it is $e^{-K/k_B T}/k_B T$, and the probability that K is at least some U_b is $e^{-U_b/k_B T}$. Its mean and variance follow as in Section 12.4, applied to the kinetic energy alone, which is independent of the positions: with $a = N_f/2$, the factor K^{a-1} that counts the velocities does not depend on β , so each derivative of the normalising integral with respect to β brings down a factor $-K$, as each derivative of Z brought down $-E_s$. The normalising integral is, with $s = \beta K$,

$$Z_K = \int_0^\infty K^{a-1} e^{-\beta K} dK = \beta^{-a} \int_0^\infty s^{a-1} e^{-s} ds,$$

and the last integral is a number, the gamma function of a , written $\Gamma(a)$ (no relation to the phase point Γ), which the function `kinetic_energy_density` evaluates through its logarithm. So $\ln Z_K = -a \ln \beta + \text{a constant}$, and

$$\langle K \rangle = -\frac{d \ln Z_K}{d\beta} = \frac{a}{\beta} = \frac{1}{2} N_f k_B T, \quad \langle K^2 \rangle - \langle K \rangle^2 = \frac{d^2 \ln Z_K}{d\beta^2} = \frac{a}{\beta^2} = \frac{1}{2} N_f (k_B T)^2.$$

The first is equipartition again. The second gives the relative fluctuation of the kinetic energy, and of the kinetic temperature, which is proportional to it:

$$\frac{\sigma_K}{\langle K \rangle} = \sqrt{\frac{2}{N_f}}. \quad (12.13)$$

For one atom it is 0.82; for 256 argon atoms, with $N_f = 765$, 0.0511. A canonical run of 256 atoms at 135 K has a kinetic temperature with a standard deviation of 6.9 K, and the larger the system, the smaller the spread.

At fixed energy, less

In a run at fixed energy E , the kinetic and the potential energy share E , as the two gases of [Section 12.3](#) did. The probability that the kinetic energy is K is proportional to the number of velocities with that kinetic energy, K^{a-1} , times $\Omega_U(E - K)$, the volume of the positions with potential energy $E - K$, which counts them as Ω counted states in [Section 12.3](#). Its width follows as in [Section 12.3](#), from the second derivative of its logarithm at the peak, which is the sum of two terms, one from each partner:

$$\begin{aligned} \frac{d^2}{dK^2} [(a-1) \ln K] &= -\frac{a-1}{K^2} \approx -\frac{a}{(ak_B T)^2} = -\frac{1}{k_B T^2 C_K}, & C_K &= ak_B = \frac{1}{2} N_f k_B, \\ \frac{d^2}{dK^2} \ln \Omega_U(E - K) &= \frac{d^2 \ln \Omega_U}{dU^2} = \frac{d}{dU} \left(\frac{1}{k_B T} \right) = -\frac{1}{k_B T^2 C_U}, & C_U &= \frac{dU}{dT}, \end{aligned}$$

using in the first $a - 1 \approx a$ for many freedoms and $K = ak_B T$ at the peak, where, as for the two gases, both partners have the same temperature. In the second, $U = E - K$, so each derivative with respect to K is minus that with respect to U and the two signs cancel; then come the definition of temperature, [Equation \(12.3\)](#), applied to the arrangements, and $d(1/k_B T)/dU = -(1/k_B T^2) dT/dU$ as in [Section 12.4](#). C_K is the kinetic energy taken up per degree and C_U the potential energy, and together they make the heat capacity, $C_V = C_K + C_U$. The logarithm of the Gaussian of [Equation \(12.1\)](#) has the second derivative $-1/\sigma_x^2$, so the variance of K , written $\langle \delta K^2 \rangle_E$ with $\delta K = K - \langle K \rangle$, is minus the reciprocal of the sum of the two curvatures:

$$\langle \delta K^2 \rangle_E = \frac{k_B T^2}{1/C_K + 1/C_U} = k_B T^2 C_K \left(1 - \frac{C_K}{C_V} \right) = \frac{1}{2} N_f (k_B T)^2 \left(1 - \frac{N_f k_B}{2C_V} \right), \quad (12.14)$$

the result of Lebowitz, Percus and Verlet⁵³. The bracket is the factor by which the variance at fixed energy falls below the canonical one, and its square root the factor for the standard deviation. An ideal gas, with no potential energy, has $C_U = 0$ and a kinetic energy that cannot fluctuate at all; a harmonic crystal, with $C_U = C_K$, fluctuates by $1/\sqrt{2}$ of the canonical amount.

[Figure 12.6\(a\)](#) shows the kinetic energy of the 256 atoms of liquid argon over 50 ps: its relative standard deviation is 0.03068, against 0.05113 for the canonical ensemble, the ratio 0.600. The heat capacity measured below, $C_V = 2.376 N k_B$, with $C_K = \frac{1}{2} N_f k_B = (765/512) N k_B = 1.4941 N k_B$, predicts $\sqrt{1 - 1.4941/2.376} = 0.609$. The ratio stays near

0.6 for all four sizes of Figure 12.6(b), from 0.564 to 0.600, while each spread falls as $1/\sqrt{N}$.

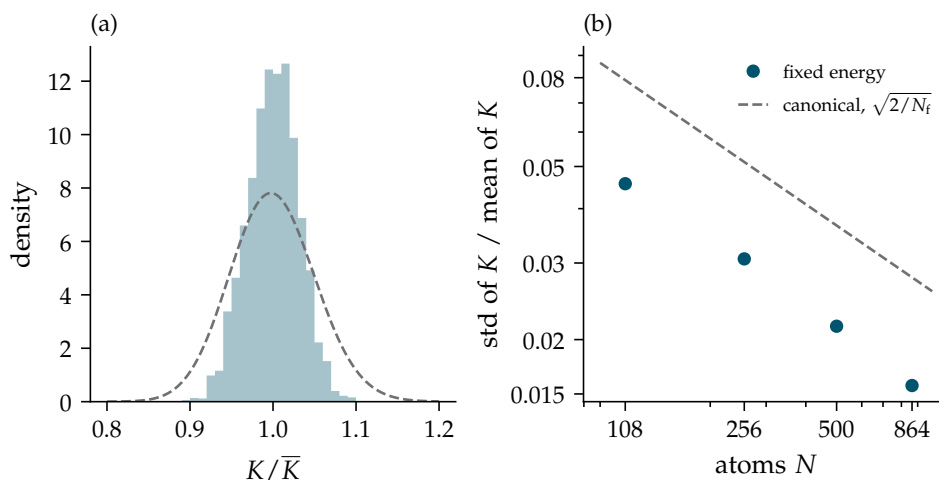


FIGURE 12.6: The kinetic energy at fixed total energy. (a) Liquid argon, 256 atoms over 50 ps: the histogram of the kinetic energy divided by its mean, against the canonical density of Equation (12.12) with $N_f = 765$ and the same mean (dashed). (b) The relative standard deviation of the kinetic energy for 108, 256, 500 and 864 atoms (circles), against the canonical $\sqrt{2/N_f}$ (dashed).

The heat capacity, two ways

Equation (12.14) solved for C_V ,

$$C_V = \frac{\frac{1}{2}N_f k_B}{1 - 2\langle \delta K^2 \rangle_E / (N_f (k_B T)^2)},$$

gives the heat capacity from one run at fixed energy; `statmech.heat_capacity_nve` computes it, with T the run's kinetic temperature. Figure 12.7 checks it against the definition, $C_V = dE/dT$. Five runs of the 256 atoms, prepared near 100, 115, 130, 145 and 160 K and run for 50 ps each, give five pairs of mean energy and mean kinetic temperature, from 99.25 K to 160.28 K. They lie on a straight line to within 0.029 eV, whose slope is $0.05242 \text{ eV K}^{-1}$, or $C_V = 2.376 N k_B$. The spreads of the five runs give between 2.249 and $2.444 N k_B$, with the mean 2.347 , and the four sizes near 130 K between 2.198 and 2.333 . The liquid lies between an ideal gas, with $\frac{3}{2} N k_B$ from its kinetic energy alone, and a harmonic crystal, with $3 N k_B$ from equal kinetic and potential shares. The single-run estimates scatter about the slope by up to 7.5%; whether that scatter is the noise of runs of 50 ps is a question Chapter 15 answers with error bars. In the canonical ensemble the total energy itself fluctuates, and Equation (12.6) gives C_V directly, which Chapter 13 measures under its thermostats.

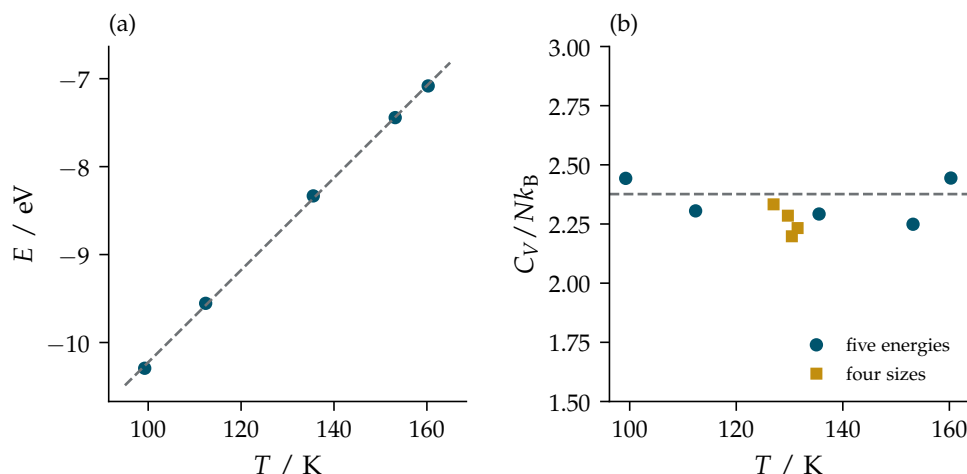


FIGURE 12.7: The heat capacity of liquid argon, 256 atoms. (a) The mean energy against the mean kinetic temperature of five runs at fixed energy, with the straight line through them, whose slope is C_V (dashed). (b) C_V per atom, in units of k_B , from the spread of the kinetic energy by Equation (12.14), for the five runs (circles) and for runs of 108 to 864 atoms near 130 K (squares), against the slope of (a) (dashed).

KEY IDEA. In the canonical ensemble the kinetic energy has the gamma distribution, $\mathcal{P}(K) \propto K^{N_f/2-1} e^{-K/k_B T}$, with mean $\frac{1}{2} N_f k_B T$ and relative spread $\sqrt{2/N_f}$. At fixed energy the kinetic and potential energies share a fixed total, and the standard deviation of the kinetic energy is $\sqrt{1 - N_f k_B / (2C_V)}$ times the canonical one, which gives the heat capacity from a single run.

NOTEBOOK 12 set the size and compare the kinetic energy at fixed energy with the canonical gamma distribution

CHECK YOUR UNDERSTANDING. By how much, relatively, does the kinetic temperature of 1000 atoms with their drift removed fluctuate in the canonical ensemble?

12.8 Other ensembles

A sample on a bench exchanges more than energy with its surroundings: it can swell against the air, and an electrode in a cell exchanges atoms and electrons with the electrolyte and the circuit. This section extends the argument of Section 12.4 to a bath that also takes volume or particles, finds the free energy each such ensemble makes least, and ends with the electrons of an electronic-structure calculation.

Volume and particles

The entropy of the bath depends on its volume and number of atoms as well as its energy, and its slopes along them, each taken with the other two held fixed (Section 7.2), define the pressure P and the chemical potential μ :

$$\frac{\partial S}{\partial E} = \frac{1}{T}, \quad \frac{\partial S}{\partial V} = \frac{P}{T}, \quad \frac{\partial S}{\partial N} = -\frac{\mu}{T}.$$

For the ideal gas of Section 12.3, $\Omega \propto V^N$ gives $\partial S/\partial V = Nk_B/V$, so $PV = Nk_B T$, the ideal gas law; Chapter 14 shows that this P is the force per unit area on the walls. The minus sign makes μ the energy that one more atom brings, as dE below shows. As Section 12.3 showed for energy, two systems that exchange volume settle where their values of P/T agree, and two that exchange atoms where their values of μ/T agree, once the atoms are counted correctly. Swapping two atoms of one kind gives back the same arrangement, so the $N!$ orderings of N identical atoms are one microstate, and Ω and Z for N such atoms are divided by $N!$, a factor that changed nothing while N was fixed. Without it the ideal gas, with $\Omega \propto V^N$, gains $\ln V$ in $\ln \Omega$ for each atom added whatever N is, and two boxes sharing atoms would not settle at equal densities, the larger box taking all the atoms; with it, one more atom changes $\ln(V^N/N!)$ by $\ln V - \ln(N+1) \approx \ln(V/N)$, and two boxes at one temperature settle at equal densities N/V .

A system that takes the volume V_s from the bath as well as the energy E_s leaves the bath $\Omega_B(E - E_s, V - V_s)$ microstates, and the Taylor series of Section 12.4, now in two variables with one term for each as in Section 3.10, gives

$$\ln \Omega_B(E - E_s, V - V_s) \approx \ln \Omega_B(E, V) - \frac{E_s}{k_B T} - \frac{PV_s}{k_B T},$$

so a state of the system with energy E_s and volume V_s has the probability proportional to $e^{-\beta(E_s + PV_s)}$: the isothermal-isobaric, or NPT , ensemble. A system that takes N_s atoms from the bath has, by the same steps with $\partial \ln \Omega_B/\partial N = -\mu/k_B T$,

$$\ln \Omega_B(E - E_s, N - N_s) \approx \ln \Omega_B(E, N) - \frac{E_s}{k_B T} + \frac{\mu N_s}{k_B T},$$

and so the weight $e^{-\beta(E_s - \mu N_s)}$: the grand canonical, or μVT , ensemble. Grouping the states by their volume or their number of atoms, the sum over each group being a canonical partition function, gives

$$Z_{NPT} = \int e^{-\beta PV} Z(N, V, T) dV, \quad \Xi = \sum_N e^{\beta \mu N} Z(N, V, T),$$

and the derivatives of their logarithms give averages and fluctuations as before: the volume of an NPT run fluctuates with the variance $-k_B T d\langle V \rangle / dP$ (Exercise 12.19), large for a soft material and small for a stiff one, which Chapter 14 uses to test a barostat's coupling.

The free energies

For a system in contact with a bath at T , the total energy is divided as in Section 12.3: when the system holds E_s it has $\Omega(E_s)$ microstates and the entropy $S = k_B \ln \Omega(E_s)$, the bath has $\Omega_B(E - E_s)$, and the most probable division has the most microstates of the two together, $\Omega(E_s) \Omega_B(E - E_s)$. Their logarithm times k_B is the total entropy, which by the expansion of Section 12.4 is

$$S + S_B(E - E_s) \approx S + S_B(E) - \frac{E_s}{T} = S_B(E) - \frac{E_s - TS}{T}.$$

$S_B(E)$ is the same for every division, so the system settles at the energy that makes $E_s - TS$ least, and for a large system that least value is the free energy F of Section 12.4. With a bath that also takes volume, it is $F + PV$ that is least, the Gibbs free energy, $-k_B T \ln Z_{NPT}$; with one that takes atoms, $F - \mu N$, the grand potential, $-k_B T \ln \Xi$. From here on E , V and N are the system's own.

For small changes dE , dV and dN , the entropy changes by one term for each, its slope times the change (Section 3.10):

$$dS = \frac{dE}{T} + \frac{P dV}{T} - \frac{\mu dN}{T},$$

$$dE = T dS - P dV + \mu dN \quad \text{multiplying by } T \text{ and rearranging.}$$

TS changes by $T dS + S dT$, by the product rule, so

$$dF = dE - T dS - S dT = -S dT - P dV + \mu dN.$$

Since T is the slope of E along S , subtracting TS exchanged S for that slope as the variable the energy depends on. By Section 7.1, $TS - E$ at the S where the slope is T is the largest gap between the line TS and the curve $E(S)$, the Legendre transform of E ; $F = E - TS$ is that transform with its sign reversed, and so the least value of $E - TS$ over S , which is why the system settles there. Adding PV or subtracting μN exchanges V for P or N for μ in the same way (Exercise 12.20). Table 12.1 collects the four ensembles and what samples each.

Molecular dynamics keeps the number of atoms, so the grand canonical ensemble of atoms is sampled by moves that insert or remove an atom at random and keep or undo

TABLE 12.1: The ensembles of this book: what is fixed, the probability of a state, the quantity that is greatest or least, and what samples the ensemble in a simulation.

ensemble	fixed	weight	extremum	sampled by
microcanonical	N, V, E	equal on the shell	S largest	Verlet (Chs 9–11)
canonical	N, V, T	$e^{-\beta E}$	F least	thermostats (Ch. 13)
isothermal-isobaric	N, P, T	$e^{-\beta(E+PV)}$	$F + PV$ least	barostats (Ch. 14)
grand canonical	μ, V, T	$e^{-\beta(E-\mu N)}$	$F - \mu N$ least	insertion and removal

each with a probability set by its weight, Monte Carlo moves, which are not dynamics; this book does not use them. The grand canonical ensemble returns for electrons.

Electrons in an orbital

In an electronic-structure calculation the electrons of a solid fill orbitals, states of definite energy ε_o that one electron can occupy. No two electrons can be in the same state (the exclusion principle), and an electron has a further two-valued property, its spin, so an orbital holds either no electron of a given spin or one. In contact with a reservoir of electrons at the chemical potential μ , the orbital with one spin is a system of two states: empty, with $N = 0$ and energy 0, and filled, with $N = 1$ and energy ε_o . Its grand partition function has two terms, $\Xi = 1 + e^{-\beta(\varepsilon_o - \mu)}$, and its mean occupation, $0 \times \mathcal{P}(\text{empty}) + 1 \times \mathcal{P}(\text{filled})$, is the probability of the second,

$$\begin{aligned}
 f &= \frac{e^{-\beta(\varepsilon_o - \mu)}}{1 + e^{-\beta(\varepsilon_o - \mu)}} \\
 &= \frac{1}{e^{(\varepsilon_o - \mu)/k_B T} + 1} \quad \text{dividing above and below by } e^{-\beta(\varepsilon_o - \mu)},
 \end{aligned}$$

the Fermi-Dirac occupation: 0.5 at $\varepsilon_o = \mu$, 0.0205 at 0.1 eV above it and 0.9795 at 0.1 eV below, at 300 K. The entropy $-k_B \sum \mathcal{P} \ln \mathcal{P}$ of its two states, with probabilities $1 - f$ and f , is $-k_B[f \ln f + (1 - f) \ln(1 - f)]$, whose slope in f , $k_B \ln((1 - f)/f)$, vanishes at $f = \frac{1}{2}$: the entropy is largest, $k_B \ln 2$, when the orbital is half filled.

FROM ELECTRONIC STRUCTURE. Mermin showed that density functional theory carries over to a finite temperature in the grand canonical ensemble, the grand potential of the electrons being least at their true density⁵⁴. A calculation of a metal that smears its occupations with this f minimises the free energy $E - TS$ of its electrons, with S the sum of the orbitals' entropies, and the force on each atom is the slope of that free energy, so a run on such forces keeps constant the sum of the atoms' kinetic energy and that free energy⁵⁵, and not the sum with E . A calculation of an electrode held at a fixed potential fixes μ of the electrons instead of their number, and makes the grand potential $E - TS - \mu N$ least: its electrons are in the grand canonical ensemble, while its atoms may be canonical.

KEY IDEA. A bath that takes volume weights a state by $e^{-\beta(E+PV)}$, and one that takes atoms by $e^{-\beta(E-\mu N)}$, with identical atoms counted once, $N!$ orderings as one microstate. In contact with a bath, the system makes a free energy least: $F = E - TS$ at fixed V , $F + PV$ at fixed P , $F - \mu N$ at fixed μ , each the Legendre transform of the energy with its sign reversed. Fermi-Dirac occupations are the grand canonical ensemble of one orbital.

NOTEBOOK 12 *set a temperature and watch the occupations and their entropy*

CHECK YOUR UNDERSTANDING. What is the occupation of an orbital $3k_B T$ below the chemical potential?

12.9 What the trajectory looks like

Section 11.8 read a run through its energy, its bonds and its centre of mass. This section reads it through its temperature and its velocities: a healthy run of liquid argon first, then six faults that show in them, each with the check that catches it.

A healthy run

Figure 12.8 follows the 256 atoms of liquid argon of **Section 12.5** for 50 ps at fixed energy. Its kinetic temperature, over 765 freedoms, averages 134.64 K over every step (the frames saved every 50 fs give the 134.63 K of **Section 12.5**), with a standard deviation of 4.13 K, 3.07% of the mean, and stays between 120.4 and 147.8 K: a band with no trend, narrower than the canonical 5.11% of **Equation (12.13)**, as **Equation (12.14)** requires of a run at fixed energy. The total energy fluctuates by 0.0037 of the kinetic energy's spread, the check of **Section 11.8**. The temperatures of the three directions, each counting one component of every velocity, have the running averages of **Figure 12.8(b)**, which end at 132.80, 135.32 and 134.20 K: the motion has no preferred direction, as

equipartition requires, to within 2% after 50 ps. Their mean, 134.11 K, is 765/768 of the kinetic temperature, since each counts 256 components while the removed drift takes one freedom from each direction (Section 12.6). The excess kurtosis of the velocities, computed on each of 1001 frames of 768 components, has the mean -0.010 and the standard deviation 0.190 , and 4.2% of the frames lie more than twice that from zero, close to the 4.55% beyond two standard deviations of a Gaussian (Section 12.1).

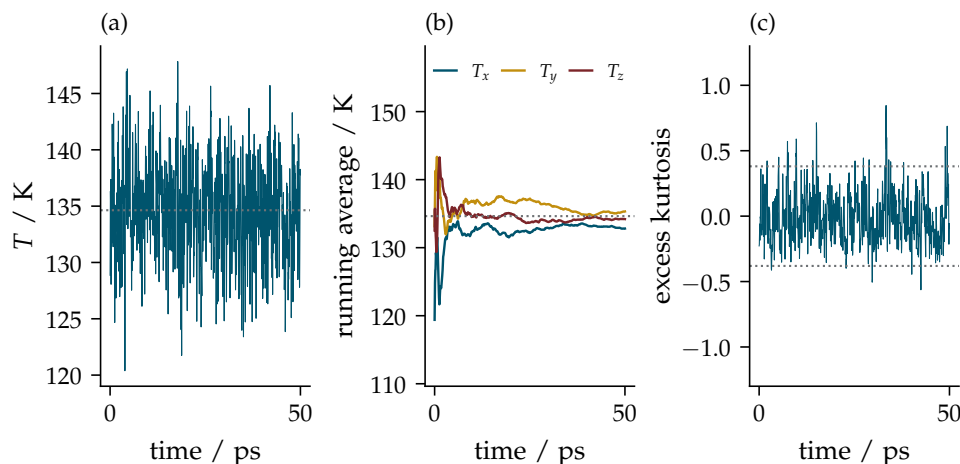


FIGURE 12.8: A healthy run: 256 atoms of liquid argon at fixed energy for 50 ps, steps of 10 fs. (a) The kinetic temperature, with its mean (dotted). (b) The running averages of the temperatures of the x , y and z components of the velocities. (c) The excess kurtosis of the velocities every 50 fs, with twice its standard deviation over the run either side of zero (dotted).

Six faults

Figure 12.9 shows six faults and Table 12.2 the check that catches each.

TABLE 12.2: The faults of Figure 12.9, the check that catches each, and its value in the healthy run (for (f), under equal sharing) and in the broken one.

fault	check	healthy	broken
(a) freedoms miscounted	T against the value asked	303.9 K	201.0 K
(b) lattice start	mean T over the last 10 ps / $T(0)$	1.042	0.506
(c) drift counted	total momentum, amu Å/fs	1.5×10^{-14}	8.857
(d) masses ignored	T of each species at the start	128, 119 K	88, 171 K
(e) not Gaussian	excess_kurtosis at the start	-0.23	-1.09
(f) no sharing	share of modes 1 to 4	0.125	0.917

(a) *Freedoms miscounted*. The rigid water of Section 11.3 was prepared with 12.93 meV for each of its 381 freedoms, $\frac{1}{2}k_B T$ at 300 K, and over its 10 ps run the kinetic temperature with that count averages 303.9 K. Divided among $3N = 576$, as if no distances were held and no drift removed, the same kinetic energy reads 201.0 K, 0.661 of it, which is 381/576. The run is sound and only the reading is wrong. A thermostat that used the

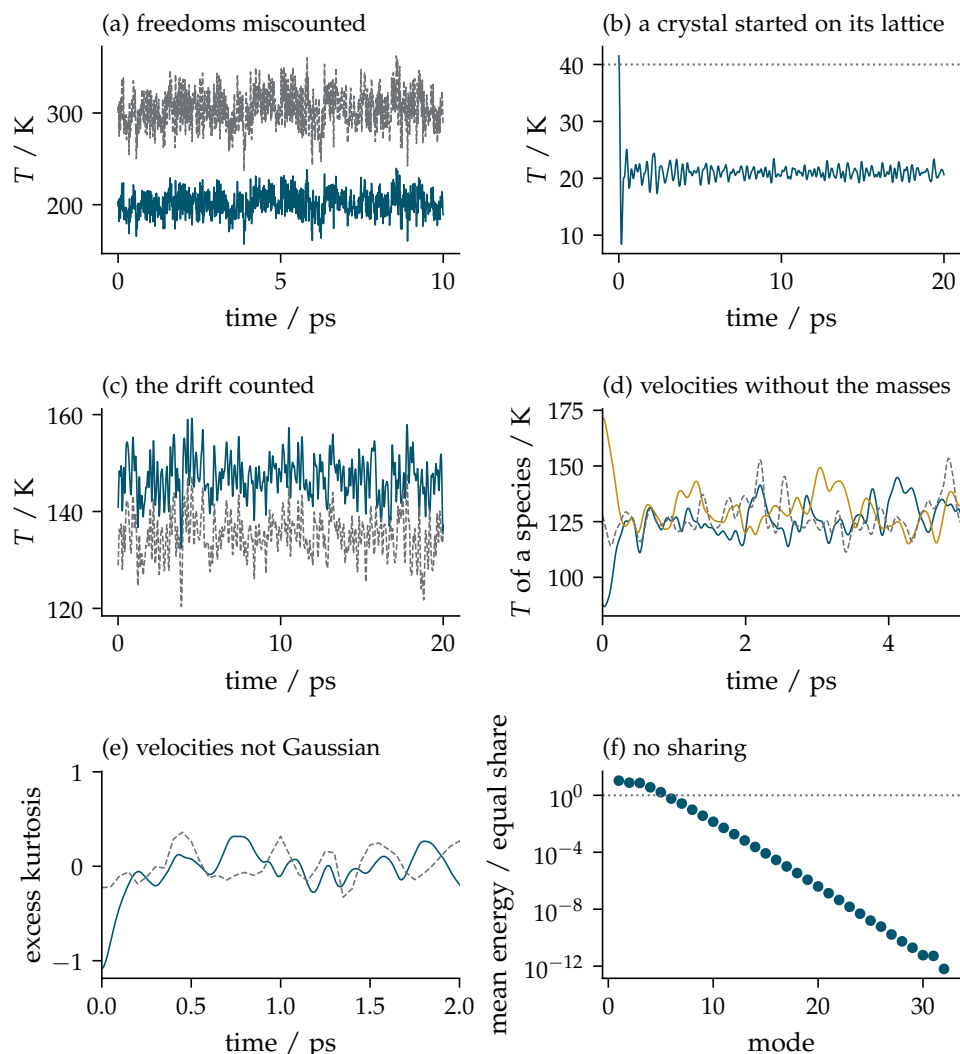


FIGURE 12.9: Faults that show in the temperature (teal), against the healthy run (grey dashed). (a) The rigid water of Chapter 11 at 300 K, its kinetic energy divided among 381 freedoms (grey) and among $3N = 576$ (teal). (b) 256 argon atoms started on their lattice with velocities drawn at 40 K (dotted). (c) The liquid started with a drift of $0.0005 \text{ \AA fs}^{-1}$ in each component of every velocity. (d) A liquid of atoms of two masses, 39.948 and 83.798 amu, started with one spread of velocity for all: the temperatures of the light (teal) and heavy (ochre) atoms, against the light atoms of the same liquid started with the masses (dashed). (e) The liquid started with velocity components spread evenly over an interval: the excess kurtosis. (f) The chain of Figure 12.2(a): the energy of each mode averaged over 20 000 units of time, divided by an equal share (dotted), on a logarithmic axis.

wrong count would heat the water until the wrong reading came up to its target, here by a factor of 1.51. The check is the kinetic energy per freedom against $\frac{1}{2}k_B T$, with the freedoms counted by `degrees_of_freedom`.

(b) *A crystal started on its lattice.* With velocities drawn at 40 K, the kinetic temperature at the start is 41.5 K; it falls below 20 K within 0.1 ps, reaches 8.4 K at 160 fs, and settles at 21.0 K, 0.506 of the start, as equipartition predicts for a nearly harmonic crystal (Section 12.6). The healthy liquid, prepared by several rounds of redrawn velocities, rises by 4% from its start. A run whose temperature moves far from its start has not reached the state asked for, and its early part is not yet the ensemble wanted; Chapter 15 finds where such a run's averages can begin.

(c) *The drift counted.* Starting velocities with a drift of $v_d = 0.0005 \text{ \AA fs}^{-1}$ in each component move the whole box, of mass $M = 256 \times 39.948 = 10\,227 \text{ amu}$, at $\sqrt{3}v_d$, with the kinetic energy $\frac{1}{2}M \times 3v_d^2 \times 103.6427 = 0.397 \text{ eV}$ and the momentum $M\sqrt{3}v_d = 8.857 \text{ amu \AA/fs}$, which momentum conservation keeps for ever. Counted as heat over 765 freedoms it reads as $2 \times 0.397 / (765k_B) = 12.1 \text{ K}$: the run shows 147.1 K, of which the motion of the atoms relative to their centre of mass gives 135.0 K. The temperature looks healthy, a steady band; the check is the total momentum, 8.857 against $1.5 \times 10^{-14} \text{ amu \AA/fs}$ in the healthy run, or `centre_of_mass_path` of Section 11.8. A thermostat acting on the total kinetic energy would cool the relative motion to make room for the drift; Chapter 13 meets the reverse, a thermostat that feeds the drift, as the flying ice cube.

(d) *Velocities drawn without the masses.* Drawing every atom's components with one spread gives the light atoms of a mixture too little kinetic energy and the heavy ones too much: 88 K and 171 K at the start, against 128 and 119 K when the masses are used, values that scatter by $\sqrt{2/384} = 7\%$, Equation (12.13) for the 384 components of 128 atoms. Collisions share the energy out, and the gap between the species first averages below 15 K about 260 fs in; over the last 10 ps the two species are at 130 and 128 K. In a liquid the fault costs only the first picosecond. The check, `part_temperature` for each species, matters most where energy is shared slowly, as in (f).

(e) *Velocities not Gaussian.* Components drawn evenly from an interval, with the right mean square, have an excess kurtosis of -1.09 at the start, near the -1.2 of an even spread, and the run reaches -0.43 after 100 fs and -0.06 after 200 fs, within the scatter of the healthy run. The temperature is right throughout; only the shape of the distribution shows the fault, and in a liquid it heals within a fraction of a picosecond. The fault is easy to make: the forecasting code of TrajCast¹, asked to draw starting velocities without naming a distribution, draws every component evenly from one interval, the same for every mass, and rescales to the temperature, faults (d) and (e) together. A kurtosis that stays away from zero shows velocities that have not yet reached the Maxwell-Boltzmann distribution.

(f) *No sharing.* The chain of Fermi, Pasta, Ulam and Tsingou of [Section 12.2](#), averaged over 20 000 units of time, keeps 0.917 of its energy in its first four modes, against $4/32 = 0.125$ for equal sharing; the first mode holds 10.7 times an equal share and no mode from the eighth on holds more than 0.097 of one. Equipartition fails because the trajectory does not explore its energy shell. The check, the energy of each mode, or more generally the temperature of each part of a system, detects a system that is not ergodic on the time scale of the run, which no single average over the whole system reveals.

IN PRACTICE. A run's temperature is read with the freedoms counted for that run, $3N - n_c - 3$ for a periodic run whose drift is removed, and compared with the value asked for only after the early part has settled. The total momentum should be zero, the temperatures of the three directions and of each species should agree to the scatter of the run, and the excess kurtosis of the velocities should be near zero. A run at fixed energy has a kinetic temperature whose spread is below $\sqrt{2/N_f}$ of its mean; a run that claims to be canonical must reach it, which is the first test of a thermostat in Chapter 13.

KEY IDEA. A healthy run's kinetic temperature, counted over the right freedoms, holds a steady band, with equal temperatures in each direction and for each species and Gaussian velocities. Miscalculated freedoms shift the reading; a lattice start halves the temperature; a kept drift adds a spurious steady part; velocities drawn without the masses or not Gaussian heal in a liquid within a picosecond; a system that does not share its energy keeps it in a few motions.

NOTEBOOK 12 *break the liquid in each of the six ways and read the checks*

CHECK YOUR UNDERSTANDING. A run of 500 argon atoms, drift removed, reports 300 K from a code that divides by $3N$. What temperature does the right count give?

What we now have

A probability is the long-run fraction of trials with an outcome; a continuous quantity has a density, the Gaussian chief among them, and variances of independent quantities add. An ensemble is a cloud of copies of a system in phase space, and the ergodic hypothesis equates its averages with the time averages of one run: it is consistent with liquid argon over 50 ps and fails for the harmonic chain. The chain of Fermi, Pasta, Ulam and Tsingou has not shared its energy equally even after millions of units of time; this is a finite-time limit on the averages we can obtain.

Counting the states of an isolated system gives its entropy $S = k_B \ln \Omega$, and two systems sharing energy settle where $d \ln \Omega / dE = 1/k_B T$ agrees. A system in contact with a

bath has the Boltzmann probabilities $e^{-E/k_B T}$, the partition function, the free energy $F = -k_B T \ln Z$ and an energy variance $k_B T^2 C_V$.

Each component of velocity is Gaussian with variance $k_B T/m$, whatever the forces, and every square term of the energy holds $\frac{1}{2}k_B T$, so the kinetic temperature $2K/(N_f k_B)$ measures T with $N_f = 3N - n_c - 3$; `mdlab` and ASE agree once they count alike.

The kinetic energy has the gamma distribution with relative spread $\sqrt{2/N_f}$ in the canonical ensemble and less at fixed energy, by an amount that gave the heat capacity of liquid argon, $2.38 N k_B$, from single runs to within 7.5%.

Baths that take volume or atoms give the NPT and grand canonical ensembles, in which $F + PV$ and $F - \mu N$ are least as F is in the canonical one, each the energy's Legendre transform with its sign reversed; Fermi-Dirac occupations are the grand canonical ensemble of an orbital.

A healthy run holds its kinetic temperature in a steady band, the same in every direction and for every species, with Gaussian velocities; six faults each leave their own mark.

Chapter 13 builds the thermostats that put a simulation in contact with a bath, and tests each against the canonical distribution of the kinetic energy derived here.

Answers to the checks

SECTION 12.1. Totals of 10, 11 and 12 come from $3 + 2 + 1 = 6$ of the 36 outcomes: $6/36 = 1/6$.

SECTION 12.2. $\omega_N = 2 \sin(N\pi/(2N+2))$, and $N\pi/(2N+2)$ approaches $\pi/2$ as N grows, so ω_N approaches 2 and the period $2\pi/\omega_N$ approaches π ; for $N = 32$ it is 3.145.

SECTION 12.3. $1/\sqrt{120} = 0.0913$.

SECTION 12.4. $k_B T = 0.008\,617\text{ eV}$ and $e^{-0.3/0.008617} = 7.6 \times 10^{-16}$.

SECTION 12.5. σ_v grows as $\sqrt{T}$: $\sqrt{2} \times 0.005995 = 0.008\,48\text{ Å fs}^{-1}$.

SECTION 12.6. 300 atoms and 300 held distances give $N_f = 900 - 300 - 3 = 597$, and $T = 2 \times 7.4/(597k_B) = 287.7\text{ K}$.

SECTION 12.7. $N_f = 2997$ and $\sqrt{2/2997} = 0.0258$, or 2.6%.

SECTION 12.8. $f = 1/(e^{-3} + 1) = 0.9526$.

SECTION 12.9. The code divides by 1500 instead of 1497, so the right count gives $300 \times 1500/1497 = 300.6 \text{ K}$.

Exercises

The exercises run from questions of understanding, through calculations, to short derivations marked ‘show that’. Full solutions are in ‘Solutions to the exercises’ at the end of the book, and Notebook 12 checks every numerical answer.

EXERCISE 12.1 (*a fair die*). Find the mean and the variance of the number shown by one fair die, and the variance of the total of two.

EXERCISE 12.2 (*the spread of a mean*). Show that the mean of n independent values, each with the variance σ_x^2 , has the variance σ_x^2/n . How many throws of a die make the standard deviation of their mean no more than 0.01?

EXERCISE 12.3 (*a chain of three*). Find the angular frequencies of the three normal modes of a chain of three atoms of unit mass between walls, joined by springs of unit stiffness, from $\omega_k = 2 \sin(k\pi/(2N+2))$, and check that their squares are the eigenvalues of the matrix with 2 on its diagonal, -1 just above and below it, and 0 elsewhere.

EXERCISE 12.4 (*entropies add*). Show that two independent systems, with Ω_1 and Ω_2 microstates, have together the entropy $S_1 + S_2$.

EXERCISE 12.5 (*how large a bath*). A system holding 0.1 eV is in contact with a bath of 10^4 atoms of an ideal gas at 300 K. What is the ratio of the term of second order in the expansion of Section 12.4 to the term of first order, $E_s/k_B T$, which is kept?

EXERCISE 12.6 (*two levels*). A system has two states, of energies 0 and E_g . Find Z , $\langle E \rangle$ and C_V . For $E_g = 0.1 \text{ eV}$, give $\langle E \rangle$ and C_V/k_B at 300 K, and find numerically the temperature at which C_V is largest.

EXERCISE 12.7 (*free energy and entropy*). Show that $F = -k_B T \ln Z$ equals $\langle E \rangle - TS$ with $S = -k_B \sum_s \mathcal{P}_s \ln \mathcal{P}_s$, and that this S is $k_B \ln \Omega$ when all Ω states are equally likely.

EXERCISE 12.8 (*a vibration's energy fluctuates*). Show from the partition function $Z = 2\pi/(\beta\omega)$ of one vibration that the variance of its energy is $(k_B T)^2$, and check Equation (12.6) with $C_V = k_B$.

EXERCISE 12.9 (*odds and temperature*). At what temperature are the odds of the 0.3 eV barrier of Section 12.4 10^{-3} ? What are the odds of a 0.5 eV barrier at 300 K?

EXERCISE 12.10 (*speeds*). Show that the speed density of Equation (12.8) is largest at $\sqrt{2}\sigma_v$ and has the mean $\sqrt{8/\pi}\sigma_v$, and give both for lithium at 300 K.

EXERCISE 12.11 (*fourth moments*). Show that the Gaussian of mean zero has $\langle x^4 \rangle = 3\sigma_x^4$, by parts as in Section 12.1, and that values spread evenly over an interval $-c$ to c have the excess kurtosis -1.2 .

EXERCISE 12.12 (*light and heavy*). In a classical simulation of lithium-intercalated graphite at 300 K, find σ_v for a lithium atom (6.94 amu) and a carbon atom (12.011 amu), and their ratio. Which has the larger mean kinetic energy?

EXERCISE 12.13 (*a turning molecule*). A molecule of two atoms with moment of inertia I turns freely; its axis points along the angles θ from z and ϕ about z , and its kinetic energy is $\frac{1}{2}I(\dot{\theta}^2 + \sin^2\theta\dot{\phi}^2)$. Show that $H = p_\theta^2/2I + p_\phi^2/(2I\sin^2\theta)$, and that its mean is $k_B T$.

EXERCISE 12.14 (*counting freedoms*). How many kinetic freedoms have (a) an isolated molecule of carbon dioxide, a linear molecule of three atoms, with its drift and turning removed; (b) 108 argon atoms in a periodic cell with the drift removed? For (b), by what factor does ASE's temperature differ from mdlab's?

EXERCISE 12.15 (*how many atoms*). How many atoms, with the drift removed, does a canonical run need for its kinetic temperature to have a standard deviation of 1% of its mean?

EXERCISE 12.16 (*the most probable kinetic energy*). Show that the gamma distribution of Equation (12.12) is largest at $K = (\frac{1}{2}N_f - 1)k_B T$. Where is it largest for one atom, and why is that not $\frac{3}{2}k_B T$?

EXERCISE 12.17 (*a harmonic crystal at fixed energy*). Show from Equation (12.14) that the kinetic energy of a harmonic crystal at fixed energy fluctuates by $1/\sqrt{2}$ of the canonical amount.

EXERCISE 12.18 (*heat capacity from one run*). A run of 500 atoms at fixed energy, drift removed, has a kinetic energy whose standard deviation is 0.025 of its mean. Find C_V per atom in units of k_B .

EXERCISE 12.19 (*volume fluctuations*). With the NPT partition function

$$Z_{NPT} = \int e^{-\beta PV} Z(N, V, T) dV,$$

show that the mean volume is

$$\langle V \rangle = -k_B T \frac{\partial \ln Z_{NPT}}{\partial P}$$

and that its variance is $-k_B T d\langle V \rangle / dP$.

EXERCISE 12.20 (*the other potentials*). From $dE = T dS - P dV + \mu dN$, show that $F + PV$ changes by $-S dT + V dP + \mu dN$ and $F - \mu N$ by $-S dT - P dV - N d\mu$.

EXERCISE 12.21 (*Fermi-Dirac*). Show that the Fermi-Dirac occupation satisfies $f(\mu + x) = 1 - f(\mu - x)$ and has the slope $-1/(4k_B T)$ at $\epsilon_o = \mu$. Find the entropy, in units of k_B , of an orbital 0.1 eV above μ at 300 K.

EXERCISE 12.22 (*a diagnosis*). A long run of argon at fixed energy reports steady direction temperatures $T_x = T_y = 295$ K and $T_z = 310$ K, each counting one component of every velocity. What fault does this suggest, which check confirms it, and what velocity of the centre of mass along z would account for the difference?

Thermostats

Chapter 12 described the canonical distribution of a system in contact with a heat bath. To reproduce that situation in a simulation, an algorithm must exchange energy with the atoms while preserving the required distribution. Such an algorithm is called a thermostat. We judge it by two questions: whether it samples the canonical ensemble, and how much it changes the motion during sampling.

We develop six approaches, beginning with direct velocity rescaling and ending with the stochastic rescaling used by TrajCast. Comparisons on liquid argon and on lithium atoms moving over the model surface of Chapter 3 show why a suitable method depends on the measurement. The chapter draws together choices for equilibration, equilibrium averages and dynamics, then examines the signs of faulty thermostating in a trajectory. Chapter 14 applies the same distinction between mean values, fluctuations and motion to pressure control.

13.1 What a thermostat must do

A real sample keeps its temperature by exchanging energy with what surrounds it, which a simulation leaves out. A thermostat stands in for those surroundings. This section sets out what it must achieve, what it should leave alone, how the chapter tests both, and the single loop into which every thermostat of the chapter fits.

Two demands

The first demand is the canonical ensemble of [Section 12.4](#). A run's time averages should equal averages with the weight $e^{-\beta H}$, which means three things that can be checked: the kinetic energy of the N_f freedoms has the gamma distribution of [Equation \(12.12\)](#), with the relative spread $\sqrt{2/N_f}$; the positions have the density $e^{-\beta U}$ ([Section 12.4](#));

and the velocities are Maxwellian and independent of the positions (Section 12.5). A thermostat that holds the mean temperature but narrows the spread of K samples the wrong ensemble, and every quantity drawn from its fluctuations, such as the heat capacity of Equation (12.6), is wrong.

The second demand is softer: the motion should stay close to the motion of the real system. In a sample the surroundings exchange energy only at its edges and only slowly, and the atoms inside move by Newton's laws. A thermostat acts on every atom at every step, so it changes the motion everywhere, and how much it changes it decides whether a diffusion coefficient, a rate of hopping or a vibrational frequency measured under it means anything. The two demands can pull apart: a thermostat that acts on each atom separately, as those of Sections 13.3 and 13.4 do, disturbs the motion more the more firmly it holds the temperature.

The tests

Every thermostat in the chapter is tested on two systems. The first is the liquid argon of Chapter 12, 256 atoms at 135 K, started from three frames of the run of Section 12.2 20 picoseconds apart and followed for 100 ps with steps of 10 fs, the first 5 ps of each set aside for the distribution of K ; the three runs give each measured quantity a mean and a spread. On it we measure the distribution of K ; the diffusion coefficient D , which measures how fast atoms wander: the square of the distance an atom moves in a time t , averaged over the atoms and the starting times, grows in proportion to t in a liquid, and D is a sixth of its slope between 2 and 20 ps, with positions measured from the centre of mass (Chapter 16 treats this properly, the six included); and the cost, the computing time of a step against that of velocity Verlet alone.

The second is lithium on the model surface of Section 3.13, standing for lithium in graphite: 1000 atoms, each on its own copy of the surface with its own thermostat, at 1000 K, started from the density $e^{-\beta U}$ and the Maxwell-Boltzmann distribution. The temperature is high so that hops are frequent: the estimate of Section 13.4 puts one hop every 1.69 ns at 300 K and 1.26 ps^{-1} at 1000 K. Each atom has only two freedoms, the hardest case for a thermostat: its kinetic energy should spread as widely as Equation (12.12) with $N_f = 2$, a relative spread of 1, and it has no other atoms to share energy with. Its exact canonical density of positions is known, so the distribution of its potential energy can be compared with the exact one, computed on a grid over one cell of the surface; and its hops can be counted.

The heat and the effective energy

A thermostat adds or removes kinetic energy at each step. Adding up what it adds over a run gives the heat it has put in, and the effective energy

$$E_{\text{eff}} = K + U - \text{heat} \quad (13.1)$$

counts every change of $K + U$ that the thermostat did not make. Between the thermostat's actions the motion is Newton's, which keeps $K + U$, so in exact dynamics E_{eff} never changes, for every thermostat. In a run it drifts only through the error of the integrator, and that drift is the thermostatted counterpart of the energy check of [Section 11.8](#).

One loop for every thermostat

Velocity Verlet ([Equation \(9.3\)](#)) is a half kick, a drift and a half kick. Splitting the drift into two halves, as [Section 9.4](#) split the Liouville operator, gives four pieces, B A A B, with B a half kick and A a half drift. A thermostat may act before the step, between the two half drifts, or after the step ([Figure 13.1](#)). `mdlab.thermostats.run` is that loop:

```

1     for step in range(n_steps + 1):
2         if step > 0:
3             v = act(th.before, v)
4             v = v + 0.5 * dt * accel * f
5             r = r + 0.5 * dt * v
6             v = act(th.middle, v)
7             r = r + 0.5 * dt * v
8             u, f = model(r)
9             v = v + 0.5 * dt * accel * f
10            v = act(th.after, v)

```

The pass with step 0 makes no step; it and every later pass end by recording the state, which the listing leaves out. `accel` is `FORCE_TO_ACCEL` divided by each mass, and `act` calls the thermostat's method and adds the change of kinetic energy it makes to the heat. With no thermostat the three calls do nothing and the loop is velocity Verlet; the thermostats of the next sections each fill one or two of the calls. ASE implements five of them independently. For four, Berendsen's, Langevin's, the Nosé-Hoover chain and CSVR, from the same start of 32 argon atoms, with the same random numbers for the two that use them, the two codes agree after 50 steps of 5 fs to within 1×10^{-7} Å in every position. ASE's version of Andersen's draws its random numbers differently, so the two are compared by what they sample: over four runs of 100 ps of the same atoms at 120 K, the mean kinetic temperature is (119.40 ± 0.55) K under ASE's and (119.95 ± 0.37) K under `mdlab`'s, each with 0.99 to 1.01 of the canonical spread.

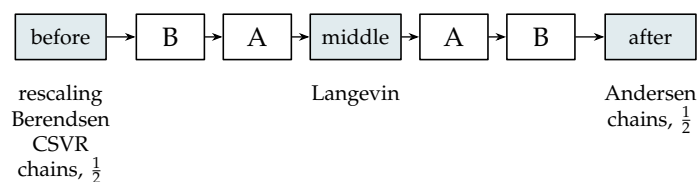


FIGURE 13.1: One step of `mdlab.thermostats.run`: velocity Verlet as a half kick (B), two half drifts (A) and a half kick, with the places where each thermostat of the chapter acts; Nosé-Hoover chains act for half a step before and half after.

KEY IDEA. A thermostat must give the canonical distributions, the gamma distribution of K with spread $\sqrt{2/N_f}$ and the density $e^{-\beta U}$ of positions, and should disturb the motion as little as it can. The effective energy, $K + U$ less the heat the thermostat has added, is kept by the exact dynamics of every thermostat, and its drift measures the integration error.

CHECK YOUR UNDERSTANDING. A thermostat removes 0.5 eV over a run in which $K + U$ falls by 0.48 eV. By how much has the effective energy changed?

13.2 Rescaling and weak coupling

The simplest way to hold a temperature is to multiply every velocity by the factor that brings the kinetic energy to its target. This section builds that, and the gentler version of Berendsen and his colleagues, measures what both do to the distribution of K , and derives two faults they share.

Rescaling

Before each step, `Rescale` multiplies every velocity by $\lambda = \sqrt{K_0/K}$, with $K_0 = \frac{1}{2}N_f k_B T_0$ the kinetic energy of the target temperature T_0 ; multiplying every velocity by λ multiplies K by λ^2 . The kinetic energy is then exactly K_0 at the start of every step, and the only fluctuation left is what the motion produces within one step: over 95 ps of liquid argon, recorded at the ends of the steps, the relative spread of K is 0.0056, against the canonical 0.0511. The canonical ensemble has a spread; rescaling all but removes it, and with it any use of the fluctuation formulae of [Section 12.7](#). The scaling factor λ is a symbol of this chapter only ([Sections 6.6](#) and [11.1](#) used it for a multiplier).

Weak coupling

Berendsen and colleagues asked the temperature instead to relax towards T_0 with a time constant τ_T ⁵⁶,

$$\frac{dT}{dt} = \frac{T_0 - T}{\tau_T}.$$

Here T is the kinetic temperature T_{kin} of Chapter 12. Over one step δt the kinetic temperature should change by $(\delta t/\tau_T)(T_0 - T)$. Scaling the velocities by λ turns T into $\lambda^2 T$, so

$$\lambda^2 T = T + \frac{\delta t}{\tau_T}(T_0 - T), \quad \lambda^2 = 1 + \frac{\delta t}{\tau_T} \left(\frac{T_0}{T} - 1 \right). \quad (13.2)$$

At 150 K, with a target of 135 K, steps of 10 fs and $\tau_T = 100$ fs, $\lambda^2 = 0.9900$ and $\lambda = 0.99499$: the velocities shrink by half a percent. With $\tau_T = \delta t$ this is rescaling; as τ_T grows, the run approaches one at fixed energy.

One thing follows from Equation (13.2). The scaling turns K into $\lambda^2 K$, so each step adds the heat $(\lambda^2 - 1)K = (\delta t/\tau_T)(K_0 - K)$, since $T_0 K/T = K_0$. Over a long run in which nothing drifts this heat averages to zero, because by Equation (13.1) the total heat is the change of $K + U$, which stays within its fluctuations however long the run; so $\langle K \rangle = K_0$ and the mean temperature is exactly the target. Over three runs it is 135.00 K with $\tau_T = 100$ fs and 135.02 K with 1 ps. But the spread of K is not the canonical one (Figure 13.2(a)): with $\tau_T = 100$ fs it is 0.0228, 0.45 of the canonical spread and smaller even than the 0.0311 of a run at fixed energy; with 1 ps, 0.0300, close to the run at fixed energy. Berendsen coupling does not sample the canonical ensemble at any τ_T : strong coupling squeezes the fluctuations, and weak coupling leaves them those of fixed energy.

What the time constant means

The temperature takes longer than τ_T to settle. Figure 13.2(b) starts the liquid at 200 K by scaling its velocities. Within 0.2 ps the kinetic temperature falls to between 165 and 170 K under every thermostat, as kinetic energy passes into potential energy, the sharing of Section 12.6. After that the thermostat removes heat at the rate $(K - K_0)/\tau_T = C_K(T - T_0)/\tau_T$, with $C_K = \frac{1}{2}N_f k_B$ the kinetic part of the heat capacity (Section 12.7). But the liquid's energy exceeds its value at T_0 by $C_V(T - T_0)$, more than the $C_K(T - T_0)$ held as kinetic energy, so it changes at the rate $C_V dT/dt$, and equating the two,

$$C_V \frac{dT}{dt} = -\frac{C_K}{\tau_T}(T - T_0),$$

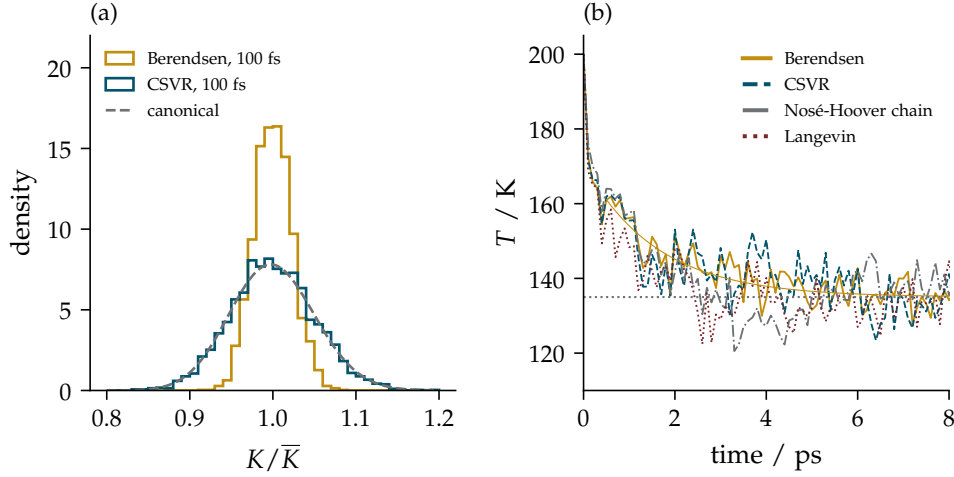


FIGURE 13.2: (a) The kinetic energy of liquid argon, divided by its mean, over three runs of 95 ps under Berendsen and under CSV (Section 13.6), both with $\tau_T = 100$ fs, against the canonical gamma density of 765 freedoms (dashed). (b) The liquid with its velocities scaled to 200 K, brought towards 135 K by Berendsen and CSV with $\tau_T = 1$ ps, a Nosé-Hoover chain with $\tau_T = 1$ ps (Section 13.5) and Langevin with $\gamma = 1 \text{ ps}^{-1}$ (Section 13.4), against the approach predicted for Berendsen (thin). The target is the dotted line.

and $T - T_0$ decays as $e^{-t/\tau_{\text{eff}}}$, as the gap w of Section 2.7 did, with $\tau_{\text{eff}} = \tau_T C_V / C_K$. With $C_V = 2.376 N k_B$ and $C_K = 1.4941 N k_B$ from Section 12.7, $C_V / C_K = 1.590$, so $\tau_{\text{eff}} = 1.59 \text{ ps}$ for $\tau_T = 1 \text{ ps}$; a fit to the run from 0.5 ps on gives 1.67 ps, uncertain by 0.19 ps in a single run.

The flying ice cube

Rescaling and Berendsen coupling multiply every velocity by the same factor, including the part of each velocity that belongs to the motion of the whole box. No force changes that motion (Section 2.10), so its kinetic energy K_{com} changes only by the thermostat, by the factor λ^2 at each step. Over n steps, $\ln K_{\text{com}}$ gains $\sum \ln \lambda^2$. Writing $T = T_0 + \delta T$, whose departure δT from the target has mean zero since $\langle T \rangle = T_0$,

$$\lambda^2 - 1 = \frac{\delta t}{\tau_T} \left(\frac{1}{1 + \delta T / T_0} - 1 \right) = \frac{\delta t}{\tau_T} \left(-\frac{\delta T}{T_0} + \left(\frac{\delta T}{T_0} \right)^2 - \dots \right),$$

$$\langle \ln \lambda^2 \rangle \approx \langle \lambda^2 - 1 \rangle \approx \frac{\delta t}{\tau_T} \left\langle \left(\frac{\delta T}{T_0} \right)^2 \right\rangle,$$

to first order in $\delta t / \tau_T$ and second in $\delta T / T_0$, using $1/(1+y) = 1 - y + y^2 - \dots$, which holds because $(1+y)(1-y+y^2) = 1 + y^3$, and $\ln(1+y) \approx y$ for small y , the Taylor series of Section 1.8 with the slope $1/(1+y)$ of the logarithm, 1 at $y = 0$. The mean is positive whatever the sign of the fluctuations, and over n steps, a time $t = n \delta t$, $\ln K_{\text{com}}$ gains $n \langle \ln \lambda^2 \rangle \approx \langle (\delta T / T_0)^2 \rangle t / \tau_T$: K_{com} grows as $e^{\langle (\delta T / T_0)^2 \rangle t / \tau_T}$, at the expense of the motion of the atoms against one another, which the thermostat cools to keep

the total at K_0 . In time the box flies through space while its atoms freeze, the flying ice cube of Harvey, Tan and Cheatham, who found velocity rescaling to break the equipartition of energy⁵⁷, revisited by Braun, Moosavi and Smit⁵⁸. For liquid argon with $\tau_T = 100$ fs, $\langle(\delta T/T_0)^2\rangle = 4.99 \times 10^{-4}$ in the first of the three runs, and the predicted rate is 0.0050 ps^{-1} ; keeping the next term of $\ln(1+y)$, $-\frac{1}{2}y^2$, lowers it by the factor $1 - \delta t/2\tau_T = 0.95$, to 0.0047 ps^{-1} , which Section 13.8 measures. A weaker form of the same flow feeds any motion that the forces barely touch, such as the slow vibrations of large molecules, held back only by its slow exchange of energy with the other motions, which is why the effect appears even when the motion of the centre of mass has been removed^{57,58}. For one lithium atom on the surface, whose two freedoms are its whole motion, Berendsen coupling with $\tau_T = 100$ fs drives the atom within a picosecond or two onto a circular orbit of radius $0.48 \text{ \AA}$ round its hollow, on which K stays at $k_B T_0$ and $K + U$ at 0.19 eV , short of the 0.3 eV barrier, and the thermostat has nothing left to correct: the 1000 atoms make 0.001 hops per atom per picosecond, against 0.93 to 1.09 under the thermostats that sample correctly.

IN PRACTICE. Rescaling and Berendsen coupling are for bringing a system near a temperature quickly. Their averages of T are right, but their fluctuations are not, and a long run under them feeds slow motions and the motion of the centre of mass. A production run, whose fluctuations or slow motions matter, uses one of the thermostats of the next sections.

KEY IDEA. Berendsen coupling scales the velocities by the factor λ of Equation (13.2), so that T relaxes towards T_0 . It holds the mean temperature exactly but narrows the spread of K below the canonical one, and it feeds motions that the forces do not touch, at the rate $\langle(\delta T/T_0)^2\rangle/\tau_T$: the flying ice cube. The temperature of a liquid relaxes with the time constant $\tau_T C_V/C_K$, longer than τ_T .

NOTEBOOK 13 set the coupling time and watch the spread of K and the drift

CHECK YOUR UNDERSTANDING. What is λ for a step of 2 fs with $\tau_T = 1 \text{ ps}$, at 310 K with a target of 300 K?

13.3 Collisions with a bath

A gas held at a temperature by the walls of its container gains its temperature in collisions: an atom that strikes the wall leaves it with a velocity typical of the wall's temperature. Andersen built a thermostat on that picture⁵⁹. This section shows why it samples the canonical ensemble and measures what it does to the motion.

The method

Each atom collides with an imagined bath at the rate ν , and a collision replaces its velocity with one drawn afresh from the Maxwell-Boltzmann distribution at T_0 , by Equation (12.7). Collisions at the rate ν , each independent of the last, leave an atom untouched over a step with the probability $e^{-\nu\delta t}$ (Exercise 13.1), so after each step Andersen redraws the velocity of each atom with the probability $1 - e^{-\nu\delta t}$. ASE's version redraws single components instead of whole velocities, after the first half kick instead of after the step, and by default removes the motion of the centre of mass; its redraws keep the same distribution, and Section 13.1 compares the two.

Why it is canonical

The canonical density $e^{-\beta H}/Z$ is kept by both parts of the motion. Between collisions the atoms follow Hamilton's equations, which keep every density that depends on the state only through H (Section 7.4). In a collision one atom's velocity is replaced; in the canonical density that velocity is independent of every other coordinate and momentum and has the Maxwell-Boltzmann distribution (Section 12.5), so replacing it with a fresh, independent draw from the same distribution leaves the joint density as it was. A density that neither part changes does not change at all: the canonical ensemble is stationary under Andersen's thermostat, and the collisions, which move the run from one energy to another, carry it through the ensemble⁵⁹.

What it does to the motion

A collision forgets the atom's velocity, so a thermostat that collides often erases the momentum by which atoms push through a liquid. Over three runs each, the diffusion coefficient of liquid argon is 0.96, 0.72 and 0.17 of its value at fixed energy, $(3.81 \pm 0.08) \times 10^{-4} \text{ \AA}^2/\text{fs}$, for $\nu = 0.1, 1$ and 10 ps^{-1} (Figure 13.3(a) of Section 13.4, with Langevin's friction). At the weakest coupling the motion is barely disturbed, but the temperature is held loosely: each atom collides once in 10 ps, the three runs average 133.66, 130.13 and 131.69 K over their last 95 ps, counting all $3N = 768$ freedoms since the collisions move the centre of mass too, and the spread of K within them is 0.85 of the canonical spread. At $\nu = 1 \text{ ps}^{-1}$ the spread is 0.98 of it and the mean $(134.5 \pm 0.6) \text{ K}$. Collisions also change the total momentum, so the box as a whole wanders, which Section 13.8 shows to matter when displacements are measured. For one lithium atom on the surface, with $\nu = 1 \text{ ps}^{-1}$, the density of its potential energy matches the exact one to within 0.011 of its peak.

KEY IDEA. Andersen's thermostat redraws each atom's velocity from the Maxwell-Boltzmann distribution at the rate ν . Hamilton's equations and the redraws both keep the canonical density, and the redraws carry a run from one energy to another, so the thermostat samples it, even for a single atom. The redraws erase momentum: the diffusion coefficient of liquid argon falls to 0.72 of its value at $\nu = 1 \text{ ps}^{-1}$ and to 0.17 at 10 ps^{-1} , while weak coupling holds the temperature only loosely.

CHECK YOUR UNDERSTANDING. With $\nu = 1 \text{ ps}^{-1}$ and steps of 2 fs, what fraction of atoms is redrawn at each step, and how many atoms of 1000 on average?

13.4 Friction and noise

A grain of pollen in water is struck by molecules from every side; on average the blows slow it down, a friction, and their irregularity jostles it, a noise. Langevin's equation keeps both and nothing else of the water. This section solves the equation for one velocity, finds the relation between friction and noise that gives the canonical distribution, builds the BAOAB step from it, and measures how the friction changes diffusion and the rate of hopping.

Random kicks

A SUM OF RANDOM KICKS. A velocity that receives, at each of many short intervals of length h , a kick $\sigma\sqrt{h}g$, with g a fresh number from the normal distribution of mean 0 and variance 1, has after n kicks the variance $n\sigma^2h = \sigma^2t$, with $t = nh$. Each kick has the variance σ^2h , since multiplying a quantity by a constant multiplies every deviation from its mean by that constant, and so its variance by the constant's square; and variances of independent kicks add (Section 12.1). The variance grows in proportion to the time, whatever the length h into which it is cut, which is why each kick is scaled by $\sqrt{h}$: kicks of $\sigma h g$ would give the variance σ^2th , which vanishes as h shrinks. A sum of independent Gaussian numbers is itself Gaussian (Exercise 13.6), so after any time the velocity has a Gaussian distribution, fixed by its mean and variance. The limit of such kicks as h shrinks is the Wiener process.

Langevin's equation for one velocity

Without forces, one component of an atom's velocity under friction γ and noise of strength σ changes over a short interval h by

$$v_{k+1} = (1 - \gamma h) v_k + \sigma\sqrt{h} g_k,$$

the change $-\gamma v_k h$ that the drag $dv/dt = -\gamma v$ of [Section 2.7](#) makes over the interval, plus a random kick. After n intervals, covering the time $\delta t = nh$, applying this n times gives

$$v_n = (1 - \gamma h)^n v_0 + \sigma \sqrt{h} \sum_{k=0}^{n-1} (1 - \gamma h)^{n-1-k} g_k.$$

As h shrinks with δt fixed, $(1 - \gamma h)^n$ tends to $e^{-\gamma \delta t}$: its logarithm is $n \ln(1 - \gamma h)$, and $\ln(1 + y)$ is y plus terms of order y^2 for small y (the Taylor series of [Section 1.8](#), with the slope $1/(1 + y)$ of the logarithm, 1 at $y = 0$), so the logarithm is $-n\gamma h = -\gamma \delta t$ plus n errors of order $\gamma^2 h^2$, together of order $\gamma^2 \delta t^2/n$, which vanish as n grows. The kick g_k enters the sum multiplied by $\sigma \sqrt{h} (1 - \gamma h)^{n-1-k}$, which multiplies its variance by the square of that factor, and the variances of the independent kicks add, so, counting them backwards with $j = n - 1 - k$, the sum is Gaussian with the variance

$$\sigma^2 h \sum_{j=0}^{n-1} (1 - \gamma h)^{2j} = \sigma^2 h \frac{1 - (1 - \gamma h)^{2n}}{1 - (1 - \gamma h)^2} \rightarrow \sigma^2 \frac{1 - e^{-2\gamma \delta t}}{2\gamma}.$$

The middle form is the geometric series: the sum $G = \sum_{j=0}^{n-1} z^j$, multiplied by z , loses its first term, 1, and gains z^n , so $zG = G - 1 + z^n$ and $G = (1 - z^n)/(1 - z)$, here with $z = (1 - \gamma h)^2$. In the limit $(1 - \gamma h)^{2n}$, the square of $(1 - \gamma h)^n$, tends to $e^{-2\gamma \delta t}$, and $h/[1 - (1 - \gamma h)^2] = 1/(2\gamma - \gamma^2 h)$ tends to $1/2\gamma$. So over any step δt , exactly,

$$v(t + \delta t) = c v(t) + \frac{\sigma}{\sqrt{2\gamma}} \sqrt{1 - c^2} g, \quad c = e^{-\gamma \delta t}, \quad (13.3)$$

the Ornstein-Uhlenbeck process⁶⁰. Over a long step c is close to 0, and the velocity is Gaussian with mean zero and variance $\sigma^2/2\gamma$, whatever it started at; [Exercise 13.7](#) shows that no other variance is kept by the step.

The fluctuation-dissipation relation

The canonical distribution needs that variance to be $k_B T/m$ ([Equation \(12.7\)](#)), so

$$\sigma^2 = \frac{2\gamma k_B T}{m} : \quad (13.4)$$

the noise and the friction are not independent. The friction removes energy at a rate set by γ , the noise supplies it, and the temperature fixes their balance. With [Equation \(13.4\)](#), [Equation \(13.3\)](#) replaces v by $c v + \sqrt{(1 - c^2) k_B T/m} g$, written $v \leftarrow c v + \sqrt{(1 - c^2) k_B T/m} g$, which takes a Maxwellian velocity to another Maxwellian velocity: the old velocity keeps the fraction c^2 of the variance and fresh noise supplies

the rest, $1 - c^2$, which grows from 0 at $\gamma = 0$ towards 1 as $\gamma\delta t$ grows. For argon at 135 K, $\sqrt{k_B T/m} = 0.001\,676\text{ Å fs}^{-1}$, and with steps of 10 fs, $\gamma = 1\text{ ps}^{-1}$ gives $c = 0.99005$.

BAOAB

With forces, Langevin's equation adds them to the drag and the noise. The splitting of [Section 9.4](#) takes each part exactly for a while: B, a half kick by the forces; A, a half drift of the positions; and O, the exact step [Equation \(13.3\)](#) of the friction and noise over the whole step. The order B A O A B, which Leimkuhler and Matthews called BAOAB⁶¹, puts O between the two half drifts of [Figure 13.1](#), where `run` calls `Langevin.middle`:

```

1     c = math.exp(-self.friction * dt)
2     sigma = np.sqrt((1 - c * c) * self._kt(self.temperature)
3                 / (m * self.mv2_to_energy))
4     return c * v + rng.standard_normal(v.shape) * sigma[:, None]
```

Leimkuhler and Matthews showed that this order samples positions with unusual accuracy; for a harmonic oscillator its positions have the exact canonical distribution at any stable step. In reduced units, with mass, spring constant, $k_B T$ and friction all 1, so that the period is 2π , 4000 oscillators followed with steps of 0.5, 1.0 and 1.5 give $\langle x^2 \rangle = 0.9986, 0.9989$ and 1.0003 , against the exact value 1, while their whole-step velocities give $\langle v^2 \rangle = 0.9367, 0.7503$ and 0.4374 , close to $1 - \delta t^2/4$. A density $e^{-\tilde{H}/[k_B T(1-\omega^2\delta t^2/4)]}$ of the whole-step pairs (x, v) , with $\tilde{H}$ the shadow energy of [Equation \(9.6\)](#) and $\omega = 1$, would give exactly these variances, the positions' canonical and the velocities' lowered by that factor. A large step thus samples the arrangements correctly while the kinetic temperature reads low; with the steps of this book the difference is small: for the lithium atom, steps of 5 fs against its period of 170.3 fs give $1 - \omega^2\delta t^2/4 = 0.991$, and its measured kinetic energy under Langevin's thermostat is 0.995 of $k_B T$.

What the friction does to the motion

Every velocity is pulled towards zero at the rate γ , so friction slows diffusion as collisions did. For liquid argon ([Figure 13.3 \(a\)](#)), $\gamma = 0.1, 1$ and 10 ps^{-1} leave 0.97, 0.71 and 0.17 of the diffusion coefficient at fixed energy, almost the same as Andersen's thermostat at the same rates. Like collisions, the noise changes the total momentum.

For an atom whose only bath is the thermostat, the friction also decides how often it hops. A hop needs the barrier's energy, which only the bath supplies, and a hop, once begun, can be turned back by the friction and the kicks. A standard yardstick is the rate of transition-state theory, which counts the atoms crossing the edges of a hollow's cell outwards, from the canonical distribution⁶². In a short time δt an atom leaves through a short piece dl of an edge if it lies within $v_\perp \delta t$ of the piece

and moves outwards across it with the speed $v_{\perp}$. The probability of lying in that strip, of area $v_{\perp} \delta t dl$, is $e^{-\beta U} v_{\perp} \delta t dl$ divided by the integral of $e^{-\beta U}$ over the cell (Section 12.4). The velocity is independent of the position (Section 12.5), so $v_{\perp}$ is averaged on its own over the Maxwellian, the atoms that move inwards contributing o: with the Maxwellian's integral over all v , $\sqrt{2\pi k_B T/m}$, as the divisor, the average is $\int_0^{\infty} v e^{-mv^2/2k_B T} dv / \sqrt{2\pi k_B T/m} = \sqrt{k_B T/2\pi m}$ (Exercise 13.10). Dividing by δt and adding the pieces round the cell gives the probability per unit time of leaving. Each hollow's cell is a hexagon whose six edges lie halfway between neighbouring hollows, pass through the bridges and are each $a/\sqrt{3}$ long, so

$$k_{\text{TST}} = \sqrt{\frac{k_B T}{2\pi m}} \frac{\oint e^{-\beta U} dl}{\iint e^{-\beta U} dA}, \quad (13.5)$$

the integrals round the edges and over the cell. For lithium at 1000 K, $\sqrt{k_B T/2\pi m} = 0.004367 \text{ Å fs}^{-1}$, the integral round the six edges is 0.2151 Å and that over the cell 0.7475 Å^2 , against the cell's area of 5.2408 Å^2 , so $k_{\text{TST}} = 0.004367 \times 0.2151/0.7475$ per femtosecond, 1.257 ps^{-1} ; at 300 K it is one hop every 1.69 ns. An atom that crosses and turns back is counted by the theory but has not hopped, so the true rate is lower.

Figure 13.3(b) counts the hops of the 1000 lithium atoms, an atom being assigned to a hollow once it comes within 0.7 Å of it. From $\gamma = 0.01$ to 3 ps^{-1} the rate lies between 0.89 and 1.02 ps^{-1} , about 0.8 of k_{TST} ; beyond it falls, to 0.715 , 0.347 and 0.149 ps^{-1} at 30 , 100 and 300 ps^{-1} , more and more nearly in proportion to $1/\gamma$, as friction turns each crossing into a slow struggle over the bridge in which the atom often turns back. That fall is one side of Kramers' theory of rates^{63,64}. The theory also predicts a fall at low friction for an atom that must gain the barrier's energy from the bath, which the circles do not show: they start from the canonical distribution, in which 0.194 of the atoms already have more energy than the barrier, and on a periodic surface such atoms are never taken away, so hops go on however weak the friction. The slow supply of energy shows when the atoms start at the bottom of a hollow with the kinetic energy of 1000 K but no potential energy: at $\gamma = 0.01 \text{ ps}^{-1}$ the thermostat supplies the missing energy so slowly that they make only 0.228 hops per picosecond over the first 100 ps , rising from 0.17 in the first 20 ps to 0.35 in the last, against 1.0 at 3 ps^{-1} .

IN PRACTICE. Langevin dynamics samples the canonical ensemble reliably, even for a single atom, and is a good thermostat for averages over arrangements and for equilibration. A friction of 1 ps^{-1} already slows the diffusion of liquid argon by more than a quarter; a diffusion coefficient or a rate measured under Langevin dynamics belongs to the friction as much as to the system, unless the friction is small against the system's own rates of losing momentum.

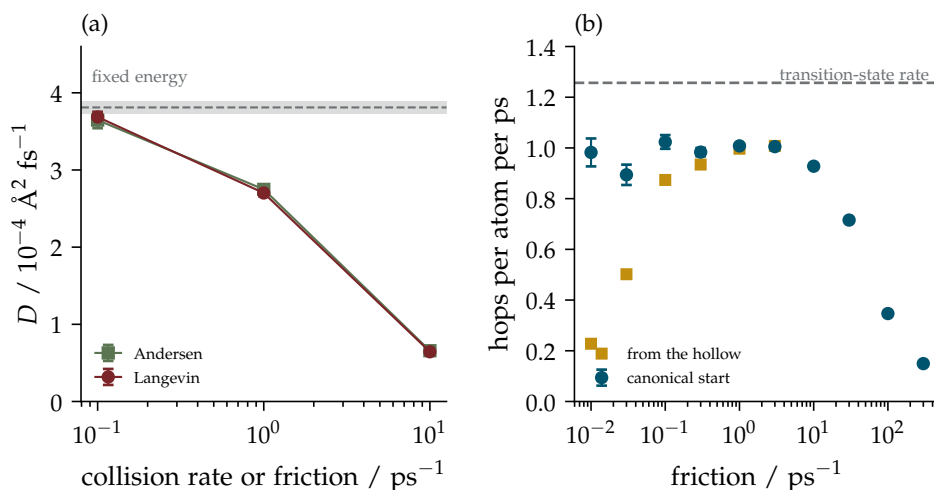


FIGURE 13.3: (a) The diffusion coefficient of liquid argon, measured from the centre of mass, against the collision rate of Andersen's thermostat and the friction of Langevin's, each the mean of three runs with its standard error, against three runs at fixed energy (line and band). (b) Hops of lithium atoms between hollows of the model surface at 1000 K, per atom and picosecond, against the friction: atoms started from the canonical distribution (circles, with the standard error over the atoms) and atoms started at the bottom of a hollow, counted over 100 ps (squares), against the rate of transition-state theory (dashed).

KEY IDEA. Friction and noise together give each velocity the exact step $v \leftarrow cv + \sqrt{(1-c^2)k_B T/m} g$, $c = e^{-\gamma \delta t}$, whose noise is tied to the friction by $\sigma^2 = 2\gamma k_B T/m$. BAOAB puts it between the half drifts of velocity Verlet and samples positions accurately even at long steps. Friction slows diffusion and, for an atom whose bath is the thermostat, sets its rate of hopping: about 0.8 of the rate of transition-state theory at low friction, falling more and more nearly in proportion to $1/\gamma$ at high.

NOTEBOOK 13 *set the friction and watch diffusion and hopping change*

CHECK YOUR UNDERSTANDING. What is c for $\gamma = 10 \text{ ps}^{-1}$ and steps of 10 fs, and what fraction of the velocity's variance is replaced by fresh noise at each step?

13.5 A friction that learns

Friction and noise disturb every atom at random. Nosé and Hoover found a deterministic alternative: a single friction, shared by all the atoms, that grows when the system is too hot and turns negative, pushing the atoms faster, when it is too cold. This section gives their equations, the quantity they conserve, why they sample the canonical ensemble when the motion is ergodic, where they fail, and the chains that cure the failure.

The equations

Hoover's form of Nosé's thermostat adds one variable, the friction ζ , with its own equation of motion^{65,66}:

$$\dot{\mathbf{r}}_i = \frac{\mathbf{p}_i}{m_i}, \quad \dot{\mathbf{p}}_i = \mathbf{F}_i - \zeta \mathbf{p}_i, \quad \dot{\zeta} = \frac{2K - N_f k_B T_0}{Q}. \quad (13.6)$$

When $2K$ exceeds its canonical mean $N_f k_B T_0$, ζ grows and the friction slows every atom; when $2K$ falls below, ζ falls, eventually below zero, and speeds them up. The thermostat's mass Q , an energy times a time squared, sets how fast ζ responds; with $Q = N_f k_B T_0 \tau_T^2$, τ_T sets the time over which it responds, a small departure of K swinging with the period $2\pi\tau_T/\sqrt{2}$ (Exercise 13.12); Q is $8.9 \times 10^4 \text{ eV fs}^2$ for 256 argon atoms at 135 K with $\tau_T = 100 \text{ fs}$.

The motion keeps a quantity. With $H = K + U$, the force $-\zeta \mathbf{p}_i$ changes H at the rate $\sum_i \mathbf{p}_i \cdot (-\zeta \mathbf{p}_i)/m_i = -2K\zeta$, while

$$\frac{d}{dt} \left(\frac{1}{2} Q \dot{\zeta}^2 \right) = Q \dot{\zeta} \ddot{\zeta} = \zeta (2K - N_f k_B T_0),$$

so

$$H + \frac{1}{2} Q \dot{\zeta}^2 + N_f k_B T_0 \int_0^t \zeta(t_1) dt_1$$

changes at the rate $-2K\zeta + 2K\zeta - N_f k_B T_0 \dot{\zeta} + N_f k_B T_0 \dot{\zeta} = 0$. Its last two terms change at the rate $2K\dot{\zeta}$, minus the rate at which the friction adds heat, so, with ζ starting at zero, they are minus the heat, and this is the effective energy of Equation (13.1). One oscillator in reduced units, followed for 1000 units of time, about 160 periods, with steps of 0.05, starts with 0.2124 of it; the integrator's error makes it swing by up to 9.4×10^{-4} over the run, but it does not drift, and it ends 4×10^{-5} from where it began.

Why it is canonical

Section 7.4 showed that the divergence of a flow is the rate at which the volume of a small blob of states grows, divided by that volume. In the space of positions, momenta and ζ , the rates $\dot{\mathbf{r}}_i$ do not depend on $\mathbf{r}_i$, $\dot{\zeta}$ does not depend on ζ , and each of the N_f free momentum components has $\partial \dot{p}/\partial p = -\zeta$, the friction keeping a zero total momentum at zero so that the motion stays among the free components; the divergence is $-N_f \zeta$ (Exercise 13.14). A blob shrinks while $\zeta > 0$, and since it keeps its copies, their density along the flow rises at the rate $N_f \zeta$. The density

$$\rho \propto e^{-\beta(H + \frac{1}{2} Q \dot{\zeta}^2)}, \quad \beta = \frac{1}{k_B T_0}.$$

changes along the flow at the rate

$$\frac{d\rho}{dt} = -\beta\rho \frac{d}{dt} \left(H + \frac{1}{2} Q \xi^2 \right) = -\beta\rho (-N_f k_B T_0 \dot{\xi}) = N_f \xi \rho,$$

using the two rates above: the rate that the shrinking of the blobs requires. The copies arriving at any state therefore have the density this formula gives there, wherever they came from, so the cloud is the same at every time, as for the Hamiltonian flow of [Section 7.4](#): the density is stationary. Integrating it over ξ , a Gaussian, leaves $e^{-\beta H}$: the canonical distribution of the atoms. The argument shows that the canonical ensemble is kept; that one trajectory explores it, the ergodic hypothesis of [Section 12.2](#), is a separate matter.

Nosé's Hamiltonian

[Equation \(13.6\)](#) came from a Hamiltonian. Nosé added a coordinate s with momentum p_s , scaled the atoms' momenta by it, and took

$$H_N = \sum_i \frac{\tilde{\mathbf{p}}_i^2}{2m_i s^2} + U + \frac{p_s^2}{2Q} + N_f k_B T_0 \ln s,$$

whose equations of motion, in a time t' of their own, are Hamilton's. Writing the real momenta as $\mathbf{p}_i = \tilde{\mathbf{p}}_i/s$, the real time by $dt = dt'/s$, and $\xi = p_s/Q$, turns them into [Equation \(13.6\)](#) ([Exercise 13.13](#)). The logarithm's slope, $N_f k_B T_0/s$, becomes the $-N_f k_B T_0$ of $\dot{\xi}$ that sets the temperature; s itself only records how the two times differ.

Where it fails, and chains

One oscillator, $H = \frac{1}{2}p^2 + \frac{1}{2}x^2$ in reduced units, under [Equation \(13.6\)](#) with $\tau_T = 1$, never fills its phase plane ([Figure 13.4\(a\)](#)): over 2000 units of time its positions have the excess kurtosis of [Section 12.5](#), $\langle x^4 \rangle / \langle x^2 \rangle^2 - 3 = -1.46$, against 0 for the canonical Gaussian. Started from 100 different positions at rest, each trajectory keeps its own time average of x^2 , from 0.78 to 1.59, where the canonical value is 1 ([Figure 13.4\(c\)](#)). The motion is not ergodic: the single variable ξ and the oscillator lock into a regular pattern. A lithium atom on the surface with its own Nosé-Hoover thermostat fails too: the density of its potential energy differs from the exact one, at the worst point, by 0.23 of the exact density's peak, and it hops 0.667 ps^{-1} , against 1.09 under the chain of three below, which samples it correctly.

Martyna, Klein and Tuckerman's cure gives ξ a thermostat of its own, and that one another, a chain⁶⁷:

$$\dot{\xi}_1 = \frac{2K - N_f k_B T_0}{Q_1} - \xi_2 \xi_1, \quad \dot{\xi}_j = \frac{Q_{j-1} \xi_{j-1}^2 - k_B T_0}{Q_j} - \xi_{j+1} \xi_j,$$

the last without the final term. The atoms feel only ξ_1 , through Equation (13.6); each later ξ_j is a friction on ξ_{j-1} , as ξ_1 is on the atoms, driven by twice the energy $\frac{1}{2} Q_{j-1} \xi_{j-1}^2$ of ξ_{j-1} against the $k_B T_0$ of one freedom. The masses are $Q_1 = N_f k_B T_0 \tau_T^2$ and, for the others, $k_B T_0 \tau_T^2$. A chain of four fills the oscillator's plane (Figure 13.4(b)) and its trajectories' averages of x^2 lie between 0.92 and 1.08; a chain of three gives the lithium atom's density to within 0.016. mdlab integrates the chain for half a step before and half after each step, the Trotter factorisation of Section 9.4 with the thermostat as a third part. Each half is made of three pieces, of w_1 , $1 - 2w_1$ and w_1 times the half step, with $w_1 = 1/(2 - 2^{1/3}) = 1.3512$, so that the middle piece runs backwards; these weights of Yoshida⁶⁸ cancel the error of third order. The scheme is the one Martyna, Tuckerman, Tobias and Klein set out⁶⁹, and ASE's. For the liquid tested here, the single thermostat and the chain both give a spread of K consistent with the canonical ensemble: the mean of three runs lies between 0.0507 and 0.0534 at $\tau_T = 0.1$ and 1 ps, against the canonical 0.0511, and the chain costs about 1.1 times a step without a thermostat.

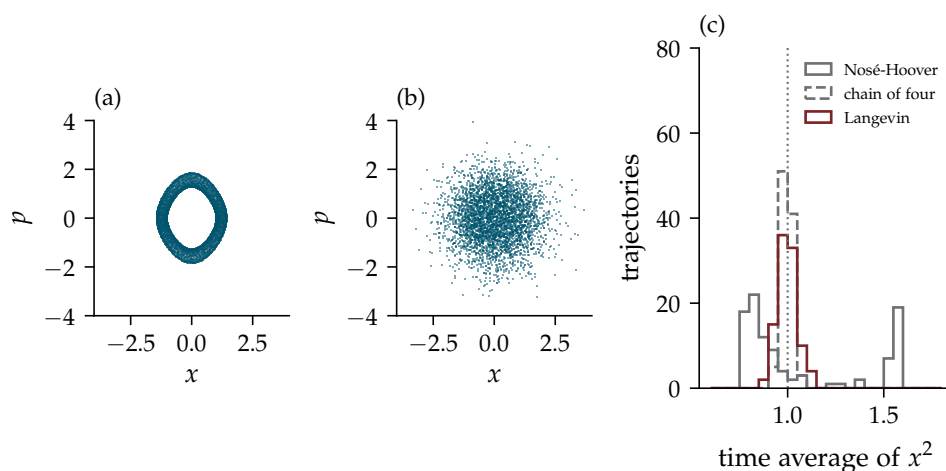


FIGURE 13.4: One oscillator ($m = k = k_B T_0 = 1$). (a) One trajectory under Nosé-Hoover ($\tau_T = 1$) over 2000 units of time, in the plane of x and p . (b) The same under a chain of four. (c) The time average of x^2 along each of 100 trajectories started at rest from random positions, under Nosé-Hoover, the chain and Langevin ($\gamma = 0.5$), against the canonical value (dotted).

KEY IDEA. Nosé-Hoover friction ζ obeys $\dot{\zeta} = (2K - N_f k_B T_0)/Q$ and keeps the effective energy. The density $e^{-\beta(H+Q\zeta^2/2)}$ is carried unchanged by the flow, so the canonical ensemble is stationary, but only an ergodic motion samples it: one oscillator, or one atom in the examples here, is not sampled correctly by the single thermostat. A chain samples these examples successfully. For the liquid tested here, both give the canonical fluctuations and disturb the motion little.

NOTEBOOK 13 follow one oscillator under Nosé-Hoover and under a chain

CHECK YOUR UNDERSTANDING. What is Q_1 for 1000 atoms whose centre of mass is held still, at 300 K with $\tau_T = 100$ fs?

13.6 Rescaling that samples

Berendsen coupling disturbs the motion little but narrows the spread of K ; Langevin's friction gives the right spread but disturbs every atom. Bussi, Donadio and Parrinello found a rescaling of all the velocities together, like Berendsen's, that samples the canonical distribution exactly⁷⁰: canonical sampling through velocity rescaling, CSVR. This section derives it from the exact step of [Section 13.4](#), measures it, and reads TrajCast's implementation, which holds its learned runs at a temperature.

The derivation

In the N_f free components of [Section 12.6](#), mass-weighted so that $K = \frac{1}{2}|\mathbf{u}|^2$, the velocities form a vector $\mathbf{u}$. With the noise of [Equation \(13.4\)](#), the exact step [Equation \(13.3\)](#) takes a velocity component v to $cv + \sqrt{(1-c^2)k_B T_0/m} g$, and so $u = \sqrt{m} v$ to $cu + \sqrt{(1-c^2)k_B T_0} g$, whatever the mass. Applied to every component with the friction $\gamma = 1/(2\tau_T)$, chosen so that the mean of K relaxes with the time constant τ_T , as shown below, and for which $c^2 = e^{-\delta t/\tau_T} = c_1$, it gives

$$\mathbf{u}' = \sqrt{c_1} \mathbf{u} + \sqrt{(1-c_1)k_B T_0} \mathbf{g},$$

with $\mathbf{g}$ a vector of N_f independent normal numbers. Its squared length is

$$|\mathbf{u}'|^2 = c_1 |\mathbf{u}|^2 + (1-c_1)k_B T_0 |\mathbf{g}|^2 + 2\sqrt{c_1(1-c_1)k_B T_0} \mathbf{u} \cdot \mathbf{g}.$$

The vector $\mathbf{g}$ can be resolved along $\mathbf{u}$ and at right angles to it. Its density, the product of the factors $e^{-g_k^2/2}$, is a constant times $e^{-|\mathbf{g}|^2/2}$, which depends only on the length of $\mathbf{g}$; turning the axes, so that the first lies along $\mathbf{u}$ and the others at right angles to it and to one another, leaves that length, and so the density, unchanged, and the

components of $\mathbf{g}$ along the new axes are again independent normal numbers. So $\mathbf{u} \cdot \mathbf{g} = |\mathbf{u}|R_1$, with R_1 the component of $\mathbf{g}$ along $\mathbf{u}$, and $|\mathbf{g}|^2 = R_1^2 + R_\perp^2$, with $R_\perp^2$ the sum of the squares of the other $N_f - 1$ components, independent of R_1 . Dividing by $|\mathbf{u}|^2 = 2K$, which turns the last term into $2R_1\sqrt{c_1(1-c_1)k_B T_0/2K}$, and writing $c_2 = (1-c_1)k_B T_0/2K = (1-c_1)K_0/(N_f K)$ (the names of Bussi and colleagues, not the components c_k of [Section 12.6](#)),

$$\frac{K'}{K} = \lambda^2 = c_1 + c_2(R_1^2 + R_\perp^2) + 2R_1\sqrt{c_1 c_2}. \quad (13.7)$$

CSVSR keeps the change of K that this step makes and throws away the change of direction: it multiplies every velocity by λ . Since λ^2 depends on the velocities only through K , the new K has, for a given K , the same distribution as after the whole Ornstein-Uhlenbeck step. That step keeps the canonical distribution, every component Gaussian with variance $k_B T_0$, and with it the gamma distribution of K ; so CSVSR keeps that distribution too, for any δt and any τ_T . In the canonical density the direction of $\mathbf{u}$ is spread evenly, independent of its length and of the positions, and the scaling changes only the length; so, as with Andersen's redraws, the whole canonical density is kept, and a run whose motion is ergodic samples it to the accuracy of the step that moves the atoms. Averaged over the noise ([Exercise 13.15](#)), $\langle K' \rangle - K = (1-c_1)(K_0 - K)$, the change that $dK/dt = (K_0 - K)/\tau_T$ makes over δt ; for a short step $1-c_1 \approx \delta t/\tau_T$, and this is the relaxation of Berendsen coupling with the same time constant. The noise supplies the canonical spread that Berendsen lacks. The sum $R_\perp^2$ of $N_f - 1$ squares need not be drawn square by square: by [Equation \(12.12\)](#) with $k_B T = 1$ ([Exercise 13.16](#)), half of it has a density proportional to $y^{(N_f-1)/2-1}e^{-y}$, the gamma distribution of shape $(N_f - 1)/2$, from which NumPy draws directly. CSVSR.before is [Equation \(13.7\)](#):

```

1      k = kinetic(m, v, self.mv2_to_energy)
2      target = 0.5 * self.n_free * self._kt(self.temperature)
3      c1 = math.exp(-dt / self.tau)
4      c2 = (1 - c1) * target / (k * self.n_free)
5      r1 = rng.standard_normal(k.shape)
6      rest = (2 * rng.standard_gamma(0.5 * (self.n_free - 1), k.shape)
7             if self.n_free > 1 else np.zeros(k.shape))
8      alpha2 = c1 + c2 * (rest + r1 * r1) + 2 * r1 * np.sqrt(c1 * c2)
9      return v * np.sqrt(alpha2)[..., None, None]
```

Bussi and colleagues wrote the factor as α , which is λ in this chapter; `alpha2` holds its square. At 150 K, with a target of 135 K, $\tau_T = 100$ fs and steps of 10 fs, $c_1 = 0.90484$ and $c_2 = 1.120 \times 10^{-4}$, and with $R_1 = 0$ and $R_\perp^2$ at its mean of 764, $\lambda^2 = 0.99037$, close to Berendsen's 0.9900.

Measured

Over three runs of liquid argon the spread of K is 0.0499 with $\tau_T = 100$ fs and 0.0508 with 1 ps, against the canonical 0.0511 (Figure 13.2(a)); the diffusion coefficient is 1.07 and 1.05 times its value at fixed energy, a gap that Section 13.7 weighs against the scatter of the runs. Acting once every three steps, over their whole length, with $\tau_T = 300$ fs, it still gives 0.0513. For a single lithium atom, the density of the potential energy is within 0.012 of the exact one, and the atoms hop 1.08 times per picosecond. An isolated molecule would show another side of the common factor: it keeps its angular momentum under the forces, and CSVr's scaling multiplies it at random, so TrajCast's authors set it to zero every hundred strides for paracetamol¹. CSVr scales every velocity by one factor, as Berendsen coupling does, but its factor is not tilted towards growth⁵⁸, and a motion of the centre of mass left in the velocities does not grow (Section 13.8).

TrajCast's thermostat

TrajCast predicts the state a stride Δt ahead, from 5 fs for water to 30 fs for quartz in the runs reported, and after each stride applies CSVr once, in the file `_thermostat.py` of its forecasting tools¹, whose lines 96-109 are Equation (13.7):

```

1      c2 = (
2          (torch.tensor(1.0) - self.c1) * e_kin_target
3          / e_kin_current / self.n_dofs
4      ).to(self.device)
5      r1 = torch.randn(1, device=self.device)
6      r2 = self._draw_sum_noises_from_gamma_dist(n_dofs_1=self.n_dofs - 1)
7      alpha_2 = (
8          self.c1
9          + c2 * (r1 * r1 + r2)
10         + torch.tensor(2.0) * r1 * (self.c1 * c2).sqrt()
11     )

```

with comments removed and one line wrapped. c_1 is $e^{-\Delta t/\tau_T}$, computed once with the stride as the step (lines 32-37); r_2 is $R_{\perp}^2$, drawn as twice a gamma number of shape $(N_f - 1)/2$, or, when $N_f - 1$ is odd, as twice one of shape $(N_f - 2)/2$ plus one more square (lines 113-123), which is the same distribution; and n_dofs is $3N$ less three for each kind of momentum, linear or angular, that the run removes or holds fixed, or less a number the run's settings give (`forecast.py`, lines 344-378). With no time constant given, τ_T is a hundred strides, so $c_1 = e^{-0.01} = 0.9900$ and the temperature is held loosely; the runs reported used ten strides for water and paracetamol, 50 and 70 fs, $c_1 = 0.9048$, and two and a half for quartz, 75 fs, $c_1 = 0.6703$. Since Equation (13.7) keeps the distribution of K for any step, the thermostat itself works at any stride, as the run acting once every three steps confirmed. The stride changes what the thermostat

corrects: a learned step may not keep the energy, and CSVr would then remove the error as heat, so in a learned run the heat may measure the model's error as well as the bath's work, a theme of the planned Part VI. TrajCast's authors found that the longer strides needed much shorter τ_T than usual to hold the temperature and its spread, and put it down to each stride standing for several steps taken before one rescaling¹.

KEY IDEA. CSVr applies the exact Ornstein-Uhlenbeck step to all N_f mass-weighted velocity components at once, keeps only its change of length, and scales every velocity by the factor λ of Equation (13.7). Its step then keeps the canonical gamma distribution of K for any step and coupling, the mean relaxes as under Berendsen coupling, and the motion is disturbed as little as by Berendsen coupling. TrajCast applies it once per learned stride, with $c_1 = e^{-\Delta t/\tau_T}$.

NOTEBOOK 13 *set the coupling and the stride, and compare CSVr with Berendsen*

CHECK YOUR UNDERSTANDING. For TrajCast's stride of 30 fs for quartz, what τ_T gives $c_1 = 0.5$?

13.7 Which thermostat, and when

The sections so far measured each thermostat on its own. This section sets the measurements side by side and draws from them which thermostat to use for the three jobs a thermostat does: bringing a system to a temperature, sampling averages at it, and leaving its dynamics fit to measure.

Side by side

Table 13.1 collects, for every thermostat and coupling, the spread of K in liquid argon as a fraction of the canonical spread, the diffusion coefficient as a fraction of its value at fixed energy, and, for one lithium atom with its own thermostat, how far the density of its potential energy misses the exact one and how often it hops. The diffusion coefficients carry the standard error of three runs, the standard deviation of their three estimates divided by $\sqrt{3}$, since the variances of the three add and their mean divides the sum by three; with so few runs it is itself rough, and Chapter 15 gives the tools to say when two such numbers differ. Three groups stand out.

- Rescaling fixes the kinetic temperature at each rescaling, and Berendsen coupling keeps its mean close to the target. Both narrow the spread, 0.11 to 0.59 of the canonical, and a single atom under Berendsen coupling stops hopping.
- Andersen and Langevin sample the canonical ensemble once the coupling is 1 ps^{-1} or more, for the liquid and for one atom alike, and cost no more than a

step without a thermostat. At that coupling they cut the diffusion coefficient to 0.71 to 0.72 of its value, and at 10 ps^{-1} to 0.17.

- Nosé-Hoover chains and CSVR give the canonical spread at both couplings and sample one atom correctly. Like Berendsen coupling and a single Nosé-Hoover thermostat, they leave the diffusion coefficient 1.01 to 1.07 of its value at fixed energy, in runs 0.4 to 1.8 K warmer than those at fixed energy. The variances of independent quantities add, so the standard error of a difference is the square root of the sum of the two squared standard errors; the chains and CSVR lie at most 2.5 of these from fixed energy, and Berendsen coupling at 1 ps 2.9. Whether so small a difference is real is a question for Chapter 15. A single Nosé-Hoover thermostat matches the chains for the liquid but not for one atom. These findings agree with Basconi and Shirts, who compared thermostats on the transport properties of liquids⁷¹.

TABLE 13.1: The thermostats measured: for liquid argon at 135 K, the relative spread of K divided by the canonical $\sqrt{2/N_f}$, and the diffusion coefficient divided by its value at fixed energy, $(3.81 \pm 0.08) \times 10^{-4} \text{ Å}^2/\text{fs}$ (three runs of 100 ps; rescaling, one); for one lithium atom on the surface at 1000 K, the largest gap between the density of its potential energy and the exact one, divided by the exact density's peak, and its hops per picosecond (1000 atoms over 100 ps). For Andersen and Langevin, which move the centre of mass, $N_f = 3N = 768$. Under Berendsen coupling the density is a narrow spike, so its gap is many times the peak.

thermostat	liquid argon		one lithium atom	
	spread	D	gap	hops/ps
fixed energy	0.61	1		
rescaling	0.11	1.01		
Berendsen, 0.1 ps	0.45	1.04 ± 0.01	15.5	0.001
Berendsen, 1 ps	0.59	1.07 ± 0.01		
Andersen, 0.1 ps^{-1}	0.85	0.96 ± 0.03		
Andersen, 1 ps^{-1}	0.98	0.72 ± 0.01	0.011	0.974
Andersen, 10 ps^{-1}	1.00	0.173 ± 0.004		
Langevin, 0.1 ps^{-1}	0.93	0.97 ± 0.02	0.024	1.024
Langevin, 1 ps^{-1}	1.01	0.71 ± 0.01	0.005	1.008
Langevin, 10 ps^{-1}	1.01	0.169 ± 0.001	0.008	0.928
Nosé-Hoover, 0.1 ps	0.99	1.02 ± 0.01	0.231	0.667
Nosé-Hoover, 1 ps	1.04	1.05 ± 0.01		
chain of 3, 0.1 ps	1.01	1.04 ± 0.03	0.016	1.087
chain of 3, 1 ps	1.01	1.01 ± 0.02		
CSV, 0.1 ps	0.98	1.07 ± 0.02	0.012	1.081
CSV, 1 ps	0.99	1.05 ± 0.01		

What to use

Table 13.2 turns the measurements into a choice for each job.

Bringing a system to a temperature. Any thermostat does it; what matters is how fast it gets there and that it settles at the target. From 200 K, Berendsen and CSVr with $\tau_T = 1$ ps and Langevin with $\gamma = 1$ ps⁻¹ all reach the target within a few picoseconds (Figure 13.2(b)). Under Berendsen and CSVr the excess decays with the time constant $\tau_T C_V / C_K = 1.59$ ps, fitted at 1.67 and 1.63 ps; under Langevin with $C_V / (2\gamma C_K) = 0.80$ ps, fitted at 0.68 ps (Exercise 13.4). A short coupling time, 0.1 to 1 ps, is right here, and the run's early part is discarded, for which Chapter 15 gives a measured criterion.

Averages at a temperature. The average and the fluctuations must both be canonical, so rescaling and Berendsen coupling are out. CSVr, Nosé-Hoover chains, Langevin and Andersen all qualify. A single Nosé-Hoover thermostat failed for one oscillator and for one atom on the surface, and is expected to fail likewise for other small or nearly harmonic systems, a single molecule or a nearly harmonic crystal⁶⁷. The heat capacity from the energy's fluctuations, Equation (12.6), shows what is at stake: it comes out at 2.28 to 2.40 Nk_B under CSVr, the chain and Langevin, against the 2.376 Nk_B of Chapter 12 from dE/dT , and at 0.41 Nk_B under Berendsen coupling with $\tau_T = 0.1$ ps.

Dynamics. The thermostat must disturb the motion as little as possible; of the motion, this chapter measured the diffusion of the liquid. A run at fixed energy started from an equilibrated state disturbs nothing, but its temperature is the one its energy gives, which in our runs fell 0.66 K short. CSVr or a Nosé-Hoover chain changes the diffusion of the liquid by at most 7%, and a coupling time of a picosecond or more keeps the thermostat's hand light. Berendsen coupling changes it no more, but it narrows the fluctuations, feeds the flying ice cube and pins a single atom. Langevin and Andersen thermostats at 1 ps⁻¹ remove more than a quarter of it.

An atom whose only bath is the thermostat. When the real bath is left out, as for an atom on a frozen surface, the thermostat's coupling becomes part of the physics: the lithium atom's hop rate fell from 1.0 to 0.35 ps⁻¹ as the friction rose from 3 to 100 ps⁻¹. Such a rate means something only if the friction stands for a real bath, and the safer course is to simulate the bath's atoms.

KEY IDEA. Rescaling and Berendsen coupling are for reaching a temperature. CSVr, Nosé-Hoover chains, Langevin and Andersen sample the canonical ensemble; of these, CSVr and chains change the diffusion of the liquid by at most 7%, while friction or collisions at 1 ps⁻¹ slow it by more than a quarter. A single Nosé-Hoover thermostat fails for small systems, and for an atom whose bath is the thermostat, the atom's rates belong to the coupling.

TABLE 13.2: Which thermostat to use, from the measurements of this chapter.

job	use	avoid
bringing to a temperature	Berendsen, CSV or Langevin, τ_T of 0.1 to 1 ps or γ of 1 to 10 ps ⁻¹ (measured at 1 ps and 1 ps ⁻¹), the early part discarded	one Berendsen thermostat on a single atom, which pins its energy
averages and fluctuations	CSV, Nosé-Hoover chains, Langevin, Andersen	rescaling, Berendsen; one Nosé-Hoover thermostat on small or nearly harmonic systems
dynamics (here, diffusion)	fixed energy after equilibrating; CSV or a chain with $\tau_T \geq 1$ ps	Langevin and Andersen at ≥ 1 ps ⁻¹ ; Berendsen, for its fluctuations and flying ice cube
learned runs (TrajCast)	CSV once per stride; its update preserves the canonical distribution of K	Berendsen

CHECK YOUR UNDERSTANDING. A run is to measure the diffusion of lithium in a liquid electrolyte at 300 K. Which thermostat, with what coupling, does this chapter recommend for the production run, and why not Langevin with $\gamma = 5$ ps⁻¹?

13.8 What the trajectory looks like

A thermostat changes what a healthy run looks like: the energy is no longer constant, and the temperature should fluctuate by a definite amount. This section shows a healthy thermostatted run of liquid argon and six faults that thermostats leave, each with the check that catches it.

A healthy run

Figure 13.5 follows the liquid under CSV with $\tau_T = 100$ fs for 100 ps. From 5 ps on, its kinetic temperature averages 135.57 K, with a relative spread of 0.0486 against the canonical 0.0511, and lies within one canonical spread of the target for 0.719 of the time, close to the 0.683 of a Gaussian (Section 12.1). The total energy is no longer kept: it wanders over 9.17 meV per atom as the thermostat exchanges heat. The effective energy of Equation (13.1) stays within 0.018 meV per atom, five hundred times less, and falls by 1.6×10^{-5} eV ps⁻¹ for the whole box: it plays the part the total energy played in Section 11.8. The total momentum stays below 4×10^{-14} amu Å/fs, since CSV scales a zero momentum to zero, and the histogram of K follows the gamma density.

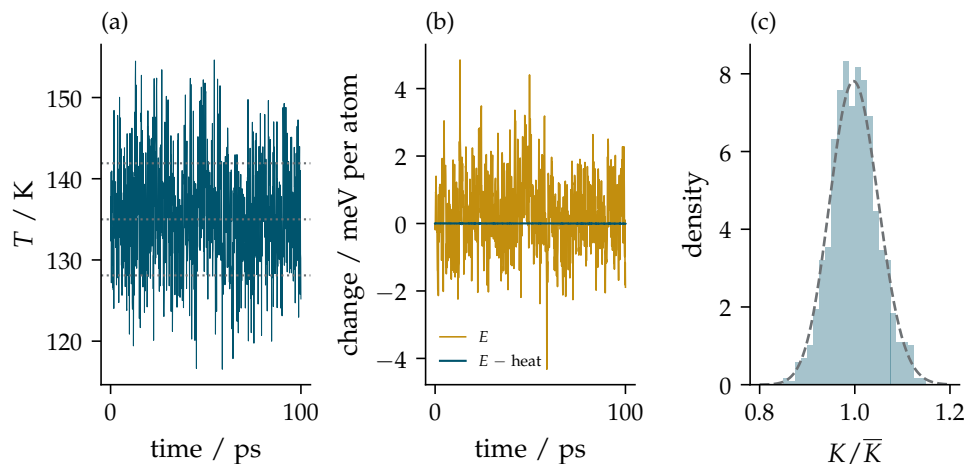


FIGURE 13.5: A healthy thermostatted run: liquid argon under CSVr with $\tau_T = 100$ fs for 100 ps. (a) The kinetic temperature every 100 fs, with the target and one canonical spread either side (dotted). (b) The total energy and the effective energy per atom, from their starting values. (c) The kinetic energy over the run, divided by its mean, against the canonical gamma density (dashed).

Six faults

Figure 13.6 shows six faults and Table 13.3 the check that catches each.

TABLE 13.3: The faults of Figure 13.6, the check that catches each, and its value in the healthy and the broken run; (a) uses the first of the three runs, so its spreads differ slightly from the three-run means of Table 13.1.

fault	check	healthy	broken
(a) coupling too strong	spread of T / canonical	0.95	0.44
(b) coupling too weak	mean T of three runs, K	135.38 ± 0.31	131.83 ± 1.02
(c) flying ice cube	growth of $\ln K_{\text{com}}$ / ps	-0.0002	0.0047
(d) one atom, one variable	gap from the exact density	0.016	0.231
(e) wrong N_f	T over the true freedoms, K	1005	1506
(f) centre as diffusion	D , 10^{-4} Å ² /fs	3.69	4.62

(a) *Coupling too strong*. Berendsen coupling with $\tau_T = 100$ fs holds the temperature in a band 0.44 as wide as the canonical one; CSVr with the same time constant gives 0.95. The average looks perfect, 135.00 K, and only the spread shows the fault: the check is the relative spread of T against $\sqrt{2/N_f}$.

(b) *Coupling too weak*. Andersen's thermostat at 0.1 ps^{-1} lets the temperature wander for tens of picoseconds: three runs average 133.66, 130.13 and 131.69 K, counting all $3N$ freedoms, against 135.58, 134.77 and 135.79 K under CSVr. Nothing in one run looks wrong; the check is the spread of the run means between independent starts, or the running mean's failure to settle near the target.

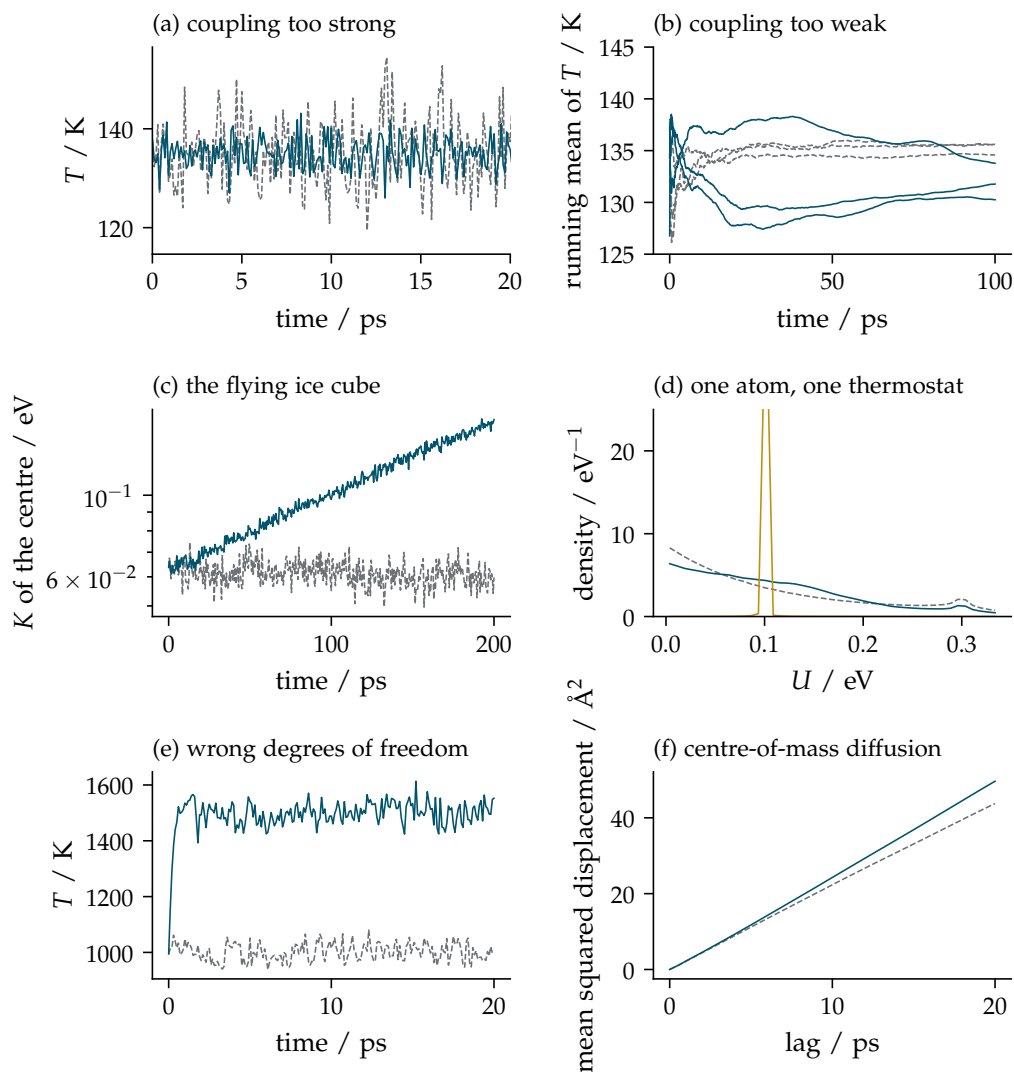


FIGURE 13.6: Thermostat faults (teal) against healthy runs (grey dashed). (a) Liquid argon under Berendsen coupling with $\tau_T = 100$ fs, against CSV. (b) The running mean of the temperature in three runs under Andersen's thermostat at 0.1 ps^{-1} , against three under CSV. (c) The kinetic energy of the motion of the centre of mass, started moving, under Berendsen coupling, against CSV, on a logarithmic axis. (d) The density of the potential energy of lithium atoms on the surface, each under its own Nosé-Hoover thermostat (teal) or Berendsen thermostat (ochre, cut off), against the exact canonical density. (e) Lithium atoms under CSV told $N_f = 3$: their kinetic temperature over the two freedoms they have, against CSV told 2. (f) The mean squared displacement in liquid argon under Langevin dynamics with $\gamma = 0.1 \text{ ps}^{-1}$, measured in the cell, against measured from the centre of mass.

(c) *The flying ice cube.* A velocity of the centre of mass of $0.0002 \text{ \AA fs}^{-1}$ in each component, left in the velocities, gives the centre of mass the kinetic energy K_{com} , whose logarithm grows under Berendsen coupling at 0.0047 ps^{-1} , against the 0.0050 ps^{-1} predicted in [Section 13.2](#): K_{com} rises from 0.064 eV to 0.161 eV in 200 ps ; under CSVR it does not grow. The check is the kinetic energy of the centre of mass, and of the slowest motions, over a long run.

(d) *One atom, one variable.* A lithium atom with its own Nosé-Hoover thermostat misses the exact density of its potential energy by 0.231 of its peak; with a chain of three, by 0.016 ; under Berendsen coupling it is pinned near one energy. The check is a distribution compared with an exact or independently sampled one, here the density of U ; for a large system, the temperatures of its parts ([Section 12.9](#)).

(e) *The wrong number of freedoms in the target.* A thermostat told three freedoms per lithium atom, which has two, drives each atom's kinetic energy to $\frac{3}{2}k_{\text{B}}T_0$, so its true temperature reads 1506 K for a target of 1000 K . The same fault in a large system is small but systematic: a thermostat that takes $3N$ freedoms for a run whose centre of mass is held still, as ASE's Berendsen, CSVR and Nosé-Hoover chain do for argon, raises the temperature of 256 atoms by the factor $768/765$, 0.5 K at 135 K , about the scatter between independent runs. For rigid water a thermostat that also ignores the held distances, as ASE's Nosé-Hoover chain does, raises it by $576/381 = 1.51$ ([Section 12.9](#)); ASE's CSVR and Berendsen subtract the held distances and miss only the three freedoms of the centre of mass, $384/381$. The check is the kinetic temperature computed with the freedoms counted for the run.

(f) *The centre of mass counted as diffusion.* Langevin and Andersen thermostats do not keep the total momentum, so the centre of mass wanders; under Langevin dynamics it diffuses, over times long against $1/\gamma$, with the coefficient $k_{\text{B}}T/(M\gamma)$, which for the whole box at $\gamma = 0.1 \text{ ps}^{-1}$ is $1.10 \times 10^{-4} \text{ \AA}^2 \text{ fs}^{-1}$ ([Exercise 13.18](#)). Displacements measured in the cell include it: the diffusion coefficient reads 4.62×10^{-4} instead of $3.69 \times 10^{-4} \text{ \AA}^2/\text{fs}$, 1.25 times too large. The check is to measure displacements from the centre of mass.

IN PRACTICE. A thermostatted run is read through its kinetic temperature, whose spread should be $\sqrt{2/N_f}$ of the target when the freedoms are counted for the run; through its effective energy, which should neither wander nor drift; through the kinetic energy of its centre of mass and of its slowest motions, which should not grow; and, for a property that depends on the motion, through the same property with the thermostat weakened, which should not change. Runs from several independent starts show whether a weakly coupled run has reached its temperature, a question Chapter 15 answers with error bars.

KEY IDEA. In a healthy thermostatted run the total energy wanders, the effective energy holds still, the temperature fluctuates by the canonical $\sqrt{2/N_f}$, and the centre of mass stays still or is measured from. Coupling too strong narrows the spread; too weak leaves the mean unsettled; rescaling and Berendsen coupling fly the ice cube; a single Nosé-Hoover variable fails on small systems; a miscounted N_f shifts the temperature; and a wandering centre of mass inflates diffusion.

NOTEBOOK 13 *break the thermostatted liquid in each of the six ways and read the checks*

CHECK YOUR UNDERSTANDING. A Langevin run of 500 argon atoms with $\gamma = 1 \text{ ps}^{-1}$ at 135 K: at what rate does its centre of mass diffuse?

What we now have

A thermostat must give the canonical distributions, the gamma distribution of K with spread $\sqrt{2/N_f}$ and the density $e^{-\beta U}$ of positions, and should disturb the motion as little as possible; the effective energy, $K + U$ less the heat it adds, is kept by the exact dynamics of every thermostat and checks the integration.

Every method of the chapter fits one loop, velocity Verlet written as B A A B, with the thermostat acting before, between the half drifts or after; four of the five that ASE also implements agree with its versions to $1 \times 10^{-7} \text{ \AA}$, and Andersen's in the temperature it holds.

Rescaling and Berendsen coupling hold the mean temperature but narrow the spread of K to 0.11 to 0.59 of the canonical, and feed motions that the forces do not touch at the rate $\langle(\delta T/T_0)^2\rangle/\tau_T$, the flying ice cube, measured at 0.0047 ps^{-1} against 0.0050 ps^{-1} predicted.

Andersen's collisions, and Langevin's friction and noise, tied by $\sigma^2 = 2\gamma k_B T/m$ and integrated exactly by the O step of BAOAB, sample the canonical ensemble even for one atom, and slow the diffusion of liquid argon by more than a quarter at 1 ps^{-1} ; for an atom whose only bath is the thermostat, Langevin's friction sets the hop rate, near 0.8 of the transition-state rate at low friction and falling more and more nearly in proportion to $1/\gamma$ at high.

Nosé-Hoover friction keeps an extended energy and carries the canonical density unchanged, but samples it only for an ergodic motion, so one oscillator or one atom fails where a chain succeeds.

CSVR applies the exact Ornstein-Uhlenbeck step to the length of the whole mass-weighted velocity, so its thermostat update preserves the canonical distribution of K

for any step and coupling. TrajCast can therefore apply that update once per learned stride, while the intervening propagation remains a separate source of sampling error.

For bringing a system to a temperature any thermostat serves; for averages, CSV, chains, Langevin or Andersen; for dynamics, a run at fixed energy, or CSV or a chain coupled weakly.

Chapter 14 does for pressure what this chapter did for temperature: derives the pressure from the virial, and builds the barostats that hold it.

Answers to the checks

SECTION 13.1. The heat is -0.5 eV, so E_{eff} changes by $-0.48 - (-0.5) = +0.02$ eV: the integration has added 0.02 eV.

SECTION 13.2. $\lambda^2 = 1 + 0.002(300/310 - 1) = 0.9999355$, so $\lambda = 0.9999677$.

SECTION 13.3. $1 - e^{-0.002} = 0.001998$, so about 2 atoms of 1000 at each step.

SECTION 13.4. $c = e^{-0.1} = 0.90484$; the old velocity keeps the fraction c^2 of its variance, so $1 - c^2 = 0.181$ is replaced by noise.

SECTION 13.5. $N_f = 2997$ and $k_B T_0 = 0.025852$ eV, so $Q_1 = 2997 \times 0.025852 \times 100^2 = 7.75 \times 10^5$ eV fs².

SECTION 13.6. $e^{-30/\tau_T} = 0.5$ gives $\tau_T = 30/\ln 2 = 43.3$ fs.

SECTION 13.7. CSV, or a Nosé-Hoover chain, with τ_T of a picosecond or more, or a run at fixed energy after equilibrating: in liquid argon the first two changed diffusion by at most 7%, the last not at all. Langevin at 5 ps^{-1} would slow it by well over a quarter, leaving between the 0.71 and 0.17 of it measured at 1 and 10 ps^{-1} .

SECTION 13.8. $k_B T/(M\gamma)$, with $k_B T = 0.011633$ eV at 135 K, $M = 500 \times 39.948 = 19974$ amu, $\gamma = 0.001 \text{ fs}^{-1}$ and the factor 103.6427 of Equation (3.15):

$$\frac{0.011633}{19974 \times 103.6427 \times 0.001} = 5.6 \times 10^{-6} \text{ \AA}^2 \text{ fs}^{-1}.$$

Exercises

The exercises run from questions of understanding, through calculations, to short derivations marked ‘show that’. Full solutions are in ‘Solutions to the exercises’ at the end of the book, and Notebook 13 checks every numerical answer.

EXERCISE 13.1 (*collisions in a step*). Collisions come at the rate ν , independently in every short interval. Show, by cutting a step δt into n intervals each with the collision probability $\nu\delta t/n$, that the probability of no collision in the step is $e^{-\nu\delta t}$. What is the probability of at least one for $\nu = 1 \text{ ps}^{-1}$ and $\delta t = 2 \text{ fs}$?

EXERCISE 13.2 (*one Berendsen step*). Find λ for a step of 1 fs , $\tau_T = 0.5 \text{ ps}$, a kinetic temperature of 290 K and a target of 300 K .

EXERCISE 13.3 (*the mean under Berendsen*). Show that a run under Berendsen coupling that has settled, so that the heat it receives averages to zero, has $\langle T \rangle = T_0$ exactly.

EXERCISE 13.4 (*how fast Langevin cools*). Show from Equation (13.3) that under Langevin dynamics the heat flows in at the rate $2\gamma(K_0 - K)$ when the forces are ignored for the moment, and hence that a liquid’s temperature relaxes with the time constant $C_V/(2\gamma C_K)$. Give it for liquid argon at $\gamma = 1 \text{ ps}^{-1}$.

EXERCISE 13.5 (*how far the ice cube flies*). Under Berendsen coupling with $\tau_T = 0.1 \text{ ps}$ and $\langle (\delta T/T_0)^2 \rangle = 5 \times 10^{-4}$, by what factor does the kinetic energy of the centre of mass grow in 1 ns ?

EXERCISE 13.6 (*sums of Gaussians*). Show that if x and y are independent and Gaussian with mean zero and variances a and b , then $x + y$ is Gaussian with variance $a + b$, by finding the density of $s = x + y$, $\int \mathcal{P}(x)\mathcal{P}(s - x) dx$, and completing the square in x .

EXERCISE 13.7 (*the variance settles*). Show from Equation (13.3) that the variance of v after one step is c^2 times the variance before plus $(\sigma^2/2\gamma)(1 - c^2)$, and that $\sigma^2/2\gamma$ is the only variance that a step leaves unchanged.

EXERCISE 13.8 (*the noise in a step*). For argon at 135 K , $\gamma = 1 \text{ ps}^{-1}$ and steps of 10 fs , find the standard deviation of the random part of one O step, $\sqrt{(1 - c^2)k_B T/m}$, in $\text{\AA fs}^{-1}$.

EXERCISE 13.9 (*no friction, no noise*). Show that BAOAB with $\gamma = 0$ is velocity Verlet.

EXERCISE 13.10 (*the outward flux*). Show that

$$\int_0^\infty v e^{-mv^2/2k_B T} dv / \sqrt{2\pi k_B T/m} = \sqrt{k_B T/2\pi m},$$

and find it for lithium at 1000 K.

EXERCISE 13.11 (*transition-state theory in one dimension*). For an atom in one dimension in a well that is harmonic near its bottom, $U \approx \frac{1}{2}m\omega^2 x^2$, with a barrier U_b on one side, show that the one-dimensional form of Equation (13.5) gives $k_{\text{TST}} \approx (\omega/2\pi) e^{-U_b/k_B T}$ when $U_b \gg k_B T$. Evaluate it for the lithium atom's period of 170.3 fs and $U_b = 0.3$ eV at 300 K, in hops per second.

EXERCISE 13.12 (*the friction's own swing*). Ignoring the forces, so that only the friction changes K , show that small departures δK of K from $K_0 = \frac{1}{2}N_f k_B T_0$ under Equation (13.6), with $Q = N_f k_B T_0 \tau_T^2$, obey

$$\frac{d^2 \delta K}{dt^2} = -\frac{2}{\tau_T^2} \delta K.$$

What is the period for $\tau_T = 100$ fs?

EXERCISE 13.13 (*from Nosé to Hoover*). Write Hamilton's equations for H_N of Section 13.5 in its time t' , and show that with $\mathbf{p}_i = \tilde{\mathbf{p}}_i/s$, $dt = dt'/s$ and $\xi = p_s/Q$ they become Equation (13.6).

EXERCISE 13.14 (*the divergence*). For one atom in three dimensions under Equation (13.6), with the coordinates $x, y, z, p_x, p_y, p_z, \xi$, find the divergence of the flow term by term.

EXERCISE 13.15 (*CSVR on average*). Show that the mean over the noise of Equation (13.7) gives $\langle K' \rangle - K = (1 - c_1)(K_0 - K)$.

EXERCISE 13.16 (*a sum of squares from a gamma number*). Using Equation (12.12) with $k_B T = 1$, show that the sum of the squares of n independent normal numbers is twice a number with the gamma distribution of shape $n/2$.

EXERCISE 13.17 (*TrajCast's coupling*). TrajCast's water runs use a stride of 5 fs. What is τ_T in femtoseconds with no time constant given, and what is c_1 ? What is τ_T for paracetamol, stride 7 fs, at ten strides?

EXERCISE 13.18 (*the wandering centre*). Under Langevin dynamics one component v_{com} of the velocity of the centre of mass of N atoms of total mass M follows Equation (13.3)

with the same γ and stationary variance $k_B T/M$. Show that its displacement x_{com} along one axis over a long time t has $\langle x_{\text{com}}^2 \rangle \approx 2Dt$ with $D = k_B T/(M\gamma)$, using $\langle v_{\text{com}}(t_1)v_{\text{com}}(t_2) \rangle = (k_B T/M)e^{-\gamma|t_1-t_2|}$. Evaluate D for 256 argon atoms at 135 K and $\gamma = 0.1 \text{ ps}^{-1}$.

EXERCISE 13.19 (*the wrong target for water*). A thermostat targeting 300 K counts three freedoms for each of the 192 atoms of 64 rigid water molecules, $3N = 576$, although their distances are held and their centre of mass is held still. What temperature does the water reach?

EXERCISE 13.20 (*a referee's objection*). A study reports the heat capacity of a liquid from the fluctuations of its energy, Equation (12.6), in a run under Berendsen coupling with $\tau_T = 0.1 \text{ ps}$. What is wrong, which way is the result wrong, and which thermostat would fix it?

EXERCISE 13.21 (*a diagnosis*). A run of 256 argon atoms under Langevin dynamics with $\gamma = 0.1 \text{ ps}^{-1}$ reports a diffusion coefficient 1.2 times that of a run at fixed energy, although friction can only slow diffusion. What explains it, and what check settles it?

Pressure and barostats

A graphite electrode swells as it charges: the carbon sheets separate from 3.34 to 3.70 Å when lithium fills the spaces between them⁷². Holding a simulation at the volume of the bare graphite would compress the lithiated solid across its sheets. To study its response at a prescribed pressure, we must allow the cell's volume and shape to change.

We first obtain pressure from the force of atoms on a wall, then use the virial to calculate it in a periodic cell without walls. The pressure tensor distinguishes the forces on different faces, allowing us to describe a layered solid whose response differs within and across its sheets. We then develop barostats and test both the mean pressure and the volume fluctuations they produce, following the approach to thermostats in Chapter 13. Their effects on motion and their characteristic faults prepare the question of Chapter 15: how long must a run be before its averages can be trusted?

14.1 Force on a wall

A bicycle pump with its outlet closed pushes back on the hand: the air inside presses on the piston, and the harder the air is squeezed the harder it presses. The push divided by the area of the piston is the pressure of the air. This section finds where that push comes from in a gas of atoms, and measures it.

Pressure and its units

The pressure on a flat wall of area A that feels the force $F_{\perp}$ at right angles to it is

$$P = \frac{F_{\perp}}{A},$$

in newtons per square metre, or pascals. The air around us presses with about 10^5 Pa, one bar; deep in the Earth and in squeezed solids pressures reach gigapascals, 1 GPa being 10^9 pascals, or 10^4 bar. In the units of this book a pressure is an energy per volume, $\text{eV } \text{\AA}^{-3}$, since a force is an energy per length and an area a length squared; one $\text{eV } \text{\AA}^{-3}$ is 160.218 GPa ([Exercise 14.1](#)).

The push of bouncing atoms

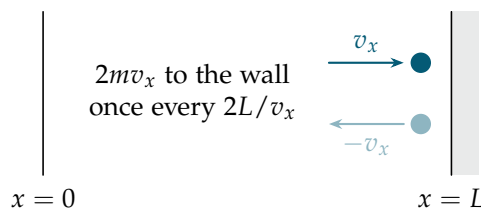


FIGURE 14.1: An atom bouncing between walls at $x = 0$ and $x = L$. Each bounce off the right-hand wall turns v_x into $-v_x$, and the atom returns to that wall after travelling $2L$ along x .

One atom of mass m moves in a box between walls at $x = 0$ and $x = L$, its velocity along x being v_x ([Figure 14.1](#)). A hard wall takes no energy from the atom, so when it strikes the wall at $x = L$ it comes back at the same speed with v_x turned to $-v_x$: its momentum along x changes by $-2mv_x$, and by Newton's third law ([Section 2.4](#)) the wall receives $+2mv_x$. The atom then crosses the box and back, a distance $2L$ at the speed v_x , before it strikes the same wall again; bounces off the walls normal to y and z turn v_y or v_z and leave v_x as it was. The bounces therefore come once every $2L/v_x$. The wall feels a short kick at each bounce and nothing between, and a force is the momentum delivered per unit time ([Section 2.3](#)), so the force on the wall, averaged over many bounces, is

$$\overline{F}_\perp = \frac{2mv_x}{2L/v_x} = \frac{mv_x^2}{L}.$$

For N atoms that do not meet one another, each contributes its own share, and the pressure on the wall, of area A , is

$$P = \frac{1}{A} \sum_i \frac{m_i v_{ix}^2}{L} = \frac{1}{V} \sum_i m_i v_{ix}^2, \quad V = AL.$$

Averaged over the velocities, each $\frac{1}{2}m_i v_{ix}^2$ is $\frac{1}{2}k_B T$ by equipartition ([Section 12.6](#)), so each $m_i v_{ix}^2$ is $k_B T$, the N of them add to $Nk_B T$, and

$$PV = Nk_B T, \tag{14.1}$$

the ideal gas law that Section 12.8 found by counting states. The pressure of a gas is the momentum its atoms deliver to the walls per unit time and area. The argument holds for molecules too, with m the mass of a molecule and v_x the velocity of its centre of mass (Section 2.9). At 300 K, where $k_B T = 0.025\,852\text{ eV}$, and one bar, $6.2415 \times 10^{-7}\text{ eV \AA}^{-3}$ (Exercise 14.1), Equation (14.1) gives $N/V = P/k_B T = 2.414 \times 10^{-5}\text{ \AA}^{-3}$: 24.1 molecules of air in a cube 10 nm, or 100 Å, on a side.

Measured

A simulation has no hard walls, but soft ones serve: a spring that pushes back an atom which has gone a distance δ beyond the wall with the force $k\delta$, from the energy $\frac{1}{2}k\delta^2$ (virial.harmonic_walls). The force the atoms exert on each wall is then the sum of these $k\delta$, recorded at every step. For 200 argon atoms that do not interact, in a cube of side 40 Å with walls of stiffness 5 eV Å⁻², followed at fixed energy for 100 ps with steps of 5 fs, the push on the six walls per unit area averages 53.47 bar (Figure 14.2(a)), and the two halves of the run give 53.27 and 53.67 bar. The kinetic energy of the run gives the temperature 123.5 K by Equation (12.11), with $N_f = 3N$ since the walls leave the total momentum free, and Equation (14.1) gives 53.29 bar, within the difference between the halves.

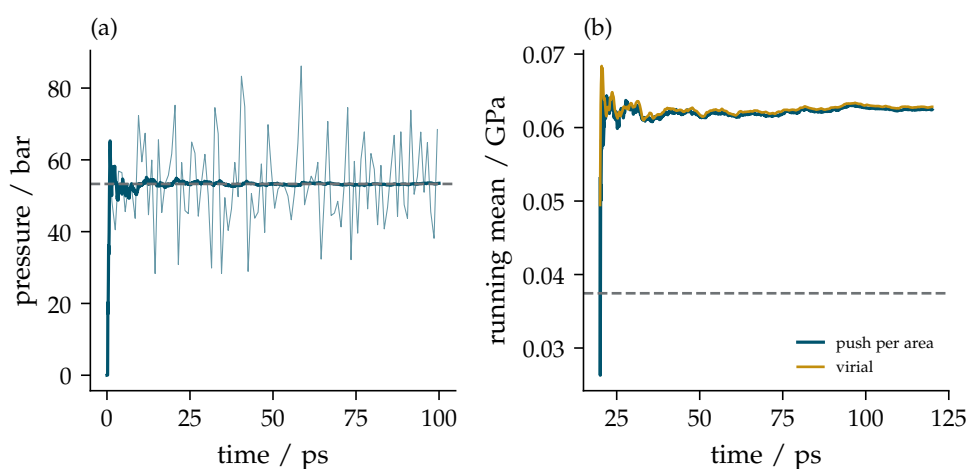


FIGURE 14.2: Pressure as the push on the walls. (a) 200 argon atoms that do not interact, in a cube of 40 Å between soft walls: the push per unit area on the six walls, averaged over each picosecond (thin) and over the run so far (thick), against $Nk_B T / V$ (dashed). (b) 256 argon atoms that interact, in a cube of 23.26 Å near 135 K: the running means of the push per unit area and of the pressure from the virial, Equation (14.2), against $Nk_B T / V$ (dashed).

Atoms that interact push differently. The 256 atoms of the liquid argon of Chapter 13, held between walls of the same stiffness in a cube of 23.26 Å under CSVr at 135 K, push on the walls with 0.0624 GPa averaged over 100 ps (0.0616 and 0.0633 GPa in the two halves). Their kinetic energy gives the temperature 133.3 K over the run, a little below the 135 K the thermostat holds on average, and with it $Nk_B T / V$ is only 0.0374 GPa

(Figure 14.2(b)). The atoms repel one another at close range and the repulsion adds to the push on the walls. The next section finds by how much.

KEY IDEA. Pressure is force per unit area. In a gas it is the momentum the atoms deliver to the walls per unit time and area, $\sum_i m_i v_{ix}^2 / V$, which averages to $Nk_B T / V$ for atoms that do not interact. One $\text{eV } \text{\AA}^{-3}$ is 160.218 GPa, and 1 GPa is 10^4 bar.

NOTEBOOK 14 *let atoms bounce between walls and read their push*

CHECK YOUR UNDERSTANDING. What pressure, in bar, do 10^4 atoms that do not interact exert at 300 K in a cube of side 100 $\text{\AA}$?

14.2 The virial

Atoms that pull on and push one another deliver momentum to the walls differently from atoms that only bounce. This section finds the pressure of interacting atoms from the equipartition theorem applied to their positions, as Chapter 12 promised, and checks it against the push measured in Section 14.1.

Equipartition for positions

Equation (12.10) says that $\langle x \partial H / \partial x \rangle = k_B T$ for any coordinate x of a system in the canonical ensemble whose range is closed by a potential that rises without limit, walls included. For atoms in a box, $H = K + U + U_w$, with U the energy of their interactions and U_w that of the walls. K depends on the momenta alone, so the slope of H along the coordinate x_i of atom i is minus the force along it, $\partial H / \partial x_i = -F_{ix} - F_{ix}^w$, the first from the other atoms and the second from the walls. The theorem for x_i then reads $\langle x_i F_{ix} \rangle + \langle x_i F_{ix}^w \rangle = -k_B T$, and likewise for y_i and z_i ; since $x_i F_{ix} + y_i F_{iy} + z_i F_{iz} = \mathbf{r}_i \cdot \mathbf{F}_i$, adding the $3N$ of them gives

$$\left\langle \sum_i \mathbf{r}_i \cdot \mathbf{F}_i \right\rangle + \left\langle \sum_i \mathbf{r}_i \cdot \mathbf{F}_i^w \right\rangle = -3Nk_B T.$$

The first sum is the virial $\mathcal{W}$ of Section 8.4, which for atoms that interact in pairs is $\mathcal{W} = -\sum_{i<j} r_{ij} \varphi'(r_{ij})$, Equation (8.4): positive when the atoms repel one another, negative when they attract.

The walls' share

The second sum comes only from atoms touching a wall. A wall pushes only at right angles to itself, so an atom touching the wall at $x = L$ has $\mathbf{r}_i \cdot \mathbf{F}_i^w = x_i F_{ix}^w$, with $x_i \approx L$ and the force pointing back into the box; the sum over those atoms is

$$\sum_i x_i F_{ix}^w \approx L \sum_i F_{ix}^w = -L F_\perp,$$

with $F_\perp$ the force the atoms exert on the wall, equal and opposite to the wall's force on them by the third law. On average $F_\perp$ is PA , so this wall contributes $-PAL = -PV$. At the wall $x = 0$ the atoms have $x_i \approx 0$ and contribute nothing; with the origin at the centre of the box the two walls share $-PV$ equally instead ([Exercise 14.5](#)). The walls normal to y and z do the same, so the walls' share is $-3PV$. That holds without error for hard walls. An atom that has gone the depth δ into a soft wall adds a further $-k\delta^2$, at $x = 0$ as at $x = L$, so soft walls enlarge the share by a fraction of about $2\delta/L$. Putting it into the sum,

$$\langle \mathcal{W} \rangle - 3PV = -3Nk_B T,$$

and so

$$PV = Nk_B T + \frac{1}{3} \langle \mathcal{W} \rangle. \quad (14.2)$$

The first term is the ideal gas's, [Equation \(14.1\)](#); the second is the interactions' share, raising the pressure when the atoms repel one another and lowering it when they attract. Since $\langle 2K \rangle = 3Nk_B T$ by equipartition for the momenta, the instantaneous value

$$P = \frac{2K + \mathcal{W}}{3V}$$

averages to the pressure, and can be computed at every step from the positions, velocities and forces the run already has.

Checked

For the walled liquid of [Figure 14.2\(b\)](#), $(2K + \mathcal{W})/3V$ averages 0.0628 GPa over the run (0.0621 and 0.0635 GPa in the halves), against 0.0624 GPa from the push on the walls: the two agree within their scatter. Of the 0.0628 GPa, 0.0374 GPa is $Nk_B T/V$ and 0.0254 GPa is $\langle \mathcal{W} \rangle/3V$: at this density the repulsion between close neighbours outweighs the pull of distant ones.

KEY IDEA. Equipartition applied to the positions, $\langle x \partial H / \partial x \rangle = k_B T$, gives $PV = Nk_B T + \frac{1}{3} \langle \mathcal{W} \rangle$: the ideal gas's pressure plus the virial's share, positive for repulsion and negative for attraction. The instantaneous pressure $(2K + \mathcal{W})/3V$ averages to it and needs only the positions, velocities and forces.

CHECK YOUR UNDERSTANDING. Two argon atoms alone in a box sit $3.0 \text{ \AA}$ apart, closer than the minimum of the Lennard-Jones potential at $2^{1/6} \sigma = 3.82 \text{ \AA}$. Does their pair raise or lower the pressure from that of two atoms that do not interact, and why?

14.3 Pressure in a periodic box

A periodic box (Section 10.1) has no walls for its atoms to push on, and Section 14.2 needed walls. This section finds the pressure of a periodic box from the free energy, shows that it is Equation (14.2) again with each pair's separation taken to its nearest copy, and shows what goes wrong when the virial is computed from the positions instead.

From the free energy

Chapter 12 found $dF = -S dT - P dV + \mu dN$ (Section 12.8), so at fixed T and N the pressure is the rate at which the free energy falls as the box grows,

$$P = -\frac{\partial F}{\partial V} = k_B T \frac{\partial \ln Z}{\partial V},$$

using $F = -k_B T \ln Z$. For a cubic box of side L , $V = L^3$, the partition function is a constant C that depends on T (from the integral over the momenta) times the integral of $e^{-\beta U}$ over the positions of every atom inside the box. Writing each position as $\mathbf{r}_i = L \mathbf{s}_i$, with $\mathbf{s}_i$ the fractional coordinates of Section 10.1, each running from 0 to 1, changes the variable of each of the $3N$ integrals and multiplies it by L (Section 10.3):

$$Z = C V^N \int e^{-\beta U(L \mathbf{s}_1, \dots, L \mathbf{s}_N)} d\mathbf{s}_1 \dots d\mathbf{s}_N.$$

Now the volume appears in two places, V^N and U , and the box's boundaries no longer depend on it. The logarithm is $\ln Z = \ln C + N \ln V + \ln \int e^{-\beta U} d\mathbf{s}$, with $d\mathbf{s}$ short for $d\mathbf{s}_1 \cdots d\mathbf{s}_N$, and its slope is

$$\begin{aligned} \frac{\partial \ln Z}{\partial V} &= \frac{N}{V} + \frac{\int (-\beta \partial U / \partial V) e^{-\beta U} d\mathbf{s}}{\int e^{-\beta U} d\mathbf{s}} && \text{the slope of } \ln, \text{ and the chain rule,} \\ &= \frac{N}{V} - \beta \left\langle \frac{\partial U}{\partial V} \right\rangle && \text{a canonical average,} \end{aligned}$$

with the slope of U taken at fixed fractional coordinates, that is, with the whole arrangement stretched evenly. For pairs, $U = \sum_{i < j} \varphi(r_{ij})$, and stretching scales every separation with the box, $r_{ij} = L |\mathbf{s}_j - \mathbf{s}_i|$ with the fractional separation taken to its nearest copy (Section 10.2). So $\partial r_{ij} / \partial L = r_{ij} / L$, and with $dL/dV = 1/(3L^2)$,

$$\frac{\partial U}{\partial V} = \sum_{i < j} \varphi'(r_{ij}) \frac{r_{ij}}{L} \frac{1}{3L^2} = \frac{1}{3V} \sum_{i < j} r_{ij} \varphi'(r_{ij}) = -\frac{\mathcal{W}}{3V}.$$

Then $P = k_B T (N/V - \beta \langle \partial U / \partial V \rangle) = N k_B T / V + \langle \mathcal{W} \rangle / 3V$, Equation (14.2) again, with $\mathcal{W}$ written with the separations of Equation (8.4), each to the nearest copy of the other atom. A periodic run keeps its total momentum at zero, which removes three of the $3N$ kinetic freedoms, so that $\langle 2K \rangle = (3N - 3)k_B T$ (Section 12.6), and holds its centre of mass still, so that only $N - 1$ atoms are placed freely and V^N becomes V^{N-1} . The pressure of such a run is $(N - 1)k_B T / V + \langle \mathcal{W} \rangle / 3V$, one part in N less in its first term, and the instantaneous $(2K + \mathcal{W}) / 3V$ averages to it. In `md.PairModel` the virial is summed over the pairs of the neighbour list from their separations, as Section 14.4 shows, and `virial.pressure` adds the kinetic part.

Not from the positions

The sum $\sum_i \mathbf{r}_i \cdot \mathbf{F}_i$ equals Equation (8.4) only when, for every pair, the difference of the positions held, $\mathbf{r}_j - \mathbf{r}_i$, is the separation from which their force was found. Positions read from the frames of a periodic run are wrapped into the cell (Section 10.1): an atom that leaves through one face comes back through the opposite one, its $\mathbf{r}_i$ jumping by a lattice vector while its force does not change. Section 8.4 showed that moving the origin by the same vector for every atom leaves $\mathcal{W}$ alone; wrapping moves single atoms, and changes $\sum_i \mathbf{r}_i \cdot \mathbf{F}_i$ by the scalar product of the lattice vector with that atom's force. Which atoms have been wrapped depends on where the cell's faces lie, so the sum depends on where the cell's origin sits. For the liquid of Chapter 13 at fixed volume, the pressure from the pair separations is 0.0877 GPa; with $\sum_i \mathbf{r}_i \cdot \mathbf{F}_i$ from the wrapped positions it is 0.0324 GPa, and 0.0354 GPa with the origin moved, a single frame's value changing by up to 0.131 GPa. Both lie near the kinetic part alone, 0.0383 GPa: on average the jumps of the wrapped atoms cancel the virial's share. Thompson, Plimpton and Mattson give

the general form of the pressure for many-body potentials in periodic cells, built in the same way from separations⁷³.

A crystal at rest

A crystal with every atom on its site has $K = 0$ and its pressure is the virial's alone. Figure 14.3(a) shows it for the face-centred cubic crystal of the switched Lennard-Jones argon against the cubic spacing a . It is zero at $a_0 = 5.26714 \text{ \AA} = 1.54916\sigma$, the spacing at which Section 10.5 found the energy least, as it must be, since $P = -dU/dV$ at $T = 0$ and the slope of U vanishes at its least value. A crystal squeezed to $a = 5.20 \text{ \AA}$ pushes outwards with 0.128 GPa; one stretched to $5.40 \text{ \AA}$ pulls inwards with 0.160 GPa, a negative pressure, a tension. The rise of the pressure for each fraction by which the volume falls is the bulk modulus,

$$B = -V \frac{dP}{dV} = -\frac{dP}{dV/V},$$

the stiffness of the material against squeezing, 2.856 GPa at a_0 ; its inverse, $\kappa_T = 1/B$ at fixed temperature, is the compressibility, the fraction by which the volume shrinks for each unit of pressure.

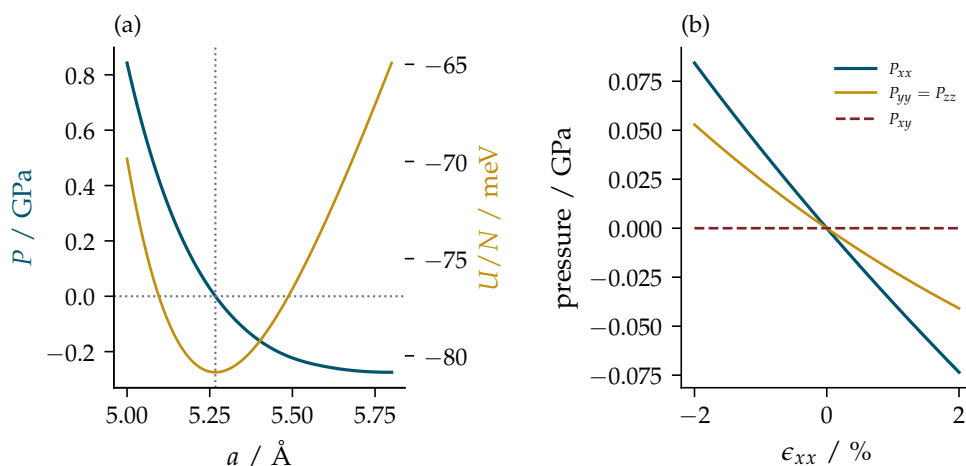


FIGURE 14.3: The pressure of the face-centred cubic crystal at rest, 256 atoms of the switched Lennard-Jones argon. (a) The pressure (teal) and the energy per atom (ochre) against the cubic spacing a , the spacing a_0 of zero pressure marked (dotted). (b) The crystal at a_0 stretched along x by the strain ϵ_{xx} : the components P_{xx} , $P_{yy} = P_{zz}$ and P_{xy} (dashed, zero) of the pressure tensor of Section 14.4.

The liquid of Chapter 13, at 135 K and $0.8\sigma^{-3}$, has the pressure 0.0877 GPa, about 880 bar, of which 0.0383 GPa is the kinetic part $2K/3V$. Between walls the same atoms gave 0.0628 GPa (Section 14.2): the walls change the liquid next to them, and only the periodic box gives the pressure of the liquid far from any wall. For the Lennard-Jones argon with its energy shifted to zero at $8.5 \text{ \AA}$ in place of the switch, mdlab and ASE's

LennardJones give the same pressure, $-0.058\,924\text{ GPa}$, for a strained crystal of 256 atoms, each displaced at random by about 0.1 Å along each axis.

KEY IDEA. In a periodic box $P = -\partial F/\partial V$ gives Equation (14.2) again, with the virial summed over pairs from their separations to the nearest copy; $\sum_i \mathbf{r}_i \cdot \mathbf{F}_i$ from wrapped positions changes with the origin and is wrong. A crystal at rest has zero pressure at the spacing of least energy, and its bulk modulus $B = -V \, dP/dV$ measures its stiffness against squeezing.

NOTEBOOK 14 *shift the origin of a periodic box and watch which pressure changes*

CHECK YOUR UNDERSTANDING. From Figure 14.3(a), the pressure falls from $+0.128$ to -0.160 GPa as a goes from 5.20 to 5.40 Å . Estimate the bulk modulus near a_0 from these two points.

14.4 The pressure tensor

A liquid at rest pushes equally on a face whichever way the face points. A solid need not: a crystal stretched along x pulls inwards harder on the faces normal to x than on the others, and a sheet of graphite is stiff along its plane and soft across it. This section generalises the pressure to a matrix that gives the push on a face of any orientation.

The push on a face

The force that the material exerts on a small flat face of area A , whose outward normal is the unit vector $\mathbf{n}$, need not point along $\mathbf{n}$, but it is linear in $\mathbf{n}$. A small wedge of material cut off by this face and by three faces normal to the axes, of areas An_x , An_y and An_z , feels forces on its faces that shrink with their area, while its mass shrinks faster, with its volume; as the wedge shrinks the forces on its four faces must therefore cancel, and the force on the slanted face is the sum of An_x , An_y and An_z times the forces per unit area on faces normal to x , y and z . A vector linear in $\mathbf{n}$ is a matrix times $\mathbf{n}$ (Section 4.9):

$$\mathbf{F} = A \mathbf{P} \mathbf{n},$$

with $\mathbf{P}$ the pressure tensor, whose component $P_{\alpha\beta}$ is the force per unit area along α on a face normal to β . For a liquid at rest $\mathbf{P} = P\mathbf{I}$, and every face is pushed straight outwards with P .

From a strain

Section 14.3 stretched the box evenly; stretching it unevenly gives every component. A small strain multiplies every vector, the lattice vectors of the cell and the position of every atom alike, by the matrix $\mathbf{I} + \epsilon$, with components $\epsilon_{\alpha\beta}$ that are small. For a cubic cell of side L the strained lattice vectors are L times the columns $\mathbf{c}_1$, $\mathbf{c}_2$ and $\mathbf{c}_3$ of $\mathbf{I} + \epsilon$, so by Equation (10.3) the volume becomes V times the triple product $\mathbf{c}_1 \cdot (\mathbf{c}_2 \times \mathbf{c}_3)$, which is the determinant $\det(\mathbf{I} + \epsilon)$, the factor by which the strain stretches volumes (Section 7.4). Written out as in Section 5.5, it is six products, each of one component of every column, and to first order in ϵ

$$\det(\mathbf{I} + \epsilon) = (1 + \epsilon_{xx})(1 + \epsilon_{yy})(1 + \epsilon_{zz}) + \cdots = 1 + \epsilon_{xx} + \epsilon_{yy} + \epsilon_{zz} + \cdots,$$

since each of the other five products contains at least two off-diagonal components and is of second order or higher (Exercise 14.7).

The strain $d\epsilon_{\alpha\beta}$ carries the face of a cubic box normal to β , at the distance L from the opposite face, a distance $L d\epsilon_{\alpha\beta}$ along α further than that opposite face. The material pushes on the face with $L^2 P_{\alpha\beta}$ along α , so it does the work $V P_{\alpha\beta} d\epsilon_{\alpha\beta}$, which its free energy loses at fixed temperature:

$$dF = -V \sum_{\alpha\beta} P_{\alpha\beta} d\epsilon_{\alpha\beta},$$

and so $P_{\alpha\beta} = -(1/V) \partial F / \partial \epsilon_{\alpha\beta}$; an even stretch, $\epsilon_{\alpha\beta} = e \delta_{\alpha\beta}$, changes the volume by $dV = 3Vde$ and gives back $dF = -P dV$ with P the mean of the diagonal. The steps of Section 14.3 carry over. In fractional coordinates the partition function carries the factor $\det(\mathbf{I} + \epsilon)^N$, whose logarithm, $N \ln(1 + \epsilon_{xx} + \epsilon_{yy} + \epsilon_{zz} + \cdots)$, has the slope $N\delta_{\alpha\beta}$ along $\epsilon_{\alpha\beta}$ at zero strain. Each pair separation $\mathbf{d}$, of length r_{ij} , becomes $(\mathbf{I} + \epsilon)\mathbf{d}$, with components $d_\alpha + \sum_\beta \epsilon_{\alpha\beta} d_\beta$; squaring and adding them, and dropping the products of two strains, gives the squared length $r_{ij}^2 + 2 \sum_{\alpha\beta} d_\alpha \epsilon_{\alpha\beta} d_\beta$, and since $\sqrt{r^2 + x} \approx r + x/2r$ for small x (Section 1.8), the length grows by $\sum_{\alpha\beta} d_\alpha \epsilon_{\alpha\beta} d_\beta / r_{ij}$. Its slope along $\epsilon_{\alpha\beta}$ is $d_\alpha d_\beta / r_{ij}$, and by the chain rule

$$\frac{\partial U}{\partial \epsilon_{\alpha\beta}} = \sum_{i < j} \phi'(r_{ij}) \frac{d_\alpha d_\beta}{r_{ij}} = -\mathcal{W}_{\alpha\beta},$$

with $\mathcal{W}_{\alpha\beta}$ the virial tensor, whose trace, the sum of its diagonal, is $\mathcal{W}$. Then, as in Section 14.3,

$$P_{\alpha\beta} = \frac{k_B T}{V} \frac{\partial \ln Z}{\partial \epsilon_{\alpha\beta}} = \frac{1}{V} (N k_B T \delta_{\alpha\beta} + \langle \mathcal{W}_{\alpha\beta} \rangle).$$

Equipartition gives $\langle m_i v_{i\alpha}^2 \rangle = k_B T$, and in the Maxwell distribution (Section 12.5) the components of a velocity are independent with mean zero, so $\langle m_i v_{i\alpha} v_{i\beta} \rangle = k_B T \delta_{\alpha\beta}$, and the instantaneous

$$P_{\alpha\beta} = \frac{1}{V} \left(\sum_i m_i v_{i\alpha} v_{i\beta} + \mathcal{W}_{\alpha\beta} \right) \quad (14.3)$$

averages to the pressure tensor, and the mean of its diagonal is the instantaneous pressure of Section 14.2. `md.PairModel` stores $\mathcal{W}_{\alpha\beta}$ at every call, from the separations $\mathbf{d}$ of the pairs and the forces between them:

```

1      r = np.asarray(positions, dtype=float)
2      i, j, d, dist = self.neighbours.pairs(r)
3      phi, dphi = self._pair(i, j, dist)
4      push = (dphi / dist)[: , None] * d # the force on i from j
5      forces = np.zeros_like(r)
6      np.add.at(forces, i, push)
7      np.add.at(forces, j, -push)
8      self.virial = -np.einsum("pa,pb->ab", d, push)
9      return float(np.sum(phi)), forces

```

$\mathbf{d}$ holds $\mathbf{r}_j - \mathbf{r}_i$ to the nearest copy and push the force on i from j , $\phi'(r_{ij}) \mathbf{d}/r_{ij}$ (Section 8.4), so `virial` is $-\sum d_\alpha F_\beta = -\sum \phi'(r_{ij}) d_\alpha d_\beta / r_{ij}$ over the pairs, $\mathcal{W}_{\alpha\beta}$. It agrees with $-\partial U / \partial \epsilon_{\alpha\beta}$ for each of the nine strains, taken by finite differences, to 2.1×10^{-9} eV, and $-\mathcal{W}_{\alpha\beta}/V$ agrees with the stress of ASE's LennardJones to 7.5×10^{-16} GPa.

Measured

The liquid's tensor is isotropic: under the cell rescaling of Section 14.6 at 0.1 GPa its diagonal averages 0.1003, 0.0995 and 0.0996 GPa and its off-diagonal components -0.0002 , 0.0000 and -0.0001 GPa, while an instantaneous diagonal component scatters by 0.023 GPa and an off-diagonal one by 0.013 GPa. The crystal at a_0 stretched along x is not (Figure 14.3(b)): at $\epsilon_{xx} = +0.02$, $P_{xx} = -0.0735$ GPa while $P_{yy} = P_{zz} = -0.0409$ GPa, and the off-diagonal components stay at zero. The slopes at zero strain are the crystal's elastic stiffnesses, $C_{11} = -\partial P_{xx} / \partial \epsilon_{xx} = 3.920$ GPa and $C_{12} = -\partial P_{yy} / \partial \epsilon_{xx} = 2.324$ GPa. An even squeeze applies all three strains at once; by the cubic symmetry P_{xx} responds to ϵ_{yy} and ϵ_{zz} as P_{yy} does to ϵ_{xx} , so each diagonal component changes by $-(C_{11} + 2C_{12})e$ while $dV/V = 3e$, and the bulk modulus is $(C_{11} + 2C_{12})/3 = 2.856$ GPa, the value Section 14.3 found from the slope of P .

ASE, like many libraries, reports the stress, which is $-P$, positive when the material is pulled; a program's sign convention is worth checking before its numbers are used. ASE's `get_stress` returns it in $\text{eV } \text{\AA}^{-3}$ as six numbers,

$$-P_{xx}, -P_{yy}, -P_{zz}, -P_{yz}, -P_{xz}, -P_{xy},$$

and leaves out the kinetic part unless it is called with `include_ideal_gas=True`; the pressure is then minus the mean of the first three.

FROM ELECTRONIC STRUCTURE. In a density-functional calculation the stress is the same derivative of the energy with respect to strain, and Nielsen and Martin showed how to compute it from the electronic structure directly⁷⁴. A basis of plane waves cut off at a fixed energy changes when the cell does, so a cell that grows during a calculation is described by a basis of different quality from the one it started with, and the stress carries a spurious part, the Pulay stress; it is removed by raising the cutoff until the stress no longer changes, or by correcting for the incomplete basis as Francis and Payne set out⁷⁵. A basis of orbitals centred on the atoms moves with them instead, and its derivatives enter the stress as Pulay terms, as they enter the forces (Section 3.12).

KEY IDEA. The pressure tensor gives the force per unit area on a face of any orientation, $\mathbf{F} = A \mathbf{P} \mathbf{n}$; it is $(\sum m v_\alpha v_\beta + \mathcal{W}_{\alpha\beta})/V$, from the response of the free energy to a strain, and the mean of its diagonal is the pressure. A liquid's is P times the unit matrix; a strained crystal's is not. Libraries report the stress, $-\mathbf{P}$.

NOTEBOOK 14 *stretch the crystal along one axis and read the tensor*

CHECK YOUR UNDERSTANDING. ASE's stress, with the kinetic part included, is $0.002 \text{ eV } \text{\AA}^{-3}$ for each of its first three numbers and zero for the others. What is the pressure in GPa, and is the material pushing on its cell or pulling on it?

14.5 What a barostat must do

A barostat lets the volume of the box change so that the run samples the ensemble at constant pressure. This section sets out what that ensemble requires, finds the exact distribution of the volume for an ideal gas and its spread for a liquid, which are the tests every barostat of the chapter must pass, and fits a barostat into the loop of Chapter 13.

The ensemble at constant pressure

A system that exchanges volume with a bath at the pressure P_0 has the weight $e^{-\beta(H+P_0V)}$ (Section 12.8). Grouping the states by their volume gives the probability of each volume,

$$\mathcal{P}(V) \propto e^{-\beta P_0 V} Z(N, V, T),$$

the canonical partition function of the box at that volume weighted by $e^{-\beta P_0 V}$. A barostat must therefore do two things that a thermostat does not: move the volume, and move it so that it takes each value with this probability. Its mean pressure should be P_0 , and the spread of its volume should be the one this distribution has.

The ideal gas exactly

For atoms that do not interact, $U = 0$ and [Section 14.3](#) gives $Z = C V^N$, so

$$\mathcal{P}(V) \propto V^N e^{-bV}, \quad b = \frac{P_0}{k_B T}.$$

The integrals $\int_0^\infty V^n e^{-bV} dV = n!/b^{n+1}$, found by integrating by parts n times ([Exercise 14.10](#)), give the mean and the variance, each the integral with one or two more powers of V divided by the integral with none:

$$\begin{aligned} \langle V \rangle &= \frac{(N+1)!/b^{N+2}}{N!/b^{N+1}} = \frac{N+1}{b} = \frac{(N+1)k_B T}{P_0}, \\ \langle V^2 \rangle &= \frac{(N+2)!/b^{N+3}}{N!/b^{N+1}} = \frac{(N+2)(N+1)}{b^2}, \quad \langle V^2 \rangle - \langle V \rangle^2 = \frac{N+1}{b^2}. \end{aligned}$$

The mean is the ideal gas law's with $N+1$ in place of N , a difference that vanishes for large N , and the relative spread is $1/\sqrt{N+1}$. For 20 atoms at 135 K and $P_0 = 1 \times 10^{-4} \text{ eV } \text{\AA}^{-3}$, the mean volume is $2443 \text{ } \text{\AA}^3$ and its spread $533 \text{ } \text{\AA}^3$: a small gas whose volume fluctuates by a fifth, a severe test (`virial.ideal_gas_volume_density` gives the whole density).

A liquid's volume

Chapter 12 found that the volume at constant pressure fluctuates with the variance $-k_B T d\langle V \rangle / dP$ ([Exercise 12.19](#)). With the compressibility of [Section 14.3](#), $d\langle V \rangle / dP = -\langle V \rangle \kappa_T$, so

$$\langle V^2 \rangle - \langle V \rangle^2 = k_B T \langle V \rangle \kappa_T. \quad (14.4)$$

A soft material swings its volume more than a stiff one. To test a barostat on the liquid this needs κ_T found without one. Three runs at fixed volume around the volume the liquid takes at 0.1 GPa, three starts each of 100 ps under CSVr at 135 K, give

$V / \text{\AA}^3$	12 112	12 301	12 477
$\langle P \rangle / \text{GPa}$	0.1109 ± 0.0004	0.1000 ± 0.0005	0.0904 ± 0.0013

with the standard error of the three starts. The parabola through the three points has, at $12\,306\text{ \AA}^3$, the mean volume of the runs at constant pressure of Sections 14.6 and 14.7, the slope dP/dV that gives $\kappa_T = -1/(V dP/dV) = (1.46 \pm 0.11)\text{ GPa}^{-1}$, the error found by redrawing the three pressures within their standard errors; the liquid's bulk modulus is a quarter of the crystal's. Equation (14.4) then predicts a spread of $183\text{ \AA}^3$, 1.5% of the volume.

The barostat in the loop

`barostats.run` is Chapter 13's loop with one addition. At the start of each step, after the thermostat has acted, the barostat reads the instantaneous pressure tensor, Equation (14.3), from the velocities and the virial the last force call left, and changes the cell, scaling the positions with it so that their fractional coordinates are kept, and, for some barostats, the velocities; the forces are then computed again in the new cell and the step continues as velocity Verlet. The neighbour list is kept across such changes while no pair can have been missed. If the strain since the list was built stretches or shrinks no separation by more than the fraction e of its length (for a box scaled by the factor $1 + e$ along every edge, e itself), and no atom has moved further than u from where the strain alone would have carried it, a pair now within r_c was within $r_c + 2u$ after the strain alone, and so within $(r_c + 2u)/(1 - e)$ when the list was built. The list of Section 10.4 therefore holds while $(r_c + 2u)/(1 - e) \leq r_c + r_{\text{skin}}$; with the box fixed, $e = 0$ and this is the rule $u \leq \frac{1}{2}r_{\text{skin}}$ of that section. The run records the work, the change of $K + U + P_0V$ that the barostat makes, beside the heat of the thermostat; the effective energy $K + U + P_0V$ less the heat and the work changes only through the integration error, as Equation (13.1) did for a thermostat.

KEY IDEA. A barostat must sample $\mathcal{P}(V) \propto e^{-\beta P_0 V} Z(N, V, T)$: the mean pressure P_0 , and the right spread of volume. For an ideal gas $\mathcal{P}(V) \propto V^N e^{-\beta P_0 V}$, with mean $(N + 1)k_B T / P_0$ and relative spread $1/\sqrt{N + 1}$; for a liquid the variance of V is $k_B T \langle V \rangle \kappa_T$, with κ_T measured at fixed volumes.

CHECK YOUR UNDERSTANDING. By what fraction does the volume of 1000 ideal-gas atoms fluctuate at constant pressure?

14.6 Weak coupling and stochastic cell rescaling

Chapter 13 built weak coupling for the temperature, found its fluctuations too narrow, and mended it with noise. The same two steps give the two first-order barostats of this section, which set the rate of change of the volume directly, with no momentum of the volume to carry it on: Berendsen's, which relaxes the pressure towards its target, and stochastic cell rescaling, which adds the noise that gives the volume its right spread.

Weak coupling

Berendsen and colleagues asked the pressure to relax towards P_0 with a time constant τ_P ⁵⁶,

$$\frac{dP}{dt} = \frac{P_0 - P}{\tau_P}.$$

The pressure responds to the volume through the compressibility, $dP = -dV/(V\kappa)$, with κ an estimate of the material's κ_T , so the volume must change at the rate

$$\frac{dV}{dt} = -V\kappa \frac{dP}{dt} = -\frac{V\kappa}{\tau_P}(P_0 - P).$$

Over one step δt the volume is multiplied by $1 - (\kappa\delta t/\tau_P)(P_0 - P)$, and each length by its cube root, which to first order in the small change, by $(1 + y)^{1/3} \approx 1 + y/3$ from the Taylor series of [Section 1.8](#), is

$$1 - \frac{\kappa\delta t}{3\tau_P}(P_0 - P);$$

barostats. Berendsen scales the cell and the positions by this factor at the start of each step, leaving the velocities as they are, and with Berendsen's thermostat follows ASE's NPTBerendsen to $7.9 \times 10^{-8} \text{ \AA}$ over 40 steps. The estimate κ need not be right. Near the mean volume $\bar{V}$, $P_0 - P \approx (V - \bar{V})/(\bar{V}\kappa_T)$ with the true κ_T , so

$$\frac{dV}{dt} = -\frac{\kappa}{\kappa_T\tau_P}(V - \bar{V}),$$

and the volume relaxes with the time constant $\tau_P\kappa_T/\kappa$, in the way the gap w of [Section 2.7](#) did: a wrong κ changes only how fast the volume settles. We gave the barostats on the liquid the rough guess $\kappa = 2.5 \text{ GPa}^{-1}$; with the measured $\kappa_T = 1.46 \text{ GPa}^{-1}$ and $\tau_P = 1 \text{ ps}$ the liquid's volume relaxes in 0.58 ps.

Like its thermostat, Berendsen's barostat holds the mean pressure and squeezes the fluctuations ([Figure 14.4](#)). On the ideal gas of [Section 14.5](#), under Langevin friction, which leaves all $3N$ momenta free so that $\langle 2K \rangle = 3Nk_B T$, its volume spreads by $192 \text{ \AA}^3$,

0.361 of the exact $533 \text{ \AA}^3$, and even its mean is off, $2328 \text{ \AA}^3$ against $2443 \text{ \AA}^3$. The barostat and the ensemble both hold the mean of the instantaneous pressure, here $2K/3V$, at P_0 , which fixes $\langle 1/V \rangle = P_0/Nk_B T$. The ensemble's wide spread makes $\langle 1/V \rangle$ larger than $1/\langle V \rangle$, by the factor $(N+1)/N$, which puts its mean at $(N+1)k_B T/P_0$; Berendsen's narrow spread leaves $\langle 1/V \rangle$ close to $1/\langle V \rangle$ and the volume near $Nk_B T/P_0 = 2327 \text{ \AA}^3$ (Exercise 14.12). On the liquid, with three runs of 300 ps at 0.1 GPa and 135 K under CSV, the mean pressure is 0.1000 GPa, but κ_T read from the volume's fluctuations by Equation (14.4) is $(0.64 \pm 0.05) \text{ GPa}^{-1}$ against the 1.46 GPa^{-1} measured at fixed volumes.

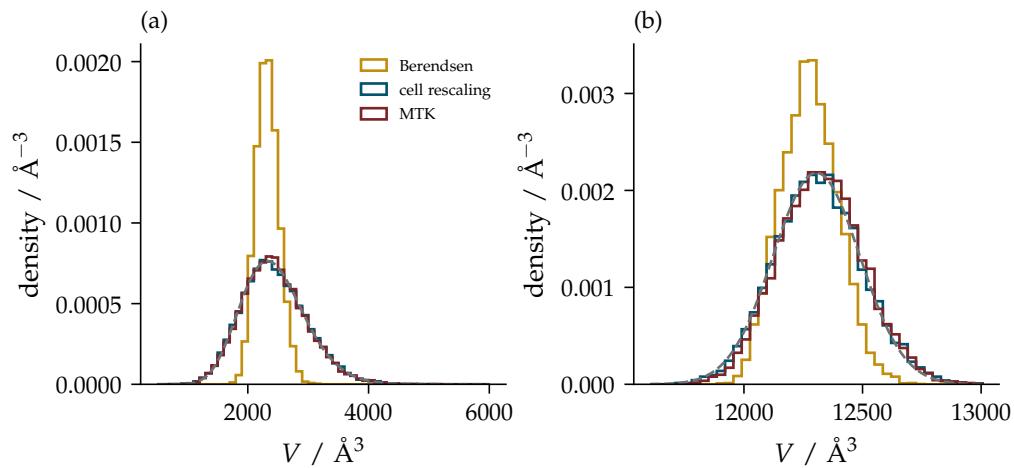


FIGURE 14.4: The volume a barostat samples. (a) 20 argon atoms that do not interact, at 135 K and $P_0 = 1 \times 10^{-4} \text{ eV \AA}^{-3}$, with $\tau_P = 0.2 \text{ ps}$, and $\kappa = 1/P_0$ and Langevin friction for the first-order barostats: the density of the volume under Berendsen's barostat, stochastic cell rescaling and the piston of Martyna, Tobias and Klein (MTK) of Section 14.7 with its chains, against the exact $V^N e^{-\beta P_0 V}$ (dashed). (b) The liquid at 135 K and 0.1 GPa, three runs of 300 ps for each: the density of the volume, against the Gaussian of variance $k_B T \langle V \rangle \kappa_T$ with κ_T from the runs at fixed volume (dashed).

Noise for the volume

Bernetti and Bussi mended the barostat as CSV mended the thermostat⁷⁶. Berendsen's drift, written for $\ln V$, is $d \ln V = dV/V = -(\kappa/\tau_P)(P_0 - P) dt$. Near the mean, $\ln V - \ln \bar{V} \approx (V - \bar{V})/\bar{V}$, so $P_0 - P \approx (\ln V - \ln \bar{V})/\kappa_T$ and

$$d \ln V = -\frac{\kappa}{\kappa_T \tau_P} (\ln V - \ln \bar{V}) dt,$$

the drag of Section 13.4 with $\gamma = \kappa/(\kappa_T \tau_P)$, acting on $\ln V - \ln \bar{V}$. Adding kicks of strength σ makes it the Ornstein-Uhlenbeck process of Equation (13.3), whose variance

settles at $\sigma^2/2\gamma$ (Exercise 13.7). The ensemble asks for the variance of $\ln V$ to be that of V over $\bar{V}^2$, $k_B T \kappa_T / \bar{V}$ by Equation (14.4), so

$$\frac{\sigma^2}{2\gamma} = \frac{k_B T \kappa_T}{\bar{V}} \implies \sigma^2 = \frac{2\kappa}{\kappa_T \tau_P} \frac{k_B T \kappa_T}{\bar{V}} = \frac{2k_B T \kappa}{\bar{V} \tau_P},$$

and the unknown κ_T cancels. Stochastic cell rescaling keeps this noise, with the current V in place of $\bar{V}$:

$$d \ln V = -\frac{\kappa}{\tau_P} (P_0 - P) dt + \sqrt{\frac{2k_B T \kappa dt}{V \tau_P}} g, \quad (14.5)$$

with g a fresh number from the normal distribution of mean 0 and variance 1 at each interval dt , the kick of Section 13.4: the fluctuation-dissipation relation again, now for the volume. Bernetti and Bussi showed that it samples the ensemble at constant pressure far from the mean as well as near it⁷⁶, provided P is the instantaneous pressure taken along the path on which the barostat moves the atoms. Their drift comes from the slope of $K + U + P_0 V$ with V along that path, so P must be $-\partial(K + U)/\partial V$ there. The virial part already is, since Section 14.3 took $\partial U/\partial V$ at fixed fractional coordinates. If every momentum is scaled by $V^{-1/3}$ as the box grows, $K \propto V^{-2/3}$ and $\partial K/\partial V = -\frac{2}{3}K/V$ (Exercise 14.15), so $-\partial K/\partial V$ is $2K/3V$, the kinetic part of the instantaneous pressure. Each length is therefore multiplied by $e^{d \ln V/3}$ and each velocity by $e^{-d \ln V/3}$. `StochasticCellRescaling.scale` takes one such step of Equation (14.5) each time step:

```

1      kappa, tau, p0 = self.compressibility, self.tau, self.pressure
2      kt_v = self.kb * self.temperature / cell_volume(h)
3      diagonal = np.diag(tensor)
4      if self.couple == "isotropic":
5          d_log_v = (-kappa / tau * (p0 - diagonal.mean()) * dt
6                     + math.sqrt(2 * kt_v * kappa * dt / tau)
7                     * rng.standard_normal())
8          stretch = np.full(3, math.exp(d_log_v / 3))

```

followed by `_scale_cell`, which multiplies the cell's lattice vectors by `stretch`, the positions with them, and the velocities by the inverse. For the liquid, a step moves $\ln V$ by noise of spread 2.75×10^{-3} , against its settled spread of 0.0149.

Measured

On the ideal gas the volume now follows the exact density closely (Figure 14.4(a)): mean $2444 \text{ \AA}^3$, 1.000 of the exact, and spread $541 \text{ \AA}^3$, 1.014 of it. On the liquid the three runs give $\kappa_T = (1.50 \pm 0.05) \text{ GPa}^{-1}$ from the fluctuations, against 1.46 GPa^{-1} at fixed

volumes, with the mean pressure 0.0988, 0.1004 and 0.1002 GPa. ASE has no stochastic cell rescaling; the ideal gas, whose answer is known exactly, is the independent check.

KEY IDEA. Berendsen's barostat scales each length by $1 - (\kappa \delta t / 3 \tau_P)(P_0 - P)$: it holds the mean pressure, relaxes the volume in $\tau_P \kappa_T / \kappa$, and squeezes the volume's fluctuations, to 0.36 of the exact spread for a small ideal gas and to less than half the compressibility for the liquid. Stochastic cell rescaling adds to $\ln V$ the noise $\sqrt{2k_B T \kappa \, dt / (V \tau_P)} g$ that the fluctuation-dissipation relation demands, and samples the volume correctly.

NOTEBOOK 14 set the barostat's time constant and compare the volume with the exact density

CHECK YOUR UNDERSTANDING. The liquid reads 0.08 GPa with the target 0.1 GPa, $\kappa = 2.5 \text{ GPa}^{-1}$ and $\delta t / \tau_P = 0.01$. By what factor does Berendsen's barostat multiply each length in this step?

14.7 A piston with mass

The barostats of [Section 14.6](#) move the volume as a heavily damped quantity, with no momentum of its own. A real piston has mass: it overshoots and swings. This section builds the barostat that treats the volume as a coordinate with inertia, finds the frequency at which it swings, and measures it.

A piston on a gas spring

A piston of mass M_p and area A rests on a column of gas of height x , so $V = Ax$, with the pressure P_0 above it. The gas pushes up with PA and the outside pushes down with $P_0 A$, so $M_p \ddot{x} = A(P - P_0)$. If the piston is moved by δx from where the two balance, the volume changes by $A \delta x$ and the gas's pressure by $\delta P = -A \delta x / (V \kappa)$, and

$$M_p \delta \ddot{x} = -\frac{A^2}{V \kappa} \delta x :$$

the oscillator of [Section 4.4](#), with the gas as its spring, of stiffness $A^2 / V \kappa$, and the frequency $\omega^2 = A^2 / (M_p V \kappa)$. A heavy piston on a soft gas swings slowly.

The volume as a coordinate

Andersen turned this into a barostat⁵⁹: the volume is given a mass and a kinetic energy of its own, the outside does the work $P_0 V$, and the atoms' positions scale with the box, so that the volume accelerates in proportion to $P - P_0$ and the sum of H , $P_0 V$ and the

piston's kinetic energy is kept. The form of Martyna, Tobias and Klein (MTK)⁷⁷, which ASE and `mdlab` follow, takes as its coordinate the strain ϵ by which every length is e^ϵ times its starting value, so $V = V_0 e^{3\epsilon}$, with the momentum p_ϵ and the mass M_ϵ . Its equations are

$$\begin{aligned}\dot{\mathbf{r}}_i &= \mathbf{v}_i + \frac{p_\epsilon}{M_\epsilon} \mathbf{r}_i, & m_i \dot{\mathbf{v}}_i &= \mathbf{F}_i - \left(1 + \frac{3}{N_f}\right) \frac{p_\epsilon}{M_\epsilon} m_i \mathbf{v}_i, \\ \dot{V} &= \frac{3V p_\epsilon}{M_\epsilon}, & \dot{p}_\epsilon &= 3V(P - P_0) + \frac{3}{N_f} \sum_i m_i v_i^2,\end{aligned}$$

with N_f the number of free momenta. Here $\mathbf{v}_i = \mathbf{p}_i/m_i$ is the velocity of atom i relative to the stretching box, which the first equation adds to the box's own motion to give $\dot{\mathbf{r}}_i$, and P is the instantaneous pressure computed with these $\mathbf{v}_i$. The first equation carries each atom along as the box grows, the second slows the atoms as it does, the third ties the volume to the strain, and the fourth pushes the piston with the difference of pressures. Martyna, Tobias and Klein showed that with the two terms in $3/N_f$ these equations, given a thermostat, sample the ensemble at constant pressure; without them the distribution of the volume is wrong by an amount that shrinks as N grows. Nosé-Hoover chains (Section 13.5) add a friction to the second and fourth equations, acting on the atoms and on p_ϵ , and the run keeps $K + U + P_0 V + p_\epsilon^2/2M_\epsilon$ plus the chains' energies. `barostats.run_mtk` integrates these equations by the Trotter splitting (Section 9.4) of Tuckerman and colleagues⁷⁸, as ASE's `IsotropicMTKNPT` does, and follows it to 5.6×10^{-7} Å over 40 steps. Both take $N_f = 3N$. A liquid whose total momentum stays at zero has only $3N - 3$ free momenta, and with $3N$ in the chain its kinetic temperature runs high by about $3/(3N - 3)$, 0.4% for 256 atoms: the three runs of Figure 14.4(b) average 135.48, 135.99 and 135.94 K against the 135 K asked for.

How fast it swings

Ignoring the chains and the term $(3/N_f) \sum_i m_i v_i^2$, which stays close to $3k_B T$ and only shifts where the piston balances, $M_\epsilon \ddot{\epsilon} = 3V(P - P_0)$. A change of the strain by $\delta\epsilon$ changes the volume by $3V\delta\epsilon$ and, taking the response at fixed temperature, the pressure by $\delta P = -3\delta\epsilon/\kappa_T$, so

$$M_\epsilon \ddot{\epsilon} = -\frac{9V}{\kappa_T} \delta\epsilon, \quad \omega^2 = \frac{9V}{M_\epsilon \kappa_T},$$

the piston on its gas spring again. ASE and `mdlab` take the mass from a time constant, $M_\epsilon = (3N + 3)k_B T \tau_P^2$, which gives the period (Exercise 14.18)

$$\frac{2\pi}{\omega} = 2\pi\tau_P \sqrt{\frac{(N+1)k_B T \kappa_T}{3V}};$$

for the liquid, with $\kappa_T = 1.46 \text{ GPa}^{-1}$, it is $0.865 \tau_P$, and $M_e = 8.97 \times 10^6 \text{ eV fs}^2$ for $\tau_P = 1 \text{ ps}$. Measured from the crossings of the volume through its mean (Figure 14.5), the periods are 0.248, 0.876, 2.450 and 8.695 ps for τ_P of 0.3, 1, 3 and 10 ps, against 0.259, 0.865, 2.594 and 8.647 ps predicted, the agreement within 6% bearing out the response taken at fixed temperature.

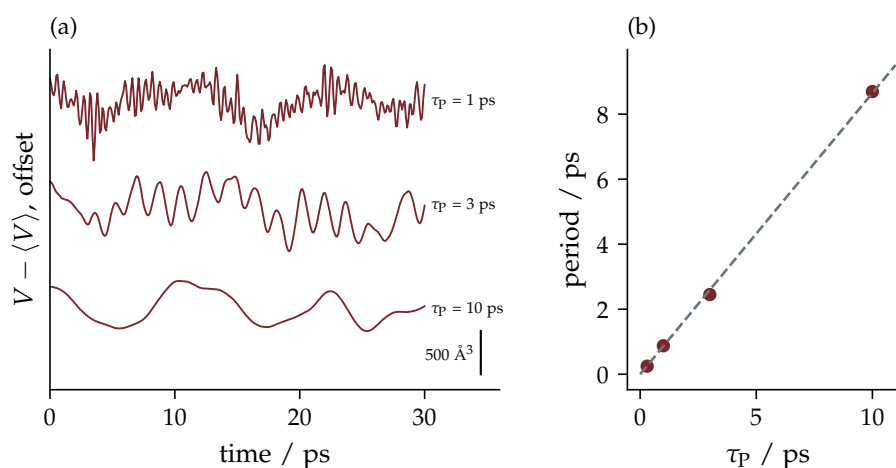


FIGURE 14.5: A piston with mass rings. The liquid at 135 K and 0.1 GPa under the MTK piston with chains. (a) The volume's departure from its mean over the first 30 ps for $\tau_P = 1, 3$ and 10 ps, each trace offset by $1200 \text{ \AA}^3$. (b) The period of the volume's swing against τ_P (points) and the period of the linearised piston (dashed).

The piston samples the right volume (Figure 14.4): on the ideal gas its mean is 0.999 and its spread 0.994 of the exact values, and on the liquid, with $\tau_P = 1 \text{ ps}$, the volume's fluctuations give $\kappa_T = (1.39 \pm 0.06) \text{ GPa}^{-1}$ against 1.46 GPa^{-1} . How heavy the piston should be is a question of time. With $\tau_P = 10 \text{ ps}$ the volume swings only 10 times in 90 ps, too few for a run of that length to average its swings away; with 1 ps it swings 103 times.

Changing shape

Parrinello and Rahman gave every component of the cell matrix $\mathbf{h}$ a mass and a momentum, so that the cell can change its shape as well as its size, each lattice vector a piston of its own, and used it to follow a crystal from one structure to another⁷⁹. Martyna, Tobias and Klein gave the same freedom to their piston. `mdlab` has only the isotropic piston; Section 14.8 changes the shape of the cell with first-order barostats instead.

KEY IDEA. A piston with mass treats the volume as a coordinate pushed by $P - P_0$; it swings like a mass on a gas spring, with $\omega^2 = 9V/(M_\epsilon \kappa_T)$, a period of $0.865 \tau_P$ for the liquid with the mass $(3N + 3)k_B T \tau_P^2$. With Nosé-Hoover chains and the corrections of Martyna, Tobias and Klein it samples the right volume; a heavy piston swings so slowly that a short run averages few swings.

NOTEBOOK 14 *set the piston's mass and watch the volume ring*

CHECK YOUR UNDERSTANDING. A piston of 1 kg and area 10 cm^2 , its weight ignored, sits on one litre of air, taken at a fixed temperature, so that $\kappa = 1/P$ with $P = 1 \text{ bar}$. What is the period of its swing?

14.8 Isotropic, semi-isotropic and anisotropic coupling

Every barostat run so far has scaled the three lengths of the box by one factor. That suits a liquid, whose pressure tensor is P times the unit matrix, and fails for a solid that wants to grow along one direction only. This section takes graphite filling with lithium as its example, sets out the ways a barostat can let the shape change, and measures what each does to a layered model solid.

Graphite opens along one axis

Graphite is a stack of sheets of carbon, each a honeycomb of strong bonds, held together by a weak attraction between the sheets. When lithium enters, it sits between the sheets over the centres of the hexagons, and the sheets, stacked A, B, A, B in graphite, each shifted so that half its atoms sit over the centres of the hexagons below, line up A, A around it, each directly over the last; in the fully lithiated LiC_6 the sheets open from $3.34 \text{ \AA}$ apart to $3.70 \text{ \AA}$, 10.8% wider⁷² (Figure 14.6). A honeycomb has two carbon atoms for each hexagon, since each of a hexagon's six corners is shared by three hexagons, so one lithium for six carbons fills every third hexagon. A barostat that scaled the three lengths together could open the gaps only by stretching the sheets by as much, and would leave the solid strained.

Three couplings

For a box whose lattice vectors lie along x , y and z , a barostat can respond to the pressure tensor in three ways.

- *Isotropic*: one factor for all three lengths, from the mean of the diagonal, P .

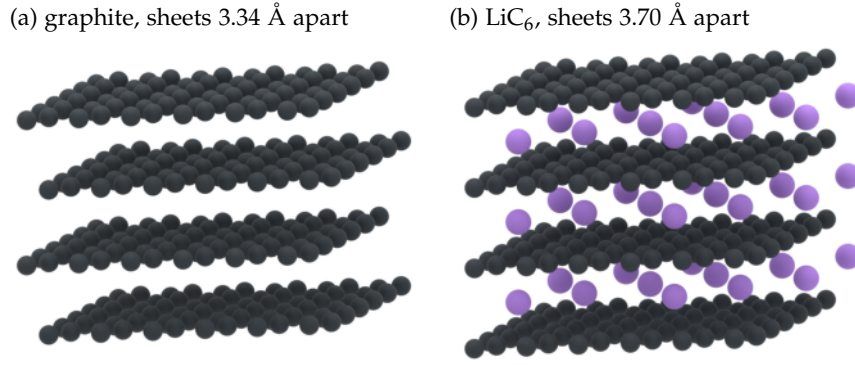


FIGURE 14.6: Graphite and LiC_6 , four sheets of carbon each, seen almost along the sheets: (a) graphite, stacked A, B, A, B, with the sheets 3.34 Å apart; (b) LiC_6 , stacked A, A, A, with lithium (purple) over every third hexagon between the sheets, 3.70 Å apart (the experimental spacings quoted by Hazrati, de Wijs and Brocks⁷²; the in-plane spacing is kept at graphite's 2.46 Å).

- *Semi-isotropic*: one factor for x and y together, from $\frac{1}{2}(P_{xx} + P_{yy})$, and another for z , from P_{zz} . This suits a layered solid or a membrane⁷⁶, whose plane keeps its own symmetry while its thickness changes.
- *Anisotropic*: a factor for each length, from its own diagonal component.

Changing the angles of the cell as well needs every component of $\mathbf{h}$ free, as in Parrinello and Rahman's barostat (Section 14.7). `barostats.Berendsen` takes each of the three couplings, scaling the length along α by $1 - (\kappa \delta t / 3\tau_P)(P_0 - P_\alpha)$ with P_α the pressure that length responds to; its anisotropic form follows ASE's `Inhomogeneous_NPTBerendsen` to 7.9×10^{-8} Å over 40 steps. Stochastic cell rescaling has a semi-isotropic form too, with its own drift and noise for the logarithm of the area A normal to z and for the height L_z ⁷⁶,

$$d \ln A = -\frac{2\kappa}{3\tau_P} \left(P_0 - \frac{P_{xx} + P_{yy}}{2} \right) dt + \sqrt{\frac{4k_B T \kappa dt}{3V\tau_P}} g_A,$$

$$d \ln L_z = -\frac{\kappa}{3\tau_P} (P_0 - P_{zz}) dt + \sqrt{\frac{2k_B T \kappa dt}{3V\tau_P}} g_z,$$

with g_A and g_z independent normal numbers. Their sum, since $\ln V = \ln A + \ln L_z$, is Equation (14.5): the drifts add to $-(\kappa/\tau_P)(P_0 - P) dt$, and the variances of the noises, $\frac{2}{3}$ and $\frac{1}{3}$ of $2k_B T \kappa dt / (V\tau_P)$, add to the whole of it, since the variances of independent numbers add (Section 12.1). The area carries two of the three lengths and the height one, hence the shares $\frac{2}{3}$ and $\frac{1}{3}$.

A layered model solid

Graphite needs a potential that a later chapter will bring; its point can be made with Lennard-Jones atoms. The model solid of Figure 14.7 is nine flat sheets of 48 argon-like atoms each, the atoms of a sheet at the corners of equilateral triangles, stacked so that each sheet's atoms sit over the hollows of the one below. Atoms within a sheet attract with eight times argon's depth and 0.95 of its σ , atoms in different sheets with argon's own, so the sheets are stiff and the gaps soft: at 40 K the sheets settle 3.151 Å apart, and the perfect solid, its atoms on their sites in that cell, has $C_{11} = -\partial P_{xx}/\partial \epsilon_{xx} = 35.81$ GPa along the sheets against $C_{33} = -\partial P_{zz}/\partial \epsilon_{zz} = 5.31$ GPa across them, 6.7 times softer. Between each pair of sheets go twelve guest atoms of lithium's mass, one for every four sheet atoms, each over a hollow of both sheets, in rows 7.25 Å apart along x and 6.28 Å along y . They interact with the sheet atoms through a Lennard-Jones potential of twice argon's depth, and with each other through one of a fifth of argon's depth and 1.2 times its σ . Their reach, the σ of their interaction with the sheets, grows from 1.6 to 2.5 Å over 10 ps, pushing the sheets apart through a potential that changes with time, as the wall pushed in of Section 3.14 did, and is then held for 30 ps more, under Berendsen's barostat at $P_0 = 0$ with each coupling in turn.

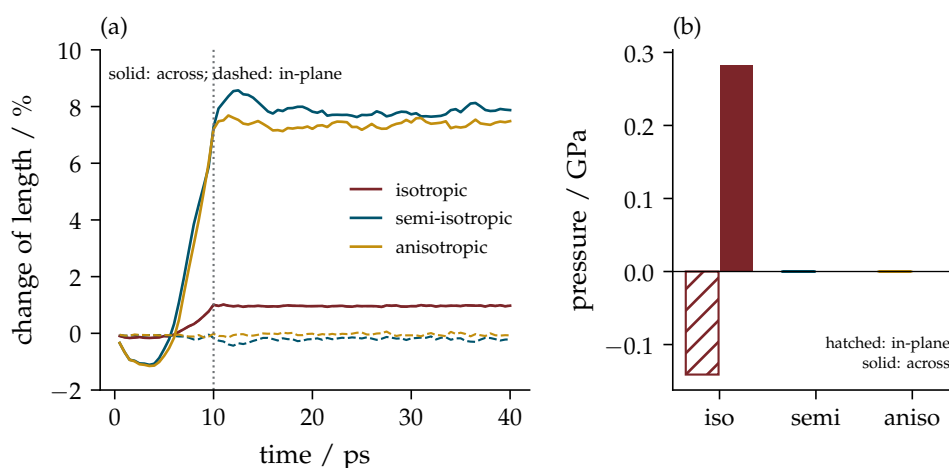


FIGURE 14.7: Coupling a layered solid. The layered model at 40 K under Berendsen's barostat at $P_0 = 0$, guests between its sheets whose reach grows over the first 10 ps (dotted) and is then held for 30 ps. (a) The change of the height c of the cell (solid) and of its in-plane width a (dashed) under isotropic, semi-isotropic and anisotropic coupling. (b) The pressure in the plane, $\frac{1}{2}(P_{xx} + P_{yy})$ (hatched), and across it, P_{zz} (solid), over the last 10 ps.

Over the last 10 ps, under semi-isotropic coupling the sheets have opened by 7.84% and the plane has shrunk by 0.20%, and the mean pressures in the plane and across it are zero. Under anisotropic coupling the sheets have opened by 7.42% and the plane has changed by less than 0.1%. Both have settled, the opening wandering by about 0.2% between successive stretches of 5 ps. The two differ because the rows of guests make the two directions of the plane unequal: semi-isotropic coupling, which scales x

and y together, leaves P_{xx} and P_{yy} at $+0.013$ and -0.014 GPa, and anisotropic coupling, which lets each length move on its own, brings both to zero. Under isotropic coupling every length grows by 0.97%: the gaps cannot open as far as the guests need without the sheets stretching equally, and the solid is left pulled along its sheets, -0.141 GPa in the plane, and squeezed across them, $+0.282$ GPa, while the mean of the three, the only number the isotropic barostat reads, is zero. The early dip of c under semi-isotropic and anisotropic coupling in Figure 14.7(a), by about 1%, comes while the guests' reach is short. Each guest is $2.60 \text{ \AA}$ from its nearest sheet atoms, which lies beyond the minimum $2^{1/6}\sigma$ of their pair energy for every σ up to $2.3 \text{ \AA}$, so until then the guests pull the sheets towards them.

KEY IDEA. A solid that wants to change shape needs a barostat that lets it: semi-isotropic coupling, one factor for the plane and one across it, for a layered solid like graphite; anisotropic coupling for each length on its own, when the directions of the plane differ too. Isotropic coupling of the layered model with guests between its sheets leaves -0.141 GPa along the sheets and $+0.282$ GPa across them while its mean pressure reads zero.

NOTEBOOK 14 *insert guests between the sheets under each coupling*

CHECK YOUR UNDERSTANDING. A graphite particle opens across its sheets by 10.8% on lithiation, and suppose it keeps its in-plane size. By what fraction does its volume grow?

14.9 What the trajectory looks like

A run at constant pressure has two more things to read than one at fixed volume: the volume itself, and a pressure whose single values mean little. This section shows a healthy run of the liquid and six faults that pressures and barostats leave, each with the check that catches it.

A healthy run

Figure 14.8 follows the liquid for 300 ps under stochastic cell rescaling at 0.1 GPa with $\tau_P = 1$ ps and CSVr at 135 K, started from the box of Chapter 13 at $12577 \text{ \AA}^3$. After the first 10 ps the volume averages $12312 \text{ \AA}^3$, a density of $0.8172\sigma^{-3}$, and spreads by $190 \text{ \AA}^3$, against the $183 \text{ \AA}^3$ of Equation (14.4); it lies within the predicted $183 \text{ \AA}^3$ of its mean for 0.667 of the time, close to the 0.683 of a Gaussian (Section 12.1). The instantaneous pressure scatters by 0.0175 GPa about its mean of 0.0988 GPa: for 256 atoms a single value of P says little, and only its average is compared with the target. The sum $K + U + P_0V$ wanders over 9.50 meV per atom as the thermostat and the

barostat exchange energy with the liquid, while the effective energy, that sum less the heat and the work, stays within 0.027 meV per atom and drifts by $-6.4 \times 10^{-6} \text{ eV ps}^{-1}$ for the whole liquid. The kinetic temperature averages 134.69 K.

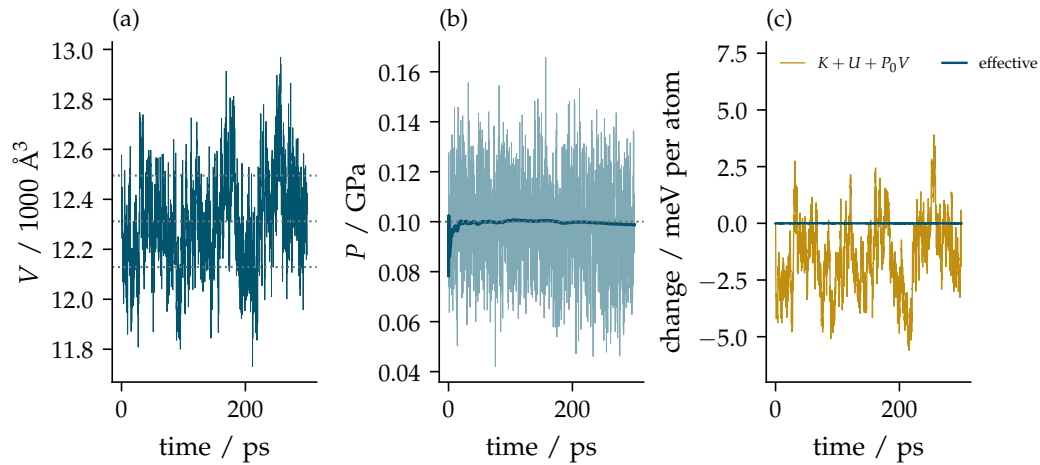


FIGURE 14.8: A healthy run at constant pressure: the liquid under stochastic cell rescaling at 0.1 GPa and CSVR at 135 K, for 300 ps. (a) The volume every 100 fs, with its mean and the spread $\sqrt{k_B T \langle V \rangle \kappa_T}$ either side (dotted). (b) The instantaneous pressure (thin) and its running mean (thick), against the target (dotted). (c) $K + U + P_0 V$ per atom and the effective energy, that less the heat and the barostat's work, from their starting values.

Six faults

Figure 14.9 shows six faults and Table 14.1 the check that catches each.

TABLE 14.1: The faults of Figure 14.9, the check that catches each, and its value in the healthy and the broken run.

fault	check	healthy	broken
(a) virial from wrapped positions	P before and after moving the origin, GPa	0.0877, 0.0877	0.0324, 0.0354
(b) kinetic part dropped	$\langle P \rangle$ with $2K$ included, GPa	0.0988	0.1411
(c) volume too narrow	κ_T from the fluctuations, GPa^{-1}	1.50	0.64
(d) piston too heavy	swings of V in 90 ps	103	10
(e) isotropic coupling of layers	$\frac{1}{2}(P_{xx} + P_{yy}), P_{zz}$, GPa	0.000, 0.000	-0.141, 0.282
(f) κ in the wrong unit	$\langle P \rangle$ after the first 50 ps, GPa	0.1001	0.0871

(a) *The virial from wrapped positions.* As Section 14.3 showed, $\sum_i \mathbf{r}_i \cdot \mathbf{F}_i$ with wrapped positions gives 0.0324 GPa for a liquid whose pressure is 0.0877 GPa, and a different value when the origin moves. The check is to move the origin: a correct pressure does not change.

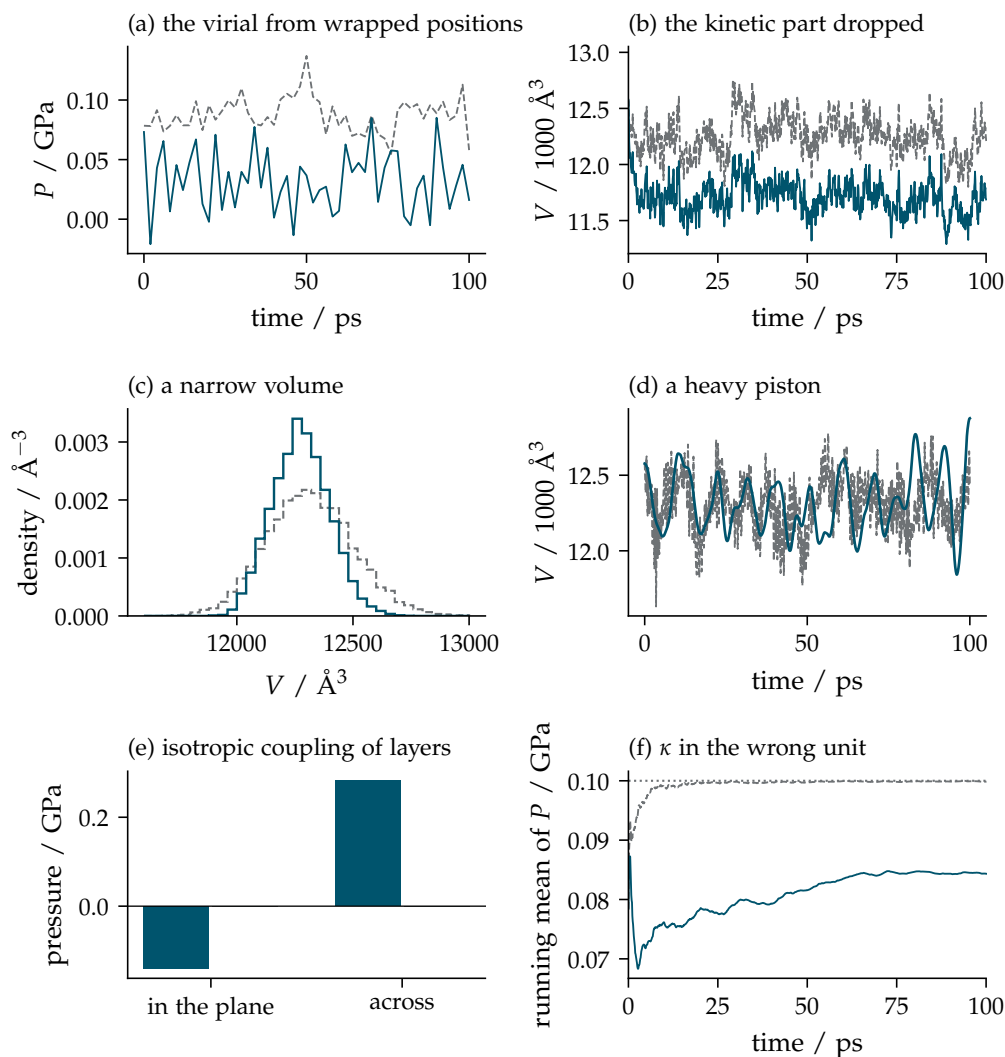


FIGURE 14.9: Pressure and barostat faults (teal) against healthy runs (grey dashed). (a) The pressure of the liquid at fixed volume from $\sum_i \mathbf{r}_i \cdot \mathbf{F}_i$ with wrapped positions, against the pair form. (b) The volume under cell rescaling given the virial's pressure without the kinetic part. (c) The density of the volume under Berendsen's barostat, against cell rescaling. (d) The volume under a piston with $\tau_p = 10$ ps, against 1 ps. (e) The layered solid with guests: the pressure in its plane and across it under isotropic coupling, against semi-isotropic coupling (grey, zero). (f) The running mean of the pressure under Berendsen's barostat given κ in bar⁻¹ where GPa⁻¹ was meant, against the right κ , with the target (dotted).

(b) *The kinetic part dropped.* A barostat handed only $\mathcal{W}/3V$, as happens when ASE's `get_stress` is called without `include_ideal_gas=True`, holds that part at P_0 and lets the true pressure rise by $Nk_B T/V$, here 0.0408 GPa: the liquid shrinks by 4.9%, to $11\,706 \text{ \AA}^3$ from $12\,312 \text{ \AA}^3$, and its full pressure reads 0.1411 GPa. The check is the mean of $(2K + \mathcal{W})/3V$ against P_0 .

(c) *A volume too narrow.* Berendsen's barostat holds the mean pressure, so nothing in one run looks wrong; the check is κ_T from the volume's fluctuations, Equation (14.4), against an independent value: 0.64 GPa^{-1} under Berendsen's barostat, 1.50 GPa^{-1} under cell rescaling, and 1.46 GPa^{-1} from the runs at fixed volume.

(d) *A piston too heavy.* With $\tau_P = 10 \text{ ps}$ the volume swings with a period of 8.7 ps, 10 times in 90 ps, so the run's mean volume rests on ten swings; with 1 ps it swings 103 times. The check is the period of the volume against the length of the run.

(e) *Isotropic coupling of a layered solid.* The mean pressure reads the target while the solid is pulled along its sheets and squeezed across them, -0.141 and $+0.282 \text{ GPa}$. The check is the diagonal of the pressure tensor, component by component.

(f) *The compressibility in the wrong unit.* The barostat's κ for the liquid, 2.5 GPa^{-1} , is 2.5×10^{-4} per bar; entered as 2.5×10^{-4} where gigapascals were meant, it makes Berendsen's barostat 10^4 times slower, a relaxation time of 5.8 ns instead of 0.58 ps. The volume stays where it started, $12\,577 \text{ \AA}^3$, and the pressure stays at 0.0871 GPa against the target 0.1. The check is the running mean of the pressure against the target, and the volume's relaxation against τ_P .

IN PRACTICE. A run at constant pressure is read through the running mean of its pressure, computed with the kinetic part and from pair separations, which should settle at P_0 ; through the spread of its volume, which should be $\sqrt{k_B T \langle V \rangle \kappa_T}$, with κ_T checked at fixed volumes; through the diagonal of its pressure tensor, which for a solid should match P_0 in every component the coupling lets relax; and through the period of its volume, or its relaxation time when the barostat has no piston, against the length of the run. Berendsen's barostat serves to bring a system to its pressure; stochastic cell rescaling or a piston with chains then samples it, and $\tau_P = 1 \text{ ps}$ sampled the liquid's volume correctly here, with 103 swings of the piston in 90 ps; a layered solid needs semi-isotropic coupling, or anisotropic coupling when the directions of its plane differ.

KEY IDEA. In a healthy run at constant pressure the instantaneous pressure scatters widely, its running mean settles at the target, the volume spreads by $\sqrt{k_B T \langle V \rangle \kappa_T}$, and the effective energy holds still. A virial from wrapped positions, a pressure without its kinetic part, Berendsen's narrow volume, a piston too heavy, isotropic coupling of a solid that wants to change shape, and a compressibility in the wrong unit each leave a mark that a single check catches.

NOTEBOOK 14 *break the run at constant pressure in each of the six ways and read the checks*

CHECK YOUR UNDERSTANDING. For a liquid of 500 atoms at 300 K in $20\,000 \text{ \AA}^3$, minus the mean of the diagonal of ASE's `get_stress()`, called with its defaults, averages -0.04 GPa . What is the liquid's pressure?

What we now have

The pressure of a gas is the momentum its atoms deliver to the walls per unit time and area, $Nk_B T/V$ for atoms that do not interact. Equipartition applied to the positions adds the virial's share, $PV = Nk_B T + \frac{1}{3} \langle \mathcal{W} \rangle$, and the free energy's response to a change of volume gives the same formula in a periodic box, with the virial summed over pairs from their separations to the nearest copy; the sum $\sum_i \mathbf{r}_i \cdot \mathbf{F}_i$ from wrapped positions changes with the origin.

Its response to a strain gives the pressure tensor, $(\sum_i m_i v_{i\alpha} v_{i\beta} + \mathcal{W}_{\alpha\beta})/V$, isotropic for a liquid, uneven for a strained solid; ASE and many libraries report the stress, its negative.

A barostat must sample $\mathcal{P}(V) \propto e^{-\beta P_0 V} Z(N, V, T)$, which for an ideal gas is $V^N e^{-\beta P_0 V}$ and for a liquid has the variance $k_B T \langle V \rangle \kappa_T$.

Berendsen's barostat holds the mean pressure but squeezes the volume's fluctuations, which then give less than half the liquid's compressibility; stochastic cell rescaling adds to $\ln V$ the noise the fluctuation-dissipation relation demands and samples the volume correctly; the MTK piston gives the volume a mass, swings with the period $2\pi\tau_P \sqrt{(N+1)k_B T \kappa_T / 3V}$, and with chains samples correctly too.

A solid that wants to change shape needs semi-isotropic or anisotropic coupling: isotropic coupling of a layered solid with guests between its sheets leaves it pulled along the sheets and squeezed across them while its mean pressure reads the target.

A healthy run shows a widely scattered instantaneous pressure whose running mean settles at P_0 , a volume that spreads by $\sqrt{k_B T \langle V \rangle \kappa_T}$, and an effective energy that holds still.

Chapter 15 asks how long a run must be before such averages, and the error bars on them, can be trusted.

Answers to the checks

SECTION 14.1. $P = Nk_{\text{B}}T/V = 10^4 \times 0.025852/100^3 = 2.585 \times 10^{-4} \text{ eV } \text{\AA}^{-3}$, which is $2.585 \times 10^{-4} \times 160.218 \times 10^4 = 414.2 \text{ bar}$.

SECTION 14.2. They raise it. Closer than the minimum, $\varphi'(r) < 0$: the atoms repel, and the pair adds $-r\varphi'(r) > 0$ to $\mathcal{W}$, and so raises $(2K + \mathcal{W})/3V$ above the $2K/3V$ of two atoms that do not interact.

SECTION 14.3. V is proportional to a^3 , and the factor cancels in $V/\Delta V$, so $B \approx -V_{\text{m}} \Delta P/\Delta V$ with V_{m} the mean of 5.20^3 and 5.40^3 : $B \approx 149.04 \times 0.288/16.86 = 2.55 \text{ GPa}$. The two spacings are centred on $5.30 \text{ \AA}$, on the stretched side of a_0 , where the crystal is softer, so the estimate lies below the 2.856 GPa at a_0 .

SECTION 14.4. $P = -0.002 \times 160.218 = -0.320 \text{ GPa}$: the stress is positive, so the material pulls on its cell.

SECTION 14.5. $1/\sqrt{1001} = 0.0316$, about 3%.

SECTION 14.6. $1 - (2.5 \times 0.01/3)(0.1 - 0.08) = 0.9998333$: each length shrinks by 17 parts in a hundred thousand, which raises the pressure towards its target.

SECTION 14.7. $\omega^2 = A^2/(M_{\text{p}}V\kappa) = (10^{-3})^2/(1 \times 10^{-3} \times 10^{-5}) = 100 \text{ s}^{-2}$, so $\omega = 10 \text{ rad s}^{-1}$ and the period is 0.628 s .

SECTION 14.8. The volume is the in-plane area times the height, and with the area kept it grows as the height does: by 10.8%.

SECTION 14.9. `get_stress` leaves out the kinetic part by default; adding $Nk_{\text{B}}T/V = 500 \times 0.025852/20000 = 6.46 \times 10^{-4} \text{ eV } \text{\AA}^{-3} = 0.1035 \text{ GPa}$ gives $-0.04 + 0.1035 = 0.0635 \text{ GPa}$.

Exercises

The exercises run from questions of understanding, through calculations, to short derivations marked ‘show that’. Full solutions are in ‘[Solutions to the exercises](#)’ at the end of the book, and Notebook 14 checks every numerical answer.

EXERCISE 14.1 (*units of pressure*). Show that $1 \text{ eV } \text{\AA}^{-3}$ is 160.218 GPa, and express one bar in $\text{eV } \text{\AA}^{-3}$.

EXERCISE 14.2 (*air*). Find the number of molecules per $\text{\AA}^3$ in air at one bar and 300 K, treated as an ideal gas, and the spacing $(V/N)^{1/3}$ between molecules.

EXERCISE 14.3 (*atoms in every direction*). Atoms of mass m at the number density ρ move in every direction, a fraction $\mathcal{P}(v_x) dv_x$ of them with velocity along x between v_x and $v_x + dv_x$ ([Section 12.5](#)). Show, by counting the atoms with $v_x > 0$ that reach a wall of area A in a time t , each delivering $2mv_x$, that the pressure is $\rho m \langle v_x^2 \rangle$, the average taken over all the atoms, and hence $\rho k_B T$.

EXERCISE 14.4 (*the sign of a pair’s share*). For the Lennard-Jones potential of argon, with $\sigma = 3.4 \text{ \AA}$, find the distance at which a pair’s contribution $-r\phi'(r)$ to the virial changes sign, and the sign of that contribution closer in.

EXERCISE 14.5 (*the origin at the centre*). Show that the walls’ share, $\sum_i \mathbf{r}_i \cdot \mathbf{F}_i^w$, is still $-3PV$ when the origin is placed at the centre of the box, with the walls at $x = \pm L/2$.

EXERCISE 14.6 (*wrapping one atom*). Show that wrapping a single atom i back into a cubic cell of side L along x changes $\sum_j \mathbf{r}_j \cdot \mathbf{F}_j$ by $\pm L F_{ix}$, while leaving [Equation \(8.4\)](#), written with nearest-copy separations, unchanged.

EXERCISE 14.7 (*a determinant to first order*). Show, by writing out the triple product $\mathbf{c}_1 \cdot (\mathbf{c}_2 \times \mathbf{c}_3)$ of the columns of $\mathbf{I} + \epsilon$ ([Section 5.5](#)), that $\det(\mathbf{I} + \epsilon) = 1 + \epsilon_{xx} + \epsilon_{yy} + \epsilon_{zz}$ plus terms of second and third order in the components of ϵ .

EXERCISE 14.8 (*the trace of the tensor*). Show that one third of the trace of [Equation \(14.3\)](#) is the instantaneous pressure $(2K + W)/3V$ of [Section 14.2](#).

EXERCISE 14.9 (*a cubic crystal’s bulk modulus*). For a cubic crystal, each diagonal component $P_{\alpha\alpha}$ of the pressure tensor falls by C_{11} for each unit of its own strain $\epsilon_{\alpha\alpha}$ and by C_{12} for each unit of the strain along either other axis. Show that $B = (C_{11} + 2C_{12})/3$, and evaluate it for the crystal of [Section 14.4](#).

EXERCISE 14.10 (*integrals of the gamma kind*). Show that $\int_0^\infty V^n e^{-bV} dV = n!/b^{n+1}$ for $b > 0$ and every whole number n , by integrating by parts and using the result for $n - 1$.

EXERCISE 14.11 (*a small ideal gas*). For 20 atoms that do not interact, at 135 K and $P_0 = 1 \times 10^{-4} \text{ eV } \text{\AA}^{-3}$, find the mean volume and its spread, and compare the mean with $Nk_B T / P_0$.

EXERCISE 14.12 (*where Berendsen's barostat leaves an ideal gas*). Berendsen's barostat drives the instantaneous pressure, $2K/3V$ for an ideal gas, to P_0 on average. Assuming the volume's spread small and K independent of V , show that the mean volume settles near $Nk_B T / P_0$ for a run whose $\langle 2K \rangle = 3Nk_B T$.

EXERCISE 14.13 (*how fast the volume settles*). For the liquid, with $\kappa_T = 1.46 \text{ GPa}^{-1}$ and $\tau_P = 1 \text{ ps}$, find the time in which the volume relaxes under Berendsen's barostat given $\kappa = 2.5 \text{ GPa}^{-1}$, and given $\kappa = 2.5 \times 10^{-4} \text{ GPa}^{-1}$.

EXERCISE 14.14 (*the spread of the logarithm*). Find the spread of $\ln V$ that stochastic cell rescaling should give the liquid at 135 K, with $\kappa_T = 1.46 \text{ GPa}^{-1}$ and $V = 12306 \text{ \AA}^3$.

EXERCISE 14.15 (*the kinetic energy of a growing box*). With each momentum written as $\mathbf{p}_i = \pi_i V^{-1/3}$, show that $\partial K / \partial V$ at fixed π_i is $-2K/3V$.

EXERCISE 14.16 (*a gas at fixed temperature*). Show that an ideal gas held at a fixed temperature has $\kappa_T = 1/P$.

EXERCISE 14.17 (*Andersen's piston*). A barostat whose coordinate is the volume itself, with the mass M_V , obeys $M_V \ddot{V} = P - P_0$. Show that it swings with $\omega^2 = 1/(M_V V \kappa_T)$ about its mean.

EXERCISE 14.18 (*the piston's period*). From $\omega^2 = 9V/(M_e \kappa_T)$ and $M_e = (3N + 3)k_B T \tau_P^2$, show that the period is $2\pi \tau_P \sqrt{(N + 1)k_B T \kappa_T / 3V}$, and evaluate the factor for the liquid of 256 atoms at 135 K, with $V = 12306 \text{ \AA}^3$ and $\kappa_T = 1.46 \text{ GPa}^{-1}$.

EXERCISE 14.19 (*graphite, isotropically*). If the 10.8% growth of volume in the check of Section 14.8, in which the in-plane size is kept, were shared equally by the three lengths, by how much would each grow?

EXERCISE 14.20 (*the kinetic part dropped*). A barostat that sees only $\mathcal{W}/3V$ holds the liquid at the true pressure $P_0 + Nk_B T/V$. Estimate from κ_T by what fraction its volume shrinks, and compare with the 4.9% of Section 14.9. Why is the estimate larger?

EXERCISE 14.21 (*a referee's objection*). A study reports the compressibility of a liquid from the fluctuations of its volume, Equation (14.4), in a run under Berendsen's barostat. What is wrong, which way is the result wrong, and which barostat would fix it?

Equilibration, convergence and uncertainty

Many results of a simulation are averages over a run of finite length. A second run from another start would give a slightly different value. Chapters 11 to 14 quoted such averages, sometimes with the spread of a few runs, but left open how accurately they were known. An error bar makes that question explicit when we compare a result with experiment, another code or another thermostat. It does not remove a bias: an early interval in which the system is still settling shifts the average, and extending the run only dilutes that shift.

We first find the error of a mean from independent samples, then account for the memory between successive samples of a trajectory. Block averages provide a second estimate and a visible check on whether the record is long enough. These tools let us choose where to begin averaging, test for drift and estimate the time needed for a chosen precision.

Convergence in time is only part of the problem. We also test the size of the periodic box and whether the thermostat samples its stated ensemble. The chapter ends with a protocol for preparing a run and a numerical criterion for each reported quantity. Chapter 16 uses these checks to measure structure and motion.

15.1 The error of an average

A run reports the average of a quantity over its samples, and that average misses the ensemble average by an amount that changes from run to run. This section finds the size of the miss when the samples are independent, on rolls of a die first, and says what an error bar built from it promises.

The mean of independent rolls

A fair die shows each face with probability $\frac{1}{6}$, so one roll x has the mean $\langle x \rangle = 3.5$ and the variance $\sigma_x^2 = 35/12$ (Exercise 12.1). Rolling it n times and averaging gives

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i,$$

the overbar marking an average over samples, as it marks an average over time (Section 12.2). Each roll has the mean $\langle x \rangle$, and the expectation of a sum is the sum of the expectations, so $\langle \bar{x} \rangle = n\langle x \rangle / n = \langle x \rangle$: many such averages centre on the right value. Their spread follows from the rule of Section 12.1 that the variances of independent quantities add, and from one more. The rolls are independent, so the variances of the n of them add, $\text{var}(\sum_i x_i) = n\sigma_x^2$. A constant factor a multiplies a variance by a^2 , since

$$\text{var}(ax) = \langle a^2 x^2 \rangle - \langle ax \rangle^2 = a^2 (\langle x^2 \rangle - \langle x \rangle^2) = a^2 \text{var}(x),$$

and with $a = 1/n$,

$$\text{var}(\bar{x}) = \frac{n\sigma_x^2}{n^2} = \frac{\sigma_x^2}{n}, \quad \sigma_{\bar{x}} = \frac{\sigma_x}{\sqrt{n}}. \quad (15.1)$$

The spread of the mean falls as the square root of the number of samples: four times as many rolls halve it, and a hundred times as many cut it to a tenth. Figure 15.1(b) rolls the die by computer, n rolls at a time, 200 000 times over: the spread of the mean is 0.8558 for 4 rolls against $\sigma_x / \sqrt{4} = 0.8539$, and 0.1067 for 256 rolls against 0.1067.

Figure 15.1(a) shows the shape as well as the spread. One roll has six equally likely values, but the mean of 4 rolls is already close to the Gaussian of Equation (12.1) with mean 3.5 and variance $\sigma_x^2/4$, and the mean of 64 is indistinguishable from its Gaussian. The mean of many independent samples approaches a Gaussian density whatever the density of one sample, provided that density has a finite variance, a result known as the central limit theorem, which the figure illustrates without proving⁸⁰.

The standard error

In a simulation σ_x is not known in advance and must be estimated from the same samples. The spread of the samples about their own mean falls short of σ_x , because $\bar{x}$

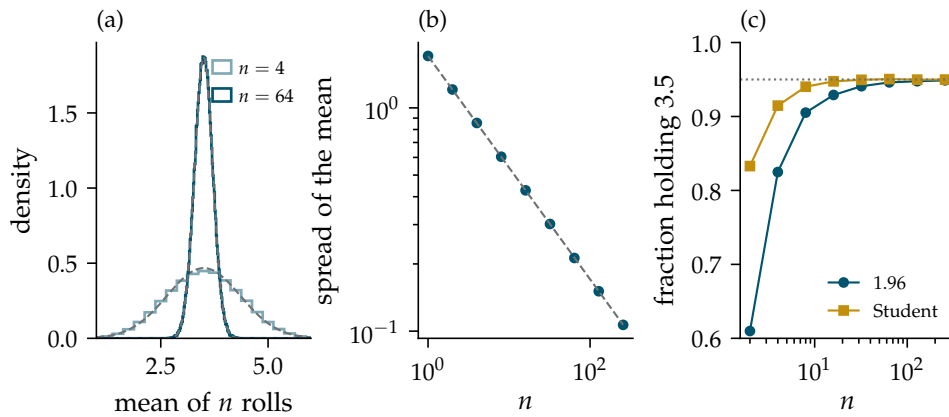


FIGURE 15.1: The error of an average of independent rolls of a die, each set of n rolls repeated 200 000 times. (a) The density of the mean of 4 rolls (light) and of 64 (dark), against the Gaussians of mean 3.5 and variance σ_x^2/n (dashed). (b) The spread of the mean against n , against $\sigma_x/\sqrt{n}$ (dashed). (c) The fraction of the intervals $\bar{x} \pm 1.96 s/\sqrt{n}$ that hold 3.5 (teal), and of those with Student's factor in place of 1.96 (ochre), against 0.95 (dotted).

sits closer to the samples than $\langle x \rangle$ does. Writing each deviation as $x_i - \bar{x} = (x_i - \langle x \rangle) - (\bar{x} - \langle x \rangle)$ and expanding the square,

$$\begin{aligned} \sum_i (x_i - \bar{x})^2 &= \sum_i (x_i - \langle x \rangle)^2 - 2(\bar{x} - \langle x \rangle) \sum_i (x_i - \langle x \rangle) + n(\bar{x} - \langle x \rangle)^2 \\ &= \sum_i (x_i - \langle x \rangle)^2 - n(\bar{x} - \langle x \rangle)^2, \end{aligned}$$

since $\sum_i (x_i - \langle x \rangle) = n(\bar{x} - \langle x \rangle)$. Averaging, the first sum gives $n\sigma_x^2$ and the second term $n \text{var}(\bar{x}) = \sigma_{\bar{x}}^2$, so the sum of squares averages to $(n-1)\sigma_x^2$. Dividing by $n-1$ rather than n therefore gives an estimate whose average is σ_x^2 ,

$$s^2 = \frac{1}{n-1} \sum_i (x_i - \bar{x})^2,$$

and the estimate of $\sigma_{\bar{x}}$ is $s/\sqrt{n}$, the standard error of the mean. The book writes a result as $\bar{x} \pm s/\sqrt{n}$, one standard error, unless a factor is stated.

What an error bar promises

When the mean is Gaussian, it lies within one $\sigma_{\bar{x}}$ of $\langle x \rangle$ in 68.27% of runs and within two in 95.45% (Section 12.1); the factor that encloses 95%, found numerically from $\text{erf}(1.960/\sqrt{2}) = 0.95$, is 1.960. An interval $\bar{x} \pm 1.96 s/\sqrt{n}$ should therefore hold the true mean in 95% of runs, and Figure 15.1(c) counts how often it does: 0.949 of the time for 256 rolls and 0.905 for 8; for 3 rolls, not drawn, it is 0.777. With few samples the estimate s is itself uncertain, and a run whose s happens to be small claims an

interval too narrow. For samples with a Gaussian density the remedy is exact: Student showed that $(\bar{x} - \langle x \rangle)/(s/\sqrt{n})$ then has a known density that depends on n only through $n - 1$, its number of degrees of freedom. The n deviations $x_i - \bar{x}$ from which s is built add to zero, which leaves $n - 1$ of them free, as holding the total momentum at zero left $3N - 3$ velocities free (Section 12.6). The density's 95% factor is 4.303 for three samples, 2.365 for 8 and 1.998 for 64, and approaches 1.960 as n grows⁸¹. Three Gaussian samples with the factor 1.96 hold their mean in 0.811 of trials, as Student's density predicts; with 4.303 in 0.950. The die is far from Gaussian, and Student's factor brings its three rolls only to 0.917. The standard errors of three runs that Chapter 13 quoted are of this kind. For the difference of two means of three runs each, Student's density has, when the two spreads are alike, $2 + 2 = 4$ degrees of freedom and the 95% factor 2.776, so the differences of up to 2.5 combined errors that Section 13.7 found for the Nosé-Hoover chains and CSVr are within what so few runs leave open, and the 2.9 of Berendsen coupling lies only just beyond it. Section 15.6 returns to those numbers with error bars from the runs themselves.

KEY IDEA. The mean of n independent samples has the spread $\sigma_x/\sqrt{n}$, and approaches a Gaussian as n grows whatever the samples' own density, provided its variance is finite. Its standard error is $s/\sqrt{n}$, with s^2 the sum of squared deviations divided by $n - 1$. The interval ± 1.96 standard errors holds the true mean in 95% of runs when n is large; with few samples it holds it less often, and Student's factor replaces 1.96.

NOTEBOOK 15 roll dice and watch the mean narrow and the error bars cover

CHECK YOUR UNDERSTANDING. A run of 100 independent samples gives a standard error of 0.05. How many samples give 0.01?

15.2 Samples that remember

The samples of a run are not independent. In one step of 10 fs an atom moves a small fraction of its distance to a neighbour, so the potential energy, the kinetic energy and the pressure change little from one step to the next, and a sample says much about the one after it. This section measures that memory, finds how it enlarges the error of a mean, and estimates it from a single run.

A series with memory

A grain of dust in still air is pushed about by the molecules that strike it, and its velocity changes over a time set by the friction of the air. Section 13.4 found the exact law of one component of such a velocity over a step: the Ornstein-Uhlenbeck process

of Equation (13.3). Measured in units of its own standard deviation, so that its variance is 1, it reads

$$x_{k+1} = c x_k + \sqrt{1 - c^2} z_k, \quad 0 \leq c < 1, \quad (15.2)$$

with z_k a fresh Gaussian number of mean 0 and variance 1 at each step (the g of Equation (13.3), renamed because g is needed below). The old value keeps the fraction c of itself, and the noise supplies the variance that this loses, $1 - c^2$. With $c = 0$ each sample is new; with c close to 1 the series wanders slowly. Figure 15.2(a) draws 400 samples with $c = 0.95$, read as samples 10 fs apart, against 400 independent ones: the first stays on one side of zero for up to 0.98 ps at a time, while the second changes sign on average every other sample.

The autocorrelation function

The memory is measured by the covariance of two quantities,

$$\text{cov}(x, y) = \langle (x - \langle x \rangle)(y - \langle y \rangle) \rangle = \langle xy \rangle - \langle x \rangle \langle y \rangle,$$

the bracket of Section 12.1 that vanished for independent quantities, which by the same expansion makes $\text{var}(x + y) = \text{var}(x) + \text{var}(y) + 2 \text{cov}(x, y)$. A run that has settled has the same statistics wherever in it they are measured, so the covariance of the samples A_i and A_{i+k} depends on their separation k alone. Divided by the variance, it is the autocorrelation function

$$\text{corr}(k) = \frac{\text{cov}(A_i, A_{i+k})}{\text{var}(A)},$$

equal to 1 at $k = 0$ and falling towards 0 as the series forgets. For the series of Equation (15.2), multiplying both sides by x_0 and averaging gives

$$\langle x_{k+1} x_0 \rangle = c \langle x_k x_0 \rangle + \sqrt{1 - c^2} \langle z_k x_0 \rangle = c \langle x_k x_0 \rangle,$$

since z_k is drawn after x_0 and independently of it, with mean zero. Applying this k times from $\langle x_0 x_0 \rangle = 1$ gives $\langle x_k x_0 \rangle = c^k$, and with the means zero and the variance 1 this is $\text{corr}(k) = c^k$, which falls by the factor c each step. Here δt is the time between samples, one integration step in this chapter's runs, and $c^k = e^{-k\delta t/\tau_c}$ with the correlation time $\tau_c = -\delta t / \ln c$, which is $1/\gamma$ for Langevin's $c = e^{-\gamma\delta t}$. For $c = 0.95$ and $\delta t = 10$ fs, $\tau_c = 195.0$ fs.

The variance of a correlated mean

The mean $\bar{A}$ of n samples has the variance of their sum divided by n^2 . The square of a sum of deviations $\delta A_i = A_i - \langle A \rangle$ is the sum of all their products, $(\sum_i \delta A_i)^2 = \sum_i \sum_j \delta A_i \delta A_j$, so

$$\text{var}(\bar{A}) = \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n \text{cov}(A_i, A_j) = \frac{\sigma_A^2}{n^2} \sum_{i=1}^n \sum_{j=1}^n \text{corr}(|i-j|),$$

the second form using the autocorrelation function at the lag $|i-j|$, since $\text{cov}(A_i, A_j) = \text{cov}(A_j, A_i)$. The pairs with $i=j$ number n , each contributing 1; those k apart, with k from 1 to $n-1$, number $n-k$ with j after i and as many with j before. Collecting them,

$$\text{var}(\bar{A}) = \frac{\sigma_A^2}{n^2} \left[n + 2 \sum_{k=1}^{n-1} (n-k) \text{corr}(k) \right] = \frac{\sigma_A^2}{n} \left[1 + 2 \sum_{k=1}^{n-1} \left(1 - \frac{k}{n}\right) \text{corr}(k) \right]. \quad (15.3)$$

For independent samples the bracket is 1 and this is Equation (15.1). A run is normally far longer than its memory, so k/n is negligible wherever $\text{corr}(k)$ is not, and the bracket settles to the statistical inefficiency⁸²

$$g = 1 + 2 \sum_{k=1}^{\infty} \text{corr}(k), \quad \text{var}(\bar{A}) = \frac{g \sigma_A^2}{n}.$$

Correlated samples spread the mean $\sqrt{g}$ times as widely as the same number of independent ones: n of them are worth $n_{\text{eff}} = n/g$ independent samples. For the series of Equation (15.2) the sum is geometric: the series of Section 13.4 gives $\sum_{k=0}^{m-1} c^k = (1 - c^m)/(1 - c)$, which tends to $1/(1 - c)$ as c^m vanishes, and without its first term, $\sum_{k \geq 1} c^k = c/(1 - c)$, so

$$g = 1 + \frac{2c}{1-c} = \frac{1+c}{1-c},$$

which is 39.0 for $c = 0.95$: a run of 10 000 such samples is worth 256 independent ones. Writing the run's length as $t_f = n\delta t$ and

$$\tau_{\text{int}} = \frac{1}{2}g\delta t, \quad \text{var}(\bar{A}) = \frac{\sigma_A^2}{t_f/(2\tau_{\text{int}})},$$

puts the result in times: a run holds one independent sample for every two integrated correlation times τ_{int} , however often it is sampled, provided the samples lie much closer together than τ_{int} . For the series above $\tau_{\text{int}} = 195.0$ fs, its τ_c to the four figures shown: $\tau_{\text{int}} = (\frac{1}{2} + \sum_{k \geq 1} c^k)\delta t$ is the area under e^{-t/τ_c} counted in trapezia of width δt , whose exact area is τ_c . Figure 15.2(c) measures the spread of the mean, 4000 series over: for $n = 1000$ it is 0.1903, against 0.1956 from Equation (15.3) and $1/\sqrt{n} = 0.0316$ for

independent samples. For runs not many times longer than the memory the bracket of Equation (15.3) has not yet reached g : 8.5 at $n = 10$, 31.4 at $n = 100$.

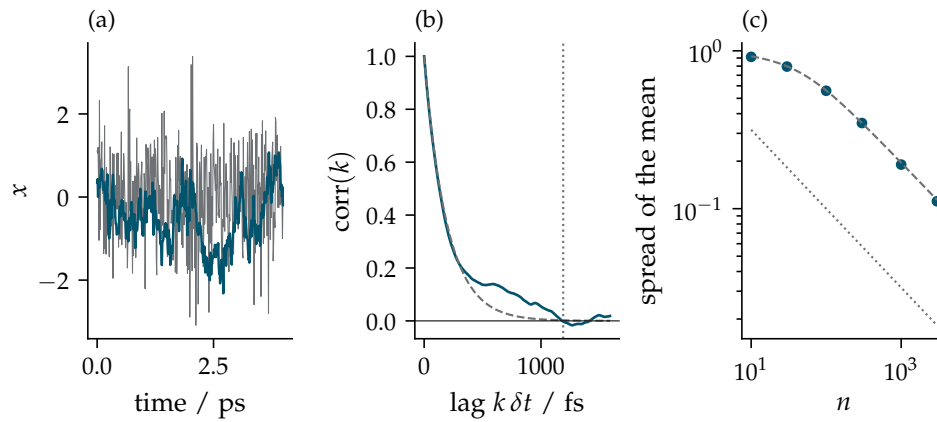


FIGURE 15.2: Samples that remember: the series of Equation (15.2) with $c = 0.95$, samples 10 fs apart. (a) 400 samples (teal) against 400 independent ones (grey). (b) Its autocorrelation function estimated from 10 000 samples, against the exact c^k (dashed); the estimate first reaches zero at the dotted line. (c) The spread of the mean of n samples, 4000 series over, against $1/\sqrt{n}$ for independent samples (dotted) and the square root of Equation (15.3) (dashed).

Estimating g from one run

A run does not know its autocorrelation function and must estimate it. With the deviations now taken from the run's own mean, $\delta A_i = A_i - \bar{A}$ in place of $A_i - \langle A \rangle$, which the run does not know, and $s'^2 = \frac{1}{n} \sum_i \delta A_i^2$, the estimate at the lag k averages the $n - k$ products available,

$$\widehat{\text{corr}}(k) = \frac{1}{(n-k)s'^2} \sum_{i=1}^{n-k} \delta A_i \delta A_{i+k},$$

the hat marking an estimate. Each value carries noise, and beyond the decay the estimates scatter about zero (Figure 15.2(b)): by 0.044 over the lags 300 to 1000 of 10 000 samples of the series above, more than $1/\sqrt{n} = 0.010$ because the products are themselves correlated. Summed over every lag with the weights $1 - k/n$, the estimates give zero in place of g , whatever the series. Multiplying out, $(1 - k/n) \widehat{\text{corr}}(k) = \sum_i \delta A_i \delta A_{i+k} / (ns'^2)$, so

$$1 + 2 \sum_{k=1}^{n-1} \left(1 - \frac{k}{n}\right) \widehat{\text{corr}}(k) = \frac{\sum_i \delta A_i^2 + 2 \sum_{k \geq 1} \sum_i \delta A_i \delta A_{i+k}}{ns'^2} = \frac{(\sum_i \delta A_i)^2}{ns'^2} = 0,$$

the middle numerator being the square of the sum written out, and the deviations from the run's own mean adding to zero. The sum must therefore stop where the signal ends. The rule of Chodera and co-workers⁸² stops it where the estimate first

reaches zero; pymbar, and the listing below, look for that zero only beyond the third lag. For the series of [Figure 15.2\(b\)](#) that is the lag 119, 1190 fs, and g comes out at 48.0 against the exact 39.0; over 200 such series the estimate averages 41.6 and spreads by 8.7. From a run worth a few hundred independent samples g is uncertain by about a fifth, and the error bar, which goes as $\sqrt{g}$, by about a tenth, since a small relative change in g changes $\sqrt{g}$ by half as much. The function `statistical_inefficiency` of `mdlab.analysis.stats` applies the rule, with the autocorrelation function computed by the fast Fourier transform for speed (a tool Chapter 16 explains); in it `c` holds the estimates $\widehat{\text{corr}}(k)$ at the lags k :

```

1 def statistical_inefficiency(series: ArrayLike, mintime: int = 3) -> float:
2     a = np.asarray(series, dtype=float)
3     n = len(a)
4     if np.all(a == a[0]):
5         raise ValueError("a constant series has no fluctuations")
6     c = autocorrelation(a)[1 : n - 1]
7     k = np.arange(1, n - 1)
8     stop = np.nonzero((c <= 0) & (k > mintime))[0]
9     end = stop[0] if len(stop) else len(c)
10    g = 1 + 2 * np.sum(c[:end] * (1 - k[:end] / n))
11    return float(max(g, 1.0))

```

It agrees with the same estimate in the `timeseries` module of the `pymbar` package to nine figures (`test_stats.py`).

The liquid

The liquid of Chapter 13, followed for 2 ns under CSVr with $\tau_T = 1$ ps and sampled every step, shows two kinds of memory ([Figure 15.3\(a\)](#)). The potential energy, the kinetic energy and the pressure lose part of their correlation within 0.1 ps, as the atoms rattle against their neighbours: `corr` is then 0.46, 0.63 and 0.66. The rest fades over picoseconds, and sets g : 146 for U , 211 for K and 188 for P , or τ_{int} of 0.73, 1.05 and 0.94 ps. The slow part comes from the thermostat. At fixed energy, from the same liquid ([Figure 15.3\(b\)](#)), U and K , whose sum cannot change, have the same function and $\tau_{\text{int}} = 77$ fs, and P has 123 fs. Under CSVr the total energy, which only the thermostat changes, has $\tau_{\text{int}} = 1.71$ ps, close to the time $\tau_T C_V / C_K = 1.59$ ps over which [Section 13.2](#) found a weak coupling to move the energy, while the shear component P_{xy} of the pressure tensor, which is not tied to the total energy, has 0.18 ps in both runs. The length of run an average needs therefore depends on the thermostat as well as on the liquid: a gentle coupling lets the energy wander slowly, and every quantity tied to the energy inherits that slowness.

[Figure 15.3\(c\)](#) shows the rule at work: under CSVr the running sum for U climbs until the estimate first reaches zero at 6.16 ps, and at fixed energy it levels off before the sum

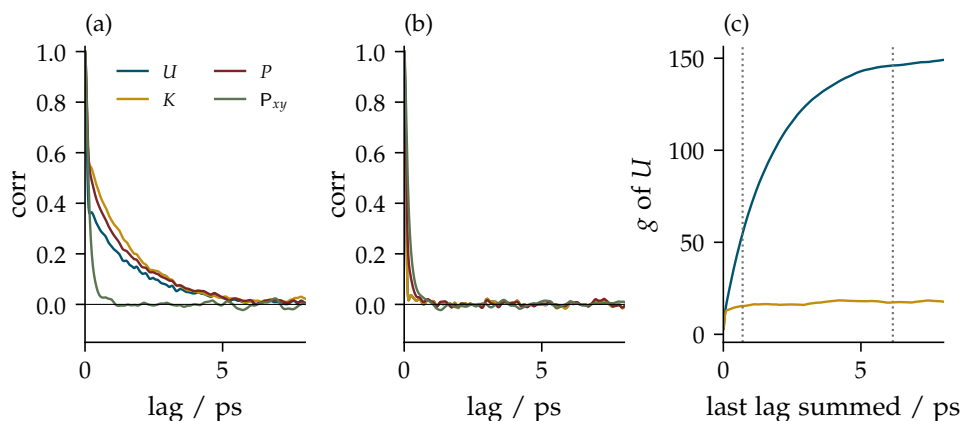


FIGURE 15.3: The memory of liquid argon at 135 K, from runs of 2 ns sampled every 10 fs. (a) Under CSVr with $\tau_T = 1$ ps: the autocorrelation functions of U , K , P and the shear component P_{xy} . (b) The same at fixed energy, where U and K have the same function. (c) The estimate of g for U summed up to each lag, under CSVr (teal) and at fixed energy (ochre), with the lags at which the rule of this section stops each sum (dotted).

stops at 0.70 ps. Over the whole run under CSVr, the potential energy of the 256 atoms is $U = (-12.8236 \pm 0.0047)$ eV, $T = (134.81 \pm 0.22)$ K and $P = (0.0849 \pm 0.0004)$ GPa. The error bars that leave out $\sqrt{g}$, $s/\sqrt{n}$ over the 200 001 samples, would be 12.1, 14.5 and 13.7 times smaller, and would claim a precision that the run does not have.

KEY IDEA. Successive samples of a run are correlated, and n of them spread the mean as widely as n/g independent ones, with the statistical inefficiency $g = 1 + 2 \sum_k \text{corr}(k)$; a run of length t_f holds $t_f / (2\tau_{\text{int}})$ independent samples, $\tau_{\text{int}} = g\delta t/2$. The error of the mean is $\sqrt{g s^2/n}$, with g estimated from the run by summing its autocorrelation function until the estimate first reaches zero.

NOTEBOOK 15 set the memory of a series and compare the naive and the correct error bars

CHECK YOUR UNDERSTANDING. A series has $\text{corr}(k) = 0.5^k$. What is g , and how many independent samples are 1000 of its samples worth?

15.3 Block averages

The estimate of g rests on a rule for where the autocorrelation function ends, and a slow tail that the rule cuts off would leave the error bar too small. Block averaging finds the error of the mean without estimating the autocorrelation function at all, and shows on a plot whether the run was long enough to find it⁸³.

Blocks of samples

We cut the n samples of a run into n_b consecutive blocks of b samples each, $n = n_b b$ (a remainder at the end is dropped), and average each block, giving the block means $\bar{A}_1, \dots, \bar{A}_{n_b}$, whose own mean is $\bar{A}$. Each block mean is the mean of b correlated samples, so by Equation (15.3) with b in place of n it has the variance $g_b \sigma_A^2 / b$, with

$$g_b = 1 + 2 \sum_{k=1}^{b-1} \left(1 - \frac{k}{b}\right) \text{corr}(k).$$

Two neighbouring blocks share only the correlation between the samples near their common edge, so blocks much longer than the run's memory have nearly independent means. The mean of n_b independent block means then has the variance of one divided by n_b (Equation (15.1)), estimated from their own spread: the error of $\bar{A}$ is $s_b / \sqrt{n_b}$, with s_b^2 the sum of the squared deviations of the block means from $\bar{A}$ divided by $n_b - 1$. No autocorrelation function enters.

How long the blocks must be follows from g_b . Splitting the sum,

$$g_b = 1 + 2 \sum_{k=1}^{b-1} \text{corr}(k) - \frac{2}{b} \sum_{k=1}^{b-1} k \text{corr}(k) \approx g - \frac{2}{b} \sum_{k=1}^{\infty} k \text{corr}(k),$$

since for blocks much longer than the memory both sums have already reached their limits. For the exponential $\text{corr}(k) = c^k$ of Equation (15.2), the second sum is $c / (1 - c)^2$ (Exercise 15.6), which is $(g^2 - 1) / 4$, since $g^2 - 1 = [(1 + c)^2 - (1 - c)^2] / (1 - c)^2 = 4c / (1 - c)^2$; for g well above 1 it is close to $g^2 / 4$, so $g_b \approx g - g^2 / (2b)$. The blocks estimate the error $\sqrt{g_b \sigma_A^2 / (n_b b)} = \sqrt{g_b \sigma_A^2 / n}$, against the true $\sqrt{g \sigma_A^2 / n}$; the part missing is the correlation between neighbouring blocks. Their ratio is $\sqrt{1 - x}$ with $x = g / (2b)$, and a small relative change changes a square root by half as much, so blocks of b samples give an error too small by the fraction

$$1 - \sqrt{g_b / g} \approx \frac{g}{4b}. \quad (15.4)$$

For the series with $g = 39$ the exact shortfall is 0.102 for $b = 100$ and 0.051 for $b = 195$, against 0.097 and 0.050 from Equation (15.4). Blocks of $5g$ samples miss about 5% of the error when the memory is exponential, a factor that the rest of the chapter uses. A memory that falls quickly at first and then fades slowly, as the liquid's does (Figure 15.3), has a larger $\sum_k k \text{corr}(k)$ for the same g , 1.7 to 2.0 times $(g/2)^2$ for U , K and P , and blocks of $5g$ samples then miss 0.107, 0.089 and 0.090 of the error, within the error of an estimate from twenty blocks.

The plateau

Flyvbjerg and Petersen double the block length at each stage, from 1 sample to as many as leave a few blocks, and plot the error against the length⁸³. Short blocks give the error of independent samples, too small by $\sqrt{g}$; as the blocks grow past the memory the estimate rises and levels off at $\sqrt{g\sigma_A^2/n}$. The level part is the plateau, and its height is the error of the mean. Its own precision is that of an estimate of a standard deviation from n_b samples: for nearly Gaussian block means the relative error of s_b is $1/\sqrt{2(n_b - 1)}$ (Exercise 15.7), 0.24 for 10 blocks and 0.16 for 20. The longest blocks, few in number, scatter. The function `block_average` of `mdlab.analysis.stats` returns the error and its own error at each length:

```

1 def block_average(series: ArrayLike, min_blocks: int = 4) -> dict:
2     a = np.asarray(series, dtype=float)
3     lengths, errors, error_errors, counts = [], [], [], []
4     b = 1
5     while len(a) // b >= min_blocks:
6         n_b = len(a) // b
7         means = a[: n_b * b].reshape(n_b, b).mean(axis=1)
8         se = means.std(ddof=1) / np.sqrt(n_b)
9         lengths.append(b)
10        errors.append(se)
11        error_errors.append(se / np.sqrt(2 * (n_b - 1)))
12        counts.append(n_b)
13        b *= 2
14    return {"length": np.array(lengths), "error": np.array(errors),
15            "error_error": np.array(error_errors),
16            "blocks": np.array(counts)}

```

Figure 15.4(a) applies it to the series of Section 15.2. With 2^{17} samples, 3361 independent ones, the error rises from 0.0028 for single samples to a plateau of 0.0164 to 0.0168 for blocks of 256 to 1024 samples, against the exact 0.0172. With 2^{10} samples, 26 independent ones, it rises to 0.182 for blocks of 64 and falls back to 0.131 for blocks of 256, of which only four remain, while the exact value is 0.193: there is no plateau, and a run that shows none is too short for its own error bar to be known. The criterion of Section 15.9 asks for at least 100 independent samples, enough for twenty blocks of 5g samples and an error known to about a sixth.

The liquid, and a test against independent runs

For the liquid of Figure 15.3, blocks of 20 ps give 1.03, 1.01 and 1.00 times the errors found from g for U , K and P , and blocks of 41 ps 0.98, 0.99 and 0.97 (Figure 15.4(b)): the rule of Section 15.2 missed no slow tail that blocks known to 7% could show. The longest blocks, 164 ps and only twelve of them, give 1.20 to 1.28, about one of their own errors, a fifth, above 1.

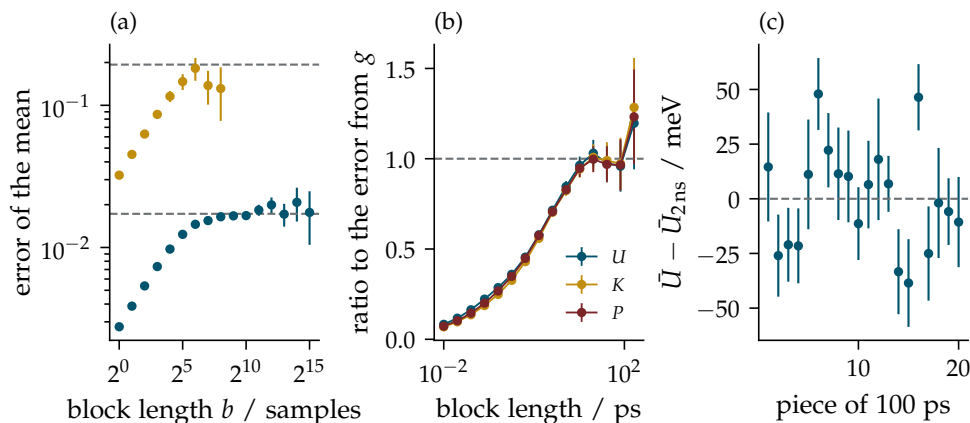


FIGURE 15.4: Block averages. (a) The series of Equation (15.2) with $c = 0.95$: the error of the mean from blocks of b samples, for 2^{17} samples (teal) and 2^{10} (ochre), with the error of each estimate, against the exact errors (dashed). (b) The liquid under CSV, 2 ns: the error of the mean of U , K and P from blocks, divided by $\sqrt{gs^2/n}$. (c) The same run cut into twenty runs of 100 ps: the mean of U in each, less that of the whole run, with its own error bar from that piece alone.

The sharper test is whether an error bar from one run predicts how far other runs fall from it. Cutting the 2 ns run into twenty pieces of 100 ps, whose means are nearly independent (successive means have the correlation 0.06), gives twenty runs to set against one another (Figure 15.4(c)). Their means of U scatter by 0.0241 eV, while the error bar that each piece finds for itself averages 0.0196 eV, from 0.0128 to 0.0278 eV; for K the figures are 0.0351 and 0.0312 eV, and for P 0.00201 and 0.001 72 GPa. Eleven of the twenty bars for U hold the mean of the whole run, where one standard error should hold it in 68% of runs, 13.7 of twenty on average, a count that itself spreads by 2.1, since each piece holds the mean or not, a yes or no of variance 0.68×0.32 , and the twenty variances add. A piece of 100 ps holds 68 independent samples of U , and the g it finds for itself ranges from 64 to 228, on either side of the 146 of the whole run: an error bar from so short a run is uncertain by about a fifth (the twenty bars for U spread by 0.20 of their mean), and on average a little low.

KEY IDEA. Block means of blocks much longer than the run's memory are nearly independent, and their spread gives the error of the mean with no autocorrelation function. Plotted against the block length, the error rises to a plateau; blocks of 5g samples miss about 5% of it for an exponential memory, and the plateau's own relative error is $1/\sqrt{2(n_b - 1)}$. A run with no plateau is too short to know its error bar.

NOTEBOOK 15 cut a series into blocks and find the plateau

CHECK YOUR UNDERSTANDING. A run of 20 000 samples has $g = 100$ and an exponential memory. How long must its blocks be to miss about 5% of the error, and how many such blocks does the run hold?

15.4 Where a run begins

A run starts from positions and velocities that the ensemble would rarely produce: a lattice for a liquid, a box of the wrong volume, a structure just built. Its early part relaxes towards the ensemble, and an average that includes it is shifted however long the run. This section finds how large that shift is, and how to choose the start of the part to keep.

A cup of coffee

Hot coffee left on a table cools at a rate proportional to how much warmer than the room it is: $dT/dt = -(T - T_{\text{room}})/\tau_r$, with τ_r set by the cup and the air. This is the decay equation of [Section 2.7](#) for $T - T_{\text{room}}$, with $1/\tau_r$ in place of γ , so

$$T(t) = T_{\text{room}} + (T(0) - T_{\text{room}})e^{-t/\tau_r}.$$

A thermometer in the cup, read from the moment the coffee is poured and averaged, gives a temperature above the room's. A run behaves the same way: a quantity A approaches its ensemble average with a decaying offset $\delta A_0 e^{-t/\tau_r}$, which is δA_0 at the start, and on which the run's own fluctuations ride. Averaged over a run of length t_f from its start, the offset contributes

$$\frac{1}{t_f} \int_0^{t_f} \delta A_0 e^{-t/\tau_r} dt = \frac{\delta A_0 \tau_r}{t_f} \left[-e^{-t/\tau_r} \right]_0^{t_f} = \frac{\delta A_0 \tau_r}{t_f} (1 - e^{-t_f/\tau_r}) \approx \frac{\delta A_0 \tau_r}{t_f}$$

for a run much longer than τ_r . The shift falls as $1/t_f$ and the error bar as $1/\sqrt{t_f}$, so a long enough run would hide the shift inside its error bar, but only slowly. Leaving out the first t_0 of the run cuts the offset's share to $\delta A_0 \tau_r e^{-t_0/\tau_r} / (t_f - t_0)$ ([Exercise 15.9](#)), at the cost of the samples discarded.

The start that leaves the most

Discarding too little leaves a shift; discarding too much throws away samples. Chodera found a way to weigh the two from the run alone⁸⁴. To the autocorrelation function a decaying offset looks like a very slow correlation, and it inflates g ; so for each candidate start t_0 , the effective number of samples left after it,

$$n_{\text{eff}}(t_0) = \frac{n - t_0/\delta t}{g(t_0)},$$

with $t_0/\delta t$ the number of samples discarded and $g(t_0)$ the statistical inefficiency of the samples from t_0 on, rises while the discard removes more of the offset's inflation of g than it costs in samples, and falls once only samples are being removed. The start kept is the one that maximises it:

```

1 def detect_equilibration(
2     series: ArrayLike, nskip: int = 1
3 ) -> tuple[int, float, float]:
4     a = np.asarray(series, dtype=float)
5     n = len(a)
6     best = (0, 1.0, 0.0)
7     for t0 in range(0, n - 1, nskip):
8         rest = a[t0:]
9         if np.all(rest == rest[0]):
10             g = float(n - t0)
11         else:
12             g = statistical_inefficiency(rest)
13         if (n - t0) / g > best[2]:
14             best = (t0, g, (n - t0) / g)
15     return best

```

The listing counts t_0 in samples. It finds the same start as the function of the same name in the `timeseries` module of `pymbar` with `fast=False` (`test_stats.py`); `pymbar`'s default estimates g more roughly and can place the start a few samples later, 550 against 534 for the series below.

Figure 15.5(a) adds the offset $3e^{-t/2\text{ps}}$, three standard deviations at the start, to 50 ps of the series of Section 15.2, and (b) shows $n_{\text{eff}}(t_0)$ rising to 158 at 5.34 ps, 2.7 decay times, and falling after. From the start the mean is 0.160 ± 0.164 , of which the offset contributes $\delta A_0 \tau_r / t_f = 0.120$; after the discard it is 0.001 ± 0.076 , the offset contributing 0.0093, and the error bar has halved. One series is one draw: over 200 such series the start falls between 0.1 and 16.5 decay times, with the median, the value with half the series on either side, at 1.4, and the offset left in the mean is at most 1.35 error bars, with the median 0.34. One error bar then holds the true mean in 0.58 of the series, where it should hold it in 0.68; without the discard, in 0.54. The method trades a shift of about a third of an error bar for an error bar half as large, and leaves part of a slow offset in place, which the test for drift of Section 15.5 looks for.

A liquid melting

Figure 15.5(c) starts eight runs of the liquid from the face-centred cubic lattice at the liquid's density, with velocities drawn at 135 K from eight seeds, under CSVr for 100 ps. The lattice has $U/N = -64.331$ meV; the energy climbs from it as the lattice melts, and the mean of the eight runs comes within 1 meV of the 2 ns run's (-50.092 ± 0.018) meV after 2.89 ps. In four of the runs the start found from U is 2.8 to 3.2 ps, at the end of

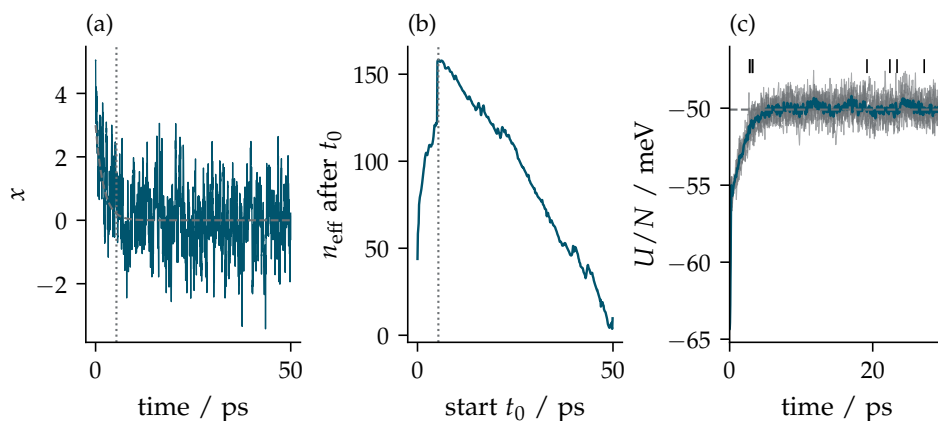


FIGURE 15.5: Where a run begins. (a) 50 ps of the series of Equation (15.2) with $c = 0.95$, with the offset $3e^{-t/2\text{ps}}$ added, the offset itself dashed, and the start found (dotted). (b) The effective number of samples after each candidate start, maximised at the dotted line. (c) The first 30 ps of eight runs of the liquid melting from the face-centred cubic lattice at 135 K: the potential energy per atom (grey), their mean (teal) and the mean of the 2 ns run (dashed), with the start found in each run from its own U (ticks).

the melting. In the other four it is 19 to 27 ps: under CSVr the energy wanders over picoseconds (Section 15.2), and a run of 100 ps cannot always tell such a wander from the tail of a transient. Discarding more than needed costs samples and, on average, does not shift the mean. Averaged from their first steps, the eight means of U/N lie at -50.169 on average, the melting's share pulling them down; after the discards they average -50.063 meV, close to the 2 ns value, and scatter by 0.111 meV against error bars of 0.046 to 0.085 meV, 0.067 on average. The error bars are low by about two fifths, though a spread from eight runs is itself uncertain by a quarter (Exercise 15.7). A run of 100 ps holds about 68 independent samples of U (Section 15.3), and the start chosen makes $(n - t_0/\delta t)/g(t_0)$ largest, which favours a start after which g happens to come out small: the eight runs find g of 45 to 138 after their starts, against the 146 of the 2 ns run, and their error bars are small with it.

The same holds for any quantity with few independent samples. Chapter 14's three runs of 300 ps at constant pressure under stochastic cell rescaling hold 35, 73 and 109 independent samples of the volume. In the second and third the start found is 1.0 and 0.0 ps, well within the first 10 ps that Section 14.9 left out of its averages. In the first, $n_{\text{eff}}(t_0)$ falls from 35 at the start, as a settled run's should, to 24 at 100 ps and 9 at 195 ps; beyond, the g estimated from the last few tens of picoseconds is so uncertain that it can come out far too small, and the largest value, 55, falls at 259 ps, a start with no meaning. The criterion of Section 15.9 therefore asks that the start lie in the first half of the run. Chapter 11 prepared its water in rounds of 1 ps and left the end of that slow change unmeasured (Section 11.3); its healthy run of 10 ps at fixed energy has its start at 0.30 ps, after which the box's kinetic energy is $K = (4.993 \pm 0.022)$ eV and the potential energy (-0.40265 ± 0.00034) eV per molecule, against 4.990 and -0.40259 over the whole run:

the preparation had all but finished, and the averages that [Section 11.8](#) quoted stand within their error bars.

KEY IDEA. A run's early part relaxes from its start, and shifts the mean of the whole run by about $\delta A_0 \tau_r / t_f$. The start to keep is the one that leaves the largest effective number of samples, $(n - t_0 / \delta t) / g(t_0)$. The method trades a shift of a fraction of an error bar for precision and needs a run with many independent samples: when the start it finds falls in the second half of the run, the run is too short to locate it.

NOTEBOOK 15 add a transient of your choosing and watch where the start is found

CHECK YOUR UNDERSTANDING. A run of 100 ps has an offset of 2 K in its temperature at the start, decaying with the time $\tau_r = 1$ ps. By how much does it shift the mean temperature of the whole run?

15.5 A slow drift

A run can settle and still not hold still. An integration step that is too long lets the total energy creep, a thermostat set to the wrong target moves the temperature slowly, and a transient that discarding the start found in [Section 15.4](#) did not remove leaves a slow slope. Each is a trend under the fluctuations, and this section fits a straight line to a run to find it, with an error that accounts for the run's memory.

A tyre that may be leaking

A bicycle tyre is checked with a gauge every morning for a month. The gauge scatters by 0.05 bar from reading to reading, and a slow leak of 0.01 bar a day hides under that scatter from one day to the next, but not over thirty days. The straight line that passes closest to the thirty readings finds it, and the toolbox gives that line and the error of its slope.

A STRAIGHT LINE THROUGH SCATTERED POINTS. Readings A_i at the times t_i , $i = 1, \dots, n$, are to be described by $A \approx a + \dot{A} (t - \bar{t})$, with $\bar{t}$ the mean time. The line of least squares chooses a and the slope $\dot{A}$ to make the sum of squared misses $S_m = \sum_i [A_i - a - \dot{A}(t_i - \bar{t})]^2$ as small as it can be, where its rates of change with a and with $\dot{A}$ both vanish. With respect to a ,

$$\frac{\partial S_m}{\partial a} = -2 \sum_i [A_i - a - \dot{A}(t_i - \bar{t})] = -2(n\bar{A} - na - \dot{A} \cdot 0) = 0,$$

since the deviations $t_i - \bar{t}$ add to zero, so $a = \bar{A}$. With respect to $\dot{A}$, multiplying each miss by $t_i - \bar{t}$,

$$\frac{\partial S_m}{\partial \dot{A}} = -2 \sum_i (t_i - \bar{t}) [A_i - \bar{A} - \dot{A}(t_i - \bar{t})] = 0, \quad \dot{A} = \frac{\sum_i (t_i - \bar{t})(A_i - \bar{A})}{S_{tt}},$$

with $S_{tt} = \sum_i (t_i - \bar{t})^2$. The slope is a weighted sum of the readings, $\dot{A} = \sum_i w_i A_i$ with $w_i = (t_i - \bar{t})/S_{tt}$, since $\sum_i w_i \bar{A} = 0$. If the misses about the true line are independent with variance σ_m^2 , the variances of the terms add, and a constant factor w_i multiplies a variance by w_i^2 (Section 15.1), so

$$\text{var}(\dot{A}) = \sigma_m^2 \sum_i w_i^2 = \frac{\sigma_m^2}{S_{tt}^2} \sum_i (t_i - \bar{t})^2 = \frac{\sigma_m^2}{S_{tt}}.$$

σ_m^2 is estimated from the misses about the fitted line, their sum of squares divided by $n - 2$, since two quantities have been fitted to the same readings; for the thousands of samples of a run the difference from n is negligible.

For thirty daily readings $S_{tt} = 2247.5 \text{ d}^2$, so the slope is known to $0.05/\sqrt{2247.5} = 0.00105 \text{ bar a day}$. One month of readings drawn by computer with the leak gives $\dot{A} = (-0.0109 \pm 0.0012) \text{ bar a day}$, nine errors from zero (the misses of this month scatter by 0.057 bar , hence 0.0012 rather than 0.00105): the leak is certain, though no single day shows it.

The slope of a correlated series

The samples of a run are correlated, and so are their misses about the line. The variance of the weighted sum is then, as in Section 15.2, a double sum over pairs,

$$\text{var}(\dot{A}) = \sigma_m^2 \sum_i \sum_j w_i w_j \text{corr}(|i - j|) \approx \sigma_m^2 \sum_i w_i^2 \left[1 + 2 \sum_{k \geq 1} \text{corr}(k) \right] = \frac{g \sigma_m^2}{S_{tt}},$$

the approximation taking $w_j = w_i$ wherever corr is not negligible, which holds when the run is much longer than its memory: across a memory of about g samples a weight changes by $g\delta t/S_{tt}$, a fraction $2g/n$ of the largest weight, $(t_f/2)/S_{tt}$. The slope's error carries the same factor $\sqrt{g}$ as the mean's, with g found from the misses. For n samples evenly spaced by δt , $S_{tt} = n(n^2 - 1)\delta t^2/12$ (Exercise 15.10), close to $nt_f^2/12$ for a run of length $t_f = n\delta t$, so with $\tau_{\text{int}} = g\delta t/2$,

$$\text{var}(\dot{A}) \approx \frac{12 g \sigma_m^2 \delta t}{t_f^3} = \frac{24 \sigma_m^2 \tau_{\text{int}}}{t_f^3}.$$

the error of a slope falls as $t_f^{-3/2}$, and a run four times as long can detect a slope eight times smaller. The ratio of the slope to its error then tests for a drift: for a series with none it is Gaussian with unit spread, and lies beyond ± 2 in 4.6% of runs (Section 12.1). Applied to 1000 series of 5000 samples of the kind in Figure 15.2, with $g = 39$ and no drift, the test `drift_test` of `mdlab.analysis.stats` finds a drift in 0.051 of them. A slope beyond two errors is evidence of drift worth investigating; within them, the run shows none that it can detect.

The time step

Chapter 9 judged a time step by the spread of the total energy at fixed energy, which should fall as δt^2 and stay small beside the spread of the kinetic energy, and by the absence of a drift (Section 9.6). The drift test makes the last of these quantitative. Figure 15.6 follows the liquid at fixed energy for 1 ns at each of six steps. The spread of the total energy over that of the kinetic energy is 3.93 times larger at 10 than at 5 fs and 4.12 times larger at 20 than at 10 fs, the factor 4 of δt^2 , but 3.93 times larger at 30 than at 20 fs, where δt^2 gives 2.25, and 6.18 times larger at 35 than at 30 fs, where it gives 1.36: beyond 20 fs the step is leaving the range in which its error is a small, smooth correction. The drift per atom per nanosecond is 0.00 ± 0.01 and -0.04 ± 0.05 μeV at 5 and 10 fs; -0.66 ± 0.17 μeV at 20 fs; -54 ± 29 μeV at 30 fs; $+642 \pm 69$ μeV at 35 fs; and at 40 fs the energy climbs by 11 meV per atom per nanosecond over the 363 ps it takes to gain 1 eV in all, and the run blows up at 557 ps.

Whether a drift matters depends on what the run measures. At fixed energy a drift $\dot{E}$ moves the temperature at the rate $\dot{E}/C_V$, and over a run of length t_f moves its mean by half of $\dot{E}t_f/C_V$. Asking that this be smaller than the mean's own error σ_T ,

$$\frac{|\dot{E}|}{N} < \frac{2\sigma_T}{t_f} \frac{C_V}{N}.$$

For this liquid $C_V/N = 2.347k_B$, the mean of the five single-run estimates of Section 12.7 (the slope of E against T gives 2.376, within the 3% errors of those runs, Section 15.6), and $\sigma_T = 0.0513$ K over 1 ns at fixed energy, so the limit for a run of 1 ns is 20.8 μeV per atom per nanosecond (dotted in Figure 15.6(b)). The drift at 20 fs is significant, 3.8 errors from zero, and harmless: it moves the temperature by 0.0033 K in a nanosecond. The drift at 30 fs lies 1.8 errors from zero and 1.1 above the limit, so the run can show it neither significant nor harmful; at 35 fs it moves the temperature by 3.2 K per nanosecond. Significance grows with the length of a run, which can detect ever smaller drifts; importance is set by the quantity measured. For this liquid 20 fs passes Chapter 9's tests of the spread, and its drift, though significant, is harmless for a run of 1 ns and at the limit for one of 10 ns (Exercise 15.11); 10 fs, the step used throughout, passes them with a wide margin and shows no drift the test can detect.

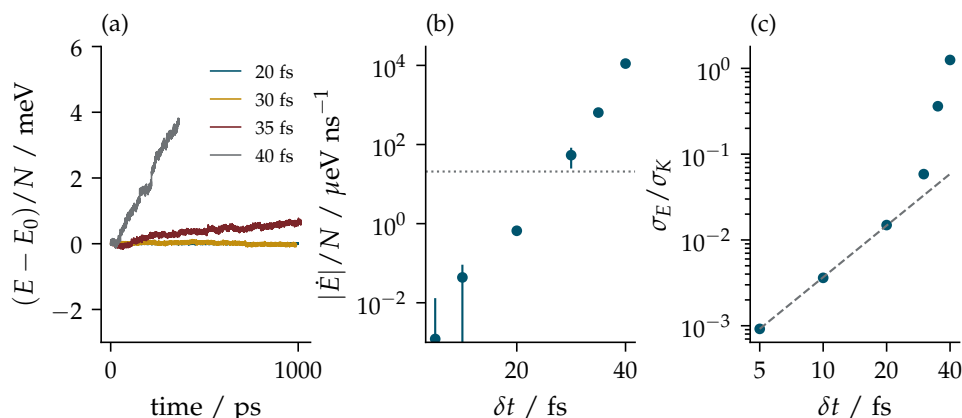


FIGURE 15.6: A slow drift: the liquid at fixed energy for 1 ns at each step. (a) The total energy per atom less its start, at 20 (teal), 30 (ochre), 35 (oxblood) and 40 fs (grey, until it passes 1 eV in all). (b) The size of the drift per atom per nanosecond, with its error, against the step, and the largest drift that moves the mean temperature of a 1 ns run by less than its error (dotted). (c) The spread of the total energy over that of the kinetic energy, against δt^2 through the point at 5 fs (dashed).

KEY IDEA. A drift is the slope of the line of least squares through a run, with the error $\sqrt{g\sigma_m^2/S_{tt}}$ that the run's memory requires; a slope beyond two errors is evidence of drift under this test. Whether it matters depends on what is measured: at fixed energy it is harmless while it moves the mean temperature by less than its error. A step is acceptable while the energy's spread grows as δt^2 and its drift is harmless.

NOTEBOOK 15 *hide a leak under noise and test for it*

CHECK YOUR UNDERSTANDING. A run of 500 ps gives a drift of (3 ± 2) in some unit. Is it a drift, and what would a run of 2 ns with the same true slope be expected to show?

15.6 How long for each quantity

Each quantity of a run has its own spread and its own memory, so each needs its own length of run for a given precision. This section measures both for the liquid, turns them into run lengths, and settles with error bars the questions that Chapters 12 to 14 left open.

The length of run for a precision

With $n_{\text{eff}} = t_f / (2\tau_{\text{int}})$ independent samples in a run of length t_f (Section 15.2), the error of the mean is $\sigma_A \sqrt{2\tau_{\text{int}}/t_f}$. Asking for an error no larger than a target σ^* and solving for t_f ,

$$t_f \geq \frac{2\tau_{\text{int}} \sigma_A^2}{\sigma^{*2}} : \quad (15.5)$$

twice the correlation time for each of the $(\sigma_A/\sigma^*)^2$ independent samples needed. Halving the error costs four times the run. Figure 15.7(a) checks the $t_f^{-1/2}$ on the 2 ns run cut into pieces of 50 to 2000 ps: for U the means of the pieces spread by 6.77, 5.14 and 3.36 times the error of the whole run for pieces of 50, 100 and 200 ps, against $\sqrt{40} = 6.32$, $\sqrt{20} = 4.47$ and $\sqrt{10} = 3.16$. The error bar that each piece finds for itself is lower, 5.38 and 4.19 for the two shortest, as Section 15.3 found, and meets the line from 200 ps on.

Table 15.1 gives the spread, the correlation time and the run needed for a stated precision, from Equation (15.5), for the quantities of the liquid. Two things stand out. The pressure has a spread of 0.0133 GPa per sample about a mean of 0.085 GPa, a sixth of its value, so with a memory like that of U and T its mean is slow to fix relative to its size: 1 ns fixes it to 0.7%, against 0.05% for U . And the runs at fixed energy need a 15th to a 38th of the time needed under CSVr for the same precision of U , T and P , because their correlation times are 8 to 14 times shorter (Section 15.2) and their spreads smaller. A run at fixed energy samples the microcanonical ensemble, at the temperature its energy fixes: here 133.70 K, 1.1 K below the CSVr run, because the energy it was given is the mean of 40 ps of CSVr, which, with the 1.71 ps memory of the energy, is uncertain by about 1.6 K of temperature. Section 13.7 recommended preparing a run under a thermostat and gathering its dynamical quantities at fixed energy; for averages too, fixed energy reaches a precision sooner, at the temperature its energy sets.

TABLE 15.1: How long each quantity needs, for liquid argon at 135 K: the spread of one sample, the integrated correlation time under CSVr with $\tau_T = 1$ ps and at fixed energy, the error of the mean after 1 ns under CSVr, and the run needed for the target error σ^* , from Equation (15.5). The volume is from Chapter 14's runs under stochastic cell rescaling at 0.1 GPa; D from blocks of 100 ps.

	spread	$\tau_{\text{int}} / \text{ps}$		error of	target	run / ns	
	CSVr / fixed E	CSVr	fixed E	1 ns	σ^*	CSVr	fixed E
U/N	0.677 meV / 0.532	0.73	0.077	0.026 meV	0.01 meV	6.7	0.44
T	6.93 K / 4.13	1.05	0.077	0.32 K	0.1 K	10	0.26
P	0.0133 GPa / 0.0096	0.94	0.123	0.000 58 GPa	0.001 GPa	0.33	0.023
P_{xy}	0.0121 GPa	0.18	0.18				
V	$185 \text{ \AA}^3$	2.5		$13 \text{ \AA}^3$	$10 \text{ \AA}^3$	1.7	
D				1.0%	1%	1.0	

Diffusion under a thermostat

Chapter 13 found the diffusion coefficient under CSV, Nosé-Hoover chains and Berendsen coupling 1.01 to 1.07 times its value at fixed energy, in runs 0.4 to 1.8 K warmer, and left open whether the difference was real (Section 13.7). Cutting each run of this chapter into blocks of 100 ps, long beside the 20 ps over which D is fitted, gives independent estimates whose spread gives the error. The 2 ns run under CSV has $D = (3.877 \pm 0.028) \times 10^{-4} \text{ Å}^2 \text{ fs}^{-1}$ at $(134.81 \pm 0.22) \text{ K}$; the 2 ns run at fixed energy has $(3.834 \pm 0.034) \times 10^{-4}$ at 133.70 K. The two are at different temperatures, and runs under CSV at 130 and 140 K give the slope $dD/dT = (0.0294 \pm 0.0071) \times 10^{-4} \text{ Å}^2 \text{ fs}^{-1} \text{ K}^{-1}$ (Figure 15.7(b)). Moved along it to 134.81 K, the run at fixed energy gives $(3.867 \pm 0.035) \times 10^{-4}$, and the ratio of the two is 1.002 ± 0.011 . Berendsen coupling with $\tau_T = 0.1 \text{ ps}$ at 130 and 140 K lies 1.3 and 0.3 errors from the line through the CSV runs. To about 2%, CSV with $\tau_T = 1 \text{ ps}$ and Berendsen coupling with $\tau_T = 0.1 \text{ ps}$ leave diffusion alone. Chapter 13 found 1.05 ± 0.01 and 1.04 ± 0.01 for these two from three runs of 100 ps each. A block of 100 ps here fixes D to 3.2%, so a mean of three such runs is uncertain by $3.2/\sqrt{3} = 1.8\%$ and a ratio of two such means by $1.8 \times \sqrt{2} = 2.6\%$: the two values lie within about two such errors of 1 once their warmth is removed, and the ± 0.01 came from the spread of only three runs, itself uncertain by half (Exercise 15.7). The other couplings of Chapter 13 were not run again.

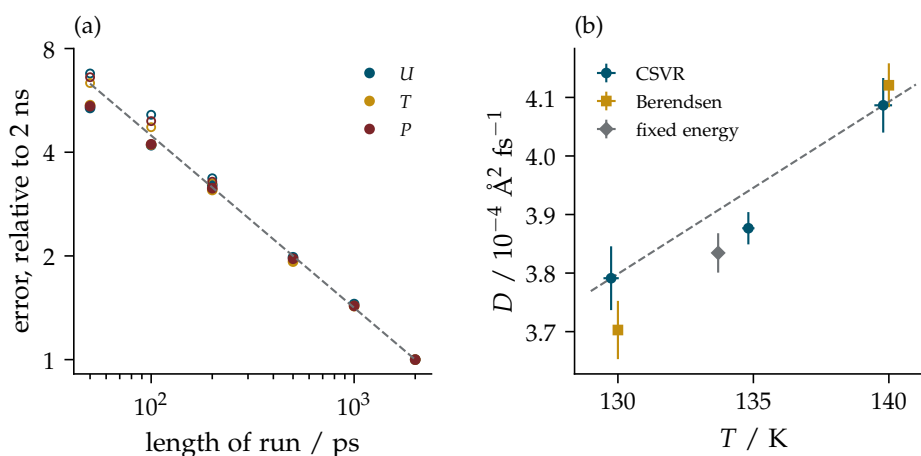


FIGURE 15.7: (a) The error of the mean of U , T and P against the length of run: the 2 ns CSV run cut into pieces, the error bar each piece finds for itself averaged over the pieces (filled) and the spread of the pieces' means (open), each divided by the error of the whole run, against the inverse square root of the length (dashed). (b) The diffusion coefficient against the temperature, with errors from blocks of 100 ps: CSV (teal), Berendsen coupling with $\tau_T = 0.1 \text{ ps}$ (ochre) and fixed energy (grey), with the line through the CSV runs at 130 and 140 K (dashed).

Questions left open

Chapter 12 found the light atoms of a mixture at 126 K and the heavy at 124 K over the last 10 ps of a run, and asked whether the gap was real (Section 12.6). Taking the difference of the two temperatures at each recorded frame as the series, its mean is (1.30 ± 2.14) K, with $g = 31.4$: 0.6 errors from zero, and equipartition between the species holds within what the run can tell. Chapter 12 also found single runs of 50 ps at fixed energy to give heat capacities scattered by up to 7.5% about the slope of E against T , $2.376 Nk_B$ (Section 12.7); the five runs of 256 atoms lie within 5.3% of it, and the 7.5% is a run of 864 atoms. A heat capacity comes from a variance, whose error the rule for a mean does not give; it is found by resampling, which Section 15.8 sets out. It gives the five runs of 256 atoms errors of 2.9 to 3.9%, $(2.292 \pm 0.067) Nk_B$ near 130 K among them, and all five lie within 1.7 of their errors of the slope: their scatter was the noise of the runs. Chapter 14 quoted the pressure of the liquid as 0.0877 GPa (Section 14.3), from 50 frames 2 ps apart, farther apart than $2\tau_{\text{int}} = 1.9$ ps, so nearly independent: its error is $0.0133/\sqrt{50} = 0.0019$ GPa. The 2 ns run gives (0.0849 ± 0.0004) GPa, 1.4 combined errors away.

KEY IDEA. A run of length t_f fixes a mean to $\sigma_A \sqrt{2\tau_{\text{int}}/t_f}$, so a target error σ^* needs $t_f \geq 2\tau_{\text{int}}\sigma_A^2/\sigma^{*2}$. Each quantity has its own σ_A and τ_{int} , and a thermostat lengthens the memory of everything tied to the energy. With error bars, the diffusion coefficient is the same under CSVr with $\tau_T = 1$ ps, Berendsen coupling with $\tau_T = 0.1$ ps and fixed energy to about 2%.

NOTEBOOK 15 find the run each quantity of the liquid needs

CHECK YOUR UNDERSTANDING. A quantity has $\sigma_A = 2$ and $\tau_{\text{int}} = 5$ ps. How long a run fixes its mean to ± 0.1 ?

15.7 Results that depend on the size of the box

An error bar measures how far a run's average scatters about the average that its own system would give in an endless run. It cannot measure a bias that every run of that system shares, and a common such bias is the size of the periodic box. This section measures the diffusion coefficient of the liquid in boxes of 256 to 2048 atoms, finds it changes by more than any error bar, and accounts for the change.

A ball in a narrow tube

A ball sinking through honey falls more slowly in a narrow tube than in a wide pot. The honey it pushes down has to flow back up past it, through the gap between the

ball and the wall, and the narrower the gap the faster that return flow and the stronger its drag. A periodic box has no walls, but it has the same return flow. An atom moving through the liquid drags its surroundings with it, and in a periodic box every image of the atom drags the same flow in every image of the box; since the liquid as a whole does not move, the fluid must flow back between them, against the atom, and the more so the closer the images are.

How strongly a fluid resists such flows is its shear viscosity η_s : a layer of fluid of thickness d between two plates, one moving past the other at the speed u , needs the force $\eta_s u/d$ per unit area of plate to keep it moving, so η_s is in Pa s. Honey has a large η_s , water a small one. The only combination of $k_B T$, η_s and the side L of the box that has the units of a diffusion coefficient is $k_B T/(\eta_s L)$ (Exercise 15.16), so the return flow should lower D by a number times $k_B T/(\eta_s L)$. Summing the flows of all the images of a cubic box gives the number, and Yeh and Hummer showed that molecular dynamics follows

$$D(L) = D_\infty - \frac{2.837297 k_B T}{6\pi\eta_s L}, \quad (15.6)$$

with D_∞ the coefficient of an endless liquid^{85,86}. The correction falls only as $1/L$, that is as $N^{-1/3}$ at a fixed density: eight times the atoms halve it.

Five boxes

The liquid at its density of $0.8\sigma^{-3}$ and 135 K was prepared as in Chapter 12 with 256, 500, 864, 1372 and 2048 atoms, boxes of side 23.26 to 46.51 Å, and followed under CSVr for 1 ns from each of two seeds. D comes from twenty blocks of 100 ps at each size, each moved to 135 K along the slope dD/dT of Section 15.6, a correction of at most $0.009 \times 10^{-4} \text{ Å}^2 \text{ fs}^{-1}$. In units of $10^{-4} \text{ Å}^2 \text{ fs}^{-1}$ it rises with the box: 3.914 ± 0.032 , 4.062 ± 0.017 , 4.166 ± 0.022 , 4.200 ± 0.015 and 4.214 ± 0.011 (Figure 15.8(a)). The change from the smallest box to the largest is nine times the larger of their two errors. A straight line in $1/L$, fitted as in Section 15.5 but with each squared miss divided by the square of that box's error, so that the precise boxes count for more, has $D_\infty = 4.503 \pm 0.032$ in the same units and the slope -12.88 ± 1.19 in $10^{-4} \text{ Å}^3 \text{ fs}^{-1}$. Their errors follow as in the toolbox of Section 15.5: with the weights $1/\sigma_i^2$, σ_i the error of box i , the slope is a weighted sum of the D of the boxes whose variance, each term's weight squared times σ_i^2 , adds up to one over $\sum_i (x_i - \bar{x})^2 / \sigma_i^2$, with $x_i = 1/L_i$ and $\bar{x}$ their weighted mean. The sum of the squared misses in units of the errors is 5.3; each such term averages 1 with the spread $\sqrt{2}$ (Exercise 15.7), and fitting two numbers to five boxes leaves three, so scatter alone gives 3 ± 2.4 . The box of 256 atoms used through Chapters 12 to 14 gives a D 13.1% below that of the endless liquid, and even 2048 atoms 6.4% below. Leaving out the smallest box moves D_∞ to 4.470, by about its error.

Through Equation (15.6), the slope gives the viscosity

$$\eta_s = \frac{2.837297 k_B T}{6\pi |\text{slope}|} = (2.18 \pm 0.20) \times 10^{-4} \text{ Pa s},$$

or 2.40 ± 0.22 in the reduced unit $\sqrt{m\epsilon}/\sigma^2 = 9.07 \times 10^{-5} \text{ Pa s}$ of the Lennard-Jones model. Chapter 16 measures η_s independently, from the fluctuations of the shear stress, which tests whether the whole of the change with size is the return flow of Equation (15.6).

The static averages change far less. The potential energy per atom and the pressure of each box lie within 2.1 combined errors of those for 2048 atoms (Figure 15.8(b)). The largest difference, the pressure of 256 atoms, is 0.0009 GPa, 1% of the pressure, where D differs by 7% between the same two boxes: the small box serves for this liquid's energy and pressure, and not for its diffusion.

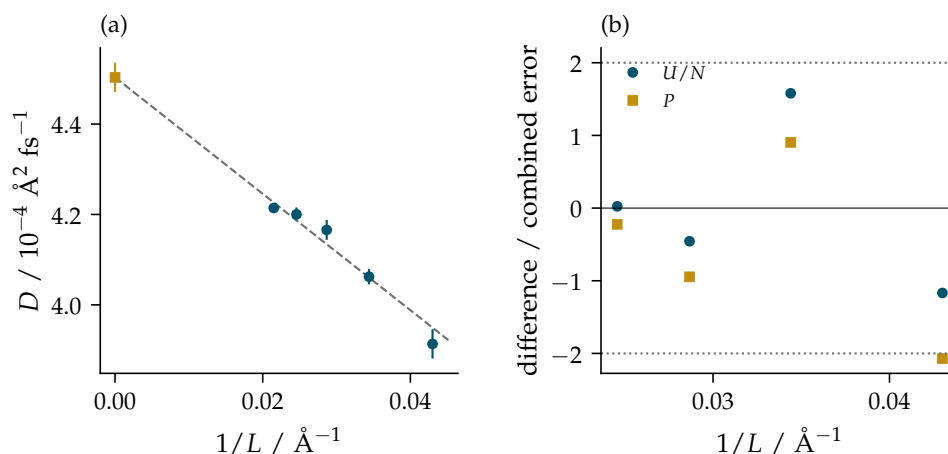


FIGURE 15.8: A result that depends on the size of the box: liquid argon at 135 K and $0.8\sigma^{-3}$ with 256 to 2048 atoms, two runs of 1 ns at each size. (a) The diffusion coefficient, with its error from blocks of 100 ps, against $1/L$, with the line of Equation (15.6) fitted to them (dashed) and its value at $1/L = 0$ (ochre). (b) The potential energy per atom and the pressure for each box less those for 2048 atoms, in units of the error of the difference, against ± 2 (dotted).

KEY IDEA. A converged average can still depend on the size of the box. The diffusion coefficient of a periodic liquid falls short of the endless liquid's by $2.837k_B T / (6\pi\eta_s L)$, as the image flows return through the box; runs at several sizes and a straight line in $1/L$ find D_∞ . Static averages of the same liquid change far less.

NOTEBOOK 15 fit the line through the five boxes and extrapolate

CHECK YOUR UNDERSTANDING. By how much should D change from a box of 500 atoms to one of 4000 at the same density, if it follows the line of this section?

15.8 Does the run sample its ensemble?

An error bar says how well a run knows its own averages. It is silent on whether they belong to the ensemble the run claims: Chapter 13 found thermostats that hold the mean temperature and get the spread of the energy wrong (Section 13.2). This section derives a test that needs no knowledge of the system, runs at two temperatures and the canonical density alone⁸⁷, and sets out the resampling that gives its error bar.

The ratio of two densities

In the canonical ensemble the probability of a microstate of energy E is $e^{-\beta E}/Z$ (Section 12.4). The microstates with energies between E and $E + \delta E$ number $\Omega(E)$, the volume of that shell (Section 12.3), and grouping them, as Section 12.7 grouped the velocities by their kinetic energy, gives the density of E at the inverse temperature β , written $\mathcal{P}(E | \beta)$:

$$\mathcal{P}(E | \beta) = \frac{\Omega(E) e^{-\beta E}}{Z(\beta) \delta E}.$$

No run knows $\Omega(E)$. At two temperatures the ratio of the densities removes it, with δE :

$$\frac{\mathcal{P}(E | \beta_2)}{\mathcal{P}(E | \beta_1)} = \frac{Z(\beta_1)}{Z(\beta_2)} e^{-(\beta_2 - \beta_1)E}, \quad \ln \frac{\mathcal{P}(E | \beta_2)}{\mathcal{P}(E | \beta_1)} = \ln \frac{Z(\beta_1)}{Z(\beta_2)} + (\beta_1 - \beta_2)E.$$

The logarithm of the ratio of the two histograms of the energy must be a straight line in E whose slope is $\beta_1 - \beta_2$, known in advance, whatever the system. The intercept, which holds the unknown Z , does not matter. For 130 and 140 K the slope is 6.376 eV^{-1} . The test needs the two histograms to overlap. With $C_V = 2.347 Nk_B$ for the liquid (Section 12.7), the mean energy rises by about $C_V \Delta T = 0.518 \text{ eV}$ (0.520 eV between the runs below), about twice the canonical spread $\sqrt{k_B T^2 C_V}$, 0.275 eV at 130 K and 0.296 eV at 140 K (Equation (12.6)), so the two share a range of energies. On 100 000 independent energies at each temperature drawn from a gamma density (Section 12.7), which has the form $\Omega(E)e^{-\beta E}$ with $\Omega \propto E^{399}$, a shape chosen only so that the two densities overlap, the fitted slope is $(6.350 \pm 0.030) \text{ eV}^{-1}$, its error from the bootstrap below, 0.9 errors from the required value.

The bootstrap

The slope is fitted to histograms, and its error is not that of a mean. Efron's bootstrap finds the error of any quantity computed from samples by recomputing it on new sets, each drawn from the samples themselves with replacement and each as large

as the original set, and taking the spread of the results⁸⁸. Drawing single samples would destroy the memory of a run and give too small an error; drawing whole blocks, longer than the memory, keeps it within each block, as in [Section 15.3](#), and makes the resampled runs nearly as correlated as the real one, losing the memory only at the joins between blocks⁸⁹ (mdlab draws non-overlapping blocks, a simpler variant of Künsch's moving blocks). The function `block_bootstrap` of `mdlab.analysis.stats` does this for several runs at once, resampling each independently. On the 2 ns run, blocks of 20 ps give the error of the mean potential energy as 0.0045 eV, against 0.0047 eV from [g. Section 15.6](#) used the same resampling for the error of a heat capacity, a variance rather than a mean.

Two thermostats

[Figure 15.9](#) runs the liquid for 1 ns at 130 and 140 K under CSVr with $\tau_T = 1$ ps, and under Berendsen coupling with $\tau_T = 0.1$ ps. After the start found from each run's total energy, the mean energies agree within 0.7 of their combined errors: -8.6409 and -8.6306 eV at 130 K, -8.1209 and -8.1118 eV at 140 K. The spreads do not: 0.273 and 0.287 eV under CSVr, close to the canonical 0.275 and 0.296 eV, and 0.112 and 0.118 eV under Berendsen coupling. The slope of the logarithm of the ratio, with its error from the block bootstrap with blocks of 20 ps, is $(6.72 \pm 0.41) \text{ eV}^{-1}$ under CSVr, 1.05 ± 0.06 times the required value and 0.8 errors from it: CSVr passes. Under Berendsen coupling it is $(35.1 \pm 3.5) \text{ eV}^{-1}$, 5.5 times the required value and 8.2 errors from it. For two Gaussian densities of the same spread σ_E whose means differ by $\Delta\bar{E}$, the logarithm of the ratio, $[(E - \bar{E}_1)^2 - (E - \bar{E}_2)^2] / 2\sigma_E^2$, is a straight line of slope $\Delta\bar{E} / \sigma_E^2$. The canonical $\Delta\bar{E} = C_V \Delta T$ and $\sigma_E^2 = k_B T^2 C_V$ make it $\Delta T / k_B T^2 \approx \beta_1 - \beta_2$; Berendsen coupling keeps $\Delta\bar{E}$ and narrows σ_E by a factor of 2.43, which makes the line $2.43^2 = 5.9$ times too steep, close to the 5.5 found. A thermostat that gives the right mean and the wrong spread fails this test at once, whatever quantity was to be measured; the physical-validation package of Merz and Shirts carries out this test and others for the common simulation codes⁹⁰.

KEY IDEA. In the canonical ensemble $\ln[\mathcal{P}(E | \beta_2) / \mathcal{P}(E | \beta_1)]$ is a straight line in E with the slope $\beta_1 - \beta_2$, whatever the system; runs at two nearby temperatures test it. The error of a fitted slope, or of any quantity that is not a mean, comes from the block bootstrap, which resamples blocks longer than the memory, recomputes the quantity and takes the spread of the results.

NOTEBOOK 15 *test a thermostat at two temperatures*

CHECK YOUR UNDERSTANDING. Why does the test not need $\Omega(E)$, and what would a run at the wrong temperature, 132 K where 130 K was claimed, do to the slope?

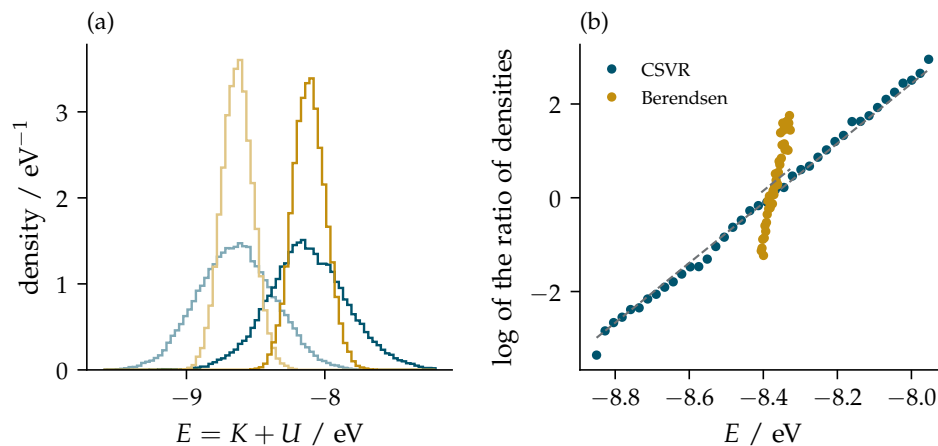


FIGURE 15.9: Does the run sample its ensemble? The liquid for 1 ns at 130 (light) and 140 K (dark), under CSV (teal) and Berendsen coupling with $\tau_T = 0.1$ ps (ochre). (a) The densities of the total energy. (b) The logarithm of the ratio of the densities at the two temperatures, against the slope $\beta_{130} - \beta_{140}$ that the canonical ensemble requires (dashed).

15.9 A protocol and a criterion

The chapter's tools combine into a procedure that takes a system from a rough start to numbers with error bars, and a criterion, with numbers, for when a quantity counts as converged. This section follows the liquid from atoms placed at random to a converged run at constant pressure, with the plot to read at each stage.

From random positions

Atoms placed at random in a box overlap: 256 argon atoms dropped into the liquid's cell start with $U = 7.91 \times 10^{10}$ eV and a largest force of $2.95 \times 10^{12} \text{ eV } \text{\AA}^{-1}$, and the first step of a run of 10 fs would carry the atom that feels it some $4 \times 10^{10} \text{ \AA}$, across more than 10^9 copies of the periodic box. The overlaps are removed first by walking downhill on the energy. Moving every atom a small distance along its own force, $d\mathbf{r}_i = h\mathbf{F}_i$ with $h > 0$, changes the energy by

$$dU = \sum_i \nabla_i U \cdot d\mathbf{r}_i = - \sum_i \mathbf{F}_i \cdot h\mathbf{F}_i = -h \sum_i |\mathbf{F}_i|^2,$$

which is negative unless every force vanishes: the forces point downhill, and a step along them lowers the energy if it is short enough for the forces not to change much along it. This is steepest descent. The function `minimise` of `mdlab.md` sets h by the atom with the largest force, which moves the distance step, so that $h = \text{step} / \max_i |\mathbf{F}_i|$; it keeps a trial that lowers the energy and lengthens the next by a fifth, and refuses one that does not and halves the distance:

```

1 def minimise(
2     model: EnergyForces,
3     positions: ArrayLike,
4     step: float = 0.1,
5     force_tolerance: float = 1e-2,
6     max_steps: int = 10000,
7 ) -> tuple[NDArray, NDArray, NDArray]:
8     r = np.array(positions, dtype=float)
9     u, f = model(r)
10    energies, largest = [u], [np.max(np.linalg.norm(f, axis=-1))]
11    for _ in range(max_steps):
12        if largest[-1] < force_tolerance:
13            break
14        trial = r + step * f / largest[-1]
15        u_trial, f_trial = model(trial)
16        if u_trial < u:
17            r, u, f = trial, u_trial, f_trial
18            energies.append(u)
19            largest.append(np.max(np.linalg.norm(f, axis=-1)))
20            step *= 1.2
21        else:
22            step *= 0.5
23    return r, np.array(energies), np.array(largest)

```

From a lattice disturbed by $0.2 \text{ \AA}$ it reaches the same minimum as ASE's FIRE optimiser, to $1 \times 10^{-3} \text{ \AA}$ in every position (`test_stats.py`). From the random start it takes 75 accepted moves to bring the largest force below $0.05 \text{ eV \AA}^{-1}$, at $U = -15.357 \text{ eV}$ (Figure 15.10(a)). The aim is only to remove the overlaps, and any minimum close to the start will do.

Temperature, then pressure

The stages that follow each settle one thing, and each has its plot. Velocities drawn at 135 K for the minimised atoms give a temperature that falls within the first picosecond to 70.1 K, about half, as the kinetic energy shares itself with the potential energy of atoms that sat at the bottom of their wells (Section 12.6); CSVr brings its average over the preceding 1 ps back above 130 K, within 5 K of the target, by 3.8 ps (Figure 15.10(b)), and over the last 25 ps of 50 ps the mean is $(136.6 \pm 1.2) \text{ K}$, 1.4 errors from the target. The running mean from the start, which lags, shows why such a stage is discarded. Stochastic cell rescaling at 0.1 GPa then lets the volume move from that of $0.8\sigma^{-3}$, $12577 \text{ \AA}^3$, to the one the liquid takes at that pressure (Figure 15.10(c)), and the run continues for 1 ns. Each quantity then goes through the criterion.

DEFINITION 15.1. A quantity A , recorded at n equal intervals δt over a run, is converged to the error σ^* when

1. the start t_0 that maximises $n_{\text{eff}}(t_0)$ lies in the first half of the run (Section 15.4);

2. the n' samples after it, with their own g and s , hold at least 100 independent samples, $n_{\text{eff}} = n'/g \geq 100$ (Section 15.3);
3. the error from blocks of $5g$ samples agrees with $\sqrt{gs^2/n'}$ within twice its own error (Section 15.3);
4. the slope of the samples after t_0 is within two of its errors of zero (Section 15.5);
5. and $\sqrt{gs^2/n'} \leq \sigma^*$, or else Equation (15.5) gives the run that would make it so (Section 15.6).

The result is reported as $\bar{A} \pm \sqrt{gs^2/n'}$, one standard error, with the number of independent samples.

The first check catches a run too short to locate its own start; the second leaves at least twenty blocks of $5g$, so that the block error the third check compares with is itself known to about a sixth, $1/\sqrt{2 \times 19}$, and keeps the bias that the search for a start brings small, as below; the third catches a slow memory that the estimate of g cut off; the fourth catches a drift or a transient left in. The function `convergence_report` of `mdlab.analysis.stats` applies the first four, and the fifth when given a target, and returns the evidence for each. The criterion says nothing about whether the run samples the right ensemble, for which Section 15.8 gives a test, or about the size of the box, which Section 15.7 shows can bias a converged number.

After 300 ps at constant pressure the report finds the volume and the potential energy not yet converged: their starts are at 16.5 and 12.0 ps, but they hold only 64.6 and 47.8 independent samples, the potential energy now having $\tau_{\text{int}} = 3.0$ ps over these 300 ps, as it follows the slow swings of the volume. A hundred independent samples need 439 and 603 ps after the start. The pressure and the temperature pass, with 813 and 139. After 1 ns all four pass: $V = (12310 \pm 10) \text{ \AA}^3$ with 316 independent samples, the volume of Section 14.5 within its error; $P = (0.0997 \pm 0.0004) \text{ GPa}$ against the 0.1 GPa asked for; $T = (135.27 \pm 0.28) \text{ K}$; and $U = (-13.049 \pm 0.014) \text{ eV}$.

Does the criterion keep its promise?

An error bar of one standard error should hold the true mean in 68% of runs. On series of known mean from Equation (15.2), 300 of each length, the error bar the report gives, whether or not the series passes, holds it in 0.55 of the series with 25 independent samples, 0.57 with 50, 0.68 with 100 and 0.70 with 200; the error bar $\sqrt{gs^2/n}$ of the whole series, without the search for a start, holds it in 0.64, 0.64, 0.71 and 0.72. The search favours a start after which g happens to come out small (Section 15.4), which makes the error bar small too, and only with about a hundred independent samples does that bias fade, which sets the threshold of the second check. With the offset of

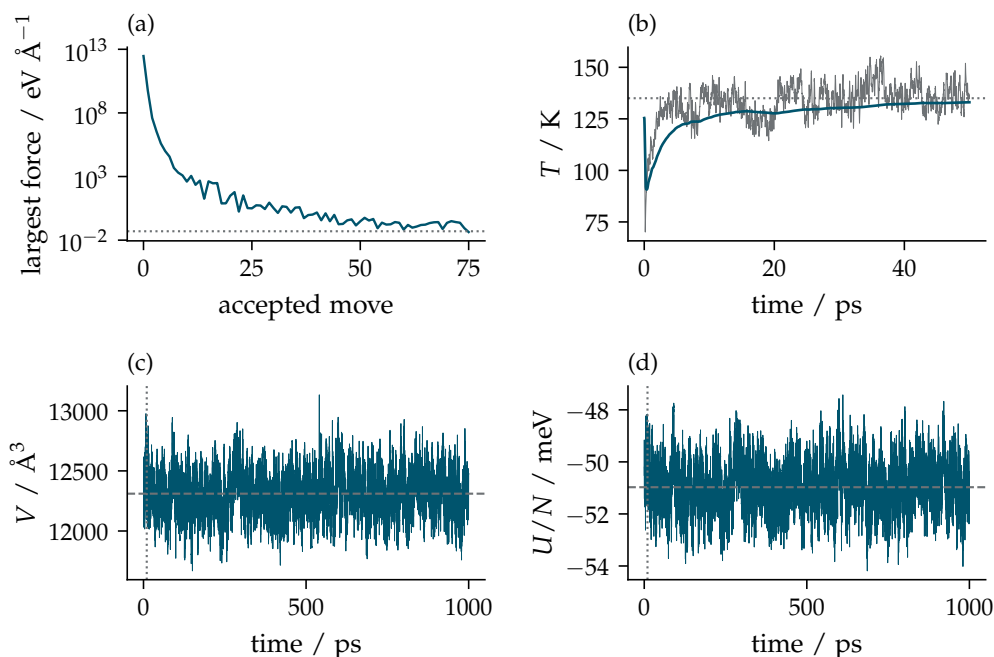


FIGURE 15.10: A protocol, stage by stage, for 256 argon atoms placed at random. (a) Steepest descent: the largest force after each accepted move, against the tolerance (dotted). (b) 50 ps of CSV at fixed volume: the temperature (grey) and its running mean (teal), against 135 K (dotted). (c) 1 ns of stochastic cell rescaling at 0.1 GPa: the volume, the start found (dotted) and the mean after it (dashed). (d) The same for the potential energy per atom.

Figure 15.5(a) added to series of 200 ps, 182 of 200 pass the criterion, and of those 0.70 hold the true mean within one error and 0.95 within two. On the liquid itself the evidence is thinner: of the sixty pieces of 100 ps of U , T and P in the 2 ns run, ten pass, and of those four hold the mean of the whole run within one error and seven within two, fewer than 68% and 95% would give, which fits pieces that pass only where g came out small.

KEY IDEA. A protocol removes overlaps by steepest descent, brings the temperature, then the pressure, to their targets, and discards each stage. A quantity is converged when its start lies in the first half of the run, at least 100 independent samples follow it, blocks confirm the error, no drift remains and the error meets its target; `convergence_report` checks all five. So reported, one error bar holds the true mean of test series in about 68% of runs; on the liquid's pieces that only just pass, less often.

NOTEBOOK 15 *run the criterion on any series, from mdlab or from ASE*

CHECK YOUR UNDERSTANDING. A run of 500 ps has $g = 300$ for a quantity sampled every 10 fs. Does it pass the second check, and if not, how long must it be?

15.10 What the trajectory looks like

Each fault of this chapter lies in the numbers drawn from a trajectory, and each shows on one of four plots of the quantity concerned, which together we call its dashboard: the series with the start found, the running mean from that start with its error band, the errors from blocks, and the autocorrelation function. This section shows the dashboard of a healthy run, then six runs or analyses that go wrong, each with the check that catches it.

A healthy run

Figure 15.11 is the dashboard of the volume in the 1 ns at constant pressure that ends the protocol of Section 15.9, drawn by dashboard of the chapter's scripts. (a) The series has settled by the start found at 10 ps and then wanders about its mean with no trend. (b) The running mean from the start swings while few independent samples are in it and then holds steady inside its error band, which narrows as one over the square root of the time since the start. (c) The error from blocks rises to a plateau, 9.6 with its own error of 1 against $9.8 \text{ \AA}^3$ from g . (d) The autocorrelation function falls to zero by 8 ps, where the estimate of g stops. The report, given no target, passes all four checks: $V = (12310.4 \pm 9.8) \text{ \AA}^3$ from 316 independent samples, with a slope 0.85 of its error.

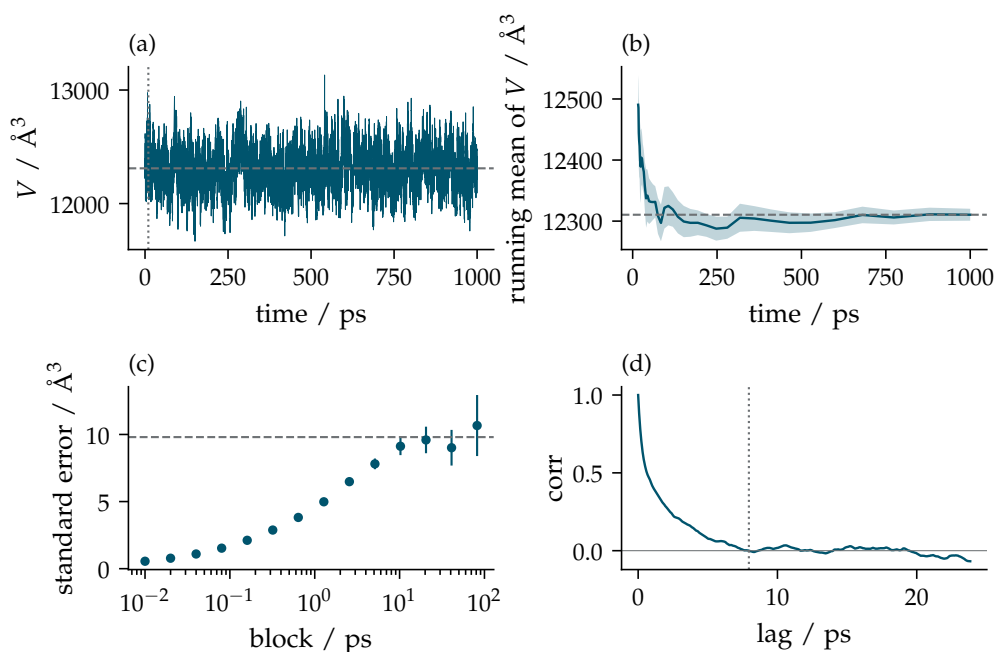


FIGURE 15.11: The dashboard of a converged quantity: the volume over 1 ns at 0.1 GPa. (a) The series, the start found (dotted) and the mean after it (dashed). (b) The running mean from the start with its error band. (c) The error of the mean from blocks, against $\sqrt{gs^2/n}$ (dashed). (d) The autocorrelation function, with the lag at which the estimate of g stops (dotted).

Six faults

Figure 15.12 and Table 15.2 set six faults against healthy runs.

TABLE 15.2: The faults of Figure 15.12, the check that catches each, and its value in the healthy and the faulty run or analysis.

fault	check	healthy	faulty
(a) averaged from the start	the start found; the mean after it	start 2.8 ps, $\bar{U}/N = (-50.060 \pm 0.085) \text{ meV}$	$\bar{U}/N = -50.173 \text{ meV}$ from $t = 0$
(b) the naive error bar	pieces whose bars hold the mean of 2 ns	11 of 20 with $\sqrt{g}$	0 of 20 with $s/\sqrt{n}$
(c) too short for a plateau	independent samples; the blocks	1369 in 2 ns	19.7 in 20 ps, no plateau
(d) a drifting energy	the drift test	-0.9 errors at 10 fs	+9.3 errors at 35 fs
(e) the wrong ensemble	slope of the ratio of densities, over the required	1.05 ± 0.06 (CSVR)	5.50 ± 0.55 (Berendsen)
(f) a box too small	$\langle \Delta r^2 \rangle / 6t$ at 20 ps, in $10^{-4} \text{ \AA}^2 \text{ fs}^{-1}$	4.177 for 2048 atoms	3.863 for 256

(a) *Averaged from the start.* A run started on the lattice and averaged from its first step includes the melting: the mean of U/N over the 100 ps is -50.173 meV , while from the start found at 2.8 ps it is $(-50.060 \pm 0.085) \text{ meV}$, against -50.092 meV over 2 ns. The running mean from $t = 0$ in Figure 15.12(a) creeps up for the whole run, the offset fading only as one over the length of the run. In this run the shift is about one error bar; over the eight runs of Section 15.4 the discard moves the mean by 0.106 meV , ± 1.6 of their error bars. Even the 2 ns run, started from an equilibrated liquid, has its start found at 30 ps: the search costs little where there is nothing to discard.

(b) *The naive error bar.* $s/\sqrt{n}$ treats the 10 000 samples of a piece of 100 ps as independent. Its bars are too short by the factor $\sqrt{g}$, about twelve, and none of the twenty holds the mean of the whole run, where the bars with $\sqrt{g}$ hold it in eleven, against the fourteen that 68% would give, within the spread of two that twenty pieces allow (Figure 15.12(b)). The check is to report g with every error bar.

(c) *Too short for a plateau.* The first 20 ps of the run hold 19.7 independent samples of U . Its block errors rise without levelling (Figure 15.12(c)), and its start falls in the second half of the run, at 12 ps: the report fails it on the start and on 100 independent samples. Its block check passes, because the 8 ps after that start give $g = 13$, a tenth of the value of the whole run, and blocks of $5g$ that short agree with an error as short; the plot of the blocks shows what the check cannot.

(d) *A drifting energy.* With steps of 35 fs the total energy of a run at fixed energy climbs by 0.6 meV per atom in a nanosecond (Figure 15.12(d)), 9.3 errors of the drift test, and

moves the temperature by 3.2 K per nanosecond; with steps of 10 fs its slope is -0.9 errors, no drift.

(e) *The wrong ensemble.* Berendsen coupling with $\tau_T = 0.1$ ps gives the right mean energies and the wrong spread, and the logarithm of the ratio of its energy densities at 130 and 140 K rises 5.5 times too steeply (Figure 15.12(e)). None of the convergence checks finds this, since each run is converged in its own narrow ensemble; the test of Section 15.8 does, as does the spread of K set against the canonical one (Section 13.7).

(f) *A box too small.* The squared displacement over $6t$, whose value at long times is D , reaches $3.863 \times 10^{-4} \text{ \AA}^2 \text{ fs}^{-1}$ at 20 ps for 256 atoms and 4.177×10^{-4} for 2048, so the smaller box gives 0.925 times the larger's (Figure 15.12(f)); these are single runs read at one time, near the 3.914 and 4.214 of Section 15.7. Each run fixes its own D to about 1%; the smaller box fixes it at a different number, for the reason Section 15.7 gives.

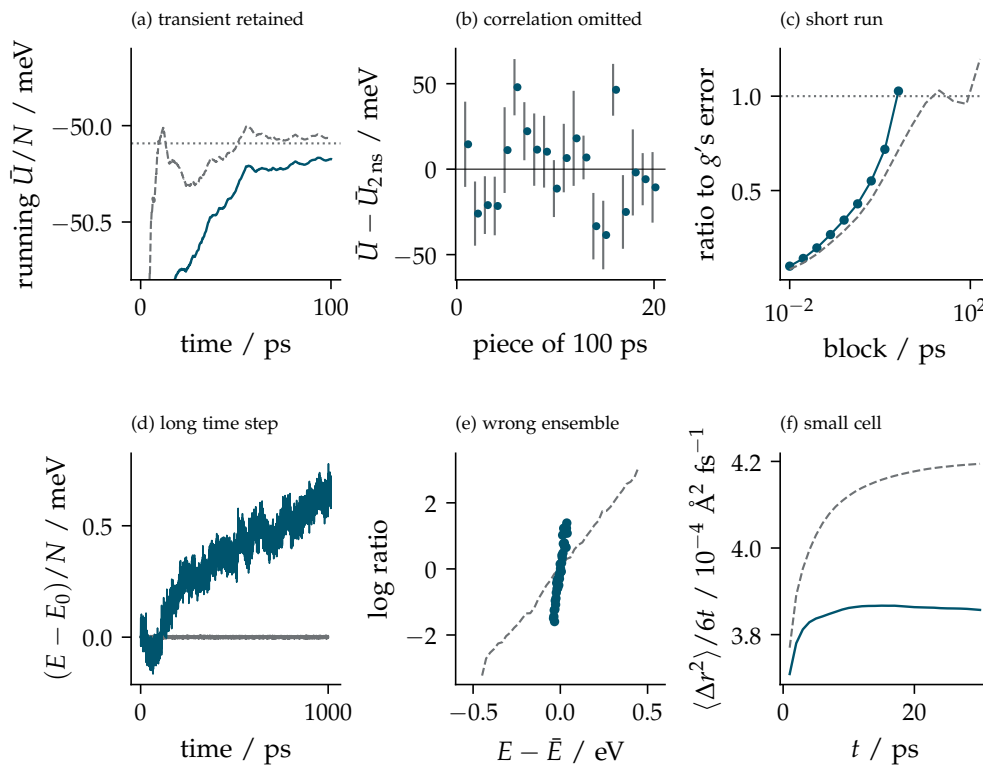


FIGURE 15.12: Faults of convergence (teal) against healthy runs (grey dashed). (a) The running mean of U/N of a melting run from $t = 0$, against that from the start found, and the mean of 2 ns (dotted). (b) Twenty pieces of 100 ps: the mean of U less that of the whole run, with $s/\sqrt{n}$ (teal, too short to see) and $\sqrt{gs^2/n}$ (grey). (c) The error of U from blocks over the first 20 ps and over 2 ns, each over its $\sqrt{gs^2/n}$. (d) The total energy per atom at fixed energy at 35 and 10 fs. (e) The logarithm of the ratio of the energy densities at 140 and 130 K under Berendsen coupling (here in teal, as the fault) and CSV, each about its mean. (f) The squared displacement over $6t$ for 256 and 2048 atoms.

IN PRACTICE. A run is read through the dashboard of each quantity to be reported: the series with its start, the running mean with its error band, the block errors with their plateau, and the autocorrelation function. Every average is reported with its error bar, its number of independent samples and the start discarded, and passes convergence_report; a run that fails is extended by the length Equation (15.5) gives. A thermostat is checked once by the test of Section 15.8, a time step by the drift test at fixed energy, and a transport coefficient by a second box size.

KEY IDEA. Each fault of convergence shows on a plot of the dashboard or on one of the chapter's tests: the start for an average taken from the first step, g for a naive error bar, the block plateau for a run too short, the drift test for a creeping energy, the ratio of densities for a thermostat that narrows the spread, and a second box for a size-dependent result.

NOTEBOOK 15 break each run in turn and watch which check catches it

CHECK YOUR UNDERSTANDING. A paper reports D from one run of 256 argon atoms as $(3.87 \pm 0.04) \times 10^{-4} \text{ \AA}^2 \text{ fs}^{-1}$. Is the error bar the error of D ?

What we now have

The mean of n independent samples spreads by $\sigma_x / \sqrt{n}$, and its error is estimated as $s / \sqrt{n}$; the samples of a run remember their predecessors, and n of them are worth n/g independent ones, with the statistical inefficiency $g = 1 + 2 \sum_k \text{corr}(k)$ estimated by summing the autocorrelation function until it first reaches zero, or a run of length t_f worth $t_f / (2\tau_{\text{int}})$.

Block averages give the same error without the autocorrelation function and show by their plateau whether the run was long enough to know it. The start of a run is found as the one that leaves the most independent samples, a drift as the slope of a straight line with its correlated error, and the run a precision needs from $t_f \geq 2\tau_{\text{int}}\sigma_A^2 / \sigma^{*2}$. For liquid argon under CSVr the energy-like quantities have τ_{int} of about 1 ps, ten times longer than at fixed energy, because the thermostat moves the energy slowly. With these error bars the diffusion coefficient is the same under CSVr with $\tau_T = 1$ ps, Berendsen coupling with $\tau_T = 0.1$ ps and fixed energy to about 2%, but depends on the size of the box as $1/L$, by 13% for 256 atoms. The ratio of the energy densities at two temperatures tests whether a thermostat samples the canonical ensemble: CSVr passes, Berendsen coupling fails by a factor of 5.5. A protocol takes atoms from random positions by steepest descent, a thermostat and a barostat to a run whose quantities pass a five-point criterion, under which one error bar holds the true mean of test series in about 68% of runs.

Chapter 16 defines the observables that describe a liquid's structure and motion, among them the radial distribution function, the mean squared displacement and the viscosity that [Section 15.7](#) inferred, and reports each with the error bar and the criterion of this chapter.

Answers to the checks

SECTION 15.1. The error falls as $1/\sqrt{n}$, so a fifth of it needs $5^2 = 25$ times as many samples: 2500.

SECTION 15.2. $g = (1 + 0.5)/(1 - 0.5) = 3$, and 1000 samples are worth $1000/3 = 333$ independent ones.

SECTION 15.3. Blocks of $5g = 500$ samples miss about 5% of the error, and the run holds $20\,000/500 = 40$ of them, which give the error to $1/\sqrt{2 \times 39} = 0.11$.

SECTION 15.4. About $\delta A_0 \tau_r / t_f = 2 \times 1/100 = 0.02$ K.

SECTION 15.5. It is 1.5 errors from zero, so the run shows no drift it can detect. A run four times as long has an error $4^{3/2} = 8$ times smaller, 0.25, and if the true slope is 3 the longer run would put it 12 errors from zero: a drift.

SECTION 15.6. $t_f \geq 2 \times 5 \text{ ps} \times 2^2 / 0.1^2 = 4000 \text{ ps}$, 4 ns.

SECTION 15.7. At the same density L grows as $N^{1/3}$, so L doubles from 29.07 Å to 58.14 Å, and D rises by $12.88 \times (1/29.07 - 1/58.14) = 0.221$ in units of $10^{-4} \text{ Å}^2 \text{ fs}^{-1}$.

SECTION 15.8. $\Omega(E)$ is the same at both temperatures and cancels in the ratio. A run at 132 K labelled 130 K has the slope $\beta_{132} - \beta_{140} = 5.024 \text{ eV}^{-1}$, 0.79 of the 6.376 eV^{-1} required, which an error of about 0.4 shows as some three errors.

SECTION 15.9. The run holds $50\,000/300 = 167$ independent samples, so it passes the second check, provided the start found leaves at least $100 \times 300 = 30\,000$ samples after it.

SECTION 15.10. No. The bar is the statistical error of D in that box. A box of 256 atoms gives a D some 13% below that of an infinite one ([Section 15.7](#)), a bias many times larger than the bar, which the error bar of a single box cannot show.

Exercises

The exercises run from questions of understanding, through calculations, to short derivations marked ‘show that’. Full solutions are in ‘[Solutions to the exercises](#)’ at the end of the book, and Notebook 15 checks every numerical answer.

EXERCISE 15.1 (*enough rolls*). How many rolls of a fair die make the interval $\bar{x} \pm 1.96 \sigma_{\bar{x}}$ no wider than ± 0.01 ?

EXERCISE 15.2 (*five samples*). With five samples, Student’s 95% factor is 2.776. By what fraction is the 95% interval wider than one built with 1.96, and why must it be?

EXERCISE 15.3 (*the variance of a difference*). Show that $\text{var}(x - y) = \text{var}(x) + \text{var}(y) - 2 \text{cov}(x, y)$, and explain why [Section 15.6](#) took the difference of the two species’ temperatures at each frame as one series, rather than combining the errors of the two temperatures.

EXERCISE 15.4 (*a series that settles*). Show that the step of [Equation \(15.2\)](#), started from $x_0 = 0$, gives $\text{var}(x_k) = 1 - c^{2k}$, and find how many steps of $c = 0.95$ bring the variance within 1% of 1.

EXERCISE 15.5 (*a slower memory*). For the step of [Equation \(15.2\)](#) with $c = 0.99$ and samples 10 fs apart, find g , τ_{int} and τ_c , and the number of independent samples in a run of 1 ns.

EXERCISE 15.6 (*a weighted geometric sum*). Show that $\sum_{k=1}^{\infty} kc^k = c/(1-c)^2$ for $0 \leq c < 1$, by multiplying the sum by $1 - c$.

EXERCISE 15.7 (*the error of an error*). For n independent Gaussian samples u_i of mean zero and variance σ_u^2 , show that $\langle u^4 \rangle = 3\sigma_u^4$, and hence that $s'^2 = \frac{1}{n} \sum_i u_i^2$ has the variance $2\sigma_u^4/n$, so that s' has the relative error $1/\sqrt{2n}$.

EXERCISE 15.8 (*blocks long enough*). A run of 2^{16} samples has $g = 60$ and an exponential memory. How long must its blocks be to miss about 5% of the error, how many such blocks does it hold, and how precisely does their spread give the error?

EXERCISE 15.9 (*what a transient leaves*). A run of length t_f carries the offset $\delta A_0 e^{-t/\tau_r}$. Show that after discarding the first t_0 , the offset adds $\delta A_0 \tau_r (e^{-t_0/\tau_r} - e^{-t_f/\tau_r}) / (t_f - t_0)$ to the mean, and evaluate it for $\delta A_0 = 3$, $\tau_r = 2$ ps, $t_f = 50$ ps and $t_0 = 5.34$ ps.

EXERCISE 15.10 (*evenly spaced times*). Show that for n times $t_i = i\delta t$, $i = 1, \dots, n$, $\sum_i (t_i - \bar{t})^2 = n(n^2 - 1)\delta t^2/12$, using $\sum_i i^2 = n(n+1)(2n+1)/6$.

EXERCISE 15.11 (*a longer run, a smaller drift*). The largest harmless drift for a run of 1 ns at fixed energy is 20.8 μeV per atom per nanosecond. Show that the limit falls as $t_f^{-3/2}$ with the run's length t_f , and evaluate it for 10 ns.

EXERCISE 15.12 (*too long a step*). For $U = \frac{1}{2}kx^2$ in one dimension, show that a step of steepest descent, $x \rightarrow x + hF$, lowers U only if $0 < h < 2/k$, and that $h = 1/k$ reaches the minimum in one step.

EXERCISE 15.13 (*the pressure to a precision*). How long a run under CSV, with the spread and τ_{int} of Table 15.1, fixes the pressure of the liquid to ± 0.0005 GPa? And at fixed energy, where a sample of the pressure spreads by 0.0096 GPa?

EXERCISE 15.14 (*a run too short*). A quantity with $\tau_{\text{int}} = 2$ ps is recorded in a run of 150 ps, whose start is found at 10 ps. Does it pass the second check of the criterion, and if not, how much longer must the run be?

EXERCISE 15.15 (*the ratio for a gamma density*). Show that for $\mathcal{P}(E | \beta) \propto E^{a-1}e^{-\beta E}$ the logarithm of the ratio of the densities at β_2 and β_1 is a straight line in E with the slope $\beta_1 - \beta_2$, and find the slope for 300 and 310 K.

EXERCISE 15.16 (*the units of the correction*). Show that $k_B T / (\eta_s L)$ has the units of a diffusion coefficient, and that no other product of powers of $k_B T$, η_s and L does.

EXERCISE 15.17 (*the size of the correction*). With $\eta_s = 2.18 \times 10^{-4}$ Pa s, find the Yeh-Hummer correction $2.837 k_B T / (6\pi\eta_s L)$ for the liquid of 256 atoms, $L = 23.26$ Å, at 135 K, in Å² fs⁻¹.

EXERCISE 15.18 (*why blocks*). Why does a bootstrap that draws single samples from a run, rather than blocks, give too small an error, and how long must the blocks be?

EXERCISE 15.19 (*a referee's question*). A study reports the heat capacity of a liquid from one run of 50 ps at fixed energy as $2.29 N k_B$, 'converged, since the running mean of K is flat'. What has it not shown, and what would show it?

Observables

A trajectory records positions and velocities. To learn from it, we need to turn those records into quantities that answer physical questions: how the atoms are arranged, how far they move, or how often they react. Such an observable may be a single number or a whole curve. It also gives us something to compare between simulations, and, where the measurement corresponds, with experiment.

We begin with structure: the distribution of neighbours, the structure factor measured by diffraction, and the bonds that identify molecules and reactions. Displacements and velocity correlations then give two routes to diffusion, followed by the transport of charge and momentum. The Fourier transform separates vibrations by frequency. Finally, histograms of selected coordinates give free-energy profiles.

The error estimates of Chapter 15 accompany these measurements. Identities between observables and comparisons with independent implementations provide checks on the analysis. The same quantities will later be used to judge learned models of motion against their reference trajectories in the planned Part VI.

16.1 How atoms arrange themselves

An orchard and a wood can hold the same number of trees on the same area and still look different. Standing at one tree of an orchard planted on a square grid, a walker finds the nearest trees at one spacing, the next at $\sqrt{2}$ spacings, then at 2, each at a fixed distance and in a fixed number. In a wood no two trees grow closer than their trunks allow, the nearest neighbours lie at roughly one distance without fixed directions, and farther out the trees thin into an even scatter. Counting the trees in rings of equal width around one tree, and comparing each count with what an even scatter would give, tells the two apart without a map. Atoms in a crystal and in a liquid differ in the same way, and the count in rings is the radial distribution function.

The radial distribution function

Around an atom, the number of others whose centres lie between the distances r and $r + dr$ would be the number density $\rho = N/V$ times the volume of that thin shell, $4\pi r^2 dr$, if the atoms were scattered evenly and independently. The radial distribution function $g(r)$ is the factor by which the actual mean number differs:

$$\text{mean number of partners between } r \text{ and } r + dr = \rho g(r) 4\pi r^2 dr. \quad (16.1)$$

Where $g(r) > 1$ partners crowd; where $g(r) < 1$ they are scarce; and $g(r) = 0$ inside the distance at which two atoms repel too strongly to meet. The mean is over every atom as the centre and over every frame of the run, so $g(r)$ describes the whole liquid or crystal. Summing [Equation \(16.1\)](#) out to a distance r counts the partners within r of an atom, the running coordination number

$$n_c(r) = \rho \int_0^r g(s) 4\pi s^2 ds,$$

whose value at the end of the first shell of neighbours is the atom's number of nearest neighbours.

A run gives $g(r)$ as a histogram ([Section 12.1](#)). The distances are cut into bins of width δr ; in each frame, every pair closer than the last edge adds one to its bin for each of its two atoms, since each is the other's partner; and the total over n frames is divided by the count an even scatter would give, $nN\rho$ times the volume of the shell between the edges r_k and r_{k+1} ,

$$\frac{4}{3}\pi(r_{k+1}^3 - r_k^3) = 4\pi r_k^2 \delta r + 4\pi r_k \delta r^2 + \frac{4}{3}\pi \delta r^3,$$

by expanding the cube of $r_k + \delta r$. The exact difference of spheres is used rather than its first term $4\pi r_k^2 \delta r$, which misses the shell's volume by a fraction of about $\delta r/r_k$, large near the origin and the whole of it in the first bin, where $r_k = 0$. In a periodic cell the separations are reduced to their nearest image ([Section 10.2](#)). That finds every pair once only out to half the cell's smallest width; beyond it a sphere around an atom would hold more than one copy of some partners while the nearest image counts one, so the histogram stops there. A partial function $g_{AB}(r)$, the partners of kind B around atoms of kind A, takes A atoms as the centres, B atoms as the partners and $\rho_B = N_B/V$ as the density.

The function `rdf` of `mdlab.analysis.structure` does this, finding the pairs through the cell list of [Section 10.4](#):

```

1 def rdf(
2     positions: ArrayLike,
3     cell: ArrayLike,
4     r_max: float,
5     n_bins: int,
6     centres: ArrayLike | None = None,
7     partners: ArrayLike | None = None,
8     per_frame: bool = False,
9 ) -> tuple[NDArray, NDArray]:
10     frames = _frames(positions)
11     h = np.asarray(cell, dtype=float)
12     in_a = _chosen(frames.shape[1], centres)
13     in_b = _chosen(frames.shape[1], partners)
14     edges = np.linspace(0.0, r_max, n_bins + 1)
15     ideal = in_a.sum() * in_b.sum() / cell_volume(h) * shell_volumes(edges)
16     out = np.empty((len(frames), n_bins))
17     for k, r in enumerate(frames):
18         i, j, _, dist = cell_list_pairs(r, h, r_max)
19         # a pair counts once for each end that is a centre whose partner
20         # is the other end
21         weight = (in_a[i] & in_b[j]).astype(float) + (in_a[j] & in_b[i])
22         out[k] = np.histogram(dist, edges, weights=weight)[0] / ideal
23     middle = 0.5 * (edges[1:] + edges[:-1])
24     return middle, (out if per_frame else out.mean(axis=0))

```

It divides by $N\rho = N^2/V$, as ASE's `get_rdf` does⁴³, and agrees with it to 3×10^{-14} on 50 frames of liquid argon. One consequence of that choice shows far from any atom. For atoms placed independently at random, an atom's partners are the other $N - 1$, spread evenly over the volume V , while the divisor assumes N , so $g(r)$ tends to $(N - 1)/N = 1 - 1/N$. For 256 atoms placed at random in the liquid's cell, 400 times over, $g(r)$ averages 0.9961 ± 0.0005 beyond $2 \text{ \AA}$, against $1 - 1/256 = 0.9961$ (Figure 16.1a). A liquid falls short of 1 by less. Each atom clears a hole around itself that holds fewer partners than an even scatter, and Section 16.2 shows that the hole takes $1 - S(0)$ of the one partner missing from ρV , with $S(0)$ about 0.06 here, leaving only $S(0)$ missing from the rest of the cell; $g(r)$ then tends to $1 - S(0)/N$, once the cell is large enough for the liquid's order to die away within it. Beyond the sphere of radius $L/2$, in the corners of the 2048-atom cell of Chapter 15, the partners lie at 0.999967 ± 0.000005 of ρ , against $1 - 1/2048 = 0.999512$. In the 256-atom cell, where $g(r)$ still ripples by a few hundredths at $L/2$, no far value is reached.

A crystal and a liquid

The fcc crystal of the model argon of Chapters 10 to 15, at its own lattice constant of $5.26714 \text{ \AA}$ and 21 K, has $g(r)$ in sharp peaks at the distances of its shells of neighbours, $(a/\sqrt{2})\sqrt{m}$, for $m = 1$ to 5: 3.724, 5.267, 6.451, 7.449 and $8.328 \text{ \AA}$ (Figure 16.1b). Between the peaks $g(r)$ is zero, and $n_c(r)$ climbs in steps of 12, 6, 24 and 12, the numbers of atoms in the first four shells of the fcc lattice, to 12, 18, 42 and 54 at the gaps between

them (Figure 16.1c). The peaks have width because the atoms vibrate, each wandering from its mean position by $0.094 \text{ \AA}$ along each axis (root mean square), and the first rises to 9.30.

The liquid at 135 K and $\rho = 0.02035 \text{ \AA}^{-3}$ (Chapter 15's run of 2 ns, every 2 ps) has none of this order beyond a few neighbours. $g(r)$ is zero inside $3 \text{ \AA}$, rises to a first peak of 2.578 at $3.684 \text{ \AA}$, falls to a first minimum of 0.663 at $5.238 \text{ \AA}$, and then oscillates about 1 with peaks that weaken shell by shell. Out to its first minimum the liquid holds 12.00 neighbours on average, as many as the crystal's first shell, though spread over a wider range of distances, at no fixed directions, and at a density of $0.02035 \text{ \AA}^{-3}$ against the crystal's $0.02737 \text{ \AA}^{-3}$.

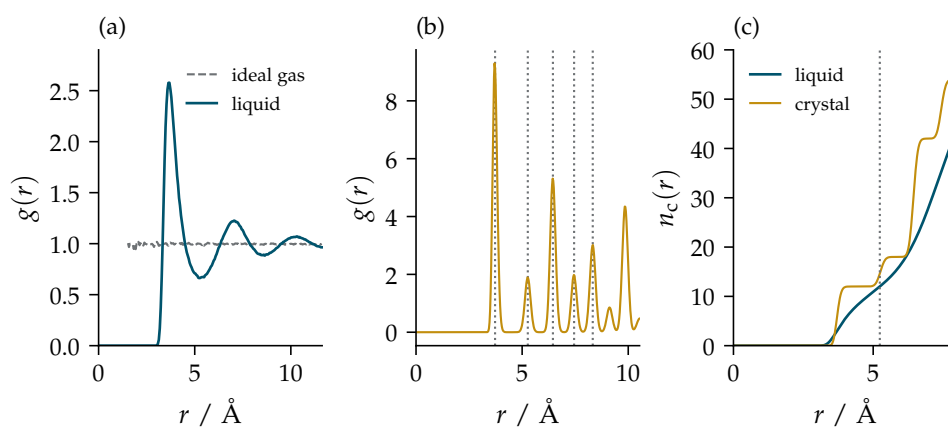


FIGURE 16.1: (a) The radial distribution function of liquid argon at 135 K (Chapter 15's run of 2 ns, every 2 ps) against 256 atoms at random in the same cell (dashed, from $1.5 \text{ \AA}$, where its bins hold enough pairs). (b) The fcc crystal at 21 K, with the distances of its first five shells (dotted). (c) The running coordination number of the liquid and the crystal, with the liquid's first minimum (dotted).

Energy and pressure from the arrangement

For a pair potential, $g(r)$ holds all that the potential energy and the virial need. The potential energy is the sum of $\varphi(r_{ij})$ over pairs. Around one atom, Equation (16.1) puts $\rho g(r) 4\pi r^2 dr$ partners at the distance r , each contributing $\varphi(r)$; summing over the N atoms counts each pair from both ends, so

$$\langle U \rangle = \frac{N}{2} \rho \int_0^\infty \varphi(r) g(r) 4\pi r^2 dr.$$

The virial of Section 14.3, $\mathcal{W} = -\sum_{\text{pairs}} r_{ij} \varphi'(r_{ij})$, follows by the same count,

$$\langle \mathcal{W} \rangle = -\frac{N}{2} \rho \int_0^\infty r \varphi'(r) g(r) 4\pi r^2 dr,$$

and with it and the run's mean kinetic energy the pressure $\langle P \rangle = (2\langle K \rangle + \langle \mathcal{W} \rangle)/3V$ of Equation (14.2). On bins of $0.01 \text{ \AA}$ and the 1000 frames, these give $U/N = -50.102 \text{ meV}$ against the run's average of $(-50.092 \pm 0.018) \text{ meV}$, 0.6 errors away, and $P = 0.0848 \text{ GPa}$ against $(0.0849 \pm 0.0004) \text{ GPa}$, 0.2 errors away. The bins must be fine: on bins of $0.05 \text{ \AA}$ the energy misses by 1.9 errors, because φ changes steeply across a bin near the core. Agreement here is a check of the normalisation of $g(r)$, of its histogram and of the run's energy together.

How long the arrangement takes

Chapter 15 deferred the convergence of $g(r)$ to this chapter. One frame already averages over 256 centres, so in bins of $0.05 \text{ \AA}$ its $g(r)$ at the first peak spreads by only 0.33 from frame to frame, and frames 500 fs apart are nearly independent, their statistical inefficiency (Section 15.2) being 1.04, so $\tau_{\text{int}} = 259 \text{ fs}$. With the run's length t_f in Equation (15.5), the error of the peak's height is 0.076 after 10 ps (2.9% of it), 0.024 after 100 ps (0.9%) and 0.005 after 2 ns, and a run of 573 ps gives it to 0.01. Pieces of the run of 5 to 200 ps scatter by 0.102 to 0.020 against the predicted 0.107 to 0.017, within the scatter expected of so few pieces, about a quarter for the 10 pieces of 200 ps, so these error bars can be trusted (Figure 16.2b). Over the whole curve, the first 10 ps lie within two of their errors of the 2 ns result in 0.95 of the bins (Figure 16.2a). The first peak of $g(r)$ is fixed to 0.9% by 100 ps, where the pressure and the diffusion coefficient need 1 ns for 0.7% and 1.0% (Section 15.6). Its memory is short: τ_{int} is 0.26 ps, against 0.94 ps for the pressure under the same thermostat.

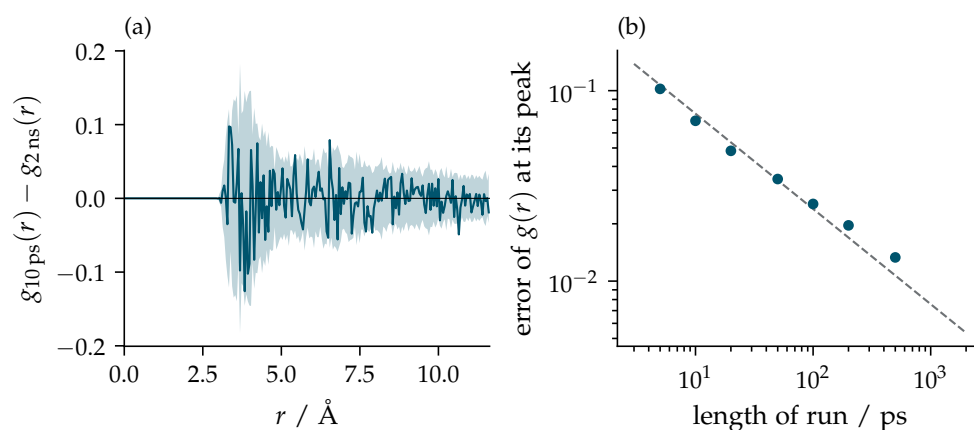


FIGURE 16.2: (a) The radial distribution function of liquid argon from the first 10 ps of Chapter 15's run, less that from all 2 ns, with twice its standard error from the 20 frames 500 fs apart (band). (b) The standard error of g at its first peak, in bins of $0.05 \text{ \AA}$, against the length of run used: predicted from one frame's spread and the statistical inefficiency (dashed), and measured as the scatter of independent pieces (points).

KEY IDEA. The radial distribution function $g(r)$ counts the partners at the distance r from an atom against an even scatter. It is a histogram of pair distances divided by $N\rho$ times the volume of each shell, out to half the cell, and tends far away to $1 - 1/N$ for atoms placed at random and to $1 - S(0)/N$, nearly 1, in a liquid in a large enough cell. Its peaks give the shells of neighbours, its integral the coordination number, and with the pair potential the energy and the pressure. Its memory is short, so it converges several times sooner than the pressure.

NOTEBOOK 16 *grow shells around an atom and watch the histogram become $g(r)$*

CHECK YOUR UNDERSTANDING. A gas of 500 atoms placed independently at random in a fixed cell is analysed with ASE's normalisation. What value should $g(r)$ approach far from any atom, and why does a larger cell bring it closer to 1?

16.2 What diffraction sees

A street lamp seen through a fine net curtain shows a pattern of bright spots: light passing between the threads is scattered from each gap, and in some directions the waves from all the gaps arrive in step and add, while in others they arrive out of step and cancel. The pattern is set by the spacing of the threads, and from it the spacing can be read without looking at the curtain. X-rays and neutrons scattered by the atoms of a liquid or a crystal do the same at the scale of ångströms, and what they measure is the structure factor, the counterpart of $g(r)$ in waves.

Waves scattered by atoms

A plane wave of wave vector $\mathbf{k}$ has the phase $\mathbf{k} \cdot \mathbf{r}$ at the point $\mathbf{r}$, its crests lying where that is a multiple of 2π . Scattered by an atom at $\mathbf{r}_j$ towards a distant detector, it leaves as a wave of the same wavelength, its wave vector $\mathbf{k}'$ of the same length as $\mathbf{k}$ and pointing at the detector. The incoming wave reaches the atom with the phase $\mathbf{k} \cdot \mathbf{r}_j$; the path on to the detector is shorter than the path from the origin by the length of $\mathbf{r}_j$ along $\mathbf{k}'$, which takes off the phase $\mathbf{k}' \cdot \mathbf{r}_j$. So the wave arrives with the phase $\mathbf{k} \cdot \mathbf{r}_j - \mathbf{k}' \cdot \mathbf{r}_j$ plus a part that is the same for every atom, and the waves from different atoms differ in phase by $-\mathbf{G} \cdot \mathbf{r}_j$, with $\mathbf{G} = \mathbf{k}' - \mathbf{k}$ the scattering vector. The wave from atom j can be drawn as an arrow of unit length turned through that angle, and the total as the sum of N such arrows. Turning every arrow the other way changes no length, so the angles may be taken as $+\mathbf{G} \cdot \mathbf{r}_j$, and the total has the components

$$\mathcal{C}(\mathbf{G}) = \sum_j \cos(\mathbf{G} \cdot \mathbf{r}_j), \quad \mathcal{S}(\mathbf{G}) = \sum_j \sin(\mathbf{G} \cdot \mathbf{r}_j),$$

the sums of [Section 10.3](#) with every charge 1. A detector measures the intensity, which grows as the square of a wave's height, here the square of the total's length; per atom it is the structure factor

$$S(\mathbf{G}) = \frac{1}{N} [\mathcal{C}(\mathbf{G})^2 + \mathcal{S}(\mathbf{G})^2]. \quad (16.2)$$

Expanding the squares and using $\cos A \cos B + \sin A \sin B = \cos(A - B)$ ([Equation \(1.6\)](#)) writes it as a double sum,

$$S(\mathbf{G}) = \frac{1}{N} \sum_i \sum_j \cos(\mathbf{G} \cdot (\mathbf{r}_j - \mathbf{r}_i)) = 1 + \frac{1}{N} \sum_i \sum_{j \neq i} \cos(\mathbf{G} \cdot \mathbf{r}_{ij}),$$

the terms with $j = i$ each being 1. Atoms placed at random give cross terms that average to zero, so $S(\mathbf{G}) = 1$; atoms of a crystal at a vector $\mathbf{G}$ that makes every $\mathbf{G} \cdot \mathbf{r}_{ij}$ a multiple of 2π give $S(\mathbf{G}) = N$, a Bragg peak. Such vectors form the crystal's reciprocal lattice, built as in [Section 10.3](#) on the crystal's own repeating unit instead of the simulation cell. A liquid has $S(\mathbf{G})$ above 1 near the repeat of its shells of neighbours and below 1 elsewhere.

In a periodic cell only waves that repeat with the cell are allowed, the vectors $\mathbf{G} = m_1 \mathbf{a}^* + m_2 \mathbf{b}^* + m_3 \mathbf{c}^*$ of [Section 10.3](#), the shortest of length $2\pi/L$ for a cube of side L . The function `structure_factor` of `mdlab` evaluates [Equation \(16.2\)](#) on each of them, averages over the frames and over the vectors of each length G , which in a liquid differ only by noise, and returns $S(G)$ against G .

The structure factor from the arrangement

The double sum can also be written with $g(r)$. Around each atom of a liquid, which has no preferred direction, [Equation \(16.1\)](#) spreads its $\rho g(r) 4\pi r^2 dr$ partners evenly over the shell, so it puts $\rho g(r) d^3r$ in each small volume d^3r at the separation $\mathbf{r}$, and the mean of the double sum is

$$S(\mathbf{G}) = 1 + \rho \int g(r) \cos(\mathbf{G} \cdot \mathbf{r}) d^3r = 1 + \rho \int [g(r) - 1] \cos(\mathbf{G} \cdot \mathbf{r}) d^3r,$$

the second form subtracting $\rho \int \cos(\mathbf{G} \cdot \mathbf{r}) d^3r$, which is zero over the cell for every allowed $\mathbf{G} \neq \mathbf{0}$ ([Section 10.3](#)), and which makes the integrand vanish far away where $g(r)$ is 1. On a shell of radius r , with θ the angle between $\mathbf{r}$ and $\mathbf{G}$, the part at angles between θ and $\theta + d\theta$ is a band of radius $r \sin \theta$ and width $r d\theta$, of area $2\pi r^2 \sin \theta d\theta$, out of $4\pi r^2$, so the average of the cosine over the shell is, with $u = \cos \theta$ and $du = -\sin \theta d\theta$, the minus sign turning the limits $u = 1$ to -1 round,

$$\frac{1}{2} \int_0^\pi \cos(Gr \cos \theta) \sin \theta d\theta = \frac{1}{2} \int_{-1}^1 \cos(Gru) du = \frac{1}{2} \left[\frac{\sin Gru}{Gr} \right]_{-1}^1 = \frac{\sin Gr}{Gr}.$$

The volume integral is then one over shells,

$$S(G) = 1 + 4\pi\rho \int_0^\infty r^2 [g(r) - 1] \frac{\sin Gr}{Gr} dr, \quad (16.3)$$

which depends only on the length G , as it must in a liquid that has no preferred direction.

Figure 16.3(a) compares the two routes for liquid argon at 135 K. From the positions, on 250 frames of the 2 ns run, $S(G)$ has its first peak of 2.271 at $G = 1.985 \text{ \AA}^{-1}$. That peak lies near the repeat of the shells of neighbours: the first three peaks of $g(r)$ lie at 3.68, 7.04 and 10.30 Å, 3.31 Å apart on average, and $2\pi/3.31 = 1.90 \text{ \AA}^{-1}$, 4% below the peak. Through Equation (16.3), with $g(r)$ known only to half the cell, the peak is 2.126, 0.15 below, and at the smallest allowed G , $0.270 \text{ \AA}^{-1}$, the positions give 0.058 and the transform 0.074, with ripples on either side that take it as low as -0.05 , which no intensity can be. Stopping the integral at $L/2$ cuts off whatever $g(r) - 1$ holds beyond. Near the peak the oscillations of $g(r)$ beyond the cut keep step with $\sin Gr$, so what is cut off adds up: the $g(r)$ of the 2048-atom runs, stopped at 11.6 Å and at 23.3 Å, gives 2.116 and 2.240 there. At small G the loss is smaller but large beside $S(G)$ itself. The direct route has no such error and is the one to trust wherever the two differ.

The long-wave limit and the compressibility

As G shrinks towards 0, $\sin(Gr)/Gr$ tends to 1 and Equation (16.3) tends to $S(0) = 1 + \rho \int [g(r) - 1] d^3r$; at $G = 0$ itself Equation (16.2) gives N , the beam that passes straight through, so $S(0)$ means this limit. The same integral measures how much the number of atoms N_r in a region of volume V_r fluctuates. The ordered pairs of distinct atoms both inside the region number $N_r(N_r - 1)$. Each atom inside has on average $\rho g(r) d^3r$ partners in each small volume at the separation r , and for a region large against the reach of $g(r) - 1$ nearly every atom is far from its surface, so

$$\langle N_r(N_r - 1) \rangle = \langle N_r \rangle \rho \int_{V_r} g(r) d^3r = \langle N_r \rangle \left[\rho \int [g(r) - 1] d^3r + \langle N_r \rangle \right],$$

since $\rho \int_{V_r} d^3r$ is $\rho V_r = \langle N_r \rangle$. Subtracting $\langle N_r \rangle^2 - \langle N_r \rangle$ from both sides leaves $\text{var}(N_r) = \langle N_r \rangle [1 + \rho \int [g(r) - 1] d^3r]$, so $S(0) = \text{var}(N_r) / \langle N_r \rangle$. That fluctuation follows from Chapter 14. A fixed number of atoms held at fixed pressure has $\text{var}(V) = k_B T \langle V \rangle \kappa_T$ (Equation (14.4)), and so has a group of atoms inside a large liquid, the liquid around it acting as the barostat. Such a group of N_r atoms has the density N_r/V , which changes by $\delta\rho/\rho = -\delta V/V$ as its volume V fluctuates; the region of fixed volume V_r has the

density N_r/V_r , which changes by $\delta\rho/\rho = \delta N_r/N_r$ as its number fluctuates. Both are the fluctuation of the liquid's density over the same volume, so

$$\frac{\text{var}(N_r)}{\langle N_r \rangle^2} = \frac{\text{var}(V)}{\langle V \rangle^2} = \frac{k_B T \kappa_T}{\langle V \rangle}, \quad S(0) = \frac{\text{var}(N_r)}{\langle N_r \rangle} = \rho k_B T \kappa_T,$$

the second following on multiplying the first by $\langle N_r \rangle = \rho \langle V \rangle$. With Chapter 14's $\kappa_T = (1.46 \pm 0.11) \text{ GPa}^{-1}$, from the slope dP/dV of runs at fixed volume near 0.1 GPa, $\rho k_B T \kappa_T = 0.0554 \pm 0.0042$.

The 256-atom cell has only four lengths of wave vector below $0.6 \text{ \AA}^{-1}$, too few to follow $S(G)$ towards $G = 0$, so Figure 16.3(b) uses Chapter 15's runs of 1372 and 2048 atoms, two of each, whose cells allow G down to $0.135 \text{ \AA}^{-1}$. Each value carries the standard error of its frames' values. A straight line in G^2 through the 29 values below $0.6 \text{ \AA}^{-1}$, fitted as in Section 15.7 with each squared miss divided by the square of that value's error, reaches $S(0) = 0.0589$ at $G = 0$. The sum of its squared misses in units of the errors is 72, where scatter alone gives 27 ± 7 for 29 values less the two fitted numbers, so the errors understate the misses by about $\sqrt{72/27} = 1.63$, and the line's error, 0.0002, enlarged by that factor becomes 0.0004. The two values, 0.0589 ± 0.0004 from the positions and 0.0554 ± 0.0042 from the change of pressure with volume, lie 0.8 combined errors apart: the long waves of the structure and the liquid's stiffness against squeezing measure the same compressibility. The test is loose in one respect: Chapter 14 took its slope at a volume 2.2% smaller than this run's, where the denser liquid is the stiffer.

The same integral fixes the far value of $g(r)$ in a fixed cell (Section 16.1). With the divisor $N\rho$, an atom's partners over the whole cell number $N - 1$, so $\rho \int_V [g(r) - 1] d^3r = -1$. In a cell large enough for $g(r) - 1$ to die away well inside it, the part of the integral near the atom is $S(0) - 1$, and the remaining $-S(0)$ is spread over the rest of the cell, where $g(r) - 1$ is then $-S(0)/\rho V = -S(0)/N$: $1 - 1/N$ for atoms at random, whose $S(0)$ is 1, and $1 - 0.059/N$ for this liquid. In the corners of the 2048-atom cell the partners fall short of ρ by a fraction $(3.3 \pm 0.5) \times 10^{-5}$, which is $S(0)/N$ for $S(0) = 0.067 \pm 0.009$, the value above within its error.

KEY IDEA. The structure factor $S(G) = [\mathcal{C}^2 + \mathcal{S}^2]/N$ is the intensity, per atom, of waves scattered with the change of wave vector $\mathbf{G}$, and what diffraction measures. It is the transform of $g(r) - 1$ by $\sin(Gr)/Gr$, which a cut at half the cell spoils, by 0.15 at the peak and by a large fraction at small G , so it is computed from the positions on the cell's own wave vectors. Its limit at $G = 0$ is $\rho k_B T \kappa_T$.

NOTEBOOK 16 compare $S(G)$ from the positions and from $g(r)$

CHECK YOUR UNDERSTANDING. For a crystal at $T = 0$, what is $S(\mathbf{G})$ at a vector $\mathbf{G}$ of its reciprocal lattice, and at a vector of the cell that is not one of them?

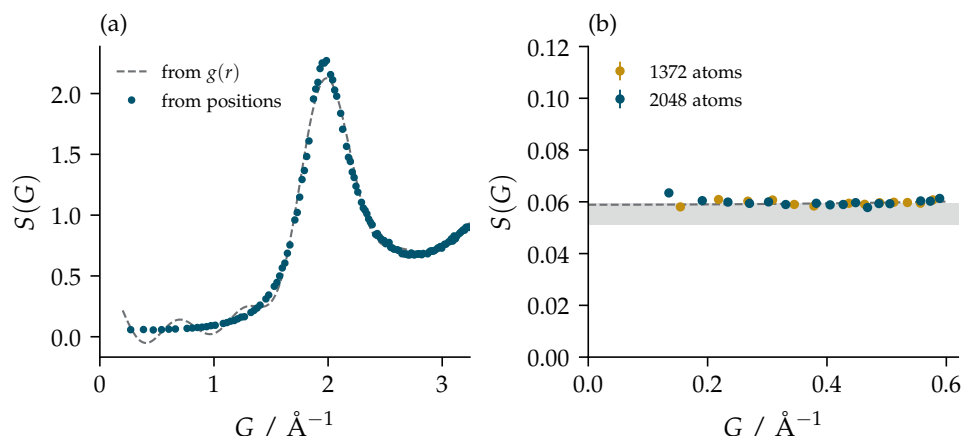


FIGURE 16.3: (a) The structure factor of liquid argon at 135 K from the positions, averaged over the wave vectors of the 256-atom cell of each length up to $12 \times 2\pi/L = 3.24 \text{\AA}^{-1}$ (points), against the transform of $g(r)$ cut at half the cell (dashed). (b) $S(G)$ at the smallest wave vectors of the 1372- and 2048-atom runs of Chapter 15, two runs of each, with their standard errors, and a straight line in G^2 fitted below $0.6 \text{\AA}^{-1}$ (dashed), against $\rho k_B T \kappa_T$ with Chapter 14's compressibility (band, its error).

16.3 Bonds and molecules

To follow a reaction, we must first decide which atoms form each molecule. We assign bonds to pairs, then collect atoms joined through those bonds into molecules. A distance cutoff gives a first rule, but thermal vibrations can carry a pair across it repeatedly without a chemical change. This section introduces a bond criterion that retains that distinction, then uses the resulting graph to identify molecules and species.

A bond from a distance

The forces of most models say nothing about bonds: they are functions of the positions. A bond is therefore assigned from the distance between two atoms, and a common yardstick is the sum of their covalent radii, $R_i + R_j$, from a table of radii for each element such as the one ASE carries⁴³: two carbon atoms ($R = 0.76 \text{\AA}$ each) are bonded within some multiple of $1.52 \text{\AA}$. Read literally, such a rule makes a bond flicker. A bonded pair vibrates, and if the far end of its vibration lies near the cutoff, each vibration carries it out and back across it, two events per period with nothing chemical happening.

The cure is a band of hysteresis, as in a thermostat that switches the heating on below one temperature and off above a higher one. A pair not yet bonded becomes bonded when it comes closer than r_{on} ; a bonded pair stays bonded until it parts beyond $r_{\text{off}} > r_{\text{on}}$. A vibration whose whole swing, from its near end to its far end, is narrower than the band cannot carry the pair across both distances, so it cannot make

a bond flicker, while a real dissociation, which carries the atoms apart for good, crosses both. `mdlab.bonds.BondTracker` takes r_{on} and r_{off} as 1.2 and 1.4 times the sum of the covalent radii unless told otherwise, factors that bracket the first minimum of $g(r)$ for carbon (Section 16.4); for a model, or for any pair of elements, the two distances can be given directly.

To measure the effect we use a model gas built to react. Two kinds of atom, A and B, of nitrogen's mass, attract each other through the Morse well of Section 8.3 with depth $D_M = 1 \text{ eV}$, width $a_M = 2 \text{ \AA}^{-1}$ and $r_0 = 1.2 \text{ \AA}$, the D and a of that section marked here to keep them apart from the diffusion coefficient and the lattice spacing of later sections; two atoms of the same kind only repel, as $D_M e^{-2a_M(r-2.6 \text{ \AA})}$, which weakens a second partner's hold: a line B-A-B with both bonds at $1.48 \text{ \AA}$ lies only 0.40 eV below AB and a free B, so such complexes form but soon part. The constants are chosen for illustration and are not fitted to any real pair of atoms. 250 molecules AB in a cubic cell of side $63 \text{ \AA}$ are held at 1500, 1750, 2000, 2250 and 2500 K by Langevin friction of 1 ps^{-1} , which stands for a buffer gas that supplies and removes energy, for 100 ps each. A bond forms where the pair's energy falls below $-D_M/2$, at $r_{\text{on}} = 1.814 \text{ \AA}$, and breaks where it rises above $-D_M/8$, at $r_{\text{off}} = 2.570 \text{ \AA}$. A molecule's small vibrations have the period $2\pi\sqrt{m_r/2D_M a_M^2} = 59.8 \text{ fs}$ (Sections 4.4 and 4.8), with $m_r = 7.0035 \text{ amu}$ the reduced mass and $2D_M a_M^2 = 8 \text{ eV \AA}^{-2}$ the stiffness of the well.

At 2000 K a single cutoff at r_{on} records 53 265 bonds formed or broken in 100 ps; the band records 16 442, 3.2 times fewer (Figure 16.4a). Many of the remaining events are not flicker either: an A and a B that meet with too much energy to stay together pass through each other's well and part again within a vibration or two. Requiring that a bond last a set time before its forming and breaking count, and likewise that a pair which parts stay apart that long, and dropping both events of any bond or gap shorter than that (`bonds.persistent`), removes these passing encounters and the flicker the band leaves. For a least lifetime of 0.5 ps the counts fall to 5501 with the single cutoff and 6548 with the band, which then keeps a fifth more. The remaining events fall more slowly as the least lifetime grows, by 40% and 36% from 0.5 to 2 ps, against a fall by 9.7 and 2.5 times from no least lifetime to 0.5 ps: they are the molecules that form and break.

Keys and set differences

Each bond is stored as one whole number, its key $iN + j$ for the atoms $i < j$. Different pairs have different keys, since $j < N$, and the atoms are recovered as the quotient and the remainder of the key divided by N . A computer stores such numbers in 64 binary digits, bits, reaching $2^{63} \approx 9.2 \times 10^{18}$ with one bit for the sign, so keys fit for N up to about 3×10^9 , and about 2×10^9 once the tracker doubles each key (below). The bonds

formed between two frames are the keys in the new set and not in the old, and those broken the keys in the old and not in the new.

SET DIFFERENCES OF SORTED LISTS. Two lists of whole numbers, each sorted and without repeats, can be compared in one pass. A pointer starts at the head of each. If the two numbers under the pointers are equal, the number is in both lists, and both pointers move on; otherwise the smaller is in its own list only, and only its pointer moves. When one list is used up, the rest of the other is in that list only. Each step moves at least one pointer, so for lists of lengths n_1 and n_2 the pass takes at most $n_1 + n_2$ steps, against $n_1 n_2$ for testing every pair. `BondTracker.update` merges this way the sorted keys within r_{off} of the new frame with the sorted bonds of the last one: a key in both is kept, a key only in the new list is a new bond if it lies within r_{on} , and a key only in the old list is a bond broken.

The candidates are the pairs that the force evaluation has already found, which the model of `md.PairModel` keeps with their distances after each call, so the tracker measures no distance of its own; this serves as long as the cutoff of the forces reaches the largest r_{off} . Most candidates lie beyond the largest r_{off} and are passed over at once; the rest are sorted, each stored as twice its key plus 1 if the pair is within r_{on} , so that one sort orders the keys, and halving, the remainder dropped, recovers each key, the remainder its flag; then they are merged:

```

1  def update(
2      self, i: ArrayLike, j: ArrayLike, dist: ArrayLike
3  ) -> tuple[NDArray, NDArray]:
4      # the keys within r_off, each carrying its "closer than r_on" flag
5      # in an extra lowest bit, so that one sort orders both
6      packed = np.sort(_jit(_held_keys)(
7          np.asarray(i, dtype=np.int64), np.asarray(j, dtype=np.int64),
8          np.asarray(dist, dtype=float), self.kind, self.r_on, self.r_off,
9          self._reach, self.n))
10     self.keys, formed, broken = _jit(_merge)(
11         packed >> 1, (packed & 1).astype(np.bool_), self.keys)
12     return formed, broken

```

The two helpers are plain loops, the filter and the merge of the toolbox, compiled by Numba as in [Section 10.7](#).

Molecules

A molecule is a set of atoms joined to one another through chains of bonds.

GRAPHS AND THEIR CONNECTED COMPONENTS. A graph is a set of nodes and a set of edges, each edge joining two nodes. Here the atoms are the nodes and the bonds the edges. A graph of N nodes can be stored as its adjacency matrix, $N \times N$, with 1

in row i and column j when i and j are joined and 0 otherwise. For atoms nearly every entry is 0, since each atom has a few bonds, so only the positions of the ones are stored, a sparse matrix whose storage grows as the number of bonds, not as N^2 . A connected component is a largest set of nodes any two of which are joined by a path of edges. It is found by a search, which starts from any node not yet assigned, adds its neighbours, then their neighbours, and so on until no new node is reached; those nodes are one component, and the search starts again from a node not yet assigned. Each edge is followed at most twice, so the work grows as the number of nodes plus edges.

`mdlab.bonds.molecules` builds the sparse matrix of the bonds and labels each atom with its component through `scipy.sparse.csgraph`:

```
1 def molecules(n_atoms: int, keys: ArrayLike) -> NDArray:
2     i, j = np.divmod(np.asarray(keys, dtype=np.int64), n_atoms)
3     graph = coo_matrix((np.ones(len(i)), (i, j)), shape=(n_atoms, n_atoms))
4     return connected_components(graph, directed=False)[1]
```

Species

Two molecules belong to the same species if they have the same atoms bonded in the same way, whatever the numbers the atoms carry in the run. The formula is the first part of the name: the number of atoms of each element, written in the Hill order: in a molecule with carbon, carbon first, then hydrogen, then the others alphabetically, and in one without, all of them alphabetically, so that a molecule of two A and one B is A_2B . The formula does not tell isomers apart, molecules of the same atoms bonded differently, such as a ring of six carbon atoms and a chain of six.

HASHING AND CANONICAL LABELS. A hash is a function that turns data of any size into a number of fixed size, such that equal data give equal numbers and different data give different numbers with high probability: a fingerprint. Two molecules should get the same number if one can be renumbered into the other. A label that does not change under renumbering is called canonical; sums over the atoms are canonical, since the order of the terms does not change a sum. `mdlab.bonds.graph_hash` builds one in rounds, after Weisfeiler and Lehman. Each atom starts with a number for its element, scrambled by a fixed function that spreads nearby numbers far apart. In each round it replaces its number by the scrambled combination of its own number and the sum of its neighbours' scrambled numbers, so that after m rounds the number describes the atom's surroundings m bonds deep, and the molecule's number is the sum over its atoms. In a ring of six every atom has two neighbours with two neighbours each, all round; at the ends of a chain an

atom has one, and its number differs from the first round on. The scheme is not perfect. Graphs in which every atom has the same number of neighbours of the same kinds, such as a triangular prism of six atoms and a hexagon of six whose opposite atoms are also joined, look alike to every round and get the same number, so two species can, rarely, share a hash.

The scrambling is the finishing step of the generator `splitmix64`, a few multiplications and shifts of 64-bit whole numbers whose sums are kept to their lowest 64 bits, dropping whatever passes 2^{64} ; four rounds are used. `ReactionLog` names each species by its formula, and a further isomer of a formula already seen by the formula with a prime, A_2B' , in order of appearance.

Checks

The tracker's first frame, with no hysteresis yet, must find the pairs closer than the sum of their radii, as ASE's `neighbor_list` does with its `natural_cutoffs`. On 80 atoms of carbon and hydrogen at random in a periodic cell, with r_{on} the sum of the radii, the two lists are identical, and both equal the list from testing every pair. Over 40 frames of atoms moving at random, the tracker's bonds equal, frame by frame, those of a direct transcription of the rule. On invented trajectories whose events are known, it finds them all: of two molecules N_2 , one pulled apart gives $N_2 \rightarrow N + N$; one of the freed atoms brought to the other molecule gives $N + N_2 \rightarrow N_3$; a ring of six carbon atoms, one of whose bonds is stretched beyond r_{off} , becomes a chain whose hash differs, $C_6 \rightarrow C'_6$. Each is a test of `mdlab`.

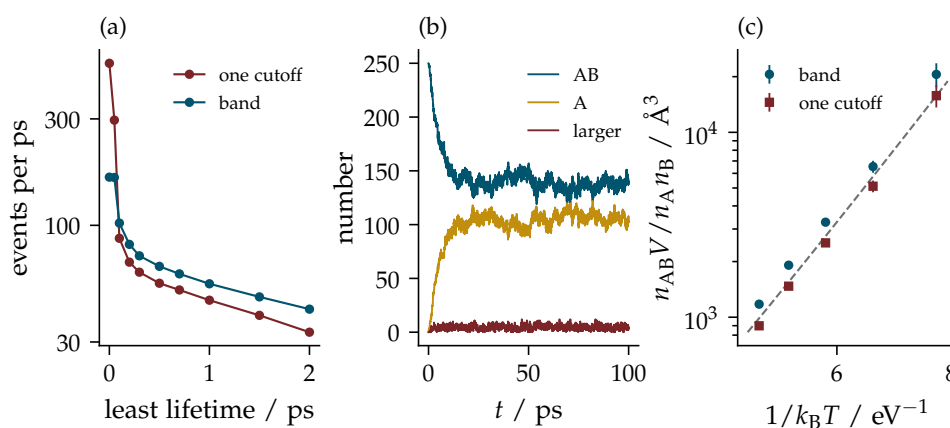


FIGURE 16.4: The A-B gas: 250 molecules in a cell of side $63 \text{ \AA}$ under Langevin friction of 1 ps^{-1} for 100 ps. (a) Bonds formed or broken per picosecond at 2000 K, counted with a single cutoff at r_{on} and with the band from r_{on} to r_{off} , against the least lifetime an event must have to count. (b) The numbers of molecules AB, of free A and of larger complexes at 2000 K against time. (c) The equilibrium constant $n_{AB}V / n_A n_B$ over the last 90 ps with the band (circles) and with the single cutoff (squares), with the standard error of nine blocks, against $\int_0^{r_{\text{on}}} 4\pi r^2 e^{-\beta\phi} dr$ (dashed).

KEY IDEA. A bond is assigned from the distance, with a band of hysteresis, forming below r_{on} and breaking above r_{off} , and events are counted only for bonds and gaps that last a least time. Bonds are whole-number keys, and those formed and broken are differences of sorted lists. Molecules are the connected components of the graph of bonds, and a species is named by its formula and a hash of its graph that does not depend on the numbering.

NOTEBOOK 16 *watch bonds flicker with one cutoff and settle with a band*

CHECK YOUR UNDERSTANDING. In a run of 1000 atoms, the bond between atoms 17 and 342 has the key 17 342. Which bond has the key 342 017?

16.4 Reactions and their rates

Once molecules have been identified, a reaction is a change in which molecules a group of atoms belongs to. We record the reactants and products, then turn repeated events into rates. The uncertainty depends on the number of independent events, rather than the number of saved frames. We also measure the cost of doing this analysis during a run.

Reactions from changes of species

Between two frames, each atom belongs to a molecule before and to a molecule after. Joining each old molecule to each new one that shares an atom with it gives a graph whose nodes are molecules, and its connected components are the reactions. A molecule that kept its atoms and its bonds is one old node joined to one new node with the same hash, and is passed over; every other component is a reaction, its old molecules the reactants and its new ones the products. A ring that opens is a component of one old and one new molecule with different hashes, $C_6 \rightarrow C'_6$; two molecules that exchange an atom, $AB + B \rightarrow B + AB$, are a component of two old and two new molecules with the same names. `mdlab.bonds.ReactionLog` does this frame by frame, recording the time, the reactants and the products of each reaction, and the counts of each species.

In the A-B gas at 2000 K the 250 molecules AB lose partners at first (Figure 16.4b), and after the first 10 ps the gas holds on average 139.3 AB, 104.1 free A and 103.9 free B, with 2.2 AB₂, 1.9 A₂B and a few larger complexes. Over 100 ps the log holds 57 kinds of reaction. The most frequent, counting every event with the band and no least lifetime, are

$AB \rightarrow A + B$	4517	$A + B \rightarrow AB$	4421
$AB + B \rightarrow AB_2$	1618	$AB_2 \rightarrow AB + B$	1602
$A + AB \rightarrow A_2B$	1556	$A_2B \rightarrow A + AB$	1566
$AB + AB \rightarrow A_2B_2$	301	$A_2B_2 \rightarrow AB + AB$	298

a reaction network: a graph whose nodes are the species and the reactions, each reaction joined to its reactants and its products and carrying its count. Each reaction and its reverse occur nearly equally often, as every process in a gas at equilibrium is balanced by its reverse: $AB \rightarrow A + B$ outnumbers its reverse by 96, about the 111 net breaks by which the molecules fell from 250 to 139.3 at the start, and the others differ by less than the $\sqrt{n}$ of the toolbox below. The complexes are brief. In a steady state the complexes present at any moment are those formed within one lifetime before it, so their number is the rate at which they form times the lifetime. AB_2 forms 16.8 times per picosecond and 2.18 are present on average, so each lasts $2.18/16.8 = 0.130$ ps; A_2B lasts 0.120 ps. They are passing encounters of the kind the least lifetime of [Section 16.3](#) removes.

RATES FROM COUNTS, AND THE POISSON DISTRIBUTION. Events that happen independently, at a steady rate ν for each of many molecules, are counted over an exposure X , the sum over the molecules of the time each was able to react. Cut into J small pieces, the exposure has in each an event with the small probability $p = \nu X/J$ and none otherwise, independently, so the count is a sum of J independent numbers each 0 or 1. Each such number has the mean p , and so does its square, which is the same 0 or 1, so its variance is $p - p^2 = p(1 - p)$. The count then has the mean $Jp = \nu X$ and, since the variances of independent numbers add ([Section 15.1](#)), the variance $Jp(1 - p)$, which tends to νX as the pieces shrink. The rate is estimated as n/X , with the standard error $\sqrt{n}/X$, a fraction $1/\sqrt{n}$ of itself: a rate from 100 events is known to a tenth, from 10 000 to a hundredth. The chance of n events is the number of ways of choosing n of the J pieces, $J(J-1) \cdots (J-n+1)/n!$, times $p^n(1-p)^{J-n}$, which with $p = \nu X/J$ is

$$\frac{J(J-1) \cdots (J-n+1)}{J^n} \frac{(\nu X)^n}{n!} \left(1 - \frac{\nu X}{J}\right)^{J-n}.$$

As J grows with n fixed, the first fraction, a product of n factors each tending to 1, tends to 1, and the logarithm of the last factor, $(J-n) \ln(1 - \nu X/J)$, tends to $-\nu X$, since $\ln(1+y)$ is y plus terms of order y^2 ([Section 13.4](#)). The chance tends to the Poisson distribution

$$\mathcal{P}(n) = \frac{(\nu X)^n}{n!} e^{-\nu X},$$

whose mean and variance are both νX , the limits of the count's.

Counted events of a run can be correlated, a bond that breaks and re-forms twice counting thrice; the least lifetime of [Section 16.3](#) removes most of these. At 2000 K, the bonds broken in 89 windows of 1 ps number 33.82 on average with variance 36.24, a ratio of 1.07, within the 0.15 by which that ratio scatters over 89 windows of a Poisson count, and their histogram lies close to the Poisson distribution of the same mean ([Figure 16.5b](#)).

Rates against temperature

With the band and a least lifetime of 0.5 ps, after the first 10 ps, the rate of breaking is the number of bonds broken divided by the bond-picoseconds of exposure, and the rate constant of forming the number formed divided by the exposure of free pairs, $\int n_A n_B / V dt$. At 1500 K 1828 bonds break over 18 168 bond-picoseconds, $k_b = (0.1006 \pm 0.0024) \text{ ps}^{-1}$, and at 2500 K 3182 over 9500, $(0.3350 \pm 0.0059) \text{ ps}^{-1}$; the rate constant of forming falls over the same range, from (1911 ± 45) to $(415 \pm 7) \text{ \AA}^3 \text{ ps}^{-1}$ ([Figure 16.5a](#)). A rate that rises as $e^{-E_a/k_B T}$ has a logarithm that falls on a straight line of slope $-E_a$ against $1/k_B T$, the Arrhenius form. The error of the logarithm of a rate from n events is its fractional error, $1/\sqrt{n}$, so each point is weighted in the fit by $\sqrt{n}$, the misses being measured in units of their errors as in [Section 15.7](#): breaking has $E_a = 0.389 \text{ eV}$ and forming -0.478 eV , a rate that falls as the temperature rises.

Neither slope is the depth of the well, 1 eV, and they need not be: each rate depends on how the buffer gas supplies and removes energy, as Chapter 13 found for hops. Their ratio does not. At equilibrium forming and breaking balance, $k_f n_A n_B / V = k_b n_{AB}$, so k_f/k_b is the equilibrium constant $K_{eq} = n_{AB} V / n_A n_B$ of the counts: 18 994 against 20 583 $\text{\AA}^3$ at 1500 K, 3406 against 3266 at 2000 and 1237 against 1177 at 2500 K, within 8%. The two are not the same average. With nearly as many bonds formed as broken, k_f/k_b is the bond-picoseconds over the exposure of free pairs, 19 162, 3416 and 1235 $\text{\AA}^3$, a ratio of time averages that counts the bonds within complexes too, while K_{eq} averages the ratio of the counts for the molecules AB alone. Since $\ln K_{eq} = \ln k_f - \ln k_b$, its slope is the difference of theirs, $0.478 - (-0.389) = 0.867 \text{ eV}$, and the counts give 0.912 eV.

That constant can be predicted for a dilute gas of A and B whose only interaction is the pair potential, with a pair bonded when closer than r_{on} . Arrangements with n_{AB} bonded pairs and with one more, made by joining a free A to a free B, are compared by the Boltzmann weights of [Section 12.4](#) integrated over the positions; the momenta, the same in both, cancel. A free pair, each atom anywhere, has the weight V^2 ; a bonded pair, its centre anywhere and its separation within r_{on} , has $V \int_0^{r_{on}} 4\pi r^2 e^{-\beta \varphi(r)} dr$. The A and the B to join can be chosen in $n_A n_B$ ways, and each result is reached from $n_{AB} + 1$

starting arrangements, one for each of its bonds, so one more bond is more probable than not by the factor

$$\frac{n_A n_B}{n_{AB} + 1} \frac{1}{V} \int_0^{r_{\text{on}}} 4\pi r^2 e^{-\beta\varphi(r)} dr.$$

At the most probable n_{AB} , large enough that $n_{AB} + 1$ differs little from n_{AB} , the factor is 1, so K_{eq} equals the integral, the mass action law. The integral falls sixteen-fold from 15756 to 987 Å³ between 1500 and 2500 K, and the constant from the single cutoff at r_{on} , with the standard error of nine blocks, follows it at 1.00, 0.89, 0.92, 0.95 and 0.91 of its value (Figure 16.4c); the band, which also keeps pairs between r_{on} and r_{off} that were once closer, gives 1.13 to 1.31. The integral treats the free atoms as an ideal gas, leaving out the attraction between unlike atoms and the repulsion between like ones beyond r_{on} .

Carbon from the melt

Carbon's bonds come and go in the melt. 216 atoms at 2.0 g cm⁻³, with the potential of Tersoff⁹² through ASE's calculator, the parameters those that ship with ASE, are held at 6000 K for 2 ps by CSVr with $\tau_T = 50$ fs, and the thermostat's target is then lowered steadily to 300 K, over 10 ps in one run and 40 ps in another, with steps of 0.5 fs. The model is carried by `io.AseModel`, which calls any ASE calculator with the positions and returns the energy and the forces as an `mdlab` model does. The default band, 1.2 to 1.4 times twice the covalent radius of carbon, 1.824 to 2.128 Å, brackets the first minimum of $g(r)$ in the melt, at 1.963 Å, and after the quench lies, all but its first 0.03 Å, in the gap from 1.85 to 2.15 Å where $g(r)$ is zero (Figure 16.6a), so the criterion counts the first shell of neighbours and nothing beyond. In the melt, between 0.5 and 2 ps, 0.764 of the atoms have three bonds, 0.161 two and 0.071 four, and the bonds form and break continually: after the first frame there are 1870 events over the faster run and 3479 over the slower, 1221 of each in the first 2 ps at 6000 K. As the melt cools the three-fold atoms grow at the expense of the others, and the last bond forms or breaks when the thermostat's target has fallen to 2335 K in the faster run and 2432 K in the slower (Figure 16.6b). At 300 K the faster quench leaves 0.852 of the atoms with three bonds, 0.065 with two and 0.083 with four; the slower leaves 0.907, 0.042 and 0.051. Every atom is in one molecule, the solid itself. These are single runs, so the difference between the two quenches has no error bar yet; the trend, fewer defects for slower cooling, is what a repeat would test.

Keeping up with the run

An analysis can read frames from a file after the run, or watch the run as it goes. `io.read_frames` reads any trajectory file ASE can read, extended XYZ or ASE's own,

converting velocities to $\text{\AA fs}^{-1}$. Watching needs no file: `md.run` and `thermostats.run` accept observers, functions called at every recorded frame with the time, the positions and the velocities, and the pairs that the force evaluation has just found are at hand in `PairModel.last_pairs`, so an observer can feed them to a `ReactionLog` without searching for neighbours again. The gas runs above did so every 10 fs.

Such analysis must cost little next to the forces. Figure 16.7 times one frame's analysis, bonds, molecules, hashes, formulas and reactions, against one force evaluation of a cheap model, liquid argon's Lennard-Jones forces with the neighbour list already built, from 864 to 97 556 atoms. The bonds are taken between atoms closer than $4.0 \text{ \AA}$, kept to $4.4 \text{ \AA}$, 4.4 per atom, a denser graph than most molecules give. With the two loops of Section 16.3 and the rounds of the hash compiled by Numba, and the pairs taken from the force evaluation, the analysis costs about 0.15 of a force evaluation at 864 atoms and about 0.10 from 10 976 atoms up, in each of two sets of timings. From 864 to 97 556 atoms, 113 times as many, both times grow roughly in proportion to N , the force evaluation's somewhat faster. Analysed at a frame saved every tenth step, the bonds cost about 1% of the run, and with any learned potential, whose forces cost far more than these, much less.

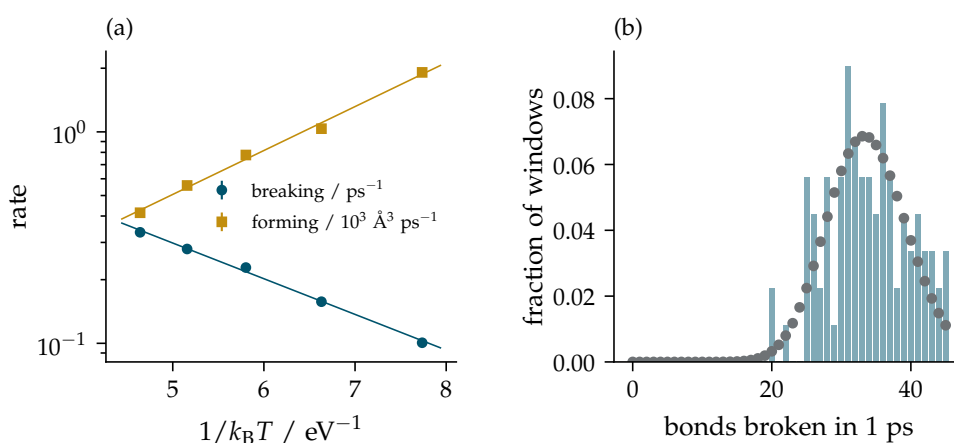


FIGURE 16.5: The A-B gas. (a) The rate of breaking per bond and picosecond (circles) and the rate constant of forming in $10^3 \text{ \AA}^3 \text{ ps}^{-1}$ (squares), counting events of the band that lived at least 0.5 ps, with the error $\sqrt{n}$ of each count, against $1/k_B T$, with straight lines fitted to their logarithms. (b) At 2000 K, the fraction of 89 windows of 1 ps in which a given number of bonds broke (bars), against the Poisson distribution of the same mean (points).

KEY IDEA. A reaction is a change of the molecules that some atoms belong to, found as a connected group of old and new molecules. Rates are counts over exposures, with the error $\sqrt{n}$ of a Poisson count; the ratio of the rate constants of a reaction and its reverse is the equilibrium constant, which statistical mechanics predicts. Observers analyse the run as it goes, reusing the force evaluation's pairs, at about a tenth of its cost.

NOTEBOOK 16 count reactions in the gas and put error bars on their rates

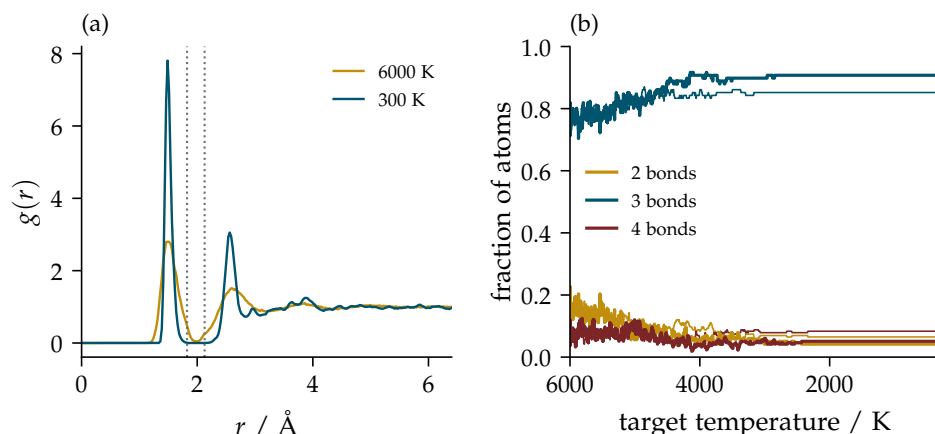


FIGURE 16.6: 216 carbon atoms at 2.0 g cm^{-3} with Tersoff's potential, quenched from 6000 K. (a) $g(r)$ of the melt and of the solid after the slow quench, with r_{on} and r_{off} of the default band (dotted). (b) The fractions of atoms with two, three and four bonds against the thermostat's target, for the quench over 10 ps (thin) and over 40 ps (thick).

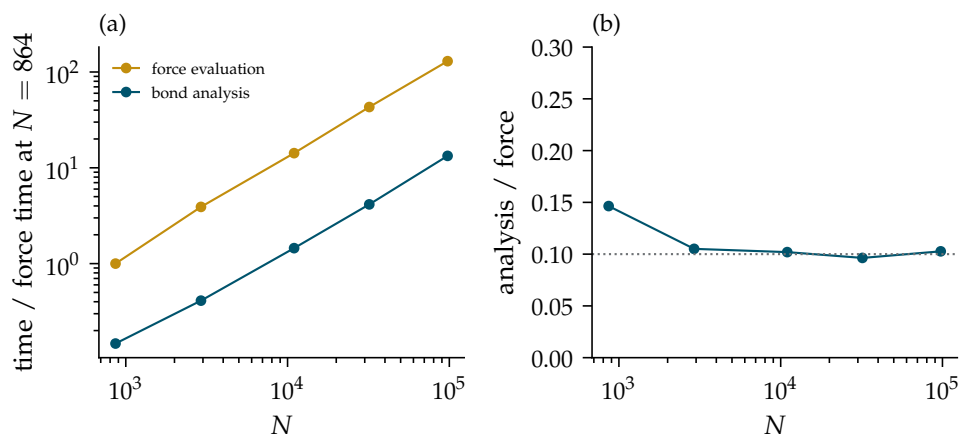


FIGURE 16.7: (a) The time of one force evaluation of liquid argon's model and of one frame's bond analysis, against the number of atoms, each relative to the force evaluation of 864 atoms, the least of three tries. (b) Their ratio, with the tenth that the analysis is meant to stay within (dotted).

CHECK YOUR UNDERSTANDING. A rate is estimated from 25 counted events. To what fraction is it known, and how many events would halve that?

16.5 How far atoms wander

A drop of ink in a glass of still water spreads without any current to carry it: each grain of dye is knocked about by the molecules around it, and the cloud of grains widens. No grain goes anywhere in particular, yet the width of the cloud grows steadily with time. Atoms in a liquid wander in the same way, and the rate at which the width grows is

the diffusion coefficient, which Chapter 13 measured as a sixth of a slope and promised to treat properly here.

A random walk

A walker who takes steps $\mathbf{s}_1, \mathbf{s}_2, \dots$, each of random direction and independent of the others, is at $\mathbf{r}_n = \mathbf{s}_1 + \dots + \mathbf{s}_n$ after n of them. The square of that sum is the sum of all products of two steps, so

$$\langle |\mathbf{r}_n|^2 \rangle = \sum_{k=1}^n \langle |\mathbf{s}_k|^2 \rangle + \sum_{k \neq l} \langle \mathbf{s}_k \cdot \mathbf{s}_l \rangle = n \langle s^2 \rangle,$$

the cross terms vanishing because independent steps of mean zero have zero covariance (Section 15.2). The mean squared distance from the start grows in proportion to the number of steps, and so to the time, while the mean position stays where it began. The diffusion coefficient D is defined by that growth: for an atom in d directions,

$$\langle |\mathbf{r}(t_0 + t) - \mathbf{r}(t_0)|^2 \rangle \rightarrow 2dDt \quad \text{as } t \text{ grows,} \quad (16.4)$$

which is $6Dt$ in three dimensions, the six of Chapter 13. The left side is the mean squared displacement, the MSD, averaged over atoms and over starting times t_0 .

The factor $2d$ makes D the coefficient of Fick's law, by which a difference of density drives a flow of atoms. In one dimension let $\rho(x, t)$ be the number density of some marked atoms and let them flow at the rate $j = -D \partial \rho / \partial x$ per unit area and time, down the gradient. Atoms are neither made nor lost, so the number between x and $x + dx$, ρdx per unit area, changes by what flows in at x less what flows out at $x + dx$, $j(x) - j(x + dx) = -(\partial j / \partial x) dx$, which gives $\partial \rho / \partial t = -\partial j / \partial x = D \partial^2 \rho / \partial x^2$. If the marked atoms all start at $x = 0$, their mean square distance $\sigma_x^2 = \int x^2 \rho dx / \int \rho dx$ changes as

$$\frac{d\sigma_x^2}{dt} = \frac{D \int x^2 \partial^2 \rho / \partial x^2 dx}{\int \rho dx} = \frac{D \int 2\rho dx}{\int \rho dx} = 2D,$$

the denominator being constant because atoms are kept. The middle step integrates by parts twice (Section 6.3), with ρ and its slope vanishing far away: $\int x^2 \rho'' dx = -\int 2x \rho' dx = \int 2\rho dx$, primes marking slopes in x . So $\sigma_x^2 = 2Dt$ in each direction, and the d directions add.

The mean squared displacement of a run

Three things make Equation (16.4) work on a trajectory. The positions must be unwrapped: an atom that leaves the cell and is put back at the far side has not jumped a cell's width, and the integrator of `md.run` keeps each path continuous for this reason (Section 10.1). The centre of mass must be taken out of every frame, so that a drift of the whole system does not count as wandering. A run started with velocities whose total momentum was not removed drifts for ever, and a thermostat that does not conserve momentum lets the centre wander: Langevin's friction and random kicks act on each atom separately, so their sums over the atoms do not cancel, and the total momentum performs a random walk of its own. And the average should run over every starting time, not the first frame alone, since every stretch of a settled run is as good a start as any other. The function `msd` of `mdlab.analysis.transport` does all three, returning the MSD along each axis; `diffusion_coefficient` fits a straight line to a window of it by least squares (Section 15.5) and divides the slope by $2d$.

At short times the MSD is not linear. Over a time short against that between collisions an atom keeps its velocity, so its displacement is $\mathbf{v}t$ and $\langle |\mathbf{v}t|^2 \rangle = (3k_B T/m) t^2$ by equipartition (Section 12.6): the motion is ballistic, the MSD growing as t^2 . The two forms meet where $(3k_B T/m)t^2 = 6Dt$, at $t = 2Dm/k_B T$, which for liquid argon is 274 fs. Figure 16.8(a) shows both regimes for the argon of Chapter 15 at fixed energy, from a run of 100 ps whose paths are summed from velocities saved every step. After a picosecond the MSD is a straight line of slope $6D$, and the fit over 2 to 20 ps gives $D = 3.763 \times 10^{-4} \text{ \AA}^2 \text{ fs}^{-1}$ at 132.1 K. A fit must start beyond the ballistic part: one over 0.02 to 0.2 ps gives a third less, while windows from t_1 to $10t_1$ with t_1 from 0.05 ps on agree within 6% (Section 16.12). The ratio $\text{MSD}/6t$ approaches D more slowly, reaching 0.9 of it only at 0.5 ps, because the straight line does not pass through the origin.

The error comes from blocks of the run, as in Chapter 15. The 2 ns run at fixed energy of Section 15.6 cut into 20 blocks of 100 ps gives $D = (3.834 \pm 0.034) \times 10^{-4} \text{ \AA}^2 \text{ fs}^{-1}$ at 133.70 K, its value there. Fits that use only the first frame of each block as the start spread 2.6 times as widely as those that use every frame, which is the reason to average over starting times. The value depends on the size of the cell as $1/L$ (Section 15.7).

Checked against ASE. ASE's `DiffusionCoefficient` fits each axis's MSD, from the first frame of each segment alone and with no centre removed, over every lag. On four segments of 50 ps of the same run, `msd` with one start per segment and no centre removed gives the same slopes to a relative 1×10^{-15} . The ASE estimate is the noisier of the two for the reason above; the comparison checks the arithmetic, and ASE's time unit, which is $1/\sqrt{\text{FORCE_T0_ACCEL}}$ femtoseconds.

Water

Chapter 11 left the oxygen atoms of its rigid water wandering by 1.41, 4.14 and 5.76 Å, root mean square from the first frame, after 1, 5 and 10 ps and promised a diffusion coefficient. Its 10 ps run of 64 molecules at 304 K, fitted over 1 to 5 ps, gives $D = (5.24 \pm 0.47) \times 10^{-4} \text{ Å}^2 \text{ fs}^{-1}$, the error from the scatter of the 64 molecules' own fits. A run so short gives D to a tenth; the 64-molecule box would also need the correction for its size of [Section 15.7](#).

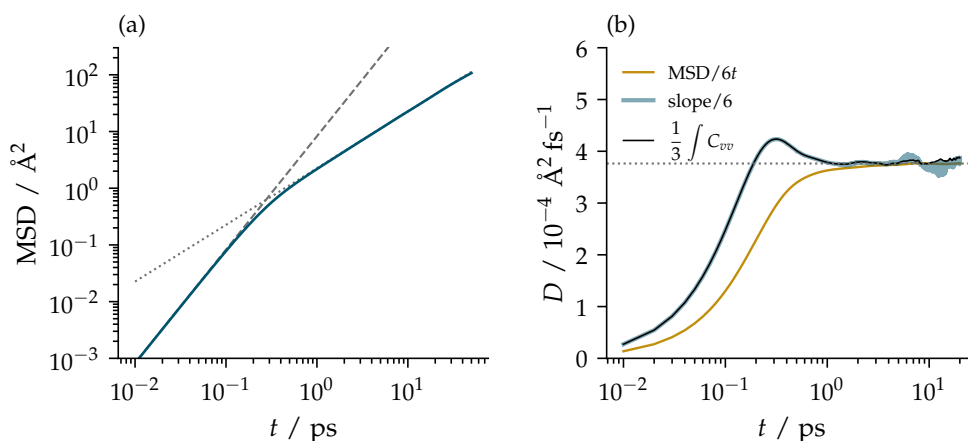


FIGURE 16.8: Liquid argon at fixed energy and 132.1 K (100 ps, paths summed from velocities saved every 10 fs). (a) The mean squared displacement against time, with the ballistic $3k_B T t^2 / m$ (dashed) and $6Dt$ (dotted). (b) Three estimates of D against the time they reach: $\text{MSD}/6t$, the slope of the MSD divided by 6 (thick, pale), and the Green-Kubo integral $\frac{1}{3} \int_0^t C_{vv}$ of [Section 16.6](#) (thin), which coincide; the dotted line is D from the fit over 2 to 20 ps.

KEY IDEA. The mean squared displacement grows as $2dDt$ once the motion forgets its velocity, and as $(3k_B T / m)t^2$ before. D is fitted to its linear part, from unwrapped paths with the centre of mass removed and averaged over every starting frame, with its error from blocks and a correction for the size of the cell.

NOTEBOOK 16 *move the fitting window across the MSD of liquid argon*

CHECK YOUR UNDERSTANDING. An atom of mass 40 amu in a liquid at 300 K has $D = 3 \times 10^{-4} \text{ Å}^2 \text{ fs}^{-1}$. Roughly when does its motion stop being ballistic?

16.6 Velocities that remember

A walk whose steps are independent spreads as $2dDt$, but an atom's velocity is not new at each instant: it changes only as collisions turn it, and for a while each velocity resembles the one before, as the samples of Chapter 15 resembled their predecessors. The memory of the velocity sets how far the atom gets, and measuring it gives the

diffusion coefficient by a second route, and other transport coefficients by the same one.

The velocity autocorrelation function

The velocity autocorrelation function is

$$C_{vv}(t) = \langle \mathbf{v}(t_0) \cdot \mathbf{v}(t_0 + t) \rangle,$$

averaged over the atoms and over the starting times t_0 , the covariance of Chapter 15 for a vector quantity. At $t = 0$ it is $\langle v^2 \rangle = 3k_B T/m$, and it falls towards zero as collisions scatter the velocity. In liquid argon ([Figure 16.9a](#)) it reaches zero at 320 fs and goes negative, to $-0.056 C_{vv}(0)$ at 420 fs: an atom that has run into the cage of its neighbours rebounds, and its velocity points, on average, backwards. In the crystal at 21 K, where the cage never opens, it oscillates, reaching $-0.62 C_{vv}(0)$ at 330 fs, the atom swinging back and forth about its site. `velocity_autocorrelation` of `mdlab` computes the average over every pair of frames by the fast Fourier transform, a tool [Section 16.9](#) explains.

From the velocity to the displacement

The displacement is the integral of the velocity, $\mathbf{r}(t_0 + t) - \mathbf{r}(t_0) = \int_0^t \mathbf{v}(t_0 + t') dt'$. Its square is a double integral, and averaging,

$$\langle |\mathbf{r}(t_0 + t) - \mathbf{r}(t_0)|^2 \rangle = \int_0^t \int_0^t \langle \mathbf{v}(t_0 + t') \cdot \mathbf{v}(t_0 + t'') \rangle dt' dt'' = \int_0^t \int_0^t C_{vv}(t'' - t') dt' dt'',$$

since in a settled run the average depends only on the separation of the two times. For the same reason C_{vv} is even: shifting the start from t_0 to $t_0 - t$ turns $\langle \mathbf{v}(t_0) \cdot \mathbf{v}(t_0 - t) \rangle$ into $\langle \mathbf{v}(t_0 - t) \cdot \mathbf{v}(t_0) \rangle$, so $C_{vv}(-t) = C_{vv}(t)$. The integrand is constant along the lines $t'' - t' = s$ across the square $0 \leq t', t'' \leq t$. The line at separation s runs from $(0, s)$ to $(t - s, t)$, a length $\sqrt{2}(t - s)$, and the line at $s + ds$ lies parallel to it at the distance $ds/\sqrt{2}$, so the strip between them has the area $(t - s) ds$; the lines at $-s$ cut out an equal strip. Collecting the strips,

$$\langle |\mathbf{r}(t_0 + t) - \mathbf{r}(t_0)|^2 \rangle = 2 \int_0^t (t - s) C_{vv}(s) ds, \quad (16.5)$$

the same counting of pairs by their separation that gave the variance of a correlated mean in Equation (15.3). Written as $2t \int_0^t C_{vv} ds - 2 \int_0^t s C_{vv} ds$, it has the slope in t

$$2 \int_0^t C_{vv}(s) ds + 2tC_{vv}(t) - 2tC_{vv}(t) = 2 \int_0^t C_{vv}(s) ds,$$

by the product rule (Section 2.7) for the first term and, for both, the fundamental theorem of calculus, by which the slope of an integral with respect to its upper limit is the integrand there (Section 1.4). Once C_{vv} has died away this slope is constant and equals $6D$ by Equation (16.4), so

$$D = \frac{1}{3} \int_0^\infty C_{vv}(t) dt, \quad (16.6)$$

a Green-Kubo formula: a transport coefficient as the integral of the autocorrelation function of a current, here the velocity of one atom⁹³.

The two routes are one identity, and on the same run they agree. Figure 16.8(b) plots the slope of the MSD divided by 6 and the running integral $\frac{1}{3} \int_0^t C_{vv}$, which lie on each other at every t ; from 1 to 10 ps the integral strays from its final value by at most 1.2%. Averaged over 2 to 10 ps, it gives $D = 3.784 \times 10^{-4} \text{ Å}^2 \text{ fs}^{-1}$, 0.5% above the fit to the MSD over 2 to 20 ps. The integral overshoots before it settles, to 1.12 times its final value at 320 fs, where C_{vv} crosses zero, because the negative part that follows takes back some of what came before: the rebound from the cage slows diffusion.

The viscosity

The same form gives the shear viscosity, the friction of one layer of a liquid sliding over another, as the integral of the autocorrelation function of the shear stress, the off-diagonal pressure P_{xy} of Section 14.4:

$$\eta_s = \frac{V}{k_B T} \int_0^\infty \langle P_{xy}(0) P_{xy}(t) \rangle dt.$$

Its derivation needs the response of a liquid to a slow shear, which this book does not develop; it is given by Allen and Tildesley⁹³. Chapter 15 recorded P_{xy} , P_{xz} and P_{yz} at every step of twelve runs of liquid argon at 133.7 to 135.3 K, the two of 2 ns and the ten of 1 ns at five sizes, 14 ns in all. Each component of each run gives one estimate, and the integrals, averaged and given the standard error of the twelve runs, reach 0.98 of their plateau by 1 ps (Figure 16.9b). Over 1 to 3 ps,

$$\eta_s = (1.853 \pm 0.038) \times 10^{-4} \text{ Pa s},$$

or 2.04 ± 0.04 in the reduced unit $\sqrt{m\epsilon}/\sigma^2$. Section 15.7 inferred $(2.18 \pm 0.20) \times 10^{-4}$ Pa s from the way D changed with the size of the box through Equation (15.6), and the two lie 1.6 combined errors apart: the change of D with size is the return flow of the hydrodynamic theory, to within its error. With this η_s the correction of Equation (15.6) for 256 atoms is $0.651 \times 10^{-4} \text{ \AA}^2 \text{ fs}^{-1}$, which added to the 256-atom value of 3.914×10^{-4} gives 4.565×10^{-4} , against the $(4.503 \pm 0.032) \times 10^{-4}$ extrapolated from five sizes.

Beyond 3 ps the integral wanders: the correlation has gone, and what is added is noise whose spread grows with the upper limit, as the integral of a sum of noisy terms does. The integral is read where it has settled and not later (Section 16.12).

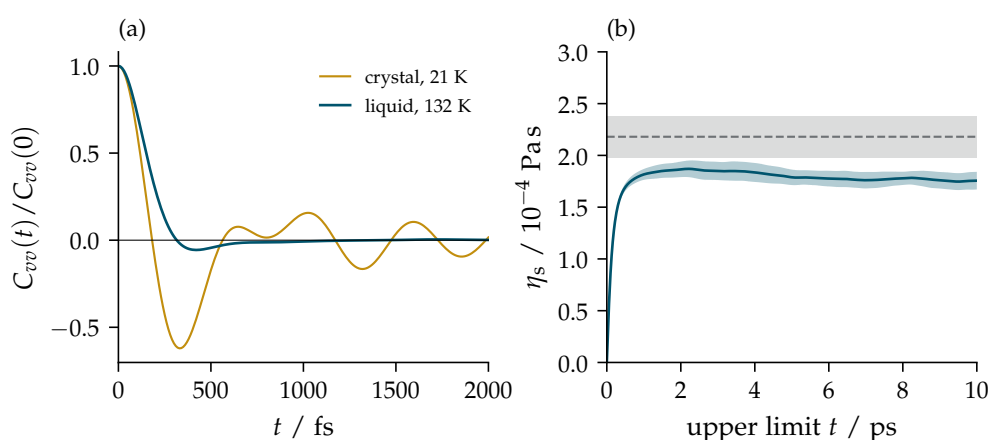


FIGURE 16.9: (a) The velocity autocorrelation function of liquid argon at fixed energy and 132 K and of the fcc crystal at 21 K, each divided by its value at $t = 0$. (b) The Green-Kubo integral of the shear stress, $(V/k_B T) \int_0^t \langle P_{xy}(0)P_{xy}(t') \rangle dt'$, from the 14 ns of Chapter 15's runs near 135 K, with twice its standard error over the twelve runs (band), against the viscosity that Equation (15.6) inferred from the change of D with size (dashed, with its error).

KEY IDEA. The mean squared displacement is twice the integral of the velocity autocorrelation function weighted by $t - s$, so D is a third of the integral of C_{vv} , a Green-Kubo formula. Both routes give the same D from the same run. The shear viscosity is the same kind of integral of the shear stress's autocorrelation, read where it has settled.

NOTEBOOK 16 *integrate the velocity autocorrelation function and watch it settle*

CHECK YOUR UNDERSTANDING. Why does the negative dip of C_{vv} in a liquid make D smaller than the integral of its first, positive part alone?

16.7 Charge transport

An electrolyte carries current through the motion of its ions. If each ion moved independently, its diffusion coefficient would determine its contribution to the conductivity. But a cation and an anion that move together as a neutral pair can diffuse without carrying net charge. We first derive the independent-ion estimate, then include correlations and measure their effect in a model molten salt.

Drift against diffusion

Suppose each ion of some kind feels a steady force F_x along x , for instance from an electric field. Its collisions with its neighbours act, on average, as a friction that grows with its velocity, so it does not speed up for ever: it drifts with a mean velocity v_d in the direction of the force, at which the friction balances the force, while diffusion spreads it. The drift and the diffusion are tied together. If the force comes from a potential energy $U(x)$, so that $F_x = -dU/dx$, the ions settle into the canonical density $\rho(x) \propto e^{-\beta U(x)}$ (Section 12.4), whose slope is $d\rho/dx = -\beta\rho dU/dx = \beta F_x \rho$. At equilibrium no net flow remains, so the drift's flow ρv_d and Fick's flow $-D d\rho/dx$ cancel:

$$\rho v_d - D\beta F_x \rho = 0, \quad v_d = \beta D F_x,$$

the Einstein relation: the drift velocity per unit force is the diffusion coefficient divided by $k_B T$. The same atom that wanders fast also drifts fast when pushed.

In an electric field $\mathcal{E}$ along x , ion i , of charge q_i , feels $F_x = q_i \mathcal{E}$ and drifts at $\beta D_i q_i \mathcal{E}$, with D_i the diffusion coefficient of its kind. The current density, the charge carried across unit area in unit time, is the sum over the ions of each charge times its drift velocity, divided by the volume, $\sum_i q_i \times \beta D_i q_i \mathcal{E} / V$. Divided by $\mathcal{E}$, this is the conductivity of independent ions, the Nernst-Einstein value

$$\sigma_{\text{NE}} = \frac{1}{V k_B T} \sum_i q_i^2 D_i \quad (16.7)$$

cations and anions add, q_i^2 being positive for both. In the units of m³mol⁻¹, e² per Å, fs and eV, a conductivity of 1 is $1.602 \times 10^6 \text{ S m}^{-1}$, the siemens being the ampere per volt.

The conductivity of the whole

Ions do not move independently. The general result has the form of [Equation \(16.4\)](#) for the charge displacement $\mathbf{M}_q(t) = \sum_i q_i \mathbf{r}_i(t)$, the sum of every ion's position weighted by its charge:

$$\sigma_e = \frac{1}{6Vk_B T} \lim_{t \rightarrow \infty} \frac{d}{dt} \langle |\mathbf{M}_q(t_0 + t) - \mathbf{M}_q(t_0)|^2 \rangle, \quad (16.8)$$

whose derivation, like the viscosity's, uses the response of the system to a weak field⁹³. Expanding the square shows how it differs from [Equation \(16.7\)](#). With $\Delta \mathbf{r}_i$ the displacement of ion i ,

$$\langle |\Delta \mathbf{M}_q|^2 \rangle = \sum_i q_i^2 \langle |\Delta \mathbf{r}_i|^2 \rangle + \sum_i \sum_{j \neq i} q_i q_j \langle \Delta \mathbf{r}_i \cdot \Delta \mathbf{r}_j \rangle.$$

The first sum grows as $\sum_i q_i^2 6D_i t$ and gives [Equation \(16.7\)](#); the second holds the correlations between different ions. A cation and an anion that move together have $q_i q_j < 0$ and $\langle \Delta \mathbf{r}_i \cdot \Delta \mathbf{r}_j \rangle > 0$, a negative term, and lower the conductivity below σ_{NE} . Their ratio is the Haven ratio $H_R = \sigma_{NE} / \sigma_e$. Like D , σ_e also follows from a Green-Kubo integral, of the autocorrelation function of the charge current $\sum_i q_i \mathbf{v}_i$, by the steps of [Equation \(16.5\)](#).

A model molten salt

The test system is 64 ions in a cubic cell of side 13.4 Å, 32 of charge $+e$ with the mass of sodium and 32 of $-e$ with that of chlorine. Every pair repels as $Ae^{-r/r_{rep}}$ with $r_{rep} = 0.3$ Å, and $A = 6231$ eV is set so that a pair of opposite charges has its least energy at 2.8 Å, where the repulsion's slope $Ae^{-r/r_{rep}}/r_{rep}$ balances the Coulomb pull $k_e e^2 / r^2$; the Coulomb energy is summed by Ewald's method ([Section 10.3](#)). The model is illustrative, chosen to resemble an alkali halide, a salt of a metal such as sodium with a halogen such as chlorine, and is not fitted to sodium chloride. Eight runs each melt the lattice at 4000 K, hold 1500 K with CSVR for 40 ps and then run 1 ns at fixed energy with steps of 2 fs. Their energies differ, since 64 ions leave the thermostatted stage with an energy that fluctuates widely, so they sit at temperatures from 1362 to 1721 K.

The liquid is ordered by charge ([Figure 16.10a](#)). Around a cation the anions crowd at 3.07 Å, where g_{+-} rises to 3.65, and other cations lie farther out, g_{++} peaking at 1.66 at 4.37 Å, g_{--} alike at 4.32 Å; each cation has 6.76 anions out to the first minimum of g_{+-} , at 4.87 Å.

Each run gives σ_e from [Equation \(16.8\)](#), with the charge displacement recorded at every step and fitted over 1 to 5 ps, and σ_{NE} from the two kinds' D over the same window. The Green-Kubo route agrees with the displacement in every run to within 0.6%. Both

conductivities rise with the run's temperature (Figure 16.10b), and the Nernst-Einstein value lies above the true one by more than half a percent in six runs of eight, within it in one and below it in one. Over the eight, at $(1487 \pm 45) \text{ K}$, $\sigma_e = (158 \pm 9) \text{ S m}^{-1}$ and $\sigma_{\text{NE}} = (168 \pm 10) \text{ S m}^{-1}$, errors that come mostly from the spread of the runs' temperatures. Their ratio, taken run by run, shows no trend with the temperature in the figure, and averages

$$H_R = 1.062 \pm 0.022,$$

2.8 errors above 1, beyond the 2.365 that Student's factor for eight runs, seven degrees of freedom, sets at 95% (Section 15.1): the ions' motions are correlated, and treating them as independent overstates the conductivity of this melt by about 6%. The diffusion coefficients of the two kinds over the eight runs, (5.19 ± 0.46) and $(5.03 \pm 0.45) \times 10^{-4} \text{ Å}^2 \text{ fs}^{-1}$ for the cations and the anions, are the same within errors that again reflect the temperatures; taken run by run, which removes the temperature, their ratio is 1.033 ± 0.008 , the cations faster in seven runs of eight.

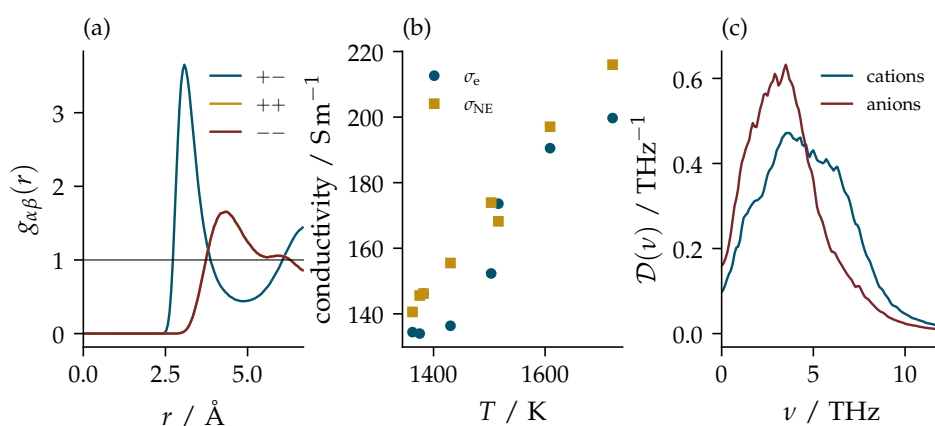


FIGURE 16.10: The model molten salt, 64 ions in eight runs of 1 ns at fixed energy. (a) The partial radial distribution functions g_{+-} , g_{++} and g_{--} . (b) The conductivity of each run from the charge displacement (circles) and from Nernst-Einstein with the ions' own diffusion coefficients (squares), against the run's temperature. (c) The vibrational density of states of the cations and of the anions in the first run (Section 16.10).

KEY IDEA. An ion pushed by a force F_x drifts at $\beta D F_x$, so independent ions conduct as $\sigma_{\text{NE}} = \sum_i q_i^2 D_i / V k_B T$. The true conductivity comes from the mean squared displacement of the total charge displacement $\sum_i q_i \mathbf{r}_i$, whose cross terms hold the correlations between ions; the Haven ratio $\sigma_{\text{NE}} / \sigma_e$ measures them.

NOTEBOOK 16 compare the charge displacement with the ions' own displacements

CHECK YOUR UNDERSTANDING. If every cation were permanently bound to one anion, moving as a neutral pair, what would σ_e be, and what would σ_{NE} still give?

16.8 Diffusion in layers

Lithium in graphite can move between the carbon sheets much more readily than it can cross them. A single diffusion coefficient hides this difference: we need the mean squared displacement in each direction. After measuring it in the layered model solid, we connect diffusion to hops between lattice sites and test the assumption that successive hops are independent.

The diffusion tensor

The MSD of Equation (16.4) is the sum of three parts, $\langle \Delta x^2 \rangle + \langle \Delta y^2 \rangle + \langle \Delta z^2 \rangle$, and each grows as $2D_{xx}t$, $2D_{yy}t$ and $2D_{zz}t$ with its own coefficient. More generally the products of the components define the diffusion tensor, $\langle \Delta r_\alpha \Delta r_\beta \rangle \rightarrow 2D_{\alpha\beta}t$, a symmetric 3×3 matrix whose eigenvectors are the directions in which the spreading is fastest and slowest (Section 4.9). In a layered solid with its sheets at right angles to z , symmetry makes z one of them and leaves the two in the plane equal, $D_{xx} = D_{yy}$, and the diffusion coefficient that a single number would give, a third of the trace, mixes two very different motions.

Section 14.8 built a model of such a solid: triangular sheets, each bound strongly within itself and weakly to its neighbours, with 108 guest atoms of lithium's mass grown into the hollows between them. Held at 150 K by CSVr in its relaxed cell for 500 ps, its guests' MSD along each axis, measured from the centre of mass, gives from five blocks of 100 ps and fits over 5 to 50 ps

$$D_{xx} = (1.16 \pm 0.11) \times 10^{-4} \text{ \AA}^2 \text{ fs}^{-1}, \quad D_{yy} = (1.02 \pm 0.05) \times 10^{-4} \text{ \AA}^2 \text{ fs}^{-1}.$$

The two in the plane agree within their errors. Along z the MSD is $0.113 \text{ \AA}^2$ at 1 ps, $0.126 \text{ \AA}^2$ at 10 and $0.132 \text{ \AA}^2$ at 50 ps (Figure 16.11a), and no guest moves more than $1.73 \text{ \AA}$ along z in the whole run: the guests rattle between their two sheets, and none crosses to the next gap, $3.40 \text{ \AA}$ away. A single crossing would add $11.6 \text{ \AA}^2$ to one guest's square, and so $11.6 \text{ \AA}^2 / 108$ to the mean, and one in the 500 ps would have meant $D_{zz} \approx 1.1 \times 10^{-7} \text{ \AA}^2 \text{ fs}^{-1}$. This is the scale of one detectable crossing, not a statistical upper bound on D_{zz} : a finite run can miss a rare event. It is a thousand times smaller than the in-plane value. A straight line fitted to the levelling MSD would only measure its slow approach to the plateau; Exercise 17.14 shows how zero events give a confidence bound.

Hops between sites

On the model surface of [Section 3.13](#) a lithium atom moves by hops between hollows a distance $a = 2.46 \text{ \AA}$ apart, and Chapter 13 counted them. If each hop goes to one of the six neighbouring hollows at random, independently of the last, the atom performs the random walk of [Section 16.5](#) with steps of length a , and after n hops its mean squared displacement is na^2 . At a rate of k_h hops per unit time, $n = k_h t$, and comparing $k_h t a^2$ with $2dDt$ for $d = 2$,

$$D = \frac{k_h a^2}{4}.$$

Counted hops thus give a diffusion coefficient without fitting an MSD, and a count has the error bar of [Section 16.4](#), $\sqrt{n}$. If the hops have various lengths ℓ but still forget their direction, the same sum gives $n\langle\ell^2\rangle$ after n hops, and $D = k_h\langle\ell^2\rangle/4$.

The relation assumes that hops forget their direction. At 1000 K, with Langevin friction of 1 ps^{-1} , 1000 atoms over 200 ps make 179 680 hops, a rate of $k_h = (0.8984 \pm 0.0021)$ per atom and picosecond, Chapter 13's rate again; $k_h a^2/4$ is then $13.6 \times 10^{-4} \text{ \AA}^2 \text{ fs}^{-1}$, but the MSD over 2 to 10 ps gives $D = 92.3 \times 10^{-4} \text{ \AA}^2 \text{ fs}^{-1}$, 6.79 times more. The ratio $f = 4D/k_h a^2$, the correlation factor, exceeds 1 for two reasons. An atom with enough energy to cross a bridge often keeps it long enough to cross the next before the friction takes it away, so that 17% of the counted hops, from one hollow where the atom is seen to the next, are longer than a , and their mean square is $1.96a^2$. And successive hops tend to keep their direction: $4D/k_h\langle\ell^2\rangle$, which would be 1 for hops that forget it, is 3.46. Raising the friction to 3, 10 and 30 ps^{-1} drains the energy faster: f falls to 3.24, 1.63 and 1.13, as the product of $\langle\ell^2\rangle/a^2$, 1.66, 1.29 and 1.07, and the memory of direction, 1.96, 1.27 and 1.05 ([Figure 16.11b](#)), while k_h stays near 0.90 until the highest friction, which slows the hops themselves to 0.708 per picosecond, as Chapter 13 found. How large f is in a real crystal depends on how fast the lattice takes away an arriving atom's energy, which only a measurement of both D and k_h settles.

KEY IDEA. In an anisotropic solid the MSD is taken along each direction, giving the diffusion tensor; in a layered solid the motion along the layers and across them differ by orders of magnitude. Diffusion by hops between sites a distance a apart gives $D = f k_h a^2/4$ in two dimensions, with $f = 1$ when successive hops are independent and one site long, and $f > 1$ when they carry the atom past the next site or keep their direction.

NOTEBOOK 16 *split the MSD of the guests into its three directions*

CHECK YOUR UNDERSTANDING. On a simple cubic lattice of spacing a in three dimensions, with independent hops at a total rate k_h , what is D ?

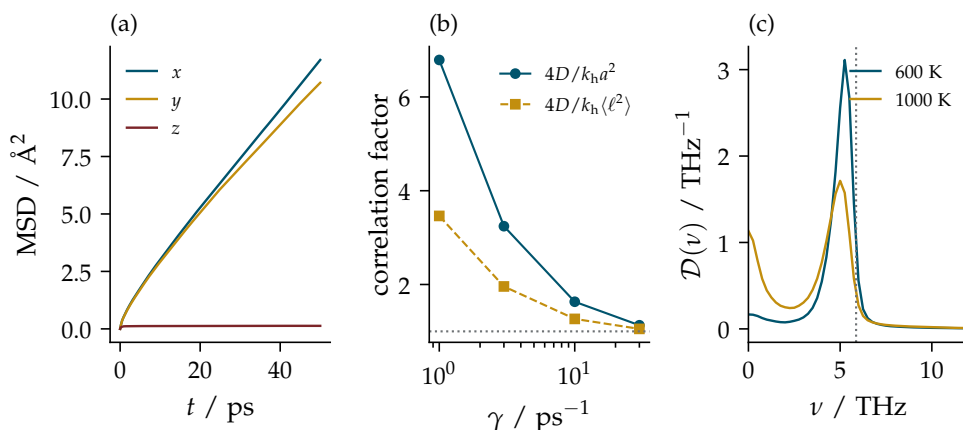


FIGURE 16.11: (a) The guests of the layered model solid at 150 K: their mean squared displacement along x , y and z , measured from the centre of mass. (b) Lithium on the model surface at 1000 K: the correlation factor $f = 4D/k_h a^2$, with D from the MSD and k_h from counted hops (circles), and $4D/k_h \langle \ell^2 \rangle$, with $\langle \ell^2 \rangle$ the mean square length of a counted hop (squares), against the Langevin friction. (c) The vibrational density of states of the lithium atoms at 600 and 1000 K (friction 1 ps^{-1}), with the frequency of small vibrations in a hollow (dotted; Section 16.10).

16.9 Frequencies in a signal

A chord struck on a piano reaches the ear as one wobbling pressure, yet a listener hears two notes, and a tuner's display shows two peaks at their pitches. Something takes a signal in time apart into the frequencies it holds. The same operation turns the velocities of a run into the frequencies its atoms vibrate at, and it has been used twice already without explanation, to compute correlation functions quickly in Chapter 15 and in Section 16.6. This section supplies it: complex numbers, the Fourier transform, its discrete form, the fast algorithm, and what sampling a signal at intervals does to the frequencies one can see.

COMPLEX NUMBERS AND EULER'S FORMULA. A complex number $z = x + iy$ is a pair of real numbers x and y combined with a symbol i whose square is -1 . They add and multiply as ordinary numbers do, replacing i^2 by -1 wherever it appears: $(a + ib)(c + id) = ac - bd + i(ad + bc)$. Drawn as the point (x, y) , z is an arrow from the origin; its length is $|z| = \sqrt{x^2 + y^2}$, and with the conjugate $\bar{z} = x - iy$, $z\bar{z} = x^2 + y^2 = |z|^2$. The Taylor series of Section 1.8 for e^x , $\cos x$ and $\sin x$ are

$$e^x = \sum_{n \geq 0} \frac{x^n}{n!}, \quad \cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots, \quad \sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots.$$

Putting $x = i\theta$ in the first, the even powers $(i\theta)^{2m} = (-1)^m \theta^{2m}$ give the series of $\cos \theta$ and the odd powers $(i\theta)^{2m+1} = i(-1)^m \theta^{2m+1}$ give i times that of $\sin \theta$:

$$e^{i\theta} = \cos \theta + i \sin \theta,$$

Euler's formula, an arrow of length 1 at the angle θ . With $-\theta$ in place of θ it gives $e^{-i\theta} = \cos \theta - i \sin \theta$, and adding and subtracting the two,

$$\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2}, \quad \sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i},$$

so any cosine or sine is a sum of two such exponentials. Exponentials multiply by adding their powers, $e^a e^b = e^{a+b}$, for complex numbers as for real ones: multiplying the two series and collecting the terms of total power n gives $\sum_l a^l b^{n-l} / l!(n-l)! = (a+b)^n / n!$ by the binomial expansion, which uses only the rules of adding and multiplying. So $e^{iA} e^{iB} = e^{i(A+B)}$, which written out is the pair of addition formulae of Equation (1.6): turning by A and then by B turns by $A+B$. In this language the sums of Equation (16.2) are the real and imaginary parts of one sum, $C + iS = \sum_j e^{i\mathbf{G} \cdot \mathbf{r}_j}$, and $S(\mathbf{G})$ is its squared length over N .

The Fourier transform

A signal $x(t)$ that repeats with period t_p is a sum of cosines and sines that repeat with it (Section 10.3), which by the toolbox are sums of $e^{2\pi i m t / t_p}$ for whole numbers m , positive and negative, each at the frequency m/t_p :

$$x(t) = \sum_m c_m e^{2\pi i m t / t_p}, \quad c_m = \frac{1}{t_p} \int_{-t_p/2}^{t_p/2} x(t) e^{-2\pi i m t / t_p} dt,$$

the coefficients found, as in Chapter 10, because the average of $e^{2\pi i(m-m')t/t_p}$ over a period is 1 for $m = m'$ and 0 otherwise. A signal that does not repeat is the limit of a long period. The frequencies $\nu_m = m/t_p$ then lie a small distance $\Delta\nu = 1/t_p$ apart, and with $\hat{x}(\nu_m) = t_p c_m$ the series reads $x(t) = \sum_m \hat{x}(\nu_m) e^{2\pi i \nu_m t} \Delta\nu$, a sum of thin strips that becomes an integral over ν as $\Delta\nu$ shrinks (Section 1.4), while the integral for $\hat{x}(\nu_m)$ comes to cover all times:

$$\hat{x}(\nu) = \int_{-\infty}^{\infty} x(t) e^{-2\pi i \nu t} dt, \quad x(t) = \int_{-\infty}^{\infty} \hat{x}(\nu) e^{2\pi i \nu t} d\nu, \quad (16.9)$$

the Fourier transform and its inverse. $\hat{x}(\nu)$ is a complex number for each frequency, whose length says how much of that frequency the signal holds and whose angle gives the wave's phase. A signal observed for a time t_f cannot separate frequencies closer

than about $1/t_f$, since two waves that differ by less than that drift apart by less than a whole cycle over the record: a chord recorded for 50 ms shows its notes only if they differ by more than 20 Hz.

The discrete Fourier transform

A run records a signal at N_s equally spaced times $t_n = n\Delta t$, $n = 0, \dots, N_s - 1$, and the integral becomes a sum,

$$X_k = \sum_{n=0}^{N_s-1} x_n e^{-2\pi i k n / N_s}, \quad x_n = \frac{1}{N_s} \sum_{k=0}^{N_s-1} X_k e^{2\pi i k n / N_s}, \quad (16.10)$$

the discrete Fourier transform, X_k belonging to the frequency $\nu_k = k/N_s\Delta t$. The inverse holds because of a geometric sum. Putting the first sum, with the index n' , into the second gives $\frac{1}{N_s} \sum_{n'} x_{n'} \sum_k w^k$ with $w = e^{2\pi i(n-n')/N_s}$, and $\sum_{k=0}^{N_s-1} w^k = (1 - w^{N_s})/(1 - w)$, where $w^{N_s} = e^{2\pi i(n-n')} = 1$; so the sum over k is 0 unless $w = 1$, that is $n' = n$, when it is N_s , and what remains is x_n . For a real signal, changing k to $N_s - k$ only turns each $e^{-2\pi i k n / N_s}$ the other way, so $X_{N_s-k} = \bar{X}_k$: the upper half of the transform mirrors the lower and belongs to the negative frequencies, and a cosine of height 1 puts $N_s/2$ at each of its two places. Summed directly, Equation (16.10) needs N_s products for each of the N_s values of k , N_s^2 in all; the function `dft` of `mlab.analysis.spectra` does this. Figure 16.12(a) shows $2|X_k|/400$, which is the height of each wave, for the notes A (440 Hz) and E (659.25 Hz) of height 1, sampled 8000 times a second for 50 ms. The 400 samples are padded with zeros to 512, the power of 2 that the fast transform below needs, whose grid of $8000/512 = 15.6$ Hz puts the two peaks at the frequencies nearest the notes, 437.5 and 656.25 Hz.

The fast Fourier transform

The sum splits into its even and odd samples, $n = 2p$ and $n = 2p + 1$:

$$X_k = \sum_{p=0}^{N_s/2-1} x_{2p} e^{-2\pi i k p / (N_s/2)} + e^{-2\pi i k / N_s} \sum_{p=0}^{N_s/2-1} x_{2p+1} e^{-2\pi i k p / (N_s/2)} = X_k^e + e^{-2\pi i k / N_s} X_k^o,$$

with X^e and X^o the transforms of the even and the odd samples, each of length $N_s/2$. These repeat after $N_s/2$ values of k , and $e^{-2\pi i(k+N_s/2)/N_s} = -e^{-2\pi i k / N_s}$, so for $k < N_s/2$

$$X_k = X_k^e + e^{-2\pi i k / N_s} X_k^o, \quad X_{k+N_s/2} = X_k^e - e^{-2\pi i k / N_s} X_k^o :$$

two transforms of half the length and N_s further operations give the whole. When N_s is a power of 2 the halving repeats down to transforms of one sample, each its own

transform, and the work $W(N_s) = 2W(N_s/2) + N_s$ adds N_s at each of $\log_2 N_s$ levels, $N_s \log_2 N_s$ in all. This is the fast Fourier transform of Cooley and Tukey⁹⁴, the one Chapter 10 promised for particle mesh Ewald. In `mdlab`:

```

1 def fft(x: ArrayLike) -> NDArray:
2     x = np.asarray(x, dtype=complex)
3     n = len(x)
4     if n == 1:
5         return x
6     if n % 2:
7         raise ValueError("the length must be a power of 2")
8     even, odd = fft(x[0::2]), fft(x[1::2])
9     twiddle = np.exp(-2j * np.pi * np.arange(n // 2) / n) * odd
10    return np.concatenate([even + twiddle, even - twiddle])

```

`dft` and `fft` agree with NumPy's FFT to 4×10^{-12} and 6×10^{-14} on 256 samples. From $N_s = 64$ to 4096, in three timings on the same machine, the direct sum takes about two thousand times as long, half the 4096 by which N_s^2 grows, since a fixed cost of each call weighs most at small N_s ; the recursive `fft` about 65 times, half the 128 of $N_s \log_2 N_s$, for the same reason; and NumPy's compiled FFT about 10 times (Figure 16.12c). Every later spectrum uses NumPy's.

Sampling and aliasing

Samples taken every Δt cannot tell the frequency ν from $1/\Delta t - \nu$: at $t_n = n\Delta t$,

$$\cos(2\pi(1/\Delta t - \nu)n\Delta t) = \cos(2\pi n - 2\pi\nu n\Delta t) = \cos(2\pi\nu n\Delta t),$$

since a whole number of turns changes no cosine (the sine changes sign). The highest frequency that samples can show without confusion is half the sampling rate, $\nu_N = 1/2\Delta t$, the Nyquist frequency; a higher one appears folded back below it, at its alias. Films show this when the spokes of a turning wheel seem to rotate slowly backwards. The E of the chord, sampled 1000 times a second, gives samples that lie equally on a wave of $1000 - 659.25 = 340.75$ Hz (Figure 16.12b), to 1×10^{-15} . For a run the consequence is direct: frames saved every Δt cannot show vibrations faster than $1/2\Delta t$, and a vibration faster than that appears at a false frequency.

Correlation functions by the transform

The products of Equation (16.10) also compute correlations. For real records x and y , the conjugate of the transform of x is $\bar{X}_k = \sum_n x_n e^{2\pi i k n / N_s}$, and

$$\frac{1}{N_s} \sum_k \bar{X}_k Y_k e^{2\pi i k j / N_s} = \sum_n \sum_m x_n y_m \frac{1}{N_s} \sum_k e^{2\pi i k (n+j-m) / N_s} = \sum_n x_n y_{n+j},$$

since by the same geometric sum the inner sum is 1 when $m = n + j$ and 0 otherwise, with $n + j$ taken round the end of the record, as if the record repeated. Padding each record with N_s zeros first makes the wrapped products zero, leaving the sum over the $N_s - j$ true pairs at the lag j ; dividing by $N_s - j$ gives the average. Three transforms of length $2N_s$ then do the work of the $N_s^2/2$ products of every pair. `stats.autocorrelation` of Chapter 15 and `transport.correlation` of this chapter work this way.

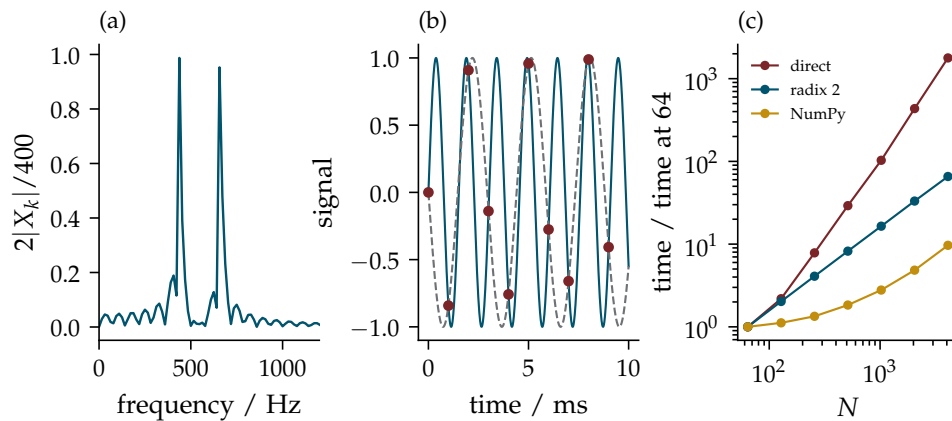


FIGURE 16.12: (a) The notes A (440 Hz) and E (659.25 Hz) of equal strength, sampled 8000 times a second for 50 ms and padded to 512 samples: $2|X_k|/400$, the height of each wave, against the frequency. (b) The E alone (line) sampled 1000 times a second (points): the samples lie on a wave of 340.75 Hz as well (dashed), its alias. (c) The time taken by the direct sum, by the recursive radix-2 fft and by NumPy's FFT against the number of samples, each relative to its own time at $N_s = 64$, the least of five tries.

KEY IDEA. The Fourier transform writes a signal as waves $e^{2\pi i \nu t}$; a record of length t_f resolves frequencies about $1/t_f$ apart. The discrete transform of N_s samples costs N_s^2 summed directly and $N_s \log_2 N_s$ by halving, the fast Fourier transform. Samples every Δt show frequencies only up to $1/2\Delta t$, and fold higher ones back below it. Correlation functions are computed through transforms of records padded with zeros.

NOTEBOOK 16 *sample a note and watch it fold back past the Nyquist frequency*

CHECK YOUR UNDERSTANDING. A vibration of 30 THz is recorded with frames every 20 fs. Where does it appear in the spectrum of the frames?

16.10 The vibrational density of states

Infrared and neutron spectra report, each weighting the atoms in its own way, at which frequencies the atoms of a material move, and so does a trajectory: an atom's velocity, taken apart into frequencies, says how much of its motion happens at each. Summed over the atoms this is the vibrational density of states, the spectrum by which a learned model of motion is first compared with the simulation it learned from¹.

The power spectrum of a signal

For a signal $x(t)$ recorded over a long time t_f , the power at the frequency ν is $|\hat{x}(\nu)|^2/t_f$, with $\hat{x}$ the transform of Equation (16.9) over the record. Writing the squared length as the transform times its conjugate,

$$|\hat{x}(\nu)|^2 = \int_0^{t_f} \int_0^{t_f} x(t) x(t') e^{-2\pi i \nu(t-t')} dt dt',$$

and averaging over many such records turns $x(t)x(t')$ into the autocorrelation function $C(t-t')$, which in a settled signal depends on the separation alone. Counting the pairs of times by their separation s , as for Equation (16.5), each s occurs over a length $t_f - |s|$, so

$$\frac{\langle |\hat{x}(\nu)|^2 \rangle}{t_f} = \int_{-t_f}^{t_f} \left(1 - \frac{|s|}{t_f}\right) C(s) e^{-2\pi i \nu s} ds \rightarrow \int_{-\infty}^{\infty} C(s) e^{-2\pi i \nu s} ds$$

once t_f is long against the time over which C dies away. The power spectrum is the Fourier transform of the autocorrelation function, the theorem of Wiener and Khinchin. Since C is even, the imaginary part of $e^{-2\pi i \nu s}$ cancels between s and $-s$, leaving $2 \int_0^{\infty} C(s) \cos(2\pi \nu s) ds$.

The density of states

Applied to the mass-weighted velocity autocorrelation function per atom, $C_m(t) = \langle m \mathbf{v}(t_0) \cdot \mathbf{v}(t_0 + t) \rangle$, the theorem gives the vibrational density of states, scaled so that its area counts the three directions of each atom:

$$\mathcal{D}(\nu) = \frac{4}{k_B T} \int_0^{\infty} C_m(t) \cos(2\pi \nu t) dt, \quad \int_0^{\infty} \mathcal{D}(\nu) d\nu = 3. \quad (16.11)$$

With $\hat{C}_m(\nu) = 2 \int_0^{\infty} C_m(t) \cos(2\pi \nu t) dt$ the transform of C_m , even in ν , $\mathcal{D} = 2\hat{C}_m/k_B T$, and the area follows from the inverse transform at $t = 0$: $C_m(0) = \int_{-\infty}^{\infty} \hat{C}_m d\nu = 2 \int_0^{\infty} \hat{C}_m d\nu$, so the area of $\mathcal{D}$ is $C_m(0)/k_B T$, and $C_m(0) = \langle m v^2 \rangle = 3k_B T$ by equipartition. For a harmonic solid the motion is a sum of normal modes (Section 4.9), each

oscillating at its own frequency ν_k with the energy $k_B T$ on average. Each mode adds $k_B T \cos(2\pi\nu_k t)/N$ to C_m per atom, a spike of area $1/N$ in $\mathcal{D}$ at ν_k , so $\mathcal{D}$ is the histogram of the mode frequencies, $3N$ spikes for N atoms making the area 3: the classical density of states. In a liquid, atoms that wander contribute at zero frequency, where Equation (16.11) and Equation (16.6) give

$$\mathcal{D}(0) = \frac{4}{k_B T} \int_0^\infty C_m(t) dt = \frac{12 m D}{k_B T},$$

so the density of states holds the diffusion coefficient too.

A run gives C_m only out to some lag t_c , and cutting the integral off sharply there adds ripples to $\mathcal{D}$, as cutting $g(r)$ at half the cell did to $S(G)$ in Section 16.2. `spectra.vdos` multiplies C_m by the window $\cos^2(\pi t/2t_c)$, which falls smoothly from 1 to 0, transforms it by the discrete transform of a record extended to negative times, and scales the result to the area 3. The window blurs each peak over about $1/t_c$; a longer t_c sharpens the peaks and needs a longer run to keep the noise down.

A crystal, with and without friction

The fcc crystal at 21 K, 50 ps at fixed energy with velocities every 10 fs and C_m taken to 10 ps, has peaks at 0.50, 0.75, 0.95, 1.15, 1.40 and 1.90 THz (Figure 16.13a). Its exact harmonic frequencies come from the Hessian of the same 256-atom cell (Section 4.9), by finite differences of the forces: three zero eigenvalues, the translations of the whole cell, and 765 modes up to 1.991 THz. A finite cell has a few distinct frequencies, and their bare histogram has spikes that the run's windowed spectrum cannot show, so the fair comparison passes the modes' own $C_m(t) \propto \sum_k \cos(2\pi\nu_k t)$ through the same window and transform. The overlap of the two spectra is 0.839 by the score defined below. They agree in the positions of the peaks, but the run's tallest peak is 3.74 per terahertz against the modes' 4.28, and the run puts 0.192 of its area of 3 above the highest harmonic frequency, against 0.011 for the modes through the same window: the run's motion is not exactly harmonic. These overlaps are taken below 3 THz, where the crystal's spectrum lies.

Friction widens every peak, as Chapter 4 found for the response of a driven spring. A mode of angular frequency ω_0 damped by Langevin friction γ swings at $\omega_d = \sqrt{\omega_0^2 - \gamma^2/4}$ (Section 4.6), and when the friction is small against ω_0 its velocity correlation decays as $e^{-\gamma|t|/2} \cos \omega_d t$, with a term in $\sin \omega_d t$ of relative size $\gamma/2\omega_d$ left out; at the crystal's lowest peak, 0.5 THz, that size is 0.32, so the form is rough there. Each half of the cosine, $\frac{1}{2}e^{\pm i\omega_d t}$, transforms into

$$\frac{1}{2} \int_{-\infty}^{\infty} e^{-\gamma|t|/2} e^{-i\delta t} dt = \frac{1}{2} \left[\frac{1}{\gamma/2 + i\delta} + \frac{1}{\gamma/2 - i\delta} \right] = \frac{\gamma/2}{\gamma^2/4 + \delta^2},$$

with $\delta = \omega \mp \omega_d$ and $\omega = 2\pi\nu$, the integral over positive times being $\int_0^\infty e^{-(\gamma/2+i\delta)t} dt$ and that over negative times its conjugate. This is a peak at $\pm\omega_d$, a Lorentzian, that falls to half its height where $\delta = \pm\gamma/2$: a full width of γ in angular frequency, or $\gamma/2\pi$ in ν . Under friction of 2 ps^{-1} that is 0.318 THz. Spreading the fixed-energy spectrum by a Lorentzian of that width reproduces the Langevin run's (Figure 16.13b): the overlap rises from 0.797 for the unspread spectrum to 0.950, and the tallest peak falls from 3.74 per terahertz to 2.01 in the Langevin run and 2.13 in the spread spectrum. A spectrum computed under a thermostat that acts on each atom belongs to the thermostat as much as to the material.

A liquid, and two kinds of atom

The argon liquid's density of states (Figure 16.13c) has no sharp peaks, a broad band near 0.5 THz, and a value at zero of 1.659 per terahertz, against $12mD/k_B T = 1.651$ with the Green-Kubo D of the same run. In the molten salt of Section 16.7 each kind of ion has its own spectrum (Figure 16.10c), computed from that kind's velocities alone: the cations' band peaks at 3.70 and the anions' at 3.50 THz, and the anions, of 1.54 times the mass and nearly the same D , have the larger value at zero, 0.160 per terahertz against 0.098. Spectra of this kind, one for each element, are how TrajCast's predictions are compared with molecular dynamics¹.

The comparison is condensed into a number by the overlap score of Schran and co-workers⁹⁵,

$$\text{overlap} = 1 - \frac{\int |\mathcal{D}_1 - \mathcal{D}_2| d\nu}{\int \mathcal{D}_1 d\nu + \int \mathcal{D}_2 d\nu},$$

which is 1 for identical curves and 0 for curves that do not overlap at all. The two ions of the salt score 0.790 against each other; the crystal's run against its harmonic modes 0.839.

A stride's Nyquist limit

A learned propagator predicts frames a stride Δt apart, and its spectrum can be computed only from those frames; so can that of the reference simulation, if it is to be compared fairly. By Section 16.9, frames every Δt show frequencies only up to $1/2\Delta t$. For the strides of TrajCast's tests, below 7 fs, that is 71.4 THz or 2383 cm^{-1} , about the 2400 cm^{-1} that the paper quotes as the range it can resolve¹. A faster vibration reappears folded below the limit.

A chain of four beads of 14.5 amu, each standing for a group CH_2 or CH_3 , shows this. Its bonds have the energy $\kappa_r(r - r_{\text{eq}})^2$ with $\kappa_r = 5\text{ eV \AA}^{-2}$, its angles likewise, and a torsion twists it about its middle bond; Section 16.11 sets the model out in full. Twenty copies

are run at fixed energy after 10 ps at 300 K, with each copy's centre-of-mass velocity removed (Figure 16.14). From every frame, 1 fs apart, the spectrum has peaks at 0.50, 4.00, 9.00 and 12.25 THz and two stretches of the bonds at 18.25 and 21.00 THz. The bond alone, a spring of stiffness $2\kappa_r$, the slope of the force $-2\kappa_r(r - r_{\text{eq}})$, between two beads of reduced mass m_r (Section 4.8), would vibrate at $\sqrt{2\kappa_r/m_r}/2\pi = 18.36$ THz. From every 20th frame, with its limit at 25 THz, the spectrum is the same, and the overlap with that from every frame, both cut at 25 THz, is 1.000. From every 30th, sampled at $1/\Delta t = 33.33$ THz with its limit at 16.67 THz, the two stretches fold to $33.33 - 18.25 = 15.08$ and $33.33 - 21.00 = 12.33$ THz, and the spectrum has peaks at 15.15 and 12.37 THz, where nothing moves. The overlap falls from 1.000 at 20 fs to 0.899 at 25 and 0.795 to 0.797 from 30 to 40 fs, the change coming as the stride passes $1/(2 \times 21.00 \text{ THz}) = 23.8$ fs. Spectra from strided frames must therefore be compared with reference spectra taken at the same stride, and peaks near or beyond the limit read with suspicion.

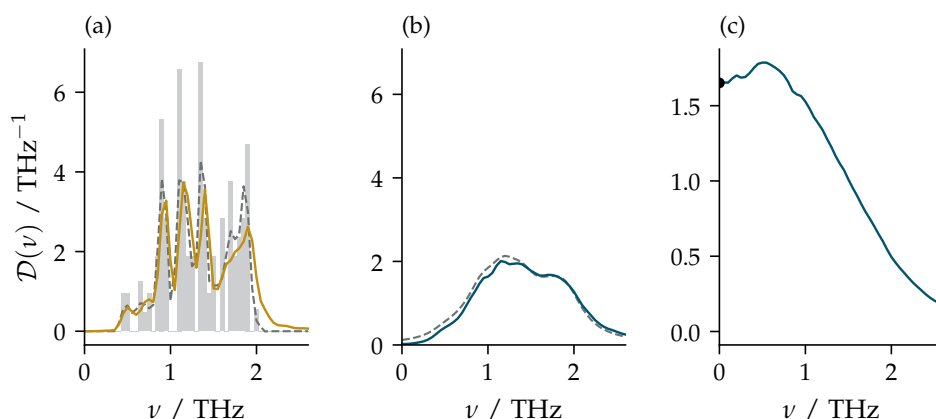


FIGURE 16.13: (a) The vibrational density of states of the fcc crystal at 21 K at fixed energy, against the histogram of the harmonic frequencies of the same cell (bars) and those frequencies passed through the same window and transform (dashed). (b) The crystal under Langevin friction of 2 ps^{-1} , against the fixed-energy spectrum spread by a Lorentzian of full width $\gamma/2\pi$ (dashed). (c) Liquid argon at fixed energy, with $12mD/k_B T$ at $\nu = 0$ (point).

How often an atom tries

Chapter 12 noted that how often an atom hops depends also on how often it tries, and left the measurement to this chapter. A lithium atom in a hollow of the model surface vibrates, and each swing towards a bridge is an attempt. Its density of states (Figure 16.11c) peaks at 5.25 and 5.00 THz at 600 and 1000 K, below the 5.87 THz of small vibrations in the hollow, because the hollow widens as the atom climbs. That frequency is $\nu_0 = \sqrt{k/m}/2\pi$, with k the curvature of the energy at the bottom of the hollow (Section 4.4). Near a hollow each of the model's three ripples (Section 3.13) has $\cos(\mathbf{g}_k \cdot \mathbf{r}) \approx 1 - (\mathbf{g}_k \cdot \mathbf{r})^2/2$, and the squared cosines of the angles between $\mathbf{r}$ and the

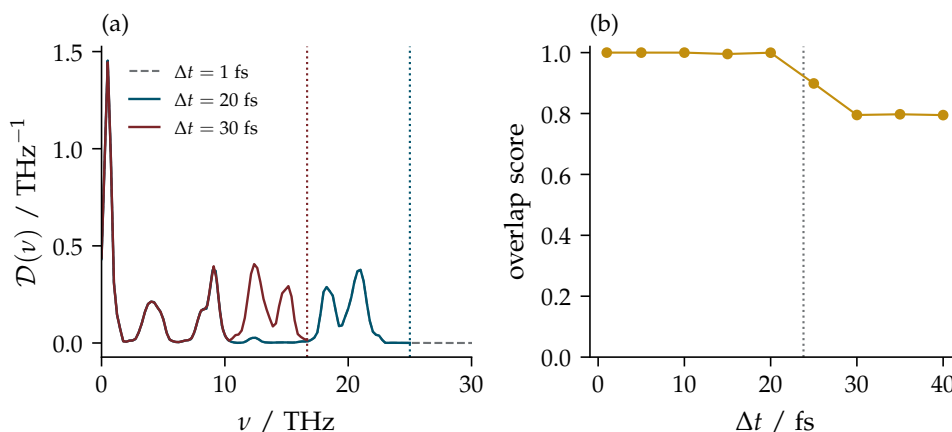


FIGURE 16.14: The four-bead chain at fixed energy, 20 copies, velocities every 1 fs with each copy's centre-of-mass velocity removed. (a) The vibrational density of states from every frame (dashed), from every 20th and from every 30th, each to its own Nyquist frequency (dotted lines). (b) The overlap score of the spectrum from every Δt -th frame with that from every frame, both cut at the Nyquist frequency of Δt , against Δt ; the dotted line marks $1/2\nu_b$, with ν_b the frequency of the highest peak.

three $\mathbf{g}_k$, 120° apart, add to $\frac{3}{2}$, so the energy rises as $\frac{3}{16}U_b|\mathbf{g}|^2r^2$ and $k = 3U_b|\mathbf{g}|^2/8 = 2\pi^2U_b/a^2 = 0.9785 \text{ eV } \text{\AA}^{-2}$. With six bridges out of each hollow, the estimate $6\nu e^{-\beta U_b}$ with this ν gives 0.095 and 0.923 hops per picosecond, against 0.134 and 0.898 counted and the transition-state rates 0.162 and 1.257 of Equation (13.5). The estimate takes the attempt frequency from the spectrum and ignores how wide the passage over each bridge is compared with the hollow, and it lies within a factor of 1.41 of the counted rates: the frequency of attempts and the Boltzmann factor of the barrier give a rate's size, and the counted hops give its value.

KEY IDEA. The vibrational density of states is the Fourier transform of the mass-weighted velocity autocorrelation function, with area three per atom; for a harmonic solid it is the histogram of the normal-mode frequencies, and at zero frequency it is $12mD/k_B T$. Friction widens each peak by $\gamma/2\pi$. Frames a stride Δt apart show frequencies only up to $1/2\Delta t$ and fold faster ones below it, so spectra are compared at the same stride, by an overlap score.

NOTEBOOK 16 set the length of the correlation window and the stride, and watch the spectrum

CHECK YOUR UNDERSTANDING. The spectrum of one atom of a liquid is computed with C_m cut at 1 ps. Roughly how close can two peaks be and still be seen as two?

16.11 Free energy from a histogram

A trajectory spends more time in some arrangements than in others. At equilibrium these populations measure free-energy differences. A histogram of a distance, angle or position therefore gives a free-energy profile along that coordinate, provided the run has sampled it adequately. We test this construction on a surface and a four-bead chain, where the answer is known, then examine why rare crossings make the uncertainty hard to estimate.

Free energy along a coordinate

In the canonical ensemble the positions have the density $\mathcal{P} \propto e^{-\beta U}$ (Section 12.4). For a coordinate x computed from the positions, the density of its values is the total weight of all arrangements with that value, and writing it as a Boltzmann factor defines the free energy along x :

$$\mathcal{P}(x) \propto e^{-\beta F(x)}, \quad F(x) = -k_B T \ln \mathcal{P}(x) + \text{constant}, \quad (16.12)$$

the free energy $-k_B T \ln Z$ of Section 12.4 for the arrangements held at that x . If x is every coordinate of the system, $F = U$ up to a constant. If it is fewer, F includes the number of arrangements that share each value, an entropy, and can differ from any energy. A free-energy profile from a run is a histogram of x , as for $g(r)$, turned into F by Equation (16.12). Its error comes from blocks, as in Chapter 15: the histogram of each block, normalised to a density; the mean and the standard error of each bin over the blocks; and, since a small change $\delta\mathcal{P}$ changes $\ln \mathcal{P}$ by $\delta\mathcal{P}/\mathcal{P}$, an error of $k_B T \delta\mathcal{P}/\mathcal{P}$ in F . `mdlab.analysis.landscape` does this for one coordinate or several:

```

1 def free_energy(
2     values: ArrayLike,
3     bins: int | ArrayLike | list,
4     temperature: float,
5     n_blocks: int = 10,
6     value_range: list | None = None,
7 ) -> tuple[list[NDArray], NDArray, NDArray]:
8     x = np.asarray(values, dtype=float)
9     x = x[:, None] if x.ndim == 1 else x
10    edges = np.histogramdd(x, bins=bins, range=value_range)[1]
11    blocks = np.array_split(x, n_blocks)
12    density = np.array(
13        [np.histogramdd(b, bins=edges, density=True)[0] for b in blocks]
14    )
15    p = density.mean(axis=0)
16    dp = density.std(axis=0, ddof=1) / np.sqrt(n_blocks)
17    kt = KB * temperature
18    with np.errstate(divide="ignore", invalid="ignore"):
```

```

19         f = -kt * np.log(p)
20         df = kt * dp / p
21     f -= np.min(f[np.isfinite(f)])
22     middles = [0.5 * (e[1:] + e[:-1]) for e in edges]
23     return middles, f, df

```

An atom on a surface

A lithium atom on the model surface of [Section 3.13](#) has two coordinates, and its position is all of them, so $F(x, y)$ must equal $U(x, y)$. 1000 atoms at 600 K under Langevin friction of 1 ps^{-1} , every 50 fs for 200 ps, folded into one cell of the hollows, give the map of [Figure 16.15\(a\)](#). A profile is tested against an exact one by the mean of its squared misses divided by their squared errors. Each such ratio has the mean 1 when the errors are right and, for misses of Gaussian density, the variance $\langle z^4 \rangle - 1 = 2$, so over n independent bins the mean scatters about 1 by $\sqrt{2/n}$; a mean of c says that the errors understate the misses by about $\sqrt{c}$. Below 0.3 eV, over 1400 bins, F less U gives 1.20, where independent bins with correct errors would give 1.00 ± 0.04 . Histogram bins are correlated, so this spread is only a guide: it is constant to within about 1.1 times its errors, and the largest miss is 15.2 meV.

The distance r_h from the nearest hollow is one coordinate of the two, and $F(r_h)$ is not U . The positions at a distance r_h fill a circle of circumference $2\pi r_h$, so the density of r_h is

$$\mathcal{P}(r_h) \propto r_h \oint e^{-\beta U(r_h, \theta)} d\theta, \quad F(r_h) = -k_B T \ln \left[r_h \oint e^{-\beta U(r_h, \theta)} d\theta \right],$$

with θ the angle around the hollow, valid out to $a/2 = 1.23 \text{ \AA}$, where the circle still lies in the region nearer that hollow than any other. The factor r_h is the number of positions at each distance, and it makes F rise as $r_h \rightarrow 0$, where U is lowest: there are few places that close to the centre, as there are few velocities with all three components near zero in Maxwell's density of speeds ([Section 12.5](#)). The run's $F(r_h)$ follows this within its errors, its squared misses over squared errors averaging 0.60 over 40 bins ([Figure 16.15b](#)), and its least value lies at $r_h = 0.225 \text{ \AA}$ from the hollow. For small r_h , $U \approx \frac{1}{2} k r_h^2$ with $k = 3U_b |\mathbf{g}|^2 / 8 = 2\pi^2 U_b / a^2$ ([Section 16.10](#)), so $\mathcal{P} \propto r_h e^{-\beta k r_h^2 / 2}$, whose largest value is at $r_h = \sqrt{k_B T / k} = 0.230 \text{ \AA}$. At the smallest r_h , F is 116 meV above its least. A free-energy profile along a coordinate carries the count of arrangements at each value, and a minimum of F need not be a minimum of U .

A dihedral angle

The free energy of a dihedral angle is how the TrajCast paper judges the slow motions of a molecule¹, and a model chain has an exact answer for it. The chain has four

beads of 14.5 amu, each standing for a CH₂ or CH₃ group, joined by springs in the force field of Equation (8.8): bonds with $\kappa_r = 5 \text{ eV } \text{\AA}^{-2}$ and $r_{\text{eq}} = 1.53 \text{ \AA}$, angles with $\kappa_\theta = 2.5 \text{ eV rad}^{-2}$ and $\theta_{\text{eq}} = 112^\circ$, and a torsion with $\kappa_1 = 0.06$, $\kappa_2 = -0.01$ and $\kappa_3 = 0.12 \text{ eV}$, which puts the lowest well at $\psi = 180^\circ$ (trans), two gauche wells 36.4 meV higher at $\pm 63.7^\circ$, barriers of 127.8 meV at $\pm 118.1^\circ$ and 180 meV at 0° . The constants are chosen for illustration. The forces of `mdlab.forcefield.BondedModel` match finite differences of its energy to $1 \times 10^{-8} \text{ eV } \text{\AA}^{-1}$.

With no other terms, the density of ψ is exactly $e^{-\beta U_t(\psi)}$, U_t the torsion energy. Building the chain bead by bead, the second bead's position relative to the first is given by a length and two angles, the third's relative to the second by r_2 , the bond angle θ_1 and an angle about the first bond, and the fourth's by r_3 , θ_2 and its angle about the middle bond, which is ψ . Each step is a set of spherical coordinates: small changes of r , of θ and of the angle ϕ about the axis move the bead by dr , $r d\theta$ and $r \sin \theta d\phi$ at right angles to one another, a small volume $r^2 \sin \theta dr d\theta d\phi$. The volume for the whole chain is a product of such factors of r and $\sin \theta$ times $d\psi$, with no factor that depends on ψ . The energy is a sum of a part in the bonds and angles and a part in ψ , so integrating over everything but ψ leaves $e^{-\beta U_t(\psi)}$ times a constant, and $F(\psi) = U_t(\psi)$ up to a constant.

Two hundred copies, each started in the trans well, are held by Langevin friction of 1 ps^{-1} at 300 K and at 200 K for 2 ns, ψ recorded every 100 fs, and the first 200 ps left out: over it the trans well holds 0.720 and 0.903 of the copies, against the 0.664 and 0.801 of equilibrium. Each bin of $F(\psi)$ is compared with $-k_B T \ln$ of the bin's average of $e^{-\beta U_t}$, since a histogram counts the whole bin. At 300 K, where each copy changes well 83.5 times per nanosecond, F matches the torsion energy within its errors (Figure 16.15c): the mean of the squared misses over squared errors is 1.02 over 72 bins with blocks in time and 0.68 with blocks of 20 copies each, and at $\psi \approx 0$ the run gives $(181 \pm 2) \text{ meV}$ against 179.

At 200 K each copy changes well only 8.0 times per nanosecond, and the profile is harder to get right. Its shape is right, $F(0) = (177 \pm 4) \text{ meV}$ against 179, but with blocks in time the squared misses over squared errors average 3.26, and they lie mostly on one side, where counts are plentiful: 7.8 between -120° and 0, 1.3 between 0 and 120° . Over the 1.8 ns kept, the gauche well near $+64^\circ$ holds 0.0158 ± 0.0088 more of the copies than the one near -64° , though by symmetry the two must hold equal shares. That is 1.8 of its errors, an ordinary fluctuation when the error is taken over the 200 independent copies; but each copy stays in a well for 125 ps on average, so the populations of consecutive blocks in time are not independent and their spread understates the error, by the mechanism of Section 15.3. Blocks of copies instead give errors 40% larger (a median of 0.92 against 0.66 meV) and a mean of 2.05, much of what remains being that single fluctuation, which moves every bin of a well together: folding ψ to $|\psi|$, which pools the two gauche wells and so cancels it, brings the mean to 1.06 with blocks of copies, while blocks in time still give 2.10. Free-energy profiles

across barriers crossed rarely need their error bars from independent runs, or from blocks far longer than the time between crossings.

IN PRACTICE. A free-energy profile is no better than the number of crossings between its wells, and counting them is the check. For a two-well process with independent, approximately exponential stays, n crossings mean each side is visited about $n/2$ times, its share of the time known to about $\sqrt{2/n}$ of itself, so the ratio of the two shares is known to about $2/\sqrt{n}$, and the difference of free energy to $2k_B T/\sqrt{n}$: two thirds of $k_B T$ for ten crossings and a tenth for 400, whatever the number of frames. Copies started in different wells, errors taken over independent copies, and the start left out, where the wells still hold the populations they began with, guard against the rest.

The distribution of the potential energy

The density of the potential energy itself is a profile of the same kind, and the TrajCast paper compares it between a learned model and the reference¹. It depends on the ensemble as well as the forces: Chapter 15's liquid argon spreads by 0.677 under CSVr and by 0.532 meV per atom at fixed energy, and the two distributions overlap by 0.882 by the score of Section 16.10, against 0.980 for the two halves of the same CSVr run. A test of a model by this distribution is a test of its forces only when both runs sample the same ensemble; with that settled, a score near the halves' own is the target.

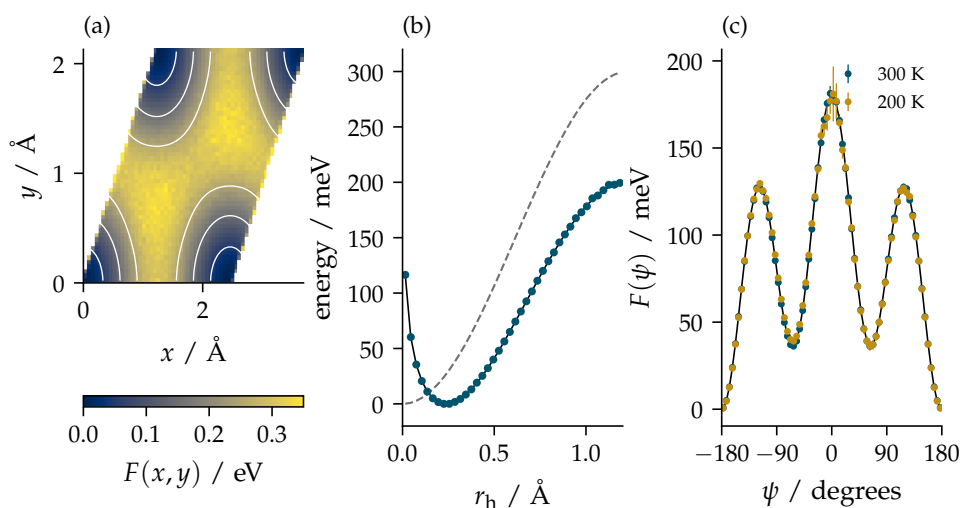


FIGURE 16.15: (a) Lithium on the model surface at 600 K, positions folded into one cell of the hollows: $F(x, y)$ (shading, 0–0.35 eV, with its additive constant aligned to U) and contours of U every 0.1 eV (lines). (b) The same atoms' free energy against the distance r_h to the nearest hollow (points, with twice their errors), against the exact $-k_B T \ln[r_h \oint e^{-\beta U} d\theta]$ (line) and U along the line from a hollow to a bridge (dashed). (c) The four-bead chain's dihedral angle at 300 and 200 K: $F(\psi)$ with twice its error from ten blocks of 20 copies, after the first 200 ps, against the torsion energy (line).

KEY IDEA. The free energy along a coordinate is $-k_B T \ln$ of the density of its values, estimated by a histogram, with errors from blocks. It equals the potential energy only when the coordinate is every coordinate; otherwise it counts the arrangements at each value. Across barriers crossed rarely, short blocks in time underestimate errors; use sufficiently long blocks or independent copies.

NOTEBOOK 16 build F from a histogram and watch the error bars at 200 K

CHECK YOUR UNDERSTANDING. Why does $F(r_h)$ for the lithium atom rise towards $r_h = 0$ although the potential energy is lowest there?

16.12 What the trajectory looks like

The observables of this chapter are computed after the run, from saved frames, and most of their faults are made then: in the frames chosen, in how they are prepared, or in where a curve is read. A healthy set of observables checks itself through the identities derived above. For the liquid argon of this chapter, $g(r)$ with the pair potential gives the run's energy and pressure, to 0.6 and 0.2 of their errors (Section 16.1); $S(G)$ from the positions agrees with the transform of $g(r)$ where the cut at half the cell does not matter and tends to $\rho k_B T \kappa_T$ at small G (Section 16.2); the slope of the MSD and the integral of C_{vv} give the same D (Section 16.6); the density of states has the value $12mD/k_B T$ at zero frequency (Section 16.10); and where the free energy is known exactly, the histogram reproduces it within its errors (Section 16.11). Each of these fails, in a recognisable way, when something has gone wrong. Figure 16.16 and Table 16.1 show six faults on liquid argon and gather those of earlier sections.

Wrapped positions. Positions put back into the cell, as most trajectory files store them, give an MSD that jumps whenever an atom crosses a face. From the run of Section 16.5 wrapped every 100 fs, it is $167.8 \text{ \AA}^2$ at 20 ps against the true 45.2, and 222.0 and $258.0 \text{ \AA}^2$ at 50 and 100 ps, levelling off towards $270 \text{ \AA}^2$, $L^2/2$, the mean squared distance between two points placed at random in the cell (Figure 16.16a). The check is the largest move between frames, which must be a small fraction of the cell (`diagnostics.largest_jump`, Section 11.8), and the cure is unwrapping (`diagnostics.unwrap`).

A drift of the centre of mass. Starting velocities drawn without removing their total momentum give the centre of mass a speed of about $\sqrt{3k_B T/Nm}$, $1.80 \times 10^{-4} \text{ \AA fs}^{-1}$ here, which it keeps for ever at fixed energy. Added to the same paths, it makes the MSD curve upwards and the fit over 2 to 20 ps give D 31% too high (Figure 16.16b); removing the centre of mass, as `msd` does by default, restores 3.764×10^{-4} . The check is the path of the centre of mass (`diagnosticscentre_of_mass_path`).

A fit in the ballistic part. A window that ends before the motion has forgotten its velocity fits a straight line to a curve: over 0.02 to 0.2 ps it gives D a third too low, while windows starting at 0.05 ps or later and ending ten times later agree with the 2 to 20 ps value within 6% (Figure 16.16c). The check is the slope of $\ln \text{MSD}$ against $\ln t$, which must be 1 over the window.

$g(r)$ past half the cell. Separations reduced to their nearest image cannot exceed half the cell's width along any axis, so beyond $L/2$ the shells are partly empty. Computed out to 16 Å in the 256-atom cell of side 23.26 Å, g falls to 0.50 at 14 Å, where the 2048-atom liquid shows 1.010 (Figure 16.16d). rdf refuses a range beyond half the smallest width, as the cell list does.

Transport under a local thermostat. Langevin friction of 10 ps^{-1} damps every velocity: C_{vv} falls to half its first value in 60 fs instead of 160, and to zero in 170 instead of 320, and its integral gives $D = 0.682 \times 10^{-4} \text{ Å}^2 \text{ fs}^{-1}$, 82% below the value at fixed energy (Figure 16.16e). Transport coefficients and spectra come from runs at fixed energy or under a global thermostat such as CSV, which Chapter 15 found leaves D unchanged within its error; under friction they belong to the friction (Section 13.4).

A Green-Kubo integral read too late. The integral of a correlation function settles when the correlation has died and then wanders as noise accumulates. For one 1 ns run of 256 atoms the shear viscosity read at 2 ps, on the plateau, is $2.18 \times 10^{-4} \text{ Pa s}$, 18% above the 1.853×10^{-4} from all 14 ns, the spread of a single run; read later, at 10, 20 and 40 ps, it is 1.91, 1.78 and 2.01×10^{-4} (Figure 16.16f), wandering by as much again. The check is to read the plateau where the correlation has died, here 1 to 3 ps, and to take the error from independent runs.

KEY IDEA. Observables check one another: $g(r)$ gives the energy and the pressure, the MSD and the velocity autocorrelation give the same D , and the density of states holds D at zero frequency. Their common faults are wrapped positions, an unremoved drift, a fit window in the ballistic part, a range past half the cell, transport under a local thermostat, an integral read too late, frames too sparse, and error bars from blocks shorter than the slowest motion.

NOTEBOOK 16 break each observable and find the check that catches it

CHECK YOUR UNDERSTANDING. An MSD from a long run levels off at a value close to $L^2/2$. What is the most likely fault?

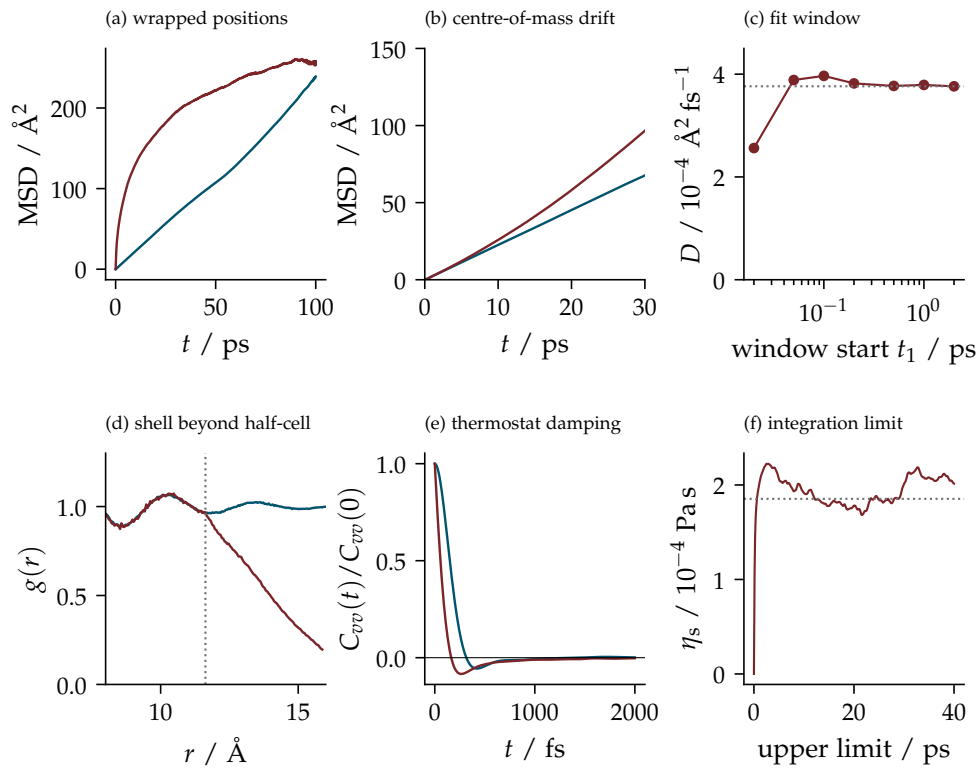


FIGURE 16.16: Six faults on liquid argon, each against the correct curve (teal). (a) The MSD from positions wrapped into the cell. (b) The MSD of paths whose centre of mass drifts at $\sqrt{3k_B T/Nm}$, not removed. (c) D fitted to windows from t_1 to $10t_1$, against t_1 , with the 2 to 20 ps value (dotted). (d) $g(r)$ of the 256-atom cell taken past half its side (dotted) with the nearest image, against the 2048-atom $g(r)$. (e) C_{vv} under Langevin friction of 10ps^{-1} . (f) The Green-Kubo integral of the shear stress from one 1 ns run, read out to 40 ps, against the value from 14 ns (dotted).

TABLE 16.1: Faults of observables, the check that finds each, and its size in this chapter's runs.

fault	check	measured
wrapped positions in an MSD	largest move between frames; unwrap	$168 \text{ \AA}^2$ at 20 ps, not 45
centre of mass drifting	path of the centre of mass; remove it	D 31% high
fit in the ballistic part $g(r)$ past half the cell	slope of $\ln \text{MSD}$ against $\ln t$ is 1 range at most half the smallest width	D 32% low $g = 0.50$ at $14 \text{ \AA}$, not 1.01
transport under friction Green-Kubo read too late bonds from one cutoff	run at fixed energy or with CSV read the plateau; errors from runs band of hysteresis; least lifetime	D 82% low at 10 ps^{-1} η_s from 1.78 to 2.18×10^{-4} 3.2 times the events (Section 16.3)
frames too sparse for a spectrum	stride below $1/2v_{\text{max}}$; same stride for both	overlap 0.795 at 30 fs (Section 16.10)
errors of F from short blocks	errors over independent copies	squared misses over squared errors 3.26, not 1 (Section 16.11)
energy distributions from two ensembles	same thermostat for both	overlap 0.882, not 0.980 (Section 16.11)

What we now have

The radial distribution function counts the partners at each distance against an even scatter, a histogram of pair distances out to half the cell that tends far away to $1 - S(0)/N$ in a fixed cell, $1 - 1/N$ for atoms placed at random; with the pair potential and the kinetic energy it gives the energy and the pressure, and with its short memory it converges several times sooner than the pressure. Its transform is the structure factor that diffraction measures, computed from the positions on the cell's own wave vectors where the cut at half the cell would spoil it, and tending to $\rho k_B T \kappa_T$ at long wavelengths. Bonds are assigned from distances with a band of hysteresis and a least lifetime, stored as whole-number keys whose sorted lists give the bonds formed and broken; molecules are the connected components of their graph, species their formula and a hash of their bonding; reactions are changes of the molecules that atoms belong to, and rates are counts over exposures with the error $\sqrt{n}$; the ratio of the rate constants of a reaction and its reverse is an equilibrium constant that statistical mechanics predicts. Observers do this as the run goes, at about a tenth of the cost of a cheap force evaluation. The mean squared displacement grows as $2dDt$, from unwrapped paths with the centre of mass removed, and equals twice the integral of the velocity autocorrelation function weighted by $t - s$, so D is also a Green-Kubo integral; the shear viscosity is another, and its value, $(1.853 \pm 0.038) \times 10^{-4}$ Pa s for liquid argon, lies 1.6 combined errors from the one that Chapter 15 inferred from the change of D with the size of the box. Independent ions conduct as $\sum_i q_i^2 D_i / V k_B T$, and the Haven ratio measures how far correlated ions fall short of it; in layers, D becomes a tensor; and hops between sites give $D = f k_h a^2 / 4$, with a correlation factor f that friction brings to 1. The Fourier transform, computed in $N_s \log_2 N_s$ operations for N_s samples, turns the velocity autocorrelation function into the vibrational density of states, whose peaks are the normal modes, whose width grows by $\gamma/2\pi$ under friction, whose value at zero holds D , and which frames a stride Δt apart show only up to $1/2\Delta t$. A histogram of any coordinate gives its free energy, which counts arrangements as well as energies, with error bars that must come from independent runs when the barriers are crossed rarely.

Chapter 17 runs these analyses on the simulations they are meant for: ASE and LAMMPS with a learned potential, lithium in graphite, and the trajectory gallery of every fault met so far.

Answers to the checks

SECTION 16.1. $1 - 1/500 = 0.998$. Each atom's partners are the other $N - 1$, while the divisor assumes N ; a larger cell at the same density holds more atoms, so $1/N$ is smaller and the far value closer to 1.

SECTION 16.2. At a vector of the crystal's reciprocal lattice every phase $\mathbf{G} \cdot \mathbf{r}_{ij}$ is a multiple of 2π , every arrow points the same way, and $S(\mathbf{G}) = N$. At an allowed vector of the cell that is not one of these, the arrows of the identical unit cells are turned through equal steps that come round to the start after a whole number of turns, so they sum to zero and $S(\mathbf{G}) = 0$.

SECTION 16.3. None: 342 017 would be $i = 342$, $j = 17$, but every key has $i < j$. The bond between atoms 17 and 342 has the key 17 342 only.

SECTION 16.4. $1/\sqrt{25} = 0.20$, to a fifth. Halving it needs four times as many events, 100.

SECTION 16.5. Near $t = 2Dm/k_B T$. With $m = 40 \text{ amu}$, $D = 3 \times 10^{-4} \text{ \AA}^2 \text{ fs}^{-1}$ and $T = 300 \text{ K}$ this is 96 fs.

SECTION 16.6. D is a third of the whole integral of C_{vv} . After the dip begins, C_{vv} is negative and its integral subtracts from what the positive part gave: on average the atom moves back after it rebounds from its neighbours, which undoes part of its displacement.

SECTION 16.7. Each pair moves $\mathbf{M}_q$ by $e(\Delta \mathbf{r}_+ - \Delta \mathbf{r}_-)$, which stays bounded while the two stay together, so the mean squared displacement of $\mathbf{M}_q$ stops growing and $\sigma_e = 0$. σ_{NE} still counts every ion's own diffusion and is not zero, so H_R would be without bound.

SECTION 16.8. Each hop moves the atom by a in one of six directions, so after $n = k_h t$ hops the MSD is $k_h t a^2 = 6Dt$, and $D = k_h a^2/6$.

SECTION 16.9. The Nyquist frequency is $1/(2 \times 20 \text{ fs}) = 25 \text{ THz}$, and 30 THz folds to $50 - 30 = 20 \text{ THz}$.

SECTION 16.10. About $1/t_c = 1 \text{ THz}$: peaks closer than that blur into one.

SECTION 16.11. The positions a distance r_h from the hollow fill a circle of circumference $2\pi r_h$, so the density of r_h carries the factor r_h , which vanishes at the centre. There

are few places that close, however low their energy, and $F = -k_B T \ln \mathcal{P}$ rises without bound as $r_h \rightarrow 0$.

SECTION 16.12. Positions wrapped into the cell: $L^2/2$ is the mean squared distance between two points placed at random in a cube of side L , which is what the MSD of wrapped positions approaches.

Exercises

The exercises run from questions of understanding, through calculations, to short derivations marked ‘show that’. Full solutions are in ‘Solutions to the exercises’ at the end of the book, and Notebook 16 checks every numerical answer.

EXERCISE 16.1 (*the volume of a shell*). For bins of $0.05 \text{ \AA}$, by what fraction does $4\pi r^2 \delta r$ fall short of the exact volume of the shell from r to $r + \delta r$, at $r = 1$ and at $5 \text{ \AA}$? Show that the fraction is close to $\delta r/r$ when $\delta r \ll r$.

EXERCISE 16.2 (*the far value of $g(r)$*). What value does $g(r)$ approach far from any atom, with ASE’s normalisation, for 256 and for 2048 atoms placed independently at random? Which normalisation would make it approach 1 for them, and why is the difference unimportant for the coordination number of the first shell?

EXERCISE 16.3 (*the shells of a face-centred cubic crystal*). In the fcc lattice of cube side a , the atoms sit at the corners of the cube and at the centres of its faces. Show that an atom’s nearest neighbours are $a/\sqrt{2}$ away and number 12, and that the next shell is at a and holds 6.

EXERCISE 16.4 (*structure without structure*). Suppose $g(r) = 0$ for $r < \sigma$ and $g(r) = 1$ beyond. Show that the Lennard-Jones energy per atom is then $-\frac{16}{9}\pi\rho\sigma^3\varepsilon$, and evaluate it for liquid argon, $\rho\sigma^3 = 0.8$ and $\varepsilon = 0.01034 \text{ eV}$. Compare it with -56.16 meV , the energy per atom that the run’s $g(r)$ gives with the full Lennard-Jones potential (the run itself switches the potential off between 2σ and 2.5σ), and say what the difference measures.

EXERCISE 16.5 (*random atoms and a perfect crystal*). Show from Equation (16.2) that $S(\mathbf{G})$ averages to 1 for atoms placed independently at random in the cell, at any allowed $\mathbf{G} \neq \mathbf{0}$, and is N for a perfect crystal at a vector of its reciprocal lattice.

EXERCISE 16.6 (*compressibility from scattering*). A liquid of number density $0.03 \text{ \AA}^{-3}$ at 300 K has $\kappa_T = 0.5 \text{ GPa}^{-1}$. What is $S(0)$? Would a liquid twice as compressible scatter long waves more strongly or less?

EXERCISE 16.7 (*flicker*). A bonded pair vibrates with a period of 60 fs, and the far end of each vibration carries it beyond a single cutoff. How many events per picosecond does the cutoff record for this one pair? With r_{on} at the old cutoff, how wide must the band of hysteresis be to record none, and how wide to record none wherever the cutoff falls within the vibration?

EXERCISE 16.8 (*bonds as keys*). Six atoms have the bonds 0-1, 1-2, 3-4 in one frame and 0-1, 2-3, 3-4, 4-5 in the next. Write both sets as keys $iN + j$ with $N = 6$, find the bonds formed and broken by the merge of sorted lists, and find the molecules of each frame by the search of the toolbox in Section 16.3.

EXERCISE 16.9 (*two graphs that look alike*). Six atoms of one element form a triangular prism (two triangles, each atom of one joined to the atom above it in the other); six others form a hexagon with each atom also joined to the opposite one. Show that every atom in both has three neighbours, and explain why every round of the hash of Section 16.3 then gives every atom of both the same number. Are the two molecules the same?

EXERCISE 16.10 (*how many events*). A rate is wanted to 2%. How many events must be counted? A run of 1 ns of 100 molecules shows 40 dissociations: give the rate per molecule with its error.

EXERCISE 16.11 (*a random walk in one dimension*). An atom on a line hops by $+a$ or $-a$ with equal probability, at the rate k_h . Show that its MSD after a time t is $k_h a^2 t$ and that $D = k_h a^2 / 2$.

EXERCISE 16.12 (*hops that keep their direction*). On a line, each step of length a repeats the direction of the one before with probability $\frac{1}{2}(1 + c)$, so that consecutive steps have the correlation c , and steps k apart c^k . Show that for many steps the MSD is $na^2(1 + c)/(1 - c)$, and so that $f = (1 + c)/(1 - c)$, the statistical inefficiency of Section 15.2. What c gives the memory of direction, $4D/k_h \langle \ell^2 \rangle = 3.46$, of the model surface at 1 ps^{-1} ?

EXERCISE 16.13 (*the end of the ballistic part*). For Chapter 11's water, $D = 5.24 \times 10^{-4} \text{ \AA}^2 \text{ fs}^{-1}$ at 304 K. Estimate when a molecule's motion stops being ballistic, taking its mass as 18.015 amu.

EXERCISE 16.14 (*an exponential memory*). Show from Equation (16.5) that $C_{vv} = C_0 e^{-t/\tau_c}$ gives $\text{MSD} = 2C_0 \tau_c [t - \tau_c(1 - e^{-t/\tau_c})]$, that this is $C_0 t^2$ for $t \ll \tau_c$ and $2C_0 \tau_c t$ for $t \gg \tau_c$, and so that $D = C_0 \tau_c / 3$.

EXERCISE 16.15 (*the viscosity on the back of an envelope*). Table 15.1 gives the shear stress of liquid argon a spread of 0.0121 GPa and an integrated correlation time of 0.18 ps.

Show that the Green-Kubo integral is $(V/k_B T)\sigma_{xy}^2 \tau_{\text{int}}$, with σ_{xy} the spread, and evaluate it for 256 atoms at $0.02035 \text{ \AA}^{-3}$ and 135 K.

EXERCISE 16.16 (*Nernst-Einstein for the salt*). From the mean diffusion coefficients of Section 16.7, 5.19 and $5.03 \times 10^{-4} \text{ \AA}^2 \text{ fs}^{-1}$ for 32 ions of each kind in a cell of side $13.4 \text{ \AA}$ at 1487 K, compute σ_{NE} in S m^{-1} .

EXERCISE 16.17 (*one number for layers*). What would a single diffusion coefficient, a sixth of the slope of the full MSD, report for the guests of Section 16.8? Why does it describe neither of their motions?

EXERCISE 16.18 (*Euler's formula at work*). Show that $|e^{i\theta}| = 1$, and recover $\cos(A+B)$ and $\sin(A+B)$ from $e^{iA}e^{iB} = e^{i(A+B)}$.

EXERCISE 16.19 (*the transform of one cosine*). Show that the discrete transform of $x_n = \cos(2\pi mn/N_s)$, for a whole number $0 < m < N_s/2$, is $N_s/2$ at $k = m$ and at $k = N_s - m$ and zero elsewhere.

EXERCISE 16.20 (*how fast is fast*). Show that $W(N_s) = 2W(N_s/2) + N_s$ with $W(1) = 0$ gives $W(N_s) = N_s \log_2 N_s$ for N_s a power of 2, and find the ratio of N_s^2 to $N_s \log_2 N_s$ for $N_s = 2^{20}$.

EXERCISE 16.21 (*a stride that hides a bond*). The chain's stretch at 18.25 THz is seen in frames every 35 fs. Where does it appear? What is the longest stride at which it appears where it is?

EXERCISE 16.22 (*friction against resolution*). A spectrum is computed from a run under Langevin friction of 1 ps^{-1} with correlations taken to 10 ps. Which widens its peaks more, the friction or the window?

EXERCISE 16.23 (*diffusion in a spectrum*). What is $\mathcal{D}(0)$, in THz^{-1} , for the oxygen atoms of Chapter 11's water, $m = 15.999 \text{ amu}$, with the D of Exercise 16.13?

EXERCISE 16.24 (*where the free energy is lowest*). Show that for $U \approx \frac{1}{2}kr_h^2$ the free energy along the distance r_h from a hollow, $F(r_h) = -k_B T \ln[r_h e^{-\beta k r_h^2/2}] + \text{constant}$, is least at $r_h = \sqrt{k_B T/k}$, and evaluate it at 300 K for $k = 0.9785 \text{ eV \AA}^{-2}$.

EXERCISE 16.25 (*crossings*). A run crosses a barrier between two wells n times. Argue that the free energy difference between the wells is known to about $2k_B T/\sqrt{n}$, and find how many crossings give it to $0.1 k_B T$.

Molecular dynamics in practice

The methods of Chapters 9 to 16 are now implemented in `mdlab`. Research codes use the same physical ideas, but their units, defaults and data formats differ. Moving a calculation between codes therefore starts with a comparison whose answer we already know. Here we run the same argon model in `mdlab`, the Atomic Simulation Environment (ASE)⁴³ and LAMMPS⁹⁶, tracing the conversions needed for their results to agree.

We then use MACE-MP-o⁹⁷, a *foundation model* trained on first-principles calculations across much of the periodic table. Its broad training makes it useful on many materials, but does not establish its accuracy on a particular system. We examine its cost and time step, and use FHI-aims⁹⁸ to measure the cost and convergence of a first-principles reference. Here *first-principles* means that the forces are calculated from the electrons (Section 8.1) rather than supplied by a fitted interatomic model.

Lithium in graphite brings these choices together in a material that the earlier chapters represented with model surfaces and layered solids. Its spacing and hops require different convergence checks. A production protocol collects those checks with the measurements that justify each setting; the final gallery links the faults of a trajectory to the checks that reveal them.

17.1 Molecular dynamics with ASE

A travel adaptor lets one plug fit the sockets of many countries: the kettle is the same, and only the connection changes. The Atomic Simulation Environment (ASE)⁴³ is such an adaptor for atomistic calculations. One Python object holds the atoms, and any of dozens of programs that compute energies and forces, from a pair potential to a learned model or a first-principles code, can be plugged into it as a *calculator*; the same integrators, thermostats, optimisers and file readers then work with every one

of them. Earlier chapters checked `mdlab`'s integrators and thermostats against ASE's (Sections 10.6 and 13.1); this section sets out how a run is written in ASE, and runs `mdlab`'s own liquid argon through it.

Atoms, calculators and units

An `ase.Atoms` object holds the chemical symbols, the positions in Å, the cell, the masses in amu and the momenta. ASE stores the lattice vectors as the *rows* of the cell, where `mdlab` stores them as the columns of `h` (Section 10.1), so one is the transpose of the other. A calculator attached to the atoms answers `get_potential_energy`, `get_forces` and `get_stress`, the last in eV Å^{-3} . The stress is the pressure tensor of Section 14.4 with its sign reversed, so that a compressed solid has a negative stress. By default it holds only the part from the forces; the kinetic part of Equation (14.3) is added only with `get_stress(include_ideal_gas=True)`, whose omission was a fault of Section 14.9, and with it the pressure is minus a third of the stress's trace.

ASE measures energy in eV, length in Å and mass in amu, and computes an acceleration as force over mass with no conversion factor. Its unit of time t_A is therefore the one for which a force of 1 eV Å^{-1} on 1 amu gives 1 Å per t_A^2 , with the joule, and so the electronvolt, being a mass times a length squared over a time squared:

$$\begin{aligned} \frac{\text{eV}}{\text{Å}} &= \text{amu} \frac{\text{Å}}{t_A^2} && F = ma \text{ with every number 1} \\ t_A^2 &= \frac{\text{amu Å}^2}{\text{eV}} && \text{multiplying by } t_A^2 \text{ Å/eV} \\ t_A &= \text{Å} \sqrt{\frac{\text{amu}}{\text{eV}}} && \text{taking the root} \\ &= 1 \times 10^{-10} \text{ m} \times \sqrt{\frac{1.6605 \times 10^{-27} \text{ kg}}{1.6022 \times 10^{-19} \text{ kg m}^2 \text{ s}^{-2}}} && \text{in SI units} \\ &= 1 \times 10^{-10} \text{ m} \times 1.018 \times 10^{-4} \text{ s m}^{-1} = 10.18 \text{ fs.} \end{aligned}$$

A time step and a thermostat's time constant are given to ASE in this unit, and a friction in its reciprocal. ASE supplies the femtosecond as `ase.units.fs`, so a step of 10 fs is written `10 * units.fs`, and a friction of 0.001 fs^{-1} `0.001 / units.fs`. Velocities are lengths per t_A : an atom that covers 1 Å in a femtosecond covers 10.18 Å in one t_A , so `mdlab`'s velocities, in Å fs^{-1} , are multiplied by 10.18 on the way in, the factor whose omission was one of the faults of Section 11.8.

Any mdlab model becomes a calculator through `mdlab.io.mdlab_calculator`, which passes ASE's positions to the model and its energy, forces and, from its virial, stress back:

```

1 def mdlab_calculator(model):
2     from ase.calculators.calculator import Calculator, all_changes
3     class MdlabCalculator(Calculator):
4         implemented_properties = ["energy", "free_energy", "forces",
5                                   "stress"]
6
7         def calculate(self, atoms=None, properties=("energy",),
8                       system_changes=all_changes):
9             super().calculate(atoms, properties, system_changes)
10            h = np.array(self.atoms.get_cell()).T
11            # any change at all, since a strain of 1e-5 must be seen
12            if hasattr(model, "set_cell") and not np.array_equal(
13                h, model.cell):
14                model.set_cell(h)
15            energy, forces = model(self.atoms.get_positions())
16            self.results = {"energy": float(energy),
17                           "free_energy": float(energy),
18                           "forces": np.array(forces)}
19            if hasattr(model, "virial"):
20                w = -np.asarray(model.virial) / self.atoms.get_volume()
21                # Voigt order: xx, yy, zz, yz, xz, xy
22                self.results["stress"] = w[[0, 1, 2, 1, 0, 0],
23                                           [0, 1, 2, 2, 2, 1]]
24
25            return MdlabCalculator()

```

The stress is $-\mathcal{W}_{\alpha\beta}/V$, the virial tensor of [Section 14.4](#) over the volume, in ASE's order xx, yy, zz, yz, xz, xy . A test strains the cell by $\pm 1 \times 10^{-5}$ in each of its six components, three stretches and three shears, and compares the change of the energy with this stress, to $1 \times 10^{-8} \text{ eV \AA}^{-3}$; the comment marks why the cell is compared exactly, since a tolerance would hide so small a strain.

Dynamics

An integrator in ASE is an object built on the atoms with a time step, whose `run(n)` takes n steps and calls any function attached with `attach(f, interval)` every so many steps, as mdlab's observers do ([Section 16.4](#)). `VelocityVerlet` is the integrator of [Section 9.2](#); `Langevin` adds the friction and random kicks of [Section 13.4](#), with the friction γ in reciprocal time units; `Bussi` is CSVR ([Section 13.6](#)); and `NoseHooverChainNVT` is the chain of [Section 13.5](#), but counts $3N$ freedoms where mdlab counts N_f : in its first mass, $3Nk_B T\tau_T^2$ against $N_f k_B T\tau_T^2$, and in its target, which raises the temperature of a run whose centre of mass is still by $3N/N_f$ ([Section 13.8](#)). For constant pressure, `IsotropicMTKNPT` is the isotropic piston of [Section 14.7](#), `MTKNPT` lets every component

of the cell move, and MaskedMTKNPT lets chosen lattice vectors change length while keeping their directions, which a layered solid needs (Section 14.8). Table 17.1 sets these beside mdLab's and LAMMPS's.

One change in ASE's own advice shows why a version should be recorded with every run. Its Langevin once held the centre of mass still with an option, `fixcm`; from version 3.28 ASE warns that this option does not sample the canonical ensemble exactly and asks for a constraint, `FixCom`, instead. The runs below follow the warning.

The same liquid in two codes

Chapter 12's liquid argon, 256 atoms of the switched Lennard-Jones model at 135 K, was run from one starting state by mdLab and by ASE with `mdlab_calculator`, with steps of 10 fs, converted with mdLab's constants rather than `units.fs`, whose older constants put a difference of 4.6 parts in 10^9 into the clock (Section 10.6). At fixed energy the two runs share their energy to the last digit printed, its spread over 5 ps being 0.469 meV in both, and their positions differ by 3.9×10^{-14} Å after 1 ps, 3.1×10^{-13} after 2 and 1.2×10^{-10} after 5 ps (Figure 17.1a). The two codes add the same forces in different orders, and the chaos of Section 10.6 amplifies the rounding about 3000 times over the four picoseconds from 1 to 5 ps, some seven times a picosecond, as it would any difference.

Under the three thermostats, at a friction of 1 ps^{-1} and time constants of 1 ps, each code was run for 200 ps. The mean temperatures lie within three standard errors of 135 K, the spreads of the kinetic temperature within 0.989 and 1.016 of the canonical $\sqrt{2/N_f} T$ (Section 12.7), and the diffusion coefficients, from the MSD over 2 to 20 ps in five blocks (Section 16.5), agree within their errors (Figure 17.1b): (3.94 ± 0.14) and $(4.11 \pm 0.19) \times 10^{-4} \text{ Å}^2 \text{ fs}^{-1}$ under CSV, (3.92 ± 0.05) and $(3.95 \pm 0.10) \times 10^{-4}$ under the chain, (2.61 ± 0.06) and $(2.66 \pm 0.08) \times 10^{-4}$ under the friction, which slows diffusion as Section 16.12 found. A step costs 1.32 ms in mdLab and 1.44 ms in ASE, since the forces are mdLab's in both.

KEY IDEA. ASE holds atoms in one object and plugs any energy-and-force program into it as a calculator, so its integrators, thermostats and barostats serve every model. Its unit of time is 10.18 fs, and its cell is the transpose of mdLab's. With the same model the two codes give the same trajectory until chaos amplifies their rounding, and the same temperatures and diffusion coefficients.

NOTEBOOK 17 *run one liquid in mdLab and in ASE and compare the paths*

CHECK YOUR UNDERSTANDING. An ASE script sets `timestep=2` without `units.fs`. What step does the run take, in femtoseconds?

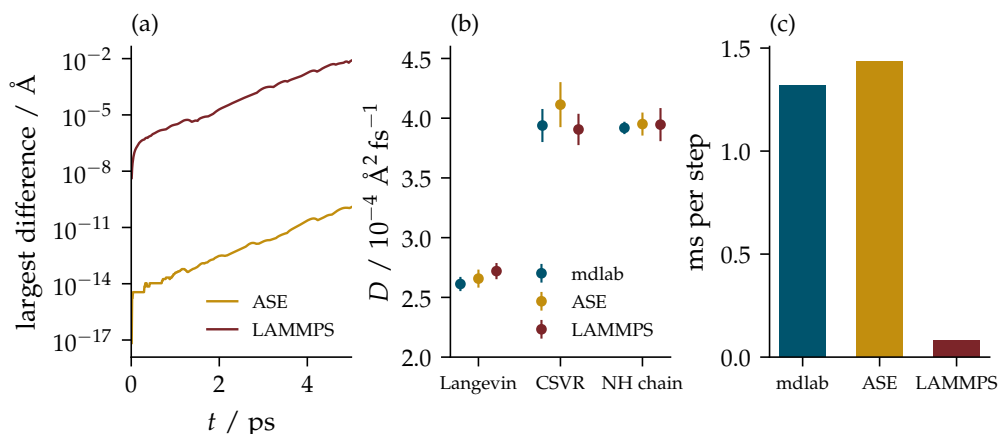


FIGURE 17.1: Chapter 12's liquid argon, 256 atoms at 135 K, in three codes with one model. (a) The largest difference of any coordinate from mdlab's, at fixed energy from one start, for ASE and for LAMMPS with its table. (b) The diffusion coefficient from the MSD over 2 to 20 ps of 200 ps runs under Langevin friction of 1 ps^{-1} , CSV, and a Nosé-Hoover chain, with time constants of 1 ps, and its standard error from five blocks. (c) The time of one step of 10 fs.

17.2 LAMMPS

A player piano plays from a punched roll: each row of holes is a command, read in order, and the piano does the rest at its own speed. LAMMPS, the molecular dynamics code of Sandia National Laboratories⁹⁶, is driven the same way, by an input script whose lines are commands read in order, and it runs them in compiled C++ spread over many processors. It is one of the codes most used for large classical and learned-potential runs. This section reads an input script line by line, sets out the units it works in, and runs the liquid of Section 17.1 in it.

An input script

The commands that the runs of Figure 17.1 pass to LAMMPS for the liquid under CSV are, with the steps written as one run (the Python script takes them ten at a time, to read the positions between),

```

1 units metal
2 atom_style atomic
3 atom_modify map array
4 read_data argon.data
5 pair_style table linear 20001
6 pair_coeff 1 1 argon.table ARGON 8.5
7 timestep 0.01
8 fix 1 all nve
9 fix 2 all temp/csvr 135.0 135.0 1.0 17
10 thermo_style custom step pe ke
11 run 20000

```

units metal fixes the units of everything that follows: lengths in Å, energies in eV, masses in grams per mole, which are numerically amu, temperatures in K, pressures in bars and times in picoseconds, so that timestep 0.01 is 10 fs and velocities are in Åps⁻¹. atom_style atomic declares atoms that carry only a type, a position and a velocity, and atom_modify map array keeps a lookup from each atom's identity number to its place in memory. read_data reads the cell, the masses, the positions and the velocities from a file that ASE writes (ase.io.write with format="lammps-data", units="metal", masses=True and velocities=True; without the last two the file holds neither). pair_style and pair_coeff set the forces between atoms of type 1 from the section ARGON of argon.table, cut off at 8.5 Å, as the next subsection describes. fix commands act on the atoms at every step: nve is velocity Verlet, and temp/csvr rescales the velocities as CSVr does, towards a target that runs from the first temperature to the second over the run (here both 135 K), with its time constant in picoseconds and a seed for its random numbers. thermo_style chooses what is printed, and run takes the steps. Table 17.1 lists the fix for each method of the book.

TABLE 17.1: The methods of Chapters 9, 13 and 14 in the three codes, with the unit of each one's time parameter: γ in fs⁻¹ and τ in fs for mdlab; for ASE, τ in its time unit and γ in its reciprocal (written tau * units.fs and gamma / units.fs); for LAMMPS, τ and the damping 1/ γ in picoseconds.

method	mdlabs	ASE	LAMMPS
velocity Verlet	md.run	VelocityVerlet	fix nve
Langevin	thermostats. Langevin, γ	Langevin, friction γ	fix langevin, damping 1/ γ , with fix nve
CSVr	thermostats.CSVr, τ	Bussi, taut	fix temp/csvr, with fix nve
Nosé-Hoover chain	thermostats. NoseHooverChain, τ	NoseHooverChainNVT, tdamp	fix nvt
Berendsen	thermostats. Berendsen, τ	NVTBerendsen, taut	fix temp/berendsen, with fix nve
MTK piston	barostats.run_mtk	IsotropicMTKNPT, pdamp	fix npt ... iso
lengths apart	barostats. StochasticCell Rescaling	MaskedMTKNPT, mask	fix npt ... aniso

The model as a table

LAMMPS has its own Lennard-Jones forms, each with its own way of ending at the cutoff, and a comparison of codes needs the model itself. Any pair potential can be given as a table: mdlab.io.lammps_table writes $\phi(r)$ and the force $F = -d\phi/dr$ at 20 001 equally spaced distances from 0.5 to 8.5 Å, 4×10^{-4} Å apart. LAMMPS does not interpolate these points as they stand: it passes a smooth curve, a spline, through them

and reads from it its own table of 20 001 entries equally spaced in r^2 , $0.0036 \text{ \AA}^2$ apart, and `pair_style` table linearly interpolates F/r along straight lines in r^2 between those. Interpolating a function y along a straight line between points a distance δx apart misses it by at most $\delta x^2 \max|y''|/8$ ([Exercise 17.5](#)). Measured from the midpoint, the parabola $y = \frac{1}{2}y''x^2$ has the height $\frac{1}{2}y''(\delta x/2)^2$ at both ends, so the chord is the flat line at that height, and at the midpoint, where the parabola is at zero, the two part by $\frac{1}{2}y''(\delta x/2)^2 = \delta x^2 y''/8$. Here y is F/r and x is r^2 . Near $3 \text{ \AA}$, about the closest that atoms of the liquid come ($2.98 \text{ \AA}$ in `mdlab`'s 200 ps under CSV), a step of $0.0036 \text{ \AA}^2$ in r^2 is $0.0036/(2 \times 3) = 6 \times 10^{-4} \text{ \AA}$ in r , and the largest miss of the force beyond $3 \text{ \AA}$, found by placing a pair of atoms at the midpoint of every interval of LAMMPS's table, is $7.6 \times 10^{-7} \text{ eV \AA}^{-1}$, 1.3×10^{-6} of the force there, and the run shows it. At fixed energy LAMMPS's energy spreads by the same 0.469 meV as `mdlab`'s, but its positions part from `mdlab`'s by $3.0 \times 10^{-6} \text{ \AA}$ after 1 ps, against ASE's 3.9×10^{-14} , and by $7.8 \times 10^{-3} \text{ \AA}$ after 5 ps ([Figure 17.1a](#)): the table's small error is a difference like any other, and the chaos amplifies it.

Temperatures and diffusion

Under the same three thermostats, given as `fix langevin`, `fix temp/csvr` and `fix nvt`, LAMMPS's runs reach mean temperatures within three errors of 135 K and spreads of 0.962 to 0.996 of the canonical, and its diffusion coefficients lie within 1.2 combined errors of the other codes' ([Figure 17.1b](#)). An option of `fix langevin` may matter as well. Over 1 ns, D is $(2.721 \pm 0.038) \times 10^{-4} \text{ \AA}^2 \text{ fs}^{-1}$ in LAMMPS and $(2.681 \pm 0.023) \times 10^{-4}$ in `mdlab`, 0.9 combined errors apart; with `zero yes`, which removes the sum of the random forces at every step so that they cannot push the centre of mass, LAMMPS gives $(2.806 \pm 0.021) \times 10^{-4}$, 2.0 combined errors above its own value without the option and 4.0 above `mdlab`'s. The option's own effect, two combined errors, is at the edge of what these runs can resolve, and a run that reports transport under a local thermostat should state such options with the friction; the transport coefficients of [Section 16.12](#) belong in any case to runs at fixed energy or under a global thermostat.

LAMMPS takes 0.08 ms a step for these 256 atoms, a sixteenth of `mdlab`'s 1.32 ms: compiled C++ throughout, against `mdlab`'s NumPy and Numba ([Section 10.7](#)).

Python and LAMMPS

LAMMPS can also be driven from Python, through its module `lammps: commands_` string passes lines of a script, `extract_atom` returns the positions, velocities, forces and other arrays of the atoms, and `get_thermo` any printed quantity. Two details of those arrays trip up a reader. LAMMPS wraps positions into the cell and keeps, for each atom, an image flag: the net number of cells it has crossed along each axis, a crossing

back counting minus one, so that the unwrapped position is the wrapped one plus the flags times the lattice vectors. And LAMMPS reorders its atoms in memory every thousand steps, to keep neighbours near one another, so the arrays must be read in the order of their identity numbers, `extract_atom("id")`, at every frame; [Section 17.8](#) measures what happens when they are not. The same module lets LAMMPS call Python for its forces, which the next section uses.

KEY IDEA. LAMMPS runs an input script of commands in order, in metal units with times in picoseconds. Thermostats and barostats are fixes, any pair potential can be a table, and its Python module reads and sets the atoms' arrays, in the order of their identities. With the same model it gives the same energies, temperatures and diffusion coefficients as `mdlab` and `ASE`, at a sixteenth of the time per step.

NOTEBOOK 17 *write a LAMMPS input from Python and check its forces*

CHECK YOUR UNDERSTANDING. A LAMMPS script in metal units sets timestep 1.0. What is the step in femtoseconds, and what would it do to liquid argon?

17.3 A learned potential in both

A translator who has read millions of sentences can render one never seen before, but only in the languages that reading covered, and the gaps in a translation show where the reading was thin. A foundation model of interatomic forces, one trained once on a wide range of materials to be used on any of them, is in the same position. MACE-MP-o⁹⁷ is a network of the kind that the planned continuation of Part V builds, trained on the calculations of the Materials Project⁹⁹, compounds of 89 elements, at every structure each calculation passed through on its way to its least energy. They use density-functional theory ([Section 8.1](#)), which needs an approximation, called the functional, for one part of the electrons' energy; theirs is PBE¹⁰⁰, with an added correction for the oxides and fluorides of some transition metals. The model gives energies and forces for structures it was never shown. This section runs it in `ASE` and in `LAMMPS`, measures what it costs, finds one of its gaps, and chooses the time step for lithium in graphite.

Running it

In `ASE` the model is a calculator like any other: MACE's package provides one that loads the trained weights, `mace_mp`. It runs on the computer's main processor, the CPU, here on five of its cores at once (five threads), or on its graphics processor, the GPU, a chip of many small cores that do the same arithmetic on many numbers at once; and it keeps its numbers in double precision (float64), the 16 significant digits of [Section 8.4](#),

or in single precision (float32), about 7. The GPU of the machine used here, an Apple M5 Pro, holds no double-precision numbers, and the version of the package used, 0.3.16, stores the energy of each atom in double precision; the chapter's scripts keep that one output in single precision on the GPU, which leaves the forces untouched, and check the result against double precision on the CPU.

LAMMPS runs MACE's own pair styles only when it is built with extra packages for them, which its standard distribution lacks. Its Python module offers another way in: the command `fix ext all external pf/callback 1 1` makes LAMMPS call a Python function at every step with the positions, and the forces that function returns are added to the atoms', its energy to the run's. The function builds an ASE Atoms from LAMMPS's positions and cell and asks MACE for the forces. On a cell of 56 atoms of LiC₆, eight copies of the seven-atom cell of [Section 17.5](#), moved at random by about 0.05 Å, LAMMPS's energy equals ASE's to the last of nine digits, and its forces equal ASE's to 1.4×10^{-13} eV Å⁻¹ once one detail is handled: LAMMPS holds every cell with its first lattice vector along *x* and its second in the *xy* plane, so ASE turns this cell to fit when it writes LAMMPS's data file, and a comparison of forces must be made with the cell as LAMMPS holds it.

A gap: dispersion

The weak attraction between the sheets of graphite comes from the correlated motion of their electrons, dispersion, which PBE leaves out. A model trained on PBE inherits the gap. Graphite's cell holds two sheets, and its height *c* is twice their spacing. With MACE-MP-o alone, graphite's energy is least at *c* = 7.818 Å, 16.5% above Trucano and Chen's 6.711 Å³³, and stretching the cell to *c* = 9 Å, the sheets 4.5 Å apart, costs only 5.2 meV per atom ([Figure 17.2c](#)). Grimme's D3 correction¹⁰¹ adds an attraction for every pair of atoms, falling as $1/r^6$ and $1/r^8$, with coefficients from reference calculations for the 94 elements from hydrogen to plutonium; Becke-Johnson damping¹⁰² keeps it finite where atoms come close. With it the least energy lies at 6.612 Å, 1.5% below the measured value, with a binding of 41.2 meV per atom against *c* = 9 Å. Relaxed with the correction, graphite's in-plane constant is 2.4647 Å, 0.03% above Trucano and Chen's 2.464 Å. The correction is added in ASE by summing its calculator with MACE's, and it costs 12 ms for 126 atoms on the CPU, a twentieth of the medium model's call on the GPU. Every run below includes it.

What it costs

MACE-MP-o comes in sizes; the medium model takes 1.20 s for one call on 126 atoms on five CPU threads in double precision, 0.62 s in single precision, and 0.25 s on the GPU, and the small model 0.40, 0.21 and 0.081 s ([Figure 17.2a](#)). Every setting grows

nearly in proportion to the number of atoms from 28 to 1008, as the neighbour list makes each atom's work local, though on the GPU the cost per atom rises by 40% from 252 to 1008 atoms; for the medium model the GPU is 1.9 to 2.7 times faster than the CPU in single precision, and 3.5 to 4.8 times faster than in double. Against the 0.08 ms that LAMMPS takes for a step of 256 argon atoms (Section 17.2), a learned potential is thousands of times dearer per atom, and cheap only beside the electronic structure it replaces (Section 17.4).

The time step

The test of Section 9.6 chooses the step. LiC_6 , 56 atoms, was started from velocities drawn at 600 K and run for 2 ps at fixed energy with steps of 0.5, 1, 2 and 3 fs in double precision (Figure 17.2b). The spread of the total energy is 0.0017, 0.0067, 0.028 and 0.082 of the kinetic energy's, growing between the shorter steps by 4.02 and 4.16, against the 4 of δt^2 , and from 2 to 3 fs by 2.92, against 2.25: at the longest step the growth outruns δt^2 . The mean total energy of the last quarter of each run differs from that of the first by less than 0.1 meV up to 2 fs, and by 17.6 meV at 3 fs. Single precision on the GPU gives the same spreads and the same small differences at 1 and 2 fs, so for this system its rounding is harmless. The longest step that passes is 2 fs; the runs below take 1 fs, a margin for runs of tens of picoseconds at up to 600 K.

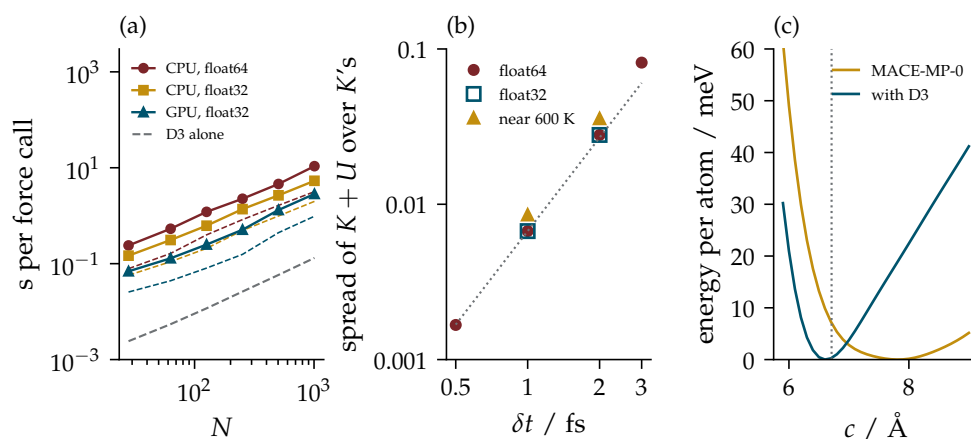


FIGURE 17.2: MACE-MP-0 with D3. (a) The time of one force call against the number of atoms, LiC_6 cells of 28 to 1008 atoms, for the medium model (solid) and the small (dashed) on five CPU threads in double and single precision and on the GPU in single precision, and for D3 alone (grey). (b) LiC_6 , 56 atoms, at fixed energy: the spread of the total energy over that of the kinetic energy against the step, in double (circles) and single (squares) precision near 245 K, and in double precision near 600 K (triangles), with lines $\propto \delta t^2$ through 1 fs (dotted). (c) Graphite's energy against the length c of its cell, with MACE-MP-0 alone and with D3, against the measured c (dotted).

KEY IDEA. A foundation model runs as an ASE calculator, and in LAMMPS through a Python callback, with the same forces. It inherits the gaps of its training functional, here dispersion, which D3 restores. It costs about a quarter of a second per call for a hundred atoms on a GPU, growing nearly in proportion to the number of atoms, and its time step is chosen by the same test at fixed energy as any other model's.

NOTEBOOK 17 *switch D3 on and off and watch graphite's sheets*

CHECK YOUR UNDERSTANDING. A learned potential trained only on bulk metals is used for a metal surface covered by water. Which part of the system needs checking first, and how?

17.4 First-principles molecular dynamics and its cost

A map drawn from a survey is only as good as the survey, and a survey of every field is too costly to repeat for each journey; the map is drawn once and then used. A learned potential is such a map, and the survey is electronic-structure theory, which gives the energy and forces of atoms from their electrons ([Section 8.1](#)). Molecular dynamics with those forces at every step, first-principles or *ab initio* molecular dynamics, needs no fitted potential, and it is what a learned potential is trained to imitate. This section measures what it costs, with FHI-aims⁹⁸, which describes each electron's wave as a sum of functions centred on the atoms, stored as tables of numbers, and can run molecular dynamics itself.

One frame of LiC₆

The 28 atoms of a $2 \times 2 \times 1$ cell of LiC₆, four copies of the seven-atom cell of [Section 17.5](#), taken from a run of MACE-MP-o at 300 K, were given to FHI-aims with the PBE functional and its standard 'light' settings, the cheapest of its standard choices of basis functions and integration grids, on four cores of an Apple M1 Max, once for each of the three grids below. The electrons are found by a self-consistent iteration: the potential each electron feels depends on the density of all of them, the number per unit volume at each point, which is built from their waves, which are in turn set by the potential; so each cycle builds a potential from the density and a density from the potential, until neither changes. This frame needed 16 or 17 cycles, 337 s to 385 s in all: 14 to 16 s for each ordinary cycle and 57 to 60 s for each of the last two, which also computed the forces. A step of molecular dynamics starts from the density of the step before, extrapolated, and needs fewer cycles. If it took half the time of this calculation from scratch, an estimate that a short run would test, a step would cost about 192 s, and a picosecond of steps of 1 fs about 53 h on these four cores; the two cycles with forces

do not shrink, so this is the cheaper end. MACE-MP-o with D₃ takes 214 ms a step of molecular dynamics for the same cell on the same machine's GPU, with other runs sharing the machine, about 900 times less (and 72 ms a call alone on the faster machine of [Figure 17.2a](#)). The gap widens with size, since the cost of the electronic structure grows faster than the number of atoms ([Section 8.1](#)).

A setting to converge

LiC₆ is a metal. In a crystal each electron's wave repeats from one cell to the next up to a shift of its phase, and the code finds the electrons separately for each of a grid of such shifts, the *k*-points, and averages over them; a metal, whose filled energy levels end abruptly, needs a fine grid. A grid of $4 \times 4 \times 10$ shifts along the three cell vectors has 160 points, but a shift and its reverse give the same energies, which leaves 84 to compute; grids of $2 \times 2 \times 5$ and $3 \times 3 \times 8$ leave 12 and 37. The forces on the carbon atoms of this frame, up to $3.1 \text{ eV } \text{\AA}^{-1}$, change by up to $0.16 \text{ eV } \text{\AA}^{-1}$ between the grids of 12 and 84 points and by up to $0.28 \text{ eV } \text{\AA}^{-1}$ between those of 37 and 84: they have not settled by 37 points, nothing yet shows that they have by 84, and a denser grid is needed before these frames can serve as a reference. A reference calculation is converged in its own settings before anything is compared with it, as a molecular dynamics run is converged in its time step and length.

What is left to check

The lithium studies of the next two sections rest on MACE-MP-o, and no calculation here has yet compared its forces, or its barrier for a lithium hop, with PBE for this system. Their numbers are therefore the model's predictions. The check, single points at a converged grid on frames of the runs and along lithium's paths between sites, is the first step of the protocol of [Section 17.7](#) for any learned potential, and the planned continuation of Part V returns to it when it trains a model of its own on these data.

KEY IDEA. First-principles molecular dynamics computes the electrons at every step and needs no fitted potential, at a cost of minutes a step for a few dozen atoms, on this estimate about nine hundred times a foundation model's here. Its settings, such as the grid of *k*-points of a metal, are converged before it serves as a reference, and a learned potential is checked against it on the system at hand before its predictions are trusted.

NOTEBOOK 17 *read the cost of a first-principles step and of a learned one*

CHECK YOUR UNDERSTANDING. A first-principles step of 28 atoms costs 192 s. If the cost grew as the cube of the number of atoms, as it does once finding the eigenvectors dominates (Section 8.1), roughly how long would a step of 112 atoms take?

17.5 LiC₆ with a foundation model

Books pushed between the shelves of a bookcase whose shelves can slide force them apart, while the shelves themselves keep their length. Lithium does this to graphite. Chapter 14 could only imitate it with a model solid (Section 14.8); with MACE-MP-o and D3 the real material can be run, and this section does so for the fully lithiated compound LiC₆ at room temperature, the study that Chapters 13 and 14 left for a real potential. Its numbers are the model's predictions (Section 17.4).

Building the cells

ASE builds the structures. Graphite's cell holds four atoms, two in each of its sheets stacked A, B, with Trucano and Chen's lattice constants as the start. In LiC₆ the sheets stack A, A and a lithium atom sits over every third hexagon, the pattern written $\sqrt{3} \times \sqrt{3} R_{30^\circ}$: its cell is the sheet's own cell made $\sqrt{3}$ times longer and turned by 30° , built in ASE as the supercell of the two-atom cell of graphene, a single sheet of graphite, whose vectors **a** and **b** have length *a* and lie 120° apart, with the vectors $2\mathbf{a} + \mathbf{b}$ and $-\mathbf{a} + \mathbf{b}$ (Exercise 17.12), and it holds six carbon atoms and one lithium. Relaxed together with its cell at 0 K, to forces below $1 \text{ meV } \text{\AA}^{-1}$ and stresses below 0.02 GPa, LiC₆'s sheets lie $3.616 \text{ \AA}$ apart, 2.3% below the measured $3.70 \text{ \AA}$ quoted by Hazrati, de Wijs and Brocks⁷², and the sheet's lattice constant *a*, the side of its two-atom cell, is $2.4864 \text{ \AA}$, 0.88% longer than the $2.4647 \text{ \AA}$ of graphite relaxed with the same model: the sheets stretch a little, as the electrons that lithium gives them are expected to make them do.

Anisotropic and isotropic coupling at 300 K

A cell of $3 \times 3 \times 1$ of these units, 63 atoms, was run at 300 K and zero pressure with ASE's MaskedMTKNPT, which lets each lattice vector change its length while keeping its direction, with $\tau_T = 100 \text{ fs}$ and $\tau_P = 1 \text{ ps}$, for 20 ps; and with IsotropicMTKNPT, which scales the three lengths together, for 10 ps (Figure 17.3). The isotropic run keeps the ratio of the lengths it started from, and the stiff sheets hold the in-plane length, and with it the spacing, at their values at 0 K: the spacing stays at $(3.6162 \pm 0.0005) \text{ \AA}$. The diagonal of the pressure tensor, computed from the model's stress with the atoms'

kinetic part added, on 25 frames of the last 5 ps, is (-1.00 ± 0.45) , (-1.11 ± 0.36) and (1.68 ± 0.10) GPa along x , y and z : the sheets push apart along z while each sheet pulls in along x and y , and only the mean is zero. With the lengths coupled apart, all three components, from 75 frames of the last 15 ps, are zero within 1.2 of their errors, (0.06 ± 0.28) , (-0.21 ± 0.18) and (0.01 ± 0.06) GPa, and the spacing opens on warming.

How long the spacing takes

The in-plane length settles at once: over the 20 ps the lattice constant a is (2.4836 ± 0.0004) Å, its quarters agreeing to 0.001 Å; convergence_report flags it only because its blocks give a smaller error, 0.0002 Å, than the one quoted. The spacing does not settle. Its means over the four quarters of the run are 3.756, 3.760, 3.766 and 3.778 Å, still rising, and convergence_report refuses it with 82 independent samples, fewer than the 100 that the criterion of Section 15.9 asks. A layered solid's soft direction relaxes slowly. These saved results do not yet establish its equilibrium spacing; a longer run is required.

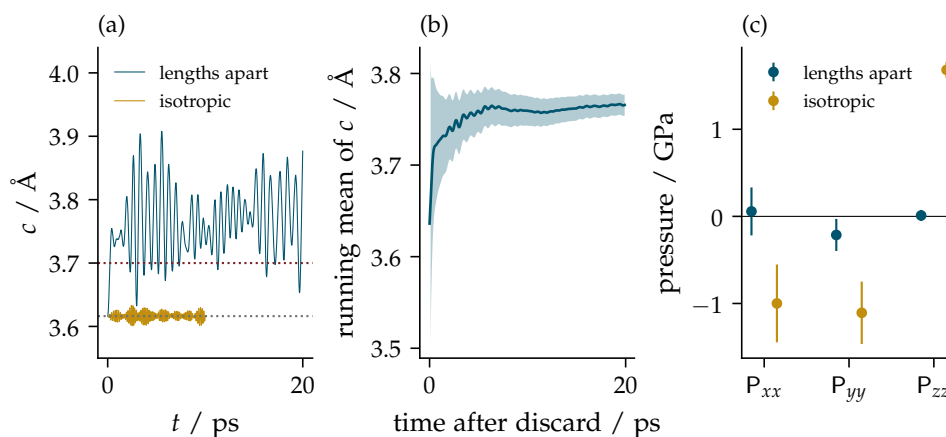


FIGURE 17.3: LiC_6 , 63 atoms, at 300 K and zero pressure with MACE-MP-o and D3. (a) The spacing of the sheets against time with the lattice lengths coupled apart (teal) and together (ochre), with the relaxed spacing at 0 K (grey dotted) and the measured 3.70 Å (red dotted). (b) The running mean of the spacing in the initial 20 ps from the start that convergence_report finds, with twice its error (band). (c) The diagonal of the pressure tensor under each coupling, with its standard error.

KEY IDEA. A foundation model with D3 relaxes LiC_6 to within a few per cent of its measured spacing, and runs it at room temperature. A layered solid needs its lattice lengths coupled apart: coupled together, the spacing is held and the diagonal of the pressure tensor shows the strain. Its soft direction relaxes over tens of picoseconds, and the convergence report decides when it has settled.

NOTEBOOK 17 couple LiC_6 's lengths together and apart, and watch the spacing

CHECK YOUR UNDERSTANDING. In LiC_6 , how many hexagons of a sheet are there for each lithium atom, and why does that give six carbon atoms?

17.6 Lithium on the move

A marble in an egg box that is being shaken rattles in its cup and now and then jumps into the next. Counting its jumps needs two decisions: what a cup is, and how long the marble must stay in a new cup for the visit to count, since a marble that touches the rim of a neighbour and falls back has gone nowhere. If the egg box itself slides about on the table, the cups move with it, and the marble's cup must be read in the frame of the box. A lithium atom in a gallery of graphite, the gap between two neighbouring sheets, is such a marble. This section counts its hops in dilute graphite, one lithium among 64 carbon atoms, and turns them into rates with error bars, taking Chapter 13's hops over a model surface (Section 13.4) and Chapter 16's counting of events (Section 16.4) to a real material.

The sites

Relaxed at 0 K, lithium sits over the centre of a hexagon of the sheet below, a hollow, and under an atom of the sheet above, since in graphite's A, B stacking a hollow of one sheet lies under an atom of the next. Graphite has an inversion centre midway between an atom of the sheet below and the atom of the sheet above that lies over it: sending every point to the point as far beyond that centre on the opposite side swaps the two sheets and leaves the crystal as it was. It carries lithium's site to one over an atom of the sheet below, a corner of lithium's hexagon, and under a hollow of the sheet above: a second family of sites with the same energy, $a/\sqrt{3} = 1.42 \text{ \AA}$ from the first, for the lattice constant $a = 2.4648 \text{ \AA}$ of the relaxed dilute cell. Each family is the hollows of one sheet, a triangular grid, and the two grids are offset by $a/\sqrt{3}$, so together they form a honeycomb on which the families alternate, as the two kinds of atom of a sheet do (Exercise 17.15).

Moved along the straight line to the nearest site of the other family, with its position in the plane held at each of eight steps while its height and the carbon atoms relax, lithium's energy rises to 73 meV halfway and falls back to zero at the far site, as the inversion requires; along the line to the next site of its own family, across the middle of a C–C bond, it rises to 107 meV (Figure 17.4a). Without D₃, at the same relaxed points, the two are 53 and 94 meV. One carbon atom of each sheet is held in the plane as well, since otherwise the sheets would slide to carry their sites back under lithium. Every path between two sites must cross the ridge that divides them, whose lowest point, the saddle, sets the barrier (Section 3.10), so the highest point of any one path is at least the

barrier, and 73 meV bounds it from above. The relaxation of the carbon atoms is part of every site. With them held where they relaxed around the first site, the other family's site lies 136 meV higher and is no minimum at all, the straight way to it passing 56, 123 and 111 meV, so a map over a frozen host finds only half the sites.

Sheets that slide

Each run, at 300, 450 and 600 K, lasted 20 ps under CSVr with $\tau_T = 1$ ps and steps of 1 fs, keeping lithium's position every 10 fs and every atom's every 100 fs. The cell was the one relaxed at 0 K, held fixed, which at these temperatures holds the sheets closer than zero pressure would, as LiC_6 's pressure along c in its own cell showed (Section 17.5); the rates below belong to that cell. In them the sheets slide as wholes. The sheet below lithium slides by as much as 0.96, 2.40 and 1.59 Å, and the sheet above by as much as 1.94, 4.76 and 2.97 Å against it, comparable with or further than the 1.42 Å between neighbouring sites, so a site fixed in the frame of the cell is soon in the wrong place. Each family is the hollows of one sheet, the first those of the sheet below and the second those of the sheet above. Lithium's position is therefore read in the frame of each sheet in turn, the sheet's slide being the mean displacement of its atoms, interpolated between the frames that hold them, and lithium is assigned to the nearest site of either family. Sites fixed in the cell where the sheets started give a count of hops that falls steadily as the least stay of the next subsection grows, from 85 to 20 at 450 K.

How far the sheets wander is a property of the small cell. Near its best stacking, sliding a sheet of N atoms by s along one direction costs about $\frac{1}{2}N\kappa s^2$, with κ the stiffness for one atom, since every atom of the sheet changes its stacking at once; equipartition gives this term $\frac{1}{2}k_B T$ on average whatever N (Section 12.6). The typical slide, $\sqrt{k_B T / N\kappa}$, therefore shrinks as the sheet grows, and the barrier between one stacking and the next, N times that for one atom, grows. A sheet of 32 atoms wanders much further than the sheets of a crystal, and a larger cell is the test of how much of what follows depends on it (Section 15.7).

Counting hops

The runs start from the relaxed cell with velocities drawn at the target, and half the kinetic energy passes at once into the potential energy, as in the crystal of Section 12.9: the first picosecond averages 200, 301 and 402 K. The test of Section 15.4 on the kinetic temperature finds the start at 2.5, 2.5 and 7.6 ps, after which the runs average 298, 447 and 592 K, and hops are counted from there, over 17.5, 17.5 and 12.4 ps.

A visit counts only if it lasts a least stay, the same device as the least lifetime of a bond (Section 16.3); a shorter visit is joined to the one before it. With both families of sites

the count barely depends on the least stay up to 50 fs: 14, 14, 12 and 12 hops at 300 K, 45, 43, 43 and 42 at 450 and 44, 43, 42 and 42 at 600 K, for least stays of 10, 20, 30 and 50 fs (Figure 17.4b). Beyond, the least stay begins to discard visits that were real, most at 600 K, where a visit lasts 0.30 ps on average. With the sites of the sheet below alone the count keeps falling at 300 K, from 28 at a least stay of 10 fs to 8 at 200 fs: a position over a site of the other family is equally far from three sites of the first, and the jitter shares it out among them, each change a hop. A count that falls steadily as the least stay grows is the sign of a wrong definition of a site; a plateau is the sign of a right one, and the least stay is chosen on it, here 50 fs. The hops then join the two families, all of them at 300 K, 98% at 450 and 95% at 600 K, as hops between neighbours on the honeycomb must; the few others join two sites of one family, the visit between them having been shorter than the least stay.

Bonds would not have served. The bond tracker of Section 16.3, with its thresholds of 2.45 and 2.86 Å for lithium and carbon, finds lithium bonded to ten carbon atoms in half the frames at 300 K and to nine in 45%: the six of the hexagon on one side, 2.30 Å away at 0 K, the atom opposite, 2.08 Å, and that atom's three neighbours, 2.45 Å, on the threshold at which a bond forms. Those three flicker in and out with the vibrations, each flicker an event of the tracker, while a hop only exchanges which sheet holds the six and which the four. The sites count the hops; the bonds would count the vibrations.

Rates

With a least stay of 50 fs the runs hold 12, 42 and 42 hops. A rate from n events is known to $1/\sqrt{n}$ (Section 16.4), and we call a rate resolved at a fifth, which needs 25 of them: at 450 K the rate is $(2.40 \pm 0.37) \text{ ps}^{-1}$ and at 600 K $(3.38 \pm 0.52) \text{ ps}^{-1}$, while at 300 K, $(0.69 \pm 0.20) \text{ ps}^{-1}$ from 12 hops, it is unresolved, and a run of 36 ps in all at that rate would resolve it. The error $\sqrt{n}$ assumes hops independent in time. Counted in ten equal windows of each run, their variance is 1.3, 1.3 and 1.8 times their mean, against 1 ± 0.47 for independent events (Exercise 15.7), within twice that scatter, so the Poisson errors stand. Over the whole runs, warming included, the ratios are 1.9, 2.2 and 2.3, beyond twice that scatter at 450 and 600 K: while a run warms its hops come in bursts, one more reason to discard the start.

The logarithm of the rate against $1/k_B T$, each point weighted by $\sqrt{n}$ as in Section 16.4, has the slope $-(80 \pm 17) \text{ meV}$ (Figure 17.4c): the rate rises 4.9 times from 300 to 600 K, where a barrier of 73 meV would raise it $e^{0.073 \times (38.68 - 19.34)} = 4.1$ times, with $1/k_B T$ in eV^{-1} . The apparent activation energy lies above the bound from the relaxed path by less than half its error, which at a fifth of its value is all that three points, one of them unresolved, can give; a barrier from rates to a few meV needs more temperatures and many more hops at each.

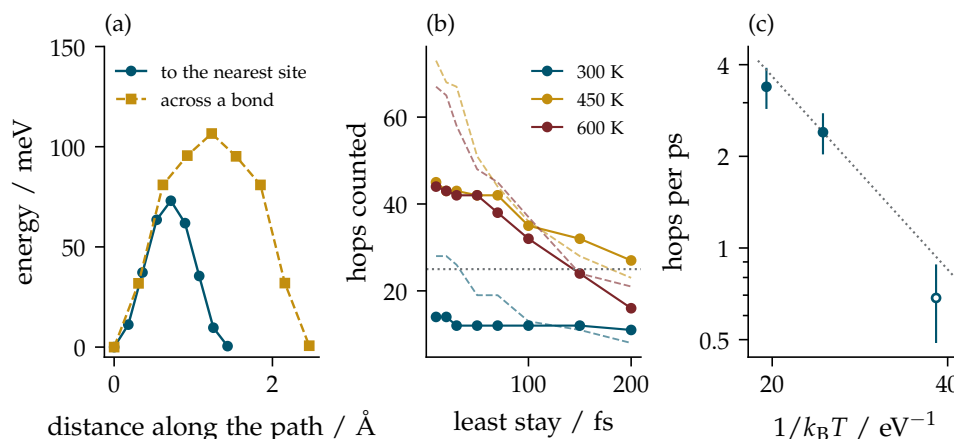


FIGURE 17.4: Lithium in dilute graphite, one lithium among 64 carbon atoms, with MACE-MP-o and D3. (a) The energy along the straight line to the nearest site of the other family (teal) and to the next site of the same family across a C–C bond (ochre), every atom relaxed but lithium and one carbon of each sheet, whose positions in the plane are held. (b) Hops counted after each run's start against the least stay a visit needs to count, with the sites of both families (solid) and of the sheet below alone (dashed), and the 25 hops that resolve a rate to a fifth (dotted). (c) The rate of hops for a least stay of 50 fs, with its Poisson error (open: unresolved), against $1/k_B T$, with the line $\propto e^{-E_a/k_B T}$ through the point at 450 K, for E_a the highest point of the teal path (dotted).

Diffusion

If each hop were independent of the last, the rate would give the diffusion coefficient, $D = k_h \ell^2 / 4$ with $\ell = 1.4231 \text{ Å}$ (Exercise 17.15): with $k_h = 2.40 \times 10^{-3}$ and 3.38×10^{-3} per femtosecond, $D = 2.40 \times 10^{-3} \times 1.4231^2 / 4 = 12.2 \times 10^{-4} \text{ Å}^2 \text{ fs}^{-1}$ at 450 K and $17.1 \times 10^{-4} \text{ Å}^2 \text{ fs}^{-1}$ at 600 K. Lithium's own mean squared displacement after the start, read in the frame midway between the two sheets and fitted from 0.5 to 2 ps in four blocks, gives (5.8 ± 2.2) , (22.4 ± 16.0) and $(36.1 \pm 12.6) \times 10^{-4} \text{ Å}^2 \text{ fs}^{-1}$, fractional errors of 0.38, 0.71 and 0.35, which miss a target of 5% by factors of seven to fourteen. Two departures from the random walk are measured. The hops are not independent: a hop returns to the site just left in 0.15 ± 0.06 of cases at both 450 and 600 K, errors $\sqrt{p(1-p)/m}$ for a fraction p of m pairs of successive hops, three errors below the 1/3 of a random walk on the honeycomb, where the site just left is one of three equally likely; at 300 K, 0.27 ± 0.13 , the difference is within its error. Lithium therefore tends to carry on the way it was going, and the correlation factor of Section 16.8 exceeds 1, as it did on the model surface. And lithium in a site is carried by its sheet, whose slides are as long as its hops. One lithium atom for 20 ps gives a rate to 15% and no diffusion coefficient: the error of D falls as one over the square root of the run's length, so 5% at 450 K needs $(0.71/0.05)^2 = 203$ times the run, or as many lithium atoms in a larger cell, which would also hold the sheets still.

The runs end with 5 ps at fixed energy from the last frame at 600 K, and another from LiC₆ at its mean cell, each keeping 501 frames with their energies and forces. These are the trajectories on which planned Chapter 28 trains a learned propagator.

KEY IDEA. A site is defined in the frame of the atoms that make it, and a hop by a least stay chosen where the count is flat; a count that keeps falling as the least stay grows means the sites are wrong. Hops are counted after the start that the test of Chapter 15 finds. A rate is reported with its Poisson error once 25 hops are seen, after a check that they are independent, otherwise as unresolved with the run that would resolve it. One atom gives a rate long before it gives a diffusion coefficient.

NOTEBOOK 17 *count lithium's hops with both families of sites and with one, and watch the count against the least stay*

CHECK YOUR UNDERSTANDING. At 450 K, how long a run of one lithium atom would resolve the rate to a tenth?

17.7 A protocol for a production run

A pilot runs through a checklist before every flight because missing one item is costly, and the list keeps the order and the checks the same each time. A production run of molecular dynamics has such a list, and every item on it has been measured in an earlier chapter. This section gathers them in the order a run meets them, with the measurement behind each and what a healthy run reads. Table 17.2 collects the readings.

Before the run

The starting structure is checked before any step is taken: the closest approach of two atoms, the largest force and the total momentum, which should be zero (Section 11.8). A structure built by hand, or with a new potential, is first relaxed (Section 15.9), so that the first steps do not turn stored strain into heat. Lattice constants come from measurement where they exist, and the model's own relaxed values are reported beside them, as Sections 17.3 and 17.5 do for graphite and LiC₆. A learned potential is checked on the system before it is trusted: against first-principles forces on configurations like those of the run (Section 17.4), and for any physics its training data lacked, such as the dispersion that holds graphite's sheets at their measured spacing (Section 17.3).

The time step

The step is measured (Section 9.6). Short runs at fixed energy with two or three steps record the fluctuation of the total energy against that of the kinetic energy, which should be small and fall in proportion to δt^2 as the step is shortened, and the drift, which should pass the test of Section 15.5 at the length of the production run. The fastest vibration sets the limit: bonds to hydrogen hold flexible water to 0.5 fs, its energy spreading by 6.7% of the kinetic energy's at 1 fs (Section 11.3), and the stiff carbon-carbon bonds of graphite to the 2 fs that Section 17.3 measures. A learned potential evaluated in single precision adds an error of its own to the forces, and the same test, run in both precisions, shows whether it matters.

Reaching the state

A run is brought to its temperature and pressure first, and only then sampled. Velocity rescaling and Berendsen coupling bring a system near a temperature quickly, but their fluctuations are wrong and a long run under them feeds slow motions (Section 13.2); Berendsen's barostat likewise serves only to approach a pressure (Sections 14.6 and 14.9). The part of the run before the averages settle is found by the search for a start of Section 15.4 and discarded.

Sampling

For averages over arrangements, any thermostat that passes the test of Section 15.8 will do: CSVr passed it with $\tau_T = 1$ ps for the liquid, Nosé-Hoover chains give the canonical spread of the temperature (Section 13.7), and Langevin dynamics samples the canonical ensemble even for a single atom (Section 13.4). For anything that depends on the motion, a diffusion coefficient, a rate, a spectrum or a viscosity, the run is made at fixed energy or under a global thermostat with a long time constant: CSVr at 1 ps left the liquid's D unchanged within its error of 1% (Section 15.6), while a Langevin friction of 1 ps^{-1} slowed it by more than a quarter (Section 13.4). For constant pressure, stochastic cell rescaling or a piston with chains samples the volume's fluctuations correctly, at τ_P near 1 ps for the liquid (Sections 14.6 and 14.7); a solid whose directions differ, such as a layered one, needs its lengths coupled separately (Sections 14.8 and 17.5).

How long

The length of a run follows from the precision asked of each quantity: $t_f \geq 2\tau_{\text{int}}\sigma_A^2/\sigma^2$ (Equation (15.5)), with τ_{int} and σ_A measured in a pilot run. For liquid argon under CSVr this ranged from 0.33 ns for the pressure to 10 ns for the temperature at the

precisions of [Table 15.1](#). Each reported average passes `convergence_report`: a start in the first half of the run, at least 100 effective samples, block errors that agree with the correlation estimate, and no drift. A rate is reported only when enough events are counted to give it an error bar, otherwise as unresolved, with the run that would resolve it ([Section 17.6](#)); a free-energy difference is known only to about $2k_B T / \sqrt{n}$ from n crossings of the barrier ([Section 16.11](#)).

After the run

Transport coefficients from periodic boxes are corrected for the size of the box, or measured at two sizes ([Section 15.7](#)). Displacements are computed from unwrapped positions, removing centre-of-mass motion where appropriate. Each observable is checked against an identity where one exists, such as the energy from $g(r)$ or D from both the MSD and the velocities ([Section 16.12](#)). Finally the record is passed through `health_report` ([Section 17.8](#)), and the run is reported with what makes it repeatable: the code and its version, the model and its weights, every option of the integrator and thermostat with its units, the seeds, and the error bar of every number.

TABLE 17.2: What a healthy run reads, and where the book measured it.

quantity	healthy reading	measured in
total energy at fixed energy	spread small against K 's, falling as δt^2 ; no drift by the test of Ch. 15	Sections 9.6 and 15.5
kinetic temperature	mean on target; spread $\sqrt{2/N_f}$ of it under a canonical thermostat, less at fixed energy	Sections 12.7 and 13.1
velocities	excess kurtosis within $3\sqrt{24/3N}$ of zero	Section 12.5
centre of mass	still, or, under a thermostat that does not keep the momentum, displacements measured from it	Sections 11.8 and 13.8
volume under a barostat	spread $\sqrt{k_B T \langle V \rangle \kappa_T}$; every coupled component of the pressure at P_0	Section 14.5
an average	passes <code>convergence_report</code>	Section 15.9
positions for displacements	continuous, every move a small fraction of the cell	Section 16.12
a rate	at least 25 events for a fifth, or reported unresolved	Section 17.6
a learned potential	forces within the error measured against its reference on this system	Section 17.4 , still to be made

KEY IDEA. A production run is relaxed, given a measured time step, brought to its state with any thermostat and barostat, sampled with one that is canonical, and, for dynamics, a global or no thermostat, run for the length its target errors need, and checked against the readings of a healthy run. Every choice in the list has a measurement behind it in this book.

CHECK YOUR UNDERSTANDING. A diffusion coefficient is to be measured, and the only thermostat at hand is Langevin. What would a careful run do?

17.8 What the trajectory looks like

Every chapter from Chapter 11 on has broken its own runs and found the check that catches each fault. Moving to production codes and a real material adds faults of two kinds: a convention of one code read with the habits of another, and physics that a model or a setting leaves out. This section measures the faults that this chapter met, and then gathers the checks that a single record allows into one function, `diagnostics.health_report`, which reads any record and says which checks it passes; Appendix F lists every fault of Chapters 11 to 17.

This chapter's faults

LAMMPS's arrays read by position. LAMMPS reorders its atoms in memory every thousand steps (Section 17.2). The Langevin run of liquid argon in LAMMPS, its positions read every 100 fs in the order of the first frame instead of by identity, gives $D = (57.7 \pm 7.0) \times 10^{-4} \text{ \AA}^2 \text{ fs}^{-1}$, 21 times the $(2.72 \pm 0.07) \times 10^{-4}$ read correctly: after each reordering every row of the array belongs to another atom, which seems to have jumped across the cell. `largest_jump` finds it at once, with a move of 3.6 cell widths between frames.

Forces compared in two orientations. LAMMPS turns a cell whose first lattice vector does not lie along x into its own orientation (Section 17.3). MACE-MP-o's forces in LAMMPS and in ASE, compared without turning one set, differ by up to $5.9 \text{ eV \AA}^{-1}$, while the energies, which do not turn, agree to nine digits. Equal energies with unequal forces point to the frame before they point to the model.

A gap in the functional. MACE-MP-o alone puts graphite's sheets 16.5% too far apart, since PBE has no dispersion (Section 17.3). The check is the lattice against measurement, and the cure D3.

Isotropic coupling of a layered solid. A barostat that scales LiC_6 's three lengths together holds the spacing of its sheets at the value of the structure it started from, and leaves the pressure along c at $(1.68 \pm 0.10) \text{ GPa}$ with the pressures in the plane near -1.0 GPa (Section 17.5): the diagonal of the pressure tensor shows it, as Section 14.9 found for the model solid.

A reference not converged. FHI-aims's forces on carbon change by up to $0.28 \text{ eV } \text{\AA}^{-1}$ between grids of 37 and 84 k -points (Section 17.4). A comparison with an unconverged reference measures the reference's error as much as the model's.

A step too long for the bonds. With steps of 3 fs, LiC_6 's total energy spreads by 0.082 of the kinetic energy's and drifts by 12.5 errors of the test of Section 15.5 in 2 ps (Section 17.3).

Sites in the wrong frame, or too few of them. Graphite's sheets slide in a small cell by more than the distance between lithium's sites (Section 17.6). With both families of sites fixed in the cell where the sheets started, the count of hops at 450 K falls from 85 to 20 as the least stay a visit needs grows from 10 to 200 fs; with the hollows of the sheet below alone, followed as it slides, half the sites are missing, and the count at 300 K falls from 28 to 8; with both families followed, it falls only from 14 to 11. A rate is reported with the definition of a site and of a hop that produced it, and the least stay is chosen where the count is flat.

A health report

`mdlab.diagnostics.health_report` takes the positions of a run and whatever else its record holds, the cell, masses, velocities, times, energies, symbols, a target temperature, the count of freedoms ($3N - 3$ unless given), the ensemble and whether the run keeps its total momentum, and runs each of the checks below that those allow, with a limit justified in its documentation. No move between frames of half the cell's smallest width or more, the most that unwrapping can undo (Section 10.1). A centre of mass that moves no more than three times the root-mean-square displacement over $\sqrt{N}$, how far N independent atoms of equal mass would carry it (Exercise 17.17), for a run that keeps the total momentum, in which it should not move at all; under Langevin dynamics the centre wanders as one free particle (Section 13.8), further than that, and `mdlab`'s healthy Langevin run of argon fails it, $4.6 \text{ \AA}$ against a limit of $3.4 \text{ \AA}$, so for such runs the check is left out. In the first and last frames, no two atoms nearer than half the sum of their covalent radii (Section 16.3). At fixed energy, a spread of the total energy below a twentieth of the kinetic energy's, the band $\omega^2 \delta t^2 / 4$ of Section 9.6 at about fourteen steps to the period of the fastest vibration, and a slope within two of its errors (Section 15.5). Under a thermostat, a mean temperature within three standard errors of the target and, for a canonical one, a spread of the kinetic temperature within three times its own relative error, $\sqrt{g/2n}$ for n samples of statistical inefficiency g (Exercise 15.7), of the canonical $\sqrt{2/N_f}$ of the target (Exercise 17.16); at fixed energy, where nothing holds it there, a mean temperature within a fifth of the target, between the 4% by which a prepared liquid moves and the half that a start on a lattice loses (Section 12.9). And, in the first and last frames, an excess kurtosis of the velocities within $3\sqrt{24/n}$ of zero, three times the scatter of a Gaussian sample of $n = 3N$ numbers

(Exercise 17.19). `io.read_run` turns any trajectory that ASE can read into the arrays it takes.

On the runs of this chapter it separates the healthy from the broken. `mdlab`'s argon at fixed energy passes all five checks its record allows, its energy spreading by 0.0036 of the kinetic energy's and drifting by 1.3 errors. The same positions wrapped into the cell fail the first check with a move of 0.999 cell widths; carried along with the centre of mass at $\sqrt{3k_B T/Nm}$, they fail the second, with a path of 0.91 Å against a limit of 0.69; the run under CSVR passes its temperature, 0.14 errors from 135 K, and its spread, within 1.6% of the canonical; the misread LAMMPS record fails the first check; and `LiC6` passes at 1 fs and fails both energy checks at 3 fs.

KEY IDEA. Production codes and real materials add faults of convention, arrays reordered in memory and cells turned into a code's own orientation, and faults of physics left out, a functional's missing dispersion, a coupling unsuited to the solid, a reference not converged and sites defined in the wrong frame. `health_report` runs the book's checks of continuity, the centre of mass, overlaps, energy, temperature and velocities on any record, and Appendix F gathers every fault with its check and its cure.

NOTEBOOK 17 read `health_report` on the chapter's records, healthy and broken

CHECK YOUR UNDERSTANDING. Two codes give the same energy for a structure but forces that differ by several eV Å^{-1} . What is the first thing to check?

What we now have

ASE and LAMMPS run the methods of the earlier chapters under their own names, units and defaults: ASE's time in units of 10.18 fs, LAMMPS's in picoseconds, a friction given as a rate in one and as its inverse in the other, chain masses from different counts of freedoms, a pair potential read from a table whose error falls as the square of its spacing, image flags packed into one number, and atoms re-sorted in memory. With these translated, the same argon gives the same spread of the total energy in all three codes, trajectories that part only as chaos amplifies the differences of rounding and of the table, and diffusion coefficients that agree within their errors under each thermostat. MACE-MP-o runs in both codes, in single precision on the graphics processor at a few tenths of a second a step for a hundred atoms; it needs D3 added to hold graphite's sheets at their measured spacing, and a step of at most 2 fs, 1 fs with a margin, for the bonds of carbon. First-principles molecular dynamics costs, on an estimate of its step, about nine hundred times as much, and its *k*-points must be converged before its forces serve as a reference.

LiC_6 needs its lattice lengths coupled apart; the first 20 ps still leave its spacing unresolved. Lithium's sites in dilute graphite are the hollows of both sheets, followed in the frame of each sheet, since the sheets of a small cell slide; hops are counted after the start that Chapter 15's test finds, with a least stay chosen where the count is flat, and rates are reported with their Poisson errors once 25 are seen and otherwise as unresolved, while one atom's diffusion coefficient stays unresolved long after its rate is known. A protocol orders the choices of a production run with the measurement behind each; `health_report` runs the book's checks of a single record on any trajectory, and Appendix F gathers every fault with its check and cure.

Part V begins to open the learned potential that this chapter used as a black box. Chapter 18 starts from the simplest learning problem, fitting a spring's constant to data by least squares, and the planned continuation builds from there to the equivariant networks of MACE.

Answers to the checks

SECTION 17.1. Two of ASE's time units, $2 \times 10.18 \text{ fs} = 20.4 \text{ fs}$: ten times the 2 fs that bonds of carbon allow (Section 17.3), and twenty to forty times the 0.5 to 1 fs that bonds to hydrogen allow (Section 11.3).

SECTION 17.2. In metal units times are picoseconds, so the step is 1000 fs, a hundred times the 10 fs that liquid argon can take; the atoms would run into one another in the first steps and the energy would explode.

SECTION 17.3. The interface: the surface and the water next to it, configurations unlike the bulk metals of the training data. Forces and energies from the reference method on frames of the interface, taken from a short run with the model, show how far the model can be trusted there.

SECTION 17.4. Four times the atoms costs $4^3 = 64$ times as much, $64 \times 192 \text{ s} = 12288 \text{ s}$, about three and a half hours a step. In this frame finding the eigenvectors took 38 s of the 385 s, so at a few dozen atoms the cost grows more slowly than this; the cube is where it ends.

SECTION 17.5. Three. Each hexagon has six corners, each shared by three hexagons, so a sheet has two carbon atoms for every hexagon; one lithium over every third hexagon gives one lithium for six carbon atoms.

SECTION 17.6. A tenth needs $1/0.1^2 = 100$ hops, which at 2.40 ps^{-1} take 42 ps after the start.

SECTION 17.7. Use Langevin dynamics only to bring the system to its temperature, then switch it off and measure D at fixed energy, as the protocol prefers. If the run must stay under friction, measure D at two or three small frictions and follow the trend to zero friction (Section 13.4 found 0.97, 0.71 and 0.17 of it at 0.1, 1 and 10 ps⁻¹), reporting the frictions with the result.

SECTION 17.8. The orientation of the cell: whether both codes hold the structure turned the same way. Equal energies with unequal forces point first to the same structure seen in two frames.

Exercises

The exercises run from questions of understanding, through calculations, to short derivations marked ‘show that’. Full solutions are in ‘Solutions to the exercises’ at the end of the book, and Notebook 17 checks every numerical answer.

EXERCISE 17.1 (*another code’s unit of time*). Some codes for biomolecules measure energy in kilocalories per mole, length in Å and mass in amu, with no conversion factor in $F = ma$. Find their unit of time in femtoseconds, with 1 kcal = 4184 J and a mole of $6.022\,140\,76 \times 10^{23}$ molecules. What number does such a code need for a step of 2 fs?

EXERCISE 17.2 (*one velocity, three codes*). An atom moves at 0.01 Å fs⁻¹ in mdlab. What number gives its velocity in ASE’s units, and what number in LAMMPS’s metal units?

EXERCISE 17.3 (*friction in two codes*). A Langevin friction of 0.5 ps⁻¹ is wanted. What damping parameter does LAMMPS’s `fix langevin` take, and what friction does ASE’s Langevin take, each in its own units?

EXERCISE 17.4 (*two masses for one chain*). ASE gives the first variable of a Nosé-Hoover chain the mass $3Nk_B T \tau_T^2$, and mdlab $N_f k_B T \tau_T^2$ with $N_f = 3N - 3$. By what factor do they differ for 256 atoms, and by what factor does the period of the thermostat’s oscillation, which goes as the square root of the mass, differ?

EXERCISE 17.5 (*the error of a table*). A smooth function $y(x)$ is replaced between two points δx apart by the straight line through its values there. Show that the line misses y by at most $\delta x^2 \max|y''|/8$.

EXERCISE 17.6 (*points for a precision*). LAMMPS’s own table of 20 001 entries misses the force beyond 3 Å by at most 7.6×10^{-7} eV Å⁻¹. How many entries over the same range would bring the miss to 1×10^{-9} eV Å⁻¹?

EXERCISE 17.7 (*image flags*). LAMMPS packs the three image counts of an atom into one whole number, $(i_x + 512) + 1024(i_y + 512) + 1024^2(i_z + 512)$, each count lying between -512 and 511 . Show that the counts are recovered from the number by remainders and whole quotients on division by 1024 , and find them for the number $540\,539\,393$.

EXERCISE 17.8 (*a nanosecond of a foundation model*). MACE-MP-o's medium model takes 0.25 s a call for 126 atoms on a GPU, in proportion to the number of atoms. How long would a nanosecond of 1000 atoms take, with steps of 1 fs?

EXERCISE 17.9 (*sheets held loosely*). Without D_3 , MACE-MP-o binds graphite's sheets by 5.2 meV per atom against $c = 9$ Å; with it, by 41.2 meV. Compare each with $k_B T$ at 300 K. The height of a cell is one number shared by all its atoms: what does stretching the 64 carbon atoms of Section 17.6 to $c = 9$ Å cost without D_3 , and what might a run at room temperature under a barostat do?

EXERCISE 17.10 (*the longest step*). LiC_6 's total energy spreads by 0.0067 of the kinetic energy's with steps of 1 fs. If the spread grows as δt^2 , at what step does it reach the limit of 0.05 that health_report uses? The run at 3 fs spreads by 0.082 : what does the excess over the estimate say?

EXERCISE 17.11 (*the price of a reference*). A first-principles step of the 28 -atom cell costs 192 s on four cores. How many core-hours, the number of cores times the hours they run, does a run of 10 ps with steps of 1 fs take, and how long on 192 cores if the work divided perfectly?

EXERCISE 17.12 (*the $\sqrt{3} \times \sqrt{3}$ cell*). With $\mathbf{a} = a(1, 0)$ and $\mathbf{b} = a(-\frac{1}{2}, \frac{\sqrt{3}}{2})$, show that $2\mathbf{a} + \mathbf{b}$ and $-\mathbf{a} + \mathbf{b}$ have length $\sqrt{3}a$, that they are 120° apart, and that the cell they span has three times the area of the sheet's own cell, and so holds six carbon atoms.

EXERCISE 17.13 (*the mean of an unequal pressure*). Under isotropic coupling LiC_6 's pressures along x , y and z are -1.00 ± 0.45 , -1.11 ± 0.36 and 1.68 ± 0.10 GPa. What is their mean, with its error if the three are independent, and why is a barostat that controls only the mean satisfied by it?

EXERCISE 17.14 (*how long to resolve a rate*). A rate is wanted to a fifth. How long a run of one atom does a rate of 2 hops per picosecond need? A run of 20 ps sees no hop at all. Using the Poisson chance of no event, $\mathcal{P}(0) = e^{-\nu X}$ (Section 16.4), find the rate above which a run would see none less than 5% of the time, an upper bound with 95% confidence, and how long a run must be before even that bound would be resolved.

EXERCISE 17.15 (*hops on a honeycomb*). Sites form a honeycomb of two families that alternate, neighbours ℓ apart, and an atom hops to one of its three neighbours at

random at the rate k_h . A hop from a site of the first family is one of three vectors ℓ_1, ℓ_2, ℓ_3 of length ℓ at 120° to one another, and a hop from the second family one of $-\ell_1, -\ell_2, -\ell_3$. Show that $\ell_1 + \ell_2 + \ell_3 = \mathbf{0}$, that the hops are therefore uncorrelated although the families alternate, and hence that $D = k_h \ell^2 / 4$ (Section 16.8). Evaluate $\ell = a / \sqrt{3}$ for lithium's sites in dilute graphite, with $a = 2.4648 \text{ \AA}$.

EXERCISE 17.16 (*the scatter of a squared ratio*). For a Gaussian number z of mean 0 and variance 1, show that z^2 has the mean 1 and the variance 2, using $\langle z^4 \rangle = 3$ (Section 12.5), so that the mean of n such squares scatters about 1 by $\sqrt{2/n}$. Hence show that the kinetic temperature of a canonical run, the target times the mean of N_f such squares, scatters by $\sqrt{2/N_f}$ of the target, the spread that health_report checks.

EXERCISE 17.17 (*the wandering centre*). N atoms of equal mass move independently, each with no mean displacement and the mean square displacement $\langle \Delta r^2 \rangle$. Show that their centre of mass has the mean square displacement $\langle \Delta r^2 \rangle / N$, the basis of health_report's limit on the centre's path.

EXERCISE 17.18 (*half a width*). Explain why no rule can recover a path from wrapped positions if an atom may move more than half a cell's width between frames, and what health_report's first check therefore guards.

EXERCISE 17.19 (*the kurtosis limit*). For n independent Gaussian numbers z_i of mean 0 and variance 1, the sample's excess kurtosis is $m_4 / m_2^2 - 3$, with m_2 and m_4 the means of z_i^2 and z_i^4 . Writing $m_2 = 1 + \epsilon_2$ and $m_4 = 3 + \epsilon_4$ and keeping the first order in the small ϵ , and using $\langle z^6 \rangle = 15$ and $\langle z^8 \rangle = 105$, show that it scatters about zero by $\sqrt{24/n}$. What limit does health_report, which allows three times this, set for the 768 velocity components of 256 atoms, and would velocities drawn evenly from an interval, of excess kurtosis -1.2 , pass it?

EXERCISE 17.20 (*a plan for a run*). A diffusion coefficient of lithium in a new electrolyte is wanted to 5% with a learned potential that has not been used on it. List, in order, what the protocol of Section 17.7 asks before the production run starts.

Learning as optimisation

A spring balance in a shop is calibrated by hanging known weights on it and marking where the pointer stops. Each reading carries a small error, from the pointer's width and the eye that reads it, so no single weight fixes the scale; the marks are drawn so that they miss the readings as little as possible, taken together. A learned interatomic potential is made the same way. Its readings are energies and forces from a first-principles calculation, its marks are the numbers inside the model, and drawing them is a search for the numbers that miss the readings least. Chapter 17 ran such a model, MACE-MP-o, as a black box; this chapter begins to open the box; the planned chapters that follow develop the network models themselves.

We begin with a loss, a number that measures how far a model misses its readings. A least-squares fit first determines one weight, the stiffness of a bond, then several weights in a curve through noisy readings of argon's pair energy. This exposes a difficulty: enough adjustable weights can fit the noise as well as the underlying curve. A penalty on large weights controls that freedom, and its interpretation as a prior leads to Gaussian processes and their uncertainty estimates.

When a fit cannot be obtained in one calculation, we lower the loss by gradient descent. Its step has a stability limit like an integrator's; mini-batches, momentum and Adam change the cost and progress of that search. The final sections separate the readings used to train a model from those used to judge it, showing why consecutive trajectory frames must not be split at random. Planned Chapter 19 builds neural networks and trains them with these optimisers.

18.1 Data, model and loss

A spring is stretched by 1, 2, ..., 10 cm, and at each stretch the force is read on a dial that errs by about 0.05 N (Figure 18.1a, readings drawn by computer for a spring of 25 N m^{-1}). Hooke's law (Section 2.8) says the force is kx for some stiffness k , and the task is to choose k . Three things make up any such task. The *data* are the pairs of stretch and force, (x_i, F_i) for $i = 1$ to n ; the *model* is the rule $F = kx$ that turns a stretch into a predicted force, with k its one adjustable number, its *weight*; and the *loss* measures how far the model misses the data. The usual loss is the sum of squared misses,

$$\mathcal{L}(k) = \sum_{i=1}^n (F_i - kx_i)^2, \quad (18.1)$$

which is zero only if every reading lies on the line, is never negative, and counts a miss twice as large four times as heavily. Learning, here, is finding the k that makes $\mathcal{L}$ least.

$\mathcal{L}$ is a parabola in k (Figure 18.1b): expanding the square, $\mathcal{L} = \sum_i F_i^2 - 2k \sum_i x_i F_i + k^2 \sum_i x_i^2$, a quadratic with a positive coefficient of k^2 . Its lowest point is where its slope vanishes:

$$\frac{d\mathcal{L}}{dk} = -2 \sum_i x_i (F_i - kx_i) = 0 \quad \text{differentiating each square by the chain rule}$$

$$\sum_i x_i F_i = k \sum_i x_i^2 \quad \text{dividing by } -2 \text{ and moving the } k \text{ terms}$$

$$\hat{k} = \frac{\sum_i x_i F_i}{\sum_i x_i^2} \quad \text{solving, the hat marking the fitted value.}$$

For the readings of the figure $\hat{k} = 25.72 \text{ N m}^{-1}$. Its error follows as in the least-squares line of Section 15.5: $\hat{k}$ is a weighted sum of the readings, $\sum_i w_i F_i$ with $w_i = x_i / \sum_j x_j^2$, so if the readings err independently by σ_m , its variance is $\sigma_m^2 \sum_i w_i^2 = \sigma_m^2 / \sum_i x_i^2$. With σ_m estimated from the misses about the fitted line, 0.056 N, the stiffness is $(25.72 \pm 0.28) \text{ N m}^{-1}$, 2.5 errors from the 25 N m^{-1} the readings were drawn from. Ten readings fix the stiffness to about 1%, and with an error estimated from only ten readings a miss this large comes in 3% of such fits, by Student's factor (Section 15.1).

The stiffness of a bond

The same fit gives the stiffness of a chemical bond from a molecular dynamics run. The A–B molecule of Chapter 16, two atoms of nitrogen's mass held by a Morse well of depth $D = 1 \text{ eV}$ and width $a = 2 \text{ \AA}^{-1}$, has the stiffness $2Da^2 = 8 \text{ eV \AA}^{-2}$ at the bottom of its well (Section 4.5). Run for 20 ps under Langevin dynamics, each 10 fs gives a

reading of the stretch $x = r - r_0$ and the force along the bond, 2001 readings in all (Figure 18.1c). At 100 K the stretch spreads by $0.0345 \text{ \AA}$, close to the $\sqrt{k_B T/k} = 0.0328 \text{ \AA}$ that equipartition gives a spring of that stiffness (Section 12.6); at 1000 K by $0.127 \text{ \AA}$, against 0.104 , reaching $0.72 \text{ \AA}$ on the soft outer side of the well.

A straight line through these readings needs two weights, an intercept θ_0 and a slope, $F \approx \theta_0 - kx$, since the bond's mean length need not sit at r_0 . Two weights are found as one was, with a little more bookkeeping, which the matrix form of the next toolbox does once for any number of weights. The line's stiffness is $7.66 \text{ eV \AA}^{-2}$ at 100 K, 7.07 at 300 and 5.26 at 1000 K, below the 8 of the well's bottom and further below the hotter the run. The fit is sound: the Morse well's own stiffness changes along it, as $2Da^2(2e^{-2ax} - e^{-ax}) \approx 8(1 - 3ax)$ for small x , by about a fifth over the range the bond explores even at 100 K, and a line fitted over a range gives the stiffness averaged over that range, which the softer outer side pulls down. A model answers the question its data pose: readings from a hot run teach the hot bond. Adding the powers x^2 , x^3 and x^4 to the model lets the fitted curve bend with the force, and its slope at $x = 0$ then gives 8.126 , 8.007 and $8.000 \text{ eV \AA}^{-2}$ at 100 K, and 8.084 at 1000 K with four powers. Choosing what goes into a model is the subject of Section 18.2.

LEAST SQUARES IN MATRIX FORM. A model whose prediction is a sum of weights times known functions, $\hat{y}(x) = \sum_{j=1}^p \theta_j \phi_j(x)$, is linear in its weights. At the n readings x_i the predictions form the vector $X\theta$, with the *design matrix* $X_{ij} = \phi_j(x_i)$, one row per reading and one column per function; for the line $\theta_0 - kx$, the columns are 1 and x_i , and $\theta = (\theta_0, -k)$. The loss is the squared length of the vector of misses,

$$\mathcal{L}(\theta) = (X\theta - \mathbf{y})^T (X\theta - \mathbf{y}) = \sum_{i=1}^n \left(\sum_{l=1}^p X_{il} \theta_l - y_i \right)^2.$$

Its rate of change with one weight θ_j , by the chain rule on each square, is

$$\frac{\partial \mathcal{L}}{\partial \theta_j} = \sum_i 2 \left(\sum_l X_{il} \theta_l - y_i \right) X_{ij} = 2 [X^T (X\theta - \mathbf{y})]_j,$$

since $\sum_i X_{ij} m_i$ is the j th entry of $X^T \mathbf{m}$ for any vector of misses $\mathbf{m}$. The p rates together are the *gradient of the loss*, $\nabla \mathcal{L} = 2X^T (X\theta - \mathbf{y})$, a vector pointing the way $\mathcal{L}$ rises fastest in the space of weights (Section 3.10). At the least loss every rate vanishes, which gives the *normal equations*

$$X^T X \theta = X^T \mathbf{y}, \quad (18.2)$$

p linear equations for p weights, solved as in Section 5.6. For one weight and the single column x_i they are $(\sum_i x_i^2)k = \sum_i x_i y_i$, the result above.

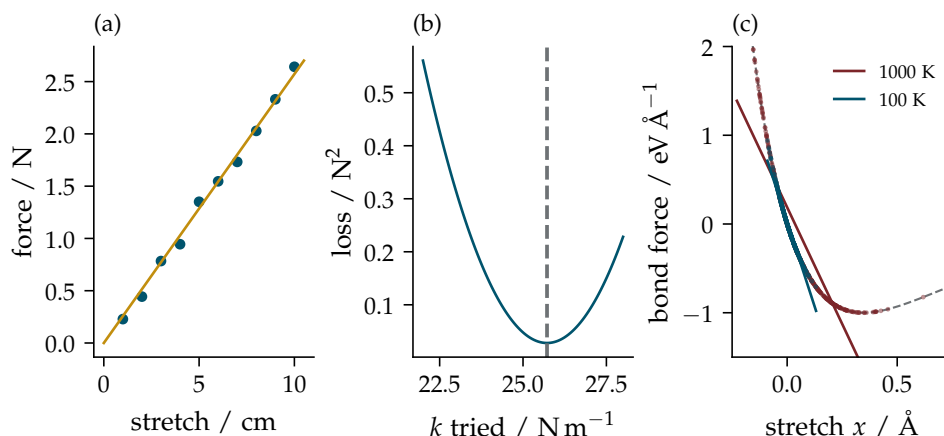


FIGURE 18.1: One weight learned by least squares. (a) A spring's force read on a dial with an error of 0.05 N at ten stretches (readings drawn by computer for 25 N m⁻¹), and the line through the origin of least squares. (b) The sum of squared misses against the stiffness tried; the fit is its lowest point (dashed). (c) The A–B molecule of Chapter 16 under Langevin dynamics: the force along the bond against its stretch, every 50 fs of 20 ps, at 100 and 1000 K, with the Morse force (grey dashed) and the straight lines of least squares.

KEY IDEA. Learning a model from data is the lowering of a loss, here the sum of squared misses. For a model linear in its weights the least loss solves the normal equations $X^T X \theta = X^T y$ at once. A fit answers the question its data pose: a line through the forces of a hot run gives the stiffness averaged over the stretches that run explores.

NOTEBOOK 18 *fit a spring and a bond, and change the temperature of the bond's run*

CHECK YOUR UNDERSTANDING. Ten readings of a spring's force at stretches x_i all lie exactly on a line through the origin. What is $\mathcal{L}(\hat{k})$, and what does the formula for the error of $\hat{k}$ then give?

18.2 Basis functions and overfitting

A curve drawn freehand through a dozen noisy points can be made to pass through every one of them, if it is allowed to wiggle; between the points it then swings far from any sensible value, and a new point measured in a gap lands nowhere near it. The curve has learned the noise. A smoother curve that passes near the points, rather than through them, predicts the gaps better. Choosing how freely a model may bend is the first decision of any fit.

A curve through noisy readings

The readings here are argon's pair energy $\varphi(r)$ between $3.3 \text{ \AA}$ and $2\sigma = 6.8 \text{ \AA}$, where the switched potential of Chapter 12 is the plain Lennard-Jones energy, read at 15 distances drawn at random with an error of 0.2 meV added, as a first-principles calculation adds its own small errors (Figure 18.2a). The model is a polynomial,

$$\hat{\varphi}(r) = \sum_{j=0}^{p-1} w_j s^j, \quad s = \frac{r - 5.05 \text{ \AA}}{1.75 \text{ \AA}},$$

in the scaled distance s , which runs from -1 to 1 over the range; the functions s^j are its *basis functions*, and the columns of the design matrix are their values at the readings. The scaling matters for the arithmetic, not for the fit: the powers of r itself range from 1 to $6.8^{14} \approx 5 \times 10^{11}$, and Section 18.5 measures what that does. The weights come from the normal equations (Equation (18.2)), and the fit is judged on 200 distances spread evenly over the range, without noise, which the fit never saw: the *test*.

Figure 18.2(b) tells the story. As the degree of the polynomial rises from 0 , the rms miss on the 15 training readings falls, from 3.38 to 0.20 meV at degree 8 , near the noise, and then further, to zero at degree 14 , where 15 weights pass a curve through 15 readings exactly. The miss on the test falls with it at first, to 0.28 meV at degree 8 , and then rises steeply: 3.9 meV at degree 9 , 140 at 12 and 41 eV at 14 , where the curve swings between the readings (Figure 18.2a). A model with too few weights misses the curve, *underfitting*; one with too many learns the noise of its readings as if it were part of the curve, *overfitting*, and the training miss falling below the noise of the readings is its sign. Only data the model has not seen can tell the two apart, since the training miss improves with every weight added.

The functions themselves can carry knowledge. The Lennard-Jones energy is a sum of $(\sigma/r)^{12}$ and $(\sigma/r)^6$, and a model built on those two functions has two weights to find. Fitted to the same 15 readings, they come out as 0.04051 and -0.04073 eV , against $4\epsilon = 0.04136 \text{ eV}$ and -4ϵ , and the test miss is 0.090 meV , a third of the best polynomial's with two weights against nine. A learned potential faces the same choice on a larger scale: a model that knows the forces between atoms depend only on their distances and on the elements needs far fewer readings than one that must discover it, which is what planned Chapter 20 builds into the description of an atom's surroundings.

KEY IDEA. A model linear in its weights is a sum of basis functions, and the number of them sets how freely it bends. Too few underfit; too many learn the noise, which shows as a training miss below the noise of the readings and a test miss that grows. Only readings the model has not seen judge it. Functions that carry the physics need fewer weights and fewer readings.

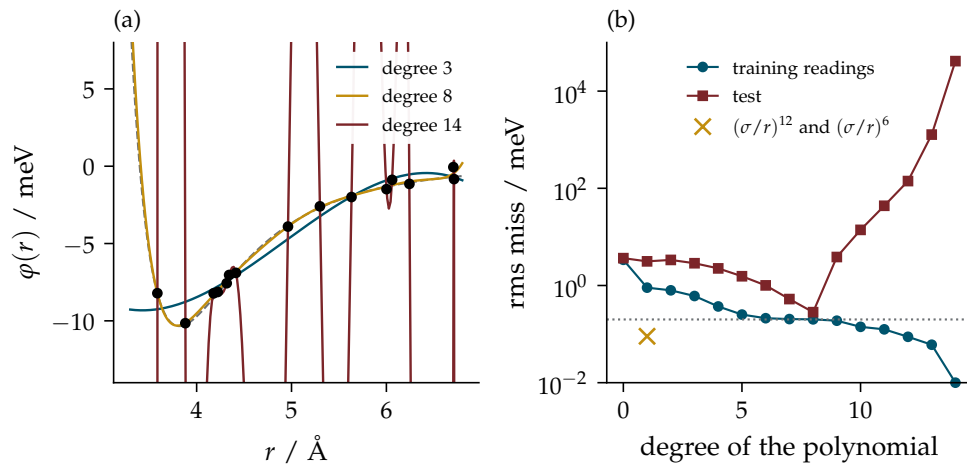


FIGURE 18.2: Basis functions and overfitting. (a) Argon's pair energy (grey dashed) read at 15 distances with an error of 0.2 meV (dots), and the polynomials of degree 3, 8 and 14 in the scaled distance fitted to the readings by least squares. (b) The rms miss on the training readings and on a test of 200 noise-free distances against the degree, with the noise (dotted) and the test miss of the two functions $(\sigma/r)^{12}$ and $(\sigma/r)^6$ (cross). At degree 14 the curve passes through every reading, and its training miss, zero, is drawn on the axis.

NOTEBOOK 18 *change the degree of the polynomial and watch the training and test misses part*

CHECK YOUR UNDERSTANDING. A polynomial of degree 14 is fitted to 15 readings. Why is its training miss zero whatever the readings, and what does that say about the training miss as a measure of a model?

18.3 Ridge regularisation and its Bayesian reading

A tailor cutting a suit from a few measurements prefers a pattern close to a standard one, and departs from it only as far as the measurements demand; a stray reading of the sleeve does not make the jacket lopsided. A fit can be given the same preference. The polynomial of degree 14 in Figure 18.2 passed through every reading with weights as large as 1×10^4 eV, which cancel one another to near zero at the readings and nowhere else. A penalty on large weights keeps the curve from paying that price for each stray reading.

The penalised loss

Ridge regularisation adds to the sum of squared misses the squared length of the weight vector, times a number $\lambda \geq 0$:

$$L(\mathbf{w}) = |\mathbf{X}\mathbf{w} - \mathbf{y}|^2 + \lambda |\mathbf{w}|^2. \quad (18.3)$$

Its gradient is that of the toolbox of [Section 18.1](#) plus the gradient of $\lambda \sum_j w_j^2$, which is $2\lambda \mathbf{w}$, so the least loss solves

$$2X^T(X\mathbf{w} - \mathbf{y}) + 2\lambda \mathbf{w} = \mathbf{0} \quad \text{setting the gradient to zero}$$

$$(X^T X + \lambda I)\mathbf{w} = X^T \mathbf{y} \quad \text{dividing by 2 and gathering the terms in } \mathbf{w},$$

with I the identity matrix. The penalty is a sum of squares too: it is the squared miss of the extra equations $\sqrt{\lambda} w_j = 0$, one for each weight, so the ridge loss is the ordinary loss of the design matrix X with $\sqrt{\lambda} I$ stacked below it, against $\mathbf{y}$ with p zeros below it, which is how `mdlab.learn.least_squares` solves it:

```

1 def least_squares(X: ArrayLike, y: ArrayLike, ridge: float = 0.0) -> NDArray:
2     X = np.asarray(X, dtype=float)
3     y = np.asarray(y, dtype=float)
4     if ridge:
5         p = X.shape[1]
6         X = np.vstack([X, np.sqrt(ridge) * np.eye(p)])
7         y = np.concatenate([y, np.zeros((p, *y.shape[1:]))])
8     return np.linalg.lstsq(X, y, rcond=None)[0]

```

NumPy's `lstsq` finds the least squared miss without forming $X^T X$, whose entries are squares and so lose half the digits that a badly scaled X still holds.

Why the penalty helps shows in the eigenvectors of $X^T X$, a symmetric matrix with eigenvalues $d_k \geq 0$ and orthonormal eigenvectors $\mathbf{v}_k$ ([Section 4.9](#)). Writing the weights as $\mathbf{w} = \sum_k c_k \mathbf{v}_k$ and taking the dot product of the normal equations with $\mathbf{v}_k$,

$$(d_k + \lambda) c_k = \mathbf{v}_k \cdot X^T \mathbf{y}, \quad c_k = \frac{\mathbf{v}_k \cdot X^T \mathbf{y}}{d_k + \lambda},$$

since $X^T X \mathbf{v}_k = d_k \mathbf{v}_k$ and the other eigenvectors are perpendicular to $\mathbf{v}_k$. Without the penalty each component is divided by d_k . A direction with a small d_k is one the readings barely constrain, a combination of basis functions that is nearly zero at every reading, and dividing by a small d_k turns the noise of the readings into a large weight along it. The penalty leaves directions with $d_k \gg \lambda$ almost as they were and shrinks those with $d_k \ll \lambda$ towards zero by the factor $d_k / (d_k + \lambda)$.

Choosing the penalty

λ is not learned from the training readings, which always prefer $\lambda = 0$. Fifteen more readings, drawn as the first were, are held back for *validation*, and λ is chosen to make the miss on them least. For the polynomial of degree 14 ([Figure 18.3a](#)), the validation miss is least at $\lambda = 10^{-3}$, where the test miss is 0.15 meV, half that of the best

polynomial without the penalty and close to the two-function fit's 0.09; the weights shrink from a length of $1.1 \times 10^4 \text{ eV}$ to 0.014 eV (Figure 18.3b). Below $\lambda \approx 10^{-6}$ the noise is learned again; above 10^{-2} the curve is held too stiff to follow the well. The test, a third set, is kept for the end: having chosen λ by the validation readings, their miss is no longer an honest estimate of the miss on new ones.

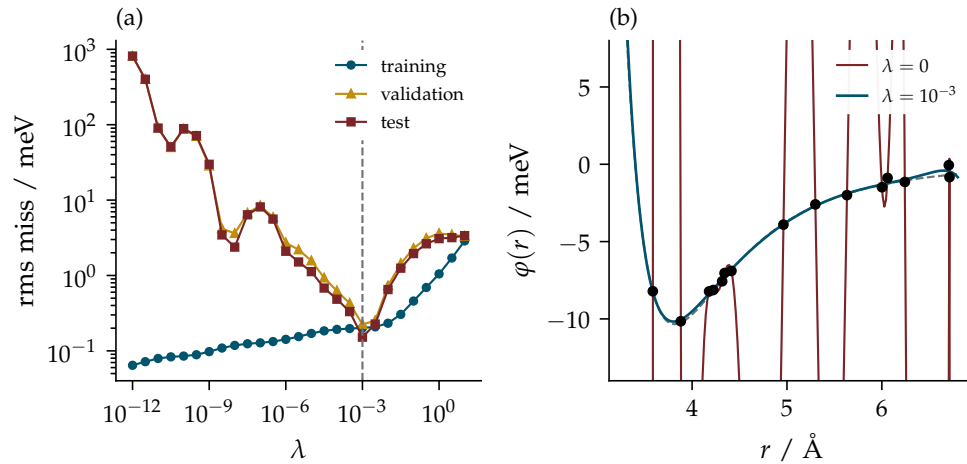


FIGURE 18.3: Ridge regularisation of the polynomial of degree 14. (a) The rms miss on the 15 training readings, on 15 validation readings and on the noise-free test against λ , with the λ of least validation miss (dashed). (b) The fit without the penalty and with $\lambda = 10^{-3}$, against the pair energy (grey dashed) and the training readings.

CONDITIONAL PROBABILITY AND BAYES'S RULE. Of N equally likely outcomes, N_A have a property A , N_B a property B and N_{AB} both. The probability of B among the outcomes with A , the *conditional probability* of B given A , is $\mathcal{P}(B | A) = N_{AB}/N_A$, and multiplying and dividing by N ,

$$\mathcal{P}(B | A) = \frac{N_{AB}/N}{N_A/N} = \frac{\mathcal{P}(A \text{ and } B)}{\mathcal{P}(A)}.$$

The same holds with A and B exchanged, so $\mathcal{P}(A \text{ and } B) = \mathcal{P}(B | A) \mathcal{P}(A) = \mathcal{P}(A | B) \mathcal{P}(B)$, and dividing by $\mathcal{P}(B)$ gives *Bayes's rule*,

$$\mathcal{P}(A | B) = \frac{\mathcal{P}(B | A) \mathcal{P}(A)}{\mathcal{P}(B)}.$$

For quantities with densities (Section 12.1) the same rule holds with densities in place of probabilities. Read with A a cause and B an observation, it turns the chance of the observation given each cause into the chance of each cause given the observation.

The Bayesian reading

The penalty has a second meaning. Suppose the readings are the model's predictions plus independent Gaussian noise of spread σ , so that the density of the readings given the weights is

$$p(\mathbf{y} | \mathbf{w}) \propto \prod_i \exp \left[-\frac{(y_i - [X\mathbf{w}]_i)^2}{2\sigma^2} \right] = \exp \left[-\frac{|\mathbf{X}\mathbf{w} - \mathbf{y}|^2}{2\sigma^2} \right],$$

a product of Gaussians (Section 12.5), and suppose that before any reading each weight was believed to lie near zero, within a spread σ_w : a *prior* density $p(\mathbf{w}) \propto \exp(-|\mathbf{w}|^2/2\sigma_w^2)$. Bayes's rule gives the density of the weights given the readings, the *posterior*,

$$p(\mathbf{w} | \mathbf{y}) = \frac{p(\mathbf{y} | \mathbf{w}) p(\mathbf{w})}{p(\mathbf{y})} \propto \exp \left[-\frac{|\mathbf{X}\mathbf{w} - \mathbf{y}|^2}{2\sigma^2} - \frac{|\mathbf{w}|^2}{2\sigma_w^2} \right],$$

where $p(\mathbf{y})$ does not depend on $\mathbf{w}$. The most probable weights make the bracket least; multiplying it by $-2\sigma^2$, they make $|\mathbf{X}\mathbf{w} - \mathbf{y}|^2 + (\sigma^2/\sigma_w^2)|\mathbf{w}|^2$ least, the ridge loss with

$$\lambda = \frac{\sigma^2}{\sigma_w^2}.$$

Ridge regression is the most probable fit under Gaussian noise and a Gaussian belief that the weights are small. The chosen $\lambda = 10^{-3}$ with the noise of 0.2 meV corresponds to $\sigma_w = \sigma/\sqrt{\lambda} = 6.3$ meV, a belief that each weight is of the size of the well's depth, which is sensible for a curve whose values are of that size and whose basis functions lie between -1 and 1 . The posterior is more than its most probable point: it gives every weight a spread, and through them the curve an error bar, which the next section obtains in another way.

KEY IDEA. Ridge regularisation adds $\lambda|\mathbf{w}|^2$ to the loss, so the least loss solves $(X^T X + \lambda I)\mathbf{w} = X^T \mathbf{y}$; it shrinks the directions the readings barely constrain. λ is chosen on validation readings and the result judged on test readings never used for any choice. It is the most probable fit under Gaussian noise of spread σ and a Gaussian prior of spread σ_w on the weights, with $\lambda = \sigma^2/\sigma_w^2$.

NOTEBOOK 18 vary the penalty and see which directions of the weights shrink

CHECK YOUR UNDERSTANDING. The readings' noise doubles while the prior belief about the weights is unchanged. What happens to λ , and to how closely the fit follows the readings?

18.4 Kernels and Gaussian processes

A forecaster who knows the temperature at a few weather stations estimates it in a village between them by leaning on the nearest stations most, since temperatures a few kilometres apart are alike, and says how sure they are: close to a station they are nearly certain, far from every station they can say little more than what is usual for the season. A Gaussian process makes this reasoning exact. It needs no basis functions chosen in advance, only a statement of how alike the values at two points are expected to be, and it returns, at every point, an estimate and an error bar.

A random function

Before any reading, the unknown curve $f(x)$ is treated as random: its values at any set of points are jointly Gaussian with mean zero, and the values at two points x and x' have the covariance

$$k(x, x') = h^2 \exp\left[-\frac{(x - x')^2}{2\ell^2}\right], \quad (18.4)$$

the *kernel*. Each value has the variance $k(x, x) = h^2$, a spread h about zero; values at points much closer than the length scale ℓ are nearly equal, their covariance almost h^2 , and values much further apart are nearly independent. The kernel is the forecaster's belief that nearby temperatures are alike, written as a number. Readings $y_i = f(x_i) + \epsilon_i$ with independent noise ϵ_i of spread σ then have the covariances $K_{ij} = k(x_i, x_j) + \sigma^2 \delta_{ij}$, the noise adding to each reading's own variance only. The value $f(x_*)$ at a new point and the readings are jointly Gaussian, with the covariances $k(x_*, x_i)$, collected in the vector $\mathbf{k}_*$. The question is what the readings say about $f(x_*)$: the Gaussian conditioned on them.

A GAUSSIAN CONDITIONED ON ANOTHER. Let a and b be jointly Gaussian with mean zero, variances σ_a^2 and σ_b^2 and covariance $c = \langle ab \rangle$. The number $e = a - (c/\sigma_b^2)b$ has the covariance with b

$$\langle eb \rangle = \langle ab \rangle - \frac{c}{\sigma_b^2} \langle b^2 \rangle = c - \frac{c}{\sigma_b^2} \sigma_b^2 = 0.$$

A linear combination of jointly Gaussian numbers is Gaussian, and two jointly Gaussian numbers with zero covariance are independent, since their joint density,

$\exp(-e^2/2\sigma_e^2 - b^2/2\sigma_b^2)$ up to a constant, is the product of the two densities (Section 12.1). So knowing b says nothing about e , and $a = (c/\sigma_b^2)b + e$ has, given b , the mean $(c/\sigma_b^2)b$ and the variance of e ,

$$\begin{aligned}\sigma_e^2 &= \langle a^2 \rangle - 2\frac{c}{\sigma_b^2}\langle ab \rangle + \frac{c^2}{\sigma_b^4}\langle b^2 \rangle && \text{expanding the square of } e \\ &= \sigma_a^2 - \frac{2c^2}{\sigma_b^2} + \frac{c^2}{\sigma_b^2} = \sigma_a^2 - \frac{c^2}{\sigma_b^2} && \text{collecting the terms in } c^2.\end{aligned}$$

The same steps with b a vector of readings, of covariance matrix K , and $\mathbf{c} = \langle a\mathbf{b} \rangle$ a vector, give the mean $\mathbf{c}^\top K^{-1}\mathbf{b}$ and the variance $\sigma_a^2 - \mathbf{c}^\top K^{-1}\mathbf{c}$, the division by σ_b^2 becoming the inverse matrix (Section 5.6). Knowing b moves the mean and always lowers the variance.

With $a = f(x_*)$, $\mathbf{b} = \mathbf{y}$ and $\mathbf{c} = \mathbf{k}_*$, the readings give $f(x_*)$ the mean and variance

$$\bar{f}(x_*) = \mathbf{k}_*^\top K^{-1}\mathbf{y}, \quad \text{var } f(x_*) = h^2 - \mathbf{k}_*^\top K^{-1}\mathbf{k}_*. \quad (18.5)$$

The mean is a weighted sum of the readings, as the forecaster's estimate is, the weights $K^{-1}\mathbf{k}_*$ larger for readings near x_* . Close to a reading the variance falls to about the noise's σ^2 ; far from all of them $\mathbf{k}_*$ vanishes and the variance returns to the prior's h^2 . This is the posterior of the weights of Section 18.3 in another form: a Gaussian process is ridge regression on as many basis functions as the kernel implies, possibly infinitely many, solved through the $n \times n$ matrix K of the readings rather than the $p \times p$ matrix $X^\top X$ of the weights¹⁰³. `mdlab.learn.GaussianProcess` solves it with the factor $K = LL^\top$, L lower triangular, by which a system with K becomes two systems with triangles, each solved one unknown at a time.

The pair energy from eight readings

Eight readings of argon's pair energy, drawn as in Section 18.2, are fitted with $h = 10$ meV, about the depth of the well, and the known noise of 0.2 meV (Figure 18.4). The length scale still has to be chosen, and the readings themselves choose it through their *marginal likelihood*, the density of the readings under the prior. Since $\mathbf{y}$ is Gaussian with covariance K , its logarithm is

$$\ln p(\mathbf{y}) = -\frac{1}{2}\mathbf{y}^\top K^{-1}\mathbf{y} - \frac{1}{2} \ln \det K - \frac{n}{2} \ln 2\pi,$$

the multivariable form of the Gaussian of Section 12.5. The first term rewards a prior under which the readings are typical; the second penalises a prior so loose, with ℓ so short, that it would have found any readings typical. Their balance peaks at $\ell = 0.70$ Å

(Figure 18.4c). There the mean follows the readings, and its standard deviation is at most 0.2 meV at the readings and 1.1 meV midway across the widest gap, from 3.36 to 4.13 Å. A length a quarter as long makes the curve sag towards zero between readings, with an honest but useless band; one four times as long makes it too stiff, with a band too narrow (Figure 18.4b).

The band must also be right. Over the test, 0.75 of the noise-free values lie within twice the standard deviation at $\ell = 0.70$ Å, against the 0.95 that a correct Gaussian band holds, and the misses gather at the steep wall below 3.8 Å and at the bottom of the well, where no reading falls. A kernel with one length scale cannot be both smooth enough for the slow tail and supple enough for the wall, and the error bar is only as good as the kernel's belief. At a quarter of the length the band holds 0.99 of the test, and at four times only 0.15. Gaussian processes on atoms, with kernels built on descriptions of each atom's surroundings, are the Gaussian Approximation Potentials¹⁰⁴, among the first learned potentials; their error bar is what lets a learned potential say where it needs another first-principles calculation, the subject of planned Chapter 23.

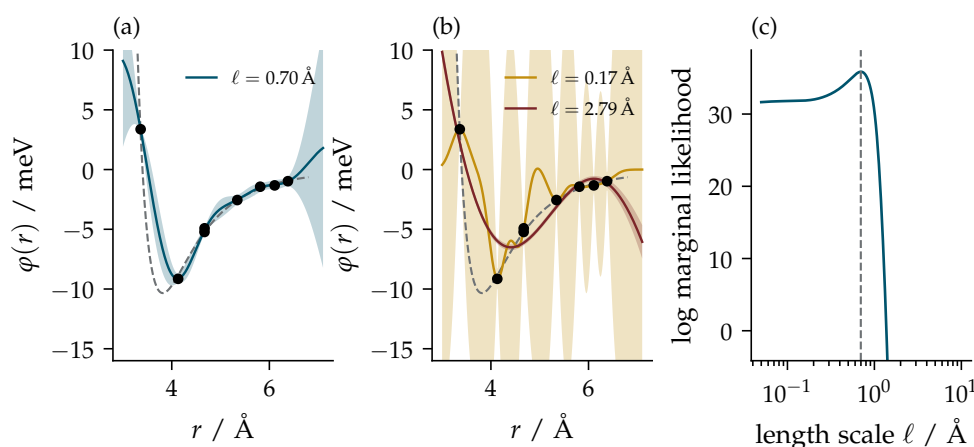


FIGURE 18.4: A Gaussian process through eight readings of argon's pair energy (dots, noise 0.2 meV), with $h = 10$ meV, against the pair energy (grey dashed). (a) The posterior mean and twice its standard deviation (band) for the length scale of greatest marginal likelihood. (b) The same for a length a quarter of that and four times it. (c) The log marginal likelihood of the readings against the length scale, with its greatest value (dashed).

KEY IDEA. A Gaussian process treats the unknown curve as a random function whose values covary by a kernel; conditioning on the readings gives the mean $\mathbf{k}_*^T K^{-1} \mathbf{y}$ and the variance $h^2 - \mathbf{k}_*^T K^{-1} \mathbf{k}_*$ at every point, an estimate with an error bar that grows away from the readings. The marginal likelihood chooses the kernel's length. The error bar is only as reliable as the kernel, and checking it against held-out values is part of the fit.

NOTEBOOK 18 move the readings and the length scale and watch the band

CHECK YOUR UNDERSTANDING. A Gaussian process is conditioned on one reading y_1 at x_1 , with no noise. What are its mean and variance at x_1 , and at a point very far from x_1 ?

18.5 Gradient descent

A walker caught in fog on a hillside, wanting the valley floor, feels the slope under their feet and steps downhill, then feels again. Short steps are safe but slow; a long step on a steep slope can carry the walker across a narrow valley and up the far side, higher than they started, and a run of such steps climbs out of the valley altogether. The normal equations solved the fits so far in one calculation, but most models, the networks of planned Chapter 19 among them, have losses that no formula minimises, and they are walked down instead.

The step and its limit

Gradient descent repeats the step

$$\mathbf{w} \leftarrow \mathbf{w} - \eta \nabla L(\mathbf{w}), \quad (18.6)$$

moving the weights against the gradient, the direction in which the loss falls fastest (Section 3.10), by an amount set by the *learning rate* η . Near a minimum with positive curvature in every direction, a smooth loss is approximated to second order by a quadratic bowl, just as a potential energy is near a stable minimum (Section 4.9). In this approximation, with $\mathbf{e} = \mathbf{w} - \mathbf{w}^*$ the departure from the minimum $\mathbf{w}^*$ and A the matrix of second derivatives there, the Hessian,

$$L = L^* + \frac{1}{2} \mathbf{e}^T A \mathbf{e}, \quad \nabla L = A \mathbf{e}.$$

For this quadratic, a step changes the departure by

$$\mathbf{e} \leftarrow \mathbf{e} - \eta A \mathbf{e} = (I - \eta A) \mathbf{e} \quad \text{subtracting } \mathbf{w}^* \text{ from both sides of the step}$$

$$c_k \leftarrow (1 - \eta \lambda_k) c_k \quad \text{along each eigenvector of } A, \text{ of eigenvalue } \lambda_k,$$

the components c_k of $\mathbf{e}$ along the eigenvectors changing independently, as the normal modes of a vibrating molecule do. Each shrinks only if $|1 - \eta\lambda_k| < 1$, that is $0 < \eta < 2/\lambda_k$, and the steepest direction, with the largest eigenvalue $\lambda_{\max}$, sets the limit:

$$\eta < \frac{2}{\lambda_{\max}}. \quad (18.7)$$

This is the stability limit of an integrator in another guise (Section 9.6): the fastest vibration of a molecule bounds the time step, and the steepest direction of a loss bounds the learning rate. Between $1/\lambda_{\max}$ and $2/\lambda_{\max}$ the factor $1 - \eta\lambda_{\max}$ is negative, and the steep component flips sign at every step while it shrinks, the zig-zag of the walker across the valley.

Figure 18.5 walks a loss of two weights whose Hessian has the eigenvalues 1 and 10, so $2/\lambda_{\max} = 0.2$. At $\eta = 0.05$ the factors are 0.95 and 0.50, and the shallow direction crawls; at 0.18 they are 0.82 and -0.80 , and the path zig-zags down the valley; at 0.205 the steep factor is -1.05 , and the path climbs out. The best fixed rate makes the slowest factor as small as it can be, which it is when the two extreme factors are equal and opposite, $1 - \eta\lambda_{\min} = -(1 - \eta\lambda_{\max})$:

$$\eta = \frac{2}{\lambda_{\min} + \lambda_{\max}}, \quad 1 - \eta\lambda_{\min} = \frac{\lambda_{\max} - \lambda_{\min}}{\lambda_{\max} + \lambda_{\min}} = \frac{\kappa - 1}{\kappa + 1},$$

with $\kappa = \lambda_{\max}/\lambda_{\min}$ the *condition number* of the Hessian. Here $\eta = 0.182$, the factor is 0.818, and since the loss falls as the square of the factor, it falls a millionfold in $\ln 10^6 / (-2 \ln 0.818) = 34$ steps.

The condition number decides

For a large condition number the factor is $1 - 2/(\kappa + 1)$, and the steps for a millionfold fall grow in proportion to κ , about $\kappa \ln 10^6 / 4$. The least-squares loss has the Hessian $2X^T X$, and the polynomial fits of Section 18.2 are badly conditioned: $X^T X$ has $\kappa = 74$ for degree 3 in the scaled distance, 4.6×10^6 for degree 8 and 5.1×10^{16} for degree 14, and for degree 8 in r itself 3.3×10^{26} , beyond what the 16 digits of double precision can even resolve. Gradient descent at its best fixed rate would need about 260 steps for degree 3 and 1.6×10^7 for degree 8. Scaling the variable, as Section 18.2 did, is the first remedy; the solvers of the normal equations are another; and the optimisers of Section 18.7 are a third, for the losses that have no normal equations.

A loss is *convex* if the straight line between any two points of its graph lies on or above the graph; a quadratic with a Hessian whose eigenvalues are all positive or zero is convex, and the least-squares loss, with Hessian $2X^T X$, is one. A convex loss has no local minima besides its least value, so gradient descent at a stable rate reaches it from

any start. The losses of neural networks are not convex, and where gradient descent ends then depends on where it began (planned Chapter 19).

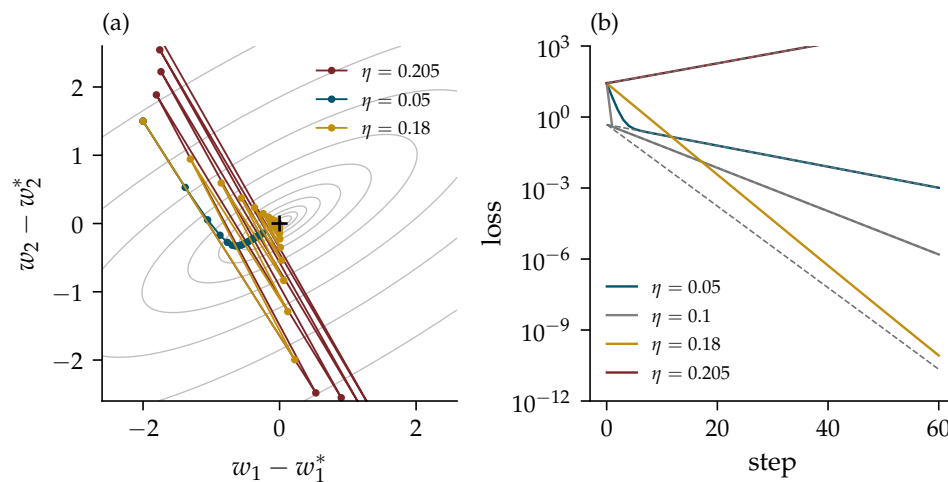


FIGURE 18.5: Gradient descent on a quadratic loss of two weights whose Hessian has the eigenvalues 1 and 10 along axes turned by 30°. (a) The paths at learning rates of 0.05, 0.18 and 0.205 over the contours of the loss (first 25 steps, 9 for the rate that diverges). (b) The loss against the step for rates of 0.05, 0.1, 0.18 and 0.205, with the term of each rate's slowest direction, $\frac{1}{2}\lambda c^2(1 - \eta\lambda)^{2t}$ for c the start's component along it (dashed).

KEY IDEA. Gradient descent steps against the gradient by the learning rate η . Near the minimum each eigen-direction of the Hessian shrinks by $1 - \eta\lambda$ a step, so $\eta < 2/\lambda_{\max}$, the stability limit of an integrator again; at the best fixed rate the slowest direction shrinks by $(\kappa - 1)/(\kappa + 1)$, and the condition number κ sets how many steps are needed. A convex loss has one least value, which descent at a stable rate reaches.

NOTEBOOK 18 *walk down a valley at different rates and condition numbers*

CHECK YOUR UNDERSTANDING. A loss has a Hessian with eigenvalues 0.5 and 50. What is the largest stable learning rate, and what factor does the slow direction shrink by at the best fixed rate?

18.6 Stochastic gradients

A poll does not ask every voter; a thousand chosen at random give the share of a party to within a few per cent, at a tiny fraction of the cost. The gradient of a loss is an average over the readings, and it too can be estimated from a few of them. A learned potential is trained on hundreds of thousands of configurations, each with hundreds of forces, and an exact gradient over all of them for every step would make training impossibly slow.

A gradient from a few readings

Written as the mean over n readings of a loss ℓ_i for each, $L = \frac{1}{n} \sum_i \ell_i(\mathbf{w})$, the loss has the gradient $\nabla L = \frac{1}{n} \sum_i \nabla \ell_i$, a mean of n numbers. A *mini-batch* of b readings chosen at random gives the estimate $\mathbf{g} = \frac{1}{b} \sum_{i \in \text{batch}} \nabla \ell_i$, whose average over the possible batches is the true gradient. If the readings' gradients scatter about their mean with the spread σ_g , each component of $\mathbf{g}$ scatters by

$$\frac{\sigma_g}{\sqrt{b}},$$

the error of a mean of b independent samples (Section 15.1). *Stochastic gradient descent* steps with $\mathbf{g}$ in place of ∇L , one step per batch, and a pass through all the readings, an *epoch*, takes n/b steps for the cost of one exact gradient.

Figure 18.6 fits the force of the A–B bond at 300 K, 2001 readings from Section 18.1, with a straight line in the scaled stretch x/σ_x , its loss the mean squared miss. At the least loss, where the exact gradient is zero, the gradient of the slope's weight from batches of ten readings scatters by 0.180, against the $\sigma_g/\sqrt{10} = 0.185$ that the readings' own spread gives (Figure 18.6b); its histogram is not yet Gaussian, since a mean of ten skewed numbers is still skewed, but its width is as predicted.

A noise floor, and a rate that falls

The noise has a cost. With the rate fixed at half the stability limit, full-batch descent reaches the least loss in under ten epochs, while batches of ten never settle: after 50 epochs their loss is still $4.7 \times 10^{-3} (\text{eV } \text{\AA}^{-1})^2$ above the least, with spikes far above it (Figure 18.6a). Near the minimum the true gradient vanishes but the estimate does not, and each step moves the weights by about $\eta \sigma_g / \sqrt{b}$ in a random direction, so the weights wander in a cloud whose size is set by the rate. A rate that falls as training proceeds, here $\eta / (1 + \text{epoch}/5)$, shrinks the cloud, and after 50 epochs the loss lies 4.4×10^{-4} above the least. The rate must fall slowly enough that the steps still add up to the distance the weights must travel, and fast enough that the wandering dies away; schedules of this kind, and the batch size, are among the settings that every training run of planned Chapter 19 and after reports.

Each run of the figure evaluated the gradient of one reading 100 050 times; the full-batch run took 50 steps with them, the mini-batch runs 10 050. When the readings are many and alike, as the frames of a long run are, a mini-batch gradient is nearly as good as the full one at a small fraction of the cost, and taking many cheap noisy steps is faster than taking a few exact ones.

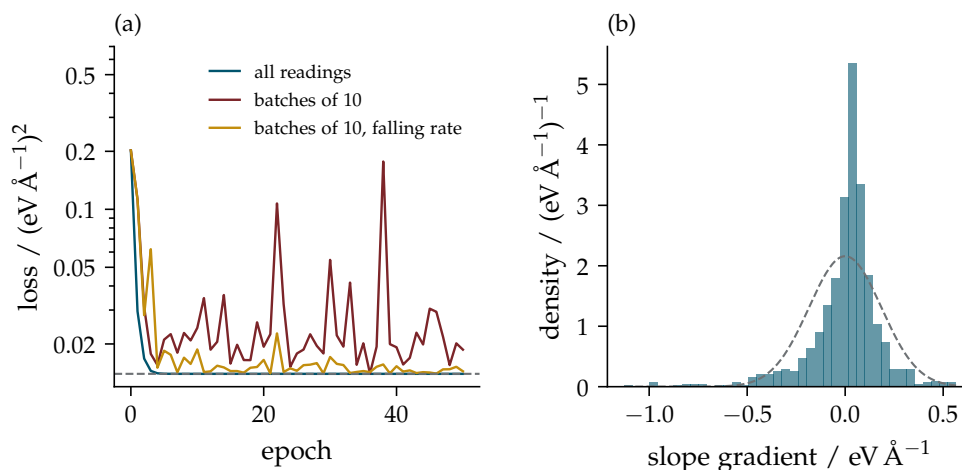


FIGURE 18.6: Gradients from mini-batches. The force of the A–B bond at 300 K (2001 readings) fitted by a straight line in the scaled stretch. (a) The mean squared miss over all the readings against the epoch, for gradient descent on all of them, for batches of ten at a fixed rate, and for batches of ten at a rate that falls, with the least loss (dashed). (b) The gradient of the slope’s weight from 1000 batches of ten at the least loss, against a Gaussian of spread $\sigma_g / \sqrt{10}$ (dashed).

KEY IDEA. A mini-batch of b readings estimates the gradient with an error that falls as $\sigma_g / \sqrt{b}$, at b/n of the cost. At a fixed rate the weights end in a cloud about the minimum whose size the rate sets; a rate that falls slowly shrinks it.

NOTEBOOK 18 *change the batch size and the schedule and watch the loss settle*

CHECK YOUR UNDERSTANDING. Batches of 10 give a gradient that scatters by 0.18. What batch size would bring the scatter to 0.06, and how many steps would an epoch of 2001 readings then take?

18.7 Momentum and Adam

A heavy block sliding down a long, gently tilted gutter with steep sides gathers speed along the gutter, since every push points the same way there, while across it the pushes alternate from one wall to the other and friction damps the swaying. The walker of Section 18.5 has no memory: each step depends only on the slope underfoot, so along a long shallow valley it creeps, and across a steep one it zig-zags. Giving the steps memory, as the block’s speed is a memory of past pushes, cures much of both.

Momentum

Gradient descent with momentum keeps a velocity $\mathbf{v}$, a running sum of past gradients that decays by a factor $\mu < 1$ a step, and moves the weights along it:

$$\mathbf{v} \leftarrow \mu \mathbf{v} + \nabla L(\mathbf{w}), \quad \mathbf{w} \leftarrow \mathbf{w} - \eta \mathbf{v}. \quad (18.8)$$

The block is more than a picture. Writing $\mathbf{w}_t$ for the weights after t steps, the second rule gives $\eta \mathbf{v}_t = \mathbf{w}_{t-1} - \mathbf{w}_t$ at every step. Multiplying the first rule by $-\eta$ and substituting gives

$$\mathbf{w}_{t+1} - \mathbf{w}_t = \mu(\mathbf{w}_t - \mathbf{w}_{t-1}) - \eta \nabla L(\mathbf{w}_t),$$

and moving $\mathbf{w}_t - \mathbf{w}_{t-1}$ to the left,

$$(\mathbf{w}_{t+1} - 2\mathbf{w}_t + \mathbf{w}_{t-1}) + (1 - \mu)(\mathbf{w}_t - \mathbf{w}_{t-1}) = -\eta \nabla L(\mathbf{w}_t).$$

The first bracket is a second difference, the change of the step from one step to the next, an acceleration, and the second a velocity: this is Newton's law with friction, $m\ddot{\mathbf{w}} + \gamma\dot{\mathbf{w}} = -\nabla L$ (Section 4.6), discretised with a step of one, the mass one, the friction $1 - \mu$ and the force η times the gradient. Momentum turns the descent into the damped motion of a particle on the loss.

On a quadratic loss each eigen-direction of the Hessian again moves on its own, the component c_t along an eigenvector of eigenvalue λ following $c_{t+1} = (1 + \mu - \eta\lambda) c_t - \mu c_{t-1}$, the line above with ∇L replaced by λc_t . Trying $c_t = z^t$, as for the oscillations of Section 9.6,

$$z^2 - (1 + \mu - \eta\lambda)z + \mu = 0.$$

The product of the two roots is μ , so when they are complex, a pair of conjugates of equal size, both have $|z| = \sqrt{\mu}$: the component swings and shrinks by $\sqrt{\mu}$ a step, whatever λ . They are complex while $(1 + \mu - \eta\lambda)^2 \leq 4\mu$, and the best choice, due to Polyak¹⁰⁵, puts both ends of the range of eigenvalues on the edge of this condition, $1 + \mu - \eta\lambda_{\min} = 2\sqrt{\mu}$ and $1 + \mu - \eta\lambda_{\max} = -2\sqrt{\mu}$:

$$\begin{aligned} (1 - \sqrt{\mu})^2 &= \eta\lambda_{\min}, & (1 + \sqrt{\mu})^2 &= \eta\lambda_{\max} && \text{as squares} \\ \frac{1 + \sqrt{\mu}}{1 - \sqrt{\mu}} &= \sqrt{\kappa} &&&& \text{dividing, and the root} \\ \sqrt{\mu} &= \frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1} &&&& \text{solving for } \sqrt{\mu}, \end{aligned}$$

the first line writing $1 + \mu \mp 2\sqrt{\mu}$ as $(1 \mp \sqrt{\mu})^2$. Adding the square roots of the first line's two equations gives $2 = \sqrt{\eta}(\sqrt{\lambda_{\min}} + \sqrt{\lambda_{\max}})$, so $\eta = 4/(\sqrt{\lambda_{\max}} + \sqrt{\lambda_{\min}})^2$. The asymptotic contraction factor in every direction is $(\sqrt{\kappa} - 1)/(\sqrt{\kappa} + 1)$ a step, against $(\kappa - 1)/(\kappa + 1)$ without momentum, and the steps needed grow as $\sqrt{\kappa}$ instead of κ . For the valley of Figure 18.7, with eigenvalues 1 and 100, the factors are 0.818 and 0.980: momentum brings the loss below 10^{-8} in 77 steps, while gradient descent at its best fixed rate has not in 300. The steep weight shows the difference (Figure 18.7a): without momentum it flips sign at every step and barely shrinks, by the factor -0.98 ; with momentum it rings and dies away, the damped swing of Chapter 4.

Adam

The networks of the later chapters are trained with Adam¹⁰⁶, which gives each weight a step of its own size. It keeps two running means of each weight's gradient, of the gradient and of its square,

$$m \leftarrow \beta_1 m + (1 - \beta_1) g, \quad s \leftarrow \beta_2 s + (1 - \beta_2) g^2,$$

with $\beta_1 = 0.9$ and $\beta_2 = 0.999$ by default, each a mean over roughly the last $1/(1 - \beta)$ steps, 10 and 1000, weighted towards the recent ones. Both start at zero, which biases them low at first: with a steady gradient g , after t steps

$$m_t = (1 - \beta_1) \sum_{k=0}^{t-1} \beta_1^k g = (1 - \beta_1) \frac{1 - \beta_1^t}{1 - \beta_1} g = (1 - \beta_1^t) g,$$

by the sum of a geometric series, so dividing by $1 - \beta_1^t$ undoes the bias, and likewise for s . The step is

$$w \leftarrow w - \eta \frac{\hat{m}}{\sqrt{\hat{s}} + \epsilon}, \quad \hat{m} = \frac{m}{1 - \beta_1^t}, \quad \hat{s} = \frac{s}{1 - \beta_2^t},$$

with $\epsilon = 10^{-8}$ guarding against a division by zero. For a steady gradient $\hat{m}/\sqrt{\hat{s}}$ is ± 1 , so each weight moves by about η a step whatever the size of its gradient; multiplying one weight's gradients by a constant multiplies $\hat{m}$ and $\sqrt{\hat{s}}$ alike and leaves its steps unchanged. In `mdlab`:

```

1 class Adam:
2     def __init__(self, rate: float = 1e-3, beta1: float = 0.9,
3                 beta2: float = 0.999, eps: float = 1e-8):
4         self.rate, self.beta1, self.beta2, self.eps = rate, beta1, beta2, eps
5         self.m = self.s = None
6         self.t = 0
7
8     def step(self, w: NDArray, g: NDArray) -> NDArray:
9         if self.m is None:
10             self.m, self.s = np.zeros_like(g), np.zeros_like(g)
11         self.t += 1
12         self.m = self.beta1 * self.m + (1 - self.beta1) * g
13         self.s = self.beta2 * self.s + (1 - self.beta2) * g * g
14         m_hat = self.m / (1 - self.beta1**self.t)
15         s_hat = self.s / (1 - self.beta2**self.t)
16         return w - self.rate * m_hat / (np.sqrt(s_hat) + self.eps)

```

A test checks that its steps, and those of Momentum, match PyTorch’s Adam and SGD with momentum on a loss of two weights to the last digit.

The scaling is Adam’s strength and its limit. A valley whose steep and shallow directions lie along the weights, as when one weight multiplies a feature a hundred times larger than another’s, is a valley of badly scaled weights, and Adam evens them out: with the rate 0.05 it reaches 10^{-8} in 189 steps (Figure 18.7b). The same valley turned by 45° , its steep direction a mixture of both weights, gives each weight a similar gradient, Adam’s scaling has nothing to even out, and after 300 steps its loss is still 0.49, worse than plain descent’s 0.0014, while momentum still needs only 85 (Figure 18.7c). Adam is robust where the scales of the weights differ widely and are not known in advance, as in a network, which is why it is the common default; a narrow valley at an angle it leaves as it found it.

KEY IDEA. Momentum adds to the step a decaying memory of past gradients; it is damped motion on the loss, and on a quadratic with Polyak’s choice it has the asymptotic contraction factor $(\sqrt{\kappa} - 1)/(\sqrt{\kappa} + 1)$ a step. Adam divides each weight’s running mean gradient by the root of its running mean square, with both corrected for their start at zero, so that every weight moves at about the rate whatever the scale of its gradient; it evens out badly scaled weights, and leaves a narrow valley at an angle as narrow as it was.

NOTEBOOK 18 *race the three optimisers down a valley, and turn the valley*

CHECK YOUR UNDERSTANDING. For a Hessian with $\kappa = 10^4$, by what factor does the slow direction shrink in a step of gradient descent at its best fixed rate, and of momentum with Polyak’s choice?

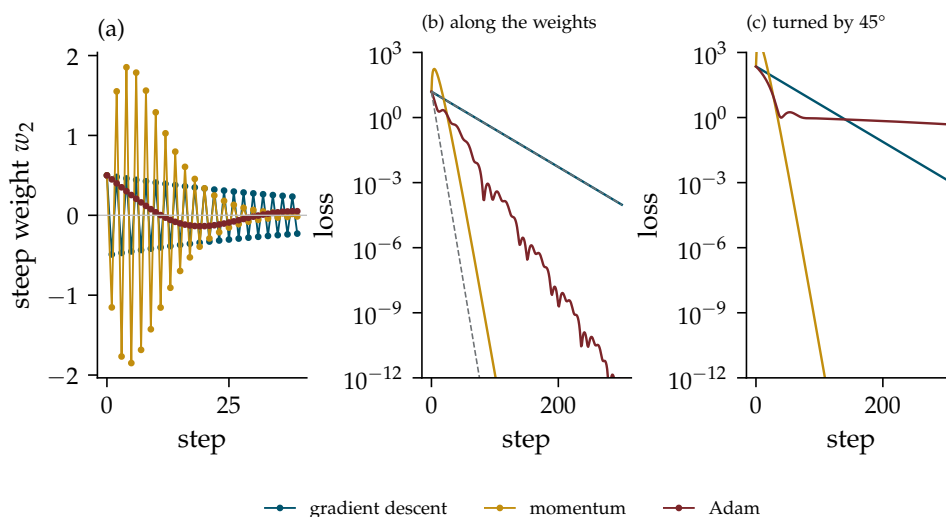


FIGURE 18.7: Gradient descent, momentum and Adam in a valley whose Hessian has the eigenvalues 1 and 100, from $(-2.5, 0.5)$: descent at its best fixed rate $2/(\lambda_{\min} + \lambda_{\max})$, momentum with Polyak's rate and μ , Adam with the rate 0.05. (a) The steep weight against the step, the valley's axes along the weights. (b) The loss against the step there, with the factors per step that theory gives descent and momentum (dashed). (c) The loss in the same valley turned by 45° .

18.8 Training, validation and test

A student who revises from past papers and then sits an exam made of the same questions scores well without having learned the subject; the score measures memory alone. A fitted model is judged the same way, on readings it has not seen, and the judgement is only as honest as the separation of those readings from the ones it learned from. With the frames of a molecular dynamics run that separation is subtler than it looks.

Three sets

The readings are divided before any fitting. The *training* set fixes the weights; the *validation* set chooses everything that the training set cannot, such as the degree of a polynomial, the penalty λ , the learning rate or the moment to stop (Section 18.3); the *test* set is opened once, at the end, to report how well the chosen model predicts. Every choice made by looking at a set spends some of that set's honesty, which is why the test is touched last.

The division assumes that the sets are independent samples of the readings the model will meet. Frames of a run are not: a frame 10 fs after another holds nearly the same positions and nearly the same energy, and Section 15.2 measured how long such a memory lasts, as the statistical inefficiency g , the number of consecutive samples worth one independent one.

Frames that leak

Two runs of Chapter 12's liquid argon were made with every 10 fs step kept, as first-principles runs often are, and a model was asked to learn each frame's potential energy U from the histogram of its 32 640 pair distances in 110 bins of 0.05 Å. The task is well posed: U is the sum of the pair energy over the pairs, so it is linear in the histogram, through the pair energy at each bin, with a miss from the binning alone of 22.5 meV against a spread of U of 175 meV. Ridge regression ($\lambda = 10^{-2}$, the 110 counts scaled by the training frames' mean and spread) learned U from the first frames of one run, with a fifth held out for validation, and was tested on frames of the other run, every 50 fs, a run it never saw (Figure 18.8).

At 10 fs the energy has $g = 109$ frames, so a training run of 2 ps, 200 frames, holds about two independent ones. Held out at random, its validation frames each have training frames 10 fs either side of them, nearly identical, and the validation miss is (30.5 ± 4.4) meV over ten random splits, while the independent run gives (128 ± 16) meV, four times more: the random split measures how well the model recalls its neighbours. Held out in whole blocks of 0.5 ps, the validation miss is (92 ± 22) meV, close to the truth. With 4 ps the random split still reports 21.7 against 37.8, the blocked one 34.8; only at 10 and 20 ps, about 9 and 18 independent frames, do all three agree near the binning's own floor, since a model with enough independent frames no longer needs the neighbours.

The cure is to split by time, into whole blocks longer than the memory g , or better whole runs, for validation and test. The same care applies to every quantity computed from the training set alone, such as the mean and spread that scale the features, which must not be computed with the validation frames in. A learned potential's reported error means little without the split that produced it.

KEY IDEA. Training readings fix the weights, validation readings choose the settings, and test readings, touched once, report the result. All three must be independent samples; consecutive frames of a run are not, and split at random they let a model look up its neighbours, reporting a miss four times too small here. Split by blocks longer than the statistical inefficiency, or by whole runs.

NOTEBOOK 18 *split a run's frames at random and in blocks and compare the misses*

CHECK YOUR UNDERSTANDING. A first-principles run of 10 ps keeps every 1 fs step, and its energy has $g = 400$ steps. About how many independent frames does it hold, and how long should each validation block be?

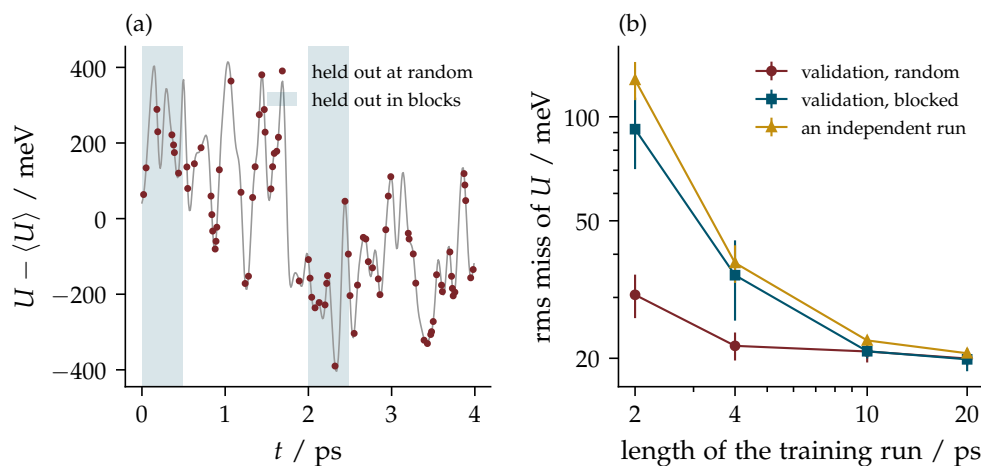


FIGURE 18.8: Consecutive frames leak between splits. Liquid argon, a frame every 10 fs; ridge regression learns each frame's potential energy from the histogram of its pair distances. (a) The energy over 4 ps, with the frames that a random split holds out (dots) and those that a split into blocks of 0.5 ps holds out (shaded). (b) The rms miss of U on validation frames held out at random and in blocks, and on an independent run, against the length of the training run, with the spread over ten splits.

18.9 What the training looks like

A run of molecular dynamics is read through its energies, and a training run through its losses. The curve of the loss against the step, for the training readings and for the validation readings, is the first thing to look at when a model is trained, and most faults show in it. Figure 18.9 trains the polynomials of Section 18.2 on the fifteen readings of argon's pair energy by lowering the mean squared miss from zero weights, once healthily and then broken in four ways, and Table 18.1 gathers the checks.

Healthy. Degree 8 in the scaled distance, Adam at the rate 10^{-4} , 40 000 steps: the training miss falls to 0.202 meV and the validation miss with it, to 0.187, both at the noise of 0.2 meV (Figure 18.9a). Two curves that fall together and level at the noise are the picture of a model that has learned what the readings hold.

The rate. Gradient descent on the same fit at 0.9 of its stability limit $2/\lambda_{\max}$ brings the training miss to 0.28 meV in 400 steps; at 1.1 of it the miss grows without bound, to 1.7×10^{32} meV; at a thousandth of it the miss has fallen only to 3.6 meV (Figure 18.9b). A loss that rises, or swings with growing amplitude, means a rate past the limit; one that barely moves, a rate far below it. Adam shows a milder form of the first: at a fixed rate of 10^{-3} its training miss jumps by up to 0.24 meV in a single step late in training, when its running mean square has decayed and the steps no longer shrink with the gradient, against 0.005 meV at 10^{-4} , the smaller rate removing the spikes.

Overfitting. Degree 14 with no penalty, Adam at 10^{-4} for 10^5 steps: the training miss keeps falling, to 0.150 meV, below the noise, while the validation miss is least, 0.38 meV,

after 160 steps and then rises to 2.95 meV (Figure 18.9c). Two curves that part, the training miss below the noise and the validation miss climbing, mean the model is learning the noise. Stopping at the step of least validation miss, *early stopping*, keeps the model of that step and works as a penalty does, since the weights have not yet grown into the directions the readings barely constrain; a penalty or fewer weights cure it outright.

Unscaled features. Degree 8 in r itself, whose design has the condition number 3.3×10^{26} against 4.6×10^6 for the scaled distance, trained by gradient descent at 0.9 of each one's stability limit: after 2000 steps the training miss is 5.3 meV with powers of r and 0.22 meV with the scaled distance (Figure 18.9d). A loss that falls a little and then crawls, with no sign of the noise floor, points to badly scaled inputs before it points to a bad model; scaling each input to a spread of about one is the cure, and the reason the readings of a learned potential are normalised before training.

A leaked split. The fault that the loss curves cannot show is the one of Section 18.8: with frames split at random, training and validation misses fall together to a healthy-looking floor, 30.5 against an independent run's 128 meV. Only a validation set separated in time, or a separate run, reveals it.

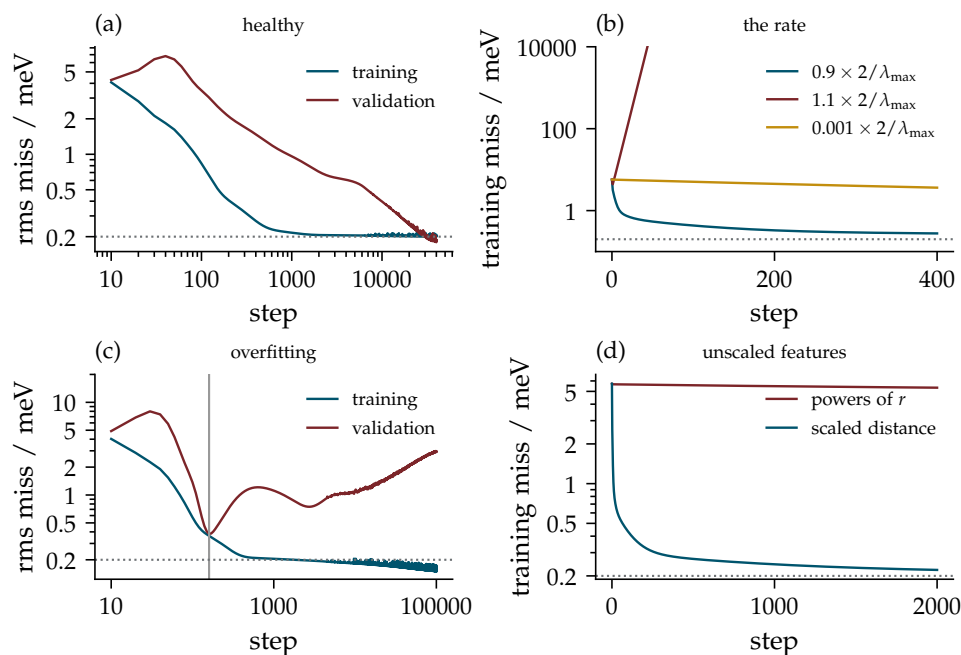


FIGURE 18.9: What the training looks like: the rms miss on the fifteen training readings of argon's pair energy, and on fifteen validation readings, against the step, from zero weights. (a) Healthy: degree 8, Adam at 10^{-4} . (b) Gradient descent at 0.9 and 1.1 of its stability limit and at a thousandth of it. (c) Overfitting: degree 14, Adam at 10^{-4} , with the step of least validation miss (grey). (d) Degree 8 in r itself and in the scaled distance, gradient descent at 0.9 of each one's limit. The noise of the readings, 0.2 meV, dotted.

TABLE 18.1: Faults of training, the check that catches each, and its value in the healthy run and in the broken one.

fault	check	healthy	broken
rate past the limit	training miss after 400 steps	0.28 meV	1.7×10^{32} meV
rate far too small	training miss after 400 steps	0.28 meV	3.6 meV
Adam's rate too large	largest jump of the training miss in a step	0.005 meV	0.24 meV
overfitting	validation miss at the end against its least	0.19 against 0.18 (degree 8)	2.95 against 0.38 meV
unscaled features	condition number of $X^T X$	4.6×10^6	3.3×10^{26}
leaked split	validation miss against an independent run's	92 against 128 (blocked)	30.5 against 128 meV

KEY IDEA. The training and validation losses against the step are read as a run's energies are: falling together to the noise is healthy; a rising or swinging loss means a rate past the limit; a crawl, a rate too small or inputs badly scaled; curves that part, overfitting, cured by early stopping, a penalty or fewer weights. A leaked split looks healthy, and only a validation set separated in time reveals it.

NOTEBOOK 18 break a training run in each of these ways and read its loss curves

CHECK YOUR UNDERSTANDING. A model's training loss falls steadily, and its validation loss falls for a while and then rises. What is happening, and which step's weights should be kept?

What we now have

Learning a model from readings is the lowering of a loss. For a model linear in its weights, a sum of basis functions, the sum of squared misses is least where the normal equations $X^T X \mathbf{w} = X^T \mathbf{y}$ hold, and a fit answers the question its readings pose: a line through the forces of a hot run gives the bond's stiffness averaged over the stretches it explores. More basis functions bend the model more freely and, beyond a point, learn the noise, which only readings the model has not seen reveal. A ridge penalty $\lambda |\mathbf{w}|^2$ shrinks the directions the readings barely constrain, is the most probable fit under Gaussian noise and a Gaussian prior with $\lambda = \sigma^2 / \sigma_w^2$, and is chosen on validation readings. A Gaussian process conditions a random function on the readings and returns an estimate with an error bar, whose kernel the marginal likelihood chooses and whose band must still be checked.

Gradient descent walks a loss down with a rate bounded by $2/\lambda_{\max}$, the stability limit of an integrator again, at a pace set by the condition number; mini-batches estimate the gradient with an error $\sigma_g / \sqrt{b}$ at a fraction of the cost; momentum is damped motion

on the loss and, on a quadratic with the optimal settings, has asymptotic contraction factor $(\sqrt{\kappa} - 1)/(\sqrt{\kappa} + 1)$ a step; Adam evens out badly scaled weights. Training, validation and test readings must be independent, and consecutive frames of a run are not. In this chapter's example, a random split makes the validation error four times too small; separating the time intervals gives a much closer estimate of the error on an independent run. The training and validation curves show the remaining faults of a training run.

Planned Chapter 19 builds models that are not linear in their weights, neural networks, computes their gradients by the chain rule run backwards, and trains them on energies and forces with these optimisers.

Answers to the checks

SECTION 18.1. Zero, since the line passes through every reading. The misses about the fitted line are then all zero, so the estimated σ_m and the error of $\hat{k}$ are zero too: ten perfect readings leave no trace of their noise, and an error estimated from the misses is only as good as the misses are typical.

SECTION 18.2. Fifteen weights and fifteen readings make the design matrix square, and for distinct distances it can be inverted, so $X\mathbf{w} = \mathbf{y}$ has an exact solution, which makes the loss zero. Any fifteen readings, noise and all, are fitted exactly, so a training miss can always be driven to zero and says nothing about prediction.

SECTION 18.3. $\lambda = \sigma^2/\sigma_w^2$ grows four times, so the penalty weighs more against the misses, and the fit follows the readings less closely, keeping nearer the smooth curve that the prior prefers: noisier readings deserve less trust.

SECTION 18.4. At x_1 , $K = h^2$ and $\mathbf{k}_* = h^2$, so the mean is $h^2 y_1 / h^2 = y_1$ and the variance $h^2 - h^4/h^2 = 0$: the curve passes through the reading with no doubt. Far from x_1 , $\mathbf{k}_*$ vanishes, the mean returns to the prior's zero and the variance to h^2 .

SECTION 18.5. The rate must lie below $2/\lambda_{\max} = 2/50 = 0.04$. With $\kappa = 100$ the slow direction shrinks by $(100 - 1)/(100 + 1) = 0.980$ a step.

SECTION 18.6. The scatter falls as $1/\sqrt{b}$, so a third of it needs nine times the batch, 90 readings; an epoch of 2001 readings is then $2001/90 = 22.2$, so 23 steps, the last batch short.

SECTION 18.7. With $\sqrt{\kappa} = 100$, gradient descent shrinks by $(10^4 - 1)/(10^4 + 1) = 0.9998$ and momentum by $(100 - 1)/(100 + 1) = 0.980$ a step: about 100 times fewer steps for the same fall.

SECTION 18.8. 10 000 steps worth one independent frame each 400 are about 25 independent frames. A validation block should be longer than the memory, several times 400 steps, say 1 to 2 ps, or better a separate run.

SECTION 18.9. The model is learning the noise of its training readings, overfitting. Keep the weights of the step where the validation loss was least, early stopping, or retrain with a penalty or fewer weights.

Exercises

The exercises run from questions of understanding, through calculations, to short derivations marked ‘show that’. Full solutions are in ‘Solutions to the exercises’ at the end of the book, and Notebook 18 checks every numerical answer.

EXERCISE 18.1 (*the least loss*). For the line through the origin, show that the least loss is $L(\hat{k}) = \sum_i F_i^2 - (\sum_i x_i F_i)^2 / \sum_i x_i^2$.

EXERCISE 18.2 (*the error of a stiffness*). A spring is stretched by 1, 2, ..., 10 cm and its force read with an error of 0.05 N each, independently. What is the error of the fitted stiffness?

EXERCISE 18.3 (*a line with an intercept*). For the model $\hat{y} = c + sx$, write $X^T X$ and $X^T \mathbf{y}$ in terms of sums over the readings, and show that the normal equations give the slope $s = \sum_i (x_i - \bar{x})(y_i - \bar{y}) / \sum_i (x_i - \bar{x})^2$ of Section 15.5.

EXERCISE 18.4 (*a stiffness that changes*). Show that the stiffness of the Morse well, $-dF/dx$ for the force $F = -dU_M/dx$, is $2Da^2(2e^{-2ax} - e^{-ax})$, and that it is $2Da^2(1 - 3ax)$ to first order in x . By what fraction does it change over $x = \pm 0.0345 \text{ \AA}$ for $a = 2 \text{ \AA}^{-1}$?

EXERCISE 18.5 (*as many weights as readings*). Show that if the design matrix X is square and has an inverse, the normal equations give $X\mathbf{w} = \mathbf{y}$ exactly, so the fit passes through every reading.

EXERCISE 18.6 (*shrinkage*). With $X^T X$ having the eigenvalues d_k , show that each component of the ridge weights along an eigenvector is that of the weights without the penalty times $d_k / (d_k + \lambda)$, for $d_k > 0$.

EXERCISE 18.7 (*the prior's spread*). With readings of noise $\sigma = 0.2 \text{ meV}$, what spread σ_w of the weights does $\lambda = 10^{-3}$ express, and what λ would express $\sigma_w = 1 \text{ meV}$?

EXERCISE 18.8 (*conditioning two Gaussians*). Two jointly Gaussian numbers a and b have mean zero, variances 4 and 1, and covariance 1.5. Given $b = 2$, what are the mean and the variance of a ?

EXERCISE 18.9 (*a process with one reading*). A Gaussian process with the kernel of Equation (18.4) is conditioned on one reading y_1 at x_1 with noise of spread σ . Show that its mean at x is $y_1 e^{-(x-x_1)^2/2\ell^2} h^2/(h^2 + \sigma^2)$, and find its variance.

EXERCISE 18.10 (*a learning rate*). A loss has a Hessian with the eigenvalues 2 and 200. Find the upper bound on a stable rate, the best fixed rate, the factor by which the slow direction shrinks at it, and the steps that bring the loss down a millionfold.

EXERCISE 18.11 (*steps against the condition number*). Show that for a large condition number the best fixed rate needs about $\kappa \ln(1/\epsilon)/4$ steps to bring the loss down by the factor ϵ , using $\ln(1+y) \approx y$ for small y .

EXERCISE 18.12 (*the batch for a precision*). The readings' gradients scatter by $\sigma_g = 0.585$. What batch gives the mini-batch gradient a scatter of 0.05?

EXERCISE 18.13 (*the memory of momentum*). With $\mu = 0.9$ and no gradient, after how many steps has the velocity fallen to $1/e$ of its value, and what is the friction $1 - \mu$ of the equivalent damped motion?

EXERCISE 18.14 (*Polyak's momentum*). For eigenvalues 1 and 100, find Polyak's $\sqrt{\mu}$, μ and η , and use the asymptotic contraction factors to estimate the steps needed to shrink the slow direction by 10^{-4} , with momentum and with gradient descent at its best fixed rate. Ignore the initial transient.

EXERCISE 18.15 (*Adam's correction*). A weight's gradient is a steady g . With $\beta_1 = 0.9$ and $\beta_2 = 0.999$, what are m and s after 10 steps, and by what factor would the step be too large if neither were corrected for its start at zero?

EXERCISE 18.16 (*Adam and scale*). Show that, with ϵ neglected, Adam's step for a weight is unchanged when all that weight's gradients are multiplied by a constant $c > 0$.

EXERCISE 18.17 (*independent frames*). A run's energy, kept every 10 fs, has the statistical inefficiency $g = 109$. How many independent frames are there in 2 and in 20 ps?

EXERCISE 18.18 (*why a neighbour helps*). A validation frame of a random split has training frames 10 fs before and after it. Explain why a model that interpolates nearby frames can predict this frame well while performing poorly on a new run.

EXERCISE 18.19 (*a plan for the splits*). Fifty first-principles runs of lithium in graphite, each 5 ps at one of ten temperatures, are to train a learned potential. Propose training, validation and test sets, and say what each must not share.

A gallery of faults

Chapters 11–17 end by breaking their own runs on purpose and finding the check that catches each fault. This appendix gathers them in one table, in the order of the chapters, each with what it looks like, the check that finds it (a function of `mdlab.diagnostics` where there is one), and the cure. `diagnostics.health_report` runs the checks of continuity, the centre of mass, overlaps, energy, temperature and velocities on a run of `mdlab` or, through `io.read_run`, of any code that ASE can read ([Section 17.8](#)), and Notebook F passes the cached runs of Chapters 11, 12, 13 and 16, healthy and broken, through it.

TABLE F.1: Faults of molecular dynamics runs, their symptoms, the checks that find them and their cures.

fault	what it looks like	check	cure
<i>Integrating and constraining (Section 11.8)</i>			
step too long	the total energy fluctuates widely, not falling as δt^2	<code>energy_fluctuation</code>	a shorter step, or held bonds
jumping cutoff	spikes of the total energy as pairs cross the cutoff	<code>energy_fluctuation</code>	the Ewald sum for charges (Section 10.3), or a potential switched off smoothly (Section 8.5)
wrong units	a kinetic energy per freedom far from $\frac{1}{2}k_B T$	<code>K</code> at the start	velocities converted between units
wrapped wrongly	held distances wrong by a cell	<code>bond_error</code>	the nearest image in the constraints
overlap	an enormous force at the start	<code>largest_force</code> , <code>closest_approach</code>	the start relaxed or rebuilt

fault	what it looks like	check	cure
loose constraints	the total energy drifts	energy_drift	a tighter tolerance
drift kept	the centre of mass travels	centre_of_mass_path	the total momentum removed
resonance	the energy drifts under multiple time steps	energy_drift	a shorter outer step
<i>Ensembles (Section 12.9)</i>			
freedoms miscounted	the temperature is wrong by a ratio of counts	T against the value asked	$3N - n_c - 3$ freedoms
lattice start	the temperature halves as energy flows into the potential	mean T against $T(0)$	a start drawn at twice the temperature, half of which flows into the potential (Section 12.6), or an equilibrated one
drift counted	a temperature that includes the flight of the whole	total momentum	the momentum removed
masses ignored	species start at different temperatures	T of each species	velocities drawn for each mass
not Gaussian	velocities spread evenly, not as Maxwell-Boltzmann	excess_kurtosis	velocities drawn from the Gaussian
no sharing	energy stays in the modes it started in	the energy of each mode	a longer run, or a thermostat acting on each atom
<i>Thermostats (Section 13.8)</i>			
coupling too strong	a temperature spread far below canonical	spread of T against $\sqrt{2/N_f} T$	CSVR or a chain, or weaker coupling
coupling too weak	runs that have not reached the temperature	mean T of several runs	longer equilibration, or stronger coupling while it lasts
flying ice cube	kinetic energy collecting in the centre of mass	growth of its kinetic energy	a thermostat that samples the canonical ensemble
one atom, one variable	a density that is not canonical	the density of U against the exact one	a chain, or Langevin
wrong N_f	a temperature on the wrong count of freedoms	T over the true freedoms	the code's count checked

fault	what it looks like	check	cure
centre as diffusion	D too large	D with the centre removed	the centre of mass removed
<i>Pressure (Section 14.9)</i>			
virial from wrapped positions	a pressure that moves with the origin	P after moving the origin	the virial from pair separations
kinetic part dropped	a pressure read too low by $Nk_B T/V$; under a barostat, a volume too small	P with $2K$	the kinetic part included
volume too narrow	a compressibility from fluctuations too small	κ_T against fixed volumes	stochastic cell rescaling or a piston
piston too heavy	few swings of the volume	swings in the run	a lighter piston, or a longer run
isotropic coupling of layers	unequal diagonal pressures	the diagonal of the pressure tensor	lengths coupled apart
κ in the wrong unit	a barostat 10^4 times too slow: the volume stays at its start and the pressure off target	mean P	the unit converted
<i>Convergence (Section 15.10)</i>			
averaged from the start	a biased mean	<code>stats.detect_equilibration</code>	the start discarded
the naive error bar	errors too small	<code>stats.statistical_inefficiency</code>	$\sqrt{gs^2/n}$
too short for a plateau	no plateau of the block errors	<code>stats.convergence_report</code>	a longer run, by Equation (15.5)
a drifting energy	a slope many errors from zero	<code>stats.drift_test</code>	a shorter step
the wrong ensemble	energy histograms whose ratio has the wrong slope	the test of Section 15.8	a canonical thermostat
a box too small	D that grows with the box	runs at two sizes	the correction of Yeh and Hummer
<i>Observables (Section 16.12)</i>			
wrapped positions in an MSD	an MSD that levels off near $L^2/2$	<code>largest_jump</code>	unwrap
centre of mass drifting	an MSD that curves upwards	<code>centre_of_mass_path</code>	the centre removed

fault	what it looks like	check	cure
fit in the ballistic part	D too small	slope of $\ln \text{MSD}$ against $\ln t$	a window after the motion forgets its velocity
$g(r)$ past half the cell	$g(r)$ falling below 1 at large r	the range against half the smallest width	the range limited
transport under friction	D too small	a run at fixed energy	fixed energy or CSVR for dynamics
Green-Kubo read too late	an integral that wanders	the plateau	read the plateau; errors from runs
bonds from one cutoff	many brief events	events with a band	a band of hysteresis and a least lifetime
frames too sparse for a spectrum	peaks where nothing vibrates	stride against $1/2v_{\max}$	a shorter stride, the same for both spectra
errors of F from short blocks	errors too small across rare barriers	errors over copies	independent copies
energy distributions from two ensembles	distributions that differ for the wrong reason	the thermostats of both	the same ensemble for both
<i>Practice (Section 17.8)</i>			
LAMMPS arrays read by position	atoms that jump cells between frames; D twenty times too large	largest_jump	arrays ordered by identity at every frame
forces compared in two frames	large force differences with equal energies	the energy, which does not turn	both codes in one orientation
a functional's gap	graphite's sheets too far apart	the lattice against measurement	D3 for dispersion
isotropic coupling of LiC_6	the spacing held at its value at 0 K; diagonal pressures of opposite sign	the diagonal of the pressure tensor	lengths coupled apart
a reference not converged	forces that change with the k -point grid	forces against a denser grid	a denser grid
a step too long for the bonds	a spread growing faster than δt^2 , and a drift	health_report at fixed energy	a shorter step
sites in the wrong frame, or too few	a count of hops that keeps falling as the least stay grows	the count against the least stay	sites fixed to the atoms that make them, and a least stay where the count is flat

Solutions to the exercises

The exercises of every chapter are solved here in full, after the main text, so that a solution is read only after a first attempt. Each is written out step by step, and the chapter's notebook checks the numerical answers and shows the same calculations in code.

Chapter 1: Motion

EXERCISE 1.1. We write L for length and T for time, and compare the two sides of each equation.

- (i) The left side is a length, L. On the right, $v_0 t^2$ has dimension $LT^{-1} \times T^2 = LT$. The two differ, so (i) does not balance.
- (ii) v^2 and v_0^2 have dimension L^2T^{-2} , and so does $2gx$, since $LT^{-2} \times L = L^2T^{-2}$; the number 2 has no dimension. Every term agrees, so (ii) balances. It is Equation (1.5) with $a = -g$ and $x_0 = 0$.
- (iii) Inside the root, x/g has dimension $L/(LT^{-2}) = T^2$, and the square root of T^2 is T, matching the left side, so (iii) balances. It is the time to fall a height x from rest.
- (iv) The right side fails twice. A function such as the exponential, built from a series of powers $1 + s + s^2/2 + \dots$, can only take a pure number s , since the terms of the series would otherwise have different dimensions; but gt has dimension $LT^{-2} \times T = LT^{-1}$. And even with a pure number in the exponent, v_0 times a number is a velocity, not a length. So (iv) does not balance.

EXERCISE 1.2. By Equation (1.4) with $a = -g$, the velocity is $v = v_0 - gt$. The ball reaches the top when $v = 0$, so

$$t_{\text{top}} = \frac{v_0}{g} = \frac{15}{9.80665} = 1.530 \text{ s.}$$

Equation (1.5) with $v = 0$, $x_0 = 0$ and $a = -g$ gives $0 = v_0^2 - 2gx$, so the greatest height is

$$x_{\text{top}} = \frac{v_0^2}{2g} = \frac{15^2}{2 \times 9.80665} = 11.47 \text{ m.}$$

Back at the hand $x = 0$ again, and Equation (1.5) gives $v^2 = v_0^2$, so $v = -v_0$: the ball returns at 15 m s^{-1} , moving downwards.

EXERCISE 1.3. The object is at rest when $v = 0$, that is when $3t^2 = 12$, so $t^2 = 4$ and, since $t \geq 0$, $t = 2 \text{ s}$. The acceleration is the derivative of the velocity, $a = dv/dt = 6t$, which at $t = 2$ is 12 m s^{-2} . The displacement is the integral of the velocity, by Equation (1.3). The function $t^3 - 12t$ has derivative $3t^2 - 12$, so

$$\int_0^3 (3t^2 - 12) dt = [t^3 - 12t]_0^3 = (27 - 36) - 0 = -9 \text{ m.}$$

The object ends 9 m behind where it started. For the distance travelled, the velocity is negative from $t = 0$ to $t = 2$ and positive from $t = 2$ to $t = 3$, so the two parts are integrated separately and both counted as positive:

$$\int_0^2 |v| dt = -[t^3 - 12t]_0^2 = -(8 - 24) = 16 \text{ m,}$$

$$\int_2^3 |v| dt = [t^3 - 12t]_2^3 = (27 - 36) - (8 - 24) = 7 \text{ m.}$$

It travels $16 + 7 = 23 \text{ m}$ in all: 16 m backwards, then 7 m forwards, ending 9 m behind its start.

EXERCISE 1.4. A whole turn is 2π radians or 360° , so one radian is $180/\pi$ degrees. Hence $1.1 \times 180/\pi = 63.03^\circ$, and $135^\circ = 135 \times \pi/180 = 3\pi/4 = 2.356 \text{ rad}$. The arc cut off by an angle θ in radians on a circle of radius R is $R\theta$, by the definition of the radian, so the arc is $0.35 \times 2.356 = 0.825 \text{ m}$.

EXERCISE 1.5. Using $\cos^2 \theta = 1 - \sin^2 \theta$ to replace the cosine in the double-angle formula,

$$\begin{aligned} \cos 2\theta &= \cos^2 \theta - \sin^2 \theta \\ &= (1 - \sin^2 \theta) - \sin^2 \theta && \text{since } \cos^2 \theta + \sin^2 \theta = 1 \\ &= 1 - 2\sin^2 \theta. \end{aligned}$$

Rearranging, $2\sin^2\theta = 1 - \cos 2\theta$, so $\sin^2\theta = \frac{1}{2}(1 - \cos 2\theta)$. With $\theta = \pi/8$, $2\theta = \pi/4$, whose cosine is $1/\sqrt{2}$ (Section 1.5), so

$$\begin{aligned}\sin^2 \frac{\pi}{8} &= \frac{1}{2} \left(1 - \frac{1}{\sqrt{2}} \right) = \frac{1}{2} - \frac{1}{2\sqrt{2}} \\ &= \frac{1}{2} - \frac{\sqrt{2}}{4} \quad \text{multiplying top and bottom of } 1/(2\sqrt{2}) \text{ by } \sqrt{2} \\ &= \frac{2 - \sqrt{2}}{4} \quad \text{writing } \frac{1}{2} \text{ as } \frac{2}{4}.\end{aligned}$$

The angle $\pi/8$ lies in the first quarter turn, where the sine is positive, so we take the positive root, $\sin(\pi/8) = \frac{1}{2}\sqrt{2 - \sqrt{2}} = 0.3827$.

EXERCISE 1.6. Comparing with $A \sin(\omega t + \phi)$ of Equation (1.8), $A = 0.2 \text{ m}$, $\omega = 5 \text{ rad s}^{-1}$ and $\phi = \pi/6$. The period is $2\pi/\omega = 2\pi/5 = 1.257 \text{ s}$, and the frequency is its inverse, $5/2\pi = 0.796 \text{ Hz}$. The velocity is $A\omega \cos(\omega t + \phi)$, whose largest size is $A\omega = 0.2 \times 5 = 1.0 \text{ m s}^{-1}$; the acceleration is $-A\omega^2 \sin(\omega t + \phi)$, whose largest size is $A\omega^2 = 0.2 \times 25 = 5.0 \text{ m s}^{-2}$. The coordinate is greatest when the sine equals 1, which first happens after $t = 0$ when $5t + \pi/6 = \pi/2$, so $5t = \pi/3$ and $t = \pi/15 = 0.209 \text{ s}$.

EXERCISE 1.7. The sine never exceeds 1, so $d = v_0^2 \sin 2\alpha / g$ is largest when $\sin 2\alpha = 1$, that is when $2\alpha = \pi/2$, or $\alpha = 45^\circ$. For a range of 10 m at 12 m s^{-1} we need

$$\sin 2\alpha = \frac{gd}{v_0^2} = \frac{9.80665 \times 10}{12^2} = 0.6810.$$

One angle with this sine is $2\alpha = \arcsin 0.6810 = 0.7492 \text{ rad} = 42.92^\circ$. Since $0 < \alpha < 90^\circ$, 2α lies between 0° and 180° , where exactly two angles have this sine (Section 1.5), and by $\sin(\pi - \theta) = \sin \theta$ the other is $2\alpha = 180^\circ - 42.92^\circ = 137.08^\circ$. Halving, the two launch angles are $\alpha = 21.46^\circ$ and $\alpha = 68.54^\circ$. They add to 90° : a low, fast throw and a high, slow one land at the same place.

EXERCISE 1.8. For uniform motion on a circle the speed is $v = R\omega$, so $\omega = v/R = 0.02/1.0 = 0.02 \text{ rad fs}^{-1}$. By Equation (1.9) the acceleration has size $R\omega^2 = v^2/R = 0.02^2/1.0 = 4.0 \times 10^{-4} \text{ Å fs}^{-2}$, directed at the centre. One turn is a distance $2\pi R$ at speed v , so the period is $2\pi R/v = 2\pi \times 1.0/0.02 = 314.2 \text{ fs}$.

EXERCISE 1.9. The cosine series is $\cos y = 1 - y^2/2! + y^4/4! - \dots$. Keeping the terms up to y^2 with $y = 0.1$,

$$\cos 0.1 \approx 1 - \frac{0.1^2}{2} = 1 - 0.005 = 0.995.$$

The cosine series has no y^3 term, so the first term left out is $y^4/4! = 0.1^4/24 = 4.167 \times 10^{-6}$, which estimates the error. The exact value is $\cos 0.1 = 0.995\,004\,165$, so the true error is 4.165×10^{-6} , within a fraction of a per cent of the estimate; the next term, $y^6/720$, is thousands of times smaller again.

EXERCISE 1.10. The forward difference is $[x(t + \tau) - x(t)]/\tau$. The Taylor series $x(t + \tau) = x + \dot{x}\tau + \frac{1}{2}\ddot{x}\tau^2 + \mathcal{O}(\tau^3)$ turns it into

$$\frac{x(t + \tau) - x(t)}{\tau} = \dot{x} + \frac{1}{2}\ddot{x}\tau + \mathcal{O}(\tau^2),$$

so its error is $\frac{1}{2}\ddot{x}\tau$ to leading order. For $x = A \sin \omega t$, $\ddot{x} = -A\omega^2 \sin \omega t$, whose largest size is $A\omega^2$, so the largest error is $\frac{1}{2}A\omega^2\tau$. Dividing by the largest speed $A\omega$ gives the relative error $\omega\tau/2$.

For the central difference, [Section 1.9](#) showed that every recovered velocity is too small by the factor $\sin(\omega\tau)/(\omega\tau)$, a relative error of about $(\omega\tau)^2/6$. Requiring 1 %,

$$\frac{(\omega\tau)^2}{6} \leq 0.01 \implies \omega\tau \leq \sqrt{0.06} = 0.245.$$

A period of 20 fs gives $\omega = 2\pi/20 = 0.3142 \text{ rad fs}^{-1}$, so $\tau \leq 0.245/0.3142 = 0.78 \text{ fs}$. Solving $1 - \sin(\omega\tau)/(\omega\tau) = 0.01$ exactly, as Notebook 01 does, changes only the third figure.

Chapter 2: Forces and Newton's laws

EXERCISE 2.1. The passenger moves at constant velocity, so by the first law the net force on them is zero: the floor pushes up exactly as hard as their weight pulls down. A frame of reference fixed to the lift moves at constant velocity relative to the ground, which is very nearly inertial, so by [Section 2.2](#) it is very nearly inertial too. While the lift speeds up, it accelerates, and a frame of reference fixed to it is no longer inertial. Seen from the ground, the passenger then accelerates upwards with the lift, so the net force on them points up: the floor pushes harder than their weight, which is why they feel heavier.

EXERCISE 2.2. By [Equation \(2.2\)](#), $v' = v - u$, and adding u to both sides, the velocity seen from the ground is $v = v' + u$, the velocity seen from the bicycle plus the velocity of the bicycle. Taking forwards as positive, $u = 6 \text{ m s}^{-1}$. Thrown forwards, $v' = +4$ and $v = 6 + 4 = 10 \text{ m s}^{-1}$. Thrown backwards, $v' = -4$ and $v = 6 - 4 = 2 \text{ m s}^{-1}$: the ball still moves forwards relative to the ground, only more slowly than the bicycle.

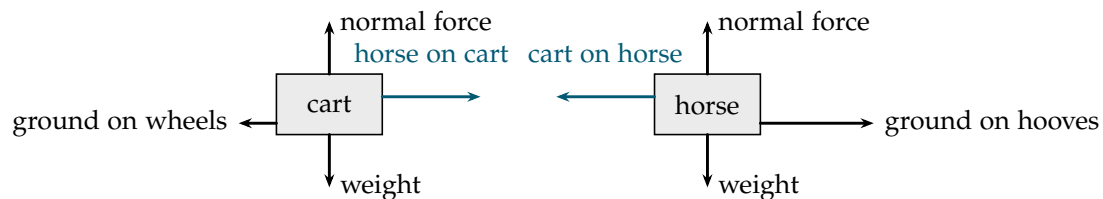
EXERCISE 2.3. Taking east as x and north as y , the net force is $\mathbf{F} = (4, 0) + (0, 3) = (4, 3)$ N. Its size is $(4^2 + 3^2)^{1/2} = 5$ N, and its direction makes an angle θ north of east with $\tan \theta = 3/4$, so $\theta = \arctan(3/4) = 36.9^\circ$. By the second law the acceleration points the same way and has size $F/m = 5/3 = 1.67 \text{ m s}^{-2}$.

EXERCISE 2.4. The same force gives accelerations in the inverse ratio of the masses, $m_2/m_1 = a_1/a_2$, so

$$m_2 = m_1 \frac{a_1}{a_2} = 2 \times \frac{0.6}{1.5} = 0.8 \text{ kg}.$$

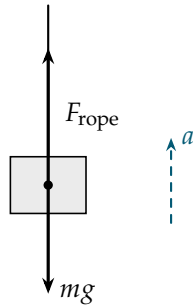
The force follows from either trolley by the second law: $F = m_1 a_1 = 2 \times 0.6 = 1.2$ N, and as a check $m_2 a_2 = 0.8 \times 1.5 = 1.2$ N.

EXERCISE 2.5. The free-body diagrams below draw the cart and the horse apart, moving to the right. The weight and normal force on each balance, the teal arrows are the third-law pair between them, and the lengths of the horizontal arrows show which forces are larger.



The two forces of the third-law pair act on different bodies, the horse's pull on the cart and the cart's pull on the horse, so they never cancel in the second law of either. Whether each body speeds up depends on all the forces on that body. On the cart, the horse's forward pull is larger than the backward friction at its wheels, so the cart accelerates forwards. On the horse, the ground pushes forwards on the hooves, the third-law partner of the hooves' backward push on the ground, and this push is larger than the cart's backward pull, so the horse accelerates forwards too. Taking horse and cart together, their pulls on each other are internal and cancel (Section 2.9); the pair speeds up because the forward push of the ground on the hooves exceeds the backward friction on the wheels.

EXERCISE 2.6. The free-body diagram of the box has two forces, the tension F_{rope} upwards and the weight mg downwards, the tension drawn longer because the box accelerates upwards.



Taking upwards as positive, the second law gives

$$F_{\text{rope}} - mg = ma \implies F_{\text{rope}} = m(g + a) = 2 \times (9.80665 + 1.5) = 22.6 \text{ N}.$$

The rope must hold up the weight, 19.6 N, and supply a further $ma = 3 \text{ N}$ to accelerate the box.

EXERCISE 2.7. Mass measures inertia and is the same everywhere, so the astronaut's mass on the Moon is 70 kg. The weight is mg with the Moon's value of g , $70 \times 1.62 = 113 \text{ N}$. On the Earth it is $70 \times 9.80665 = 686 \text{ N}$, so on the Moon the astronaut weighs $1.62/9.80665 = 0.165$ of their weight on the Earth, about a sixth.

EXERCISE 2.8. Measuring the height x upwards from the ground, the state at $t = 0$ is $x_0 = 3 \text{ m}$, $v_0 = -2 \text{ m s}^{-1}$, and by Equation (1.4) with $a = -g$,

$$x(t) = 3 - 2t - \frac{1}{2}gt^2.$$

The ball reaches the ground when $x = 0$. Multiplying by -1 to make the leading coefficient positive, $\frac{1}{2}gt^2 + 2t - 3 = 0$, a quadratic $At^2 + Bt + C = 0$ with $A = \frac{1}{2}g = 4.9033$, $B = 2$ and $C = -3$. Its roots are $t = [-B \pm (B^2 - 4AC)^{1/2}]/2A$, and the positive one is

$$t = \frac{-2 + \sqrt{4 + 4 \times 4.9033 \times 3}}{2 \times 4.9033} = \frac{-2 + 7.927}{9.807} = 0.604 \text{ s},$$

the red curve of Figure 2.6(a).

EXERCISE 2.9. One step of length τ from $x = 0$, $v = v_0$ gives $x = v_0\tau$, while the true height is $v_0\tau - \frac{1}{2}g\tau^2$ by Equation (1.1). The step overshoots by $v_0\tau - (v_0\tau - \frac{1}{2}g\tau^2) = \frac{1}{2}g\tau^2$. Two steps of $\tau/2$ go as follows. The first gives $x_1 = v_0\tau/2$ and $v_1 = v_0 - g\tau/2$; the second starts from this state, so

$$\begin{aligned} x_2 &= x_1 + v_1 \frac{\tau}{2} = \frac{v_0\tau}{2} + \left(v_0 - \frac{g\tau}{2}\right) \frac{\tau}{2} \\ &= v_0\tau - \frac{1}{4}g\tau^2 \end{aligned} \quad \text{expanding; the two } v_0\tau/2 \text{ add.}$$

The overshoot is now $(v_0\tau - \frac{1}{4}g\tau^2) - (v_0\tau - \frac{1}{2}g\tau^2) = \frac{1}{4}g\tau^2$, half that of the single step. Halving the step halves the error over a fixed time, though each step's own error falls by four.

EXERCISE 2.10. With $\gamma = 5 \text{ s}^{-1}$ the terminal velocity is $v_\infty = 9.80665/5 = 1.961 \text{ m s}^{-1}$, and the starting gap is $v_0 - v_\infty = 4 - 1.961 = 2.039 \text{ m s}^{-1}$. At $t = 0.2 \text{ s}$, $\gamma t = 1$, so by [Equation \(2.7\)](#) the gap has shrunk by the factor $e^{-1} = 0.368$:

$$v = 1.961 + 2.039 \times 0.368 = 1.961 + 0.750 = 2.711 \text{ m s}^{-1}.$$

The bead slows towards v_∞ from above.

EXERCISE 2.11. From rest, $v = v_\infty(1 - e^{-\gamma t})$, so 99% of v_∞ is reached when $e^{-\gamma t} = 0.01$. Taking the natural logarithm of both sides, $-\gamma t = \ln 0.01 = -4.605$, so $t = 4.605/5 = 0.921 \text{ s}$. Each further factor of ten closer to v_∞ needs $e^{-\gamma t}$ to shrink by another factor of 10. Since $e^{-\gamma(t+\Delta t)} = e^{-\gamma t} e^{-\gamma \Delta t}$, this takes a further Δt with $e^{-\gamma \Delta t} = 0.1$, that is $\Delta t = 2.303/\gamma = 0.461 \text{ s}$, as in [Section 2.7](#).

EXERCISE 2.12. Differentiating [Equation \(2.8\)](#) term by term, the derivative of $v_\infty t$ is v_∞ , and the derivative of $1 - e^{-\gamma t}$ is $\gamma e^{-\gamma t}$ by the chain rule, so

$$\frac{dx}{dt} = v_\infty - \frac{v_\infty}{\gamma} \gamma e^{-\gamma t} = v_\infty(1 - e^{-\gamma t}),$$

the velocity of a bead released from rest. Differentiating again, $d^2x/dt^2 = \gamma v_\infty e^{-\gamma t}$. The right side of [Equation \(2.6\)](#) is

$$\begin{aligned} g - \gamma \frac{dx}{dt} &= g - \gamma v_\infty + \gamma v_\infty e^{-\gamma t} && \text{multiplying out the bracket} \\ &= \gamma v_\infty e^{-\gamma t} && \text{since } \gamma v_\infty = g, \end{aligned}$$

which equals d^2x/dt^2 , as required.

EXERCISE 2.13. By [Equation \(2.11\)](#), $\omega = \sqrt{k/m} = \sqrt{80/0.2} = \sqrt{400} = 20 \text{ rad s}^{-1}$ and the period is $2\pi/\omega = 2\pi/20 = 0.314 \text{ s}$. Released from rest at 3 cm, the amplitude is $A = 0.03 \text{ m}$. The largest speed, at the centre, is $A\omega = 0.03 \times 20 = 0.6 \text{ m s}^{-1}$, and the largest acceleration, at the turning points, is $A\omega^2 = 0.03 \times 400 = 12 \text{ m s}^{-2}$.

EXERCISE 2.14. Here $x_0 = 0$ and $v_0 = 0.5 \text{ m s}^{-1}$, so

$$A = \sqrt{x_0^2 + (v_0/\omega)^2} = \sqrt{0 + 0.05^2} = 0.05 \text{ m},$$

with $\cos \phi = x_0/A = 0$ and $\sin \phi = -v_0/(\omega A) = -0.5/(10 \times 0.05) = -1$. The angle with cosine 0 and sine -1 is $\phi = -\pi/2$. The motion is therefore $x = 0.05 \cos(10t - \pi/2)$. By $\cos(-\theta) = \cos \theta$ and $\cos(\pi/2 - \theta) = \sin \theta$ from [Section 1.5](#), $\cos(\omega t - \pi/2) = \cos(\pi/2 - \omega t) = \sin \omega t$, so $x = 0.05 \sin 10t$: the block starts at the centre and moves outwards.

EXERCISE 2.15. Let the spring be stretched by s when the block hangs at rest. The spring then pulls up with ks and the weight pulls down with mg , and since the block does not accelerate, $ks = mg$, so $s = mg/k$. Now let the block be a distance x below this resting place, so that the spring is stretched by $s + x$. Taking downwards as positive, the net force is

$$\begin{aligned} mg - k(s + x) &= (mg - ks) - kx && \text{multiplying out the bracket} \\ &= -kx && \text{since } ks = mg, \end{aligned}$$

and the second law gives $m d^2x/dt^2 = -kx$, or $d^2x/dt^2 = -(k/m)x$. This is [Equation \(2.9\)](#), so the block oscillates about its new resting place with the same $\omega = \sqrt{k/m}$ and period as on the table. Gravity only moves the centre down by mg/k .

EXERCISE 2.16. On the level, smooth track no external force acts along the track, so the total momentum is the same before and after the carts couple:

$$2 \times 3 + 1 \times 0 = (2 + 1)v \implies v = \frac{6}{3} = 2 \text{ m s}^{-1}.$$

EXERCISE 2.17. No horizontal external force acts on the person and boat together, and they start at rest, so their centre of mass stays where it is. Let the person move Δx_p and the boat Δx_b relative to the water, positive in the direction the person walks. The centre of mass is at $X = (70 x_p + 140 x_b)/210$, and when the person moves by Δx_p and the boat by Δx_b it moves by $(70 \Delta x_p + 140 \Delta x_b)/210$. Keeping it fixed therefore requires $70 \Delta x_p + 140 \Delta x_b = 0$. Walking 3 m along the boat means $\Delta x_p = \Delta x_b + 3$. Substituting,

$$70(\Delta x_b + 3) + 140 \Delta x_b = 0$$

$$70 \Delta x_b + 210 + 140 \Delta x_b = 0 \quad \text{multiplying out}$$

$$210 \Delta x_b = -210 \quad \text{collecting } \Delta x_b, \text{ subtracting } 210$$

$$\Delta x_b = -1 \text{ m} \quad \text{dividing by } 210.$$

The boat moves 1 m backwards, and the person 2 m forwards, relative to the water.

EXERCISE 2.18. By Section 2.10, a force in $\text{eV } \text{\AA}^{-1}$ divided by a mass in amu and multiplied by $9.648\,533 \times 10^{-3}$ gives the acceleration in $\text{\AA fs}^{-2}$:

$$a = \frac{0.5}{15.999} \times 9.648\,533 \times 10^{-3} = 3.015 \times 10^{-4} \text{\AA fs}^{-2}.$$

From rest, after 5 fs the velocity is $at = 5 \times 3.015 \times 10^{-4} = 1.508 \times 10^{-3} \text{\AA fs}^{-1}$. Since $1 \text{\AA fs}^{-1} = 10^{-10} \text{ m}/10^{-15} \text{ s} = 10^5 \text{ m s}^{-1}$, this is 151 m s^{-1} .

EXERCISE 2.19. Write $\mathbf{u} = \mathbf{P}/M$ for the drift and $\mathbf{v}'_i = \mathbf{v}_i - \mathbf{u}$ for the new velocities. The new total momentum is

$$\begin{aligned} \sum_i m_i \mathbf{v}'_i &= \sum_i m_i (\mathbf{v}_i - \mathbf{u}) && \text{substituting } \mathbf{v}'_i \\ &= \sum_i m_i \mathbf{v}_i - \sum_i m_i \mathbf{u} && \text{multiplying out each term} \\ &= \sum_i m_i \mathbf{v}_i - \left(\sum_i m_i \right) \mathbf{u} && \mathbf{u} \text{ is common to every term} \\ &= \mathbf{P} - M \frac{\mathbf{P}}{M} = \mathbf{0}. \end{aligned}$$

Subtracting the same constant $\mathbf{u}$ from every velocity is the change to a frame of reference moving at $\mathbf{u}$ (Section 2.2), in which $\mathbf{a}' = \mathbf{a}$, so no acceleration changes. Every atom's position is shifted by the same amount $\mathbf{u}t$, so the distances between atoms, on which their forces depend, are unchanged as well.

Chapter 3: Work and energy

EXERCISE 3.1. No. On a circle the velocity is perpendicular to the line to the centre (Section 1.7), and the string pulls along that line, towards the peg at the centre. The pull is therefore perpendicular to the velocity at every instant, its power $\mathbf{F} \cdot \mathbf{v}$ is zero, and it does no work. The weight and the push of the table are vertical, across the horizontal motion, and do no work either. By the work-energy theorem the kinetic energy does not change, which agrees with the constant speed; the pull changes only the direction of the motion.

EXERCISE 3.2. Only the component of the pull along the floor does work. By Equation (3.1),

$$W = F d \cos \theta = 60 \times 25 \times \cos 50^\circ = 60 \times 25 \times 0.6428 = 964.2 \text{ J}.$$

EXERCISE 3.3. The weight is $mg = 70 \times 9.80665 = 686.47 \text{ N}$, and over a rise of 3 m it does $-686.47 \times 3 = -2059.4 \text{ J}$ of work. The climber moves at a steady speed, so the kinetic energy does not change, and the 2059.4 J that the weight takes must be supplied by the climber's muscles: the stairs push on a foot that does not move while it pushes, and so do no work. Done in 5 s, this is an average power of $2059.4/5 = 411.9 \text{ W}$.

EXERCISE 3.4. In metres per second,

$$50 \text{ km h}^{-1} = \frac{50 \times 1000}{3600} \text{ m s}^{-1} = \frac{125}{9} \text{ m s}^{-1} = 13.889 \text{ m s}^{-1},$$

so

$$K = \frac{1}{2} \times 1200 \times \left(\frac{125}{9}\right)^2 = 115\,741 \text{ J},$$

about 116 kJ. The braking force does the work $-6000 d$ over a distance d , and the car stops when this equals $-K$, so $d = 115741/6000 = 19.29 \text{ m}$. At twice the speed the kinetic energy is four times larger, and so is the distance, 77.16 m.

EXERCISE 3.5. The hand pulls against the spring, so its work from x_A to x_B is Equation (3.4) with the signs reversed, $\frac{1}{2}k(x_B^2 - x_A^2)$. From 0 to 0.1 m: $\frac{1}{2} \times 200 \times 0.1^2 = 1 \text{ J}$. From 0.1 m to 0.2 m: $\frac{1}{2} \times 200 \times (0.04 - 0.01) = 3 \text{ J}$, three times as much, because the spring pulls harder over the second stretch.

EXERCISE 3.6. The spring stores $\frac{1}{2}kx^2 = \frac{1}{2} \times 500 \times 0.08^2 = 1.6 \text{ J}$, and on a smooth level table all of it becomes kinetic energy, so

$$v = \sqrt{\frac{2 \times 1.6}{0.1}} = 5.657 \text{ m s}^{-1}.$$

On the smooth ramp only the weight does work, so the puck rises until its potential energy mgh equals 1.6 J: $h = 1.6/(0.1 \times 9.80665) = 1.632 \text{ m}$. The angle of the ramp does not matter.

EXERCISE 3.7. The bob moves on a circle about the point of support, so its velocity is perpendicular to the string, which pulls along its own length; as for the bead on its wire, the pull does no work. Only the weight does work, and energy is conserved. The bob starts at a height $L \cos 40^\circ$ below the support and ends at L below it, so it falls through

$$L(1 - \cos 40^\circ) = 1.2 \times (1 - 0.766044) = 0.28075 \text{ m},$$

and at the bottom $\frac{1}{2}mv^2 = mg \times 0.28075$, giving

$$v = \sqrt{2 \times 9.80665 \times 0.28075} = 2.347 \text{ m s}^{-1}.$$

EXERCISE 3.8. The bead's energy is mgh_E with $\frac{1}{2}mv^2 = mg(h_E - h)$ at the valley bottom, so $h_E = h + v^2/2g = 0.183 + 3^2/(2 \times 9.80665) = 0.642$ m. The hill is 1.009 m high, so the bead cannot cross it. In [Figure 3.5\(a\)](#) the line $E/mg = 0.642$ m meets the deep valley at about $x = -2.0$ and -0.7 m; the notebook finds the turning points -1.973 m and -0.693 m. The line also meets the shallow valley, at 1.147 m and 1.520 m, but the bead started in the deep valley and cannot get there.

EXERCISE 3.9. Write $s = x/a$, so that $U = U_0(s^2 - 1)^2$ and, by the chain rule with $ds/dx = 1/a$,

$$U'(x) = U_0 \times 2(s^2 - 1) \times 2s \times \frac{1}{a} = \frac{4U_0}{a} s(s^2 - 1),$$

$$U''(x) = \frac{4U_0}{a} (3s^2 - 1) \frac{1}{a} = \frac{4U_0}{a^2} (3s^2 - 1),$$

the second line differentiating $s^3 - s$ to $3s^2 - 1$ and multiplying by ds/dx again. $U' = 0$ where $s = 0$ or $s^2 = 1$, that is at $x = 0$ and $x = \pm a$. At $x = 0$, $U'' = -4U_0/a^2 < 0$: unstable, a hill of height U_0 . At $x = \pm a$, $U'' = 8U_0/a^2 > 0$: stable, the bottoms of two wells where $U = 0$. For $E = U_0/2$ the turning points satisfy $U_0(s^2 - 1)^2 = U_0/2$, so $s^2 - 1 = \pm 1/\sqrt{2}$ and $s^2 = 1 \pm 1/\sqrt{2}$, giving $x = \pm a\sqrt{1 \pm 1/\sqrt{2}}$, that is $\pm 0.5412a$ and $\pm 1.3066a$. Since $E < U(0) = U_0$, the body cannot cross the middle: there are two separate allowed regions, $0.5412a \leq |x| \leq 1.3066a$, one in each well.

EXERCISE 3.10. As for the box in [Section 3.7](#), the friction $\mu_k mg$ removes all the kinetic energy over the distance $d = v_0^2/(2\mu_k g) = 3^2/(2 \times 0.05 \times 9.80665) = 9.177$ m.

EXERCISE 3.11. The ramp is $L = 2/\sin 25^\circ = 2/0.42262 = 4.732$ m long. As in [Section 3.7](#), the normal force is $mg \cos 25^\circ$ and the friction $\mu_k mg \cos 25^\circ$, which does the work $-\mu_k mg \cos 25^\circ L$; the weight does $mg \times 2$. Dividing the work-energy theorem by m ,

$$\begin{aligned} \frac{1}{2}v^2 &= g \times 2 - \mu_k g \cos 25^\circ L \\ &= 19.613 - 0.15 \times 9.80665 \times 0.9063 \times 4.732 = 19.613 - 6.309 = 13.304 \text{ J kg}^{-1}, \end{aligned}$$

so $v = \sqrt{2 \times 13.304} = 5.158 \text{ m s}^{-1}$. The mass does not enter.

EXERCISE 3.12. The spring stores $\frac{1}{2} \times 200 \times 0.05^2 = 0.25$ J. Momentum gives $1 \times v_1 + 2 \times v_2 = 0$, so $v_1 = -2v_2$, and energy gives

$$\frac{1}{2} \times 1 \times (-2v_2)^2 + \frac{1}{2} \times 2 \times v_2^2 = 2v_2^2 + v_2^2 = 3v_2^2 = 0.25,$$

so $v_2 = +\sqrt{0.25/3} = +0.2887 \text{ m s}^{-1}$, taking the positive root because the spring pushes the cart on the right forwards, and $v_1 = -2v_2 = -0.5774 \text{ m s}^{-1}$.

EXERCISE 3.13. Before, $K = \frac{1}{2} \times 2 \times 3^2 = 9 \text{ J}$; after, $K = \frac{1}{2} \times 3 \times 2^2 = 6 \text{ J}$. 3 J has gone into the coupling: the latch and buffers are deformed and warmed, and some energy goes into sound. The forces between the carts during the coupling are a third-law pair, so they cancel in the total momentum whatever they are. They are not the forces of a spring that stores and returns energy, so nothing makes them conserve the kinetic energy, and [Section 3.8](#) showed that a pair of forces does work whenever the carts move relative to each other.

EXERCISE 3.14. Along the straight line, $\mathbf{r}(s) = (2s, s)$ for s from 0 to 1, so $d\mathbf{r}/ds = (2, 1)$ and $\mathbf{F} = a(s, 2s)$, giving

$$\int_0^1 a(s \times 2 + 2s \times 1) ds = a \int_0^1 4s ds = 2a = 1 \text{ eV}.$$

Along the x axis $y = 0$, so $F_x = ay = 0$ and that leg does no work. Going up the line $x = 2$, $F_y = ax = 2a$ and the displacement is 1, so the work is $2a = 1 \text{ eV}$ again. The curl is $\partial F_y/\partial x - \partial F_x/\partial y = a - a = 0$ in the whole plane, so the force is conservative, and by [Equation \(3.12\)](#) with $\mathbf{r}_0 = \mathbf{0}$, $U(x, y)$ is minus the work along the second kind of route to (x, y) : none along the x axis, then ax times the rise y . Hence $U = -axy$, and indeed $-\partial U/\partial x = ay$ and $-\partial U/\partial y = ax$. The work is $U(\mathbf{0}) - U(2, 1) = 0 - (-2a) = 2a$, as found.

EXERCISE 3.15. $\partial U/\partial x = kx$ and $\partial U/\partial y = 4ky$, so $\mathbf{F} = -\nabla U = -k(x, 4y)$, which at $(1, 1)$ is $-(2, 8) \text{ eV \AA}^{-1}$. The contour through $(1, 1)$ is $x^2 + 4y^2 = 5$. A small step $(\Delta x, \Delta y)$ along it keeps $x^2 + 4y^2$ fixed, so by the gradient toolbox of [Section 3.10](#) $2x \Delta x + 8y \Delta y \approx 0$, with an error small compared with the step; at $(1, 1)$ this is $2\Delta x + 8\Delta y = 0$, met by the direction $(4, -1)$. Its scalar product with the force is $-(2 \times 4 - 8 \times 1) = 0$, so the force is perpendicular to the contour. The direction to the origin is $-(1, 1)$, which is not parallel to $-(2, 8)$: the force points across the contour, more steeply in y , where the well is stiffer.

EXERCISE 3.16. The three curl components are

$$\frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z} = a(2z - 2z) = 0,$$

$$\frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x} = a(0 - 0) = 0,$$

$$\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} = a(2x - 2x) = 0.$$

To find U , integrate $\partial U/\partial x = -F_x = -2axy$ with respect to x , holding y and z fixed: $U = -ax^2y + f(y, z)$, where the constant of integration f may depend on the fixed y and z but not on x . Then $-\partial U/\partial y = ax^2 - \partial f/\partial y$ must equal $F_y = a(x^2 + z^2)$, so $\partial f/\partial y = -az^2$ and $f = -ayz^2 + \chi(z)$. Then $-\partial U/\partial z = 2ayz - \chi'(z)$ must equal $F_z = 2ayz$, so $\chi' = 0$, and $U(0) = 0$ makes $\chi = 0$:

$$U = -a(x^2y + yz^2).$$

At $(1, 2, 3)$, $U = -(1 \times 2 + 2 \times 9) = -20 \text{ eV}$.

EXERCISE 3.17. Side by side, anticlockwise:

- bottom, $y = 0$ from $x = 0$ to L : $F_x = -cy = 0$, work 0;
- right, $x = L$ from $y = 0$ to L : $F_y = cL$, work $cL \times L = cL^2$;
- top, $y = L$ from $x = L$ back to 0: $F_x = -cL$ and the displacement is $-L$, work $(-cL)(-L) = cL^2$;
- left, $x = 0$ from $y = L$ down to 0: $F_y = 0$, work 0.

The total is $2cL^2$, which is $2c$ times the area L^2 , as Green's theorem with curl $2c$ requires.

EXERCISE 3.18. Write $\rho^2 = x^2 + y^2$ and $q = 1/\rho^2$, so $F_y = bxq$ and $F_x = -byq$. Differentiating $\rho^2q = 1$ with respect to x by the product rule gives $2xq + \rho^2\partial q/\partial x = 0$, so $\partial q/\partial x = -2xq/\rho^2 = -2x/\rho^4$; in the same way $\partial q/\partial y = -2y/\rho^4$. By the product rule again,

$$\frac{\partial F_y}{\partial x} = b \left[q + x \frac{\partial q}{\partial x} \right] = b \left[\frac{1}{\rho^2} - \frac{2x^2}{\rho^4} \right] = b \frac{\rho^2 - 2x^2}{\rho^4} = b \frac{y^2 - x^2}{\rho^4},$$

putting both terms over ρ^4 and using $\rho^2 = x^2 + y^2$; and in the same way, differentiating $-byq$ with respect to y ,

$$\frac{\partial F_x}{\partial y} = -b \left[q + y \frac{\partial q}{\partial y} \right] = -b \left[\frac{1}{\rho^2} - \frac{2y^2}{\rho^4} \right] = -b \frac{x^2 - y^2}{\rho^4} = b \frac{y^2 - x^2}{\rho^4}.$$

The two are equal, so the curl is zero wherever $\mathbf{F}$ is defined. On the circle $\mathbf{r}(s) = R(\cos s, \sin s)$, $\mathbf{F} = (b/R)(-\sin s, \cos s)$ and $d\mathbf{r}/ds = R(-\sin s, \cos s)$, so $\mathbf{F} \cdot d\mathbf{r}/ds = b$ and the work once round is $2\pi b$, for every radius. This does not contradict the curl test, because the force is not defined at the origin: the plane with the origin removed has a hole, and Green's theorem needs the curl to be zero over the whole area inside the loop, origin included. A circle that does not enclose the origin encloses no hole, so by Green's theorem the work around it is zero.

EXERCISE 3.19. Write $g = |\mathbf{g}| = 4\pi/(\sqrt{3}a)$. A vector of length g at an angle α to the x axis is $g(\cos \alpha, \sin \alpha)$ (Section 1.6), so, with $\cos -30^\circ = \sqrt{3}/2$, $\sin -30^\circ = -\frac{1}{2}$, $\cos 210^\circ = -\sqrt{3}/2$ and $\sin 210^\circ = -\frac{1}{2}$,

$$\mathbf{g}_1 = g \left(\frac{\sqrt{3}}{2}, -\frac{1}{2} \right), \quad \mathbf{g}_2 = g (0, 1), \quad \mathbf{g}_3 = g \left(-\frac{\sqrt{3}}{2}, -\frac{1}{2} \right).$$

With $\mathbf{r} = (a/2, a/(2\sqrt{3}))$ and the scalar product in components,

$$\begin{aligned} \mathbf{g}_1 \cdot \mathbf{r} &= g a \left(\frac{\sqrt{3}}{4} - \frac{1}{4\sqrt{3}} \right) = g a \frac{3-1}{4\sqrt{3}} = \frac{g a}{2\sqrt{3}} && \text{writing } \frac{\sqrt{3}}{4} \text{ as } \frac{3}{4\sqrt{3}} \\ &= \frac{4\pi}{\sqrt{3}a} \times \frac{a}{2\sqrt{3}} = \frac{4\pi}{6} = \frac{2\pi}{3}, && \text{since } \sqrt{3} \times \sqrt{3} = 3, \\ \mathbf{g}_2 \cdot \mathbf{r} &= \frac{g a}{2\sqrt{3}} = \frac{2\pi}{3}, \\ \mathbf{g}_3 \cdot \mathbf{r} &= -g a \left(\frac{\sqrt{3}}{4} + \frac{1}{4\sqrt{3}} \right) = -g a \frac{3+1}{4\sqrt{3}} = -\frac{g a}{\sqrt{3}} = -\frac{4\pi}{3}. \end{aligned}$$

Each cosine is $-\frac{1}{2}$: $\cos(2\pi/3) = \cos(\pi - \pi/3) = -\cos(\pi/3) = -\frac{1}{2}$ by the reflection rule $\cos(\pi - \theta) = -\cos \theta$ of Section 1.5, and $\cos(-4\pi/3) = \cos(-4\pi/3 + 2\pi) = \cos(2\pi/3) = -\frac{1}{2}$ by the period 2π . Hence $U = (U_b/4)(3 + \frac{3}{2}) = 9U_b/8$.

EXERCISE 3.20. By Pythagoras' theorem, $r_{12}^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2$. Differentiating both sides with respect to x_2 , the left by the chain rule with outer function u^2 and the right term by term, of which only the first depends on x_2 ,

$$2r_{12} \frac{\partial r_{12}}{\partial x_2} = 2(x_2 - x_1), \quad \text{so} \quad \frac{\partial r_{12}}{\partial x_2} = \frac{x_2 - x_1}{r_{12}}$$

on dividing by $2r_{12}$, and likewise for y_2 and z_2 , so $\nabla_2 r_{12} = (\mathbf{r}_2 - \mathbf{r}_1)/r_{12}$. Differentiating with respect to x_1 instead, the right side becomes $-2(x_2 - x_1)$, by the chain rule with inner function $x_2 - x_1$, whose derivative with respect to x_1 is -1 ; so $\nabla_1 r_{12} = -(\mathbf{r}_2 - \mathbf{r}_1)/r_{12}$. By the chain rule again,

$$\mathbf{F}_2 = -\nabla_2 \varphi(r_{12}) = -\varphi'(r_{12}) \frac{\mathbf{r}_2 - \mathbf{r}_1}{r_{12}}, \quad \mathbf{F}_1 = -\nabla_1 \varphi(r_{12}) = +\varphi'(r_{12}) \frac{\mathbf{r}_2 - \mathbf{r}_1}{r_{12}} = -\mathbf{F}_2.$$

Both are multiples of $\mathbf{r}_2 - \mathbf{r}_1$, so they act along the line joining the atoms. Where $\varphi' > 0$, the energy rises with distance and the atoms pull together.

EXERCISE 3.21. By Equation (3.15), $K = \frac{1}{2} \times 15.999 \times 0.01^2 \times 103.6427 = 0.08291 \text{ eV}$. The opposing force does the work $-0.5d$ over a distance d in ångströms, and the atom stops when this equals $-K$: $d = 0.08291/0.5 = 0.1658 \text{ Å}$.

EXERCISE 3.22. With no potential energy, $E = K$, and Equation (3.18) gives $dK/dt = -\gamma m \mathbf{v} \cdot \mathbf{v} = -\gamma m v^2 = -2\gamma K$. By the exponential toolbox of Section 2.7, $K(0)e^{-2\gamma t}$ has derivative -2γ times itself and the right value at $t = 0$, and by the uniqueness of solutions (Section 2.6) it is the only such function, so $K(t) = K(0)e^{-2\gamma t}$. For the lithium atom, $K(0) = \frac{1}{2} \times 6.94 \times 0.02^2 \times 103.6427 = 0.14386 \text{ eV}$, and $2\gamma t = 2 \times 0.005 \times 100 = 1$, so $K = 0.14386 e^{-1} = 0.05292 \text{ eV}$.

Chapter 4: Oscillations

EXERCISE 4.1. At the angle π the point P is at $(-A, 0)$, the left end of the circle's horizontal diameter, so the block is at $x = A \cos \pi = -A$, a turning point, momentarily at rest. Going anticlockwise, P is moving straight down there, and next swings towards the right, so the block is about to move back towards the centre, in the positive direction.

EXERCISE 4.2. $E = \frac{1}{2}kA^2 = \frac{1}{2} \times 80 \times 0.03^2 = 0.036 \text{ J}$. With $\omega = \sqrt{80/0.2} = 20 \text{ rad s}^{-1}$, the largest speed is $A\omega = 0.03 \times 20 = 0.6 \text{ m s}^{-1}$. At $x = 0.015 \text{ m}$, $U = \frac{1}{2} \times 80 \times 0.015^2 = 0.009 \text{ J}$, so $K = 0.036 - 0.009 = 0.027 \text{ J}$ and

$$v = \sqrt{\frac{2 \times 0.027}{0.2}} = 0.520 \text{ m s}^{-1}.$$

EXERCISE 4.3. Rearranging $\cos 2\theta = 1 - 2\sin^2 \theta$ gives $\sin^2 \theta = \frac{1}{2} - \frac{1}{2} \cos 2\theta$. With $\theta = \omega t$, the average of $\cos 2\omega t$ over a period $2\pi/\omega$ is zero: $\sin 2\omega t / (2\omega)$ has derivative $\cos 2\omega t$ by the chain rule, so the integral is $[\sin 2\omega t / (2\omega)]_0^{2\pi/\omega} = (\sin 4\pi - \sin 0) / (2\omega) = 0$. The integral of a sum is the sum of the integrals, and a constant averages to itself, so the average of $\sin^2 \omega t$ is $\frac{1}{2} - \frac{1}{2} \times 0 = \frac{1}{2}$.

EXERCISE 4.4. $E = \frac{1}{2}mv^2 + \frac{1}{2}kx^2 = \frac{1}{2} \times 0.5 \times 0.4^2 + \frac{1}{2} \times 50 \times 0.03^2 = 0.04 + 0.0225 = 0.0625 \text{ J}$. The semi-axes are $\sqrt{2E/k} = \sqrt{0.0025} = 5 \text{ cm}$ and $\sqrt{2E/m} = \sqrt{0.25} = 0.5 \text{ m s}^{-1}$. The velocity is negative, so x is about to shrink; the state lies in the lower half of the ellipse, moving left.

EXERCISE 4.5. From $2\pi\sqrt{L/g} = 2$, squaring and solving for L ,

$$L = g \left(\frac{2}{2\pi} \right)^2 = \frac{9.80665}{\pi^2} = 0.9936 \text{ m}.$$

EXERCISE 4.6. By the argument of Equation (4.6), the time from $x = 0$ to $x = A/2$ is the integral of dx/v , with $v = \omega\sqrt{A^2 - x^2}$ for the spring (Section 4.5), so by the toolbox on the derivative of an inverse function

$$t = \frac{1}{\omega} \int_0^{A/2} \frac{dx}{\sqrt{A^2 - x^2}} = \frac{1}{\omega} \left[\arcsin \frac{x}{A} \right]_0^{A/2} = \frac{1}{\omega} \left(\arcsin \frac{1}{2} - 0 \right) = \frac{\pi}{6\omega},$$

since $\sin(\pi/6) = \frac{1}{2}$: an equilateral triangle of side 1, cut in half, has an angle of $\pi/6$ opposite a side of $\frac{1}{2}$. For $\omega = 10 \text{ rad s}^{-1}$ this is $\pi/60 = 0.0524 \text{ s}$, a twelfth of the period $2\pi/\omega$. The whole way out, from the centre to the turning point, takes a quarter period, $\pi/(2\omega) = 3\pi/(6\omega)$, so the second half of the way takes $3\pi/(6\omega) - \pi/(6\omega) = 2\pi/(6\omega)$, twice as long as the first, because the block slows down.

EXERCISE 4.7. With $s = x/a$, the solution of Exercise 3.9 gives $U''(x) = (4U_0/a^2)(3s^2 - 1)$, which at $s = \pm 1$ is $8U_0/a^2$. By Equation (4.4), $\omega_0 = \sqrt{8U_0/(ma^2)}$.

EXERCISE 4.8. On the line $x = 0$, $\mathbf{r} = (0, y)$. Writing g for $|\mathbf{g}|$, the solution of Exercise 3.19 gives $\mathbf{g}_1 = g(\frac{\sqrt{3}}{2}, -\frac{1}{2})$, $\mathbf{g}_2 = g(0, 1)$ and $\mathbf{g}_3 = g(-\frac{\sqrt{3}}{2}, -\frac{1}{2})$, so the products $\mathbf{g}_k \cdot \mathbf{r}$ are $-\frac{1}{2}gy$, gy and $-\frac{1}{2}gy$. The cosine is even, so

$$U(0, y) = \frac{1}{4}U_b [3 - 2\cos(\frac{1}{2}gy) - \cos(gy)].$$

Differentiating twice by the chain rule, $-2\cos(\frac{1}{2}gy)$ gives $2(\frac{1}{2}g)^2 \cos(\frac{1}{2}gy)$, which is $\frac{1}{2}g^2$ at $y = 0$, and $-\cos(gy)$ gives $g^2 \cos(gy)$, which is g^2 . So $U''(0) = \frac{1}{4}U_b(\frac{1}{2}g^2 + g^2) = \frac{3}{8}U_b g^2$. Squaring $g = 4\pi/(\sqrt{3}a)$ gives $g^2 = 16\pi^2/(3a^2)$, and

$$\frac{3}{8}U_b \frac{16\pi^2}{3a^2} = \frac{2\pi^2 U_b}{a^2},$$

the curvature along x : the well is equally stiff in both directions, and by a longer argument in every direction.

EXERCISE 4.9. By Section 4.5, $U''_{\text{LJ}} = (4\epsilon/r^2)[156(\sigma/r)^{12} - 42(\sigma/r)^6]$, which is zero when $156(\sigma/r)^6 = 42$, after dividing by $4\epsilon/r^2$ and by $(\sigma/r)^6$. So $(\sigma/r)^6 = 42/156 = 7/26$ and $r = (26/7)^{1/6}\sigma = 1.2445\sigma$. The bracket is $(\sigma/r)^6 [156(\sigma/r)^6 - 42]$, positive at smaller r , where $(\sigma/r)^6$ is larger, and negative at larger r , so inside this distance the slope of U steepens with r and outside it flattens. Beyond r_{min} the force $-U'$ pulls the atoms together with a size equal to that slope, so here the attractive force is greatest.

EXERCISE 4.10. Write $w = e^{-a(r-r_0)}$, so that $dw/dr = -aw$ by the chain rule, and $U_M = D(1-w)^2$. Then

$$\begin{aligned}\frac{dU_M}{dr} &= 2D(1-w)\left(-\frac{dw}{dr}\right) = 2Da w (1-w) = 2Da(w-w^2), \\ \frac{d^2U_M}{dr^2} &= 2Da(1-2w)\frac{dw}{dr} = -2Da^2 w(1-2w),\end{aligned}$$

and at $r = r_0$, where $w = 1$, this is $-2Da^2 \times 1 \times (-1) = 2Da^2$.

EXERCISE 4.11. $\omega_d = \sqrt{10^2 - 6^2} = 8 \text{ rad s}^{-1}$, a period of $2\pi/8 = 0.785 \text{ s}$. The swing shrinks by $e^{-\gamma t/2} = e^{-6t}$, which is $\frac{1}{10}$ when $6t = \ln 10$, that is, taking the natural logarithm of $e^{6t} = 10$, at $t = 2.303/6 = 0.384 \text{ s}$.

EXERCISE 4.12. Write $y = c_1 + c_2 t$ and $z = e^{-\omega_0 t}$ for short, so that $\dot{x} = yz$, $\dot{y} = c_2$ and $\dot{z} = -\omega_0 z$. By the product rule,

$$\dot{x} = z(c_2 - \omega_0 y), \quad \ddot{x} = z(-\omega_0 c_2) - \omega_0 z(c_2 - \omega_0 y) = z(\omega_0^2 y - 2\omega_0 c_2).$$

With $\gamma = 2\omega_0$, $\ddot{x} + \gamma\dot{x} + \omega_0^2 x = z[\omega_0^2 y - 2\omega_0 c_2 + 2\omega_0 c_2 - 2\omega_0^2 y + \omega_0^2 y] = 0$, which is Equation (4.7) with its right side moved to the left.

EXERCISE 4.13. By Equation (4.9), at $\omega = 5$,

$$A = \frac{1}{\sqrt{(100-25)^2 + (2 \times 5)^2}} = \frac{1}{75.66} \text{ m} = 1.32 \text{ cm}, \quad \tan \delta = \frac{10}{75}.$$

Here $\omega_0^2 - \omega^2 = 75$ is positive, so $\cos \delta$ and $\sin \delta$ are both positive and δ lies between 0 and $\pi/2$, inside the range of the arctangent: $\delta = \arctan(10/75) = 0.133 \text{ rad}$. At $\omega = 15$, $A = 1/\sqrt{125^2 + 30^2} = 1/128.55 \text{ m} = 0.778 \text{ cm}$ and $\tan \delta = 30/(-125)$. Now $\omega_0^2 - \omega^2 = -125$, so $\cos \delta$ is negative while $\sin \delta$ is still positive, and δ lies between $\pi/2$ and π , outside the range of the arctangent. By the rule of Section 4.7, $\delta = \pi - \arctan(30/125) = \pi - 0.236 = 2.906 \text{ rad}$. Below resonance the block nearly follows the push; above it, it nearly opposes it.

EXERCISE 4.14. Let $x = c_1 x_1 + c_2 x_2$, and write Equation (4.7) as $\ddot{x} + \gamma\dot{x} + \omega_0^2 x = 0$, adding $\omega_0^2 x + \gamma\dot{x}$ to both sides. Derivatives of sums are sums of derivatives, and constants pass through, so $\ddot{x} + \gamma\dot{x} + \omega_0^2 x = c_1(\ddot{x}_1 + \gamma\dot{x}_1 + \omega_0^2 x_1) + c_2(\ddot{x}_2 + \gamma\dot{x}_2 + \omega_0^2 x_2) = c_1 \times 0 + c_2 \times 0 = 0$, each bracket vanishing because x_1 and x_2 are motions.

EXERCISE 4.15. The reduced masses are $m_{\text{r,OH}} = 0.9483 \text{ amu}$ (Section 4.8) and $m_{\text{r,OD}} = 2.014 \times 15.999/18.013 = 1.7888 \text{ amu}$. With the same stiffness, $\omega = \sqrt{k/m_{\text{r}}}$ for each bond, and dividing one by the other cancels k :

$$\frac{\omega_{\text{OD}}}{\omega_{\text{OH}}} = \sqrt{\frac{m_{\text{r,OH}}}{m_{\text{r,OD}}}} = \sqrt{\frac{0.9483}{1.7888}} = 0.728 :$$

the O–D vibration has 0.728 times the angular frequency, and its period is longer by the factor $1/0.728 = 1.37$.

EXERCISE 4.16. The middle spring is stretched by $x_2 - x_1$, so $m\ddot{x}_1 = -kx_1 + k'(x_2 - x_1) = -(k + k')x_1 + k'x_2$ and $m\ddot{x}_2 = -k'(x_2 - x_1) - kx_2 = k'x_1 - (k + k')x_2$. Written as $m\ddot{\mathbf{x}} = -\Phi\mathbf{x}$, the matrix Φ has rows $(k + k', -k')$ and $(-k', k + k')$. Acting on $(1, 1)$ it gives $(k, k) = k(1, 1)$, so by Equation (4.12) $m\omega^2 = k$; acting on $(1, -1)$ it gives $(k + 2k', -k - 2k') = (k + 2k')(1, -1)$, so $m\omega^2 = k + 2k'$.

EXERCISE 4.17. Write $\rho = 2m_{\text{O}}/m_{\text{C}}$, so $\mathbf{u} = (1, -\rho, 1)$. Row by row,

$$\Phi\mathbf{u} = k(1 + \rho, -1 - 2\rho - 1, \rho + 1) = k(1 + \rho)(1, -2, 1),$$

and, since $\rho m_{\text{C}} = 2m_{\text{O}}$, $\mathbf{Mu} = (m_{\text{O}}, -\rho m_{\text{C}}, m_{\text{O}}) = m_{\text{O}}(1, -2, 1)$. Hence $\Phi\mathbf{u} = [k(1 + \rho)/m_{\text{O}}]\mathbf{Mu}$, and $\omega^2 = (k/m_{\text{O}})(1 + 2m_{\text{O}}/m_{\text{C}})$.

EXERCISE 4.18. Dividing each entry by $\sqrt{m_a m_b}$, the mass-weighted Hessian has rows $(k/m_1, -k/\sqrt{m_1 m_2})$ and $(-k/\sqrt{m_1 m_2}, k/m_2)$. Its characteristic equation is

$$\left(\frac{k}{m_1} - \lambda\right)\left(\frac{k}{m_2} - \lambda\right) - \frac{k^2}{m_1 m_2} = \lambda^2 - \lambda k\left(\frac{1}{m_1} + \frac{1}{m_2}\right) = \lambda\left(\lambda - \frac{k}{m_{\text{r}}}\right) = 0,$$

multiplying out, cancelling the two terms $k^2/(m_1 m_2)$, taking out the factor λ and using $1/m_{\text{r}} = 1/m_1 + 1/m_2$. The eigenvalues are the squared angular frequencies, so $\lambda = 0$ is the pair moving together without stretching the spring, and $\lambda = k/m_{\text{r}}$, $\omega = \sqrt{k/m_{\text{r}}}$, is the vibration of Section 4.8.

EXERCISE 4.19. $\omega = 2\pi \times 2.99792458 \times 10^{-5} \times 3657 = 0.6889 \text{ rad fs}^{-1}$. For a spring $\omega^2 = k/m_{\text{r}}$, which, with k in $\text{eV } \text{\AA}^{-2}$, m_{r} in amu and ω in rad fs^{-1} , reads $\omega^2 = k \times 9.648533 \times 10^{-3}/m_{\text{r}}$, as for lithium in Section 4.4: the factor of Section 2.10 turns a force divided by a mass into an acceleration. Solving for k ,

$$k = \frac{0.9483 \times 0.6889^2}{9.648533 \times 10^{-3}} = 46.6 \text{ eV } \text{\AA}^{-2},$$

nearly fifty times stiffer than the lithium atom's hollow of Section 4.4.

EXERCISE 4.20. The shortest period in Table 4.1 for water is 8.88 fs, so the step could be at most 0.888 fs, and a picosecond, 1000 fs, would take $1000/0.888 = 1126.1$ steps of that length; 1126 steps fall just short, so it takes 1127.

Chapter 5: Rotation and angular momentum

EXERCISE 5.1. The torque is the distance from the hinge times the part of the force across the door, $G = rF_\theta$, so the same push turns the door more the further from the hinge it acts. At 30° to the face of the door, the push splits, as the weight on the ramp of Section 3.7, into the part $10 \cos 30^\circ$ along the door, which points at the hinge and turns nothing, and the part $10 \sin 30^\circ = 5$ N across it, so $G = 0.8 \times 5 = 4$ N m, half the torque of the push at right angles.

EXERCISE 5.2. $L = 0.12 \text{ kg m}^2 \text{ s}^{-1}$ is conserved, so at 0.1 m the puck goes round at $v_\theta = 0.12/(0.2 \times 0.1) = 6 \text{ m s}^{-1}$. Its kinetic energy is $\frac{1}{2} \times 0.2 \times 6^2 = 3.6$ J, and since it started with $\frac{1}{2} \times 0.2 \times 1.2^2 = 0.144$ J, the hand has done the work $3.6 - 0.144 = 3.456$ J (Section 3.2).

EXERCISE 5.3. With no force, $\mathbf{r} = \mathbf{r}_0 + \mathbf{v}t$ with $\mathbf{v}$ constant. About the point $\mathbf{a}$, putting in $\mathbf{r}$,

$$m(\mathbf{r} - \mathbf{a}) \times \mathbf{v} = m(\mathbf{r}_0 - \mathbf{a}) \times \mathbf{v} + mt \mathbf{v} \times \mathbf{v} = m(\mathbf{r}_0 - \mathbf{a}) \times \mathbf{v},$$

a sum splitting and $\mathbf{v} \times \mathbf{v} = \mathbf{0}$: the angular momentum is constant. In equal times the body covers equal lengths $|\mathbf{v}| \Delta t$ of its line, and the triangles with these as bases and $\mathbf{a}$ as the third corner all have the same height, the distance from $\mathbf{a}$ to the line, so they have equal areas, half base times height.

EXERCISE 5.4. By the components,

$$(2, 0, 1) \times (1, 3, -1) = (0 \times (-1) - 1 \times 3, 1 \times 1 - 2 \times (-1), 2 \times 3 - 0 \times 1) = (-3, 3, 6).$$

Its scalar products with the two vectors are $-6 + 0 + 6 = 0$ and $-3 + 9 - 6 = 0$. The area is its length, $\sqrt{9 + 9 + 36} = \sqrt{54} = 7.348$, and also, by the second toolbox of Section 5.2, with $\mathbf{a} = (2, 0, 1)$ and $\mathbf{b} = (1, 3, -1)$, so that $|\mathbf{a}|^2 = 4 + 0 + 1 = 5$, $|\mathbf{b}|^2 = 1 + 9 + 1 = 11$ and $\mathbf{a} \cdot \mathbf{b} = 2 + 0 - 1 = 1$, $\sqrt{|\mathbf{a}|^2 |\mathbf{b}|^2 - (\mathbf{a} \cdot \mathbf{b})^2} = \sqrt{5 \times 11 - 1^2} = \sqrt{54}$.

EXERCISE 5.5. $\mathbf{L} = m \mathbf{r} \times \mathbf{v}$ is at right angles to $\mathbf{r}$ at every instant, so $\mathbf{r} \cdot \mathbf{L} = 0$ always. With $\mathbf{L}$ a fixed vector that is not zero, the points $\mathbf{r}$ with $\mathbf{r} \cdot \mathbf{L} = 0$ form the plane through the origin at right angles to $\mathbf{L}$, and the body never leaves it.

EXERCISE 5.6. The sum splits, and the constant $\mathbf{a}$ comes outside it:

$$\sum_i (\mathbf{r}_i - \mathbf{a}) \times \mathbf{p}_i = \sum_i \mathbf{r}_i \times \mathbf{p}_i - \mathbf{a} \times \sum_i \mathbf{p}_i = \mathbf{L} - \mathbf{a} \times \mathbf{P}.$$

It is the same about every point when, and only when, $\mathbf{P} = \mathbf{0}$: if $\mathbf{P} \neq \mathbf{0}$, any $\mathbf{a}$ not along $\mathbf{P}$ gives $\mathbf{a} \times \mathbf{P}$ the length $|\mathbf{a}||\mathbf{P}|\sin\theta \neq 0$.

EXERCISE 5.7. Put the rod along the x axis from $-\ell/2$ to $\ell/2$. A short piece dx has the mass $(M/\ell) dx$ at the distance $|x|$ from the axis, so

$$I = \int_{-\ell/2}^{\ell/2} x^2 \frac{M}{\ell} dx = \frac{M}{\ell} \left[\frac{x^3}{3} \right]_{-\ell/2}^{\ell/2} = \frac{M}{\ell} \left(\frac{(\ell/2)^3}{3} - \frac{(-\ell/2)^3}{3} \right) = \frac{M}{\ell} \left(\frac{\ell^3}{24} + \frac{\ell^3}{24} \right) = \frac{M\ell^2}{12}.$$

About one end the rod runs from 0 to ℓ , and $I = (M/\ell)[x^3/3]_0^\ell = M\ell^2/3$.

EXERCISE 5.8. Put the origin at the centre of mass, take the axis through it along z , and the parallel axis through (s_x, s_y) , with $s^2 = s_x^2 + s_y^2$. Then, expanding the squares and writing $\sum_i m_i = M$,

$$\begin{aligned} I &= \sum_i m_i [(x_i - s_x)^2 + (y_i - s_y)^2] \\ &= \sum_i m_i (x_i^2 + y_i^2) - 2s_x \sum_i m_i x_i - 2s_y \sum_i m_i y_i + Ms^2 \\ &= I_{\text{cm}} + Ms^2, \end{aligned}$$

since $\sum_i m_i x_i = \sum_i m_i y_i = 0$, the positions being measured from the centre of mass. For the rod, $s = \ell/2$ and $M\ell^2/12 + M\ell^2/4 = M\ell^2/3$, as in Exercise 5.7.

EXERCISE 5.9. $I = 2 + 2 \times 3 \times 0.7^2 = 4.94 \text{ kg m}^2$ before and $2 + 2 \times 3 \times 0.3^2 = 2.54 \text{ kg m}^2$ after. Conserving $L_z = I\omega = 4.94 \times 1.5 = 7.41 \text{ kg m}^2 \text{ s}^{-1}$, $\omega = 7.41/2.54 = 2.917 \text{ rad s}^{-1}$. The kinetic energy rises from $\frac{1}{2} \times 4.94 \times 1.5^2 = 5.5575 \text{ J}$ to $\frac{1}{2} I\omega^2 = L_z^2/(2I) = 7.41^2/(2 \times 2.54) = 10.8087 \text{ J}$, putting in $\omega = L_z/I$, so the arms do $10.8087 - 5.5575 = 5.251 \text{ J}$ of work.

EXERCISE 5.10. With $\kappa = \frac{2}{5}$, $1 + \kappa = 1.4$ and $V = \sqrt{2 \times 9.80665 \times 1/1.4} = 3.743 \text{ m s}^{-1}$. On the slope, with $\sin 30^\circ = 0.5$, $\dot{V} = 9.80665 \times 0.5/1.4 = 3.502 \text{ m s}^{-2}$, and from $s = \frac{1}{2} \dot{V} t^2$ the 2 m take $\sqrt{2 \times 2/3.502} = 1.069 \text{ s}$. The ball comes second, after the sliding block (0.903 s) and before the disc and the hoop: the smaller κ , the less energy goes into turning.

EXERCISE 5.11. Along the slope the weight pulls down with $Mg \sin \alpha$, split as on the ramp of Section 3.7, and the friction f pushes up, so $M\dot{V} = Mg \sin \alpha - f$. With $\dot{V} = g \sin \alpha / (1 + \kappa)$ from Section 5.4,

$$f = Mg \sin \alpha - \frac{Mg \sin \alpha}{1 + \kappa} = \frac{(1 + \kappa)Mg \sin \alpha - Mg \sin \alpha}{1 + \kappa} = \frac{\kappa Mg \sin \alpha}{1 + \kappa},$$

putting the two terms over the common denominator $1 + \kappa$. For the disc, $\kappa = \frac{1}{2}$, $f = \frac{1}{3}Mg \sin \alpha = \frac{1}{3} \times 1 \times 9.80665 \times 0.5 = 1.634 \text{ N}$. A sliding block, $\kappa = 0$, needs none.

EXERCISE 5.12. With every mass 1 and $z = 0$ for every atom, $I_{xx} = \sum(y^2 + z^2) = 4 \times 1 = 4$, $I_{yy} = 4$ likewise, and $I_{zz} = \sum(x^2 + y^2) = 4 \times 2 = 8$, in $\text{amu} \text{ \AA}^2$; $I_{xy} = -\sum xy = -(1 - 1 - 1 + 1) = 0$, and $I_{xz} = I_{yz} = 0$. The tensor is diagonal, with principal moments 4, 4 and 8. Any vector $(a, b, 0)$ in the plane gives $\mathbf{l}(a, b, 0) = (4a, 4b, 0) = 4(a, b, 0)$, so it is an eigenvector: every axis in the plane is a principal axis, because the two moments in the plane are equal.

EXERCISE 5.13. With $z_i = 0$, $I_{xx} = \sum m_i y_i^2$, $I_{yy} = \sum m_i x_i^2$ and $I_{zz} = \sum m_i (x_i^2 + y_i^2) = I_{xx} + I_{yy}$, while $I_{xz} = -\sum m_i x_i z_i = 0$ and $I_{yz} = 0$. Then $\mathbf{l}_{\mathbf{e}_z} = (I_{xz}, I_{yz}, I_{zz}) = I_{zz} \mathbf{e}_z$, so $\mathbf{e}_z$ is an eigenvector, a principal axis. For water, $0.8188 + 0.9538 = 1.7726$ from the entries of Section 5.5.

EXERCISE 5.14. With the centre of mass at the origin, the atoms are at $(0, 0, -d_1)$ and $(0, 0, d_2)$, with d_1 and d_2 as in Equation (5.9). Every x_i and y_i is zero, so $I_{zz} = 0$ and every entry off the diagonal, each containing x_i or y_i , is zero, while $I_{xx} = I_{yy} = m_1 d_1^2 + m_2 d_2^2 = m_r d^2$ by Equation (5.9).

EXERCISE 5.15. The first equation gives $\omega_y = 6 - 4\omega_x$ and the third $\omega_z = (14 - \omega_y)/4$. Putting this ω_z into the middle equation and multiplying by 4, $4\omega_x + 16\omega_y + 14 - \omega_y = 48$, or $4\omega_x + 15\omega_y = 34$; then putting in $\omega_y = 6 - 4\omega_x$, $4\omega_x + 90 - 60\omega_x = 34$, that is $90 - 56\omega_x = 34$, so $56\omega_x = 90 - 34 = 56$ and $\omega_x = 1$. Hence $\omega_y = 2$ and $\omega_z = (14 - 2)/4 = 3$.

EXERCISE 5.16. The masses are equal, so $\mathbf{V}$ is the average velocity, $\frac{1}{2}((0.01, 0.01, 0) + (-0.01, 0.03, 0)) = (0, 0.02, 0)$, leaving $\mathbf{v}'_1 = (0.01, -0.01, 0)$ and $\mathbf{v}'_2 = (-0.01, 0.01, 0)$. Then

$$\begin{aligned} \mathbf{L}' &= (-1, 0, 0) \times (0.01, -0.01, 0) + (1, 0, 0) \times (-0.01, 0.01, 0) \\ &= (0, 0, 0.01) + (0, 0, 0.01) = (0, 0, 0.02). \end{aligned}$$

The atoms lie on the x axis, about their centre of mass at the origin, so $I_{xx} = \sum(y^2 + z^2) = 0$, $I_{yy} = I_{zz} = 1 \times 1^2 + 1 \times 1^2 = 2$, and every entry off the diagonal is zero.

Then $\mathbf{l}\omega = (0, 2\omega_y, 2\omega_z) = (0, 0, 0.02)$ gives $\omega_y = 0$ and $\omega_z = 0.01 \text{ rad fs}^{-1}$, and, the pair lying on a line, we take $\omega_x = 0$ (Section 5.6). The turning velocity of atom 1 is $(0, 0, 0.01) \times (-1, 0, 0) = (0, -0.01, 0)$, so $\mathbf{u}_1 = (0.01, 0, 0)$, and likewise $\mathbf{u}_2 = (-0.01, 0, 0)$: the atoms approach each other along the line joining them, so what remains is the vibration of the pair, caught as the bond shortens. In $\text{amu } \text{\AA}^2 \text{ fs}^{-2}$, the kinetic energy is $\frac{1}{2}(0.01^2 + 0.01^2 + 0.01^2 + 0.03^2) = 0.0006$; the drift has $\frac{1}{2} \times 2 \times 0.02^2 = 0.0004$, the turning $\frac{1}{2} \times 2 \times 0.01^2 = 0.0001$ and the rest $\frac{1}{2}(0.01^2 + 0.01^2) = 0.0001$, which add up to 0.0006. Multiplying by 103.6427 (Section 3.12), the kinetic energy is 0.0622 eV.

EXERCISE 5.17. Each has 4 atoms and is not linear, so $12 - 6 = 6$ internal degrees of freedom.

EXERCISE 5.18. With $\mathbf{e} = (0, 0, d)/d = (0, 0, 1)$, the blocks $\pm k e_a e_b$ have a single non-zero entry, $\pm k$, in the zz place. Ordering the coordinates $x_1, y_1, z_1, x_2, y_2, z_2$,

$$\Phi = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & k & 0 & 0 & -k \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -k & 0 & 0 & k \end{pmatrix}.$$

A small turning about the x axis moves atom 1, at the origin, by $\theta \mathbf{e}_x \times \mathbf{0} = \mathbf{0}$ and atom 2 by $\theta \mathbf{e}_x \times (0, 0, d) = \theta(0, -d, 0)$. The displacement $(0, 0, 0, 0, -\theta d, 0)$ has zero third and sixth entries, the only ones the Hessian uses, so the Hessian turns it into zero. A turning about the z axis moves neither atom, since $\mathbf{e}_z \times \mathbf{r} = \mathbf{0}$ for both: it is not a motion of the molecule, which is why a linear molecule has only two turnings.

EXERCISE 5.19. With $\mathbf{F}$ the force of the bond on the second atom and $-\mathbf{F}$ that on the first, dividing each second law by its mass and subtracting, as in Section 4.8, the separation $\mathbf{r} = \mathbf{r}_2 - \mathbf{r}_1$ has $\ddot{\mathbf{r}} = \mathbf{F}/m_2 + \mathbf{F}/m_1 = \mathbf{F}/m_r$, so $m_r \ddot{\mathbf{r}} = \mathbf{F}$. Going round a circle of radius r at ω needs the inward pull $m_r \omega^2 r$ (Section 1.7), so $k(r - d) = m_r \omega^2 r$ and

$$r - d = \frac{m_r \omega^2}{k} r = \left(\frac{\omega}{\omega_0} \right)^2 r \approx d \left(\frac{\omega}{\omega_0} \right)^2,$$

replacing r by d on the right, which changes it only by $(r - d)(\omega/\omega_0)^2$, the small stretch times the small ratio. With $\omega = 0.006335 \text{ rad fs}^{-1}$ from Section 5.8, and $\omega_0 = 2\pi/36.64 = 0.17148 \text{ rad fs}^{-1}$, the ratio squared is 1.365×10^{-3} , and the stretch is $1.5639 \times 1.365 \times 10^{-3} = 0.00213 \text{ \AA}$.

EXERCISE 5.20. $\omega = \sqrt{2K/I}$ grows as the square root of the energy, so the period $2\pi/\omega$ shrinks as one over that square root: $992 \times \sqrt{0.025852/0.08617} = 543$ fs. Directly, with K in $\text{amu } \text{\AA}^2 \text{ fs}^{-2}$,

$$K = 0.08617 \times 9.648533 \times 10^{-3} = 8.314 \times 10^{-4},$$

$$\omega = \sqrt{\frac{2 \times 8.314 \times 10^{-4}}{12.432}} = 0.01157 \text{ rad fs}^{-1},$$

and $2\pi/\omega = 543$ fs.

Chapter 6: Lagrangian mechanics

EXERCISE 6.1. (a) One, the angle of the bead round the hoop. (b) Three: two Cartesian coordinates of each atom in the plane, four, less one for the fixed distance; for instance the centre of mass in the plane and the angle of the bond. (c) Six: nine Cartesian coordinates less three fixed distances, which leaves the three coordinates of the centre of mass and three angles for the turning (Section 5.7).

EXERCISE 6.2. With $f = x$ and $g = e^x$, so that $g' = e^x$ and $f' = 1$,

$$\int_0^1 x e^x dx = [x e^x]_0^1 - \int_0^1 e^x dx = (1 \times e - 0 \times 1) - [e^x]_0^1 = e - (e - 1) = 1,$$

using $e^0 = 1$ and the fundamental theorem of calculus (Section 1.4), since e^x is its own derivative (Section 2.7).

EXERCISE 6.3. With $\mathcal{L} = \frac{1}{2}m\dot{x}^2 - \frac{1}{2}kx^2$ and the path $x + \zeta\eta$, expanding the squares as in Section 6.2,

$$\mathcal{S}(\zeta) = \mathcal{S}(0) + \zeta \int_0^\tau (m\dot{x}\dot{\eta} - kx\eta) dt + \zeta^2 \left(\frac{1}{2}m \int_0^\tau \dot{\eta}^2 dt - \frac{1}{2}k \int_0^\tau \eta^2 dt \right).$$

The term in ζ is zero on the true path, by Section 6.3. With $\eta = \sin(n\pi t/\tau)$ and, by the chain rule, $\dot{\eta} = (n\pi/\tau) \cos(n\pi t/\tau)$, the squares are $\frac{1}{2} - \frac{1}{2} \cos(2n\pi t/\tau)$ and $\frac{1}{2} + \frac{1}{2} \cos(2n\pi t/\tau)$ (Section 4.2, with $\sin^2 = 1 - \cos^2$), and $\cos(2n\pi t/\tau)$ makes n whole turns from 0 to τ , so it integrates to zero. The two integrals are therefore $\tau/2$ and $(n\pi/\tau)^2 \tau/2$, and the term in ζ^2 is $\zeta^2 (\frac{1}{2}m(n\pi/\tau)^2 - \frac{1}{2}k) \tau/2 = \zeta^2 (\tau/4) (m(n\pi/\tau)^2 - k)$. With $\tau = 0.5$, $n\pi/\tau = 2n\pi$: for $n = 1$, $(2\pi)^2 = 39.478$ and $0.125 \times (0.5 \times 39.478 - 50) = -3.783$; for $n = 2$, $(4\pi)^2 = 157.91$ and $0.125 \times (0.5 \times 157.91 - 50) = +3.620$, in J s m^{-2} .

EXERCISE 6.4. $\partial\mathcal{L}/\partial\dot{x} = m\dot{x}$, whose rate of change is $m\ddot{x}$, and $\partial\mathcal{L}/\partial x = -mg$, so $m\ddot{x} = -mg$ and $\ddot{x} = -g$.

EXERCISE 6.5. $\partial\mathcal{L}/\partial\dot{x} = m\dot{x}$ and $\partial\mathcal{L}/\partial x = -\partial U/\partial x$, so $m\ddot{x} = -\partial U/\partial x$, the x component of $-\nabla U$; the y equation is the same with y in place of x .

EXERCISE 6.6. By the Taylor series of h' about x_0 (Section 1.8), with $h'(x_0) = 0$, $h'(x) = h''(x_0)(x - x_0) + \mathcal{O}((x - x_0)^2)$. In a small swing $x - x_0$ and $\dot{x}$ are both small, so in the numerator of Equation (6.4) $h''\dot{x}^2$ is small beside g , and in the denominator h'^2 is small beside 1. Keeping the largest term, $\ddot{x} \approx -g h''(x_0)(x - x_0)$, the equation of a spring with $\omega_0^2 = g h''(x_0)$ (Section 2.8), about x_0 .

EXERCISE 6.7. With $x = R \sin \theta$, for $|\theta| < 90^\circ$, so that $\cos \theta > 0$, the chain rule gives $\dot{x} = R \cos \theta \dot{\theta}$, and

$$h = R - \sqrt{R^2(1 - \sin^2 \theta)} = R - \sqrt{R^2 \cos^2 \theta} = R(1 - \cos \theta).$$

The slope follows from the chain rule with x a function of θ , $dh/d\theta = h' dx/d\theta$, that is $R \sin \theta = h' R \cos \theta$, so $h' = \tan \theta$ and $1 + h'^2 = 1/\cos^2 \theta$ (Section 6.4). Then

$$\mathcal{L} = \frac{1}{2}m \frac{1}{\cos^2 \theta} R^2 \cos^2 \theta \dot{\theta}^2 - mgR(1 - \cos \theta) = \frac{1}{2}mR^2 \dot{\theta}^2 - mgR(1 - \cos \theta),$$

the Lagrangian of Section 6.1 for a pendulum of length R .

EXERCISE 6.8. Multiplying $\ddot{\theta} = -(g/L) \sin \theta$ by $\dot{\theta}$, the left side is $d(\frac{1}{2}\dot{\theta}^2)/dt$ and the right side $-(g/L) \sin \theta \dot{\theta} = d((g/L) \cos \theta)/dt$, both by the chain rule. So $\frac{1}{2}\dot{\theta}^2 - (g/L) \cos \theta$ does not change; multiplying by mL^2 and adding the constant mgL , neither does $\frac{1}{2}mL^2\dot{\theta}^2 + mgL(1 - \cos \theta)$.

EXERCISE 6.9. Along the motion $\mathcal{L}$ changes, by the chain rule along a path (Section 3.10), at the rate $(\partial\mathcal{L}/\partial q)\dot{q} + (\partial\mathcal{L}/\partial\dot{q})\ddot{q}$, and $\dot{q} \partial\mathcal{L}/\partial\dot{q}$ by the product rule at the rate $\ddot{q} \partial\mathcal{L}/\partial\dot{q} + \dot{q} d(\partial\mathcal{L}/\partial\dot{q})/dt$. Subtracting, the terms in $\ddot{q}$ cancel and

$$\frac{d}{dt} \left(\dot{q} \frac{\partial\mathcal{L}}{\partial\dot{q}} - \mathcal{L} \right) = \dot{q} \left(\frac{d}{dt} \frac{\partial\mathcal{L}}{\partial\dot{q}} - \frac{\partial\mathcal{L}}{\partial q} \right) = 0$$

by Equation (6.3). For the bead, $\dot{x} \partial\mathcal{L}/\partial\dot{x} = m(1 + h'^2)\dot{x}^2 = 2K$, so the quantity is $2K - (K - U) = K + U$, the energy.

EXERCISE 6.10. With $M = m_1 + m_2$, putting $x_2 = x_1 + r$ into $MX = m_1x_1 + m_2x_2$ gives $MX = Mx_1 + m_2r$, so $x_1 = X - (m_2/M)r$, and then $x_2 = x_1 + r = X + (m_1/M)r$, since

$1 - m_2/M = m_1/M$. Then $\dot{x}_1 = \dot{X} - (m_2/M)\dot{r}$ and $\dot{x}_2 = \dot{X} + (m_1/M)\dot{r}$, and expanding the squares,

$$\begin{aligned} K &= \frac{1}{2}m_1\dot{x}_1^2 + \frac{1}{2}m_2\dot{x}_2^2 = \frac{1}{2}M\dot{X}^2 + \dot{X}\dot{r}\left(-\frac{m_1m_2}{M} + \frac{m_2m_1}{M}\right) + \frac{1}{2}\frac{m_1m_2^2 + m_2m_1^2}{M^2}\dot{r}^2 \\ &= \frac{1}{2}M\dot{X}^2 + \frac{1}{2}m_r\dot{r}^2, \end{aligned}$$

since $m_1m_2(m_2 + m_1)/M^2 = m_1m_2/M = m_r$. So $\mathcal{L} = \frac{1}{2}M\dot{X}^2 + \frac{1}{2}m_r\dot{r}^2 - \frac{1}{2}k(r - \ell)^2$. It does not contain X , so $p_X = M\dot{X} = m_1\dot{x}_1 + m_2\dot{x}_2$, the total momentum, is conserved; and the r equation is $m_r\ddot{r} = -k(r - \ell)$, as in [Section 4.8](#).

EXERCISE 6.11. $L = 0.2 \times 0.4 \times 0.3 = 0.024 \text{ kg m}^2 \text{ s}^{-1}$ and $E = \frac{1}{2} \times 0.2 \times 0.3^2 + \frac{1}{2} \times 2 \times 0.4^2 = 0.009 + 0.16 = 0.169 \text{ J}$. The turning points solve $\frac{1}{2}kr^2 + L^2/(2mr^2) = E$; with $\frac{1}{2}k = 1$, multiplying by r^2 gives $r^4 - 0.169r^2 + 0.00144 = 0$, since $L^2/(2m) = 0.000576/0.4 = 0.00144$. So $r^2 = \frac{1}{2}(0.169 \pm \sqrt{0.028561 - 0.00576}) = \frac{1}{2}(0.169 \pm 0.151) = 0.16$ or 0.009 , and the puck swings between 0.0949 m and 0.4 m .

EXERCISE 6.12. A circle has r constant, so $\ddot{r} = 0$ and the r equation of [Section 6.5](#) gives $L^2/(mr^3) = U'(r) = kr$, that is $r^4 = L^2/(mk)$, the bottom of U_{eff} . Then $r^2 = L/\sqrt{mk}$, and $\dot{\theta} = L/(mr^2) = L\sqrt{mk}/(mL) = \sqrt{mk}/\sqrt{m^2} = \sqrt{k/m}$. For $L = 0.05$, $r^4 = 0.0025/(0.2 \times 2) = 0.00625$ and $r = 0.2812 \text{ m}$.

EXERCISE 6.13. With $f = xy$ and $\chi = x^2 + y^2 - 1$, $\nabla f = \lambda \nabla \chi$ reads $y = 2\lambda x$ and $x = 2\lambda y$. Putting the first into the second, $x = 4\lambda^2 x$, so $\lambda = \pm \frac{1}{2}$ (with $x = 0$ the first gives $y = 0$, off the circle). With $\lambda = \frac{1}{2}$, $y = x$, and $x^2 + y^2 = 1$ gives $x = y = \pm 1/\sqrt{2}$, where $xy = \frac{1}{2}$, the largest value; $\lambda = -\frac{1}{2}$ gives $y = -x$ and the smallest, $-\frac{1}{2}$.

EXERCISE 6.14. The bob is released at $\theta_0 = 60^\circ$, where $\cos \theta_0 = \frac{1}{2}$, so the force of the rod, $F_{\text{rod}} = mg(3 \cos \theta - 2 \cos \theta_0)$, is $mg(3 \cos \theta - 1)$, whatever the length of the rod. At the bottom it is $2mg = 2 \times 0.5 \times 9.80665 = 9.807 \text{ N}$; at the release point, $\cos 60^\circ = \frac{1}{2}$, it is $\frac{1}{2}mg = 2.452 \text{ N}$, the part of the weight along the rod.

EXERCISE 6.15. During the swing $|\theta| \leq \theta_0$, so $\cos \theta \geq \cos \theta_0$, and the force $mg(3 \cos \theta - 2 \cos \theta_0)$ is smallest at the turning points, where it is $mg \cos \theta_0$. A string can only pull, so it stays taut throughout only if $\cos \theta_0 \geq 0$, that is $\theta_0 \leq 90^\circ$.

EXERCISE 6.16. By Pythagoras, $\chi = \frac{1}{2} \sum_a ((\mathbf{r}_j)_a - (\mathbf{r}_i)_a)^2 - \frac{1}{2}d^2$, summing over the components $a = x, y, z$. Only the term a of the sum contains $(\mathbf{r}_j)_a$, and d is a constant, so differentiating with respect to $(\mathbf{r}_j)_a$ by the chain rule gives $\frac{1}{2} \times 2((\mathbf{r}_j)_a - (\mathbf{r}_i)_a) = (\mathbf{r}_j)_a - (\mathbf{r}_i)_a$; with respect to $(\mathbf{r}_i)_a$ the inner derivative is -1 , which gives its negative. So $\nabla_j \chi = \mathbf{r}_j - \mathbf{r}_i$ and $\nabla_i \chi = -(\mathbf{r}_j - \mathbf{r}_i)$, and the forces $\lambda \nabla_i \chi = -\lambda(\mathbf{r}_j - \mathbf{r}_i)$ and $\lambda \nabla_j \chi = \lambda(\mathbf{r}_j - \mathbf{r}_i)$ are equal, opposite and along the line of the bond.

EXERCISE 6.17. The bead starts at $x = -2.042$ m, where $h = 0.8$ m, swings through the deep valley, turns at $x = -0.4765$ m, the other point at height 0.8 m (Section 3.6), since the top of the hill, 1.009 m high, is out of its reach, and is back at its start after 2.200 s.

Chapter 7: Hamiltonian mechanics and phase space

EXERCISE 7.1. The slope is $f'(v) = e^v$, which grows with v and is positive, so each $p > 0$ is reached at the one v with $e^v = p$, that is $v = \ln p$ (Section 2.7). Then

$$f^*(p) = pv - f(v) = p \ln p - e^{\ln p} = p \ln p - p.$$

The slope of $\ln p$ is $1/p$, by the derivative of an inverse function (Section 4.5): $e^v = p$ has $dp/dv = e^v = p$, so $dv/dp = 1/p$. By the product rule, the slope of f^* is then $\ln p + p \times (1/p) - 1 = \ln p = v$.

EXERCISE 7.2. U does not contain the rates, so $p_1 = \partial K / \partial \dot{q}_1 = a\dot{q}_1 + b\dot{q}_2$ and $p_2 = \partial K / \partial \dot{q}_2 = b\dot{q}_1 + c\dot{q}_2$, the factor $\frac{1}{2}$ cancelling the 2 that differentiating each square brings and the 2 of the middle term. Then

$$p_1\dot{q}_1 + p_2\dot{q}_2 = a\dot{q}_1^2 + b\dot{q}_1\dot{q}_2 + b\dot{q}_1\dot{q}_2 + c\dot{q}_2^2 = 2K,$$

and $H = 2K - (K - U) = K + U$.

EXERCISE 7.3. $K = \frac{1}{2}m\dot{q}^2$ gives $p = m\dot{q}$ and $H = p^2/(2m) + \frac{1}{2}kq^2 - mgq$. Hamilton's equations are $\dot{q} = p/m$ and $\dot{p} = -kq + mg$. The state does not change when both rates are zero: $p = 0$ and $q = mg/k$, where the spring holds up the weight.

EXERCISE 7.4. $p_x = \partial \mathcal{L} / \partial \dot{x} = m\dot{x}$ and $p_y = m\dot{y}$, so $H = (p_x^2 + p_y^2)/(2m) + mgy$. Hamilton's equations are $\dot{x} = p_x/m$, $\dot{y} = p_y/m$, $\dot{p}_x = -\partial H / \partial x = 0$ and $\dot{p}_y = -\partial H / \partial y = -mg$. H does not contain x , so p_x is conserved: nothing pushes the ball sideways.

EXERCISE 7.5. $p_r = \partial \mathcal{L} / \partial \dot{r} = m\dot{r}$ and $p_\theta = \partial \mathcal{L} / \partial \dot{\theta} = mr^2\dot{\theta}$. Then

$$\begin{aligned} H &= p_r\dot{r} + p_\theta\dot{\theta} - \mathcal{L} = m\dot{r}^2 + mr^2\dot{\theta}^2 - \frac{1}{2}m\dot{r}^2 - \frac{1}{2}mr^2\dot{\theta}^2 + U \\ &= \frac{1}{2}m\dot{r}^2 + \frac{1}{2}mr^2\dot{\theta}^2 + U = \frac{p_r^2}{2m} + \frac{p_\theta^2}{2mr^2} + U(r), \end{aligned}$$

putting $\dot{r} = p_r/m$ and $\dot{\theta} = p_\theta/(mr^2)$ in the last step. H does not contain θ , so $\dot{p}_\theta = -\partial H/\partial \theta = 0$. For r , $\dot{r} = p_r/m$ and

$$\dot{p}_r = -\frac{\partial H}{\partial r} = \frac{p_\theta^2}{mr^3} - U'(r) = -\frac{d}{dr}\left(U + \frac{L^2}{2mr^2}\right),$$

with $L = p_\theta$, the derivative of $p_\theta^2/(2mr^2)$ being $-2 \times p_\theta^2/(2mr^3)$ by the power rule. With $\dot{p}_r = m\ddot{r}$, this is the motion in the effective potential.

EXERCISE 7.6. Write $A = 1 + h'(x)^2$, so that $H = p^2/(2mA) + mgh$ and $dA/dx = 2h'h''$. Hamilton's equations give $\dot{x} = p/(mA)$ and

$$\begin{aligned}\dot{p} &= -\frac{\partial H}{\partial x} = -\frac{p^2}{2m}\left(-\frac{1}{A^2}\frac{dA}{dx}\right) - mgh' \\ &= \frac{p^2}{2mA^2}2h'h'' - mgh' = m\dot{x}^2h'h'' - mgh',\end{aligned}$$

the derivative of A^{-1} being $-A^{-2}dA/dx$ by the power rule and the chain rule, and $p = mA\dot{x}$ being put in at the last step. Differentiating $p = mA\dot{x}$ with respect to time instead, by the product rule, with $dA/dt = 2h'h''\dot{x}$,

$$\dot{p} = mA\ddot{x} + 2mh'h''\dot{x}^2.$$

Setting the two equal and taking $2mh'h''\dot{x}^2$ to the right, $mA\ddot{x} = -mh'h''\dot{x}^2 - mgh'$, and dividing by mA , $\ddot{x} = -h'(g + h''\dot{x}^2)/(1 + h'^2)$, [Equation \(6.4\)](#).

EXERCISE 7.7. $p = 2mL^2\omega_0 = 2 \times 0.72 \times 2.8587 = 4.117 \text{ kg m}^2 \text{ s}^{-1}$. The bob's speed is $L|\dot{\theta}| = p/(mL) = 4.117/0.6 = 6.861 \text{ m s}^{-1}$. In symbols, $p/(mL) = 2L\omega_0 = 2L\sqrt{g/L} = 2\sqrt{gL}$, and $\frac{1}{2}mv^2 = mg \times 2L$ gives the same $v = \sqrt{4gL}$, the speed that the fall of $2L$ from the top gives by energy.

EXERCISE 7.8. By $\dot{\phi} = -\omega_0\phi$, the gap is $\phi = \phi_0 e^{-\omega_0 t}$ ([Section 2.7](#)), so it falls from 10° to 1° , by the factor 10, when $e^{\omega_0 t} = 10$, that is after $t = \ln 10/\omega_0 = 2.3026/2.8587 = 0.805 \text{ s}$. Only the ratio of the angles enters, so they need not be turned into radians.

EXERCISE 7.9. By the product rule, $\partial(BC)/\partial p = (\partial B/\partial p)C + B\partial C/\partial p$, and the same for q . Putting these into [Equation \(7.3\)](#),

$$\begin{aligned}\{A, BC\} &= \frac{\partial A}{\partial q}\left(\frac{\partial B}{\partial p}C + B\frac{\partial C}{\partial p}\right) - \frac{\partial A}{\partial p}\left(\frac{\partial B}{\partial q}C + B\frac{\partial C}{\partial q}\right) \\ &= \left(\frac{\partial A}{\partial q}\frac{\partial B}{\partial p} - \frac{\partial A}{\partial p}\frac{\partial B}{\partial q}\right)C + B\left(\frac{\partial A}{\partial q}\frac{\partial C}{\partial p} - \frac{\partial A}{\partial p}\frac{\partial C}{\partial q}\right) = \{A, B\}C + B\{A, C\}.\end{aligned}$$

EXERCISE 7.10. Differentiating $r^2 = x^2 + y^2$ with respect to x gives $2r \partial r / \partial x = 2x$, so $\partial r / \partial x = x/r$ (Section 5.3), and by the chain rule $\partial U / \partial x = U'(r) x/r$; likewise $\partial U / \partial y = U'(r) y/r$. With one pair of terms for each of (x, p_x) and (y, p_y) ,

$$\begin{aligned}\{L_z, H\} &= \frac{\partial L_z}{\partial x} \frac{\partial H}{\partial p_x} - \frac{\partial L_z}{\partial p_x} \frac{\partial H}{\partial x} + \frac{\partial L_z}{\partial y} \frac{\partial H}{\partial p_y} - \frac{\partial L_z}{\partial p_y} \frac{\partial H}{\partial y} \\ &= p_y \frac{p_x}{m} - (-y) \frac{U'x}{r} + (-p_x) \frac{p_y}{m} - x \frac{U'y}{r} = 0,\end{aligned}$$

the first and third terms cancelling, and the second and fourth.

EXERCISE 7.11. $\hat{L}p = \dot{p} = -kq$, then $\hat{L}^2 p = -k \hat{L}q = -kp/m = -\omega^2 p$, so each pair of steps again brings $-\omega^2$, and $\hat{L}^3 p = -\omega^2(-kq)$. Putting these into Equation (7.4) at the start,

$$p(t) = p_0 - kq_0 t - \omega^2 p_0 \frac{t^2}{2!} + \omega^2 kq_0 \frac{t^3}{3!} + \cdots,$$

and gathering the terms in p_0 and in q_0 , with $kq_0 t = (kq_0/\omega) \omega t$,

$$p(t) = p_0 \left(1 - \frac{(\omega t)^2}{2!} + \cdots \right) - \frac{kq_0}{\omega} \left(\omega t - \frac{(\omega t)^3}{3!} + \cdots \right) = p_0 \cos \omega t - m\omega q_0 \sin \omega t,$$

since $k/\omega = m\omega^2/\omega = m\omega$. This is m times the rate of $q(t)$ found in Section 7.3.

EXERCISE 7.12. The flow is $q = q_0 + p_0 t/m = q_0 + 2p_0$ (in metres, with p_0 in kg m s^{-1}) and $p = p_0$. The corners $(0,0)$, $(0.1,0)$, $(0.1,0.1)$ and $(0,0.1)$ go to $(0,0)$, $(0.1,0)$, $(0.3,0.1)$ and $(0.2,0.1)$: a parallelogram with a base of 0.1 m and a height of 0.1 kg m s^{-1} , so its area is 0.01 Js, that of the square.

EXERCISE 7.13. $e^{-\gamma t} = \frac{1}{2}$ gives $t = \ln 2 / \gamma = 0.6931 / 0.5 = 1.386$ s.

EXERCISE 7.14. Moving both carts by α , $x_1 \rightarrow x_1 + \alpha$ and $x_2 \rightarrow x_2 + \alpha$, leaves the rates and $x_2 - x_1$ unchanged, and so $\mathcal{L}$. In Noether's theorem this is the change with $s_1 = s_2 = 1$, and $p_1 = \partial \mathcal{L} / \partial \dot{x}_1 = m_1 \dot{x}_1$, $p_2 = m_2 \dot{x}_2$, so $p_1 s_1 + p_2 s_2 = m_1 \dot{x}_1 + m_2 \dot{x}_2$ is conserved.

EXERCISE 7.15. U_{int} is unchanged by every move and turn. Moving the group along x or y leaves every z_i , and so U , unchanged, so P_x and P_y are conserved; moving it along z by c changes U by $\sum_i m_i g c = M g c$, with M the total mass, so P_z is not. Turning about a vertical axis leaves every height unchanged, so the component of $\mathbf{L}$ along a vertical axis, L_z , is conserved. Turning about a horizontal axis raises some atoms and lowers others, so it changes U for most arrangements of the group: it is not a symmetry, and the horizontal components of $\mathbf{L}$ are not conserved in general. The energy is conserved too, since U does not contain the time.

EXERCISE 7.16. About the point $\mathbf{a}$ the positions are measured from $\mathbf{a}$, so the torque is

$$\sum_i (\mathbf{r}_i - \mathbf{a}) \times \mathbf{F}_i = \sum_i \mathbf{r}_i \times \mathbf{F}_i - \mathbf{a} \times \sum_i \mathbf{F}_i = \mathbf{G},$$

the second sum being zero. About the centre of the box, $(x_1 - 5) - (x_2 - 5) = x_1 - x_2$ and likewise for y , so the calculation of Section 7.5 is unchanged and $G_z = -1.056 \text{ eV}$.

EXERCISE 7.17. The argument needs $H(q, -p) = H(q, p)$. H_1 has it, since $(-p)^2 = p^2$. $H_2(q, -p) = (p + c)^2 / (2m)$, which differs from $(p - c)^2 / (2m)$ by $2pc/m$, and $H_3(q, -p)$ differs from $H_3(q, p)$ by $-2bqp$, so neither has it: only H_1 .

EXERCISE 7.18. $\omega \delta t = 10 \times 0.005 = 0.05$, so each step multiplies the energy by $1 + 0.05^2 = 1.0025$, and n steps by $1.0025^n = e^{n \ln 1.0025}$, since $1.0025 = e^{\ln 1.0025}$ (Section 2.7). This is $2 = e^{\ln 2}$ when $n = \ln 2 / \ln 1.0025 = 277.6$. After 278 steps, 1.39 s, the energy has doubled, a little over two periods of 0.628 s.

EXERCISE 7.19. $q' = q + \delta t p/m$ and $p' = p - \delta t k(q + \delta t p/m) = -k \delta t q + (1 - \omega^2 \delta t^2) p$. So

$$\mathbf{J} = \begin{pmatrix} 1 & \delta t/m \\ -k \delta t & 1 - \omega^2 \delta t^2 \end{pmatrix}, \quad \det \mathbf{J} = 1 - \omega^2 \delta t^2 + \frac{\delta t}{m} k \delta t = 1.$$

It is a drift followed by a kick, each a shear.

EXERCISE 7.20. Write everything in terms of q and the new momentum p' , using $p = p' + k \delta t q$ and $q' = q + \delta t p'/m$. Before the step,

$$\begin{aligned} \tilde{H}(q, p) &= \frac{(p' + k \delta t q)^2}{2m} + \frac{1}{2} k q^2 - \frac{k \delta t}{2m} q (p' + k \delta t q) \\ &= \frac{p'^2}{2m} + \frac{k \delta t}{m} q p' + \frac{k^2 \delta t^2 q^2}{2m} + \frac{1}{2} k q^2 - \frac{k \delta t}{2m} q p' - \frac{k^2 \delta t^2 q^2}{2m} \quad \text{expanding} \\ &= \frac{p'^2}{2m} + \frac{1}{2} k q^2 + \frac{k \delta t}{2m} q p'. \end{aligned}$$

After it,

$$\begin{aligned} \tilde{H}(q', p') &= \frac{p'^2}{2m} + \frac{1}{2} k \left(q + \frac{\delta t p'}{m} \right)^2 - \frac{k \delta t}{2m} \left(q + \frac{\delta t p'}{m} \right) p' \\ &= \frac{p'^2}{2m} + \frac{1}{2} k q^2 + \frac{k \delta t}{m} q p' + \frac{k \delta t^2 p'^2}{2m^2} - \frac{k \delta t}{2m} q p' - \frac{k \delta t^2 p'^2}{2m^2} \quad \text{expanding} \\ &= \frac{p'^2}{2m} + \frac{1}{2} k q^2 + \frac{k \delta t}{2m} q p', \end{aligned}$$

the same. $\tilde{H}$ differs from H only by the small term $(k\delta t/(2m))qp$, which is why the energy stays in a band.

Chapter 8: Potential energy surfaces and interatomic models

EXERCISE 8.1. With the same stiffness, $\omega_0 = \sqrt{k/m}$ is inversely proportional to $\sqrt{m}$, so the ratio is $\sqrt{6.94/5.486 \times 10^{-4}} = \sqrt{12650} = 112.5$.

EXERCISE 8.2. $T = \hbar\omega/k_B$, with $k_B = 8.617 \times 10^{-5} \text{ eV K}^{-1}$. For the hollow $\hbar\omega = 0.6582 \times 2\pi/170.3 = 0.02428 \text{ eV}$, so $T = 0.02428/8.617 \times 10^{-5} = 282 \text{ K}$, just below room temperature; for the O–H stretch $T = 0.4534/8.617 \times 10^{-5} = 5262 \text{ K}$.

EXERCISE 8.3. The pull is $\varphi'(r) = 4\varepsilon(-12\sigma^{12}/r^{13} + 6\sigma^6/r^7)$, by the power rule for negative powers (Section 4.5). It is greatest where its own slope, $\varphi''(r) = 4\varepsilon(156\sigma^{12}/r^{14} - 42\sigma^6/r^8)$, is zero, that is where $156\sigma^{12}/r^{14} = 42\sigma^6/r^8$; multiplying by $r^{14}/(42\sigma^6)$ gives $r^6 = (156/42)\sigma^6 = (26/7)\sigma^6$, so $r = (26/7)^{1/6}\sigma = 1.2445\sigma$. There $(\sigma/r)^6 = 7/26$ and $(\sigma/r)^{12} = 49/676$, so

$$\varphi' = \frac{4\varepsilon}{r} \left(-12 \times \frac{49}{676} + 6 \times \frac{7}{26} \right) = \frac{4\varepsilon}{1.2445\sigma} (-0.8698 + 1.6154) = 2.396 \frac{\varepsilon}{\sigma},$$

a pull of $2.396\varepsilon/\sigma$ towards the other atom.

EXERCISE 8.4. With $y = r - r_0$ and the chain rule, $\varphi' = D(-2a e^{-2ay} + 2a e^{-ay}) = 2aD e^{-ay}(1 - e^{-ay})$, zero only where $1 - e^{-ay} = 0$, since e^{-ay} is never zero, that is at $y = 0$, where $\varphi = D(1 - 2) = -D$. Differentiating again, $\varphi'' = D(4a^2 e^{-2ay} - 2a^2 e^{-ay})$, which at $y = 0$ is $2Da^2 > 0$: a minimum, with that stiffness.

EXERCISE 8.5. $\varphi' = -(A/\rho) e^{-r/\rho} + 6C/r^7$. At $r = r_m$, $r/\rho = \alpha$, so $A e^{-\alpha} = 6\varepsilon/(\alpha - 6)$, and

$$\varphi'(r_m) = -\frac{\alpha}{r_m} \frac{6\varepsilon}{\alpha - 6} + \frac{6}{r_m^7} \frac{\alpha \varepsilon r_m^6}{\alpha - 6} = 0, \quad \varphi(r_m) = \frac{6\varepsilon}{\alpha - 6} - \frac{\alpha \varepsilon}{\alpha - 6} = -\varepsilon.$$

Differentiating again, $\varphi'' = (A/\rho^2) e^{-r/\rho} - 42C/r^8$, which at r_m is

$$\frac{\alpha^2}{r_m^2} \frac{6\varepsilon}{\alpha - 6} - \frac{42\alpha\varepsilon}{(\alpha - 6)r_m^2} = \frac{6\alpha(\alpha - 7)\varepsilon}{(\alpha - 6)r_m^2} > 0 \quad \text{for } \alpha > 7:$$

a minimum.

EXERCISE 8.6. With $k = 72\epsilon/r_m^2$ (Section 8.3) and the reduced mass $m/2$, $\omega_0 = \sqrt{2k/m} = \sqrt{144\epsilon/(mr_m^2)} = (12/r_m)\sqrt{\epsilon/m}$, and Equation (8.2) gives $\sqrt{\epsilon/m} = \sigma/\tau$, so $\omega_0 = 12\sigma/(r_m\tau) = 12/(2^{1/6}\tau) = 10.69/\tau$. The period is $2\pi/10.69 = 0.5877\tau$, which for argon is $0.5877 \times 2151 = 1264$ fs, and it spans $0.5877/0.005 = 117.5$, about 118 steps.

EXERCISE 8.7. In $\sum_i \mathbf{F}_i = \sum_i \sum_{j \neq i} \phi'(r_{ij}) \mathbf{r}_{ij}/r_{ij}$ each pair appears twice, as (i, j) and as (j, i) . The distances are equal and $\mathbf{r}_{ji} = -\mathbf{r}_{ij}$, so the two terms cancel, and the sum is zero.

EXERCISE 8.8. $\phi'(1.5\sigma) = 4\epsilon(-12/1.5^{13} + 6/1.5^7)/\sigma = 4\epsilon(-0.0617 + 0.3512)/\sigma = 1.158\epsilon/\sigma$, so $\mathcal{W} = -1.5 \times 1.158 = -1.737\epsilon$. $\mathcal{W} = -r\phi'(r)$ is positive where $\phi' < 0$, for $r < r_m$, where the pair repels.

EXERCISE 8.9. $h = (3\epsilon_m|U|/|U'''|)^{1/3} = (3 \times 2.2 \times 10^{-16} \times 10.75/10^4)^{1/3} = 8.9 \times 10^{-7} \sigma$, close to the $10^{-6}\sigma$ measured.

EXERCISE 8.10. By the Taylor series, $U(x+h) = U + hU' + \frac{1}{2}h^2U'' + \mathcal{O}(h^3)$, so $[U(x+h) - U(x)]/h = U' + \frac{1}{2}hU'' + \mathcal{O}(h^2)$, an error of $\frac{1}{2}h|U''|$, to which rounding adds $\epsilon_m|U|/h$ as for central differences. The slope of the total with respect to h , $\frac{1}{2}|U''| - \epsilon_m|U|/h^2$, is zero at $h = \sqrt{2\epsilon_m|U|/|U''|}$. If U changes by about its own size over a length L , so that $|U''| \approx |U|/L^2$ and the forces are of size $F \approx |U|/L$, this step is $\sqrt{2\epsilon_m}L$, of order $\sqrt{\epsilon_m} = 1.5 \times 10^{-8}$ times L , and there each error is $\frac{1}{2}h|U''| = \sqrt{\epsilon_m/2}F$, about $10^{-8}F$. For central differences, $|U'''| \approx |U|/L^3$ and $h \approx \epsilon_m^{1/3}L$ in Equation (8.5) give an error of about $\epsilon_m^{2/3}F \approx 4 \times 10^{-11}F$.

EXERCISE 8.11. $S(0) = 1$ and $S(1) = 1 - 10 + 15 - 6 = 0$. $S'(x) = -30x^2 + 60x^3 - 30x^4 = -30x^2(1 - 2x + x^2) = -30x^2(1 - x)^2$, zero at both ends, and $S''(x) = -60x + 180x^2 - 120x^3 = -60x(1 - x)(1 - 2x)$, zero at both ends too, as expanding $(1 - x)(1 - 2x) = 1 - 3x + 2x^2$ confirms.

EXERCISE 8.12. $u(1) = 1 - 28 + 48 - 21 = 0$. Differentiating term by term, $u'(x) = -168x^5 + 336x^6 - 168x^7$ and $u''(x) = -840x^4 + 2016x^5 - 1176x^6$, so $u'(1) = -168 + 336 - 168 = 0$ and $u''(1) = -840 + 2016 - 1176 = 0$.

EXERCISE 8.13. The atoms between r and $r + dr$ number $4\pi r^2 n dr$, and each pair is shared by two atoms, so the energy left out per atom is

$$\begin{aligned} \frac{1}{2} \int_{r_c}^{\infty} 4\pi r^2 n \varphi_{\text{LJ}}(r) dr &= 8\pi n \epsilon \int_{r_c}^{\infty} \left(\frac{\sigma^{12}}{r^{10}} - \frac{\sigma^6}{r^4} \right) dr \\ &= 8\pi n \epsilon \left(\frac{\sigma^{12}}{9r_c^9} - \frac{\sigma^6}{3r_c^3} \right) \\ &= 8\pi n \epsilon \sigma^3 \left[\frac{1}{9} \left(\frac{\sigma}{r_c} \right)^9 - \frac{1}{3} \left(\frac{\sigma}{r_c} \right)^3 \right] \quad \text{taking out } \sigma^3 \\ &= \frac{8}{3} \pi n \epsilon \sigma^3 \left[\frac{1}{3} \left(\frac{\sigma}{r_c} \right)^9 - \left(\frac{\sigma}{r_c} \right)^3 \right], \end{aligned}$$

since $r^{-9}/(-9)$ has the derivative r^{-10} and falls to zero far out, so $\int_{r_c}^{\infty} r^{-10} dr = r_c^{-9}/9$, and likewise $\int_{r_c}^{\infty} r^{-4} dr = r_c^{-3}/3$. With $n = 0.8\sigma^{-3}$ and $\sigma/r_c = 0.4$, $\frac{8}{3}\pi \times 0.8 \times (\frac{1}{3} \times 0.000262 - 0.064) = -0.428\epsilon$.

EXERCISE 8.14. The motion is that of the shifted potential, whose total energy is conserved, and the truncated total differs from it by $\varphi(r_c) = -0.0163\epsilon$ for each pair inside the cutoff. A rise of 0.49ϵ is therefore $0.49/0.0163 = 30$ pairs fewer inside than at the start.

EXERCISE 8.15. By the chain rule, $\nabla_k(-\sqrt{\rho_i}) = -\frac{1}{2}\rho_i^{-1/2}\nabla_k\rho_i$, and the force is minus this. ρ_i depends on $\mathbf{r}_k$ only through its term $j = k$, so, with $\nabla_k r_{ik} = (\mathbf{r}_k - \mathbf{r}_i)/r_{ik}$ (Section 5.3, with k in the place of i and i in the place of j),

$$\nabla_k \rho_i = -\frac{2\zeta}{r_0} \zeta^2 e^{-2\zeta(r_{ik}/r_0-1)} \frac{\mathbf{r}_k - \mathbf{r}_i}{r_{ik}},$$

which points from k towards i : the term pulls k towards i with a strength divided by $\sqrt{\rho_i}$, weaker when i has many neighbours.

EXERCISE 8.16. With a pair potential and bonds of energy $\varphi < 0$, each atom has z bonds, each shared with one other atom, so the energy holding one atom is $\frac{1}{2}z|\varphi|$. Removing an atom breaks z bonds, costing $z|\varphi|$; putting it back gains $\frac{1}{2}z|\varphi|$; the vacancy costs $\frac{1}{2}z|\varphi|$, the same. In the second-moment model each atom has the bond energy $-\zeta\sqrt{z}$, so the energy holding one atom is $\zeta\sqrt{z}$. Removing an atom removes its own term, costing $\zeta\sqrt{z}$, and each of its z neighbours goes from $-\zeta\sqrt{z}$ to $-\zeta\sqrt{z-1}$, costing $z\zeta(\sqrt{z} - \sqrt{z-1})$; putting the atom back gains $\zeta\sqrt{z}$. The vacancy costs $z\zeta(\sqrt{z} - \sqrt{z-1})$, which divided by $\zeta\sqrt{z}$ is $z(\sqrt{z} - \sqrt{z-1})/\sqrt{z} = z(1 - \sqrt{(z-1)/z}) = z(1 - \sqrt{1 - 1/z})$: for $z = 12$, $12(1 - \sqrt{11/12}) = 0.511$.

EXERCISE 8.17. $\mathbf{b}_1 = (-1, 0, 0)$, $\mathbf{b}_2 = (0, 0, 1)$, $\mathbf{b}_3 = (0, 1, 0)$. Then $\mathbf{n}_1 = \mathbf{b}_1 \times \mathbf{b}_2 = (0, 1, 0)$ and $\mathbf{n}_2 = \mathbf{b}_2 \times \mathbf{b}_3 = (-1, 0, 0)$, whose dot product is 0, so $\psi = \pm 90^\circ$; $\mathbf{n}_1 \times \mathbf{n}_2 =$

$(0, 0, 1)$ points along $\mathbf{b}_2$, so $\psi = +90^\circ$. There $\cos \psi = 0$, $\cos 2\psi = -1$ and $\cos 3\psi = 0$, so the brackets of Equation (8.8) are $1 + \cos \psi = 1$, $1 - \cos 2\psi = 2$ and $1 + \cos 3\psi = 1$, and the energy is $\frac{1}{2} \times 0.1 + \frac{1}{2} \times 0.02 \times 2 + \frac{1}{2} \times 0.05 = 0.095$ eV.

EXERCISE 8.18. $k_e/r = k_B T$ gives $r = 14.3996/0.02585 = 557$ Å.

EXERCISE 8.19. At $\sqrt{3}d$: the 8 sites $(\pm 1, \pm 1, \pm 1)$, with $a + b + c$ odd, so opposite in charge. At $2d$: the 6 sites such as $(\pm 2, 0, 0)$, with $a + b + c$ even, so the same charge. Each opposite ion adds $+1/(\text{distance}/d)$ to $\mathcal{M}$ and each like ion subtracts that: up to d , $\mathcal{M} = 6$; up to $\sqrt{2}d$, $6 - 12/\sqrt{2} = -2.485$; up to $\sqrt{3}d$, $-2.485 + 8/\sqrt{3} = 2.134$; up to $2d$, $2.134 - 6/2 = -0.866$.

EXERCISE 8.20. The cube of half-side d holds the 6 face ions at d (opposite, weight $\frac{1}{2}$), the 12 edge ions at $\sqrt{2}d$ (like, weight $\frac{1}{4}$) and the 8 corner ions at $\sqrt{3}d$ (opposite, weight $\frac{1}{8}$): $\mathcal{M} = 6 \times \frac{1}{2} - 12 \times \frac{1}{4}/\sqrt{2} + 8 \times \frac{1}{8}/\sqrt{3} = 3 - 2.121 + 0.577 = 1.456$, already within 17% of 1.7476.

Chapter 9: Integrating the equations of motion

EXERCISE 9.1. Each step changes the velocity by $-g\delta t$, so $v_j = v_0 - jg\delta t$, and the position by $\delta t v_j$, so

$$x_n = x_0 + \delta t \sum_{j=0}^{n-1} (v_0 - jg\delta t) = x_0 + v_0 n\delta t - g\delta t^2 \frac{n(n-1)}{2},$$

using $0 + 1 + \dots + (n-1) = n(n-1)/2$, since the sum written forwards plus the sum written backwards is n pairs that each add to $n-1$. The true position at $t_f = n\delta t$ is $x_0 + v_0 n\delta t - \frac{1}{2} g n^2 \delta t^2$, so the computed position exceeds it by $\frac{1}{2} g \delta t^2 n^2 - \frac{1}{2} g \delta t^2 (n^2 - n) = \frac{1}{2} g n \delta t^2 = \frac{1}{2} g t_f \delta t$.

EXERCISE 9.2. $16 = 2^4$, so the global error is of order δt^4 : a method of order 4, with a local error of order δt^5 .

EXERCISE 9.3. The δt^4 terms of the two series of Section 9.2 are both $\frac{1}{24} \ddot{a} \delta t^4$, since an even power does not change sign, and add to $\frac{1}{12} \ddot{a} \delta t^4$. So $x(t + \delta t) = 2x - x(t - \delta t) + \delta t^2 a + \frac{1}{12} \ddot{a} \delta t^4 + \mathcal{O}(\delta t^6)$, and Equation (9.1), which drops the δt^4 term, is wrong by $\frac{1}{12} \ddot{a} \delta t^4$ for a correct start.

EXERCISE 9.4. Subtracting the two series instead, the even powers cancel: $x(t + \delta t) - x(t - \delta t) = 2v\delta t + \frac{1}{3} \dot{a} \delta t^3 + \mathcal{O}(\delta t^5)$. Dividing by $2\delta t$ gives $v + \frac{1}{6} \dot{a} \delta t^2 + \mathcal{O}(\delta t^4)$.

EXERCISE 9.5. The true position is $x + v\delta t + \frac{1}{2}a\delta t^2 + \frac{1}{6}\dot{a}\delta t^3 + \mathcal{O}(\delta t^4)$, and Equation (9.2) gives $x + v\delta t + \frac{1}{2}a\delta t^2$, which lacks $\frac{1}{6}\dot{a}\delta t^3$.

EXERCISE 9.6. $x_{n+1} - x_n = \delta t v_{n+\frac{1}{2}}$ and $x_n - x_{n-1} = \delta t v_{n-\frac{1}{2}}$. Subtracting, $x_{n+1} - 2x_n + x_{n-1} = \delta t (v_{n+\frac{1}{2}} - v_{n-\frac{1}{2}}) = \delta t^2 a_n$; adding $2x_n - x_{n-1}$ to both sides gives Equation (9.1).

EXERCISE 9.7. The step, with $a = -kq/m = -q$, gives $q = 1 + \delta t \times 0 = 1$ and $v = 0 + \delta t \times (-1) = -\delta t$. Reversed, $(1, \delta t)$, and a second step gives $q = 1 + \delta t^2$ and $v = \delta t - \delta t \times 1 = 0$: it lands at $(1 + \delta t^2, 0)$, not at the start $(1, 0)$.

EXERCISE 9.8. The products are taken two at a time, by the toolbox of Section 9.3. With $\mathbf{J} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, $\mathbf{\Omega} = \begin{pmatrix} c & d \\ -a & -b \end{pmatrix}$, and

$$\mathbf{J}^T \mathbf{\Omega} \mathbf{J} = \begin{pmatrix} a & c \\ b & d \end{pmatrix} \begin{pmatrix} c & d \\ -a & -b \end{pmatrix} = \begin{pmatrix} ac - ca & ad - cb \\ bc - da & bd - db \end{pmatrix} = (ad - bc) \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},$$

so $\mathbf{J}^T \mathbf{\Omega} \mathbf{J} = \mathbf{\Omega}$ when $ad - bc = \det \mathbf{J} = 1$, and only then.

EXERCISE 9.9. $e^{\delta t X} e^{\delta t Y} = (1 + \delta t X + \frac{1}{2}\delta t^2 X^2)(1 + \delta t Y + \frac{1}{2}\delta t^2 Y^2) + \mathcal{O}(\delta t^3) = 1 + \delta t(X + Y) + \delta t^2(\frac{1}{2}X^2 + XY + \frac{1}{2}Y^2) + \mathcal{O}(\delta t^3)$, while $e^{\delta t(X+Y)}$ has $\frac{1}{2}\delta t^2(X^2 + XY + YX + Y^2)$ at second order. The difference is $\delta t^2(XY - \frac{1}{2}XY - \frac{1}{2}YX) = \frac{1}{2}\delta t^2(XY - YX)$.

EXERCISE 9.10. The right-hand exponential acts on q first (Section 9.4): $e^{\delta t \hat{L}_r} q = q + \delta t p/m$. Applying $e^{\delta t \hat{L}_p}$ to this function, $\hat{L}_p = F(q) \partial/\partial p$ gives $\hat{L}_p(q + \delta t p/m) = \delta t F/m$ and $\hat{L}_p^2(q + \delta t p/m) = 0$, so the result is $q + \delta t p/m + \delta t \cdot \delta t F/m = q + \delta t(p + \delta t F)/m$.

EXERCISE 9.11. $\hat{L}_r q^3 = 3q^2 p/m$, $\hat{L}_r^2 q^3 = 6qp^2/m^2$, $\hat{L}_r^3 q^3 = 6p^3/m^3$ and $\hat{L}_r^4 q^3 = 0$, so the series is $q^3 + 3q^2(\delta t p/m) + \frac{1}{2} \times 6q(\delta t p/m)^2 + \frac{1}{6} \times 6(\delta t p/m)^3 = (q + \delta t p/m)^3$, by the binomial expansion.

EXERCISE 9.12. With $p_{\frac{1}{2}} = p' + \frac{1}{2}\delta t k q'$,

$$\begin{aligned} \frac{p_{\frac{1}{2}}^2}{2m} &= \frac{p'^2}{2m} + \frac{\delta t k}{2m} q' p' + \frac{\delta t^2 k^2 q'^2}{8m}, \\ -\frac{\delta t k}{2m} q' p_{\frac{1}{2}} &= -\frac{\delta t k}{2m} q' p' - \frac{\delta t^2 k^2 q'^2}{4m}. \end{aligned}$$

Adding $\frac{1}{2}kq'^2$, the terms in $q'p'$ cancel, and those in q'^2 combine as $\frac{1}{8} - \frac{1}{4} = -\frac{1}{8}$:

$$\frac{p'^2}{2m} + \frac{1}{2}kq'^2 - \frac{\delta t^2 k^2 q'^2}{8m} = \frac{p'^2}{2m} + \frac{1}{2}kq'^2 \left(1 - \frac{\omega^2 \delta t^2}{4}\right) = \tilde{H}(q', p'),$$

since $\delta t^2 k^2 / (8m) = \frac{1}{2}k \times \omega^2 \delta t^2 / 4$.

EXERCISE 9.13. With $p' = p - \delta t k q$ and $q' = q + \delta t p'/m$, the last two terms of the quantity at (q', p') are

$$\begin{aligned} \frac{1}{2}kq'^2 - \frac{\delta t k}{2m} q' p' &= \frac{1}{2}kq^2 + \frac{\delta t k}{m} qp' + \frac{\delta t^2 k p'^2}{2m^2} - \frac{\delta t k}{2m} qp' - \frac{\delta t^2 k p'^2}{2m^2} \\ &= \frac{1}{2}kq^2 + \frac{\delta t k}{2m} qp', \end{aligned}$$

and adding $p'^2/(2m)$,

$$\frac{p'^2}{2m} + \frac{\delta t k}{2m} qp' = \frac{p'(p' + \delta t k q)}{2m} = \frac{p'p}{2m} = \frac{p^2}{2m} - \frac{\delta t k}{2m} qp,$$

so the quantity at (q', p') equals $p^2/(2m) + \frac{1}{2}kq^2 - (\delta t k/(2m)) qp$ at (q, p) . With $q = A \cos \phi$ and $p = m\omega A \sin \phi$, $p^2/(2m) = \frac{1}{2}m\omega^2 A^2 \sin^2 \phi = \frac{1}{2}kA^2 \sin^2 \phi$, so $H = \frac{1}{2}kA^2$, and $(\delta t k/(2m)) qp = \frac{1}{2}kA^2 \omega \delta t \sin \phi \cos \phi = \frac{1}{2}H \omega \delta t \sin 2\phi$, by Equation (1.6). The kept quantity is therefore $H(1 - \frac{1}{2}\omega \delta t \sin 2\phi)$, and as $\sin 2\phi$ runs from -1 to 1 , H runs between $1/(1 + \omega \delta t/2)$ and $1/(1 - \omega \delta t/2)$ times it.

EXERCISE 9.14. From Section 9.5, with $k/m = \omega^2$,

$$q' = q + \frac{\delta t}{m} \left(p - \frac{1}{2}\delta t k q \right) = \left(1 - \frac{1}{2}\omega^2 \delta t^2 \right) q + \frac{\delta t}{m} p,$$

$$\begin{aligned} p' &= p - \frac{1}{2}\delta t k q - \frac{1}{2}\delta t k \left[\left(1 - \frac{1}{2}\omega^2 \delta t^2 \right) q + \frac{\delta t}{m} p \right] \quad \text{putting in } q' \\ &= -k\delta t \left(1 - \frac{1}{4}\omega^2 \delta t^2 \right) q + \left(1 - \frac{1}{2}\omega^2 \delta t^2 \right) p \quad \text{collecting, with } k\delta t^2/m = \omega^2 \delta t^2. \end{aligned}$$

The step's matrix holds these coefficients,

$$\begin{pmatrix} 1 - \frac{1}{2}\omega^2 \delta t^2 & \delta t/m \\ -k\delta t (1 - \frac{1}{4}\omega^2 \delta t^2) & 1 - \frac{1}{2}\omega^2 \delta t^2 \end{pmatrix}.$$

Its trace is $2 - \omega^2 \delta t^2$, and its determinant, the product of the diagonal entries minus the product of the other two (Section 7.4), is $(1 - \frac{1}{2}\omega^2 \delta t^2)^2 + (\delta t/m) k\delta t (1 - \frac{1}{4}\omega^2 \delta t^2) = 1 - \omega^2 \delta t^2 + \frac{1}{4}\omega^4 \delta t^4 + \omega^2 \delta t^2 - \frac{1}{4}\omega^4 \delta t^4 = 1$.

EXERCISE 9.15. The trace is $2 - 2.05^2 = -2.2025$, and by the formula for the roots of a quadratic, $\lambda = \frac{1}{2}(-2.2025) \pm \sqrt{2.2025^2/4 - 1} = -1.10125 \pm 0.46125$, that is -1.5625 or -0.64 . The motion grows by the factor 1.5625 per step, changing sign each time.

EXERCISE 9.16. $\omega = 2\pi/8.88 = 0.7076 \text{ rad fs}^{-1}$, and $\omega^2 \delta t^2/4 \leq 0.03$ gives $\omega \delta t \leq \sqrt{0.12} = 0.346$, so $\delta t \leq 0.346/0.7076 = 0.49 \text{ fs}$.

EXERCISE 9.17. $\omega = 2\pi/170.3 = 0.0369 \text{ rad fs}^{-1}$. A band of 1% needs $\omega^2 \delta t^2/4 = 0.01$, so $\omega \delta t = \sqrt{0.04} = 0.2$ and $\delta t = 0.2/0.0369 = 5.42 \text{ fs}$, ten times below the stability limit $2/\omega = 54.2 \text{ fs}$.

EXERCISE 9.18. $\omega \delta t = 0.2$ (Exercise 9.17), and a period $2\pi/\omega$ holds $2\pi/(\omega \delta t) = 2\pi/0.2 = 31.4$ steps.

EXERCISE 9.19. $q'^2 + u'^2 = (cq + su)^2 + (-sq + cu)^2 = (c^2 + s^2)(q^2 + u^2)$, the cross terms $2csqu$ and $-2csqu$ cancelling, and the energy is $\frac{1}{2}m\omega^2(q^2 + u^2)$. Squaring each term of c and adding twice each product of two, $c^2 = 1 + \frac{1}{4}x^4 + \frac{1}{576}x^8 - x^2 + \frac{1}{12}x^4 - \frac{1}{24}x^6$, and $s^2 = x^2 - \frac{1}{3}x^4 + \frac{1}{36}x^6$; in the sum $\frac{1}{4} + \frac{1}{12} - \frac{1}{3} = 0$ and $\frac{1}{36} - \frac{1}{24} = -\frac{1}{72}$, leaving $1 - \frac{1}{72}x^6 + \frac{1}{576}x^8$. At $x = 0.2$, $x^6/72 = 8.89 \times 10^{-7}$ and $x^8/576 = 4.4 \times 10^{-9}$, so the factor is $1 - 8.84 \times 10^{-7}$, and $(1 - 8.84 \times 10^{-7})^{10000} = 0.9912$: the energy loses 0.88%.

EXERCISE 9.20. $\omega \delta t = 0.7076 \times 5 = 3.54$, beyond the limit of 2: velocity Verlet would blow up, so a learned propagator cannot be velocity Verlet with a long step.

EXERCISE 9.21. $\omega = 2\pi/8.88 = 0.7076 \text{ rad fs}^{-1}$, $\omega \delta t = 0.7076 \times 0.7 = 0.4953$, and $\cos \tilde{\omega} \delta t = 1 - 0.4953^2/2 = 0.8773$ gives $\tilde{\omega} \delta t = 0.5005$, so $\tilde{\omega}/\omega = 1.0105$ and the stretch appears at $3756 \times 1.0105 = 3795 \text{ cm}^{-1}$.

Chapter 10: Building a molecular dynamics code

EXERCISE 10.1. The fraction is below 1% when $(1 - 2/n)^3 > 0.99$, that is $1 - 2/n > 0.99^{1/3} = 0.99665549$, or $n > 2/0.00334451 = 597.995$. So $n = 598$, at which the fraction is 0.0099999, just below 1%, and $N = 598^3 = 2.14 \times 10^8$ atoms. The estimate $6/n$ would give $n = 600$.

EXERCISE 10.2. $\mathbf{r} = s_1\mathbf{a} + s_2\mathbf{b} + s_3\mathbf{c}$ reads $as_1 - \frac{1}{2}as_2 = \frac{1}{2}a$, $\frac{\sqrt{3}}{2}as_2 = \frac{\sqrt{3}}{2}a$ and $cs_3 = \frac{1}{4}c$. The second gives $s_2 = 1$, then the first $s_1 = 1$, and the third $s_3 = \frac{1}{4}$. Wrapping removes the whole numbers, leaving $(0, 0, \frac{1}{4})$, the position $(0, 0, \frac{1}{4}c)$: the atom is a copy of the one at fractional coordinates $(0, 0, \frac{1}{4})$, the first of graphite's four (Section 10.5).

EXERCISE 10.3. $\mathbf{b} \times \mathbf{c} = (\frac{\sqrt{3}}{2}ac, \frac{1}{2}ac, 0)$, of length ac , and $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = \frac{\sqrt{3}}{2}a^2c$. Likewise $\mathbf{c} \times \mathbf{a} = (0, ac, 0)$, of length ac , and $\mathbf{a} \times \mathbf{b} = (0, 0, \frac{\sqrt{3}}{2}a^2)$. The widths $V/|\mathbf{b} \times \mathbf{c}|$, $V/|\mathbf{c} \times \mathbf{a}|$ and $V/|\mathbf{a} \times \mathbf{b}|$ are $\frac{\sqrt{3}}{2}a$, $\frac{\sqrt{3}}{2}a$ and c : $V = 35.29 \text{ \AA}^3$, and the widths are 2.134, 2.134 and 6.711 \AA.

EXERCISE 10.4. With the factor $\frac{1}{2}a_0$ taken out of each vector, $(1, 0, 1) \times (1, 1, 0) = (-1, 1, 1)$, of length $\sqrt{3}$, and $(0, 1, 1) \cdot (-1, 1, 1) = 2$, so $V = 2(\frac{1}{2}a_0)^3 = \frac{1}{4}a_0^3$ and the

width is $2(\frac{1}{2}a_0)^3/(\sqrt{3}(\frac{1}{2}a_0)^2) = a_0/\sqrt{3}$, the same in all three directions by symmetry. For $a_0 = 5.267 \text{ \AA}$: $V = 36.53 \text{ \AA}^3$, widths $3.041 \text{ \AA}$, and a cutoff of at most $1.520 \text{ \AA}$.

EXERCISE 10.5. $\mathbf{s} = (0.9, -0.967, 0.025)$, which rounds to $(1, -1, 0)$, so the copy is $\mathbf{d} - (2, -3, 0) = (-0.2, 0.1, 0.1) \text{ \AA}$. Each component can be changed only by a whole side, and $-0.2, 0.1$ and 0.1 are already within half a side of zero, so no copy is shorter.

EXERCISE 10.6. With $u = (x - x_0)/(s\sqrt{2})$, $dx = s\sqrt{2} du$ and the limits stay infinite, so the integral is $s\sqrt{2} \int e^{-u^2} du = s\sqrt{2}\sqrt{\pi} = s\sqrt{2\pi}$.

EXERCISE 10.7. $e^{-u^2} = 1 - u^2 + \frac{1}{2}u^4 - \dots$, from the series of e^y (Section 2.7) with $y = -u^2$. Integrating term by term from 0 to x gives $x - x^3/3 + x^5/10 - \dots$, so $\text{erf}(\alpha r)/r = (2/\sqrt{\pi})(\alpha - \alpha^3 r^2/3 + \dots)$, which tends to $2\alpha/\sqrt{\pi}$ as $r \rightarrow 0$.

EXERCISE 10.8. The product is $\frac{1}{2}[\cos(2\pi(m - m')x/L) + \cos(2\pi(m + m')x/L)]$. For a whole number $k \neq 0$, $\cos(2\pi kx/L)$ has the antiderivative $L \sin(2\pi kx/L)/(2\pi k)$, which takes the same value at $x = 0$ and $x = L$, so its average over a period is zero; for $k = 0$ the cosine is 1. If $m = m'$, then $m + m' \neq 0$ and the average is $\frac{1}{2}(1 + 0)$; if $m \neq m'$, neither $m - m'$ nor $m + m'$ is zero and the average is 0.

EXERCISE 10.9. With $\mathbf{a} = (a, 0, 0)$, $\mathbf{b} = (-\frac{1}{2}a, \frac{\sqrt{3}}{2}a, 0)$ and $\mathbf{c} = (0, 0, c)$: $\mathbf{a} \cdot \mathbf{a}^* = 2\pi$, $\mathbf{b} \cdot \mathbf{a}^* = \frac{2\pi}{a}(-\frac{1}{2}a + \frac{\sqrt{3}}{2}a \cdot \frac{1}{\sqrt{3}}) = 0$, $\mathbf{a} \cdot \mathbf{b}^* = 0$, $\mathbf{b} \cdot \mathbf{b}^* = \frac{2\pi}{a} \cdot \frac{\sqrt{3}}{2}a \cdot \frac{2}{\sqrt{3}} = 2\pi$, and $\mathbf{c}^*$ is at right angles to $\mathbf{a}$ and $\mathbf{b}$ with $\mathbf{c} \cdot \mathbf{c}^* = 2\pi$, while $\mathbf{a}^*$ and $\mathbf{b}^*$ have no z component. The lengths are $|\mathbf{a}^*| = |\mathbf{b}^*| = \frac{2\pi}{a}\sqrt{1 + \frac{1}{3}} = 4\pi/(\sqrt{3}a) = 2.9445 \text{ \AA}^{-1}$ and $|\mathbf{c}^*| = 2\pi/c = 0.9363 \text{ \AA}^{-1}$.

EXERCISE 10.10. Only $\mathcal{C}$ and $\mathcal{S}$ depend on $\mathbf{r}_i$, through the terms $q_i \cos(\mathbf{G} \cdot \mathbf{r}_i)$ and $q_i \sin(\mathbf{G} \cdot \mathbf{r}_i)$, whose gradients (Section 3.10) are $-q_i \sin(\mathbf{G} \cdot \mathbf{r}_i) \mathbf{G}$ and $q_i \cos(\mathbf{G} \cdot \mathbf{r}_i) \mathbf{G}$. The gradient of $\mathcal{C}^2 + \mathcal{S}^2$ is $2\mathcal{C}$ times the first plus $2\mathcal{S}$ times the second, $2q_i \mathbf{G} [\mathcal{S}(\mathbf{G}) \cos(\mathbf{G} \cdot \mathbf{r}_i) - \mathcal{C}(\mathbf{G}) \sin(\mathbf{G} \cdot \mathbf{r}_i)]$, and $\mathbf{F}_i$ is minus $2\pi k_e/V$ times the sum of $e^{-G^2/(4\alpha^2)}/G^2$ times that; the minus sign reverses the bracket and the 2 makes 4π , which is the formula. For the real-space term, $\frac{d}{dr} \text{erfc}(\alpha r) = -(2/\sqrt{\pi}) \alpha e^{-\alpha^2 r^2}$ by the fundamental theorem of calculus and the chain rule, and the product rule gives the slope of $\text{erfc}(\alpha r) \cdot r^{-1}$ as $-\text{erfc}(\alpha r)/r^2 - (2\alpha/\sqrt{\pi}) e^{-\alpha^2 r^2}/r$.

EXERCISE 10.11. The cell is $2\mathbf{I}$, so the reciprocal vectors are π along the axes and $\mathbf{G} = \pi(m_1, m_2, m_3)$. For a cation at $\mathbf{p}$, $\mathbf{G} \cdot \mathbf{p} = \pi \mathbf{m} \cdot \mathbf{p}$ is a whole multiple of π , and for its anion πm_1 more, so every sine is zero and $\mathcal{S} = 0$. By the addition formula, $\cos(x + \pi m_1) = (-1)^{m_1} \cos x$, so $\mathcal{C} = (1 - (-1)^{m_1}) \sum_{\mathbf{p}} \cos(\pi \mathbf{m} \cdot \mathbf{p})$, and, since $\cos k\pi = (-1)^k$ for a whole number k , the sum over the four cations is $1 + (-1)^{m_2+m_3} + (-1)^{m_1+m_3} + (-1)^{m_1+m_2}$. If the m_k are all even or all odd, every exponent is even and the sum is 4; otherwise one of them differs in parity from the other two, the two sums that contain

it are odd and the third even, and the sum is $1 + 1 - 1 - 1 = 0$. The factor $1 - (-1)^{m_1}$ is 2 for odd m_1 and 0 for even, so $\mathcal{C} = 8$ when all are odd and 0 otherwise.

EXERCISE 10.12. The copies within r_c number about $\frac{4}{3}\pi r_c^3 \rho \propto \rho/\alpha^3$; the waves within G_c number about $\frac{4}{3}\pi G_c^3$ divided by the volume per wave, $(2\pi)^3/V$, the cube of the spacing $2\pi/L$ for a cube of side L , so $\propto V\alpha^3$. The cost is $T = C_1 N \rho/\alpha^3 + C_2 N V \alpha^3$ for constants C_1 and C_2 , and $dT/d\alpha = -3C_1 N \rho/\alpha^4 + 3C_2 N V \alpha^2 = 0$ gives $\alpha^6 = C_1 \rho/(C_2 V) = (C_1/C_2) N/V^2$. There the two terms are equal, and $T = 2C_2 N V \alpha^3 \propto N V (N/V^2)^{1/2} = N^{3/2}$.

EXERCISE 10.13. $\sqrt{-\ln 10^{-6}} = 3.717$, so $\alpha = 3.717/15 \text{ \AA} = 0.2478 \text{ \AA}^{-1}$ and $G_c = 2 \times 0.2478 \times 3.717 = 1.842 \text{ \AA}^{-1}$. The waves form a cubic grid of spacing $2\pi/30 = 0.2094 \text{ \AA}^{-1}$, one per $(2\pi/30)^3$ of volume, so the sphere of radius G_c holds about $\frac{4}{3}\pi(1.842)^3/(0.2094)^3 = 2850$; counting the grid points gives 2896.

EXERCISE 10.14. $\frac{4}{3}\pi(10)^3 \times 0.0334 = 139.9$, about 140 molecules.

EXERCISE 10.15. $\lfloor 40/9.5 \rfloor = 4$, so $4^3 = 64$ small cells; $\lfloor 30/9.5 \rfloor = 3$, 27 small cells; and $\lfloor 25/9.5 \rfloor = 2$, fewer than three, so all pairs are tested.

EXERCISE 10.16. $\sqrt{2K/m} = \sqrt{2 \times 0.0104/(39.948 \times 103.64)} = 0.00224 \text{ \AA fs}^{-1}$, with the factor $MV2_T0_EV$ of Equation (3.15), or $0.0224 \text{ \AA}$ a step, so $0.5 \text{ \AA}$ takes 22 steps. The run rebuilds every 10.7 steps because it rebuilds as soon as any one of 500 atoms has moved half the skin, and some atoms move faster than the root-mean-square speed.

EXERCISE 10.17. $\sum_i m_i(\mathbf{v}_i - \mathbf{P}/M) = \mathbf{P} - M\mathbf{P}/M = \mathbf{0}$, and $\sum_i m_i(c\mathbf{v}_i) = c \sum_i m_i \mathbf{v}_i = \mathbf{0}$.

EXERCISE 10.18. Started at the minimum, $x = A \sin \omega t$ and $K = \frac{1}{2}m\omega^2 A^2 \cos^2 \omega t$, with $K_0 = \frac{1}{2}m\omega^2 A^2$. The average of $\cos^2$ over a period is $\frac{1}{2}$ (Section 4.2), so $\bar{K} = \frac{1}{2}K_0$.

EXERCISE 10.19. The run in ASE's units lasts 1000 of its femtoseconds, $1000 \times 4.6 \times 10^{-9} = 4.6 \times 10^{-6} \text{ fs}$ longer than `mdlab`'s, and starts with velocities smaller by the same fraction. For an atom in free flight the two cancel; a vibrating atom keeps its frequency whatever its speed, so the extra time carries it on by about its speed times $4.6 \times 10^{-6} \text{ fs}$, $0.0031 \times 4.6 \times 10^{-6} = 1.4 \times 10^{-8} \text{ \AA}$: the size of the difference found.

Chapter 11: Constraints and multiple time steps

EXERCISE 11.1. Six atoms have $3 \times 6 = 18$ freedoms, and each of the four held bonds removes one: 14 per molecule. For 100 molecules, $1800 - 400 = 1400$, and removing the drift of the centre of mass leaves 1397.

EXERCISE 11.2. $d^2 = 2.25$, $\mathbf{r}_{ij} \cdot \mathbf{r}_{ij} = 1.6 \times 1.5 = 2.4$ and $|\mathbf{r}_{ij}|^2 = 2.56 + 0.09 = 2.65$, so Equation (11.1) reads

$$2.25 \eta^2 + 4.8 \eta + 0.40 = 0,$$

and by the quadratic formula

$$\eta = \frac{-4.8 \pm \sqrt{4.8^2 - 4 \times 2.25 \times 0.40}}{2 \times 2.25} = \frac{-4.8 \pm 4.4091}{4.5},$$

giving -0.08687 or -2.0465 . The root near zero is the correction; the other turns the bond round. With $m_r = 2 \times 5/7 = 1.4286$ amu, atom j moves by $\eta m_r / m_j \mathbf{r}_{ij} = -0.08687 \times 1.4286/5 \times (1.5, 0, 0) = (-0.03723, 0, 0)$ Å and atom i by $-\eta m_r / m_i \mathbf{r}_{ij} = (+0.09308, 0, 0)$ Å. The new bond, $\mathbf{r}_j - \mathbf{r}_i$, has x component $1.6 - 0.03723 - 0.09308 = 1.4697$ and y component 0.3 , so its length is $\sqrt{2.1600 + 0.09} = 1.5000$ Å.

EXERCISE 11.3. $f'(y) = 2y$, so $y_{n+1} = y_n - (y_n^2 - 3)/(2y_n)$: $y_1 = 2 - 1/4 = 1.75$, $y_2 = 1.75 - 0.0625/3.5 = 1.7321428571$, $y_3 = 1.7320508100$, with errors 1.8×10^{-2} , 9.2×10^{-5} and 2.4×10^{-9} , against 0.27 for y_0 : each is the square of the one before divided by $2y_n$, which is 4 , then 3.5 , then 3.46 (Exercise 11.4).

EXERCISE 11.4. With $f'(y) = 2y$,

$$y_{n+1} = y_n - \frac{y_n^2 - a}{2y_n} = \frac{2y_n^2 - y_n^2 + a}{2y_n} = \frac{1}{2} \left(y_n + \frac{a}{y_n} \right).$$

Writing $y_{n+1} = (y_n^2 + a)/(2y_n)$ and $\sqrt{a} = 2y_n \sqrt{a}/(2y_n)$,

$$y_{n+1} - \sqrt{a} = \frac{y_n^2 + a - 2y_n \sqrt{a}}{2y_n} = \frac{(y_n - \sqrt{a})^2}{2y_n},$$

since $(y_n - \sqrt{a})^2 = y_n^2 - 2y_n \sqrt{a} + a$.

EXERCISE 11.5. The centre of mass of the pair is $(m_i \mathbf{r}_i + m_j \mathbf{r}_j)/(m_i + m_j)$. The correction moves $\mathbf{r}_j$ by $(\eta m_r / m_j) \mathbf{r}_{ij}$ and $\mathbf{r}_i$ by $-(\eta m_r / m_i) \mathbf{r}_{ij}$, so the numerator changes by $m_j(\eta m_r / m_j) \mathbf{r}_{ij} - m_i(\eta m_r / m_i) \mathbf{r}_{ij} = \eta m_r \mathbf{r}_{ij} - \eta m_r \mathbf{r}_{ij} = \mathbf{0}$.

EXERCISE 11.6. Solving $M\mathbf{V} = m_i\mathbf{v}_i + m_j\mathbf{v}_j$ and $\mathbf{v}_{ij} = \mathbf{v}_j - \mathbf{v}_i$ for the two velocities gives $\mathbf{v}_i = \mathbf{V} - (m_j/M)\mathbf{v}_{ij}$ and $\mathbf{v}_j = \mathbf{V} + (m_i/M)\mathbf{v}_{ij}$. Expanding the squares by $|\mathbf{a} + \mathbf{b}|^2 = |\mathbf{a}|^2 + 2\mathbf{a} \cdot \mathbf{b} + |\mathbf{b}|^2$, the kinetic energy $\frac{1}{2}m_i|\mathbf{v}_i|^2 + \frac{1}{2}m_j|\mathbf{v}_j|^2$ has three kinds of term: $\frac{1}{2}(m_i + m_j)|\mathbf{V}|^2 = \frac{1}{2}M|\mathbf{V}|^2$; the cross terms

$$\mathbf{V} \cdot \mathbf{v}_{ij} \left(-\frac{m_i m_j}{M} + \frac{m_j m_i}{M} \right) = 0;$$

and

$$\frac{1}{2} \left(\frac{m_i m_j^2}{M^2} + \frac{m_j m_i^2}{M^2} \right) |\mathbf{v}_{ij}|^2 = \frac{1}{2} \frac{m_i m_j}{M} |\mathbf{v}_{ij}|^2,$$

since $m_i m_j^2 + m_j m_i^2 = m_i m_j (m_i + m_j) = m_i m_j M$. Their sum is the result, with $m_i m_j / M = m_r$. The correction changes $m_j \mathbf{v}_j$ by $m_r \zeta \mathbf{r}_{ij}$ and $m_i \mathbf{v}_i$ by the opposite, so $M\mathbf{V}$ and $\mathbf{V}$ are unchanged, and $\mathbf{v}_{ij}$ becomes $\mathbf{v}_{ij} + \zeta \mathbf{r}_{ij}$. With $\mathbf{r}_{ij} = d \mathbf{e}$ and $\zeta = -\mathbf{v}_{ij} \cdot \mathbf{e} / d$, this is $\mathbf{v}_{ij} - (\mathbf{v}_{ij} \cdot \mathbf{e}) \mathbf{e}$, the relative velocity without its part along $\mathbf{e}$, so

$$|\mathbf{v}_{ij} - (\mathbf{v}_{ij} \cdot \mathbf{e}) \mathbf{e}|^2 = |\mathbf{v}_{ij}|^2 - 2(\mathbf{v}_{ij} \cdot \mathbf{e})^2 + (\mathbf{v}_{ij} \cdot \mathbf{e})^2 = |\mathbf{v}_{ij}|^2 - (\mathbf{v}_{ij} \cdot \mathbf{e})^2,$$

using $\mathbf{e} \cdot \mathbf{e} = 1$, and the kinetic energy falls by $\frac{1}{2} m_r (\mathbf{v}_{ij} \cdot \mathbf{e})^2$.

EXERCISE 11.7. $\mathbf{v}_{ij} = \mathbf{v}_H - \mathbf{v}_O = (0.019, 0.01, 0)$ and $\mathbf{r}_{ij} = (0.96, 0, 0)$, so $\zeta = -0.019 \times 0.96 / 0.96^2 = -0.019792 \text{ fs}^{-1}$. With $m_r = 16/17 = 0.94118 \text{ amu}$, the hydrogen's x -velocity changes by $m_r \zeta \times 0.96 / 1 = -0.017882$ and the oxygen's by $-m_r \zeta \times 0.96 / 16 = +0.0011176$: $\mathbf{v}_O = (0.0021176, 0, 0)$ and $\mathbf{v}_H = (0.0021176, 0.01, 0) \text{ Å/fs}$, moving together along the bond. The momentum along x is 0.036 before and after. The kinetic energy falls from 2.580×10^{-4} to $8.812 \times 10^{-5} \text{ amu Å}^2/\text{fs}^2$, by 1.6988×10^{-4} , which is $\frac{1}{2} \times 0.94118 \times 0.019^2$, as Exercise 11.6 predicts.

EXERCISE 11.8. $6 \times 216 - 3 = 1293$ freedoms, and $1293 \times 12.93 \text{ meV} = 16.72 \text{ eV}$.

EXERCISE 11.9. With $n = 1$ the outer step is a half kick by $\mathbf{F}^s$, a half kick by $\mathbf{F}^f$, a drift of δt , a half kick by the new $\mathbf{F}^f$ and a half kick by the new $\mathbf{F}^s$. Two kicks in a row, with no drift between, happen at the same positions and add: the first two make a half kick by $\mathbf{F}^f + \mathbf{F}^s$ and the last two another, which with the drift between is velocity Verlet with the summed force.

EXERCISE 11.10. Velocity Verlet calls both forces at every step: $10^6 / 0.41 = 2.44 \times 10^6$ steps of $24 + 0.04 = 24.04 \text{ ms}$, 58600 s , about 16 hours. RESPA: $10^6 / 0.72 = 1.39 \times 10^6$ outer steps of 24 ms plus three calls of the springs at 0.04 ms , together 33500 s , about 9 hours.

EXERCISE 11.11. $\mathbf{M}^2$ has rows $(1 + 6, 2 + 8) = (7, 10)$ and $(3 + 12, 6 + 16) = (15, 22)$. The trace is 5 and the determinant $4 - 6 = -2$, so $5\mathbf{M} + 2\mathbf{I}$ has rows $(5 + 2, 10) = (7, 10)$ and $(15, 20 + 2) = (15, 22)$: the same.

EXERCISE 11.12. With $\theta = 2\pi - \epsilon$ and ϵ small and positive, $\cos(2\pi - \epsilon) = \cos \epsilon$ and $\sin(2\pi - \epsilon) = -\sin \epsilon$, so $\cos \theta \approx 1 - \frac{1}{2}\epsilon^2$ and $\sin \theta \approx -\epsilon$, so with $\omega_s^2 \delta t / \omega_f = b\theta \approx 2\pi b$,

$$\text{tr } \mathbf{M} \approx 2 - \epsilon^2 + 2\pi b \epsilon,$$

which exceeds 2 when $\epsilon^2 < 2\pi b \epsilon$, that is $0 < \epsilon < 2\pi b$: θ from $2\pi(1 - b)$ to 2π . Since $\delta t = \theta / \omega_f = \theta \mathcal{T}_f / (2\pi)$, the band runs from $(1 - b)\mathcal{T}_f$ to $\mathcal{T}_f$, of width $b \mathcal{T}_f$.

EXERCISE 11.13. $b = 0.09$. Below half the period the band runs from $\frac{1}{2}(1 - b) = 0.455$ to 0.5, against 0.459 to 0.500 from the full trace; below the period from $1 - b = 0.910$ to 1, against 0.919 to 1.000. The upper edges are exact, since $\sin \theta = 0$ there. The lower ones come out a little low because the slow term was taken at $\theta = \pi$, or 2π , rather than at $\pi - \epsilon$: keeping it, the condition $\frac{1}{2}\epsilon^2 < \frac{1}{2}b(\pi - \epsilon)\epsilon$ gives $\epsilon < b\pi / (1 + b)$ and an edge at $\frac{1}{2} / (1 + b) = 0.4587$, and below the period $1 / (1 + b) = 0.9174$. The approximation holds less well as b grows.

EXERCISE 11.14. At 0.25 fs the inner steps already follow the springs closely: velocity Verlet with all forces at that step has a fluctuation of only 0.36%. The error at an outer step of 2 fs comes from the slow forces being applied as kicks once per outer step, which a finer inner step does not change; the resonances of Section 11.5 depend on the outer step alone. That 10.4% is no better than 8.8% shows that the inner step was never the limit.

EXERCISE 11.15. The oxygen keeps $15.999 - 2 \times 1.008 = 13.983$ amu. The reduced mass is $13.983 \times 2.016 / (13.983 + 2.016) = 1.7620$ amu, 1.858 times the ordinary 0.9483, so the stretch slows by $\sqrt{1.858} = 1.363$.

EXERCISE 11.16. Each hydrogen lies $0.9572 \times \cos 52.26^\circ = 0.5859$ Å from the oxygen along the bisector, their parts across the bisector cancel, and the oxygen's own term is zero, so the centre of mass is $2m_H \times 0.5859 / 18.015$ from the oxygen: 0.0656 Å with $m_H = 1.008$ and 0.1967 Å with 3.024.

EXERCISE 11.17. The forces and the potential energy do not depend on the masses, and the probability of an arrangement depends on the potential energy alone (Chapter 12), so the mean potential energy and the distribution of O–O distances are unchanged. The diffusion rate and the vibrational frequencies depend on how fast the atoms respond to the forces, and change.

EXERCISE 11.18. $|\mathbf{P}| = \sqrt{0.2817^2 + 0.2936^2 + 0.4033^2} = 0.5729 \text{ amu } \text{\AA}/\text{fs}$ and $M = 64 \times 18.015 = 1152.96 \text{ amu}$, so the centre of mass moves at $4.969 \times 10^{-4} \text{ \AA fs}^{-1}$, or $4.969 \times 10^{-4} \times 10^6 = 497 \text{ \AA}$ in a nanosecond of 10^6 fs , 40 times the side of the cell. Its kinetic energy is $|\mathbf{P}|^2/(2M)$ times the factor 103.64 of Equation (3.15), 0.01475 eV.

EXERCISE 11.19. The step is $2 \times 10.18 = 20.36 \text{ fs}$. The O–H stretch has $\omega = 2\pi/8.88 = 0.708 \text{ rad fs}^{-1}$, so $\omega\delta t = 14.4$, far beyond the limit 2 of Section 9.6: the run blows up.

EXERCISE 11.20. An error of the integration would fall as δt^2 ; a drift that does not shrink when the step is halved, with the bonds and the centre of mass healthy, lies in the forces, which are not the slopes of a smooth energy, as with a cutoff that jumps (Table 11.2b). Comparing the model's forces with differences of its energy (Section 8.4), and the energy as a pair crosses the cutoff, would confirm it.

Chapter 12: Ensembles

EXERCISE 12.1. The mean is $(1 + 2 + \cdots + 6)/6 = 21/6 = 3.5$ and $\langle x^2 \rangle = (1 + 4 + 9 + 16 + 25 + 36)/6 = 91/6$, so the variance is $91/6 - 3.5^2 = (182 - 147)/12 = 35/12 = 2.9167$. Two independent dice have twice that, $35/6 = 5.8333$.

EXERCISE 12.2. Multiplying a quantity by c multiplies its deviation from its mean by c and its variance by c^2 . The sum of n independent values has the variance $n\sigma_x^2$, since variances of independent quantities add and the values can be added one at a time, each independent of the sum of those before it, so the mean, the sum times $1/n$, has $n\sigma_x^2/n^2 = \sigma_x^2/n$. For a die, $(35/12)/n = 0.01^2$ gives $n = 29\,167$ throws, rounding up.

EXERCISE 12.3. $\omega_k = 2 \sin(k\pi/8)$ gives 0.7654, 1.4142 and 1.8478, with squares $2 - \sqrt{2}$, 2 and $2 + \sqrt{2}$. The force on atom j is $q_{j-1} - 2q_j + q_{j+1}$, minus the j th entry of the matrix times $\mathbf{q}$, so a mode needs the matrix times its pattern $\mathbf{u}$ to be $\omega^2 \mathbf{u}$. The patterns $u_j = \sin(jk\pi/4)$ are, up to a factor, $(1, \sqrt{2}, 1)$, $(1, 0, -1)$ and $(1, -\sqrt{2}, 1)$, and multiplying each by the matrix,

$$(2 - \sqrt{2}, 2\sqrt{2} - 2, 2 - \sqrt{2}) = (2 - \sqrt{2}) (1, \sqrt{2}, 1),$$

$$(2, 0, -2) = 2 (1, 0, -1),$$

$$(2 + \sqrt{2}, -2\sqrt{2} - 2, 2 + \sqrt{2}) = (2 + \sqrt{2}) (1, -\sqrt{2}, 1),$$

so $2 - \sqrt{2}$, 2 and $2 + \sqrt{2}$ are eigenvalues of the matrix, one for each of the chain's three modes.

EXERCISE 12.4. Every microstate of the first combines with every microstate of the second, so together they have $\Omega_1\Omega_2$, and $k_B \ln(\Omega_1\Omega_2) = k_B \ln \Omega_1 + k_B \ln \Omega_2 = S_1 + S_2$.

EXERCISE 12.5. The bath holds $E = \frac{3}{2}Nk_BT = 1.5 \times 10^4 \times 0.025852 = 387.8 \text{ eV}$, and the ratio is $E_s/2E = 0.1/775.6 = 1.29 \times 10^{-4}$.

EXERCISE 12.6. $Z = 1 + e^{-\beta E_g}$. Writing $x = \beta E_g$,

$$\langle E \rangle = -\frac{d \ln Z}{d\beta} = \frac{E_g e^{-x}}{1 + e^{-x}} = \frac{E_g}{e^x + 1},$$

$$C_V = \frac{d\langle E \rangle}{dT} = -\frac{E_g e^x}{(e^x + 1)^2} \frac{dx}{dT} = k_B \frac{x^2 e^x}{(e^x + 1)^2},$$

by the chain rule, with $dx/dT = -E_g/k_BT^2$. At 300 K, $x = 0.1/0.025852 = 3.868$: $\langle E \rangle = 0.00205 \text{ eV}$ and $C_V = 0.300 k_B$. Maximising $x^2 e^x / (e^x + 1)^2$ numerically gives $x = 2.399$, so C_V is largest, $0.439 k_B$, at $k_BT = 0.4168 E_g$, which is 484 K for $E_g = 0.1 \text{ eV}$.

EXERCISE 12.7. With $\mathcal{P}_s = e^{-\beta E_s}/Z$, $\ln \mathcal{P}_s = -\beta E_s - \ln Z$, so

$$S = -k_B \sum_s \mathcal{P}_s (-\beta E_s - \ln Z) = \frac{\langle E \rangle}{T} + k_B \ln Z = \frac{\langle E \rangle - F}{T},$$

using $\sum_s \mathcal{P}_s = 1$; so $F = \langle E \rangle - TS$. With $\mathcal{P}_s = 1/\Omega$ for each of Ω states,

$$S = -k_B \Omega \times \frac{1}{\Omega} \ln \frac{1}{\Omega} = k_B \ln \Omega.$$

EXERCISE 12.8. $\ln Z = -\ln \beta + \ln(2\pi/\omega)$, so

$$-\frac{d \ln Z}{d\beta} = \frac{1}{\beta} = k_B T, \quad \frac{d^2 \ln Z}{d\beta^2} = \frac{1}{\beta^2} = (k_B T)^2.$$

With $C_V = d(k_B T)/dT = k_B$, Equation (12.6) gives $k_B T^2 \times k_B = (k_B T)^2$, the same.

EXERCISE 12.9. $0.3/k_B T = \ln 1000 = 6.908$ gives $k_B T = 0.04343 \text{ eV}$, $T = 504 \text{ K}$. At 300 K, $e^{-0.5/0.025852} = e^{-19.34} = 4.0 \times 10^{-9}$.

EXERCISE 12.10. The density is a constant times $v^2 e^{-v^2/2\sigma_v^2}$, whose derivative $(2v - v^3/\sigma_v^2)e^{-v^2/2\sigma_v^2}$ vanishes at $v = \sqrt{2}\sigma_v$. The mean needs $\int_0^\infty v^3 e^{-v^2/2\sigma_v^2} dv$. With $w = v^2/2\sigma_v^2$ and $dw = v dv/\sigma_v^2$,

$$v^3 dv = v^2 \times v dv = 2\sigma_v^2 w \times \sigma_v^2 dw, \quad \int_0^\infty 2\sigma_v^4 w e^{-w} dw = 2\sigma_v^4,$$

since $\int_0^\infty w e^{-w} dw = 1$ by parts. With $(m/2\pi k_B T)^{3/2} = (2\pi\sigma_v^2)^{-3/2}$,

$$\langle v \rangle = \frac{4\pi \times 2\sigma_v^4}{(2\pi)^{3/2}\sigma_v^3} = \frac{4\sigma_v}{\sqrt{2\pi}} = \sqrt{\frac{8}{\pi}} \sigma_v.$$

For lithium at 300 K, $\sigma_v = 0.005995 \text{ Å fs}^{-1}$: the most probable speed is $0.00848 \text{ Å fs}^{-1}$ and the mean $0.00957 \text{ Å fs}^{-1}$.

EXERCISE 12.11. With $u = x$ and $s = \sigma_x$, as in Section 12.1, and by parts, with u^3 and $u e^{-u^2/2s^2}$, whose integral is $-s^2 e^{-u^2/2s^2}$,

$$\int u^4 e^{-u^2/2s^2} du = \left[-s^2 u^3 e^{-u^2/2s^2} \right]_{-\infty}^{\infty} + 3s^2 \int u^2 e^{-u^2/2s^2} du = 3s^2 \times s^2 \times s\sqrt{2\pi},$$

using the integral of Section 12.1; dividing by $s\sqrt{2\pi}$, $\langle x^4 \rangle = 3\sigma_x^4$. For an even spread over $-c$ to c , with the density $1/2c$, $\langle x^2 \rangle = \int_{-c}^c x^2 dx / 2c = c^2/3$ and $\langle x^4 \rangle = c^4/5$, so $\langle x^4 \rangle / \langle x^2 \rangle^2 = 9/5$ and the excess kurtosis is $9/5 - 3 = -1.2$.

EXERCISE 12.12. $\sigma_v = \sqrt{k_B T / m}$ with m times MV2_T0_EV: $0.00600 \text{ Å fs}^{-1}$ for lithium and $0.00456 \text{ Å fs}^{-1}$ for carbon, in the ratio $\sqrt{12.011/6.94} = 1.316$. Their mean kinetic energies are equal, $\frac{3}{2}k_B T$ each, by equipartition; in real graphite the stiff carbon vibrations are quantum at 300 K (Section 8.2), and carbon's mean kinetic energy exceeds $\frac{3}{2}k_B T$.

EXERCISE 12.13. $p_\theta = \partial \mathcal{L} / \partial \dot{\theta} = I\dot{\theta}$ and $p_\phi = \partial \mathcal{L} / \partial \dot{\phi} = I \sin^2 \theta \dot{\phi}$, so $\dot{\theta} = p_\theta / I$ and $\dot{\phi} = p_\phi / (I \sin^2 \theta)$, and

$$H = p_\theta \dot{\theta} + p_\phi \dot{\phi} - \mathcal{L} = \frac{p_\theta^2}{I} + \frac{p_\phi^2}{I \sin^2 \theta} - \frac{1}{2} \left(\frac{p_\theta^2}{I} + \frac{p_\phi^2}{I \sin^2 \theta} \right) = \frac{p_\theta^2}{2I} + \frac{p_\phi^2}{2I \sin^2 \theta}.$$

Equation (12.10) with $x = p_\theta$ gives $\langle p_\theta^2 / I \rangle = k_B T$, and with $x = p_\phi$, at each θ , $\langle p_\phi^2 / (I \sin^2 \theta) \rangle = k_B T$; each term of H is half of these, so $\langle H \rangle = k_B T$.

EXERCISE 12.14. (a) $3 \times 3 = 9$, less 3 for the drift and 2 for the turning of a linear molecule: 4. (b) $3 \times 108 - 3 = 321$; ASE divides the same kinetic energy by 324, so its temperature is $321/324 = 0.9907$ of mdlab's.

EXERCISE 12.15. $\sqrt{2/N_f} = 0.01$ needs $N_f = 20000$, and $3N - 3 \geq 20000$ needs $N \geq 6667.7$: 6668 atoms.

EXERCISE 12.16. $\ln \mathcal{P} = (a - 1) \ln K - K/k_B T$ with $a = \frac{1}{2}N_f$, whose derivative $(a - 1)/K - 1/k_B T$ vanishes at $K = (a - 1)k_B T$. For one atom, $a = \frac{3}{2}$ and the peak is at $\frac{1}{2}k_B T$; the mean is $\frac{3}{2}k_B T$ because the density has a long tail of high energies, which pulls the mean above the peak.

EXERCISE 12.17. Above its minimum the potential energy of a harmonic crystal is a sum of squares of its normal coordinates, one for each of the N_f freedoms (the three drifts carry none), so by equipartition $\langle U \rangle = \frac{1}{2}N_f k_B T = \langle K \rangle$ and $C_U = C_K$. Then $C_V = 2C_K = N_f k_B$ and the bracket of Equation (12.14) is $1 - \frac{1}{2} = \frac{1}{2}$: the variance is half the canonical one, and the standard deviation $1/\sqrt{2}$ of the canonical one.

EXERCISE 12.18. $N_f = 1497$, and $\langle K \rangle = \frac{1}{2}N_f k_B T$ gives $k_B T = 2\langle K \rangle / N_f$. With $\langle \delta K^2 \rangle_E = (0.025\langle K \rangle)^2$,

$$\frac{2\langle \delta K^2 \rangle_E}{N_f(k_B T)^2} = \frac{2 \times 0.025^2 \langle K \rangle^2}{4\langle K \rangle^2 / N_f} = \frac{0.025^2 N_f}{2} = 0.4678.$$

Then $C_V = 748.5 k_B / (1 - 0.4678) = 1406 k_B$, or $2.81 k_B$ per atom.

EXERCISE 12.19. $\partial \ln Z_{NPT} / \partial P = Z_{NPT}^{-1} \int (-\beta V) e^{-\beta P V} Z dV = -\beta \langle V \rangle$, which is the first result. Differentiating again, by the product rule as in Section 12.4, $\partial^2 \ln Z_{NPT} / \partial P^2 = \beta^2 (\langle V^2 \rangle - \langle V \rangle^2)$, and $d\langle V \rangle / dP = -k_B T \partial^2 \ln Z_{NPT} / \partial P^2 = -\beta (\langle V^2 \rangle - \langle V \rangle^2)$, so the variance is $-k_B T d\langle V \rangle / dP$.

EXERCISE 12.20. $dF = -S dT - P dV + \mu dN$, so $d(F + PV) = dF + P dV + V dP = -S dT + V dP + \mu dN$, and $d(F - \mu N) = dF - \mu dN - N d\mu = -S dT - P dV - N d\mu$.

EXERCISE 12.21. $f(\mu + x) = 1/(e^{\beta x} + 1)$ and $1 - f(\mu - x) = 1 - 1/(e^{-\beta x} + 1) = e^{-\beta x} / (e^{-\beta x} + 1) = 1/(1 + e^{\beta x})$, the same. The slope is $-\beta e^{\beta(\epsilon_0 - \mu)} / (e^{\beta(\epsilon_0 - \mu)} + 1)^2$, which at $\epsilon_0 = \mu$ is $-\beta/4 = -1/(4k_B T)$. At 0.1 eV above μ , $f = 0.02047$, and $-[f \ln f + (1 - f) \ln(1 - f)] = 0.0999$: the entropy is $0.0999 k_B$, against $\ln 2 = 0.693$ for a half-filled orbital.

EXERCISE 12.22. A steady excess in one direction suggests a drift of the centre of mass along z , fault (c) of Section 12.9; the total momentum, or `centre_of_mass_path`, confirms it. With N atoms of mass m , $T_z = \sum m v_z^2 / (N k_B)$. Writing each $v_z = u_z + v_d$ with $\sum m u_z = 0$,

$$\sum m v_z^2 = \sum m u_z^2 + 2v_d \sum m u_z + N m v_d^2 = \sum m u_z^2 + N m v_d^2,$$

so T_z gains $m v_d^2 / k_B$ over the 295 K of the relative motion. For 15 K, with m in amu and the factor 103.6427 of Equation (3.15), $v_d = \sqrt{1.293 \times 10^{-3} / (39.948 \times 103.6427)} = 5.6 \times 10^{-4} \text{ \AA fs}^{-1}$.

Chapter 13: Thermostats

EXERCISE 13.1. The intervals are independent, so the probability of no collision in any of them is $(1 - \nu\delta t/n)^n$, whose logarithm $n \ln(1 - \nu\delta t/n)$ tends to $-\nu\delta t$ as n grows, by $\ln(1 + y) \approx y$ (Section 13.4): the probability tends to $e^{-\nu\delta t}$. At least one collision then has the probability $1 - e^{-0.002} = 0.001998$.

EXERCISE 13.2. $\lambda^2 = 1 + (1/500)(300/290 - 1) = 1.00006897$, so $\lambda = 1.00003448$: the velocities grow by 34 parts in a million.

EXERCISE 13.3. Each step adds the heat $(\delta t/\tau_T)(K_0 - K)$ (Section 13.2). Its average over the settled run is zero, so $\langle K \rangle = K_0$, and since the kinetic temperature is proportional to K , $\langle T \rangle = T_0$.

EXERCISE 13.4. Averaging the square of Equation (13.3) with $\sigma^2 = 2\gamma k_B T_0/m$ (Equation 13.4)), the noise having mean 0 and variance 1 and being independent of v , $\langle v'^2 \rangle = c^2 \langle v^2 \rangle + (1 - c^2)k_B T_0/m$, so each component's kinetic energy changes over the step by $(1 - c^2)(\frac{1}{2}k_B T_0 - \frac{1}{2}m\langle v^2 \rangle)$. For a short step $1 - c^2 = 1 - e^{-2\gamma\delta t} \approx 2\gamma\delta t$, so the rate is $2\gamma(\frac{1}{2}k_B T_0 - \frac{1}{2}m\langle v^2 \rangle)$, and over the $3N$ components, all of which the noise reaches, $2\gamma(K_0 - K)$ with $K_0 = \frac{3}{2}Nk_B T_0$; the three freedoms of the centre of mass change C_K by 0.4 %, which is negligible here. As in Section 13.2, $C_V dT/dt = -2\gamma C_K(T - T_0)$, with the time constant $C_V/(2\gamma C_K) = 1.590/(2 \times 1 \text{ ps}^{-1}) = 0.795 \text{ ps}$; the fit in Figure 13.2(b) gives 0.68 ps, uncertain by 0.22 ps in a single run, so the two agree.

EXERCISE 13.5. $\langle (\delta T/T_0)^2 \rangle / \tau_T = 0.005 \text{ ps}^{-1}$, so in 1000 ps the factor is $e^5 = 148$.

EXERCISE 13.6. The density of s adds up, over every x , the chance that the first number is x and the second $s - x$:

$$\mathcal{P}(s) = \frac{1}{2\pi\sqrt{ab}} \int e^{-x^2/2a - (s-x)^2/2b} dx.$$

The exponent is $-(a+b)x^2 - 2asx + as^2/2ab$, and completing the square,

$$(a+b)x^2 - 2asx + as^2 = (a+b)\left(x - \frac{as}{a+b}\right)^2 + \frac{ab s^2}{a+b},$$

so the exponent is $-(a+b)(x - x_0)^2/2ab - s^2/2(a+b)$, with $x_0 = as/(a+b)$. The Gaussian integral over x gives $\sqrt{2\pi ab/(a+b)}$, and

$$\mathcal{P}(s) = \frac{1}{\sqrt{2\pi(a+b)}} e^{-s^2/2(a+b)},$$

the Gaussian of variance $a + b$.

EXERCISE 13.7. The two terms of Equation (13.3) are independent, so their variances add, and multiplying a quantity by c multiplies its variance by c^2 : the variance after the step is $c^2\sigma_v^2 + (\sigma^2/2\gamma)(1 - c^2)$, with σ_v^2 the variance of v before. It equals σ_v^2 only if $\sigma_v^2(1 - c^2) = (\sigma^2/2\gamma)(1 - c^2)$, that is, since $c < 1$, if $\sigma_v^2 = \sigma^2/2\gamma$.

EXERCISE 13.8. With $c = e^{-0.01} = 0.99005$,

$$\sqrt{1 - c^2} \sqrt{k_B T / m} = 0.14072 \times 0.001676 \text{ Å fs}^{-1} = 2.36 \times 10^{-4} \text{ Å fs}^{-1},$$

a seventh of the thermal spread.

EXERCISE 13.9. With $\gamma = 0$, $c = 1$ and the noise vanishes, so O leaves the velocities unchanged; the two half drifts then make one drift, and B A A B is a half kick, a drift and a half kick, Equation (9.3).

EXERCISE 13.10. With $w = mv^2/2k_B T$, $dw = mv dv/k_B T$, so

$$\int_0^\infty v e^{-mv^2/2k_B T} dv = \frac{k_B T}{m} \int_0^\infty e^{-w} dw = \frac{k_B T}{m}.$$

Dividing by $\sqrt{2\pi k_B T / m}$, with $k_B T / m = \sqrt{k_B T / m} \sqrt{k_B T / m}$, gives $\sqrt{k_B T / m} / \sqrt{2\pi} = \sqrt{k_B T / 2\pi m}$. For lithium at 1000 K, with the factor 103.6427 of Equation (3.15),

$$\sqrt{\frac{0.086173}{2\pi \times 6.94 \times 103.6427}} = 0.00437 \text{ Å fs}^{-1}.$$

EXERCISE 13.11. In one dimension the edge is a point and the cell an interval, so $k_{\text{TST}} = \sqrt{k_B T / 2\pi m} e^{-U_b/k_B T} / \int e^{-U/k_B T} dx$. When $U_b \gg k_B T$ the integral comes almost wholly from the bottom of the well, where it is the Gaussian integral $\int e^{-m\omega^2 x^2/2k_B T} dx = \sqrt{2\pi k_B T / m\omega^2}$; dividing,

$$k_{\text{TST}} = \sqrt{\frac{k_B T}{2\pi m}} \sqrt{\frac{m\omega^2}{2\pi k_B T}} e^{-U_b/k_B T} = \frac{\omega}{2\pi} e^{-U_b/k_B T},$$

the frequency of attempts times the Boltzmann factor of the barrier. With $\omega/2\pi = 1/170.3 \text{ fs}$ and $e^{-0.3/0.025852} = 9.12 \times 10^{-6}$, $k_{\text{TST}} = 5.36 \times 10^{-8} \text{ fs}^{-1}$, or 5.4×10^7 hops per second. It is eleven times below the rate of Equation (13.5) for the surface at 300 K, 5.9×10^8 per second: a hollow has six bridges to leave by, and across the path each bridge has a third of the hollow's curvature, which widens the way out by a further $\sqrt{3}$, and $6\sqrt{3} = 10.4$.

EXERCISE 13.12. Without forces $\dot{K} = -2\zeta K$, so $\ddot{K} = -2\dot{\zeta}K - 2\zeta\dot{K}$. For small departures ζ is small and $K \approx K_0$, so to first order $d^2 \delta K / dt^2 = -2K_0 \dot{\zeta} = -2K_0 \cdot 2 \delta K / Q$, since $\dot{\zeta} = (2K - 2K_0) / Q$. With $K_0 = \frac{1}{2} N_f k_B T_0$ and $Q = N_f k_B T_0 \tau_T^2$, $4K_0 / Q = 2 / \tau_T^2$. The swing has the angular frequency $\sqrt{2} / \tau_T$ and the period $2\pi \tau_T / \sqrt{2} = 444$ fs for $\tau_T = 100$ fs.

EXERCISE 13.13. Hamilton's equations in t' are $d\mathbf{r}_i / dt' = \tilde{\mathbf{p}}_i / m_i s^2$, $d\tilde{\mathbf{p}}_i / dt' = \mathbf{F}_i$, $ds / dt' = p_s / Q$ and $dp_s / dt' = \sum_i \tilde{\mathbf{p}}_i^2 / m_i s^3 - N_f k_B T_0 / s$. With $d/dt = s d/dt'$:

$$\frac{d\mathbf{r}_i}{dt} = \frac{\tilde{\mathbf{p}}_i}{m_i s} = \frac{\mathbf{p}_i}{m_i},$$

$$\frac{d\mathbf{p}_i}{dt} = s \frac{d}{dt'} \frac{\tilde{\mathbf{p}}_i}{s} = \frac{d\tilde{\mathbf{p}}_i}{dt'} - \frac{\tilde{\mathbf{p}}_i}{s} \frac{ds}{dt'} = \mathbf{F}_i - \zeta \mathbf{p}_i,$$

$$\frac{d\zeta}{dt} = \frac{s}{Q} \frac{dp_s}{dt'} = \frac{s}{Q} \left(\sum_i \frac{\tilde{\mathbf{p}}_i^2}{m_i s^3} - \frac{N_f k_B T_0}{s} \right) = \frac{1}{Q} \left(\sum_i \frac{\mathbf{p}_i^2}{m_i} - N_f k_B T_0 \right) = \frac{2K - N_f k_B T_0}{Q},$$

the second line by the product rule, with $\zeta = p_s / Q$.

EXERCISE 13.14. $\dot{x} = p_x / m$ does not depend on x , and likewise for y and z : three zeros. $\dot{p}_x = F_x - \zeta p_x$ gives $\partial \dot{p}_x / \partial p_x = -\zeta$, and likewise for p_y and p_z . $\dot{\zeta} = (2K - 3k_B T_0) / Q$ does not depend on ζ . The divergence is -3ζ .

EXERCISE 13.15. R_1 has mean 0 and R_1^2 mean 1, and $R_\perp^2$ has the mean $N_f - 1$, so $\langle \lambda^2 \rangle = c_1 + c_2 N_f = c_1 + (1 - c_1) K_0 / K$. Then $\langle K' \rangle = \langle \lambda^2 \rangle K = c_1 K + (1 - c_1) K_0$, and $\langle K' \rangle - K = (1 - c_1)(K_0 - K)$.

EXERCISE 13.16. In the canonical ensemble each mass-weighted velocity component u_k has a density proportional to $e^{-u_k^2 / 2k_B T}$ (Section 12.5), which with $k_B T = 1$ is the normal distribution of variance 1; so n independent normal numbers are such components, and $K = \frac{1}{2} \sum u_k^2$ has the density Equation (12.12) with $N_f = n$, proportional to $K^{n/2-1} e^{-K}$: the gamma distribution of shape $n/2$. The sum of squares is $2K$.

EXERCISE 13.17. A hundred strides of 5 fs is $\tau_T = 500$ fs, and $c_1 = e^{-5/500} = 0.9900$; for paracetamol, ten strides of 7 fs make 70 fs.

EXERCISE 13.18. $x_{\text{com}} = \int_0^t v_{\text{com}}(t_1) dt_1$, and its square is the product of two copies, $\int_0^t v_{\text{com}}(t_1) dt_1 \int_0^t v_{\text{com}}(t_2) dt_2$, a sum over every pair of times; the average of a sum is the sum of the averages, so

$$\begin{aligned} \langle x_{\text{com}}^2 \rangle &= \int_0^t \int_0^t \langle v_{\text{com}}(t_1) v_{\text{com}}(t_2) \rangle dt_1 dt_2 \\ &= \frac{2k_B T}{M} \int_0^t dt_1 \int_0^{t_1} e^{-\gamma(t_1-t_2)} dt_2 \quad \text{pairs with } t_2 > t_1 \text{ repeat those with } t_2 < t_1 \\ &= \frac{2k_B T}{M} \int_0^t \frac{1 - e^{-\gamma t_1}}{\gamma} dt_1 \quad \text{since } \int_0^{t_1} e^{\gamma t_2} dt_2 = (e^{\gamma t_1} - 1)/\gamma \\ &= \frac{2k_B T}{M\gamma} \left(t - \frac{1 - e^{-\gamma t}}{\gamma} \right). \end{aligned}$$

For $\gamma t \gg 1$ the bracket is $t - 1/\gamma \approx t$, so $D = k_B T / (M\gamma)$; this is one component, and the three together give the $6Dt$ of [Section 13.1](#). For 256 argon atoms, $M = 10\,227$ amu, with $k_B T = 0.011\,633$ eV at 135 K, $\gamma = 1 \times 10^{-4} \text{ fs}^{-1}$ and the factor 103.6427 of [Equation \(3.15\)](#): $0.011633 / (10227 \times 103.6427 \times 10^{-4}) = 1.10 \times 10^{-4} \text{ \AA}^2 \text{ fs}^{-1}$.

EXERCISE 13.19. The true count is $3N - n_c - 3 = 576 - 192 - 3 = 381$ ([Section 12.6](#)). The thermostat drives $2K / (576k_B)$ to 300 K, while the true temperature is $2K / (381k_B)$, larger by $576/381 = 1.512$: 453.5 K.

EXERCISE 13.20. Berendsen coupling with $\tau_T = 0.1$ ps narrows the fluctuations of the kinetic energy to 0.45 of the canonical spread, and those of the total energy with them, while [Equation \(12.6\)](#) holds only for canonical fluctuations. The heat capacity, which goes as the square of the energy's spread, comes out too small: in this chapter's runs, $0.41Nk_B$ against $2.376Nk_B$. CSVR, a Nosé-Hoover chain or Langevin dynamics give canonical fluctuations, and with them 2.28 to $2.40Nk_B$.

EXERCISE 13.21. The noise does not keep the total momentum, so the centre of mass diffuses, at $k_B T / (M\gamma) = 1.10 \times 10^{-4} \text{ \AA}^2 \text{ fs}^{-1}$ for this box over times long against $1/\gamma = 10$ ps ([Exercise 13.18](#)), and displacements measured in the cell include it. Over the fitted lags of 2 to 20 ps, γt is only 0.2 to 2, and the formula of [Exercise 13.18](#) rises there with 0.64 of its long-time slope, so the centre of mass should add about $0.64 \times 1.10 = 0.70$: $3.69 + 0.70 \approx 4.4$, against the 4.62 measured, 1.2 times the 3.81 at fixed energy. The centre of mass is one particle, so its share scatters widely, 0.55, 1.30 and 0.95 in the three runs. Measuring displacements from the centre of mass gives 3.69, 0.97 of the value at fixed energy, as friction requires.

Chapter 14: Pressure and barostats

EXERCISE 14.1. One electronvolt is $1.602\,177 \times 10^{-19}$ J and one cubic ångström is 10^{-30} m³, so $1 \text{ eV } \text{Å}^{-3}$ is $1.602\,177 \times 10^{11} \text{ Pa} = 160.218 \text{ GPa}$. One bar, 10^5 Pa , is then $10^5 / (1.602177 \times 10^{11}) = 6.2415 \times 10^{-7} \text{ eV } \text{Å}^{-3}$.

EXERCISE 14.2. $N/V = P/k_B T$ with $P = 6.2415 \times 10^{-7} \text{ eV } \text{Å}^{-3}$ and $k_B T = 0.025\,852 \text{ eV}$: $2.414 \times 10^{-5} \text{ Å}^{-3}$. Each molecule has $V/N = 41\,419 \text{ Å}^3$ to itself, a cube of side 34.6 Å .

EXERCISE 14.3. Of the atoms with a given $v_x > 0$, those within $v_x t$ of the wall reach it in the time t : the box of area A and depth $v_x t$ holds $\rho A v_x t$ atoms, of which the fraction $\mathcal{P}(v_x) dv_x$ have that velocity. Each delivers $2m v_x$, so the momentum delivered in t by all the atoms moving towards the wall is

$$\int_0^\infty 2m v_x \rho A v_x t \mathcal{P}(v_x) dv_x = 2\rho A t m \int_0^\infty v_x^2 \mathcal{P}(v_x) dv_x.$$

$\mathcal{P}(v_x)$ is symmetric about zero, and so is v_x^2 , so the integral over $v_x > 0$ is half that over all v_x , which is $\langle v_x^2 \rangle$; the momentum is $\rho A t m \langle v_x^2 \rangle$. Dividing by $A t$, $P = \rho m \langle v_x^2 \rangle$, and $m \langle v_x^2 \rangle = k_B T$ by equipartition, so $P = \rho k_B T$.

EXERCISE 14.4. With $\varphi = 4\epsilon[(\sigma/r)^{12} - (\sigma/r)^6]$, the slope is $\varphi' = 4\epsilon[-12\sigma^{12}/r^{13} + 6\sigma^6/r^7]$, and multiplying by $-r$,

$$-r\varphi'(r) = 4\epsilon[12(\sigma/r)^{12} - 6(\sigma/r)^6].$$

This is zero where $12(\sigma/r)^{12} = 6(\sigma/r)^6$, that is $(\sigma/r)^6 = \frac{1}{2}$, at $r = 2^{1/6}\sigma = 2^{1/6} \times 3.4 = 3.816 \text{ Å}$, the minimum of φ . Closer in, σ/r is larger, the first term wins and the contribution is positive: the pair pushes outwards and raises the pressure.

EXERCISE 14.5. Let $F_\perp$ be the total force the atoms exert on one wall, so that the wall pushes them back with $-F_\perp$ in all. The atoms touching the wall at $x = L/2$ have $x \approx L/2$ and feel a total force $-F_\perp$ along x , contributing $-(L/2)F_\perp$; those touching the wall at $x = -L/2$ have $x \approx -L/2$ and feel $+F_\perp$, contributing $-(L/2)F_\perp$ too. On average $F_\perp = PA$, so the pair of walls gives $-LPA = -PV$, and the three pairs $-3PV$, as before.

EXERCISE 14.6. Wrapping moves $\mathbf{r}_i$ by $\pm L$ along x and changes no force, since the forces depend on separations to the nearest copy, so $\sum_j \mathbf{r}_j \cdot \mathbf{F}_j$ changes by $\pm L F_{ix}$, which is not zero in general. Equation (8.4) uses only the separations r_{ij} to the nearest copy, which wrapping does not change.

EXERCISE 14.7. The columns of $\mathbf{I} + \epsilon$ are $\mathbf{c}_1 = (1 + \epsilon_{xx}, \epsilon_{yx}, \epsilon_{zx})$, $\mathbf{c}_2 = (\epsilon_{xy}, 1 + \epsilon_{yy}, \epsilon_{zy})$ and $\mathbf{c}_3 = (\epsilon_{xz}, \epsilon_{yz}, 1 + \epsilon_{zz})$. The x component of $\mathbf{c}_2 \times \mathbf{c}_3$ is $(1 + \epsilon_{yy})(1 + \epsilon_{zz}) - \epsilon_{zy}\epsilon_{yz}$, which is $1 + \epsilon_{yy} + \epsilon_{zz}$ plus products of two components; its y and z components, $\epsilon_{zy}\epsilon_{xz} - \epsilon_{xy}(1 + \epsilon_{zz})$ and $\epsilon_{xy}\epsilon_{yz} - (1 + \epsilon_{yy})\epsilon_{zx}$, are each one component plus products of two. In the dot product with $\mathbf{c}_1$, the x components give $(1 + \epsilon_{xx})(1 + \epsilon_{yy} + \epsilon_{zz} + \dots) = 1 + \epsilon_{xx} + \epsilon_{yy} + \epsilon_{zz}$ plus products of two or three components, while the y and z components are multiplied by ϵ_{yx} and ϵ_{zx} and give only products of two or more. To first order the determinant is $1 + \epsilon_{xx} + \epsilon_{yy} + \epsilon_{zz}$.

EXERCISE 14.8. The trace of $\sum_i m_i v_{i\alpha} v_{i\beta}$ is $\sum_i m_i v_i^2 = 2K$, and that of $\mathcal{W}_{\alpha\beta}$ is $\mathcal{W}$, so one third of the trace of Equation (14.3) is $(2K + \mathcal{W})/3V$.

EXERCISE 14.9. An even squeeze $\epsilon_{xx} = \epsilon_{yy} = \epsilon_{zz} = e$ changes P_{xx} by $-C_{11}e$ from its own strain and by $-C_{12}e$ from each of the other two, $-(C_{11} + 2C_{12})e$ in all, and likewise each diagonal component, so $dP = -(C_{11} + 2C_{12})e$. The volume changes by $dV = 3Ve$ (Exercise 14.7), so $B = -V dP/dV = (C_{11} + 2C_{12})/3 = (3.920 + 2 \times 2.324)/3 = 2.856$ GPa.

EXERCISE 14.10. For $n = 0$, $\int_0^\infty e^{-bV} dV = 1/b$. By parts, with $f = V^n$ and $g' = e^{-bV}$,

$$\int_0^\infty V^n e^{-bV} dV = \left[-\frac{V^n e^{-bV}}{b} \right]_0^\infty + \frac{n}{b} \int_0^\infty V^{n-1} e^{-bV} dV = \frac{n}{b} \frac{(n-1)!}{b^{n-1}} = \frac{n!}{b^{n+1}},$$

using the result for $n - 1$. The bracket vanishes at both ends: at $V = 0$ because $V^n = 0$ for $n \geq 1$, and as $V \rightarrow \infty$ because the exponential falls faster than any power grows.

EXERCISE 14.11. $k_B T/P_0 = 0.011633/10^{-4} = 116.3 \text{ \AA}^3$, so $\langle V \rangle = 21 \times 116.3 = 2443 \text{ \AA}^3$ and the spread is $\sqrt{21} \times 116.3 = 533 \text{ \AA}^3$. $Nk_B T/P_0 = 2327 \text{ \AA}^3$, 4.8% less.

EXERCISE 14.12. Each step changes $\ln V$ in proportion to $P_0 - P$, so once the volume has settled the average of $P_0 - P$ is zero: $\langle P_0 - 2K/3V \rangle = 0$. With K independent of V , $\langle 2K/3V \rangle = \frac{2}{3} \langle K \rangle \langle 1/V \rangle$, and for a narrow volume $\langle 1/V \rangle \approx 1/\langle V \rangle$. So $P_0 = \frac{2}{3} \langle K \rangle / \langle V \rangle = Nk_B T / \langle V \rangle$, and $\langle V \rangle \approx Nk_B T / P_0$.

EXERCISE 14.13. The volume relaxes in $\tau_P \kappa_T / \kappa$ (Section 14.6): $1 \times 1.46/2.5 = 0.58$ ps with $\kappa = 2.5 \text{ GPa}^{-1}$, and 10^4 times longer, 5.8 ns, with κ entered 10^4 times too small.

EXERCISE 14.14. Since 1 GPa^{-1} is $160.218 \text{ \AA}^3 \text{ eV}^{-1}$ (Exercise 14.1), $\kappa_T = 1.46 \times 160.218 = 233.9 \text{ \AA}^3 \text{ eV}^{-1}$, and the variance of $\ln V$ is $k_B T \kappa_T / V = 0.011633 \times 233.9/12306 = 2.21 \times 10^{-4}$, a spread of 0.0149, 1.5% of the volume.

EXERCISE 14.15. $K = \sum_i p_i^2 / 2m_i = V^{-2/3} \sum_i \pi_i^2 / 2m_i$, and at fixed π_i only the power of V changes: $\partial K / \partial V = -\frac{2}{3} V^{-5/3} \sum_i \pi_i^2 / 2m_i = -2K/3V$.

EXERCISE 14.16. At fixed temperature $V = Nk_B T/P$, so $dV/dP = -Nk_B T/P^2 = -V/P$ and $\kappa_T = -(1/V) dV/dP = 1/P$.

EXERCISE 14.17. About the mean $\bar{V}$, $P - P_0 \approx -(V - \bar{V})/(\bar{V}\kappa_T)$, so with $\delta V = V - \bar{V}$

$$M_V \delta \ddot{V} = -\frac{\delta V}{\bar{V}\kappa_T},$$

an oscillator with $\omega^2 = 1/(M_V \bar{V}\kappa_T)$.

EXERCISE 14.18. $\omega^2 = 9V/[(3N+3)k_B T \tau_P^2 \kappa_T] = 3V/[(N+1)k_B T \kappa_T \tau_P^2]$, so $2\pi/\omega = 2\pi \tau_P \sqrt{(N+1)k_B T \kappa_T / 3V}$. With $N = 256$, $k_B T = 0.011633$ eV, $\kappa_T = 1.46$ GPa⁻¹ = 233.9 Å³ eV⁻¹ (Exercise 14.14) and $V = 12306$ Å³,

$$\frac{(N+1)k_B T \kappa_T}{3V} = \frac{257 \times 0.011633 \times 233.9}{3 \times 12306} = 0.01894,$$

and the factor is $2\pi\sqrt{0.01894} = 0.865$.

EXERCISE 14.19. $1.108^{1/3} = 1.0348$: each length would grow by 3.48%. The sheets, 6.7 times stiffer along their plane than across it in the model of Section 14.8, would be stretched by 3.48%, and the gaps would open by only about a third of the 10.8%.

EXERCISE 14.20. The true pressure rises by $Nk_B T/V$, which is 0.0388 GPa at the healthy volume of 12312 Å³, so the volume shrinks by about $\kappa_T Nk_B T/V = 1.46 \times 0.0388 = 0.057$, 5.7%. The run shrinks by only 4.9%, to 11706 Å³, because the liquid grows stiffer as it is squeezed: between 11706 and 12112 Å³ its pressure rises by 7.46×10^{-5} GPa for each Å³ lost, against 5.76×10^{-5} GPa between 12112 and 12301 Å³ (the runs at fixed volume of Section 14.5), so κ_T taken at 0.1 GPa overestimates the response to a larger squeeze.

EXERCISE 14.21. Berendsen's barostat squeezes the fluctuations of the volume, so Equation (14.4) gives a compressibility too small, 0.64 against 1.46 GPa⁻¹ for the liquid of this chapter. Stochastic cell rescaling, or a piston with Nosé-Hoover chains, samples the volume's fluctuations correctly.

Chapter 15: Equilibration, convergence and uncertainty

EXERCISE 15.1. The half-width is $1.96 \sigma_x / \sqrt{n}$ with $\sigma_x^2 = 35/12$, so

$$n \geq \frac{1.96^2 \times 35/12}{0.01^2} = 112\,046.7,$$

which is 112 047 rolls.

EXERCISE 15.2. $2.776/1.96 = 1.416$: the interval is 41.6% wider. With five samples s is itself uncertain, and a run whose s happens to be small would otherwise claim too narrow an interval; the wider factor restores 95% for Gaussian samples (Section 15.1).

EXERCISE 15.3. With $\delta x = x - \langle x \rangle$ and $\delta y = y - \langle y \rangle$, the deviation of $x - y$ from its mean is $\delta x - \delta y$, and

$$\text{var}(x - y) = \langle (\delta x - \delta y)^2 \rangle = \langle \delta x^2 \rangle + \langle \delta y^2 \rangle - 2\langle \delta x \delta y \rangle,$$

expanding the square and averaging term by term; the last average is $\text{cov}(x, y)$. In the run of Section 12.6, at fixed energy, kinetic energy passes back and forth between the two species, so when one is warmer the other tends to be cooler: over the last 10 ps the two temperatures have the correlation -0.50 , and their difference fluctuates more than $\text{var}(x) + \text{var}(y)$ suggests, 145.1 K^2 against 47.2 and 49.6 K^2 . Combining the two errors as for independent quantities gives 1.73 K , too small against the 2.14 K of the gap's own series; taking the gap at each frame as the series lets its own spread and its own g measure it, whatever the sign of the covariance.

EXERCISE 15.4. z_k is independent of x_k , so the variances of the two terms add, and a factor a multiplies a variance by a^2 :

$$\text{var}(x_{k+1}) = c^2 \text{var}(x_k) + (1 - c^2), \quad 1 - \text{var}(x_{k+1}) = c^2 [1 - \text{var}(x_k)],$$

the second by subtracting the first from 1. Starting from $\text{var}(x_0) = 0$, the shortfall from 1 is multiplied by c^2 at each step, so $1 - \text{var}(x_k) = c^{2k}$. It falls below 0.01 when $2k \ln(1/c) > \ln 100$, $k > 44.9$ for $c = 0.95$: 45 steps, 450 fs, a little over two correlation times.

EXERCISE 15.5. $g = (1 + c)/(1 - c) = 1.99/0.01 = 199$ and $\tau_{\text{int}} = g\delta t/2 = 995 \text{ fs}$, equal to $\tau_c = -\delta t/\ln c = 995.0 \text{ fs}$ to the figures shown. A run of 1 ns holds 10^5 samples, worth $10^5/199 = 502.5$ independent ones.

EXERCISE 15.6. Let $S = \sum_{k \geq 1} kc^k$. Then

$$(1 - c)S = \sum_{k \geq 1} kc^k - \sum_{k \geq 1} kc^{k+1} = \sum_{k \geq 1} kc^k - \sum_{k \geq 2} (k-1)c^k = \sum_{k \geq 1} c^k = \frac{c}{1-c},$$

renaming $k+1$ as k in the second sum, which then runs from $k=2$, or from $k=1$ since that term is zero; at each power the two leave $kc^k - (k-1)c^k = c^k$. Dividing by $1-c$, $S = c/(1-c)^2$.

EXERCISE 15.7. Integrating by parts as in Section 12.1, with u^3 as the factor differentiated and $u e^{-u^2/(2\sigma_u^2)}$ as the factor integrated, to $-\sigma_u^2 e^{-u^2/(2\sigma_u^2)}$,

$$\begin{aligned} \int u^4 e^{-u^2/(2\sigma_u^2)} du &= \left[-\sigma_u^2 u^3 e^{-u^2/(2\sigma_u^2)} \right]_{-\infty}^{\infty} + 3\sigma_u^2 \int u^2 e^{-u^2/(2\sigma_u^2)} du \\ &= 0 + 3\sigma_u^2 \times \sigma_u^2 \sigma_u \sqrt{2\pi}, \end{aligned}$$

and dividing by $\sigma_u \sqrt{2\pi}$, $\langle u^4 \rangle = 3\sigma_u^4$. The u_i^2 are independent, each with mean σ_u^2 and variance $\langle u^4 \rangle - \langle u^2 \rangle^2 = 2\sigma_u^4$, so their mean s'^2 has the variance $2\sigma_u^4/n$ (Equation (15.1)), a relative spread of $\sqrt{2/n}$. A small relative change of s'^2 changes its square root by half as much, since $d\sqrt{y}/\sqrt{y} = \frac{1}{2}dy/y$, so s' has the relative error $1/\sqrt{2n}$. Measuring the deviations from the samples' own mean leaves $n-1$ free deviations, as for Student's density (Section 15.1), and for Gaussian samples n becomes $n-1$, a result quoted here without proof and confirmed by the blocks of Figure 15.4.

EXERCISE 15.8. Blocks of at least $5g = 300$ samples miss about 5% (Equation (15.4)). The run of 65 536 samples holds 218 of them, and their spread gives the error to $1/\sqrt{2 \times 217} = 0.048$. With the doubled lengths of Flyvbjerg and Petersen the first length long enough is 512, giving 128 blocks and 0.063.

EXERCISE 15.9. The mean of the offset over the samples from t_0 to t_f is

$$\frac{1}{t_f - t_0} \int_{t_0}^{t_f} \delta A_0 e^{-t/\tau_r} dt = \frac{\delta A_0 \tau_r}{t_f - t_0} \left[-e^{-t/\tau_r} \right]_{t_0}^{t_f} = \frac{\delta A_0 \tau_r (e^{-t_0/\tau_r} - e^{-t_f/\tau_r})}{t_f - t_0}.$$

With the values given, $3 \times 2 \times (e^{-2.67} - e^{-25})/44.66 = 0.0093$, as Section 15.4 found.

EXERCISE 15.10. $\bar{t} = (n+1)\delta t/2$, since $\sum_i i = n(n+1)/2$ (the first and last terms pair to $n+1$, in $n/2$ pairs), and $\sum_i (t_i - \bar{t})^2 = \sum_i t_i^2 - n\bar{t}^2$, by the expansion of Section 15.1 with 0 in place of $\langle x \rangle$. So

$$\begin{aligned} \frac{1}{\delta t^2} \sum_i (t_i - \bar{t})^2 &= \frac{n(n+1)(2n+1)}{6} - \frac{n(n+1)^2}{4} \\ &= \frac{n(n+1)[2(2n+1) - 3(n+1)]}{12} = \frac{n(n+1)(n-1)}{12}, \end{aligned}$$

since $2(2n+1) - 3(n+1) = n-1$; this is $n(n^2-1)/12$.

EXERCISE 15.11. The limit is $2\sigma_{\bar{T}}C_V/(Nt_f)$, and $\sigma_{\bar{T}}$ falls as $t_f^{-1/2}$, so the limit falls as $t_f^{-3/2}$: for 10 ns it is $20.8 \times 10^{-3/2} = 0.658 \mu\text{eV}$ per atom per nanosecond.

EXERCISE 15.12. $F = -kx$, so the step gives $x' = x - hkx = (1 - hk)x$ and $U' = (1 - hk)^2 U$. This is below U when $(1 - hk)^2 < 1$, that is $-1 < 1 - hk < 1$, or, subtracting 1 and multiplying by -1 , $0 < hk < 2$; for $h = 1/k$, $x' = 0$, the minimum. A longer step overshoots, and beyond $h = 2/k$ each step lands farther up the other side.

EXERCISE 15.13. From Equation (15.5), under CSV and at fixed energy,

$$t_f \geq \frac{2 \times 940 \text{ fs} \times 0.01328^2}{0.0005^2} = 1.33 \text{ ns}, \quad t_f \geq \frac{2 \times 123 \text{ fs} \times 0.009613^2}{0.0005^2} = 0.091 \text{ ns}.$$

EXERCISE 15.14. After the start 140 ps hold $140/(2 \times 2) = 35$ independent samples, short of 100. A hundred need 400 ps after the start, a run of 410 ps, 260 ps longer. Its start, at 10 ps, passes the first check.

EXERCISE 15.15. With $Z_i = \int_0^\infty E^{a-1} e^{-\beta_i E} dE$ the normalising integral at β_i , $\ln[\mathcal{P}(E | \beta_2)/\mathcal{P}(E | \beta_1)] = \ln(Z_1/Z_2) + (a-1)\ln E - (a-1)\ln E - \beta_2 E + \beta_1 E$, in which the factors E^{a-1} cancel: a straight line with the slope $\beta_1 - \beta_2$. For 300 and 310 K it is $(1/300 - 1/310)/k_B = 1.2478 \text{ eV}^{-1}$.

EXERCISE 15.16. $k_B T$ is an energy, $\text{kg m}^2 \text{s}^{-2}$; a viscosity is a force per area, a pascal or $\text{kg m}^{-1} \text{s}^{-2}$, divided by the rate u/d , s^{-1} , which gives $\text{kg m}^{-1} \text{s}^{-1}$; so $k_B T/(\eta_s L)$ has the units $\text{kg m}^2 \text{s}^{-2}$ over kg s^{-1} , which is $\text{m}^2 \text{s}^{-1}$, those of D . A product $(k_B T)^a \eta_s^b L^c$ has the units $\text{kg}^{a+b} \text{m}^{2a-b+c} \text{s}^{-2a-b}$; matching $\text{m}^2 \text{s}^{-1}$ needs $a+b=0$, $2a-b+c=2$ and $2a+b=1$, so $a=1$, $b=-1$, $c=-1$, and no other choice.

EXERCISE 15.17. $k_B T = 0.011633 \times 1.602177 \times 10^{-19} = 1.8638 \times 10^{-21} \text{ J}$, so the correction is

$$\frac{2.837297 \times 1.8638 \times 10^{-21}}{6\pi \times 2.18 \times 10^{-4} \times 23.26 \times 10^{-10}} = 5.53 \times 10^{-10} \text{ m}^2 \text{s}^{-1},$$

and since $1 \text{ \AA}^2 \text{ fs}^{-1}$ is $10^{-5} \text{ m}^2 \text{ s}^{-1}$, that is $0.553 \times 10^{-4} \text{ \AA}^2 \text{ fs}^{-1}$, 14% of the D of that box.

EXERCISE 15.18. Single samples drawn at random lose the order of the run, and the resampled runs have no memory: the spread of their statistic is that of n independent samples, too small by about $\sqrt{g}$. Blocks keep the memory within each; they must be several correlation times long, as for block averages, so that neighbouring blocks are nearly independent, $5g$ samples or more.

EXERCISE 15.19. A heat capacity is a variance, and a flat running mean of K says nothing about the error of its variance. It has shown neither the error of C_V nor that the run has enough independent samples to know it. The block bootstrap of the heat capacity gives it, 2.9 to 3.9% for a run of 50 ps (Section 15.6), and $(2.292 \pm 0.067) Nk_B$ for the run near 130 K; the criterion applied to $(K - \bar{K})^2$, whose mean is the variance, and a second run would show whether that error is to be trusted.

Chapter 16: Observables

EXERCISE 16.1. The exact volume is $\frac{4}{3}\pi[(r + \delta r)^3 - r^3] = 4\pi r^2 \delta r(1 + \delta r/r + \delta r^2/3r^2)$, so the first term falls short by the fraction $1 - 1/(1 + x) = x/(1 + x)$, with $x = \delta r/r + \delta r^2/3r^2$, which is close to x , and so to $\delta r/r$, when $\delta r \ll r$. Numerically, at $1 \text{ \AA}$ it is 0.0484 and at $5 \text{ \AA}$ 0.0099, against $\delta r/r = 0.05$ and 0.01.

EXERCISE 16.2. $1 - 1/256 = 0.99609$ and $1 - 1/2048 = 0.99951$. Dividing by $N(N - 1)/V$ instead of N^2/V makes it approach 1. The coordination number counts partners, $\rho g(r)$ times the volume of each shell; the new divisor scales $g(r)$ up by $N/(N - 1)$ and goes with the partners' density $(N - 1)/V$, smaller by the same factor, so the count is unchanged. Used with $\rho = N/V$ it would add $12/(N - 1) = 0.05$ to the 12 of the first shell for 256 atoms. In a liquid in a large enough cell the far value under ASE's normalisation is $1 - S(0)/N$ instead, within 3×10^{-5} of 1 for 2048 atoms of argon.

EXERCISE 16.3. With an atom at a corner as the origin, every atom of the lattice sits at $(a/2)(n_1, n_2, n_3)$ with whole numbers whose sum is even: all three even at the corners, two odd and one even at the face centres. Its squared distance is $(a^2/4)(n_1^2 + n_2^2 + n_3^2)$, and the sum of squares is then even. The smallest value, 2, needs two of the n equal to ± 1 and the third 0: three places for the 0 times four choices of signs, 12 atoms at $\sqrt{2}a/2 = a/\sqrt{2}$. The next, 4, cannot come from $(\pm 1, \pm 1, \pm 1)$, whose sum is odd, so it needs one of them ± 2 and the others 0: 6 atoms at the distance a .

EXERCISE 16.4. With $g = 1$ beyond σ ,

$$\frac{U}{N} = \frac{\rho}{2} \int_{\sigma}^{\infty} 4\epsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right] 4\pi r^2 dr = 8\pi\rho\epsilon\sigma^3 \left(\frac{1}{9} - \frac{1}{3} \right) = -\frac{16}{9}\pi\rho\sigma^3\epsilon,$$

since $\int_{\sigma}^{\infty} \sigma^{12} r^{-10} dr = \sigma^3/9$ and $\int_{\sigma}^{\infty} \sigma^6 r^{-4} dr = \sigma^3/3$. With $\rho\sigma^3 = 0.8$ it is -46.20 meV, 9.97 meV above the value from the real $g(r)$. The difference measures the liquid's structure: the real $g(r)$ crowds partners into the first shell, where the potential is most negative. The run's own -50.09 meV lies higher than -56.16 meV because its switched potential leaves out the attraction beyond 2σ ; with that potential the mean field itself gives -40.07 meV, and the structure again about 10 meV.

EXERCISE 16.5. $S(\mathbf{G}) = 1 + (1/N) \sum_i \sum_{j \neq i} \cos(\mathbf{G} \cdot \mathbf{r}_{ij})$. For independent positions spread evenly over the cell, each cosine averages over the cell to zero for $\mathbf{G} \neq \mathbf{0}$ (Section 10.3), leaving $S(\mathbf{G}) = 1$. In a perfect crystal at a reciprocal lattice vector every $\mathbf{G} \cdot \mathbf{r}_{ij}$ is a multiple of 2π , every cosine is 1 , and $S(\mathbf{G}) = 1 + N(N-1)/N = N$.

EXERCISE 16.6. $S(0) = \rho k_B T \kappa_T$ with $\kappa_T = 0.5 \times 160.2 = 80.1 \text{ \AA}^3 \text{ eV}^{-1}$, so $S(0) = 0.03 \times 0.02585 \times 80.1 = 0.0621$. Twice as compressible, twice $S(0)$: a softer liquid lets its density fluctuate more, and scatters long waves more strongly.

EXERCISE 16.7. Each period carries the pair out across the cutoff and back, two events, so $2 \times 1000/60 = 33.3$ events per picosecond. With r_{on} at the old cutoff the bond, once formed, breaks only if the pair passes r_{off} , so the band records none if it is wider than the part of the swing beyond the cutoff. Wherever the cutoff falls within the vibration, a band wider than the whole swing, from its near end to its far end, records none, since no swing can then carry the pair across both distances.

EXERCISE 16.8. With $N = 6$ the first frame's keys are $0 \cdot 6 + 1 = 1$, $1 \cdot 6 + 2 = 8$ and $3 \cdot 6 + 4 = 22$; the second's are 1 , 15 , 22 and 29 . Merging the sorted lists $(1, 8, 22)$ and $(1, 15, 22, 29)$: 1 is in both; 8 is smaller than 15 and only in the first, broken; 15 is smaller than 22 and only in the second, formed; 22 is in both; 29 is left in the second, formed. So $2-3$ and $4-5$ form and $1-2$ breaks. The search starting at atom o finds $\{0, 1, 2\}$, then $\{3, 4\}$, then $\{5\}$ in the first frame, and $\{0, 1\}$ and $\{2, 3, 4, 5\}$ in the second.

EXERCISE 16.9. In the prism each atom is joined to the two others of its triangle and to one atom of the other; in the hexagon each is joined to its two neighbours round the ring and to the opposite atom. Every atom has three neighbours, all of the same element. In the first round every atom starts with the same number, and its neighbours' sum is three times the same scrambled number, so every atom of both gets the same new number; the same holds in every later round, and the sums over the molecules are equal. The molecules differ: the prism has triangles of bonds, and the hexagon with its cross-links has none. Its atoms alternate round the ring between two sets of three, and

each bond, round the ring or across it, joins one set to the other, so any closed path of bonds returns to its start after an even number of steps.

EXERCISE 16.10. The relative error is $1/\sqrt{n}$, so 2% needs $1/0.02^2 = 2500$ events. Forty events over $100 \times 1000 \text{ ps} = 1 \times 10^5$ molecule-picooseconds give $(4.0 \pm 0.6) \times 10^{-4}$ per picosecond, the error $\sqrt{40}/10^5$.

EXERCISE 16.11. After $n = k_{\text{h}}t$ hops $x = \sum_k s_k$ with $s_k = \pm a$ independent, so $\langle x^2 \rangle = \sum_k \langle s_k^2 \rangle = na^2 = k_{\text{h}}a^2t$. Comparing with $2Dt$ for $d = 1$, $D = k_{\text{h}}a^2/2$.

EXERCISE 16.12. The probability $\frac{1}{2}(1+c)$ of repeating and $\frac{1}{2}(1-c)$ of reversing give $\langle s_k s_{k+1} \rangle = a^2[\frac{1}{2}(1+c) - \frac{1}{2}(1-c)] = ca^2$, and each step depends on the previous alone, so $\langle s_k s_{k+m} \rangle = c^m a^2$. Then, as for Equation (15.3),

$$\langle x^2 \rangle = \sum_k \sum_l \langle s_k s_l \rangle = a^2 \left[n + 2 \sum_{m \geq 1} (n-m) c^m \right] \rightarrow na^2 \left[1 + \frac{2c}{1-c} \right] = na^2 \frac{1+c}{1-c}$$

for many steps, so $f = (1+c)/(1-c)$. Solving $(1+c)/(1-c) = 3.46$ gives $c = (3.46 - 1)/(3.46 + 1) = 0.552$.

EXERCISE 16.13. The motion stops being ballistic near $t = 2Dm/k_{\text{B}}T$. With $1 \text{ amu} = 103.64 \text{ eV fs}^2 \text{ \AA}^{-2}$, $m = 18.015 \times 103.64$ in those units, and $k_{\text{B}}T = 0.02620 \text{ eV}$, this is about 75 fs.

EXERCISE 16.14. Writing τ for τ_c in this solution, with $C_{vv}(s) = C_0 e^{-s/\tau}$,

$$\begin{aligned} 2 \int_0^t (t-s) C_0 e^{-s/\tau} ds &= 2C_0 [t\tau(1 - e^{-t/\tau}) - \tau^2 + (\tau^2 + t\tau)e^{-t/\tau}] \\ &= 2C_0 \tau [t - \tau(1 - e^{-t/\tau})], \end{aligned}$$

using $\int_0^t s e^{-s/\tau} ds = \tau^2 - (\tau^2 + t\tau)e^{-t/\tau}$, by parts. For $t \ll \tau$, $1 - e^{-t/\tau} = t/\tau - t^2/2\tau^2 + \dots$, and the bracket is $t^2/2\tau$, so the MSD is $C_0 t^2$, the ballistic form with $C_0 = 3k_{\text{B}}T/m$. For $t \gg \tau$ the bracket is $t - \tau$, the MSD grows as $2C_0 \tau t$, and $6D = 2C_0 \tau$ gives $D = C_0 \tau/3$, which is also $\frac{1}{3} \int_0^\infty C_{vv}$.

EXERCISE 16.15. The integral of $\langle P_{xy}(0)P_{xy}(t) \rangle$ is the variance σ_{xy}^2 times the integral of the normalised autocorrelation function, which is τ_{int} by its definition (Section 15.2). With $V = 256/0.02035 = 12580 \text{ \AA}^3$, $k_{\text{B}}T = 0.01163 \text{ eV}$, $\sigma_{xy} = 0.0121/160.2 = 7.55 \times 10^{-5} \text{ eV \AA}^{-3}$ and $\tau_{\text{int}} = 180 \text{ fs}$, the product is $1.11 \text{ eV fs \AA}^{-3}$, and with $1 \text{ eV \AA}^{-3} = 1.602 \times 10^{11} \text{ Pa}$ and $1 \text{ fs} = 10^{-15} \text{ s}$ it is $1.78 \times 10^{-4} \text{ Pa s}$, near the 1.853×10^{-4} of Section 16.6.

EXERCISE 16.16. $\sigma_{\text{NE}} = e^2 \sum_i D_i / V k_{\text{B}}T$ with the sum $32 \times (5.19 + 5.03) \times 10^{-4} \text{ \AA}^2 \text{ fs}^{-1} = 3.27 \times 10^{-2} \text{ \AA}^2 \text{ fs}^{-1} = 3.27 \times 10^{-7} \text{ m}^2 \text{ s}^{-1}$, $V = 2.406 \times 10^{-27} \text{ m}^3$ and $k_{\text{B}}T = 2.053 \times$

10^{-20} J, and $e^2 = 2.567 \times 10^{-38} \text{ C}^2$: 170 S m^{-1} , close to the mean of the runs' own values, 168, since each run's temperature differs.

EXERCISE 16.17. A sixth of the slope of the full MSD is a third of the trace, $(1.16 + 1.02 + 0.00)/3 = 0.73 \times 10^{-4} \text{ Å}^2 \text{ fs}^{-1}$. It is neither the fast motion in the plane, 1.1×10^{-4} , nor the motion across it, for which no crossing was observed. A single number hides this distinction; the one-crossing scale is about 10^{-7} .

EXERCISE 16.18. $|e^{i\theta}|^2 = \cos^2 \theta + \sin^2 \theta = 1$. Multiplying $(\cos A + i \sin A)(\cos B + i \sin B) = \cos A \cos B - \sin A \sin B + i(\sin A \cos B + \cos A \sin B)$ and equating its real and imaginary parts with those of $\cos(A + B) + i \sin(A + B)$ gives both formulae.

EXERCISE 16.19. Writing N for N_s in this solution, $\cos(2\pi mn/N) = \frac{1}{2}[e^{2\pi i mn/N} + e^{-2\pi i mn/N}]$, so

$$X_k = \frac{1}{2} \sum_n e^{2\pi i(m-k)n/N} + \frac{1}{2} \sum_n e^{-2\pi i(m+k)n/N}.$$

By the geometric sum of [Section 16.9](#) the first is $N/2$ when $k = m$ and zero otherwise, and the second is $N/2$ when $m + k$ is a multiple of N , that is $k = N - m$, and zero otherwise.

EXERCISE 16.20. Writing N for N_s , if $W(N/2) = (N/2) \log_2(N/2)$ then $W(N) = N[\log_2 N - 1] + N = N \log_2 N$, and $W(1) = 1 \times 0 = 0$ starts the chain. For $N = 2^{20}$ the ratio $N^2/N \log_2 N = N/20 = 52\,429$.

EXERCISE 16.21. The Nyquist frequency of 35 fs is 14.29 THz, and 18.25 THz folds to $1/\Delta t - \nu = 28.57 - 18.25 = 10.32 \text{ THz}$. It appears where it is for strides below $1/(2 \times 18.25 \text{ THz}) = 27.4 \text{ fs}$.

EXERCISE 16.22. The friction widens each peak to $\gamma/2\pi = 0.159 \text{ THz}$; the window of 10 ps blurs it over about $1/t_c = 0.10 \text{ THz}$. The friction does more, though the two are comparable.

EXERCISE 16.23. $12mD/k_B T = 12 \times 15.999 \times 103.64 \times 5.24 \times 10^{-4}/0.02620 = 398 \text{ fs}$, which is 0.398 per terahertz.

EXERCISE 16.24. $F = \frac{1}{2}kr_h^2 - k_B T \ln r_h$ up to a constant, so $dF/dr_h = kr_h - k_B T/r_h$, zero at $r_h^2 = k_B T/k$, and $d^2F/dr_h^2 = k + k_B T/r_h^2 > 0$, a minimum. At 300 K, $r_h = \sqrt{0.02585/0.9785} = 0.163 \text{ Å}$.

EXERCISE 16.25. The free energy difference is $-k_B T \ln$ of the ratio of the times spent in the two wells. In the two-state approximation, each crossing starts a new, independent stay, so with n crossings each well is visited about $n/2$ times. The stays have a spread

about as large as their mean, as the waits between Poisson events do (Section 16.4), so each well's total time is known to about $1/\sqrt{n/2} = \sqrt{2/n}$ of itself, and their ratio, the two errors combined, to $\sqrt{2}\sqrt{2/n} = 2/\sqrt{n}$. A relative change δ in the ratio changes its logarithm by δ , so the difference is known to about $2k_B T / \sqrt{n}$, and $0.1 k_B T$ needs 400 crossings.

Chapter 17: Molecular dynamics in practice

EXERCISE 17.1. As in Section 17.1, a force of one energy unit per Å on 1 amu must give one Å per t^2 , so $t = \text{Å}\sqrt{\text{amu}/E}$ with E the energy unit. One kilocalorie per mole is $4184/6.022\,140\,76 \times 10^{23} = 6.9477 \times 10^{-21}$ J for one molecule, so

$$\begin{aligned} t &= 1 \times 10^{-10} \text{ m} \sqrt{\frac{1.660\,54 \times 10^{-27} \text{ kg}}{6.9477 \times 10^{-21} \text{ kg m}^2 \text{ s}^{-2}}} && \text{in SI units} \\ &= 1 \times 10^{-10} \times 4.8888 \times 10^{-4} \text{ s} = 48.888 \text{ fs} && \text{as } \sqrt{\text{s}^2/\text{m}^2} = \text{s m}^{-1}. \end{aligned}$$

A step of 2 fs is $2/48.888 = 0.04091$ of these units.

EXERCISE 17.2. ASE's velocity unit is Å per 10.1805 fs, so 0.01 Å fs^{-1} is $0.01 \times 10.1805 = 0.1018$ of them. LAMMPS's metal unit is Å ps⁻¹, so the number is $0.01 \times 1000 = 10$.

EXERCISE 17.3. LAMMPS's damping parameter is the time $1/\gamma = 2$ ps. ASE's friction is γ in reciprocal ASE time units, $0.0005 \text{ fs}^{-1} \times 10.1805 \text{ fs} = 0.00509$.

EXERCISE 17.4. $3N/(3N-3) = 768/765 = 1.00392$, and the period differs by its square root, 1.00196: a fifth of a per cent, far below anything a run could notice.

EXERCISE 17.5. Let the points be 0 and δx , $\tilde{y}(x)$ the straight line through y 's values there, and $\Delta(x) = y(x) - \tilde{y}(x)$ its miss, zero at both points. Fix a point x_0 between them. The parabola $\Delta(x_0) x(x - \delta x) / [x_0(x_0 - \delta x)]$ is zero at both points and equals $\Delta(x_0)$ at x_0 , so

$$g(x) = \Delta(x) - \Delta(x_0) \frac{x(x - \delta x)}{x_0(x_0 - \delta x)}$$

is zero at 0, at x_0 and at δx . A smooth function that is zero at two points has a zero slope somewhere between them (Rolle's theorem: to return to zero it must turn). Between 0 and x_0 , and between x_0 and δx , this gives two points where $g' = 0$, and between those

a point x_1 where $g'' = 0$. Since $\tilde{y}'' = 0$ for a straight line and $x(x - \delta x) = x^2 - \delta x x$ has the second derivative 2,

$$0 = g''(x_1) = y''(x_1) - \frac{2\Delta(x_0)}{x_0(x_0 - \delta x)}$$

$$\Delta(x_0) = \frac{1}{2}y''(x_1) x_0(x_0 - \delta x) \quad \text{solving for } \Delta(x_0).$$

$|x_0(x_0 - \delta x)| = x_0(\delta x - x_0)$ is a parabola in x_0 , zero at both ends and so largest midway, $\delta x^2/4$ at $x_0 = \delta x/2$; hence $|\Delta| \leq \delta x^2 \max|y''|/8$.

EXERCISE 17.6. The miss goes as the square of the spacing, so the spacing must fall by $\sqrt{7.6 \times 10^{-7}/10^{-9}} = 27.6$ times, and the 20 000 intervals become $20\,000 \times 27.6 \approx 5.5 \times 10^5$: about 5.5×10^5 entries.

EXERCISE 17.7. The second and third terms are whole multiples of 1024 and $i_x + 512$ lies between 0 and 1023, so on division by 1024 the remainder is $i_x + 512$ and the quotient $(i_y + 512) + 1024(i_z + 512)$; the same argument on the quotient gives the remainder $i_y + 512$ and the quotient $i_z + 512$. For 540 539 393: the remainder is 513, so $i_x = 1$; the quotient 527 870 has the remainder 510, so $i_y = -2$, and the quotient 515, so $i_z = 3$.

EXERCISE 17.8. A call costs $0.25 \times 1000/126 = 1.98$ s for 1000 atoms, and a nanosecond is 10^6 steps: 1.98×10^6 s, about 23 days. The measured call at 1008 atoms, 2.84 s (Figure 17.2a), gives about 33 days, since on the GPU the cost per atom grows by 40% from 252 to 1008 atoms. Runs of this length are why the cost of a learned potential matters, and why The planned Part VI turns to learned propagators.

EXERCISE 17.9. $k_B T$ at 300 K is 25.9 meV; per atom, 5.2 meV is a fifth of it and 41.2 meV 1.6 times it. But the sheets move together: stretching 64 carbon atoms costs $64 \times 5.2 \text{ meV} = 0.33 \text{ eV}$, 13 times $k_B T$, so even without D3 the sheets do not come apart. What fails is the spacing, 16.5% too wide under a barostat, and its stiffness: in so shallow and broad a well the cell's height swings much further than with D3, and lithium sits in a gallery too wide.

EXERCISE 17.10. $0.0067(\delta t/1 \text{ fs})^2 = 0.05$ at $\delta t = \sqrt{0.05/0.0067} = 2.73 \text{ fs}$. At 3 fs the estimate is $0.0067 \times 9 = 0.060$, already over the limit, and the run's 0.082 is more again: the spread grows faster than δt^2 as the step nears the stability limit of the fastest vibrations (Section 9.6), the point where it also drifts. Part of the excess is that drift, which a standard deviation counts as spread: with a straight line through the energy removed, the run spreads by 0.074, still above the estimate.

EXERCISE 17.11. 10^4 steps of 192 s on four cores are $10^4 \times 192 \times 4/3600 = 2133$ core-hours; on 192 cores, divided perfectly, 11.1 h.

EXERCISE 17.12. The two vectors are

$$2\mathbf{a} + \mathbf{b} = a\left(\frac{3}{2}, \frac{\sqrt{3}}{2}\right), \quad -\mathbf{a} + \mathbf{b} = a\left(-\frac{3}{2}, \frac{\sqrt{3}}{2}\right),$$

each of length $a\sqrt{9/4 + 3/4} = \sqrt{3}a$. Their dot product is $a^2(-\frac{9}{4} + \frac{3}{4}) = -\frac{3}{2}a^2$, and dividing by the product of their lengths, $3a^2$, gives the cosine of the angle between them, $-\frac{1}{2}$: 120° . The cell they span has the area $3a^2 \sin 120^\circ = \frac{3\sqrt{3}}{2}a^2$ and the sheet's own cell $a^2 \sin 120^\circ = \frac{\sqrt{3}}{2}a^2$, a ratio of 3, which is also the determinant $2 \times 1 - 1 \times (-1)$ of the vectors' coefficients in $\mathbf{a}$ and $\mathbf{b}$ (Section 7.4); the sheet's cell holds two carbon atoms, so this one holds six.

EXERCISE 17.13. The mean and its error are

$$\frac{-1.00 - 1.11 + 1.68}{3} = -0.14 \text{ GPa}, \quad \frac{\sqrt{0.45^2 + 0.36^2 + 0.10^2}}{3} = 0.20 \text{ GPa},$$

zero within its error. An isotropic barostat compares only the mean with the target, so a solid strained in opposite senses along different directions satisfies it.

EXERCISE 17.14. A fifth needs 25 hops, which at 2 per picosecond take 12.5 ps. At a rate ν the chance of no hop in 20 ps is $e^{-\nu \times 20 \text{ ps}}$, below 0.05 once ν exceeds $-\ln 0.05/20 \text{ ps} = 0.150 \text{ ps}^{-1}$, so the rate is below this with 95% confidence. 25 hops at that rate need 167 ps, the shortest run that could resolve it, since the true rate is probably lower.

EXERCISE 17.15. With $\ell_1 = \ell(1, 0)$, the other two are turned by 120° and 240° : $\ell_2 = \ell(-\frac{1}{2}, \frac{\sqrt{3}}{2})$ and $\ell_3 = \ell(-\frac{1}{2}, -\frac{\sqrt{3}}{2})$, and adding the three gives $\ell(1 - \frac{1}{2} - \frac{1}{2}, \frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{2}) = \mathbf{0}$. A hop from either family therefore has the mean $\pm(\ell_1 + \ell_2 + \ell_3)/3 = \mathbf{0}$, whatever the hops before it, so the mean of its product with any earlier hop is zero: after n hops the cross terms of the squared displacement vanish and its mean is $n\ell^2$ (Section 16.5). With $n = k_{\text{h}}t$, setting $k_{\text{h}}t\ell^2 = 2dDt$ with $d = 2$ gives $D = k_{\text{h}}\ell^2/4$. For lithium, $\ell = 2.4648/\sqrt{3} = 1.4231 \text{ \AA}$.

EXERCISE 17.16. $\langle z^2 \rangle = 1$ by definition of the variance, and $\text{var}(z^2) = \langle z^4 \rangle - \langle z^2 \rangle^2 = 3 - 1 = 2$. The mean of n independent such squares has the variance $2/n$ (Section 15.1), so it scatters about 1 by $\sqrt{2/n}$. In a canonical run each freedom contributes $mv^2/k_{\text{B}}T_0 = z^2$ (Section 12.5), so T/T_0 is the mean of N_{f} such squares and scatters by $\sqrt{2/N_{\text{f}}}$.

EXERCISE 17.17. The centre moves by $\frac{1}{N} \sum_i \Delta \mathbf{r}_i$. Its square is $\frac{1}{N^2} \sum_i \sum_j \Delta \mathbf{r}_i \cdot \Delta \mathbf{r}_j$, every product of one atom's move with another's, and the N terms with $i = j$ are the squares Δr_i^2 . For $i \neq j$ independent atoms give $\langle \Delta \mathbf{r}_i \cdot \Delta \mathbf{r}_j \rangle = \langle \Delta \mathbf{r}_i \rangle \cdot \langle \Delta \mathbf{r}_j \rangle = 0$, since neither has a mean displacement, leaving $\frac{1}{N^2} N \langle \Delta r^2 \rangle = \langle \Delta r^2 \rangle / N$. In a run that keeps the total momentum the centre does not move at all; under Langevin dynamics it moves further,

since the forces between the atoms cancel in its motion and only the friction slows it (Section 13.8).

EXERCISE 17.18. A wrapped position fixes the atom only up to a whole number of cell vectors. A move of $\mathbf{d}$ and one of $\mathbf{d}$ less a cell vector leave the same wrapped positions, and the shorter of the two is taken; if a true move can exceed half a width, the shorter is not always the true one, and the path is lost. The first check therefore asks that the record be continuous already: a wrapped record fails it and must be unwrapped, which is safe only if its frames are close enough that no true move nears half a width; a record whose atoms jump further, from frames too far apart or rows out of order, has lost its paths.

EXERCISE 17.19. To first order, $1/m_2^2 = (1 + \epsilon_2)^{-2} \approx 1 - 2\epsilon_2$, so

$$\begin{aligned} \frac{m_4}{m_2^2} - 3 &\approx (3 + \epsilon_4)(1 - 2\epsilon_2) - 3 && \text{the first order of } (1 + \epsilon_2)^{-2} \\ &\approx \epsilon_4 - 6\epsilon_2 && \text{dropping } \epsilon_4\epsilon_2, \text{ of second order.} \end{aligned}$$

Its variance is $\text{var}(\epsilon_4) - 12\text{cov}(\epsilon_2, \epsilon_4) + 36\text{var}(\epsilon_2)$, and each is that of one number over n (Section 15.1): $\text{var}(z^4) = \langle z^8 \rangle - \langle z^4 \rangle^2 = 105 - 9 = 96$, $\text{cov}(z^2, z^4) = \langle z^6 \rangle - \langle z^2 \rangle \langle z^4 \rangle = 15 - 3 = 12$ and $\text{var}(z^2) = 2$. The variance is therefore $(96 - 144 + 72)/n = 24/n$, and the scatter $\sqrt{24/n}$. For $n = 768$ the limit is $3\sqrt{24/768} = 0.530$. Velocities spread evenly, with -1.2 , lie far outside it and fail.

EXERCISE 17.20. Check the potential on the new system against a converged reference, on frames like those of the run; build and relax the starting structure and check its closest approach, largest force and total momentum; choose the step by runs at fixed energy; bring the system to its temperature and pressure and find the start; measure D with its error from blocks in a pilot run, and lengthen the run by the square of the ratio of that error to the 5% asked (Section 15.6); run at fixed energy or under CSVr with a long time constant; and plan two box sizes for the correction of Section 15.7.

Chapter 18: Learning as optimisation

EXERCISE 18.1. With $S_{xx} = \sum_i x_i^2$, $S_{xF} = \sum_i x_i F_i$ and $S_{FF} = \sum_i F_i^2$, the expanded loss is $L = S_{FF} - 2kS_{xF} + k^2S_{xx}$, and at $\hat{k} = S_{xF}/S_{xx}$

$$L(\hat{k}) = S_{FF} - 2\frac{S_{xF}^2}{S_{xx}} + \frac{S_{xF}^2}{S_{xx}} = S_{FF} - \frac{S_{xF}^2}{S_{xx}}.$$

EXERCISE 18.2. The stretches in metres are 0.01 to 0.10, and the sum of their squares is $385 \times 10^{-4} = 0.0385 \text{ m}^2$, so the error is $0.05/\sqrt{0.0385} = 0.255 \text{ N m}^{-1}$.

EXERCISE 18.3. The columns of X are $\mathbf{1}$ and x_i , so

$$X^T X = \begin{pmatrix} n & \sum_i x_i \\ \sum_i x_i & \sum_i x_i^2 \end{pmatrix}, \quad X^T \mathbf{y} = \begin{pmatrix} \sum_i y_i \\ \sum_i x_i y_i \end{pmatrix}.$$

The first equation, $nc + s \sum_i x_i = \sum_i y_i$, divided by n gives $c = \bar{y} - s\bar{x}$. Substituting into the second, $(\bar{y} - s\bar{x}) \sum_i x_i + s \sum_i x_i^2 = \sum_i x_i y_i$, and with $\sum_i x_i = n\bar{x}$,

$$s \left(\sum_i x_i^2 - n\bar{x}^2 \right) = \sum_i x_i y_i - n\bar{x}\bar{y}.$$

Expanding $\sum_i (x_i - \bar{x})^2 = \sum_i x_i^2 - 2\bar{x} \sum_i x_i + n\bar{x}^2 = \sum_i x_i^2 - n\bar{x}^2$, and likewise $\sum_i (x_i - \bar{x})(y_i - \bar{y}) = \sum_i x_i y_i - n\bar{x}\bar{y}$, gives the slope of [Section 15.5](#).

EXERCISE 18.4. With $U_M = D(1 - e^{-ax})^2$, the chain rule gives $F = -2Da e^{-ax}(1 - e^{-ax}) = -2Da(e^{-ax} - e^{-2ax})$, and

$$-\frac{dF}{dx} = 2Da(-ae^{-ax} + 2ae^{-2ax}) = 2Da^2(2e^{-2ax} - e^{-ax}).$$

To first order $e^{-2ax} \approx 1 - 2ax$ and $e^{-ax} \approx 1 - ax$, so the bracket is $2 - 4ax - 1 + ax = 1 - 3ax$. At $x = \pm 0.0345 \text{ \AA}$, $3ax = 0.207$: the stiffness is 21% below its value at the bottom on the outer side and 21% above it on the inner.

EXERCISE 18.5. If X has an inverse, so does X^T , and multiplying the normal equations $X^T X \mathbf{w} = X^T \mathbf{y}$ by $(X^T)^{-1}$ on the left gives $X \mathbf{w} = \mathbf{y}$.

EXERCISE 18.6. As in [Section 18.3](#), the component along $\mathbf{v}_k$ is $c_k = \mathbf{v}_k \cdot X^T \mathbf{y} / (d_k + \lambda)$ with the penalty and $c_k^0 = \mathbf{v}_k \cdot X^T \mathbf{y} / d_k$ without it, so $c_k = c_k^0 d_k / (d_k + \lambda)$.

EXERCISE 18.7. $\sigma_w = \sigma / \sqrt{\lambda} = 0.2 / \sqrt{10^{-3}} = 6.32 \text{ meV}$; and $\lambda = \sigma^2 / \sigma_w^2 = (0.2/1)^2 = 0.04$.

EXERCISE 18.8. The mean is $(c/\sigma_b^2) b = (1.5/1) \times 2 = 3$, and the variance $\sigma_a^2 - c^2/\sigma_b^2 = 4 - 2.25 = 1.75$.

EXERCISE 18.9. With one reading, K is the number $h^2 + \sigma^2$ and $\mathbf{k}_* = h^2 \mathbf{e}^{-(x-x_1)^2/2\ell^2}$, so [Equation \(18.5\)](#) gives the mean $\mathbf{k}_* y_1 / K$, the expression asked for, and the variance

$$h^2 - \frac{h^4 \mathbf{e}^{-(x-x_1)^2/\ell^2}}{h^2 + \sigma^2},$$

which at x_1 is $h^2\sigma^2/(h^2 + \sigma^2)$, below both h^2 and σ^2 , and far away returns to h^2 .

EXERCISE 18.10. The rate must be below the stability boundary $2/200 = 0.01$; the best fixed rate $2/(2 + 200) = 0.0099$; with $\kappa = 100$ the factor is $99/101 = 0.980$; and the loss falls as its square, so a millionfold takes $\ln 10^6 / (-2 \ln 0.980) = 345$ steps.

EXERCISE 18.11. The factor is $(\kappa - 1)/(\kappa + 1)$ and the loss falls as its square, so t steps bring the factor ε when

$$\begin{aligned} t &= \frac{\ln(1/\varepsilon)}{2 \ln[(\kappa + 1)/(\kappa - 1)]} && \text{taking logarithms} \\ &= \frac{\ln(1/\varepsilon)}{2 \ln[1 + 2/(\kappa - 1)]} \approx \frac{(\kappa - 1) \ln(1/\varepsilon)}{4} && \text{writing } (\kappa + 1)/(\kappa - 1) = 1 + 2/(\kappa - 1), \end{aligned}$$

by $\ln(1 + y) \approx y$ for the small $y = 2/(\kappa - 1)$; for large κ this is $\kappa \ln(1/\varepsilon)/4$.

EXERCISE 18.12. $\sigma_g/\sqrt{b} = 0.05$ gives $b = (0.585/0.05)^2 = 137$.

EXERCISE 18.13. With no gradient $\mathbf{v}$ is multiplied by μ each step, so after t steps by $0.9^t = e^{-1}$ when $t = -1/\ln 0.9 = 9.5$ steps. The friction is $1 - \mu = 0.1$ per step.

EXERCISE 18.14. $\sqrt{\kappa} = 10$, so $\sqrt{\mu} = 9/11 = 0.818$, $\mu = 0.669$ and $\eta = 4/(10 + 1)^2 = 0.0331$. Using the asymptotic factors, shrinking by 10^{-4} takes approximately $\ln 10^{-4} / \ln 0.818 = 46$ steps with momentum and $\ln 10^{-4} / \ln 0.980 = 461$ without.

EXERCISE 18.15. $m = (1 - 0.9^{10})g = 0.651g$ and $s = (1 - 0.999^{10})g^2 = 0.00996g^2$, so that $\sqrt{s}$ is $0.0998|g|$. The corrected step is $\eta g/|g|$; uncorrected it would be $\eta \times 0.651/0.0998 = 6.5\eta$ in size, six and a half times too large, since s starts far more biased than m .

EXERCISE 18.16. Each m is a weighted sum of the gradients, so it becomes cm ; each s a weighted sum of their squares, so c^2s , and $\sqrt{c^2s} = c\sqrt{s}$ for $c > 0$. The corrections divide both by the same numbers, and the step $\eta \hat{m}/\sqrt{\hat{s}}$ is unchanged.

EXERCISE 18.17. 2 ps is 200 frames, about $200/109 = 1.8$ independent ones; 20 ps, about 18.

EXERCISE 18.18. The frames on either side differ from it by 10 fs of motion, much less than the memory of the run, so their features and energies are nearly its own. A model that fits its training frames and interpolates smoothly between their features can therefore predict this validation frame with a similarly small miss. A new run need not supply such close neighbours. The random split then tests local interpolation more strongly than transfer to new configurations.

EXERCISE 18.19. Split by whole runs, never by frames: for instance, if there are five runs at each of the ten temperatures, put three at each temperature in training, one in validation and one in test. This gives 30, 10 and 10 runs, with every temperature represented in each set. No two sets may share a run, and the scaling of the features, the choice of settings and the early stop come from the training and validation sets alone; the test runs are used once, at the end.

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